

General Chemistry

Part I

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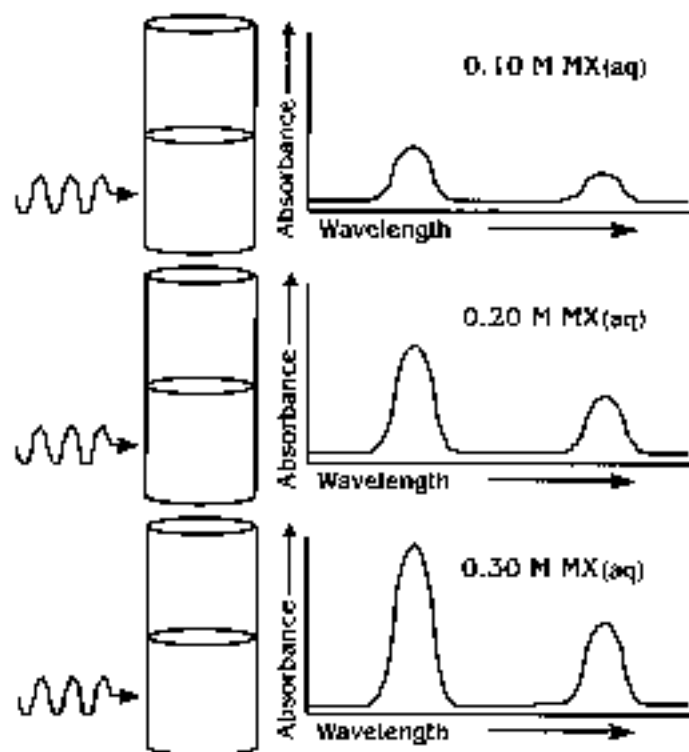
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Section I

Stoichiometry

by Todd Bennett



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Stoichiometry Section Goals



Know how to convert one kind of concentration unit into another.

The concentration of a solution can be measured in terms of molarity, molality, and density. You must know the definitions of each unit and how they differ from one another. Although the test does not feature a great deal of math, you should have an idea of how to convert between units.



Understand the difference between empirical and molecular formulas.

Know the difference between the molecular formula (actual ratio of atoms in a molecule) and the empirical formula (simplest whole number ratio of the atoms in a molecule). Be familiar with the experiments and information needed to determine both of the formulas.



Know the effect of standard conditions.

Standard conditions are defined as 1 atm. and 298 K for thermodynamics, but STP (standard temperature and pressure) is defined as 1 atm. and 273 K. Many calculations of gas volume use the ideal gas assumption that at STP, one mole of gas occupies 22.4 liters.



Understand dilution and its effect on concentration of a solute.

Dilution involves a reduction in the concentration of a solute in solution by the addition of solvent to the mixture. The addition of solvent therefore dilutes the concentration, but does not change the moles of solute. The equation that you must recall is based on the constant number of solute moles: $M_{\text{initial}} \cdot V_{\text{initial}} = M_{\text{final}} \cdot V_{\text{final}}$.



Recognize standard reactions from general chemistry.

The most commonly recurring reactions in general chemistry that you are expected to know include combustion, single replacement, double displacement, and proton transfer, to name just a few. You must recognize these reactions and have a basic understanding of them.



Recognize the limiting reagent in a reaction and know its effect on the reaction.

The limiting reagent dictates the amount of product that can be formed and consequently the percent yield for a reaction. Using only starting values and the stoichiometric equation, you must be able to determine which reactant is the limiting reagent in the reaction.



Understand the stoichiometric ratios in combustion reactions.

In the combustion of both hydrocarbons and carbohydrates, there is a consistent relation between the number of oxygen molecules on the reactant side, and the number of water and carbon dioxide molecules that form on the product side. Know each reaction so that you may easily balance the coefficients.

Stoichiometry

The perfect spot to start any review of general chemistry is the basics, which traditionally include stoichiometry and chemical equations. The most fundamental perspective of a chemical reaction, where bonds are broken so that new bonds can be formed, is at the atomic and molecular levels. Due to the minute size of atoms, we can never actually view a chemical reaction (so states Heisenberg's uncertainty principle). We must therefore rely upon developing models that can account for changes in all of the atoms and molecules involved in a chemical reaction or physical process. At the molecular level, we consider molecules. At the macroscopic level, we consider moles. Stoichiometry allows us to convert one into the other and to shift between these two perspectives. The number of molecules is converted into the number of moles using Avogadro's number (6.022×10^{23}). The concept of a mole is based upon the amount of carbon-12 that is contained in exactly 12.0 grams of carbon, a quantity determined by knowing the volume of a 12.0-g carbon sample, the type of molecular packing in it, and the dimensions of the carbon atom. This task of quantifying atoms in a mole is similar to guessing the number of peas that are contained in an aquarium. It is important that you utilize the mole concept to understand, and later to balance and manipulate, chemical equations.

In the stoichiometry section, we focus on those skills needed to solve ratio questions. Stoichiometry is most commonly thought of as the mathematical portion of general chemistry. The MCAT, however, has relatively few calculations. It is a conceptual test, emphasizing logical thought process rather than calculations. For some of you, this is great news. But before celebrating too much, consider where the mathematical aspects of general chemistry fit into a conceptual exam. The MCAT does involve some math, but it is not too complicated. Math-related calculations required for MCAT questions involve making approximations, determining ratios, setting up calculations, and estimating the effect of errors on results. The initial problems presented in this section involve slightly more calculations than you should expect to see on the MCAT. Some of them may look familiar to you from your general chemistry courses and should stimulate your recall. As the section proceeds, less emphasis is placed on calculating and estimating, and more emphasis is placed on the art of quickly determining ratios and approximating values.

The focus of the stoichiometry section is problem-solving, with special attention to the idiosyncrasies of each type of problem. Definitions of important terms are presented with sample questions and their solutions. Answer solutions discuss test strategy and the information needed to obtain the correct answer. Each problem in the stoichiometry section represents what we might call the "book-keeping" of reactions in general chemistry, and it offers an ideal opportunity to begin work on fast math skills as well. As you do each of the questions, learn the definitions and develop an approach that works well for you. You may want to consider multiple pathways to arrive at the correct answer. It is important that you be able to solve questions in several different ways and to get into the mindset of the test writers. As you read a passage, think about the questions that could be asked about it. If a passage gives values for various masses and volumes, there will probably be a question about density. If it gives values for moles and solution volume, there will probably be questions on concentration and dilution. Use your intuition and common sense as much as you can, and make every effort to develop your test-taking logic.

Unit Conversion

Dimensional Analysis

Dimensional analysis is a mathematical conversion from one set of units into another. It involves multiplying a given value by a conversion factor or a series of conversion factors until the value is finally expressed in the desired units. Converting from one set of units to another is a critical skill needed to eliminate incorrect answer choices in the physical sciences section. Be systematic when converting between units. The standard measurements that can be expressed in a variety of units are distance (1 m = 1.094 yd, 2.54 cm = 1.00 in, and 1.609 km = 1.00 mile), mass (1.00 kg = 2.205 lb and 453.6 g = 1.00 lb), volume (3.79 L = 1.00 gal and 1.00 L = 1.06 qt), and time (3600 s = 1.00 hr). Always convert units as they appear in a problem into the units indicated in the answer choices (the so-called "target units").

Example 1.1

Sprinters can run 100 meters in just under 10 seconds. At what average speed in miles per hour must a runner travel to cover 100 meters in 10.0 seconds?

- A. 3.7 miles/hour
- B. 11.2 miles/hour
- C. 22.4 miles/hour
- D. 36.0 miles/hour

Solution

The first task is to determine the given units and the target units. From there, convert the given units into the target units. We are given 100 meters in 10 seconds, but the answer choices are expressed in miles per hour. Use the correct conversion factors, as follows:

$$\text{Given: } \frac{100 \text{ m}}{10.0 \text{ s}} \times ? \times ? \times ? = \frac{\text{miles}}{\text{hour}}$$

$$\text{Conversion of distance: } \frac{100 \text{ m}}{10.0 \text{ s}} \times \frac{\text{miles}}{\text{m}} = \frac{\text{miles}}{\text{s}}$$

$$\text{Conversion of time: } \frac{100 \text{ m}}{10.0 \text{ s}} \times \frac{\text{s}}{\text{hr}} = \frac{\text{m}}{\text{hour}}$$

$$\text{Overall: } \frac{100 \text{ m}}{10.0 \text{ s}} \times \frac{1 \text{ km}}{1000 \text{ m}} \times \frac{1 \text{ mile}}{1.609 \text{ km}} \times \frac{3600 \text{ s}}{1.00 \text{ hr}} = \frac{100 \times 3600}{10 \times 1000 \times 1.609} \frac{\text{miles}}{\text{hour}}$$

$$\frac{100 \times 3600}{10 \times 1000 \times 1.609} = \frac{3600}{10 \times 10 \times 1.609} = \frac{36}{1.609} \frac{\text{miles}}{\text{hour}}$$

$$\frac{36}{2} < \frac{36}{1.609} < \frac{36}{1}, \text{ where } \frac{36}{2} = 18 \text{ and } \frac{36}{1} = 36. \text{ So } 18 < \frac{36}{1.609} < 36$$

Only choice C falls within the range of 18 to 36.

A frequent task in chemistry is the conversion between various types of temperature units, volume units, pressure units, and concentration units. Chemists, like most scientists, employ the MKS system, so the conversion from conventional units less commonly used in science to MKS units is routine. Example 1.2 demonstrates the interconversion between the conventional Fahrenheit unit and the scientific Celsius unit of temperature.

Example 1.2

At what temperature is the numerical value the same, whether the units are in Celsius or Fahrenheit?

- A. 32°
- B. 0°
- C. -40°
- D. -273°

Solution

The formulas for conversion between Celsius and Fahrenheit are as follows:

$$T_F = \frac{9}{5} T_C + 32 \quad \therefore T_C = \frac{5}{9} (T_F - 32)$$

Answering the question requires setting $T_F = T_C$.

$$T_F = \frac{9}{5} T_C + 32 \text{ becomes: } T = \frac{9}{5} T + 32, \text{ so } T = 1.8 T + 32$$

$$T = 1.8 T + 32 \Rightarrow -0.8 T = 32 \therefore T = -40^\circ$$

Density Determination

The density of a material or solution is the mass of the sample divided by the volume of the sample. Density is a measured quantity, determined experimentally. Understand the techniques used to measure density. The term *specific gravity* refers to the density of a material relative to the density of water, and may be used in a question in lieu of density. For our purposes, specific gravity means the same thing as density, but it has no expressed units. Determining density is a typical example of dimensional analysis.

Example 1.3

Exactly 10.07 mL of an unknown non-volatile liquid is poured into an empty 25.41-gram open flask. The combined mass of the unknown non-volatile liquid and the flask is 34.12 grams. What is the density of the unknown liquid?

- | | |
|---|--|
| A. $\frac{34.13 \text{ g}}{10.07 \text{ mL}}$ | B. $\frac{8.71 \text{ g}}{10.07 \text{ mL}}$ |
| C. $\frac{10.07 \text{ g}}{8.71 \text{ mL}}$ | D. $\frac{8.71 \text{ mL}}{10.07 \text{ g}}$ |

Solution

The density of the liquid is found by dividing the mass of the liquid by the volume of the liquid. This results in units of grams per milliliter, which eliminates choice D. The volume of the liquid is 10.07 mL, so 10.07 should be in the denominator. This eliminates choice C. The mass of the liquid is the difference between the final mass of the flask and liquid combined, and the mass of the flask (34.12 - 25.41), which is equal to 8.71 grams. This means that the numerator should be 8.71. The correct answer is choice B. Incidentally, the question did not state the reason for using a non-volatile liquid. The liquid must be non-volatile, to prevent any loss due to evaporation from the open flask. In the event the liquid evaporates away, then the mass you determine is too small, due to the loss of vapor molecules.

Density questions may on occasion involve a more intricate conversion of units. For any density question, keep in mind that the target units are mass solution divided by volume solution. The mass percent of a solute is mass solute divided by mass solution. The product of the density and mass percent is mass solute divided by volume solution. Converting the mass of solute into moles of solute yields molarity. Example 1.4 shows this.

Example 1.4

What is the molarity of a 3% NaCl solution with a density of 1.05 grams/mL?

- A. 0.497 M NaCl
- B. 0.504 M NaCl
- C. 0.539 M NaCl
- D. 0.724 M NaCl

Solution

The first step is to determine the units you are looking for, which in this example is moles solute per liter solution. You must find both moles solute and liters solution. The density of the solution is 1.05 grams/mL which means that one liter of the solution weighs 1050 g. Three percent (3%) of the solution is sodium chloride, so the mass of sodium chloride is 0.03×1050 g. This is the same as 3% of 1000 g + 3% of 50 g, which is 30 g + 1.5 g = 31.5 g of NaCl per liter solution. The grams of sodium chloride are converted to moles by dividing by the molecular weight of NaCl (58.6 grams/mole). The unit factor method is shown below:

$$\begin{aligned} & 1.05 \frac{\text{g solution}}{\text{mL solution}} \times \frac{1000 \text{ mL}}{\text{L}} \times \frac{3 \text{ g NaCl}}{100 \text{ g solution}} \times \frac{1 \text{ mole NaCl}}{58.6 \text{ g NaCl}} \\ &= \frac{1.05 \times 1000 \times 3 \text{ moles NaCl}}{100 \times 58.6 \text{ L solution}} = \frac{31.5 \times 10 \text{ moles NaCl}}{58.6 \text{ L solution}} \end{aligned}$$

On the MCAT, you will not have time to solve for values precisely, so you must make an approximation. Select the answer that is closest to that approximation.

$$\frac{31.5}{58.6} > \frac{30}{60} = \frac{1}{2}, \text{ so the value is greater than } 0.500 \text{ M}$$

$$\frac{31.5}{58.6} < \frac{33}{60} = \frac{11}{20} = \frac{55}{100}, \text{ so the value is less than } 0.550 \text{ M}$$

The value falls between 0.500 M and 0.550 M, so choices A and D are eliminated. Next you must choose between 0.504 M and 0.539 M. The value is not close enough to 0.504 M, so you should choose C, and be a wise student! Wise students are a good thing. Some questions on the MCAT may present mathematical set-ups, without solving for an exact number.

The physical sciences section of the exam incorporates physics and general chemistry, so from the beginning of your review, make a conscious effort to consider physics when working on general chemistry and to consider general chemistry when working on physics. Determining density is a problem common to both disciplines. Example 1.5 shows an approach to the concept of density that is more typical of what is found in a physics problem.

Example 1.5

What can be concluded about the density of a metal object which, when placed in a beaker of water at room temperature, sinks to the bottom?

- A. The density of the metal is less than the density of either water or ice.
- B. The density of the metal is less than the density of water, but greater than the density of ice.
- C. The density of the metal is greater than the density of water, but less than the density of ice.
- D. The density of the metal is greater than the density of either water or ice.

Solution

When an object floats in a liquid medium, its density is less than that of the medium surrounding it. The fact that it floats means the buoyant force pushing upward against it ($\rho_{\text{medium}} V_{\text{object}} g$) is greater than gravitation force pushing downward (weight = $mg = \rho_{\text{object}} V_{\text{object}} g$). Thus, an object floats when $\rho_{\text{medium}} > \rho_{\text{object}}$. Because the metal object sinks in water, it must be denser than water. Ice floats in water, meaning that ice is less dense than water and thus less dense than the metal object. The density of the metal must be greater than the density of either water or ice. The correct answer is therefore choice D.

Typical Conversions

In chemistry, conversions between products and reactants are common, so the mole concept is frequently employed. The mole concept is pertinent in the interconversion between moles and mass, using either atomic mass (for elements) or molecular mass (for compounds). These calculations involve using the unit factor method (also known as dimensional analysis.)

Example 1.6

How many moles of NaHCO_3 are contained in 33.6 grams NaHCO_3 ?

- A. 0.20 moles NaHCO_3
- B. 0.40 moles NaHCO_3
- C. 0.50 moles NaHCO_3
- D. 0.60 moles NaHCO_3

Solution

The first step in determining the number of moles is to determine the molecular mass of NaHCO_3 . The mass is $23 + 1 + 12 + 48 = 84$ grams. The number of moles of NaHCO_3 is found by dividing 33.6 by 84, which is less than 0.50. This eliminates choices C and D. The number is greater than 0.25 (21 over 84) and thus greater than 0.20, so choice A is eliminated. The only value left is choice B, 0.40 moles.

Beyond determining the moles from grams for the same compound are questions where the moles of products are determined from the grams of reactants. These questions require converting from grams of a given substance to moles of the given substance, and then expressing the quantity of a final substance in terms of moles, grams, or liters. By balancing the reaction, the mass of a selected product that is formed in the reaction can be calculated based on the mass of a selected reactant (which must be the limiting reagent). Examples 1.7 and 1.8 involve determining moles, mass, and volume from the given values.

Example 1.7

Based on the following reaction, how many grams of water would form from 0.33 moles $\text{C}_4\text{H}_{10}\text{O}$ reacting with an unlimited amount of oxygen gas?



- A. 18.00 grams
- B. 24.00 grams
- C. 30.00 grams
- D. 36.00 grams

Solution

With an excess of oxygen, the limiting reagent in this reaction is $\text{C}_4\text{H}_{10}\text{O}$. The amount of water formed is determined by the 0.33 moles of $\text{C}_4\text{H}_{10}\text{O}$ reactant. Using the balanced equation, the ratio of H_2O to moles $\text{C}_4\text{H}_{10}\text{O}$ is 5 : 1, so 1.667 moles of water are formed. At 18 grams per mole, this means that fewer than 36 grams but more than 27 grams are formed. This makes choice C the best answer.

$$0.33 \text{ moles } \text{C}_4\text{H}_{10}\text{O} \times \frac{5 \text{ moles } \text{H}_2\text{O}}{1 \text{ mole } \text{C}_4\text{H}_{10}\text{O}} \times \frac{18 \text{ g } \text{H}_2\text{O}}{1 \text{ mole } \text{H}_2\text{O}} = 1 \times 5 \times 18 \text{ g } \text{H}_2\text{O} = 30 \text{ g } \text{H}_2\text{O}$$

Example 1.8

How many liters of $\text{CO}_2(\text{g})$ result from the complete decomposition of 10.0 grams of $\text{CaCO}_3(\text{s})$ to carbon dioxide and calcium oxide at STP?



- A. 1.12 liters
- B. 2.24 liters
- C. 3.36 liters
- D. 4.48 liters

Solution

You are asked to determine the amount of product from a known quantity of reactant. The first step in problems of this type is to make sure the reaction is balanced. In this case, it is already balanced. The mole ratio of the two compounds is 1 : 1. The required conversion involves changing from mass reactant, to moles reactant, to moles product, and finally to volume product. This is one variation of unit conversion via mole ratio calculation. In addition, there is the "g · m · m · g" conversion and the "v · m · m · g" conversion. You need three steps to go from grams reactant to the target (liters product). Units are important here. The units for the mass of reactant is grams. You need to multiply mass by moles and divide by grams. This is the same as dividing by the MW. The second step is to read the mole ratio from the balanced equation. In this reaction, the mole ratio is 1 : 1 (the units of both numerator and denominator are moles). The third and final step is to convert from moles product into liters product (i.e., multiply by liters and divide by moles.) This is done by multiplying by the molar volume of the product gas, which at STP (standard temperature and pressure) is 22.4 liters.

$$\begin{aligned} 10 \text{ grams } \text{CaCO}_3 &\times \frac{1 \text{ mole } \text{CaCO}_3}{100 \text{ grams } \text{CaCO}_3} \times \frac{1 \text{ mole } \text{CO}_2}{1 \text{ mole } \text{CaCO}_3} \times \frac{22.4 \text{ liters } \text{CO}_2}{1 \text{ mole } \text{CO}_2} \\ &= \frac{10 \times 22.4}{100} = 2.24 \text{ liters } \text{CO}_2 \end{aligned}$$

Elemental Analysis

Elemental analysis determines the atomic composition of an unknown molecule. It is based on the idea that all molecules of the same substance combine atoms of that substance in the same way. In other words, water always has two hydrogen atoms and one oxygen atom. Because of this feature of structural uniformity, it is possible to determine the atomic composition of any molecule. The fundamental process of elemental analysis involves oxidizing an unknown completely and collecting the products. The amount of each element that was present in the unknown compound can be determined from the amount of oxidized product. These mass values can be converted to mass percent and mole ratio values. In the determination of the empirical formula, the mass percent is converted to a relative mass value and then a relative mole value. The mass percent of an element within a compound must be determined prior to determining the empirical formula for an unknown compound. An empirical formula, you may recall, is the simplest whole number ratio of the atoms in a molecule.

Mass Percent (Percent Composition by mass)

The mass percent of a particular element within a compound is found by dividing the mass of that element by the mass of the compound and then converting the fraction to a percentage. This is shown in Equation 1.1.

$$\text{mass percent} = \frac{\text{mass atoms}}{\text{mass compound}} \times 100\% \quad (1.1)$$

Mass percent can never exceed 100% for any component element. Determining the mass percent of an element from the molecular formula is a straightforward task, although the math may be challenging. Mass percent questions can be asked in a conceptual or mathematical manner. Mass percent is independent of the total mass of the sample of compound.

Table 1.1 shows the relative masses of oxygen and carbon from different samples of carbon dioxide. This demonstrates the law of multiple proportions. Atoms combine in a fixed ratio in terms of mass and moles. Note that the outcome is the same in all four trials measuring the ratio of oxygen gas that reacts with a known mass of carbon. The experiment involves oxidizing a known amount of carbon and collecting the product gas. The mass of this product gas is determined, and the mass of oxygen is assumed to be the difference between the initial and final weighed masses of the carbon sample.

Mass Carbon	Mass Oxygen	Mass O / Mass C
1.33 g	3.53 g	$3.53 / 1.33 = 2.65$
1.07 g	2.87 g	$2.87 / 1.07 = 2.68$
1.11 g	2.96 g	$2.96 / 1.11 = 2.67$
1.27 g	3.39 g	$3.39 / 1.27 = 2.67$

Table 1.1

The mass ratio of oxygen to carbon in the four trials averages out to be 2.67 : 1, which is roughly 8 : 3. This means that for the oxidation product of carbon, the ratio of oxygen to carbon is 8 grams to 3 grams, equivalent to 2 moles to 1 mole.

Example 1.9

How much calcium metal combines with one gram of oxygen to form CaO ?

- A. 1.00 g Ca
- B. 1.25 g Ca
- C. 1.67 g Ca
- D. 2.50 g Ca

Solution

From the molecular formula, the mole ratio of calcium to oxygen is 1 : 1. The atomic mass of calcium (Ca) is 40.08, while the atomic mass of oxygen (O) is 16.00. The mass ratio for the compound is 40.08 to 16.00, which reduces to 2.505 : 1, which rounds to 2.50 to 1. This means that 2.50 grams of calcium combine with 1.00 grams of oxygen to form CaO . Choice D is best.

Example 1.10

What is the mass ratio of iron to oxygen in Fe_2O_3 ?

- A. 1.08 g Fe to 1.00 g O
- B. 1.63 g Fe to 1.00 g O
- C. 2.33 g Fe to 1.00 g O
- D. 3.49 g Fe to 1.00 g O

Solution

From the molecular formula, the mole ratio of iron to oxygen is 2 : 3. The atomic mass of iron (Fe) is 55.85, while the atomic mass of oxygen (O) is 16.00. The mass ratio of iron to oxygen for the compound is 2(55.85) to 3(16.00), which equals 111.7 : 48.0. This ratio is approximately equal to 116 : 50, or 232 : 100, which reduces to 2.32 : 1. Both numbers must be increased proportionally to keep the ratio the same. Choice C is a ratio of 2.33 to 1, which is the closest of the choices. This means that 2.33 grams of iron combine with 1.00 grams of oxygen to form Fe_2O_3 . Choice C is best. You should note that iron and oxygen can combine to make other compounds (with different molecular formulas). One of these compounds is FeO , with a mass ratio of 55.85 to 16.00, which reduces to a ratio of 3.49 : 1.00. The mass ratio (and mole ratio) of an oxide can be used to identify a specific compound. This process is known as combustion analysis.

Examples 1.9 and 1.10 demonstrate how mass percent questions can be mathematical. Mass percent questions can also be asked in a conceptual manner, where the relative mass percentage of a specific element is compared for several compounds. Examples 1.11, 1.12, and 1.13 demonstrate some different forms of this type of question, starting with typical examples and graduating to more abstract ways of asking for mass percent.

Example 1.11

What is the mass percent of oxygen in carbon dioxide?

- A. 27.3%
- B. 57.1%
- C. 62.5%
- D. 72.7%

Solution

The mass of carbon in CO_2 is 12 grams, and the mass of oxygen in CO_2 is 32 grams. The total mass of CO_2 is 44 grams, so the mass percent of oxygen is the ratio of 32 to 44. This ratio reduces to 8 over 11.

$$\text{Mass percent O} = \frac{32 \text{ g O}}{44 \text{ g CO}_2} \times 100\% = \frac{32}{44} \times 100\% = \frac{8}{11} \times 100\%$$

Quick Calculation Technique:

Quick calculations require knowing the values of selected fractions. One-eleventh is equal to 0.091; therefore, eight-elevenths is equal to $8(0.091) = 0.728$. This method gets an exact value and is very fast, if you know how to do it.

$$\frac{8}{11} \approx 8 \times \frac{1}{11} = 8 \times 0.091 = 0.728 = 72.8\%$$

Narrowing-Down-Choices Technique:

On a multiple-choice exam, you can eliminate answers by narrowing down the range into which the answer fits. 8 over 11 is less than 9 over 12, but greater than 7 over 10. A range has been established between $\frac{9}{12}$ and $\frac{7}{10}$. 9 over 12 is 75%, and 7 over 10 is 70%, so the correct answer falls between 70% and 75%.

$$\frac{9}{12} > \frac{8}{11} > \frac{7}{10}, \text{ where } \frac{9}{12} = 75\% \text{ and } \frac{7}{10} = 70\%. \text{ So: } 75\% > \frac{8}{11} > 70\%$$

Choice D is the best answer.

Example 1.12

What is the mass percent of nitrogen in NH_4NO_3 ?

- A. 28%
- B. 35%
- C. 42%
- D. 50%

Solution

The total mass of the nitrogen in the compound is 28 g/mole, because there are two nitrogen atoms in the compound at 14 g/mole each. The mass of the compound is $28 + 4 + 48 = 80$ g/mole. You must divide 28 by 80 quickly. The common denominator of both is 4. Reducing by 4 yields a fraction of $\frac{7}{20}$.

$$\text{Mass percent N} = \frac{28 \text{ g N}}{80 \text{ g NH}_4\text{NO}_3} \times 100\% = \frac{28}{80} \times 100\% = \frac{7}{20} \times 100\%$$

Quick Calculation Technique:

Quick calculations may involve getting a denominator to some easy-to-use number, such as 10, 100, or 1000. It is easy to convert a fraction into decimals or percents when the denominator is either 10, 100, or 1000. For this question, a denominator of 100 works well. To convert 20 to 100, we must multiply by 5. Multiply both numerator and denominator by 5, to change the fraction $\frac{7}{20}$ to $\frac{35}{100}$.

The percentage is 35%, so choice B is correct.

$$\frac{28}{80} = \frac{7}{20} = \frac{7 \times 5}{20 \times 5} = \frac{35}{100} = 0.35 = 35\%$$

Narrowing-Down-Choices Technique

Narrowing down the range into which the answer fits can be applied to any multiple-choice math question. 28 over 80 reduces to 7 over 20. The value of 7 over 20 is greater than 7 over 21, but less than 8 over 20. A range has been established between $\frac{7}{21}$ and $\frac{8}{20}$. The value of 7 over 21 is 33.3%, and the value of 8 over 20 is 40%, so the correct answer falls between 33.3% and 40%. Only choice B fits in this range, so choice B must be the correct answer.

$$\frac{7}{21} < \frac{7}{20} < \frac{8}{20}, \text{ where } \frac{7}{21} = 33.3\% \text{ and } \frac{8}{20} = 40\%. \text{ So } 33.3\% < \frac{7}{20} < 40\%$$

Example 1.13

Which of the following samples yields the MOST moles of sodium cation?

- A. 1.0 g NaCl
- B. 1.0 g NaBr
- C. 1.0 g NaNO₃
- D. 1.0 g Na₂CO₃

Solution

This question is a subtle way of asking, "Which salt has the greatest mass percent of sodium?" All choices are 1.0 g of compound, so the most moles of sodium are found in the compound with the greatest mass percentage of sodium. Choices A, B, and C have the same number of sodium atoms in the compound (one), so they each have the same numerator in the mass percent formula. The best choice of those three salts is the lightest compound (resulting in the smallest denominator when calculating mass percent). The lightest compound of the three choices A, B, and C is the salt with the lightest anion, which is choice A, NaCl. Now the question involves comparing the mass percent of sodium in NaCl to the mass percent of sodium in Na₂CO₃, choice D.

$$\text{Mass percent Na in NaCl} = \frac{23.0 \text{ g Na}}{58.5 \text{ g NaCl}} \times 100\% = \frac{23}{58.5} \times 100\%$$

$$\text{Mass percent Na in Na}_2\text{CO}_3 = \frac{46.0 \text{ g Na}}{106 \text{ g Na}_2\text{CO}_3} \times 100\% = \frac{46}{106} \times 100\% = \frac{23}{53} \times 100\%$$

$$\frac{23}{53} > \frac{23}{58.5}, \text{ so choice D is the best answer.}$$

Empirical Formula

An empirical formula for molecules uses the smallest whole number ratio of the atoms in a compound. It is the formula that gives the relative numerical values for each element in the molecule in such a way that the numbers in the ratio cannot be reduced without involving fractions. An empirical formula may or may not be the actual formula of the molecule. It is calculated from the mass percentage of each element within a compound. You may recall from your general chemistry courses that we start by assuming a 100-gram sample, so that the percentages can be changed easily into mass figures. From this point, it is a matter of converting from mass into moles, using the atomic masses for each element. The empirical formula is a whole number ratio of these mole values. Empirical formulas must include whole number quantities as subscripts.

Example 1.14

Which of the following is an empirical formula?

- A. C_2H_6
- B. C_3H_8
- C. C_4H_{10}
- D. C_6H_6

Solution

An empirical formula for a molecule is defined as the formula in which the constituent atoms are in their smallest possible whole number ratio. Choice A, C_2H_6 , can be reduced to C_1H_3 (normally written as CH_3), so choice A is not an empirical formula. Choice C, C_4H_{10} , can be reduced to C_2H_5 , so choice C is not an empirical formula. Choice D, C_6H_6 , can be reduced to C_1H_1 (normally written as CH), so choice D is not an empirical formula. This eliminates all of the choices except choice B, C_3H_8 . The ratio of 3 : 8 cannot be reduced any further, so C_3H_8 is an empirical formula, making choice B the correct answer. In the case of C_3H_8 , the empirical formula and the molecular formula are the same, because the compound is completely saturated with hydrogens. C_6H_{16} has too many hydrogens and is not a possible formula. Organic chemistry rules can help to save time on formula questions.

Example 1.15

What is the empirical formula for a compound that is 72% C, 12% H, and composed solely of carbon, hydrogen, and oxygen?

- A. C_3H_6O
- B. $C_6H_{12}O$
- C. $C_6H_{14}O$
- D. $C_7H_{14}O$

Solution

For empirical formula calculations, assume a 100-gram sample of the compound. A 100-gram sample in this case would contain 72 grams C, 12 grams H, and 16 grams O. The 16 grams of oxygen are determined from the difference between the mass of carbon plus hydrogen and the 100 grams of sample. Next, you must convert the grams of each element into the corresponding moles of each element. To go from grams to moles, divide the mass of the element by its atomic mass. In this case, 72 grams of C is equivalent to 6 moles of C, 12 grams of H is equivalent to 12 moles of H, and 16 grams of O is equivalent to 1 mole of O. This is a particularly easy example, because the ratios turn out to be whole numbers. In cases where they don't come out whole, you must divide the mole quantity of each element in the compound by the lowest mole quantity for any of the elements. However, for this example the best answer is choice B. Drawn below is a useful layout of empirical formula calculations. It is often easier just to plug terms into an equation such as this, because you don't have to show your work on the MCAT. Do the questions as quickly and carefully as you can, emphasizing organization in your path to a solution.

$$\frac{\text{C percentage carbon}}{\text{molar mass carbon}} \frac{\text{H percentage hydrogen}}{\text{molar mass hydrogen}} \frac{\text{O percentage oxygen}}{\text{molar mass oxygen}} = \frac{C_{72}}{12} \frac{H_{12}}{1} \frac{O_{16}}{16} = C_6H_{12}O_1$$

Example 1.16

What is the empirical formula of an oxide of sulfur that is 60% oxygen by weight?

- A. SO
- B. SO₂
- C. S₂O₅
- D. SO₃

Solution

Here again you should assume a 100-gram sample. A 100-gram sample would have 60 grams of oxygen and 40 grams of sulfur. The moles of O = $\frac{60}{16}$, and the moles of S = $\frac{40}{32}$. 60 is more than double $\frac{40}{32}$, so there are more than two oxygen atoms per sulfur atom. This eliminates choices A and B. Upon reducing $\frac{40}{32}$ to $\frac{20}{16}$, we see that the ratio of oxygen to sulfur is 3 : 1, making choice D the best answer.

$$\frac{\text{S percentage sulfur}}{\text{molar mass sulfur}} \div \frac{\text{O percentage oxygen}}{\text{molar mass oxygen}} = \frac{40}{32} \div \frac{60}{16} = \frac{40}{32} \times \frac{16}{60} = \frac{20}{16} = \frac{5}{4} = \text{S}_1\text{O}_3$$

The test emphasizes ratios, so the more numerical intuition you develop, the better. To make problem-solving less mathematical, focus on eliminating choices by comparing relative ratios. An alternative way to ask an empirical formula question with reduced math is shown in Example 1.17.

Example 1.17

If a molecule is composed of only two elements (X and Y), and if X and Y combine in equal mass quantities, and if Y is less than twice as heavy as X, which of the following molecular formulas is NOT possible?

- A. XY
- B. XY₂
- C. X₃Y₂
- D. X₃Y

Solution

If Y were exactly twice as heavy as X, then equal masses of X and Y would result in exactly twice as many moles of X as Y, a 2 : 1 ratio of X to Y. Because Y is less than twice as heavy as X, there are fewer than twice as many moles of X as Y. Thus, the ratio of X : Y must be 2 : 1 or smaller. This limiting ratio is true of all the answers except choice D. The wording of this question allows for the possibility that the molecular mass of X is equal to or greater than Y.

Molecular Formula

The molecular formula is the actual mole ratio of the elements within the compound. The molecular formula is found by multiplying the empirical formula by the whole number ratio of the molecular mass to the empirical mass (including 1, in some cases). Therefore, conversion from the empirical formula to the molecular formula requires knowing the molecular mass of the compound. If the molecular mass is double the empirical mass, then all of the elements in the empirical formula are doubled to get the molecular formula.

Example 1.18

What is the molecular formula for a compound that is 82.76% C, has a molecular mass of 58.1 grams per mole, and is composed solely of carbon and hydrogen?

- A. CH_2
- B. C_2H_5
- C. C_3H_8
- D. C_4H_{10}

Solution

Choice B is eliminated immediately, because a hydrocarbon cannot have an odd number of hydrogens. For neutral C_2H_5 to exist, it would have to be a free radical. Continuing to use organic chemistry logic, choice A is not physically possible. One carbon requires four bonds to hydrogen atoms to form a stable molecule (methane). C_1H_2 would be a carbene ($:\text{CH}_2$), which is not stable due to its lack of an octet. To decide between choices C and D, you must first find the empirical formula, and then convert that into the molecular formula.

$$\frac{\text{C percentage carbon}}{\text{molar mass carbon}} \frac{\text{H percentage hydrogen}}{\text{molar mass hydrogen}} = \frac{\text{C } 82.76}{12} \frac{\text{H } 17.24}{1} = \text{C}_{6.93}\text{H}_{17.24}$$

$$\text{empirical formula: } \frac{\text{C}_{6.93}\text{H}_{17.24}}{6.93 \quad 6.93} = \text{C}_1\text{H}_{2.5} = \text{C}_1\text{H}_{2.5} = \text{C}_2\text{H}_5$$

An empirical formula of C_2H_5 when multiplied by a whole number cannot yield C_3H_8 , so choice C is eliminated. This leaves only choice D. The correct ratio of molecular mass to empirical mass confirms that choice D is the best answer. The molecular formula is found using the molecular mass of 58.1 grams per mole. The empirical mass is $2(12) + 5 = 29$. This value is only half of the molecular mass, so the formula must be doubled to yield C_4H_{10} . This question could have been solved in seconds by seeing that only choice D has a molecular mass of 58.

Example 1.19

An unknown stable gas is composed of 13.10% H, 52.23% C, and the remainder O. A 0.10-mole sample weighs 4.61 grams. What is the molecular formula for the compound?

- A. $\text{C}_2\text{H}_6\text{O}$
- B. $\text{C}_3\text{H}_8\text{O}$
- C. $\text{C}_3\text{H}_9\text{O}$
- D. $\text{C}_4\text{H}_8\text{O}$

Solution

A stable compound made of carbon, hydrogen, and oxygen cannot have an odd number of hydrogens, so choice C is eliminated. Neutral $\text{C}_3\text{H}_9\text{O}$ would have to be a free radical. The remaining choices obey the octet rule. The molecular mass is 46.1, so choice A is the correct answer. That is the method you should use on a multiple-choice exam. Now let's confirm that by using the molecular formula. First you must assume a 100-gram sample, resulting in 13.10 g H, 52.23 g C, and 34.67 g O. Next, the numbers are converted into moles, and then the values are reduced to a whole number ratio. The calculation of the empirical formula is shown below, where a formula arrangement is used to help keep track of the values.

$$\frac{\text{C } 52.23}{12} \frac{\text{H } 13.10}{1} \frac{\text{O } 34.67}{16} = \text{C}_{4.4}\text{H}_{13.1}\text{O}_{2.2} = \frac{\text{C}_{4.4}\text{H}_{13.1}\text{O}_{2.2}}{2.2 \quad 2.2 \quad 2.2} = \text{C}_2\text{H}_6\text{O}_1$$

The empirical mass is $2(12) + 6 + 16 = 46$. This value is equal to the molecular mass, so the empirical formula is the molecular formula.

Combustion Analysis (an Experimental Procedure)

Combustion analysis entails determining the mass percent of each component element in an unknown compound. It is accomplished by oxidizing the unknown with excess oxygen (to ensure complete combustion), followed by the separation and collection of all of the oxidized products. This is an experimental procedure, which makes it a likely topic on a conceptual exam such as the MCAT. When a hydrocarbon is oxidized, carbon dioxide and water are formed. Carbon dioxide and water can be separated using various methods. One method involves passing the CO_2 gas and H_2O vapor across a hygroscopic salt of known mass. The hygroscopic salt absorbs the water, and thus increases in mass.

The hygroscopic salt must not react with carbon dioxide. A good choice for the hygroscopic salt is either calcium chloride or magnesium sulfate (both of which you should have used as drying agents in your organic chemistry lab). Once the water is absorbed, the remaining gas is passed across a sample of KOH of known mass. KOH undergoes a combination reaction with carbon dioxide to form potassium bicarbonate (KHCO_3). The potassium hydroxide salt absorbs the carbon dioxide, and thus increases in mass. Knowing the masses of CO_2 and H_2O , we can determine the masses of carbon and hydrogen by multiplying the mass percent of each element by the mass of its respective oxide product that was collected. Upon dividing these numbers by the mass of the original sample, the mass percents of hydrogen and carbon in the original sample are determined.

The carbon dioxide and water can be separated and collected using a different method than passing the gases over salts that bind the products. By lowering the temperature, carbon dioxide and water can be converted to their solid states. Because solids do not flow, they can be collected easily. The math is the same, once the quantity of each product has been established.

Figure 1-1 shows a typical apparatus used in a combustion analysis. The oxygen tank serves to provide excess oxygen to the system constantly. The pressure valve is a one-way valve designed so that oxygen can flow into the sample chamber, but no gases can flow back. The resistor in the base of the sample chamber provides heat to initiate the oxidation. The tube to the right of the sample chamber is connected to a vacuum, to generate a low pressure. Once the reaction is complete, Valve #3 is closed so that no gases are lost to the environment, and Valve #1 is opened. Gases flow into the region above the magnesium sulfate. Magnesium sulfate absorbs moisture from the gases. After a time, Valve #2 is opened so that gases flow into the region above sodium hydroxide. Sodium hydroxide absorbs carbon dioxide gas. Oxygen gas is then used to flush any remaining gas in the sample chamber through the system.

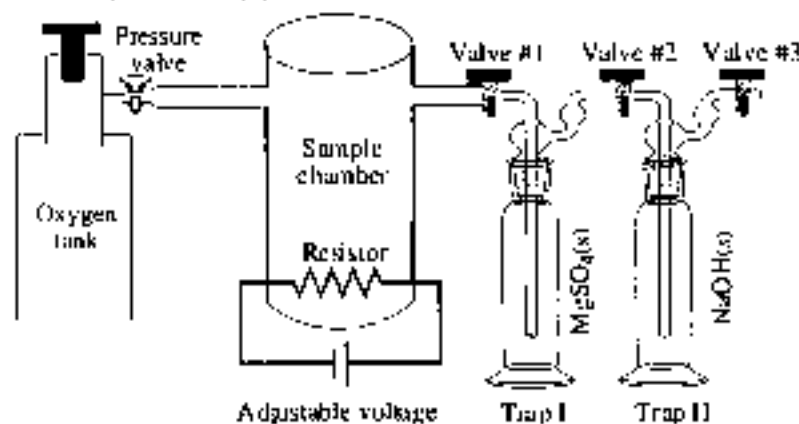


Figure 1-1

Solution Concentration

Units and Terminology

Solutions are mixtures formed by the addition of a solute to a solvent. A solution may contain several different solutes. The amount of solute is measured relative to the amount of solvent, which results in a certain concentration for the solution. Concentration units include molarity (moles solute per liter of solution), molality (moles solute per kilogram of solvent), mass percent (mass solute per mass of solution), and density (mass solution per volume solution). The concentration of a solute can be changed by changing the amount of solvent. Addition of solvent to solution is referred to as **dilution** and results in a lower concentration by any measurement. Paramount to solving problems involving concentrations and dilution is an understanding of the different units.

Molarity

Molarity (M) is the concentration of a fluid solution defined as the moles of a solute per volume of solution, where the volume is measured in liters (L). To determine the molarity of a solution, the moles of solute are divided by the liters of solution.

Example 1.20

What is the molarity of 500.0 mL of solution containing 20.0 grams of $\text{CaCO}_3(\text{s})$?

- A. 0.15 M $\text{CaCO}_3(\text{aq})$
- B. 0.20 M $\text{CaCO}_3(\text{aq})$
- C. 0.33 M $\text{CaCO}_3(\text{aq})$
- D. 0.40 M $\text{CaCO}_3(\text{aq})$

Solution

Molarity is defined as moles of solute per liter of solution. In this question, you must convert from grams CaCO_3 into moles CaCO_3 by dividing by the molecular mass of CaCO_3 , and then dividing this value by the liters of solution:

$$\frac{20 \text{ grams } \text{CaCO}_3}{100 \text{ grams/mole}} = 0.20 \text{ moles } \text{CaCO}_3$$

$$\frac{0.20 \text{ moles } \text{CaCO}_3}{0.50 \text{ L solution}} = \frac{0.20}{0.50} \text{ M } \text{CaCO}_3 = \frac{0.40}{1} \text{ M } \text{CaCO}_3 = 0.40 \text{ M } \text{CaCO}_3, \text{ choice D}$$

These questions can be trickier if the units are milligrams, milliliters, or millimolar. The question uses similar math, but there are more opportunities to make an error. A common error to avoid is the "factor of a thousand" error. To become more conscious of possible trick questions, ask yourself: "If I were writing this test question, what would I ask?" If you consider questions from the test writer's point of view, the tricks become more apparent.

Molality

Molality (m) is the concentration of a fluid solution defined as the moles of a solute per kilogram of solvent. The molality of a solution does not change with temperature, so it is often used to determine a change in the solution's temperature when the change depends on concentration. Notable examples of this include boiling-point elevation and freezing-point depression. To determine the molality of a solution, the moles of solute are divided by the kilograms of the solvent.

Example 1.21

What is the molality of a solution made by adding 7.46 g KCl to 250 g of water?

- A. 0.20 m KCl(aq)
- B. 0.33 m KCl(aq)
- C. 0.40 m KCl(aq)
- D. 0.50 m KCl(aq)

Solution

Molality is defined as moles of solute per kilogram of solvent. In this question, you must convert from grams KCl into moles KCl by dividing by the molecular mass of KCl, and then dividing this value by the kilograms of solvent:

$$\frac{7.46 \text{ grams KCl}}{74.6 \frac{\text{grams}}{\text{mole}}} = 0.10 \text{ moles KCl}$$

$$\frac{0.10 \text{ moles KCl}}{0.25 \text{ kg H}_2\text{O}} = \frac{0.10}{0.25} \text{ m KCl} = \frac{0.40}{1} \text{ m KCl} = 0.40 \text{ m KCl, choice C}$$

Mass Percent (in Solution)

Mass percent is the concentration of a fluid solution defined as the mass of solute per mass of solution multiplied by one hundred percent. The mass percent of a solution remains constant as temperature changes. To determine the mass percent of a solution, the mass of solute is divided by the mass of the solution (where both are measured in grams). Mass percent is a unitless value, because mass is divided by mass.

Example 1.22

What is the mass percent of a 1.0 m NaCl(aq) solution?

- A. 5.53% NaCl by mass
- B. 5.85% NaCl by mass
- C. 6.22% NaCl by mass
- D. 9.50% NaCl by mass

Solution

Mass percent is defined as grams of solute per grams of solution. In this question, you must convert from moles NaCl into grams NaCl by multiplying by the molecular mass of NaCl, and then dividing this value by the total mass of solution. The total mass of solution is the sum of the mass of solute and the mass of solvent. It is easy to forget to consider the mass of solute in the total mass, which leads to the incorrect answer choice B.

$$1.0 \text{ moles NaCl(aq)} \times 58.5 \frac{\text{g}}{\text{mole}} = 58.5 \text{ g NaCl}$$

$$\text{Total mass solution} = 58.5 \text{ g NaCl} + 1000 \text{ g H}_2\text{O} = 1058.5 \text{ g solution}$$

$$\text{Mass \% NaCl} = \frac{58.5 \text{ g NaCl}}{1058.5 \text{ g solution}}, \text{ where } \frac{58.5}{1058.5} < \frac{58.5}{1000} = 5.85\%$$

Only choice A is less than 5.85%, so choice A is the best answer. Sometimes questions that would normally require a calculator for a precise answer can be determined well enough without one to answer a multiple-choice test question.

Density

Density (ρ) is the concentration of a fluid solution defined as the mass of solution per volume of solution. The density of a solution varies with temperature. The density of a solution is uniform throughout, so a small sample of the solution has the same density as the entire solution. To determine the density of a solution, the mass of a sample of the solution is divided by the volume of the sample (usually measured in milliliters). Examples 1.3 and 1.5 addressed the topic of density.

Dilution

Dilution involves the addition of solvent to a solution, thus resulting in an increase in the volume of the solution and a decrease in the concentration of the solute in solution. Equation 1.2 below describes simple dilution where a solvent is added to solution. Determining the concentration when two solutions are mixed requires more work than simple dilution.

$$M_{\text{initial}} \cdot V_{\text{initial}} = M_{\text{final}} \cdot V_{\text{final}} \quad (1.2)$$

When working with dilution questions, be aware of a common twist that the writers can employ. Their question may ask for *volume added* rather than asking for the final total volume. Percent dilution may also be discussed. Multiple containers are used in standard dilution procedure, so rinsing ensures that the concentration of solution on the walls of the new containers are equilibrated with the contents they will hold. You may recall filling a volumetric pipette with a solution in general chemistry lab, then draining the pipette before filling it with the sample to be transferred. This is done to ensure that any residual liquid in the pipette has the same concentration as the solution being transferred and that any water in the pipette is rinsed away.

Example 1.23

What is the molarity of a solution made by mixing 200 mL pure water with 100 mL 0.75 M KCl(aq)?

- A. 0.25 M KCl(aq)
- B. 0.50 M KCl(aq)
- C. 1.50 M KCl(aq)
- D. 2.25 M KCl(aq)

Solution

Because water has been added to the solution, the concentration must decrease, so choices C and D are eliminated. Solving this question involves using Equation 1.2 to determine the effect of dilution on the molarity. The initial molarity (M_{initial}) is 0.75 M, the initial volume (V_{initial}) is 100 mL, and the final volume (V_{final}) is 300 mL. The question requires solving for the final molarity (M_{final}).

$$M_{\text{initial}} \cdot V_{\text{initial}} = M_{\text{final}} \cdot V_{\text{final}} \quad \therefore 0.75 \text{ M} \cdot 100 \text{ mL} = M_{\text{final}} \cdot 300 \text{ mL}$$

$$M_{\text{final}} = \frac{M_{\text{initial}} \cdot V_{\text{initial}}}{V_{\text{final}}} \quad \therefore M_{\text{final}} = \frac{0.75 \text{ M} \cdot 100 \text{ mL}}{300 \text{ mL}} = 0.75 \text{ M} \left(\frac{1}{3} \right) = 0.25 \text{ M}$$

The final molarity is 0.25 M, so choice A is the best answer. Because the molarity is decreased by a factor of three, the dilution process in this example is referred to as a *threefold dilution*. That is, when two parts solvent are added to one part solution, the volume is tripled and the dilution is threefold. This terminology may be unfamiliar at first, but in a short time it should make sense.

Example 1.24

How many milliliters of water are needed to dilute 80 mL 5.00 M $\text{KNO}_3(\text{aq})$ to 1.00 M?

- A. 160 mL H_2O
- B. 320 mL H_2O
- C. 400 mL H_2O
- D. 480 mL H_2O

Solution

Again, Equation 1.2 should be employed to calculate the change in concentration of a solution after dilution from the addition of solvent. First we must solve for the final volume. You are provided with values for M_{initial} , M_{final} , and V_{initial} , so you can manipulate the equation to solve for V_{final} .

$$V_{\text{final}} = \frac{M_{\text{initial}} \cdot V_{\text{initial}}}{M_{\text{final}}} = \frac{5.00 \text{ M} \times 80 \text{ mL}}{1.00 \text{ M}} = 5 \times 80 \text{ mL} = 400 \text{ mL}$$

The question asks how much water is *added*, not the final volume. The volume added is the difference between the initial volume (80 mL) and the final volume (400 mL). The difference between the two values is 320 mL, so 320 mL of water must be added to 80 mL of 5.00 M $\text{KNO}_3(\text{aq})$ to dilute it from 5.00 M $\text{KNO}_3(\text{aq})$ to 1.00 M $\text{KNO}_3(\text{aq})$. Choice B is the best answer.

Example 1.25

Which dilution converts 6.00 M $\text{HCl}(\text{aq})$ to 0.30 M $\text{HCl}(\text{aq})$?

- A. 11 parts water to 1 part 6.00 M $\text{HCl}(\text{aq})$
- B. 19 parts water to 1 part 6.00 M $\text{HCl}(\text{aq})$
- C. 20 parts water to 1 part 6.00 M $\text{HCl}(\text{aq})$
- D. 21 parts water to 1 part 6.00 M $\text{HCl}(\text{aq})$

Solution

Hydrochloric acid goes from 6.00 M to 0.30 M, which is a twenty fold dilution. This means that the final volume is twenty (20) times the initial volume. When dealing with answer choices that present the dilution in terms of parts, the ratio is volume of solvent added to volume of original solution. For the final volume to be twenty times greater than the initial volume, nineteen parts must be added.

$$\frac{V_{\text{final}}}{V_{\text{initial}}} = \frac{M_{\text{initial}}}{M_{\text{final}}} \Rightarrow \frac{V_{\text{final}}}{V_{\text{initial}}} = \frac{6.00 \text{ M}}{0.30 \text{ M}} = 20 \therefore V_{\text{final}} = 20 (V_{\text{initial}})$$

$$V_{\text{added}} = V_{\text{final}} - V_{\text{initial}} = 20 V_{\text{initial}} - V_{\text{initial}} = 19 (V_{\text{initial}})$$

The ratio of the volume added to the volume of solution initially present is 19 : 1, so the best answer is choice B. A part can be any set volume. A 20 : 1 dilution would result in a final volume that is 21 times the initial volume, so the final concentration would be less than 0.30 M.

A solution can be diluted by adding solvent or another solution to it. The addition of pure solvent is known as a *simple dilution*. Mixing two solutions is more complicated than a simple dilution where pure solvent is added, because solute from both initial solutions must be considered. The final concentration lies somewhere between the two initial concentration values before mixing. The final concentration is a weighted average of the initial concentrations.

Example 1.26

What is the final concentration of Cl^- ions after mixing equal volumes of 0.20 M KCl(aq) with 0.40 M $\text{CaCl}_2\text{(aq)}$?

- A. 0.20 M $\text{Cl}^-\text{(aq)}$
- B. 0.30 M $\text{Cl}^-\text{(aq)}$
- C. 0.40 M $\text{Cl}^-\text{(aq)}$
- D. 0.50 M $\text{Cl}^-\text{(aq)}$

Solution

The salt KCl yields one chloride ion when it dissociates in water, so the chloride concentration is 0.20 M. The salt CaCl_2 yields two chloride ions when it dissociates in water, so the chloride concentration is 0.80 M. The final concentration equals the total Cl^- ions from both solutions divided by the new total volume. Because equal volumes are mixed, the final concentration will be an average of the two initial concentrations.

$$\frac{0.20 \text{ M Cl}^- + 0.80 \text{ M Cl}^-}{2} = \frac{1.00 \text{ M Cl}^-}{2} = 0.50 \text{ M Cl}^-$$

If the volumes are not equal, then a weighted average yields the final concentration. For this example, choice D is the best answer.

Example 1.27

What is the concentration of K^+ ions in solution after 25.0 mL of 0.10 M $\text{K}_2\text{SO}_4\text{(aq)}$ is added to 50.0 mL of 0.40 M KOH(aq) ?

- A. 0.25 M $\text{K}^+\text{(aq)}$
- B. 0.30 M $\text{K}^+\text{(aq)}$
- C. 0.33 M $\text{K}^+\text{(aq)}$
- D. 0.50 M $\text{K}^+\text{(aq)}$

Solution

The salt KOH yields one potassium ion when it dissociates in water, so the K^+ concentration is 0.40 M. The salt K_2SO_4 yields two potassium ions when it dissociates in water, so the K^+ concentration is 0.20 M. The final concentration equals the total K^+ ions from both solutions divided by the new total volume. Because unequal volumes are mixed, the final concentration is a weighted average of the two initial concentrations. The mixture involves 25 mL 0.20 M K^+ with 50 mL 0.40 M K^+ , so the final concentration must fall between 0.20 M and 0.40 M. This eliminates choice D. If the two volumes were equal, the final concentration would be 0.30 M K^+ , the average of the two concentrations. But because there is more of the more concentrated solution, the final concentration is greater than 0.30 M, so choices A and B are eliminated. Only choice C remains.

$$\begin{aligned} \frac{25 \text{ mL} \times 0.20 \text{ M K}^+}{75 \text{ mL}} + \frac{50 \text{ mL} \times 0.40 \text{ M K}^+}{75 \text{ mL}} &= \frac{1}{3} (0.20 \text{ M K}^+) + \frac{2}{3} (0.40 \text{ M K}^+) \\ &= \frac{2}{3} + \frac{8}{3} \text{ MK}^+ = \frac{10}{3} \text{ MK}^+ = 0.33 \text{ MK}^+, \text{ choice C} \end{aligned}$$

Beer's Law

When electromagnetic radiation is passed through a solution, the solute may absorb some of the light. The light absorbed is in a specific wavelength range, and the intensity of the absorbance varies with the concentration of solute. A generic absorbance spectrum for a hypothetical solute is shown in Figure 1-2.

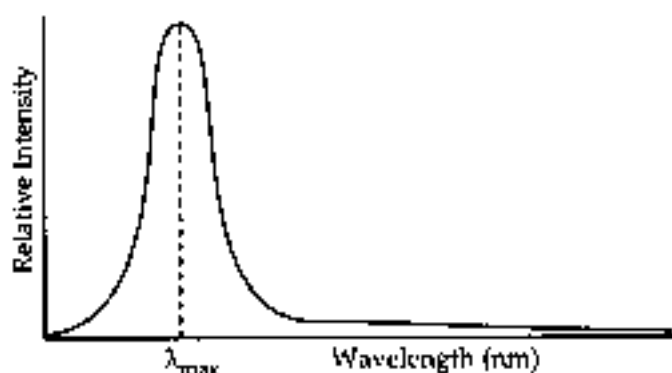


Figure 1-2

Because the absorbance of light varies with concentration, absorbance can be used to determine the concentration of a solute. This is the essence of Beer's law. Beer's law is expressed in Equation 1.3, where ϵ is a constant for the solute at $\lambda_{\max}$ (the wavelength of greatest absorbance), C is the solute concentration ($C = [\text{Solute}]$), and l is the width of the cuvette (length of the pathway through which the light passes).

$$\text{Absorbance} = \epsilon Cl \quad (1.3)$$

The key feature of this equation is its expression of the principle that absorbance is proportional to concentration. By knowing the absorbance for solutions of known concentration, the concentration of an unknown solution can be determined by comparing its absorbance value to the known values.

Example 1.28

For 100 mL of a solution with an absorbance of 0.511, what amount of water must be added to reduce the absorbance to 0.100?

- A. 389 mL H_2O
- B. 411 mL H_2O
- C. 488 mL H_2O
- D. 511 mL H_2O

Solution

For this question, a hybrid of Equations 1.2 and 1.3 should be employed to determine the volume that must be added to dilute the solution. Because absorbance is directly proportional to concentration, Equation 1.2 can be re-written as follows:

$$\text{Abs}_{\text{initial}} \cdot V_{\text{initial}} = \text{Abs}_{\text{final}} \cdot V_{\text{final}}$$

First, we must solve for the final volume. You are provided with values for $\text{Abs}_{\text{initial}}$, $\text{Abs}_{\text{final}}$, and V_{initial} , so you can solve for V_{final} :

$$V_{\text{final}} = \frac{\text{Abs}_{\text{initial}} \cdot V_{\text{initial}}}{\text{Abs}_{\text{final}}} = \frac{0.511 \times 100 \text{ mL}}{0.100} = \frac{5.11 \times 100 \text{ mL}}{1} = 511 \text{ mL}$$

The question asks for how much water is *added*, which is the difference between the initial volume (100 mL) and the final volume (511 mL). The difference between the two values is 411 mL; therefore, 411 mL of water must be added. The best answer is choice B.

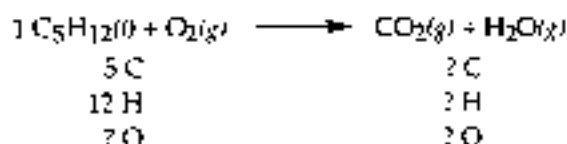
Balancing Reactions

Standard Balancing

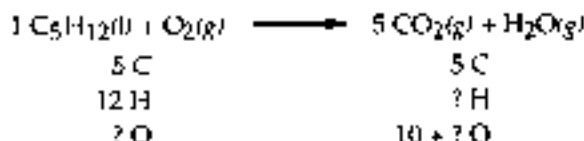
Let us briefly address the process of balancing chemical reactions. Reactions are written from reactants to products. Because of the law of conservation of matter, the number of atoms must be identical on each side of the reaction. The two sides of the reaction are separated by an arrow drawn from left to right.



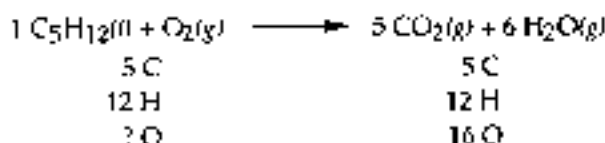
There are carbon atoms, hydrogen atoms, and oxygen atoms on both sides of the reaction. To balance the reaction, keep track of the atoms on each side of the reaction. Start with the compound whose atoms are least present in the reaction (carbon and hydrogen are present in only two compounds each, so we start with C_5H_{12}). Starting with one C_5H_{12} , the atoms must be balanced step by step:



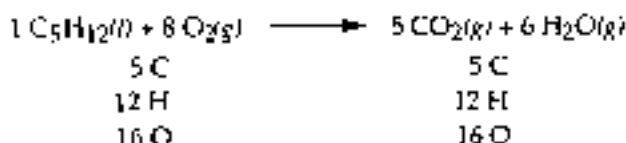
Balance carbon atoms by multiplying CO_2 by five:



Balance hydrogen atoms by multiplying H_2O by six:



Balance oxygen atoms by multiplying O_2 by eight:



Example 1.29

What are the correct coefficients needed to balance the following reaction?



- A. 2 : 3 : 1 : 6
- B. 3 : 2 : 1 : 3
- C. 2 : 3 : 1 : 3
- D. 3 : 2 : 1 : 6

Solution

Balancing equations requires that you keep track of each atom. In this case, because of cobalt, the ratio of $\text{Co}(\text{OH})_3(\text{s})$ to $\text{Co}_2(\text{SO}_4)_3(\text{aq})$ must be 2 : 1, which eliminates choices B and D. The correct answer is found using water. Two moles of $\text{Co}(\text{OH})_3(\text{s})$ and three moles of $\text{H}_2\text{SO}_4(\text{aq})$ have a total of 12 H atoms and 18 O atoms, making choice A the best answer.

Balanced equations can be used to determine the amount of a product from a given amount of reactant. As we saw in Example 1.7, balanced equations can be used to determine how much product is formed from a given mass of a reactant. We refer to these questions as gram-to-mole-to-mole-to-gram conversions, where the overall conversion process is from grams reactant to grams product.

Example 1.30

How many grams of water are formed when 25.0 grams of pentane (C_5H_{12}) reacts with oxygen?

- A. 18.0 g H_2O
- B. 25.0 g H_2O
- C. 37.5 g H_2O
- D. 75.0 g H_2O

Solution

Step 1: Convert grams reactant to moles reactant by dividing by molecular mass of the reactant:

$$\text{grams reactant} \times \frac{\text{mole reactant}}{\text{grams reactant}} = \text{moles reactant}$$

Step 2: Convert moles reactant to moles product using the coefficient ratio from the balanced reaction:

$$\text{moles reactant} \times \frac{\text{moles product}}{\text{moles reactant}} = \text{moles product}$$

Step 3: Convert moles product to grams product by multiplying by molecular mass of the product:

$$\text{moles product} \times \frac{\text{grams product}}{\text{moles product}} = \text{grams product}$$

The overall conversion is as shown below:

$$25.0 \text{ g } C_5H_{12} \times \frac{1 \text{ mole } C_5H_{12}}{72 \text{ g } C_5H_{12}} \times \frac{6 \text{ mole } H_2O}{1 \text{ mole } C_5H_{12}} \times \frac{18 \text{ g } H_2O}{1 \text{ mole } H_2O} = 37.5 \text{ g } H_2O$$

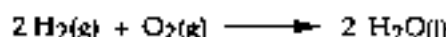
The ratio of water to pentane comes from the balanced oxidation reaction. The product of 6×18 is 108, which is greater than 72. This means that the original 25.0 grams is multiplied by a number greater than 1, which in turn means that the final number is greater than 25.0 g. This eliminates choices A and B. 108 over 72 is less than two, so the final value is less than 50.0 grams, so choice D is eliminated. The only answer that remains is choice C, 37.5 g.

Limiting Reagents

Determining the limiting reagent in a reaction requires comparing the number of moles of each of the reactants. The limiting reagent is the reactant that is exhausted first, not necessarily the reactant with the lowest number of moles. When the limiting reagent is completely consumed, the reaction stops, regardless of the amount of the other reactant. To determine the limiting reagent, the amount of all reactants and the mole ratio of the reactants must be known. If the ratio of the moles of Reactant A to Reactant B is greater than the ratio of Reactant A to Reactant B from the balanced equation, then Reactant B is the limiting reagent. If the ratio of the moles of Reactant A to Reactant B is less than the ratio of Reactant A to Reactant B from the balanced equation, then Reactant A is the limiting reagent.

Example 1.31

Assuming that the following reaction between oxygen and hydrogen goes to completion, which statement is true if 10.0 grams of hydrogen are mixed with 64.0 grams of oxygen?



- A. The limiting reagent is oxygen.
- B. 74.0 grams of water will form.
- C. 3.0 moles of hydrogen will be left over following the reaction.
- D. 68.0 grams of water will form.

Solution

In limiting reagent reactions, you must decide which reactant is depleted first. Limiting reagent questions often look like ordinary stoichiometric questions. The rule is simple: If they give you quantities for all reactants, it is probably a limiting reagent problem. In this question, you are given quantities for both hydrogen and oxygen. 10 grams of H is equal to 5 moles of H_2 , and 64 grams of oxygen is 2 moles of O_2 (remember your diatomic elements!) From the balanced equation, we learn that twice as many moles of hydrogen as oxygen are needed. The number of moles indicates there is a 5 : 2 ratio of hydrogen to oxygen, which is greater than a 2 : 1 reaction ratio, so oxygen is depleted first. The correct choice is answer A. The question gives you opposing choices in A and B. One of these two choices must be true. The correct choice is A.

Example 1.32

What is the limiting reagent when 22.0 grams C_3H_8 are mixed with 48.0 grams O_2 ?



- A. Oxygen is the limiting reagent.
- B. Propane is the limiting reagent.
- C. Water is the limiting reagent.
- D. There is no limiting reagent.

Solution

This question is more difficult than the previous question, but you are still deciding which reactant is depleted first. Because the limiting reagent is a reactant, choice C (a product) is eliminated. To solve the question, stick to this simple rule: Compare the actual ratio of the two reactants to the balanced equation ratio of the two reactants. In this question, you are given unequal mass quantities of C_3H_8 and O_2 and a mole ratio that is not 1 : 1. Good luck. Remember, the first step is to balance the reaction.



22 grams of C_3H_8 is $\frac{22}{44}$ moles of C_3H_8 , which is 0.50 moles C_3H_8 . 48 grams of O_2 is $\frac{48}{32}$ moles of O_2 (remember your diatomic elements!), which is 1.50 moles of O_2 . From the balanced equation, we see that we need 5 moles of O_2 for 1 mole of C_3H_8 . The number of moles indicates there is a 1.50 : 0.50 ratio of O_2 to C_3H_8 , which is less than 5 : 1. This means that oxygen (O_2) is depleted first. Oxygen gas is the limiting reagent, so choice A is the best answer.

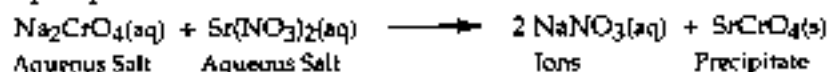
Reaction Types

Common Reactions

There are some reaction types that are standard reactions in inorganic chemistry. Included among the common reaction types are the following five: 1) precipitation reactions (also known as double-displacement reactions), 2) acid-base reactions (also known as neutralization reactions), 3) composition reactions, 4) decomposition reactions, and 5) oxidation-reduction reactions (electron-transfer reactions). Oxidation-reduction reactions can be categorized as either single replacement or combustion. Each reaction type will be addressed in more detail in later sections, so let us consider each type of reaction in minimal detail in this section.

Precipitation Reactions

A reaction that involves two aqueous salts being added together to form spectator ions and a solid salt precipitate that drops out of solution is known as a *precipitation reaction*. It may also be referred to as a *double-displacement reaction*, although that term is not as useful in describing the chemistry. Drawn below is a sample precipitation reaction:

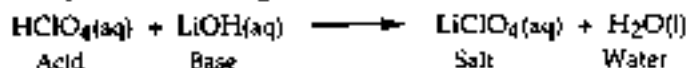


Precipitation reactions can be recognized by the solid salt on the product side of the equation. Recognition of the type of reaction is useful for predicting the product. Recognizing a precipitate is highly beneficial in identifying a double-displacement reaction. The following solubility rules can be helpful in identifying the likelihood of a precipitate's forming:

1. Most salts containing alkali metal cations (Li^+ , Na^+ , K^+ , Cs^+ , Rb^+) and ammonium (NH_4^+) are water-soluble.
2. Most nitrate (NO_3^-) salts are water-soluble.
3. Most salts containing halide anions (Cl^- , Br^- , I^-) are water-soluble (with heavy metal exceptions such as Ag^+ and Pb^{2+}).
4. Most salts containing sulfate anions (SO_4^{2-}) are water-soluble (with exceptions such as Ba^{2+} , Pb^{2+} , Hg^{2+} , and Ca^{2+}).
5. Most hydroxide anion (OH^-) salts are only slightly water-soluble. KOH and NaOH are substantially soluble, while $\text{Ca}(\text{OH})_2$, $\text{Sr}(\text{OH})_2$, and $\text{Ba}(\text{OH})_2$ are fairly soluble in water.
6. Most carbonate anion (CO_3^{2-}), chromate anion (CrO_4^{2-}), phosphate anion (PO_4^{3-}), and sulfide anion (S^{2-}) salts are only slightly water-soluble.

Acid-Base Reactions

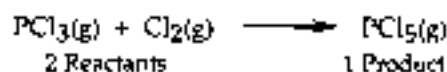
A reaction between an acid (a proton donor) and a base (a proton acceptor) forms a neutral salt and water. For now, recognize that proton donors (acids) must have a proton on an acid ($\text{H}-\text{Acid}$). Acids to recognize are HCl , HBr , HI , HNO_3 , H_2SO_4 , and NH_4^+ . Bases to recognize are NaOH , KOH , LiOH , and CaCO_3 .



Acid-base reactions can be recognized by the formation of a salt and water on the product side of the equation. Aqueous acid-base reactions can be identified by the transfer of an H from the acid to the hydroxide of the base on the reactant side of the equation.

Composition Reactions

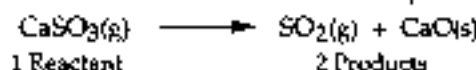
A composition reaction involves the combining of reactants to form a product. The number of reactants exceeds the number of products in a composition reaction. Entropy decreases and more bonds are formed than are broken in composition reactions.



Composition reactions may fall into other reaction categories as well. In the sample reaction, when PCl_3 reacts with Cl_2 , PCl_3 is oxidized and Cl_2 is reduced.

Decomposition Reactions

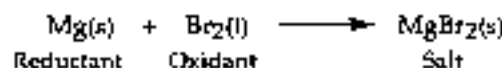
A decomposition reaction is the opposite of a composition reaction. It involves reactants decomposing to form multiple products. The number of reactants is less than the number of products in a decomposition reaction. Entropy increases and more bonds are broken than are formed in decomposition reactions.



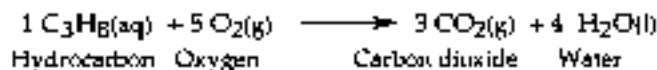
Like composition reactions, decomposition reactions can also fall into other reaction categories as well.

Oxidation-Reduction Reactions

An oxidation-reduction reaction involves the transfer of electrons from one atom to another. Loss of electrons is defined as oxidation, while gain of electrons is defined as reduction (LeoGer). The atom (or compound) losing electrons is causing reduction, so it is referred to as the *reductant* (reducing agent), while the atom (or compound) gaining electrons is causing oxidation, so it is referred to as the *oxidant* (oxidizing agent). The oxidation states must change in an oxidation-reduction reaction. A sample reaction (below) shows how magnesium is losing electrons (thus being oxidized and having an increase in oxidation state) to bromine (which is being reduced and having a decrease in oxidation state):

**Combustion Reactions**

Combustion reactions are a special case of oxidation-reduction reactions, where the oxidizing agent is oxygen gas, and the products are oxides. Typical examples of combustion reactions include the oxidation of organic compounds, such as hydrocarbons and carbohydrates, into carbon dioxide and water.

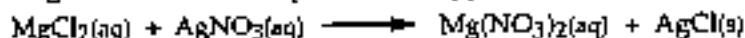


Combustion reactions of both hydrocarbons and monosaccharides balance in a predictable manner, as shown below:

Hydrocarbon CombustionMonosaccharide Combustion

Example 1.33

The following reaction is an example of which type of reaction?



- A. Cation-crossover
- B. Oxidation-reduction
- C. Neutralization
- D. Double-displacement

Solution

Of the generic reactions with which you are familiar, there are typical features to note. In this example, you have two salts undergoing an exchange reaction to yield a precipitate. This makes it a precipitation reaction. No proton was transferred (eliminating neutralization), and no oxidation states changed (eliminating oxidation-reduction). Cation-crossover is a fabricated name, so choice A is eliminated. Choice D, double-displacement, is another name for a precipitation reaction. In double-displacement reactions, you have two salts undergoing a reaction where they exchange counterions, and one of the new combinations forms a precipitate. This is shown in a generic fashion below:

**Oxidation States**

Assigning an oxidation state to an atom is a matter of distributing electrons within a bond based on which atom is more electronegative. The oxidation state of an atom can be determined by assigning it a value of positive one (+1) for every bond it forms with a more electronegative atom and assigning it a value of negative one (-1) for every bond it forms with a less electronegative atom. The oxidation state is a sum of all these bonding values. In general chemistry, it is often easiest to say oxygen is -2 (except in molecular oxygen and peroxides), hydrogen is +1 (except in molecular hydrogen and hydrides), and halides are -1 (except when they are a central atom in an oxyacid). The sum of the oxidation states of the elements in the compound must equal the overall charge, so the oxidation state of any remaining atom can be determined by finding the difference between its charge and the sum of the known oxidation states.

Example 1.34

What is the oxidation state of manganese in KMnO_4 ?

- A. +1
- B. +3
- C. +5
- D. +7

Solution

To simplify this example, we will consider that O = -2, and alkali metals = +1. In this case, K = +1, and there are 4 Os valued at -2 each, for a net oxidation state of -8. Summing oxygen and potassium yields a total of -7. This means that for the molecule to be neutral, the Mn (the only atom remaining) must cancel out that -7 by being +7. In other words, the sum of the oxidation states equals the molecule's formal charge (zero). So in this case, the oxidation state of manganese is +7. The correct answer is choice D.

Test-Taking Tips

General Advice

The stoichiometry section of this review course is best learned by trial and error from examples (i.e., practice with many problems.) For the most part, to be successful in stoichiometry requires being fast at math and being able to see immediately what a question is asking for. These are skills that are acquired and not necessarily memorized. Keep in mind that on a multiple-choice exam, the math has been done for you, so all you need to do is approximate the answer. There is no universal shortcut that works in every situation, but finding a range within which only one answer choice fits is a good approach to most questions.

Intuition will also prove useful on the MCAT. Traditional testing at major universities generally rewards memorization over intuitive skills, but preparing for the MCAT forces you to hone your analytical and intuitive skills as you recall certain facts from memory. You should try to emphasize this thought process early and regularly throughout your review studies.

In the examples above, several topics and styles of questions were presented. Before you move on to the practice questions in the passages, make sure that you understand the basic principle of each topic and the math typically used to answer these questions. Math tricks may prove helpful, even for the conceptual questions in stoichiometry. Keep in mind that you are *not* graded for showing your work on the MCAT, so don't solve every problem to the last decimal place. Analyze each question only well enough to eliminate three wrong answers. Be concise and efficient in your problem-solving, not exhaustive. Generally, the questions ask you to decide which fraction (or ratio) is larger. This can be done easily by converting all the fractions to values over the same denominator, and looking for an answer choice that falls within a range. Keep it simple.

Mathematical Tips and Shortcuts

While the MCAT does not require elaborate calculations, you still *must* be able to deal with ratios and percentages. Do not use a calculator when practicing for the MCAT. The following strategies are useful ways to calculate a value quickly and to a fair approximation without tables or a calculator:

Addition and Subtraction

Splitting numbers and adding common terms is a useful way to make addition and subtraction easier. To split a number, consider how you would round it, and then split it into the rounded number and the difference between the original and rounded numbers. 193 rounds up to 200, so 193 can be thought of as $200 - 7$. 826 rounds down to 800, so think of 826 as $800 + 26$. Adding and subtracting requires linking like terms. Thus, adding 193 to 826 can be thought of as:

$$193 + 826 = 200 - 7 + 800 + 26 = 200 + 800 + 26 - 7 = 1000 + 19 = 1019$$

Subtraction is done in a similar way. $826 - 193$ can be thought of as:

$$826 - 193 = 800 + 26 - 200 + 7 = 800 - 200 + 26 + 7 = 600 + 33 = 633$$

This approach may seem awkward at first, but it is effective and easy when adding and/or subtracting several numbers at once. For instance, consider adding 213 to 681, then subtracting 411.

$$\begin{aligned} 213 + 681 - 411 &= 200 + 13 + 700 - 19 - 400 - 11 = 200 + 700 - 400 + 13 - 19 - 11 \\ &= 500 - 17 = 483 \end{aligned}$$

Averaging Terms

Averaging terms involves estimating a mean value, and then keeping a running tally of the differences between the actual values and the estimated average. To find the average difference, the running tally is divided by the number of values being averaged. For instance, the average of 25, 33, 21, 28, and 30 can be thought of as being around 28 (the median value), so the actual average is $28 \pm$ the average difference:

$$\text{The total difference is } -3 + 5 - 7 + 0 + 2 = -3$$

When the total difference is divided by 5, it yields an average difference of -0.6 .

$$\text{The average of the five values is thus } 28 - 0.6 = 27.4$$

Multiplication

Multiplication can also be made easier by splitting numbers as you would round them. For instance, 97 is $100 - 3$. Only one number need be split in multiplication. Thus, multiplying 97 by 121 can be thought of as:

$$\begin{aligned} 97 \times 121 &= (100 - 3) \times 121 = (100 \times 121) - (3 \times 121) = 12,100 - 363 \\ 12,100 - 363 &= 11,700 + 400 - 363 = 11,700 + 37 = 11,737 \end{aligned}$$

Division

If you memorize the following set of fraction-to-decimal conversions, then problems involving division will be far easier:

$$\frac{1}{5} = 0.200, \frac{1}{6} = 0.166, \frac{1}{7} = 0.143, \frac{1}{8} = 0.125, \frac{1}{9} = 0.111, \frac{1}{11} = 0.091, \frac{1}{12} = 0.083$$

Memorizing these decimal values can be useful in several ways. For instance, the decimal equivalent of the fraction $18/66$ can be found in the following manner:

$$\frac{18}{66} = \frac{3}{11} = 3 \times \frac{1}{11} = 3 \times (0.091) = 0.273$$

Knowing these decimal values is also useful for estimating in decimal terms fractions that are just less than 1. For instance, the decimal equivalent of the fraction $11/12$ can be found in the following manner:

$$\frac{11}{12} = \frac{12-1}{12} = \frac{12}{12} - \frac{1}{12} = 1 - \frac{1}{12} = 1 - 0.083 = 0.917$$

These decimal values are also useful in deciding what to multiply a denominator by to convert it to some number close to 10, 100, or 1000. For instance, the numerator and denominator of the fraction $47/142$ should be multiplied by 7, because $0.143 = 1/7$ so 7×143 is nearly 1000 (actually, it's 1001). The decimal equivalent of the fraction $47/142$ can be found in the following manner:

$$\frac{47}{142} = \frac{7 \times 47}{7 \times 142} = \frac{329}{994} = \frac{329}{1000} + \text{a little} = 0.329 + \text{a little}$$



Stoichiometry Passages

12 Passages
100 Questions

Suggested Stoichiometry Passage Schedule:

- I:** After reading this section and attending lecture: Passages I - III & VI - VIII
Grade passages immediately after completion and log your mistakes.
- II:** Following Task I: Passages IV, V, & IX, (20 questions in 26 minutes)
Time yourself accurately, grade your answers, and review mistakes.
- III:** Review: Passages X - XII & Questions 87 - 100
Focus on reviewing the concepts. Do not worry about timing.



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Stoichiometry Study Passages

I. Density Experiment	(1 - 7)
II. Combustion Analysis	(8 - 13)
III. Empirical Formula Determination	(14 - 20)
IV. Molar Volume of a Gas	(21 - 26)
V. Elemental Analysis	(27 - 33)
VI. Dilution Experiment	(34 - 40)
VII. Solution Concentrations and Dilution	(41 - 47)
VIII. Beer's Plot and Light Absorption	(48 - 54)
IX. Beer's Law Experiment	(55 - 61)
X. Reaction Types	(62 - 68)
XI. Calcium-Containing Bases	(69 - 78)
XII. Industrial Chemicals	(79 - 86)
Questions Not Based on a Descriptive Passage	(87 - 100)

Stoichiometry Scoring Scale

Raw Score	MCAT Score
84 - 100	13 - 15
66 - 83	10 - 12
47 - 65	7 - 9
34 - 46	4 - 6
1 - 33	1 - 3

Passage I (Questions 1 - 7)

A student fills a 50-mL graduated cylinder exactly halfway with water, adds a previously weighed sample of an unknown solid, and records the new water level indicated by the markings on the side of the graduated cylinder. After recording the volume, she removes the unknown solid and adds water to the cylinder to raise the volume back to precisely 25 mL, replacing any water that may have adhered to the solid. This procedure is repeated for a total of five unknown solids, and it is discovered that each of the solids sinks to the bottom of the graduated cylinder. Table 1 shows the data for all five trials.

Unknown	Mass	Volume Reading
1	9.63 g	31.42 mL
2	12.38 g	31.19 mL
3	14.85 g	29.95 mL
4	8.22 g	28.00 mL
5	5.64 g	26.41 mL

Table 1

A second experiment is conducted with liquids, using a 10.00-mL volumetric cylinder (one that holds exactly 10.00 mL of solution) that weighs 42.61 grams when empty. In four separate trials, unknown liquids are poured into the cylinder exactly to the 10.00-mL mark on the cylinder each time, and the combined mass of the cylinder and the liquid is recorded. Table 2 shows the results of the second experiment.

Unknown	Mass of Cylinder with Liquid
6	51.33 g
7	58.72 g
8	53.21 g
9	49.03 g

Table 2

1. Given that all of the unknown liquids are immiscible (will not dissolve) in water, how many of the unknown liquids can float on water?

A. 1
B. 2
C. 3
D. 4

2. Which of these unknown solids is the DENSEST?

A. Solid #2
B. Solid #3
C. Solid #4
D. Solid #5

3. What would be the volume of a 20.0-gram piece of unknown Solid #1?

A. 13.3 mL
B. 15.0 mL
C. 25.0 mL
D. 30.0 mL

4. How many of the unknown solids can float on Liquid #7?

A. 0
B. 1
C. 2
D. 3

5. Which of the following sequences does NOT accurately reflect the relative densities of the unknown liquids?

A. Liquid #7 > Liquid #6 > Liquid #8
B. Liquid #8 > Liquid #6 > Liquid #9
C. Liquid #7 > Liquid #8 > Liquid #9
D. Liquid #7 > Liquid #8 > Liquid #6

6. How would the results in Experiment 2 differ from the actual results, if a heavier graduated cylinder had been used?

A. Both the mass of the cylinder with the liquid and the density of the liquid would increase.
B. The mass of the cylinder with the liquid would increase, while the density of the liquid would decrease.
C. The mass of the cylinder with the liquid would increase, while the density of the liquid would remain the same.
D. The mass of the cylinder with the liquid would decrease, while the density of the liquid would increase.

7. Which of the following is NOT a unit of density?

A. $\frac{\text{g}}{\text{mL}}$
B. $\frac{\text{oz}}{\text{cm}^3}$
C. $\frac{\text{kg}}{\text{L}}$
D. $\frac{\text{lbs}}{\text{in}^2}$

Passage II (Questions 8 - 13)

Elemental analysis is often a preliminary study in structural analysis. A sample compound is placed into a chamber with a positive pressure of oxygen gas flowing in. The chamber has an ignition coil that is heated by a current. As the reaction proceeds, the pressure in the chamber builds up. After a short time, Valve #1 is opened to allow the product gas mixture from the reaction to flow into an evacuated tube containing some magnesium sulfate, which absorbs water vapor from the product gas mixture. Then Valve #2 is opened, allowing the gas to flow into a second evacuated tube containing some sodium hydroxide, which absorbs carbon dioxide from the product gas mixture. The apparatus is shown in Figure 1. The oxygen tank provides oxygen in excess throughout the process. Valve #3 is connected to a line that can either evacuate the system or supply nitrogen to the system.

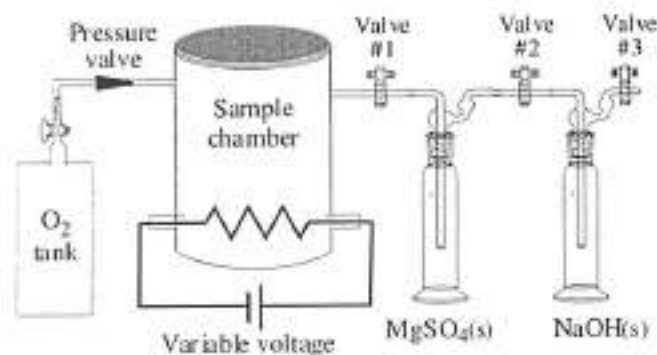


Figure 1

Four different samples are analyzed. The sample mass of each unknown substance is approximately two grams. Table 1 shows the sample mass placed into the reaction chamber, and the initial and final masses of magnesium sulfate and sodium hydroxide in the side tubes.

Unknown	Sample Mass	MgSO ₄ Tube	NaOH Tube
I	2.011 g	Init: 40.00 g Fin: 41.21 g	Init: 30.00 g Fin: 32.94 g
II	1.995 g	Init: 40.01 g Fin: 41.26 g	Init: 30.00 g Fin: 34.39 g
III	2.003 g	Init: 40.00 g Fin: 42.00 g	Init: 30.00 g Fin: 34.89 g
IV	2.001 g	Init: 40.00 g Fin: 41.99 g	Init: 30.00 g Fin: 36.53 g

Table 1

8. Which of the following relationships best describes the relative mass percent of carbon in the four unknown compounds used in the experiment?
- A. I > II > III > IV
 B. I > III > II > IV
 C. IV > II > III > I
 D. IV > III > II > I

9. Ideally, the properties of sodium hydroxide should include being:
- A. hydrophobic and semi-reactive with CO₂.
 B. hydrophobic and highly reactive with CO₂.
 C. hygroscopic and semi-reactive with CO₂.
 D. hygroscopic and highly reactive with CO₂.
10. Why is the oxygen tank attached to a pressure valve?
- A. It absorbs excess oxygen.
 B. It helps cool the reaction chamber.
 C. It ensures that oxygen gas is in excess.
 D. It ensures that oxygen is limiting.
11. Which of the four unknown compounds is LEAST likely to contain oxygen?
- A. Compound I
 B. Compound II
 C. Compound III
 D. Compound IV
12. Why are the U-tubes containing the two salts arranged in the order that they are?
- A. To ensure that water is absorbed before the gases interact with the NaOH chamber
 B. To ensure that carbon dioxide is absorbed before the gases interact with the NaOH chamber
 C. To enhance the reaction between water and carbon dioxide
 D. To absorb any excess oxygen gas before it reacts with NaOH
13. What solid is being formed in the second tube after the product gas mixture interacts with the salt?
- A. Magnesium bicarbonate
 B. Magnesium carbonate
 C. Sodium bicarbonate
 D. Sodium carbonate

Passage III (Questions 14 - 20)

Exactly 10.0 grams of an unknown organic compound is poured into a flask. The compound is then exposed to excess oxygen gas to oxidize it to CO_2 gas and H_2O gas. The oxidized vapor flows through a tube filled first with copper oxide to ensure complete oxidation. The vapor then flows through 100.00 grams of powdered anhydrous sodium sulfate, which binds water vapor to form 112.16 grams of hydrated salt. The vapor continues to flow through 100.00 grams of powdered anhydrous sodium hydroxide, which binds carbon dioxide vapor to form 123.79 grams of bicarbonate salt.

The unknown compound contains only oxygen, carbon, and hydrogen. The mass percent of carbon in the compound is determined to be greater than 50%. In a subsequent experiment, the compound is found to have a molecular mass somewhere between 70 and 80 grams per mole. When the bottle containing the unknown compound is left uncapped, its contents slowly evaporate.

The information from the combustion reaction can be converted into mass percent for both carbon and hydrogen. By multiplying the grams of CO_2 times the mass of one carbon atom and dividing by the mass of carbon dioxide, the mass of the carbon in the original sample can be determined. The mass of hydrogen in the original sample can be found in a similar manner. These two mathematical procedures convert the grams of product molecules into the grams of each atom. The final numbers are the grams of carbon and hydrogen, respectively, in the unknown compound. To determine the mass percent, the mass of the atom is divided by the mass of the sample. The mass percent of oxygen in the unknown compound is determined by difference.

The information from the mass percents of the component atoms can be used to determine the empirical formula (formula of the lowest coefficients) for the unknown compound. To determine the molecular formula from the empirical formula, the compound's molecular weight must be known. For compounds containing only carbon, hydrogen, and oxygen, the molecular formula must always have an even number of hydrogens. Molecular formulas with an odd number of carbons and oxygens, however, are possible.

14. If 20.0 grams of the unknown compound described in the passage were oxidized, what would be observed?
- A. The moles of CO_2 would double, while the percentage of carbon in the sample would remain the same.
 - B. The moles of CO_2 would double, and the percentage of carbon in the sample would also double.
 - C. The moles of CO_2 would remain the same, and the percentage of carbon in the sample would also remain the same.
 - D. The moles of CO_2 would remain the same, while the percentage of carbon in the sample would double.

15. The percentage of carbon by mass in the unknown compound can be calculated as:

- A. $23.79 \times \frac{12}{44} \times 10 \times 100\%$
- B. $23.79 \times \frac{44}{12} \times 10 \times 100\%$
- C. $23.79 \times \frac{12}{44} \times \frac{1}{10} \times 100\%$
- D. $23.79 \times \frac{44}{12} \times \frac{1}{10} \times 100\%$

16. What can be said about the boiling point (b.p.) and melting point (m.p.) of the unknown compound relative to ambient temperature (T_a)?

- A. m.p. $> T_a$, and b.p. $> T_a$
- B. m.p. $> T_a$, and b.p. $< T_a$
- C. m.p. $< T_a$, and b.p. $< T_a$
- D. m.p. $< T_a$, and b.p. $> T_a$

17. Which of the following formulas CANNOT be a molecular formula?

- A. $\text{C}_2\text{H}_4\text{O}$
- B. $\text{C}_2\text{H}_5\text{O}$
- C. $\text{C}_3\text{H}_6\text{O}$
- D. $\text{C}_4\text{H}_8\text{O}_2$

18. How many moles of water are formed from the combustion of 10.0 grams of the unknown compound?

- A. 0.31 moles $\text{H}_2\text{O}()$
- B. 0.69 moles $\text{H}_2\text{O}()$
- C. 1.10 moles $\text{H}_2\text{O}()$
- D. 1.38 moles $\text{H}_2\text{O}()$

19. What is the empirical formula for the unknown compound?

- A. $\text{C}_3\text{H}_6\text{O}_2$
- B. $\text{C}_4\text{H}_8\text{O}_2$
- C. $\text{C}_4\text{H}_{10}\text{O}$
- D. $\text{C}_8\text{H}_{10}\text{O}$

20. What is the mass percent of carbon in $\text{C}_3\text{H}_{12}\text{O}_2$?

- A. 26.4%
- B. 57.7%
- C. 60.8%
- D. 68.2%

Passage IV (Questions 21 - 26)

A researcher completely oxidizes exactly 1.00 grams of an unknown liquid hydrocarbon in a containment vessel to yield carbon dioxide and water vapor. The two gases thus formed are collected and analyzed for quantity. The water vapor is collected by passing the gas through a tube containing anhydrous calcium chloride. The carbon dioxide gas is collected by passing the remaining gas through a tube containing anhydrous sodium hydroxide. The mass of the carbon dioxide gas thus collected is 3.045 grams at STP. The carbon dioxide gas is regenerated upon heating the sodium carbonate and this gas is found to occupy a volume of 1.55 liters at STP. The experimental apparatus is shown in Figure 1.

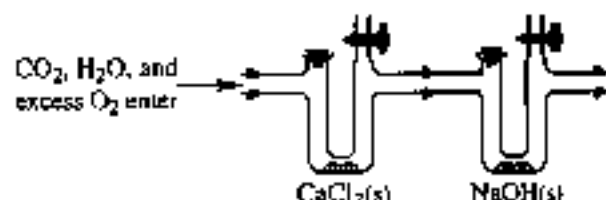


Figure 1

In a second experiment, the researcher places a 5.0-mL aliquot of the unknown liquid into a capped 1.00-liter flask. The cap has a tiny hole in the top, and the empty flask with cap weighs exactly 120.00 grams. The compound is heated until it reaches a gentle boil. The vapor escapes through the tiny pore in the cap. The liquid continues boiling at 31°C, until none of it remains visible in the flask. The heat source is removed from the flask, and the contents are allowed to cool back to ambient temperature. As the flask cools, the vapor in it condenses into a small pool of liquid at the base of the flask.

The flask and cap are then massed with the condensed liquid present. The entire system is found to have a mass of exactly 122.32 grams. This means that the mass of the liquid is 2.32 grams. It is assumed that at the moment when the heat source was removed, the flask was completely filled with vapor from the liquid and that all of the air originally in the flask was displaced. Table 1 lists the molar volume for an ideal gas at selected temperatures.

Temperature (K)	Molar Volume
273	22.41 L
288	23.64 L
298	24.46 L
304	24.96 L
313	25.69 L
323	26.51 L

Table 1

21. What is the mass percent of carbon in CO_2 gas?
 - A. 25.0%
 - B. 27.3%
 - C. 31.4%
 - D. 35.0%
22. How can the molecular weight of this unknown liquid be determined?
 - A. $\frac{22.41 \text{ grams}}{2.32 \text{ mole}}$
 - B. $\frac{24.96 \text{ grams}}{2.32 \text{ mole}}$
 - C. $(2.32 \times 22.41) \frac{\text{grams}}{\text{mole}}$
 - D. $(2.32 \times 24.96) \frac{\text{grams}}{\text{mole}}$
23. If the mass percent of carbon in the unknown compound is found to be 82.9%, what is the empirical formula of the unknown hydrocarbon?
 - A. CH_2
 - B. C_2H_5
 - C. CH_3
 - D. C_2H_7
24. Using the data from the first experiment, how can the mass percent of carbon in the unknown compound be determined?
 - A. $\frac{1.55 \times 12.011}{22.41 \times 1.00} \times 100\%$
 - B. $\frac{1.55 \times 22.41 \times 12.011}{1.00} \times 100\%$
 - C. $\frac{22.41 \times 12.011}{1.55 \times 1.00} \times 100\%$
 - D. $\frac{22.41 \times 1.00}{1.55 \times 12.011} \times 100\%$
25. How many moles of CO_2 gas were formed in the first experiment?
 - A. $\frac{1.55}{22.41}$ moles CO_2
 - B. $\frac{1.55}{23.64}$ moles CO_2
 - C. $\frac{22.41}{1.55}$ moles CO_2
 - D. $\frac{1.00}{22.41 \times 1.55}$ moles CO_2

26. If in the second experiment the organic vapor had not fully displaced all of the air from the flask by the time the heat was removed from the flask, how would the results have been affected?

- A. The mass of unknown liquid collected would be too great, so the calculated molecular mass would be too high.
- B. The mass of unknown liquid collected would be too small, so the calculated molecular mass would be too high.
- C. The mass of unknown liquid collected would be too great, so the calculated molecular mass would be too low.
- D. The mass of unknown liquid collected would be too small, so the calculated molecular mass would be too low.

Passage V (Questions 27 - 33)

The empirical formula for a compound can be determined using the technique of elemental analysis. For hydrocarbons and carbohydrates, the process involves trapping and removing water vapor and carbon dioxide gas and then quantifying the amounts collected. The products can be trapped in many different ways. In this experiment, the trapping of the gases is accomplished by passing the product gas through a series of low-temperature gas traps. By lowering the temperature, the gas can be converted into solids, which cannot flow and thus are easily collected at the bases of their respective temperature traps. The system is evacuated completely and then flushed with nitrogen gas to remove any remaining air. The vapor from the reaction vessel is then allowed to enter one trap at a time. The gas remains isolated in the region of each trap for a short interval, to allow any gases to form a solid or freeze into a liquid. The apparatus is shown in Figure 1.

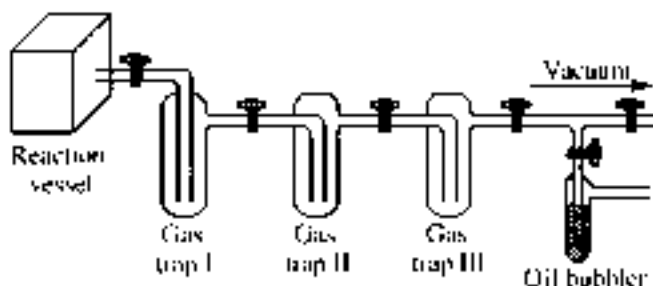


Figure 1

It is important to allow the excess oxygen gas to flow out of the system. To accomplish this, the line is fitted with a one-way oil bubbler. The oil bubbler maintains the closed system by not allowing air to flow into the system, while allowing the pressure to equilibrate with the environment through venting.

27. At what temperature should the first trap be held in order to isolate water vapor?
- A. 25°C (standard temperature)
 - B. 0°C (melting point of ice)
 - C. -33°C (boiling point of Freon refrigerant)
 - D. -196°C (boiling point of liquid nitrogen)
28. The temperatures of the successive traps (i.e., Trap I, Trap II, and Trap III) should be set in what manner?
- A. The temperatures should gradually increase, so that each gas is selectively removed one trap at a time.
 - B. The temperatures should gradually increase, so that each gas can be trapped into all three traps, allowing one to determine the moles by difference.
 - C. The temperatures should gradually decrease, so that each gas is selectively removed one trap at a time.
 - D. The temperatures should gradually decrease, so that each gas can be trapped into all three traps, allowing one to determine the moles by difference.

29. What additional piece of information is necessary to determine the molecular formula for the experimental compound?
- A. The volume of CO_2 collected
 - B. The volume of the water collected
 - C. The volume of the hydrocarbon before the reaction was carried out
 - D. The molecular mass of the hydrocarbon
30. If an unknown compound were combusted in the presence of excess oxygen, what by-product of the combustion would be collected to determine the percentage of sulfur within that compound?
- A. SO_2
 - B. CS_2
 - C. H_2S
 - D. S_8
31. The mass percent of oxygen within a compound cannot be determined directly using elemental analysis. Which of the following is NOT an explanation for this?
- A. Oxygen gas does not exist in the solid phase at any temperature.
 - B. When a carbohydrate is oxidized, the oxygen of the unknown carbohydrate can be found in both water and carbon dioxide.
 - C. The procedure requires adding excess oxygen, so the oxygen atoms from the carbohydrate cannot be distinguished from the oxygen reactant.
 - D. Oxygen in the carbohydrate, being fully reduced, does not react with oxygen gas.
32. Why is the bubbler filled with mineral oil?
- A. The oil traps out any unreacted organic vapor.
 - B. The oil can transfer heat to warm the gas rapidly.
 - C. The oil prevents back-flow of gas from the outside environment.
 - D. The oil filters out any liquid products from the reaction.

33. Which of the following is NOT associated with an increasing mass percent of carbon in a hydrocarbon?
- I. An increase in the mass of carbon per gram of the compound
 - II. An increase in the mass of water formed upon oxidation of one gram of the compound
 - III. An increase in the mass of hydrogen per gram of the compound
- A. II only
 - B. III only
 - C. I and II only
 - D. II and III only
-

Passage VI (Questions 34 - 40)

Dilution reduces the concentration of a solute by adding more solvent to the solution. The addition of solvent increases the volume of solution while having no effect on the moles of solute. Molarity is defined as moles solute per liter solution, so the denominator is increased by the addition of solvent, while the numerator is unaffected. To determine the concentration, use:

$$M_i V_i = M_f V_f$$

Equation 1

where M_i is the initial molarity, V_i is the initial volume, M_f is the final molarity, and V_f is the final volume.

Dilution can be described by the relative concentration of the initial and final solutions. For instance, a fifty percent dilution involves a reduction of the molarity by fifty percent. This would result from doubling the volume of the solution, achieved by mixing one part solvent with one part solution.

A solution is diluted as a solvent is added to it in a volumetric flask, until the desired volume is reached. To ensure complete transfer of the solute, the original flask is often rinsed with the new solvent, and then the contents are poured into the volumetric flask. The laboratory instructions for a tenfold dilution are:

- I: Fill a volumetric pipette with a sample of solution from a beaker and then discard the solution. Repeat this procedure two additional times to equilibrate the concentration of the solution on the walls of the pipette with the concentration of the solution in the beaker.
- II: Using the treated volumetric pipette, transfer ten milliliters of solution to a 100-mL volumetric flask.
- III: Rinse pure water through the pipette and into the volumetric flask until the flask is roughly eighty percent full.
- IV: Set the pipette aside, and continue to add water to the flask until the base of the meniscus is flush with the 100-mL line on the volumetric flask.

34. Which of the following solutions has the GREATEST molarity?
- A. 4% by mass KBr in water
 - B. 4% by mass KCl in water
 - C. 4% by mass NaBr in water
 - D. 4% by mass NaCl in water
35. To convert 300 milliliters of 0.150 M solution to a solution with a concentration of 0.0075 M, how much water must be added?
- A. 6.00 liters
 - B. 5.70 liters
 - C. 4.30 liters
 - D. 4.20 liters

36. What is the final Cl^- concentration after you mix 50.00 mL 0.25 M HCl with 25.00 mL 0.50 M NaOH?
- A. 0.33 M
 - B. 0.25 M
 - C. 0.17 M
 - D. 0.15 M
37. Why in step III is water passed through the volumetric pipette?
- A. To ensure complete transfer of solution
 - B. To measure the volume of the water added
 - C. To cool the volumetric pipette
 - D. To warm the water prior to mixing
38. Which of the following mixtures results in a 10-fold dilution?
- A. 9 parts solvent with 1 part solution
 - B. 10 parts solvent with 1 part solution
 - C. 10% solvent with 90% solution
 - D. 91% solvent with 9% solution
39. Addition of water to an aqueous salt solution would do all of the following EXCEPT:
- A. lower the molality.
 - B. lower the molarity.
 - C. increase the density.
 - D. increase the mass percent of solvent.
40. Which of the following would MOST dilute 0.10 M LiCl(aq) ?
- A. The addition of 100 mL $\text{H}_2\text{O(l)}$ to 25 mL 0.10 M LiCl(aq)
 - B. The addition of 200 mL $\text{H}_2\text{O(l)}$ to 60 mL 0.10 M LiCl(aq)
 - C. The addition of 50 mL $\text{H}_2\text{O(l)}$ to 15 mL 0.10 M LiCl(aq)
 - D. The addition of 150 mL $\text{H}_2\text{O(l)}$ to 50 mL 0.10 M LiCl(aq)

Passage VII (Questions 41 - 47)

There are many ways in which the concentration of a solution can be expressed, including:

Molarity: The concentration of a solution as measured in moles solute per liter solution.

$$M = \frac{\text{moles solute}}{\text{liters solution}}$$

Molality: The concentration of a solution as measured in moles solute per kilogram solvent.

$$m = \frac{\text{moles solute}}{\text{kilogram solvent}}$$

Percent solution: The percent of solute in a solution by mass or moles.

$$\% \text{ Solution by mass} = \frac{\text{mass solute}}{\text{mass solution}} \times 100\%$$

$$\% \text{ Solution by moles} = \frac{\text{moles solute}}{\text{total moles in solution}} \times 100\%$$

Density: The mass of the solution divided by the volume of the solution.

$$\rho = \frac{\text{mass solution}}{\text{volume solution}}$$

The concentration of a solution can be expressed in any of these units, which can be converted into one another as long as the molecular mass of the solute and solvent are known. For instance, when the percent solution by mass is multiplied by the density, the result is mass of solute per volume of solution. When the mass of solute is converted into the moles of solute (which requires knowing the molecular mass), the molarity can be determined. The percent solution by mass can be converted into molality by subtracting the mass of solute from the mass of solution to find the mass of solvent. That determines the denominator. To get the numerator, the mass of solute is converted into moles solute, and solving for the molality becomes a simple division problem.

Adding solvent to a solution dilutes the solution and thus reduces the concentration of the solute in the solution. Addition of solvent to the solution decreases all of the above measurements of concentration, with the exception of the density. The density change of a solution depends on the relative density of the solvent and solution.

41. Adding water to an aqueous solution of known concentration always decreases all of the following EXCEPT:

- A. density.
- B. molarity.
- C. molality.
- D. mass percent of the solute.

42. When 1.0 grams of a salt are dissolved into 100 mL of water, the volume of the solution is greater than 100 mL but less than 101 mL. What can be said about the concentration of the solution?

- A. The molality of the solution is greater than the molarity of the solution; the density of the solution is greater than that of pure water.
- B. The molarity of the solution is greater than the molality of the solution; the density of the solution is greater than that of pure water.
- C. The molality of the solution is greater than the molarity of the solution; the density of the solution is less than that of pure water.
- D. The molarity of the solution is greater than the molality of the solution; the density of the solution is less than that of pure water.

43. An organic compound with a density that is less than 1.00 g/mL is added to an organic liquid, also with a density that is less than 1.00 g/mL. What can be said about the concentration of the solution?

- A. The molality of the solution is greater than the molarity of the solution; the density of the solution is greater than that of pure organic liquid.
- B. The molarity of the solution is greater than the molality of the solution; the density of the solution is greater than that of pure organic liquid.
- C. The molality of the solution is greater than the molarity of the solution; the relative densities of the solution and the organic liquid cannot be determined without more information.
- D. The molarity of the solution is greater than the molality of the solution; the relative densities of the solution and the organic liquid cannot be determined without more information.

44. Given that a solute is denser than the solvent into which it dissolves, what is TRUE of the concentration measurements of different solutions made up solely of the two components?

- A. The solution with the greatest density also has the greatest molarity and molality.
- B. The solution with the greatest density also has the greatest molarity, but the molality is the same for all of the solutions.
- C. The solution with the greatest density also has the lowest molarity and molality.
- D. The solution with the greatest density also has the lowest molarity, but the molality is the same for all of the solutions.

45. How many milliliters of pure water must be added to 100 mL 0.25 M KBr to dilute it to 0.10 M?

- A. 100
- B. 150
- C. 250
- D. 500

46. To achieve the same chloride ion concentration as 1.0 grams NaCl(s) dissolved into 100 mL solution, how many grams of $\text{MgCl}_2(\text{s})$ must be added to enough water to make 100 mL of solution?

- A. $1.0 \times \frac{1}{2} \times \frac{58.4}{94.9}$ g $\text{MgCl}_2(\text{s})$
- B. $1.0 \times 2 \times \frac{58.4}{94.9}$ g $\text{MgCl}_2(\text{s})$
- C. $1.0 \times \frac{1}{2} \times \frac{94.9}{58.4}$ g $\text{MgCl}_2(\text{s})$
- D. $1.0 \times 2 \times \frac{94.9}{58.4}$ g $\text{MgCl}_2(\text{s})$

47. Given two compounds, Compound A and Compound B, and the fact that B has a higher molecular mass than A, but A is denser than B, which of the following mixtures would have the greatest mole fraction of A?

- A. The mixture of 1.0 grams Compound A with 1.0 grams Compound B
- B. The mixture of 1.0 moles Compound A with 1.0 moles Compound B
- C. The mixture of 1.0 mL Compound A with 1.0 mL Compound B
- D. The mixture of 1.0×10^{23} molecules Compound A with 1.0×10^{23} molecules Compound B

Passage VIII (Questions 48 - 54)

The absorbance of visible light by colored aqueous solutions is directly proportional to the concentration of solute in the solution. Based on this fact, the concentration of a solution can be determined by monitoring the absorbance at one wavelength of light. For best results, the detector should be focused on the wavelength of highest absorbance (known as λ_{max}). The relationship between absorbance and solute concentration is expressed as

$$\text{Absorbance} = \epsilon[C]l$$

Equation 1

where ϵ is the molar absorptivity constant of the solute, $[C]$ is the concentration of solute, and l is the path length of the light passing in through the cuvette.

A student measures the absorbance for a series of standard solutions. Once enough data points are collected, the molarity of another solution using the same solute in an unknown concentration is analyzed by comparing its properties with the experimental data. The molar absorptivity constant and cuvette path length remain constant throughout all the trials, so any difference in absorbance between the unknown and the reference compounds can be attributed to differences in solute concentration. Figure 1 is a graph of the student's data collected for the standard solutions.

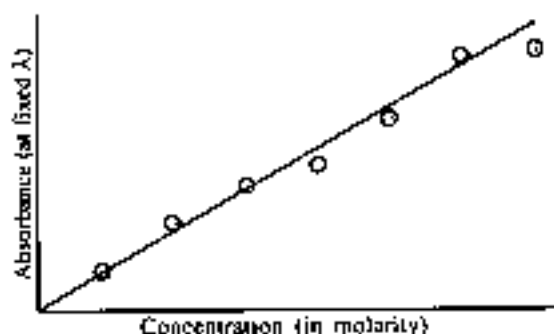


Figure 1

Table 1 lists the same data summarized graphically above.

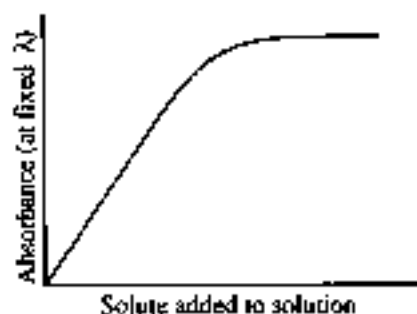
Molarity	Absorbance
0.10	0.093
0.20	0.188
0.30	0.278
0.40	0.363
0.50	0.456
0.60	0.560
0.70	0.636

Table 1

48. If the concentration of a solute were doubled, what would happen to the absorbance of the solution?

- A. It would increase by a factor of four.
- B. It would double.
- C. It would be cut in half.
- D. It would decrease by a factor of four.

49. How could the plateau of the following curve be explained?



- A. The solvation catalyst in solution has become saturated.
 B. The reverse reaction is favored at higher solute concentration.
 C. As more solute is added to the solution, the solute that is already dissolved begins to repel the solvent.
 D. A maximum solute concentration has been reached, because no more solute molecules can dissolve into solution.
50. Adding 50 mL of pure water to a 10.0-mL sample of aqueous salt solution with an absorbance of 0.518 would yield a new absorbance of:
- A. 0.518.
 B. 0.104.
 C. 0.086.
 D. 0.259.
51. This concentration can be found according to which of the following equations?
- A. $[C] = \frac{Abs}{\epsilon \cdot l}$
 B. $[C] = \frac{\epsilon \cdot l}{Abs}$
 C. $[C] = \frac{Abs \cdot l}{\epsilon}$
 D. $[C] = \frac{Abs \cdot \epsilon}{l}$
52. According to the data from the experiment, what is the concentration of an unknown solution, if it has an absorbance of 0.242?
- A. 0.197 M
 B. 0.240 M
 C. 0.258 M
 D. 0.289 M

53. What are the units of ϵ ?

- A. M-cm
 B. M-cm⁻¹
 C. cm-M⁻¹
 D. $\frac{l}{M \cdot cm}$

54. Which of the following relationships may be TRUE?

- I. As the molarity increases, the absorbance increases.
 II. If Compound X has a lower molar absorptivity constant (ϵ) than Compound Y, then to have equal absorbance readings for separate solutions of X(aq) and Y(aq), the concentration of Y must be greater than the concentration of X.
 III. Absorbance = $\epsilon[C]$ at all λ where absorbance of light can occur.
- A. I only
 B. II only
 C. I and III only
 D. II and III only

Passage IX (Questions 55 - 61)

An experiment to ascertain the effects of solution concentration on the absorbance of visible light studies solutions of varying concentration for three different compounds. The solutions are analyzed in cuvettes using a UV-VIS spectrometer set at fixed values specific for each compound. The goal is to maximize the absorbance, so the wavelength of maximum absorbance is used for each compound. For Compound M, the spectrometer was set at 561 nm; for Compound Q, the spectrometer was set at 413 nm; and for Compound T, the spectrometer was set at 697 nm. Table 1 shows concentration and corresponding absorbance for each solution.

Trial	Contents	Absorbance
I	0.10 M Compound M	0.362
II	0.10 M Compound Q	0.299
III	0.10 M Compound T	0.511
IV	0.06 M Compound M	0.217
V	0.06 M Compound Q	0.180
VI	0.06 M Compound T	0.307
VII	0.03 M Compound M	0.109
VIII	0.03 M Compound Q	0.090
IX	0.03 M Compound T	0.153

Table 1

The absorbance of each solution was compared to the absorbance of a sample of distilled water, which remained in place in the spectrometer for the duration of the study. Because all three compounds have absorbance bands in the visible range, they are all observed to have a distinct color when they are exposed to white light. The absorbed color is the complementary color of the observed color. The visible spectrum ranges from a wavelength of 400 nm to 700 nm.

55. Why is the spectrophotometer set at 561 nm for the trial involving Compound M?

- A. 561 nm is the average wavelength of absorbance for the complementary color of what is absorbed.
- B. 561 nm is the average wavelength of absorbance for the color that is absorbed.
- C. 561 nm is the wavelength of maximum absorbance for the complementary color of what is absorbed.
- D. 561 nm is the wavelength of maximum absorbance for the color that is absorbed.

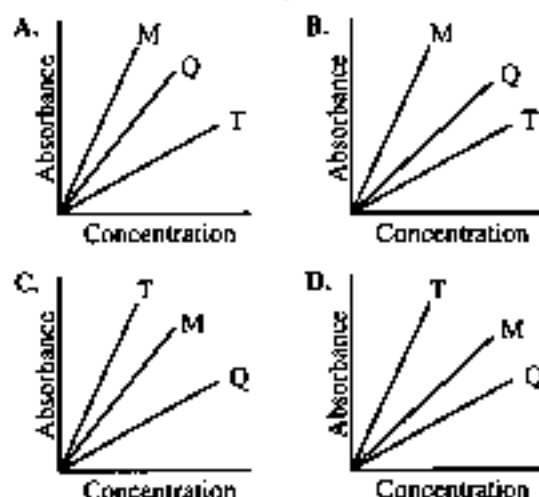
56. How can a compound's molar absorptivity constant be obtained, if absorbance varies with cuvette length?

- A. $\epsilon = \text{Abs} \cdot [\text{Compound}] \cdot l$
- B. $\epsilon = \frac{\text{Abs}}{[\text{Compound}] \cdot l}$
- C. $\epsilon = \frac{[\text{Compound}] \cdot l}{\text{Abs}}$
- D. $\epsilon = \frac{\text{Abs}}{l}$

57. Which solution has the GREATEST concentration?

- A. Compound M solution, with an absorbance 0.400
- B. Compound Q solution, with an absorbance 0.250
- C. Compound T solution, with an absorbance 0.500
- D. 0.10 M KCl(aq)

58. Which graph accurately shows absorbance as a function of concentration for Compounds M, Q, and T?



59. What can be expected for other solutions?

- I. 0.05 M Compound M has an absorbance of $A = 0.181$ at $\lambda = 561$ nm.
 - II. A solution of Compound Q with an absorbance of $A = 0.225$ at $\lambda = 413$ nm has a concentration of 0.075 M.
 - III. 0.11 M Compound T has an absorbance of $A = 0.611$ at $\lambda = 710$ nm.
- A. I only
 - B. I and II only
 - C. I and III only
 - D. II and III only

60. To form a solution of Compound T with an absorbance of 0.250 at $\lambda = 697$ nm, what must be done?

- A. Mix 10.0 mL 0.10 M T with 5.0 mL H_2O
- B. Mix 20.0 mL 0.10 M T with 20.0 mL H_2O
- C. Mix 19.0 mL 0.10 M T with 20.0 mL H_2O
- D. Mix 10.0 mL 0.10 M T with 9.0 mL H_2O

61. What are the observed colors for each solution?

- A. M: green; Q: violet; T: red
- B. M: red; Q: violet; T: green
- C. M: green; Q: yellow; T: red
- D. M: red; Q: yellow; T: green

Passage X (Questions 62 - 68)

Stoichiometric reactions can be classified into six categories:

Combination Reaction: Occurs with the combination of reactants to form one product.



Decomposition Reaction: Occurs with the decomposition of one reactant to form two products.



Single-Replacement Reaction: Occurs with the exchange of either the cations or the anions in a salt, but not both. A single-replacement reaction is also referred to as an *oxidation-reduction reaction*.



Metathesis Reaction: Occurs when two cations exchange their anions. At least one precipitate falls out of solution. A metathesis reaction is also referred to as a *double-displacement reaction*.



Combustion Reaction: Occurs with the addition of oxygen to a reactant to form oxidized products (usually carbon dioxide and water, when dealing with hydrocarbons and carbohydrates).

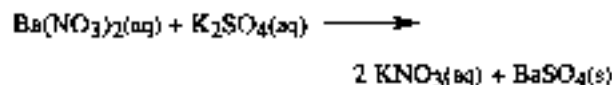


Neutralization Reaction: Occurs with the reaction of an acid with a base to form water and a salt.



Each reaction is unique from a stoichiometric perspective. When a solid is formed by the reaction of ions (in the metathesis example above), it is referred to as a *precipitate*. Gases can also be formed from reactions. The neutralization of sodium bicarbonate (NaHCO_3) yields carbon dioxide gas. It is possible to categorize all inorganic chemistry reactions by reaction type.

62. How should the following reaction be classified?



- A. Combination reaction
- B. Decomposition reaction
- C. Single-replacement reaction
- D. Metathesis reaction

63. A precipitate is LEAST likely to occur as a result of a.

- A. combustion reaction.
- B. decomposition reaction.
- C. single-replacement reaction.
- D. metathesis reaction.

64. What is the precipitate formed when aqueous sodium iodide reacts with aqueous calcium nitrate?

- A. CaI
- B. NaNO_3
- C. CaI_2
- D. Na_2NO_3

65. What is the gas formed when magnesium carbonate is treated with hydrobromic acid?

- A. Hydrogen gas (H)
- B. Hydrogen gas (H_2)
- C. Carbon dioxide gas (CO_2)
- D. Magnesium bromide gas (MgBr_2)

66. What type of reaction is LEAST likely to form carbon dioxide gas?

- A. Combustion reaction
- B. Metathesis reaction
- C. Decomposition reaction
- D. Neutralization reaction

67. How should the following reaction be classified?



- A. Combination reaction
- B. Decomposition reaction
- C. Single-replacement reaction
- D. Metathesis reaction

68. What type of reaction is MOST likely to have a negative value for the change in entropy (ΔS)?

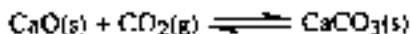
- A. Combination reaction
- B. Decomposition reaction
- C. Single-replacement reaction
- D. Combustion reaction

Passage XI (Questions 69 - 78)

Calcium carbonate is found in many everyday products, such as marble, chalk, and antacids. Commercially, it can be excavated as either calcium oxide or calcium carbonate. Calcium oxide converts to calcium carbonate upon exposure to carbon dioxide under high pressure. At room temperature, calcium carbonate is relatively insoluble in water. The solubility of calcium carbonate increases as the pH of the aqueous solution decreases, because calcium carbonate is basic. Reaction I can be combined with Reaction II to convert calcium oxide into calcium carbonate in water.

**Reaction I****Reaction II**

Ca(OH)_2 and CaCO_3 both readily form a relatively insoluble white solid precipitate in water. Because of this low solubility, the products of both Reaction I and Reaction II are easy to isolate from solution. Fortunately, calcium hydroxide is more soluble in water than is calcium carbonate, which allows for the selective precipitation of calcium carbonate in Reaction II. Because of this, industrial processes for isolating calcium carbonate are primarily water-based. Calcium carbonate can also be formed according to the following equilibrium reaction:

**Reaction III**

69. Ca(OH)_2 is considered to be which of the following?

- A. An amphoteric salt
- B. A non-metal hydroxide
- C. An Arrhenius acid
- D. An Arrhenius base

70. Which of the following molecules are NOT held together by ionic bonds?

- I. CO_3^{2-}
- II. CO_2
- III. CaO

- A. I only
- B. II only
- C. III only
- D. I and II only

71. The mass percent of calcium is LEAST in which of the following molecules?

- A. CaO(s)
- B. $\text{Ca(OH)}_2\text{(s)}$
- C. $\text{CaCO}_3\text{(s)}$
- D. $\text{CaCl}_2\text{(g)}$

72. If 10.00 grams of $\text{Ca(OH)}_2\text{(s)}$ produces 5.00 grams of $\text{CaCO}_3\text{(s)}$, what is the percent yield for the reaction?

- A. 37%
- B. 68%
- C. 74%
- D. 100%

73. What is the mass percent of calcium in CaO ?

(Ca = 40 g/mole O = 16 g/mole)

- A. 28.6%
- B. 50.0%
- C. 66.7%
- D. 71.4%

74. 28.0 grams of CaO(s) when reacted with 10.00 grams of $\text{H}_2\text{O(l)}$ would yield which of the following?

(Ca = 40 g/mole O = 16 g/mole H = 1 g/mole)

- A. 37.00 grams Ca(OH)_2 with leftover water
- B. 37.50 grams Ca(OH)_2 with leftover CaO
- C. Exactly 38.00 grams Ca(OH)_2 with no leftover
- D. Exactly 42.60 grams Ca(OH)_2 with no leftover

75. Which of the following is required to neutralize 5.00 mL of 0.20 M $\text{CaCO}_3\text{(aq)}$?

- A. 10.0 mL 0.10 N NaOH(aq)
- B. 10.0 mL 0.10 N $\text{HNO}_3\text{(aq)}$
- C. 10.0 mL 0.30 N $\text{H}_3\text{PO}_4\text{(aq)}$
- D. 10.0 mL 0.20 N HCl(aq)

76. What is the final concentration of $\text{Ca(OH)}_2\text{(aq)}$ after 50.0 mL of pure water are added to 5.0 mL of 0.50 M $\text{Ca(OH)}_2\text{(aq)}$?

- A. 0.055 M $\text{Ca(OH)}_2\text{(aq)}$
- B. 0.050 M $\text{Ca(OH)}_2\text{(aq)}$
- C. 0.046 M $\text{Ca(OH)}_2\text{(aq)}$
- D. 0.025 M $\text{Ca(OH)}_2\text{(aq)}$

77. If 20.0 mL of 0.20 M $\text{CaCl}_2(\text{aq})$ were mixed with 30.0 mL of 0.30 M $\text{CaCO}_3(\text{aq})$, what would the final Ca^{2+} concentration be?

A. 0.233 M $\text{Ca}^{2+}(\text{aq})$
B. 0.250 M $\text{Ca}^{2+}(\text{aq})$
C. 0.260 M $\text{Ca}^{2+}(\text{aq})$
D. 0.267 M $\text{Ca}^{2+}(\text{aq})$

78. If 50.0 grams of $\text{CaCO}_3(\text{s})$ are completely neutralized with HCl , how many liters of $\text{CO}_2(\text{g})$ form at STP, knowing that one CaCO_3 yields one CO_2 ?

A. 4.48 L
B. 10.00 L
C. 11.20 L
D. 13.56 L

Passage XII (Questions 79 - 96)

Every year, a substantial amount of the countless tons of chemicals produced worldwide is used in the manufacture of fertilizers and plastics. Described below, according to a common element each one contains, are some of the typical chemical fertilizers used in America.

Potassium:

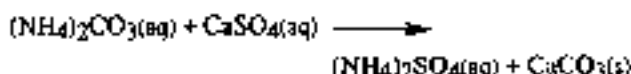
Salts containing potassium are referred to as *potash*. The most common forms are K_2SO_4 , $2 \text{MgSO}_4 \cdot \text{K}_2\text{SO}_4$, and KCl . The amount of potassium per gram of salt is important to know when determining the quantity of fertilizer needed for a job. The compound richest in potassium (by mass percent) is potassium oxide, K_2O . The potassium content of any potash is expressed as a fraction of the potassium in potassium oxide (K_2O).

Nitrogen:

Salts containing nitrogen are very useful as fertilizers. The most common forms are NH_4NO_3 , $(\text{NH}_4)_2\text{SO}_4$, and $\text{NH}_4\text{H}_2\text{PO}_4$. A very common organic fertilizer containing nitrogen is urea, H_2NCONH_2 , which is made from ammonia and carbon dioxide. Ammonium sulfate is also made from ammonia by combining Reaction I with Reaction II:



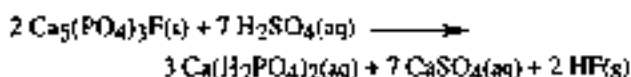
Reaction I



Reaction II

Phosphorus:

Salts containing phosphorus are also very useful as fertilizers. The most common form is $\text{Ca}(\text{H}_2\text{PO}_4)_2$. Calcium bisdihydrogenphosphate is produced by Reaction III:



Reaction III

Fluorapatite ($\text{Ca}_5(\text{PO}_4)_3\text{F}$) is added as the limiting reagent in Reaction III. This is done to maximize the percent yield of phosphorus in the reaction. Industrially, the percent yield of a reaction and the mass percent of its product are critical in terms of profit margin for the fertilizer producer.

79. Which of the following compounds has the GREATEST amount of potassium per gram?
- A. K_2O
B. K_2SO_4
C. KCl
D. KNO_3

80. Which of the following compounds has the LOWEST mass percent of nitrogen?

- A. H_2NCONH_2
- B. NH_4NO_3
- C. $(\text{NH}_4)_2\text{SO}_4$
- D. $\text{NH}_4\text{H}_2\text{PO}_4$

81. What is the mass percent of nitrogen in urea?

- A. 23.3% N
- B. 31.9% N
- C. 46.6% N
- D. 66.7% N

82. Why is $\text{Ca}_5(\text{PO}_4)_3\text{F}$ the limiting reagent in the synthesis of $\text{Ca}(\text{H}_2\text{PO}_4)_2$?

- A. It prevents leftover $\text{Ca}_5(\text{PO}_4)_3\text{F}$ from being wasted, so that P-containing compounds are conserved.
- B. It allows leftover $\text{Ca}_5(\text{PO}_4)_3\text{F}$ to be wasted, so that P-containing compounds are conserved.
- C. It prevents leftover $\text{Ca}_5(\text{PO}_4)_3\text{F}$ from being wasted, so that P-containing compounds are not conserved.
- D. It allows leftover $\text{Ca}_5(\text{PO}_4)_3\text{F}$ to be wasted, so that P-containing compounds are not conserved.

83. If 10.0 grams of $(\text{NH}_4)_2\text{CO}_3$ are used in Reaction II to obtain 10.0 grams of $(\text{NH}_4)_2\text{SO}_4$, then what is the percent yield for the reaction?

- A. Less than 50%
- B. Greater than 50%, but less than 75%
- C. Greater than 75%, but less than 100%
- D. Greater than 100%

84. The molarity of potassium is GREATEST for which of the following solutions?

- A. 10.0 g KCl in enough water to form 100 mL of solution
- B. 10.0 g K_2SO_4 in enough water to form 100 mL of solution
- C. 10.0 g K_2CO_3 in enough water to form 100 mL of solution
- D. 10.0 g KNO_3 in enough water to form 100 mL of solution

85. From which of the following reactions is it EASIEST to isolate the desired product?

- A. A reaction yielding the product as a precipitate
- B. A reaction yielding the product as a liquid
- C. A reaction yielding the product as a gas
- D. A reaction yielding the product as an aqueous solute

86. Which of the following relationships must be TRUE?

- I. $\text{p}K_{a1}(\text{H}_3\text{PO}_4)$ is less than $\text{p}K_a(\text{HF})$
- II. $\text{p}K_{a2}(\text{H}_3\text{PO}_4)$ is less than $\text{p}K_{a2}(\text{H}_2\text{SO}_4)$
- III. $\text{p}K_a(\text{HF})$ is less than $\text{p}K_{a1}(\text{H}_2\text{SO}_4)$

- A. I only
 - B. II only
 - C. I and II only
 - D. II and III only
-

Questions 87 through 100 are NOT based on a descriptive passage.

87. If 25 grams of oxygen are combined with 20 grams of propane gas, then which of the following statements would be TRUE after the mixture is ignited?
- A. 45.0 grams of carbon dioxide forms.
 - B. 38.0 grams of water vapor forms.
 - C. Oxygen is the limiting reagent.
 - D. Propane is the limiting reagent.
88. The hemoglobin in red blood corpuscles of most mammals contains approximately 0.33% iron by weight. If osmotic pressure measurements show that the molecular weight is 68,000 for hemoglobin, how many iron atoms must be present in each molecule of hemoglobin?
- A. 1
 - B. 4
 - C. 224
 - D. 400
89. 9.00 grams of a sugar are burned in a containment vessel, and all the CO_2 is collected. The volume occupied by the CO_2 at STP is 6.72 liters. If the molecular weight of the sugar is 180 g/mole, what is the ratio of O_2 to CO_2 in the balanced equation?
- A. 3 : 3
 - B. 3 : 6
 - C. 6 : 3
 - D. 6 : 6
90. What is the molecular formula for an unknown gas with the empirical formula $\text{C}_2\text{H}_3\text{O}$, if 1.00 grams of the unknown gas occupies 260 mL at STP?
- A. $\text{C}_2\text{H}_3\text{O}$
 - B. $\text{C}_4\text{H}_6\text{O}_2$
 - C. $\text{C}_6\text{H}_9\text{O}_3$
 - D. $\text{C}_8\text{H}_{12}\text{O}_4$
91. What volume of $\text{O}_2(\text{g})$ is produced from 1.0 g BaO upon its decomposition to $\text{Ba}(\text{s})$ and $\text{O}_2(\text{g})$ at STP?
- A. 0.074 L
 - B. 0.100 L
 - C. 0.148 L
 - D. 0.166 L
92. For the following reaction, calculate the mass of $\text{Mg}_2\text{P}_2\text{O}_7(\text{s})$ that is formed from the decomposition of 2.0 grams $\text{MgNH}_4\text{PO}_4(\text{s})$.
- $$2 \text{MgNH}_4\text{PO}_4(\text{s}) \longrightarrow \text{Mg}_2\text{P}_2\text{O}_7(\text{s}) + 2 \text{NH}_3(\text{g}) + \text{H}_2\text{O}(\text{g})$$
- A. 0.8 grams
 - B. 1.7 grams
 - C. 2.2 grams
 - D. 2.8 grams
93. A compound containing 50% by weight of Element X (atomic weight = 40) and 50% by weight of Element Z (atomic weight = 80) is one in which:
- A. the molecular formula is XZ or ZX .
 - B. the simplest formula is XZ or ZX .
 - C. the simplest formula is XZ_2 or Z_2X .
 - D. the simplest formula is X_2Z or ZX_2 .
94. 11.89 grams of hot iron are exposed to a continuous stream of pure oxygen for ten minutes. At the end of this time, the completely oxidized sample weighs 16.99 grams. The empirical formula for the compound thus formed is MOST accurately written as:
- A. Fe_3O_2
 - B. FeO
 - C. Fe_2O_3
 - D. FeO_3
95. A stable compound consisting of 53.4% C, 11.0% H, and the remainder O has a molecular weight of 90 grams/mole. The molecular formula for the compound is:
- A. $\text{C}_5\text{H}_6\text{O}_2$
 - B. $\text{C}_3\text{H}_6\text{O}_3$
 - C. $\text{C}_4\text{H}_{10}\text{O}$
 - D. $\text{C}_4\text{H}_{10}\text{O}_2$
96. In reducing $\text{CrO}_4^{2-}(\text{aq})$ to $\text{Cr}_2\text{O}_3(\text{s})$, how does the oxidation state of chromium change?
- A. From +6 to +3
 - B. From +4 to +3
 - C. From +3 to +4
 - D. From +3 to +6

97. An unknown metal is found to combine with oxygen in a 2 : 3 ratio in the molecular formula. The metal oxide is approximately 53% metal by mass, and the remainder is oxygen. What is the MOST probable identity of the metal?
- Calcium
 - Iron
 - Chromium
 - Aluminum

98. Which of the following organic compounds has the GREATEST mass percent of carbon?
- Acetic acid ($\text{CH}_3\text{CO}_2\text{H}$)
 - Ethanol ($\text{CH}_3\text{CH}_2\text{OH}$)
 - Methyl acetate ($\text{CH}_3\text{CO}_2\text{CH}_3$)
 - Glucose ($\text{C}_6\text{H}_{12}\text{O}_6$)

99. When one gram of each of the following organic compounds is burned (oxidized), which one yields the GREATEST amount of carbon dioxide (by mass or moles)?
- Acetic acid ($\text{CH}_3\text{CO}_2\text{H}$)
 - Ethanol ($\text{CH}_3\text{CH}_2\text{OH}$)
 - Methyl acetate ($\text{CH}_3\text{CO}_2\text{CH}_3$)
 - Glucose ($\text{C}_6\text{H}_{12}\text{O}_6$)

100. What is the mass percentage of chlorine in $\text{Mg}(\text{ClO}_4)_2$?
- 71.0%
 - 40.6%
 - 31.8%
 - 15.9%



"Count on minutes to make your day!"

1. B	2. D	3. A	4. B	5. A	6. C
7. D	8. D	9. B	10. C	11. D	12. A
13. C	14. A	15. C	16. D	17. B	18. B
19. C	20. B	21. B	22. D	23. B	24. A
25. A	26. D	27. C	28. C	29. D	30. A
31. A	32. C	33. D	34. D	35. B	36. C
37. A	38. A	39. C	40. A	41. A	42. A
43. C	44. A	45. B	46. C	47. C	48. B
49. D	50. C	51. A	52. C	53. D	54. C
55. D	56. B	57. A	58. D	59. B	60. C
61. D	62. D	63. A	64. C	65. C	66. B
67. C	68. A	69. D	70. D	71. D	72. A
73. D	74. A	75. D	76. C	77. C	78. C
79. A	80. D	81. C	82. A	83. B	84. C
85. A	86. A	87. C	88. B	89. D	90. B
91. A	92. B	93. D	94. C	95. D	96. A
97. D	98. B	99. B	100. C		

Stoichiometry Passage Answers

Passage I (Questions 1 - 7)

Density Experiment

- Choice B is correct.** In order for a liquid to float on water, the liquid must be both immiscible in water and less dense than water. All of the liquids are assumed to be immiscible in water, according to the question, so the only stipulation that remains to be considered is the density of each liquid. The density of water is defined as 1.00 grams per milliliter. We know that the mass of the empty volumetric cylinder is 42.61 grams and its capacity is 10.00 mL. For the density of any liquid to be less than 1.00, a 10.00-mL sample of the liquid must have a mass of less than 10.00 grams. This means that when 10.00 mL of a liquid is added to the volumetric cylinder, the liquid and cylinder must have a combined mass of less than 52.61 grams. If the combined mass is less than that, then the density of the liquid must be less than 1.00. According to Table 1, unknown Liquid #6 (with a combined mass with the volumetric cylinder of 51.33 grams) and unknown Liquid #9 (with a combined mass with the volumetric cylinder of 49.03 grams) are the only two unknown liquids with combined masses of less than 52.61 grams. Only Liquid #6 and Liquid #9 can float on water. The best answer is therefore choice B.
- Choice D is correct.** This question requires evaluating the density for unknown Solids #2, #3, #4, and #5. Density is defined as mass per volume. In this case, the volume of the solid is obtained by taking the volume reading from the chart for each unknown and subtracting 25.00 mL for the volume of the water already in the cylinder. This method of measurement is known as the "volume by displacement technique." The following table shows the values of mass, volume, and density obtained for each solid:

Unknown	Mass (g)	Volume (mL)	Density
2	12.38	$31.19 - 25.00 = 6.19$	$\frac{12.38 \text{ g}}{6.19 \text{ mL}} = 2.00 \frac{\text{g}}{\text{mL}}$
3	14.85	$29.95 - 25.00 = 4.95$	$\frac{14.85 \text{ g}}{4.95 \text{ mL}} = 3.00 \frac{\text{g}}{\text{mL}}$
4	8.22	$28.00 - 25.00 = 3.00$	$\frac{8.22 \text{ g}}{3.00 \text{ mL}} = 2.74 \frac{\text{g}}{\text{mL}}$
5	5.64	$26.41 - 25.00 = 1.41$	$\frac{5.64 \text{ g}}{1.41 \text{ mL}} = 4.00 \frac{\text{g}}{\text{mL}}$

The unknown solid with the greatest density is unknown Solid #5. The correct answer is thus choice D. The numbers could have been compared to one another to obtain the relative values.

- Choice A is correct.** The density of unknown Solid #1 must be determined first. It is found by dividing its mass by its volume (found by difference):

$$\rho = \frac{9.63 \text{ g}}{31.42 - 25.00 \text{ mL}} = \frac{9.63 \text{ g}}{6.42 \text{ mL}} = 1.50 \frac{\text{g}}{\text{mL}}$$

Because the density is 1.50 grams per milliliter, a 20.0-gram piece of the solid must have a volume of 13.3 mL, because 20.0 grams divided by 1.50 mL is equal to 13.3 mL. The correct answer is choice A. Choices C and D should have been eliminated immediately, because when the volume of an object is greater than its mass, then the density of the object is less than 1.00. Solid #1 sinks when placed into water, indicating that its density is greater than that of water, which is 1.00 grams per milliliter. Distinguishing between choice A and choice B requires looking more closely at the ratios.

- Choice B is correct.** In order for a solid to float on unknown Liquid #7, it must have a density less than that liquid. The density for unknown Liquid #7 is:

$$\rho = \frac{58.72 - 42.61 \text{ g}}{10.00 \text{ mL}} = \frac{16.11 \text{ g}}{10.00 \text{ mL}} = 1.611 \frac{\text{g}}{\text{mL}}$$

Only Solid #1, with a density of only 1.50 grams per milliliter, has a density less than 1.611 grams per milliliter. The best answer is choice B. The relative densities of Liquid #7 and Solid #1 are shown below:

$$\rho_{\text{Liquid \#7}} = \frac{16.11 \text{ g}}{10.00 \text{ mL}} = \frac{32.22 \text{ g}}{20.00 \text{ mL}}; \rho_{\text{Solid \#1}} = \frac{9.63 \text{ g}}{6.42 \text{ mL}} = \frac{28.89 \text{ g}}{19.26 \text{ mL}} \Rightarrow \frac{32.22 \text{ g}}{20.00 \text{ mL}} > \frac{28.89 \text{ g}}{19.26 \text{ mL}}$$

5. **Choice A is correct.** The following table shows how to solve for the densities of the four liquids:

Unknown	Mass (g)	Volume (mL)	Density ($\frac{g}{mL}$)
6	$51.33 - 42.61 = 8.72$	10.00	$\frac{8.72 \text{ g}}{10.00 \text{ mL}} = 0.872 \frac{\text{g}}{\text{mL}}$
7	$58.72 - 42.61 = 16.11$	10.00	$\frac{16.11 \text{ g}}{10.00 \text{ mL}} = 1.611 \frac{\text{g}}{\text{mL}}$
8	$53.21 - 42.61 = 10.60$	10.00	$\frac{10.60 \text{ g}}{10.00 \text{ mL}} = 1.060 \frac{\text{g}}{\text{mL}}$
9	$49.03 - 42.61 = 6.42$	10.00	$\frac{6.42 \text{ g}}{10.00 \text{ mL}} = 0.642 \frac{\text{g}}{\text{mL}}$

From the calculations, the relative densities in descending order are: Liquid #7 > Liquid #8 > Liquid #6 > Liquid #9. The answer choice that does not follow this pattern is choice A: Liquid #7 > Liquid #6 > Liquid #8. This question could also have been solved by comparing the masses in the chart for the cylinder and liquid combined. All of the liquids have the same volume (10.00 mL) and were in the same cylinder, so the same mass is subtracted from each in determining the liquid's mass. This would have saved much time.

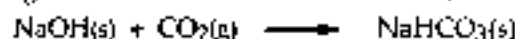
6. **Choice C is correct.** If the mass of the cylinder were heavier than 42.61 grams, then the reading for the mass of the liquid and cylinder combined would be greater than it was in Experiment 2. However, the mass of liquid in a fixed volume is the same, so its density does not change. No matter what container is chosen to hold it, the density of a liquid is an invariant property of that liquid. This is best reflected in answer choice C.
7. **Choice D is correct.** Density is defined as a measure of mass per volume. The units for density should therefore reflect a mass unit divided by a volume unit. In answer choice A, the mass is measured in grams and the volume is measured in milliliters, which makes choice A acceptable. In answer choice B, the mass is measured in ounces and the volume is measured in centimeters cubed, which makes choice B acceptable. You should recall that a milliliter is a centimeter cubed. In answer choice C, the mass is measured in kilograms and the volume is measured in liters, which makes choice C acceptable. In answer choice D, the weight (and not necessarily the mass) is measured in pounds and is divided by an area (dimension squared) and not by volume. Choice D is thus a measure of pressure, not density. The correct answer is choice D.

Passage II (Questions 8 - 13)

Combustion Analysis

8. **Choice D is correct.** Given that all of the samples were of nearly equal mass (between 1.995 g and 2.011 g), the greatest mass percent of carbon is in the compound that has the greatest mass of carbon. The compound with the greatest mass of carbon produces the greatest mass of carbon dioxide gas. This means that the easiest way to solve this question is to compare the amount of carbon dioxide collected for each sample, as listed in column 4 (the NaOH tube column) in Table 1. From the data in the NaOH tube column, the greatest mass of CO_2 is collected from Compound IV ($36.53 - 30.00 = 6.53$), eliminating choices A and B. Because a greater mass of CO_2 is produced by oxidizing Compound III than by oxidizing Compound II, choice D is the best answer.
9. **Choice B is correct.** The role of the sodium hydroxide salt is to bind CO_2 , not to bind H_2O . This means that the salt should be both hydrophobic (non-water-binding) and reactive with carbon dioxide, making choice B the best answer. The term "hygroscopic" refers to a compound with a high affinity for binding moisture in the air.
10. **Choice C is correct.** The pressure valve is designed to allow oxygen gas to flow from the tank into the system when a threshold pressure is maintained. The oxygen partial pressure can be controlled and maintained at a high level. The oxygen tank is left open with a positive pressure of oxygen to ensure that oxygen gas is continually flowing into the system, so choice A is eliminated. Nothing was mentioned about the temperature of the oxygen gas, so choice B is eliminated. Oxygen gas is always present, so it is in excess and is not a limiting reagent. This eliminates choice D and makes choice C the correct answer.
11. **Choice D is correct.** The greatest mass percent of carbon is found in the compound with the smallest number of oxygen atoms in its formula. Thus, the compound least likely to contain oxygen is the one that produces the most CO_2 upon combustion. That is Compound IV. In addition, pure carbon when oxidized yields 7.33 g CO_2 . Compound IV yields 6.53 grams CO_2 , so it is close enough to pure carbon to assume no oxygen is in it, choice D.

12. **Choice A is correct.** When the first valve is opened, the gas is exposed to magnesium sulfate, which binds water, but not carbon dioxide. This means that water is bound first, leaving an atmosphere of excess oxygen gas and carbon dioxide. This is important, because the sodium hydroxide compound can bind both carbon dioxide and water. Using magnesium sulfate first to remove the water vapor ensures that all water is removed from the gas when the second valve is opened, exposing the gas to the sodium hydroxide salt. Any increase in the mass of magnesium sulfate is due to the binding of water. Any increase in the mass of sodium hydroxide is due to the binding of carbon dioxide. If the gases were first exposed to sodium hydroxide, the increase in mass would be due to both water and carbon dioxide. Choice A is the best answer. Excess oxygen leaves the system as a free gas.
13. **Choice C is correct.** The solid formed in the second tube results from the reaction of sodium hydroxide (NaOH) with carbon dioxide (CO₂), so it must be a sodium salt. This eliminates choice A and B. The following reaction confirms that the best answer of the given choices is sodium bicarbonate, choice C:



Passage III (Questions 14 - 20)

Empirical Formula Determination

14. **Choice A is correct.** If 20.0 grams of the unknown were oxidized, instead of 10.0 grams, then the amount of carbon dioxide and water formed as products would double. This eliminates choices C and D. The mass percent of carbon should remain the same, because the mass of carbon dioxide formed and the mass of the compound both doubled. The best answer is choice A. The mass percent of carbon is constant, because the molecular formula of the compound is constant.
15. **Choice C is correct.** To determine the mass percent of carbon in the unknown compound, the grams of carbon in carbon dioxide (CO₂) are divided by the total number of grams of original compound. The mass of carbon dioxide formed is found by subtracting 100.00 grams of original sodium hydroxide from the 123.79 grams of bicarbonate salt. The difference is the mass of carbon dioxide that binds the salt.

$$\frac{23.79 \text{ grams CO}_2 \times \frac{12 \text{ g C}}{44 \text{ g CO}_2}}{10.0 \text{ grams unknown}} \times 100\% = 23.79 \times \frac{12}{44} \times \frac{1}{10} \times 100\%$$

This makes choice C the best answer.

16. **Choice D is correct.** In the first sentence of the passage, we read that the unknown organic compound is poured into a flask. The term "poured" implies that the compound flows, which makes it a fluid. The fact that it can be poured into a flask means that it is flowing down, which defines it more specifically as a liquid. The compound is a liquid at ambient temperature (room temperature). This means that the melting point is less than ambient temperature, because at room temperature it has already melted into a liquid. The boiling point is greater than ambient temperature, because at room temperature it has not yet boiled into a gas. Because it readily evaporates, the boiling point may be close to ambient temperature; but because it is a liquid at ambient temperature, the boiling point must be greater than ambient temperature. The best answer is choice D.
17. **Choice B is correct.** As stated at the end of the passage, the molecular formula for a compound with just hydrogen, oxygen, and carbon *cannot* have an odd number of hydrogens. An odd number of hydrogens results in an odd number of bonding electrons (electrons present in bonds). Considering that there are two electrons per bond, an odd number of bonding electrons results in only half of a bond somewhere in the compound (which equates to a single electron, or free radical.) A half-bond is not stable (possible to isolate physically), so choice B, with five hydrogens in the formula, is not possible. Knowledge from organic chemistry can prove useful in solving general chemistry questions. You must incorporate information from many sources to excel at this exam.
18. **Choice B is correct.** There are 12.16 grams of water produced from the oxidation of 10.0 grams of the unknown. This can be determined by subtracting 100.00 grams for the mass of the anhydrous sodium sulfate from the final mass of 112.16 grams for the hydrated sodium sulfate salt. The moles of water formed are greater than 0.5 moles, because 12.16 grams divided by 18 is greater than 9 grams divided by 18. The moles of water formed is less than 1.0 moles, because 12.16 grams divided by 18 is less than 18 grams divided by 18. The best answer is therefore choice B.

$$0.5 \text{ moles} = \frac{9 \text{ g}}{18 \text{ g/mole}} < \frac{12.16 \text{ g}}{18 \text{ g/mole}} < \frac{18 \text{ g}}{18 \text{ g/mole}} = 1.0 \text{ moles}$$

19. **Choice C is correct.** The question asks for the empirical formula, but because information is given about the molecular mass, you may wish to consider the molecular formula first, then eliminate choices based on that information. According to the passage, the molecular mass has a value between 70 and 80 grams per mole, so choice B (MW = 48 + 8 + 32 = 88) and choice D (MW = 96 + 10 + 16 = 122) are eliminated. The only choices with a mass in the range of 70 grams to 80 grams are choice A (which has a mass of 74 grams per mole) and choice C (which also has a mass of 74 grams per mole). Choice A is $\frac{36}{74}$ percent carbon, while choice C is $\frac{48}{74}$ percent carbon. The mass percent of carbon is greater than fifty percent, so choice A is eliminated, leaving choice C as the best answer. The exact numerical value for the mass percent of carbon can be approximated as follows:

$$\frac{23.79 \text{ grams CO}_2 \times \frac{12 \text{ g C}}{44 \text{ g CO}_2}}{10.0 \text{ grams unknown}} \times 100\% = 23.79 \times \frac{12}{44} \times \frac{1}{10} \times 100\% = \frac{23.79 \times 12}{44 \times 10} \times 100\% = \frac{23.79 \times 3}{11 \times 10} \times 100\%$$

$$= \frac{71.37}{110} \times 100\%, \text{ where } \frac{71.37}{110} \times 100\% > 50\%$$

Choice C is both the molecular formula and the empirical formula.

20. **Choice B is correct.** The mass percent of carbon in $\text{C}_5\text{H}_{12}\text{O}_2$ is found by dividing the mass of five carbon atoms (60 amu) by the molecular mass of $\text{C}_5\text{H}_{12}\text{O}_2$ (60 + 12 + 32 = 104 amu). The value determined when 60 is divided by 104 is less than 60%, because it is less than 60 divided by 100. The best choice for a value close to, but less than, 60% is choice B, 57.7%. You really should choose answer choice B, if you know what's best for you.

$$\text{mass percent of carbon} = \frac{5 \times 12}{60 + 12 + 32} = \frac{60}{104} < \frac{60}{100} = 60\%$$

Passage IV (Questions 21 - 26)

Molar Volume of a Gas

21. **Choice B is correct.** The mass percent of carbon in carbon dioxide is the mass of carbon (12.011) divided by the mass of carbon dioxide (44.009) $\times 100\%$. The fraction $\frac{12}{44}$ reduces to $\frac{3}{11} = 3 \times \frac{1}{11} = 3 \times 0.091 = 0.273 \times 100\% = 27.3\%$. To make math easier, you should memorize the following set of fraction-to-decimal conversions. A demonstration of how these fractions are useful is found on page 30.

$$\frac{1}{3} = 0.333, \frac{1}{4} = 0.250, \frac{1}{5} = 0.200, \frac{1}{6} = 0.166, \frac{1}{7} = 0.143, \frac{1}{8} = 0.125, \frac{1}{9} = 0.100, \frac{1}{10} = 0.091, \frac{1}{11} = 0.091, \frac{1}{12} = 0.083$$

This trick works well, once learned. It may seem awkward at first, but try it. The correct answer is choice B.

22. **Choice D is correct.** The second experiment was conducted to determine the molecular mass of the unknown liquid. At 304 K, the boiling point of the liquid (and thus, the temperature of the gas), the volume of gas is exactly 1.00 liters, assuming that the flask is completely filled with vapor from the liquid. Molar volume at this temperature is 24.96 liters per mole. At 304 K, the gas has a density of 2.32 grams per liter. When this value is multiplied by molar volume at 304 K, the liters cancel out and the remaining units are grams per mole. The molecular mass is $2.32 \text{ g/L} \times 24.96 \text{ L/mole} = 2.32 \times 24.96 \text{ g/mole} = 58 \text{ g/mole}$. The best answer is choice D. You should always consider units when looking at math questions. The question asks for a molecular weight, which has units of grams per mole. The numbers in the answer choices are 2.32 (which is in grams per liter), 22.41 (which is in liters per mole), and 24.96 (which is in liters per mole.) To get the target units, grams per liter are multiplied by liters per mole, choices C or D. Choice D has the correct molar volume of the unknown liquid at 304 K.
23. **Choice B is correct.** You know the molecular mass of the unknown compound (58 grams per mole) and the mass percent (82.9%), so you can make an intuitive determination of the molecular formula. The mass of carbon in the compound must equal either 12, 24, 36, or 48 (multiples of 12). Given that 82.9% of 58 is closest to 48 of the multiples of 12, it can be assumed that the molecule has four carbons. This leaves ten hydrogens (to round out the 10 grams in 58 grams per mole not accounted for by the four carbons); so the molecular formula is C_4H_{10} , which has an empirical formula of C_2H_5 . All of this intuition should lead you to pick B.

If you want to solve for the empirical formula exactly, here is the math using the compact formula, where a 100-gram sample is assumed. Do lengthy math calculations only if they are absolutely necessary.

$$\frac{\text{C}_{82.9}\text{H}_{17.1}}{12} = \text{C}_{6.9}\text{H}_{1.71} = \frac{\text{C}_{6.9}\text{H}_{1.71}}{6.9} = \text{C}_1\text{H}_{2.5} = \text{C}_2\text{H}_5$$

24. **Choice A is correct.** The mass percent of carbon in the unknown compound must be less than 100%. This automatically eliminates choices B, C, and D. If you didn't notice that fact, then the problem can be solved knowing that the mass of carbon in the CO_2 is equal to the mass of carbon in the unknown compound. The mass of carbon in CO_2 will be the moles of CO_2 ($1.55/22.41$) multiplied by the mass of carbon (12.011). When this is divided by the mass of the sample (1.00), the result is answer choice A.
25. **Choice A is correct.** The key fact in the passage is that the 3.045 grams of carbon dioxide gas formed occupies 1.55 L at STP. The question asks for moles of carbon dioxide, so the liters must be converted to moles. The conversion is given in the chart as 22.4 liters per mole. To convert to moles, either the 3.045 CO_2 grams must be divided by the molecular mass of 44 grams per mole, or the 1.55 liters CO_2 must be divided by the 22.41 liters per mole. The second option is answer choice A, so pick that. If you wish to get an exact value for the moles, you must get the denominator close to 100. This can be done by scaling the numerator and the denominator (i.e., adding a value to each that is equivalent to multiplying both by the same value).
- $$\frac{1.55 \times 4}{22.4 \times 4} = \frac{6.20}{89.6} = \frac{6.20 + 0.775}{89.6 + 11.2} = \frac{6.975}{100.8} = 0.069 \text{ moles } \text{CO}_2$$
- This is fairly precise, but unnecessary on the MCAT. The addition of 0.775 to the numerator and addition of 11.2 to the denominator accomplish the goal of making the denominator close to 100 while scaling proportionally.
26. **Choice D is correct.** If the organic vapor had not displaced all of the air in the flask at the time the heat was removed, then the flask would not have been filled with pure organic vapor at that time (as was assumed). The actual amount of organic vapor in the flask would be less than the assumed value (100%), so the actual amount of liquid collected would be too low. If the measured mass of the liquid is too small, then the value in the numerator is too small for the calculation of the molecular mass, so the calculated molecular mass is too low as well. Pick D to feel that happy sensation of correctness.

Passage V (Questions 27–33)

Elemental Analysis

27. **Choice C is correct.** Water must be frozen out of solution, so the temperature of the trap should be less than the freezing point of water (0°C). This eliminates choices A and B. The temperature of the trap cannot be lower than the sublimation point of carbon dioxide, however; otherwise, the carbon dioxide gas would solidify in the trap along with the water vapor. The trap's temperature must be greater than the sublimation point of carbon dioxide, but that temperature is neither given in the passage nor is it common knowledge, unless you happen to own the Jeopardy™ Chemistry game. To play it safe, choose a temperature just less than the freezing point of water. The value closest to 0°C without exceeding it is -33°C . The best answer is therefore choice C.
28. **Choice C is correct.** The successive traps should be aligned in a manner to isolate each gas separately. The moles are not determined by difference, so choices B and D are eliminated. The trick here is determining the relative temperature sequence. If the traps were set first to isolate carbon dioxide by deposition (converting from gas into solid), then water vapor would freeze out along with the carbon dioxide gas, and thus the compounds would not be separated. By freezing water vapor first, the carbon dioxide can pass to a later trap where it can in turn be isolated free of water. Because the freezing point of water is 0°C and the sublimation point of carbon dioxide is -78°C , the temperatures of successive traps should gradually decrease. Choose C.
29. **Choice D is correct.** To determine the molecular formula from the empirical formula, the molecular mass of the compound must be known. This fact makes choice D the best answer. The volume of the products and reactants is dependent on the mole ratio in the reaction, but not exclusively on the molecular formula. The volume may change depending on other conditions, such as temperature. This eliminates choices A, B, and C. Pick D. Determination of the molecular formula from the empirical formula involves comparing the empirical mass to the molecular mass. If the empirical mass is equal to the molecular mass, then the empirical formula is the molecular formula. Choose D for the satisfaction of knowing you got this correct.
30. **Choice A is correct.** Combustion analysis involves oxidizing compounds and collecting their oxide gases. Sulfur gas can be oxidized like carbon (given that carbon and sulfur are roughly equivalent in electronegativity). This means that sulfur oxide can be collected and analyzed like the oxide of carbon. The oxide of sulfur that is formed is uncertain, but the only answer choice showing oxidized sulfur is choice A, where sulfur carries a positive four (+4) oxidation state. Sulfur can also be oxidized into sulfur trioxide (SO_3), but that is not listed as a choice.

31. **Choice A is correct.** If oxygen gas were trapped out from the gas mixture, there would be no way to know whether the oxygen gas collected was from the original hydrocarbon, or from the excess oxygen gas needed to ensure the complete combustion. This makes choice C valid. It is true that oxygen from the carbohydrate is found in both the carbon dioxide product and the water product; thus, the oxygen from the compound cannot be isolated. This makes choice B valid.

Oxygen in the carbohydrate is fully reduced (has a negative two (-2) oxidation state when bonded to carbon and hydrogen). When looking at complete combustion reactions, remember that oxygen atoms from the compound do not react with oxygen gas at all. This makes choice D valid. Oxygen gas has a sublimation point around -180°C (so oxygen gas undergoes deposition when the temperature drops below -180°C); thus, oxygen gas can be collected out of the air using a liquid nitrogen trap. Nitrogen liquid has a boiling point of -196°C , so the temperature of the trap must be -196°C . The best choice is therefore choice A, because oxygen can exist as a solid. The correct answer was a double negative.

32. **Choice C is correct.** As stated in the passage, the bubbler vents any pressure buildup within the system while keeping the system closed. The bubbler is a one-way valve that allows gas to effuse from the system, but does not let gas infuse from the environment. The oil in the bubbler is intended to interact only with the gas, so no unreacted organic vapor should dissolve into the oil. Choice A is a valid possibility in the reaction (a true statement), but it is not relevant to the goals of the experiment. Answer choices like choice A are difficult to eliminate sometimes. The oil in the bubbler is not serving as an oil bath, so choice B is eliminated. The bubbler keeps the system closed by preventing infusion of gaseous compounds from the outside. The best answer is choice C. Liquids cannot flow into the oil in the bubbler, so choice D is eliminated.

33. **Choice D is correct.** An increase in the mass percent of carbon is defined as an increase in the amount of carbon per gram of the compound, so statement I is valid.

The amount of water formed depends on the number of hydrogens in the compound, which depends on the mass percent of hydrogen. As the mass percent of carbon within a hydrocarbon increases, the mass percent of hydrogen in the hydrocarbon decreases. This reduces the amount of water formed from one gram of compound, so it is *not* true that the mass of water formed increases upon oxidation. The decrease in mass percent of hydrogen reduces the mass of hydrogen formed per gram of compound. This means that neither statements II nor III is associated with an increasing mass percent of carbon within a hydrocarbon. Choice D is the best answer.

Passage VI (Questions 34 - 40)

Dilution Experiment

34. **Choice D is correct.** The four salts listed as choices are all present in the same mass quantity in solution (4% of their respective solution.) All of the solutions have 96% water solvent, so the salt with the greatest number of moles has the highest molarity. Given that all of the salts have an equal mass, the greatest number of moles belongs to the salt with the lowest molecular weight. Sodium (Na) is lighter than potassium (K), and chlorine (Cl) is lighter than bromine (Br), so NaCl is the lightest salt of the choices. Pick D, and feel fresh.
35. **Choice B is correct.** The trick here lies in the wording: "...how much water must be *added*?" This question requires that you use the dilution equation, $M_{\text{initial}}V_{\text{initial}} = M_{\text{final}}V_{\text{final}}$, where M is molarity and V is volume.

$$0.15 \text{ M} \times 300 \text{ mL} = 0.0075 \text{ M} \times V_{\text{final}} \therefore V_{\text{final}} = \frac{0.15 \text{ M} \times 300 \text{ mL}}{0.0075 \text{ M}} = \frac{15}{0.75} \times 300 \text{ mL} = \frac{15 \times 4}{3} \times 300 \text{ mL} \\ = 15 \times 4 \times 100 \text{ mL} = 6000 \text{ mL} = 6.0 \text{ L}$$

This value is V_{final} , *not* the answer to the question (volume of water added)! To finish with 6.0 liters of solution from an original volume of 0.3 liters, 5.7 liters must be added. Choice B is the correct answer.

36. **Choice C is correct.** As far as chloride ion concentration is concerned, it doesn't matter whether you add pure water or NaOH(aq) to the HCl(aq) solution, because Cl^- is just a spectator ion in the acid-base reaction. Neither NaOH(aq) nor pure water increases the moles of chloride anion in the solution. Solve the question by using the dilution equation, $M_{\text{initial}}V_{\text{initial}} = M_{\text{final}}V_{\text{final}}$, where M is molarity and V is volume:

$$0.25 \text{ M} \times 50 \text{ mL} = M_{\text{final}} \times 75 \text{ mL} \therefore M_{\text{final}} = \frac{0.25 \text{ M} \times 50 \text{ mL}}{75 \text{ mL}} = \frac{50}{75} \times 0.25 \text{ M} = \frac{2}{3} \times 0.25 \text{ M} = 0.167 \text{ M}$$

The final molarity is 0.167 M, which makes choice C, 0.17M, the best answer.

37. Choice A is correct. Pure water is passed through the volumetric pipette to flush any residue out of the volumetric pipette. Not all of the contents of the pipette may come free because of the adhesion of water to the glass, so rinsing ensures that the residual solution in the pipette is forced from the pipette into the new solution. The rinsing is not quantitative, so it does not measure a volume. This eliminates choice B. No temperature is mentioned, so the water used to rinse the pipette is not responsible for heating or cooling anything. This eliminates choices C and D. The best answer is choice A.
38. Choice A is correct. A ten-fold dilution is defined as a dilution that results in a final concentration of a solution that is ten percent of its original concentration. For this to occur, the final volume of the solution must be ten times the initial volume. The final volume is the sum of the initial volume and the added volume, so the following math can be applied:
- $$V_{\text{final}} = V_{\text{initial}} + V_{\text{added}} \quad \text{and} \quad V_{\text{final}} = 10V_{\text{initial}}, \text{ so } 10V_{\text{initial}} = V_{\text{initial}} + V_{\text{added}}$$
- $$9V_{\text{initial}} = V_{\text{added}}, \text{ so 1 part solution is mixed with 9 parts solvent.}$$
- Ten parts solvent mixed with one part solution yields a final volume that is eleven times larger; thus, the final concentration would be one-eleventh, which is less than ten percent. This eliminates choice B. Choice C is the reverse of a ten-fold dilution. Choice C is a dilution to ninety percent of the original concentration. Choice D is the same as choice B. Pick choice A to feel that jovial tingle of correctness.
39. Choice C is correct. The addition of water to an aqueous salt solution increases the mass of the solvent without changing the moles of solute, so the molality decreases (given that the denominator increases.) This makes choice A valid. Addition of water to an aqueous salt solution increases the volume of the solution without changing the moles of solute, so the molarity decreases (given that the denominator increases.) This makes choice B valid. The salt solution is denser than the pure water, so the addition of water (a less dense solution) to the salt solution decreases the density of the solution, making choice C invalid. Addition of water to an aqueous salt solution increases the mass of the solvent without changing the mass of solute, so the mass percent of solvent increases. This makes choice D valid. The correct answer is choice C.
40. Choice A is correct. The greatest dilution results from the greatest relative addition of solvent. The greatest dilution involves the greatest ratio of solvent added to initial solution present, and it does not depend on the total volume of the initial or final solution. The ratio of solvent added to the initial volume of solution in choice A is 100 : 25, which reduces to 4 : 1. The ratio of solvent added to the initial volume of solution in choice B is 200 : 60, which reduces to 10 : 3. The ratio of solvent added to the initial volume of solution in choice C is 50 : 15, which reduces to 10 : 3. The ratio of solvent added to the initial volume of solution in choice D is 150 : 50, which reduces to 3 : 1. The greatest ratio of solvent added to initial volume of solution is found in choice A, the best answer.

Passage VII (Questions 41–47)

Solution Concentrations and Dilution

41. Choice A is correct. If the density of the solution is less than the density of water, then the addition of water to the solution may actually increase the density of the solution. The molarity and the molality of the solution always decrease as solvent is added, because the denominator in both molarity and molality increase while the numerator remains the same. As solvent is added to the solution, the mass percent of solute in solution decreases, because the mass of solute in solution remains constant while the mass of solution is increasing. Only the density does not always decrease upon the addition of water to an aqueous solution; thus, the best answer is choice A. In rare cases where the aqueous solution has a density of less than 1.00, the density increases when water is added.
42. Choice A is correct. Calculating the molarity and the molality of the solution involves the moles of salt in the numerator. The numerator is the same in both the molarity and molality, so the difference between the two values depends solely on the denominator. The mass of water is 0.100 kg, while the volume of solution is greater than 0.100 liters. This means that the denominator in the molarity calculation is greater than the denominator in the molality calculation. The larger the denominator, the smaller the value. This means that the molality is greater than the molarity, which eliminates choices B and D. The density of pure water is 1.00 grams per mL. The density of the aqueous salt solution is the mass (101 grams due to salt and water) divided by the volume. The volume of the solution is less than 101 mL, so the density of the solution is greater than 1.00 (given that the numerator is 101 and the denominator is less than 101). The density of the solution is greater than the density of water, so the best answer is choice A.

43. **Choice C is correct.** Because the density of the solvent is less than 1.0 grams per mL, the volume in liters exceeds the mass in kilograms for the solvent. When comparing molality and molarity, both values have the same numerator. You are looking for the value with the smaller denominator. The mass of solvent is less than the volume of solution, so molality is greater than molarity. This eliminates choices B and D. The density is the mass of the solution divided by the volume of the solution. The density values are less than 1, but are not known specifically. In addition, there is no information about the mixing process. The density cannot be determined without information about the density of the components. The best answer is choice C.

44. **Choice A is correct.** All of the solutions are made up of one solute and solvent, but in different ratios. The solute is denser than the solvent, so it can be concluded that the solution with the greatest density has the highest concentration of solute. As solute concentration is reduced, the density is also reduced. Increasing the amount of solute relative to solvent increases both molality and molarity. The best answer is choice A.

45. **Choice B is correct.** This question requires the use of the dilution equation, $M_{\text{initial}}V_{\text{initial}} = M_{\text{final}}V_{\text{final}}$, where M is the molarity and V is the volume. Plugging the given values into the equation yields:

$$0.25 \text{ M} \times 100 \text{ mL} = 0.10 \text{ M} \times V_{\text{final}} \therefore V_{\text{final}} = \frac{0.25 \text{ M} \times 100 \text{ mL}}{0.10 \text{ M}} = \frac{0.25}{0.10} \times 100 \text{ mL} = 2.5 \times 100 \text{ mL} = 250 \text{ mL}$$

The final volume (V_{final}) is 250 mL, but V_{final} is *not* the answer. The question asks for the volume of water added. To start with 100 mL of solution and finish with 250 mL of solution, 150 mL of water must be added.

46. **Choice C is correct.** The units of molarity are moles solute per liter solution. Because both solutions have the same final volume (100 mL), the question reduces to asking how many grams of MgCl_2 yield the same number of moles of Cl^- as 1.0 grams NaCl . There are two chlorides per magnesium chloride molecule, so only half the number of moles of MgCl_2 as moles of NaCl is required. This accounts for the factor of one-half in the calculation. The grams of NaCl must be converted into moles of NaCl , which in turn are converted into grams of MgCl_2 after accounting for the mole ratio of the two salts. This calculation requires knowing the molecular mass for both salts. Because the magnesium chloride is heavier than sodium chloride, intuition tells us that more grams of the magnesium chloride are needed to form a mole quantity equal to that for the sodium chloride. This means that in the calculation, the molecular mass ratio term should be greater than 1, making choice C the best answer. The unit factor method solution is as follows:

$$1.0 \text{ grams NaCl} \times \frac{1 \text{ mole NaCl}}{58.4 \text{ grams NaCl}} \times \frac{1 \text{ MgCl}_2}{2 \text{ NaCl}} \times \frac{94.9 \text{ grams MgCl}_2}{1 \text{ mole MgCl}_2} = 1.0 \times \frac{1}{2} \times \frac{94.9}{58.4} \text{ grams MgCl}_2$$

The final answer is less than 1.0 grams MgCl_2 .

47. **Choice C is correct.** The mole fraction of a compound in a solution is found by dividing the moles of the given compound by the total moles of the solution. Choice B has a mole fraction of 0.50 for both Compound A and Compound B, because there is one mole of each compound, and the total number of moles is two. In an equal-mass mixture (one gram each), the greater number of moles is present in the compound with the lower molecular mass. Compound A is lighter than Compound B; therefore, in equal masses of Compound A and Compound B, there are more moles of Compound A than Compound B. The mole fraction of Compound A is therefore greater than 0.50 in choice A. This eliminates choice B. For an equal volume solution (in choice C there is one mL each), the relative masses can be determined by the densities. Because Compound A is denser than Compound B, equal volumes of Compound A and Compound B result in the mass of Compound A being greater than the mass of Compound B. The mass percent of Compound A is greater than 50%, as it is in answer choice A; thus, choice C results in an even larger mole fraction of Compound A than in the equal mass solution of choice A. The equal-molecules solution (choice D) has the same mole fraction as the equal-mole solution (0.50 for each). Because choice B and choice D are the same answer, they are both eliminated. The greatest mole fraction of Compound A is found in the equal-volume solution, choice C.

Passage VIII (Questions 48 - 54)

Beer's Plot and Light Absorption

48. **Choice B is correct.** According to the absorbance equation (and the data and the graph), when the concentration of solute is doubled, the absorbance of the solution containing it doubles. This occurs because twice as many molecules are present to absorb light. The best answer is choice B. Choices C and D should have been eliminated immediately, because the absorbance increases as the concentration of solute increases. Choice A is eliminated, because the absorbance does not increase as the square of an increase in molarity.

49. **Choice D is correct.** At low concentrations, the relationship between concentration and absorbance is linear, as predicted from the absorbance equation. At higher concentrations, deviation begins to occur, until the absorbance reaches a maximum. There is no reaction occurring, so choice B can be eliminated. No solute would dissolve into the solvent if the two repelled one another, so choice C can also be eliminated. Choice A may sound tempting, if you are blindly choosing an amount on the basis of words you recall from biochemistry. The saturation of the solvation catalyst sounds like a good answer (especially considering that Michaelis-Menten kinetics shows a similar graph when saturation of some sort occurs). The problem here is that "saturation of solvation catalyst" means nothing, as there is no solvation catalyst. Also, catalysts affect the rate (not the concentration) of reactants, products, solutes, or other components of a solution. Choice A should be eliminated. Once water (the solvent) has interacted with as much solute as it can, no more solute can dissolve into solution. The molarity reaches a maximum, so the absorbance reaches a maximum. The graph would be a line that just ends at the saturated concentration. The best answer is choice D.
50. **Choice C is correct.** The addition of 50 mL of water to 10 mL of solution results in a dilution to one-sixth of the original concentration value. The absorbance of light should also be reduced by a factor of one-sixth. One-sixth of 0.518 is a value less than 0.100, so the only possible answer choice is choice C.
51. **Choice A is correct.** Dividing the equation $\text{Absorbance} = \epsilon[C]l$ by ϵ and l isolates the value of solute concentration ($[C]$). The result is that choice A is the best answer. This question may have seemed too simple to you. On occasion, there are some simple questions about the MCAT, so don't try to find tricks that aren't there.
52. **Choice C is correct.** From the data in Table 1, we know that the absorbance of a solution with a concentration of 0.20 M is 0.188, while the absorbance of a solution with a concentration of 0.30 M is 0.278. If the absorbance of the unknown solution is 0.242, then the molarity of the solution must lie between 0.20 M and 0.30 M. The average of 0.20 M and 0.30 M is 0.25 M, and the average of 0.188 and 0.278 is 0.233, so the absorbance for a 0.250 M solution should be roughly 0.233. Because 0.242 is slightly greater than 0.233, an absorbance value of 0.242 is associated with a concentration value that is slightly greater than 0.250 M. The best answer is choice C.
53. **Choice D is correct.** Isolating the value for ϵ in the absorbance equation yields absorbance divided by (concentration-pathlength). Absorbance is unitless, while concentration is in molarity, and pathlength is in some fraction of meters (like centimeters). The units of ϵ therefore must involve inverse molarity times inverse centimeters. The best answer is choice D.
54. **Choice C is correct.** As solute concentration (molarity) increases, the absorbance increases. This makes statement I valid. At this point, the answer choices narrow to either choice A or choice C. If you are strapped for time, just look at statement III. Beer's law applies at all wavelengths (λ_{max} is chosen because it is the greatest value, and thus is the easiest wavelength at which to obtain an accurate measure of absorbance). Because statement III is valid, choice C is the best answer. The absorbance depends on both the molar absorptivity constant and solute concentration, so if Compound X has a lower constant than Compound Y, then the solution with Compound X must be more concentrated than the solution with Compound Y, in order to have an equal absorbance. This makes statement II invalid. The best answer is choice C.

Passage IX (Questions 55 - 61)

Beer's Law Experiment

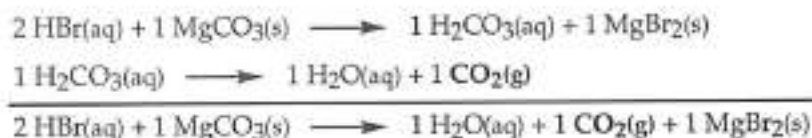
55. **Choice D is correct.** The spectrophotometer measures the absorbance of light, so the wavelength setting corresponds to an absorbed color, not a complementary color. This eliminates choices A and C, both of which refer to the reflected (or complementary) color. To obtain the most accurate value, the spectrophotometer is set at the wavelength corresponding to maximum absorbance. This makes choice D the best answer. Choice B is coincidentally correct in some cases, where the absorbance band is symmetric. This does not make choice B a better answer than choice D, but it does raise an important point: Remember that the test writers reward you for choosing the best answer, not just a correct answer.
56. **Choice B is correct.** The question states that the absorbance varies with cuvette length. From the passage, we know that absorbance varies with the concentration of solute in the solution. It makes sense that the absorbance depends on the molar absorptivity constant (ϵ), so the Beer's law relationship of $\text{Abs.} = \epsilon[C]l$ can be deduced. Dividing both sides of the equation by $[\text{Compound}] \cdot l$ yields choice B. From the term "molar absorptivity," it can be inferred that absorbance is in the numerator and molarity in the denominator, which makes choice B the only possible answer. Units may not always be the best route to the correct answer.

57. **Choice A is correct.** This question requires converting absorbance to concentration. A 0.10 M solution of Compound M has an absorbance of 0.362, so if the absorbance is 0.400, the molarity is greater than 0.10 M. This eliminates choice D. Compound Q with an absorbance of 0.250 is less than 0.10 M (where the absorbance is 0.299), so choice B is eliminated. Compound T with an absorbance of 0.500 is less than 0.10 M (where the absorbance is 0.511), meaning choice C is eliminated, leaving choice A as the best answer.
58. **Choice D is correct.** From the data in Table 1, we see that Compound T has the greatest absorbance of the three compounds, so choices A and B are eliminated. The absorbance values for Compound M at a concentration equal to that of the other compounds are closer to the values of Compound Q than Compound T. That means that the better graph is choice D. This is a question where it is easy to make a mistake by choosing answer choice C without examining the other choices more closely. A brief survey of former students showed that roughly one-third of you will make a careless mistake on this question and choose C.
59. **Choice B is correct.** Compound M has an absorbance of 0.217 at 561 nm when the concentration is 0.06 M. Therefore, when the concentration is reduced to 0.05 M (a value that is about 17% less), the absorbance should be reduced to a value that is five-sixths of 0.217, which is roughly 0.181. Statement I is valid. A solution of Compound Q with an absorbance of 0.225 at 413 nm must have a concentration between 0.06 M (where the absorbance is 0.180) and 0.10 M (where the absorbance is 0.299). Because 0.225 is a little closer to 0.180 than 0.299, the concentration should be a little closer to 0.06 M than 0.10 M, which makes statement II valid, and makes choice B the only possible choice. Statement III is invalid, because at 697 nm, 0.10 M Compound T has an absorbance of 0.511. This means that at the same wavelength, with a molarity of 0.11 M, the absorbance is about 0.56+. Given that the wavelength is not 697 nm (the wavelength where maximum absorbance is observed), the absorbance is actually less. This makes statement III invalid, agreeing with the selection of choice B. Pick B to gain the prestige, honor, and pride that goes with knowing you got another question right.
60. **Choice C is correct.** Compound T has an absorbance of 0.511 when it has a concentration of 0.10 M. An absorbance of 0.250 is just less than half of the 0.511 value, so the concentration must be just less than 0.05 M (half of 0.10 M). The question is then, "Which solution results in a concentration of Compound T that is just less than 0.05 M?" The starting solutions are all 0.10 M Compound T; so to achieve a solution that is just under 0.05 M, the amount of water added should be just a tiny amount greater than the amount of Compound T solution. This is observed with choice C. The absorbance for each choice is listed below.
- Choice A: $\frac{10}{15} (0.511) > \frac{1}{2} (0.511) > 0.25$; therefore, choice A has too high an absorbance.
- Choice B: $\frac{20}{40} (0.511) = \frac{1}{2} (0.511) > 0.25$; therefore, choice B has too high an absorbance.
- Choice C: $\frac{19}{39} (0.511) < \frac{1}{2} (0.511) \therefore \frac{19}{39} (0.511)$ may equal 0.25; therefore, choice C is good.
- Choice D: $\frac{10}{19} (0.511) > \frac{1}{2} (0.511) > 0.25$; therefore, choice D has too high an absorbance.
- The value of $\frac{19}{39} (0.511)$ should be close enough to 0.250 to convince you to choose C.
61. **Choice D is correct.** Although colors were not reviewed in this section, it is a topic that frequently is discussed along with the absorbance of light. One of the goals of these review passages is to expose you to this kind of integrated material. In each section, there are some questions on topics that have yet to be discussed. Such is the case here. The color observed for the solution is the complementary color of what is absorbed. The visible spectrum is from roughly 700+ nm to 400 nm. Red light is the least energetic of visible light, so red light has a broad wavelength of approximately 700 nm. Violet light is the most energetic of visible light, so violet light has a narrow wavelength of approximately 400 nm. Compound Q absorbs light with a wavelength of maximum absorbance (λ_{max}) of 413 nm. This corresponds to violet light, so the color observed for the solution is yellow (the complementary color of violet). This eliminates choices A and B. Compound T absorbs light with a wavelength of maximum absorbance (λ_{max}) of 697 nm. This corresponds to red light, so the color observed for the solution is green (the complementary color of red). This eliminates choice C, and makes choice D the best answer. You should be able to answer this with the background knowledge that 400 nm is at the violet end of the visible spectrum, 700 nm is at the red end of the visible spectrum, and a working knowledge of complementary colors. Compound M is irrelevant in solving this question. Appearing red (reflecting red light) means that green light (the complementary color of red) was absorbed. Green light is in the middle of the visible spectrum, so it could have a wavelength of 561 nm. There is no reason for you to know the exact λ values for each color, but you must be able to deduce the best answer to this question. Absorbed light, reflected light, complementary colors, the color wheel, and the visible spectrum will all be addressed in the next chapter.

62. **Choice D is correct.** It is not a combination reaction, because no compounds combined to form a single compound, so choice A is eliminated. It is not a decomposition reaction, because no compound broke apart to form multiple compounds, so choice B is eliminated. It is not a single-replacement reaction, because no metal compound was formed or consumed, so choice C is eliminated. Because the barium cation (Ba^{2+}) and the potassium cation (K^+) have exchanged their respective anions in this reaction, the reaction is classified as a metathesis reaction (known also as a double-displacement reaction). The best answer is choice D. By picking choice D, you can feel satisfied knowing you picked a correct answer. Go ahead, feel good!
63. **Choice A is correct.** A precipitate is a solid that forms from solution and deposits on the bottom of the flask as a reaction proceeds. The reaction least likely to form a precipitate is the reaction that is least likely to form a solid product. In choice A, the combustion reaction forms water liquid and carbon dioxide gas, but does not form a solid of any kind. For this reason, choice A is the best choice. In a decomposition reaction, a solid can be formed, as in the example offered in the passage. In the single-replacement and metathesis reactions, salts are formed as products, so precipitates are both likely and probable.
64. **Choice C is correct.** The reaction of aqueous sodium iodide (NaI(aq)) with aqueous calcium nitrate ($\text{Ca(NO}_3)_2\text{(aq)}$) is a double-displacement (metathesis) reaction, so the precipitate formed in the reaction must be one of the two possible product salts. Calcium is a dication (carries a +2 charge), so choice A is eliminated, because the calcium salt is not neutral when bonded to only one iodide anion. Sodium and nitrate are both highly soluble in water, so neither sodium nor nitrate is found in the precipitates. Both choice B and choice D are therefore eliminated. The best choice is answer C. The reaction is given below:



65. **Choice C is correct.** The reaction of aqueous hydrobromic acid (HBr) with magnesium carbonate (MgCO_3) is a neutralization reaction. It is stated in the passage that carbonates, when neutralized, yield carbon dioxide gas. The best choice is answer C. The reaction is drawn below:



You should recall that baking soda (sodium bicarbonate) effervesces when an acid is added. The bubbles that form are carbon dioxide gas. Hydrogen gas cannot exist as H (molecular hydrogen is a diatomic gas), and magnesium bromide is an ionic salt (which is definitely not a gas at room temperature), meaning that choices A and D should be discarded immediately. To eliminate choice B, you had to recognize that magnesium carbonate is a base, and not a reducing agent. Magnesium metal is rich in electrons, so it would make a strong reducing agent, but not magnesium cation (Mg^{2+}). The addition of HBr to Mg(s) would yield $\text{H}_2\text{(g)}$ by way of an oxidation-reduction reaction.

66. **Choice B is correct.** Carbon dioxide gas is a common product in combustion reactions, so choice A is eliminated. The example reaction for decomposition shows calcium carbonate (CaCO_3) decomposing to form calcium oxide and carbon dioxide. Choice C is thus eliminated. The same reaction is possible with calcium sulfite (CaSO_3), so sulfur dioxide can be formed in the decomposition reaction of calcium sulfite. The passage states that the neutralization of sodium bicarbonate (NaHCO_3) forms carbon dioxide gas; therefore, choice D is eliminated. The only choice left is answer choice B, the metathesis reaction, where cations are interchanged, but no carbon dioxide gas is formed. If the anion begins as carbonate, it finishes as carbonate, and not carbon dioxide.
67. **Choice C is correct.** In the reaction, the chlorine gas is reduced to chloride anion and exchanged for the bromide anion, which is oxidized into bromine liquid. The product is not the result of coupling (combining) of the reactants, so it is not a combination reaction. This eliminates choice A. The products are not the result of the breakdown (decomposition) of a reactant, so it is not a decomposition reaction. That eliminates choice B. The anion in the salt is exchanged, while the cation is not. This describes a single-replacement reaction, meaning choice C is valid. Because the cation remains the same, the reaction cannot be a metathesis reaction, which eliminates choice D. The best answer is choice C.

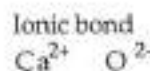
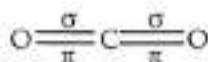
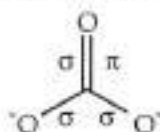
68. Choice A is correct. A negative value for ΔS results when the reaction gains order (or loses randomness). In a combination reaction, the number of molecules decreases from reactant to product, which is an increase in order. As such, a combination reaction is likely to have a negative value for ΔS . This makes choice A the best answer. In a decomposition reaction, the number of molecules increases from reactant to product, which is an increase in disorder. A decomposition reaction is likely to have a positive value for ΔS . This eliminates choice B.

In a single-replacement reaction, the number of molecules remains the same on each side of the reaction. Both sides of the equation have a precipitate and a solute, which leads to no conclusion about the change in order for the system. A single-replacement reaction can have either a positive or negative value for ΔS . Although a single-replacement reaction may have a negative ΔS , choice C is not as good an answer as choice A. In a combustion reaction, there is an increased number of molecules from reactants to products. The products are gases, so there is a large positive value for ΔS . This eliminates choice D.

Passage XI (Questions 69 - 78)

Calcium-Containing Bases

69. Choice D is correct. $\text{Ca}(\text{OH})_2$ yields a hydroxide anion (OH^-) when added to water, which makes $\text{Ca}(\text{OH})_2$ an Arrhenius base. $\text{Ca}(\text{OH})_2$ cannot donate a proton, so it is not an Arrhenius acid. This eliminates choice C. Choice A is also eliminated, because it cannot be an amphoteric species, unless it can act as both an acid and a base. Calcium is a metal, so $\text{Ca}(\text{OH})_2$ is a metal hydroxide, not a non-metal hydroxide. Choice B is eliminated. The correct answer is choice D.
70. Choice D is correct. Although CO_3^{2-} is charged and thus an ion, the anion itself is held together by covalent bonds (three σ -bonds and a resonating π -bond). Carbon dioxide is held together by covalent bonds between the central carbon and the two adjoining oxygens atoms. Only CaO (III) has an ionic bond, and it is considered to be partially ionic. Calcium carries a +2 charge, and oxygen carries a -2 charge. This makes D the correct choice. The structures and bonds are shown below:



71. Choice D is correct. All of the compounds contain only one calcium per molecule, so the smallest mass percent of calcium is found in the compound with the greatest molecular mass. The mass percent is found by dividing the mass of calcium by the mass of the compound. The molecular mass for CaO is 56 grams per mole; for $\text{Ca}(\text{OH})_2$ it is 74 grams per mole; for CaCO_3 it is 100 grams per mole; and for CaCl_2 it is 111 grams per mole. The heaviest compound is CaCl_2 (choice D), meaning choice D has the greatest denominator and therefore the smallest mass percent of calcium. This makes choice D the best answer.
72. Choice A is correct. To calculate the percent yield, the actual yield (in moles) is divided by the theoretical yield (in moles). The actual yield in moles is $5 \text{ g}/100 \text{ g}\cdot\text{moles}^{-1}$, while the theoretical yield in moles is $10 \text{ g}/74 \text{ g}\cdot\text{moles}^{-1}$. The percent yield is found as follows:

$$\text{Percent yield} = \frac{\frac{5 \text{ g}}{100 \text{ g}\cdot\text{mole}^{-1}}}{\frac{10 \text{ g}}{74 \text{ g}\cdot\text{mole}^{-1}}} \times 100\% = \frac{5 \times 74}{100 \times 10} \times 100\% = \frac{74}{100 \times 2} \times 100\% = \frac{37}{100} \times 100\% = 37\%$$

The percent yield is 37%, which makes choice A the correct answer.

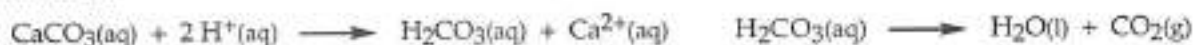
73. Choice D is correct. The mass percent of calcium in calcium oxide is the mass of calcium (40 grams per mole) divided by the mass of calcium oxide (56 grams per mole) and multiplied by 100%. $40/56 \times 100\%$ is greater than 50%, which eliminates choices A and B, and is greater than 66.7%, which eliminates choice C. Only choice D is greater than 66.7%, so choice D must be the correct answer.

$$\text{Range method: } \frac{40 \text{ g}}{56 \text{ g}} \times 100\% > \frac{40}{60} \times 100\% = \frac{2}{3} \times 100\% = 66.7\%$$

$$\text{Calculation method: } \frac{40 \text{ g}}{56 \text{ g}} = \frac{5}{7} = 5 \times \frac{1}{7} = 5 \times 0.143 = 0.715 = 71.5\%$$

74. **Choice A is correct.** Because matter can be neither created nor destroyed, the total mass of reactants must equal the total mass of the products. There are 38.0 grams of reactants, so choice D is eliminated for having more than 38.0 grams of products. The question from here becomes a limiting reagent problem. We must decide whether there is leftover water (is CaO the limiting reagent?), leftover calcium oxide (is water the limiting reagent?), or an exact mixture (there is no leftover and thus no limiting reagent.) The moles of CaO equal 28/56 (which is one-half) and the moles of H₂O equal 10/18 (which is greater than one-half.) Because the calcium oxide reacts with the water in a one-to-one ratio, there is leftover water. This means CaO is the limiting reagent in the reaction, so the reaction does result in leftover H₂O. The only choice with leftover water is A.

75. **Choice D is correct.** The N in the answer choices stands for *normality*, which is the molarity of the protons that can be generated by the acid. Normality is the molarity of an acid times the equivalents of hydronium per molecule. CaCO₃(aq) reacts according to the equations shown below. A 2 : 1 ratio is needed to reach full neutralization.



5 mL $\times$ 0.2 moles/liter = 1 mmole CaCO₃, so 2 mmoles of a strong acid (which yield H⁺) are needed to reach equivalence. Choice A is a strong base, so eliminate it. Choice B yields only 1 mmole of H⁺, and choice C yields 3 mmole of H⁺, so eliminate both choices B and C. Choice D yields the 2 mmole of H⁺ that are needed. The best answer is choice D. For a monoprotic acid, normality and molarity are the same. However, for a diprotic acid, the normality is double the molarity; and for a triprotic acid, the normality is triple the molarity.

76. **Choice C is correct.** This is a case where Equation 1.2 in the text, $M_{\text{initial}}V_{\text{initial}} = M_{\text{final}}V_{\text{final}}$, must be used.

$$0.500\text{ M} \times 5.0\text{ mL} = M_{\text{final}} \times 55.0\text{ mL} \therefore M_{\text{final}} = \frac{5}{55} \times 0.50 \therefore M_{\text{final}} = \frac{1}{11} \times 0.50 < \frac{1}{10} \times 0.50 = 0.050$$

The final molarity is therefore less than 0.05 M. Without pounding numbers, you should see that choice C is the best choice (the choice just less than 0.050 M). Choice D is too far below 0.050 M to be reasonable. Do not solve math questions exactly, but find a range into which only one answer choice fits. In this case, the range is from just less than 0.050 (maybe 0.040 or so) to 0.050. Only choice C fits into that range.

77. **Choice C is correct.** The final concentration can be determined by taking a weighted average of the concentrations for the two initial solutions before they are mixed. A quick observation you should apply to this question is that if the mixture were a fifty-fifty mixture by volume (the mixture of equal volume solutions), then the final concentration would lie exactly between (be the mean value of) the two values. In this particular question, the mean value would be 0.250 M. Because there is an excess of the higher-concentration solution (30 mL at 0.30 M, with only 20 mL at 0.20 M), the answer should be closer to 0.30 M than 0.20 M (making it greater than 0.250 M). This eliminates choices A and B. The exact value can be solved for as follows:

$$\frac{\text{Total moles}}{\text{Total volume}} = \frac{(20\text{ mL} \times 0.20\text{ M}) + (30\text{ mL} \times 0.30\text{ M})}{50\text{ mL}} = \frac{4 + 9}{50} = \frac{13}{50} = 0.26, \text{ choice C.}$$

78. **Choice C is correct.** 50 grams CaCO₃(s) is equal to 0.50 moles CaCO₃(s), because the molecular mass of CaCO₃(s) is 100 grams per mole. Because there is only one carbon in CaCO₃(s), 0.50 moles CaCO₃(s) yield 0.50 moles of CO₂. 0.50 moles of CO₂ at STP occupies a volume of: 0.50 moles $\times$ 22.4 liters/mole = $\frac{1}{2} \times 22.4 = 11.2\text{ L}$. Pick C.

Passage XII (Questions 79 - 86)

Industrial Chemicals

79. **Choice A is correct.** The greatest amount of potassium per gram of compound is found in the compound with the greatest mass percent of potassium. The greatest mass percent of potassium is found in K₂O, as was mentioned in the passage. This means that the best choice is answer A. The mass percents for the four salts are as follows:

$$\begin{array}{ll} \text{K}_2\text{O}: \frac{2 \times 39.1}{(2 \times 39.1) + 16} = \frac{78.2}{94.2} = \frac{39.1}{47.1} & \text{K}_2\text{SO}_4: \frac{2 \times 39.1}{(2 \times 39.1) + 32 + (4 \times 16)} = \frac{78.2}{174.2} = \frac{39.1}{87.1} \\ \text{KCl}: \frac{39.1}{39.1 + 35.5} = \frac{39.1}{74.6} & \text{KNO}_3: \frac{39.1}{39.1 + 14 + (3 \times 16)} = \frac{39.1}{101.1} \end{array}$$

The relative mass percents when comparing the four values are: $\frac{39.1}{47.1} > \frac{39.1}{74.6} > \frac{39.1}{97.1} > \frac{39.1}{101.1}$

The greatest mass percent of the choices is found with K₂O (choice A).

80. Choice D is correct. The compounds in choices A, B, and C all have two nitrogen atoms per compound, while the compound in choice D has only one nitrogen atom in the compound. The lowest mass percent of nitrogen is in choice D, because the four compounds are roughly comparable in molecular mass, but they have different masses of nitrogen. The mass percents for the four choices are shown below:

$$\begin{aligned} \text{A. } \text{H}_2\text{NCONH}_2 &\Rightarrow \frac{28\text{g N}}{60\text{g H}_2\text{NCONH}_2} = \frac{28}{60} \approx \frac{7}{15} < \frac{1}{2} \\ \text{B. } \text{NH}_4\text{NO}_3 &\Rightarrow \frac{28\text{g N}}{80\text{g NH}_4\text{NO}_3} = \frac{28}{80} = \frac{7}{20} = 35\% \\ \text{C. } (\text{NH}_4)_2\text{SO}_4 &= \frac{28\text{g N}}{132\text{g } (\text{NH}_4)_2\text{SO}_4} = \frac{28}{132} = \frac{7}{33} = 21\% \\ \text{D. } \text{NH}_4\text{H}_2\text{PO}_4 &\Rightarrow \frac{14\text{g N}}{115\text{g NH}_4\text{H}_2\text{PO}_4} = \frac{14}{115} < 14\% \\ \frac{28}{60} &> \frac{28}{80} > \frac{28}{132} > \frac{14}{115} \quad (= \frac{14}{115}) \end{aligned}$$

81. Choice C is correct. The mass percent of nitrogen in urea is just less than 50%:

$$\text{Mass percent urea} = \frac{28\text{g N}}{60\text{g H}_2\text{NCONH}_2} = \frac{28}{60} = \frac{7}{15} < \frac{1}{2}$$

The only answer choice that is just less than 50% is choice C.

82. Choice A is correct. According to the passage, the phosphorus content of the compound is critical, so no phosphorus should be wasted in unreacted reactant. This can be prevented by adding the reactants in a ratio that avoids the presence of any leftover phosphorus-containing reactant. The goal is to conserve the phosphorus-containing compound, which makes choice A the best answer.

83. Choice B is correct. The percent yield for a reaction is defined as the actual quantity of product isolated from the reaction mixture, divided by the theoretical quantity of product that should form, as determined from the limiting reagent of the reaction. According to the balanced equation, for every one mole of $(\text{NH}_4)_2\text{CO}_3$ that reacts, one mole of $(\text{NH}_4)_2\text{SO}_4$ forms. This means that 10 grams of $(\text{NH}_4)_2\text{CO}_3$ (which is $10/96$ moles $(\text{NH}_4)_2\text{CO}_3$) should produce $10/96$ moles of $(\text{NH}_4)_2\text{SO}_4$, which is $(10/96 \times 132)$ grams $(\text{NH}_4)_2\text{SO}_4$. Dividing the mass of $(\text{NH}_4)_2\text{SO}_4$ obtained (10 grams) by the theoretical mass of $(\text{NH}_4)_2\text{SO}_4$ that should have formed (13.75 grams) determines the percent yield. The calculation of the yield is shown below:

$$\begin{aligned} \text{Theoretical mass: } \frac{10}{96} \times 132 &= \frac{1320}{96} = \frac{330}{24} = \frac{110}{8} > \frac{104}{8} + \frac{6}{8} = 13\frac{3}{4} = 13.75 \\ \text{Range for the percent yield: } 50\% &< \frac{5}{7} \approx \frac{10}{14} < \frac{10.00}{13.75} < \frac{3}{4} = \frac{10}{13.33} = 75\% \end{aligned}$$

The value for the percent yield lies somewhere between 50% and 75%, which makes choice B the best answer.

84. Choice C is correct. All of the choices have the same volume (100 mL) and the same mass of salt, so the greatest molarity of potassium is present in the solution with the greatest number of moles of potassium in solution. Hence, the greatest molarity results from using the compound with the greatest mass percent of potassium, which corresponds to the salt with the lowest molecular mass. Choice D is eliminated, because NO_3 is heavier than Cl , so KCl has a lower molecular mass than KNO_3 , and thus a greater mass percent of potassium than in KNO_3 . Choice B is eliminated, because SO_4 is heavier than CO_3 , so K_2CO_3 has a lower molecular mass than K_2SO_4 , and thus a greater mass percent of potassium than K_2SO_4 . The question boils down to determining the relative mass percents of potassium in KCl and K_2CO_3 . The molecular mass of KCl is 74.5 grams per mole, and the molecular mass of K_2CO_3 is 138.2 grams per mole. Potassium carbonate (K_2CO_3) yields two potassium ions per molecule, so to have the same mass percent of potassium as KCl , its molecular mass would have to be double that of KCl (which equals 149 grams per mole.) Because K_2CO_3 is less than twice as massive as KCl , the relative denominator is smaller when calculating the mass percent of K_2CO_3 than when calculating the mass percent of KCl . The greatest mass percent of potassium, and thus the most grams of potassium in a 10.0-gram salt sample, is found in K_2CO_3 . This means that the highest concentration of potassium is present in the aqueous K_2CO_3 solution, so choose answer choice C for best results.

85. **Choice A is correct.** It is easiest to isolate a solid (precipitate) from solution, because the solid can be filtered from solution rather easily. Equally, a liquid and gas can flow, while a solid cannot, so removing liquid and gas to leave solid behind is easy. This is the crux of many lab techniques. A liquid must be distilled away from solution, a solute must be either distilled, extracted, or removed by chromatography, and a gas must be collected in a gas-trap free of air. The best answer is thus choice A.
86. **Choice A is correct.** This is another case where the question focuses on a topic we have yet to discuss. There are a few things you should recall. First, the lower the pK_a value is, the stronger the acid is. Second, when the pH of the solution exceeds the pK_a of an acid, that site exists predominantly in the deprotonated state. Statement III can immediately be eliminated, because the first proton of sulfuric acid is strong, meaning it has a very low (negative) pK_a value associated with it. From the products of the last equation in the passage, it can be seen that $H_2PO_4^-$, SO_4^{2-} , and HF all simultaneously exist in solution. Sulfuric acid has lost both of its protons, phosphoric acid has lost its first proton, and hydrofluoric acid has yet to lose a proton. The first proton of phosphoric acid is lost more readily than the proton of HF, which means that pK_{a1} of H_3PO_4 is lower than the pK_a value of HF. Statement I is thus true. Because the second proton of sulfuric acid has been lost, while the second proton of phosphoric acid has not been lost, the second proton of sulfuric acid is more acidic than the second proton of phosphoric acid. The result numerically is that pK_{a2} of H_2SO_4 is less than pK_{a2} of H_3PO_4 . Statement II is thus false. Only statement I is true, so the best answer is choice A.

Questions 87 - 100

Not Based on a Descriptive Passage

87. **Choice C is correct.** This is a limiting reagent question. The balanced equation for the reaction is as follows:
- $$1 C_3H_8(g) + 5 O_2(g) \longrightarrow 3 CO_2(g) + 4 H_2O(g)$$
- There are 25/32 moles $O_2(g)$ and 20/44 moles $C_3H_8(g)$ as reactants. According to the mole ratio from the balanced equation, the moles of $O_2(g)$ must be five times that of $C_3H_8(g)$. The moles of $O_2(g)$ needed to react with 20/44 moles of $C_3H_8(g)$ is 100/44. The value of 25/32 is less than 1, so there is less than one mole of $O_2(g)$ present. The value of 100/44 is greater than 2, so more than two moles of $O_2(g)$ are needed. This means that oxygen ($O_2(g)$) is depleted before $C_3H_8(g)$, making $O_2(g)$ the limiting reagent and choice C the best answer.
88. **Choice B is correct.** If you know this answer from your biology classes, trust your knowledge and don't waste time doing the calculation. You should know that hemoglobin contains four iron atoms. That is the point of this question. The math is time-consuming, so save time wherever you can. If you didn't know that fact about hemoglobin, then doing the math was necessary. The mass of iron in hemoglobin is the mass percent of iron times the total mass of hemoglobin. $0.33\% \times 68,000 = .0033 \times 68,000 = 3.3 \times 68 = 204 + 20.4 = 224.4$ grams Fe. The number of irons is therefore $224.4/55.8$, which makes 4 (choice B) the best answer choice.
89. **Choice D is correct.** The first method which probably comes to your mind is the method learned from general chemistry classes. In this method, the 9 grams of carbohydrate are converted to 0.050 moles of carbohydrate by dividing by the molecular mass of 180. The 6.72 liters of carbon dioxide at STP are then divided by 22.4 liters per mole to yield 0.30 moles CO_2 . The ratio of CO_2 to O_2 in the combustion of a monosaccharide is 1 : 1. Therefore, 0.30 moles of oxygen reacted as well, and thus the mole ratio of carbohydrate to O_2 to CO_2 is 0.05 : 0.30 : 0.30, which equals 1 : 6 : 6. This is a long-winded, but valid, solution. A shorter method is as follows:
- In the combustion of a monosaccharide, the mole ratio of oxygen reactant to carbon dioxide to water is always 1 : 1 : 1. A monosaccharide has the molecular formula $C_nH_{2n}O_n$, so it has a molecular mass of $30n$ ($12n + 2n + 16n$) grams per mole. For a molecular mass of 180 grams per mole, the value of n is 6, so the formula for the monosaccharide is $C_6H_{12}O_6$. The mole ratio of O_2 to CO_2 is therefore 6 : 6. Pick D and feel relaxed.
90. **Choice B is correct.** Choices A and C are eliminated, because they cannot be a molecular formula. For a compound with just carbon, oxygen, and hydrogen, the number of hydrogens must be an even number. With an odd number of hydrogens, the bonding does not work out. To decide between choices B and D, you must determine the molecular mass of the compound, by dividing the mass by the volume and then multiplying this value by the molar volume:

$$\frac{1 \text{ gram}}{0.26 \text{ liters}} \times \frac{22.4 \text{ liters}}{1 \text{ mole}} = \frac{22.4 \text{ grams}}{0.26 \text{ moles}} = \frac{22.4 \text{ grams}}{0.25 \text{ moles}} = 4 \times \frac{22.4 \text{ grams}}{\text{mole}} = 89.6 \frac{\text{grams}}{\text{mole}}$$

Only $C_4H_6O_2$ has a molecular mass just under 90 g/mole, so choice B is the best answer.

91. Choice A is correct. This is a question testing your knowledge of the unit factor method. Barium oxide (BaO) can form barium metal and oxygen gas (O₂) upon decomposition. These are the only two products possible from the decomposition of barium oxide. Hence, for every two moles of BaO, one mole of O₂ forms. To solve this problem, the 1.0 grams of BaO must be converted to moles, and the moles of BaO in the 1.0-gram sample must be multiplied by one-half. This gives the moles of O₂ formed. Multiplying the moles of O₂ by 22.4 liters per mole (the volume for one mole of gas at standard temperature and pressure), gives liters of oxygen gas produced by decomposition:

$$1.0 \text{ grams BaO} \times \frac{1 \text{ mole BaO}}{153 \text{ grams BaO}} \times \frac{1 \text{ mole O}_2}{2 \text{ moles BaO}} \times \frac{22.4 \text{ liters}}{1 \text{ mole}} = \frac{22.4}{153 \times 2} = \frac{22.4}{306} \approx 0.073 \text{ liters}$$

Only choice A is less than 0.1 liters, so pick choice A and make yourself one point wiser.

92. Choice B is correct. To determine the grams of product, the conversion must go through moles. This question is a standard grams reactant-to-moles reactant-to-moles product-to-grams product conversion.

$$\frac{2.0 \text{ grams MgNH}_4\text{PO}_4}{137.3 \text{ grams/mole}} \times \frac{1 \text{ mole Mg}_2\text{P}_2\text{O}_7}{2 \text{ moles MgNH}_4\text{PO}_4} \times \frac{222.6 \text{ g Mg}_2\text{P}_2\text{O}_7}{1 \text{ mole Mg}_2\text{P}_2\text{O}_7} = \frac{2 \times 222.6}{137.3 \times 2} = \frac{222.6}{137.3} \text{ g Mg}_2\text{P}_2\text{O}_7$$

$$2 > \frac{222.6}{137.3} > 1$$

The mass of Mg₂P₂O₇(s) is greater than 1, but less than 2. The only answer that falls between 1 and 2 is answer choice B.

93. Choice D is correct. We know nothing of the molecular mass, so any formula we determine is the empirical formula (simplest formula). This eliminates choice A. Because the elemental mass of Z is twice the elemental mass of X, a compound with equal mass quantities of X and Z contains twice as many moles of X as moles of Z. The mole ratio of X to Z is therefore 2 : 1; thus, the empirical formula is either X₂Z or X₂Z₁. Pick choice D to feel correct.

94. Choice C is correct. The math here is beyond the MCAT level, but the point of such a question is to encourage you to use intuition and simplify the question. You can always eliminate some of the answer choices, and at least narrow your guess from random choice to a fifty-fifty chance of success. This question is simplified by knowing the possible oxidation states of iron. From your experience with hemoglobin, you know that iron has an oxidation state of either +2 or +3, and that oxygen has an oxidation state of -2 when it is coupled with a metal. If oxygen has an oxidation state of -2, then iron cannot have an oxidation state of +2 or +3 in either Fe₃O₂ or FeO₃. Choices A and D are eliminated based on outside knowledge. This problem can now be solved using your mathematical intuition. In the iron oxide sample, there are 11.89 grams of iron and 5.10 grams of oxygen (found by subtracting 11.89 from 16.99). The molecular weight of iron (Fe) is 55.85 grams per mole, and the molecular weight of oxygen (O) is 16.0 grams per mole. An empirical formula is based on mole ratios, so the elemental masses must be converted into mole quantities. 5.1 over 16 is just less than one-third, so there are roughly 0.30 moles of oxygen in the iron oxide. 11.89 over 55.85 is just over one-fifth, so there is roughly 0.20 moles of iron in the iron oxide. There are more moles of oxygen than iron, which eliminates choice B (FeO has equal moles of iron and oxygen) and leaves choice C as the correct answer by default. Minimize doing math by using your background knowledge and logic.

$$\frac{\text{Fe}_{11.89} \text{O}_{5.10}}{55.85 \quad 16.0} = \frac{\text{Fe}_{1.0} \text{O}_{0.33}}{5 \quad 3} = \text{Fe}_{0.2} \text{O}_{0.33} \approx \text{Fe}_2\text{O}_3$$

95. Choice D is correct. It is not possible to have 26 hydrogens and only 4 carbons in a stable compound. You may recall from organic chemistry that the maximum number of hydrogens in a hydrocarbon or carbohydrate is 2n + 2, where n is the number of carbons in the compound. This eliminates choice C. Perhaps it is easiest to solve this question by first determining the number of carbons in the molecule. The mass percent of carbon in the molecule is 53.4%. When this percentage is multiplied by 90 grams/mole, it tells us that just over half the mass is carbon. Carbon has a mass of 12 grams per mole, so the value must be a multiple of 12. The mass due to carbon is 48 grams, which is the mass of four carbon atoms. The molecular formula therefore contains four carbons, eliminating choices A and B. The only choice left is answer D, which does in fact have roughly 11% hydrogen. Again, the goal in preparing for your exam is to be able to solve these questions quickly and using as much intuition as possible. You are not rewarded only for being thorough; you are also rewarded for being fast. Take logical short cuts whenever they present themselves. Do not blindly repeat the calculation techniques you learned in your general chemistry courses.

96. **Choice A is correct.** Because this is a reduction reaction, the oxidation state must decrease, which eliminates choices C and D. By assuming oxygen has a -2 oxidation state, it is possible to reduce this problem to simple algebra. In the reactant, the sum of the oxidation states for the four oxygen atoms is -8. In order for the overall charge on $\text{CrO}_4^{2-}(\text{aq})$ to be -2, the chromium must have an oxidation state of +6. The sum of +6 and -8 is the ionic charge of -2. In the product $\text{Cr}_2\text{O}_3(\text{s})$, each chromium must be +3 to cancel out the three oxygen atoms (at -2 each) to make the overall molecule neutral. For this reason, choice A is the best answer.

97. **Choice D is correct.** The metal combines in a 2 : 3 ratio with oxygen to form an oxide that is 53% metal by mass. Assuming a 100-gram sample implies that there are 53 grams of metal and 47 grams of oxygen in the 100-gram sample of the metal oxide. This means that the ratio of (53 divided by the molecular mass of the unknown metal) to (47 divided by the molecular mass of oxygen, which is 16) is equal to 2 : 3. This can be solved long-hand or by intuition. Rather than solve the math exactly, it is a good idea to plug the values for all four answer choices into the setup, and get a rough estimate:

$$\frac{53 \text{ g metal}}{\text{MW}_{\text{metal}}} \cdot \frac{47 \text{ g O}}{16 \frac{\text{g}}{\text{mole}}} = 2 : 3 \Rightarrow \text{Given that } \frac{47}{16} \text{ is roughly } 3, \frac{53 \text{ g metal}}{\text{MW}_{\text{metal}}} \text{ must be about } 2$$

The molecular masses of the four answer choices are 40.0, 55.9, 52.0, and 27.0 respectively. 47 divided by 16 is slightly less than 3, so 53 divided by the molecular mass of the metal must be slightly less than 2. This eliminates the first three choices (40.0, 55.9, and 52.0), leaving 27.0 (aluminum) as the best answer. The correct answer is choice D. You could also have eliminated choice A from knowing that calcium cannot achieve a +3 oxidation state (or charge), so calcium cannot combine with oxygen in a 2 : 3 ratio.

98. **Choice B is correct.** The mass percent of carbon is the mass of carbon in each compound, divided by the total mass of each compound, multiplied by 100%. For each compound, the mass percent can be determined as follows:

$$\begin{array}{llll} \text{Acetic acid: } \frac{24}{60} \times 100\% & \text{Ethanol: } \frac{24}{46} \times 100\% & \text{Methyl acetate: } \frac{36}{74} \times 100\% & \text{Glucose: } \frac{72}{180} \times 100\% \\ \frac{24}{60} < \frac{30}{60} = \frac{1}{2} & \frac{24}{46} > \frac{23}{46} = \frac{1}{2} & \frac{36}{74} < \frac{37}{74} = \frac{1}{2} & \frac{72}{180} < \frac{90}{180} = \frac{1}{2} \end{array}$$

Of the choices, only choice B (ethanol) has a fraction greater than one-half. This implies that the greatest mass percent of carbon is found in ethanol. The best answer is therefore choice B.

99. **Choice B is correct.** The MCAT test writers can ask about the same concept in many ways. It is more important that you walk away from this question knowing the concept than it is getting the question correct. The most carbon dioxide results from the compound with the greatest mass percent of carbon, as long as there are equal masses of each sample present initially. There was one gram of each compound initially, before oxidation, so our only concern is with the mass percent of carbon. The greatest mass percent of carbon is found in ethanol, so the greatest amount of carbon dioxide results from the oxidation of ethanol. The best answer is choice B.

100. **Choice C is correct.** Because there are two Cl in the $\text{Mg}(\text{ClO}_4)_2$ molecule, the mass percent of chlorine is the mass of two Cl over the mass of $\text{Mg}(\text{ClO}_4)_2$.

$$\frac{\text{Mass chlorine}}{\text{Mass compound}} = \frac{2 \times 35.5}{24.3 + (2 \times 35.5) + (8 \times 16)} = \frac{71}{24.3 + 71 + 128} = \frac{71}{223.3}$$

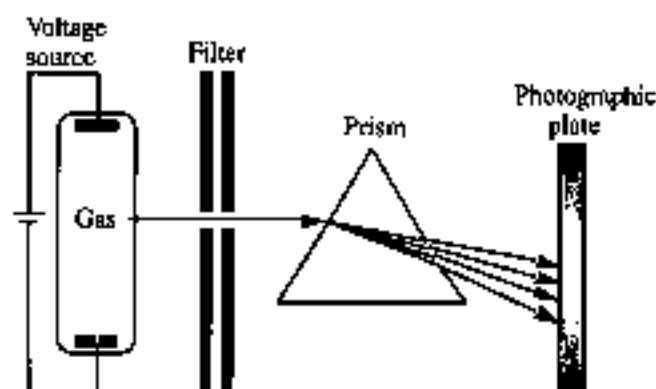
$$\frac{71}{213} > \frac{71}{223.3} > \frac{71}{284}, \text{ where } \frac{71}{213} = \frac{1}{3} = 33.3\%, \text{ and } \frac{71}{284} = \frac{1}{4} = 25\%. \text{ So } 33.3\% > \frac{71}{223.3} > 25\%$$

The mass percent is a little less than 33.3%, so choices A and B are eliminated. The value is just a little less than 33.3%, because the denominator (223.3) is just a little greater than 213. The best answer is the choice that is slightly less than 33.3%, which is 31.8%, choice C. Choice D can be eliminated, because it is less than 25%.

Section II

Atomic Theory

by Todd Bennett



Atomic Structure

- a) Subatomic Particles
- b) Isotopes
- c) Average Atomic Mass
- d) Classical Experiments and Machinery
 - i. Thomson Experiment
 - ii. Mass Spectroscopy
 - iii. Millikan Oil Drop Experiment
 - iv. Rutherford Experiment
- e) Heisenberg's Uncertainty Principle
- f) Atomic Model
- g) Hydrogen Energy Levels

Electronic Structure

- a) Electronic Theory
- b) Effective Nuclear Charge
- c) Electronic Spin Pairing
- d) Spin Pairing and Magnetism
- e) Electronic Density and Orbitals
- f) Collective Orbital View of Energy Levels
- g) Electronic Configuration
- h) Quantum Numbers

Periodic Trends

- a) The Periodic Table
- b) General Elemental Periodic Trends
 - i. Atomic Radius
 - ii. Ionization Energy
 - iii. Electron Affinity
 - iv. Electronegativity
- c) Periodic Families (Groups)

Light Absorption and Emission

- a) Excitation and Relaxation
- b) Atomic Spectrum of Hydrogen
- c) Electromagnetic Spectrum
- d) Visible Spectrum and Colors
 - iii. Emitted Color
 - iv. Reflected Color and the Color Wheel
- e) Fluorescence
- f) Photoelectric Effect

Nuclear Chemistry

- a) Nuclear Particles
- b) Nuclear Decay and Capture
- c) Half-Life



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Atomic Structure Section Goals

1.  **Be familiar with the location, mass, and charge of sub-atomic particles.**
An atom, from the chemist's perspective, is composed of a nucleus made up of protons and neutrons, surrounded by orbiting electrons. You must know the Bohr model and the properties of each particle with regard to its charge, mass, and effect on the atom. For instance, additional electrons result in the formation of an anion, while a decrease in electrons results in the formation of a cation.
2.  **Be familiar with energy levels and energy transitions.**
The electrons orbit in distinct, quantized levels. There is a base (lowest-energy) level, and excited states above that to which electrons jump when energy is added to the atomic system. Energy is absorbed when an electron is elevated from the ground state to an excited state. Energy, in the form of a photon, is released when an electron drops to the ground state from an excited state.
3.  **Understand electronic configurations.**
Electrons fill the orbitals according to a defined sequence, described by the Aufbau principle. The keys facts to recall about the principle are that there are two electrons per orbital, and there is one s-orbital per level, three p-orbitals per level, five d-orbitals per level, and seven f-orbitals per level. The filling order has exceptions in the transition metals, lanthanides, and actinides. There are exceptions that allow for half-filled shell stability, as is seen with chromium.
4.  **Understand quantum numbers.**
Quantum numbers are used to define the orientation and location of an electron. There are four quantum numbers: n , l , m_l , and m_s . You must be familiar with the quantum numbers and how to determine them for a given electron within an atom.
5.  **Be familiar with periodic trends.**
Knowing the periodic trends for the main-group elements is essential. The trends that you must be familiar with include ionization energy, atomic radius, electron affinity, and electronegativity (although electronegativity of the noble gases need not be committed to memory). It is important that you be familiar with each basic trend and the reason for that trend. Be able to explain any deviations from standard periodic behavior.
6.  **Understand isotopes, average atomic mass, and isotopic labeling.**
Isotopes are atoms with the same number of protons, but a different number of neutrons in their nuclei. For some elements, there are several isotopes. The elemental mass (average atomic mass) found in the periodic table is a weighted average of all of the isotopes of that given element. Isotopes are chemically equivalent, so they are hard to detect without using a mass spectrometer. Isotopes, because of their chemical similarity, are substituted for one another in some reactions to serve as labels.
7.  **Understand nuclear decay and half-life associated with first-order decay.**
It is essential that you understand the breakdown and buildup of nuclei through the gain and loss of nuclear particles. It is also important to be able to determine the concentration of a species by combining the initial concentration with half-life information. The half-life is the period of time required for half of a given sample to decay to some product. Most examples will involve determining the concentration at a given time for a reaction (or process) that follows first-order decay kinetics.

Atomic Theory

From the work of Thomson, Millikan, Rutherford, Bohr, Pauling, and others, we have a modern view of the atom and its fundamental structure. The core of the atom by this view is a nucleus composed of protons and neutrons held together by an unbelievably strong force. The details of the nucleus are not well understood, but using a simplistic model, the majority of the mass and all of the positive charge of the atom is found at the core. The nucleus is surrounded by orbiting electrons that stay in distinct orbits, no two of which are exactly the same (Hund's rule). Coulomb's law explains the mutual attraction of the orbiting electrons and the nucleus. Electron energy levels are based on Coulomb's law, although quantum mechanics is invoked to explain the overall behavior of the orbiting electron. The closer an electron is on average to the nucleus, the more tightly it is held, and the greater the energy required to remove that electron from the atom. The space in which an electron is believed to orbit is referred to as an *orbital*. Each orbital is distinct from all other orbitals, and electrons have the option of spinning clockwise or counterclockwise as they occupy the orbital. Only two electrons may occupy each orbital at the same time, and they must have opposite spins to do so. These distinct orbitals are quantized energy levels where the electrons are said to reside. This fundamental idea is the basis of all atomic theory.

Atomic behavior and electronic configurations are explained by the concept of electronic bookkeeping. The ease or difficulty of gaining, removing, or sharing an electron are determined by the location of the electron. An electronic configuration is an account of all of the electrons in an atom. Quantum numbers are specific for each individual electron. Questions about electronic configuration and quantum numbers on the MCAT should be some of the simpler questions you see, so be sure to get them correct, as they are worth the same amount of points as the more difficult topics. Understanding the experiments and applications associated with the absorption and emission of light is critical to performing well on the MCAT. Absorption spectroscopy, lasers, and fluorescent tubes are just a few examples of devices that utilize light that you are expected to understand for the MCAT.

Equally important as understanding the structure of an atom and the energetics of its orbiting electrons is seeing the effect the structure has on the reactivity of the atom. There are distinct trends in atomic radius, ionization energy, electron affinity, and electronegativity that can be traced back to the filling of electrons in the atom. Knowing the reasoning behind the periodic trends is more important than memorizing the direction of each trend and notable exceptions to the trend.

The last of the topics that fits under the heading of atomic structure is nuclear chemistry. This section should be one of the easier sections, as scientists do not understand the topic in enough detail to expect you to have a deep understanding of it. Nuclear chemistry is made easy by knowing the particles and the processes of decay and capture, and your ability to do algebra. Know the definitions of isotopes, nuclear decay, nuclear capture, nuclear particles, and half-life. This chapter starts with an elementary look at subatomic particles and finishes with an overview of the atom, the energy states of its particles, and transitions between energy levels.

Atomic Structure

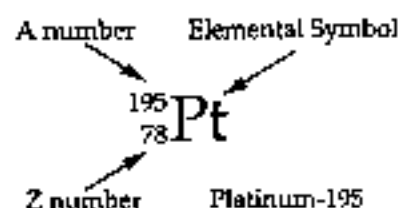
Subatomic Particles

Our most fundamental view of matter is that atoms are composed of three subatomic particles. These particles are the proton, neutron, and electron. Each subatomic particle is unique in its properties and position. A proton and neutron are comparable in mass, while an electron is roughly $1/1800$ of the mass of a proton. The proton and electron carry the same magnitude of charge, but with opposite sign. Physical properties associated with each particle are listed in Table 2.1

Particle	Mass (kg)	Mass (amu)	Charge (C)	Charge (e)
Electron	9.11×10^{-31}	5.49×10^{-4}	-1.602×10^{-19}	-1
Neutron	1.67×10^{-27}	1.0087	0	0
Proton	1.67×10^{-27}	1.0073	1.602×10^{-19}	+1

Table 2.1

The nucleus is made up of neutrons and protons. Mass and charge are the two measurements repeatedly used to describe subatomic particles and whole atoms. By convention, the mass of an atom is said to be due only to the protons and neutrons, because the electrons are essentially massless. An atom normally carries a neutral charge, unless it has lost or gained electrons. In a neutral atom, the number of protons equals the number of electrons. A convenient shorthand notation is used to describe every element. There is the mass number (A) used to describe the number of protons and neutrons in an atom. There is also the atomic number (Z) used to describe the number of protons in an atom (and thus the number of electrons in a neutral atom). The notation used to represent each atom is shown in Figure 2-1.



The A number tells us that the combined number of neutrons and protons is 195. The Z number tells us that the number of protons is 78. Combining these facts tells us that there are 117 neutrons and 78 protons in this atom.

Figure 2-1

Isotopes

Isotopes are atoms of the same element that contain a different number of neutrons but the same number of protons within their nuclei. Isotopes, having the same number of protons, are chemically similar but have different atomic masses. Isotopes react the same way chemically and thus can be distinguished only by mass separation techniques (such as mass spectroscopy). Isotopes are often used as markers in chemical labeling experiments and as tags in nuclear magnetic spectroscopy studies. An isotope can be traced from the source (where it is added) to the endpoint in a physiological system, a biochemical pathway, or a reaction mechanism. Typical examples of isotopes include ^1H (standard hydrogen), ^2H (deuterium), and ^3H (tritium). Common isotopes used in labeling studies include deuterium, tritium, carbon-13, carbon-14, phosphorus-32, and iodine-121. These isotopes can be monitored either by the radioactive decay they emit or by nuclear magnetic resonance imaging (known as NMR spectroscopy).

Example 2.1

How do the subatomic particles in ^{26}Al and ^{27}Al differ?

- A. The two are isotopes with a different number of neutrons.
- B. The two are isotopes with a different number of protons.
- C. The two are ions with a different number of electrons.
- D. The two are ions with a different number of protons.

Solution

Because the symbol is Al, the atomic number is always 13, implying that there are 13 protons. This eliminates choices B and D. In either case there is no charge, so there are thirteen electrons present and the species is not an ion. This eliminates choice C and leaves choice A as the correct answer. In ^{26}Al , there are 13 neutrons (because 13 and 13 sum to 26), while in ^{27}Al , there are 14 neutrons (because 14 and 13 sum to 27). In ^{26}Al , there are 13 neutrons, 13 protons, and 13 electrons, while in ^{27}Al , there are 14 neutrons, 13 protons, and 13 electrons. The two differ by one neutron, making them isotopes. Choice A is good.

Example 2.2

What symbol represents the neutral atom with fifteen electrons and sixteen neutrons?

- A. $^{31}_{15}\text{Ga}$
- B. $^{16}_{15}\text{P}$
- C. $^{31}_{15}\text{P}$
- D. $^{31}_{16}\text{S}$

Solution

Because there are fifteen electrons, there are also fifteen protons in the neutral atom. This makes the atomic number 15, which is associated with the element phosphorus, and eliminates choices A and D. The mass is 31 (15 protons and 16 neutrons), so the symbol is $^{31}_{15}\text{P}$, choice C.

Example 2.3

Which of the following is an isotope of element 35 containing 44 neutrons?

- A. $^{81}\text{Br}^-$
- B. $^{79}\text{Br}^+$
- C. ^{81}Br
- D. ^{79}Kr

Solution

The atomic number of bromine is 35, regardless of which isotope is being observed. Because the element in this question is bromine, choice D is eliminated immediately. When the 35 protons of bromine are coupled with 44 neutrons, the mass of the bromine isotope is 79 amu. The answer which shows a mass of 79 amu is choice B. Choice A is an anion, which is formed upon the addition of an electron.

Average Atomic Mass

The average atomic mass of an element is a weighted average of the masses of all of the isotopes, an average that takes into account abundance. The reference for all isotopic masses is ^{12}C , which is assigned a mass of 12.000 amu. All isotopic masses are measured relative to ^{12}C .

Example 24

When a sample of magnesium is subjected to mass spectroscopy, it is found that there are three detectable isotopes. What is the average atomic mass of magnesium, given that the relative isotopic abundance is 79% ^{24}Mg , 10% ^{25}Mg , and 11% ^{26}Mg ?

- A. 23.71 g/mole
- B. 24.31 g/mole
- C. 25.83 g/mole
- D. 26.92 g/mole

Solution

The math associated with this question is time-consuming, so before diving into it, review the answers to see what can be eliminated. The average mass has to fall within the range between the lightest and heaviest isotopes. In other words, the average is in the middle somewhere. This eliminates choices A and D. The most abundant isotope is ^{24}Mg , so the average atomic mass should be close to 24 grams per mole. Because the heavier isotopes are 21% the total mass of the sample, the average atomic mass is a little more than 0.21 grams above 24. The best answer is choice B. The mathematical solution is shown below, but in preparation for this exam, use rigorous math only to confirm intuitively obvious answers.

$$\begin{aligned}\text{Average atomic mass} &= 79\%(24) + 10\%(25) + 11\%(26) \\ &= (0.79 \times 24) + (0.10 \times 25) + (0.11 \times 26) \\ &= (0.79 \times 24) + (0.10 \times (24 + 1)) + (0.11 \times (24 + 2)) \\ &= (0.79 \times 24) + (0.10 \times 24) + (0.10 \times 1) + (0.11 \times 24) + (0.11 \times 2) \\ &= (0.79 \times 24) + (0.10 \times 24) + (0.11 \times 24) + (0.10 \times 1) + (0.11 \times 2) \\ &= (1.00 \times 24) + (0.10 \times 1) + (0.11 \times 2) = 24 + 0.10 + 0.22 = 24.32\end{aligned}$$

The math is shown in an intuitive, step-wise fashion that does not require a calculator. The number used as the center point, 24, was chosen knowing that the average value was around 24.

Classical Experiments and Machinery

Classical experiments in chemistry are studies that determined the fundamental features of matter. These experiments are important, because they define the nature of matter and the smallest units of matter. Of interest when looking at matter are charge, mass, location, and composition. There are three classical experiments: the Thomson experiment (used to determine the sign of charges), the Millikan oil drop experiment (used to determine the magnitude of charge), and the Rutherford experiment (used to determine the location of dense particles). The experiments as listed here are modified from their original form to emphasize the rationale, rather than the procedure. Of equal significance today is the mass spectrometer (used to determine the charge to mass ratio for a particle), which was developed to support the classical experiments.

Thomson Experiment

The Thomson experiment demonstrated the existence of opposite charges in an atom and that charge is a fixed quantity. Thomson deflected a stream of charged particles (electrons) using an external electric field (the plates of a capacitor). Because the stream of particles bent in a uniform fashion, Thomson concluded that there was a consistent charge-to-mass ratio for the particles.

Procedure and Apparatus

In the Thomson experiment, a beam of electrons was generated traveling left to right, as shown in Figure 2-2. Thomson observed that when he applied an electric field (a positively charged plate on one side and a negatively charged plate on the other) perpendicular to the electron beam, he could deflect it by an exact amount each time. The magnitude of deflection depends on the strength of the field (charge on the plates) and the mass of the electron. Reversing the plates of the external field gets the opposite deflection.

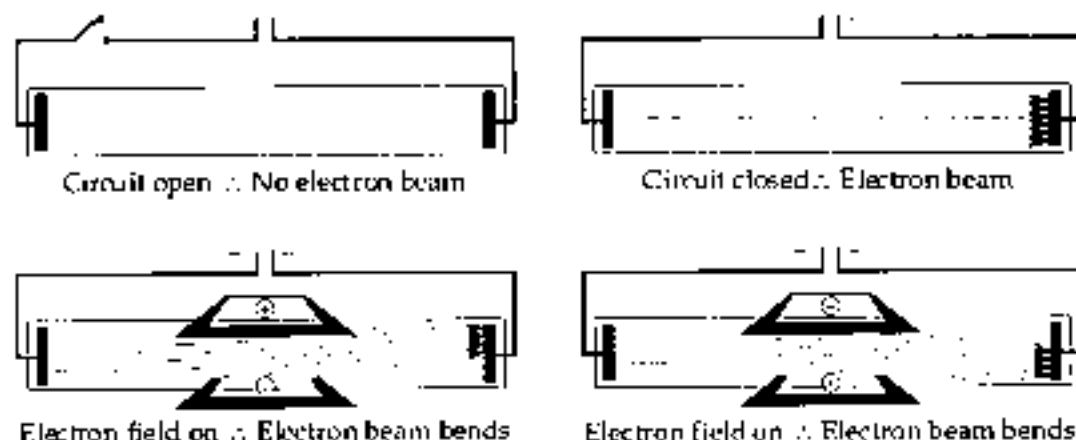


Figure 2-2

Results and Conclusions

Because the direction of the deflection changed when the orientation of the field changed, Thomson concluded that there must be two types of charge that oppose one another. Because the arc of the deflection was constant, Thomson concluded that electrons have a fixed charge-to-mass ratio, measured to be 1.76×10^8 C/g.

Adapted Thomson Experiment (in MCAT testing style)

What is presented here is a spin-off of the Thomson experiment. The Thomson experiment concept can be applied to various particle beams besides the electron beam. A simplified version of the apparatus has an accelerating field (not shown), which accelerates particles to the left, a double filter to ensure uniform linear trajectory, and an electric field perpendicular to the vector of entry. Charged particles are deflected according to their sign of charge. Four particles are considered. Their pathways are shown in Figure 2-3.

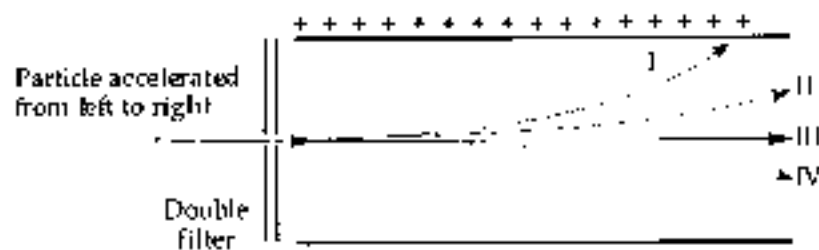


Figure 2-3

Pathway I is attributed to an electron beam, while Pathways II, III, and IV are not identified. The MCAT test writers are more interested in ascertaining your intuitive understanding of problems than your superficial knowledge of facts. Questions are likely to introduce obscure particles and test you on their properties, which must be derived from the experiment. Consider the following question based on the data in Figure 2-3.

Example 2.5

Pathway II is taken by which of the following?

- A. A positron
- B. A neutrino
- C. A gamma ray
- D. A muon

Solution

The particle following Pathway II must be negatively charged, based on the direction of its deflection. It deflects in the same direction as the electron. Although some of the particles in the answer choices may seem unfamiliar, the question still can be answered. Based on the name, you should deduce that a positron is positively charged. This eliminates choice A. A positron is an anti-electron, in that it has the same mass as an electron, but the opposite charge. Based on the name, you should deduce that a neutrino is neutrally charged. This eliminates choice B. A neutrino is essentially massless and carries no charge. They are difficult to detect. Detection of neutrinos is done through collision and scintillation. Choice C is eliminated, because a gamma ray is a photon. A photon does not bend in an electric field. Photons may be refracted, but that requires a change in medium. The only choice left is a muon, choice D. In all likelihood, you do not know what a muon is. This question is not testing your knowledge of particles; it is testing your reasoning abilities. A muon traces a different pathway than the electron, because it is about 200 times as massive as an electron.

Mass Spectrometry

A mass spectrometer is designed to measure the charge-to-mass ratio for a charged particle. This is accomplished by sending a particle into a perpendicular magnetic field and observing the degree to which it curves. The degree of arcing (radius of curvature) for a particle can vary with mass, initial velocity, magnitude of charge, and the strength of the magnetic field. As momentum increases (either mass or initial velocity), the particle deflects less, so the radius of curvature increases. As the charge magnitude increases, the force causing deflection increases, so the particle deflects more, causing the radius of curvature to decrease. By comparing the curvature for an atomic or molecular ion to a known standard, the mass of the unknown ion can be determined. The mass spectrometer is used in general chemistry to determine isotopic abundance. Recall that isotopes are the same element with a different number of neutrons. In organic chemistry, the mass spectrometer is used to determine molecular mass and fragmentation behavior to help elucidate the structure of an unknown compound. Figure 2-4 shows a basic schematic design for a mass spectrometer. Note that some mass spectrometers may also have a velocity selector situated between the accelerating region and the deflecting region.

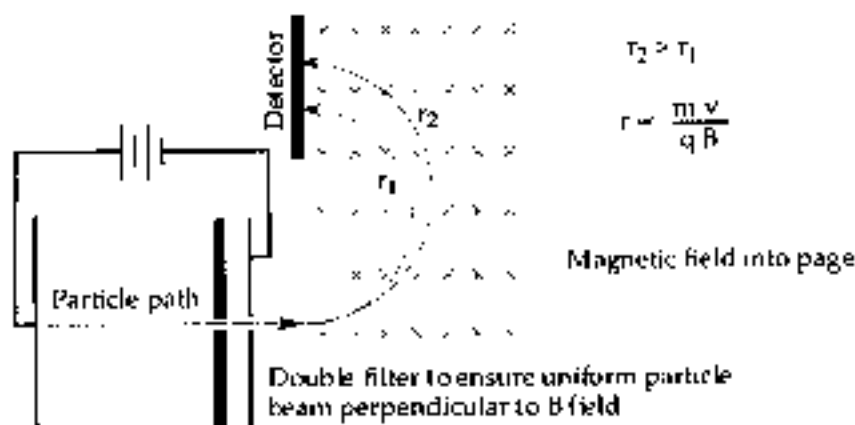


Figure 2-4

Procedure and Apparatus

The mass spectrometer embodies a simple concept. Force depends on mass, so when an equal force is applied to different masses, they accelerate at different rates. The mass spectrometer takes advantage of this by accelerating charged particles in motion using a magnetic field. The procedure is as follows:

1. An element or molecule is ionized using high-energy electron impact or incident electromagnetic radiation. The ionized particle is then accelerated to the cathode plate. The strength of the accelerating region (voltage of the electric field) is adjustable, so the velocity of the particle can be set to any desired value.
2. The particle passes through the double filter to ensure a uniform perpendicular beam.
3. The particle leaves the accelerating region and enters a perpendicular magnetic field (oriented into the page as drawn in Figure 2-4) where it is deflected in a counterclockwise, radial fashion by the perpendicular magnetic force.
4. The radius is ascertained from the strike point against a collision detector. The mass-to-charge ratio is calculated from the radius of the arc.

Example 2.6

Which of the changes to the system would increase the radius of curvature in the mass spectrometer?

- A. Using a doubly ionized rather than singly ionized element
- B. Using an isotope with fewer neutrons
- C. Increasing the accelerating voltage
- D. Increasing the magnetic field

Solution

The trick here is translating what the answer choices mean in terms of the physics of the apparatus. Using a doubly ionized element (+2 cation) rather than a singly ionized element (+1 cation) results in a greater q value, so a greater degree of deflection is observed. This reduces the radius of curvature (r), so choice A is invalid. Using an isotope with fewer neutrons results in a reduced value for mass, so a greater degree of deflection is observed. This reduces the radius of curvature (r), so choice B is invalid. Increasing the accelerating voltage results in a greater velocity, so a lessened degree of deflection is observed. This increases the radius of curvature (r), so choice C is the best answer. Increasing the magnetic field results in a greater force, so a greater degree of deflection is observed. This reduces the radius of curvature (r), so choice D is invalid. This question can be answered from a conceptual perspective or based on the relationship of variables described in a formula, with equal success. Whether you use equations or intuition is a matter of personal preference and timing.

The mass spectrometer can be used to determine the charge-to-mass ratio for the electron and the proton. The mass of a neutron is obtained by looking at the mass difference between known isotopes. For instance, the mass difference between $^1\text{H}^+$ (a proton) and $^2\text{H}^+$ (deuterium ion) is the mass of one neutron. The mass spectrometer can also be used to determine the isotopic abundance for the component atoms of each element. Common isotopes that should be memorized include: ^{12}C (the most abundant isotope of carbon), ^{13}C (used in carbon NMR), ^{14}C (used in carbon dating, because it undergoes decay), ^1H (the most abundant isotope of hydrogen), ^2H (deuterium, used in proton NMR solvents), ^3H (tritium, used in radio-labeling experiments), ^{235}U (used in nuclear fission), and ^{238}U (the most abundant isotope of uranium).

Millikan Oil Drop Experiment

The Millikan oil drop experiment is a difficult experiment to perform. Its aim is to suspend a charged oil drop in an electric field. To do this, an electron must be added to the oil drop or the oil drop must be ionized by impact, before it is placed into the electric field. We will consider adding the electron here. This cannot be accomplished easily, given that the oil drop is neutral and has no affinity for the negatively charged electron. Falling oil drops pass through a beam of electrons where some oil drops are penetrated at random by an electron. Ideally, the electron penetrates the core of the oil drop and comes to rest, due to the viscosity of the oil. The suspension of an electron in the oil drop produces a charged oil drop. Enough charged oil drops continue to fall, one of which eventually passes through a pore in the upper plate of a capacitor. The two plates of the capacitor are separated, but both lie within the walls of a glass cylinder. Figure 2-5 is a basic schematic of the apparatus used in the Millikan oil drop experiment.

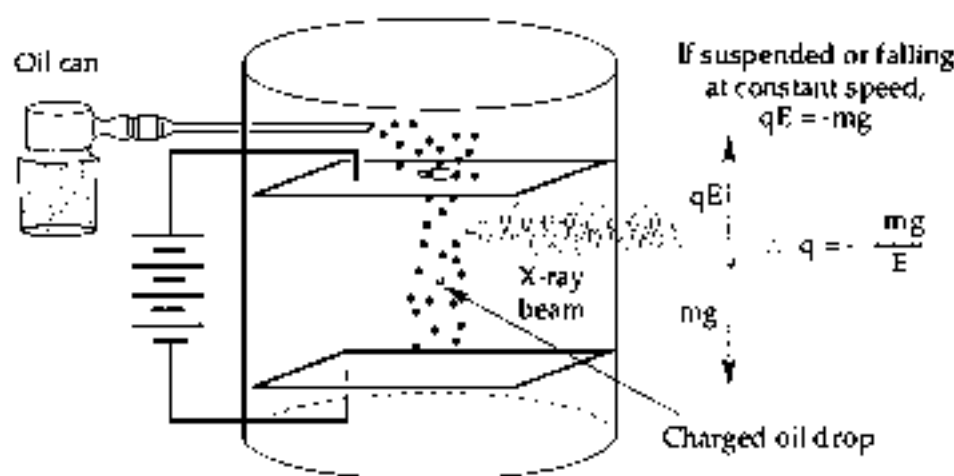


Figure 2-5

Procedure and Apparatus

In the Millikan experiment, a fine mist of oil droplets is allowed to fall through a tiny pore in the upper capacitor plate into a region where a uniform electric field exists. The oil droplets fall because of gravity, so Millikan set out to apply a force that could stop the droplets from falling. If the droplets are suspended, then the force applied equals the gravitational force (a known quantity that is dependent on the mass of the falling object). The applied force is formed by charging the oil droplets by exposure to either an electron beam or an x-ray beam (only one of the beams is applied in different trials of the experiment). An electric field is applied to suspend the charged droplets. The uncharged oil drops fall unaffected by the field. Despite the difficulty of this experiment, Millikan obtained enough valid data so that an average measurement was put forth and accepted by the scientific community at large.

- 1 The falling oil drop gains a charge by either losing or engulfing an electron as it falls (depending on the version of the experiment). On rare occasions, the falling oil drop both gains an electron after passing through the pore in the upper plate. The electric field strength is adjustable, so that the oil droplet can be suspended.
- 2 If the particle is suspended (or falling at a constant velocity), then the net force is zero, so $mg = -qE$. Because we know g and we can set the electric field strength (thus we know E), by plugging in the average mass of an oil drop (we know an average m), we can solve for q , the charge of the electron.

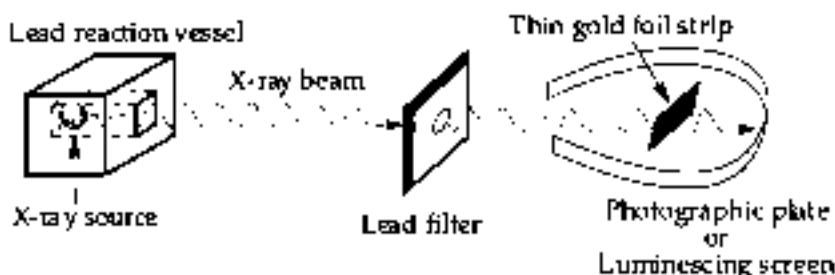
Results and Conclusions

The charge of an electron has a fixed numerical value that is the same for all electrons. The value for this fundamental unit of charge is 1.6×10^{-19} C. The charge of a proton is found to have the same magnitude, but opposite sign of the electron. When a proton is combined with an electron, there is no net charge. The Millikan oil drop experiment was difficult to carry out, so even after many trials, there were only two significant figures in the final number.

The Thomson experiment, the Millikan oil drop experiment, and the use of the mass spectrometer generally address questions about the subatomic particles in terms of *what they are*. The *where they are* questions are answered by interactions with incident light and particle beams. The Rutherford experiment is the most significant location experiment we will consider.

Rutherford Experiment

The Rutherford experiment determined that atoms have dense nuclei with nearly all of the atomic mass centrally concentrated, and that metals have uniformly spaced atoms in their microscopic composition. The Rutherford experiment relies upon a technique that can be employed to find a ball in the bushes in the dark. "What does finding a ball in the bushes at night have to do with the MCAT?" you may ask. Well, when a ball is lost in the bushes, it is easier to find its location by looking for its shadow than by hunting for the ball itself. By shining light through the bushes and observing the shadow cast against the background of a wall, a circular shadow can give hints as to the location of the spherical ball. Depth in the bushes can be obtained by first moving the flashlight closer to the bushes, then moving it farther away. The size of the shadow varies with position of the light source, so the relative position of the ball to the light source is determined from shadow dimensions. In essence, the ball can be located and its dimensions can be found without ever seeing the ball. This same principle was used by Rutherford to find subatomic particles. A subatomic particle is smaller than a ball, so a light source of significantly shorter wavelength must be employed. In this example, we use x-rays as the light source. In the analogy, the bushes had to be thin for the light to pass through them, so a thin strip of gold foil is used for the study. Rather than looking for a shadow on the wall, the experiment uses photographic paper to collect the x-rays that pass through. Figure 2-6 shows the basic design of the experiment.

**Figure 2-6****Procedure and Apparatus**

An incident beam (x-rays, alpha particles or electrons may be used) is focused and aimed at a thin slice of gold metal, thin enough that the beam is able to penetrate and pass through the gold foil. Gold is chosen, because it has a large nucleus and its atoms pack in such a manner where light can pass through its lattice more easily than other metals. If alpha particles are used, a luminescent screen is placed around the gold foil to detect where the particles pass through the foil and strike the luminescent screen (which glows when struck by an alpha particle). If x-rays are used, a photographic plate is placed around the gold foil to detect where the photons pass through the foil and strike the film. In the alpha-particle version of the experiment, some particles are deflected by the gold sample, resulting in parts of the luminescing screen never illuminating.

Results and Conclusions

Because the incident beam mostly passes straight through the sample, with deflection (ricocheting particles) being observed in only a few cases, it is concluded that the atom is made up predominantly of empty space. The mass associated with the atom occupies very little space and is not spread uniformly through the material. Atoms are composed of a nucleus holding the mass (the protons and neutrons.) This dense nucleus carries all of the mass besides that of the electrons. But given that the electrons are of such low mass, it is not possible to discern their whereabouts from the Rutherford experiment.

The Rutherford experiment disproved the diffuse particle model (referred to as the *plum pudding model*.) In the version of this experiment using the x-ray beam, the photographic screen displayed the output shown in Figure 2-7.

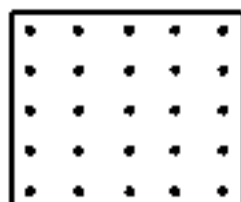


Figure 2-7

Dark spots represent areas where no x-rays struck the film. In essence, the spots are the shadows of the sub-atomic particles. Based on the distribution of the dark spots, it is concluded that the mass of gold is not evenly distributed in the gold foil, but is in fact found in concentric, dense nuclei. Because the nuclei (spots) are evenly spaced, the gold atoms must be arranged in a lattice structure. This is because the shadow pattern on the film mimics the distribution of particles in the material. Gold is chosen because it is malleable and it has a massive nucleus, thus it diffracts x-rays more readily than lighter elements, like aluminum. This is why x-rays in medical imaging show bones and teeth (rich in calcium nuclei) rather than tissue (rich in the light carbon, oxygen, and nitrogen nuclei). In order to analyze blood using x-rays, a heavy salt must be added to the solution, often barium iodide.

Example 2.7

What hypothesis did the Rutherford experiment support?

- A. Atoms combine in definite proportions.
- B. Atoms contain subatomic particles.
- C. Protons and electrons carry opposite charges.
- D. Solids are made of atoms with a dense nucleus and vast empty space between nuclei.

Solution

Choice A is Dalton's law of definite proportions. Choice B is particle theory. Choice C is the Thomson experiment, conducted by observing the deflection of the beam in a cathode ray tube when an external electric field is applied. The Rutherford experiment involves the bombardment of a thin piece of metal foil with either high-energy photons (x-rays) or a beam of electrons (Rutherford did both in separate experiments.) The photons pass through and strike an illuminating screen, which forms a shadow pattern indicative of the material's shape. Most of the beam passes through, with a minimal amount being either reflected or diffracted. The conclusion is that the material is essentially empty space, with a few dense nuclei scattered throughout the material. The spots on the screen are uniformly spaced, so the nuclei must also be evenly spaced within the element. This makes choice D the best answer.

The location of the electrons is not determined by Rutherford experiment. Electrons are too small and moving too fast to locate precisely. This is described by Heisenberg's uncertainty principle.

Heisenberg's Uncertainty Principle

The Heisenberg uncertainty principle quantifies the idea that it is not possible simultaneously to identify a particle's position (where something is) and velocity (how fast and where it's going). Equation 2.1 is the mathematical version of the Heisenberg uncertainty principle, where Δx is the uncertainty in position and $\Delta(mv)$ is the uncertainty in momentum.

$$\Delta x \Delta(mv) \geq h/4\pi \quad (2.1)$$

The basic premise here is that you can know either where something is, or how fast it is going, but not both at the same time. Think of using a camera to focus on a moving ball. If your aperture is small and the shutter speed is fast, then the picture of the ball shows you where it is, but you don't know where and how fast it's going. If your aperture is large and the shutter speed is slow, then the picture of the ball is a streak that shows you where and how fast it's going, but you don't know exactly where the ball is. Because we cannot locate an electron's precise position, we settle for a view where the electron is observed over time. This results in orbitals as a model for the orbiting electron.

Atomic Model

The Bohr model presents a simplified picture that explains the quantization of light and the reproducibility of spectra. The basic premise is that electrons occupy specific circular orbits about the nucleus, and thus the electrons have specific energy levels (associated with each orbit). Electrons can exist only in specified orbits (electronic shells), so each energy level of an atom is quantized. Figure 2-8 shows this:

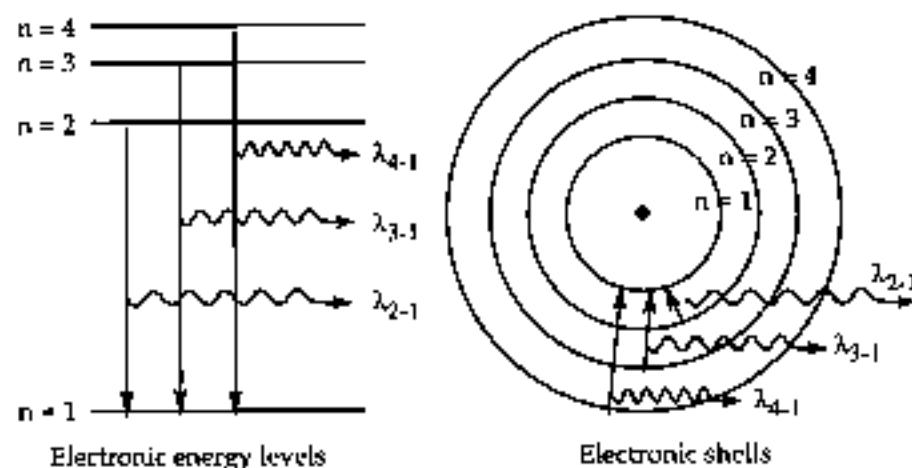


Figure 2-8

The energy levels are spaced according to the energetics of transition between the levels. More energy is required to carry out transitions when the electron is nearest to the nucleus. The electrons are situated in various energy levels (known more accurately as *orbitals*). These are quantized states that electrons occupy. Principle energy levels are numbered 1 to ∞ , where $n = 1$ is the lowest electronic energy level. Energy must be absorbed by the atom for an electron to elevate to a higher energy level. This is referred to as both *excitation* of an electron and *absorption* of energy. Conversely, energy is emitted when an electron drops from a higher energy level (excited state) to the lowest energy level (ground state). A good analogy to electrons climbing energy levels is a

rocket ship escaping Earth's gravitational pull. It takes less energy to increase distance as the rocket ship pulls farther away from the Earth. Likewise, it takes less energy for an electron to increase energy levels as the electrons increase to higher energy levels (farther from the nucleus). For instance, it requires more energy for an electron to go from the $n = 2$ level to the $n = 3$ level, than for the electron to go from the $n = 3$ level to the $n = 4$ level in any atom. This is why the energy levels for electron states are drawn closer and closer as the principle quantum number (n) increases in Figure 2-8. This picture becomes a little more complicated if the rotational energy levels associated with an atom are combined with the electronic energy levels. At the level of understanding needed for answering MCAT questions, we ignore the rotational energy levels when we look at electronic energy levels. We shall consider only the principal energy level when considering electrons. Equation 2.2 is used to determine the energy of an electron in its principal energy level.

$$E = \frac{2\pi^2 m Z^2 e^4}{n^2 h^2} \quad (2.2)$$

E = energy (principal energy level) m = mass of an electron (9.11×10^{-31} kg)
 Z = nuclear charge e = the charge of an electron (1.6×10^{-19} C)
 n = the electronic energy level h = Planck's constant (6.63×10^{-34} J·sec)

The mass of an electron, m , the charge of an electron, e , and Planck's constant, h , are all constants, so when Equation 2.2 is considered as a proportionality, it becomes Equation 2.3.

$$E \propto \frac{Z^2}{n^2} \quad (2.3)$$

Example 2.8

According to Figure 2-8, how would the photon from an $n = 4$ to $n = 2$ transition compare to the photon from an $n = 2$ to $n = 1$ transition?

- The $n = 4$ to $n = 2$ transition is twice as energetic as the $n = 2$ to $n = 1$ transition.
- The $n = 4$ to $n = 2$ transition is more than twice as energetic as the $n = 2$ to $n = 1$ transition.
- The $n = 2$ to $n = 1$ transition is twice as energetic as the $n = 4$ to $n = 2$ transition.
- The $n = 2$ to $n = 1$ transition is more than twice as energetic as the $n = 4$ to $n = 2$ transition.

Solution

The distance between the $n = 4$ and $n = 2$ levels is less than the distance between $n = 2$ and $n = 1$, so transition energy is greater from the $n = 2$ level to the $n = 1$ level. The photon released from an $n = 2$ level to $n = 1$ level transition has more energy than the photon released from an $n = 4$ level to $n = 2$ level transition. This eliminates choices A and B. The transition energy from the $n = 2$ level to the $n = 1$ level is more than twice the transition energy from $n = \infty$ to $n = 2$, so it is definitely more than twice the energy of the $n = 4$ level to $n = 2$ level transition. Choice D is the best answer. This energy difference is shown in Figure 2-8. On the MCAT, you should assume that the diagrams are drawn to scale, unless otherwise noted in the question or passage.

The energy required to excite an electron from a lower level (orbital) to a higher level is often in the ultraviolet (UV) to visible range of electromagnetic radiation, so the energy given off as the electron drops back down to the lower level (orbit) is emitted as light energy. This is the basic principle behind spectroscopy. Lower energy levels have less absolute energy and thus are more stable states in which an electron can exist. The smaller the gap between energy levels, the less energy that is given off, and therefore the longer the wavelength of light that is emitted. One formula is important for understanding the relationship between the speed of light, the frequency and wavelength of light, and light energy. The energy of a photon and its wavelength of light are inversely proportional. Equation 2.4 sums this up, where E is the energy of the photon, ν is frequency, c is speed of the wave, and λ is wavelength.

$$E = h\nu = \frac{hc}{\lambda} \quad (2.4)$$

Hydrogen Energy Levels

Hydrogen is the simplest atom to study, because it has only one electron and one proton. Much of our atomic theory is extrapolated from what we know about hydrogen. Because energy is quantized and the energy of the electron depends on features of the hydrogen atom, the energy levels can be calculated. Equation 2.5 represents the energy of the different levels of hydrogen.

$$E = -2.178 \times 10^{-18} \left(\frac{Z^2}{n^2} \right) \quad (2.5)$$

Energy levels are defined as being negative relative to a free electron. If the electron is in the $n = \infty$ energy level, then $E = 0$, and the electron is free from a nucleus. Considering that photons are absorbed and emitted when electrons change energy levels, the more useful application of the energy equation involves transition energy. Equations 2.6 and 2.7 show the relationship between transition energy and the corresponding wavelength of the photon involved.

$$\Delta E = E_{\text{final}} - E_{\text{initial}} \therefore \Delta E = -2.178 \times 10^{-18} \left(\frac{1}{n_{\text{final}}^2} - \frac{1}{n_{\text{initial}}^2} \right) \quad (2.6)$$

$$\lambda = \frac{hc}{\Delta E} \quad (2.7)$$

The ionization energy of hydrogen from its ground state (from the $n = 1$ level) is 1312 kJ/mole. Because of the squaring of the principle energy level, the ionization of an electron in hydrogen from the $n = 2$ level is one-fourth of that value (328 kJ/mole). The transition energy from the $n = 1$ level to $n = 2$ level is the difference between the two values, 984 kJ/mole.

Electronic Structure

Electronic Theory

In atoms, the electrons orbit in distinct shells. Not all shells can hold the same number of electrons. Shells farther from the nucleus have a greater radius, and thus a greater capacity to hold electrons. Equation 2.8 gives the maximum occupancy of electrons in a shell, where n is the principle quantum number.

$$\text{Number of electrons in shell} = 2(n)^2 \quad (2.8)$$

Figure 2-9 shows electrons of the lithium atom in their respective shells. The first shell holds two electrons, so the third electron must occupy the second shell. The first shell is the *core shell*, while the outermost shell is the *valence shell*.

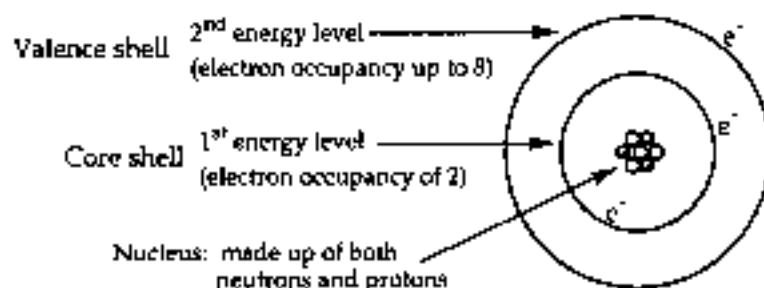


Figure 2-9

Effective Nuclear Charge (Nuclear Attraction)

Orbiting electrons are held in their orbits by an attractive electrostatic force to the nucleus. In addition to nuclear attraction, electrons are also repelled by other electrons. The net force is responsible for holding the valence electrons in place. The net charge exerted upon the valence electrons is referred to as the *effective nuclear charge*. The effective nuclear charge accounts for attraction to the nucleus, repulsion from core electrons, and minimal repulsion by other valence electrons. When approximating the effective nuclear charge (Z_{eff}), the nuclear charge is added to the core electron charge (a negative term). Figure 2-10 shows effective nuclear charge increasing while moving left to right across the periodic table.

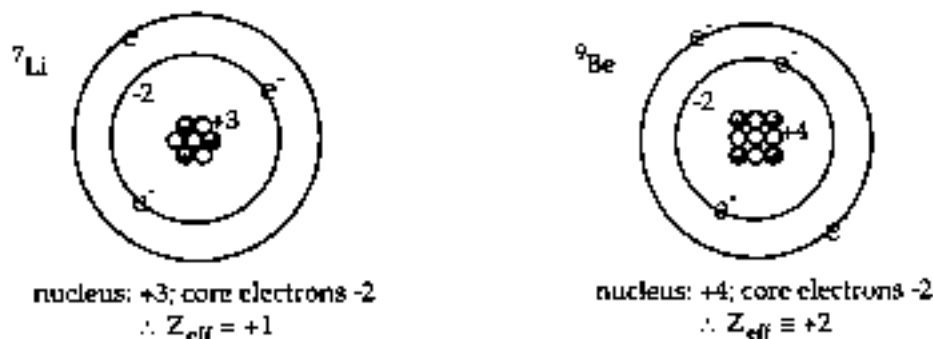


Figure 2-10

When we move from left to right in the periodic table, the nucleus of each succeeding atom adds a proton and the valence shell adds an electron. The effect of the extra valence electron is not as significant as the effect of the additional proton. As a result, the effective nuclear charge increases as the periodic table is scanned from left to right.

Each electron in an element travels a unique pathway, dictated by several principles. Electrons are difficult to track, so we have theories to explain electronic behavior, but they are just models to explain observed behavior. According to the Heisenberg uncertainty principle we cannot see an electron, but we can study its pathway over time. Charged particles in motion create magnetic fields, so by studying the magnetic field generated by a moving electron, it is possible to learn about the pathway and position of the moving electron. This is why two of the four quantum numbers associated with an electron refer to magnetism that results from a moving electron. We shall blend the many ideas about the electron that have evolved over time, starting with the most simplistic model, the Bohr model.

Electron Spin Pairing

Electrons fill orbitals in a pre-determined sequence, filling evenly into orbitals of equal energy with like spin (all orbitals get a single electron, said to be "spin up"), before placing a second electron with opposite spin into each orbital. The physical reality is that electrons may spin either clockwise or counterclockwise about their axis. Spinning charged particles generate magnetic moments, so the two opposite spins produce opposite magnetic fields. The magnetic fields generated by electrons revolving about their axis are referred to as either *spin up* (implying that the spin produces a magnetic field vector oriented upward) or *spin down*. By convention, electrons are said to fill orbitals spin up first, before filling spin down. Figure 2-11 shows the electron filling of lithium-7 and beryllium-9, where arrows represent electrons, and the orientation of the arrow implies spin.

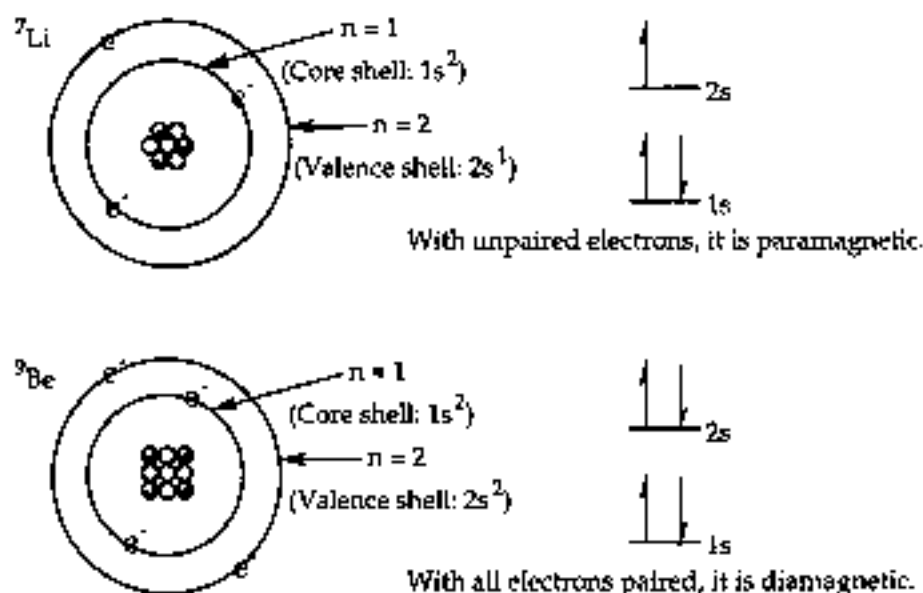


Figure 2-11

The shells represent energy levels an electron can occupy, while orbitals represent the region in which the electron is likely to be found. An s-orbital has spherical electron density. The difference between the 1s and 2s orbitals lies in their dimensions. The 1s has a smaller radius and has no nodal shells (regions where the electron has zero probability of existing.)

Spin Pairing and Magnetism

A *paramagnetic species* is defined as an atom or molecule that contains at least one unpaired electron. In organic chemistry, paramagnetic compounds are referred to as *radicals*. An unpaired electron is an electron that has no second electron spin paired with it. By convention, the first electron into an orbital is said to enter in a spin up fashion, thus an unpaired electron is a spin up electron in an orbital that has no spin down electron. Because the electron is unpaired, it is susceptible to magnetic fields. If an external magnetic field is applied to a paramagnetic species, the electron spins align with the field. This induces a magnetic moment into the compound, thus making it magnetic. This is to say that paramagnetic species can have magnetism induced into them.

A *diamagnetic species* is defined as an atom or molecule that contains no unpaired electrons. All electrons in the atom or molecule are spin-paired, meaning that every electron that is spin up will have a spin down electron sharing its orbital. By convention, the first electron into an orbital is said to enter in a spin up fashion, so the second electron is a spin down electron. Because all of the electrons are spin-paired, diamagnetic compounds are not susceptible to magnetic fields. If a magnetic field is applied to a diamagnetic species, half of the electron spins align with the field, forcing the other half to align against the field. No magnetic moment is induced into the compound. This is to say that diamagnetic species cannot have magnetism induced into them.

Electron Density and Orbitals

Atomic orbitals are three-dimensional pictorial representations of the region where an electron is likely to be found. Because we observe electrons over time, we look at where the electron usually is, and draw a probability map of the electron distribution. It's like look at a spinning fan. You cannot see each blade as they turn, but you can see over time the area where they spin. Figure 2-12 represents the electron density of an electron in an s-orbital over time, the orbital representation, and the probability map based on distance from the nucleus.

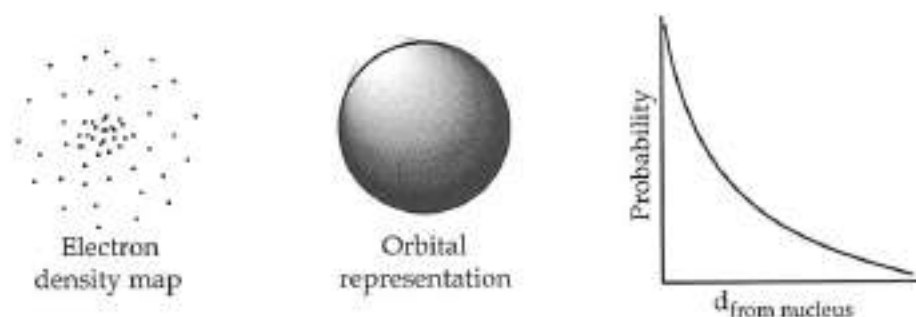


Figure 2-12

The electron density map shows that electrons are found most often near the nucleus. This is also represented by the graph of the probability of finding an electron as a function of its distance from the nucleus. The orbital representation is typically used by chemists to depict the s-orbital. The size of the sphere varies with the electron density map, depending on the atom. The shape of an orbital is defined by the distribution of electrons about the nucleus. Ninety-five percent of the time, the electron can be found within the boundaries of the orbital. We will look at the s-, p-, and d-orbitals in substantial detail, while f-orbitals will be considered, but in minimal detail. Most common elements do not have electrons occupying the f-orbitals.

S-orbitals

S-orbitals result from spherical distribution of the electrons about the nucleus. Figure 2-13 shows three different s-orbitals, where the principle quantum number represents the energy level and the average distance from the nucleus.



Figure 2-13

P-orbitals

P-orbitals result from barbell-like distribution of the electrons about the nucleus. Figure 2-14 shows the three different p-orbitals, each oriented about a different axis. Electrons are not found at the nucleus in p-orbitals. Absence of electron density at any point is referred to as a node. P-orbitals have one node at the nucleus that is part of a nodal plane between the two lobes.

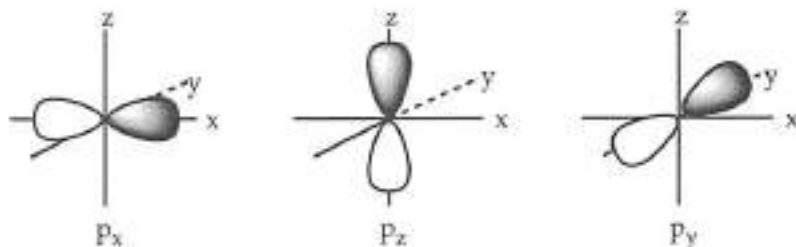


Figure 2-14

D-orbitals

D-orbitals result from double barbell-like distribution of the electrons about the nucleus. Figure 2-15 shows the five different d-orbitals, each oriented differently. D-orbitals have two nodal planes, and electrons are not found at the nucleus.

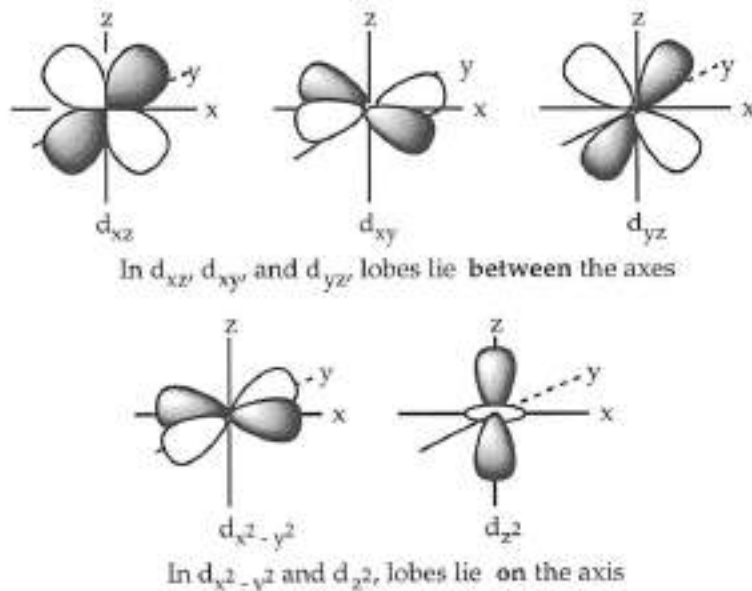


Figure 2-15

F-orbitals

F-orbitals result from triple barbell-like distribution of the electrons about the nucleus. There are the seven different f-orbitals, each oriented about a different plane or axis. Electrons are not found at the nucleus. F-orbitals have three nodal planes. Little chemistry is carried out with the f-orbitals, so they are uncommon.

Collective Orbital View of Energy Levels

Orbitals result from probability calculations, where energetics is considered. Different orbitals are associated with different energies. Conceptually, we use orbitals to show the energy and most frequent location of an electron. Figure 2-16 shows orbitals with relative size emphasized, from lowest energy levels to the higher energy levels. Levels are spaced according to energetics. Arrows represent electrons and their spin orientation. Because there are twelve electrons shown, the element represented is magnesium.

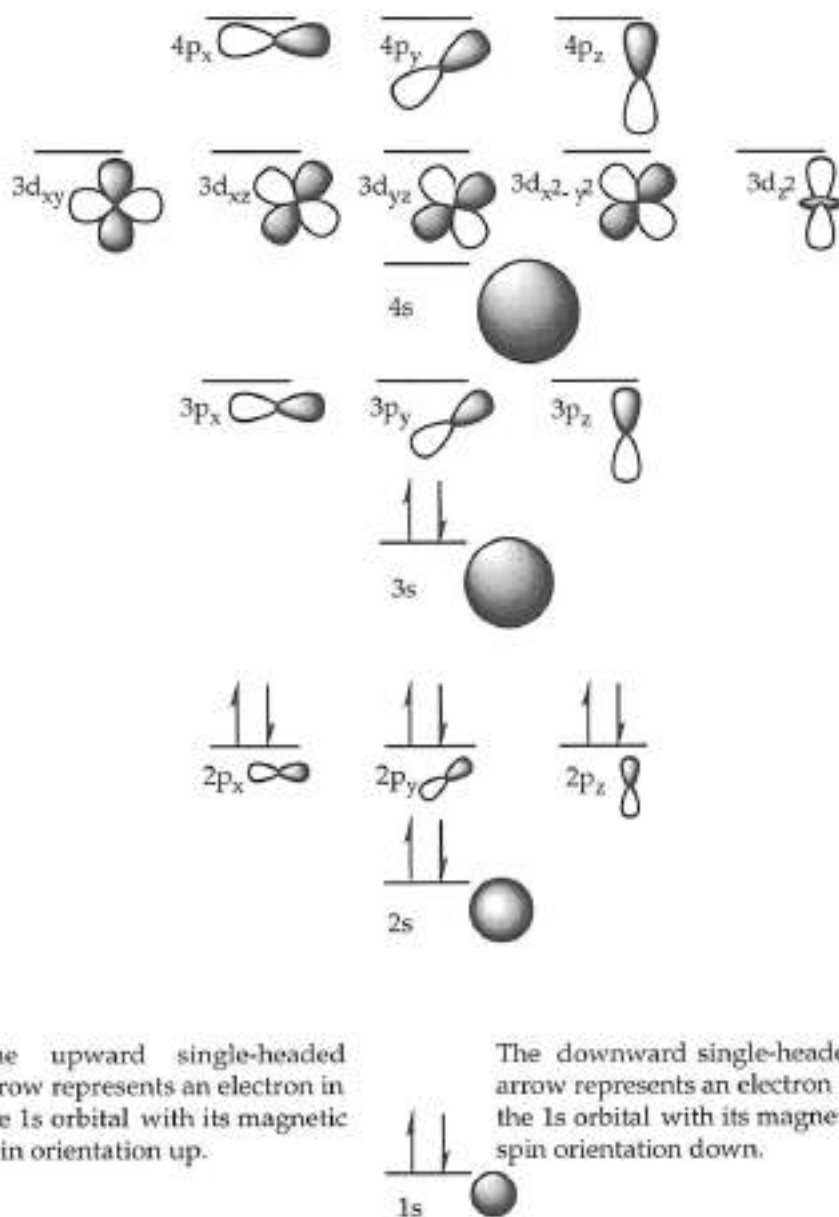


Figure 2-16

Electrons within an element fill the energy levels starting from the lowest energy. This means that the electrons within an element follow a specific filling order. There are some rules to consider when looking at electronic configurations.

Pauli's exclusion principle: No two electrons can have the same set of quantum numbers (n , l , m_l , m_s).

Hund's rule: Electrons completely fill lower energy levels before starting to fill higher energy levels. In a degenerate set of orbitals, electrons singly occupy each orbital before a second electron pairs up within the same orbital. Figure 2-17 demonstrates Hund's rule.

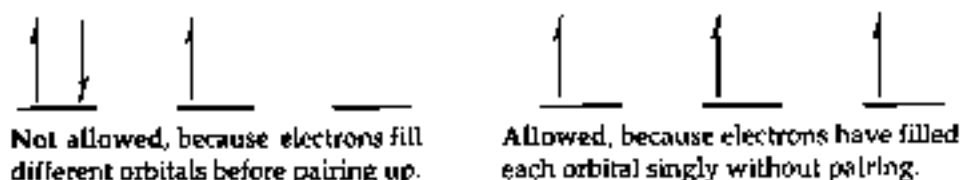


Figure 2-17

Aufbau principle: Electrons are added one by one to the shells, starting with the lowest energy level, and then into sequentially increasing energy levels. The numbers in Figure 2-18 represent the sequence of addition for the electrons.

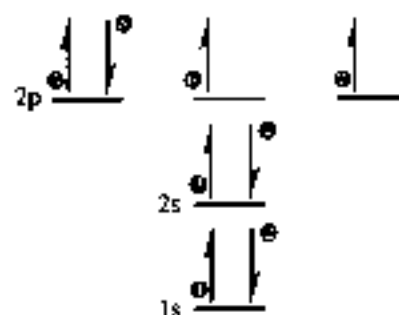


Figure 2-18

Electronic Configuration

Electronic configurations are shorthand notation for the electrons present in an atom and their energy levels. Electrons fill according to a set pattern, one that is derived from the Aufbau principle chart shown in Figure 2-19. By drawing the table and then sequentially following the arrows, the orbital filling sequence is generated. Couple this information with the orbital occupancy, and electronic configurations are seen to be systematic. For instance, the first line shows that the 1s level fills first, to an occupancy of two electrons. The next arrow crosses through the 2s level, so the 2s orbital is filled next. From here the third arrow shows that the 2p level followed then by the 3s level are filled. It continues down the chart. The first break from numerical sequencing comes when the 4s level is filled before the 3d level, despite the fact that the perimeter of the 3d level is closer to the nucleus than the perimeter of the 4s orbital. The reason for the apparent discrepancy is that the energy of the level is based on an average position of the electron, not the extreme position. Ionizing electrons are not removed from the atom in reverse order, however. Outer shell electrons are always removed first when forming cations. Figure 2.8 shows only the first five arrows, but the pattern continues. You should also be able to deduce the electronic configurations for neutral atoms, cations, anions, excited states, and any exceptions to the rules.

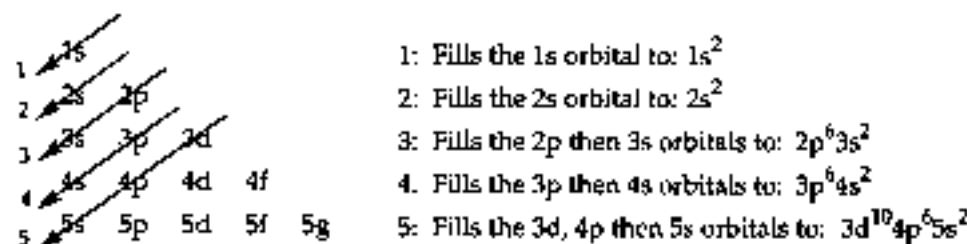


Figure 2-19

Figure 2-19 shows that the electrons fill according to the orbitals listed by the sequential arrows. It works by following the arrows sequentially. Arrow 1 goes through 1s, so the 1s orbital is filled first. Arrow 2 goes through 2s, so the 2s orbital is filled next. Arrow 3 goes first through 2p, then through 3s, so the 2p orbital is filled after the 2s orbital, followed by the filling of the 3s orbital. The process is repeated arrow after arrow until all of the electrons have been accounted for. Although g-, h-, and j-orbitals exist in theory, the periodic table contains no elements that have electrons in either g-, h-, or j-orbitals.

Example 2.9

The electronic configuration for manganese is which of the following?

- A. $1s^2 2s^2 2p^6 3s^2 3p^6 3d^7$
- B. $1s^2 2s^2 2p^6 3s^2 3p^6 3d^5$
- C. $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^5$
- D. $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^7$

Solution

Manganese (Mn) is element number 25, so a neutral manganese atom must contain 25 electrons. This eliminates choice B (only 23 electrons) and choice D (containing 27 electrons). Because the 4s orbital is filled before the 3d orbital, choice A is eliminated. This leaves only choice C.

Example 2.10

An element in which column of the periodic table is diamagnetic?

- A. Column 1 (alkali metals)
- B. Column 2 (alkaline earth metals)
- C. Column 6 (chalcogens)
- D. Column 7 (halogens)

Solution

A diamagnetic compound has all of its electrons spin-paired. This means that there must be an even number of electrons in the element. Based on the even number constraint, choices A and D are eliminated. Column 6 elements (the chalcogens) have a valence electronic configuration of $ns^2 np^4$, which results in two p-orbitals having only one electron each. This means that chalcogens are paramagnetic, eliminating choice C. This means that the alkaline earth metals in column 2 are diamagnetic, with a valence electronic configuration of ns^2 . The alkaline earth metals and the noble gases are diamagnetic. The best answer is choice B.

Electronic configurations may include abbreviations based on filled core shells. A filled core is represented by the noble gas that contains those same electrons. For example, aluminum (Al) is $1s^2 2s^2 2p^6 3s^2 3p^1$, which is equivalent to drawing it as $[\text{Ne}]3s^2 3p^1$. This shorthand is typical. In addition, you must be aware of some common exceptions to the Aufbau principle. Half-filled d-shell and filled d-shell stability results when a single electron is elevated from a lower energy level that is paired (usually the s-orbital) to yield even distribution of electrons in the d-level. Half-filled d-shell stability is seen with chromium, molybdenum, and tungsten. Filled d-shell stability is seen with copper, silver, gold, and some say platinum. Figure 2-20 shows the electronic configurations for chromium and copper, exceptions to the Aufbau filling order.

Half-filled d-shell stability in chromium: $[\text{Ar}]4s^1 3d^5$ rather than $[\text{Ar}]4s^2 3d^4$

Filled d-shell stability in copper: $[\text{Ar}]4s^1 3d^{10}$ rather than $[\text{Ar}]4s^2 3d^9$

Figure 2-20

Elements in the same column of the periodic table have similar valence shells and electronic configurations, with the notable difference being the shell number. For instance, Na is $[\text{Ne}]3s^1$ and potassium is $[\text{Ar}]4s^1$. This means that alkali metals are s^1 metals, and exhibit similar chemical behavior, given their common tendency to lose one electron. Blocks in the periodic table are named after the last electron in the electronic configuration. Alkali metals fall into the s-block by virtue of their last electron in an s-orbital. So far, we have viewed ground state electronic configurations. Ground state electronic configurations occur when the electrons occupy the orbitals in the exact predicted order, starting from least energetic and filling orbitals that are progressively of higher energy. An excited state electronic configuration occurs when any electron absorbs energy and moves to a higher energy level than it normally occupies in the ground state. The absorption and emission of energy, usually in the form of a photon, is associated with the excitation and relaxation of an electron, as it moves between the ground and excited states.

Example 2.11

Which electronic configuration represents an excited state?

- A. F: $1s^2 2s^2 2p^6$
- B. N: $1s^2 2s^2 2p^3$
- C. He: $1s^2$
- D. Li: $1s^2 2p^1$

Solution

An excited state electronic configuration does not follow energetic sequence. An excited state has at least one electron in an energy level higher than what is drawn as standard for the ground state. Be sure not to confuse an ion (either cation or anion) with an excited state. A cation is an atom that has a deficit of at least one electron and thus carries a positive charge. An anion is an atom that has an excess of at least one electron and thus carries a negative charge. In this question, choice A is a fluorine anion (it contains an extra electron), and choices B and C are normal. For Li, it should have $1s^2 2s^1$ as a ground state. The electronic configuration given in the answer choice has the last electron in a 2p-orbital, which is of higher energy than the ground state 2s. This makes choice D the correct answer, because it is an excited state.

Quantum Numbers

Quantum numbers are a set of four numbers that uniquely describe an electron within an atom. Four factors describe an electron: the shell, the orbital, the orientation of the orbital, and the alignment of the magnetic field resulting from the precessing electron. These terms translate into the quantum numbers. A fundamental rule with quantum numbers is that no two electrons can have the same set of quantum numbers (this is the Pauli exclusion principle). Quantum numbers are used to describe the motion and location of each electron in an element. Quantum numbers describe the shape of an electron's cloud, the momentum of the electron, the orientation of the electron density, and the rotation of the electron about its axis. There are four quantum numbers used to describe an electron; n , l , m_l , and m_s . There are also rules that must be followed when assigning quantum numbers to an element. Each number has its specific guidelines, which often depend on the other quantum numbers. Table 2.2 lists the rules for assigning quantum numbers.

Quantum #	Rules
n	Principle (n): Describes the shell (average radius of the electron from the nucleus and its energy level) in which the electron resides. It can be any integer greater than zero.
l	Angular Momentum (l): Describes the orbital (shape of the electron cloud formed by the orbiting electron) in which the electron resides. It must be less than the value of n . It can be a positive value or zero.
m_l	Magnetic (m_l): Describes the orientation of the orbital about a plane or axis. It can be any value in the range from negative l to positive l , including zero.
m_s	Spin (m_s): Describes the rotation (counterclockwise or clockwise) of the electron about its axis. It can be either positive or negative one-half (spin up or spin down).

Table 2.2

Table 2.3 shows the correlation of quantum numbers to electrons within an atom.

n	l	m_l	m_s	orbital	total electrons	Description
1	0	0	$\pm \frac{1}{2}$	1s	2	1s $\uparrow$ 1s $\downarrow$
2	0	0	$\pm \frac{1}{2}$	2s	2	2s $\uparrow$ 2s $\downarrow$
2	1	$\pm 1, 0$	$\pm \frac{1}{2}$	2p	6	2p $_x\uparrow$ 2p $_y\uparrow$ 2p $_z\uparrow$ 2p $_x\downarrow$ 2p $_y\downarrow$ 2p $_z\downarrow$
3	0	0	$\pm \frac{1}{2}$	3s	2	3s $\uparrow$ 3s $\downarrow$
3	1	$\pm 1, 0$	$\pm \frac{1}{2}$	3p	6	3p $_x\uparrow$ 3p $_y\uparrow$ 3p $_z\uparrow$ 3p $_x\downarrow$ 3p $_y\downarrow$ 3p $_z\downarrow$
4	0	0	$\pm \frac{1}{2}$	4s	2	4s $\uparrow$ 4s $\downarrow$
3	2	$\pm 2, \pm 1, 0$	$\pm \frac{1}{2}$	3d	10	3d $_{xy}\uparrow$ 3d $_{xz}\uparrow$ 3d $_{yz}\uparrow$ 3d $_{x^2-y^2}\uparrow$ 3d $_{z^2}\uparrow$ 3d $_{xy}\downarrow$ 3d $_{xz}\downarrow$ 3d $_{yz}\downarrow$ 3d $_{x^2-y^2}\downarrow$ 3d $_{z^2}\downarrow$

Table 2.3

When assigning quantum numbers to an electron, you must keep in mind that "no two electrons orbiting the nucleus of the same element have the same set of quantum numbers." To assign quantum numbers, you must first describe the electron in words. Consider an electron in the third shell. There are eighteen electrons held in the third shell, so we need to be more specific. An electron in the third shell can be in either an s-orbital, a p-orbital, or a d-orbital. For sake of argument, let's consider a p-orbital. There are three p-orbitals, each capable of holding two electrons, so there are six electrons that can be found in the 3p level. We need to be more specific. Each p-orbital has a different orientation in space (p_x along the x-axis, p_y along the y-axis, and p_z along the z-axis). For sake of argument, let's consider the p-orbital aligned on the x-axis. This is the p_x -orbital. Two electrons can be found within a $3p_x$ -orbital, one with a magnetic spin moment upward and the other with a magnetic spin moment downward. By describing the electron as spin up, the electron is unique. There is only one electron that can be spin up within a $3p_x$ -orbital. It took four terms to narrow it down to a unique electron, hence there are four quantum numbers. Let's consider that same electron:

A $3p_x$ -spin up electron has the following description in words and therefore these corresponding quantum numbers:

Third shell	therefore, the principle quantum number (n) is 3
P-orbital	therefore, the angular momentum quantum number (l) is 1
X-axis orientation	therefore, the magnetic quantum number (m_l) is -1
Spin up	therefore, the magnetic spin quantum number (m_s) is $+\frac{1}{2}$

The quantum numbers for the electron are: $n = 3$, $l = 1$, $m_l = -1$, and $m_s = +\frac{1}{2}$. The skill you must (re)develop is getting the four numbers quickly. Determining n and l is relatively easy. The n -value is the shell number, so it is pretty much a given. The l -value is the orbital. Orbitals increase from s to p to d to f, and so on, and the l -values increase from 0 to 1 to 2 to 3 and so on. An l -value of 0 corresponds to an s-orbital, an l -value of 1 corresponds to a p-orbital, and so forth. The challenging part is finding the m_l and m_s values. In all likelihood, you have no idea why x-axis orientation leads to the conclusion that $m_l = -1$. That's just the way they do it. Just as x comes before y and z in the alphabet, -1 comes before 0 and +1 numerically. To get the m_l and m_s values, you must draw out the energy levels. The assignment of m_l values for the p-orbitals is shown in Figure 2-21.

p_x	p_y	p_z
$m_l = -1$	$m_l = 0$	$m_l = +1$

Figure 2-21

Chemistry convention tells us to number the different orbitals from - l to $+l$ sequentially. The middle orbital always has an m_l value of 0. This is true for all orbitals. The assignment of m_l values for the d-orbitals is shown in Figure 2-22.

d_{xy}	d_{xz}	d_{yz}	$d_{x^2 - y^2}$	d_{z^2}
$m_l = -2$	$m_l = -1$	$m_l = 0$	$m_l = +1$	$m_l = +2$

Figure 2-22

Electrons are filled into orbitals one at a time from left to right, with spin up going in first, followed by spin down once each degenerate orbital has one electron. Filling spin up first is a convention and does not represent the physical reality of electrons. When an electron is spin up in an orbital, it has a spin quantum number (m_s) of $+1/2$. When an electron is spin down in an orbital, it has $m_s = -1/2$. For the $3p_x$ spin up electron, the filling is shown in Figure 2-23.

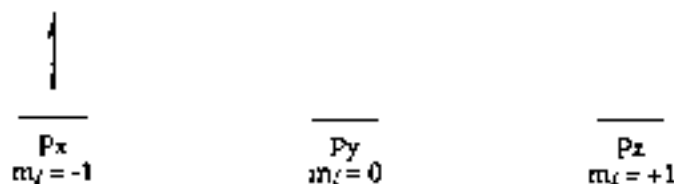


Figure 2-23

The electron falls into the first p-orbital, so m_l is -1. The electron is spin up, so m_s is $+1/2$.

To be able to apply these quantum numbers, keep in mind that each electron within an element has a unique set of quantum numbers. An electron can be described in terms of words (such as an electron in the second energy level in a p-orbital with x-orientation and spin up is a $2p_x \uparrow$) or in terms of numbers ($n = 2$, $l = 1$, $m_l = -1$, $m_s = +1/2$). Quantum numbers follow arbitrary guidelines. For instance, the electron in a p-orbital oriented on the z-axis in the third shell with spin up has the quantum numbers $n = 3$ (for the third shell), $l = 1$ (because the electron is in a p-orbital), $m_l = +1$ (for the z-axis), and $m_s = +1/2$ (for spin up).

Example 2.12

All of the following are true of an electron EXCEPT:

- A. electrons in a lower energy level can absorb energy and elevate to a higher energy.
- B. exactly the same amount of energy is emitted when an electron relaxes between the same states.
- C. there are many energy absorptions possible, but they are always of a precisely known energy.
- D. an electron in the $n = 1$ energy level can be found at an infinite number of distances from the nucleus.

Solution

Our model of atoms is a positive concentric nucleus surrounded by orbiting electrons. These electrons may occupy only specific orbits, which have distinct energies and pathways. According to this description, electrons in a lower energy level can absorb energy and elevate to a higher energy, so choice A is valid and thus eliminated. According to this description, exactly the same amount of energy is emitted when an electron relaxes between the same states, so choice B is valid and thus eliminated. According to this description, there are many possible energy absorptions possible, but they are always of an exact energy, so choice C is valid and thus eliminated. According to this description, an electron in the $n = 1$ energy level is found at only one distance from the nucleus, not at an infinite number of distances. This makes choice D invalid, and thus the best answer.

Periodic Trends

The Periodic Table

The periodic table is organized according to valence electrons. Figure 2-24 shows the shell of the periodic table. The blocks are named after the last electron that fills each respective atom in that section of the periodic chart. For instance, an element in the D-block has its last electron (a valence electron) in a d-orbital. Understanding the blocks helps to understand the periodic trends. The S-block comprises only metals. The D-block houses the transition metals, where periodic behavior is not necessarily an obvious trend. The P-block includes metals and metalloids in its left side and non-metals in its right side.

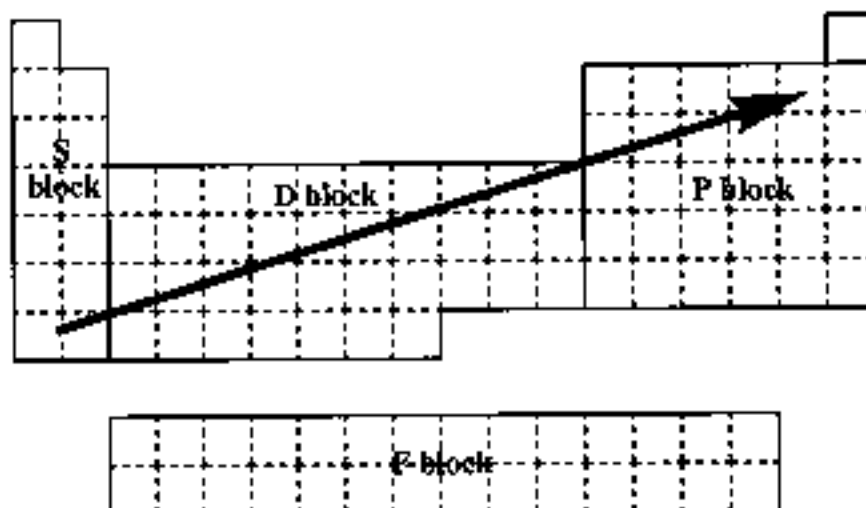


Figure 2-24

General Elemental Periodic Trends

Periodic trends refer to any chemical behavior that can be matched to a trend within the periodic table. All chemical properties depend on the valence electrons, so periodic trends ultimately grow out of valence electron trends. Any feature of an atom that affects how tightly a valence electron is held contributes to periodic trends. The two major factors are the effective nuclear charge and the valence shell, both of which support periodic trends. They are listed below:

- ❶ As you move from left to right across a period in the periodic table, the effective nuclear charge increases.
- ❷ As you descend a family in the periodic table, the valence shell increases, so the distance of the valence electron from the nucleus of the atom increases.

The effective nuclear charge (Z_{eff}) is the net charge exerted upon the outermost electrons (valence electrons). This value is empirically determined and takes into account attraction due to the protons, shielding due to the neutrons, and repulsion due to the core electrons. It is generally approximated as the proton charge minus the electron repulsion. The effective nuclear charge affects how tightly the electrons are held, which affects the ionization energy, the electron affinity, and the atomic radius. The effective nuclear charge increases across a row in the periodic table. Although we generally approximate the effective nuclear charge, it can be derived from the ionization energy. This procedure assumes that ionization energy is purely related to the effective nuclear charge, and fails to account for shell stability, particularly filled-octet stability.

Example 2.13

Which of the following atoms has the **GREATEST** effective nuclear charge?

- A. Carbon
- B. Fluorine
- C. Sodium
- D. Sulfur

Solution

Effective nuclear charge increases from left to right in the periodic table, so the element in the column that is farthest to the right has the greatest effective nuclear charge. Fluorine is to the right of carbon within the same period, so it has a greater effective nuclear charge (Z_{eff}). Choice A is eliminated. Sodium is in the first column of the periodic table, so it has the smallest effective nuclear charge. Choice C is eliminated. The correct answer is choice B.

Periodic trends depend on both the effective nuclear charge (affecting the strength with which valence electrons are held) and the valence shell (affecting the distance between electrons and the nucleus.) Periodic trends as we move from left to right across a row of the periodic table (period) are attributed to increasing effective nuclear charge. Periodic trends as we move up through a column of the periodic table (family) are attributed to decreasing valence shells. The net result of these two effects is represented by the bold arrow shown in Figure 2-24. As we move along the pathway of the bold arrow, the following general atomic trends are observed:

- ➊ The atomic size decreases (the radius of the atom is defined as the distance from the center of the nucleus to the exterior of the valence electron cloud).
 - ➋ The ionization energy increases (the energy required to remove the outermost electron from the atom).
 - ➌ The electron affinity increases (the energetics associated with an atom gaining an electron)
 - ➍ The electronegativity increases (the tendency to share an electron with another atom within a bond).
-

Example 2.14

Which sequence accurately lists increasing ionization energy of the atoms?

- A. $\text{Br} > \text{F} > \text{Cl} > \text{Te}$
- B. $\text{O} > \text{S} > \text{P} > \text{Br}$
- C. $\text{Br} < \text{F} < \text{Cl} < \text{Te}$
- D. $\text{O} < \text{S} < \text{P} < \text{Br}$

Solution

The appearance of fluorine in the middle of a trend should get your attention as a wrong answer, eliminating choices A and C. Oxygen is directly above sulfur in the periodic table, so oxygen has a greater ionization energy than sulfur. The correct answer is choice B.

None of the trends is uniform or perfect. The effective nuclear charge does not uniformly increase when we scan across a period. There are deviations in some trends. One of the more common deviations is seen with electron affinity and ionization energy, due to half-filled stability and filled-shell stability. For instance, nitrogen has a greater ionization energy than oxygen, because upon ionization, nitrogen loses its half-filled p-shell. On the contrary, oxygen gains half-filled stability upon being ionized. The test writers may not prey on these exceptions, but they certainly can emphasize the conceptual aspects by looking at the factors that affect periodicity. For instance, rather than ask about atomic radius, they may ask about ionic radius.

As a general rule, cations are smaller than neutral atoms, because the loss of electrons allows the atom to compact more tightly, given the diminished repulsion associated with the missing electrons. As a general rule, anions are larger than neutral atoms, because the gain of electrons causes the atom to expand, given the enhanced repulsion associated with the additional electrons. Valence electrons account for the size of anions, neutral atoms, and cations. Extra electrons repel and thus increase the atomic size of an anion, while a loss of electrons results in less repulsion and a smaller radius for a cation.

Example 2.15

When strontium (Sr) becomes an ion, what is observed?

- A. It forms a +1 cation that is smaller than Sr.
- B. It forms a +1 cation that is larger than Sr.
- C. It forms a +2 cation that is smaller than Sr.
- D. It forms a +2 cation that is larger than Sr.

Solution

Choice C is correct. Strontium (Sr) is found in the second column of the periodic table. Alkaline earth metals lose two electrons to gain octet stability. As such, strontium carries a +2 charge, so choices A and B are eliminated. Cations are smaller than neutral species, because there are fewer electrons and thus less repulsion. This makes choice C the best answer.

Example 2.16

Which of the following ions is the LARGEST?

- A. Cl^-
- B. Na^+
- C. K^+
- D. Br^-

Solution

Within a period, anions are larger than cations, so chloride (Cl^-) is larger than sodium cation, and bromide (Br^-) is larger than potassium cation. This eliminates choices B and C. Because Br is lower in the periodic table than Cl, Br is larger as a neutral atom than Cl. This same trend holds true, if both Br and Cl pick up the same number of electrons. In this case, both bromide and chloride picked up one electron each, so bromide, with its electrons in a higher valence shell, is larger than chloride. The answer is choice D.

Atomic Radius

The atomic radius is the distance from the center of the nucleus to the edge of the valence cloud of electrons. However, most of the empirical measurements of atomic radii that exist are not from electron density maps, but instead are determined by dividing bond distances between like atoms in half. Because of overlapping electron clouds, this method does not generate a true atomic radius, but rather a covalent bonding radius. However, it does lead to internally consistent values. Atomic radii are measured in units of picometers. The radius of an atom decreases as a family in the periodic table is ascended, because the number of electronic shells decreases. The radius of an atom decreases as a period in the periodic table is scanned from left to right, because the effective nuclear charge increases. The trend is fairly uniform from left to right, with no distinct exceptions due to half-filled stability.

Figure 2-25 lists the atomic radii of the first twenty elements. As a general observation, within a period, atomic radius decreases as the atomic number increases.

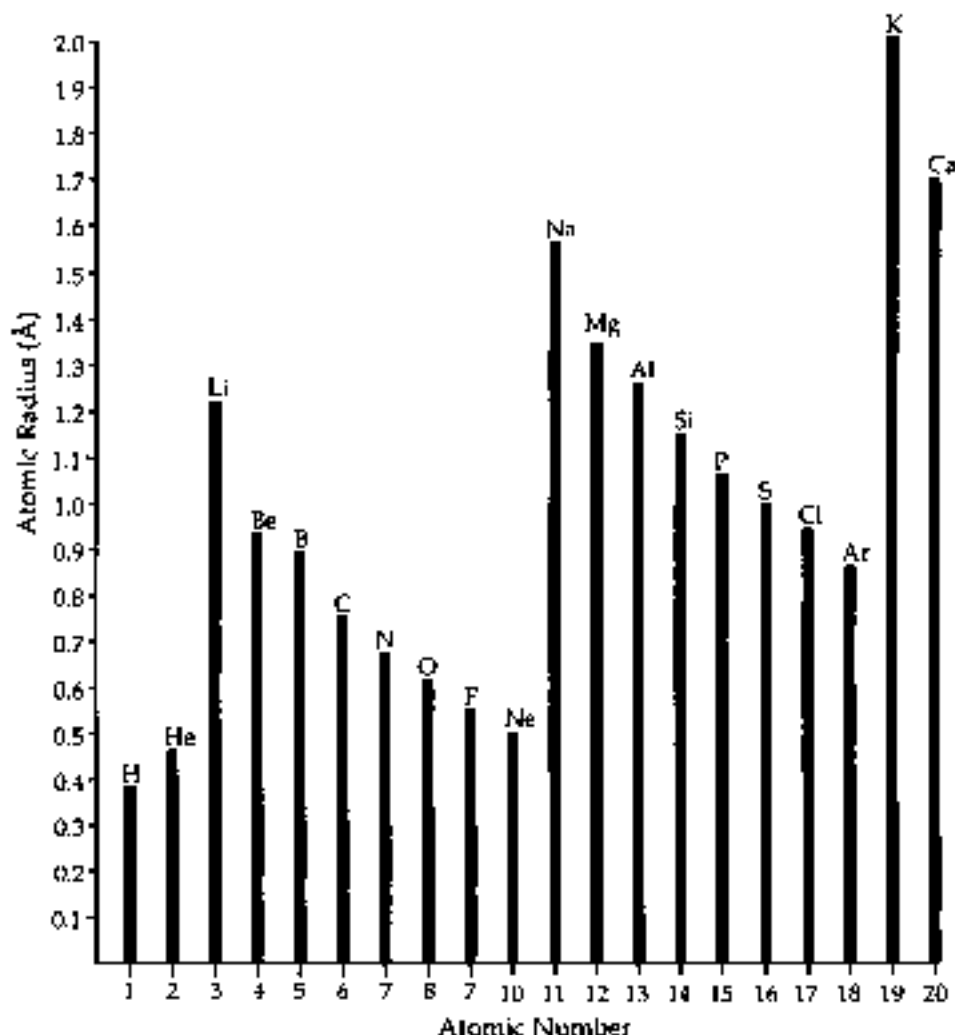


Figure 2-25

The sudden increase in size from He to Li, Ne to Na, and Ar to K is attributed to the expanded valance shell associated with the additional electron. For instance, the electronic configuration of Ar is $1s^2 2s^2 2p^6 3s^2 3p^6$, while K is $1s^2 2s^2 2p^6 3s^2 3p^6 4s^1$. The fourth shell ($n = 4$ quantum level) has a larger radius, so the potassium atom is larger than the argon atom. Table 2-25 terminates at element 20, because from element 21 to 30 the radius stays roughly equal, since the electrons are being added to the third quantum level (3d orbitals). This does not affect the radius of the electron shell drastically. The transition metals have very similar atomic radii, although they are not exactly equal.

The trend is consistent through the elements listed, with the exception of helium and hydrogen. The larger atomic radius of helium when compared to hydrogen goes against the discussed trend in effective nuclear charge. The best explanation for this deviation involves both the shielding effect of the two neutrons in the helium nucleus and the electron repulsion experienced by electrons in the first quantum level where they are closer together than in any other quantum shell. In other words, the electrons in the $n = 1$ level repel one another more than electrons in the $n = 2$ level, because they have the smallest interelectronic distance. This repulsion forces the electrons away from one another, resulting in a greater area being occupied by the orbiting electrons. Keep in mind that the electrons, not protons or neutrons, define the radius of an atom. The atomic radii of atoms may be used to predict the bond length within molecules. The smaller the atomic radius of the atom, the shorter the bond it forms when sharing electrons with another atom. Shorter bonds are stronger bonds, so there exists a correlation between an element's location in the periodic table and the strength of the bonds that element can form.

Example 2.17

Which of the following elements has the **LARGEST** atomic radius?

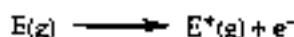
- A. O
- B. F
- C. Ne
- D. Na

Solution

This is just a simple case of reading from the periodic table. The element in the lowest and furthest left position is sodium, Na, so choice D is the best answer.

Ionization Energy

Ionization is the process of losing an electron from the valence shell. When an atom is ionized, it becomes a cation. The energy required to remove the outermost electron from the valence shell is known as the *ionization energy*. A generic reaction for ionization is shown below, where E represents any element.



The energy required to carry out ionization depends on the attraction of the electron to the nucleus, its distance from the nucleus, and the stability of its electronic configuration. Because several factors influence ionization energy, it is too difficult to calculate, and thus it is generally evaluated in a qualitative sense.

Figure 2-26 lists the ionization energies of the first twenty elements in the periodic table. Within a row in the periodic table, ionization energy increases as the atomic number increases. This is a general trend, but with some exceptions.

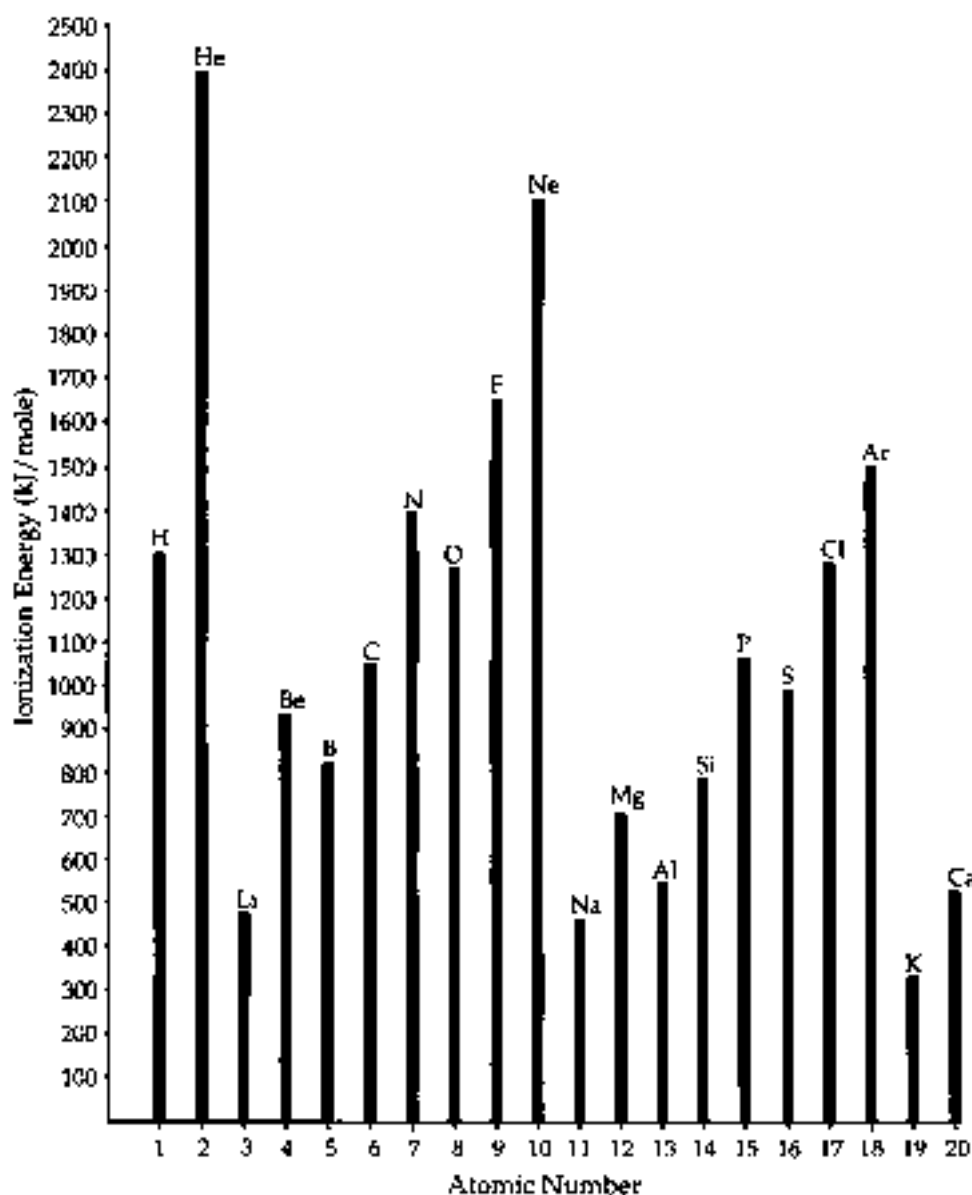


Figure 2-26

Ionization energy for an element generally increases as you move left to right in the periodic table. Notable exceptions occur when there is half-filled stability of the energy level and when there is an s^2 -shell. Ionization energy for an element increases as you ascend a column in the periodic table, with the element higher up the column having the greater ionization energy. This is because as the number of electronic shells decreases, the proximity of an electron to the nucleus increases, and thus the attraction to the nucleus increases. The sudden decrease in ionization energy from He to Li, Ne to Na, and Ar to K is attributed to the expanded valance shell (and thus reduced attraction) associated with the additional electron. For instance, the electronic configuration of Ne is $1s^2 2s^2 2p^6$, while for Na it is $1s^2 2s^2 2p^6 3s^1$. The third shell ($n = 3$ quantum level) has a larger radius, so the sodium atom can more easily lose an electron than the neon atom (with its outermost electron being more attracted by the nucleus). As with atomic radius, from element 21 to element 30 the ionization energy remains roughly equal, because the electrons are being removed from the same 4s-orbital. The effective nuclear charge on the 4s-electrons does not change drastically. The exceptions in the transition metals are also due to half-filled and filled d-shell stability.

Ionization energy may be used to predict the oxidation and reduction potentials of an atom. The easier it is to ionize an atom, the easier it is to oxidize that atom by one electron. This leads to a larger (more positive) value for the oxidation potential. A low ionization energy for an atom correlates to a smaller (or more negative) value for the reduction potential of the cation that is formed.

Example 2.18

Why is the ionization energy of beryllium greater than the ionization energy of lithium?

- A. Be has a larger principal quantum number than Li.
- B. Li has a greater density than Be.
- C. Be has a larger effective nuclear charge than Li.
- D. Li has a bigger proton count than Be.

Solution

Both beryllium and lithium have their last electron (the electron lost upon ionization) in a 2s-orbital. This eliminates choice A, because the principle quantum number (valence shell) is the same for both. Lithium is less massive and larger than beryllium, so it is less dense. This eliminates choice B. Lithium has three protons, while beryllium has four, so choice D is a false statement. Only choice C remains. The difference between lithium and beryllium lies in the effective nuclear charge. The beryllium nucleus has four protons, while the lithium nucleus has only three protons. The greater number of protons increases the attractive pull on the electron. Because the pull is greater, the effective nuclear charge is greater. The greater the effective nuclear charge, the greater the ionization energy; therefore beryllium has a greater ionization energy than lithium, because it has a greater effective nuclear charge than lithium.

The MCAT test writers design questions to encompass logical analysis. To test a concept and generate the illusion of unfamiliarity, they can make subtle changes to a question. The second ionization energy can be tested in a question. Consider the periodic trends for elements, but remember that when you evaluate ions, reading from the periodic table does not always give the right answer.

Example 2.19

How does the ionization energy of sodium compare to the ionization energy of magnesium?

- A. Na has a greater ionization energy, because it has the greater effective nuclear charge.
- B. Mg has a greater ionization energy, because it has the greater effective nuclear charge.
- C. Na has a greater ionization energy, because it has the greater electronic shell.
- D. Mg has a greater ionization energy, because it has the greater electronic shell.

Solution

This is just a simple case of reading from the periodic table. Both magnesium and sodium are in the same period (third), so they have the same valence shell. This eliminates choices C and D. The element that is furthest to the right in the periodic table is magnesium, so magnesium has the greater ionization energy. Choice B is the best answer, because Na and Mg are in the same row of the periodic table, where effective nuclear charge is the reasoning behind periodic differences.

Example 2.20

How does the second ionization energy of sodium compare to the second ionization energy of magnesium?

- A. Na has a greater second ionization energy, because it has the greater effective nuclear charge.
- B. Mg has a greater second ionization energy, because it has the greater effective nuclear charge.
- C. Na has a greater second ionization energy, because it has the smaller electronic shell.
- D. Mg has a greater second ionization energy, because it has the smaller electronic shell.

Solution

The second ionization energy is the energy associated with losing the second electron, which takes the element from +1 to +2. For sodium, an octet is obtained by losing the first electron, thus the second electron lost drastically destabilizes the electron cloud. This makes the second ionization energy very high. For magnesium, an octet is obtained by losing the first and second electrons, thus the second electron lost stabilizes the electron cloud. This makes the second ionization energy very low for magnesium, eliminating choices B and D. Because we are talking about shell stability, the best (albeit not perfect) answer is choice C.



The electronic configurations show that sodium loses octet stability upon its second ionization, while magnesium gains octet stability upon its second ionization.

By asking about the second ionization energy, the concept rather than the memorization of periodic trends is tested.

Electron Affinity

Electron affinity measures the tendency of an element to gain an electron. It is a measurement of the energy absorbed or released when an electron is added into the valence shell. Electron affinity can be either negative or positive, meaning that gaining an electron can be either exothermic or endothermic. The generic reaction for electron affinity is shown below, where E represents any element:



Figure 2-27 lists the electron affinity for the first twenty elements in the periodic table. Within a row in the periodic table, electron affinity correlates with atomic number, but there are some extreme spikes in the trend.

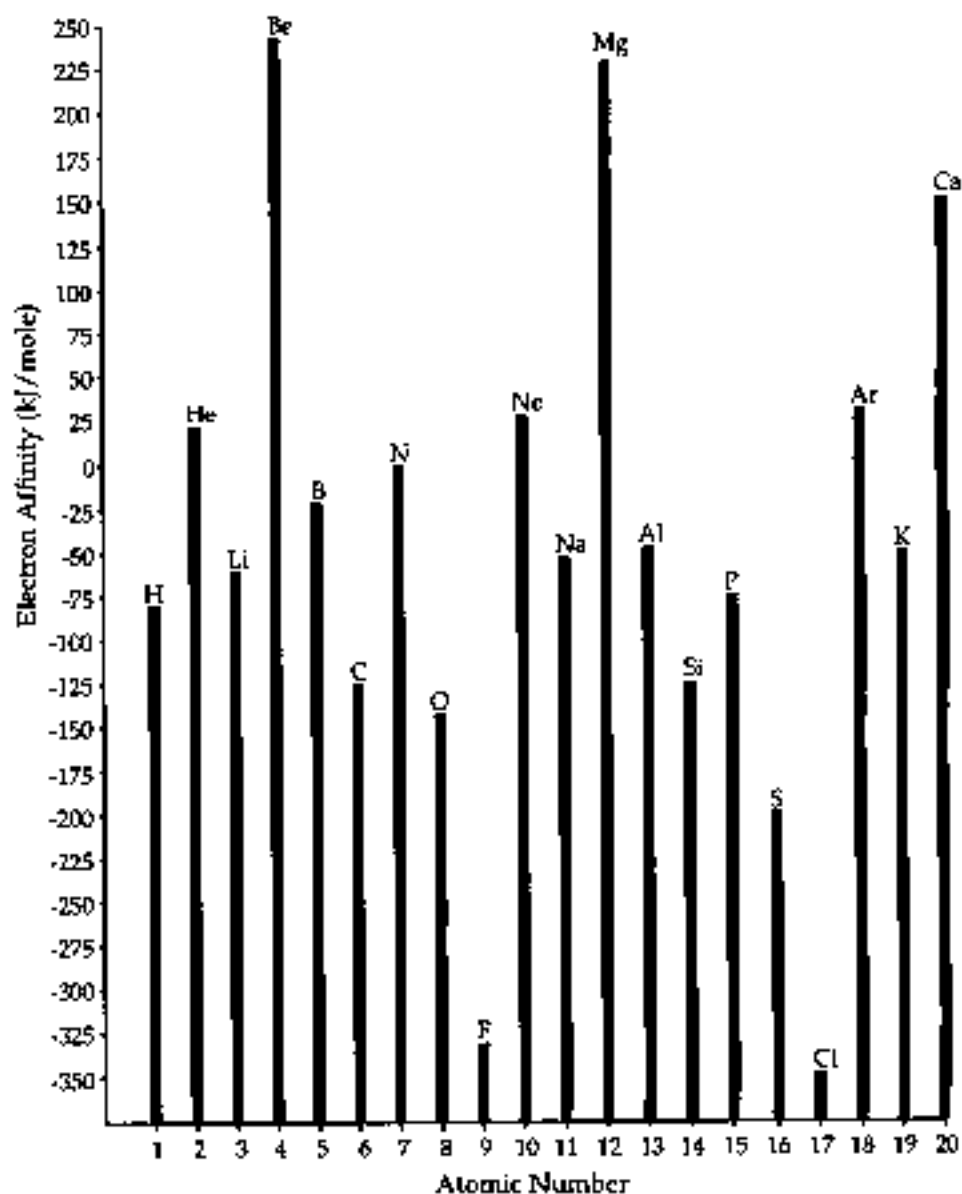


Figure 2-27

The biggest deviations are attributed to the stability associated with a filled s-shell. The graph may seem confusing at first, because the lowest numbers have the greatest electron affinity. The numbers listed are energy released upon gaining an electron, so a negative number refers to an element with a high electron affinity. The sudden increase in electron affinity (energy released upon gaining an electron) from Be to B, Mg to Al, and Ca to Ga is attributed to the instability of one electron in the p-level. For instance, upon gaining an electron, the electronic configuration of Mg goes from $1s^2 2s^2 2p^6 3s^2$ to $1s^2 2s^2 2p^6 3s^2 3p^1$, which creates a new energy level, and is unfavorable. Upon gaining an electron, the electronic configuration of Na goes from $1s^2 2s^2 2p^6 3s^1$ to $1s^2 2s^2 2p^6 3s^2$, which fills the s-shell and generates stability. From element 21 to element 30, electron affinity is erratic, because the d-shell stability is changing. No trend for electron affinity is evident in the transition metals.

Like ionization energy, the energy associated with electron affinity depends on the attraction of an electron to the nucleus, its distance from the nucleus, and the stability of its electronic configuration. Because several factors influence electron affinity, the trend across a period is erratic. In general, an element releases more energy upon gaining an electron as you move left to right in the periodic table. Drastic exceptions occur when there is half-filled stability of the energy level and when there is an s²-shell. In general, an element also releases more energy upon gaining an electron as you ascend a column in the periodic table. This is because as the number of electronic shells decreases, the new electron is closer to the nucleus, and thus the attraction to the nucleus increases.

Example 2.21

The electron affinity of an element is MOST similar to which of the following properties?

- A. Electronegativity
- B. Ionization energy
- C. Oxidation potential
- D. Reduction potential

Solution

The electron affinity for an element measures the energy associated with the gain of one electron. Choices B and C are out, because both of them deal with losing an electron. Electronegativity is not the best choice, because it deals with the sharing of electrons in a bond, not the gaining of an electron. The best answer is choice D, reduction potential, because reduction is the gain of an electron. An element with a high reduction potential has a high electron affinity.

Electronegativity

Electronegativity is a measure of an atom's tendency to gain and retain an electron from a neighboring atom *within a bond*. It is formally defined as the ability of an atom to attract towards itself the electrons in a chemical bond. Electronegativity is related to both ionization energy and electron affinity. This is to say that the electronegativity of an atom depends on both the electron affinity and ionization energy of that atom. Linus Pauling generated a method to measure electronegativity, and created a scale, referred to as the Pauling scale. Electronegativity is measured on a relative scale, with the values measured from the electron distribution within a bond. The standards are 0.9 for sodium and 4.0 for fluorine, and all other values are based on dipole moments associated with bonds to these atoms.

The electronegativity of an atom increases as the periodic table is ascended, because as the number of electronic shells decreases, causing the attraction to the nucleus to increase. The electronegativity of an atom increases as the periodic table is scanned from left to right, because the effective nuclear charge increases. The trend in electronegativity is very clean, showing no exceptions. Figure 2-28 reflects these trends.

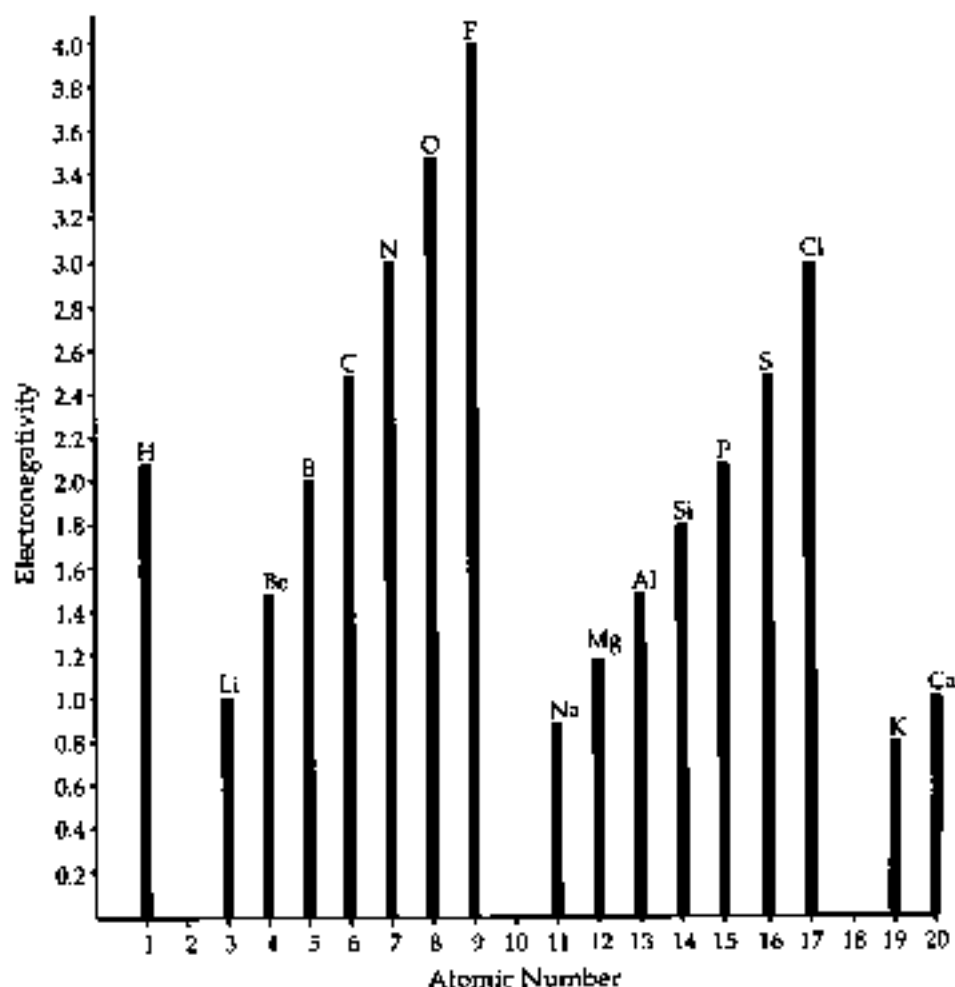


Figure 2-28

When the electronegativity values of the two atoms within a bond are close, the bond is covalent. When the electronegativity difference exceeds 2.0, then the bond is ionic. Figure 2-29 represents the electron clouds and shows the relative electronegativity of the atoms involved in an ionic bond and a covalent bond.



Figure 2-29

Example 2.22

Electronegativity difference between bonded atoms is BEST determined by:

- A. measuring the bond length.
- B. measuring the dipole moment.
- C. calculating the difference in electron affinity between the two elements.
- D. calculating the difference in ionization energy between the two elements.

Solution

Because electronegativity measures the tendency to share an electron, and the dipole moment represents the degree of sharing between two atoms in a bond, the best answer is choice B. Electronegativity is related to both electron affinity and ionization energy, so electronegativity can be estimated knowing both ionization energy and electron affinity, but not just one of them. Choices C and D are eliminated, because you need both to approximate the electronegativity difference. Choice A is eliminated, because bond length dictates bond strength, but not necessarily the relative electronegativity. This question may seem challenging, because the terms are interconnected.

Example 2.23

The trend in electronegativity increases with which of the following?

- A. Ionization energy
- B. Atomic radius
- C. Atomic number
- D. Number of valence electrons

Solution

This question is close to verbatim in reproducing a question from a recent MCAT exam. Electronegativity follows a cleanly predictable trend, so choice A is eliminated, because ionization energy follows an erratic trend. Electronegativity increases as atomic radius decreases, so choice B is eliminated. Increasing atomic number sounds tempting; but when a new shell is formed, electronegativity drops, while atomic number increases. This can be seen in going from fluorine to neon to sodium. This eliminates choice C. As the number of valence electrons increases, we are moving from left to right across a period of the periodic table. Increasing valence electrons does not affect shells. From left to right in a period, electronegativity increases. This makes choice D the best answer.

Periodic Families (Groups)

Columns in the periodic table (families) contain elements that have the same valence electron count and thus show similar chemical reactivity. They are referred to as groups of elements. Some of the groups you should know are the alkali metals (first column of H through Fr), the alkaline earth metals (second column of Be through Ra), the chalcogens (sixth column of O through Po), the halogens (the seventh column of F through At), and the noble gases (last column of He through Rn). The last column of elements is observed to be generally non-reactive. They are the inert gases. The MCAT test-writers have been known to include a great deal of excess information in the passage. It is up to you to recognize what is pertinent and what is not.

Alkali Metals (Group I)

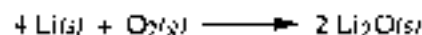
Alkali metals are in the first column of the periodic table. Included are lithium, sodium, potassium, rubidium, cesium, francium, and to some extent hydrogen. Hydrogen can act as both a halogen and an alkali metal. The common feature is that their valence shell is ns^1 , where n is any integer greater than one. As neutral elements, they are strong reducing agents, because they readily lose an electron to become a +1 cation (which gives them a complete octet). They are some of the strongest reducing agents (most favorably oxidized). They react with any compound or element that has even a slight electron affinity. Their reactivity increases as you descend the column, mostly because it is easier to lose an s -electron from a shell that is further out (greater n quantum number). Their cation form is very soluble in water with almost any anion.

All alkali metals react favorably with water to form the metal hydroxide and hydrogen gas. Reaction 2.1 is the generic reaction:

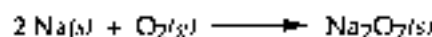


Reaction 2.1

The oxides they form are variable with the metal. Lithium forms an oxide (M_2O), sodium forms a peroxide (M_2O_2), and potassium, rubidium, and cesium form superoxides (MO_2). Reactions 2.2, 2.3, 2.4, 2.5, and 2.6 show the oxidation reactions of the alkali metals:



Reaction 2.2



Reaction 2.3



Reaction 2.4



Reaction 2.5



Reaction 2.6

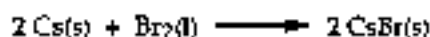
Besides reacting with oxygen, alkali metals are oxidized by halogens, nitrogen, and hydrogen. Reactions 2.7, 2.8, and 2.9 are a sampling of these reactions.



Reaction 2.7



Reaction 2.8



Reaction 2.9

Example 2.24

What happens to sodium metal when it is added to water?

- A. It is oxidized to yield sodium hydroxide, which is insoluble in water.
- B. It is oxidized to yield sodium hydroxide, which is soluble in water.
- C. It is reduced to yield sodium hydride, which is insoluble in water.
- D. It is reduced to yield sodium hydride, which is soluble in water.

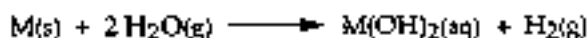
Solution

As shown generically in Reaction 2.1, a metal hydroxide is formed upon addition of an alkali metal to water. Because sodium is going from neutral to +1 when losing an electron to oxygen, sodium is oxidized. This eliminates choices C and D. You may recall from your acid-base chemistry experience that sodium hydroxide is a strong base, and it readily dissociates in water. This makes NaOH very soluble in water. The best answer is choice B.

Alkaline Earth Metals (Group II)

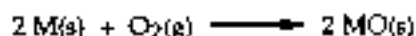
Alkaline earth metals are metals from the second column of the periodic table. Included are beryllium, magnesium, calcium, strontium, and barium. Most beryllium complexes are covalent in nature. The common feature is that their valence shell is ns^2 , where n is any integer greater than one. As neutral elements, they are strong reducing agents, because they readily lose two electrons to become a +2 cation with a filled octet. However, they are not as reactive as alkali metals. Like alkali metals, they are strong reducing agents. They too react with any compound or element that has a high electron affinity. Their reactivity increases as you descend the column, because the first and second ionization energies both decrease. Their cation form is not as soluble in water as are the alkali metals, primarily due to their +2 charge and smaller radius.

Alkaline earth metals, except beryllium, react favorably with water to form a metal hydroxide and hydrogen gas. Reaction 2.10 is the generic reaction:



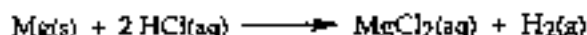
Reaction 2.10

The alkaline earth metals all form oxides (MO) when oxidized by oxygen gas. The generic reaction is shown in Reaction 2.11.



Reaction 2.11

Besides reacting with oxygen, alkaline earth metals can also be oxidized by halogens, nitrogen, and hydrogen. Reactions 2.12, 2.13, 2.14, and 2.15 are a random sampling of these reactions:



Reaction 2.12



Reaction 2.13



Reaction 2.14



Reaction 2.15

Chalcogens (Group VI)

Chalcogens are metalloids and non-metals from the sixth column of the periodic table. Included are oxygen, sulfur, selenium, tellurium, and polonium. Oxygen, sulfur, and selenium are non-metals, while tellurium and polonium are metalloids. The common feature is that their valence shell is ns^2np^4 . They form several covalent molecules with non-metals. As neutral elements, they are oxidizing agents, because they gain two electrons to become a -2 anion with a filled octet. However, their reactivity decreases as you descend the column, because the first and second electron affinities are not as great. They are often insoluble, although it varies with their counterion (cation). Oxygen exists as a diatomic molecule (O_2), sulfur and selenium exist as octatomic molecules (S_8 and Se_8), and tellurium and polonium exist in vast molecular matrices.

Halogens (Group VII)

Halogens are non-metals from the seventh column of the periodic table. Included are fluorine, chlorine, bromine, iodine, and astatine. The common feature is that their valence shell is ns^2np^5 . They form covalent molecules with non-metals and ionic compounds with metals. As neutral elements, they are strong oxidizing agents, because they readily gain an electron to become a -1 anion with a filled octet. However, their reactivity decreases as you descend the column, because the electron affinity is not as great. They are often soluble, although it varies with their counterion (cation). They all exist as a diatomic molecules (X_2), although little is known of astatine due to its radioactivity.

Noble Gases (Group VIII)

Noble gases are non-metals from the eighth (and last) column of the periodic table. Included are helium, neon, argon, krypton, xenon, and radon. The common feature is that their valence shell is complete at ns^2np^6 . For the most part, they form no bonds and exist as monatomic atoms. Thanks to the work of Neil Bartlett of U.C. Berkeley, xenon and krypton are known to form compounds with halogens. The compounds with fluorine show more electron density around fluorine, implying that fluorine is more electronegative than either xenon or krypton.

Light Absorption and Emission

Excitation and Relaxation

It requires energy to ionize an atom, because one of its electrons is removed from a stable environment and placed into a less stable environment. However, when a quantum of energy less than what is required to ionize the element is absorbed by the element, then an electron can be excited to a higher energy state (known as an *excited state*). Different forms of energy may be absorbed to excite an electron up to a higher energy level. For instance, heat energy can be absorbed to excite the electron. When the electron relaxes back toward its ground state, the energy is released as a photon. This is seen with fireworks. Some different forms of energy include light energy, thermal energy (usually vibrational kinetic energy), mechanical energy (usually translational kinetic energy), and electrical energy. It is important to consider different energy forms, because there are many things capable of absorbing one form of energy and emitting a different form.

Absorption (Excitation): The gain of energy by an element or molecule, resulting in the excitation of an electron from a lower energy state (often the ground state) to a higher energy state (an excited state). The form of energy absorbed can vary.

Emission (Relaxation): The loss of energy by an element or molecule, resulting in the relaxation of an electron from a higher energy state (which must be an excited state) to a lower energy state (which can be the ground state or another excited state, but of lower energy). The form of energy emitted can vary.

Absorption of energy is defined as any process in which a photon is absorbed by a compound to excite an electron in the compound to an elevated electronic energy level (excited state). This means that the electron goes to a higher energy state. The period of time that the electron remains in this elevated energy level is referred to as the *lifetime* of the excited state. A compound can store energy by maintaining a high population of electrons in elevated energy states. When the electron relaxes back down to the ground state, energy is emitted in the form of a photon. Absorption and emission are therefore opposite processes. Figure 2-30 shows the absorption and emission processes for a theoretical pair of energy levels. Because the energy levels are equal, the energy of the photon absorbed for excitation is equal in energy to the photon released upon relaxation.

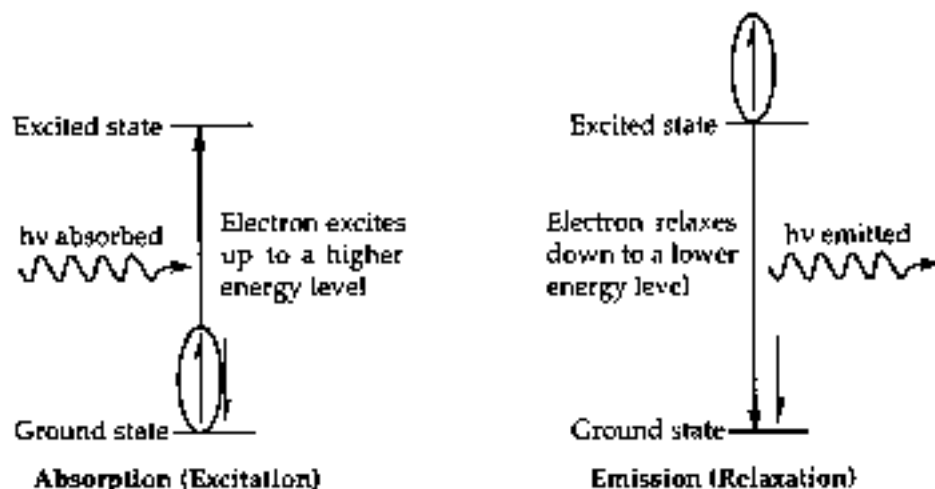


Figure 2-30

In this generic example, the energy of the photon absorbed is equal to the energy of the photon emitted. In actuality, there is more than one singular energy level for the ground state and the excited state due to the coupling of electrical energy levels and rotational energy levels associated with the atom. Transitions involve a change in electronic state as well as a change in the rotational energy of the atom. In molecules, there are vibrational energy levels to consider, in addition to the rotational and electronic energy levels. The result is that the ground state and excited state exists as a band of energy levels, not a single level. This means that multiple possibilities exist for the energy of transition. Rather than a single line absorption or emission spectrum being formed, a range is formed. This is why Figure 1-2 is a broad peak and not a sharp line. Figure 2-31 shows different types of emission and absorption spectra where the color range is shown. Absorption spectra show all light except what was absorbed, which appears as a black line, due to the absence of light. They are black lines in a rainbow. Emission spectra show only the emitted light. They are colored lines against a dark background (due to only selected frequencies being emitted). They are stripes of color.

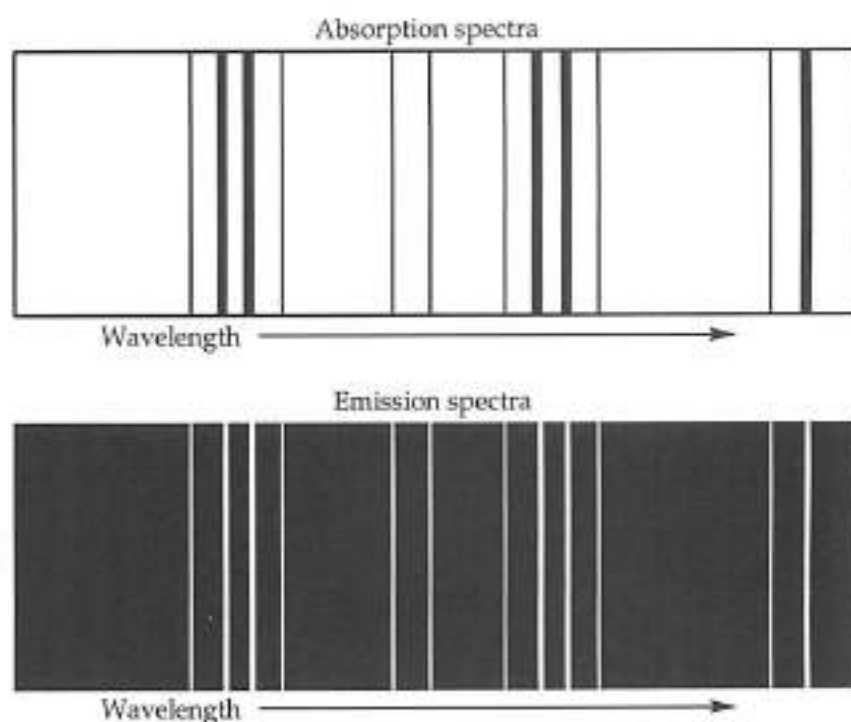


Figure 2-31

The lifetime of an excited state is often in the picosecond to nanosecond range, so the interval of combined excitation and relaxation is very fast. The conversion between energy forms is exploited in many devices. Incandescent bulbs absorb thermal energy resulting from the resistance associated with electrical flow and emit light energy (although some energy is dissipated in the form of heat through conduction and convection). The operation of tube lighting involves this conversion process. Plate charges build up at each end of the tube, creating an electric field. A cation in the gas tube is accelerated by the electric force. As it collides with other gas molecules, its kinetic energy is transferred to the other molecule, which absorbs the energy to excite an electron. Upon relaxing, the energy is released as light. In reality, the tubes must run on alternating current (AC), so that the cation never reaches a plate. This is why tube lights, such as fluorescent tubes, are actually high-frequency strobe lights.

Atomic Spectrum of Hydrogen

Hydrogen is highly studied in terms of its absorption of light. Separate samples of hydrogen gas exhibit exactly the same emission and absorption spectra. This means that all hydrogen molecules absorb the same frequency of photons. The repeatability of the hydrogen spectrum is attributed to the quantization of the energy transitions. What is meant by "quantized energy transitions" is that the absorption and emission of light by hydrogen occurs in distinct increments (or quantities). Because hydrogen has distinct transitions between energy levels, it must also have distinct energy levels. However, with the large number of energy levels, there are a large number of transitions that are possible. For hydrogen, groups of transitions are named for the energy level to which they relax. For instance, all transitions down to the $n = 1$ level are categorized into the Lyman series and emit photons in the ultraviolet region of the EM spectrum. Selected energy levels and transition series for hydrogen are shown in Figure 2-32.

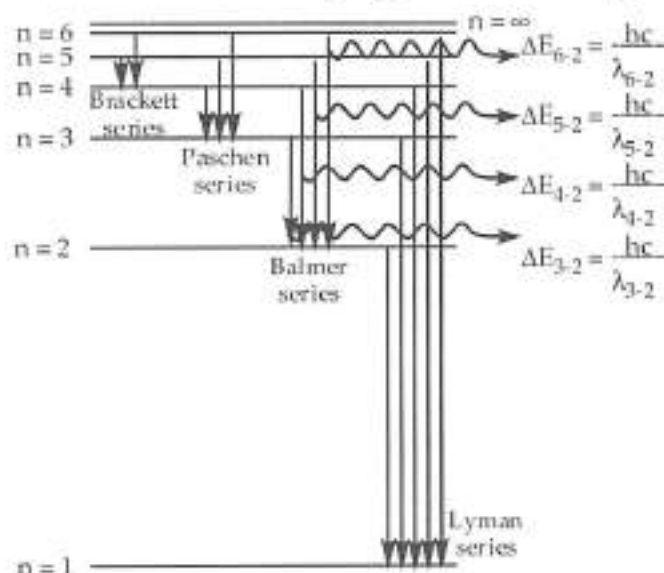


Figure 2-32

These electronic transition series names are specific to hydrogen. Of the four series shown in Figure 2-32, only the Balmer series emits photons in the visible range. When a sample of hydrogen gas glows, it is because electrons are relaxing down to the $n = 2$ level, which is still an excited state. Figure 2-33 shows the line spectrum associated with the Balmer series:

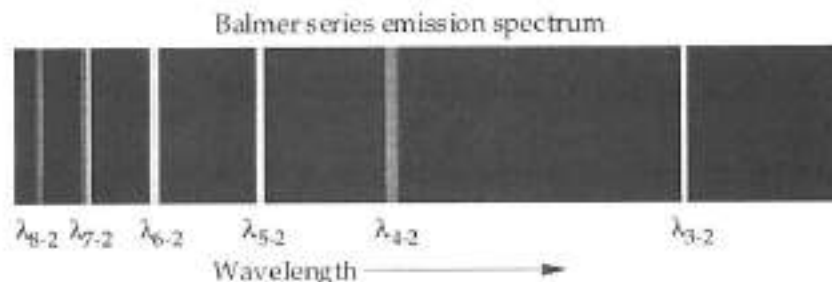


Figure 2-33

The Balmer series emits photons in the visible range. In the Balmer series, λ_{3-2} is 656 nm, λ_{4-2} is 486 nm, λ_{5-2} is 380 nm and λ_{6-2} is 410 nm, so they fall in the visible range of the spectrum. This also means that λ_{4-2} and λ_{5-2} fall in the

visible range. The Paschen series involves photons in the infrared region of the EM spectrum. The Brackett series involves photons in the low infrared and microwave regions.

Example 2.25

Given that $\lambda_{1,2}$ is 656 nm, which of the following is NOT true?

- A. All Balmer series transitions are less than 800 nm.
- B. All Lyman series transitions are less than 400 nm.
- C. The lowest energy Paschen transition is greater than 800 nm.
- D. The highest energy Brackett transition is less than 800 nm.

Solution

This requires looking closely at Figure 2-32.

Choice A: Transitions to the $n=2$ level fall into the Balmer series. The lowest energy transition in the Balmer series is from $n=3$ to $n=2$ and is given as 656nm. All other transitions in the Balmer series are more energetic, so these transitions emit photons with a wavelength less than 656nm. Choice A is a valid statement.

Choice B: Transitions to the $n=1$ level fall into the Lyman series. The lowest energy transition in the Lyman series is from $n=2$ to $n=1$, and it is of significantly higher energy than the $n=3$ to $n=2$ transition. This means that all photons in the Lyman series are of higher energy and shorter wavelength than the transitions in the Balmer series. According to Figure 2-33, the Balmer series emits photons near 400nm, so transitions in the Lyman series emit photons with a wavelength less than 400nm. Choice B is a valid statement.

Choice C: Transitions to the $n=3$ level fall into the Paschen series. The lowest energy transition in the Paschen series is from $n=4$ to $n=3$, and it is of significantly lower energy than the $n=3$ to $n=2$ transition. This means that the lowest energy photon in the Paschen series is of lower energy and higher frequency than the transitions in the Balmer series. The Balmer series emits a photon at 656nm, so the lowest energy transition in the Paschen series emits a photon with a wavelength greater than 656nm, and according to Figure 2-32, higher than 800nm. Choice C is a valid statement.

Choice D: Transitions to the $n=4$ level fall into the Brackett series. The highest energy transition in the Brackett series is from $n=\infty$ to $n=4$, and it is of much lower energy than the $n=3$ to $n=2$ transition. This means that the highest energy photon in the Brackett series is of lower energy and higher frequency than the transitions in the Balmer series. The Balmer series emits a photon at 656nm, so the lowest energy transition in the Brackett series emits a photon with a wavelength greater than 656nm, and according to Figure 2-32, higher than 800nm. Choice D is an invalid statement, and thus is the correct answer choice.

Electromagnetic Spectrum

Photons have associated with them a distinctive frequency, energy amount, and wavelength. The relative energetics of various radiation must be known. If you ever have difficulty recalling the electromagnetic radiation spectrum, think about the fact that you wear sunglasses to protect your eyes from ultraviolet radiation, not visible radiation. This means that ultraviolet radiation is of higher energy than visible light. Also, lead shields are used to filter out x-rays, so they must be of even higher energy than ultraviolet photons. Conversely, we are fairly safe from radio waves considering the fact that millions of radio waves pass through us every second. Common sense can be applied to recall the relative spectrum. Table 2.4 lists the common EM ranges in terms of frequency and wavelength.

EM Radiation	Wavelength (m)	Frequency (Hz)	Common Usage
AM	$10^3 - 10^4$	10^5 to 10^6	Cheaper radio communication
FM	$1 \cdot 10^2$	10^7 to 10^8	Expensive radio communication
Microwave	$10^{-4} - 1$	10^8 to 10^{12}	Satellite, cell phones, radar, heating water
Infrared	$10^{-6} - 10^{-3}$	10^{11} to 10^{14}	Line-of-sight com, molecular ID, heat
Visible	$4 \times 10^{-7} - 7 \times 10^{-7}$	$4.3 \times 10^{14} - 7.5 \times 10^{14}$	Vision, fiber optic communication
Ultraviolet	$10^{-8} - 10^{-7}$	10^{15} to 10^{16}	Bond-breaking, exciting electrons
X-ray	$10^{-12} - 10^{-8}$	10^{16} to 10^{20}	Nucleus detection, core e ⁻ ionization
Gamma	$10^{-15} - 10^{-10}$	10^{18} to 10^{23}	Nuclear excitation, world destruction

Table 2.4

Common practical uses of EM radiation are something that should be learned, because of the tradition of test writers asking about such devices. It should be common knowledge that radio waves and low-energy microwaves are used in communications. Microwave communications include satellite transmissions, cable television, cellular phone networks, and airport landing systems. Microwaves are also used for cooking and heating things that contain water, because the frequency required to rotate a water molecule is found in the microwave region. Remote control units (also known as line-of-sight communicators) often use infrared signals to send information. The frequency of the EM radiation used in these common tools and appliances to which we are directly exposed cannot be near 1700 cm^{-1} , 3000 cm^{-1} , or 3500 cm^{-1} , because those are the bond-stretching frequencies associated with molecular bonds common to all animal life. Ultraviolet light is used to break bonds, and x-rays are used for imaging atomic structure (including bones and teeth).

Visible Spectrum and Colors

The visible spectrum runs from red (around 700 nm), orange, yellow, green, blue, through violet (around 400 nm). The relative wavelength, frequency, and energy of photons in the visible spectrum of light (by decreasing wavelength, increasing frequency, and increasing energy) is given in Figure 2-34.

Red	<	Orange	<	Yellow	<	Green	<	Blue	<	Violet
700 to 627 nm		627 to 594 nm		594 to 561 nm		561 to 477 nm		477 to 438 nm		438 to 400 nm

Figure 2-34

The order of the colors of the visible spectrum is often recalled using the mnemonic ROY G. BiV, an acronym for the spectrum of visible light in sequence of increasing energy. It is also important to know that the visible range of light is from approximately 700 nm for red light to approximately 400 nm for violet light. Ultraviolet spectroscopy ranges from 20 nm to 400 nm.

Example 2.26

For some substance X, ΔE_1 yields yellow light, while ΔE_3 yields blue light. What must be true of the photon associated with ΔE_2 ?

- A. It is probably orange.
- B. It has a wavelength of 700 nm.
- C. It is probably green.
- D. It has a wavelength of 400 nm.

Solution

ΔE_2 falls between ΔE_1 and ΔE_3 , so the photon emitted must be between yellow and blue light in the visible spectrum. According to the spectrum mnemonic (ROYGBIV), green light falls between yellow light and blue light. This eliminates choice A and makes choice C the best answer. A wavelength of 700 nm corresponds to red light, and a wavelength of 400 nm corresponds to violet light, eliminating both choice B and choice D.

Color is a phenomenon associated with vision. Light in the 400 nm to 700 nm range can be detected by the eye and processed in the brain. Color is the result of light within this range being absorbed by the cones of the eye. Rods also detect light (in the green range), but this is for the purpose of intensity analysis of details in the image. Consider the eye to function like a visible light spectrophotometer, analyzing stimuli for intensity and wavelength. There are three kinds of cones, each one responsible for detecting different frequency ranges of light. Color, in a photon sense, can result from one of two phenomena. Color is perceived from either the emission of light at a specific frequency or the reflection of light at a specific frequency. Hence, there are two types of color to consider: emitted and reflected.

Emitted Color

Colors is emitted from a source that radiates visible photons. Atoms emit light energy when an electron relaxes from an excited state to a lower level. Figure 2-35 illustrates the concept of emitted color.

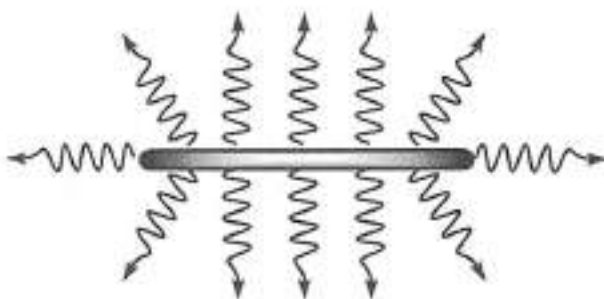


Figure 2-35

An electron relaxes from an excited state and releases a photon of some wavelength. All of the photons in Figure 2-35 are of essentially the same wavelength. The color that is observed is the color emitted. Simply put, with emitted color, "What you get is what you see." Emitted colors can be seen at night, because their source is emitting light. A good test to determine whether a color is an emitted color is to ask whether it can be seen in the dark. If the answer is yes, then it is an emitted color. Examples of emitted light sources include: neon light tubes, television screens, fireworks, and glow-in-the-dark ink.

Reflected Color and the Color Wheel

Reflected colors are produced when incident light (we'll consider it to be white light, for the sake of simplicity) strikes a surface, and the surface absorbs certain frequencies, thus reflecting only a portion of the incident light to the eye. The reflected radiation appears as a color that is a combination of the reflected photons. The color we observe is the complementary color of the frequency that had the highest intensity of absorption. This is where the color wheel can help. Complementary colors are on opposite sides in the color wheel. Given that white light is a combination of all colors, it is a combination of all complementary pairs. This is a over-simplification, because the primary colors of light are different from the primary pigment colors, but we shall use the two interchangeably to aid in answering chemistry questions. Complementary pairs include red and green, blue and orange, and yellow and violet. A color wheel is shown in Figure 2-36



The color wheel is used to determine complementary colors. Red and green are opposing on the color wheel, so they are complementary colors. Moving around the wheel, violet is the complementary color of yellow and blue is the complementary color of orange.

Figure 2-36

Chlorophyll appears green in the presence of white light. The conclusion is that chlorophyll contains a pigment that has a maximum intensity absorbance of red light, the complementary color of green. One type of chlorophyll has an absorbance maximum at 740 nm, while another has an absorbance maximum at 680 nm. Both of these values correspond to red light. Figure 2-37 illustrates the concept of emitted color as it relates to chlorophyll.

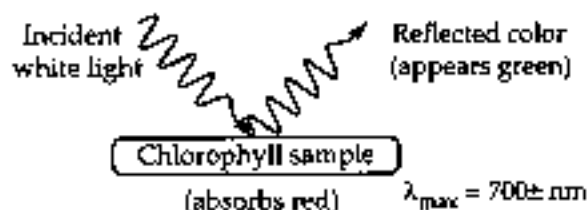


Figure 2-37

Reflected colors can be seen only in the presence of an external light source. If a color cannot be seen in the dark, then it is a reflected color. Examples of reflected light sources include paint pigments, pen inks, and fabric dyes.

Example 2.27

Which of the following statements is NOT true?

- A. A yellow paint pigment absorbs light with a λ_{max} of 411 nm.
- B. A green light bulb emits light with a λ_{max} of 522 nm.
- C. A blue shirt has a dye that absorbs light with a λ_{max} of 611 nm.
- D. A television screen glowing violet emits light with a λ_{max} of 573 nm.

Solution

This question entails comparing reflected and emitted colors. The first thing you should ask yourself is whether the color can be seen at night. If the answer is yes, then it is an emitted color and what you see is the color involved in the electronic transition (emission). If the answer is no, then it is a reflected color and what you see is the complementary color of the light involved in the electronic transition (absorption). Choice A is a paint pigment, which cannot be seen at night. The yellow color is a reflected color, so according to the color wheel, violet light has been absorbed. Violet light has a wavelength near 400 nm, so choice A is viable. Choice B is a light bulb, which can be seen at night. The green color is an emitted color, so green light has been emitted. Green light has a wavelength between 561 nm and 477 nm, so choice B is valid. Choice C is a fabric dye, which cannot be seen at night. The blue color is a reflected color, so according to the color wheel, orange light has been absorbed. Orange light has a wavelength between 627 nm and 594 nm, so choice C is valid. Choice D is a radiating screen, which can be seen at night. The violet color is an emitted color, so violet light has been emitted. Violet light has a wavelength near 400 nm, so choice D is invalid, and thus the best answer to this question.

The absorption of energy followed by the emission of a photon of light is a useful principle that can be applied to many forms of spectroscopy. Infrared spectroscopy can be used in organic chemistry to identify bond types and functional groups. It is also involved in UV-visible spectroscopy, where a compound is subjected to incident light at a specific frequency, and the relative intensity of a transmitted beam compared to a reference beam is measured. Conclusions about the concentration and the kind of compound itself can be inferred from the results. Classical experiments that involve measuring quantities of components in a gaseous mixture often focus on analysis of the emission and absorbance spectra of the gases. This is at the heart of atomic absorption spectroscopy, used to analyze amongst other things planetary gases.

The MCAT is not printed in color, so questions involving color will present numerical data for wavelengths or graphs of absorbance spectrum. Figure 2-38 shows an absorbance spectrum and its analysis.

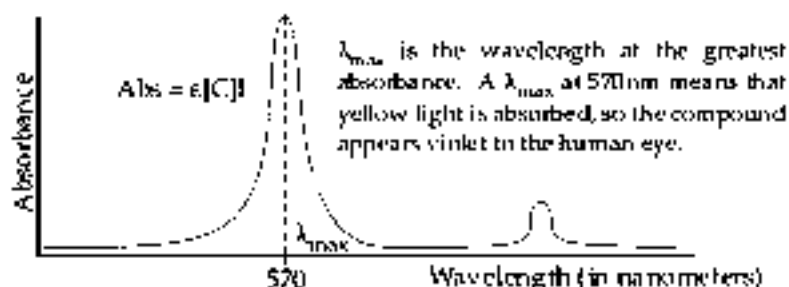


Figure 2-38

Fluorescence

Fluorescence at the simplest level is the conversion of ultraviolet light into visible light. A material with a semi-stable excited state can absorb an ultraviolet photon, relax a slight amount giving off infrared photons, and then relax completely back to its original state. The visible photon emitted is of slightly less energy than the ultraviolet photon absorbed, because of the small loss of energy when the infrared photon was lost. The next energy range down from ultraviolet is violet light in the visible range. This is why fluorescing materials appear purple. The term "black light" refers to an ultraviolet light (it emits light outside of the visible range) that when shined upon fluorescing material appears purple. You may have seen this in organic chemistry lab when you analyzed TLC plates or at social gatherings with fluorescent hand stamps. This concept does not need to be interpreted at any higher level, so this rough explanation is adequate. Fluorescence will be addressed in a passage.

Photoelectric Effect

The photoelectric effect is exactly what the name implies. An incident photon causes the release of an electron. It is a fancy way of saying that a compound can be ionized with a photon, as long as the photon has energy greater than the ionization limit of the material. The energy required to remove an electron from the surface of the solid material is referred to as the work function, ϕ . Excess energy (energy in excess of the work function) becomes kinetic energy for the electron that is emitted. The photoelectric effect is really quite simple, but confusion sometimes arises from the different terminology used in chemistry and physics. Don't be fooled; just say no to physics (just kidding, but I'm a chemistry nerd and just had to say that). Equation 2.9 is the equation for the photoelectric effect.

$$h\nu = \phi + \frac{1}{2}mv^2 \quad (2.9)$$

$$h\nu = \text{energy to eject electron} + \text{kinetic energy of ejected electron} \quad (2.10)$$

The photoelectric effect is the principle at work in solar panels. Photons of all different wavelengths and energy are emitted by the Sun. This energy can be collected at the Earth's surface. Some solar photons are sufficiently energetic to eject an electron when they strike a metal surface. To do so, the photon must be of an energy high enough to remove the electron from the valence shell of the metal. The loss of an electron from the metal creates electrical flow (electricity), which can be converted to either potential energy (stored in an electrochemical cell) or mechanical energy (used to power a device).

As the energy of the incident photon increases, the kinetic energy of the discharged electron increases. Figure 2-39 shows an apparatus devised to measure the photoelectric effect, while Figure 2-40 is a graph of the electron's kinetic energy as a function of the frequency of the incident photon.

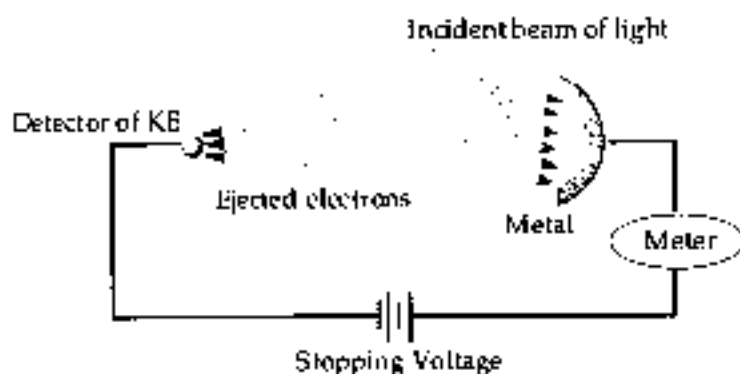


Figure 2-39

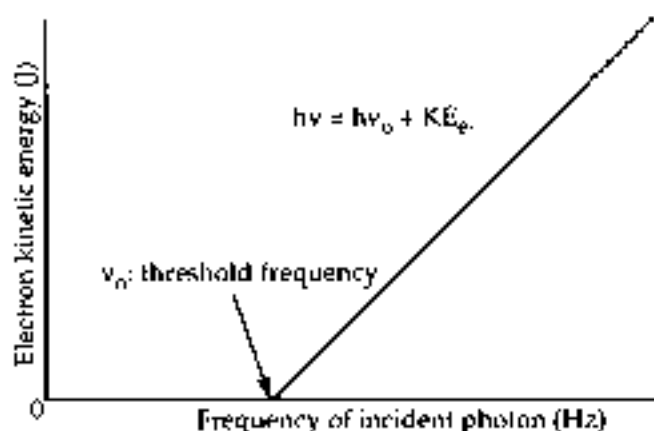


Figure 2-40

The threshold frequency corresponds to the photon equal in energy to the ionizing energy. Any photon with a frequency less than the threshold frequency does not have enough energy to ionize the material, so no electron is ejected.

Example 2.28

Which material probably has the LOWEST threshold frequency for the photoelectric effect?

- A. Boron
- B. Carbon
- C. Sodium
- D. Sulfur

Solution

The threshold frequency corresponds to the work function of the material ($\phi = h\nu_0$). Each atom holds onto its valence electrons with a different work function, therefore each atom requires a different amount of energy to undergo ionization. The lowest threshold frequency for the photoelectric effect corresponds to the material that is easiest to ionize. This question is asking for the material with the lowest work function. Of the choices, only one is a metal, so only one has a low work function. The best answer is choice C, sodium. This answer corresponds with lowest ionization energy as well.

Nuclear Chemistry

Nuclear Particles

Simply put, nuclear chemistry is the chemistry that the nucleus of an atom can undergo. Instead of referring to it as *reactivity*, it is referred to as *instability* of the nucleus. The science community knows little of nuclear behavior at this point in history, but there are theories. From our perspective, however, all we will concern ourselves with is *nuclear decay* and *nuclear capture*. The process of particle loss from the nucleus that results in a different nucleus is referred to as nuclear decay. In physics, it may also be referred to as *fission*. The process of particle gain by the nucleus which results in a different nucleus is referred to as nuclear capture. In physics, it may also be referred to as *fusion*. As a general rule, particles with mass less than 56 amu undergo fusion, while particles with mass greater than 56 amu undergo fission. Particles that we consider to be lost or gained are alpha particles (helium nucleus), beta particles (an electron), positrons (a positively charged particle with the mass of an electron), neutrons, and neutrinos (an uncharged particle with the mass of an electron). The nucleus also has ground states and excited states associated with it (just as the orbiting electrons have energy levels), so nuclei can also undergo photon absorption and emission. The high-energy photon is a gamma ray. Table 2.5 lists some of the common nuclear processes. You should treat these reactions as exercises in simple algebra. The total mass and number of protons should be equal on both sides of the reaction. This is because mass and charge are conserved (along with momentum) in each process.

Process	Reaction Tracking	Notes
α -Decay	${}^{120}_{50}\text{X} \rightarrow {}^4_2\alpha + {}^{116}_{48}\text{Y}$	The mass works: $120 = 4 + 116$. The proton number works: $50 = 2 + 48$.
β -Decay	${}^{120}_{50}\text{X} \rightarrow {}^0_{-1}\text{e} + {}^{120}_{51}\text{Z}$	The mass works: $120 = 0 + 120$. The proton number works: $50 = -1 + 51$.
β^+ -Decay	${}^{120}_{50}\text{X} \rightarrow {}^0_1\text{e} + {}^{120}_{49}\text{Q}$	The mass works: $120 = 0 + 120$. The proton number works: $50 = 1 + 49$.
γ -Emission	${}^{120}_{50}\text{X}^* \rightarrow h\nu + {}^{120}_{50}\text{X}$	Mass and proton number do not change. The nucleus relaxes to emit a photon.
α -Capture	${}^{120}_{50}\text{X} + {}^4_2\alpha \rightarrow {}^{124}_{52}\text{A}$	The mass works: $120 + 4 = 124$. The proton number works: $50 = 2 + 48$.
β -Capture	${}^{120}_{50}\text{X} + {}^0_{-1}\text{e} \rightarrow {}^{120}_{49}\text{Q}$	The mass works: $120 + 0 = 120$. The proton number works: $50 + -1 = 49$.
β^+ -Capture	${}^{120}_{50}\text{X} + {}^0_1\text{e} \rightarrow {}^{120}_{51}\text{Z}$	The mass works: $120 + 0 = 120$. The proton number works: $50 + 1 = 51$.
γ -Absorption	${}^{120}_{50}\text{X} + h\nu \rightarrow {}^{120}_{50}\text{X}^*$	Mass and proton number do not change. The nucleus absorbs a photon to excite.

Table 2.5

Nuclear Decay and Capture

A heavy element undergoes a decay process to increase its nuclear stability. A light element undergoes a capture process when struck by a high-energy particle to increase its nuclear stability. There are a few points you should note for these processes. With β -decay, the electron may or may not be added to the electrical shell of the atom. If it is, the atom remains neutral. If the electron escapes, the

compound becomes a cation. Usually the particle escapes, and an ion is formed, although we typically ignore this fact in nuclear chemistry. The representation of the beta particle as having a -1 for its Z term is not accurate, but it works for the algebra of the equation. Beta capture and positron decay both result in the same element being formed. Likewise, beta decay and positron capture both result in the same element being formed. The positron behaves just as a beta particle does, only it carries a positive charge and is an anti-electron. This makes it antimatter, but for our purpose that doesn't matter.

The reactions in Table 2.5 are simplified. We have ignored conservation of energy and conservation of momentum. To balance these more accurately, the mass of the electron (beta particle) and positron (anti-electron) cannot be treated as zero. A neutrino is required in some cases to balance the equation. Solving decay and capture questions in the manner that these examples are presented will work fine on test questions you may see. The questions you may see will involve determining the particle that was lost in a process, or determining the final elemental product after a nuclear reaction has taken place. It is sort of odd that if you reverse the first two letters in nuclear chemistry the phrase becomes unclear chemistry. Given the science community's level of understanding at the present time, *unclear* is an accurate depiction of the subject.

Example 2.29

Which of the following elements results from two consecutive alpha decays of ^{210}At ?

- A. ^{210}Fr
- B. ^{206}Bi
- C. ^{202}Tl
- D. ^{202}Bi

Solution

Alpha particles are helium nuclei (alpha-helium sounds like alpha-helix, so remember the phrase "alpha helium"). An alpha particle has a mass of 4 amu, thus losing two alpha particles results in the loss of 8 amu. The element that remains after two consecutive alpha decays has four fewer protons than the original element (from 85 to 81) and has a mass eight less than the original element (from 210 to 202 amu). At is element number 85, so the final element must have atomic number 81. Consulting a periodic table shows that element 81 is Tl. ^{202}Tl is the best answer, so pick C.

Example 2.30

Electron capture by a nucleus results in which of the following?

- A. An increase in atomic number by one, and a mass increase of one amu.
- B. An increase in atomic number by one, and no change in atomic mass.
- C. A decrease in atomic number by one, and a mass decrease of one amu.
- D. A decrease in atomic number by one, and no change in atomic mass.

Solution

Electron capture involves the gain of an electron by the nucleus. An electron has a -1 charge and no mass. This means that the element should experience a decrease in its positive charge by one, and no change in its mass. This is best described in choice D.

Example 2.31

Which of the following processes would NOT result in the formation of tritium?

- A. Alpha decay of ${}^7\text{Li}$
- B. Positron emission by ${}^3\text{He}$
- C. Beta decay of ${}^4\text{He}$
- D. Gamma emission from ${}^3\text{H}$

Solution

An alpha particle is a helium nucleus (${}^4\text{He}$), so when ${}^7\text{Li}$ loses a helium nucleus, the mass decreases by four to three and the number of protons decreases by two to one. The final product is tritium (${}^3\text{H}$). Choice A is valid. A positron is a massless positively charged particle that when lost, converts a proton into a neutron. The mass remains the same, but the ${}^3\text{He}$ convert into tritium (${}^3\text{H}$). Choice B is valid. Beta decay involves the loss of an electron from the nucleus. This converts a neutron into a proton. The mass does not change and thus remains at four. The number of protons increases by one to three. The product is ${}^4\text{Li}$, which is NOT tritium. The best answer is choice C. Gamma emission does not change the particle, thus choice D is valid.

Example 2.32

In the conversion from ${}^{165}\text{Ho}$ to ${}^{157}\text{Gd}$, at least two alpha particles were emitted. What else was emitted?

- A. A third alpha particle
- B. One neutron
- C. One beta particle
- D. One positron

Solution

In converting from ${}^{165}\text{Ho}$ to ${}^{157}\text{Gd}$, the mass decreases by eight, which equates to the mass of two alpha particles. This means that any other particles emitted are massless. This eliminates choices A and B. The atomic number has decreased by three. Loss of two alpha particles decreases the atomic number by four, so the other particle emitted must increase the atomic number by one. Loss of a positron decreases the atomic number by one, so choice D is eliminated. A beta particle must also have been emitted in addition to the two alpha particles. Choice C is correct.

Half-Life

The half-life of a sample of material is the period of time required for 50% of the concentration of material to decay to a different (possibly stabler) form. It is most common to see half-lives associated with first-order decay processes. The graphs shown in Figures 2-41 and 2-42 show the concentration of a component over time for first-order decay (exponential decay) and zero-order decay (linear decay). The graphs shown in Figures 2-43 and 2-44 are the derivative graphs of Figures 2-41 and 2-42, which represent the change in rate as a function of change in time.

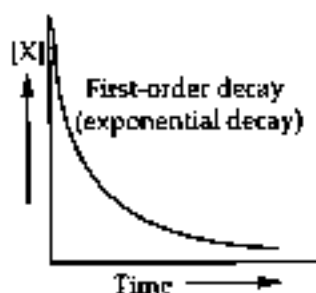


Figure 2-41

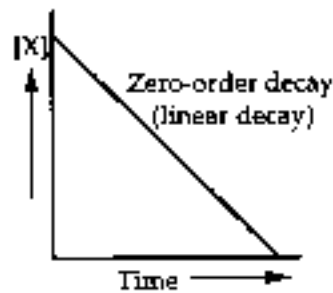


Figure 2-42

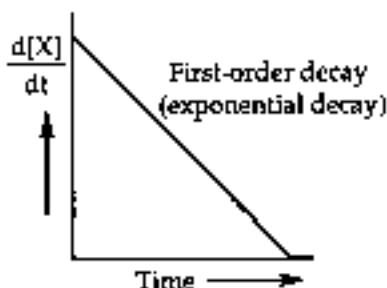


Figure 2-43

$$\text{rate} = \frac{d[X]}{dt}$$

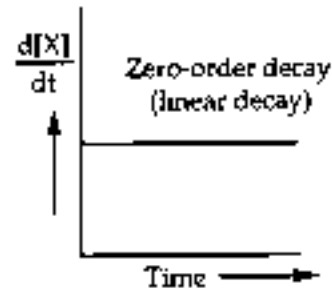


Figure 2-44

In Figures 2-45 and 2-46, the graphs are marked at 50% and 25% of the initial concentration values. These are the various concentrations after consecutive half-lives. It takes one half-life to cut the concentration in half, and a second half-life to cut that concentration in half again, which results in one-quarter of the original concentration. Figure 2-45 shows half-life as a function of time for a first-order decay. Figure 2-46 shows half-life as a function of time for a zero-order decay.

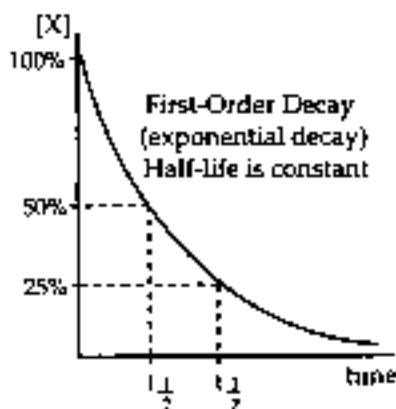


Figure 2-45

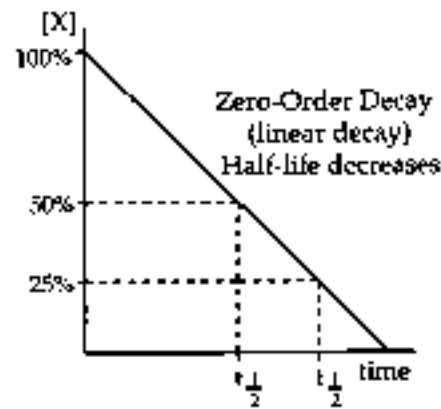


Figure 2-46

Figure 2-45 demonstrates that for a first-order process, the half-life is constant, regardless of concentration. As the concentration decreases, the duration of the half-life remains the same. Figure 2-46 demonstrates that for a zero-order process, the half-life is directly proportional to the concentration of the compound. As the concentration decreases, so does the duration of the half-life. The second half-life is half as long as the first half-life, because the concentration change is half as large. As a point of interest, for a second-order process, the half-life is indirectly proportional to the concentration of the compound. As the concentration decreases, the length of the half-life increases.

These graphs should look familiar from biology and physics. In biology, bacterial growth is exponential, and enzyme kinetics can be either exponential or linear. In physics, displacement, velocity and acceleration follow similar patterns. The point is that you must understand what these graphs tell you about some values increasing or decreasing as a function of time. Half-life analysis is most commonly evaluated using the upper graphs.

On the MCAT, you can expect to have to solve some half-life questions. As far as half-life problems are concerned, there are two methods by which they are solved. One method is to use Equation 2.11, the first-order decay equation:

$$C_t = C_0 e^{-kt} \quad (2.11)$$

where C_t is the concentration at a given time, C_0 is the initial concentration, k is the rate constant for the decay process, and t is the elapsed time. This works but requires the use of natural logs, which can be time-consuming.

The second method involves the sequential summing of half-lives required to reach a specific concentration. The test traditionally uses numbers that work well with this method. It is the most time-efficient method. When a problem asks how much time passes until a certain percentage of the original quantity remains, it is easiest to figure how many half-lives are required to reach that percentage and then convert the quantity of half-lives to total time. When using this method, use reasonable approximations. For instance, if the half-life is 110 years, then 335 years is three half-lives plus a little bit. There is 50% remaining after the first half-life, 25% remaining after the second half-life, and 12.5% remaining after the third half-life. Thus, there is just under 12.5% remaining after 335 years.

Example 2.33

How many half-lives are required to decompose 93.75% of some material?

- A. 2
- B. 3
- C. 4
- D. 5

Solution

Decomposing (decaying) 93.75% of the initial material results in 6.25% remaining. The quick and easy method is to cut the percentage in half continually until 6.25 is reached.

$$100\% \rightarrow 50\% \rightarrow 25\% \rightarrow 12.5\% \rightarrow 6.25\%$$

It requires a total of four half-lives to reach a point where 6.25% remains, which is a decomposition of 93.75% of the material. The best answer is choice C.

Example 2.34

If the initial concentration of a toxic material is 1500 parts per million and the half-life is 4.5 days, how long will it take for the level to reach a concentration of 100 ppm?

- A. 13.6 days
- B. 17.8 days
- C. 18.2 days
- D. 22.3 days

Solution

We must sequentially halve 1500 until a number close to 100 is reached.

$$1500 \text{ ppm} \rightarrow 750 \text{ ppm} \rightarrow 375 \text{ ppm} \rightarrow 187.5 \text{ ppm} \rightarrow 93.75 \text{ ppm}$$

After four half-lives, there is a little less than the target value of 100 ppm remaining. This means that just under four half-lives are needed to reach 100 ppm. Four half-lives are eighteen days, so the best answer is slightly less than 18 days. The best answer is choice B.

Sequential halving of the concentration in first-order decay questions is far easier than using Equation 2.11 or other rate equations. The test-writers may give you the appropriate equations needed to solve the question in longhand, but this method should be your method of choice. Just because the test writers provide you with a piece of information, does not mean that you need to use it.

This technique can also be applied to exponential growth problems, as seen in biology with population genetics that obey exponential growth behavior. With growth, you continually double the concentration rather than cut it in half. For instance, if a bacterial population doubles in 12 seconds, then in one minute it increases 32 times (2^5).

$$X \rightarrow 2X \rightarrow 4X \rightarrow 8X \rightarrow 16X \rightarrow 32X$$

Example 2.35

If the half-life of a particular atom is 20.4 minutes and the initial concentration is 1200 parts per million, what will the concentration be after exactly one hour?

- A. 582 ppm
- B. 291 ppm
- C. 155 ppm
- D. 145 ppm

Solution

One hour is not exactly equal to three half-lives (three half-lives = 1 hour and 1.2 minutes). After three half-lives, we have the following:

$$1200 \text{ ppm} \rightarrow 600 \text{ ppm} \rightarrow 300 \text{ ppm} \rightarrow 150 \text{ ppm}$$

We haven't quite used a full three half-lives, so the concentration should not have decreased all the way down to 150 ppm. This means that the correct answer should be just over 150 ppm (150+ ppm). The only answer choice that is slightly greater than 150 ppm is 155 ppm, choice C.

Atomic Structure Passages

14 Passages

100 Questions

Suggested Atomic Structure Passage Schedule:

- I:** After reading this section and attending lecture: Passages I, IV - VI, & VIII
Grade passages immediately after completion and log your mistakes.
- II:** Following Task I: Passages II, III, VII, IX, & XIV. (34 questions in 44 minutes)
Time yourself accurately, grade your answers, and review mistakes.
- III:** Review: Passages X - XIII & Questions 96 - 100
Focus on reviewing the concepts. Do not worry about timing.



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Atomic Theory Study Passages

I. Classical Experiments	(1 - 8)
II. Isotopic Abundance and Average Atomic Mass	(9 - 14)
III. Bohr Model of Hydrogen	(15 - 21)
IV. Paramagnetism and Liquid Crystal Displays	(22 - 28)
V. Migration through a Membrane	(29 - 35)
VI. Ionization Energy	(36 - 42)
VII. Transition Metal Trends	(43 - 49)
VIII. Lasers	(50 - 56)
IX. Paint Pigments	(57 - 63)
X. Fluorescence and Phosphorescence	(64 - 69)
XI. Flame Test	(70 - 76)
XII. Glyphosate	(77 - 81)
XIII. Technetium Decay	(82 - 88)
XIV. Cold Fusion	(89 - 95)
Questions Not Based on a Descriptive Passage	(96 - 100)

Atomic Structure Scoring Scale

Raw Score	MCAT Score
84 - 100	13 - 15
66 - 83	10 - 12
47 - 65	7 - 9
34 - 46	4 - 6
1 - 33	1 - 3

Passage 1 (Questions 1 - 8)

Two of the more famous classical experiments in establishing the properties of subatomic particles are the Thomson experiment and the Millikan oil drop experiment. The Thomson experiment determined that the mass-to-charge ratio for an electron is $5.686 \times 10^{-12} \text{ kg/C}$. The Thomson experiment involved the application of a perpendicular electric field to a beam of electrons produced by a cathode ray tube. The Thomson apparatus is shown in Figure 1.

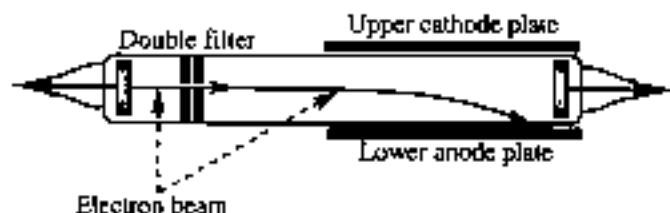


Figure 1

The double filter ensures that the electron beam is in a single direction. The electron beam, when exposed to the electric field generated by the two plates, is bent towards the positively charged plate. The curvature is constant as long as the field strength is constant. The bending of the electron beam is attributed to both the attractive force of the oppositely charged plate and the repulsive force of the like charged plate. A magnetic field may also be used to bend the electron beam. When using a magnetic field to deflect the electrons, the radius of the arc of the electron beam can be employed to determine the mass-to-charge ratio using Equation 1.

$$\frac{m}{q} = \frac{B r}{v}$$

Equation 1

In Equation 1, the mass of the electron is m , the charge of the electron is q , the strength of the magnetic field is B , the radius of the arc of the electron pathway is r , and the velocity of the electron is v .

In the Millikan oil drop experiment, after ten years of data collection, it was determined that the charge of an electron is $1.6 \times 10^{-19} \text{ C}$. The experiment involved charging an oil drop by bombarding it with an ionizing electron beam. Several oil drops passed through the ionizing beam, but only a few gained a charge. The oil drops then fell into an electric field generated by two charged plates in a glass cylinder. The field was aligned so that the lower plate repelled the oil drop upward, while the upper plate pulled the oil drop upward.

The force upward on the oil drop is thus qV , where q is the charge of the oil drop and V is the voltage of the field. The force pulling the oil drop downward is mg , where m is the mass of the oil drop, and g is the gravitational force constant. If the oil drop is suspended, then the two forces are equal, and it is possible to determine the charge of the oil drop. The charge of the oil drop must be a whole number multiple of the charge of an electron. The apparatus used in the Millikan oil drop experiment is shown in Figure 2.

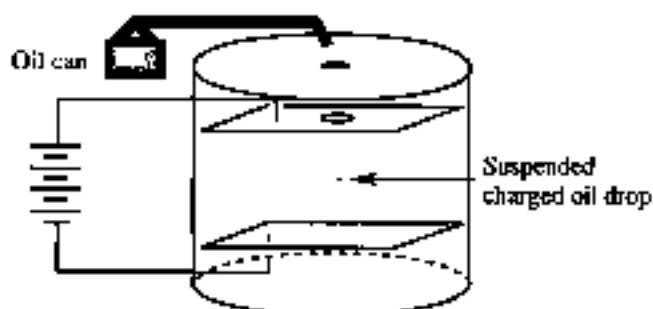


Figure 2

The hole in the top plate allows for oil droplets to fall into the chamber. The field between the two plates is assumed to be uniform, so that the force on the oil droplet is equal at any position between the two charged plates. The field is adjustable, so that droplets of variable mass can be suspended. Not all oil droplets are of equal mass, so an average value is assumed.

- An assumption of the Millikan oil drop experiment is which of the following?
 - Oil drops are naturally charged.
 - The pull of gravity is equal in all directions at the microscopic level.
 - The mass of electrons is negligible compared to the mass of the oil drop.
 - The magnitude of charge for an electron can vary.
- The electron beam bends downward as it passes between the two charged plates in the Thomson experiment. What can we expect for a beam of neutrons and protons?
 - The proton beam would arc upwards as it travels, while the neutron beam would not arc at all.
 - The neutron beam would arc upwards as it travels while the proton beam would not arc at all.
 - Both the neutron and the proton beams would arc upwards as they travel between the plates.
 - Neither the neutron nor the proton beam would arc as they travel between the plates.
- How could a value of $3.2 \times 10^{-19} \text{ C}$ in an oil drop in the Millikan oil drop experiment be explained?
 - The oil drop was doubly ionized.
 - The oil drop gained an electron rather than lost an electron.
 - The oil drop lost a proton rather than lost an electron.
 - The oil drop gained a proton rather than gained an electron.

4. The Thomson experiment determined which of the following?
- The mass of an electron
 - The mass of a proton
 - The mass-to-charge ratio of an electron
 - The mass-to-charge ratio of a proton
5. The Millikan oil drop experiment requires knowing all of the following EXCEPT:
- the mass of the oil droplet.
 - the voltage of the electric field.
 - the gravitational force constant.
 - the temperature of the chamber.
6. In a neutral atom, what can be said about the number of electrons relative to the number of protons and neutrons?
- The number of electrons must equal both the number of protons and the number of neutrons.
 - The number of electrons must equal the number of protons only, and the number of neutrons is irrelevant.
 - The number of electrons must equal the number of neutrons only, and the number of protons is irrelevant.
 - The number of electrons must equal the sum of the protons and neutrons combined.
7. The STRONGEST attractive electrostatic force is felt between:
- a proton and a neutron.
 - a neutron and an electron.
 - an electron and an electron.
 - an electron and a proton.
8. If an oil drop were slowly descending once it was in the electric field between the two charged plates, then which of the following changes would NOT stop it from continuing to fall?
- Increase the number of electrons on the drop.
 - Decrease the number of electrons on the drop.
 - Increase the voltage difference between the two plates.
 - Offset the gravitational pull with an electric force.

Passage II (Questions 9 - 14)

To determine the relative abundance of the isotopes for an element, a mass spectrometer is employed. The element is first vaporized into its gaseous state, if it is not already in gas form. Because this is carried out in a vacuum, a high temperature may not be required to vaporize the material. The cationic compound is generated by high-energy electron impact with the neutral atoms. A neutral atom loses a valence electron to ionization upon impact. The cationic species formed is then accelerated in an electric field until it reaches a set velocity (v). The particle beam produced is projected through a double filter to ensure that no deflected particles pass through to the detector.

Once particles are passed through the double filter, they are subjected to a magnetic field perpendicular to the original electric field. This curves the pathway of the particles. The force on charged particles moving through a magnetic field is found using Equation 1.

$$F = qvB$$

Equation 1

where q is the charge of the particle (assumed to be $+1$) and B is the magnetic field strength. According to Newton's second law, the force on the particle is $F = ma$, where m is mass and a is acceleration. The particle is moving along a curved path, so it has angular acceleration, which is found using Equation 2.

$$a = \frac{v^2}{R}$$

Equation 2

where R is the radius of the arc. Substituting ma for F in Equation 1 and manipulating the variables yields Equation 3.

$$qB = \frac{mv}{R}$$

Equation 3

The equations support the intuitive perception of how the path is affected by mass, charge, velocity, and external field strength. Figure 1 shows a basic schematic of a mass spectrometer.

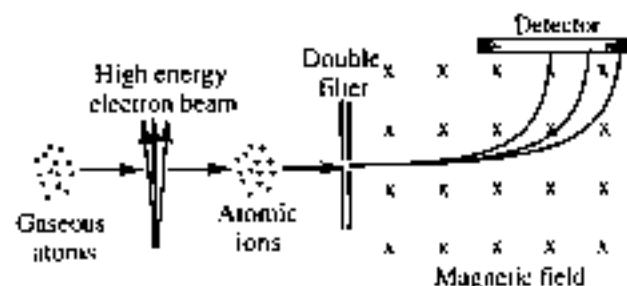


Figure 1

Different isotopes have different atomic masses, so each isotope has a unique radius (R). The detector determines the relative quantities of each isotope. This information is then used to determine the average atomic mass, a value that takes into account all of the isotopes for a particular element.

Table 1 lists the isotopic abundance for the common isotopes of boron, magnesium, and silver.

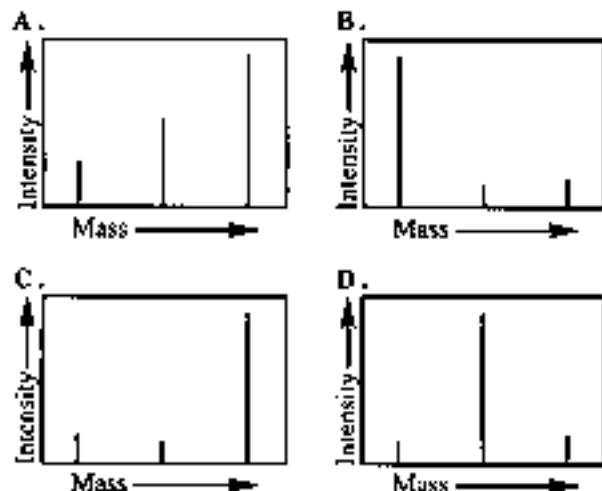
Element	Atomic Mass	Percentage
Boron-10	10.013 amu	19.78%
Boron-11	11.009 amu	80.22%
Magnesium-24	23.985 amu	78.70%
Magnesium-25	24.986 amu	10.13%
Magnesium-26	25.983 amu	11.17%
Silver-107	106.91 amu	51.80%
Silver-109	108.90 amu	48.20%

Table 1

9. Why is carbon added to a sample before it is vaporized in a mass spectrometer?

A. Carbon-12 is used as a standard for mass reference.
 B. Carbon-13 is used as a standard for mass reference.
 C. Carbon helps to lubricate the atoms at an atomic level.
 D. Carbon absorbs heat from the atoms and prevents the sample from overheating.

10. Which of the following plots represents the output from the mass spectrometer for magnesium?



11. Which of the following calculations accurately determines the average atomic mass for silver?

A. $0.787(23.985) + 0.1013(24.986) + 0.1117(25.983)$
 B. $0.1117(23.985) + 0.1013(24.986) + 0.787(25.983)$
 C. $0.482(106.91) + 0.518(108.90)$
 D. $0.518(106.91) + 0.482(108.90)$

12. When a mass spectrum is to be obtained for a solid compound, the first step in the process is which of the following?

A. Vaporization
 B. Condensation
 C. Ionization
 D. Sublimation

13. Identify the TRUE statement(s) from the following:

I. The average atomic mass of an element is always greater than the mass of the heaviest isotope for that element.
 II. The average atomic mass of an element is always greater than the mass of the lightest isotope for that element.
 III. When there are two isotopes for an element, the average atomic mass is closer to the more abundant isotope than the less abundant isotope.

A. I only
 B. II only
 C. I and III only
 D. II and III only

14. The difference between the two isotopes of boron is:

A. the isotope of boron with mass 11.009 amu has one proton less than the isotope of boron with mass 10.013 amu.
 B. the isotope of boron with mass 11.009 amu has one proton more than the isotope of boron with mass 10.013 amu.
 C. the isotope of boron with mass 11.009 amu has one neutron less than the isotope of boron with mass 10.013 amu.
 D. the isotope of boron with mass 11.009 amu has one neutron more than the isotope of boron with mass 10.013 amu.

Passage III (Questions 15 - 20)

The Bohr model for hydrogen proposes that there are quantized energy levels in which an orbiting electron may be found. The theory is derived from the observation that exact frequencies of light are absorbed by hydrogen gas, no matter what the concentration of the sample or intensity of the incident light source. Given that light frequency directly relates to light energy, the idea is that quantized amounts of energy are absorbed and emitted as an electron moves between energy levels. Figure 1 shows the Bohr representation of hydrogen.

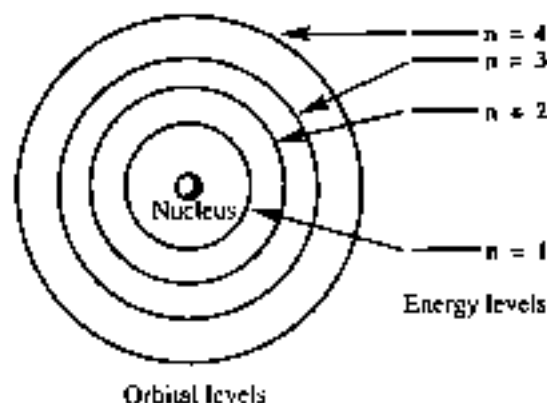


Figure 1

Due to Coulombic attraction, the most stable energy levels for the electron correspond to the closest proximity to the nucleus. This means that the energetics of the orbiting electron depend on the charge nucleus and the distance from the nucleus (or quantum level). The energy level for the electron of hydrogen is found using Equation 1.

$$E = -2.178 \times 10^{-18} \left(\frac{Z^2}{n^2} \right) \text{ J}$$

Equation 1

15. What is the ionization energy for hydrogen?

- A. $1.634 \times 10^{-18} \text{ J}$
- B. $2.178 \times 10^{-18} \text{ J}$
- C. $4.356 \times 10^{-18} \text{ J}$
- D. $8.712 \times 10^{-18} \text{ J}$

16. The deBroglie equation relates the energy of a photon to its equivalent mass according to $E = mc^2$. How can the deBroglie wavelength of an electron be found?

- A. $\lambda = \frac{h}{mc}$
- B. $\lambda = \frac{mc}{h}$
- C. $\lambda = \frac{2h}{mc^2}$
- D. $\lambda = \frac{mc^2}{2h}$

17. Which of the following transitions relates to the GREATEST energy of absorption?

- A. From the $n = 2$ level to the $n = 4$ level
- B. From the $n = 2$ level to the $n = 1$ level
- C. From the $n = 1$ level to the $n = 2$ level
- D. From the $n = 2$ level to the $n = 5$ level

18. Which statements are valid regarding the energy levels of hydrogen?

- I. As the value of n increases, the electron is farther from the nucleus, therefore the electron is present in a higher energy state.
 - II. As the value of n increases, the difference between adjacent energy levels (with n values differing by one), gets smaller.
 - III. A larger nuclear charge has no effect on transition energies, although it lowers all of the energy levels.
- A. I only
 - B. II only
 - C. I and II only
 - D. II and III only

19. For some compound X, which emits orange light for a transition from the $n = 4$ level to the $n = 3$ level, what must be TRUE?

- A. A transition from the $n = 5$ level to the $n = 4$ level emits green light.
- B. A transition from the $n = 4$ level to the $n = 2$ level emits red light.
- C. A transition from the $n = 6$ level to the $n = 4$ level emits violet light.
- D. A transition from the $n = 5$ level to the $n = 3$ level emits yellow light.

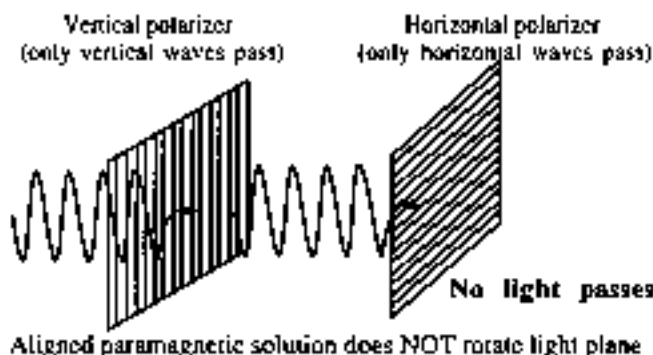
20. Which of the following observation leads to the conclusion that energy levels are quantized?

- A. A sample always absorbs the same amount of light.
- B. A sample always absorbs the same color of light.
- C. A sample always emits different frequencies at different temperatures.
- D. A sample always emits different frequencies at different concentrations.

Liquid crystal displays (LCDs) utilize the properties of paramagnetic species. When plane-polarized light passes through a solution containing a paramagnetic species, it rotates the plane-polarized light. If the paramagnetic species is subjected to a magnetic field, then the particles align in the field, and the plane-polarized light is not rotated. The display commonly seen in calculators and digital watches uses this principle to filter or pass light through templates.

The basic design of the display board involves passing light through a polarizing plate into a cell containing the paramagnetic species and then having the light pass through a second polarizing plate that is rotated 90° relative to the first plate. If the field is off, the light will rotate and thus be passed through the second polarizer. This results in a bright spot on the display board. If the field is on, the light will not rotate and thus will not pass through the second polarizer. This results in a dark spot on the display board. Figure 1 shows a simple display circuit.

Circuit on (Solution is aligned)



Circuit off (Solution is random)

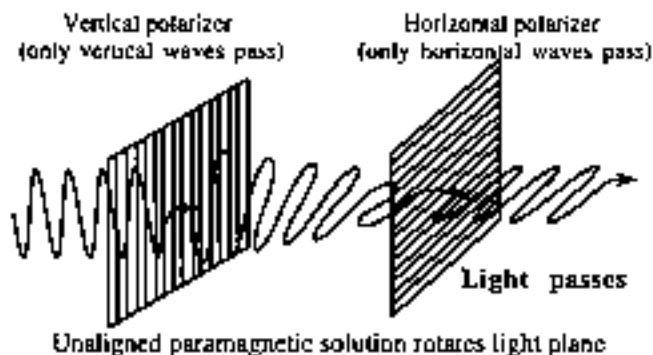


Figure 1

A paramagnetic species is defined as a compound that has unpaired electrons. Many common examples involve transition metals with a d-shell count of d^1 to d^9 . Organic compounds that are paramagnetic are referred to as *radicals*. Oxygen gas has two unpaired electrons; thus, when exposed to a magnetic field, oxygen molecules will align with the field.

21. Which of the following can be used in an LCD?

- A. CdCl_2
- B. F_2
- C. $\text{CoCl}_2 \cdot 6 \text{H}_2\text{O}$
- D. H_2

22. What are the quantum numbers associated with the LAST electron in the nitrogen atom?

- A. $n = 2$ $l = 1$ $m_l = +1$ $m_s = +\frac{1}{2}$
- B. $n = 2$ $l = 1$ $m_l = +1$ $m_s = -\frac{1}{2}$
- C. $n = 2$ $l = 1$ $m_l = -1$ $m_s = +\frac{1}{2}$
- D. $n = 2$ $l = 1$ $m_l = -1$ $m_s = -\frac{1}{2}$

23. What is the shape of $\text{Fe}(\text{CN})_6^{3-}$?

- A. Octahedral
- B. Square planar
- C. Tetrahedral
- D. Trigonal bipyramidal

24. Which of the following compounds is polar?

- A. FeCl_6^{3-}
- B. $\text{Ni}(\text{CN})_5^{3-}$
- C. $\text{cis-Pt}(\text{NH}_3)_2\text{Cl}_2$
- D. $\text{trans-Pt}(\text{NH}_3)_2\text{Cl}_2$

25. Which ion has the MOST unpaired electrons?

- A. Cr^{3+}
- B. Mn^{2+}
- C. Cu^+
- D. Ni^{2+}

26. Which of the following statements is NOT true?
- A. Cr has a higher formal charge in $\text{Cr}(\text{CN})_6$ than $\text{Cr}(\text{NH}_3)_6^{3+}$.
 - B. $\text{Cu}(\text{H}_2\text{O})_6^{3+}$ has ten d-electrons.
 - C. $\text{Fe}(\text{NH}_3)_6^{3+}$ has the same shape as FeCl_6^{3-} .
 - D. Scandium is more likely to have +4 formal charge than titanium.
27. Which of the following pairs of molecules does NOT have the central atom (transition metal) with the same number of d-electrons in both compounds?
- A. $\text{Fe}(\text{NH}_3)_6^{3+}$ and FeCl_6^{3-}
 - B. $\text{Co}(\text{H}_2\text{O})_6^{3+}$ and MnCl_6^{4-}
 - C. $\text{Cr}(\text{NH}_3)_6^{3+}$ and MoCl_6^{3-}
 - D. $\text{Os}(\text{NH}_3)_4^{2+}$ and $\text{RhCl}_3(\text{PR}_3)_3$

Passage V (Questions 28 - 34)

In vivo, semi-permeable membranes are used by the body to control diffusion and osmosis (entropically favorable occurrences). In the kidney, Bowman's capsule functions in such a manner. Semipermeable membranes have microscopic pores that allow the passage of some molecules or, in certain cases such as the sodium-potassium pump, ions. They function by allowing only selected ions to pass from the solution on one side of the membrane to the solution on the other side of the membrane. Semi-permeable membranes can distinguish by size, charge, or both.

Semipermeable membranes can be applied to bench top chemistry (*in vitro*) as well. This application can increase a reaction's selectivity. Semipermeable membranes used *in vitro*, that separate by size, are often referred to as molecular and atomic sieves. While some molecular sieves differentiate by atomic radius, others differentiate by charge. To study the effectiveness of a semipermeable membrane, two solutions of different solute concentrations are separated by the membrane. Figure 1 shows a U-tube with a semipermeable membrane separating the two solutions.

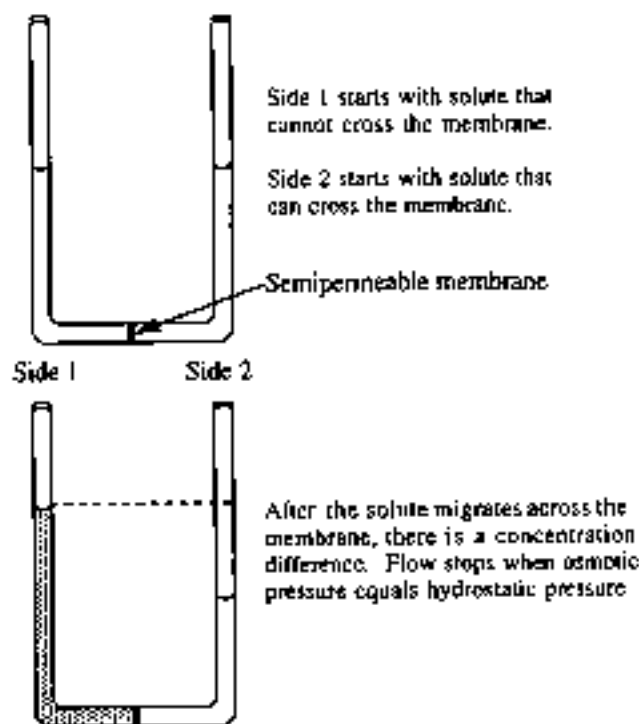


Figure 1

Table 1 lists the rates of effusion from Solution I through a synthetic membrane into Solution II for a generic study.

Cation	Rate (M/s)
Li^+	4.7×10^{-2}
Na^+	1.1×10^{-2}
Mg^{2+}	2.9×10^{-2}
K^+	3.5×10^{-3}
Ca^{2+}	8.3×10^{-3}

Table 1

Note: The synthetic membrane used in the experiment represented in Table 1 is believed to be aprotic and discriminates by ionic size. It is modeled after a protein that is found in the lipid bilayer.

28. Based on the data in Table 1, which of the following conclusions is valid?

- A. As cationic size increases, the rate of migration across the membrane decreases.
- B. As cationic size increases, the rate of migration across the membrane increases.
- C. The cationic size has no direct effect on the rate of migration across the membrane.
- D. Because cations are larger than neutral atoms, the effect of cationic size on the rate of migration cannot be determined.

29. The BEST explanation for why potassium (K) has a lower ionization potential than sodium (Na) is that:

- A. potassium is more metallic than sodium.
- B. potassium has a larger enthalpy of reaction with water than does sodium.
- C. the 4s-orbital is larger than the 3s-orbital.
- D. potassium has a larger electron affinity than does sodium.

30. The BEST explanation for why fluorine is smaller than carbon is which of the following?

- A. Fluorine's electrons are at a lower quantum level than those of carbon.
- B. Filled p-orbitals contract more than half-filled p-orbitals.
- C. Fluorine has a larger effective nuclear charge than carbon.
- D. Fluorine is heavier than carbon.

31. Which of the following is the SMALLEST?

- A. F⁻
- B. Ne
- C. Na⁺
- D. Cl⁻

32. What is the charge on the MOST stable ion of magnesium?

- A. +2
- B. +1
- C. -1
- D. -2

33. Which of the following atoms is LARGEST?

- A. F
- B. Cl
- C. Br
- D. I

34. If the pore of the membrane used in the experiment associated with Table 1 were to distinguish by both size and charge, the SLOWEST rate would occur with which cation?

- A. Li⁺
 - B. K⁺
 - C. Mg²⁺
 - D. Ca²⁺
-

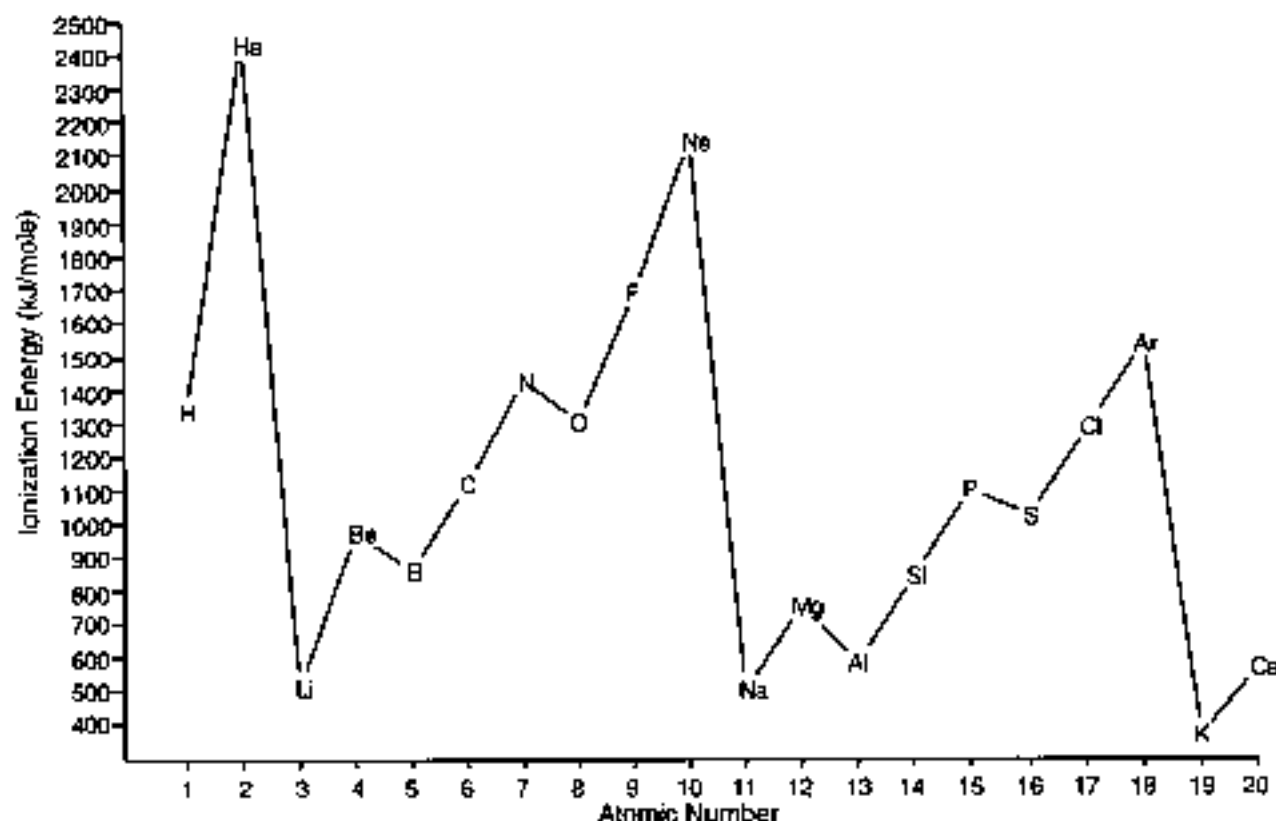
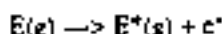


Figure 1

Figure 1 shows the first ionization energy for the first twenty elements in the periodic table. The first ionization energy is defined as the energy required to remove the first electron from the valence shell of an element to form a monocation. The general reaction, where E symbolizes any element from the periodic table, is shown in Reaction 1.



Reaction 1

To negate solvent effects, ionization reactions are done in the gas phase. The first ionization energy is proportional to the square of the effective nuclear charge of the element. The effective nuclear charge is the charge exerted upon the valence electrons by all other charged species in the atom. The effective nuclear charge takes into account the number of protons in the nucleus and the number of core electrons (electrons in principal quantum shells lower than the valence electrons). Ionization energy depends on the valence shell principal quantum number. Ionization energy decreases as you descend a family in the periodic table. A larger principle quantum number corresponds to a lower ionization energy.

Figure 1 shows deviations from the behavior predicted from effective nuclear charge data. Deviations are seen with elements having valence electronic configurations of ns^2np^1 and ns^2np^4 , which is attributed to the filled s-level and the half-filled p-level associated with the cations that are formed. There is additional stability when the p-orbitals have equal occupancy of electrons. The symmetry associated with the half-filled state results in cation stability.

35. Which of the following BEST explains the high ionization energy of helium?

- A. Helium has no valence electrons to remove.
- B. Helium has an effective nuclear charge greater than two.
- C. The electron must be removed from the first quantum level, which experiences the greatest nuclear charge.
- D. Helium requires a great amount of energy to be converted first into the gas phase.

36. What can be expected for the second ionization energy of lithium, sodium, and potassium?

- A. The second ionization energies are only slightly larger than the first ionization energies, because the Z_{eff} has not changed drastically.
- B. The second ionization energies are only slightly larger than the first ionization energies, because the valence shell has changed.
- C. The second ionization energies are substantially larger than the first ionization energies, because the Z_{eff} has not changed drastically.
- D. The second ionization energies are substantially larger than the first ionization energies, because the valence shell has changed.

37. How can the lower ionization energy for oxygen than nitrogen BEST be explained?
- A. Nitrogen is more electronegative than oxygen, so it does not share electrons as readily.
 - B. Nitrogen has more protons than oxygen, so it carries a greater effective nuclear charge.
 - C. Nitrogen is larger than oxygen, so it is harder to remove a valence electron.
 - D. Nitrogen has a half-filled p-level, so it does not lose an electron as readily. Oxygen, when it loses one electron, attains half-filled p-level stability.
38. How can the trend in first ionization energy between aluminum, silicon, and phosphorus BEST be explained?
- A. The effective nuclear charge increases as you move from Al to P in the periodic table.
 - B. The effective nuclear charge decreases as you move from Al to P in the periodic table.
 - C. The number of core electrons increases as you move from Al to P in the periodic table.
 - D. The number of core electrons decreases as you move from Al to P in the periodic table.
39. Why is the first ionization energy of fluorine greater than the first ionization energy of chlorine?
- A. Fluorine is less electronegative than chlorine, so it shares electrons more readily than chlorine.
 - B. Chlorine has more protons than fluorine, so it carries a greater effective nuclear charge.
 - C. Chlorine is losing an electron from a shell that is farther from the nucleus than fluorine, so chlorine can lose an electron more easily.
 - D. Chlorine has no electrons in the p-level, so it does not lose an electron. Fluorine, when it loses one electron, completes its valence shell.
40. Which of the following elements can be oxidized MOST easily?
- A. Sulfur
 - B. Magnesium
 - C. Boron
 - D. Argon
41. What can be expected for the first ionization energies of bromine, krypton, and selenium?
- A. $I.E._{Br} > I.E._{Kr} > I.E._{Se}$
 - B. $I.E._{Kr} > I.E._{Br} > I.E._{Se}$
 - C. $I.E._{Se} > I.E._{Br} > I.E._{Kr}$
 - D. $I.E._{Se} > I.E._{Kr} > I.E._{Br}$
42. The first ionization energy for sodium is 495 kJ/mole and the second ionization energy is 4558 kJ/mole. The first ionization energy of magnesium is 732 kJ/mole, and the second ionization energy is 1451 kJ/mole. How can the change in second ionization energy be explained?
- A. The first and second ionizations of sodium are from different quantum levels, while the first and second ionizations of magnesium are from the same quantum level.
 - B. Sodium is less electronegative than magnesium, which implies that the second ionization energy must be greater.
 - C. Once one electron has been lost from magnesium, the second electron is repelled by the positive charge of the magnesium cation.
 - D. Sodium metal gains half-filled stability by losing one electron, while magnesium requires losing two electrons to attain half-filled stability.

Passage VII (Questions 43 - 49)

Transition metals do not follow the same periodic trends associated with the main-group elements. The reasoning behind their separate trend is rooted in their valence electrons. The d -electrons of transition metals are not the outermost electrons, despite the fact that the d -level is being filled as you move left to right through any row of the transition metals. When we look at first-row transition metals, we see electrons fill the levels according to the Aufbau principle, so they fill the $4s$ -level prior to the $3d$ -level. However, when electrons are lost (when the element undergoes ionization), the electrons are lost from the $4s$ -level before the $3d$ -level. Table 1 shows the ionization energy data associated with the first row of transition metals. The first three ionization energy values are listed for each of the transition metals.

	M	M^{2+}	M^{3+}	1st IE	2nd IE	3rd IE
Sc	$4s^2 3d^1$	—	[Ar]	631	1235	2389
Ti	$4s^2 3d^2$	$3d^2$	$3d^1$	658	1309	2650
V	$4s^2 3d^3$	$3d^3$	$3d^2$	650	1413	2828
Cr	$4s^1 3d^5$	$3d^4$	$3d^3$	652	1591	2986
Mn	$4s^2 3d^5$	$3d^5$	$3d^4$	717	1509	3250
Fe	$4s^2 3d^6$	$3d^6$	$3d^5$	759	1561	2956
Co	$4s^2 3d^7$	$3d^7$	$3d^6$	760	1645	3231
Ni	$4s^2 3d^8$	$3d^8$	$3d^7$	736	1751	3393
Cu	$4s^1 3d^{10}$	$3d^9$	$3d^8$	745	1958	3578
Zn	$4s^2 3d^{10}$	$3d^{10}$	—	922	1772	—

Table 1

Periodic trends also include atomic radius, ionic radius, and atomic density at 25°C . Transition metals also show a deviation from main group elements in terms of trends in atomic size. Table 2 lists radius data for selected metals.

	Radius			Physical Properties		
	M	M^{2+}	M^{3+}	M.P.	B.P.	Density
K	227	—	—	63.7	760	0.86
Cu	197	99	—	838	1441	1.54
Sc	162	—	81	1539	2738	3.0
Ti	147	90	77	1668	3259	4.51
V	134	88	74	1903	3447	6.1
Cr	130	85	64	1874	2667	7.19
Mn	135	80	66	1241	2152	7.43
Fe	126	77	60	1536	3001	7.86
Co	125	75	64	1495	2908	8.9
Ni	124	69	—	1453	2731	8.92
Cu	128	72	—	1083	2597	8.96
Zn	138	74	—	419.5	906	7.14

Table 2

Trends are predictable when based on the effective nuclear charge and the outermost electron shell (valence shell). However, the effective nuclear charge and valence cloud of the different transition metals do not follow the same pattern as main block elements. Similar reasons can be employed to explain deviations in trends. Half-filled d -shell stability is associated with chromium, and filled d -shell stability is associated with copper, which accounts for deviations in their electronic configurations from what is expected according to the Aufbau principle.

43. What is the density of manganese at 1000°C ?

- A. Between 1.00 and 2.00 g per cm^3 .
- B. Between 2.00 and 7.43 g per cm^3 .
- C. Between 7.43 and 9.00 g per cm^3 .
- D. Between 9.00 and 12.00 g per cm^3 .

44. What is true of the relative second ionization energy values of Mo, Tc, Pd, and Ag?

- A. $\text{Mo} > \text{Tc} > \text{Pd} > \text{Ag}$
- B. $\text{Ag} > \text{Pd} > \text{Tc} > \text{Mo}$
- C. $\text{Ag} > \text{Tc} > \text{Mo} > \text{Pd}$
- D. $\text{Ag} > \text{Pd} > \text{Mo} > \text{Tc}$

45. How can the lower melting point and boiling point for zinc compared to other transition metals be explained?

- A. Increased interatomic forces with the filled $3d$ -shell.
- B. Reduced interatomic forces with the filled $3d$ -shell.
- C. Increased interatomic forces with the half-filled $3d$ -shell.
- D. Reduced interatomic forces with the half-filled $3d$ -shell.

46. What can be concluded about the ease of removing a $3d$ -electron compared to a $4s$ -electron?

- A. The $4s$ is more easily removed because it is closer to the nucleus.
- B. The $4s$ is less easily removed because it is closer to the nucleus.
- C. The $4s$ is more easily removed because it is farther from the nucleus.
- D. The $4s$ is less easily removed because it is farther from the nucleus.

47. What is TRUE about changes in density and atomic radius as one scans across the 3d row of the periodic table?
- As atomic radius increases, density increases.
 - As atomic radius increases, density decreases.
 - As atomic radius increases, density increases, except for elements with filled d-shell stability.
 - As atomic radius increases, the density does not vary in a predictable fashion.
48. How can the reduced radii of transition metal cations be explained?
- Transition metals lose their outermost electrons from the 4s-orbital, when becoming a cation.
 - Transition metals lose their outermost electrons from the 3d-orbital, when becoming a cation.
 - Transition metals lose their highest-energy electrons from the 4s-orbital, when becoming a cation.
 - Transition metals lose their highest-energy electrons from the 3d-orbital, when becoming a cation.
49. If the reduction potential of Mn^{2+} is -1.18 V and Co^{2+} is -0.28 V , what can be concluded about the reduction potential of Fe^{2+} ?
- $E^\circ_{\text{reduction of Fe}^{2+}}$ is less than -1.18 V .
 - $E^\circ_{\text{reduction of Fe}^{2+}}$ is between -1.18 and -0.73 V .
 - $E^\circ_{\text{reduction of Fe}^{2+}}$ is between -0.73 and -0.28 V .
 - $E^\circ_{\text{reduction of Fe}^{2+}}$ is greater than -0.28 V .

Passage VIII (Questions 50 - 56)

The basic principle behind the operation of a laser is the excitation of electrons from the ground state of a compound to some meta-stable excited state. The term *meta-stable* refers to an excited state where an electron can exist for a sustained period of time before it falls back to its ground state. By constantly supplying energy to the compound, the electrons can be "held" in the excited state. This preference for a higher energy level is in accordance with Boltzmann's distribution law of energy for any chemical or physical system. Boltzmann's distribution law predicts that the energy of a system will be distributed throughout the different levels according to a standard probability function that follows a bell curve. At higher total energy for the system, more electrons are present in an excited state than at lower total energy for the system. Figure 1 shows a molecular system before and after energy has been added. The excited and ground states are represented as levels capable of holding multiple electrons.

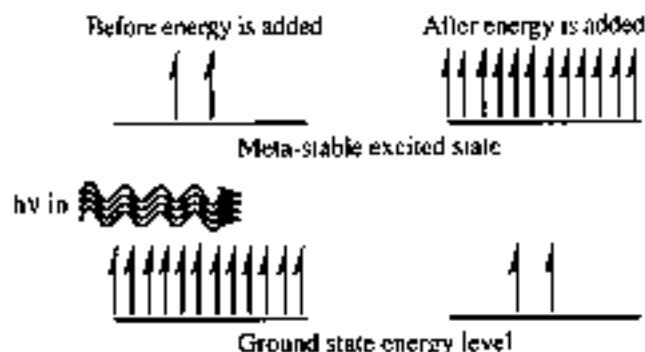


Figure 1

Energy is most easily introduced into the system by incident light (photons). A system analogous to the laser is the electrical capacitor, which can be filled with electrical energy and then instantly drained, releasing a pulse of energy. Lasers can also store energy (light energy) and then release a pulse of photons. A laser emits light energy corresponding to the transition-level changes. Once the crystal or gas in the laser releases photons, the light is collected and reflected between mirrors, before it is released through an aperture emitting the laser beam. Light emitted from a laser is close to being monochromatic (that is, light of one wavelength), but not precisely monochromatic. To date, absolutely monochromatic light has yet to be observed. This is attributed to the fact that the ground state and excited states are not single energy levels, but are in fact a band of quantized levels with identical electronic energy, but varying vibrational and rotational energies.

50. Which of the following electronic configurations is the ground state electronic configuration for aluminum?
- $1s^2 2s^2 2p^6 3p^3$
 - $1s^2 2s^2 2p^6 3s^3$
 - $1s^2 2s^2 2p^6 3s^2 3p^1$
 - $1s^2 2s^2 2p^6 3s^1 3p^2$

51. Monochromatic light is NOT observed for which of the following reasons?

- A. Lasers are not precise in their emissions.
- B. Crystals always have some imperfections.
- C. The laser re-absorbs some of the energy emitted.
- D. Each electronic energy level has many rotational sub-levels.

52. What is the energy in joules associated with a photon of light with a wavelength of 330 nanometers?

($E = \frac{hc}{\lambda}$, where $h = 6.6 \times 10^{-34}$, λ = wavelength in nm)

- A. 6.0×10^{-28} J
- B. 6.0×10^{-19} J
- C. 6.0×10^{-17} J
- D. 6.0×10^{15} J

53. Which of the following sets of quantum numbers are associated with the last electron in neutral vanadium (element # 23)?

- A. $n = 3$; $l = 2$; $m_l = 0$; $m_s = +\frac{1}{2}$
- B. $n = 3$; $l = 2$; $m_l = 0$; $m_s = -\frac{1}{2}$
- C. $n = 4$; $l = 0$; $m_l = 0$; $m_s = +\frac{1}{2}$
- D. $n = 4$; $l = 0$; $m_l = +1$; $m_s = -\frac{1}{2}$

54. Which of the following elements would have an electronic configuration of $ns^1(n-1)d^5$ due to half-filled stability?

- A. Copper
- B. Silicon
- C. Manganese
- D. Molybdenum

55. According to Figure 1, the photons emitted compare in what way to the photons absorbed by the transition between energy levels in the laser system?

- A. The light emitted is of higher wavelength.
- B. The light emitted is of equal wavelength.
- C. The light emitted is of lesser wavelength.
- D. The light emitted is incomparable.

56. For a laser to emit visible light, the material used must have an emission in what energy range, knowing that the visible range is approximately 400 nm to 750 nm?

($E = \frac{1.24 \times 10^{-6} \text{ eV}}{\lambda}$, where λ is measured in meters)

- A. 6.25 eV to 9.90 eV
- B. 3.10 eV to 6.25 eV
- C. 1.65 eV to 3.10 eV
- D. 1.24 eV to 1.65 eV

Long before the advent of modern technology, colored paints existed. The pigments for these paints were extracted from natural sources, such as plants and mineral deposits. In Europe and the Middle East, the majority of the pigments used for painting were metal oxides. In the Orient, more organic pigments were used. By tracing the origins of pigments, art historians have been able to determine the source not only of paintings, but of stylistic influences as well. Listed in Table 1 are some of the more common sources of colors used in the early centuries A.D.

Pigment	Chemical Formula	Color
Alizarin	1,2-dihydroxyanthraquinone	Yellow
Azurite	$2\text{CuCO}_3 \cdot \text{Cu}(\text{OH})_2$	Blue
Cinnabar	HgS	Red
Massicot	PbO	Yellow
Minium	Pb_3O_4	Scarlet
Orpiment	As_2S_3	Yellow
Realgar	As_2S_2	Orange red
Rust	$\text{Fe}_2\text{O}_3 \cdot n\text{H}_2\text{O}$	Brick red
Ultramarine blue	$\text{SiS}_2/\text{Al}_2\text{S}_3$	Blue
Verdigris	$\text{Cu}(\text{C}_2\text{H}_3\text{O}_2)_2 \cdot \text{Cu}(\text{OH})_2$	Green/blue
White lead	$2\text{PbCO}_3 \cdot \text{Pb}(\text{OH})_2$	White

Table 1

As a rule, inorganic paint pigments last for a far longer period of time than the organic paint pigments. For this reason, inorganic pigments are a more reliable source used to identify the time and region from which many paintings originated. The colors in inorganic paint pigments are caused by the absorption of light in the visible range of the light spectrum. The absorption is due to transitions between the d-orbitals, which because of the asymmetry of the ligands are not degenerate. The splitting of the d-orbitals results in different energy levels within the d-level. Figure 1 shows the energetic splitting of the d-orbitals in an octahedral complex.

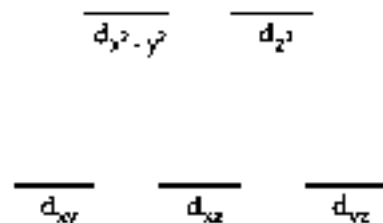


Figure 1

The color we see is the reflected color, the complementary color of the color that is absorbed. If a paint appears blue, it is because of a paint pigment is present in it that is absorbing orange light (the complementary color of blue). Figure 2 shows an artist's color wheel, which can be used to determine complementary colors. Complementary colors oppose one another on the color wheel. Blue opposes orange, so they are complementary colors of one another. Red and green are also a complementary pair.



Figure 2

White light is a combination of all colors, so when one color is absorbed from white light by a pigment, the other colors reflect. All the colors are no longer present, so at least one of the colors has no complementary color present. The result is that the reflected light takes on a hue of the color that has no complementary color to cancel it. When white light reflects off of a pigment, the reflected light can be analyzed for intensity versus wavelength after a prism splits the light via refraction. The output is an absorption spectrum, which appears as a rainbow band with vertical black lines, due to the absence of the light absorbed. Absorption spectra are the opposite of emission spectra, although both require a prism to refract (grate) the light.

57. Which of the following pigments absorbs light of the **SHORTEST** wavelength?
- A. Alizarin
B. Azurite
C. Cinnabar
D. Verdigris
58. When a sample is heated by flame, and the emitted light is put through a prism, what result is produced?
- A. An emission spectrum of dark background with some colored bands.
B. An absorption spectrum of dark background with some colored bands.
C. An emission spectrum of colored background with some black bands.
D. An absorption spectrum of colored background with some black bands.
59. In the absorption spectrum of realgar, what color band would contain black lines?
- A. Orange and red
B. Orange and green
C. Blue and red
D. Blue and green

60. Given that the wavelength of visible light has a range from about 740 nm to 390 nm, what is the approximate wavelength of light absorbed by the copper dication in $\text{Cu}(\text{C}_2\text{H}_3\text{O}_2)_2 \cdot \text{Cu}(\text{OH})_2$?

- A. 450 nm
- B. 550 nm
- C. 650 nm
- D. 750 nm

61. White lead absorbs which colors?

- A. All colors
- B. Green, blue, and violet
- C. Red, orange, and yellow
- D. No colors in the visible range

62. All of the following exhibit reflected color EXCEPT:

- A. Candle wax
- B. Fabric dyes
- C. Pen ink
- D. Gas-filled light tubes

63. Which of the following pigments will decompose first?

- A. Alizarin
- B. Azurite
- C. Rust
- D. Verdigris

Passage X (Questions 64 - 69)

Fluorescence occurs with the absorption of a photon by a molecule, exciting one of its electrons to a higher energy state. Because the material is molecular and not atomic, there are also vibrational and rotational energy levels associated with it. The molecule can change its vibrational energy level (by changing its bond-stretching frequency) and give off energy in the form of IR photons or heat. Heat is released during collisions with less energetic molecules. When the electron falls back to the ground state, a photon of less energy than was initially absorbed is given off. A fluorescing compound can thus absorb high-energy light (such as ultraviolet) and give off lesser-energy light (such as visible light).

Phosphorescence also begins with the absorption of a photon by a molecule, exciting one of its electrons to a higher energy state. The electron can flip its spin in this excited state, dissipating some of the extra energy. When the electron relaxes, because it shares the same spin as the ground state electron, it cannot fall back to its ground state. It must relax to an intermediate state, so a lower-energy photon is emitted than was initially absorbed. Figure 1 represents what goes on during both fluorescence and phosphorescence. The X_0 potential well represents the ground electronic state, and X_1 represents the excited state. The lines within the well represent the different stretching modes (vibrational energy levels) of the molecule.

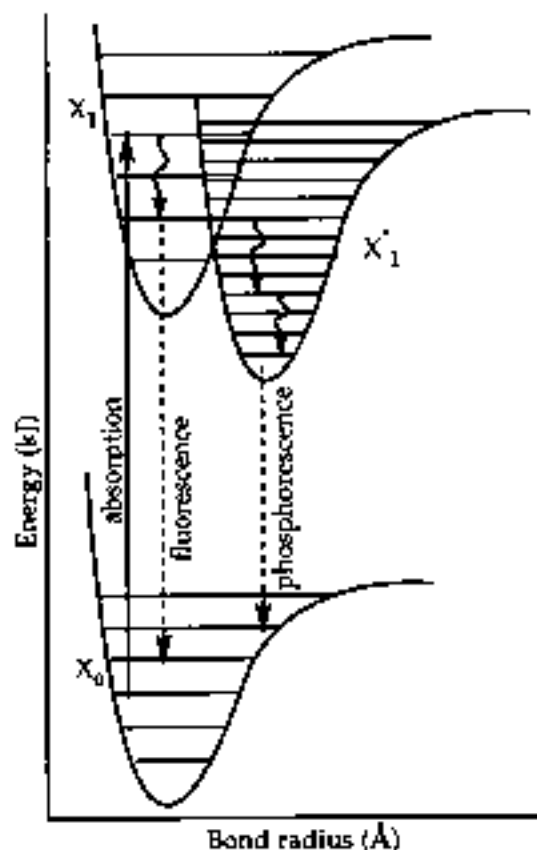


Figure 1

The wiggly arrows in Figure 1 represent energy that is dissipated as an excited electron drops in vibrational energy, which is released in the form of an IR photon. The vibrational energy thereby released can be measured as heat given off from the system. The amount of heat emitted varies with different molecules, because the electronic energy levels are not singular levels (they have both vibrational and rotational energy levels associated with them). The transition energy is thus the result of a random combination of energy levels. Because the electronic energy levels are not singular levels and because the transition is random, it is impossible for a molecule to absorb or emit light in such a manner that all of the photons simultaneously have the same frequency (or wavelength). For this reason, monochromatic light is not physically possible.

64. A spin flip in the excited state is associated with which of the following processes?
- Fluorescence only
 - Phosphorescence only
 - Both fluorescence and phosphorescence
 - Neither fluorescence and phosphorescence
65. What can be said about the photon absorbed relative to the photon emitted by a fluorescing diatomic molecule?
- The energy of the photon absorbed is greater than the energy of the photon emitted, while the wavelength of the photon absorbed is less than the wavelength of the photon emitted.
 - The energy of the photon absorbed is less than the energy of the photon emitted, while the wavelength of the photon absorbed is greater than the wavelength of the photon emitted.
 - Both the energy and wavelength of the photon absorbed are greater than the energy and wavelength of the photon emitted.
 - Both the energy and wavelength of the photon absorbed are less than the energy and wavelength of the photon emitted.
66. What is TRUE about monochromatic light?
- Monochromatic light is not possible, because light is not quantized.
 - Monochromatic light is possible, because light is quantized.
 - Monochromatic light is not possible, because there are vibrational energy levels of lower energy difference than the electronic energy levels with which they are associated.
 - Monochromatic light is possible, because there are vibrational energy levels of greater energy difference than the electronic energy levels with which they are associated.

67. According to the graph, how do the photons emitted from fluorescence and phosphorescence compare?
- The photons emitted from fluorescence have a longer wavelength than the photons emitted from phosphorescence.
 - The photons emitted from fluorescence have a shorter wavelength than the photons emitted from phosphorescence.
 - The photons emitted from fluorescence have the same wavelength as the photons emitted from phosphorescence.
 - The photons emitted from phosphorescence are of higher energy than the photons emitted from fluorescence.
68. Which of the following transitions emits the SHORTEST wavelength of light?
- $X_1 V = 1$ to $X_0 V = 0$, where V is the vibrational energy level.
 - $X_1 V = 0$ to $X_0 V = 0$, where V is the vibrational energy level.
 - $X_0 V = 1$ to $X_1 V = 0$, where V is the vibrational energy level.
 - $X_0 V = 0$ to $X_1 V = 0$, where V is the vibrational energy level.
69. Which of the following statements about fluorescence is FALSE?
- Fluorescence is possible with molecules.
 - Fluorescence converts visible light into ultraviolet light.
 - Fluorescence can dissipate energy in the form of heat.
 - There are some atoms for which fluorescence is not possible.

Passage XI (Questions 70 - 76)

A student conducts an experiment where a series of aqueous solutions of group I cations (alkali metals) or group II cations (alkaline earth metals) are each heated by a flame. The heat of the flame, if hot enough, can stimulate electrons in the cations to leave their ground state. When they relax to the ground state, light is emitted. Because the light is a narrow band of photons with predominantly one wavelength, it appears colored. The color is used to identify the cation in solution. Table 1 shows the student's result.

Barium	Light-green ring around flame
Lithium	Brilliant crimson flame
Potassium	Light purple ring around flame
Sodium	Orange-yellow flame
Strontium	Red flame

Table 1

The student then conducts the same experiment using transition metals rather than main-group metals. The trend in color is predictable with main-group cations of the same column (family) in the periodic table. The color of the transition metal cations depends more on the ligands attached to the metal cation. Table 2 also shows the colors obtained.

Vanadium (II)	Violet flame
Copper (I)	Orange flame
Nickel (II)	Red flame
Chromium (III)	Red flame
Cobalt (II)	Green flame

Table 2

With transition metals, the color emitted by the flame is often the color of the light absorbed when the cation in solution is exposed to white light. The color of the aqueous solution that is detected by the eye is the complementary color of the color absorbed. The color that is associated with transition metals is due to an electronic transition within the d-orbitals (between the d-levels, which split due to the presence of ligands). These electronic transitions are referred to as *d-d transitions*, and they occur only with transition metals that are not d^0 or d^{10} in their electronic configuration.

70. Which cation is paramagnetic?

- A. Na^+
- B. Sr^{2+}
- C. Cu^+
- D. Cr^{3+}

71. What is the electronic configuration of chromium?

- A. $[\text{Ar}]4s^23d^4$
- B. $[\text{Ar}]4s^13d^5$
- C. $[\text{Ar}]3d^6$
- D. $[\text{Ar}]3d^3$

72. Which of the following transition metals has the GREATEST d-d transition energy, given that the color observed in the flame test is due to a d-d electronic transition?

- A. Ni^{2+}
- B. Cr^{3+}
- C. Co^{2+}
- D. V^{2+}

73. Identify the TRUE statement(s) from the following.

- I. As ionic size increases, the energy of the transition increases.
 - II. The transition energy of K^+ is greater than the transition energy of Cr^{3+} .
 - III. Cu^+ and Ni^{2+} have the same electronic configuration.
- A. I only
 - B. II only
 - C. I and II only
 - D. II and III only

74. What are the quantum numbers associated of the last electron of nickel dication?

- A. $n = 3$ $l = 2$ $m_l = -1$ $m_s = +\frac{1}{2}$
- B. $n = 3$ $l = 2$ $m_l = 0$ $m_s = -\frac{1}{2}$
- C. $n = 3$ $l = 2$ $m_l = +1$ $m_s = -\frac{1}{2}$
- D. $n = 3$ $l = 2$ $m_l = +1$ $m_s = +\frac{1}{2}$

75. Given that lithium cation emits a red flame, sodium cation emits an orange-yellow flame, and potassium cation emits a violet flame, what color flame will be emitted by rubidium cation?

- A. Yellow
- B. Green
- C. Blue
- D. Colorless (and thus the normal orange)

76. An emission spectrum of lithium would appear in what manner?

- A. It would be a rainbow with a black line in the red region of the spectrum.
- B. It would be a rainbow with a black line in the green region of the spectrum.
- C. It would have a black background with a red line.
- D. It would have a black background with a green line.

Passage XII (Questions 77 - 81)

Glyphosate, shown in Figure 1, is a synthetic compound that is currently the world's largest selling commercial herbicide. Glyphosate kills vegetation by binding metals within a plant, thus starving the organism of essential metallic nutrients needed for the transport of other nutrients and waste, such as carbon dioxide and oxygen. When Glyphosate leeches metals from the plant, its respiration is shut down. Glyphosate is shown in Figure 1 in its salt form, as extracted from pH = 7 aqueous solution.

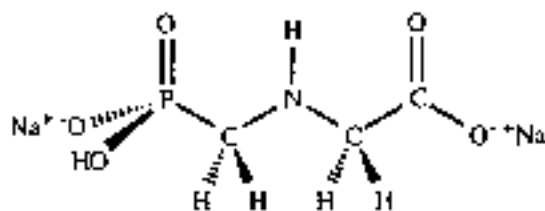


Figure 1

After the plant has died, Glyphosate decomposes in the presence of moisture into phosphate, carbon dioxide, and ammonia. The decomposition products are environmentally safe, and some are useful nutrients for the soil. In a study of the decomposition of Glyphosate *in vivo*, isotopically labeled Glyphosate was applied to vegetation, and the isotopic abundance was monitored at regular intervals over a twenty-day period. Based on kinetic data associated with hydrolysis and decomposition, the conclusion was reached that the decay time of Glyphosate to its inorganic products is approximately fourteen days. The compound was radio-labeled in three separate experiments, using a different radio-label in each experiment. The first experiment employed tritium (^3H), the second experiment employed ^{14}C , and the third experiment employed ^{32}P to monitor the half-life and decay rates. Carbon-14 was used as a marker, but not for kinetic purposes. The half-life of carbon-14 is far greater than the lifetime of a typical lab experiment. The isotopic marker helps to study the migration of Glyphosate through the environment.

77. Which of the following is an isotope of ^{32}P ?
- A. ^{32}S
 - B. $^{32}\text{P}^-$
 - C. ^{31}P
 - D. ^{31}Si
78. The molecular shape about the nitrogen in Glyphosate is BEST described as:
- A. trigonal planar.
 - B. trigonal pyramidal.
 - C. square planar.
 - D. bent

79. The half-life of ^{32}P is approximately 14 days. If the concentration of a ^{32}P radio-labeled Glyphosate sample ($\text{C}_3\text{H}_5\text{NO}_6\text{P}$) is found to be 188 parts per million initially, how long will it take until the concentration is 25 ppm?
- A. 27 days
 - B. 33 days
 - C. 41 days
 - D. 43 days

80. Which of the following molecules have ionic bonding?
- I. NH_3
 - II. CO_2
 - III. $\text{Na}_2\text{PO}_4\text{H}$
- A. I only
 - B. II only
 - C. III only
 - D. I and II only
81. As ^{32}P decays, it changes into a product that is still phosphorus. This is necessary in a marker so that the chemical behavior of the labeled species does not change. What is the MOST likely particle given off in the decay of phosphorus-32?
- A. A neutron
 - B. An alpha particle
 - C. A beta particle
 - D. A positron

Passage XIII (Questions 82 - 88)

In order to image vital organs by photon emission, technetium-99 is ingested into the body so that its low-level gamma radiation (140-keV) may be detected. For brain imaging, ^{99}Tc is introduced via the bloodstream, and it subsequently diffuses throughout the body, including to the brain. Technetium-99 has a half-life of approximately six hours, so scanning and analysis must closely follow ingestion. For ^{99}Tc to migrate to all of the target organs in a concentration high enough to be detectable requires roughly two hours. After two hours, the concentration of active technetium-99 is approximately eighty percent (80%) of its original value. Technetium-99 decays through gamma emission, meaning that its nucleons simply drop in one step from an excited nuclear state to a ground nuclear state when they emit photons. The nucleus of technetium does not change upon the emission of the gamma photon.

For brain imaging, the skull is encompassed by a detection device that collects and records the gamma radiation at arbitrary sites. Radiation escapes more readily from the areas of the brain that have a lower tissue density and fluid than those that have a greater density, but a complication in the analysis of brain images is that the radiation dissipates in liquid. This means that slight variations in the texture and composition of brain matter can produce different imaging patterns. The display pattern collected can be converted into a contour map of the brain. This technique work well with organs that have a relatively low water content.

82. Beta decay of ^{210}Po results in which of the following?

- A. ^{210}At
- B. ^{210}Bi
- C. ^{206}Pb
- D. ^{209}At

83. Alpha decay is observed in all of the following transformations EXCEPT:

- A. ^{254}Es to ^{250}Bk
- B. ^{238}U to ^{234}Th
- C. ^{223}Fr to ^{221}Bi
- D. ^{247}Cm to ^{243}Pu

84. If the initial dosage of ^{99}Tc gave a reading of 120 μCi , what will the reading be after 12 hours?

- A. 60 μCi
- B. 40 μCi
- C. 30 μCi
- D. 20 μCi

85. Electron capture by carbon produces the same element as which of the following processes?

- A. Positron capture by boron
- B. Positron decay of carbon
- C. Beta capture by nitrogen
- D. Alpha decay of oxygen

86. Gamma radiation is considered to:

- A. weigh 4 amu and have a nuclear charge of +2.
- B. be massless with a nuclear charge of -1.
- C. weigh 4 amu and have no nuclear charge.
- D. be massless and have no nuclear charge.

87. The electronic configuration for ^{99}Tc is which of the following?

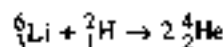
- A. $[\text{Ar}] 5s^2 3d^5$
- B. $[\text{Kr}] 5s^2 4d^5$
- C. $[\text{Kr}] 5s^1 4d^6$
- D. $[\text{Kr}] 5s^2 4d^7$

88. When ^{99}Tc undergoes gamma decay, the final product is which of the following?

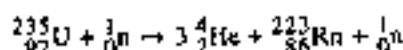
- A. ^{99}Tc
 - B. ^{99}Ru
 - C. ^{95}Nb
 - D. ^{99}Mo
-

Passage XIV (Questions 89 - 95)

As of March 1989 physicists and chemists had yet to carry out a nuclear fusion reaction that did not require more energy input than output. Fusion reactions require high temperature and strong magnetic fields to be carried out. This is the reason that nuclear power generators are fission reactors. Reaction 1 is an example of a fusion reaction, while Reaction 2 is an example of a fission reaction.



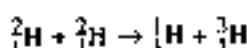
Reaction 1



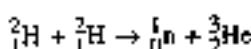
Reaction 2

Then in March of 1989, B. Stanley Pons and Martin Fleischmann announced that they had achieved a nuclear fusion reaction in a test tube at room temperature. Pons and Fleischmann set up a simple electrochemical cell with a palladium cathode in 0.1 M LiOD(D₂O) and a platinum anode. The design of the cell was to form D₂ gas by passing an electron flow from the platinum anode to the palladium cathode. The cell generated more heat than expected, so it was proposed that a nuclear reaction must have transpired.

The researchers believed that the deuterium migrated into the palladium metal of the cathode and gathered in the pockets of the lattice, so they hypothesized that the deuterium fused together in the palladium electrode by either Reaction 3 or Reaction 4.



Reaction 3



Reaction 4

Pons and Fleischmann in their original paper stated that both tritium and neutrons were observed in the cathode of their electrochemical cell, but the concentrations were too low to account for the extra heat generated. Since that time, the hypothesis of a "cold fusion" nuclear reaction taking place under these experimental conditions has been dismissed as improbable by most scientists in the field.

89. What is NOT a product when two deuterium atoms undergo a fusion reaction?

- A. Tritium
- B. Helium-4
- C. Helium-3
- D. Hydrogen

90. What evidence did Pons and Fleischmann point to that would indicate that a nuclear reaction had transpired in the electrochemical cell?

- A. Less heat was released than expected.
- B. More heat was released than expected.
- C. That electrons were emitted by the cell.
- D. That electrons were absorbed by the cell.

91. Cold fusion is BEST described as:

- A. an endothermic nuclear reaction capable of rapidly cooling the environment.
- B. an exothermic nuclear reaction capable of rapidly cooling the environment.
- C. an endothermic nuclear reaction capable of being carried out at room temperature.
- D. an exothermic nuclear reaction capable of being carried out at room temperature.

92. Nuclear power plants employ what reaction to generate power?

- A. Fission, because fusion requires such a great energy input that it is inefficient, and the net energy change is unfavorable.
- B. Fusion, because fusion requires such a great energy input that it is inefficient, and the net energy change is unfavorable.
- C. Fission, because fission requires such a great energy input that it is inefficient, and the net energy change is unfavorable.
- D. Fusion, because fission requires such a great energy input that it is inefficient, and the net energy change is unfavorable.

93. Which of the following is a fusion reaction?

- A. ${}^{235}_{92}\text{U} + {}^1_0\text{n} \rightarrow {}^{232}_{90}\text{Th} + {}^4_2\text{He}$
- B. ${}^{239}_{94}\text{Pu} \rightarrow {}^{235}_{92}\text{U} + {}^4_2\text{He}$
- C. ${}^{235}_{92}\text{U} + {}^1_0\text{n} \rightarrow {}^{246}_{97}\text{Cf}$
- D. ${}^{99}_{43}\text{Tc}^* \rightarrow {}^{99}_{43}\text{Tc} + \text{hf}$

94. What is formed when ${}^{238}_{92}\text{U}$ emits an alpha particle and a deuterium nucleus?

- A. ${}^{234}_{90}\text{Ac}$
- B. ${}^{232}_{90}\text{Ac}$
- C. ${}^{236}_{94}\text{U}$
- D. ${}^{232}_{94}\text{Ra}$

95. Capture of which of the following particles would NOT result in a change in the atomic number?

- A. Alpha
- B. Beta
- C. Neutron
- D. Tritium nucleus

Questions 96 - 100 are NOT based on a descriptive passage.

96. Which of the following supports the conclusion that electrons have quantized energy levels?

- A. The existence of a nucleus
- B. Scattering of X-rays by a thin sheet of a material
- C. The bending of nuclear radiation particles by a magnetic field
- D. Distinct lines in an electromagnetic radiation spectra

97. The LONGEST wavelength of light would be associated with:

- A. X-rays.
- B. violet light.
- C. green light.
- D. infrared light.

98. The ionization energy of $H(g)$ is $1312 \frac{kJ}{mole}$. A good approximation for the second ionization potential of helium (g) is:

- A. $5248 \frac{kJ}{mole}$
- B. $2624 \frac{kJ}{mole}$
- C. $656 \frac{kJ}{mole}$
- D. $328 \frac{kJ}{mole}$

99. Which of the following is the electronic configuration for an excited state of Na^+ ?

- A. $1s^2 2s^2 2p^5 3s^2$
- B. $1s^2 2s^2 2p^6 3s^1$
- C. $1s^2 2s^2 2p^6$
- D. $1s^2 2s^2 2p^5 3s^1$

100. What is the MOST common shape for a transition metal with five ligands attached?

- A. Square planar
- B. Trigonal bipyramidal
- C. Pentagonal planar
- D. Hexahedral



1. C	2. A	3. A	4. C	5. D
6. B	7. D	8. B	9. A	10. B
11. D	12. D	13. D	14. D	15. B
16. A	17. C	18. C	19. D	20. B
21. C	22. A	23. A	24. C	25. B
26. D	27. B	28. A	29. C	30. C
31. C	32. A	33. D	34. D	35. C
36. D	37. D	38. A	39. C	40. B
41. B	42. A	43. B	44. D	45. B
46. C	47. B	48. A	49. C	50. C
51. D	52. B	53. A	54. D	55. B
56. C	57. A	58. A	59. D	60. C
61. D	62. D	63. A	64. B	65. A
66. C	67. B	68. A	69. B	70. D
71. B	72. D	73. C	74. B	75. D
76. C	77. C	78. B	79. C	80. C
81. A	82. A	83. C	84. C	85. B
86. D	87. B	88. A	89. B	90. B
91. D	92. A	93. C	94. B	95. C
96. D	97. D	98. A	99. D	100. B

Atomic Theory Passage Answers

Passage I (Questions 1 - 8)

Classical Experiments

- Choice C is correct.** If oil drops were naturally charged, then the experiment could not work, because the total charge on the droplet would not be generated by the electron beam alone. The charge determined from the experiment would not necessarily reflect the charge of an electron. This eliminates choice A, because it is assumed in the experiment that the oil droplet is initially uncharged before exposure to the electron beam. Because the gravitational pull on the oil droplet is calculated as opposing the electric field, it must be assumed to be in one direction (downward). If the gravitational pull were in all directions, the experiment would not be possible. This eliminates choice B. The mass used in the calculation is the average mass of an oil droplet, so the mass of the electrons is in fact ignored. This can be assumed knowing that protons and neutrons are far more massive than electrons. The change in mass by gaining or losing electrons is negligible compared to the rest of the atomic mass. This makes choice C the best answer; an answer to be chosen by many, like you. The charge of an electron is assumed to be constant, if you are solving for an exact value. The experiment could be solved for an average value, but there is no reasoning behind one electron being of a different charge than another electron. This is to say that charge is quantized (has an exact value).
- Choice A is correct.** In the tube in Figure 1, the electron beam is negatively charged, so it bends downwards due to its attraction to the positively charged plate below and its repulsion from the negatively charged plate above. It may seem odd that the cathode plate (plate above) is negatively charged and the anode plate (plate below) is positively charged, but the plates form a charged capacitor, not a discharging battery. For a discharging battery and charging capacitor, the plate charges are opposite. Because a proton has the charge opposite from an electron (protons carry a positive charge), protons exhibit behavior opposite from an electron. Proton beams therefore bend upwards. A neutron is neutral, so its pathway would not arc at all between the two charged plates. The neutron does not arc, because it is unaffected by the charged plates. The best answer is choice A.
- Choice A is correct.** The value of the charge is twice as large as expected. This can be explained in terms of magnitude of charge rather than sign of charge. A value that is twice as large as expected is attributed to a doubly charged oil droplet. Choice A, doubly ionized, would explain this. Choices B, C, and D all address sign of charge and thus are eliminated.
- Choice C is correct.** The Thomson experiment, as stated in the passage, determined the mass-to-charge ratio of the electron. It can be used to determine the mass-to-charge ratio for any charged particle, but in this particular experiment, the particle was an electron. The best answer is choice C. A mass spectrometer is a mechanical variation of the Thomson experiment theme.
- Choice D is correct.** The charge of an electron is determined by equating the mass of the droplet and the gravitational force constant (mg) with the voltage of the electric field and charge of the oil droplet (qV). To solve for q , the other three variables must be known. This means that choices A, B, and C are all values that must be known. There is no temperature factor in the equation (mass, charge, gravity, and voltage do not vary with temperature), so choice D represents the factor that is least important in the calculations.
- Choice B is correct.** An atom is neutral when it carries no net charge. Electrons carry a negative charge, while protons carry a positive charge. Neutrons are uncharged. In order for an atom to be neutral, the number of electrons must be equal to the number of protons. The best answer is choice B. Choice A is not false per se, but because there exist isotopes whose number of electrons does not equal the number of neutrons, no conclusion can be drawn about the number of electrons and neutrons within a neutral atom.
- Choice D is correct.** The strongest electrostatic attractive force is felt between particles of opposite charge. This is known as Coulomb's law. An electron and proton carry opposite charges, so the best answer is choice D. A proton and neutron exhibit no Coulombic attraction, but because they are held tightly in the core of the nucleus, there must be some attractive force between them. Without the electrostatic caveat, this question could not have a proton and a proton listed as an answer choice because of the strong force associated with the nucleus that holds protons together. This is a more advanced topic that you may see in courses in nuclear chemistry (atomic physics) but as it is presented in the MCAT, the ambiguity presented by nuclear attractions (the uncertainty of the source of the strong force) is not a probable topic for questions.

8. **Choice B is correct.** If an electron is descending, a net upward force must be applied to accelerate the particle in the opposite direction. To increase the net upward force, the force upward must be increased or the force downward reduced. This would slow the particle and eventually stop it from dropping. This means that either the mass must be decreased, the gravity must be decreased, the charge must be increased, or the voltage of the field must be increased. The easiest thing to do is increase the voltage of the field. In the actual experiment, this is what is done. Increasing the number of electrons in the drop increases the charge, which should stop (or at least slow) the descent or change the direction of its path to an ascent. Decreasing the gravitational pull, which requires placing the apparatus in an anti-gravity environment, would also alleviate the descending difficulty. The only choice that would definitely not stop the descent, but would in fact increase the descent, is to increase the mass of the oil droplet. The best answer is choice B.

Passage II (Questions 9 - 14)

Isotopic Abundance and Average Atomic Mass

9. **Choice A is correct.** All atomic masses in the periodic table are referenced against carbon-12. Carbon-12 is the standard traditionally used. The carbon-12 isotope is defined as having a mass of exactly 12.0000 amu. Carbon is added to a sample to standardize the mass of the peaks given in the spectrum. Not all elements have exact whole number masses, so a reference is necessary. You are wise to pick A.
10. **Choice B is correct.** According to Table 1, magnesium has three major isotopes. The isotope with the lowest atomic mass is the most abundant (78.70%). This means that the lowest peak by mass (the first peak from left to right) must be the largest peak. This is true only in the spectrum of choice B.
11. **Choice D is correct.** The average atomic mass for any element is the weighted average of all of the isotopes of the given element. Choices A and B are eliminated, because Table 1 lists only two isotopes for silver, and the calculation shows three values being summed. Choice A is the calculation of average atomic mass for magnesium. The correct answer is choice D, because that has the correct percentages multiplied by the correct isotopic masses. You would be a very wisdom-laden soul, if you were to select D as your answer.
12. **Choice D is correct.** In order to send the isotopes of the element through the mass spectrometer, the element must be in the gas phase. This means that any sample not in the gas phase initially must be converted to the gas phase to be analyzed using mass spectroscopy. The process of converting a solid into a gas is referred to as sublimation, thus making choice D correct.
13. **Choice D is correct.** Because the average atomic mass is an average of the masses of all of the isotopes, the average atomic mass must fall within the range of the isotopic masses, meaning that the average atomic mass is greater than the lightest isotope but less than the most massive isotope. When there are two isotopes only, then the average atomic mass is the weighted average of the two. In a fifty-fifty mixture of two isotopes, the average atomic mass lies perfectly between the two masses of the isotopes. If the mixture favors one of the isotopes, then the average atomic mass would be closer to the more abundant isotope than the less abundant isotope. The true statements are II and III, making the best answer choice D.
14. **Choice D is correct.** The difference between isotopes of the same element is found in the number of neutrons. If two atoms have a different number of protons, then they are not the same element. Choices A and B are eliminated, because they allude to a different number of protons. The greater mass associated with the boron-11 isotope is attributed to an extra neutron being present in the Boron-11 isotope as compared to the boron-10 isotope. The best answer is choice D.

Passage III (Questions 15 - 20)

Bohr Model of Hydrogen

15. **Choice B is correct.** The ionization of hydrogen involves an electronic transition from the $n = 1$ energy level to the $n = \infty$ energy level. Using Equation 1 to determine the transition energy yields the following:

$$2.178 \times 10^{-18} \left(Z^2 \left[\frac{1}{n_i^2} - \frac{1}{n_f^2} \right] \right)$$

where $Z = 1$, $n_i = 1$, and $n_f = \infty$. Having ∞ in the denominator makes the number zero, which results in a value of $2.178 \times 10^{-18} (Z^2)$. With a value of $Z = 1$, the ionization energy is $2.178 \times 10^{-18} \text{ J}$. The ionization energy of hydrogen from its valence level is $2.178 \times 10^{-18} \text{ J}$, which makes the best answer choice B.

16. **Choice A is correct.** This solution requires relating the formula for the energy of a photon to the energy equation given in the question. The relationship is as follows:

$$E = mc^2 \text{ and } E = \frac{hc}{\lambda} \therefore mc^2 = \frac{hc}{\lambda} \Rightarrow mc = \frac{h}{\lambda} \Rightarrow \lambda = \frac{h}{mc}$$

which is choice A.

17. **Choice C is correct.** Choice B is eliminated, because from the $n = 2$ energy level to the $n = 1$ energy level, there is a drop in energy, which corresponds to emission of light, not absorption. The best answer is choice C, because the energy levels get closer together as the n value increases, so the transition from the $n = 1$ energy level to the $n = 2$ energy level is always of greater energy than any transition starting at an energy state where $n > 1$. However, to be certain, the following mathematical relationship can be derived:

$$\text{Given that the energy level of an electron is } E = -2.178 \times 10^{-18} \left(\frac{Z^2}{n^2} \right) \text{ J,}$$

$$\text{the transition energy is } \Delta E = -2.178 \times 10^{-18} (Z^2) \left(\frac{1}{n_f^2} - \frac{1}{n_i^2} \right) \text{ J} = 2.178 \times 10^{-18} (Z^2) \left(\frac{1}{n_i^2} - \frac{1}{n_f^2} \right) \text{ J.}$$

The relative energies are therefore comparable by the value of the $\left(\frac{1}{n_f^2} - \frac{1}{n_i^2} \right)$ term.

$$\left(\frac{1}{1^2} - \frac{1}{2^2} \right) > \left(\frac{1}{2^2} - \frac{1}{5^2} \right) > \left(\frac{1}{2^2} - \frac{1}{4^2} \right) = \left(\frac{1}{1} - \frac{1}{4} \right) > \left(\frac{1}{4} - \frac{1}{25} \right) > \left(\frac{1}{4} - \frac{1}{16} \right)$$

$$\left(1 - \frac{1}{4} \right) > \left(\frac{25}{100} - \frac{4}{100} \right) > \left(\frac{4}{16} - \frac{1}{16} \right) \Rightarrow \frac{3}{4} > \frac{21}{100} > \frac{3}{16}, \text{ therefore choice C is the best answer.}$$

18. **Choice C is correct.** Statement I is valid, because as an electron's principle quantum number (n) increases, the electron is in an energy level (orbit) that is farther from the nucleus. Statement II can be viewed from the pictorial representation of the energy levels in Figure 1. As the value of n is increasing, the energy levels become closer to one another. This makes statement II a valid statement. According to the answer choices, this makes choice C the best answer. A larger nuclear charge (Z) affects the energy at each level, and thus affects the transition energies as well. This makes statement III incorrect, and confirms that choice C is the best answer. Choose C and feel a little giddy.
19. **Choice D is correct.** The electronic transition from the $n = 5$ energy level to the $n = 4$ energy level is of lower energy than the electronic transition from the $n = 4$ energy level to the $n = 3$ energy level. Therefore, the photon emitted is of lower energy than a photon corresponding to orange light. Because green light is of higher energy than orange light, choice A is eliminated. The electronic transition from the $n = 4$ energy level to the $n = 2$ energy level is of higher energy than the electronic transition from the $n = 4$ energy level to the $n = 3$ energy level. Therefore, the photon emitted is of higher energy than a photon corresponding to orange light. Because red light is of lower energy than orange light, choice B is eliminated. The electronic transition from the $n = 6$ energy level to the $n = 4$ energy level is of lower energy than the electronic transition from the $n = 4$ energy level to the $n = 3$ energy level. Therefore, the photon emitted is of lower energy than a photon corresponding to orange light. Because violet light is of higher energy than orange light, choice C is eliminated. The electronic transition from the $n = 5$ energy level to the $n = 3$ energy level is of higher energy than the electronic transition from the $n = 4$ energy level to the $n = 3$ energy level. Therefore, the photon emitted is of higher energy than a photon corresponding to orange light. Because yellow light is of higher energy than orange light, choice D is the best answer.
20. **Choice B is correct.** Choice A is an invalid statement, because according to Beer's law, the absorbance of light is proportional to sample concentration. Because samples can have varying concentrations, they can in fact absorb different amounts of light. The color of light (or frequency) corresponds to transition energy. Because the sample, independent of concentration, absorbs the same frequency of light each time, the transition energy must be a fixed value. This leads to the conclusion that energy levels are fixed, and thus quantized. Choice B is the best answer. The sample emits light of the same frequency, regardless of solution's temperature and concentration. Temperature can affect the intensity of the light that is emitted, but not the frequency. The frequency of the light remains constant. This eliminates choice C (temperature effects) and choice D (concentration effects). Choice B is your answer choice.

21. **Choice C is correct.** A liquid crystal display uses a paramagnetic compound in its polarizing cells to rotate (or not rotate) light. A paramagnetic species has at least one unpaired electron. Because chlorides are -1 each, cadmium in CdCl_2 carries a +2 charge. This is the result of losing two 5s electrons from elemental cadmium, which results in an electronic configuration of $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^{10} 4p^6 4d^{10}$ for Cd^{2+} . Each level is full, so there are no unpaired electrons. Choice A is eliminated. In elemental fluorine, the two fluorine atoms share their unpaired electrons in the form of a bond. This means that the fluorine molecule has no unpaired electrons. Choice B is eliminated. Because chlorides are -1 each, cobalt in $\text{CoCl}_2 \cdot 6\text{H}_2\text{O}$ carries a +2 charge. This is the result of losing two 4s-electrons from elemental cobalt, which results in an electronic configuration for Co^{2+} of $1s^2 2s^2 2p^6 3s^2 3p^6 3d^7$. There is an *odd number* of electrons, so the species *must* be paramagnetic. The correct choice is answer C. Just as was the case with F_2 , molecular hydrogen (H_2) has no unpaired electrons.

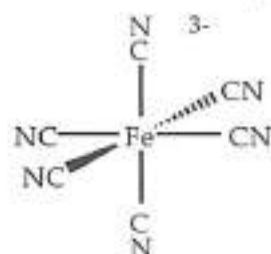
22. **Choice A is correct.** Atomic nitrogen (N) has the electronic configuration $1s^2 2s^2 2p^3$. The last electron is the third electron to enter the 2p orbital. The principle quantum number is given as 2, and because it is a p-orbital, the angular momentum quantum number (l) is equal to 1. All of the answer choices contain these two values, so nothing is eliminated. To obtain the m_l and m_s values, the electrons must be filled into their respective p-orbitals. This is drawn below:



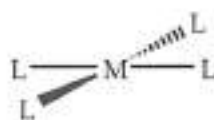
Last electron is in the third p-orbital, so $m_l = +1$;
Last electron is spin up, so $m_s = +\frac{1}{2}$

The correct answer is therefore choice A.

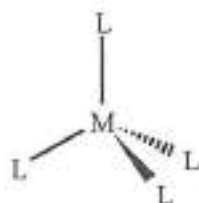
23. **Choice A is correct.** Having six ligands attached results in an octahedral shape. The correct answer is choice A. Drawn below are generic structures for the other answer choices.



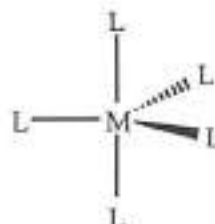
Octahedral



Generic square planar

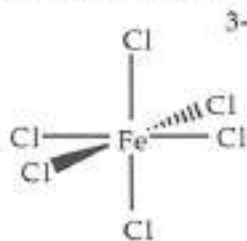
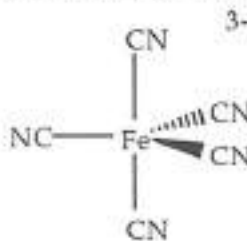


Generic tetrahedral



Generic trigonal bipyramidal

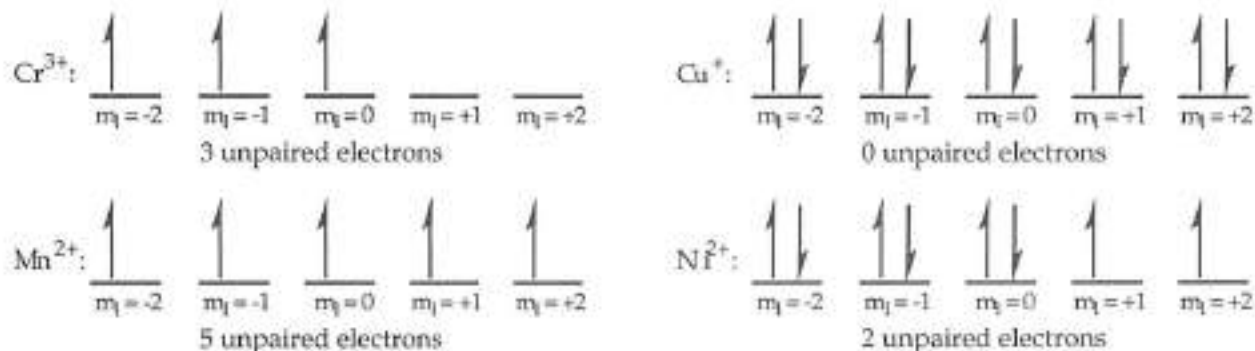
24. **Choice C is correct.** Polarity results from the asymmetric distribution of electron density. Cis compounds are always polar. The correct answer is choice C. The structures for the four choices are drawn below.


 FeCl_6^{3-}

 Fe(CN)_5^{3-}

 cis- $\text{Pt(NH}_3)_2\text{Cl}_2$

 trans- $\text{Pt(NH}_3)_2\text{Cl}_2$

25. **Choice B is correct.** To determine the number of unpaired electrons (as well as which one has the most), the d-electron count for each transition metal cation first must be worked out. To determine the d-electron count for the four cations, you must consider the electronic configuration. Neutral chromium is $[\text{Ar}]4s^13d^5$, so when it is Cr^{3+} , it has lost the 4s-electron and two 3d-electrons, leaving it with an electronic configuration of $[\text{Ar}]3d^3$. We refer to Cr^{3+} as a d^3 -cation. Neutral manganese is $[\text{Ar}]4s^23d^5$, so when it is Mn^{2+} , it has lost both 4s-electrons, leaving it with an electronic configuration of $[\text{Ar}]3d^5$. We refer to Mn^{2+} as a d^5 -cation. Neutral copper is $[\text{Ar}]4s^13d^{10}$, so when it is Cu^+ , it has lost its 4s-electron, leaving it with an electronic configuration of $[\text{Ar}]3d^{10}$. We refer to Cu^+ as a d^{10} -cation. Finally, neutral nickel is $[\text{Ar}]4s^23d^8$, so when it is Ni^{2+} , it has lost both 4s-electrons, leaving it with an electronic configuration of $[\text{Ar}]3d^8$. We refer to Ni^{2+} as a d^8 -cation. From here, it is a matter of placing the d electrons into their respective orbitals. The electron filling is as follows:



The most unpaired electrons is found with Mn^{2+} , so choice B is the best answer.

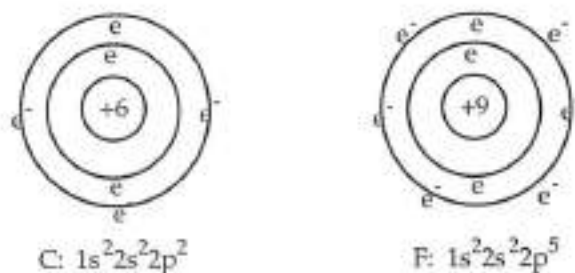
26. **Choice D is correct.** The cyano ligand carries a -1 charge, so chromium must have a +6 charge in order for $\text{Cr}(\text{CN})_6$ to be neutral. The amino ligand is neutral, so chromium must have a +3 charge in order for $\text{Cr}(\text{NH}_3)_6$ to have an overall positive three charge. The charge of chromium is in fact greater in $\text{Cr}(\text{CN})_6$ than $\text{Cr}(\text{NH}_3)_6$, so choice A is valid. The water ligand is neutral, so copper must have a +1 charge in order for $\text{Cu}(\text{H}_2\text{O})_6$ to have an overall +1 charge. When copper is neutral its electronic configuration is $[\text{Ar}]4s^13d^{10}$, so when copper carries a +1 charge, it has electronic configuration $[\text{Ar}]3d^{10}$. This gives copper ten d-electrons which makes choice B valid. In both $\text{Fe}(\text{NH}_3)_6^{3+}$ and FeCl_6^{3-} , iron has six ligands attached, so the shape of both molecules is the same. Choice C is valid. Scandium has the electronic configuration $[\text{Ar}]4s^23d^1$, so it can lose only three electrons. It is not possible for scandium to have a +4 charge (at most it is +3). This means that choice D is *not* true. Pick choice D, and feel a little peppier because of it.
27. **Choice B is correct.** To determine the number of electrons on the central metal, the formal charge of the metal first must be determined. From the charge, the electronic configuration is found, so the d-electron count is found. Fe carries a +3 charge in both $\text{Fe}(\text{NH}_3)_6^{3+}$ and FeCl_6^{3-} , so choice A is eliminated, because they both must have the same d-count. Co carries a +3 charge in $\text{Co}(\text{H}_2\text{O})_6^{3+}$, and Mn carries a +2 charge in MnCl_6^{4-} . Neutral Co is $[\text{Ar}]4s^23d^7$, so Co^{3+} has a d-electron count of 6 ($3d^6$). Neutral Mn is $[\text{Ar}]4s^23d^5$, so Mn^{2+} has a d-electron count of 5 ($3d^5$). The d-electron counts are not equal, so choice B must be the correct answer. Cr carries a +3 charge in $\text{Cr}(\text{NH}_3)_6^{3+}$, and Mo carries a +3 charge in MoCl_6^{3-} . Neutral Cr is $[\text{Ar}]4s^13d^5$, so Cr^{3+} has a d-electron count of 3 ($3d^3$). Neutral Mo is $[\text{Kr}]5s^14d^5$, so Mo^{3+} has a d-electron count of 3 ($4d^3$). The d-electron counts are equal so choice C is eliminated. Finally, Os carries a +2 charge in $\text{Os}(\text{NH}_3)_4^{2+}$ and Rh carries a +3 charge in $\text{RhCl}_3(\text{PR}_3)_3$. Neutral Os is $[\text{Xe}]6s^24f^{14}5d^6$, so Os^{2+} has a d-electron count of 6 ($5d^6$). Neutral Rh is $[\text{Kr}]5s^24d^7$, so Rh^{3+} has a d-electron count of 6 ($4d^6$). The d-electron counts are equal so choice D is eliminated.

Passage V (Questions 28 - 34)

Migration through a Membrane

28. **Choice A is correct.** The relative sizes of ions from smallest to largest radius is: $\text{Li}^+ < \text{Mg}^{2+} < \text{Na}^+ < \text{Ca}^{2+} < \text{K}^+$. The rate of effusion is: $\text{Li}^+ > \text{Mg}^{2+} > \text{Na}^+ > \text{Ca}^{2+} > \text{K}^+$. Li^+ (the smallest) is the fastest while, K^+ (the largest) is the slowest. The best answer is A. Choice D is eliminated, because cations are smaller than neutral atoms.
29. **Choice C is correct.** Every trend in the periodic table comes back to the nuclear pull on the electrons. The ionization potential is the energy required to remove an electron from the outermost shell, which in the case of sodium and potassium is the 3s and the 4s, respectively. It requires less energy to remove an electron from the larger 4s-orbital, because the electron is farther away from the nucleus than the 3s-orbital. This makes choice C the best choice. Choice D is eliminated, because potassium has a lower electron affinity than sodium.

30. **Choice C is correct.** The size of an ion or element is a result of the nuclear pull on the electrons orbiting the nucleus. This is best explained by choice C. Because C (carbon) has six protons, F (fluorine) has nine protons, and both have the same principle quantum number for their valence shell, the nuclear pull of fluorine is greater than that of carbon. Because the electrons are pulled closer to the nucleus in fluorine, and because size of an atom is determined by the electron cloud, fluorine is smaller than carbon.



31. **Choice C is correct.** F⁻, Ne, and Na⁺ all have 10 electrons total, so Cl⁻ (with 18 electrons) is eliminated. Of the three choices left, the largest nuclear charge is found on Na⁺, making it the smallest (the one with the electrons held most tightly). Choose C to choose correctness. The following chart of the protons and electrons for the three choices shows that the greater the proton-to-electron ratio, the smaller the species, assuming that the outermost electrons are in the same valence shell.

Element	Protons	Electrons	Observation	Radius
Na ⁺	11	10	protons exceed electrons, therefore it contracts	65 pm
Ne	10	10	protons equal electrons	70 pm
F ⁻	9	10	electrons exceed protons, therefore it expands	136 pm

32. **Choice A is correct.** The electronic configuration for magnesium is $1s^2 2s^2 2p^6 3s^2$. Magnesium must lose 2 electrons (the two 3s electrons) to have a filled outer valence shell (the $n = 2$ shell). This would make magnesium a +2 cation. Answer choice A is a fine selection if your goal is to be correct.
33. **Choice D is correct.** According to periodic trends, the size of an atom increases as you descend a column in the periodic table. I (iodine) is the lowest in the periodic table of the halogen choices, so I is the largest of the halogen choices. Trust periodic trends and choose D.
34. **Choice D is correct.** If the pore were to distinguish by charge, then the greater the charge of the cation, the slower the rate of migration for the cation. For example, if the pore were capable of forming attractive interactions with the cations such as polar attraction or hydrogen bonding, then the pore would distinguish by charge. The cations with the greatest charges are Mg^{2+} and Ca^{2+} . Because Ca^{2+} is larger than Mg^{2+} , Ca^{2+} would migrate more slowly than Mg^{2+} through a pore which distinguishes by both size and charge. This means that overall, Ca^{2+} would have the slowest migration rate. This makes choice D the best choice.

Passage VI (Questions 35 - 42)

Ionization Energy

35. **Choice C is correct.** The electronic configuration for helium is $1s^2$. The two electrons of helium are both valence electrons, so choice A is eliminated. Helium cannot have an effective nuclear charge greater than 2, because it contains only two protons. Choice B is thus eliminated. The ionization energy does not include any vaporization energy. Helium is a gas at room temperature, so there is no need to add energy to vaporize helium. This eliminates choice D. The electron must be removed from the first quantum level. The first quantum level experiences the greatest nuclear attraction, so the 1s-electrons are hardest to remove. It is in your best interest to choose C.
36. **Choice D is correct.** The second ionization energy of the alkali metals (lithium, sodium, and potassium) is substantially larger than the first ionization energy, because the second electron is being removed from a filled octet. After the first electron has been lost and the alkali metals are cations, their electronic configuration is $ns^2 np^6$, and they each have a filled valence shell. To remove the second electron would be like removing an electron from a noble gas, only harder, because the nuclear charge is greater for the alkali cation than the neutral noble gas. Given the answer selections, choice D is the best choice.

37. **Choice D is correct.** As stated in the passage, there is half-filled stability for the p-level when each of the p-orbitals contains one electron. Nitrogen as a neutral element has half-filled stability, so when it is ionized, it loses its half-filled stability. Oxygen, on the other hand, has one electron beyond the half-filled state; therefore, when oxygen is ionized, it attains half-filled stability. Losing half-filled stability raises the ionization energy of nitrogen (ionization is less favorable), while gaining half-filled stability lowers the ionization energy of oxygen (ionization is more favorable). Nitrogen is less electronegative than oxygen, so choice A is eliminated. Nitrogen has seven protons, while oxygen has eight, so choice B is eliminated. Nitrogen is in fact larger than oxygen, but a larger radius implies that the electrons are farther from the nucleus on average. Being farther from the nucleus, the electrons are not as tightly held, so the larger radius of nitrogen would imply a lower ionization energy for nitrogen than oxygen. The best answer is choice D.
38. **Choice A is correct.** The ionization energies for aluminum, silicon, and phosphorus follow an increasing linear trend. All three elements are in the same row (period) of the periodic table, so the number of core electrons for all three is the same. This eliminates choices C and D. As you move from left to right across a period of the periodic table, the atomic number increases, so the number of protons increases, and ultimately the effective nuclear charge increases. The increasing nuclear charge best explains the trend in first ionization energy between aluminum, silicon, and phosphorus. Be a hero or heroine by choosing A. Note that there is no half-filled or unfilled p-orbital stability affecting the observed trend.
39. **Choice C is correct.** Fluorine and chlorine are in the same column (family) of the periodic table. Fluorine has its valence electrons in the $n = 2$ quantum level, while chlorine has its valence electrons in the $n = 3$ quantum level. The larger the quantum number, the easier it is to remove the electrons and thus the lower the ionization energy. Fluorine is more electronegative than chlorine, so choice A is wrong and eliminated. The effective nuclear charge is found from both the number of protons and the number of core electrons. It is true that chlorine has a greater nuclear charge than fluorine, but chlorine has more core electrons than fluorine. This implies that the effective nuclear charge is approximately equal for the two halogens, eliminating choice B. The larger radius of chlorine implies that the valence electrons are farther from the nucleus on average than the valence electrons of fluorine. This makes the ionization energy of fluorine greater than that of chlorine and consequently makes choice C the best choice. Choice D is nonsensical, so ignore it.
40. **Choice B is correct.** The term "oxidation" refers to the loss of an electron. The lower the ionization energy, the easier it is for an element to lose an electron. The chart lists the first ionization energies, therefore it can be inferred that the element with the lowest first ionization energy is MOST easily oxidized. Of the choices, magnesium has the lowest first ionization energy. Choice B is the choice of winners... be a winner.
41. **Choice B is correct.** The relative ionization energies of krypton, bromine, and selenium are predictable, because they are in the same row (period) of the periodic table. By comparing the three to chlorine, argon, and sulfur, a trend can be determined. The ionization energies of argon, chlorine, and sulfur follow: $IE_{Ar} > IE_{Cl} > IE_S$, so the first ionization energies of bromine, krypton, and selenium should be $IE_{Kr} > IE_{Br} > IE_{Se}$, choice B.
42. **Choice A is correct.** When sodium loses its first electron, it gains a filled octet and thus stability in its valence shell. If it were to lose the second electron, the octet would be lost and thus it would become an unstable cation. This explains the drastic difference between the first and second ionization energies for sodium. For magnesium to have a full octet, it must lose two electrons resulting in relatively low first and second ionization energies. This answer is best explained in answer choice A. The following illustrates the point:
- 1st ionization: $Na(g) \longrightarrow Na^+(g) + e^-$ Na: $1s^2 2s^2 2p^6 3s^1$ and Na^+ : $1s^2 2s^2 2p^6$ Na^+ has a filled octet.
- 2nd ionization: $Na^+(g) \longrightarrow Na^{2+}(g) + e^-$ Na^+ : $1s^2 2s^2 2p^6$ and Na^{2+} : $1s^2 2s^2 2p^5$. Na^{2+} has lost filled octet.
- 1st ionization: $Mg(g) \longrightarrow Mg^+(g) + e^-$ Mg: $1s^2 2s^2 2p^6 3s^2$ and Mg^+ : $1s^2 2s^2 2p^6 3s^1$. Mg^+ has no filled octet.
- 2nd ionization: $Mg^+(g) \longrightarrow Mg^{2+}(g) + e^-$ Mg^+ : $1s^2 2s^2 2p^6 3s^1$ and Mg^{2+} : $1s^2 2s^2 2p^6$. Mg^{2+} has a filled octet.

Passage VII (Questions 43–49)

Transition Metal Trends

43. **Choice B is correct.** Table 1 lists room temperature density values of the transition metals. As a metal is heated, it expands. The density of manganese decreases with increasing temperature, because volume increases as mass remains constant. The density is slightly less than 7.43 grams per cm^3 , so the answer is choice B.

44. **Choice D is correct.** The first, second, and third rows of the transition metals follow similar trends as you move left to right across any particular row. The metals from the answer choices are all in the second-row of the transition metals. No information is given for the second-row transition metals, so their relationship must be extrapolated from the information given for the first-row transition metals. The first-row transition metals follow the trend $\text{Cu} > \text{Ni} > \text{Cr} > \text{Mn}$ for the second ionization energy. According to periodic behavior trends, Mo and Cr should exhibit similar properties, Tc and Mn should exhibit similar properties, Pd and Ni should exhibit similar properties, and Cu and Ag should exhibit similar properties. This means that the correct relationship between the 4d-transition metals is found by substituting the second-row transition metals into the relationship for the first-row transition metals. The relationship is $\text{Ag} > \text{Pd} > \text{Mo} > \text{Tc}$, making choice D the best answer.
45. **Choice B is correct.** The boiling point and melting point of an element increase as the forces holding the atoms together increase. The greater the forces, the greater the energy required to break the forces. Because zinc has a lower boiling and melting point than other transition metals, it can be concluded that there are weaker forces holding zinc atoms together than the other first-row transition metals. Because zinc has a filled d-shell ($[\text{Ar}]4s^23d^{10}$), there should be no covalent interactions between the zinc atoms. This makes choice B the correct choice.
46. **Choice C is correct.** All of the transition metals listed in Table 1 lose their 4s-electrons prior to losing their 3d-electrons, implying that it is easier to lose the 4s-electrons. Because the 4s-level fills prior to the 3d-level, it can also be concluded that 4s-electrons are at a lower energy level. In theory, the 3d-electrons should require less energy to remove (since the 3d-level is of higher energy than the 4s-level). However, the 4s-electrons are more exposed (further from the nucleus), so they are lost more easily than 3d-electrons. This makes choice C the best answer.
47. **Choice B is correct.** Within a row of the periodic table, atomic radius decreases as you move left to right in main-group elements. Table 2 shows that it is nearly true for the first row of transition metals, with zinc showing the only notable deviation. As you move left to right across a row of the periodic table, the mass of the elements increases. The density of an element is measured in terms of mass per volume. As the atomic radius increases, the volume increases, so moving left to right across a row in the periodic table results in greater mass and reduced volume. This means that the density of the element increases from left to right in the periodic table. Data in Table 2 confirms this inverse relationship, except in the case of manganese, where the radius and density both increase. This eliminates choices A and C. Manganese is an exception, because it has a larger radius due to half-filled stability, but it follows the same trend with density because its mass is greater than chromium (the previous transition metal). Zinc follows the trend of greater radius associated with lower density, so it is not an exception to that rule. Choice D is eliminated, because a general trend is observable, despite the deviation due to manganese. Choice B is the best choice.
48. **Choice A is correct.** When a transition metal element becomes a cation, it loses electrons from the 4s-level. As electrons are lost from the outer shell (valence shell), the radius must become smaller. As a note, the effective nuclear charge is increasing, because the cation experiences less valence electron repulsion with the absence of valence electrons. The best answer is choice A.
49. **Choice C is correct.** A negative reduction potential implies that it is less favorable to reduce the dication than it is to reduce hydrogen ion (a proton). The unfavorable nature of the reduction correlates to a low ionization energy. The easier it is to ionize (lose two electrons to become the dication), the easier it is to oxidize the element. The easier it is to oxidize the element, the harder it is to reduce the dication that forms. The sum of the first and second ionization energies of Mn is 2226 kJ/mole, and the reduction potential is -1.18 volts. The sum of the first and second ionization energies of Co is 2405 kJ/mole, and the reduction potential is -0.28 volts. The sum of the first and second ionization energies of Fe is 2320 kJ/mole, which is closer to the value of cobalt (differs by 85 kJ/mole) than the value of Mn (differs by 94 kJ/mole). The reduction potential should be between -1.18 volts and -0.28 volts, but closer to -0.28 volts. The best answer is choice C. (The actual value for any of you electromotive force trivia buffs is -0.44 volts.) The data are given below:

$\text{Mn} \longrightarrow \text{Mn}^{2+} + 2e^-$	$\Sigma \text{ 1}^{\text{st}} \text{ and } 2^{\text{nd}} \text{ I.E.} = 2226 \text{ kJ/mole}$	$E^{\circ} \text{ oxidation} = 1.18 \text{ V} \therefore E^{\circ} \text{ reduction} = -1.18 \text{ V}$
$\text{Fe} \longrightarrow \text{Fe}^{2+} + 2e^-$	$\Sigma \text{ 1}^{\text{st}} \text{ and } 2^{\text{nd}} \text{ I.E.} = 2320 \text{ kJ/mole}$	$E^{\circ} \text{ oxidation} = ??? \text{ V} \therefore E^{\circ} \text{ reduction} = ??? \text{ V}$
$\text{Co} \longrightarrow \text{Co}^{2+} + 2e^-$	$\Sigma \text{ 1}^{\text{st}} \text{ and } 2^{\text{nd}} \text{ I.E.} = 2405 \text{ kJ/mole}$	$E^{\circ} \text{ oxidation} = 0.28 \text{ V} \therefore E^{\circ} \text{ reduction} = -0.28 \text{ V}$

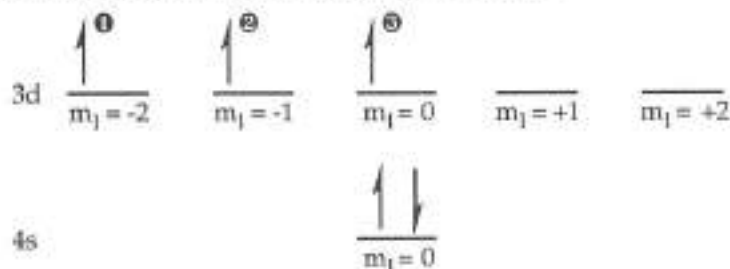
50. **Choice C is correct.** It is grunt work when it comes to figuring out electronic configurations! You can save yourself some time by eliminating choices with either too many or too few electrons. Aluminum has thirteen electrons, as do all of the choices. This means that no choices are eliminated. Choice A is out, because an s-orbital can hold no more than two electrons. Choices B and D are eliminated, because the s-orbital fills completely before the p-orbital begins to fill. The correct answer is choice C. If it is needed, the Aufbau principle can be applied to determine the filling order.
51. **Choice D is correct.** Monochromatic light is light that is composed of photons with exactly *one* wavelength (or frequency). This would occur if every photon emitted came from exactly the same electronic transition between energy levels. However, not all electrons are at exactly the same energy level, due to the close proximity of rotational and vibrational levels. As a result, not every electron undergoes the exact same energy transition, and thus not every photon that is emitted has the same energy (and thus neither the same frequency nor wavelength). This is stated near the end of the passage. Choose D to be a stellar student.

52. **Choice B is correct.** This question requires determining the photon's energy from its wavelength

$$E = \frac{hc}{\lambda} = \frac{(6.6 \times 10^{-34} \text{ J} \cdot \text{sec} \times 3.0 \times 10^8 \frac{\text{m}}{\text{sec}})}{330 \times 10^{-9} \text{ m}} = \frac{19.8 \times 10^{-26} \text{ J} \cdot \text{m}}{3.3 \times 10^{-7} \text{ m}} \Rightarrow E = \frac{19.8}{3.3} \times 10^{-19} \text{ J}$$

The more important part of the answer is the power of ten. From that, you must select choice B.

53. **Choice A is correct.** The shorthand for the electronic configuration of vanadium is $[\text{Ar}]4s^23d^3$. For the quantum numbers of vanadium, we are concerned only with the last electron in vanadium (V). We are therefore concerned with the third 3d-electron. Drawn below is the 3d level:



Last electron is in the third d orbital, so $m_l = 0$; Last electron is spin up, so $m_s = +\frac{1}{2}$

The last electron is in the $n = 3$ level, because that is given by $3d^3$. The electron is in the $l = 2$ level, because that is given by d-orbital in $3d^3$. The m_l and m_s values must be derived by filling electrons into their respective levels. As listed in the box, $m_l = 0$, $m_s = +\frac{1}{2}$, which is choice A.

54. **Choice D is correct.** Half-filled stability would come into play for atoms that can promote *one* electron from the lower energy s-orbital up to the higher energy d-orbital to yield a d^5 species. The term "half-filled stability" is derived from the half-filled d-level (the d-level has a maximum occupancy of ten electrons). Half-filled stability is possible for only chromium (Cr), molybdenum (Mo), and tungsten (W), because they are all in the column that should be s^2d^4 . To be a quality chemistry student, pick choice D. Copper can excite *one* s-electron to fill the d-level completely (to make the species a d^{10} atom).
55. **Choice B is correct.** Whatever goes in, must come out for normal behavior of light. There are situations where a high-energy photon is absorbed that results in the excitation of an electron through multiple levels. From here, the electron may relax (fall back to the ground state) by a variety of pathways (either gradually dissipating its energy or releasing the energy all at once). As a general rule, the energy that is absorbed is also emitted. Pick choice B. The exception to this rule is phosphorescence. With a phosphorescing species, light energy is absorbed, and the compound undergoes a change in its electronic structure. The excited state is therefore a different complex than the ground state. When the new complex gives off energy to fall to its ground state, the photon released is not of the same energy as the original photon absorbed. Phosphorescence and fluorescence were passage topics on a previous MCAT. Quantum numbers were also a passage topic.

56. **Choice C is correct.** Visible light is found approximately in the range 390×10^{-9} m to 740×10^{-9} m. To be used in the formula provided, the wavelength must be in terms of 10^{-6} m. Visible light has a range from 0.39×10^{-6} m to 0.74×10^{-6} m. The value for this range in terms of eV is $1.24/0.74$ eV to $1.24/0.39$ eV. This is a range from some number just over 1.5 to some number barely greater than 3.0, a range that fits choice C.

Passage IX (Questions 57 - 63)

Paint Pigments

57. **Choice A is correct.** The reflected color of the four pigments are yellow for alizarin, blue for azurite, red for cinnabar, and blue-green for verdigris. To determine the light absorbed, you must take the complementary color of the color observed. The absorbencies are thus violet for alizarin, orange for azurite, green for cinnabar, and orange-red for verdigris. The shortest wavelength is associated with the highest-energy light. Violet has the highest energy in the visible spectrum, so it has the shortest wavelength. The correct answer is choice A.
58. **Choice A is correct.** When a sample is heated, its electrons are thermally excited to a higher energy state. When they relax back to their ground state, light is emitted at an exact wavelength. This results in light of a specific color, best described by choice A.
59. **Choice D is correct.** The absorption spectrum contains black lines where the complementary colors of orange-red (the colors reflected by realgar) should be. The complementary colors are blue-green, so the best answer selection is choice D.
60. **Choice C is correct.** The reflected color of verdigris ($\text{Cu}(\text{C}_2\text{H}_3\text{O}_2)_2 \cdot \text{Cu}(\text{OH})_2$) is green-blue, so the absorbed color is red-orange. Red and orange lie at the low end of the visible spectrum as far as the energy is concerned, so the wavelength lies at the high end of the visible spectrum. The spectrum goes red-orange-yellow-green-blue-violet, so the wavelength of orange light falls below 700 nm, and the wavelength of red is around 700 nm. This means that the wavelength is less than 700 nm by a small amount. The best answer is 650 nm, which makes the best answer choice C. The range of visible light in the EM spectrum is given on page 113.
61. **Choice D is correct.** Because white lead appears white in color, no light has been absorbed in the visible range. When no light is absorbed, white light (the incident light) is reflected. The d-d transition associated with white lead must lie outside of the visible range. The best answer is thus choice D.
62. **Choice D is correct.** A reflected color can be seen only when white light is reflected off of it. This means that reflected colors cannot be seen in the dark (absence of white light). Candles cannot be seen in the dark, so a colored candle is made from a wax that contains a dye that exhibits reflected color. Clothes cannot be seen in the dark, so a fabric dye exhibits reflected color. If you wish to argue that there are certain glow-in-the-dark fabric dyes, you're absolutely right. Glow-in-the-dark dyes exhibit emitted color. You are wise in the ways of trivia. Unfortunately, you get zero credit on this question, because it is not the best answer. Knowing about special cases like that is a great way to impress your peers, but it hurts you on a standardized exam. Ink from pens cannot be seen in the dark, so ink exhibits reflected color. Gas-filled light tubes (i.e., neon lights) can be seen in the dark, so a glowing gas-filled light bulb exhibits emitted color. The best answer is choice D.
63. **Choice A is correct.** The fastest decomposition is observed in the organic pigment. This is stated in the passage. The inorganic pigments are already oxidized, so they should remain air-stable for some time. The organic pigments can oxidize in air, so they do not last as long. The only organic compound among the choices (and in Table 1) is 1,2-dihydroxyanthraquinone, which is the pigment of alizarin. This makes choice A correct. Note that organic pigments contain conjugated π -networks and are found in oil-based paints.

Passage X (Questions 64 - 69)

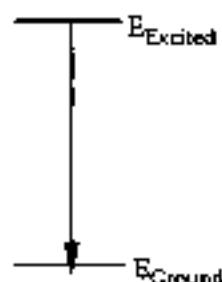
Fluorescence and Phosphorescence

64. **Choice B is correct.** As stated in the passage, a spin flip is associated with the process of phosphorescence. In the ground state, two electrons in the same orbital must be spin-paired (have opposite spins). Once an electron has been excited to a higher electronic level where it occupies the orbital alone, it is free to flip its spin. Equally, the electron that remains in the ground state may also flip its spin. A spin flip changes the total energy of the system. The excited electron may not be able to relax back to its original level, since it shares the same spin as the lower level electron now. When the electron falls back to a lower level, it falls to a different energy level, which emits a different frequency of light than it absorbed. Whether or not you know what phosphorescence is, the answer is given in the passage. The best answer is choice B.

65. **Choice A is correct.** A diatomic molecule can gain and lose energy in several ways, including changing electronic levels, vibrating at different frequencies, and rotating at different frequencies. In a diatomic molecule, when an electron is excited, the valence shell increases, which affects the bond length and bond strength. The diatomic molecule is more susceptible to changing its vibrational frequency. This is the physical cause of fluorescence. When an electron in the diatomic molecule is excited, a photon is absorbed. Because energy is dissipated from the excited state in the form of rotational and vibrational energy, the molecule is in a lower energy state. When it finally relaxes back to its ground state, the energy of the photon emitted is less than the energy of the photon absorbed. The energy of the photon absorbed is equal to the energy of the photon emitted plus the dissipated energy (vibrational and rotational transitions). This eliminates choices B and D. As the wavelength of a photon increases, it has less energy. This implies that the photon emitted (being of less energy) has a longer wavelength than the photon absorbed. This makes choice A correct.
66. **Choice C is correct.** "Because the electronic energy levels are not singular levels and because the transition is random, it is impossible for a molecule to absorb or emit light in such a manner that all of the photons simultaneously have the same frequency (or wavelength). For this reason, monochromatic light is not physically possible." These last two sentence from the passage state that monochromatic light (light of one wavelength) is not possible, and give the reasoning for that. Because monochromatic light is not possible, choices B and D are eliminated. The reasoning has to do with energy levels, not the quantization of light, so choice C is the best answer. This question about a difficult concept is actually an easy question to answer, if you don't get intimidated. Learning to manage the intimidation factor associated with seeing new information is a part of your MCAT preparation.

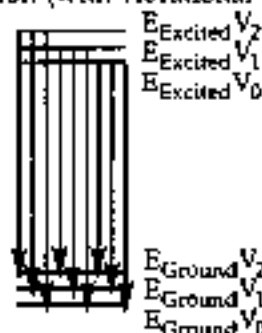
For the sake of learning the concept, we shall look at what choice C is stating. The diagram below shows two scenarios, one where electronic transitions are not coupled with vibrational transitions (on the left) and the other one where electronic transitions are coupled with vibrational transitions (on the right).

Electronic Transition (no vibrational transitions)



Electronic transitions occur without vibrational transitions, so a single energy transition is possible. Only one photon is emitted, which would result in monochromatic light.

Electronic Transition (with vibrational transitions)



Electronic transitions occur with vibrational transitions, so a multiple energy transitions are possible. Multiple photons are emitted, so the emission is polychromatic light.

From the diagram, it can be seen that when vibrational energy levels are closer together than electronic energy levels, the transitions can couple. Single transitions between electronic levels are not possible, although single transitions between vibrational levels appear to be possible. However, vibrational transitions couple with rotational levels, so infrared emissions are not of single wavelength. The best answer is choice C.

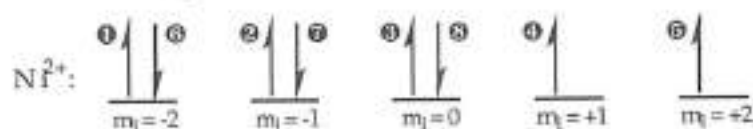
67. **Choice B is correct.** The arrow in Figure 1 that represents emission due to fluorescence is longer than the arrow that represents emission due to phosphorescence, so the energy associated with fluorescence is greater than the energy associated with phosphorescence. This means that the light from fluorescence is of shorter wavelength than the light from phosphorescence. Choice B is the best answer. Choices A and D are the same answer worded differently (if fluorescence emission is of longer wavelength than phosphorescence, then phosphorescence must have emission of higher energy than fluorescence), so both choices should have been eliminated (assuming as we do that there is only one best answer per question).
68. **Choice A is correct.** The shortest wavelength of light belongs to light of the greatest energy. This occurs with the transition from the highest excited state relaxing to the lowest ground state. Choices C and D are eliminated immediately, because the transitions they represent are increases in energy, which absorb light, not emit light. The best answer is choice A, because the excited state is the highest of the choices left (A and B), and both of the choices left drop energy to the same ground state.

69. **Choice B is correct.** Fluorescence is possible with molecules, because molecules can exhibit vibrational energy transitions. This allows for the dissipation of energy via of heat. Choices A and C true, so they are eliminated. Because atoms do not have vibrational energy transitions (they have no bonds, so they have no bending and stretching modes of their bonds), atoms may not necessarily exhibit fluorescence. It is possible to convert ultraviolet light to the lesser-energy visible light by fluorescence, but it is not possible to convert visible light to the higher-energy ultraviolet light by fluorescence. This makes choice B the false statement.

Passage XI (Questions 70 - 76)

Flame Test

70. **Choice D is correct.** The definition of paramagnetic is having at least one unpaired electron. The electronic configuration for Na^+ is $1s^2 2s^2 2p^6$. All electrons are paired, because the octet is complete, eliminating choice A. The electronic configuration for Sr^{2+} is $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^{10} 4p^6$. All electrons are paired, because the octet is complete, eliminating choice B. The electronic configuration for Cu^+ is $1s^2 2s^2 2p^6 3s^2 3p^6 3d^{10}$ (copper has filled d-shell stability, and it loses its 4s-electron before its 3d-electrons). All electrons are paired, because each level is filled, eliminating choice C. The electronic configuration for Cr^{3+} is $1s^2 2s^2 2p^6 3s^2 3p^6 3d^3$ (first-row transition metals lose their 4s-electrons before losing their 3d-electrons). Not all electrons can be paired, because there is an odd number of electrons. Because Cr^{3+} is paramagnetic, choice D is the best answer.
71. **Choice B is correct.** Chromium has half-filled d-shell stability, giving it an electronic configuration of $1s^2 2s^2 2p^6 3s^2 3p^6 4s^1 3d^5$. The correct choice is answer choice B. Without the half-filled stability, the answer would have been choice A. Molybdenum (Mo) and tungsten (W) also exhibit half-filled stability.
72. **Choice D is correct.** The largest transition is associated with the greatest energy. Vanadium dication (V^{2+}) produces violet light, therefore the highest energy is associated with vanadium dication. Pick choice D.
73. **Choice C is correct.** It can be observed from the data in Table 1 that as the first or second columns in the periodic table are descended, the light emitted from the transition is of progressively higher energy. This makes statement I a true statement. Because statement I is not included in choices B and D, choices B and D are eliminated. It can be deduced from the answer choices that remain that statement III must be false. To verify this, copper has the electronic configuration $[\text{Ar}]4s^1 3d^{10}$, so Cu^+ has the electronic configuration $[\text{Ar}]3d^{10}$. Nickel has the electronic configuration $[\text{Ar}]4s^2 3d^8$, so Ni^{2+} has the electronic configuration $[\text{Ar}]3d^8$. Statement III is in fact a false statement, because Cu^+ and Ni^{2+} do not have the same electronic configuration. The transition for potassium cation (K^+) emits purple light, while the transition for chromium trication (Cr^{3+}) yields red light. The transition for K^+ is of higher energy than the transition for Cr^{3+} , making statement II true and choice C correct.
74. **Choice B is correct.** Nickel dication (Ni^{2+}) has the electronic configuration $1s^2 2s^2 2p^6 3s^2 3p^6 3d^8$. The last electron is the eighth electron in the 3d-orbital. The principle quantum number is given as 3 and being in a d-orbital makes l equal to 2. From the answer choices, this is already known. To obtain the m_l and m_s values, the electrons must be filled into their respective d-orbitals. This is drawn below:



Last electron is in the third d orbital, so $m_l = 0$; Last electron is spin down, so $m_s = -\frac{1}{2}$

The correct answer is therefore choice B.

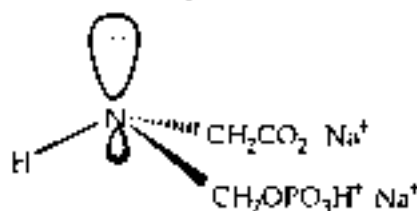
75. **Choice D is correct.** Because violet light is higher in energy than orange-yellow light, which is in turn higher energy than red light, the electronic transitions for the group I cations increase as the column is descended. The transition for rubidium should therefore be of greater energy than violet light, which makes the transition emit ultraviolet light. Ultraviolet light is not detected by the human eye, so the flame from heating rubidium should appear colorless. The best answer is choice D.

76. Choice C is correct. An emission spectrum shows just the color emitted by the compound after it has been excited. The color observed in the flame test is an emitted color, so it is present in the emission spectrum. Crimson was emitted by lithium cation, so the emission spectrum is simply a bright red line. The best answer is choice C. Other minor emissions may be seen, but they won't be as intense as the red emission.

Passage XII (Questions 77–81)

Glyphosate

77. Choice C is correct. Two isotopes (of neutral elements) have the same number of protons and electrons, but a different number of neutrons. Choice A has one more proton than phosphorus-32, choice B has one more electron than phosphorus-32, and choice D has one electron less than phosphorus-32. This eliminates choices A, B, and D. Choice C has a mass that is one less with the same number of protons as phosphorus-32, so it must have one neutron less. This makes choice C the correct answer.
78. Choice B is correct. Nitrogen in neutral molecules makes three bonds total (in this case, the three bonds are all sigma bonds), and has one lone pair of electrons. The lone pair of electrons repels the electrons in the three sigma bonds to form a trigonal pyramidal orientation about the nitrogen. This can be confirmed when looking at the hybridization of nitrogen (sp^3). The best choice is therefore answer B. Drawn below is a three-dimensional picture of Glyphosate with the nitrogen isolated:

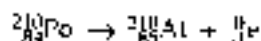


79. Choice C is correct. The concentration of ^{32}P label at consecutive half-life points along the first-order decay is: $188 \rightarrow 94 \rightarrow 47 \rightarrow 23.5$. Each arrow represents one half-life, so after three half-lives the concentration is less than 25 ppm (it has decayed to 23.5 ppm). To reach a concentration of 25 ppm, it takes a little less than three half-lives. The best answer is a little less than 42 (3×14) minutes. Choice C, 41 minutes, is the best answer.
80. Choice C is correct. An ionic compound is made up of ions. The quickest way, without just knowing the answer, is to look for metals such as sodium. Ammonia is held together by three covalent bonds (sigma bonds). Carbon dioxide is held together by two covalent bonds (two double bonds made up of one sigma bond and one π -bond each). It is only in compound III, $\text{Na}_2\text{PO}_3\text{H}$, that we find ionic bonds. Choice C, III only, is the best answer.
81. Choice A is correct. Because ^{32}P remains chemically equivalent after nuclear decay, it is phosphorus. This means that a proton was neither gained nor lost in the process. When a neutron is lost, a phosphorus-32 isotope becomes a phosphorus-31 isotope. This makes choice A the best choice listed. An alpha particle contains two protons and two neutrons, so the loss of an alpha particle would form aluminum, eliminating choice B. The loss of a beta particle converts a neutron into a proton, which would form sulfur, eliminating choice C. The loss of a positron converts a proton into a neutron which would form silicon, eliminating choice D. The answer choices did not list a gamma ray. A gamma ray is a high-energy photon that, when given off, does not change the chemical behavior, either. This was not listed as a choice, but it is food for thought.

Passage XIII (Questions 82–88)

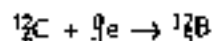
Technetium Decay

82. Choice A is correct. Beta decay is the loss of an electron from the nucleus. No mass is lost, therefore the mass should not change. Choices C and D are eliminated. Charge must be conserved, so losing a negative charge should increase the atomic number by 1. This makes choice A the best answer. The reaction is shown below:



83. Choice C is correct. An alpha particle is a helium nucleus (mass = 4 amu and $z = 2$), so the loss of an alpha particle decreases the mass by 4 and the atomic number by 2. ^{250}Bk is 4 mass units less than ^{254}Es , so choice A is eliminated. ^{234}Th is 4 mass units less than ^{238}U , so choice B is eliminated. ^{221}Bi is only 2 mass units less than ^{223}Fr , so choice C is the best answer. ^{243}Pu is 4 mass units less than ^{247}Cm , so choice D is eliminated. What else but berkelium could be right?

84. Choice C is correct. From the passage, the half-life is given as six hours. The twelve-hour duration is therefore a total of two half-lives. The initial concentration should therefore be cut in half two successive times to determine the final concentration. The math is as follows: $120\ \mu\text{Ci} \rightarrow 60\ \mu\text{Ci} \rightarrow 30\ \mu\text{Ci}$. The final reading is $30\ \mu\text{Ci}$, so the correct choice is choice C.
85. Choice B is correct. Electron capture by a nucleus decreases the positive charge (converting a proton into a neutron), which reduces z by 1. This converts element # 6 (carbon) into element # 5 (boron).



The question is asking for a nuclear process that forms boron. Positron capture increases the nuclear charge by one, so positron capture by boron cannot yield boron. It yields carbon, one atomic number higher, which eliminates choice A. Positron decay decreases the nuclear charge by one, so positron decay by carbon yields boron, one atomic number lower. This makes choice B the best answer. Beta capture decreases the nuclear charge by one, so beta capture by nitrogen yields carbon, one atomic number lower. This eliminates choice C. Alpha decay decreases the nuclear charge by two, so alpha decay by oxygen yields carbon, two atomic numbers lower. This eliminates choice D. The best answer is choice B, the only choice that didn't form carbon. The processes in choices A, B, C, and D are shown below.

Choice A: ${}^{12}_5\text{B} + {}^0_{+1}\text{e} \rightarrow {}^{12}_6\text{C}$; Choice B: ${}^{12}_6\text{C} \rightarrow {}^{12}_5\text{B} + {}^0_{+1}\text{e}$; Choice C: ${}^{14}_7\text{N} + {}^0_{-1}\text{e} \rightarrow {}^{14}_6\text{C}$; Choice D: ${}^{16}_8\text{O} \rightarrow {}^{12}_6\text{C} + {}^4_2\text{He}$

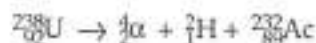
86. Choice D is correct. A gamma ray is high-energy electromagnetic radiation, not a particle. The energy associated with a gamma ray is greater than the energy associated with an x-ray. Because it is a photon (energy) and not a particle, a gamma ray is massless and without charge. When the nucleus of what emits a gamma ray, it drops from a nuclear excited state to a lower (and possibly ground) state, as mentioned in the passage. No mass is lost by gamma emission. Choice D is the best answer. Choice A describes an alpha particle, and choice B sort of describes a beta particle (the charge is negative one, but it is not necessarily the nuclear charge). Choice C is not a common nuclear particle and is probably a conglomeration of subatomic particles.
87. Choice B is correct. The electronic configuration for ${}^{99}\text{Tc}$ is no different than for any other isotope of technetium. Although isotopes have a different number of neutrons, they have an identical number of protons and an identical number of electrons in their neutral state. Technetium is element number 43, so it has 43 protons and 43 electrons as a neutral element. The filling of electrons follows standard Aufbau principle rules, which makes answer B the right choice. It is sometimes a shortcut to look at the periodic table and see where the last electron falls. In the case of technetium, it is directly below manganese, so its last electron should be a d^5 electron (like manganese). The last electron is in a $4d$ -orbital, so choice B is best.
88. Choice A is correct. Gamma decay just involves the loss of nuclear energy, and not a particle. Technetium goes from an excited nuclear state to a lower nuclear state after it undergoes gamma decay. No nuclear particles are lost or gained by technetium, so its mass and atomic number remain the same. This means that the nuclear composition of the element remains the same, so the element remains the same. The correct choice is thus A.

Passage XIV (Questions 89–91)

Cold Fusion

89. Choice B is correct. Reaction 3 and Reaction 4 show fusion of two deuterium atoms. The fusion of two deuterium atoms can generate either a tritium (${}^3\text{H}$) and hydrogen (${}^1\text{H}$) (as shown in Reaction 3), or one neutron and helium isotope (${}^3\text{He}$) (as shown in Reaction 4). The only particle in the answer choices that is not shown as a product is helium-4, so the correct answer is choice B.
90. Choice B is correct. We see in the passage, that the major piece of evidence for believing a nuclear reaction had transpired was the release of more heat than can be explained by the electrochemical cell reaction. The expected nuclear products were observed only in low concentrations however. The best answer is choice B. The absorption or emission of an electron (beta particle) had no effect on the reaction.
91. Choice D is correct. As described in the passage, cold fusion is a nuclear fusion reaction that can be carried out at room temperature. The example in the passage takes place in an electrode within a test tube. Because more energy is released than expected, it can be assumed that the fusion reaction is exothermic (because of the excess heat that was released). The correct answer is choice D. Pick D, and you'll smile brightly.

92. **Choice A is correct.** In the passage, we read that nuclear power plants employ the fission reaction, because to date, fusion reactions require both high temperature and strong magnetic fields. Fusion reactions require more input energy than the fusion reaction releases, making them endothermic and unfavorable as a source of energy. The best answer is choice A.
93. **Choice C is correct.** A fusion reaction results in the combination of the nuclei of the two particles undergoing fusion. Answer choice C involves the combining of two particles to form one product (with an atomic number greater than the two incident particles). This defines fusion. Choices A and B are both fission, and choice D involves the emission of a gamma photon via the drop from a nuclear excited state to the nuclear ground state.
94. **Choice B is correct.** An alpha particle has a mass of four and two protons, and a deuterium nucleus has a mass of two and one proton. After losing an alpha particle and deuterium nucleus, the mass of an element drops by six and the number of protons drops by three. Choices A and C are not six mass units less, so they are eliminated. Choice D is not three atomic numbers less, so it is eliminated. The process is shown below.

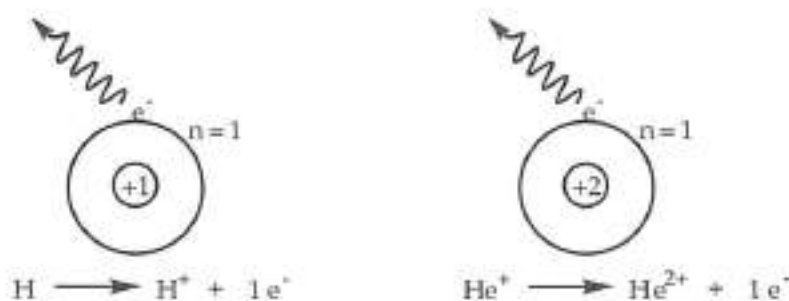


95. **Choice C is correct.** A change in the atomic number results from a change in the number of protons in the nucleus. The alpha particle has two protons, so the capture of an alpha particle increases the atomic number by two. This eliminates choice A. The capture of a beta particle converts a proton into a neutron, so the atomic number decreases by one. This eliminates choice B. The tritium nucleus carries one proton, so capture of a tritium nucleus increases the atomic number by one. This eliminates choice D. A neutron capture increases the mass by one, but does *not* affect the atomic number. Choice C is the best answer.

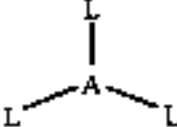
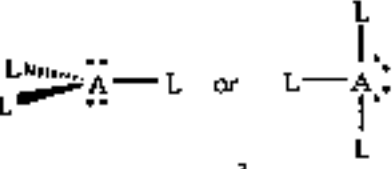
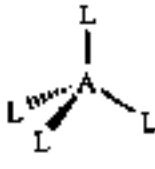
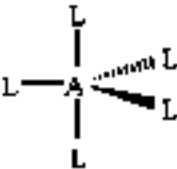
Questions 96 - 100

Not Based on a Descriptive Passage

96. **Choice D is correct.** "Quantized energy levels" refer to states of finite energy where electrons may exist. For this exam, you should know conceptually what behavior is expected. The existence of a neutron or proton at the nucleus may exert Coulombic forces on an orbiting electron, but it does not have any bearing on the quantization of energy levels. Choice A is therefore eliminated. The scattering of x-rays by thin sheets of material (metal foil in the Rutherford experiment) shows that matter is mostly empty space, with dense uniformly spaced nuclei. Choice B is eliminated. The bending of any particle when moving through a magnetic field simply indicates that the particle in motion has a net charge of some kind, and that the direction of motion is not in line with the field. Choice C is eliminated. Distinct lines (which can be reproduced in separate trials) show that the same amount of energy is absorbed when an electron is excited. If the transition between levels is a quantized value (an exact quantity), then it seems logical that the energy levels are also quantized. This makes choice D the correct answer.
97. **Choice D is correct.** As the wavelength of a photon increases, the energy of the photon decreases. This question is testing your recall of relative energetics of electromagnetic radiation. The lowest energy of the choices given is associated with infrared light. This makes D the best choice.
98. **Choice A is correct.** The value of Z_{eff} for H is +1. The value of Z_{eff} for He^+ , the starting point for the second ionization of helium, is +2. Because the ionization energy is proportional to Z^2 , the second ionization energy of helium should be four times as great as the ionization energy of hydrogen. Both electrons in question are being ionized from the 1s-orbital, so $n = 1$ for both ionization energies. You need consider only the effective nuclear charge. To feel the sensation of correctness, pick choice A. Drawn below are the respective ionization reactions:



99. **Choice D is correct.** Sodium cation has ten electrons, which eliminates choices A and B, both with eleven electrons. Choice C is the ground state (all electrons fill sequentially) for sodium cation, given that it has the ten electrons filled in order. In choice D, there are ten electrons and an electron has been excited from the 2p-level to the 3s-level. This leaves choice D to be the best answer.
100. **Choice B is correct.** The most common shape for a transition metal with five ligands is trigonal bipyramidal. The best answer is choice B. Square planar has only four ligands attached, so choice A is eliminated. Hexahedral does not exist; and if it did, hex is Greek for six, and only five ligands are attached. This would also eliminate choice D. Drawn below is a chart for determining molecular shapes:

Coordination Number = 2 L — A — L no lone pairs (sp hybr.) Linear Bond Angle = 180°	Coordination Number = 2  one lone pair (sp² hybr.) Bent Bond Angle < 120°	Coordination Number = 2  two lone pairs (sp³ hybr.) Bent Bond Angle < 109.5°
Coordination Number = 3  no lone pairs (sp² hybr.) Trigonal Planar Bond Angles = 120°	Coordination Number = 3  one lone pair (sp³ hybr.) Trigonal Pyramidal Bond Angles < 109.5°	Coordination Number = 3  two lone pairs (dsp³ hybr.) Trigonal Planar or T-shaped Bond Angles = 90°, 120°, 180°
Coordination Number = 4  no lone pairs (sp³ hybr.) Tetrahedral Bond Angles = 109.5°	Coordination Number = 5  no lone pairs (dsp³ hybr.) Trigonal Bipyramidal Bond Angles = 90° & 120°	Coordination Number = 6  no lone pairs (d²sp³ hybr.) Octahedral Bond Angles = 90°

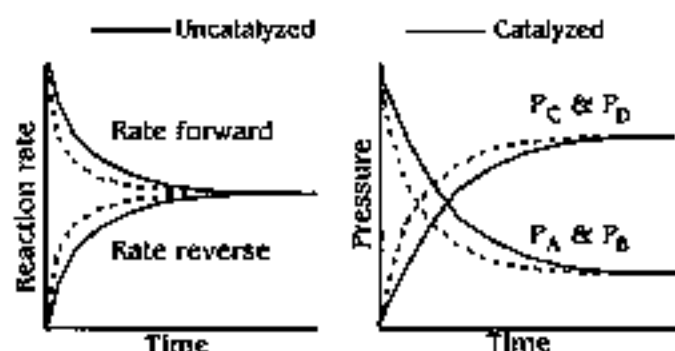
Section III

Equilibrium

by Todd Bennett



$$K_{eq} = \frac{\text{Products}}{\text{Reactants}} = \frac{(P_C)(P_D)}{(P_A)(P_B)}$$



Fundamentals of Equilibrium

- Definitions and Terminology
- Equilibrium Constant (K_{eq})
- Reaction Quotient (Q_{rx})
- Case Specific K-Values
- K_{eq} Calculations
- Using K_{eq} to Calculate Shifts
- Complex Equilibrium
- Experimental Determination of K

Le Châtelier's Principle

- Effect of Stress
- Perturbations and Shifts
 - Direction of Shift

Solubility

- Definitions
- Solubility Rules
 - Nomenclature of Salts
 - Polyatomic Ions
- Solubility Product and Molar Solubility
- Relative Solubility
- Solubility Experiments
- Common Ion Effect
- Separation by Precipitation
- Ion Exchange Columns



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Equilibrium Section Goals

-  **Know how to determine an equilibrium constant from experimental data.**
There are experiments that are designed to determine a change in total pressure over time for a gas equilibrium system. The change in pressure (from the time the reactants are mixed until the reaction reaches a steady equilibrium pressure) can be used to calculate an equilibrium constant. The difference in pressure is the shift required to reach equilibrium.
-  **Understand Le Châtelier's principle and its effects on equilibrium systems.**
Le Châtelier's principle predicts certain behavior in an equilibrium system once a stress is applied to the system. Because changing one variable can affect another variable (for instance, changing the pressure may also change the volume), you must evaluate how the chemical reaction will adjust to alleviate the applied stress in order to reestablish equilibrium.
-  **Understand the mathematical arrangement of the equilibrium expression.**
The equilibrium expression, simply put, is the ratio of products over reactants. The mathematical rules require that you do not put pure liquids (solvents) or solids into the expression. If more than one mole of product or reactant are involved in the reaction, then the stoichiometric coefficient in the balanced equation becomes an exponent in the equilibrium expression. The value of the equilibrium constant changes only with temperature.
-  **Know the effects of the system variables P, V, n, and T on reaction equilibrium.**
It is not possible to change just one variable in an equality. Changing the conditions of the system shifts an equilibrium, but it does not necessarily change the equilibrium constant. The numerical value of the equilibrium constant changes only with temperature.
-  **Know the relationship between molar solubility and solubility product.**
The molar solubility of a salt is the concentration of the salt (as measured in molarity) needed to solvate an aqueous solution completely. The solubility product is an equilibrium constant for the dissociation reaction. What makes solubility products unique is that the reactant is always a solid, so the equilibrium expression has no denominator. Molar solubility is a more useful quantity to know than solubility product, because it measures the amount of salt in solution.
-  **Understand separation by precipitation and the chelation effect.**
Chelating is the formation of a Lewis acid-base bond between a lone pair-donor (ligand) and a lone-pair acceptor (central atom). Chelation changes the solubility of a salt by changing the concentration of free ions in solution. When a ligand binds a central metal, there is a formation constant that measures the strength of the chelation. This allows for specific ions to be removed from solution by binding them to form a more soluble complex ion. Ions also can be removed from solution by adding counter-ions that form an insoluble salt.
-  **Understand the common ion effect.**
The common ion effect is a twist on Le Châtelier's principle as it applies to solubility. The addition of products to an equilibrium mixture shifts the reaction in the reverse direction, which in the case of solubility results in precipitation. This means that addition of an ion to solution or the presence of an ion in solution reduces the solubility of a salt.

Equilibrium

Equilibrium, as you have learned it, involves the balancing of a chemical reaction between reactants and products. Equilibrium is a state that is achieved when the reactants go on to form products at the same rate that the products go back to form reactants. Equilibrium reactions take place in both the gas phase and solution state, where reactant and product molecules are free to migrate and collide. The kinetic theory model states that molecules must collide to react. Equilibrium occurs only in a closed system, although steps in a pathway (an open system) may be an equilibrium reaction.

Equilibrium is an odd yet obvious thing. It is the essence of nature and the foe of permanence. We can't beat it...EVER! The concept of equilibrium pervades not only chemistry, but politics, economics, sociology, health...even our relationships. A bank represents a good illustration of this concept. There are deposits and withdrawals continually going on, but on the whole, the amount of money in the bank remains essentially constant (except following a bank robbery, after which a new equilibrium will be established). Everything lies in a balance, and all one has to do is read the scale. Fortunately, in chemistry it is easy to read the scale. To understand equilibrium better, consider the following saga:

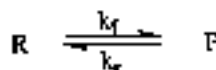
In a certain house with a large backyard lived an elderly man with a plum tree. Next-door to him lived a young boy who also had a plum tree. Given that neither of these two were farmers by any means, the fruit generally fell to the ground and rotted. One day, while out in his back yard cleaning up the plums, the elderly man was struck by a flash of insight; instead of using all his energy to consolidate the plums in a trash container, he would instead simply fling them over the fence. The boy next-door saw this and decided that two could play at this game. The next time he was out in the back yard, he flung a few plums over the fence. This soon evolved into a daily contest with the advantage going to the little boy, for he was younger and quicker and could move around the yard faster. He was therefore able to fling the unlikely projectiles over the fence at three times the rate of the old man. At first, there were roughly equal amounts of plums in each yard, but due to the boy's greater flinging prowess, the old man's yard gradually accumulated more. Finally, when the point was reached where the old man's yard had three times as much as the boy's, the overall amount in each yard became stable and didn't change. Because the boy had so few plums in his yard, he spent the majority of his time running around collecting them. The man, however, could simply stand in one place, scooping and flinging. From the neighbors' perspective, for every one that would sail through the air to the left, there was one that would sail through the air to the right. The plum-flinging had reached its equilibrium. The ratio of the plums in each yard was equal to the ratio of the rates at which the two could fling plums (known as the "plum-flinging rate"). This in essence is as deep as equilibrium gets.

This example may not be the most eloquent, but it serves the purpose of generating a memorable analogy to chemical reactions. It is the external conditions that affect the equilibrium between products and reactants, and a skilled chemist knows how to manipulate this relationship. We will address how equilibrium plays out in a solvent environment and in the gas phase. We shall view the effect of factors governing the system, such as volume, pressure, temperature, and concentration. We will finish by looking at the equilibrium of a salt dissociating into aqueous solution, and the factors that are involved. Throughout all of the discussion, Le Châtelier's principle will play a roll.

Fundamentals of Equilibrium

Definitions and Terminology

Equilibrium occurs in a closed system when the rate of the forward reaction is equal to the rate of the reverse reaction (rate forward = rate reverse). The result is that from a macroscopic perspective, the system appears not to change. However, a net change of zero does not automatically mean the system is in equilibrium. Reactions may stop when there is no reactant present, which is a case of no reaction, not equilibrium. As an introduction to equilibrium, consider Reaction 3.1, where R = reactant, P = product, k_f = forward rate constant, and k_r = reverse rate constant.



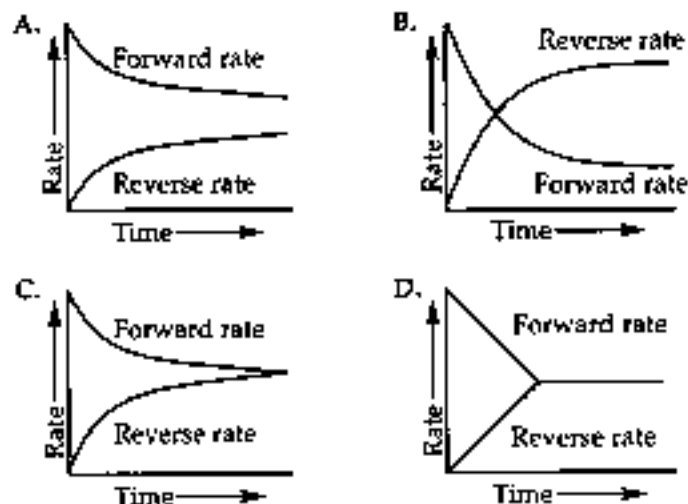
Reaction 3.1

The forward rate for this reaction is based on the amount of reactants and the forward rate constant (rate forward = $k_f[\text{Reactants}]$). The reverse rate for this reaction is based on the amount of products and the reverse rate constant (rate reverse = $k_r[\text{Products}]$). Equilibrium is achieved when:

$$k_f[\text{Reactants}] = k_r[\text{Products}] \quad (3.1)$$

Example 3.1

Which of the following graphs represents what is observed over time for a reaction starting with all reactants?



Solution

Choices A and B are eliminated, because the rates must be equal at equilibrium, and neither graph reflects this. Choice D is eliminated, because straight lines are not very common for graphs in chemistry. A straight line would imply that the reaction abruptly stopped, once equilibrium was reached. The reality is that the reaction gradually slows until it reaches equilibrium. The best answer is choice C, which shows equal rates after time and constantly changing rates until equilibrium is reached.

Equilibrium Constant (K_{eq})

The *equilibrium constant* is a mathematical quantity that is calculated for a reaction at equilibrium. By definition, the equilibrium constant is the concentration of products at equilibrium divided by the concentration of reactants at equilibrium. Equation 3.2 shows this relationship.

$$K_{eq} = \frac{[\text{Products}]}{[\text{Reactants}]} \quad (3.2)$$

By applying Equation 3.2 to Reaction 3.1, we find that the equilibrium constant is equal to the forward rate constant divided by the reverse rate constant.

Example 3.2

For Reaction 3.1, if the forward rate constant is four times the reverse rate constant, what is K_{eq} after a catalyst has been added that doubles the rate of the forward reaction?

- A. $K_{eq} = 2$
- B. $K_{eq} = 4$
- C. $K_{eq} = 8$
- D. $K_{eq} = 16$

Solution

Adding a catalyst lowers the activation energy, so the reaction speeds up. In this example, the forward rate is doubled, because the forward rate constant is doubled. However, the activation energy is lowered for the reverse reaction as well. The reverse reaction rate is also doubled. The ratio of the forward rate to the reverse rate remains the same. This means that equilibrium is the same, so the equilibrium constant (K_{eq}) is the same. Equilibrium is achieved sooner, but the same equilibrium conditions are reached. The forward rate is four times the reverse rate, so the equilibrium constant is equal to 4, choice B.

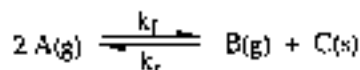
Table 3.1 lists some rules about the equilibrium constant you must know. Among them is the rule that the numerical value of K_{eq} changes only with temperature. Knowing this rule would have made Example 3.2 easier to solve.

1	Stoichiometric values from the balanced equation become exponents in the K_{eq} expression.
2	Do not include solids or pure liquids in the K_{eq} expression, only solutes (for K_c) and gases (for K_p).
3	The numerical value of K_{eq} varies only with changing temperature, not with catalysts, pressure, volume, or moles.

Table 3.1

When multiple reactants or multiple products are present in the reaction, which is usually the case, rule # 1 applies. Because the concentration of a solid or pure liquid (solvent) does not change, their values are constant. For this reason, they are ignored in the K_{eq} determination. As a rule of thumb, only molecules that are free to move and are rarely in contact with other reacting molecules affect the equilibrium expression. Lastly, because the equilibrium constant is a measure of energy distribution, only a change in temperature (a measure of the system's energy) changes the value of K_{eq} .

Consider Reaction 3.2,



Reaction 3.2

Applying the three rules to Reaction 3.2 generates the correct expression for K_{eq} .

We start by plugging in products over reactants: $K_{\text{eq}} = \frac{[\text{B}][\text{C}]}{[\text{A}]}$

Invalid, because stoichiometric coefficients must be exponents. $\therefore K_{\text{eq}} = \frac{[\text{B}][\text{C}]}{[\text{A}]^2}$

Invalid, because K_{eq} does not include solids or liquids. $\therefore K_{\text{eq}} = \frac{[\text{B}]}{[\text{A}]^2}$

K_{eq} can be written as either $K_c = \frac{[\text{B}]}{[\text{A}]^2} \text{M}^{-1}$ or $K_p = \frac{P_{\text{B}}}{(P_{\text{A}})^2} \text{atm.}^{-1}$

Reaction Quotient (Q_{rx})

When a reaction is not at equilibrium, the expression of products over reactants is said to be the *reaction quotient* (Q_{rx}). The relationship between K_{eq} and Q_{rx} dictates the direction in which a reaction proceeds to reach equilibrium. When $K > Q$, the denominator of Q (reactants) is too large and the numerator of Q (products) is too small. To establish equilibrium, the reaction must shift to the right. The reaction shifts to the right to increase the products (numerator) and decrease the reactants (denominator). When $K < Q$, the denominator of Q (reactants) is too small and the numerator of Q (products) is too large. To establish equilibrium, the reaction must shift to the left. The reaction shifts to the left to decrease the products (numerator) and increase the reactants (denominator). A shortcut to determine the direction the reaction proceeds to reach equilibrium involves drawing the relationship of K and Q alphabetically, and then converting the $<$ or $>$ sign into an arrow. For example: $K > Q$ becomes $K \rightarrow Q$, so the reaction moves right to reach equilibrium, because the arrow is pointing to the right.

Example 3.3

When the reaction quotient is greater than the equilibrium constant, which of the following is NOT true?

- A. The system has too many products and too few reactants.
- B. The reaction is displaced from equilibrium.
- C. The reaction must shift in the forward direction to reach equilibrium.
- D. The reverse reaction rate is greater than the forward reaction rate.

Solution

When the reaction quotient is greater than the equilibrium constant, the system has an excess of products and shortage of reactants, relative to equilibrium. Choice A is a valid statement. The system is not at equilibrium, so choice B is a valid statement. To reach equilibrium, the reaction must have a net shift in the reverse direction to reduce the amount of products and increase the amount of reactants. This means that the reverse reaction rate is greater than the forward reaction rate, making choice D valid. The system cannot have a net shift in the forward direction, so choice C is an invalid statement.

Case Specific K-values

All equilibrium constants obey the same rules, but depending on the reaction, there may be special features that recur. Different reactions have special K-values. Table 3.2 lists the six types of K_{eq} values we shall address.

K type	Type of reaction to which the K applies
K_p	K_{eq} for the reaction of gases. Values are in pressure units.
K_c	K_{eq} for the reaction of solutes. Values are in concentration units.
K_{sp}	K_{eq} for salts dissociating into ions. Measures the solubility.
K_a	K_{eq} for acids dissociating in water. Measures the acidity.
K_b	K_{eq} for bases hydrolyzing in water. Measures the basicity.
K_w	K_{eq} for autoionization of water into hydronium and hydroxide.

Table 3.2

Understanding that the rules are the same for all types of K-values will enhance your journey through the wonderful world of equilibrium. Knowing that the common ion effect is nothing more than Le Châtelier's principle applied to solubility systems is a perfect example of how the rules apply to all equilibrium systems. There are just different names to describe the reaction conditions.

 K_{eq} Calculations

Now comes the math part, which we will handle through practice! Getting reacquainted with equilibrium math is a matter of repetition and practice. Once you feel sufficiently familiarized, then move on to the next topic.

Example 3.4

At 650K, the partial pressures of the component gases were determined for the following reaction:



$$P_{\text{H}_2} = 0.20 \text{ atm.}, P_{\text{I}_2} = 1.50 \text{ atm.}, \text{ and } P_{\text{HI}} = 3.00 \text{ atm.}$$

What is the K_p for this reaction?

- A. 10.0 atm.
- B. 15.0 atm.
- C. 30.0 atm.
- D. 45.0 atm.

Solution

Possible errors with this problem stem from forgetting to square the numerator or ignoring I_2 , because you have seen it as a solid before (note that this reaction is at 650K). In this case, forgetting to square the numerator would yield an answer that is too small by a factor of 3. As long as you don't forget these things, the problem merely involves doing your math quickly. The math setup is as follows:

$$K_p = \frac{(P_{\text{HI}})^2}{(P_{\text{H}_2})(P_{\text{I}_2})} = \frac{(3)^2}{(0.2)(1.5)} = \frac{9}{0.3} = \frac{30}{1} = 30$$

Choice C is the correct answer. To make this a more conceptual type of question, they may give the answer choices as ranges rather than exact numbers.

Example 3.5

At 323°C, there are 0.10 moles $\text{H}_2(\text{g})$, 0.20 moles $\text{HClO}_4(\text{g})$, 0.10 moles $\text{H}_2\text{O}(\text{g})$, and 0.36 moles $\text{HCl}(\text{g})$ at equilibrium in a 400-mL flask. What is K_{eq} for this equilibrium mixture?



- A. 0.0362
- B. 0.0724
- C. 1.8000
- D. 44.100

Solution

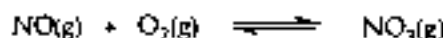
In this problem, the system is already at equilibrium, so no determination of changes (x-values) is required. This problem is of the plug-'n'-chug (simple substitution) type. Moles can be used directly in the equilibrium expression, because although it is technically correct to use concentrations, in this case volume cancels out of both the denominator and the numerator, leaving just the moles. This is true any time that the number of reactants is equal to the number of products in either the gas phase or as solutes. The question is really just a math problem, solved in the following way:

$$K_{\text{eq}} = \frac{(P_{\text{HCl}})(P_{\text{H}_2\text{O}})^4}{(P_{\text{HClO}_4})(P_{\text{H}_2})^4} = \frac{(0.36)(0.10)^4}{(0.20)(0.10)^4} = \frac{0.36}{0.20} = \frac{0.18}{0.10} = 1.8$$

This makes choice C the correct answer.

Example 3.6

At STP, the partial pressure of NO is 152 torr and the partial pressure of O_2 is 228 torr. If the mixture is at equilibrium, what is the K_{eq} at STP for the following reaction?



- A. 8.333 atm⁻¹
- B. 4.167 atm⁻¹
- C. 1.000 atm⁻¹
- D. 0.240 atm⁻¹

Solution

The key piece of information in this question is STP (standard temperature and pressure). This implies that the total pressure of the systems is 760 torr. The sum of the partial pressures is the total pressure, so $760 = P_{\text{NO}_3} + P_{\text{NO}} + P_{\text{O}_2}$. By substitution, $760 = P_{\text{NO}_3} + 152 + 228$, so $P_{\text{NO}_3} = 380$ torr. Because the answer is listed in atmospheres, the values in torr must be converted to atmospheres before they are useful. The conversion is 760 torr per atmosphere. The equation for calculating K_{eq} is:

$$K_{\text{eq}} = \frac{P_{\text{NO}_3}}{(P_{\text{NO}})(P_{\text{O}_2})} = \frac{380/760}{(152/760)(228/760)} = \frac{1/2}{(1/5)(3/10)} = \frac{1/2}{3/50} = \frac{50}{6} > 8$$

Of the answer choices given, only choice A is greater than 8. If you pick A, you will definitely be a star!

Example 3.7

For the following reaction, 0.20 moles $\text{Bi}_2\text{S}_3(\text{s})$ are mixed with 0.50 moles $\text{H}_2(\text{g})$. Once equilibrium is established, 0.225 moles of $\text{H}_2(\text{g})$ remain. What is the value of K_{eq} for this reaction?



- A. 0.57
B. 0.82
C. 1.22
D. 1.81

Solution

This question is best solved by determining the equilibrium concentrations for both of the gases. Solids do not affect the equilibrium, so be sure not to include any solids in the equilibrium expression. The equilibrium expression involves partial pressures, but the answer is the same whether you use the partial pressures of the gases or the concentrations of the gases. This is because there is the same number of gas molecules on each side of the reaction. The solids are ignored altogether as long as they are not the limiting reagent. Because you need three times as much $\text{H}_2(\text{g})$ as $\text{Bi}_2\text{S}_3(\text{s})$ and you have only 2.5 times as much, hydrogen is the limiting reagent in this reaction, if it were to go to completion. This means that to determine the value of K_{eq} , you must determine the ratio of the two gases. The values are found in the following way:

Reaction:	$\text{Bi}_2\text{S}_3(\text{s})$	$3 \text{ H}_2(\text{g})$	$\rightleftharpoons$	$2 \text{ Bi}(\text{s})$	$3 \text{ H}_2\text{S}(\text{g})$
Initially:	0.2	0.5		0	0
Shift:	$-x$	$-3x$	$\longrightarrow$	$+2x$	$+3x$
Equilibrium:	$0.2 - x$	$0.5 - 3x$		$2x$	$3x$

In this case, we can solve for the value of $3x$ from the information given. At equilibrium, there are 0.225 moles of hydrogen gas remaining, so $0.5 - 3x = 0.225$. This means that $3x = 0.275$. There is no need to solve for x , because $3x$ is present in the gas terms, and solids are not going to be considered. Plugging 0.275 in for $3x$ into the equilibrium line of the reaction chart yields:

Reaction:	$\text{Bi}_2\text{S}_3(\text{s})$	$3 \text{ H}_2(\text{g})$	$\rightleftharpoons$	$2 \text{ Bi}(\text{s})$	$3 \text{ H}_2\text{S}(\text{g})$
Equilibrium:	don't care	0.225		don't care	0.275

These numbers work nicely in determining the equilibrium constant. The math is shown below:

$$K_{\text{eq}} = \frac{(P_{\text{H}_2\text{S}})^3}{(P_{\text{H}_2})^3} = \frac{(\text{moles}_{\text{H}_2\text{S}})^3}{(\text{moles}_{\text{H}_2})^3} = \frac{(0.275)^3}{(0.225)^3} = \left(\frac{0.275}{0.225}\right)^3 = \left(\frac{11}{9}\right)^3 = 1.22^3 = 1.22^*$$

Only choice D is a value that is greater than 1.22, so that is the best answer. If you forget to cube the value, it is easy to choose answer choice C by mistake. Choices A and B are eliminated, because there are more products than reactants, so the value of K_{eq} must be greater than 1.0. Be sure to use common sense to eliminate incorrect answer choices. Developing intuition and learning to trust your common sense is more important in MCAT preparation than honing your algebra and multiplication skills.

Using K_{eq} to Calculate Shifts

The equilibrium constant is used to calculate the quantity of the products and the reactants present at equilibrium. This is achieved by following the moles of each species from initial conditions to equilibrium conditions, specifically by setting up a table to keep track of the components systematically over the course of the reaction. There are three stages to consider: the initial stage, the shift, and equilibrium. Consider Reaction 3.3:

**Reaction 3.3**

At 375°C, the equilibrium constant for Reaction 3.3 is 2.51×10^1 . The equilibrium constant is unitless when the number of products equals the number of reactants. Consider the reaction to start with 1.0 atm of CO and 1.0 atm. of H₂O. Figure 3-1 shows the setup for determining the shift and final pressures in Reaction 3.3.

Reaction:	CO(g)	H ₂ O(g)	$\rightleftharpoons$	CO ₂ (g)	H ₂ (g)
Initially:	1.00	1.00		0	0
Shift:	-x	-x	$\longrightarrow$	+x	+x
Equilibrium:	1.00 - x	1.00 - x		x	x

Figure 3-1

As mentioned, there are three considerations in the setup. The first line is what is initially given. The second line shows the direction of the shift and the stoichiometric consequences. You must be able to determine the direction of the shift by comparing the initial concentrations to the equilibrium distribution. In this case it was easy, because there are no products, so the reaction must shift in the forward direction. The third line accounts for what is present once equilibrium is established. Values from line 3 are plugged into the equilibrium expression, to solve for x, the shift in the reaction. The solution is as follows:

$$K_{eq} = \frac{(P_{\text{CO}_2})(P_{\text{H}_2})}{(P_{\text{CO}})(P_{\text{H}_2\text{O}})} = \frac{(x)(x)}{(1-x)(1-x)} = 25.1 \Rightarrow 25.1 = \frac{x^2}{(1-x)^2}$$

$$25.1 = \frac{x^2}{(1-x)^2} \Rightarrow 5 = \frac{x}{1-x} \Rightarrow 5 \cdot 5x = x \Rightarrow 5 = 6x \therefore x = \frac{5}{6} = 0.83$$

The math was simpler than it first appeared in this case. The MCAT does not seek to test your algebra skills, as much as it tests your reasoning ability. Being able to estimate the magnitude of the x (the shift) relative to the initial values is important.

Example 3.8

At 773K, the K_p for the following reaction is 3.0×10^{-5} . If the partial pressure of N₂(g) is initially 3.75 atm., and the partial pressure of H₂(g) is initially 2.0 atm., what is the partial pressure of NH₃(g) once equilibrium is established, assuming there is no ammonia in the system initially?



- A. 0.0900 atm.
- B. 0.0300 atm.
- C. 0.0100 atm.
- D. 0.0010 atm.

Solution

This is a straightforward example. Because the reaction starts with all reactants, it shifts forward to reach equilibrium. The reaction has a very small K_{eq} and starts with all reactants, so the shift is small. The x -term can be ignored when it is subtracted from or added to numerical values. The setup is as follows:

Reaction:	$N_2(g)$	$3 H_2(g)$	$\rightleftharpoons$	$NH_3(g)$
Initially:	3.75	2.00		0
Shift:	$-x$	$-3x$	$\longrightarrow$	$+2x$
Equilibrium:	$3.75 - x$	$2.00 - 3x$		$2x$

Ignoring the x and $3x$ portions of the reactant quantities yields the following:

$$K_{eq} = \frac{(P_{NH_3})^2}{(P_{N_2})(P_{H_2})^3} = \frac{(2x)^2}{(3.75)(2)^3} = 3 \times 10^{-5} \Rightarrow 3 \times 10^{-5} = \frac{4x^2}{3.75 \times 8} = \frac{4x^2}{30}$$

$$x^2 = \frac{30}{4} \times 3 \times 10^{-5} = 22.5 \times 10^{-5} \Rightarrow x^2 = 2.25 \times 10^{-4} \therefore x = \sqrt{2.25} \times 10^{-2} = 1.5 \times 10^{-2}$$

The concentration of NH_3 at equilibrium is $2x$, which is $3.0 \times 10^{-2} M$. The best answer is choice B. This is considerably more math than the MCAT requires you to use, so think of this example as merely a step towards getting re-acquainted.

A concern you may recall from your general chemistry class involves whether it is safe to ignore x when it is either added to or subtracted from a numerical term. The x -value is ignored when the initial conditions are like the equilibrium conditions, because the shift is minimal and the value of x is trivial. In Example 3.8, K_{eq} is less than 1.0, so there are fewer products than reactants at equilibrium. This means that hardly anything shifts over to the product side, so x can be ignored. Table 3.3 shows cases when x can and cannot be ignored.

Initial Conditions	Equilibrium Constant	Shift	x -term?
All Reactants	$K_{eq} < 10^{-3}$	Q and K are SIMILAR $\therefore$ SMALL x SMALL shift in the forward direction	Ignore
All Products	$K_{eq} < 10^{-3}$	Q and K are DIFFERENT $\therefore$ LARGE x LARGE shift in the reverse direction	Consider
All Reactants	$K_{eq} > 10^3$	Q and K are DIFFERENT $\therefore$ LARGE x LARGE shift in the forward direction	Consider
All Products	$K_{eq} > 10^3$	Q and K are SIMILAR $\therefore$ SMALL x SMALL shift in the reverse direction	Ignore
Reactants & Products	$K_{eq} = 1$	Q and K are SIMILAR $\therefore$ SMALL x SMALL shift in either direction	Ignore
Reactants & Products	$K_{eq} \gg$ or $\ll 1$	Q and K are DIFFERENT $\therefore$ LARGE x LARGE shift in either direction	Consider

Table 3.3

Understanding the math associated with equilibrium is important. K_{eq} can be used to find concentration values at equilibrium, and equilibrium concentration values can be used to find K_{eq} . This fits nicely into an MCAT experiment, because equilibrium constants result from experiments conducted to determine equilibrium concentrations, in several trials with different starting conditions.

Example 3.9

At 92.2°C, the K_p for the following reaction is 0.2000 atm^{-1} . If you were to place exactly 0.200 atm. of $\text{N}_2\text{O}_4(\text{g})$ into a 1.00 liter vessel, what would the partial pressure of $\text{NO}_2(\text{g})$ be once equilibrium was established?



- A. 0.025 atm. $\text{NO}_2(\text{g})$
- B. 0.200 atm. $\text{NO}_2(\text{g})$
- C. 0.350 atm. $\text{NO}_2(\text{g})$
- D. 0.400 atm. $\text{NO}_2(\text{g})$

Solution

To solve this precisely, it is necessary to use the quadratic equation. On the MCAT, it is unlikely that you will need to use the quadratic equation. This question emphasizes the technique of eliminating answer choices through approximation. First, you must use the equilibrium expression to estimate the magnitude of x . In this case, K_{eq} is less than 1.0, and the reaction starts with all products. The value of x is going to be significant (more than half shifts over). If half of the 0.200 atm. of N_2O_4 shifts over, then the partial pressure of nitrogen dioxide is 0.200 atm. Considering that more than half of the $\text{N}_2\text{O}_4(\text{g})$ is going to shift, the value of $\text{NO}_2(\text{g})$ is greater than 0.200 atm. However, not all of the $\text{N}_2\text{O}_4(\text{g})$ can shift over (which would result in 0.400 atm. of NO_2), so the answer must be less than 0.400 atm. Only choice C falls within the range of 0.200 to 0.400. In a multiple-choice format, this question is rather easy to answer.

In the interest of developing an alternative method to answer the question, substitute one of the four answer choices (the one closest to the x you approximate) into the equilibrium expression and then compare the answer you get to the given value of K_{eq} . Your answer either will equal K_{eq} or it won't equal K_{eq} . If it equals K_{eq} , you picked well (go to Las Vegas or Atlantic City with those skills). If it does not equal K , the error can be used to zero in on the correct answer, depending on whether your value is too high or too low. For this reason, always start by substituting a middle value.

Reaction:	$2 \text{NO}_2(\text{g})$	$\rightleftharpoons$	$\text{N}_2\text{O}_4(\text{g})$
Initially:	0		0.200
Shift:	$+2x$	$\leftarrow$	$-x$
Equilibrium:	$2x$		$0.20 - x$

These numbers should be plugged into the equilibrium expression.

$$K_p = \frac{P_{\text{N}_2\text{O}_4}}{(P_{\text{NO}_2})^2} = \frac{0.2 - x}{(2x)^2} = \frac{0.2 \cdot x}{4x^2}$$

Be sure not to substitute the answer choice values directly for x , because the values in the answer choices are for P_{NO_2} which is $2x$. Choices B and C fall in the middle of the range, so choose either one to plug in. For numerical ease, let's use choice B. If P_{NO_2} is 0.200 atm., then x is 0.100.

$$K_p = \frac{0.2 - x}{(2x)^2} = \frac{0.2 - 0.1}{(0.20)^2} = \frac{0.10}{0.04} = \frac{10}{4} = 2.5$$

This number (2.5) is too large (greater than 0.2000), which means that the x we chose was too small, so we need to choose a larger value for x . Choice C is best.

Complex Equilibrium

A complex equilibrium is a balance between two separate reactions that share a common reagent. In essence, it is two reactions whose equilibrium states depend on one another. In a complex equilibrium, a product in one reaction is a reactant in the other reaction. As a result, if one reaction is displaced from equilibrium, then the other reaction is also affected. When adding the component reactions, the common reagent is absent in the overall reaction and is considered to be an intermediate. The equilibrium constant for the overall reaction is found by multiplying the individual equilibrium constants of the component reactions. This is because the reagent that disappears when you add the reactions is in the numerator of one reaction and in the denominator of the other reaction. To eliminate it, K_{eq} s are multiplied. Figure 3-2 is a sample of complex equilibrium:

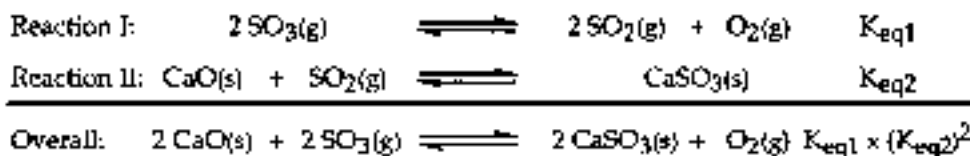
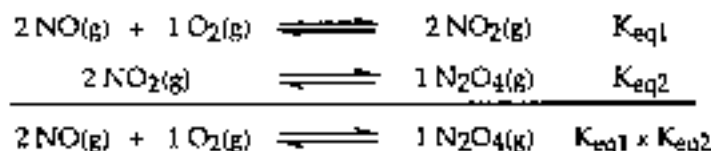


Figure 3-2

Reaction II is multiplied by 2 to balance SO_2 when you add the reactions. Because Reaction II is multiplied by 2, its K_{eq} value is squared. The addition of calcium oxide to an equilibrium mixture of O_2 , SO_2 , and SO_3 reduces the partial pressure of SO_3 . This occurs despite the fact that sulfur trioxide and calcium oxide do not react directly. Complex equilibrium questions are essentially just manipulations of the overall equilibrium and of Le Châtelier's principle.

Example 3.10

Consider the following complex equilibrium:



When N_2O_4 gas is removed, how are the partial pressures of NO gas and NO_2 gas affected?

- A. P_{NO} and P_{NO_2} both decrease.
- B. P_{NO} and P_{NO_2} both increase.
- C. P_{NO} decreases, while P_{NO_2} remains the same.
- D. P_{NO} increases, while P_{NO_2} remains the same.

Solution

The second reaction shifts to the right to compensate for the loss of N_2O_4 , so the partial pressure of NO_2 decreases. A decrease in NO_2 causes the first reaction to shift to the right as well, resulting in a decrease in the partial pressure of NO gas and an increase in the partial pressure of NO_2 . However, the increase in NO_2 from the forward shift of the first reaction is less significant than the decrease in NO_2 caused by the forward shift of the second reaction. This is because the shift in the first reaction cannot completely replenish the lost NO_2 without losing so much NO that the reaction is beyond equilibrium. A shift never regenerates as much as was lost. Both NO and NO_2 decrease, making choice A the best answer.

Experimental Determination of K

In this experiment, a 500-mL closed glass flask is employed, so the system is isochoric (has V_{constant}). The pressure of the system is monitored continually over the course of the experiment. The researcher tests Reaction 3.4 by observing the total pressure of the system during a fixed interval.

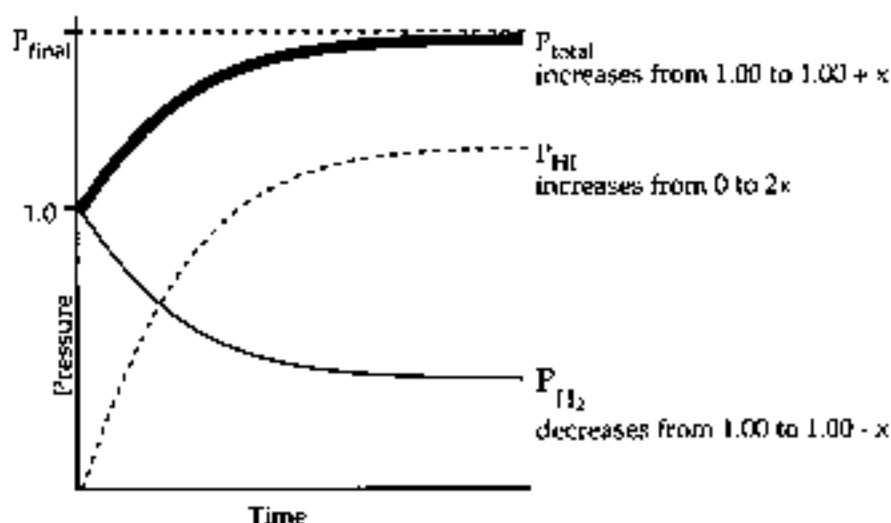
**Reaction 3.4**

The researcher mixes together 1.00 atm of hydrogen gas (H_2) with excess iodine solid (I_2). The total pressure of the system is monitored by a small detector in the wall of the flask throughout the reaction, which is said to be in equilibrium once the total pressure of the system stabilizes. Ample time is given to allow the system to reach equilibrium. The setup for the reaction is shown in Figure 3-3.

Reaction:	$\text{H}_2(\text{g})$	$\text{I}_2(\text{s})$	$\rightleftharpoons$	$2 \text{HI}(\text{g})$
Initially:	1.00	excess		0
Shift:	$-x$	$-x$	$\longrightarrow$	$+2x$
Equilibrium:	$1.00 - x$	who cares?		$2x$

Figure 3-3

The reaction proceeds forward to reach equilibrium. It is unknown how much the reaction proceeds, so the shift value is assigned the term x . Hydrogen gas decreases by x , while hydroiodic acid gas increases by $2x$, because of the stoichiometry. Because the number of gas molecules increases, so does the total pressure of the system. Figure 3-4 shows the total pressure of the system as a function of time. The pressures for H_2 and HI are inferred from the total pressure graph.

**Figure 3-4**

K_{eq} is calculated from P_{total} , because the shift in the reaction (x) is equal to ΔP . For instance, if P_{total} is 1.60 atm. and P_{initial} is 1.00 atm., then ΔP is 0.60 atm. If ΔP is 0.60 atm., then at equilibrium P_{HI} is 1.20 atm. and P_{H_2} is 0.40 atm. The equilibrium expression for the reaction does not include iodine, because it is a solid. Plugging into the equilibrium expression yields a numerical value for K_{eq} .

Determination of K

$$K_{eq} = \frac{(P_{I_2})^2}{P_{H_2}} = \frac{(2x)^2}{1.0-x} = \frac{(1.2)^2}{0.4} = \frac{(1.2)(1.2)}{0.4} = \frac{1.2(1.2)}{0.4} = (3)(1.2) = 3.6$$

This experiment determines the value of K_{eq} under the ambient reaction conditions. Changes to the system are made to study equilibrium further.

Effect of the Addition of Reactant

Once the pressure stabilizes at 1.60 atm., the researcher alters the system by adding 0.10 atm. of isotopically rich hydrogen gas (predominantly molecular deuterium, $^2\text{H}-^2\text{H}$). The total pressure of the system increases as hydrogen gas is added to the system. After the addition is complete, the total pressure continues to increase, gradually slowing until it reaches a stable value of 1.76 atm. P_{total} for the system increases by 0.16 atm., more than the 0.10 atm. added. Analysis using infrared spectroscopy confirms that both $^2\text{H}-\text{I}$ and $^2\text{H}-\text{H}$ are formed. From these observations, it is concluded that addition of a reactant gas shifted the reaction in the forward direction to a new distribution of compounds. Substituting the values for P_{H_2} and P_{I_2} in the equilibrium expression reveals that K_{eq} remains the same, although the partial pressure of each gas has changed. This means that addition of the reactant gas displaces the reaction from equilibrium, following which the reaction shifts in such a way as to reestablish equilibrium. In addition, the equilibrium is *dynamic*, given that the mixed hydrogen ($^2\text{H}-\text{H}$) forms by the reverse reaction.

Following the addition of the labeled hydrogen, the researcher makes a second addition to the system, after the pressure stabilizes at 1.76 atm., by adding 10.0 g of isotopically rich iodine (predominantly $^{127}\text{I}-^{127}\text{I}$). The system is continually monitored, but no change in pressure is detected. Again, the gases in the system are analyzed by infrared spectroscopy. The labels appear to be completely scrambled, with $\text{H}-^{127}\text{I}$, $^2\text{H}-^{127}\text{I}$, $\text{H}-\text{I}$, and $^2\text{H}-\text{I}$ all being observed. From this, it is concluded that the addition of a solid reactant does not shift the reaction in either direction. It reconfirms that the equilibrium is *dynamic*, given that the mixture of isotopically labeled compounds.

A major point of this experiment is to support the concept of *dynamic equilibrium*, a state where the system is continually reacting in both the forward and reverse directions. On the macroscopic level, there is no net change. This implies that the forward and reverse reaction rates are equal and that the concentration of the reactants and products remains constant. The scrambling of the isotopic labels supports this idea. If this were a case of *static equilibrium*, the isotope would not be incorporated into the product or back-react to form the mixed isotope reactant. Any set of questions on the MCAT that accompanies the description of an experiment like this, would include a question on dynamic equilibrium.

Such an experiment is typical for an MCAT physical sciences passage. It is to your advantage to ponder what types of questions might be asked. Questions could involve total pressure and its relationship to equilibrium through partial pressure. The shift can be rationalized using Le Châtelier's principle, so questions on Le Châtelier's principle are probable. To round out the questions, there is the possibility of pH questions, solubility questions, and the relationship between the reaction quotient (Q) and the equilibrium constant (K). It is more than worth your time to make up some multiple-choice questions to accompany this passage. If you know how to write a multiple-choice test, you will understand better how to take one.

Le Châtelier's Principle

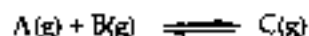
Le Châtelier's principle is justified both mathematically and theoretically. The basic rule is that whatever change you make to a system that is in equilibrium, the reaction mixture will react in a way to undo the change and reestablish equilibrium. The formal definition of the principle is as follows:

If an external stress is applied to a system at equilibrium, the system will shift itself in such a way that the stress is partially relieved and equilibrium is reestablished.

Effects of Stress

An external stress on the system includes changing moles (concentration of a component), pressure, volume, and temperature. In all cases, except changing temperature, the numerical value of K remains constant, and the system will continue to react until the K value is obtained once again. When a reaction system is not in a state of equilibrium, the same calculation for the equilibrium expression (K) is used, but it is referred to as the reaction quotient (Q). If K is not equal to Q , then the system is not in equilibrium. It is important to note that changing an equilibrium system by adding a pure solid or a pure liquid does not disrupt the equilibrium nor change the equilibrium constant.

This is why in the sample experiment on the preceding page, when iodine crystals were added (a solid), the system was not moved from equilibrium. However, when molecular hydrogen was added (a gas), the system was moved from equilibrium, and it shifted accordingly, to reestablish equilibrium. It is important that this conceptual view of equilibrium makes sense to you, because the MCAT is a conceptual exam that evaluates your understanding at this level. We shall look at Le Châtelier's principle in terms of mathematics and intuitive problem-solving. We shall address the mathematical relationships only to the extent necessary to support our observations. In generic Reaction 3.5, the mathematical aspects of Le Châtelier's Principle are considered.



Reaction 3.5

Let's assume that K_{eq} for the reaction is 1.00 and that initially, the partial pressure of all three components is 1.00 atm. By substitution into the equilibrium expression, we see that the value is 1.0, meaning that the system is in equilibrium. Upon doubling the external pressure (i.e., by applying a stress to the equilibrium), the system is displaced from equilibrium. According to Le Châtelier's principle, it will shift to reduce the pressure of the system, which we predict will be in the forward direction. By shifting in the forward direction, the reaction goes from the side with two molecules to the side with one molecule, thereby reducing the number of molecules and thus reducing the pressure. The math to support this prediction is as follows:

Reaction:	A(g)	+	B(g)	$\rightleftharpoons$	C(g)	$C/A \cdot B$	State
Initially:	1.0		1.0		1.0	1.00	equilibrium
After Stress:	2.0		2.0		2.0	0.5	not equilibrium
Shift:	-x		-x	$\longrightarrow$	+x		reacting
Final:	2 - x		2 - x		2 + x	1.00	new equilibrium

The external pressure increased, causing internal pressures to increase. This generated a system where $K < Q$, so to reestablish equilibrium, the reaction shifts forward until once again $Q = K$.

The numerical value of K does not change with the change in pressure, although the equilibrium distribution does change.

$$K_{eq} = \frac{P_C}{(P_A)(P_B)} = \frac{2+x}{(2-x)^2} = \frac{x+2}{x^2-4x+4} = 1 \Rightarrow x^2-4x+4 = x+2 \Rightarrow x^2-5x+2 = 0$$

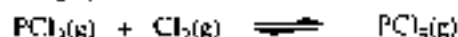
Using the quadratic equation yields a value of roughly 0.44, which means that the new partial pressures are $P_A = 1.56$ atm., $P_B = 1.56$ atm., and $P_C = 2.44$ atm. Table 3.4 shows different equilibrium distributions for Reaction 3.5. You should notice that the ratio of the compounds varies, but the value of K_{eq} is constant.

P_A	P_B	P_C	P_C/P_A	$P_C/P_A \cdot P_B$
0.80	0.80	0.64	0.80	1.000
0.90	0.90	0.81	0.90	1.000
1.00	1.00	1.00	1.00	1.000
1.10	1.10	1.21	1.10	1.000
1.25	1.25	1.56	1.25	1.000
1.50	1.50	2.25	1.50	1.000
1.56	1.56	2.44	1.56	1.003
0.80	1.25	1.00	1.25	1.000
1.20	1.50	1.80	1.50	1.000
1.33	0.50	0.67	0.50	1.000

Table 3.4

Example 3.11

What is the observed result of increasing the total pressure under isothermal conditions in the following system initially at equilibrium?



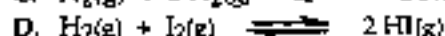
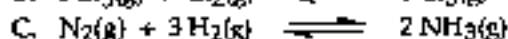
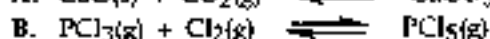
- A. No change in the P_{PCl_3} to P_{PCl_5} ratio
- B. An increase in the P_{Cl_2} to P_{PCl_5} ratio
- C. A decrease in the ratio of P_{PCl_5} to $(P_{PCl_3}) \times (P_{Cl_2})$
- D. An increase in the P_{PCl_5} to P_{PCl_3} ratio

Solution

Increasing the external pressure shifts the reaction to the right, so it can reduce the pressure to counteract the stress associated with increasing the pressure of the system. An alternative way to look at this is to say that when the total pressure increases, the number of collisions between molecules increases, forcing the reaction to proceed in the forward direction (only the forward reaction depends on collisions). If the system shifts to the right, the partial pressure of PCl_3 decreases, the partial pressure of Cl_2 decreases, and the partial pressure of PCl_5 increases. This eliminates choice A, because choice A could be true only if the reaction did not shift. Choice B is eliminated, because an increase in the Cl_2 -to- PCl_5 ratio results from a reaction shifting in the reverse direction. Choice C is eliminated, because the equilibrium constant does not change unless the temperature changes. The change is said to be isothermal, so temperature did not change during the shift. Choice D results from the reaction shifting in the forward direction, so it is the best answer.

Example 3.12

Which of the following reactions at equilibrium would NOT shift as the result of an increase in pressure?

**Solution**

The equilibrium reaction does not shift after an increase in pressure when there is an equal number of gas molecules on each side of the reaction. This is observed in choice D. As a point of interest, CaCO_3 cannot be stored in an open space for this reason, because it will fully dissociate into CaO and CO_2 over time, if the carbon dioxide partial pressure is not high enough. Also take note that I_2 is a gas in choice D, which means that the reaction must be at a temperature above ambient conditions.

Perturbations and Shifts

Before conducting an experiment, it is important to consider the properties of the container in which a reaction transpires. Two common containers are the closed flask and the closed piston. Flasks are made of a rigid material, so their volume is fixed (V_{constant}). The pressure varies when using a flask (P_{variable}). Pistons are flexible (a wall of the piston is free to move), so the volume can vary (V_{variable}). In piston reactions, the initial pressure equals the final pressure ($P_{\text{initial}} = P_{\text{final}}$), if the volume of the piston exhibits no instantaneous change. This is to say that for a piston that starts with a stationary lid, once the lid is stationary again, the internal pressure equals the external pressure. As a result, reactions are susceptible to environmental changes (perturbations) depending on the container. Temperature may change in each container, but its effect on pressure and volume depend on the container itself. Le Châtelier's principle should be applied taking into account the features of the reaction container.

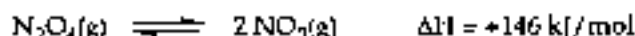
Le Châtelier's principle deals with changes to a system that starts in equilibrium. When you have a system in balance, a change in environment results in the disturbance of equilibrium. To compensate, the system will shift either left or right (increasing and decreasing concentrations in doing so) to reestablish equilibrium. The general rule is that the system will do whatever it takes to undo or compensate for what you have done to disturb it. Table 3.5 lists the generic shifts a reaction undergoes to alleviate an applied stress.

Applied Stress	System's Adjustment	Direction of Shift
Add reactant	Remove reactant	To the Right
Add product	Remove product	To the Left
Decrease volume	Decrease gas volume	To the side with fewer molecules
Increase volume	Increase gas volume	To the side with more molecules
Decrease temperature	React to generate heat	In the exothermic direction
Increase temperature	React to absorb heat	In the endothermic direction

Table 3.5

Direction of Shift

To observe how these shifts work, each is applied to Reaction 3.6.



Reaction 3.6

Situation #1 Add $\text{N}_2\text{O}_4(\text{g})$: This perturbation results in a system with too many reactants. The value of K is greater than Q . To reestablish equilibrium, the system reacts in the forward direction to absorb some of the excess reactants and form more products. The system shifts to the right. The same result would have occurred had $\text{NO}_2(\text{g})$ been removed.

Situation #2 Add $\text{NO}_2(\text{g})$: This perturbation results in a system with too many products. The value of K is less than Q . To reestablish equilibrium, the system reacts in the reverse direction to form more reactants and absorb some of the excess products. The system shifts to the left. The same result would have occurred had $\text{N}_2\text{O}_4(\text{g})$ been removed.

Situation #3 Increase the external pressure: This perturbation results in a system where the partial pressures are too high. This means that the space in which the gases coexist has been reduced, so their molecules are more crowded and collide more often. To reduce this crowding, the system goes from products (two molecules) to reactants (one molecule) and reduce the total moles of gas in the container. Also, if they collide more often, this forces the NO_2 molecules to form bonds and thus dimerize to N_2O_4 . The system shifts to the left.

Situation #4 Increase the volume: This perturbation results in a system where the concentrations are too low. This means that the space in which the gases coexist has increased, so they are less crowded and collide less often. Using the inverse of the reasoning from situation #3, the system reacts to make two molecules of $\text{NO}_2(\text{g})$ from $\text{N}_2\text{O}_4(\text{g})$. The system shifts to the right.

Situation #5 Heat the system: This perturbation results in a system where there is too much free energy. By heating the system, we have added energy to the reaction. To consume most of this additional energy, the system reacts in the endothermic direction, which for Reaction 3.6 is the forward direction. The system shifts to the right.

Situation #6 Cool the system: This perturbation results in a system where there is too little free energy. By cooling the system we have taken energy away from the reaction. To regenerate energy to balance this loss of energy to some degree, the reaction moves in the exothermic direction, which for Reaction 3.6 is the reverse direction. The system shifts to the left.

These scenarios all address a gas-phase equilibrium. The shifts are similar for a solution-phase equilibrium, except that concentrations are considered, rather than volume changes of the container. Changes in the concentration can result from changes in the volume (quantity) of solvent. The only effect is that situations #3 and #4 are now dilution and evaporation of solvent. The system still reacts by asserting the inverse of the stress done upon it. When diluted, it reacts to increase its concentration. When solvent is removed, increasing the concentration, it reacts to reduce the concentration. The rules of Le Châtelier's principle work very well, if you apply them correctly.

Example 3.13

Which of the following is the result of cooling an endothermic reaction that starts at equilibrium?

- A. The amount of products decreases.
- B. The amount of reactants decreases.
- C. The equilibrium constant increases.
- D. All partial pressures increase.

Solution

Cooling the system causes it to move in the direction that generates heat. Heat is released when a reaction that is endothermic as written goes from products to reactants. Endothermic reactions are viewed as follows:



Heat can be treated as a reactant. As the reaction shifts to the left, the amount of products decrease, the amount of reactants increase, and the equilibrium constant decreases. This eliminates choices B and C, and makes choice A the correct answer. Cooling a system decreases the pressure, so in all likelihood, all partial pressures would decrease, not increase. Even if the reaction shifts enough to offset the pressure decrease due to reduced temperature, that is true only for reactants. Partial pressure of products must decrease. This eliminates choice D. You should recall that ΔH is positive for an endothermic reaction.

Example 3.14

If the following reaction represents a system at equilibrium, indicate which statement is NOT true.



- A. Increasing the pressure would cause a decrease in PCl_3 .
- B. Adding PCl_5 would cause an increase in PCl_3 .
- C. Increasing the volume would cause an increase in PCl_3 .
- D. Removing Cl_2 would cause a decrease in PCl_3 .

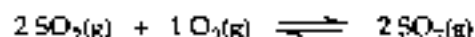
Solution

The term "NOT true" means false. Do not forget halfway through the problem that you're looking for the *false* answer choice. A common mistake on "NOT" problems is to forget that the correct answer is a false statement. To avoid this, write either "T" or "F" next to each answer choice as you run through them. Then choose the answer choice with the unique letter.

To compensate for increasing external pressure, the reaction shifts to the right, so it can reduce the pressure to counteract the increase in pressure felt from the change to the system. If the system shifts to the right, PCl_3 decreases, so choice A is true. Adding a product to an equilibrium mixture shifts the reaction to the reactant side in order to counteract the increase in PCl_5 (the product). If the system shifts to the left, PCl_3 increases, so choice B is true. Increasing the volume decreases the crowding in of the system. To compensate for increasing volume, the reaction shifts to the left, so it fills in the empty space created by the volume increase. If the system shifts to the left, PCl_3 increases, so choice C is true. Removing a reactant shifts the reaction to the reactant side in order to make up for the lost reactant (Cl_2). If the system shifts to the left, PCl_3 increases, so choice D is a false statement, and is thus our answer choice.

Example 3.15

For the following equilibrium reaction, what is the effect of increasing the external pressure?



- A. A decrease in the moles of sulfur dioxide
- B. An increase in the moles of oxygen
- C. A decrease in the moles of sulfur trioxide
- D. An increase the equilibrium constant

Solution

Increasing the pressure crowds the molecules in a reaction and forces the reaction to shift to the side of the equilibrium with fewer gas molecules (in this case, the product side). This means that the moles of products (sulfur trioxide) increase and the moles of reactants (sulfur dioxide and oxygen) decrease. This eliminates choices B and C, and makes choice A the best answer. The equilibrium constant does not change with pressure changes, so choice D is eliminated.

Example 3.16

NO_2 is a brown gas, while N_2O_4 (the product of dimerization) is a clear gas. The two are in equilibrium in a 1.00 liter flask. Upon heating, the contents of the flask became darker brown. What can be said about the following reaction as written?



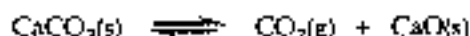
- A. The reaction is endothermic as written.
- B. The reaction is exothermic as written.
- C. The reaction is isothermal as written.
- D. The reaction is adiabatic as written.

Solution

Because the reaction produced a darker mixture with the addition of heat, the reaction must have shifted to the left, since it formed more of the brown gas ($\text{NO}_2(\text{g})$). This means that heat is acting as a product in this reaction. If heat is a product, then the reaction is exothermic as written. This makes choice B correct. The term *isothermal* means that there was no change in temperature during the reaction, and the term *adiabatic* means that there was no change in heat during the reaction. Both of these statements (choices C and D) are incorrect.

Example 3.17

What is the effect of adding $\text{CaO}(\text{s})$ to the following equilibrium mixture?



- A. The products will decrease.
- B. The reactants will increase.
- C. The equilibrium constant will increase.
- D. There is no change in the equilibrium

Solution

Adding a solid to a reaction mixture that is already at equilibrium has no effect on the equilibrium. This means that the correct answer is choice D. A solid affects a reaction mixture only if the reaction is not yet at equilibrium. This is to say that the solid is involved only when it is the limiting reagent for the reaction.

Solubility

Solubility is a popular topic on the MCAT. However, for many people, it was a topic covered only briefly during their general chemistry course. For this reason, we shall look thoroughly at solubility. Conceptually, solubility is the breaking apart of a lattice to allow particles to move freely in solvent. In this section, we focus on the dissociation of a salt into water. When a salt dissociates into water, the ionic forces holding the charged species together break, and interactions with the dipoles of the water form. The fundamental rationale behind solubility involves lattice energy (the interionic forces in the crystal form), the solvation energy (the strength of the attraction between solvent and dissociated ions), and entropy (the solute form is more disordered than the crystalline form). Whether a salt remains in its crystalline form or dissociates into solution depends completely on the relative energetics of the three key features. Figure 3-5 shows the process of dissociation into solvent (water) for sodium chloride.

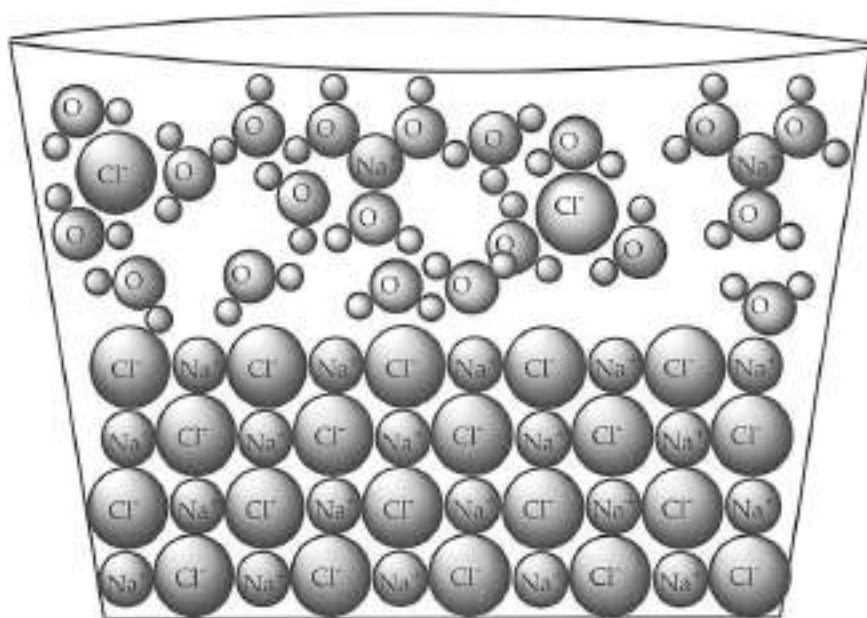


Figure 3-5

The base of the container holds the undissociated salt, while above the salt are ions in solution. Salts dissociate from the surface, where they have the least ionic interactions with neighboring ions in the lattice, and thus are held least tightly. Surface ions have the greatest contact with solvent, so they are more susceptible to solvation. Corners dissolve away fastest, followed by edges and then faces.

Definitions

For every solute, there is a maximum amount that dissolves into a given volume of solvent at a set temperature. The solubility of most salts has been measured at 25°C and are compiled in tables (like Table 3.8). You should know how to extract information from a table. Information listed in solubility tables measures the degree to which a salt dissociates into water. As a rule, some salts have a very low solubility, leading to a low numerical value for the equilibrium constant associated with dissociation (K_{sp}). The equilibrium constant associated with dissociation is known as the *solubility product*. However, the relative K_{sp} values for a group of salts is not always a good indicator of their relative solubility. This is because the units of solubility product (K_{sp}) vary with the number of ions.

The subject of solubility requires being familiar with several terms. Defined below are eleven that are regularly encountered in this subject.

Dissolving: The breakdown of intermolecular forces between molecules as a solid becomes a solute within a solvent. The molecule remains intact when dissolving into solution. An example is the dissolving of sucrose into water, where atoms in the sucrose molecule remain covalently bonded, but the forces between sucrose molecules are eliminated.

Dissociation: The breakdown of ionic bonds between atoms within a lattice structure as a salt turns into a solute within the solvent. The crystal lattice of the salt breaks apart when it dissociates into solution. An example is sodium chloride dissociating into water. The ionic bonds between sodium cations and chloride anions break, and the ions are stabilized by the partial charges of water.

Solvent: The species in greatest concentration into which the solute dissolves, or salt dissociates. A solvent must be a fluid (have the ability to flow).

Solute: The species not in highest concentration that dissolves into the solvent, or in the case of a salt, dissociates.

Solubility: A measurement of the degree of dissolving that a solute undergoes within a particular solvent. The driving force for solubility is a preference for solvation of molecules (or ions) over the lattice strength of the solid. In addition, entropy favors the dissolving process. As the solubility of a compound increases, it is deduced that either the lattice energy of the solid is decreasing, the solvation energy of the solute form is increasing, or both effects are taking place.

Saturated: Describes the state of a solution at the point where no more solid (solute) can dissolve into solution. When an aqueous salt solution is saturated, the rate of dissociation of the salt equals the rate of precipitation.

Supersaturated: Describes the state of a solution where the amount of solid (solute) that is dissolved into solution is beyond the maximum amount at a given temperature. The solution is actually a suspension that when disturbed can form a precipitate rapidly. This state can be achieved by first heating a solvent, then adding solute to the solution until the solution is saturated at that temperature. Slowly cooling this solution causes the amount of solute in it to exceed what should dissolve at the reduced temperature.

Solubility product (K_{sp}): The equilibrium constant for a dissociation reaction, determined from the molar solubility according to standard rules for calculating equilibrium constants.

Molar solubility: The quantitative measurement of the maximum number of moles of solid (solute) that can dissolve into enough solvent to make one liter of solution under standard conditions. For all practical purposes, the solvent is always water in inorganic chemistry and thus the calculations are similar in nearly every example. Molar solubility can be thought of as the x -value in the calculation of the solubility product (K_{sp}).

Gram solubility: The quantitative measurement of the maximum number of grams of solid (solute) that can dissolve into enough solvent to make one hundred milliliters of solution under standard conditions.

Common ion effect: This results in a reduction in the amount of solid (solute) that can dissolve into solution due to the presence in the solution of an ion that is also present in the solid. This concept is similar to Le Châtelier's principle, except that with Le Châtelier's principle, the addition of one of the products (ions) causes precipitation (reduced solubility). With the common ion effect, the ion causing the reduced solubility is present in solution at the beginning of the reaction, rather than being added once the solution has reached a solubility equilibrium.

Solubility Rules

You may recall from general chemistry a list of general rules for predicting solubility. We will emphasize using data from tables to determine solubility, but here is a paraphrased version of the solubility rules. The rules as they are typically written in general chemistry textbooks are listed on page 26 of this book. These rules may be used to give a good estimate of solubility and should be used when no solubility data are available.

1. Most salts composed of a +1 cation (excluding transition metals) and -1 anion are soluble in water at room temperature.
2. Nitrate (NO_3^-) is a large anion that forms weak lattice interactions and forms strong hydrogen bonds with water, so most nitrate salts are water-soluble.
3. Most salts containing sulfate anions (SO_4^{2-}) with +1 cations (excluding transition metals) are water-soluble.
4. Most salts with -2 or -3 anions are insoluble in water, excluding the sulfate salts.
5. Most oxide (O^{2-}) and hydroxide anion (OH^-) salts are only slightly water-soluble. KOH and NaOH are notable exceptions that are substantially soluble.

Ionic Structures

Composed of ions (charged species) held together by ionic bonds (electrostatic forces). In ionic structures, electrons are not shared; they are transferred between atoms, so that atoms with a deficiency of electrons (cation) and atoms with an excess of electrons (anions) are formed. A typical example of an ionic compound is sodium chloride (NaCl).

Cation: An atom in which the number of protons exceeds the number of electrons, thereby resulting in an excess of positive charge. A cation is a positively charged atom. Potassium cation (K^+) carries a +1 charge, implying that there is one more proton than the number of electrons in the atom. Potassium has nineteen protons, so potassium cation has eighteen electrons.

Anion: An atom in which the number of electrons exceeds the number of protons, thereby resulting in an excess of negative charge. An anion is a negatively charged atom. Fluoride anion (F^-) carries a -1 charge, implying that there is one more electron than the number of protons in the atom. Fluorine has nine protons, so fluoride anion has ten electrons. Note that the name changes from "fluorine" (used when the atom is neutral) to "fluoride" (used when the atom carries a negative charge). Negatively charged species are given the "-ide" suffix.

Nomenclature of Salts

Scientific convention says that when you name a salt, the name of the cation precedes the name of the anion. That is why we refer to NaCl as sodium chloride, rather than chloride sodium. The rules for naming a simple binary salt are as follows:

1. Name the cation before the anion in the salt.
2. The cation name is derived from the element forming the cation (often ending in "-ium")
3. The anion name is derived from the element forming the anion with an "-ide" suffix added.

Example 3.18

What is the proper chemical name for the salt BaF_2 ?

- A. Barium difluorine
- B. Barium fluoride
- C. Difluorobarane
- D. Fluorinium baride

Solution

The cation of this salt is derived from barium, and the anion is derived from fluorine. Thus, the name begins with barium and ends with the "-ide" form of fluorine (fluoride): barium fluoride, which makes choice **B** the correct answer. The subscript two in the formula (showing that there are two fluoride anions) need not be named, given that fluoride can be only -1 and barium is +2. The ratio is implied. It was fun to write wrong answers for this question. Granted, I am a chemistry-loving geek, so my credibility when it comes to determining what is cool and what isn't is dubious, but nevertheless, choice C is a cool wrong answer.

Polyatomic Ions

Over the course of your academic science career, you may have seen the same ions recurring in a variety of contexts in different subjects. In biology, the interactions of bicarbonate with carbonate and dihydrogen phosphate with hydrogen phosphate are essential in blood buffering and in the action of the kidney. Kidney stones result from the precipitation of calcium with various polyatomic ions. Because of the frequency with which molecular ions appear as a factor in many problems, the MCAT test writers expect you to be familiar with some common ones. Table 3.6 lists a few that you should know.

-1 Anions		-2 Anions	
Acetate	$\text{C}_2\text{H}_3\text{O}_2^-$	Carbonate	CO_3^{2-}
Bicarbonate	HCO_3^-	Chromate	CrO_4^{2-}
Bisulfite (hydrogen sulfite)	HSO_3^-	Dichromate	$\text{Cr}_2\text{O}_7^{2-}$
Bisulfate (hydrogen sulfate)	HSO_4^-	Hydrogen phosphate	HPO_4^{2-}
Dihydrogen phosphate	H_2PO_4^-	Oxalate	$\text{C}_2\text{O}_4^{2-}$
Hypochlorite	ClO^-	Oxide	O^{2-}
Chlorite	ClO_2^-	Peroxide	O_2^{2-}
Chlorate	ClO_3^-	Sulfite	SO_3^{2-}
Perchlorate	ClO_4^-	Sulfate	SO_4^{2-}
Cyanide	CN^-		
Hydroxide	OH^-	-3 Anions	
Superoxide	O_2^-	Phosphate	PO_4^{3-}
Nitrite	NO_2^-		
Nitrate	NO_3^-	+1 Cations	
Permanganate	MnO_4^-	Ammonium	NH_4^+

Table 3.6

You should know these twenty-six ions by charge, structure, and nomenclature. Knowing a little about their common applications may help, too.

Solubility Product and Molar Solubility (Mathematical Applications)

With solubility, calculations typically involve converting between molar solubility and solubility product (K_{sp}). Molar solubility and solubility product show up together in calculations, but they are very different terms. If one salt has a lower K_{sp} value than another salt, it does not mean that it is less soluble (has a lower molar solubility). Solubility is determined by the molar solubility of a salt, not the solubility product. K_{sp} values are calculated as products over reactants, but because solids are ignored, there is no denominator. Table 3.7 shows different ion combinations for salts, where x represents molar solubility.

Solubility Reaction	K_{sp} Expression	K_{sp} Calculation
$MX(s) \rightleftharpoons M^+(aq) + X^-(aq)$	$K_{sp} = [M^+][X^-]$	$K_{sp} = (x)(x) = x^2$
$MX_2(s) \rightleftharpoons M^{2+}(aq) + 2 X^-(aq)$	$K_{sp} = [M^{2+}][X^-]^2$	$K_{sp} = (x)(2x)^2 = 4x^3$
$MX_3(s) \rightleftharpoons M^{3+}(aq) + 3 X^-(aq)$	$K_{sp} = [M^{3+}][X^-]^3$	$K_{sp} = (x)(3x)^3 = 27x^4$
$M_2X(s) \rightleftharpoons 2 M^+(aq) + X^{2-}(aq)$	$K_{sp} = [M^+]^2[X^{2-}]$	$K_{sp} = (2x)^2(x) = 4x^3$
$M_3X(s) \rightleftharpoons 3 M^+(aq) + X^{3-}(aq)$	$K_{sp} = [M^+]^3[X^{3-}]$	$K_{sp} = (3x)^3(x) = 27x^4$

Table 3.7

Example 3.19

What is the molar solubility for the hypothetical compound MX , if it is known to have $K_{sp} = 4.0 \times 10^{-10} M^2$ in water?

- A. $1.0 \times 10^{-5} M$
 B. $2.0 \times 10^{-5} M$
 C. $4.0 \times 10^{-5} M$
 D. $2.0 \times 10^{-10} M$

Solution

For an MX salt, $K_{sp} = [M^+][X^-] = x^2$, where x is the molar solubility. This means that the molar solubility is the square root of the K_{sp} value.

$$K_{sp} = x^2 \therefore x = \sqrt{K_{sp}} = \sqrt{4.0 \times 10^{-10}} = \sqrt{4.0} \times \sqrt{10^{-10}} = 2.0 \times 10^{-5}$$

This makes choice **B** the best answer.

Example 3.20

What is the molar solubility for the hypothetical compound MX_2 , if it is known to have $K_{sp} = 1.08 \times 10^{-7} M^3$ in water?

- A. $1.65 \times 10^{-4} M$
 B. $2.10 \times 10^{-3} M$
 C. $3.00 \times 10^{-3} M$
 D. $4.80 \times 10^{-3} M$

Solution

For an MX_2 salt, $K_{sp} = [M^{2+}][X^-]^2 = 4x^3$, where x is the molar solubility. This means that the molar solubility is the cube root of one-fourth of the K_{sp} value.

$$x = \sqrt[3]{\frac{K_{sp}}{4}} = \sqrt[3]{\frac{1.08 \times 10^{-7}}{4}} = \sqrt[3]{\frac{108 \times 10^{-9}}{4}} = \sqrt[3]{27} \times \sqrt[3]{10^{-9}} = 3.0 \times 10^{-3}$$

This makes choice **C** the best answer.

It is good to understand the hypothetical examples, because actual salts are done the same way, except the numbers are not as clean. The key fact to recognize is found in the units. If K_{sp} is M^2 , then square root must be found. Square roots are easier to solve if the power of ten is divisible by two. If K_{sp} is M^3 , then cube root is necessary. Cube roots are easier to solve if the power of ten is divisible by three. From that point it requires approximating square roots and cube roots by ranging the values based on squares and cubes you know.

Example 3.21

What is the molar solubility of lead(II) iodide, PbI_2 , if it is known to have $K_{sp} = 1.42 \times 10^{-8} M^3$ in water?

- A. $1.19 \times 10^{-4} M$
- B. $8.83 \times 10^{-4} M$
- C. $1.52 \times 10^{-3} M$
- D. $2.42 \times 10^{-3} M$

Solution

For PbI_2 , $K_{sp} = [Pb^{2+}][I^-]^2 = 4x^3$, where x is the molar solubility. This means that the molar solubility is the cube root of one-fourth of the K_{sp} value.

$$\begin{aligned}
 K_{sp} = 4x^3 \therefore x &= \sqrt[3]{\frac{K_{sp}}{4}} = \sqrt[3]{\frac{1.42 \times 10^{-8}}{4}} = \sqrt[3]{\frac{14.2 \times 10^{-9}}{4}} \\
 &= \sqrt[3]{\frac{14.2}{4}} \times \sqrt[3]{10^{-9}} = \sqrt[3]{3.55} \times 10^{-3} \approx 1.5 \times 10^{-3} M
 \end{aligned}$$

The cube root of 3.55 is not a common piece of knowledge. However, the cube root of 1 is 1 and the cube root of 8 is 2. This means that the cube root of 3.55 falls between 1 and 2. This makes the best answer choice C.

In the previous three questions, you were asked to derive the molar solubility from the solubility product. An alternative way to pose this question is to ask you to derive the solubility product from the molar solubility. Before you attack these problems, it is important that you make a conscious note of potential mistakes. Remember to multiply the ions by the coefficient from the balanced equation! This is the most common error on these problems. The second trick is to remember cube roots by thinking of common numbers cubed, such as $2^3 = 8$, $3^3 = 27$, and $4^3 = 64$.

Example 3.22

What is the solubility product of an M_2X salt with molar solubility $= 5.0 \times 10^{-2} M$?

- A. $1.25 \times 10^{-4} M^3$
- B. $2.50 \times 10^{-4} M^3$
- C. $5.00 \times 10^{-4} M^3$
- D. $1.25 \times 10^{-3} M^3$

Solution

For an M_2X salt, $K_{sp} = [M^+]^2[X^{2-}] = 4x^3$, where x is the molar solubility. This means that the K_{sp} value is four times the molar solubility cubed.

$$K_{sp} = 4x^3 = 4 \times (5.0 \times 10^{-2})^3 = 4 \times 125 \times 10^{-6} = 500 \times 10^{-6} = 5.0 \times 10^{-4}$$

This makes choice C the best answer.

Relative Solubility

It is difficult to predict relative solubility. The only way to compare is through experiments or data analysis. We'll start by comparing the solubilities of calcium carbonate (CaCO_3) and calcium fluoride (CaF_2). When a question asks for the highest solubility, it refers to the compound that produces the greatest amount of dissociated salt, which refers to greatest molar solubility, not solubility product!

The solubility product of CaCO_3 is $8.7 \times 10^{-9} \text{ M}^2$, and the solubility product for CaF_2 is $4.0 \times 10^{-11} \text{ M}^3$. Which of the two salts is more soluble?

With questions like these, keep in mind that you never compare numbers with unlike units. Solubility products are used to determine the molar solubility. CaCO_3 is an MX salt, so $K_{sp} = x^2$, while CaF_2 is an MX_2 salt, so $K_{sp} = 4x^3$. The determinations of molar solubility for both calcium salts is shown below.

$$\begin{aligned}\text{CaCO}_3: x &= \sqrt{K_{sp}} = \sqrt{8.7 \times 10^{-9}} = \sqrt{87} \times \sqrt{10^{-10}} = 9.3 \times 10^{-5} \text{ M} \\ \text{CaF}_2: x &= \sqrt[3]{\frac{K_{sp}}{4}} = \sqrt[3]{\frac{4.0 \times 10^{-11}}{4}} = \sqrt[3]{\frac{40 \times 10^{-12}}{4}} = \sqrt[3]{10} \times \sqrt[3]{10^{-12}} = 2.2 \times 10^{-4} \text{ M}\end{aligned}$$

The molar solubility for CaCO_3 is $9.3 \times 10^{-5} \text{ M}$ and for CaF_2 it is 2.2×10^{-4} , but exact values are not necessary, because $9.3 \times 10^{-5} \text{ M}$ is less than $2.2 \times 10^{-4} \text{ M}$. CaF_2 is more soluble than CaCO_3 , even though it has a smaller solubility product. Relative solubility questions are asked in many ways such as: "Which salt exhibits greater solvation?" and "Which salt precipitates first?". Use molar solubility to answer questions that address relative solubility. Solubility product is employed only for calculation-based questions.

Example 3.23

Which of the following salts yields the LEAST amount of silver ion in water?

- A. AgBr(s) $K_{sp} = 7.9 \times 10^{-13} \text{ M}^2$
- B. AgCl(s) $K_{sp} = 1.6 \times 10^{-10} \text{ M}^2$
- C. $\text{Ag}_2\text{CrO}_4\text{(s)}$ $K_{sp} = 9.0 \times 10^{-12} \text{ M}^3$
- D. $\text{Ag}_3\text{PO}_4\text{(s)}$ $K_{sp} = 1.8 \times 10^{-18} \text{ M}^4$

Solution

You must recognize that the choices are not all the same type of salt. Choices A and B are MX salts, so their solubility products may be compared. AgBr has a lower solubility product than AgCl , so AgBr also has a lower molar solubility. This eliminates choice B. Choice C is an M_2X salt, and choice D is an M_3X salt. To solve the question, you need molar solubility values. The calculation of the molar solubility for the three remaining salts is shown below.

$$\begin{aligned}\text{Choice A: } x &= \sqrt{K_{sp}} = \sqrt{7.9 \times 10^{-13}} = \sqrt{79} \times \sqrt{10^{-14}} \approx 8.9 \times 10^{-7} \text{ M} \\ \text{Choice C: } x &= \sqrt[3]{\frac{K_{sp}}{4}} = \sqrt[3]{\frac{9.0 \times 10^{-12}}{4}} = \sqrt[3]{2.25} \times \sqrt[3]{10^{-12}} \approx 1.3 \times 10^{-4} \text{ M} \\ \text{Choice D: } x &= \sqrt[4]{\frac{K_{sp}}{27}} = \sqrt[4]{\frac{1.8 \times 10^{-18}}{27}} = \sqrt[4]{6.7} \times \sqrt[4]{10^{-20}} \approx 1.6 \times 10^{-5} \text{ M}\end{aligned}$$

AgBr , choice A, has the lowest molar solubility. However, the question asked for lowest $[\text{Ag}^+]$, not lowest molar solubility. For choice C, Ag_2CrO_4 , there are two silver cations, and choice D, Ag_3PO_4 , has three silver cations, so the $[\text{Ag}^+]$ is actually double and triple their respective molar solubilities, making them much too high to be the actual answer.

Solubility Experiments

The molar solubility of a salt is an empirical value, determined from experiments involving saturated solutions. While forming a saturated solution is easy to do, determining the ion concentration is not necessarily an easy task. To form a saturated solution, enough salt is added to water so that precipitate remains on the bottom of the container. Determining the amount of dissolved salt may be accomplished in many ways, of which we shall discuss three. The first method involves spectroscopy, where the amount of absorbed light at a fixed wavelength depends on the concentration of ions, according to Beer's law (Equation 1.3). This method works only if one of the ions absorbs electromagnetic radiation. The second method involves measuring the amount of salt added and then collecting, drying, weighing, and then subtracting the mass of precipitate from this value. This method works only if the salt is highly soluble. The third method involves using an ion exchange column to exchange the cation in solution for hydronium. The concentration of hydronium is determined via pH. This method works only if the salt does not exhibit acid-base properties and has a cation concentration of at least 10^{-5} M. You may encounter passages about all three methods.

Example 3.24

What is the molar solubility of calcium carbonate in water given that the addition of 4.00 mg of $\text{CaCO}_3(\text{s})$ to enough water to form exactly 500.0 mL of an aqueous CaCO_3 solution yields 0.55 mg of anhydrous precipitate?

- A. 6.90×10^{-3} M CaCO_3
B. 3.45×10^{-3} M CaCO_3
C. 6.90×10^{-5} M CaCO_3
D. 3.45×10^{-5} M CaCO_3

Solution

You must keep in mind the units of the question and then solve for each unit. In other words, the answer is in terms of moles/liter, which implies that you must find both moles and liters to solve the question. The liters of solution are simple, in that 500.0 mL is equal to 0.500 liters. The amount of calcium carbonate that dissociates into solution is 3.45 mg. The amount of calcium carbonate in milligrams is converted to grams by multiplying by 10^{-3} , and then to moles by dividing by the molecular mass of calcium carbonate (100 grams per mole). The mathematical set-up is shown below

$$\text{Moles: } \frac{3.45 \times 10^{-3} \text{ grams CaCO}_3}{100 \text{ grams/mole}} = 3.45 \times 10^{-5} \text{ moles CaCO}_3$$

$$\text{Volume: } 500.0 \text{ mL} \times \frac{1.00 \text{ L}}{1000 \text{ mL}} = 0.500 \text{ L CaCO}_3(\text{aq})$$

$$\text{Concentration: } \frac{3.45 \times 10^{-5} \text{ moles CaCO}_3}{0.500 \text{ L solution}} = 6.90 \times 10^{-5} \text{ M CaCO}_3(\text{aq})$$

The correct answer is choice C

Solubility looks daunting, but is actually quite simple conceptually. Don't be intimidated by difficult-looking topics. They're usually pretty easy. Some applications of solubility include ion exchange (ion exchange columns), selective solubility (precipitation of a selected cation or anion), and the common ion effect. Of ion exchange, selective solubility, and the common ion effect, only common ion effect involves heavy mathematics.

Common Ion Effect

The common ion effect is the reduction in solubility of an ionic solid (salt) in solution, because one of the ions in the salt is already present in solution. In a way, the concept is similar to Le Châtelier's principle. According to Le Châtelier's principle, when you add one of the ions to a saturated salt solution, the reaction shifts to the left (reactant side). This results in the formation of a precipitate (more solid), so the salt is less soluble. Addition of an ion to solution reduced the solubility. According to the common ion effect, if one of the ions is already present in solution, then less salt is capable of dissociating, so the solubility is reduced. The results are the same when an ion is added to solution; the difference between Le Châtelier's principle and the common ion effect is the timing of when they are added. Consider Reaction 3.7.

**Reaction 3.7**

With the addition of either Ag^+ or Cl^- to solution in Reaction 3.7, there is a reduction in the amount of AgCl that can dissociate into it, due to the presence in the solution of one of the ions constituting the salt (Ag^+ or Cl^-). This is similar to Le Châtelier's principle, except that either Ag^+ or Cl^- is present in solution in the beginning rather than being added once the solution has reached a solubility equilibrium. Questions on the MCAT may be asked from either a common ion perspective or a Le Châtelier's principle's perspective.

Le Châtelier's style question: Which way will the reaction shift when $\text{Ag}^+(aq)$ is added to a saturated aqueous silver chloride solution? According to Le Châtelier's principle, adding a product shifts the reaction to the reactant side, so a precipitate forms. This means that the salt becomes less soluble.

Common ion effect style question: Is silver chloride more soluble in pure water or a 0.10 M $\text{NaCl}(aq)$ solution? According to the common ion effect, because the sodium chloride solution has chloride ions (also found in silver chloride) already present in solution, the solubility of the salt is reduced. We shall look at this question from a mathematical perspective, later in the discussion of this topic.

The common ion effect, simply put, says that the presence of a mutual ion already dissolved into solution reduces the solubility of a second salt sharing that mutual ion. For instance, stannous fluoride is less soluble in a solution of sodium fluoride than distilled water because of the presence of the fluoride anion. Common ion effect questions can incorporate pH and hydroxide concentration.

Example 3.25

Magnesium hydroxide is MOST soluble in which of the following solutions?

- A. Vinegar (pH = 2.5)
- B. Citric acid (pH = 4.5)
- C. Los Angeles tap water (pH = 5.5)
- D. Distilled water (pH = 7.0)

Solution

This question involves the common ion effect, where the common ion is hydroxide anion. Magnesium hydroxide is most soluble in the solution with the least hydroxide anion, which corresponds to the most acidic solution. The most acidic solution has the lowest pH, which makes choice A the best answer. This is the reason that acid rain is such a serious problem: it dissolves basic salts.

Example 3.26

At what pH is $\text{Ca}(\text{OH})_2(s)$ most soluble in water?

- A. 3
- B. 5
- C. 9
- D. 11

Solution

The pH at which $\text{Ca}(\text{OH})_2(s)$ is most soluble is the pH at which there is the least common ion (OH^-) present in solution. There is the least OH^- present at lower pH values, where the solution is acidic. The best answer is choice A. Note that basic salts are most soluble in acidic solutions.

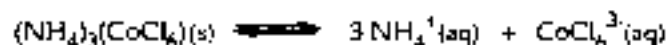
Example 3.27

$(\text{NH}_4)_3(\text{CoCl}_6)(s)$ is MOST soluble at which pH?

- A. 3
- B. 5
- C. 9
- D. 11

Solution

The pH at which $(\text{NH}_4)_3(\text{CoCl}_6)(s)$ is most soluble is the pH at which there is the most OH^- in solution. This is because of a complex equilibrium. The OH^- in solution can deprotonate the ammonium cation (NH_4^+) to form ammonia, which reduces the amount of ammonium present in solution. As the amount of NH_4^+ in solution decreases, the $(\text{NH}_4)_3(\text{CoCl}_6)(s)$ equilibrium shifts to products to generate more NH_4^+ . The most OH^- is present at higher pH values, where the solution is basic. The best answer is choice D. Note here that acidic salts are most soluble in basic solutions. The two equilibrium equations are as follows:

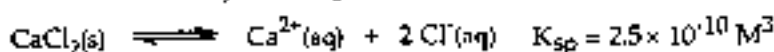


As the second reaction shifts to the right, the first reaction is displaced from equilibrium, so it too shifts to the right. This is a complex equilibrium. The two reactions are dependent on one another.

Be able to make qualitative predictions about solubility based on both the common ion effect and complex equilibrium. Example 3.25 and Example 3.26 are examples of the common ion effect. Example 3.27 is an example of increased solubility due to the presence of a complexing ion in solution. Both of these concepts play a role in determining solubility. When doing calculations with the common ion effect, be sure not to plug variables into the solubility product formula blindly. In the case of an MX salt, for instance, K_{sp} still is equal to $[\text{M}^+][\text{X}^-]$, but the values of $[\text{M}^+]$ and $[\text{X}^-]$ are not just x with the common ion effect. The concentration of the ion that is already present in the solution is found by summing the initial concentration and the additional ion formed from the dissociation of the salt (x). This is $[\text{X}^-]_{\text{initial}} + x$. In most cases, the x will be insignificant relative to the $[\text{X}^-]_{\text{initial}}$, so it can be ignored. Most common ion effect questions should be simple and fun after you have done enough of them.

Example 3.28

What is the molar solubility of $\text{CaCl}_2(\text{s})$ in 0.01 M $\text{NaCl}(\text{aq})$ solution?



- A. $2.5 \times 10^{-4} \text{ M}$
- B. $2.5 \times 10^{-6} \text{ M}$
- C. $2.5 \times 10^{-8} \text{ M}$
- D. $2.5 \times 10^{-10} \text{ M}$

Solution

This question is purely mathematical. The key fact is recognizing that there is 0.01 M Cl^{-} in solution initially. The setup and solution are as follows:

Reaction:	$\text{CaCl}_2(\text{s})$	$\rightleftharpoons$	$\text{Ca}^{2+}(\text{aq})$	+	2Cl^{-}
Initially:	excess		0		0.01
Shift:	$-x$	$\longrightarrow$	$+x$		$+2x$
Equilibrium:	who cares?		x		$0.01 + 2x$

$$K_{\text{sp}} = [\text{Ca}^{2+}][\text{Cl}^{-}]^2 \therefore K_{\text{sp}} = (x)(0.01 + 2x)^2$$

$$\text{If we ignore } 2x: K_{\text{sp}} = 2.5 \times 10^{-10} = (x)(0.01)^2 = 0.0001(x) \therefore x = 2.5 \times 10^{-6} \text{ M}$$

The correct answer is choice B.

Example 3.29

What is the molar solubility of $\text{Ca}(\text{OH})_2$ in an aqueous solution at $\text{pH} = 13$, given that the K_{sp} for $\text{Ca}(\text{OH})_2$ is $6.5 \times 10^{-6} \text{ M}^3$?

- A. $6.5 \times 10^{-6} \text{ M}$
- B. $6.5 \times 10^{-4} \text{ M}$
- C. $1.2 \times 10^{-2} \text{ M}$
- D. $6.5 \times 10^{20} \text{ M}$

Solution

This question is purely mathematical as well. You must recognize that there is 0.10 M OH^{-} present in a $\text{pH} = 13$ solution, because $\text{pH} + \text{pOH} = 14$ for an aqueous solution and $[\text{OH}^{-}] = 10^{-\text{pOH}}$. The equation, set-up, and solution are as follows:

Reaction:	$\text{Ca}(\text{OH})_2(\text{s})$	$\rightleftharpoons$	$\text{Ca}^{2+}(\text{aq})$	+	2OH^{-}
Initially:	excess		0		0.10
Shift:	$-x$	$\longrightarrow$	$+x$		$+2x$
Equilibrium:	who cares?		x		$0.10 + 2x$

$$K_{\text{sp}} = [\text{Ca}^{2+}][\text{OH}^{-}]^2 \therefore K_{\text{sp}} = (x)(0.10 + 2x)^2$$

$$\text{Upon ignoring } 2x: K_{\text{sp}} = 6.5 \times 10^{-6} = (x)(0.10)^2 = 0.01(x) \therefore x = 6.5 \times 10^{-4} \text{ M}$$

The correct answer is choice B.

Calculation questions involving the common ion effect are actually rather simple, but in most cases the average student is not familiar with the process. Simply plug in the value for the preexisting concentration, and solve for x from the K_{sp} expression.

Separation by Precipitation

Removal of certain ions (cations or anions) from solution can be done in a selective fashion, by precipitation of cations with various anions, according to their relative solubility values. The least soluble salt precipitates from solution first. Questions about selective precipitation involve comparing molar solubility values. Be careful not to fall into the trap of thinking that the smallest K_{sp} is the least soluble. You must consider what type of ionic system it is. The compound with the smallest value for x (molar solubility) precipitates from solution first.

Table 3.8 lists the solubility products for some common salts at standard temperature. The values are listed without units, as is conventional in general chemistry textbooks, but you should always consider the units when they are provided.

Salt	K_{sp}	Salt	K_{sp}	Salt	K_{sp}
BaF ₂	2.4×10^{-5}	CaF ₂	4.0×10^{-11}	MgF ₂	6.4×10^{-9}
PbF ₂	4.1×10^{-8}	SrF ₂	7.9×10^{-10}	ZnF ₂	2.5×10^{-8}
BaSO ₄	1.5×10^{-9}	CaSO ₄	6.1×10^{-5}	MgSO ₄	7.2×10^{-5}
PbSO ₄	4.1×10^{-8}	SrSO ₄	7.9×10^{-10}	ZnSO ₄	8.3×10^{-9}
BaCO ₃	1.6×10^{-9}	CaCO ₃	6.1×10^{-9}	MgCO ₃	1.1×10^{-15}
PbCO ₃	1.4×10^{-15}	SrCO ₃	8.4×10^{-10}	ZnCO ₃	2.2×10^{-12}
Ba(OH) ₂	5.3×10^{-3}	Ca(OH) ₂	1.4×10^{-6}	Mg(OH) ₂	1.2×10^{-11}
Pb(OH) ₂	1.3×10^{-15}	Sr(OH) ₂	4.1×10^{-4}	Zn(OH) ₂	2.1×10^{-16}

Table 3.8

Figure 3-6 shows a precipitation flow chart from a qualitative analysis experiment designed to identify cations in solution. The cations are selected from Table 3.8. All of the cations are +2 cations, so they all form the same type of salts. As such, solubility products can be directly compared. Each step shows the salt that precipitates.

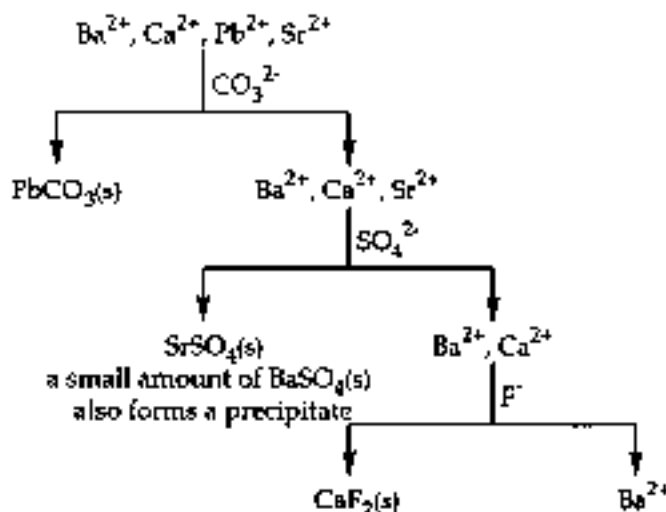


Figure 3-6

Example 3.30

According to data in Table 3.8, to remove strontium by precipitation from a solution with Ca^{2+} , Mg^{2+} , and Sr^{2+} , it is BEST to add:

- A. F^-
- B. SO_4^{2-}
- C. CO_3^{2-}
- D. OH^-

Solution

The best anion to add to solution to remove a cation is the one that forms a salt less soluble than the salts of the other cations in solution. Of the three cations (Ca^{2+} , Mg^{2+} , and Sr^{2+}), the least soluble fluoride salt is CaF_2 , eliminating choice A. Of the three cations, the least soluble sulfate salt is SrSO_4 , so choice B is the best answer. Of the three cations, the least soluble carbonate salt is MgCO_3 , and the least soluble hydroxide salt is Mg(OH)_2 , eliminating choices C and D.

Example 3.31

Mercury II cations are MOST soluble in which of the following solutions?

- A. 0.10 M NaCl
- B. 0.10 M Na_2S
- C. 0.10 M NaI
- D. 0.10 M NaNO_3

Solution

This is a question of relative solubilities. This question is asking what the relative solubilities of HgCl_2 , HgS , HgI_2 , and $\text{Hg(NO}_3)_2$ are. The key fact to keep in mind is that nitrates are infinitely soluble in water (one of the solubility rules). Of the anions, only nitrate can form hydrogen bonds with water, so it has the greatest solvation energy of the four anions. It is also the largest of the anions, so it has the weakest electrostatic forces in its lattice structure. The conclusion is that nitrate salts are the most soluble, so the best answer is choice D. This is purely a qualitative argument for the solubility. There is mathematical evidence to back this up as well, but the numbers were not given in the problem.

Example 3.32

Given a solution that contains $\text{Ag}^+(\text{aq})$, $\text{Pb}^{2+}(\text{aq})$, $\text{Sr}^{2+}(\text{aq})$, and $\text{Zn}^{2+}(\text{aq})$, what is the sequence of precipitation when $\text{NaCl}(\text{aq})$ is added to solution?

$$K_{\text{sp}} \text{AgCl} = 2 \times 10^{-10}, K_{\text{sp}} \text{PbCl}_2 = 2 \times 10^{-5}, K_{\text{sp}} \text{SrCl}_2 = 9 \times 10^{-3}, K_{\text{sp}} \text{ZnCl}_2 = 1 \times 10^{-4}$$

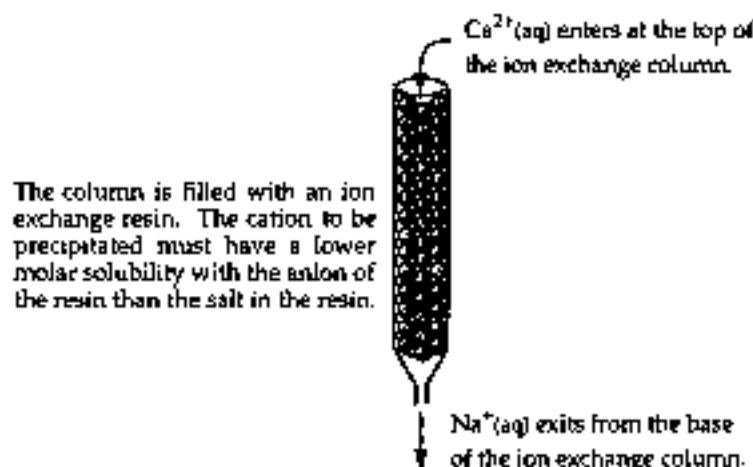
- A. 1st: AgCl , 2nd: SrCl_2 , 3rd: ZnCl_2 , 4th: PbCl_2
- B. 1st: AgCl , 2nd: PbCl_2 , 3rd: ZnCl_2 , 4th: SrCl_2
- C. 1st: SrCl_2 , 2nd: ZnCl_2 , 3rd: PbCl_2 , 4th: AgCl
- D. 1st: PbCl_2 , 2nd: ZnCl_2 , 3rd: SrCl_2 , 4th: AgCl

Solution

The salt with the lowest molar solubility forms a precipitate first. AgCl has the lowest solubility product, and the molar solubility is the square root of K_{sp} . For the other three salts, use the cube root of K_{sp} to obtain molar solubility. This makes AgCl the least soluble, and the first to precipitate, eliminating choices C and D. The remaining salts have +2 cations, so the order of precipitation is from the lowest K_{sp} value to the highest K_{sp} value. This makes choice B correct.

Ion Exchange Column

An ion exchange column exchanges one ion in solution for another (which is initially bound to the column), by precipitating out the ion that forms the less soluble salt and releasing a more soluble ion into solution to replace it. A water softener is an ion exchange column. Hard water (rich in $\text{Ca}^{2+}(\text{aq})$) travels down the column where the anion in the ion exchange resin binds calcium cations to form an insoluble salt while releasing sodium cations ($\text{Na}^{+}(\text{aq})$) into solution. Aqueous sodium cation is referred to as *soft water*. As calcium precipitates through the ion exchange column, it is filtered from the water, preventing it from forming precipitates in the plumbing lines of appliances (such as a washer or water heater). To evaluate the viability of an ion exchange column, the molar solubility values are compared. Figure 3-7 shows an ion exchange column.

**Figure 3-7**

In the water softening example, the removal of calcium ions from tap water is accomplished by passing the water across rock salt (large chunks of NaCl) and then filtering out the calcium chloride. Reaction 3.8 is the exchange reaction in a water softener.

**Reaction 3.8**

Rock salt is employed to minimize surface area, and to prevent sodium chloride from dissociating into solution too rapidly. Sodium chloride is used because its ions are non-toxic, and it does not change the pH of the water.

Selective Precipitation and Ion Exchange: The addition of an ion to solution with the intention of precipitating an existing ion out from the solution. This is achieved when the salt being precipitated is less soluble (has a lower molar solubility) than any other salt combination in solution.

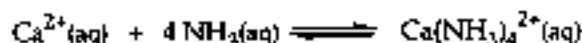
**Reaction 3.9**

Because mercury chloride is less soluble than silver chloride, the addition of silver chloride to the mercury solution precipitates mercury out of water. This is the same principle applied in ion exchange columns and can also be applied in qualitative analysis when we look for ions in solution.

Complex Ion Formation: Complex equilibrium occurs when two reactions are summed to form one overall reaction. This is possible when the product of one reaction is a reactant in a second reaction. A great example involves the addition of ammonium to salt buildup in sinks in order to remove the deposit. To check this, take a small whiff of a bathroom cleaner, and you will detect ammonia.



Reaction 3.10



Reaction 3.11

Because ammonia forms hydrogen bonds with water, the calcium-ammonia complex is more soluble than the calcium cation. By the addition of ammonia to solution, free calcium cation forms a complex ion (the result of Reaction 3.11 shifting in the forward direction), which reduces the amount of free calcium and forces the Reaction 3.10 to proceed in the forward direction. As mentioned prior, when reactions are added together, the equilibrium constant for the overall reaction is the product of the equilibrium constants of the component reactions.

Complex Ions (Coordination complexes): Complex ions form when a ligand (lone pair donor or Lewis base) donates a pair of electrons to a central atom (Lewis acid), which is typically a metal, to form a coordinate covalent bond. This process is referred to as *chelation*, where the Lewis base is a chelating agent. A prime example is hemoglobin whose porphyrin ring serves as a polydentate ligand by chelating the central iron.

Example 3.33

Silver chloride is MOST soluble in which of the following solutions at 25°C?

- A. 0.10 M $\text{HgNO}_3(\text{aq})$
- B. Pure water
- C. $\text{AgNO}_3(\text{aq})$
- D. $\text{NaCl}(\text{aq})$

Solution

Silver chloride ($\text{AgCl}(\text{s})$) is most soluble in a solution where Hg^+ is present. This is because of a complex equilibrium. The Hg^+ in solution can bind the chloride anion (Cl^-) to form a precipitate. As the amount of Cl^- in solution decreases, more AgCl dissolves into solution to replace it. The best answer is choice A. Both choices C and D are eliminated, because of the common ion effect.



Equilibrium Passages

14 Passages

100 Questions

Suggested Equilibrium Passage Schedule:

- I:** After reading this section and attending lecture: Passages I, II & VI - IX
Grade passages immediately after completion and log your mistakes.
- II:** Following Task I: Passages III - V, X & XI (34 questions in 44 minutes)
Time yourself accurately, grade your answers, and review mistakes.
- III:** Review: Passages XII - XIV & Questions 97 - 100
Focus on reviewing the concepts. Do not worry about timing.



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Equilibrium Study Passages

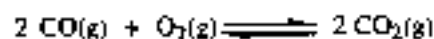
I. Carbon Dioxide and Carbon Monoxide Equilibrium	(1 - 7)
II. Hydrogen and Bromine Equilibrium	(8 - 15)
III. Equilibrium Constant Magnitude	(17 - 22)
IV. Gas-Phase Equilibrium Constant Experiment	(23 - 29)
V. Equilibrium Reactions	(30 - 36)
VI. Le Châtelier's Principle	(37 - 42)
VII. Solubility and Qualitative Analysis	(43 - 49)
VIII. Solubility Experiment	(50 - 56)
IX. Qualitative Analysis Experiment	(57 - 62)
X. Solubility Chart	(63 - 69)
XI. Calcium Salts Solubility	(70 - 75)
XII. Complex Equilibrium	(76 - 82)
XIII. Hemoglobin and Acclimation	(83 - 89)
XIV. Equilibrium Reaction of NO_2 and N_2O_4	(90 - 96)
Questions Not Based on a Descriptive Passage	(97 - 100)

Equilibrium Scoring Scale

Raw Score	MCAT Score
84 - 100	13 - 15
66 - 83	10 - 12
47 - 65	7 - 9
34 - 46	4 - 6
1 - 33	1 - 3

Passage 1 (Questions 1 - 7)

A researcher studies the equilibrium of carbon monoxide with carbon dioxide in the presence of oxygen gas at various temperatures. The relationship is shown in Reaction 1.

**Reaction 1**

This is accomplished by monitoring the total pressure of the system as it depends on temperature. There are different stoichiometric amounts of gases on each side of Reaction 1, so the reaction can shift with changes in pressure or volume. To quantify the contents in the flask at equilibrium, the pressure for the system is analyzed. It is assumed that pressure changes are due to shifts in equilibrium and changes in the system that obey the ideal gas equation ($PV = nRT$). Changes in the total pressure are evaluated for the system and the equilibrium constant is derived from the values for pressure. The reaction is monitored for three trials at each temperature over a wide range of temperatures. The results are summarized in Table 1.

Temperature (K)	K_p (atm ⁻¹)
375	09.0
425	20.8
500	32.2
600	60.5

Table 1

Data in Table 1 demonstrate that the value of the equilibrium increases with increasing temperature. Although the equilibrium system shifts upon changes in the moles of reactant, moles of product, external pressure, and volume, the equilibrium constant remains constant.

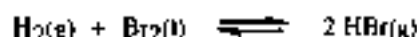
- At 102°C, $P_{\text{CO}} = 2.0$ atm, and $P_{\text{O}_2} = 1.0$ atm. What is the partial pressure of CO_2 at this temperature?
 - 2.2 atm.
 - 6.0 atm.
 - 9.0 atm.
 - 18.0 atm.
- Addition of $\text{CO}_2\text{(g)}$ to the equilibrium mixture results in which of the following?
 - A decrease in the moles of $\text{O}_2\text{(g)}$.
 - A decrease in the moles of CO(g) .
 - An increase in the value of K_p .
 - No change in the value of K_p .

- According to Table 1, the reaction is which of the following?
 - Exothermic as written
 - Endothermic as written
 - Adiabatic as carried out
 - Isothermal upon reaction
- The equilibrium constant (K) varies only with which of the following?
 - A change in the pressure of the system
 - A change in the volume of the system
 - A change in the concentration of the system
 - A change in the temperature of the system
- If a flask were filled with pure $\text{CO}_2\text{(g)}$ to a total pressure of 1.00 atm., then once equilibrium is reached, the total pressure of the system is which of the following?
 - Less than 1.00 atm.
 - Exactly 1.00 atm.
 - Between 1.00 atm. and 1.50 atm.
 - Exactly 1.50 atm.
- According to Reaction 1, compressing the reaction vessel leads to which of the following?
 - An increase in the partial pressure of $\text{CO}_2\text{(g)}$
 - An increase in the moles of $\text{O}_2\text{(g)}$
 - An increase in the mole fraction of CO(g)
 - An increase in the equilibrium constant
- Based on the following reaction, the addition of $\text{Ca(OH)}_2\text{(aq)}$ to Reaction 1 at equilibrium results in which of the following?

$$\text{Ca(OH)}_2\text{(aq)} + \text{CO}_2\text{(g)} \rightleftharpoons \text{CaCO}_3\text{(s)} + \text{H}_2\text{O(l)}$$
 - No change in the moles of $\text{CO}_2\text{(g)}$
 - A decrease in the concentration of $\text{O}_2\text{(g)}$
 - An increase in partial pressure of CO(g)
 - An increase in the total pressure.

Passage II (Questions 8 - 15)

Hydrogen bromide gas is one of the strongest acids available in gas form. In aqueous solution, HBr has a pK_a value of approximately -12. A pK_a value of -12 is indicative of a very strong acid. By comparison, HCl has a pK_a of approximately -9. Hydrobromic acid (HBr) can be produced by Reaction 1.



Reaction 1

A student decides to study the equilibrium distribution for Reaction 1. Under ambient conditions, Reaction 1 reaches equilibrium at some temperature T and pressure P . Once at equilibrium, measurements are taken for the partial pressure of each gas and the mass of bromine liquid. The values are listed in Table 1.

$Br_2(l)$	10.0 grams
$H_2(g)$	1.36 atm
$HBr(g)$	2.72 atm

Table 1

The student takes notice that Reaction 1 shifts with changes in the conditions of the system, such as the addition of bromine liquid, the removal of hydrobromic acid, and increases in volume. The student irreversibly increases the volume of the system by opening a valve on the reaction flask that is connected to an evacuated column. Not all changes shift the reaction. Changes in the equilibrium concentrations are not accurately recorded.

8. What is the value of K at ambient temperature, T ?

- A. 0.20 atm.
- B. 0.544 atm.
- C. 2.00 atm.
- D. 5.44 atm.

9. Addition of $HBr(g)$ to an equilibrium mixture of Reaction 1 results in which of the following?

- A. An increase in the partial pressure of $H_2(g)$
- B. A decrease in the partial pressure of $H_2(g)$
- C. An increase in the concentration of $Br_2(l)$
- D. A decrease in the concentration of $Br_2(l)$

10. Which of the following starting conditions results in the GREATEST amount of $H_2(g)$ at equilibrium?

- A. 0.80 atm $H_2(g)$ and 20 g $Br_2(l)$
- B. 1.00 atm $H_2(g)$ and 20 g $Br_2(l)$
- C. 0.80 atm $H_2(g)$ and 30 g $Br_2(l)$
- D. 1.00 atm $HBr(g)$ and 30 g $Br_2(l)$

11. Subsequent experiments show that as the temperature increases, the mole percent of the hydrogen bromide gas increases. Which of the following is TRUE about the equilibrium reaction as written?

- A. It is endothermic.
- B. It is exothermic.
- C. It is isobaric.
- D. It is not an equilibrium mixture.

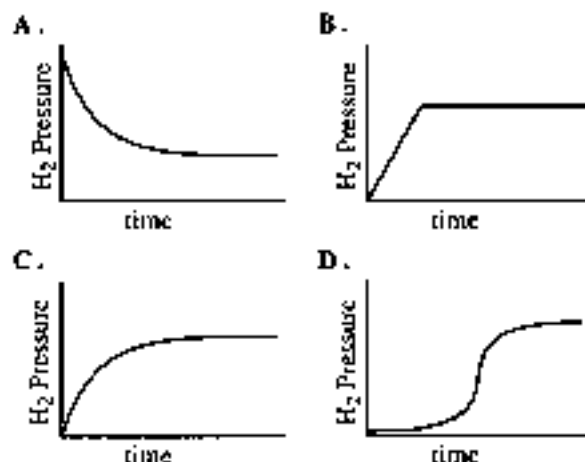
12. How can it be explained that a reaction mixture of 1.0 atm $HBr(g)$ and 1.0 atm $H_2(g)$ results in no change in hydrogen gas partial pressure?

- A. The reaction is in equilibrium.
- B. The reaction cannot move in the forward direction due to a limiting reagent in zero concentration.
- C. The reaction cannot move in the reverse direction due to a limiting reagent in zero concentration.
- D. The reaction is moving forward and backward at the same rate.

13. Mixing 3.0 atm $HBr(g)$ with 1.5 atm $H_2(g)$ over $Br_2(l)$ shows which of the following in time?

- A. A large increase in the partial pressure of $HBr(g)$
- B. A small increase in the partial pressure of $HBr(g)$
- C. A large increase in the partial pressure of $H_2(g)$
- D. A small increase in the partial pressure of $H_2(g)$

14. A flask is initially filled with pure $HBr(g)$. Which of the following graphs represents the partial pressure of $H_2(g)$ over time?

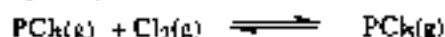


15. Addition of which of the following to the equilibrium mixture will NOT affect the partial pressure of $H_2(g)$?

- A. $NaOH(aq)$
- B. $Br_2(l)$
- C. $H_2(g)$
- D. $HBr(g)$

Passage III (Questions 16 - 22)

Chlorine gas has a high reduction potential (large positive voltage for its reduction half-reaction). Chemically speaking, chlorine gas is very easily reduced. Chlorine gas is a strong oxidizing agent that can dissolve away most metals in short periods of time. The dissolving is the result of an oxidation-reduction reaction. The relative ease with which metals are oxidized accounts for this strong tendency to lose electrons to chlorine. Chlorine gas is not as successful at oxidizing non-metals as it is at oxidizing metals. This is attributed to the decreased desire for non-metals to be oxidized relative to metals. Non-metals have a higher ionization energy and electronegativity than metals. With non-metals, chlorine may only partially oxidize the corresponding reducing agent. Reactions 1 and 2 are prime examples of the oxidation capability of chlorine on non-metal compounds.



Reaction 1



Reaction 2

The equilibrium constant for Reaction 1 is 1.95 atm^{-1} at 100°C , while the equilibrium constant for Reaction 2 is $2.1 \times 10^4 \text{ atm}^{-1}$ at 100°C . Both reactions favor products as written, but not to the same degree.

In these examples, the non-metals (phosphorus in Reaction 1 and nitrogen in Reaction 2) are already in a positive oxidation state in the reactants. The greater the value of the equilibrium constant, the more favorable the reaction, and thus the more product that is formed.

16. According to the equilibrium constants associated with Reaction 1 and Reaction 2, which compound is most readily oxidized by chlorine?
- PCl_3
 - PCl_5
 - NO
 - NOCl
17. If a flask were filled with PCl_3 and Cl_2 to a total pressure of 1.50 atm. such that the mole fraction of PCl_3 is twice that of Cl_2 , then what is the total pressure of the system once at equilibrium?
- Less than 1.00 atm.
 - Between 1.00 atm. and 1.25 atm.
 - Between 1.25 atm. and 1.50 atm.
 - Greater than 1.50 atm.

18. What is the equilibrium constant for the following reaction?



- $4.9 \times 10^{-17} \text{ atm}$.
 - $4.9 \times 10^{-5} \text{ atm}$.
 - $1.4 \times 10^2 \text{ atm}$.
 - $2.1 \times 10^4 \text{ atm}$.
19. If there is initially 1.00 atm. of NOCl in a flask, what is the partial of NO gas, once at equilibrium?
- 0.0230 atm.
 - 0.0460 atm.
 - 0.0025 atm.
 - 0.0050 atm.
20. The GREATEST concentration of $\text{Cl}_2(\text{g})$ is in which of the following systems, once equilibrium is reached?
- A system initially with 1.0 atm. PCl_3
 - A system initially with 1.0 atm. PCl_5
 - A system initially with 1.0 atm. NO
 - A system initially with 1.0 atm. NOCl
21. Which of the following statements is true about the effect of adding chlorine gas to each equilibrium?
- It will cause the greatest change in Reaction 1.
 - It will cause the greatest change in Reaction 2.
 - It will cause equal changes in both reactions.
 - It will change the equilibrium constant.
22. A flask at 100°C is charged with 0.40 atm. each of PCl_3 , PCl_5 , and Cl_2 . In which direction will it react to reach equilibrium, according to Reaction 1?
- It is already at equilibrium.
 - To the left (reactant side)
 - To the right (product side)
 - It varies with a catalyst.

Passage IV (Questions 23 - 29)

The equilibrium constant for a reaction involving gases can be determined by experimentation, using a glass column as a closed system. The apparatus used in equilibrium experiments, shown in Figure 1, is connected to a vacuum pump at one end and is sealed at the other. On the column are three stopcocks with air tight fittings, to which flasks containing gases may be attached. At the sealed end of the tube is a pressure gauge. The volume of the glass column is exactly 1.00 liters, when all stopcocks are closed. Each flask and attached stopcock contain volumes of exactly 250 mL. This means that when all three stopcocks are open in a manner where gas is free to flow between the glass column and any of the attached flasks, the total volume for the closed system is exactly 1.75 liters. Figure 1 shows the apparatus.

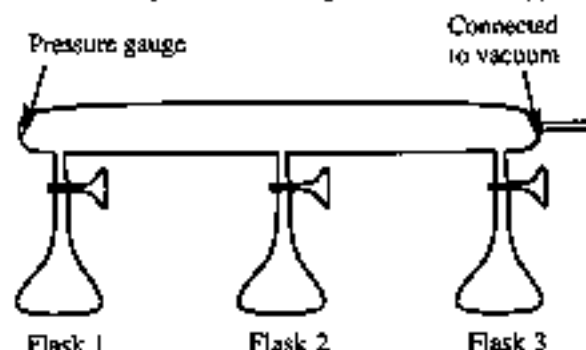


Figure 1

During an experiment, the closed glass cylinder is filled with 1.00 atm. of hydrogen (H_2) according to the gauge in the cylinder. Flask 1 is filled with 1.0 atm. carbon disulfide (CS_2). The reaction starts when the stopcock is opened, allowing the two gases to mix. The temperature of the glass column is maintained at $25^\circ C$ using an external heat sink. The internal pressure is monitored until it stays constant. Figure 2 shows the internal pressure over time, where $t = 0$ represents the time at which the two gasses were mixed.

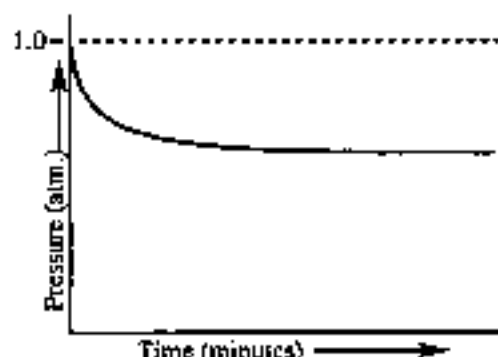


Figure 2

The pressure of each gas in the reaction mixture can be calculated from the change in internal pressure. The initial partial pressure of hydrogen gas is 0.8 atmospheres in the 1.25 liter closed system. The decrease in partial pressure of hydrogen gas is double the decrease in the internal pressure, based on the stoichiometry of the Reaction 1, which shows the reactivity of the compounds.



Reaction 1

The final internal pressure is the sum of the partial pressures. The total pressure can be found using Equation 1.

$$P_{\text{total}} = P_{CS_2} + P_{H_2} + P_{CH_4} + P_{H_2S}$$

Equation 1

23. Which of the following is the **BEST** explanation as to why the reaction temperature must be held at $25^\circ C$?
 - A. A temperature change can cause the carbon disulfide to decompose to carbon dioxide.
 - B. The reaction will not proceed unless the temperature is exactly $25^\circ C$.
 - C. The glass will begin to melt at temperatures in excess of $25^\circ C$.
 - D. The equilibrium constant can change with temperature.
24. The temperature of the closed reaction vessel was increased. This resulted in the pressure of the closed system decreasing, rather than increasing as expected. Which of the following explanations **BEST** explains this observation?
 - A. The pressure of a closed system decreases as the temperature increases according to Charles's law.
 - B. The reaction is endothermic as written.
 - C. The reaction is exothermic as written.
 - D. Heavier molecules are being formed causing the number of collisions to increase.
25. How can the partial pressures of CH_4 and H_2S at equilibrium be determined from the experiment?
 - A. At equilibrium, the partial pressure of CH_4 equals the value for the change in the pressure of the system (ΔP_{total}), while the partial pressure of H_2S equals twice the value for the change in the pressure of the system ($2\Delta P_{\text{total}}$).
 - B. At equilibrium, the partial pressure of H_2S equals the value for the change in the pressure of the system (ΔP_{total}), while the partial pressure of CH_4 equals twice the value for the change in the pressure of the system ($2\Delta P_{\text{total}}$).
 - C. At equilibrium, the partial pressure of CH_4 equals the value for the change in the pressure of the system (ΔP_{total}), while the partial pressure of H_2S equals half the value for the change in the pressure of the system ($\frac{1}{2}\Delta P_{\text{total}}$).
 - D. At equilibrium, the partial pressure of H_2S equals the value for the change in the pressure of the system (ΔP_{total}), while the partial pressure of CH_4 equals half the value for the change in the pressure of the system ($\frac{1}{2}\Delta P_{\text{total}}$).

26. Which of the following relationships between partial pressure for the gases in the mixture at equilibrium will ALWAYS be true for the reaction as it is set up?

A. $P_{CS_2} > P_{CH_4}$
 B. $P_{CH_4} > P_{H_2S}$
 C. $P_{H_2} > P_{CS_2}$
 D. $P_{H_2S} > P_{H_2}$

27. Which of the following equations represents the equilibrium expression for the reaction?

A. $K_p = \frac{(P_{CH_4})(P_{H_2S})^2}{(P_{CS_2})(P_{H_2})^4}$
 B. $K_p = \frac{(P_{CS_2})(P_{H_2})^4}{(P_{CH_4})(P_{H_2S})^2}$
 C. $K_p = \frac{(P_{CH_4})(P_{H_2S})}{(P_{CS_2})(P_{H_2})}$
 D. $K_p = \frac{(P_{CS_2})(P_{H_2})}{(P_{CH_4})(P_{H_2S})}$

28. The final total pressure of the system must be:

A. greater than 1.6 atmospheres.
 B. greater than 1.0, but less than 1.6 atmospheres.
 C. greater than 0.6, but less than 1.0 atmospheres.
 D. less than 0.6 atmospheres.

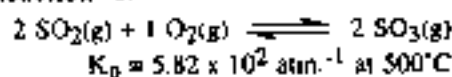
29. Which of the following stresses on the system will change the value of the equilibrium constant?

A. Opening the stopcock to Flask 2, which was filled with hydrogen gas.
 B. Turning on the vacuum pump to remove some of the gas in the column.
 C. Opening the stopcock to Flask 3, which was filled with hydrogen sulfide (H_2S) gas.
 D. Heating the gas in the glass column by heating the glass with a hot air blower.

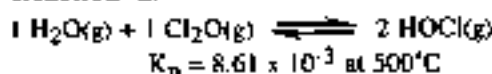
Passage V (Questions 30 - 36)

Chemical equilibrium obeys the rules of Le Châtelier's principle. Le Châtelier's principle states that when a system at equilibrium is stressed, the reaction will shift in a manner to partially relieve the stress and reestablish equilibrium. In a gas phase equilibrium, stresses that can be applied include changes in pressure, volume, temperature, and moles of either reactant or product. It is not possible to change just one of the variables, so a change in one variable will lead to a change in another variable. For instance, if the volume is increased, the pressure must decrease. Written below are three reactions and their respective equilibrium constants at $500^\circ C$:

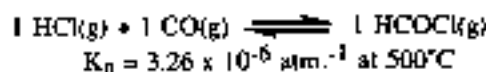
Reaction 1:



Reaction 2:

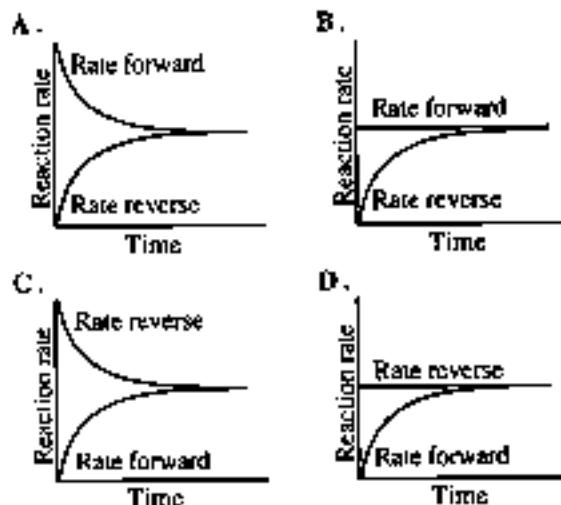


Reaction 3:



The equilibrium constant is derived by dividing the partial pressures of the product gases by the partial pressures of the reactant gases. In Reaction 1 and Reaction 3, the equilibrium constants were determined using a piston system where the total pressure and volume were monitored, and any change in pressure or volume can be attributed to a shift in the reaction equilibrium. Because the number of products equals the number of reactants, the total pressure and volume cannot change in reaction 2, so equilibrium concentrations were determined using infrared spectroscopy.

30. Given the initial partial pressure for PH_2O is 0.5 atm and for PCl_2O is 0.5 atm, which graph accurately shows the rate as a function of time for the reaction?



31. Which reactions will shift when the pressure changes?

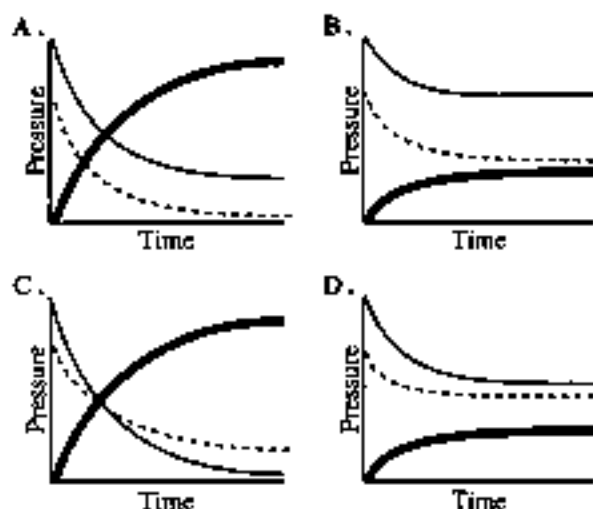
- A. Reaction 1 only
- B. Reactions 1 and 2 only
- C. Reactions 1 and 3 only
- D. Reactions 1, 2, and 3

32. Upon increasing the volume at constant temperature of a vessel holding Reaction 3, how are the partial pressures affected?

- A. $\frac{P_{\text{HCOCl}}}{P_{\text{HCl}} \cdot P_{\text{CO}}}$ increases; $\frac{P_{\text{HCOCl}}}{P_{\text{HCl}}}$ increases
- B. $\frac{P_{\text{HCOCl}}}{P_{\text{HCl}} \cdot P_{\text{CO}}}$ decreases; $\frac{P_{\text{HCOCl}}}{P_{\text{HCl}}}$ decreases
- C. $\frac{P_{\text{HCOCl}}}{P_{\text{HCl}} \cdot P_{\text{CO}}}$ remains constant; $\frac{P_{\text{HCOCl}}}{P_{\text{HCl}}}$ increase
- D. $\frac{P_{\text{HCOCl}}}{P_{\text{HCl}} \cdot P_{\text{CO}}}$ remains constant; $\frac{P_{\text{HCOCl}}}{P_{\text{HCl}}}$ decreases

33. Which graph depicts partial pressures as a function of time after 0.75 atm SO_2 is mixed with 0.50 atm O_2 ?

— SO_2 - - - O_2 — SO_3



34. Which K_{eq} value does NOT match the reaction?

- A. $2 \text{HOCl(g)} \rightleftharpoons 1 \text{H}_2\text{O(g)} + 1 \text{Cl}_2\text{O(g)}$
 $K_p = 8.61 \times 10^{-3}$ at 500°C
- B. $\frac{1}{2} \text{H}_2\text{O(g)} + \frac{1}{2} \text{Cl}_2\text{O(g)} \rightleftharpoons 1 \text{HOCl(g)}$
 $K_p = 9.28 \times 10^{-2}$ at 500°C
- C. $\frac{1}{2} \text{HCOCl(g)} \rightleftharpoons \frac{1}{2} \text{HCl(g)} + \frac{1}{2} \text{CO(g)}$
 $K_p = 5.54 \times 10^2$ at 500°C
- D. $2 \text{HCl(g)} + 2 \text{CO(g)} \rightleftharpoons 2 \text{HCOCl(g)}$
 $K_p = 1.06 \times 10^{-11}$ at 500°C

35. To convert a K_p value in terms of torr^{-1} to a K_p value in terms of atm^{-1} , you should:

- A. multiply the K_p value by 760.
- B. divide the K_p value by 760.
- C. multiply the K_p value by 0.76.
- D. divide the K_p value by 0.76.

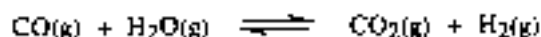
36. Cooling an endothermic equilibrium reaction results in which of the following?

- A. The mole fraction of products will decrease.
- B. The mole fraction of reactants will decrease.
- C. The mole fraction of products will increase.
- D. The ratio of products to reactants will increase.

Passage VI (Questions 37 - 42)

Le Châtelier's principle is invoked to explain reaction shifts following the disruption of equilibrium. A system in dynamic equilibrium is a closed system where the forward reaction rate and reverse reaction rate are equal. Components in the mixture maintain fixed concentration. Equilibrium is different from steady state. In steady state systems, an intermediate has a static concentration, because the formation reaction and the consumption reaction have equal rates.

A system can be disturbed from equilibrium by applying a stress, such as changing the temperature, volume, pressure, or moles of a reagent. According to Le Châtelier's principle, the system reacts in a manner to partially alleviate the stress and reestablish equilibrium. Reaction 1 starts with three different starting concentrations on its way to equilibrium.



Reaction 1

Table 1 lists the starting and equilibrium concentrations for each of the three trials. The reactions are observed at 1000°C and the equilibrium constant is found to be 0.569.

	Trial I		Trial II		Trial III	
Molarity	Initial	Final	Initial	Final	Initial	Final
CO	1.00	0.57	0.10	0.014	0.50	0.373
H ₂ O	1.00	0.57	1.00	0.914	0.50	0.373
CO ₂	0	0.43	0	0.086	0.50	0.627
H ₂	0	0.43	0	0.086	0	0.127

Table 1

Once at equilibrium, a separate stresses were applied to each system. To reaction System I, an additional 0.10 moles of CO was added to the 1.00 liter vessel in which the reaction was housed. To reaction System II, all of the H₂O was removed from the 1.00 liter vessel in which the reaction was housed. The 1.00 liter vessel containing reaction System III was cooled to 500°C. Once equilibrium was reestablished in the three reaction vessels, the partial pressure of CO₂ had increased in reactions systems I and III and decreased in II.

37. All of the following affect the reactant/product ratios of the equilibrium mixture EXCEPT:

- A. increasing the volume of the reaction vessel.
- B. heating the reaction vessel.
- C. cooling the reaction vessel.
- D. adding reactant to the reaction vessel.

38. If 1.00 moles of both CO₂ and H₂ were mixed in a 1.00 liter container at 1000°C, what will the equilibrium concentrations be for both CO₂ and H₂?

- A. [CO₂] = 0.57 M; [H₂] = 0.57 M
- B. [CO₂] = 0.57 M; [H₂] = 0.43 M
- C. [CO₂] = 0.43 M; [H₂] = 0.57 M
- D. [CO₂] = 0.43 M; [H₂] = 0.43 M

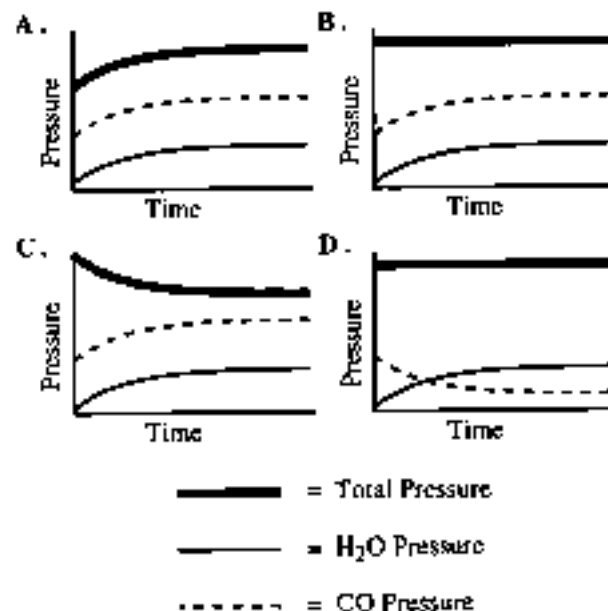
39. As heat is added to Reaction 1, what occurs?

- A. Both the equilibrium constant (K_{eq}) and the partial pressure of CO₂ decrease.
- B. Both the equilibrium constant (K_{eq}) and the partial pressure of CO₂ increase.
- C. The equilibrium constant (K_{eq}) remains constant while the partial pressure of CO₂ increases.
- D. The equilibrium constant (K_{eq}) remains constant while the partial pressure of CO₂ decreases.

40. What effect would an increase in pressure have on the equilibrium system of Reaction 1?

- A. The reaction would shift to the right and the equilibrium constant would increase.
- B. The reaction would shift to the left and the equilibrium constant would decrease.
- C. The reaction would shift to the right and the equilibrium constant would remain constant.
- D. The reaction would not shift in either direction and the equilibrium constant would remain constant.

41. Which of the following graphs represents what would be observed over time if equal parts of CO(g), CO₂(g), and H₂(g) were mixed in a closed system?



42. For an exothermic reaction at 100°C, the equilibrium concentration of the products equals the equilibrium concentration of the reactants. What is true at 25°C?

- A. $K_{eq} > 1$; [Products] > [Reactants]
- B. $K_{eq} > 1$; [Products] < [Reactants]
- C. $K_{eq} < 1$; [Products] > [Reactants]
- D. $K_{eq} < 1$; [Products] < [Reactants]

Passage VII (Questions 43 - 49)

The solubility of a salt in water is measured by its molar solubility. The *molar solubility* is defined as the moles of salt that dissociate into one liter of aqueous solution. The solubility product (K_{sp}) is the equilibrium constant for the dissociation reaction. Table 1 lists solubility data for various salts of silver, lead, strontium, and zinc.

Compound	K_{sp}	Molar solubility
AgCl	$1.7 \times 10^{-10} \text{ M}^2$	$1.3 \times 10^{-5} \text{ M}$
PbCl_2	$1.6 \times 10^{-5} \text{ M}^3$	$1.2 \times 10^{-2} \text{ M}$
SrCl_2	$8.8 \times 10^{-3} \text{ M}^3$	$1.3 \times 10^{-1} \text{ M}$
ZnCl_2	$1.1 \times 10^{-4} \text{ M}^3$	$3.1 \times 10^{-2} \text{ M}$
Ag_2CO_3	$8.1 \times 10^{-12} \text{ M}^3$	$1.2 \times 10^{-4} \text{ M}$
PbCO_3	$8.3 \times 10^{-11} \text{ M}^2$	$9.1 \times 10^{-6} \text{ M}$
SrCO_3	$7.1 \times 10^{-10} \text{ M}^2$	$2.6 \times 10^{-5} \text{ M}$
ZnCO_3	$2.0 \times 10^{-10} \text{ M}^2$	$1.4 \times 10^{-5} \text{ M}$
AgOH	$2.0 \times 10^{-8} \text{ M}^2$	$1.4 \times 10^{-4} \text{ M}$
Pb(OH)_2	$1.2 \times 10^{-15} \text{ M}^3$	$6.5 \times 10^{-6} \text{ M}$
Sr(OH)_2	$3.2 \times 10^{-4} \text{ M}^3$	$4.3 \times 10^{-2} \text{ M}$
Zn(OH)_2	$2.1 \times 10^{-16} \text{ M}^3$	$3.8 \times 10^{-6} \text{ M}$
Ag_2S	$1.6 \times 10^{-49} \text{ M}^3$	$3.4 \times 10^{-17} \text{ M}$
PbS	$2.6 \times 10^{-27} \text{ M}^2$	$5.1 \times 10^{-14} \text{ M}$
SrS	$4.0 \times 10^{-6} \text{ M}^2$	$2.0 \times 10^{-3} \text{ M}$
ZnS	$2.5 \times 10^{-22} \text{ M}^2$	$1.6 \times 10^{-11} \text{ M}$
Ag_2SO_4	$1.4 \times 10^{-5} \text{ M}^3$	$1.3 \times 10^{-2} \text{ M}$
PbSO_4	$1.7 \times 10^{-8} \text{ M}^2$	$1.3 \times 10^{-4} \text{ M}$
SrSO_4	$3.2 \times 10^{-7} \text{ M}^2$	$5.6 \times 10^{-4} \text{ M}$
ZnSO_4	$7.6 \times 10^{-8} \text{ M}^3$	$2.7 \times 10^{-4} \text{ M}$

Table 1

The relative molar solubility values can be used to determine which salts precipitate out from solution when various cations or anions are added. For instance, PbSO_4 precipitates first when sodium sulfate is added to a solution containing lead, strontium, and zinc cations, because lead sulfate is the least soluble (has the lowest molar solubility) of the sulfate salts.

43. Which of the following statements are valid?
- Sulfates are more soluble than carbonates.
 - Insoluble cations can be dissolved into solution by adding a complexing agent such as EDTA.
 - The solubility of chloride salts show pH dependence, while the solubility of sulfide salts show no pH dependence.
- A. I and II only
 B. I and III only
 C. II and III only
 D. I, II, and III

44. The K_{sp} value for which of the following salts is the MOST accurate in terms of smallest percent error?
- A. Ag_2SO_4
 B. Pb(OH)_2
 C. SrCO_3
 D. ZnS
45. Which anion should be added to selectively remove Zn^{2+} from an aqueous solution that contains Ag^+ , Sr^{2+} , and Zn^{2+} ?
- A. Cl^-
 B. CO_3^{2-}
 C. OH^-
 D. SO_4^{2-}
46. If a precipitate forms when NaCl is added to an aqueous salt solution, what conclusion can be drawn?
- A. There is no $\text{Zn}^{2+}(\text{aq})$ present in solution.
 B. There is $\text{Ag}^+(\text{aq})$ present in solution.
 C. There is both $\text{Sr}^{2+}(\text{aq})$ or $\text{Pb}^{2+}(\text{aq})$ present in solution.
 D. There is no $\text{Sr}^{2+}(\text{aq})$ present in solution.
47. If an aqueous solution containing Ag^+ , Pb^{2+} , Sr^{2+} , and Zn^{2+} is treated first with chloride anion, secondly with sulfide dianion, and lastly with hydroxide anion, then what cation is most likely still present in solution?
- A. Ag^+
 B. Pb^{2+}
 C. Sr^{2+}
 D. Zn^{2+}
48. Adding acid to an aqueous solution containing a basic salt has what effect?
- A. It will increase the solubility.
 B. It will decrease the solubility.
 C. It will have no effect on the solubility.
 D. It will lower the temperature of the solution.
49. Silver hydroxide is MOST water-soluble at which of the following pH values?
- A. 3
 B. 5
 C. 9
 D. 11

The solubility of salts is often measured in terms of moles solute per liter solution in a saturated solution. It may also be measured in terms of grams solute per 100 mL solution. The term saturated refers to a solution in which the maximum amount of salt that can dissociate into solution is dissociated. The more salt that dissociates into solution, the more soluble the salt. Solubility measurements cannot be used interchangeably when comparing the solubility of two salts without first considering the molecular mass of each salt. This can be observed in the following experiment.

Experiment 1

Water at 25°C is added in a calibrated burette, to the 25-mL mark. A salt is added in 1.0 grams increments, until the salt no longer dissociates into solution. At this point, a precipitate should settle to the bottom of the burette. The temperature of the solution in the burette is slowly increased until the salt completely dissociates. The solution is slowly cooled until the first signs of precipitate are observed. At this point, the temperature and volume of the solution are recorded.

Experiment 2

The solution continues cooling until it reaches 25°C. At this point, pure water at 25°C is added quantitatively until the last detectable sign of the precipitate dissociate into solution. The volume of the solution is recorded as precisely as the burette reads.

Table 1 lists data for this experiment using hypothetical salts MX, MY, and MZ. Column 3 lists data from Experiment 1 and column 4 lists data from Experiment 2.

Salt	Mass	Volume when heated	Volume w/ H ₂ O added
MX	7.0 g	26.2 mL (@ 31.6°C)	28.9 mL (@ 25.0°C)
MY	9.0 g	25.3 mL (@ 27.4°C)	26.1 mL (@ 25.0°C)
MZ	4.0 g	25.7 mL (@ 30.3°C)	28.7 mL (@ 25.0°C)

Table 1

The molecular mass for MX is 100 grams per mole, for MY is 120 grams per mole, and for MZ is 150 grams per mole. The concentration of each solution is measurable in any standard units, including molarity (moles solute per liter solution), grams per mL, mass percent, and density. Concentration units can be inter-converted. For instance, the percent solution by mass can be multiplied by the density to determine the mass solute per volume of solution, which can be converted to molarity using molecular mass.

Adding solvent to a solution dilutes the solution and thus reduces the concentration of the solution. All of the measurements of concentration decrease upon the addition of solvent, with the exception of the density. The density change depends on the relative density of the solvent and solution. In these experiments, water is the solvent for each solution, and the density of water at 25°C is found to be 0.9971 grams per mL.

50. How do the solubility products for the three salts in the passage compare to one another?

- A. $K_{sp}MX > K_{sp}MY > K_{sp}MZ$
- B. $K_{sp}MY > K_{sp}MX > K_{sp}MZ$
- C. $K_{sp}MZ > K_{sp}MX > K_{sp}MY$
- D. $K_{sp}MZ > K_{sp}MY > K_{sp}MX$

51. Which of the following solutions has the GREATEST molarity?

- A. Saturated MX at 31.6 °C
- B. Saturated MX at 25.0 °C
- C. Saturated MY at 27.4 °C
- D. Saturated MY at 25.0 °C

52. Roughly how many grams of MX will dissociate into 25 mL of water at 25.0°C?

- A. Less than 4.0 grams.
- B. Greater than 4.0 grams, but less than 7.0 grams.
- C. Greater than 7.0 grams, but less than 9.0 grams.
- D. Greater than 9.0 grams.

53. What is the molarity of a saturated solution of MY at 25°C?

- A. $\frac{9 \text{ grams}}{120 \frac{\text{grams}}{\text{mole}}} \times 0.0261 \text{ liters}$
- B. $\frac{9 \text{ grams}}{120 \frac{\text{grams}}{\text{mole}}} \times 0.0253 \text{ liters}$
- C. $\frac{9 \text{ grams}}{120 \frac{\text{grams}}{\text{mole}}} \times 0.0261 \text{ liters}$
- D. $\frac{9 \text{ grams}}{120 \frac{\text{grams}}{\text{mole}}} \times 0.0253 \text{ liters}$

54. As temperature increases, what happens to the various concentration measurements of a solution, assuming no evaporation of solvent and no addition of solute?

- A. The density and molarity both increase.
- B. The density increases, while the molarity decreases.
- C. The density decreases, while the molarity increases.
- D. The density and molarity both decrease.

55. Which of the following relationships at 25°C accurately shows the relative density values of saturated solutions of MX(aq), MY(aq), and MZ(aq)?

- A. $1.00 > \text{MX(aq)} > \text{MY(aq)} > \text{MZ(aq)}$
- B. $1.00 > \text{MY(aq)} > \text{MX(aq)} > \text{MZ(aq)}$
- C. $\text{MX(aq)} > \text{MY(aq)} > \text{MZ(aq)} > 1.00$
- D. $\text{MY(aq)} > \text{MX(aq)} > \text{MZ(aq)} > 1.00$

56. Which solution has the **HIGHEST** boiling point?

- A. 1.0 grams MX with 10 mL water at 25°C
- B. 1.0 grams MY with 10 mL water at 25°C
- C. 1.0 grams MX with 10 mL water at 50°C
- D. 1.0 grams MY with 10 mL water at 50°C

Passage IX (Questions 57 - 62)

The qualitative analysis of an aqueous salt solution involves the systematic addition of reagents designed to identify certain component ions. For instance, mercury can be distinguished from alkaline earth metals by adding sulfide (S^{2-}) to solution by the formation of an insoluble precipitate with mercury dication. Silver can be distinguished from alkali metals by the addition of either chloride, bromide, or iodide. The precipitate that forms between chloride and silver can be re-dissolved by adding ammonia to solution.

Once a precipitate is formed, it can be removed from solution by centrifuging the mixture and decanting away the supernatant or by filtering away the filtrate. The solution can then be further analyzed for other ions. Table 1 shows the results of a matrix involving the mixture of four cation nitrate solutions mixed with five potassium anion solutions mixed one at a time. All solutions are 0.05 M, with the exception of KOH(aq) , which has a concentration of 0.10 M. Any compound that forms a precipitate is assumed to have a molar solubility less than 0.001.

	$\text{Ba(NO}_3)_2$	$\text{Sr(NO}_3)_2$	$\text{Ca(NO}_3)_2$	$\text{Mg(NO}_3)_2$
K_2CO_3	Pcpt	Pcpt	Pcpt	No pcpt
KOH	Pcpt	No pcpt	No pcpt	Pcpt
K_2CrO_4	Pcpt	No pcpt	No pcpt	No pcpt
K_2SO_4	Pcpt	Pcpt	No pcpt	No pcpt
$\text{K}_2\text{C}_2\text{O}_4$	Pcpt	Pcpt	Pcpt	No pcpt

Table 1

Table 2 shows the results of a similar experiment where anions were exposed to a sequence of multiple reactions. The anions selected for testing are chloride, iodide, carbonate, and sulfate. The test solutions are silver nitrate, barium nitrate, ammonia, and nitric acid. The matrix of Table 2 lists the observations of sixteen different test tubes. Nitric acid reacts with carbonate anion to form carbon dioxide gas, which then bubbles out of solution.

	HNO_3	1. Ba^{2+} 2. HNO_3	1. Ag^+ 2. HNO_3	1. Ag^+ 2. NH_3 3. HNO_3
Cl^-	No Rx	1. No Rx 2. No Rx	1. Wht ppt 2. No Rx	1. Wht ppt 2. Dissolves 3. No Rx
I^-	No Rx	1. No Rx 2. No Rx	1. Yell ppt 2. No Rx	1. Yell ppt 2. No Rx 3. No Rx
CO_3^{2-}	Gas forms (Bubbles)	1. Pcpt 2. Bubbles	1. Wht ppt 2. Bubbles	1. Wht ppt 2. No Rx 3. Bubbles
SO_4^{2-}	No Rx	1. Pcpt 2. No Rx	1. No Rx 2. No Rx	1. No Rx 2. No Rx 3. No Rx

Table 2

57. What anion should be added to a solution in order to identify any barium cation present in the solution?

- A. CrO_4^{2-}
- B. SO_4^{2-}
- C. $\text{C}_2\text{O}_4^{2-}$
- D. CO_3^{2-}

58. What reaction took place when nitric acid was added to the precipitate formed from mixing Ba^{2+} and CO_3^{2-} ?

- A. $\text{Ba}_2\text{CO}_3(\text{s}) + 2 \text{HNO}_3(\text{aq}) \rightarrow 2 \text{BaNO}_3(\text{aq}) + \text{CO}_2(\text{g}) + \text{H}_2\text{O}(\text{l})$
- B. $\text{BaCO}_3(\text{s}) + 2 \text{HNO}_3(\text{aq}) \rightarrow \text{Ba}(\text{NO}_3)_2(\text{aq}) + \text{CO}_2(\text{g}) + \text{H}_2\text{O}(\text{l})$
- C. $\text{Ba}_2\text{CO}_3(\text{s}) + \text{HNO}_3(\text{aq}) \rightarrow \text{BaHCO}_3(\text{aq}) + \text{BaNO}_3(\text{aq})$
- D. $\text{BaCO}_3(\text{s}) + 2 \text{HNO}_3(\text{aq}) \rightarrow \text{Ba}(\text{NO}_3)_2(\text{aq}) + \text{CO}_2(\text{g}) + \text{H}_2(\text{g}) + \text{O}_2(\text{g})$

59. Why is it NOT possible to use hydrochloric acid instead of nitric acid in experiment II?

- A. Chloride anion will interfere with the reactions.
- B. Nitrate anion will interfere with the reactions.
- C. Hydrochloric acid is too weak to react.
- D. Hydrochloric acid is too strong to use.

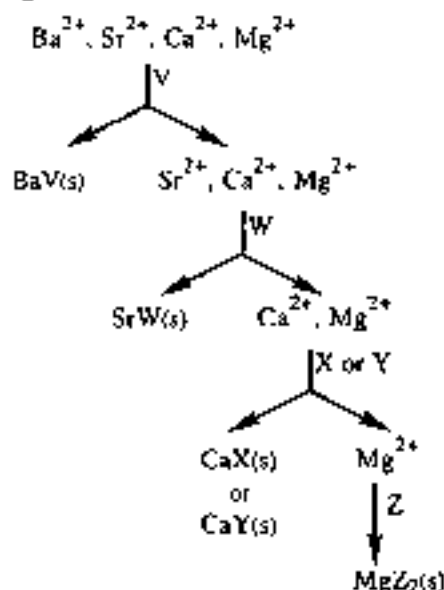
60. If an unknown mixture was treated with K_2SO_4 and a precipitate formed, what can be concluded?

- A. The unknown contained no calcium or magnesium cations.
- B. The unknown contained a barium cation.
- C. The unknown contained a strontium cation.
- D. The unknown contained either a strontium cation, a barium cation, or both.

61. Which of the following anions does form a white precipitate with silver cation but does NOT react with nitric acid?

- A. Sulfate
- B. Carbonate
- C. Chloride
- D. Iodide

62. The information from Table 1 was used to derive the following flow chart:



Which of the following choices represents an INCORRECT anion-to-letter correlation?

- A. $\text{V} = \text{CO}_3^{2-}$
- B. $\text{W} = \text{SO}_4^{2-}$
- C. $\text{X} = \text{C}_2\text{O}_4^{2-}$
- D. $\text{Z} = \text{OH}^-$

Water can be analyzed for various inorganic components using a series of qualitative tests for metals in their various oxidation states. This process is referred to as *qualitative analysis*. Many of the tests involve the formation of an insoluble salt. The formation of a precipitate results from the addition of an anion to an aqueous cation solution. Some anions may complex with more than one metal, so multiple tests are necessary. Molar solubility is different for each salt, so occasionally an anion may precipitate with one cation over another and the second metal can be "masked". For this reason a schematic of the series of tests is evaluated. Insoluble to the eye is defined as less than one mg dissociating into one mL of solution. By looking at the combination of precipitates formed, the cations can be narrowed down until one is chosen with reasonable accuracy.

A researcher analyzed an aqueous sample believed to contain roughly equal concentrations of Ag^+ , Ca^{2+} , Zn^{2+} , and Na^+ . Table 1 lists solubility data useful in developing a schematic for the tests. An ideal schematic sequentially identifies one cation at a time, leaving the remaining cations in solution.

Compound	K_{sp}	Molar solubility
Ag_2S	$1.6 \times 10^{-49} \text{ M}^3$	$3.4 \times 10^{-17} \text{ M}$
ZnS	$2.5 \times 10^{-22} \text{ M}^2$	$1.6 \times 10^{-11} \text{ M}$
CaS	$1.1 \times 10^{-31} \text{ M}^2$	$3.5 \times 10^{-6} \text{ M}$
CaCO_3	$8.7 \times 10^{-9} \text{ M}^2$	$9.3 \times 10^{-5} \text{ M}$
Ag_2CO_3	$8.1 \times 10^{-12} \text{ M}^3$	$1.2 \times 10^{-4} \text{ M}$
ZnCO_3	$2.0 \times 10^{-10} \text{ M}^2$	$1.4 \times 10^{-5} \text{ M}$
AgCl	$1.6 \times 10^{-10} \text{ M}^2$	$1.2 \times 10^{-5} \text{ M}$
ZnCl_2	$1.1 \times 10^{-4} \text{ M}^3$	$3.1 \times 10^{-2} \text{ M}$

Table 1

The values listed in Table 1 are all determined in distilled water at 25°C . It is occasionally necessary to conduct tests at an elevated temperature, in order to carry out qualitative analysis and the separation of cations. Sodium salts are not listed in the table, because sodium salts are infinitely soluble with nearly all anions, compared to the other metals listed in the data. To identify sodium cation, a flame test is often applied. By applying a flame to a small sample of the solution, electrons in sodium are excited. When they relax back to their ground state, photons are emitted. In the case of sodium, the light emitted is orange in color.

63. When 0.1 gram ZnCO_3 is added to 100 mL of water at 25°C , which of the following statements is true?
- It dissolves completely.
 - It dissolves almost completely, with only a small portion not dissolving.
 - It dissolves slightly, with most of the ZnCO_3 remaining not dissolving.
 - None of the sample dissolves.

64. It is best to visually distinguish a sample tube containing Zn^{2+} from a separate sample tube containing Ag^+ by the addition of:
- 0.10 M Na_2S
 - 0.10 M NaCl
 - 0.10 M NaNO_3
 - 0.10 M Na_2CO_3

65. In which solution is Ag_2S MOST soluble?

- 0.10 M $\text{Ag}^+(\text{aq})$
- 0.01 M $\text{Ag}^+(\text{aq})$
- 0.10 M $\text{S}^{2-}(\text{aq})$
- 0.01 M $\text{S}^{2-}(\text{aq})$

66. What is the molar solubility of ZnCO_3 in 0.010 M ZnCl_2 ?

- $2.0 \times 10^{-12} \text{ M}$
- $2.0 \times 10^{-10} \text{ M}$
- $2.0 \times 10^{-8} \text{ M}$
- $1.4 \times 10^{-8} \text{ M}$

67. Which of the following salts is MOST soluble in water?

- CaCO_3
- Ag_2CO_3
- ZnCO_3
- AgCl

68. Which of the following values accurately depicts the solubility product of $\text{Ca}_3(\text{PO}_4)_2$ if the molar solubility of $\text{Ca}_3(\text{PO}_4)_2$ is represented by x ?

- $18x^3$
- $27x^4$
- $54x^5$
- $108x^5$

69. To remove a cation from solution, the solution can be flowed through an ion exchange column where it will precipitate the cation while releasing a more soluble cation. Which of the following filters will NOT work?

- $\text{Ag}^+(\text{aq})$ through a column containing $\text{NaCl}(\text{s})$
- $\text{Zn}^{2+}(\text{aq})$ through a column containing $\text{CaS}(\text{s})$
- $\text{Ca}^{2+}(\text{aq})$ through a column containing $\text{ZnCO}_3(\text{s})$
- $\text{Ca}^{2+}(\text{aq})$ through a column containing $\text{Na}_2\text{S}(\text{s})$

Passage XI (Questions 70 - 75)

Calcium salts are generally insoluble in water at standard temperature. Biologically, this is beneficial in that the Ca^{2+} ion is soluble enough to be transported through aqueous medium, yet insoluble enough to be a major component of structural features in the human body, such as bones. Listed in Table 1 are some K_{sp} values at 27°C for various calcium salts.

Calcium Salt	K_{sp} (@ 27°C)
CaF_2	$4.8 \times 10^{-11} \text{ M}^3$
CaSO_4	$6.1 \times 10^{-6} \text{ M}^2$
CaCO_3	$9.1 \times 10^{-9} \text{ M}^2$
$\text{Ca}(\text{OH})_2$	$4.3 \times 10^{-6} \text{ M}^3$
$\text{Ca}_3(\text{PO}_4)_2$	$1.3 \times 10^{-32} \text{ M}^5$

Table 1

The solubility products of the given salts can be compared only when the anions bonded to calcium carry the same negative charge. For instance, the solubility products of CaF_2 and $\text{Ca}(\text{OH})_2$ can be directly compared to determine their relative solubilities. To enhance the solubility of a calcium salt, an anion that complexes calcium can be added to solution. The anion competes for calcium in a complex equilibrium. A complex equilibrium is defined as the coupling of at least two equilibrium reactions where the product of one reaction is the reactant in another equilibrium reaction. Reaction 1 and Reaction 2 combine to form a complex equilibrium involving calcium cation.



Reaction 1



Reaction 2

A change in calcium cation concentration affect both reactions. If the Reaction 2 shifts to the right, then Reaction 1 is forced to shift to the right in order to maintain its equilibrium.

70. What is observed when sodium carbonate is added to a saturated calcium hydroxide solution with undissolved calcium hydroxide on the bottom of the flask?

- A. The pH of the solution will increase more than it would increase had it been added to an unsaturated solution of calcium hydroxide.
- B. The pH of the solution will increase less than it would increase had it been added to an unsaturated solution of calcium hydroxide.
- C. The pH of the solution will decrease more than it would decrease had it been added to an unsaturated solution of calcium hydroxide.
- D. The pH of the solution will decrease less than it would decrease had it been added to an unsaturated solution of calcium hydroxide.

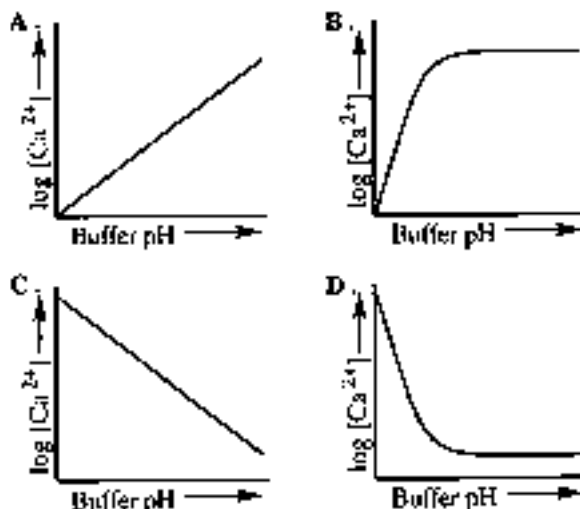
71. The maximum $[\text{Ca}^{2+}]$ in a $\text{pH} = 14$ solution would be which of the following values?

- A. $4.3 \times 10^8 \text{ M Ca}^{2+}$
- B. $4.3 \times 10^{-2} \text{ M Ca}^{2+}$
- C. $4.3 \times 10^{-6} \text{ M Ca}^{2+}$
- D. $4.3 \times 10^{-10} \text{ M Ca}^{2+}$

72. Enough CaCl_2 is added to water so that not all of it dissolves, and thus some CaCl_2 remains as a solid on the bottom of the flask. The $[\text{Cl}^-]$ is then measured. Addition of which of the following to the solution will increase the chloride ion concentration $[\text{Cl}^-]$?

- A. Water
- B. Silver nitrate
- C. Calcium fluoride
- D. Sodium phosphate

73. Which of the following graphs depicts the $\log [\text{Ca}^{2+}]$ versus the buffered pH of an aqueous buffer solution as calcium hydroxide is dissolved into different solutions?



74. If addition of salt to water makes the water warmer once the salt has dissolved, which of the following are true for solution of that salt?

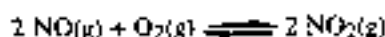
- A. ΔH is positive, ΔS is negative.
- B. ΔH is negative, ΔS is positive.
- C. ΔH and ΔS are both positive.
- D. ΔH and ΔS are both negative.

75. Which of the following, when added to water at 300 K , will yield the highest $[\text{Ca}^{2+}]$?

- A. CaF_2
- B. CaSO_4
- C. CaCO_3
- D. $\text{Ca}(\text{OH})_2$

Passage XII (Questions 76 - 82)

Smog is a collection of several gas pollutants. Most prevalent in the conglomeration of impurities are the sulfur and nitrogen oxides. The brown color so familiar in smoggy skies is due to the presence of nitrogen dioxide (NO_2). This, along with nitric oxide (NO), make up the majority of the airborne nitrogen oxides. The nitrogen oxides can be inter-converted by a series of equilibrium reactions.



Reaction 1



Reaction 2



Reaction 3

Reactions 1 and 2 are both exothermic while the Reaction 3 is endothermic. Reaction 3 requires the addition of light to transpire. As the day progresses, sunlight carries out Reaction 3, resulting in the highest ozone concentration in the early afternoon. Figure 1 depicts the concentration of pollutants during a hypothetical day.

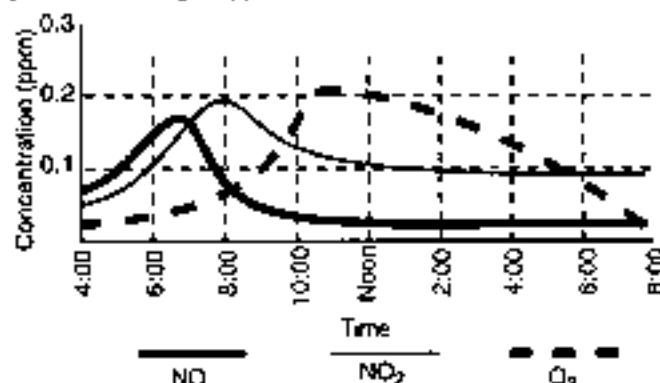


Figure 1

The amount of the nitrogen oxides in the air can be reduced by bubbling air through an aqueous transition metal halide solution such as $\text{MnCl}_2\text{(aq)}$. Nitrogen dioxide (NO_2) reacts with the water to form nitric acid (HNO_3) and nitrous acid (HNO_2), both of which undergo deprotonation and bind the manganese, forming a complex ion. Reactions 4 and 5 make up a complex equilibrium, as do Reactions 4 and 6.



Reaction 4



Reaction 5



Reaction 6

When reactions are added to produce an overall reaction, the equilibrium constants for the reactions are multiplied to obtain the equilibrium constant for the overall reaction. This is true for any type of equilibrium reaction.

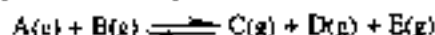
76. Which of the reactions has a negative entropy change?

- A. Reaction 2
- B. Reaction 3
- C. Reaction 5
- D. Reaction 6

77. Which of the following will NOT increase the $[\text{N}_2\text{O}_4]$?

- A. Decreasing the temperature.
- B. Increasing the pressure.
- C. Addition of water vapor to the air.
- D. Addition of NO_2 to the air.

78. If the following reaction is carried out in a closed steel reaction vessel, what will be observed for the pressure of the system if the temperature is doubled?



$$\Delta H = -112 \text{ kJ/mole}$$

- A. The pressure will remain the same.
- B. The pressure will increase by less than 100%.
- C. The pressure will exactly double.
- D. The pressure will increase by more than 100%.

79. What can be concluded about the change in enthalpy for Reaction 2?

- A. Because a bond is broken, the reaction is endothermic in the forward direction.
- B. Because a bond is broken, the reaction is exothermic in the forward direction.
- C. Because a bond is formed, the reaction is endothermic in the forward direction.
- D. Because a bond is formed, the reaction is exothermic in the forward direction.

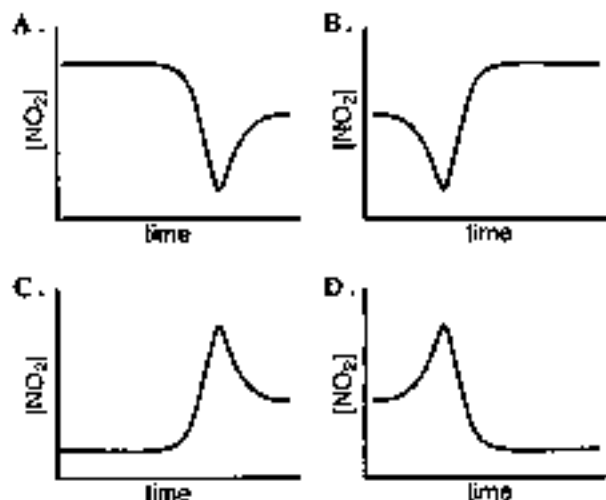
80. If Reaction 1 is carried out in a closed piston system with an external pressure of 1 atm, what occurs when 0.1 atm of NO gas is added to the system?

- A. The volume decreases by more than ten percent.
- B. The volume decreases by less than ten percent.
- C. The volume increases by less than ten percent.
- D. The volume increases by more than ten percent.

81. Which will shift reaction 4 to the left?

- A. Addition of sodium hydroxide to the solution
- B. Addition of manganese(II) chloride to the solution
- C. Removal of nitrate from the solution
- D. Removal of water from the solution

82. Which of the following graphs accurately depicts the NO_2 concentration in a flask containing reaction 2 before and after it has been bubbled through $\text{MnCl}_2(\text{aq})$?



Passage XII (Questions 83 - 89)

Physiological response is correlated with climate. This can be proven by observing the shortness of breath and/or headaches experienced by mountain climbers and skiers when they rapidly change altitudes. These symptoms are associated with a disorder referred to as *hypoxia*, a deficiency in the amount of oxygen that reaches body tissue. Hypoxia is a temporary disorder that in some cases can be fatal in the short term. After time, the disorder will disappear because the body becomes acclimated to the new environment.

The cause of hypoxia is a result of the decrease in oxygen in the environment caused by the increase in altitude. The partial pressure of oxygen gas at sea level in most parts of the world is just over 0.20 atm. At an altitude of one mile above sea level, the partial pressure of oxygen gas is roughly 0.16 atm. Over time, the body adjusts to the lower oxygen content by increasing the production of hemoglobin (Hb), the molecule that binds oxygen. Reaction 1 expresses the binding of oxygen by hemoglobin.



Reaction 1

The equilibrium expression for Reaction 1 is:

$$K_c = \frac{[\text{Hb}(\text{O}_2)_4]}{[\text{Hb}][\text{O}_2]^4}$$

Figure 1

Hemoglobin is composed of four separate polypeptide strands, each capable of binding a ferrous (Fe^{2+}) cation, held together as one molecule. By producing more hemoglobin, the body can shift the equilibrium to the right (forward direction). This results in more oxygen diffusing across the cell membrane. Once in the cell, myoglobin transports one oxygen molecule from the interior of the cell membrane to the mitochondria.

In acclimation, the amount of myoglobin does not change. Acclimation takes a period of time that ranges from a week or two to months. For the 1968 Mexico City Olympics, several athletes trained in high altitude to approximate the environment of Mexico City (elevation 7500 feet above sea level). For those athletes that trained closer to sea level, many did not perform well in the "thinner" air. Conversely, many current runners train in the mountains for sea-level races so that their blood can provide extra oxygen to starving muscle cells.

83. Once hemoglobin transfers oxygen to the cell membrane where it is absorbed, how many myoglobin molecules should be present per hemoglobin?

- A. 1
- B. 2
- C. 4
- D. 8

84. Which of the following changes will NOT increase the amount (in moles) of free oxygen in the body?

- A. Increased ventilation, resulting in an increased amount of air transferred through the lungs.
- B. An increase in the production of Hb
- C. Increasing the partial pressure of CO consumed in a normal breath.
- D. Increasing the total amount of blood.

85. What will be observed when a long-time mountain resident travels back to sea level?

- A. They will experience hypoxia due to the lower partial pressure of oxygen at sea level.
- B. They will experience hypoxia due to the higher partial pressure of oxygen at sea level.
- C. They will experience an increase in vitality due to the lower partial pressure of oxygen at sea level.
- D. They will experience an increase in vitality due to the higher partial pressure of oxygen at sea level.

86. In the presence of carbon monoxide, hemoglobin preferentially binds CO over O₂ in a ratio close to 200 : 1. Over time, what occurs in the blood of someone moving to a carbon monoxide rich environment?

- A. The increased CO reduces the amount of Hb that binds oxygen, so the Hb(O₂)₄ decreases. To compensate, the body produces more Hb and increases its rate of respiration.
- B. The increased CO reduces the amount of Hb that binds oxygen, so the Hb(O₂)₄ increases. To compensate, the body produces more Hb and increases its rate of respiration.
- C. The increased CO reduces the amount of Hb that binds oxygen, so the Hb(O₂)₄ decreases. To compensate, the body produces less Hb and decreases its rate of respiration.
- D. The increased CO reduces the amount of Hb that binds oxygen, so the Hb(O₂)₄ increases. To compensate, the body produces less Hb and decreases its rate of respiration.

87. How does hypoxia affect the amount of myoglobin?

- A. Myoglobin increases initially and then decreases.
- B. Myoglobin increases initially and then remain at higher concentration.
- C. Myoglobin decreases initially and then remain at lower concentration.
- D. Myoglobin remains constant.

88. What activity will most likely result in hypoxia?

- A. Repelling down a mountain
- B. SCUBA diving
- C. Water skiing on the ocean
- D. Snow skiing

89. Why should an athlete choose to train at high altitudes?

- A. At higher altitudes, the partial pressure of oxygen is greater, thus the athlete's body will increase its hemoglobin count.
 - B. At higher altitudes, the partial pressure of oxygen is less, thus the athlete's body will increase its hemoglobin count.
 - C. At higher altitudes, the partial pressure of oxygen is greater, thus the athlete's body will decrease its hemoglobin count.
 - D. At higher altitudes, the partial pressure of oxygen is less, thus the athlete's body will decrease its hemoglobin count.
-

Passage XIV (Questions 90 - 96)

The equilibrium of a gas-phase reaction can vary with the dimensions of the container. This is observed when a reaction has a different number of gas molecules on the two sides of the reaction equilibrium. Although the equilibrium constant remains constant when pressure, volume, and moles are altered, the ratio of product to reactant can vary. The following reaction is a typical reaction which shows a dependence on the dimensions of the container:



Reaction 1

The reaction may be carried out in a glass flask, where the volume of the container remains constant while the internal pressure may vary. The reaction may also be carried out in an enclosed piston, where the internal pressure remains constant while the volume of the container may vary. Le Châtelier's principle predicts that when the internal pressure rises, Reaction 1 will shift to the left to alleviate the increase in pressure. Le Châtelier's principle also predicts that when the volume of the container is increased, Reaction 1 will shift to the right to fill the volume. The equilibrium constant for Reaction 1 at 25°C is 4.72×10^{-3} atm.

Independent of the starting conditions and presence or absence of an inert gas, the ratio of the square of the partial pressure of nitrogen dioxide to the partial pressure of dinitrogen tetroxide is constant, given that the temperature does not change. The ratio of nitrogen dioxide to dinitrogen tetroxide increases as the temperature is increased.

90. The reaction as written is:

- A. endothermic with $\Delta S > 0$.
- B. endothermic with $\Delta S < 0$.
- C. exothermic with $\Delta S > 0$.
- D. exothermic with $\Delta S < 0$.

91. The reaction will shift to the right with all of the following changes EXCEPT:

- A. addition of $\text{N}_2\text{O}_4(\text{g})$.
- B. an increase in volume at constant pressure.
- C. a decrease in pressure at constant volume.
- D. addition of He gas to the system at constant volume.

92. Which of the following accurately shows K_{eq} in terms of ΔG ?

- A. $K_{\text{eq}} = -\frac{\Delta G}{\ln RT}$
- B. $K_{\text{eq}} = \frac{\Delta G}{\ln RT}$
- C. $K_{\text{eq}} = e^{\Delta G/RT}$
- D. $K_{\text{eq}} = e^{-\Delta G/RT}$

93. What is NOT true for a reaction with an equilibrium constant of 1.0×10^{-5} ?

- A. There is mostly product present at equilibrium.
- B. When starting with all reactants, the shift to reach equilibrium is insignificant.
- C. The K_{eq} for the reverse reaction is 1.0×10^{-5} .
- D. The same reaction when carried out with a catalyst will have the same equilibrium constant.

94. If the value of K_{eq} at a certain temperature is greater than one, what will be observed after radiolabeled $^{15}\text{N}_2\text{O}_4$ is added to an equilibrium mixture of NO_2 and N_2O_4 in a rigid container?

- A. The amount of radio-labeled NO_2 increases, while the amount of radio-labeled N_2O_4 decreases.
- B. The amount of radio-labeled NO_2 decreases, while the amount of radio-labeled N_2O_4 increases.
- C. The amount of radio-labeled NO_2 increases, while the amount of radio-labeled N_2O_4 remains constant.
- D. The amount of radio-labeled NO_2 decreases, while the amount of radio-labeled N_2O_4 remains constant.

95. As the handle of a piston container filled with an equilibrium mixture of NO_2 and N_2O_4 is lifted, what occurs?

- I. The $\frac{P_{\text{NO}_2}}{P_{\text{N}_2\text{O}_4}}$ ratio increases.
 - II. The piston cools down.
 - III. The mole percent of N_2O_4 increases.
- A. I only
 - B. II only
 - C. I and II only
 - D. II and III only

96. The K_{eq} for the reaction as written is:

- A. $K_{\text{eq}} = \frac{P_{\text{NO}_2}}{P_{\text{N}_2\text{O}_4}}$
- B. $K_{\text{eq}} = \frac{(P_{\text{NO}_2})^2}{P_{\text{N}_2\text{O}_4}}$
- C. $K_{\text{eq}} = \frac{P_{\text{N}_2\text{O}_4}}{P_{\text{NO}_2}}$
- D. $K_{\text{eq}} = \frac{P_{\text{N}_2\text{O}_4}}{(P_{\text{NO}_2})^2}$

Questions 97 - 100 are NOT based on a descriptive passage.

97. If the molar solubility of an MX type of salt is defined as y , then the solubility product is:

- A. y
- B. $2y$
- C. y^2
- D. $4y^2$

98. $\text{MgF}_2(\text{s})$ would be most soluble in which of the following solutions?

- A. 0.10 M $\text{CaF}_2(\text{aq})$
- B. 0.10 M $\text{NaF}(\text{aq})$
- C. 0.10 M $\text{NaCl}(\text{aq})$
- D. 0.10 M $\text{MgCl}_2(\text{aq})$

99. If the forward rate constant for a one-step reaction is four times the reverse rate constant, then which of the following is true?

- A. $K_{\text{eq}} = 0.0625$
- B. $K_{\text{eq}} = 0.25$
- C. $K_{\text{eq}} = 4.00$
- D. $K_{\text{eq}} = 16.00$

100. The amount of a salt that dissociates into water is ALWAYS increased by which of the following changes?

- A. Increasing the temperature.
- B. Decreasing the temperature.
- C. Increasing the amount of water.
- D. Decreasing the amount of water.



1. B	2. D	3. B	4. D	5. C
6. A	7. B	8. D	9. A	10. B
11. A	12. B	13. D	14. C	15. B
16. C	17. B	18. B	19. B	20. B
21. B	22. B	23. D	24. B	25. D
26. C	27. A	28. C	29. D	30. A
31. C	32. D	33. C	34. A	35. A
36. A	37. A	38. D	39. A	40. D
41. B	42. A	43. A	44. A	45. C
46. B	47. C	48. A	49. A	50. B
51. C	52. B	53. A	54. D	55. D
56. C	57. A	58. B	59. A	60. D
61. C	62. A	63. C	64. B	65. D
66. C	67. B	68. D	69. C	70. A
71. C	72. D	73. C	74. B	75. D
76. A	77. C	78. B	79. D	80. C
81. D	82. A	83. C	84. B	85. D
86. A	87. D	88. D	89. B	90. A
91. D	92. D	93. B	94. A	95. C
96. B	97. C	98. C	99. C	100. C

Equilibrium Passage Answers

Passage I (Questions 1 - 7)

Carbon Dioxide and Carbon Monoxide Equilibrium

1. **Choice B is correct.** This question involves first determining the value of the equilibrium constant (K_{eq}) at 102°C , and then solving for the partial pressure of CO_2 at 102°C . Converting 102°C into Kelvin yields 375 K . According to Table 1, at 375 K , $K_{eq} = 9\text{ atm}^{-1}$. The solution is as follows:

$$K_{eq} = 9 = \frac{(P_{\text{CO}_2})^2}{(P_{\text{CO}})^2(P_{\text{O}_2})} = \frac{(P_{\text{CO}_2})^2}{(2)^2(1)} = \frac{(P_{\text{CO}_2})^2}{4} = 9 \therefore (P_{\text{CO}_2})^2 = 36 \therefore P_{\text{CO}_2} = 6\text{ atm.}, \text{ choice B}$$

2. **Choice D is correct.** Adding CO_2 (a product) to an equilibrium mixture forces the reaction to the left. When the reaction shifts to the left, reactants ($\text{O}_2(\text{g})$ and $\text{CO}(\text{g})$) increase. This eliminates choices A and B. Because K_{eq} changes only with temperature, and the temperature did not change in this case, choice C is invalid. By default, (the elimination of three wrong choices), choice D is the correct answer.
3. **Choice B is correct.** Because the value of K_{eq} increases as the temperature increases, products increase upon the addition of heat to the reaction. This means that heat lies on the reactant side of the reaction and thus the reaction is endothermic as written. The ΔH is positive for an endothermic reaction. Pick B and feel warm.
4. **Choice D is correct.** The equilibrium constant is just that, a constant. It is specific for a given temperature, so it varies only with a change in temperature. The equilibrium may shift with changes in either the pressure, volume or concentration, but the value of K remains constant. This makes choice D the best answer.
5. **Choice C is correct.** The following chart summarizes the partial pressures during the course of the reaction:

Reaction:	$2\text{ CO}(\text{g})$	$\text{O}_2(\text{g})$	$\rightleftharpoons$	$2\text{ CO}_2(\text{g})$
Initially:	0	0		1.00
Shift:	$+2x$	$+x$	$\longrightarrow$	$-2x$
Equilibrium:	$2x$	x		$1 - 2x$

The total pressure of the system is the sum of the partial pressures of each component at any given time during the reaction. The question here asks for the total pressure at equilibrium, so the equilibrium partial pressures must be added. To solve this question, use the equation:

$$P_{\text{total}} = P_{\text{CO}_2} + P_{\text{CO}} + P_{\text{O}_2} = (1 - 2x) + 2x + x = 1 + x.$$

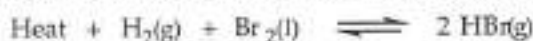
The value of x can be no larger than 0.5 atm. , because you can lose no more $\text{CO}_2(\text{g})$ than you start out with (1.00 atm.). The value of x must be somewhere between 0 and 0.5 , because some of the carbon dioxide is lost, but not all of it. This makes the following relationship $1.5 > P_{\text{total}} > 1$ true, which is choice C, your best choice.

6. **Choice A is correct.** Compressing the reaction vessel results in reduced volume and increased pressure. Upon decreasing the container volume, prior to any shifting of the reaction, the partial pressure and concentration of each gas increases, but the mole ratio is the same until the reaction shifts. This is why no answer choices contained the term *concentration*. Compressing the container decreases the free area in which the gases can exist. The more crowded environment favors the formation of CO_2 , because carbon dioxide lies on the less crowded side of the reaction. This means that moles of CO_2 increase, moles of CO decrease, and moles of O_2 decrease. This eliminates choice B. When moles of CO decrease, the mole fraction of CO decreases, so choice C is eliminated. The temperature did not change, so the equilibrium constant remains the same, so choice D is eliminated. The best answer is choice A. As a note of interest, all of the partial pressures increase, but carbon dioxide increases by the greatest amount. The shift to re-establish equilibrium never completely offsets the stress, so the result is that CO and O_2 have slight increases in partial pressure overall, while CO_2 shows a significant increase in partial pressure.
7. **Choice B is correct.** Addition of $\text{Ca}(\text{OH})_2(\text{aq})$ to an equilibrium mixture of Reaction 1 serves to remove $\text{CO}_2(\text{g})$ from the reaction mixture. To compensate for lost $\text{CO}_2(\text{g})$, Reaction 1 proceeds in a way so as to regenerate $\text{CO}_2(\text{g})$, but it is not completely regenerated. This eliminates choice A. To produce $\text{CO}_2(\text{g})$, the reaction must shift right, which consumes both $\text{O}_2(\text{g})$ and $\text{CO}(\text{g})$. This results in a decrease of both $\text{O}_2(\text{g})$ and $\text{CO}(\text{g})$. The best answer is choice B. Total moles of gas decrease, so the total pressure of the system decreases.

8. **Choice D is correct.** K_{eq} for the reaction is determined by plugging the values from Table 2 into the equilibrium expression. Upon doing so, the value is found to be 5.44 atm., making choice D the best answer.

$$K_{eq} = \frac{(P_{HBr})^2}{P_{H_2}} = \frac{(2.72)^2}{1.36} = \frac{(2.72)(2.72)}{1.36} = 2.72 \times \frac{2.72}{1.36} = 2.72 \times 2 = 5.44 \text{ atm.}$$

9. **Choice A is correct.** Addition of the gaseous HBr (a product in Reaction 1) to the equilibrium mixture pushes the reaction to the reactant side (right), and thus *increases* both the partial pressure and mole percent of H_2 . If you are of sound mind and body, then you should pick choice A. As long as the temperature remains constant, the concentration of $Br_2(l)$ (or any other pure liquid) remains constant. The concentration of a liquid is measured as density, which changes with temperature. This eliminates choices C and D, so don't pick them.
10. **Choice B is correct.** The *greatest* amount of $H_2(g)$ forms from the reaction that generates the greatest back reaction or smallest forward reaction. Choices A and C are eliminated, because less $H_2(g)$ is present initially than in choice B, therefore less $H_2(g)$ exists at equilibrium. In addition, the difference between choices A and C is the amount of $Br_2(l)$, and liquids do not affect the equilibrium distribution, so both yield the same amount of $H_2(g)$. There is only one correct answer per question, so identical answer choices should both be eliminated. If choice D went one hundred percent in the reverse direction, then 0.50 atm. of $H_2(g)$ would be generated. This value is half of the amount of $H_2(g)$ in choice B, so choice D is eliminated. To generate an equivalent amount of $H_2(g)$ at equilibrium as starting with 1.00 atm. $H_2(g)$ and excess $Br_2(l)$, requires starting with 2.00 atm. HBr(g).
11. **Choice A is correct.** An increase in temperature results in the addition of heat to the equilibrium mixture. Because the mole percent of product (HBr) increases, heat must be viewed as a reactant in the reaction.



When heat is viewed as a reactant, the reaction is defined as being endothermic, so choice A is the correct answer. An endothermic reaction has ΔH value that is a *positive* number. Choice C is eliminated, because the pressure changed with both the influx of heat and the shift of the reaction, so it cannot be an isobaric (constant pressure) process. Choice D is eliminated, because the passage states that Reaction 1 is an equilibrium mixture.

12. **Choice B is correct.** The definition of equilibrium is a state where the forward and back reactions have equal rate. Choices A and D are equivalent, so they are both eliminated. The reaction lacks $Br_2(l)$, and despite liquids not affecting equilibrium, keep in mind that the system is not in equilibrium. Liquids can be a limiting reagent. Because there is no $Br_2(l)$ present, the reaction cannot move in the forward direction, even if it wanted to. This makes choice B the best answer. At this point, we are uncertain it is correct, but it is the best answer choice. To determine the direction in which a reaction proceeds, it is necessary to compare the value of the reaction quotient (Q_{rx}) to the equilibrium constant (K_{eq}). The two possible scenarios are: $Q_{rx} > K_{eq}$, in which case the numerator is too large in the equilibrium quotient (products are in excess of reactants), making the reaction shift to the left (reactant side) to reach equilibrium. The second scenario is where $Q_{rx} < K_{eq}$, in which case the denominator is too large in the equilibrium quotient (reactants are in excess of products), making the reaction shift to the right (product side) to reach equilibrium. We solve for Q as follows:

$$Q_{rx} = \frac{(P_{HBr})^2}{P_{H_2}} = \frac{1^2}{1} = 1 < \frac{(2.72)^2}{1.36} \therefore Q_{rx} < K_{eq}$$

Because $Q_{rx} < K_{eq}$, Reaction 1 should shift right (to the product side). The problem here is that there is no Br_2 present, so there can be no reaction. This confirms that Br_2 is the *limiting reagent*. To quickly determine the direction of a reaction, put K_{eq} and Q_{rx} alphabetically and turn the greater than or less than sign into an arrow (i.e., the $K_{eq} > Q_{rx}$ becomes $K_{eq} \rightarrow Q_{rx}$, meaning the reaction proceeds forward to reach equilibrium).

13. **Choice D is correct.** All reagents are present, so a reaction is possible. To answer this question, solve for Q_{rx}

$$Q_{rx} = \frac{(P_{HBr})^2}{P_{H_2}} = \frac{3^2}{1.5} > \frac{(2.72)^2}{1.36} \therefore K_{eq} < Q_{rx} \text{ by a small amount}$$

Because $K_{eq} < Q_{rx}$, the reaction proceeds to the left (reactant side). This decreases HBr(g) and increases $H_2(g)$, so choices A and B are eliminated. The shift is small, because Q and K are close, so choice D is the best answer.

14. **Choice C is correct.** If the flask is initially charged with only HBr(g) , then the graph of $\text{H}_2\text{(g)}$ over time starts at 0, eliminating choice A. Choice D is a titration curve, so it is an incorrect answer choice. The graphs depict the progression of a reaction. Reactions proceed with a gradual tapering off (choice C), not an instantaneous shut-off (choice B). Pick C, because $\text{H}_2\text{(g)}$ increases most rapidly initially, then levels off until equilibrium.
15. **Choice B is correct.** "...NOT affect the partial pressure of $\text{H}_2\text{(g)}$?" Addition of NaOH(aq) removes HBr and indirectly shifts the Reaction 1 equilibrium to the right, through a complex equilibrium. Addition of $\text{H}_2\text{(g)}$ and HBr(g) directly affect the equilibrium, shifting the equilibrium to the opposite side of the compound added. If the reaction is already at equilibrium, then neither a solid nor a liquid (such as $\text{Br}_2\text{(l)}$ in this case) affects the equilibrium. For this reason, choice B is a particularly swell answer choice. A solid or liquid can only affect the net results (equilibrium conditions) if the system is not at equilibrium and the compound being added (the solid or liquid) is the limiting reagent. When added in, the system can then achieve equilibrium.

Passage III (Questions 16 - 22)

Equilibrium Constant Magnitude

16. **Choice C is correct.** According to the two reactions, only PCl_3 and NO react with Cl_2 , so PCl_5 and NOCl are not oxidized by Cl_2 . This eliminates answer choices B and D. By looking at the oxidation states of the reactants and products, PCl_3 ($P = +3$), PCl_5 ($P = +5$), NO ($N = +2$), and NOCl ($N = +3$), it shows that the forward direction in both Reaction 1 and Reaction 2 as drawn are oxidation half-reactions for the non-metal compounds. The reactants get oxidized by chlorine in the forward reaction as written. Comparing the two values of K_{eq} shows that reaction 2 lies further to the right (product side) than does Reaction 1. Thus, Reaction 2 favors the formation of products more so than Reaction 1. This implies that NO reacts with Cl_2 more readily than PCl_3 reacts with Cl_2 . The greater the reaction with Cl_2 means more oxidation by Cl_2 . Choose C to feel good.
17. **Choice B is correct.** Initially, there is twice as much PCl_3 as Cl_2 , and their sum pressure is 1.50 atm. When the reaction shifts in the forward direction, total pressure decreases, so choice D is eliminated. According to the mole ratio, the initial partial pressure of phosphorus trichloride is 1.0 atm. and the initial partial pressure of chlorine gas is 0.5 atm. The chart below summarizes the partial pressures during the course of the reaction.

Reaction:	$\text{PCl}_3\text{(g)}$	$\text{Cl}_2\text{(g)}$	$\rightleftharpoons$	$\text{PCl}_5\text{(g)}$
Initially:	1.0	0.5		0
Shift:	$-x$	$-x$	$\longrightarrow$	$+x$
Equilibrium:	$1-x$	$0.5-x$		x

The total pressure of the system is a sum of the partial pressures of each component at any given time during the reaction. The question here asks for the total pressure at equilibrium, so the equilibrium partial pressures must be added. To solve this question, use the formula $\sum P_i = P_{\text{total}}$. In this case the following holds true:

$$P_{\text{total}} = P_{\text{PCl}_3} + P_{\text{Cl}_2} + P_{\text{PCl}_5} = (1-x) + (0.5-x) + x = 1.5-x.$$

If everything reacted to the limiting reagent capacity, then x is 0.5. But the reaction does not go to completion, so the value of x falls between 0 and 0.5. This is because some, but not all, of the reactant is lost. Choice A is eliminated, because $1.5 > P_{\text{total}} > 1$. To solve this question, you must estimate the magnitude of x . The value of x is greater than 0.25, because over half shifts to products to reach an equilibrium constant of 1.95. This makes choice B the best answer. To solve it more precisely, assume P_{total} at equilibrium is 1.25 (the borderline between answer choices B and C). If P_{total} at equilibrium is 1.25 atm., then $x = 0.25$ atm., and at equilibrium, $P_{\text{PCl}_3} = 0.75$ atm., $P_{\text{Cl}_2} = 0.25$ atm., and $P_{\text{PCl}_5} = 0.25$ atm. The following Q would result:

$$Q_{\text{rx}} = \frac{P_{\text{PCl}_5}}{(P_{\text{PCl}_3})(P_{\text{Cl}_2})} = \frac{0.25}{(0.75)(0.25)} = \frac{1}{0.75} = 1.33 \therefore K_{\text{eq}} > Q_{\text{rx}}, \text{ so more must shift over.}$$

This means that x is between 0.25 and 0.50 atm., and thus P_{total} at equilibrium is between 1.00 and 1.25.

18. **Choice B is correct.** The reaction listed in the question is the reverse reaction of Reaction 2. When a reaction is reversed, the equilibrium expression is inverted, because reactants and products are inter-converted. The resulting equilibrium constant is the inverse of the original value. The inverse of something times 10^4 is something times 10^{-5} , making choice B the best answer. Choice A is an incorrect answer that could have been reached if you inadvertently decided the values must multiply to 1.0×10^{-14} (a common value in acid-base chemistry). Choice C was the result of taking the square root, which is wrong in this case. Choice D is eliminated because the forward and back reactions are different.

19. **Choice B is correct.** Welcome to the wonderful world of math. The following chart summarizes the changes in partial pressures as the system goes to equilibrium. You must solve for the x -value. Keep in mind that the answer you seek is actually $2x$, but you must first solve for x before you determine $2x$.

Reaction:	2 NO(g)	$\text{Cl}_2\text{(g)}$	$\rightleftharpoons$	2 NOCl(g)
Initially:	0	0		1.0 atm.
Shift:	$+2x$	$+x$	$\leftarrow$	$-2x$
Equilibrium:	$2x$	x		$1 - 2x$

From here, it is a matter of plugging the equilibrium values into the equilibrium expression.

$$K_{\text{eq}} = \frac{(P_{\text{NOCl}})^2}{(P_{\text{NO}})^2(P_{\text{Cl}_2})} = \frac{(1.0 - 2x)^2}{(2x)^2(x)} = \frac{1^2}{(2x)^2(x)} = \frac{1}{4x^3} = 2.1 \times 10^4$$

The easiest thing to do here is to take the reciprocal of each side and then solve for x , keeping in mind that the answer choices are $2x$.

$$4x^3 = \frac{1}{2.1 \times 10^4} = 4.9 \times 10^{-5} = 49 \times 10^{-6} \therefore x^3 = 12.25 \times 10^{-6} \therefore x = \sqrt[3]{12.25 \times 10^{-6}}$$

The cube root of 12.25 lies between 2 and 3 (because $2^3 = 8$ and $3^3 = 27$), so let's call it 2.7. $P_{\text{NO}} = 2(2.7 \times 10^{-2}) = 4.7 \times 10^{-2}$, which is choice B. The math is not that bad, if you approximate.

20. **Choice B is correct.** Immediately, choices A and C are eliminated, because there is no reaction possible, when the reaction has only one reactant (and lacks Cl_2). From here, it comes down to interpreting the magnitude of K_{eq} values. The reaction that forms the greatest amount of Cl_2 is the reaction with the K_{eq} that most favors formation of reactants (because Cl_2 is a reactant in Reaction 1 and Reaction 2). Reaction 1, with the smaller K_{eq} , most readily favors the formation of reactants out of the two reactions. Because of the squared term in k_{eq} for Reaction 2, we need to look more closely in most cases, but the K_{eq} values are so drastically different for the two reactions, that you are safe ignoring the squared terms in the K_{eq} of Reaction 2. The larger K_{eq} is associated with Reaction 2, so choice D is out and the best answer is choice B.
21. **Choice B is correct.** This is a rehash of an ongoing theme in this passage. Addition of Cl_2 shifts both Reaction 1 and Reaction 2 in the forward direction. The question here entails the magnitude of the shift. Reaction 1, with a K_{eq} value of approximately 1, shifts the equilibrium to the product side to use roughly half of the Cl_2 gas that was added. Adding Cl_2 to reaction 2, with a K_{eq} that is extremely large, will shift the equilibrium to the product side to use nearly all of the Cl_2 gas that was added. Choice B is the best answer.
22. **Choice B is correct.** To determine the direction of a shift requires first solving for the equilibrium quotient (Q) and then comparing that value to the K_{eq} .

$$Q_{\text{rx}} = \frac{P_{\text{PCl}_5}}{(P_{\text{PCl}_3})(P_{\text{Cl}_2})} = \frac{0.40}{(0.40)(0.40)} = \frac{1}{0.40} = 2.5 \therefore K_{\text{eq}} < Q_{\text{rx}}, \text{ so the reaction shifts left.}$$

When $K < Q$, the Q is too large, so the products must decrease and the reactants must increase ($K \leftarrow Q$). This forces the reaction to move to the left (reactant side). You really should pick choice C. Choice D is eliminated, because a catalyst affects only the kinetics (rate), and not the thermodynamics of a reaction. A catalyst simply lowers the activation energy, but has no effect on the heat of reaction (ΔH), the free energy change (ΔG) for the reaction, or the equilibrium constant (K) for the reaction.

Passage IV (Questions 23 - 29)

Gas-Phase Equilibrium Constant Experiment

This is an experiment that is used to determine the equilibrium constant for the gas phase reaction listed in the passage. In interpreting the apparatus in Figure 1, you must note the two reactants are being reacted once the valve is opened between the gas line and the flask. The reactants are mixed in a ratio of 4 H_2 to 1 CS_2 , and they just-so-happen to react in a ratio of 4 : 1. This means that the reactants always exist in a 4 : 1 ratio and the products always exist in a 2 : 1 ratio. The reactants are never completely depleted, thus the total pressure ($P_{\text{init}} - 2x$) of the system can never reach a value as low as 0.60 atm. Overall, the number of molecules of gas are decreasing as the reaction proceeds, which accounts for the decrease in pressure experienced by the system as it approaches equilibrium. The final total pressure is not given on the graph, but a ΔP is a negative value. This passage seems hard, but it is really testing your organization skills.

23. **Choice D is correct.** The purpose of the experiment is to determine the equilibrium constant (K) for Reaction 1. The value of the equilibrium constant varies with the temperature, so the temperature **must** be held constant to accurately evaluate the value of the equilibrium constant. This means that the best answer is choice D. Hopefully the other answer choices did not seem that tempting.
24. **Choice B is correct.** Upon heating a closed, isochoric system, the internal pressure increases according to the ideal gas law ($PV = nRT$). This in essence is Charles' law, eliminating choice A, which has Charles' law listed incorrectly. Because the pressure decreases rather than increases, another variable must be changing. The system is in a flask, so the volume can't change. This means that moles must be decreasing, causing the pressure to decrease. The only way for moles to change in a closed system, is for a chemical reaction to take place. Reaction 1 shifted from the reactant side (five gas molecules) to the product side (three gas molecules). This means that the reaction proceeds in the forward direction to consume the heat added to the system. This means that the reaction as written must be endothermic, making choice B correct.

25. **Choice D is correct.** This question requires setting up the equilibrium relationship that transpires during the course of the reaction. The reaction goes as follows:

Reaction:	$\text{CS}_2(\text{g})$	$4 \text{H}_2(\text{g})$	$\rightleftharpoons$	$\text{CH}_4(\text{g})$	$2 \text{H}_2\text{S}(\text{g})$
Initially:	0.20	0.80		0	0
Shift:	$-x$	$-4x$	$\longrightarrow$	$+x$	$+2x$
Equilibrium:	$0.2 - x$	$0.8 - 4x$		x	$2x$

At any point, including equilibrium, the total pressure of the system is found according to Equation 1.

$$P_{\text{total}} = P_{\text{CS}_2} + P_{\text{H}_2} + P_{\text{CH}_4} + P_{\text{H}_2\text{S}} = (0.2 - x) + (0.8 - 4x) + x + 2x = 1.0 - 2x$$

The initial pressure of the system was 1.0 atmospheres when the valve was first opened to start the reaction. The pressure of the system at equilibrium is $1 - 2x$ atm., therefore the change in internal pressure from the start of the reaction until equilibrium is $2x$. This means that the partial pressure at equilibrium due to H_2S (which is also $2x$) equals the change in internal pressure during the reaction. The partial pressure due to CH_4 at equilibrium is one-half of the change in internal pressure. This makes choice D the best answer.

26. **Choice C is correct.** This question requires that you deduce that the value of x can never exceed 0.2. This upper limit of 0.2 is due to the initial pressure of CS_2 being only 0.2 atm. You can lose no more CS_2 than you have available as a reactant. Because of this, no matter what value for x is used (any value between 0 and 0.2), $0.8 - 4x$ will always be greater than $0.2 - x$. This means that P_{H_2} is always greater than P_{CS_2} because $4x$ must always be less than 0.8. The best answer is choice C. If the reaction goes completely in the forward direction, then products will exceed reactants and $P_{\text{CS}_2} < P_{\text{CH}_4}$, eliminating choice A. Because there is no CH_4 or H_2S in the initial reaction mixture, the only CH_4 or H_2S present during the reaction or at equilibrium, is the result of the reaction proceeding in the forward direction. For every one CH_4 that forms, two molecules of H_2S form, so $P_{\text{H}_2\text{S}}$ is always greater (not less) than P_{CH_4} , eliminating choice B. If the reaction barely proceeds in the forward direction, then the reactants will exceed the products and $P_{\text{H}_2} > P_{\text{H}_2\text{S}}$. This eliminates choice D.
27. **Choice A is correct.** The equilibrium constant is the pressure of the products divided by the pressure of the reactants. Answer choices B and D (which both have reactants divided by products) are eliminated. The coefficients from the balanced equation **must** be included as exponents in the equilibrium expression, so choice A is the correct choice. All of the reactants and products are gases, thus they all belong in the K_{eq} expression. Keep in mind that solids and pure liquids are not included in the equilibrium expression. The equilibrium constant for this reaction is referred to as K_p , because it is a gas system and the values are partial pressures.
28. **Choice C is correct.** The total pressure of the system at equilibrium is a sum of the partial pressure of each component gas at equilibrium. The total pressure of the system is found as follows:

$$P_{\text{total}} = P_{\text{CS}_2} + P_{\text{H}_2} + P_{\text{CH}_4} + P_{\text{H}_2\text{S}} = (0.2 - x) + (0.8 - 4x) + x + 2x = 1.0 - 2x$$

Choices A and D are eliminated, because the total pressure is less than 1.0 atm. The final pressure of the system at equilibrium is $1 - 2x$. The value of x cannot be greater than 0.20 atm because the CS_2 gas is the limiting reagent and only 0.20 atm. of CS_2 is initially present. Because the value of $2x$ is subtracted, the final pressure drops, but it cannot drop to a value less than 0.6 atmospheres. The best answer is choice C.

29. **Choice D is correct.** The equilibrium constant (a numerical value), changes with the temperature, but it does not change with volume, pressure, moles of reactants, or moles of products. Choice A involves adding hydrogen gas (a reactant). The reaction shifts to the right to reestablish equilibrium, but the equilibrium constant remains constant. This is to say that the equilibrium position may change, but the equilibrium constant does not. Choice B involves removing all four gases. To determine which escapes most readily involves molecular mass and effusion rates, but that is irrelevant in this question. Losing both products and reactants causes the reaction to shift to reestablish equilibrium, but the equilibrium constant remains constant. Choice B is eliminated. Choice C involves adding hydrogen sulfide gas (a product). The reaction shifts to the left to reestablish equilibrium, but the equilibrium constant will remain constant. Choice D involves increasing the temperature. The equilibrium position and the equilibrium constant both change with a change in temperature. The value of the equilibrium constant can only be changed by temperature.

Passage V (Questions 30 - 36)

Equilibrium Reactions

30. **Choice A is correct.** Given that the initial partial pressures of H_2O and Cl_2O are both 0.5 atmospheres, and both are reactants, it is possible initially for the reaction to proceed in the forward direction. As such, the initial forward rate cannot start at zero. This eliminates choices C and D. There is no HOCl present initially, therefore the reverse reaction rate must start at zero. This reaffirms that choices C and D are invalid. As the reaction proceeds to equilibrium, the reactant pressures decrease, thus the forward reaction slows gradually until equilibrium is reached. Equilibrium occurs when the rate forward is equal to the rate reverse. Because the forward rate diminishes with time, the best answer is choice A.
31. **Choice C is correct.** Reactions that can shift when the pressure of the system changes are reactions that have an unequal number of gas molecules on the reactant and product sides of the reaction. In Reaction 1, there are three reactant gas molecules and two product gas molecules, so Reaction 1 shifts in the forward direction if the total pressure increases on the equilibrium system. In Reaction 2, there are two reactant gas molecules and two product gas molecules, so Reaction 2 cannot shift in either the forward or reverse direction if the total pressure increases. In Reaction 3, there are two reactant gas molecules and one product gas molecule, so Reaction 3 shifts in the forward direction if the total pressure increases on the equilibrium system. This means that both Reaction 1 and Reaction 3 shift when the total pressure of the system changes. The best answer is choice C.
32. **Choice D is correct.** The equilibrium constant of a reaction remains constant unless the temperature of the system is changed. Changing the volume of the system does not change the equilibrium constant (although the reaction may shift and the equilibrium ratios change). This eliminates choices A and B, which show the K_{eq} in its equation form. As the volume of the system increases, the reaction shifts to the side of the reaction with more molecules (in this case the reactant side), thus the ratio of HCOCl (a product) to HCl (a reactant) decreases as the reaction reestablishes equilibrium following a change in volume. The best answer is choice D.
33. **Choice C is correct.** Sulfur dioxide and oxygen are both reactants, according to Reaction 1. When mixing 0.75 atm. SO_2 with 0.50 atm. O_2 , Reaction 1 proceeds in the forward direction to establish equilibrium. The value of K_{eq} for this reaction (582), as listed in the passage, is well above one. At equilibrium, there is significantly more SO_3 than both SO_2 and O_2 . This eliminates choices B and D. The SO_2 should start higher than the O_2 , but it diminishes twice as fast as O_2 , thus the SO_2 line should drop below the O_2 line at equilibrium. If SO_2 is completely depleted (all 0.75 atm. are consumed), 0.375 atm. of O_2 would be consumed, leaving 0.125 atm. O_2 . This is because SO_2 is the limiting reagent, if the reaction goes to completion. This makes choice C the best answer.
34. **Choice A is correct.** The equilibrium constant for reaction 2 is 8.61×10^{-3} at 500°C . Choice A is the reverse reaction from Reaction 2, so the equilibrium constant is the reciprocal of 8.61×10^{-3} at 500°C . This leads to a value with 10^2 , not 10^{-3} in it. This makes choice A the mismatch of K_{eq} and reaction. Choice B is the same as Reaction 2 except the values have been cut in half. The result is that the K_{eq} as written should be the square root of 8.61×10^{-3} . This leads to a value with 10^{-2} in it, therefore choice B seems reasonable and thus valid. The equilibrium constant for Reaction 3 is 3.26×10^{-6} at 500°C . Choice C is the reverse reaction from Reaction 3, where all the values have been cut in half. The equilibrium constant as written should be the square root of the reciprocal of 3.26×10^{-6} at 500°C . This leads to a value with 10^2 in it, so choice C seems reasonable and valid. Choice D is Reaction 3, where all the values have been doubled. The equilibrium constant as written should be the square of 3.26×10^{-6} at 500°C . This leads to a value with 10^{-11} or 10^{-12} in it, so choice D seems reasonable and valid. The only mismatch is found with choice A.

35. **Choice A is correct.** The conversion from the pressure unit of torr to the pressure unit of atmospheres involves dividing the value in torr by 760 (the conversion factor from torr to atm.). Because the K_p value is given in terms of torr^{-1} , the value must be multiplied by 760 in order to convert it into a value in terms of atm^{-1} . The best answer is choice A.
36. **Choice A is correct.** An endothermic reaction is set up as follows:



The cooling of the reaction can be treated as the removal of heat. When heat is removed from an endothermic reaction, because the heat is acting like a reactant, the reaction shifts to the left (reactant side) to regenerate the lost heat. This shift causes an increase in reactants and a decrease in products. The product to reactant ratio therefore decreases. Choices B, C, and D all account for the same shift, so the three choices can be eliminated. The best choice is thus choice A.

Passage VI (Questions 37–42)

Le Châtelier's Principle

The interesting feature of this reaction is that the ratio of the reagents is 1 : 1 : 1 : 1, so the total pressure of the system never changes. The reactants always decrease by the same amount, and the products always increase by the same amount as the reaction proceeds from the initial conditions to the equilibrium conditions. The purpose of this experiment is to prove that equilibrium is pathway independent, implying that whether the reaction starts with all reactants or all products, it will reach the exactly the same equilibrium.

37. **Choice A is correct.** Increasing the volume of the reaction vessel has no effect on the equilibrium, because there are equal moles of gas molecules on both sides of the reaction. Adding moles of either reactant or product will shift the equilibrium to the opposite side. Because the reaction is exothermic, it will shift forward with a decrease in temperature and reverse with an increase in temperature. Thus, the best answer is choice A.
38. **Choice D is correct.** The equilibrium constant is less than 1.00, so the reactants are more abundant than the products. This means at equilibrium, both CO_2 and H_2 have concentrations less than half of their initial concentration (1.00 M). The only answer that shows this is choice D. Note that the final concentrations are the same as in trial I. Regardless of the starting concentrations, the equilibrium concentrations will be the same if the total pressure and the temperature are the same.
39. **Choice A is correct.** When reaction vessel III was cooled from 1000°C to 500°C , the partial pressure of carbon dioxide gas (CO_2) increased. This increase in partial pressure for carbon dioxide indicates that the reaction shifted to the right (product side). The reaction must therefore be exothermic, because as heat is removed from the system, the reaction shifts to the product side to produce heat. Any reaction that releases heat in the forward direction is exothermic. This means that as heat is added to the reaction, the reaction shifts to the left (to the reactant side). This results in a decrease in the partial pressure of CO_2 and a decrease in the equilibrium constant. The correct answer is choice A.
40. **Choice D is correct.** A change in pressure will never change the equilibrium constant. This eliminates choices A and B. Because there is the same number of gas molecules on both sides of the equation, a change in pressure has no effect on the equilibrium. This means that the equilibrium does not shift so the best choice is D.
41. **Choice B is correct.** Because there is an equal number of gas molecules on both sides of the reaction (two molecules on each side), the total pressure of the system will not change with shifts in the reaction. This eliminates choices A and C. Because there is no water present initially, the reaction must shift left to reach equilibrium. This results in an increase in both water and carbon monoxide, therefore both slopes should increase equally. This is shown only in choice B.
42. **Choice A is correct.** When the equilibrium constant (K_{eq}) is greater than 1, the concentration of products is greater than the concentration of reactants. As such, choices B and C are incorrect statements, and can be eliminated. For an exothermic reaction, the equilibrium of the system shifts to the right (product side) as the temperature of the system decreases. This means that the products increase and the reactants decrease as the temperature of the system decreases. Both choices B and D are eliminated because of this, leaving only choice A still standing. If the products outnumber the reactants, then the equilibrium constant is greater than 1.0, so the best answer is choice A.

43. **Choice A is correct.** The molar solubility values of the sulfate salts are greater than the molar solubility values of the carbonate salts, as shown by the values in Table 1. This makes statement I valid. The addition of complexing agents helps to dissolve cations into aqueous solution by forming ligand bonds to the metal. The most common example is hemoglobin, which serves to make iron dication and trication more water soluble. This makes statement II valid. Given that both statements I and II are true, choices B and C are both eliminated. Hydrochloric acid is a strong acid, therefore chloride anion is not affected by the presence of hydronium ion. As a result, chloride salts show no pH dependence. Sulfide is a weak base, therefore sulfide shows a pH dependence. This makes statement III invalid, so the best answer is choice A.
44. **Choice A is correct.** The most accurate value for K_{sp} is found when the most accurate measurement of the molar solubility is used in the calculation. The most accurate measurement of the molar solubility is found with the salt having the greatest molar solubility. As a general rule, the larger the value, the less significant the error in its measurement. The highest molar solubility is found with Ag_2SO_4 , at 1.3×10^{-2} M. The best answer is choice A. This question is just a different way of asking which compound has the greatest molar solubility.
45. **Choice C is correct.** The anion chosen to remove Zn^{2+} selectively is the anion that forms the least soluble salt with zinc dication, rather than with silver cation or strontium dication. According to molar solubility values, the least soluble chloride salt is AgCl , so chloride anion is not a good choice. According to molar solubility values, the least soluble carbonate salt is ZnCO_3 , so carbonate anion is a possible choice. According to molar solubility values, the least soluble hydroxide salt is Zn(OH)_2 , by a substantial margin, thus hydroxide anion is the best choice so far. According to molar solubility values, the least soluble sulfate salt is ZnSO_4 , so sulfate anion is a possible choice. The greatest difference between zinc and the next most soluble cation comes with hydroxide anion. The best answer is thus choice C.
46. **Choice B is correct.** Silver chloride is the only chloride salt that is highly insoluble. A precipitate forming when chloride is added to solution indicates that silver cation must be present. This makes choice B the best answer. Choices A and D cannot be concluded, because no precipitate is expected between chloride and either zinc or strontium. This means that zinc or strontium may or may not be present.
47. **Choice C is correct.** When the cation mixture is treated with chloride anion, the least soluble chloride salt is silver chloride, therefore chloride will remove the Ag^+ from solution. When the solution with the three remaining cations is treated with sulfide dianion, the least soluble sulfide salt is lead sulfide, therefore sulfide will remove the Pb^{2+} from solution. When the solution with the two remaining cations is treated with hydroxide anion, the least soluble hydroxide salt is zinc hydroxide, therefore hydroxide will remove the Zn^{2+} from solution. The cation that will remain in solution is strontium (Sr^{2+}), thus the best answer is choice C.
48. **Choice A is correct.** An acid can protonate a basic salt in solution (which in some cases serves to form the more soluble ionic form of the product). The acid serves to reduce the hydroxide concentration which allows more of the basic salt to dissociate. This is a complex equilibrium that develops as a result. A complex equilibrium usually enhances the solubility in aqueous solution. Pick A to feel perky.
49. **Choice A is correct.** Because the solution has OH^- present already, the solubility of AgOH will be hindered by the common ion effect where hydroxide is the common ion. AgOH will be most soluble in the solution with the least hydroxide anion present. The least hydroxide is found in the solution with the most H^+ present. This solution consequently has the lowest pH. Pick A to feel spunky.

50. **Choice B is correct.** Because all three salts dissociate into one cation and one anion, the compound with the greatest molar solubility also has the greatest value for its solubility product (K_{sp}). The MZ salt is the heaviest and has the least mass dissociate into solution, so MZ has the lowest moles of salt in solution and thus has the lowest solubility product. This eliminates choices C and D. Deciding between MX and MY requires more work. More mass of MY dissociates than MX, but MX has a lower molecular mass. The moles of MX equal 7 over 100 while the moles of MY equal 9 over 120. This means that there are 0.07 moles MX and 0.075 moles MY. MY is present in a smaller volume of solution than MX, so MY has a greater molarity and solubility product than MX at 25°C . The correct answer is choice B.

51. **Choice C is correct.** Because the solubility of a salt increases with temperature, answer choices B and D are eliminated. It can also be read from the third column in Table 1 that more MX and MY dissociate into water at a temperature greater than 25°C than what dissociates at 25.0°. This also eliminates choices B and D. The molarity of MX at 31.6°C and the molarity of MY at 27.4°C are found as follows:
- $$M_{MX @ 31.6^\circ C} = \frac{7 \text{ g}}{0.0262 \text{ L}} \cdot \frac{100 \text{ g} \cdot \text{mole}^{-1}}{0.0262 \text{ L}} = \frac{0.070 \text{ mol}}{0.0262 \text{ L}} = \frac{7}{2.62} \text{ M} \quad M_{MY @ 27.4^\circ C} = \frac{9 \text{ g}}{0.0253 \text{ L}} \cdot \frac{120 \text{ g} \cdot \text{mole}^{-1}}{0.0253 \text{ L}} = \frac{0.075 \text{ mol}}{0.0253 \text{ L}} = \frac{7.5}{2.53} \text{ M}$$
- There are more moles of MY than MX in a smaller volume of solution, therefore saturated MY at 27.4°C is more concentrated than saturated MX at 31.6°C. The best answer is choice C.
52. **Choice B is correct.** Exactly 7.0 grams of MX dissociates into 28.9 mL of water at 25.0°C according to Table 1. The mass that will dissolve into 25.0 mL is proportionally less. The volume ratio is just under ninety percent, so roughly ninety percent of 7.0 grams MX will dissociate. The best answer is choice B.
53. **Choice A is correct.** As with any question like this, first consider units. liters must be in the denominator, so choices C and D are eliminated. At 25°C, 9.0 grams of MY requires that enough water be added to reach 26.1 mL of solution when fully dissociated. This makes choice A the best answer. To determine the molarity of the solution, the 9.0 grams must first be converted into moles, by dividing 9.0 grams by the molecular mass of MY (120 grams per mole). This value is divided by the volume of the solution as measured in liters (0.0261 L). The value of 0.0253 preys upon the possibility you read the wrong column. The best answer is choice A.
54. **Choice D is correct.** As the temperature rises, the volume of the solution increases according to Table 1. Liquids, as a general rule, expand with increasing temperature. As the volume increases, both the molarity and the density of a solution decrease. The best answer is therefore choice D.
55. **Choice D is correct.** Density is defined as mass per volume. The densities of all of the solutions are greater than pure water (1.00), so choices A and B are eliminated. It is easy to forget to consider the mass of the water when determining the density. At 25°C there are 7.0 grams of MX and 25.0 grams of H₂O in 28.9 mL of solution, 9.0 grams of MY and 25.0 grams of H₂O in 26.1 mL of solution, and 4.0 grams of MZ and 25.0 grams of H₂O in 28.1 mL of solution. The densest solution (greatest mass in the least volume) is found with the saturated MY solution. This makes choice D the correct choice.
56. **Choice C is correct.** The solution with the highest boiling point is the solution with the highest molality of impurities. Molality is defined as moles solute per kg solvent. MX has a lower molecular mass than MY, thus 1.0 grams of MX generates a greater moles of solute than 1.0 grams of MY. Because the density of water decreases as the temperature of water increases, the mass of 10 mL water is greater at the lower temperature. This means that the largest molality is found with 10 mL water at 50°C, because the mass solvent is least. The best answer is therefore choice C.

Passage IX (Questions 57–62)

Qualitative Analysis Experiment

57. **Choice A is correct.** An anion can be used to identify a cation when it reacts differently with that cation than the other cations. Of the five anions used in table I, all five form a precipitate with barium cation. This means that to identify barium (distinguish barium from the other cations) the anion chosen must precipitate exclusively with barium. Only chromate CrO_4^{2-} forms a precipitate with barium and no other cation. To distinguish barium cation from the others, chromate should be added to solution. If a precipitate forms, then barium was present in the solution. If no precipitate forms, then barium was not present in the solution. The best answer is choice A.
58. **Choice B is correct.** Nitric acid protonated carbonate anion to form carbonic acid (H_2CO_3), which subsequently decomposed to form water and carbon dioxide. Because the carbonate anion carries a -2 charge, two equivalents of nitric acid are required. The two nitrate anions and barium cation are spectator ions. The best answer is therefore choice B. This can also be inferred from the passage, where it is stated that nitric acid is added to react with carbonate anion to form carbon dioxide gas. The barium cation carries a +2 charge and carbonate anion carries a -2 charge, therefore the two must be in a one-to-one ratio in the salt. This eliminates choices A and C.

59. **Choice A is correct.** Hydrochloric acid is not used, because it would introduce the chloride anion, which according to table II, forms a precipitate with silver cation. This means that chloride interferes, making choice A the correct choice.
60. **Choice D is correct.** The salt K_2SO_4 delivers the anion SO_4^{2-} (sulfate) to solution. From table I, sulfate precipitates with both barium and strontium cations. The conclusion reached upon observing the formation of a precipitate should be that either strontium or barium is present in solution. It cannot be determined whether or not calcium or magnesium cations are present because they do not form any detectable precipitate with sulfate. The best answer is choice D.
61. **Choice C is correct.** Looking at the third reactivity column of table II shows that a white precipitate forms upon the addition of silver cation with both carbonate and chloride anions. Addition of nitric acid dissolves the carbonate away, but it does not react with the chloride. The best choice is the chloride anion, choice C.
62. **Choice A is correct.** The first anion added precipitates with only barium. This according to the chart is chromate (CrO_4^{2-}). Answer choice A shows this to be carbonate (CO_3^{2-}), making choice A an incorrect choice. Just to be certain, anion W precipitates with strontium but not calcium and magnesium. According to the chart, this is sulfate (SO_4^{2-}) so choice B is valid. Anion X precipitates with calcium but not magnesium. According to Table 1, it is either carbonate (CO_3^{2-}) or oxalate ($C_2O_4^{2-}$), so choice C (oxalate) is valid. Anion Z precipitates with magnesium. According to the chart, this is hydroxide (OH^-), so choice D (hydroxide) is valid.

Passage X (Questions 63 - 69)

Solubility Chart

The focus of this passage is on the solubility of different cations in the presence of anions. Solubility is one of the more favored topics on the MCAT so far, so be certain to make sense of this passage. **The key fact to remember is that when comparing solubility, look at the molar solubility, not the K_{sp} value.**

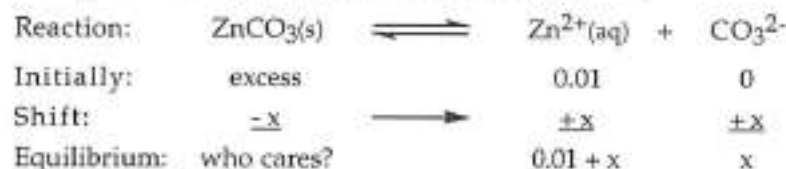
63. **Choice C is correct.** This question boils down to whether Q_{sp} or K_{sp} is larger. When $Q > K$, there are too many products, in which case the salt cannot completely dissociate. When $Q < K$, there is room for more products, in which case the salt completely dissociates. When $Q = K$, it is exactly saturated. In this question, it is far easier to decide whether the molarity of $ZnCO_3$ in the event it fully dissociates exceeds the molar solubility conveniently listed in Table 1. The molarity if $ZnCO_3$ fully dissociates is calculated as follows:

$$\frac{0.1 \text{ grams}}{125.4 \text{ grams/mole}} = \frac{0.1}{125} \text{ mole} = \frac{0.8}{1000} \text{ mole} = 0.0008 \text{ mole} \Rightarrow \frac{0.0008 \text{ mole}}{0.10 \text{ L}} = 0.008 \text{ M}$$

$0.008 \text{ M} > 0.000014$ by a large amount. This means that very little of the $ZnCO_3$ dissociates. On this question, you must go even further. It will barely dissociate according to the numbers which implies choice C. Do what is expected of you: be kind to strangers, brush after every meal, and pick choice C.

64. **Choice B is correct.** To distinguish one thing from another visually, you must be able to see a difference in their behaviors. The behavior here is the formation of a precipitate. To visually distinguish a tube with Zn^{2+} from a tube with Ag^+ , an anion that precipitates with only one of them must be used. The passage defines the cut-off for seeing a precipitate as one mg per mL solution. Assuming the average salt to have a molecular mass around 100 grams per mole, means that the cut-off for visual detectability is roughly 0.010 M. You must locate the anion that shows a greater molar solubility than 0.01 M for one cation, but less for the other. Sulfides show the biggest ratio of numerical values, but in a practical sense, neither dissolves to a detectable extent to visually distinguish. The numbers are different in a multiplicative manner, but they are close in an additive sense. Choice A is eliminated. With the chloride salts ($[Ag^+] = 1.2 \times 10^{-5}$, while $[Zn^{2+}] = 3.1 \times 10^{-2}$), there is a case where the silver chloride precipitates while the zinc chloride dissociates. The zinc concentration is detectable to the eye. In other words, we can see the quantity of salt that dissociates. Choose B, and feel nice.
65. **Choice D is correct.** This question addresses the common ion effect from a conceptual perspective. Silver sulfide is most soluble in a solution with the least amount of common ion. This eliminates choices A and C. The question is whether or not silver cation or sulfur dianion more affects the calculation. In this case, we are deciding which leads to a greater x value, $K_{sp} = (.01)^2(x)$ or $K_{sp} = (2x)^2(0.01)$. In a solution with 0.01 M Ag^+ , the solubility is $10^4 \times K_{sp}$. In a solution with 0.01 M S^{2-} , the solubility is $5 \times$ square root of K_{sp} . Because the K_{sp} for Ag_2S is so small, the square root of K_{sp} is a larger number than K_{sp} . The best answer is choice D.

66. **Choice C is correct.** This question is purely mathematical. The key fact is recognizing that there is 0.01 M Cl^- present in solution initially. The set-up and solution are as follows:

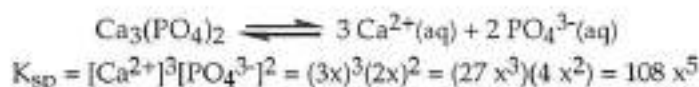


$$K_{\text{sp}} = [\text{Zn}^{2+}][\text{CO}_3^{2-}] \therefore K_{\text{sp}} = (0.01 + x)(x)$$

$$\text{Upon ignoring } x: K_{\text{sp}} = 2.0 \times 10^{-10} = (0.01 + x)(x) = 0.01(x) \therefore x = 2.0 \times 10^{-8} \text{ M}$$

The correct answer is choice C.

67. **Choice B is correct.** The most soluble salt is the salt with the highest molar solubility (not necessarily the highest solubility product). This is a "read-the-chart" type of question. According to data in Table 1, the largest molar solubility of the choices is found with Ag_2CO_3 ($1.2 \times 10^{-4} \text{ M}$). The trick here is picking the correct column in the table to reference (molar solubility rather than K_{sp}). To be more than you can be, pick B.
68. **Choice D is correct.** To calculate the solubility product for the calcium phosphate salt, the solubility reaction is needed:



The best answer is choice D.

69. **Choice C is correct.** An ion exchange column involves competition by two cations for an anion bound to the column. The cation of the less soluble salt will precipitate with the anion, which is bound to the column. The column works if the cation in solution is less soluble with the anion in the column than the cation originally coupled with the anion in the column. Silver chloride (AgCl) has a lower molar solubility than sodium chloride (NaCl), therefore silver cation precipitates with chloride anion preferentially over sodium cation. Choice A is a valid statement. Zinc sulfide (ZnS) has a lower molar solubility than calcium sulfide (CaS), therefore zinc cation precipitates with sulfide anion preferentially over calcium cation. Choice B is a valid statement. Calcium carbonate (CaCO_3) has a higher molar solubility than zinc carbonate (ZnCO_3), therefore calcium cation does *not* precipitate with carbonate anion preferentially over zinc cation. The column of choice C will not work, thus choice C is the best answer. Calcium sulfide (CaS) has a lower molar solubility than sodium sulfide (Na_2S), therefore calcium cation precipitates with sulfide anion preferentially over sodium cation. Even though the two salts have different stoichiometry, they can be compared directly through their molar solubility values. Choice D is a valid statement.

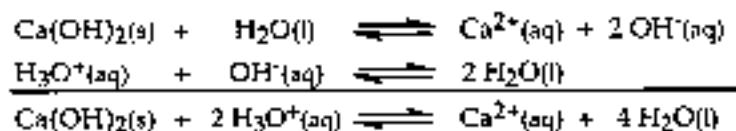
Passage XI (Questions 70 - 75)

Calcium Salts Solubility

70. **Choice A is correct.** Because there is solid calcium hydroxide at the base of the flask, it can safely be assumed that the solution is saturated. As sodium carbonate is added to the flask, calcium carbonate will begin to precipitate. This reduces the calcium ion concentration, causing more of the calcium hydroxide solid can begin to dissociate in order to regenerate calcium ion in solution. As calcium hydroxide dissociates into solution, the hydroxide concentration increases. As the hydroxide ion increases, the solution's pH increases, eliminating choices C and D. Had the solution not been saturated, there would not be any solid calcium hydroxide present to dissociate into solution. This means that the pH increases more in the solution with solid calcium hydroxide present than the solution with no calcium hydroxide present. The best choice is therefore answer A.
71. **Choice C is correct.** The maximum calcium ion concentration at pH = 14 can be determined using the solubility product of calcium hydroxide ($K_{\text{sp}} = [\text{Ca}^{2+}][\text{OH}^-]^2$). Because the pH is 14, the pOH is 0, so the hydroxide anion concentration is 1.00 M. This means that when plugging into the solubility product to determine the calcium ion concentration, it can be determined that the calcium ion concentration equals the numerical value of the solubility product ($K_{\text{sp}} = [\text{Ca}^{2+}]$). The solubility product is $4.3 \times 10^{-6} \text{ M}^3$. The best choice is C, $4.3 \times 10^{-6} \text{ M}$.

72. **Choice D is correct.** In order to increase the chloride concentration in solution, an anion must be added that precipitates calcium cation out from solution. This allows the calcium chloride solid present on the bottom of the flask to dissociate into solution to regenerate the lost calcium cation. As the calcium chloride dissociates, the chloride anion concentration increases. Choice C should be eliminated immediately, because the solubility of calcium chloride is reduced by the common ion effect. Choice B is eliminated, because the silver cation will precipitate chloride anion from solution. This is one of the solubility rules you may want to know. Silver halides are insoluble in water. The answer is either choice A or D. The sodium phosphate adds phosphate anion into solution, which then precipitates out of solution as calcium phosphate and thus decreases the calcium cation concentration. The solid calcium chloride dissociates to counteract the decrease in calcium concentration and in doing so increases the chloride ion concentration. The best answer is choice D.
73. **Choice C is correct.** It is important to realize that the solution is buffered, so the calcium hydroxide that dissociates does not drastically affect the pH. The buffer absorbs the hydroxide that is released as calcium hydroxide dissociates. The amount of calcium hydroxide that dissociates is controlled by the pH of the solution. At a high pH, there is a large amount of hydroxide present in solution, so only a small amount of calcium hydroxide dissociates. As the pH decreases, the hydroxide anion concentration decreases, resulting in an increasing calcium ion concentration. The overall effect is that as the pH of the buffer increases, calcium hydroxide is less soluble due to the common ion effect. This results in an inverse relationship, so choices A and B are eliminated.

The question may also be addressed from a complex equilibrium perspective. As the buffer pH decreases, the amount of hydronium in solution increases. Increasing the hydronium concentration forces the second reaction in the forward direction. This reduces the hydroxide concentration. To compensate, the first reaction shifts forward to regenerate hydroxide. In doing so, calcium cation is released into solution. This confirms that the calcium ion concentration increases as the pH decreases.



The net result is that as the $[\text{H}_3\text{O}^{+}]$ increases, the $[\text{Ca}^{2+}]$ increases. As the calcium cation concentration increases, the log of the calcium cation concentration increases, therefore the log of the calcium ion concentration increases as the pH decreases. Because both the x-axis and y-axis are based on a log scale, the relationship is linear, but not necessarily with a slope of one. The slope is negative (due to the negative sign in determining pH). The best answer is choice C.

74. **Choice B is correct.** If the dissociating of a salt into water increases the temperature of the water, then dissociation (the solvation process) releases heat. The release of heat makes the solvation process exothermic which makes the enthalpy change (ΔH) a negative number. This eliminates choices A and C. The entropy increases when a salt dissociates into solution, because the ions have more freedom to randomize within the solution than within the lattice. The change in entropy (ΔS) for dissociation is therefore positive. The correct answer is choice B.
75. **Choice D is correct.** This question requires viewing Table 1. The solubility of calcium fluoride and calcium hydroxide are directly comparable using the solubility product, because fluoride and hydroxide carry the same anionic charge. The solubility of calcium sulfate and calcium carbonate are directly comparable using the solubility product, because sulfate and carbonate carry the same anionic charge. Choices A and C are eliminated due to their lower solubility product values. Because calcium hydroxide forms three ions when dissociating into solution, the solubility product is $4x^3$, where x is the calcium ion concentration ($[\text{Ca}^{2+}]$). The value of x ($[\text{Ca}^{2+}]$) is some number times 10^{-2} for Ca(OH)_2 . For calcium sulfate, the solubility product is x^2 , where x is the calcium ion concentration ($[\text{Ca}^{2+}]$). The value of x ($[\text{Ca}^{2+}]$) is some value times 10^{-3} for calcium sulfate. The value is greater for calcium hydroxide, so the best answer is calcium hydroxide, choice D.

This is a difficult passage because as one reaction shifts, it affects the other reactions. It is easiest to view the reactions as independent, and observe how specific reagents are changing during each step. For instance, the shift of reaction 1 will affect the partial pressure of NO_2 which will in turn affect reactions 2, 3, and 4. The result is a complex equilibrium.

76. **Choice A is correct.** A negative change in entropy results when the system becomes more ordered. The system becomes more ordered when it decreases in volume or changes phase to a more ordered phase. Reaction 2 goes from two gas molecules to one gas molecule, therefore it has lost in entropy. Reaction 3 goes from two gas molecules to two gas molecules, therefore entropy has not changed. Reaction 5 goes from three ions in solution to three ions in solution, therefore entropy has not changed. Reaction 6 also goes from three ions in solution to three ions in solution, therefore entropy has not changed. The best answer is therefore choice A.
77. **Choice C is correct.** Decreasing the temperature shifts reaction 2 in the forward direction, because it is exothermic. This results in an increase in $[\text{N}_2\text{O}_4]$. Increasing the pressure shifts reaction 2 in the forward direction. This results in an increase in $[\text{N}_2\text{O}_4]$. Water vapor in the air reacts with the NO_2 in the air (according to reaction 4), and thus reduces the amount of NO_2 . Reaction 2 shifts in the reverse direction to re-establish equilibrium. This decreases (NOT increases) the $[\text{N}_2\text{O}_4]$. Choice C is the correct choice. Addition of NO_2 shifts reaction 2 in the forward direction which increases the $[\text{N}_2\text{O}_4]$.
78. **Choice B is correct.** According to the ideal gas law, when the temperature of a gas system is doubled, the pressure doubles, as long as the volume of the system and moles of gas remain the same. The reaction vessel is a closed steel container, so the volume of the system cannot change. The problem here is that the moles of gas change with the increase in temperature. Because the reaction is exothermic, an increase in temperature pushes the reaction in the reverse direction. This increases the reactants and decreases the products. There are more products than reactants in the balanced equation, so a shift in the reverse direction results in fewer moles of gas in the system. The increase in pressure due to the ideal gas law is not be as great as expected, due to the decrease due to the shift in equilibrium. The ideal gas law predicts that change in temperature doubles the pressure, while the shift in reaction (reduction in moles of gas) predicts that the pressure decreases. The change in moles is a smaller factor than the increase temperature, because at the least, the moles of gas would be two-thirds of their original value. The moles cannot be cut in half according to the balanced equation. Therefore, the overall change in pressure is a little less than double. This is best described as choice B.
79. **Choice D is correct.** A bond must be formed in reaction 2, because the two NO_2 molecules are combining to form one N_2O_4 molecule. This eliminates choices A and B. Bond formation is an exothermic process, therefore choice D is the correct answer.
80. **Choice C is correct.** The total pressure of reaction 1 at equilibrium is 1.00 atmospheres, because the internal pressure can equilibrate with the external pressure in a piston system. The pressures equilibrate by having the piston plunger either rise up (increasing the volume) or drop down (decreasing the volume). The addition of 0.10 atmospheres of NO gas initially increases the total pressure of the system to 1.10 atmospheres, before the system can equilibrate. The piston rises to accommodate the change in pressure (and equilibrate the internal and external pressures), thus increasing the volume of the piston. The addition of NO gas displaces the reaction from equilibrium. The reaction is no longer at equilibrium, so it reacts to re-establish equilibrium. Excess reactant is present, so the reaction shifts in the forward direction (to use up the excess reactant). There are three moles of reactants and only two moles of products, so the number of moles decreases, causing the volume to decrease slightly. The overall result is a slight increase in volume, making choice C correct. If there was no shift in the reaction, the volume would increase by exactly ten percent, therefore the increase in volume must be less than ten percent.
81. **Choice D is correct.** Addition of sodium hydroxide to solution deprotonates HNO_2 (HNO_3 is a strong acid and has already fully dissociated), and thus shifts reaction 4 to the product side to re-establish equilibrium. Addition of manganese(II) chloride (MnCl_2) to solution removes both NO_2^- and NO_3^- from solution through complexing of the ligands. To re-establish equilibrium in reaction 4, the reaction must shift right to make more NO_2^- and NO_3^- . Removal of nitrate (NO_3^-) from solution results in a shift in the product direction to re-establish equilibrium. Removal of water (a reactant) results in a shift in the reverse direction (left) to re-establish equilibrium. Choice D is the best answer available to you.

82. **Choice A is correct.** At equilibrium, the NO_2 concentration is constant. When bubbled through water, it reacts to form nitrous acid (HNO_2) and nitric acid (HNO_3), which can both deprotonate and form ligand bonds to the manganese cation. This means that the $\text{NO}_2(\text{g})$ is depleted when it is bubbled through an aqueous solution of manganese chloride. Graphs C and D are eliminated, because they show that the concentration of NO_2 increases with the addition of aqueous manganese chloride solution. The $\text{NO}_2(\text{g})$ is gradually regenerated by both the reverse reaction of reaction 2 and the forward reaction of reaction 1. This means NO_2 is regenerated (and thus increases), but not to the level it was initially at. This is best shown in graph A.

Passage XIII (Questions 83 - 89)

Hemoglobin and Acclimation

83. **Choice C is correct.** The passage states that hemoglobin binds four oxygen molecules while myoglobin binds only one oxygen molecule. To transfer all of the oxygen, there must be four myoglobin molecules per hemoglobin. This should be common knowledge from biology. Pick choice C, and start the passage off on the right foot.
84. **Choice B is correct.** Increasing the rate of respiration increases the uptake of oxygen. This results in an increase in free oxygen, which lessens slightly by the shift in equilibrium to the oxygenated hemoglobin. Overall, the amount of free oxygen increases, so choice A is eliminated. An increase in the amount of hemoglobin results in a shift in the equilibrium in the forward direction. This reduces the amount of free oxygen. Choice B is therefore the best answer. An increase in CO consumed results in more bound sites on the iron of hemoglobin, so less oxygen can bind. Less oxygen bound results in more free oxygen. Choice C is eliminated. More blood results in more moles of all components in the equilibrium including oxygen. Choice D is thus eliminated. Select choice B.
85. **Choice D is correct.** A long-time mountain resident has more hemoglobin in their blood than a long-time sea-level resident. When the long-time mountain resident descends to sea level where there is a higher abundance of oxygen, they experience increased vitality due to the increased partial pressure of oxygen gas. Pick D.
86. **Choice A is correct.** Because CO binds the iron of hemoglobin preferentially over O_2 , any CO present in the air bind iron cation, and thus reduce the amount of O_2 that can bind. To compensate, respiration increases, in order to increase the amount of air consumed. Over time, the body produces more Hb (this is a result of acclimation). The best answer is choice A.
87. **Choice D is correct.** Because hypoxia does not directly affect the cellular uptake rate of oxygen, once it is absorbed from the lungs, myoglobin (present in cells) is not affected directly by hypoxia. Without considering any other factors, the myoglobin should remain constant. The best answer is choice D.
88. **Choice D is correct.** Hypoxia results from a drastic decrease in the amount of oxygen present in the air. This decrease is associated with an increase in elevation. Repelling down a mountain and scuba diving both involve increases in the oxygen present. In scuba tanks, the gas may be mixed with helium (an inert gas) to compensate for the greater amount of air consumed per breath (due to the increased pressure under water). If the tanks were not partially filled with helium, too much nitrogen and oxygen would enter the body. Of the last two choices, only snow skiing involves a high elevation, so choice D is the best answer.
89. **Choice B is correct.** According to the passage, there is less oxygen present at higher elevations. By training at higher elevations, where the air has less moles of oxygen, the body acclimates by producing more Hb. Once an athlete returns to a lower elevation, the increased amount of Hb remains for a short time, before the body can re-acclimate. For a short period of time, an athlete can increase their oxygen carrying capacity (and thus increase their cellular metabolism). The best answer is choice B.

This passage mimics a passage that appeared on the MCAT years ago. The reaction is a typical example of gas phase equilibrium, and is found in most every general chemistry text book. The reaction is often used to demonstrate Le Châtelier's principle. In this passage, the reaction is chosen because it has an unequal number of reactant molecules as product molecules. This means that changes in the condition of the system (volume and pressure) can shift the equilibrium reaction, but the equilibrium constant remains the same, as long as a temperature change does not accompany the volume or pressure change. The passage also points out that an inert gas does not disturb an equilibrium, as long as the container is rigid (which a glass container is assumed to be). You may wish to note that if inert gas is added to an expandable container, it can disrupt the equilibrium, because the partial pressures of the component gases are changed.

90. Choice A is correct. Because the reaction involves the breaking of a bond, it must be an endothermic reaction. This is confirmed in the last sentence of the passage, which states that the ratio product to reactant increases as the temperature increases. Because there are two molecules formed from just one, the ΔS for the reaction as written is greater than zero (positive). The best answer is therefore choice A.
91. Choice D is correct. Addition of reactant (N_2O_4) shifts the reaction to the right (product side), thus choice A is valid. An increase in volume disrupts equilibrium, and results in the reaction shifting from the side with one molecule to the side with two molecules. This results in a shift to the right, so choice B is valid. A decrease in pressure disrupts equilibrium, and results in the reaction shifting from the side with fewer molecules (one) to the side with more molecules (two). This results in a shift to the right, thus choice C is valid. Choices B and C should both have been eliminated, because they are the same answer and cannot both be correct. When adding helium gas (an inert gas) to the reaction at constant volume, the partial pressures of N_2O_4 and NO_2 do not change. Because the partial pressures do not change, the reaction is not displaced from equilibrium, thus it does not shift in either direction. This makes choice D the correct answer.
92. Choice D is correct. The equation relating ΔG and K_{eq} is $\Delta G = -RT \ln K_{\text{eq}}$. To isolate K_{eq} , both sides of the equation are first divided by $-RT$. To eliminate the natural log function, the two sides of the equation must be exponents of e . This makes choice D the correct answer. The derivation is shown below.

$$\Delta G = -RT \ln K_{\text{eq}} \quad \therefore \ln K_{\text{eq}} = -\frac{\Delta G}{RT} \quad \therefore K_{\text{eq}} = e^{-\Delta G/RT}$$

93. Choice B is correct. When the equilibrium constant has 10^5 associated with it, it is said to be large. As such, the reaction distribution at equilibrium is almost exclusively products, making choice A valid. If the reaction starts as mostly reactants, it shifts nearly one hundred percent to form products, making the shift significant. Choice B is NOT true. The equilibrium constant for the reverse reaction is the reciprocal of the equilibrium constant for the forward reaction. This makes choice C valid. A catalyst increases the reaction rate (in both the forward and the reverse directions), but it does not affect the equilibrium constant. This makes choice D valid. The best answer is choice B.
94. Choice A is correct. Because the equilibrium is dynamic, the forward and reverse reactions are continually transpiring. When the equilibrium is disturbed by the addition of N_2O_4 (labeled or not), the equilibrium is lost and the reaction must undergo a net shift forward to compensate for the excess reactant. The reverse reaction continues as well, but not to the degree of the forward reaction. The result is that the amount of N_2O_4 decreases and the amount of NO_2 increases. The radiolabeled nitrogen will eventually be evenly distributed between the products and reactants, once equilibrium has been re-established. This makes choice A the best answer. This is referred to as scrambling of the label.
95. Choice C is correct. Because there are more molecules on the product side than the reactant side, the reaction shifts to products as the volume of the piston increases. This makes statement I a valid statement. The reaction is shifting in the forward direction, which is endothermic, therefore heat is absorbed by the reaction. The result is that the temperature decreases. This makes statement II a valid statement. Do not mistakenly think of $PV = nRT$, because the volume changes and the pressure changes, the temperature was not changed to cause the volume or pressure change. As the reaction shifts to the right, N_2O_4 decreases and the amount of NO_2 increases, causing the mole fraction of N_2O_4 to decrease. This makes statement III an invalid statement. The best answer is choice C, both statements I and II.

96. **Choice B is correct.** In this case, the equilibrium constant is the product squared (because of the stoichiometric coefficient) divided by the reactant. This makes choice B the best answer.

Questions 97 - 100

Not Based on a Descriptive Passage

97. **Choice C is correct.** When an MX salt dissociates into water, it forms M^+ cation and X^- anion. The solubility product (K_{sp}) is equal to $[M^+][X^-] = y^2$. You really have no alternative but to pick C.
98. **Choice C is correct.** This is a question involving the common ion effect. Because F^- is present in solution in choices A and B, and Mg^{2+} is present in choice D, all of the choices except C are eliminated due to the common ion effect. It is only in choice C that the compound does not have a common ion (either Mg^{2+} or F^-) present.
99. **Choice C is correct.** Equilibrium is the state in which the forward reaction rate equals the reverse reaction rate. For a one-step reaction at equilibrium, $k_f[R] = k_r[P]$. K_{eq} is defined as products over reactants, which is manipulated as follows:

$$\text{Given } k_f[R] = k_r[P] \text{ and } K_{eq} = \frac{[P]}{[R]}, K_{eq} = \frac{k_f}{k_r}$$

Because the forward rate is four times the reverse rate, $4k_r$ can be substituted for k_f and thus K_{eq} is $4k_r$ divided by k_r , which is 4. Choose with dignity; choose C.

100. **Choice C is correct.** Choice D is eliminated, because less solvent reduces the amount of a salt that can dissociate into solution. Because a solvation reaction can be either exothermic or endothermic, the effect on the system by a change in temperature varies, and is thus unpredictable. This eliminates choices A and B. In most cases, an increase in temperature results in an increase in the amount of salt that dissociates into solution, but there are some exceptions. Only choice C, increasing the solvent, always increases the amount of a salt that dissociates into solution. As more solvent is added, more compound can dissociate. Be sure that you realize that this question is asking about the amount (in mass or moles) of salt dissociated, not the concentration (molar solubility).

Section IV

Acids and Bases

by Todd Bennett

Strong Acid:

$$\text{pH} = -\log [\text{HX}]$$

Strong Base:

$$\text{pOH} = -\log [\text{MOH}]$$

Weak Acid:

$$\text{pH} = \frac{1}{2} \text{pK}_a - \frac{1}{2} \log [\text{HA}]$$

Weak Base:

$$\text{pOH} = \frac{1}{2} \text{pK}_b - \frac{1}{2} \log [\text{A}^-]$$

Buffer:

$$\text{pH} = \text{pK}_a + \log \frac{[\text{A}^-]}{[\text{HA}]}$$

or

$$\text{pH} = \text{pK}_a + \log \frac{\text{moles Conjugate Base}}{\text{moles Conjugate Acid}}$$

Terminology

- a) Fundamental Definitions
- b) Water-based Acid-Base Chemistry
 - i. Acid Dissociation
 - ii. Base Hydrolysis
- c) Determination of Reagent Strength
 - i. Strong Acids
 - ii. Weak Acids
 - iii. Very Weak Acids
 - iv. Strength and the pK Scale
 - v. Strong Bases
 - vi. Weak Bases
 - vii. Very Weak Bases

Types of Acids and Bases

- a) Haloacids
- b) Oxyacids
- c) Metal Oxides and Metal Hydroxides
- d) Organic Acids
- e) Polyprotic Acids

Calculating pH

- a) Determining pH
- b) Log Review
- c) pH for Strong Reagents
- d) pH for Weak Reagents

Conjugate Pairs

- a) Typical Conjugate Pairs
- b) Relationship of pK_a and pK_b
- c) Henderson-Hasselbalch Equation



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Specializing in MCAT Preparation

Acids & Bases Section Goals



Know the definitions for an acid and some common examples of acids.

An acid can be defined as either a proton donor, an electron-pair acceptor, or a compound that yields H_3O^+ when added to water. Typical examples include acetic acid ($\text{H}_3\text{CCO}_2\text{H}$), hydrochloric acid (HCl), and sulfuric acid (H_2SO_4). You are expected to recognize the common acids.



Know the definitions for a base and some common examples of bases.

A base can be defined as either a proton acceptor, an electron-pair donor, or a compound that yields OH^- when added to water. Typical examples include sodium hydroxide (NaOH), ammonia (NH_3), and potassium tert-butoxide ($(\text{CH}_3)_3\text{COK}$). You are expected to recognize the common bases.



Be able to calculate the pH of aqueous solutions of base or acid.

The pH of a solution is defined as the negative log of the hydronium ion concentration in the solution. You must be able to determine the pH of the solution, knowing the concentration and strength of the species. The pH of a weak acid is greater than the pH of a strong acid when the two are in equal molar concentrations.



Understand what is meant by the strength of a reagent.

The strength of a reagent is the measure of its degree of dissociation in water. A compound that fully dissociates in water is said to be strong, while a reagent that only partially dissociates in water is said to be weak. You should be able to determine the relative strengths of acid from chemical features and its pH in the aqueous solution.



Know how conjugate pairs and buffers work.

A buffer is formed when a weak acid and its conjugate base are combined in an aqueous solution. An equilibrium exists between the two species, so as long as both are present in solution, the hydronium ion concentration will remain fairly constant, and the pH will also remain constant. This effect is known as "buffering." You must understand buffers and how pH is determined using the Henderson-Hasselbalch equation.



Recognize and specific types of compounds.

You should understand why metal oxides and metal hydroxides are basic. You should understand why non-metal oxides and non-metal hydroxides are acidic. Be familiar with typical examples such as acid rain and soil pH. Recognize that the conversion from a non-metal oxide (Lewis acid form) to a non-metal hydroxide (Bronsted-Lowry form) involves hydration of the acid. The strength of the reagent is not affected by this conversion.



Understand the terms associated with polyprotic acids.

Polyprotic acids have multiple pK_a and pK_b values. You must understand the conceptual and mathematical relationships of the variables to one another. You must be familiar with terms such as "normality" and "equivalents." Know the typical examples of polyprotic acids and their formulae.

Acids and Bases

Acid-and-base chemistry is a foundation for understanding organic chemistry, biochemistry, and physiology. As such, we shall address it from several perspectives, observing what effect the varying of a solvent has on the nature of a reaction. Water and other protic solvents solvate charged species, so acid-and-base chemistry can be monitored by the gain and loss of charge, associated specifically with the gain and loss of an ionized hydrogen atom (H^+). The terms "protonated" and "deprotonated" are derived from the ionized hydrogen atom, which is just a proton. In a lipid environment, charges cannot be stabilized, so acid-and-base chemistry is typically considered from an electron-transfer perspective, where bonds are broken and formed. The different definitions of acids and bases are the result of differences in solvent, not chemical reactivity.

Once the definitions are established, reactivity will be considered. There are two common misconceptions to clear up, if we are to understand acid and base chemistry better. The first common misconception is that the pH scale has fixed limits. The second common misconception is that weak acids automatically have strong conjugate bases. Let us address both of these before anything else.

Acidity is typically measured in terms of $[H^+]$ on the pH scale. The pH value of a solution is determined by taking the negative log of the hydronium concentration. Because neutral water has a hydronium concentration of 10^{-7} M, due to autoionization of water, neutral water has a pH = 7.0. Acidic solutions have pH values less than 7.0, while basic solutions have pH values greater than 7.0. The only limits to the value of pH are associated with concentration and solubility, so the pH scale is limitless in theory. In practice, there are limits.

The pH scale does not range from 0 to 14. The pH of a compound can be negative for highly concentrated strong acids and greater than 14 for highly concentrated strong bases (e.g., 10 M HCl has pH = -1, and 10 M NaOH has pH = 15).

The relative strength of an acid can be determined from the relative strength of its conjugate base. This relation is reversible: The relative strength of a base can also be determined from the relative strength of its conjugate acid. Using the relationship between pK_a and pK_b for conjugate pairs in water at 25°C, either pK value can be determined from knowing the other. For instance, a weak acid with a pK_a of 7 has a conjugate base with a pK_b of 7. Note that the conjugate base of the weak acid is a weak base. This is not a typo! It is the relative reactivities that are compared, not the absolute reactivities. If acid HA is stronger than acid HB, then base A^- is weaker than base B^- . This can be demonstrated by comparing Reaction 4.1 with Reaction 4.2:



Reaction 4.1



Reaction 4.2

The lower pK_a value indicates that HA is a stronger acid than HB while the larger pK_b value indicates that A^- is a weaker base than B^- .

The stronger the acid, the weaker its conjugate base. It is not necessarily the case that weak acids have strong conjugate bases. A prime example is a weak acid with $pK_a = 7$. The pK_b of the conjugate base is 7, which is not strong.

Terminology

Fundamental Definitions

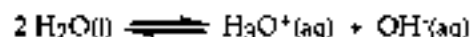
A perfect place to start a discussion of acids and bases is with the definitions of an acid, a base, and strength. There are three definitions proposed for both acids and bases, although each considers an acid as the opposite of a base. The first definition is the Arrhenius definition, which states that an acid yields H_3O^+ when added to water, while a base yields OH^- when added to water. An acidic aqueous solution therefore has a higher hydronium concentration than hydroxide. This definition is useful when doing calculations of pH and pK_a . The second definition is the Brønsted-Lowry definition, which states that an acid is a proton (H^+) donor and that a base is a proton (H^+) acceptor. This definition is based on what occurs in a protic solvent, where reactions are viewed as proton-transfer reactions. The third definition is the Lewis definition, which states that an acid is an electron-pair acceptor while a base is an electron-pair donor. This definition is based on what occurs in an aprotic solvent, and is typically used in organic chemistry. For this section, only the Arrhenius definition will be emphasized. Table 4.1 lists the different definitions of acids and bases.

Term	Definition	Example
Arrhenius acid	yields H_3O^+ when added to H_2O	aq: $[\text{H}_3\text{O}^+] > [\text{OH}^-]$
Arrhenius base	yields OH^- when added to H_2O	aq: $[\text{OH}^-] > [\text{H}_3\text{O}^+]$
Brønsted-Lowry acid	Proton donor	HX in protic solvent
Brønsted-Lowry base	Proton acceptor	KOH in protic solvent
Lewis acid	Electron pair acceptor	BF_3 in aprotic solvent
Lewis base	Electron pair donor	NH_3 in aprotic solvent

Table 4.1

Water-based Acid-Base Chemistry

For our purposes, acid and base chemistry is to be considered as reactions that only occur in aqueous solution. Therefore, to understand water-based acidity and basicity, it is vital to understand the properties of water. Water is amphoteric, meaning that it may act as either an acid or a base. Neutral water naturally dissociates into hydronium and hydroxide, according to Reaction 4.3.



Reaction 4.3

Only a small fraction of water dissociates into solution. Water at 25°C , in the absence of an acid or a base, dissociates enough to generate a solution with $10^{-7} \text{ M } \text{H}_3\text{O}^+(\text{aq})$ and $10^{-7} \text{ M } \text{OH}^-(\text{aq})$. This means that neutral water has both a little hydronium and a little hydroxide. A neutral aqueous solution is defined as one in which $[\text{H}_3\text{O}^+] = [\text{OH}^-]$, and therefore $\text{pH} = \text{pOH}$. If the solution is at 25°C , then the pH and pOH are both equal to 7.0. For an acidic aqueous solution, $[\text{H}_3\text{O}^+] > [\text{OH}^-]$, $\text{pH} < \text{pOH}$, and the pH of the solution is less than 7.0. Equally, for a basic aqueous solution, $[\text{OH}^-] > [\text{H}_3\text{O}^+]$, $\text{pH} > \text{pOH}$, and the pH of the solution is greater than 7.0. There are several perspectives according to which a solution may be deemed acidic or basic, and regardless of the reasoning behind each one, you must know how all of these different perspectives relate to one another. Any acidic solution has a pH of less than 7.0, hydronium in it being more abundant than hydroxide, and it can turn blue litmus paper red.

Acid Dissociation

When an acid is mixed with water, it is said to dissociate. This is the reason we use the term *acid dissociation*, expressed quantitatively with an acid dissociation constant (K_a). The acid dissociation constant is nothing more than the equilibrium constant for the dissociation reaction of an acid in water. A molecule of an acid when added to water, dissociates to form hydronium ion (H_3O^+) and its conjugate base (expressed generically as A^-) upon reaction with one water molecule. Reaction 4.1 is an acid dissociation reaction in water.

**Reaction 4.1**

Equation 4.1 is the equilibrium expression (used to solve for the equilibrium constant) for the acid dissociation reaction shown in Reaction 4.1.

$$K_a = \frac{[\text{H}_3\text{O}^+][\text{A}^-]}{[\text{HA}]} \quad (4.1)$$

Equations 4.2 and 4.3 are the equations for converting between $\text{p}K_a$ and K_a .

$$\text{p}K_a = -\log K_a \quad (4.2)$$

$$K_a = 10^{-\text{p}K_a} \quad (4.3)$$

As the relative strength of an acid increases, its K_a increases and its $\text{p}K_a$ decreases. This means that stronger acids have higher K_a values and lower $\text{p}K_a$ values. The K_a and $\text{p}K_a$ of an acid depend on the strength of the acid, but *not* its concentration.

Base Hydrolysis

When a base is mixed with water, it is said to undergo hydrolysis. This is the reason we use the term *base hydrolysis*, expressed quantitatively with a base hydrolysis constant (K_b). The base hydrolysis constant is nothing more than the equilibrium constant for the hydrolysis reaction of a base in water. Each molecule of a base when added to water, hydrolyzes one water molecule to form hydroxide ion (OH^-) and its conjugate acid (expressed generically as HA). Reaction 4.4 is a base hydrolysis reaction in water.

**Reaction 4.4**

Equation 4.4 is the equilibrium expression (used to solve for the equilibrium constant) for the base hydrolysis reaction shown in Reaction 4.4.

$$K_b = \frac{[\text{HA}][\text{OH}^-]}{[\text{A}^-]} \quad (4.4)$$

Equations 4.5 and 4.6 are the equations for converting between $\text{p}K_b$ and K_b .

$$\text{p}K_b = -\log K_b \quad (4.5)$$

$$K_b = 10^{-\text{p}K_b} \quad (4.6)$$

As the relative strength of a base increases, its K_b increases and its $\text{p}K_b$ decreases. This means that stronger bases have higher K_b values and lower $\text{p}K_b$ values. As seen with the acids, the K_b and $\text{p}K_b$ of a base depend on the strength of the base, but *not* its concentration.

Tying everything together for a conjugate set, we observe that as the acid gets stronger, its conjugate base gets weaker. The overall correlation is shown in Figure 4-1.

Overall Relationship

As acid strength $\uparrow$, $K_a \uparrow$, $pK_a \downarrow$, conjugate base strength $\downarrow$, $K_b \downarrow$, $pK_b \uparrow$

Figure 4-1

Determination of Reagent Strength

The strength of a reagent is determined strictly from the K_a (or pK_a) and the K_b (or pK_b). Strength is a measure of the completeness of a reaction in water. Terms that are sometimes used instead of "dissociation" are *ionization* and *electrolytic nature*. The stronger the acid, the more electrolytic it is, because it conducts electricity better due to the greater number of ions in solution. For water to conduct electricity, there must be ions in solution to transfer the electron charge.

Because the equilibrium expressions are comparable, the values can be correlated for a conjugate pair. At 25°C, Equation 4.7 describes the relationship between pK_a and pK_b .

$$pK_a (HA) + pK_b (A^-) = 14 \quad (4.7)$$

The strength of a reagent is measured by its ability to carry out a reaction in water. The stronger an acid, the more readily it dissociates into water. The stronger the base, the more readily it undergoes hydrolysis when mixed with water. Be careful not to confuse the concentration of a reactant with its strength. A highly concentrated weak acid may have a lower pH than a strong acid in low concentration.

Example 4.1

What can be said of the pK_b associated with the conjugate base of the more electrolytic acid of a pair of acids?

- A. The pK_b associated with the conjugate base of the more electrolytic acid is greater than the pK_b associated with the conjugate base of the less electrolytic acid.
- B. The pK_b associated with the conjugate base of the more electrolytic acid is smaller than the pK_b associated with the conjugate base of the less electrolytic acid.
- C. The pK_b associated with the conjugate base of the more electrolytic acid is equal to the pK_b associated with the conjugate base of the less electrolytic acid.
- D. The pK_b values associated with the conjugate bases of two electrolytic acids cannot be compared.

Solution

The acid that is more electrolytic is the acid that forms more ions, and thus is better able to conduct electricity. The stronger acid dissociates more, and in doing so, produces a greater ion concentration. The more electrolytic acid is the stronger acid, and according to Figure 4-10, the stronger acid has a conjugate base with a higher pK_b (associated with the weaker conjugate base.) This makes choice A the best answer.

Strong Acids

Strong acids are acids that dissociate fully when mixed with water. That is to say that strong acids ionize completely into hydronium and a conjugate base when added to water. Reactants convert completely into products, so the K_a value of a strong acid is very large. It is safe to assume that an acid that does not have a measurable K_a (i.e., it has a K_a that is too large to measure, making it greater than 1.0) is a strong acid. Reaction 4.5 shows the dissociation of a strong acid, HX.

**Reaction 4.5**

The equilibrium constant for the acid dissociation reaction shown in Reaction 4.5 has a large numerator and a minuscule denominator. As such, K_a is significantly greater than 1.0, and $\text{p}K_a$ is negative for HX. This is summarized in Figure 4-2.

$$K_a = \frac{[\text{H}_3\text{O}^+][\text{X}^-]}{[\text{HX}]} \gg 1 \therefore \text{p}K_a < 0$$

Figure 4-2

It happens that all of the strong acids can be classified as either haloacids or oxyacids. Some typical strong acids are listed in Table 4.2.

Acid	Name	$\text{p}K_a$
HCl	Hydrochloric acid	-7
HBr	Hydrobromic acid	-7
HI	Hydroiodic acid	-9
H_2SO_4	Sulfuric acid	-9 ($\text{p}K_{a1}$)
HNO_3	Nitric acid	-2
HClO_4	Perchloric acid	-10

Table 4.2**Weak Acids**

Weak acids are acids that only dissociate partially when dissolved into water. They do not fully ionize into conjugate base and hydronium ion in water. Reactants are converted partially into products, so dividing products by reactants indicates a small value for K_a (a value that is less than 1.0). Any acid with K_a less than 1.0 is a weak acid. Reaction 4.1 shows the dissociation of a weak acid.

**Reaction 4.1**

The equilibrium constant for the acid dissociation reaction shown in Reaction 4.1 is less than 1.0. However, because our definitions must apply to biochemistry as well as general chemistry, weak acids are classified as weak or very weak. For instance, aspartic acid is said to have an acidic side chain, while the side chain of leucine is considered to be neutral. We will define any acid with a K_a less than 10^{-14} (and thus a $\text{p}K_a$ greater than 14) as a very weak acid, meaning that the $\text{p}K_a$ range for a weak acid is from 0 to 14. This is summarized in Figure 4-3.

$$K_a = \frac{[\text{H}_3\text{O}^+][\text{A}^-]}{[\text{HA}]}; \text{ where } 1 > K_a > 10^{-14} \therefore 0 < \text{p}K_a < 14$$

Figure 4-3

Weak acids are members of a number of different classifications, including oxyacids of low oxidation state, one haloacid, carboxylic acids, alkyl ammoniums, and phenols. Table 4.3 lists some typical weak acids and their corresponding pK_a values.

Acid	Name	pK_a
Cl_3CCO_2H	Trichloroacetic acid	0.64
Cl_2HCCO_2H	Dichloroacetic acid	1.27
H_2SO_3	Sulfurous acid	1.82 (pK_{a1})
$HClO_2$	Chlorous acid	1.90
ClH_2CCO_2H	Chloroacetic acid	2.82
HF	Hydrofluoric acid	3.15
HNO_2	Nitrous acid	3.41
HCO_2H	Formic acid	3.74
H_3CCO_2H	Acetic acid	4.74
$2,4-(H_3C)_2C_6H_3NH_3^+$	2,4-dimethylanilinium	5.08
$4-H_2NC_6H_4NH_3^+$	4-aminoanilinium	6.18
H_2CO_3	Carbonic acid	6.36 (pK_{a1})
$4-O_2NC_6H_4OH$	4-nitrophenol	7.15
$HClO$	Hypochlorous acid	7.46
$HBrO$	Hypobromous acid	8.72
NH_4^+	Ammonium	9.26
HCN	Hydrogen cyanide	9.32
HIO	Hypoiodous acid	10.66

Table 4.3

Very Weak Acids

Very weak acids dissociate less than water. The K_a value of a very weak acid is less than 10^{-14} , because products divided by reactants is less than 1.0×10^{-14} (K_w). Given that water is thought to be neutral, an acid with a K_a less than 10^{-14} (and pK_a greater than 14) is a very weak acid. This is summarized in Figure 4-4.

$$K_a = \frac{[H_3O^+][A^-]}{[HA]}, \text{ where } K_a < 10^{-14} \therefore pK_a > 14$$

Figure 4-4

Strength and the pK Scale

The same rules that apply to acids also apply to bases, except that hydrolysis, rather than dissociation, is considered, and K_b replaces K_a . As the pK value for a compound decreases, its strength increases (this is true for both acids and bases). As a rule, the stronger the acid, the weaker its conjugate base. Strong acids have very weak conjugate bases, and strong bases have very weak conjugate acids. The odd sounding relationship is that weak acids have weak conjugate bases. It may seem peculiar, but the conjugate base of a weak acid is most often a weak base. You have seen this relationship before with buffers, although it is unlikely that it was emphasized in your general chemistry courses.

Table 4.4 defines the strength of an acid and a base in relative terms.

Compound	Numerical Data	Conjugate	Numerical Data
Strong acid (Fully dissociates in H_2O) e.g., HCl ($pK_a = -7$)	$K_a \gg 1$ $pK_a < 0$	Very weak base (Hydrolyzes less than H_2O) e.g., Cl^- ($pK_b = 21$)	$K_b < 10^{-14}$ $pK_b > 14$
Weak acid (Partly dissociates in H_2O) e.g., RCO_2H ($pK_a = 3-5$)	$10^{-14} < K_a < 1$ $0 < pK_a < 14$	Weak base (Partly hydrolyzes in H_2O) e.g., RCO_2^- ($pK_b = 9-11$)	$10^{-14} < K_b < 1$ $0 < pK_b < 14$
Very weak acid (Dissociates less than H_2O) e.g., CH_4 ($pK_a = 49$)	$K_a < 10^{-14}$ $pK_a > 14$	Strong base (Fully hydrolyzes in H_2O) e.g., CH_3^- ($pK_a = -35$)	$K_b \gg 1$ $pK_b < 0$

Table 4.4

The strength of a reagent is measured by its ability to react in water. The more readily an acid dissociates into water, the stronger it is. The more readily a base undergoes hydrolysis, the stronger it is. Do not confuse the concentration of a reactant with its strength. Both affect pH. Within conjugate pairs, to calculate the strength of one reagent from the strength of its conjugate, use Equation 4.7.

On the exam, one of the required skills will be determining what the question is asking. In terms of acid and base chemistry, many of the questions will be more complicated versions of the basic question of "which acid is stronger?"

Example 4.2

Which of the following acids would yield the highest pH in water?

- A. 0.10 M $HCl(aq)$
- B. 0.10 M $HClO_4(aq)$
- C. 0.10 M $HBr(aq)$
- D. 0.10 M $HCO_2H(aq)$

Solution

The highest pH in water results from the lowest hydronium concentration. This is associated with the weakest acid, which in this case is carboxylic acid, choice D.

Choices A, B, and C are all strong acids that fully dissociate in water.

Example 4.3

Which of the following is the BEST choice to titrate 0.10 M $H_3CNH_2(aq)$?

- A. 0.50 M $KOH(aq)$
- B. 0.05 M $NH_3(aq)$
- C. 0.10 M $HBr(aq)$
- D. 0.10 M $HCO_2H(aq)$

Solution

To ensure complete reaction, the titrant must always be a strong reagent. This eliminates choice B (a weak base) and choice D (a weak acid). In this case, the solution being titrated is a weak base solution (H_3CNH_2 is a weak base), so strong acid is added. Choice A is a strong base and choice C is a strong acid.

Example 4.4

Which of the following acids dissociates to the GREATEST extent when added to water?

- A. HNO_3
- B. HNO_2
- C. H_3PO_4
- D. H_2CO_3

Solution

The most complete dissociation is associated with the strongest acid. The strongest acid in this case is nitric acid (HNO_3), choice A. These questions show a few of the many ways relative acidity can be compared. In addition to dissociation, acidities can also be compared by pK_a values (lower is more acidic), K_a values (larger is more acidic), pH values (lower is more acidic for equal acid concentrations), electrolytic strength (more electrolytic is more acidic), and reactivity with bases (stronger acids react with weaker bases).

Strong Bases

Strong bases, like strong acids, react completely when they are added to water. Strong bases fully hydrolyze water, so they completely ionize when dissolved into water. Reactants are completely converted into products; thus, the K_b value of a strong base is much greater than 1.0, because products divided by reactants is very large. Any base that does not have a measurable K_b (or has a K_b that is much greater than 1.0) is a strong base. Reaction 4.6 shows the hydrolysis of the strong base, KOH :



Reaction 4.6

The equilibrium constant for the base hydrolysis reaction shown in Reaction 4.6 has a large numerator and a minuscule denominator. Its value, therefore, is significantly greater than 1.0. This results in a negative pK_b value for a strong base. This is summarized in Figure 4-5.

$$\text{K}_\text{b} = \frac{[\text{M}^+][\text{OH}^-]}{[\text{MOH}]} \gg 1 \therefore \text{pK}_\text{b} < 0$$

Figure 4-5

It happens that all of the strong bases are either hydrides, hydroxides, alkoxides, amides, or carbides. Some typical strong bases are listed in Table 4.5. No pK_b values are listed, because all of these bases when added to water will form hydroxide.

Base	Name
KH	Potassium Hydride
NaOH	Sodium Hydroxide
KOCH ₃	Potassium Methoxide
NaNH ₂	Sodium Amide
Li(CH ₂) ₃ CH ₃	Butyl Lithium

Table 4.5

Weak Bases

Weak bases, like weak acids, partially react when added to water. Weak bases partially hydrolyze water, so they only partially ionize when dissolved into water. Reactants are only partially converted into products, so dividing products by reactants is placing a small number over a large number, leading to a small value for K_b (less than one). As with the weak acids, because water is considered to be neutral, we will define any base with a K_b less than 10^{-14} (and thus pK_b greater than 14) to be a base that is too weak to consider. This means that the pK_b range for a weak base is from 0 to 14. It should be safe to assume that any base on the test for which you are given a K_b value (or pK_b value), is likely to be a weak base. A base with less than 100% hydrolysis is a weak base. Reaction 4.7 shows the hydrolysis of a weak base, A^- .

**Reaction 4.7**

The equilibrium constant for the base hydrolysis reaction shown in Reaction 4.7 has a small numerator and a larger denominator, making it less than 1.0. This results in a small positive pK_b value for a weak base. This is summarized in Figure 4-6.

$$K_b = \frac{[HA][OH^-]}{[A^-]}; \text{ where } 10^{-14} < K_b < 1 \therefore 14 > pK_b > 0$$

Figure 4-6

There are many different types of weak bases. Some typical weak bases are the conjugate bases of the acids listed in Table 4.3. In addition, other common weak bases include the carboxylates (RCO_2^-), the alkyl amines (RNH_2), bicarbonate (HCO_3^-), carbonate (CO_3^{2-}), phosphate (PO_4^{3-}), and phenoxides ($C_6H_5O^-$).

Very Weak Bases

Very weak bases do not undergo any significant (or detectable) hydrolysis in water. The K_b value for a very weak base is less than 10^{-14} , because the products divided by reactants is less than 1.0×10^{-14} (K_w). Given that water is considered to be neutral, a base with a K_b less than 10^{-14} (and pK_b greater than 14) is a very weak base. This is summarized in Figure 4-7.

$$K_b = \frac{[HA][OH^-]}{[A^-]}; \text{ where } K_b < 10^{-14} \therefore pK_b > 14$$

Figure 4-7

Example 4.5

Which of the following compounds when added to water does NOT yield a solution where hydroxide ion is in greater concentration than hydronium ion?

- A. Na_2CO_3
- B. HCO_2Na
- C. HNO_3
- D. $\text{C}_6\text{H}_5\text{ONa}$

Solution

A base is a compound that hydrolyzes water to generate hydroxide, and thus form a solution where hydroxide is more concentrated than hydronium. This question is in essence asking, "Which compound is NOT a base?" Choice C, nitric acid, is NOT a base, making choice C the answer.

Example 4.6

Which of the following bases has the STRONGEST conjugate acid?

- A. 0.50 M KOH
- B. 0.50 M $\text{LiN}(\text{CH}_3)_2$
- C. 0.50 M NaH
- D. 0.50 M NaHCO_3

Solution

The concentration does not affect the strength. No matter what concentration is added to solution, the value of K_b is constant. This means that this question simply addresses the strength of the compound. The strongest conjugate acid is associated with the *weakest* base. In this question, choice A is a metal hydroxide, choice B is a metal amide, and choice C is a metal hydride. Choices A, B, and C are strong bases. The only weak base is the bicarbonate base, choice D.

Example 4.7

Which of the following is the BEST choice to titrate 25 mL of 0.10 M $\text{HCO}_2\text{H}(\text{aq})$?

- A. 0.10 M KOH
- B. 0.05 M H_3CNH_2
- C. 0.10 M HBr
- D. 0.05 M $\text{H}_3\text{CCO}_2\text{H}$

Solution

To ensure complete reaction, the titrant must be a strong reagent. This eliminates choice B (a weak base) and choice D (a weak acid). In this case, the solution being titrated is a weak acid solution (HCO_2H is a weak acid), so strong base must be added. Choice C is a strong acid, so the only strong base is choice A, KOH.

Types of Acids and Bases

Haloacids

The haloacids are the series of HX acids, whose X represents a halogen. The acidity of haloacids is predictable based on the size of the halide ion. The larger the halide, the more acidic the haloacid. This means that $HI > HBr > HCl > HF$. This trend in acidity is attributed to the increased stability of the conjugate base as it increases in size. The negative charge is more diffuse on the larger anion. The more diffuse, the less basic the anion, and thus the more acidic its conjugate acid. A second, and perhaps easier, way to view the acidity of haloacids involves estimating their bond strengths. The longer the bond, the weaker the bond, as a general rule in chemistry. As you descend the halogen column in the periodic table, halogen size increases. As halogen size increases, the bond length of an $H-X$ bond must also increase. As the bond length increases, the bond strength decreases, and the H^+ can be removed more readily. This approach works even though bond dissociation energies are determined from homolytic bond breaking, as opposed to the heterolytic cleavage associated with acid-base chemistry. Ions (H^+ and A^-) are the result of heterolytic bond breaking, but the correlation between bond strength and acidity still holds for haloacids. Figure 4-8 summarizes the effect of halide size on acidity.

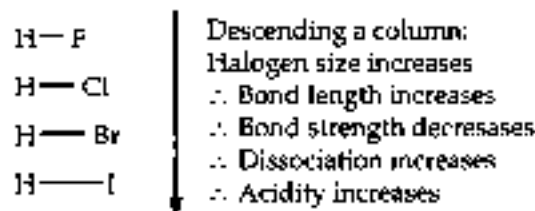


Figure 4-8

Example 4.8

Which of the following conclusions can be made concerning the relative acidities of haloacids?

- A. Acid strength increases with increasing electron affinity of the halide.
- B. Acid strength increases with increasing electronegativity of the halide.
- C. Acid strength increases with increasing ionic radius of the halide.
- D. Acid strength increases with decreasing isotopic abundance of the halide.

Solution

As you descend a column in the periodic table (such as the halogen column), acidity increases due to the increase in atomic size. The anion formed upon deprotonation is more stable as it increases in size, so acidity increases as the column is descended. Electron affinity and electronegativity decrease as the column is descended, and they have no bearing on acidity. Isotopic abundance and atomic mass do not affect acidity, either, so choice C is the best answer.

An important fact to recall about haloacids is that they are all strong acids, except for hydrofluoric acid (HF). HF has a pK_a of roughly 3.3, so it does not fully dissociate when added to water. As we have observed in studying other topics related to the properties of this chemical family, this weaker acidity is attributed to the smaller atomic radius of fluorine relative to the other halogens.

Example 4.9

Which of the following acids has the largest K_a value?

- A. HF
- B. HCl
- C. HBr
- D. H_2S

Solution

The largest K_a value is associated with the strongest acid. Because Br is the largest anion of the halogens listed, it dissociates from the proton most readily. This makes HBr the most acidic, so you should pick choice C. Choices D (H_2S) can be eliminated, because S is of roughly comparable size with, but less electronegative than, Cl. This means that HCl is a stronger acid than H_2S .

Within a period (row) of the periodic table, it is electronegativity that dictates the strength of an acid, not atomic radius. A prime example of this idea is the relationship between ammonia ($H-NH_2$), water ($H-OH$), and hydrofluoric acid ($H-F$). The strongest acid of the three compounds is the hydrofluoric acid, because fluorine is more electronegative than both nitrogen and oxygen. The atomic size does not change that noticeably between N, O, and F, because they all have the same valence level ($n = 2$). The periodic trend that most changes is the electronegativity.

Oxyacids (Non-metal Hydroxides and Non-metal Oxides)

Oxyacids are also an acid type to be familiar with. They differ from haloacids, in that the hydrogen dissociates from an oxygen rather than a halide. In an oxyacid, the acidic hydrogen is bonded to an oxygen, which in turn is bonded to a central atom (which in some cases can be a halogen). The simplest rule is that the more oxygen atoms there are bonded to the central atom, the more the oxygen atoms withdraw electron density from the central atom, and thus the more acidic the oxyacid. This can be summarized as the resonance effect (which is typically classified as an organic chemistry concept). Unlike the haloacids, where size is important, in oxyacids, the electronegativity of the central atom is most important. As a general rule, for every additional oxygen on the central atom, the pK_a of the acid will drop by approximately 5 pK_a units. Table 4.6 lists some typical oxyacids.

Oxyacid	Name	Oxyacid	Name
HNO_2	Nitrous acid	HNO_3	Nitric acid
H_2SO_3	Sulfurous acid	H_2SO_4	Sulfuric acid
H_3PO_3	Phosphorous acid	H_3PO_4	Phosphoric acid
H_2CO_3	Carbonic acid	HO_2CCO_2H	Oxalic acid
$HOClO$	Hypochlorous acid	$HOClO_2$	Chlorous acid
$HOClO_2$	Chloric acid	$HOClO_3$	Perchloric acid
$HOBrO$	Hypobromous acid	$HOBrO_2$	Bromous acid
$HOBrO_2$	Bromic acid	$HOBrO_3$	Perbromic acid
$HOIO$	Hypoiodous acid	$HOIO_2$	Iodous acid
$HOIO_2$	Iodic acid	$HOIO_3$	Periodic acid

Table 4.6

When comparing the relative strength of oxyacids, the number of excess oxygen atoms and the electronegativity of the central atom must both be considered. For carbonic acid, there is an excess oxygen count of one. What is meant by excess oxygen count is the number of oxygen atoms exceeding the number of hydrogen atoms. The excess oxygen count often turns out to be the number of oxygen atoms double-bonded to central atom. The more π -bonds to oxygen from the central atom, the more resonance withdrawal from the central atom, and thus the more acidic the compound. This explains why H_2SO_4 is a stronger acid than H_2SO_3 . When two compounds have the same excess oxygen count, the next factor to consider is the electronegativity of the central atom. The more electronegative the central atom, the more it withdraws electron density from the acidic proton, increasing the acidity. This is in essence the inductive effect. Sulfur is more electronegative than carbon, so H_2SO_3 is a stronger acid than H_2CO_3 . As you learned in organic chemistry, the resonance effect is greater than the inductive effect.

Example 4.10

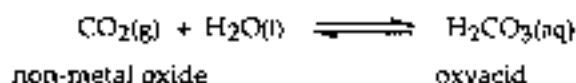
Which of the following acids has the LARGEST pK_a value?

- A. HClO_2
- B. HIO_2
- C. HClO_3
- D. HIO_3

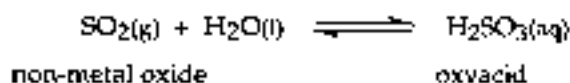
Solution

The largest pK_a value is associated with the weakest oxyacid, which is associated with the acid having the smallest number of excess oxygens and the least electronegative central atom. Choices C and D are eliminated, because they contain two excess oxygens each. Choice B is better than choice A, because they both have one excess oxygen, but iodine is less electronegative than chlorine. This question could have easily asked for the strongest acid. It is important to realize that most questions will be asking for either the strongest acid or the weakest acid among the answer choices. The wording may be in reference to conjugate bases, electrolytic nature, ionizability, K_a values, pK_a values, pH values, or reactivity. The secret is to do enough practice questions so that you encounter all the different possibilities at least a few times before your test.

As a point of interest, oxyacids result from the hydration of non-metal oxides. This is the cause of acid rain, where most often nitrogen oxides and sulfur oxides (Lewis acids) react with moisture in the air to form Brønsted-Lowry acids. A lowering in pH for rainfall may also be observed in environments rich in carbon dioxide (which hydrates to become carbonic acid). Reaction 4.8 shows the hydration of carbon dioxide, while Reaction 4.9 shows the hydration of sulfur trioxide.



Reaction 4.8

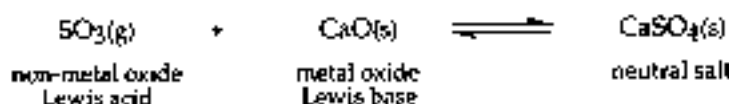


Reaction 4.9

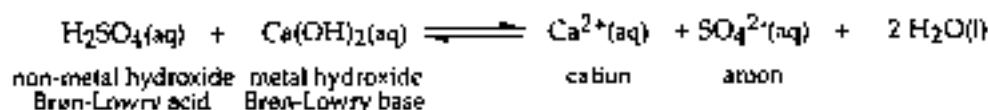
Non-metal oxides act as Lewis acids (electron-pair acceptors), while their hydrated counterparts (non-metal hydroxides, commonly known as oxyacids) act as Brønsted-Lowry acids. Both SO_3 and H_2SO_4 react with hydroxide to form HSO_4^- , so they are equivalent in terms of strength. The difference is that SO_3 is hydrated to become H_2SO_4 . By a similar reaction, sulfur dioxide (SO_2) forms H_2SO_3 (sulfurous acid) and nitrogen dioxide (NO_2) forms both HNO_2 (nitrous acid) and HNO_3 (nitric acid). These are major components of acid rain. Acid rain is a combination of rain and air-borne pollutants, such as sulfur oxides and nitrogen oxides. It is often treated with steam to convert from the Lewis acid form (non-metal oxide) into the Brønsted-Lowry form (non-metal hydroxide) and then neutralized with calcium oxide.

Metal Hydroxides and Metal Oxides

Just as non-metal oxides are Lewis acids and non-metal hydroxides are Brønsted-Lowry acids, metal oxides are Lewis bases and metal hydroxides are Brønsted-Lowry bases. You should be familiar with these general classifications. Metal oxides are basic and will form metal hydroxides when treated with water. A prime example is calcium oxide (CaO), which forms calcium hydroxide (Ca(OH)_2) when hydrated. Reaction 4.10 shows the Lewis acid-base reaction of a metal oxide and a non-metal oxide, while Reaction 4.11 shows the Brønsted-Lowry acid-base reaction of a metal hydroxide and a non-metal hydroxide. The reactants in Reaction 4.11 are the hydrated form of the reactants in Reaction 4.10.



Reaction 4.10



Reaction 4.11

Organic Acids

Let us consider three types of organic acids: carboxylic acids, phenols, and alkyl ammonium salts. It is important that you recognize these functional groups and know their pK_a ranges. For carboxylic acids and alkyl ammoniums, you should know their organic pK_a range and their range in amino acids. In carboxylic acids and phenols, the proton comes off of an oxygen that is involved in resonance. In an alkyl ammonium cation, the proton comes off of a nitrogen. Figure 4-9 shows a generic carboxylic acid and its pK_a range, a generic phenol and its pK_a range, and a generic alkyl ammonium cation and its pK_a range. The AA in parenthesis refers to amino acid terminals, while the R designation refers to an alkyl group.

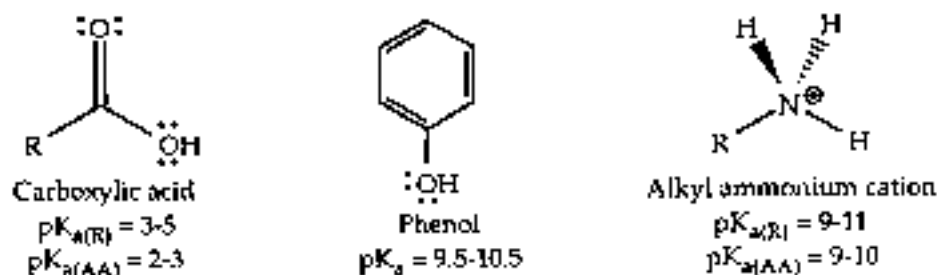


Figure 4-9

Polyprotic Acids

Polyprotic acids are acids which yield multiple equivalents of hydronium (H_3O^+) when treated with a base. The three most common examples from general chemistry are carbonic acid (H_2CO_3), sulfuric acid (H_2SO_4), and phosphoric acid (H_3PO_4). In addition to these three acids, there are also common amino acids that qualify as either diprotic (no active proton on the side chain) or triprotic (an active proton on the side chain). Concentrations of polyprotic acids are often given in terms of normality (N). Normality is defined as moles of equivalents per liter solution. This is to say that a 1.0 molar diprotic acid solution would be listed as "2.0 normal," because there are two equivalents of acid.

Example 4.11

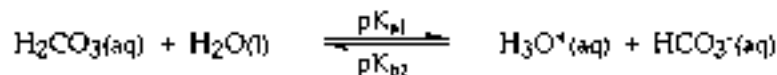
Which of the following acid solutions is 3.0 N?

- A. 1.00 M alanine
- B. 1.50 M carbonic acid
- C. 1.50 M phosphoric acid
- D. 1.25 M sulfuric acid

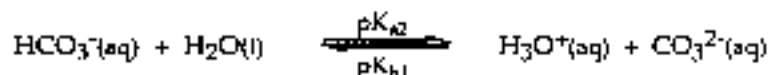
Solution

Normality (N) is found by multiplying the molarity of an acid by the number of protons per molecule. In choice A, alanine ($\text{H}_3\text{NCH}(\text{CH}_3)\text{CO}_2\text{H}$) is diprotic, so the normality is 1.00×2 , which equals 2.0 N. Choice A is eliminated. In choice B, carbonic acid (H_2CO_3) is diprotic, so the normality is 1.50×2 , which equals 3.0 N. Choice B is the correct answer. In choice C, phosphoric acid (H_3PO_4) is triprotic, so the normality is 1.50×3 , which equals 4.5 N. Choice C is eliminated. In choice D, sulfuric acid (H_2SO_4) is diprotic, so the normality is 1.25×2 , which equals 2.5 N. Choice D is eliminated.

Polyprotic acids have multiple pK_a values, one for each dissociable proton. By definition, the first proton removed is more acidic than the second one removed, so pK_{a1} is always lower than pK_{a2} . Some difficulty may arise when you consider the pK_a values of a polyprotic acid and its conjugate base. For a diprotic acid such as H_2CO_3 , the first proton removed corresponds to the second proton gained by the conjugate base. H_2CO_3 and HCO_3^- are a conjugate pair with pK_{a1} and pK_{b2} summing to 14. The full dissociation of carbonic acid is shown in Reaction 4.12 and Reaction 4.13.



Reaction 4.12



Reaction 4.13

Determining the amount of base needed to neutralize a polyprotic acid is a typical question from general chemistry. While the MCAT does not emphasize solving mathematical problems, understanding the setup is still important, because understanding equivalents can help to determine the pH of mixtures.

Example 4.12

How many mL of 0.40 M NaOH are required to neutralize 100 mL 0.25 M H_2SO_4 ?

- A. 62.5 mL
- B. 80.0 mL
- C. 100.0 mL
- D. 125.0 mL

Solution

In order to neutralize an acid, an equal mole quantity of hydroxide must be added to the hydronium source. The general relationship is shown in Equation 4.8.

$$\text{moles OH}^- = \text{moles H}^+ \therefore (M_{\text{OH}^-})(V_{\text{OH}^-}) = (M_{\text{H}^+})(V_{\text{H}^+}) \quad (4.8)$$

Substituting into Equation 4.8 yields the following results:

$$\begin{aligned} (M_{\text{OH}^-})(V_{\text{OH}^-}) &= (M_{\text{H}^+})(V_{\text{H}^+}) \\ (0.40 \text{ M})(V_{\text{OH}^-}) &= 2 \times (0.25 \text{ M})(100 \text{ mL}) \\ V_{\text{OH}^-} &= 2 \times \left(\frac{0.25 \text{ M}}{0.40 \text{ M}} \right) \times 100 \text{ mL} = \left(\frac{0.50}{0.40} \right) \times 100 \text{ mL} = 125 \text{ mL} \end{aligned}$$

Choice D is the best answer. Choice A would have been the result of your calculation, if you had forgotten to multiply the molarity by the number of equivalents, a common mistake with these types of questions.

Example 4.13

How many milliliters of 0.60 M HCl are required to neutralize 3.0 grams CaCO_3 ?

- A. 50 mL
- B. 100 mL
- C. 200 mL
- D. 300 mL

Solution

According to the balanced equation, two molecules of HCl are required for every one molecule of CaCO_3 . The balanced equation is shown below:



The mathematical setup is:

$$\begin{aligned} \text{moles OH}^- &= 2 \times \text{moles CO}_3^{2-} = (M_{\text{H}^+})(V_{\text{H}^+}) \\ 2 \times \frac{3.0 \text{ g}}{100 \frac{\text{g}}{\text{mole}}} &= (0.60 \text{ M})(V_{\text{H}^+}) \\ 0.06 \text{ moles} &= (0.60 \text{ M})(V_{\text{H}^+}) \therefore V_{\text{H}^+} = 0.10 \text{ L} = 100 \text{ mL} \end{aligned}$$

Choice B is the best answer. Choice A would have been the result of your calculation, if you had forgotten to multiply the molarity by the number of equivalents, a common mistake with these types of questions.

Example 4.14

Citric acid ($\text{C}_6\text{H}_8\text{O}_7$) has three dissociable protons with pK_a values of 3.14, 4.79, and 5.20. Which of the following solutions would have the lowest pH values?

- A. 50 mL 0.10 M citric acid(aq) + 75 mL 0.20 M NaOH(aq)
- B. 25 mL 0.10 M citric acid(aq) + 50 mL 0.10 M NaOH(aq)
- C. 25 mL 0.20 M citric acid(aq) + 75 mL 0.20 M NaOH(aq)
- D. 50 mL 0.20 M citric acid(aq) + 50 mL 0.10 M NaOH(aq)

Solution

For questions involving mixtures, it is important to think of the reagents in terms of equivalents. The lowest pH belongs to the solution that is most acidic. The most acidic solution is the solution where the fewest equivalents of base relative to citric acid have been added. In choice A, the NaOH(aq) solution is 1.5 times the volume and twice the concentration of the citric acid solution, so there are three equivalents of NaOH(aq). In choice B, the NaOH(aq) solution is double the volume of the citric acid solution and of equal concentration, so there are two equivalents of NaOH(aq). In choice C, the NaOH(aq) solution is three times the volume of the citric acid solution and of equal concentration, so there are three equivalents of NaOH(aq). In choice D, the NaOH(aq) solution is of equal volume and half the concentration of the citric acid solution, so there is only one-half of an equivalent of NaOH(aq). The fewest equivalents are found in choice D, so choice D is the best answer.

Calculating pH

Determining pH

A solution's acidity is measured in terms of hydronium concentration ($[\text{H}_3\text{O}^+]$) using the pH scale. The pH value of a solution is determined using Equation 4.9.

$$\text{pH} = -\log [\text{H}_3\text{O}^+] \quad (4.9)$$

By manipulating Equation 4.9 so as to isolate hydronium concentration, Equation 4.10 can be derived.

$$[\text{H}_3\text{O}^+] = 10^{-\text{pH}} \quad (4.10)$$

Because neutral water has an H_3O^+ concentration of 10^{-7} M, due to the autoionization of water, neutral water has a pH of 7.0. Acidic solutions have pH values less than 7.0, while basic solutions have pH values greater than 7.0. There are no limits to the pH scale other than those imposed by the strength and concentration of the acid or base in solution.

On the test, you can almost assume that they will have questions about both pH and pOH for an aqueous solution. Just as Equation 4.9 defines pH, Equation 4.11 defines pOH.

$$\text{pOH} = -\log [\text{OH}^-] \quad (4.11)$$

This means that in order to calculate pH or pOH, it is necessary to determine the H_3O^+ concentration or the OH^- concentration. Once this is accomplished, it is simply log math (negative logs, actually). There are no calculators allowed on this test. The volume of a solution does not matter in determining pH; only the concentration is important. Equation 4.12 may be used to interconvert between pH and pOH in an aqueous solution at 25°C.

$$\text{pH} + \text{pOH} = 14 \quad (4.12)$$

Example 4.15

What is the pH of a solution where the hydroxide concentration is 10^6 times greater than the hydronium concentration?

- A. 4
- B. 6
- C. 10
- D. 13

Solution

Because hydroxide is in greater concentration than hydronium, the solution is basic, so the pH must be greater than 7. This eliminates choices A and B. The trick here is to use the log scale correctly. Because the concentrations differ by a factor of 10^6 , the pH and pOH differ by $\log 10^6$, which is 6. If the pH is 10, then the pOH is 4, which are different by 6. If the pH is 13, then the pOH is 1, and the difference is 12. This means that the best answer is choice C. This is a tricky question where understanding the conversion between the log scale and concentration scale is pertinent. Most students choose D, because they add 6 to the neutral pH of 7. It is important to recognize why choice D is a typical mistake.

Log Review

Because pH is based on logs, it is helpful to do a quick review of logs. When you **multiply** numbers, you **add** their logs. When you **divide** numbers, you **subtract** their logs. You can solve for any log, given that $\log 2 = 0.3$ and $\log 3 = 0.48$. Drawn in Table 4.7 are examples of how to solve for a log value.

Number	Mathematical Calculation	Log Value
2	Given value	0.301
3	Given value	0.477
4	$\log 4 = \log (2 \times 2) = \log 2 + \log 2 = .301 + .301 = .602$	0.602
5	$\log 5 = \log (10 \div 2) = \log 10 - \log 2 = 1.00 - .301 = .699$	0.699
6	$\log 6 = \log (3 \times 2) = \log 3 + \log 2 = .477 + .301 = .778$	0.778
7	Log 7 is approximated as being closer to log 8 than log 6	0.845 $\pm$.01
8	$\log 8 = \log (2 \times 2 \times 2) = 3(\log 2) = 3(.301) = .903$	0.903
9	$\log 9 = \log (3 \times 3) = \log 3 + \log 3 = .477 + .477 = .954$	0.954
0.33	$\log 0.33 = \log (1 \div 3) = \log 1 - \log 3 = 0 - .477$	-0.477
0.50	$\log 0.50 = \log (1 \div 2) = \log 1 - \log 2 = 0 - .301$	-0.301
1.20	$\log 1.2 = \log (6 \div 5) = \log 6 - \log 5 = .778 - .699 = .079$	0.079
1.25	$\log 1.25 = \log (5 \div 4) = \log 5 - \log 4 = .699 - .602 = .097$	0.097
1.33	$\log 1.33 = \log (4 \div 3) = \log 4 - \log 3 = .602 - .477 = .125$	0.125
1.40	$\log 1.40 = \log (7 \div 5) = \log 7 - \log 5 = .845 - .699 = .146$	0.146
1.50	$\log 1.50 = \log (3 \div 2) = \log 3 - \log 2 = .477 - .301 = .176$	0.176
1.60	$\log 1.60 = \log (8 \div 5) = \log 8 - \log 5 = .903 - .699 = .204$	0.204
1.67	$\log 1.67 = \log (5 \div 3) = \log 5 - \log 3 = .699 - .477 = .222$	0.222
1.75	$\log 1.75 = \log (7 \div 4) = \log 7 - \log 4 = .845 - .602 = .243$	0.243
1.80	$\log 1.80 = \log (9 \div 5) = \log 9 - \log 5 = .954 - .699 = .255$	0.255

Table 4.7

Log calculations should be carried out only to the level of approximation. Do not forget that $\log$ of $10^x = x$, so knowing the power of 10 that some quantity is can be a useful hint when selecting the correct log value. Often, only one answer choice relates to the correct power of 10, which saves a great deal of time spent in calculating. All sorts of shortcuts and tricks will be presented during the following sample problems. Before moving on in this section, however, make a concerted effort to work through Table 4.7 and understand thoroughly how to estimate log values.

Determining pH for Strong Reagents

Calculating the pH for strong acid and strong base solutions is based on the concept of full dissociation. The calculation of the pH for a strong reagent follows an easy pattern. For a strong acid, Equation 4.9 is employed, where the concentration of the strong acid ($[HX]$) is substituted for the concentration of hydronium ($[H_3O^+]$). For a strong base, Equation 4.11 is employed, where the concentration of the strong base ($[MOH]$) is substituted for the concentration of hydroxide ($[OH^-]$). Calculating pH and pOH from concentrations is most easily done when the concentration is written in scientific notation. This is because taking the log of a number in scientific notation is more convenient in terms of bookkeeping than taking the log of a number in standard notation.

Example 4.16

What is the pH of 0.0020 M HCl(aq)?

- A. 2.00
- B. 2.70
- C. 3.00
- D. 7.00

Solution

Determining an exact numerical value involves calculations. The pH for any solution is defined as $\text{pH} = -\log [\text{H}_3\text{O}^+]$, so for a strong acid, the pH is $-\log [\text{HX}]$. It is best to use scientific notation for the concentration.

$$\begin{aligned}\text{pH} &= -\log (2 \times 10^{-3}) \\ \text{pH} &= -(\log 2 + \log 10^{-3}) = -\log 2 - \log 10^{-3} \\ \text{pH} &= -\log 2 - (-3) = 3 - \log 2 \\ \text{pH} &= 3 - 0.3 = 2.7\end{aligned}$$

The correct answer is choice B. Choice D should have been eliminated, because the solution is acidic, so pH is less than 7.00. Choices A and C could also have been eliminated, if you noted that the log values ended in ".00". For a log to be a whole number, the concentration must be a power of ten. The fact that the concentration was .002 tells us that the log could not be a whole number. From this example, you should derive a shortcut for use in the future. You might take notice that the $-\log$ of 2×10^{-3} is equal to $3 - \log 2$, so why not remember this and skip a few steps in the future. Use the relationship: $-\log (x \times 10^y) = y - \log x$.

This shortcut applies to all negative log calculations, including the conversion from K_a to $\text{p}K_a$. For instance, the $\text{p}K_a$ for a weak acid with K_a equal to 4.1×10^{-6} is $6 - \log 4.1$. This value can be estimated to be greater than 5.0 (which is equal to $6 - \log 10$), but less than 5.5 (which is equal to $6 - \log 3$). A range of 5.0 to 5.5 should be good enough to choose the correct answer from four choices. With exact numerical questions on the MCAT, your goal should be to narrow the answer choice range enough so that three wrong answers may be eliminated.

Example 4.17

What is the pH of 100 mL of 0.030 M HBr(aq)?

- A. 1.30
- B. 1.52
- C. 2.30
- D. 7.00

Solution

The volume of the solution does not affect the pH, unless another solution is added. Using the shortcut, the pH is found as follows:

$$\begin{aligned}\text{pH} &= -\log (3 \times 10^{-2}) = 2 - \log 3 \\ \text{pH} &= 2 - 0.48 = 1.52\end{aligned}$$

The correct answer is choice B. Choice D should have been eliminated, because the solution is acidic, so pH is less than 7.00. Choices A and C could also have been eliminated, if you noted that the log values ended in ".30". This would come from logarithmic insights.

Example 4.18

What is the pOH of 0.050 M KOH(aq)?

- A. 1.30
- B. 1.70
- C. 12.30
- D. 12.70

Solution

The pOH of a basic solution is found in a manner similar to getting the pH of an acidic solution. Using the shortcut, the pOH is found as follows:

$$\begin{aligned}\text{pOH} &= -\log(5 \times 10^{-2}) = 2 - \log 5 \\ \text{pOH} &= 2 - 0.7 = 1.30\end{aligned}$$

The correct answer is choice A. Choices C and D should have been eliminated, because the solution is basic, so pH is greater than 7.00, and therefore pOH is less than 7.00. The pH of the solution can be found using $\text{pH} = 14 - \text{pOH}$, where $\text{pOH} = 14 - 1.30 = 12.70$.

Example 4.19

What is the pH of 200 mL of 0.00391 M KOH(aq)?

- A. 2.41
- B. 2.61
- C. 11.39
- D. 11.59

Solution

This question would seem to be quite difficult at first glance, but if you follow the rules, it is easy. Because KOH is a strong base, it will fully dissociate when added to water. Plugging values into our shortcut method yields the following:

$$\begin{aligned}\text{pOH} &= -\log(3.91 \times 10^{-3}) = 3 - \log 3.91 \\ 3 - \log 10 &< 3 - \log 3.91 < 3 - \log 3 \quad \therefore 2 < \text{pOH} < 2.5 \\ \text{If } 2 < \text{pOH} < 2.5, \text{ then } 12 > \text{pH} > 11.5\end{aligned}$$

The correct answer is choice D. Choices A and B should have been eliminated, because the solution is basic, so pH is greater than 7.00. Choice C is eliminated, because it does not fit into the range for the correct number. Some of you may have chosen to approximate 3.91 as 4, and solved accordingly. This method is fine, too.

You should be able to determine pH or pOH for strong compounds in less than fifteen seconds. While the MCAT does not offer up many calculation questions, if you are fortunate enough to get a pH calculation question, you should finish it quickly, and carry the time you save over to more difficult questions.

Determining pH for Weak Reagents

Because weak acids and weak bases do not fully dissociate, it is necessary to use the K_a and K_b values to determine their $\text{H}_3\text{O}^+(\text{aq})$ and $\text{OH}^-(\text{aq})$ concentrations. K_a and K_b are equilibrium constants for the respective compounds in water. Equation 4.7 shows the relationship between the $\text{p}K_a$ and $\text{p}K_b$ for an acid and its conjugate base. Equation 4.13 is based on the relationship of equilibrium constants to K_w (the dissociation constant for water).

$$K_a \times K_b = 10^{-14} \quad (4.13)$$

According to Reaction 4.1, weak acids dissociate into hydronium and conjugate base when added to water. Equal parts of conjugate base and hydronium ion form. To determine the $[\text{H}_3\text{O}^+]$, the dissociation reaction and the dissociation constant (K_a) must be employed. This too, like the strong acids, is a systematic process to master, in the course of which we shall discover another shortcut. To understand the process, consider some concentration of a weak acid (HA) with a $\text{p}K_a$ between 2 and 12. The setup for the reaction is shown in Figure 4-10.

Reaction:	$\text{HA}(\text{aq})$	$\text{H}_2\text{O}(\text{l})$	$\rightleftharpoons$	$\text{H}_3\text{O}^+(\text{aq})$	$\text{A}^-(\text{aq})$
Initially:	$[\text{HA}]_{\text{init}}$	excess		negligible	0
Shift:	$-x$	$-x$	$\longrightarrow$	$+x$	$+x$
Equilibrium:	$[\text{HA}]_{\text{init}} - x$	irrelevant		x	x

Figure 4-10

Substituting values into the acid dissociation expression leads to Equation 4.14, which can be applied if the concentration is greater than the K_a , and if $\text{p}K_a$ falls between 2 and 12. Equation 4.14 does not generate precise answers, but it does give a very close approximation.

$$\begin{aligned} \text{Given: } [\text{A}^-] &= [\text{H}_3\text{O}^+], \text{ then } K_a = \frac{[\text{A}^-][\text{H}_3\text{O}^+]}{[\text{HA}]} = \frac{[\text{H}_3\text{O}^+][\text{H}_3\text{O}^+]}{[\text{HA}]} = \frac{[\text{H}_3\text{O}^+]^2}{[\text{HA}]} \\ K_a &= \frac{[\text{H}_3\text{O}^+]^2}{[\text{HA}]} \therefore [\text{H}_3\text{O}^+]^2 = K_a \times [\text{HA}] \therefore [\text{H}_3\text{O}^+] = \sqrt{K_a \times [\text{HA}]} \end{aligned}$$

Plugging this value into Equation 4.9 for $[\text{H}_3\text{O}^+]$, yields Equation 4.14

$$\text{pH} = -\log \sqrt{K_a \times [\text{HA}]} \quad (4.14)$$

Equation 4.14 can be further manipulated to generate Equation 4.15 (the shortcut equation), which will save you time, once it is understood and mastered.

$$\begin{aligned} \text{pH} &= -\log \sqrt{K_a \times [\text{HA}]} = -\log \sqrt{K_a} + (-\log \sqrt{[\text{HA}]}) \\ \text{pH} &= -\log \{K_a\}^{1/2} + \{-\log [\text{HA}]\}^{1/2} = -\frac{1}{2} \log K_a - \frac{1}{2} \log [\text{HA}] \\ \text{pH} &= \frac{1}{2} \log K_a + \frac{1}{2} \log [\text{HA}] = \frac{1}{2} \text{p}K_a - \frac{1}{2} \log [\text{HA}] \\ \text{pH} &= \frac{1}{2} \text{p}K_a - \frac{1}{2} \log [\text{HA}] \quad (4.15) \end{aligned}$$

To ensure that Equation 4.15 makes conceptual sense, let's consider the pH of a weak acid solution. If you add more acid, the pH should decrease. According to Equation 4.15, increasing $[\text{HA}]$ lowers the pH. Stronger acids have more dissociation, so they should form solutions of a lower pH. Equation 4.15 shows this by including the $\text{p}K_a$ term. Getting the pH of a weak acid should be easy.

Example 4.20

What is the pH of 1.00 M HF(aq) with $pK_a = 3.32$?

- A. 0.50
- B. 1.66
- C. 2.00
- D. 6.64

Solution

This question is made easy by applying Equation 4.15. Two requirements for Equation 4.15 to work are that: 1) the weak acid concentration must be greater than the K_a and 2) the pK_a must lie between 2 and 12. Both of these criteria are met, so Equation 4.15 may be applied.

$$pH = \frac{1}{2} pK_a - \frac{1}{2} \log [HA] = \frac{1}{2} (3.32) - \frac{1}{2} \log 1.0 = 1.66 - \frac{1}{2}(0) = 1.66$$

The correct answer is choice B. Because $[HA]$ is 1.0 M, the pH is half of the pK_a . If the acid concentration is 0.10 M, then the pH is half of the $pK_a + 0.5$. This means that the pH of a weak acid can be estimated quickly.

Example 4.21

What is the pH of 0.07562 M HCO_2H with a pK_a of 3.642?

- A. 2.259
- B. 2.383
- C. 2.739
- D. 2.883

Solution

The pH can be estimated from one-half of the pK_a . If the acid concentration were 1.00 M, then the pH would be 1.821 (half of the pK_a). If the acid concentration were 0.10 M, then the pH would be 2.321 (half of the $pK_a + 0.5$). If the acid concentration were 0.010 M, then the pH would be 2.821 (half of the $pK_a + 1.0$). The concentration falls between 0.10 M and 0.010 M, so the pH must fall between 2.321 and 2.821. This eliminates choices A and D. Because the concentration is just less than 0.10 M, the pH should be slightly higher than 2.321, making choice B the best answer.

You should not expect numbers to be this difficult. But given that these numbers now can be handled without difficulty, any weak acid pH calculation can be made easy. The answer to a question like this one in a multiple-choice exam can be approximated quickly and simply using this technique. The answer choices may be considered as other expressions of

- A. $1.821 + 0.438$
- B. $2.321 + 0.062$
- C. $2.321 + 0.438$
- D. $2.821 + 0.062$

Equation 4.16 is the equivalent equation for the pOH of a weak base solution. It is derived in the same fashion as Equation 4.15, but K_b replaces K_a , $[A^-]$ replaces $[HA]$, and base hydrolysis is considered instead of acid dissociation.

$$pOH = \frac{1}{2} pK_b - \frac{1}{2} \log [A^-] \quad (4.16)$$

Example 4.22

What is the pH of 0.20 M sodium propionate, if it has $K_b = 7.2 \times 10^{-10}$?

- A. Less than 3
- B. Between 3 and 7
- C. Between 7 and 11
- D. Greater than 11

Solution

The solution is basic, so the pH is greater than 7.0, eliminating choices A and B. The weak base concentration is greater than K_b , and pK_b falls between 2 and 12, so Equation 4.16 can be applied. The pK_b for the base is $10 - \log 7.2$, which is slightly more than 9. A good estimate is 9.2.

The pOH can be estimated from one-half of the pK_b . If the base concentration were 1.00 M, then the pOH would be 4.6 (half of the pK_b). If the base concentration were 0.10 M, then the pOH would be 5.1 (half of the $pK_b + 0.5$). The concentration is 0.20 M, so the pOH lies between 4.6 and 5.1. This means that pH lies between 8.9 and 9.4, making choice C the best answer.

Conjugate Pairs

A conjugate pair consists of an acid and a base that exchange one proton; one gains a proton and the other member of the pair loses that proton. An acid when deprotonated forms its conjugate base, and a base when protonated forms its conjugate acid. Reaction 4.1 and Reaction 4.4 show that HA and A^- are a generic conjugate pair. The charge difference between the acid and the base in a conjugate pair is +1.

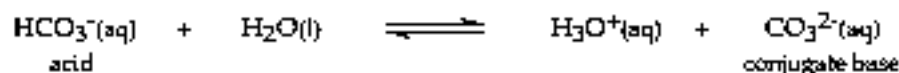
Example 4.23

The conjugate base of HCO_3^- is which of the following?

- A. CO_3^{2-}
- B. H_2CO_3
- C. CO_2
- D. HCO_2^-

Solution

A conjugate base is formed when an acid loses one proton. This eliminates choice B, which happens to be the conjugate acid of HCO_3^- . The conjugate base of HCO_3^- is CO_3^{2-} , so choice A is the best answer. The dissociation reaction of bicarbonate into hydronium and carbonate is shown below:

**Example 4.24**

Which of the following pairs of compounds is NOT a conjugate pair?

- A. $\text{NH}_3/\text{NH}_4^+$
- B. $\text{H}_2\text{CO}_3/\text{HCO}_3^-$
- C. $\text{H}_2\text{SO}_3/\text{HSO}_3^-$
- D. $\text{PO}_4^{3-}/\text{H}_2\text{PO}_4^-$

Solution

In choice A, NH_4^+ is formed when a proton is added to NH_3 , so NH_4^+ and NH_3 differ by one proton. This makes them a conjugate pair, and it eliminates choice A. In choice B, H_2CO_3 is formed when a proton is added to HCO_3^- , so H_2CO_3 and HCO_3^- differ by one proton. This makes them a conjugate pair, and it eliminates choice B. In choice C, H_2SO_3 is formed when a proton is added to HSO_3^- , so H_2SO_3 and HSO_3^- differ by one proton. This makes them a conjugate pair, and it eliminates choice C. Choice D is the best answer because, PO_4^{3-} and H_2PO_4^- differ by two protons, so they do not constitute a conjugate pair.

Recognizing conjugate pairs should be effortless. Knowing how to apply the concept of a conjugate pair to solving problems is the more useful ability.

Typical Conjugate Pairs

Typical conjugate pairs either are observed in biological examples or are common laboratory buffer systems. There are many weak acids and conjugate bases, so it is a good idea for you to recall classes of compounds, rather than specific compounds. Common conjugate pairs where both components are weak include the carboxylic acids/carboxylates ($\text{RCO}_2\text{H}/\text{RCO}_2^-$), the alkyl ammoniums/alkyl amines ($\text{RNH}_3^+/\text{RNH}_2$), and phenols/phenoxides ($\text{C}_6\text{H}_5\text{OH}/\text{C}_6\text{H}_5\text{O}^-$). Specific pairs that are likely to appear on the MCAT with great frequency are carbonic acid/bicarbonate ($\text{H}_2\text{CO}_3/\text{HCO}_3^-$) and phosphoric acid/dihydrogen phosphate ($\text{H}_3\text{PO}_4/\text{H}_2\text{PO}_4^-$), due to their presence in physiological systems. It is a good idea to know the acid-base properties of compounds that are common in physiology.

The distribution within a conjugate pair is dictated by the pH of the solution. The conjugate pair favors the conjugate acid form in the presence of hydronium. The conjugate pair favors the conjugate base form in the presence of hydroxide. The exact distribution is determined by the relationship between pH of the solution and the pK_a of the conjugate acid. Figure 4-11 shows this relationship.

If $\text{pH} > \text{pK}_a$, the solution is basic relative to the compound, so it is deprotonated.

If $\text{pH} < \text{pK}_a$, the solution is acidic relative to the compound, so it is protonated.

Figure 4-11

The pH refers to the surrounding solution (environment) in which the compounds exist. The pK_a refers to the conjugate acid that exists in solution. The compound responds to the pH of the solution. Because of the importance of relating pH to pK_a , common pK_a ranges should be known. Some common pK_a ranges are shown in Figure 4-9. Carboxylic acids have pK_a values of 2 to 5. In amino acids, the carboxyl terminal has a pK_a between 2 and 3. Because of the low pK_a value relative to physiological pH, the physiological form of carboxylic acids is the deprotonated form. Ammonium and alkyl ammoniums have pK_a values of 9 to 11. In amino acids, the amino terminal has a pK_a between 9 and 10. Because of the high pK_a value relative to physiological pH, the physiological form of amines is the protonated form. Figure 4-12 summarizes some common physiological compounds and their natural states:

Acid	Conjugate Base
RCO_2H ($\text{pK}_a = 3-5$)	RCO_2^- ($\text{pK}_b = 9-11$) physiological form, because $7.4 > 5$
RNH_3^+ ($\text{pK}_a = 9-10$) physiological form, because $7.4 < 9$	RNH_2 ($\text{pK}_b = 4-5$)
H_2PO_4^- ($\text{pK}_{a2} = 7.2$)	HPO_4^{2-} ($\text{pK}_{b2} = 6.8$) physiological form, because $7.4 > 7.2$
H_2CO_3 ($\text{pK}_{a1} = 6.4$)	HCO_3^- ($\text{pK}_{b2} = 7.6$) physiological form, because $7.4 > 6.4$

Figure 4-12

Relationship of pK_a and pK_b

Because the members of a conjugate pair exchange one proton, their respective equilibrium constants are related. An acid when deprotonated forms its conjugate base, and a base when protonated forms its conjugate acid. Equation 4.7 shows the relationship of the pK_a for the acid and the pK_b for its conjugate base. This relationship is useful for interconverting between pK values. However, when polyprotic acids are involved, it gets a little more complicated. The relationship is emphasized in Reaction 4.12 and Reaction 4.13. Also keep in mind that Equation 4.7 applies only at 25°C in water.

Example 4.25

What is the pK_a for ammonia, given that the pK_b for ammonia is 4.7?

- A. 4.7
- B. 7.0
- C. 9.3
- D. 33

Solution

It is not 9.3! The compound with a pK_a of 9.3 is ammonium (NH_4^+), the conjugate acid of ammonia. Ammonia is a weaker acid than ammonium, so it has a pK_a greater than 9.3. The best answer is choice D, 33, indicating that ammonia is such a weak acid that when added to water, there is no detectable dissociation. The mistake of choosing 9.3 is easy to make, one that most students make routinely. But the equation $pK_a (HA) + pK_b (A^-) = 14$ is for conjugate pairs, not for the same compound.

The test-writers are more likely to apply this concept to carbonic acid (H_2CO_3), which has two dissociable protons, and thus has both pK_{a1} and pK_{a2} . The correct relationships between pK_a and pK_b for the two respective conjugate pairs are $pK_{a1} + pK_{b2} = 14$ and $pK_{a2} + pK_{b1} = 14$. Also note that pK_{a1} is always less than pK_{a2} , because the first proton is more acidic than the second one, by definition.

Henderson-Hasselbalch Equation

The pH of a solution comprised of both components in a weak conjugate pair can be determined using the Henderson-Hasselbalch equation, which is shown below as Equation 4.17. For the Henderson-Hasselbalch equation to hold true, both the acid and its conjugate base must be present in appreciable concentration in solution.

$$\text{pH} = \text{pK}_a + \log \frac{[\text{Conjugate base}]}{[\text{Conjugate acid}]} \quad (4.17)$$

The equation shows that as the conjugate base concentration ($[\text{conjugate base}]$) increases, the pH of the buffer increases. It also shows that as the conjugate acid concentration ($[\text{conjugate acid}]$) increases, the pH of the buffer decreases. Equation 4.17 is derived from the K_a equation.

$$K_a = \frac{[A^-][H_3O^+]}{[HA]} \quad \therefore [H_3O^+] = K_a \frac{[HA]}{[A^-]}$$

$$\text{taking } -\log \text{ of everything yields: } -\log [H_3O^+] = -\log K_a - \log \left(\frac{[HA]}{[A^-]} \right)$$

$$-\log [H_3O^+] = -\log K_a - \log \left(\frac{[HA]}{[A^-]} \right) \Rightarrow \text{pH} = \text{pK}_a - \log \left(\frac{[HA]}{[A^-]} \right) = \text{pK}_a + \log \left(\frac{[A^-]}{[HA]} \right)$$

Equation 4.17 can be used with either concentration units or mole quantities for HA and A^- . This means that Equation 4.17 can be rewritten as Equation 4.18.

$$\text{pH} = \text{pK}_a + \log \frac{\text{moles Conjugate Base}}{\text{moles Conjugate Acid}} \quad (4.18)$$

According to Equation 4.17, the addition of water to a conjugate pair mixture has no effect on the pH of the solution. The compounds are diluted, and thus less concentrated, but the pH remains the same because the $[A^-] : [HA]$ ratio remains the same.

Example 4.26

Which of the following solutions has the GREATEST pH?

- A. 10 mL 0.10 M $\text{NH}_3(aq)$ with 15 mL 0.10 M $\text{NH}_4^+(aq)$
- B. 15 mL 0.10 M $\text{NH}_3(aq)$ with 10 mL 0.10 M $\text{NH}_4^+(aq)$
- C. 10 mL 0.10 M $\text{HCO}_2\text{Na}(aq)$ with 15 mL 0.10 M $\text{HCO}_2\text{H}(aq)$
- D. 15 mL 0.10 M $\text{HCO}_2\text{Na}(aq)$ with 10 mL 0.10 M $\text{HCO}_2\text{H}(aq)$

Solution

The pH of a conjugate mixture can be determined using Equation 4.18. To make the pH high, the pK_a must be high, and the mixture must be rich in conjugate base. The pK_a of ammonium is greater than the pK_a of a carboxylic acid (in this case, formic acid), so choices C and D are eliminated. Choice A has more acid than conjugate base, so the pH is less than the pK_a for the acid. Choice B has more base than conjugate acid, so the pH is greater than the pK_a for the acid. This means that choice B has the greatest pH.

Acids and Bases Passages

13 Passages

100 Questions

Suggested Acids and Bases Passage Schedule:

- I:** After reading this section and attending lecture: Passages I, III, V, VII, & IX
Grade passages immediately after completion and log your mistakes.
- II:** Following Task I: Passages II, IV, VI, & VIII (27 questions in 35 minutes)
Time yourself accurately, grade your answers, and review mistakes.
- III:** Review: Passages X - XIII & Questions 92 - 100
Focus on reviewing the concepts. Do not worry about timing



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Acids and Bases Study Passages

I. Acid and Base Definitions	(1 - 7)
II. Dissociation and Colligative Properties	(8 - 13)
III. Oxyacids	(14 - 21)
IV. Acidity of Thiols and Alcohols	(22 - 28)
V. Organic Acids	(29 - 34)
VI. Electron-Withdrawing Effect and Acidity	(42 - 48)
VII. Weak Acid pH Equation	(35 - 41)
VIII. Aspirin and Antacids	(49 - 55)
IX. Household Acids and Bases	(56 - 62)
X. Stomach Acid and pH	(63 - 70)
XI. Tooth Decay and pH	(71 - 76)
XII. Acid Rain and Scrubbers	(77 - 84)
XIII. Amino Acids pK_a Values	(85 - 91)
Questions Not Based on a Descriptive Passage	(92 - 100)

Acids and Bases Scoring Scale

Raw Score	MCAT Score
84 - 100	13 - 15
66 - 83	10 - 12
47 - 65	7 - 9
34 - 46	4 - 6
1 - 33	1 - 3

Passage I (Questions 1 - 7)

A base can be defined in three different ways, depending on its solvent. The definitions are:

1. The Arrhenius definition states that a base yields $\text{OH}^-(\text{aq})$ when added to water.
2. The Brønsted-Lowry definition states that a base is a proton acceptor.
3. The Lewis definition states that a base is an electron pair donor.

The same base can fit the description of all three definitions. The perfect example of a base that fits the three definitions is ammonia (NH_3), which can donate its lone pair of electrons to accept a proton from water, to yield an aqueous hydroxide anion ($\text{OH}^-(\text{aq})$).

Although a base may simultaneously fit the three definitions, each definition has its own unique application. For the calculation of the pH of water-based solutions, the Arrhenius definition is the most applicable. The Brønsted-Lowry definition is applicable in a more general sense, because it accounts for acid-base chemistry that takes place in a protic solvent other than water. The Lewis definition is most commonly applied to organic chemistry where bases donate their lone pairs to empty p-orbitals. A Lewis base can react in any solvent, including aprotic ones.

A Lewis base also may be referred to as a *nucleophile*. The electrons of the nucleophile are donated to the partially positive site on an electrophile (which also may be a proton in addition to a partially positive carbon). The partial or full cationic charge is often the result of an excess of protons on the electrophile. The term "nucleophile" is derived from the strong affinity of the substances for positive charge (the charge of protons in an atomic nucleus). All three definitions can account for the nucleophilic nature of a base.

1. If the $[\text{OH}^-]$ of a $\text{pH} = 8.0$ solution is tripled, the new pH will be:
A. 5.0.
B. 7.5.
C. 8.5.
D. 11.0.
2. Which of the following bases would have the **LARGEST** pK_b value?
A. A base that undergoes 20% hydrolysis in water
B. A base that undergoes 15% hydrolysis in water
C. A base that undergoes 10% hydrolysis in water
D. A base that undergoes 5% hydrolysis in water

3. Which of the following is an Arrhenius base?

A. HClO
B. HBrO_2
C. Li_2CO_3
D. HNO_2

4. Which of the following would be the **BEST** choice to neutralize 25 mL 0.10 M HClO_3 ?

A. 25 mL 0.10 M $\text{NaOH}(\text{aq})$
B. 250 mL 0.010 M $\text{NH}_3(\text{aq})$
C. 25 mL 0.10 M $\text{HCl}(\text{aq})$
D. 250 mL 0.010 M $\text{HCO}_2\text{H}(\text{aq})$

5. As the conjugate acid for a base gets stronger, the base:

A. exhibits a decreasing pK_b value.
B. can react with weaker acids.
C. acquires a higher pOH value in water.
D. requires more moles of acid to be neutralized.

6. Which of the following definitions does **NOT** describe an acid?

A. A electron-pair acceptor
B. A proton donor
C. A compound that produces hydronium ion in water
D. A nucleophilic molecule

7. As the strength of an acid increases, which of the following does **NOT** happen?

A. The acid becomes more electrolytic.
B. The acid dissociates more.
C. The acid has a lower pK_a .
D. The acid has a lower K_a .

Passage II (Questions 8 - 13)

The strength of an acid in water is defined by its ability to dissociate into hydronium and conjugate base. Strong acids generate more ions in solution. As the number of ionic impurities in the water increases, so does the solution's electrical conductivity. Current does not readily pass through distilled water, so the degree of an acid's dissociation can be estimated by a solution's ability to conduct electricity.

To determine the correlation between acid strength and electrical conductivity, a researcher places the two ends of an open circuit into a container of water so that the solution becomes part of a closed circuit. Two volumetric tubes are poised above the solution, allowing for an exact quantity of solution to be added to the circuit solution. Figure 1 shows the apparatus used in the experiment.

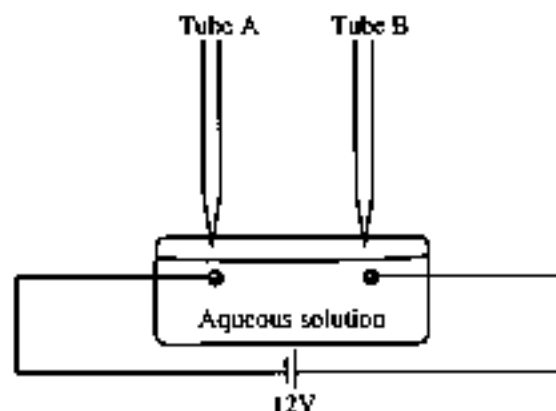


Figure 1

In different trials, 25 mL aliquots of acid are added, and the current is measured at different points in the wire. An average current was recorded. Any deviation in current at different sites can be attributed to errors in measurement, as the circuit is a single loop, so current should be uniform throughout. Table 1 lists the average current in the wire and the contents of each tube in six separate trials.

Trial	Tube A	Tube B	Current
1	0.10 M HCl	Nothing	5.94 amps
2	0.10 M HClO	Nothing	0.42 amps
3	0.10 M HCl	0.10 M KOH	6.03 amps
4	0.10 M HClO	0.10 M KOH	5.98 amps
5	0.10 M KOH	Nothing	5.89 amps
6	0.10 M KOAc	Nothing	5.92 amps

Table 1

The voltage of the battery is constant for the duration of the experiment.

8. What current would be expected, if 25.0 mL of 0.10 M HF were added to the aqueous solution?

- A. 0.02 amps
- B. 0.48 amps
- C. 5.42 amps
- D. 6.17 amps

9. How can it be explained that there is no difference between the current readings in Trial 5 and Trial 6?

- A. Both KOH and KOAc are strong bases.
- B. Both KOH and KOAc are weak bases.
- C. Neither compound dissociates into ions in water.
- D. The number of ions in solution does not depend on the base strength.

10. In Trial 2, the water is acting as:

- A. a Brønsted-Lowry base.
- B. an Arrhenius acid.
- C. a Lewis acid.
- D. an amphoteric species.

11. Which of the following is NOT true?

- A. The solution in Trial 1 has a higher boiling point than the solution in Trial 2.
- B. The solution in Trial 1 has a higher freezing point than the solution in Trial 2.
- C. The solution in Trial 1 has a higher osmotic pressure than the solution in Trial 2.
- D. The solution in Trial 1 is more electrolytic than the solution in Trial 2.

12. All of the following would show a current of roughly 6 amps when added to the solution EXCEPT:

- A. 0.10 M HNO₃.
- B. 0.10 M H₂SO₄.
- C. 0.10 M NaOH.
- D. 0.10 M NH₃.

13. What might be the reason for the similarities between the results in Trial 3 and Trial 4?

- A. A complete reaction transpires only in Trial 3.
- B. A complete reaction transpires only in Trial 4.
- C. After reaction, whether it is a strong acid with a strong base or a weak acid with a strong base, the same concentration of spectator ions remains.
- D. Only anions conduct electricity, and both solutions have the same amount of anions after reaction.

Passage III (Questions 14 - 21)

The term *oxyacid* is coined from the oxygen, the key component of this type of acid. Oxyacids contain a non-metal atom (highly electronegative) bonded to oxygens, and the acidic hydrogen is bonded to an oxygen. A typical example is nitric acid (HNO_3), where nitrogen is the electronegative atom that is bonded to the oxygens, one of which is also bonded to a hydrogen. Some oxyacids of interest are those containing halides, sulfur, phosphorus, and nitrogen.

The primary rule for predicting the strength of an oxyacid is that the greater the number of oxygens attached to the central atom, the more acidic the compound. For example, HClO_4 is more acidic than HClO_3 . The secondary rule for predicting the strength of oxyacids is that it increases as the electronegativity of the central atom increases when the number of oxygens is equal between two oxyacids. Table 1 shows the trend for the halide acids:

Acid	Formula	K_a value	$\text{p}K_a$
Perchloric	HClO_4	1.4×10^9	-8.8
Perbromic	HBrO_4	2.6×10^5	-4.6
Periodic	HIO_4	1.5×10^2	-1.8
Chloric	HClO_3	8.9×10^{-1}	0.1
Chlorous	HClO_2	1.3×10^{-2}	1.9
Hypochlorous	HOCl	5.4×10^{-8}	7.3
Hypobromous	HOBr	2.3×10^{-9}	8.6
Hypoiodous	HOI	1.7×10^{-11}	10.8

Table 1

Table 1 shows that the more dominant of the two causes of strengthened acidity is the increase in oxygens attached to the central atom of the acid. This exceeds the effect of changing the halide central atom. A rough approximation is that each additional oxygen will lower the $\text{p}K_a$ of the acid by approximately 5. Fluorine cannot expand its octet to accommodate multiple oxygens, so it is not among the elements that form halogen-based oxyacids.

14. The acidity of halide-containing oxyacids increases directly with which of the following?
- The increasing electronegativity of the halide
 - The increasing bond length of the H-O bond
 - The increasing size of the halide
 - The increasing bond angle of H-O-X
15. Which sequence accurately describes the relative strength of oxyacids?
- $\text{HIO}_4 > \text{HClO}_4 > \text{HClO}_3 > \text{HBrO}_3$
 - $\text{HClO}_4 > \text{HBrO}_3 > \text{HIO}_4 > \text{HIO}_3$
 - $\text{HIO}_4 > \text{HClO}_4 > \text{HBrO}_3 > \text{HClO}_3$
 - $\text{HClO}_4 > \text{HClO}_3 > \text{HBrO}_3 > \text{HIO}_2$

16. If HClO has a $\text{p}K_a = 7.26$, then the pH of a 0.10 M HClO solution is which of the following?
- 3.63
 - 4.13
 - 7.26
 - 8.26
17. If $\text{p}K_{a1}$ for a diprotic acid is 7.8, then the BEST choice for $\text{p}K_{a2}$ for its conjugate base, formed when the acid loses a proton, is which of the following?
- 3.9
 - 6.2
 - 7.8
 - 12.9
18. If the dissociation of an acid is exothermic, then (assuming that entropy is negligible):
- K_a should increase as the temperature increases.
 - K_a should remain constant as temperature increases.
 - K_a should decrease as the temperature increases.
 - K_a never changes with varying temperature.
19. Which of the following values MOST accurately describes the $\text{p}K_a$ for HBrO_2 , based on Table 1?
- 0.63
 - 1.44
 - 1.92
 - 2.85
20. Which of the following acids would show the GREATEST dissociation in water?
- H_3PO_4
 - HNO_2
 - HIO_2
 - HClO_4
21. What is the pH for a 125-mL sample of 0.10 M HClO_4 , given that HClO_4 is a strong acid?
- 1
 - 0.1
 - 1
 - 7

Passage IV (Questions 22 - 28)

The acidity of thiols (RSH) is observed to be greater than the acidity of alcohols (ROH). This is determined by comparing the relative pK_a values for corresponding alkyl groups attached to thiols and alcohols. Table 1 lists pK_a values for various alcohols and thiols with comparable alkyl substituents. From these values, it is possible to determine the relative acidities of the two classes of compounds.

Alcohol	pK_a	Thiol	pK_a
H_3COH	16.1	H_3CSH	10.3
H_3CCH_2OH	16.3	H_3CCH_2SH	10.6
$(H_3C)_2CHOH$	17.0	$(H_3C)_2CHSH$	11.0
$(H_3C)_3COH$	17.8	$(H_3C)_3CSH$	11.4

Table 1

The difference in acidity between alcohols and thiols is attributed to the polarizability of their respective conjugate bases. The larger the anion (more correctly, the atom carrying the negative charge in the conjugate base), the more diffuse the electrons will be, and thus the more polarizable the electron cloud of the anion. The result is that the electron cloud is spread over more area, increasing the stability of the anion (conjugate base).

As the stability of the conjugate base increases, the basicity of the conjugate base decreases. The final correlation is that as the basicity of the conjugate base decreases, the acidity of the conjugate acid increases. This leads ultimately to the conclusion that as the size of the atom to which the acidic hydrogen is attached increases, the acidity of the compound increases. This can also be correlated to the bond length of the bond between the acidic hydrogen and the atom. The longer the bond, the weaker that bond will be.

22. Which of the following compounds would be MOST acidic?

- A. H_3CSCH_3
- B. H_3COCH_3
- C. H_3CNHCH_3
- D. $H_3CCH_2CH_3$

23. From the table of pK_a values, what conclusion can be drawn about the role of alkyl groups?

- A. Alkyl groups are electron-withdrawing and decrease acidity.
- B. Alkyl groups are electron-donating and decrease acidity.
- C. Alkyl groups are electron-withdrawing and increase acidity.
- D. Alkyl groups are electron-donating and increase acidity.

24. Which of the following halogen-containing acids is the STRONGEST acid?

- A. HF
- B. HCl
- C. HBr
- D. HI

25. HCl is considerably more acidic than H_2S , because the relative acidities of the compounds formed by:

- A. atoms in the same row of the periodic table depend on the electronegativity of the non-hydrogen atom in the compound.
- B. atoms in the same column of the periodic table depend on the electronegativity of the non-hydrogen atom in the compound.
- C. atoms in the same row of the periodic table depend on the size of the non-hydrogen atom in the compound.
- D. atoms in the same column of the periodic table depend on the size of the non-hydrogen atom in the compound.

26. How does ethanol compare with its corresponding ethyl thiol?

- A. Ethanol has a greater value of K_a .
- B. Ethanol dissociates more completely in water.
- C. $H_3CCH_2O^-$ is a stronger base than $H_3CCH_2S^-$.
- D. Ethanol yields a greater $[H_3O^+]$.

27. When an atom in question is not directly attached to the acidic hydrogen, then the acidity of that compound correlates to the electronegativity of that atom, not its size. This is known as the *inductive effect*. According to the inductive effect, which of these acids is the STRONGEST?

- A. $H_3CCH_2CO_2H$
- B. $H_3CCl_2CO_2H$
- C. H_3CCSCO_2H
- D. $H_3CCl_2CO_2H$

28. Which sequence accurately describes the relative strength of these acids?

- A. $H_3CCH_2SH > (H_3C)_3CSH > H_3CCO_2H$
- B. $(H_3C)_3CSH > H_3CCH_2SH > H_3CCO_2H$
- C. $H_3CCO_2H > H_3CCH_2SH > (H_3C)_3CSH$
- D. $H_3CCO_2H > (H_3C)_3CSH > H_3CCH_2SH$

Passage V (Questions 29 - 34)

Common organic acids include carboxylic acids (RCO_2H) and phenols ($\text{C}_6\text{H}_5\text{OH}$). Electron-withdrawing groups along the backbone of an acid increase its acidity, and thus lower its pK_a . Chloro, fluoro, and nitro groups are among the common electron-withdrawing groups. Table 1 lists some organic acids with their respective pK_a values. They are all monoprotic organic acids.

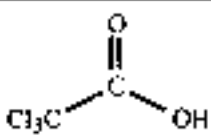
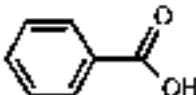
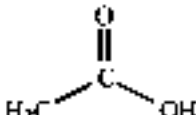
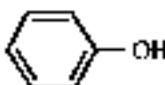
Structure	Formula	pK_a
	$\text{Cl}_3\text{CCO}_2\text{H}$	0.64
	$\text{p-O}_2\text{NC}_6\text{H}_4\text{CO}_2\text{H}$	3.40
	$\text{C}_6\text{H}_5\text{CO}_2\text{H}$	4.21
	$\text{H}_3\text{CCO}_2\text{H}$	4.74
	$\text{p-O}_2\text{NC}_6\text{H}_4\text{OH}$	7.18
	$\text{C}_6\text{H}_5\text{OH}$	10.01

Table 1

When a weak acid dissolves into water, it partially dissociates according to its K_a . Equation 1 can be applied to calculate the pH of an aqueous solution of a weak acid.

$$\text{pH} = \frac{1}{2} \text{pK}_a - \frac{1}{2} \log [\text{HA}]$$

Equation 1

Equation 1 is derived from the equilibrium expression for the dissociation of a weak acid in water. It applies only if the acid concentration exceeds K_a by 100 fold and if the pK_a of the weak acid is between 2 and 12.

$$\log 2 = 0.3 \quad \log 3 = 0.48$$

29. What is the pH for an acid solution with a concentration of 0.50 M and a $K_a = 8 \times 10^{-8}$?

A. $0 < \text{pH} < 3.0$
 B. $3.0 \leq \text{pH} \leq 3.5$
 C. $3.5 < \text{pH} \leq 7.0$
 D. $\text{pH} > 7.0$

30. What is the mass percent of carbon in phenol ($\text{C}_6\text{H}_5\text{OH}$)?

A. 66.7%
 B. 72.0%
 C. 76.6%
 D. 82.4%

31. Which of the following acids would yield the lowest pH value once completely neutralized by strong base?

A. Acetic acid ($\text{H}_3\text{CCO}_2\text{H}$)
 B. Trichloroacetic acid ($\text{Cl}_3\text{CCO}_2\text{H}$)
 C. p-Nitro benzoic acid ($\text{O}_2\text{NC}_6\text{H}_4\text{CO}_2\text{H}$)
 D. Benzoic acid ($\text{C}_6\text{H}_5\text{CO}_2\text{H}$)

32. When comparing equal molar aqueous solutions of acids from Table 1, what is TRUE?

A. 1.0 M acetic acid has a higher boiling point than 1.0 M p-nitro benzoic acid.
 B. 1.0 M trichloroacetic acid has a higher boiling point than 1.0 M p-nitrophenol ($\text{p-O}_2\text{NC}_6\text{H}_4\text{OH}$).
 C. 1.0 M trichloroacetic acid has a higher freezing point than 1.0 M benzoic acid.
 D. 1.0 M acetic acid has a higher freezing point than 1.0 M phenol.

33. Which of the following acids has the STRONGEST conjugate base?

A. Phenol ($\text{C}_6\text{H}_5\text{OH}$)
 B. p-Nitrophenol ($\text{p-O}_2\text{NC}_6\text{H}_4\text{OH}$)
 C. Acetic acid ($\text{H}_3\text{CCO}_2\text{H}$)
 D. Trichloroacetic acid ($\text{Cl}_3\text{CCO}_2\text{H}$)

34. After 0.10 moles of an unknown weak acid HA are dissolved into 100 mL of H_2O , the solution's pH is 3.7. What is the concentration of the conjugate base (A^-)?

A. $[\text{A}^-] > 2 \times 10^{-4} \text{ M}$
 B. $[\text{A}^-] = 2 \times 10^{-4} \text{ M}$
 C. $[\text{A}^-] < 2 \times 10^{-4} \text{ M}$
 D. The K_a of the weak acid must be known to determine the value of $[\text{A}^-]$.

Passage VI (Questions 35 - 41)

The strength of an organic acid varies directly with the strength of any electron-withdrawing groups attached to its organic backbone. For instance, the acidity of acetic acid can be increased by adding an electron-withdrawing group to the methyl carbon. The acidity of acetic acid can be decreased by adding an electron donating group to the methyl carbon. The effect on the strength of the acid varies with the location and electronegativity of the electron-withdrawing group and is most substantial when the functional group withdraws by way of resonance. The following set of phenols demonstrates the effect of these substituents on acidity:

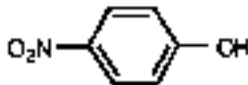
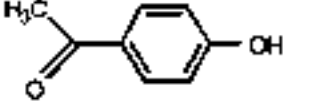
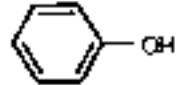
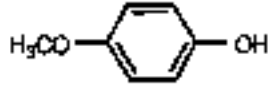
Organic Acid	pK_a
	7.2
	8.4
	10.0
	10.4
	11.2

Figure 1

Increased acidity can be attributed to the withdrawal of electron density (through either the inductive effect or resonance) from the O-H bond of phenol. The loss of electron density weakens the bond, causing it to break in a heterolytic fashion more easily. This makes the compound more acidic. Resonance withdrawal is most pronounced when the substituent is in the ortho or para position, so the strongest acids are those with electron-withdrawing groups attached to the phenyl ring at the ortho or para positions. Resonance is substantially weaker from the meta position. The inductive effect is weaker than resonance from any position.

35. The LOWEST pH value would be associated with:
- 0.10 M $O_2NC_6H_4OH$ (p-nitrophenol).
 - 0.50 M $O_2NC_6H_4OH$ (p-nitrophenol).
 - 0.10 M C_6H_5OH (phenol).
 - 0.50 M C_6H_5OH (phenol).

36. A 100 mL 1.0 M phenol solution has a pH of:
- 10.0.
 - 7.0.
 - 5.0.
 - 1.0.

37. According to the pK_a data in Figure 1, which compound is the STRONGEST electron-withdrawing group?
- O_2N-
 - H_3CO-
 - H_3C-
 - H_3CCO-

38. The conjugate base of p-nitrophenol has a pK_b of:
- 10.4.
 - 7.2.
 - 6.8.
 - 3.6.

39. How many grams of methoxy phenol must be added to 100 mL of pure water to reach a pH of 6.1?
- 12.4 grams
 - 2.48 grams
 - 1.24 grams
 - 0.62 grams

40. What is the K_a associated with p-methyl phenol?
- 4.0×10^{-10}
 - 2.5×10^{-10}
 - 4.0×10^{-11}
 - 2.5×10^{-11}

41. What is the pH of a solution made by mixing 20 mL 0.15 M phenol with 10 mL water?
- 10.0
 - 7.0
 - 5.8
 - 5.5

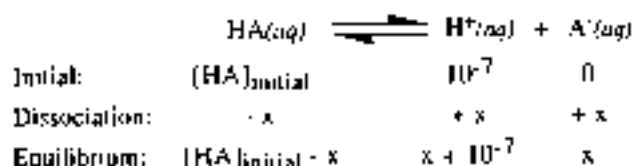
Passage VII (Questions 42–48)

A strong acid fully dissociates when dissolved in water, and the pH of the resulting solution can be found by taking the negative log of the strong acid concentration:

$$\text{pH} = -\log [\text{strong acid}]$$

Equation 1

For a weak acid, the determination of the pH is not as simple as it is with a strong acid, because weak acids do not dissociate completely when added to water. To find the pH of a weak acid in water, the $[\text{H}_3\text{O}^+]$ must first be determined from the weak acid equilibrium:



10^{-7} is assumed to be negligible compared to x , and x is assumed to be negligible compared to $(\text{HA})_{\text{initial}}$.

The $[\text{H}_3\text{O}^+]$ can be determined from the K_a value. The following is a derivation of Equation 2, used to determine the pH for an aqueous solution of a weak acid.

$$K_a = \frac{[\text{H}_3\text{O}^+][\text{A}^-]}{[\text{HA}]} = \frac{[\text{H}_3\text{O}^+]^2}{[\text{HA}]} \quad \therefore K_a \times [\text{HA}] = [\text{H}_3\text{O}^+]^2$$

$$[\text{H}_3\text{O}^+] = \sqrt{K_a \times [\text{HA}]}$$

$$\text{pH} = -\log [\text{H}_3\text{O}^+] = -\log \sqrt{K_a \times [\text{HA}]}$$

$$-\log \sqrt{K_a \times [\text{HA}]} = -\log \sqrt{K_a} - \log \sqrt{[\text{HA}]}$$

$$= -\frac{1}{2} \log K_a - \frac{1}{2} \log ([\text{HA}]) = \frac{1}{2} \text{p}K_a - \frac{1}{2} \log ([\text{HA}])$$

$$\therefore \text{pH} = \frac{1}{2} \text{p}K_a - \frac{1}{2} \log ([\text{HA}])$$

Equation 2

Equation 2 works only when the value of K_a is less than $[\text{HA}]$, and when $\text{p}K_a$ is between 2 and 12.

42. For which of the following acid solutions does Equation 2 NOT work?
- A. HClO
 B. H_2CO_3
 C. HI
 D. $\text{H}_3\text{C}_6\text{CO}_2\text{H}$
43. What assumption is inherent in the derivation of Equation 2?
- A. $[\text{HA}] < [\text{H}_3\text{O}^+]$
 B. $[\text{H}_3\text{O}^+] = [\text{A}^-]$
 C. $[\text{HA}] = [\text{A}^-]$
 D. $[\text{HA}] < [\text{A}^-]$

44. If the pH of a 0.10 M aqueous weak acid solution is 2.70, what is the concentration of the dissociated conjugate base in solution?
- A. 0.10 M
 B. 2.0×10^{-2} M
 C. 2.0×10^{-3} M
 D. 2.0×10^{-4} M
45. Which of the following equations can be used to determine the pOH of an aqueous weak base solution?
- A. $\text{pOH} = -\log [\text{OH}^-]$
 B. $\text{pOH} = \frac{1}{2} \text{p}K_b - \frac{1}{2} \log ([\text{A}^-])$
 C. $\text{pOH} = \frac{1}{2} \text{p}K_b - \frac{1}{2} \log ([\text{A}^-])$
 D. $\text{pOH} = 14 - \left(\frac{1}{2} \text{p}K_a - \frac{1}{2} \log ([\text{HA}]) \right)$
46. If a weak acid is titrated with enough strong base so that $[\text{A}^-] > [\text{HA}]$, then for the resulting compound:
- A. $[\text{H}_3\text{O}^+] < K_a$
 B. $\text{pH} < \text{p}K_a$
 C. $[\text{H}_3\text{O}^+] > [\text{A}^-]$
 D. $[\text{HA}] < [\text{H}_3\text{O}^+]$
47. Which acid yields the GREATEST conjugate base concentration when 0.010 moles HA are added to enough water to make the final volume 100 mL?
- A. HF
 B. HNO_3
 C. HCN
 D. $\text{H}_3\text{CCO}_2\text{H}$
48. A series of 0.10 M weak acid solutions, each containing a different acid are compared. What is true for the acid in the series that has the GREATEST K_a value?
- A. Its aqueous solution has the lowest conjugate base concentration.
 B. Its aqueous solution has the lowest pOH.
 C. Its aqueous solution has the highest $[\text{H}_3\text{O}^+]$.
 D. Its aqueous solution has the highest ratio of HA to A^- .

Passage VIII (Questions 49 - 55)

When the pH of the stomach drops to a value less than 2.0, the excess acidity may be reduced by consuming an antacid. The role of the antacid is to neutralize the HCl in the gastric fluid. Some common bases found in antacids are CaCO_3 , MgCO_3 , $\text{Mg}(\text{OH})_2$, and NaHCO_3 . Calcium carbonate, magnesium carbonate, and magnesium hydroxide neutralize acid in a two-to-one ratio. Sodium bicarbonate neutralizes only in a one-to-one ratio.

The gas buildup experienced after consuming an antacid is due to carbon dioxide gas, which forms in the stomach when carbonic acid (formed upon the full protonation of carbonate) decomposes into water and carbon dioxide. The principal antacid ingredients of some common, commercially available products are listed in Table 1.

Alka-Seltzer	Aspirin, NaHCO_3 , citric acid
Bufferin	Aspirin, MgCO_3 , aluminum glycinate
Milk of Magnesia	$\text{Mg}(\text{OH})_2$
Rolaids	Dihydroxyaluminum sodium carbonate
Tums	Calcium carbonate

Table 1

Aspirin (acetylsalicylic acid) can irritate the stomach lining. In an acidic environment, aspirin does not deprotonate, so the molecule remains neutral and thus lipid soluble. As a result, aspirin is able to work its way into the membrane of the stomach lining and then into hydrophilic pores within the membrane. It deprotonates inside the hydrophilic pore, causing ion levels to build up in the pocket, until osmotic pressure forces water into the interior region of the membrane, where parietal cells swell and rupture. The net effect is to eat away the stomach lining. Figure 1 shows the structure of aspirin and its ionization equilibrium.

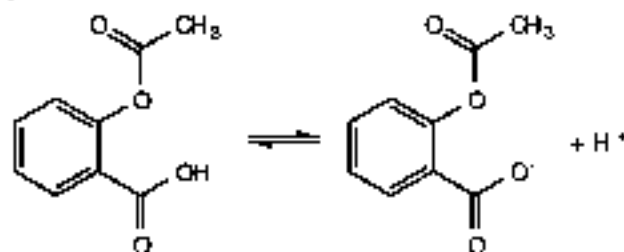


Figure 1

49. At what pH is aspirin MOST soluble in water?

- A. 1.5
- B. 3.4
- C. 7.4
- D. 9.2

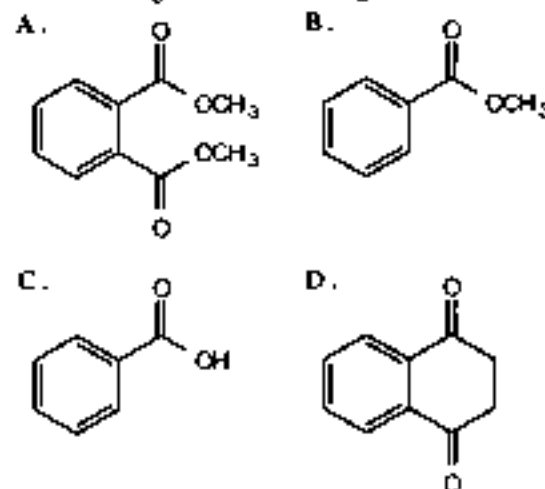
50. How many acidic sites are there in acetylsalicylic acid?

- A. 1
- B. 2
- C. 3
- D. 4

51. Which antacid gives the MOST neutralizing strength per gram?

- A. Magnesium carbonate (84.3 grams per mole)
- B. Calcium carbonate (100 grams per mole)
- C. Magnesium hydroxide (58.3 grams per mole)
- D. Sodium bicarbonate (83.9 grams per mole)

52. Which of the following compounds would cause the MOST damage to stomach lining?



53. How many acidic protons can be neutralized per molecule of dihydroxyaluminum sodium carbonate ($\text{Al}(\text{OH})_2\text{NaCO}_3$)?

- A. 1
- B. 2
- C. 3
- D. 4

54. The K_{eq} for the ionization reaction of acetylsalicylic acid in water is:

- A. much greater than 1.00.
- B. barely greater than 1.00.
- C. barely less than 1.00.
- D. much less than 1.00.

55. Which of the following mixtures results in a buffer solution?

- A. 1 equiv. acetylsalicylic acid + 1 equiv. HCl
- B. 2 equiv. acetylsalicylic acid + 1 equiv. HCl
- C. 1 equiv. acetylsalicylate + 1 equiv. HCl
- D. 2 equiv. acetylsalicylate + 1 equiv. HCl

Passage IX (Questions 56 - 62)

Many common products used every day have acidic and basic properties. An example is the antacid taken by many heartburn sufferers to neutralize excess stomach acid (HCl). Antacids often contain hydroxide anion, carbonate anion, or both in their salt forms. The contents of a normally functioning stomach can reach a pH as low as 1.7 (highly acidic) and in extreme cases can reach a pH as low as 1.0. Because the pH scale is a log scale, a decrease of 0.7 pH units represents a hydronium ion (H_3O^+) concentration that is 500% greater. Equation 1 is used to determine the pH of a solution.

$$\text{pH} = -\log [\text{H}_3\text{O}^+]$$

Equation 1

Table 1 lists some common household acids.

Acids	Formula	Product
Acetic acid	$\text{H}_3\text{CCO}_2\text{H}$	Vinegar
Ascorbic acid	$\text{C}_6\text{H}_8\text{O}_6$	Vitamin C
Hypochlorous acid	HClO	Bleach
Acetyl salicylate	$\text{H}_3\text{C}_2\text{OC}_6\text{H}_4\text{CO}_2\text{H}$	Aspirin
Sulfuric acid	H_2SO_4	Battery acid

Table 1

Table 2 lists some common household bases.

Bases	Formula	Product
Ammonia	NH_3	Windex
Sodium Bicarbonate	NaHCO_3	Baking Soda
Sodium Hydroxide	NaOH	Drano
Sodium Lauryl sulfate	$\text{H}_3\text{C}(\text{CH}_2)_{16}\text{SO}_4\text{Na}$	Shampoo

Table 2

Bases when added to neutral water raise the pH of the solution to a value greater than 7.0. Acids when added to neutral water lower the pH of the solution to a value less than 7.0. A pH of 7.0 is considered to be neutral, because pure (distilled) water has an $[\text{H}_3\text{O}^+] = 1.0 \times 10^{-7}$.

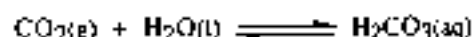
Acid in solution may be represented as either H_3O^+ or H^+ , depending on the solvent. Both representations are equivalent ways of describing an acid.

56. A hydroxide anion is formed when water loses a proton (H^+) to another water (known as autoionization). If the concentration of dissociated hydroxide anion (OH^-) in distilled water is $1.0 \times 10^{-6} \text{ M}$, the pH of the solution must be:
- 7.0, because $[\text{H}_3\text{O}^+] = [\text{OH}^-]$.
 - 7.0, because water always has a pH of 7.0.
 - 6.0, because $[\text{H}_3\text{O}^+] = [\text{OH}^-]$.
 - 8.0, because the solution is rich in base (OH^-).

57. What is the hydronium ion concentration ($[\text{H}_3\text{O}^+]$) of an aqueous solution with a pH value of 1.7?
- $1.7 \text{ M H}_3\text{O}^+$
 - $2.0 \times 10^{-2} \text{ M H}_3\text{O}^+$
 - $1.0 \times 10^{-3} \text{ M H}_3\text{O}^+$
 - $5.0 \times 10^{-13} \text{ M H}_3\text{O}^+$
58. If 10 mL of an aqueous solution of a strong acid with $\text{pH} = 2.0$ were mixed with 100 mL of pure water, then the final pH value would be:
- less than 2.0.
 - between 2.0 and 3.0.
 - exactly 3.0.
 - greater than 3.0.
59. How many grams of CaCO_3 are needed to neutralize 50 mL of stomach acid at $\text{pH} = 2.0$ completely, if the following equation represents the neutralization reaction?
- $$\text{CaCO}_3(\text{aq}) + 2 \text{H}^+(\text{aq}) \rightarrow \text{Ca}^{2+}(\text{aq}) + \text{CO}_2(\text{g}) + \text{H}_2\text{O}(\text{l})$$
- 25 mg
 - 50 mg
 - 100 mg
 - 1.0 grams
60. Which of the following household products would NOT undergo an acid-base reaction with Windex?
- Vinegar
 - Aspirin
 - Drano
 - Bleach
61. Which of the following when added to an aqueous solution at $\text{pH} = 6.0$ would NOT raise the pH of the solution?
- Distilled water
 - Shampoo
 - Antacid
 - Vitamin C
62. How many milliliters of $0.20 \text{ M H}_3\text{O}^+$ are required to neutralize 1.68 grams of baking soda completely?
- $$\text{NaHCO}_3(\text{aq}) + \text{H}^+(\text{aq}) \rightarrow \text{Na}^+(\text{aq}) + \text{CO}_2(\text{g}) + \text{H}_2\text{O}(\text{l})$$
- 10 mL
 - 25 mL
 - 50 mL
 - 100 mL

The average human being produces between two and three liters of gastric fluid in the normal day. Gastric fluid is highly acidic (due to the presence of hydrochloric acid). It is secreted by the mucous membrane of the stomach lining to aid in the digestion of food. The average pH of this fluid is around 1.5. The acid concentration necessary for this pH is 0.030 M, if the acid is a strong acid. The acid in gastric fluid is strong enough and concentrated enough to dissolve (oxidize) metals with a positive oxidation potential.

The lining of the stomach includes parietal cells that are tightly fused to form junctions in the stomach wall. These cells have a cell membrane that is permeable to neutral molecules (such as water), but not to ions (such as Na^+ and Cl^-). Hydronium ion, responsible for the acidity of the stomach, is a byproduct of metabolism. Carbon dioxide, a byproduct of metabolism, is a non-metal oxide that converts to an oxyacid when combined with water. The hydration of carbon dioxide (CO_2) to carbonic acid (H_2CO_3) is shown in Reaction 1. Carbonic acid is a weak acid that partially dissociates into bicarbonate anion and a proton, as shown in Reaction 2.



Reaction 1



Reaction 2

The proton, along with a chloride anion, is carried across the membrane into the stomach via active transport. Enzymes assist the migration of both the proton and chloride anion from the blood plasma into the interior of the stomach. These ions remain in the stomach until removed, because of the impermeability of the cell membrane to ions.

Eating stimulates the production of the enzyme responsible for active transport and thus the release of acid into the stomach to hydrolyze food molecules. Some protons are absorbed by the mucous lining of the stomach wall, resulting in small localized hemorrhages. About 30 million cells are destroyed per hour in normal stomach activity. As a result, the stomach's entire lining is regenerated roughly once every 72 hours. Excess acid production increases this hemorrhaging and, in the worst case scenario, an ulcer develops.

64. Which solution has the LOWEST pH?

- A. Carbonated water
- B. Distilled water
- C. Salt water
- D. Lime water ($\text{CaO}(\text{aq})$)

65. Which of the following metals can be oxidized by gastric fluid?

- A. Copper
- B. Gold
- C. Silver
- D. Zinc

66. Which of the following reactions is NOT catalyzed by acid?

- A. Hydrolysis of disaccharides
- B. Hydrolysis of polypeptides
- C. Hydrolysis of esters
- D. Hydrolysis of alkanes

67. Which will NOT reduce the hydronium ion (H_3O^+) concentration in the stomach?

- A. The consumption of water
- B. The consumption of baking soda (NaHCO_3)
- C. The consumption of aluminum metal
- D. The consumption of solid food

68. What is the pH of 0.030 M $\text{HCl}(\text{aq})$ solution?

- A. 0.7
- B. 1.5
- C. 3.0
- D. 7.0

69. Which of the following beverages when consumed does NOT promote the decay of the stomach lining?

- A. Orange juice (citric acid)
- B. Lemonade (citric acid)
- C. Milk (lactose)
- D. Coca-Cola (phosphoric acid)

70. Which of the following has the LOWEST pH?

- A. 0.03 M $\text{HCl}(\text{aq})$
- B. 0.01 M $\text{HCl}(\text{aq})$
- C. 0.03 M $\text{H}_2\text{CO}_3(\text{aq})$
- D. 0.01 M $\text{H}_2\text{CO}_3(\text{aq})$

Passage XI (Questions 71 - 76)

As a general rule, the solubility for basic salts (conjugate bases of weak acids) increases as the pH of the solution decreases. This is to say that basic salts are more soluble in acidic solution than in neutral solution. The calcium salts such as calcium hydroxide (found in cement), calcium carbonate (marble), and hydroxyapatite (tooth enamel) are perfect examples. Hydroxyapatite is $\text{Ca}_5(\text{PO}_4)_3\text{OH}$ but can also be written as $3\text{Ca}_3(\text{PO}_4)_2 \cdot \text{Ca}(\text{OH})_2$. Cavities result when the hydroxyapatite dissolves away. When small crevices are formed in teeth, acid-producing bacteria build up and further dissolve away the worn region in the enamel. The acid that dissolves the enamel is formed when aldoses (aldehyde sugars, such as glucose) are oxidized to their respective aldonic acid compounds. To reduce the dissolving of the enamel, toothpaste and mouthwash containing fluoride are recommended. The fluoride will substitute for hydroxide in the salt to form the less soluble fluorapatite $\text{Ca}_5(\text{PO}_4)_3\text{F}$ (also known as $3\text{Ca}_3(\text{PO}_4)_2 \cdot \text{CaF}_2$). This lengthens the lifetime of tooth enamel.

The solubility product (K_{sp}) for calcium fluoride (CaF_2) is $3.4 \times 10^{-11} \text{ M}^3$, for pure calcium phosphate ($\text{Ca}_3(\text{PO}_4)_2$) is $1.4 \times 10^{-26} \text{ M}^5$, and for calcium hydroxide ($\text{Ca}(\text{OH})_2$) free of calcium oxide is $2.3 \times 10^{-8} \text{ M}^3$. The lower solubility of calcium fluoride compared to calcium hydroxide is mimicked in the reduced solubility of fluorapatite compared to hydroxyapatite.

The following chart lists the molar solubility of calcium hydroxide at varying pH along with the gram solubility per 100 mL solution at varying pH. The temperature was held constant at 25°C for all values:

pH	Molar solubility	Gram solubility
1	2.3×10^{18}	1.7×10^{19}
2	2.3×10^{16}	1.7×10^{17}
3	2.3×10^{14}	1.7×10^{15}
7	2.3×10^6	1.7×10^7
11	2.3×10^{-2}	1.7×10^{-1}
12	2.3×10^{-4}	1.7×10^{-3}

Table 1

Table 1 shows that the effect of pH on the solubility of basic salts is drastic. This is because pH is measured on a log scale, while solubilities are measured on a linear scale.

71. As sodium fluoride (NaF) is added to an aqueous solution with undissolved hydroxyapatite in it, what happens to the solution?
- The pH of the solution increases.
 - The pH of the solution decreases.
 - The pH of the solution remains constant at a value greater than 7.0.
 - The pH of the solution remains constant at a value less than 7.0.

72. Calcium hydroxide is BEST described as which of the following?

- A strong acid
- A strong base
- A weak acid
- A weak base

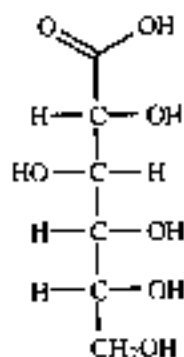
73. If the solubility of enamel in a 1.00 M solution of acid HA is greater than the solubility of enamel in a 1.00 M solution of acid HB, which of the following conclusions can be drawn concerning the two acids?

- HA when dissolved into 100 mL pure water has a higher pH than HB.
- The conjugate base of HA is a stronger base than the conjugate base of HB.
- HA has a higher K_a value than HB.
- HA has a higher pK_a value than HB.

74. Which of the following changes will ALWAYS increase the amount of hydroxyapatite that dissolves into solution?

- Adding a strong acid to the solution
- Adding a strong base to the solution
- Increasing the pH of the solution
- Lowering the amount of water in the solution by evaporation

75. An example of an aldonic acid is drawn below. With which of the answer selections listed should the aldonic acid share a similar pK_a value?



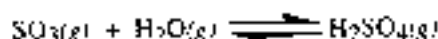
- Acetic acid ($\text{H}_3\text{CCO}_2\text{H}$)
- Hydrochloric acid (HCl)
- Nitric acid (HNO_3)
- Sodium carbonate (Na_2CO_3)

76. In which of the following solutions does tooth enamel dissolve the FASTEST?

- 100 mL 0.10 M HCl
- 100 mL 0.0010 M HCl
- 100 mL 0.10 M HF ($K_a = 6.8 \times 10^{-4}$)
- 100 mL 0.0010 M HF ($K_a = 6.8 \times 10^{-4}$)

Normal rainfall has a pH between 5.6 and 6.0, depending on the amount of carbon dioxide in the air. Excessive amounts of atmospheric carbon dioxide lower the pH of rain. However, even with high carbon dioxide concentration, the pH of rain does not drop much below 5.4. While rain rich in carbon dioxide is more acidic than normal rain, it is not considered to be *acid rain*.

Acid rain is attributed to nonmetal oxides present in the air, like sulfur trioxide, and the nitrogen oxides. These nonmetal oxides combine with atmospheric moisture to form nonmetal hydroxides, known as *oxyacids*. Reaction 1 shows the hydration of sulfur trioxide, converting it from a nonmetal oxide into a nonmetal hydroxide.



Reaction 1

Sulfur oxides are common by-products produced during the combustion of coal. A *scrubber* (a chamber at the base of an exhaust stack filled with high pressure steam and calcium oxide) can help neutralize the non-metal oxides in the industrial emissions of coal-burning factories. As the exhaust passes through the chamber, steam hydrates the nonmetal oxide and converts it from an airborne Lewis acid into an aqueous Brønsted-Lowry acid. Then as this nonmetal hydroxide solution flows across the calcium oxide in the scrubber, it is converted into calcium sulfate and water. This is shown in Reaction 2.



Reaction 2

If the acidic industrial waste generated by burning fossil fuel is not treated in this manner, it can produce rain with a pH as low as 4.0. Acid rain is known to damage bodies of water by changing the bacteria population. Acid rain damages plants by changing soil pH. Changes in pH affect the structure of plant enzymes needed for the uptake and transport of nutrients. The enzymes become ineffective at lower pH because of their denatured structures.

77. What chemical reaction is indicated in Reaction 1?

- A. The neutralization of sulfur trioxide
- B. The acidification of sulfur trioxide
- C. The conversion of a Brønsted-Lowry acid into a Lewis acid.
- D. The conversion of a Lewis acid into a Brønsted-Lowry acid.

78. A nonmetal oxide can be characterized as:

- A. amphoteric
- B. an Arrhenius base
- C. a Brønsted-Lowry acid.
- D. a Lewis acid.

79. Which of the following compounds would create a solution with the LOWEST pH when added to water?

- A. SO_2
- B. CO_2
- C. N_2O_5
- D. MgO

80. What is the pH of 0.10 M $\text{HOCl}(aq)$, which has a pK_a of 7.46?

- A. 1.00
- B. 4.23
- C. 6.46
- D. 7.46

81. Which of the following could be a component of acid rain?

- A. Na_2O
- B. N_2
- C. NO_2
- D. H_2

82. All of the following qualifications affect the acidity of an oxyacid EXCEPT:

- A. the oxidation state of the central atom
- B. the electronegativity of the central atom
- C. the size of the central atom
- D. the number of π -bonds to the central atom

83. All of these statements about H_2SO_4 , H_2SO_3 , and their constituents are true EXCEPT:

- A. SO_3^{2-} has a lower pK_{a1} than SO_4^{2-} .
- B. 1.0 M $\text{H}_2\text{SO}_4(aq)$ has a lower pH than 1.0 M $\text{H}_2\text{SO}_3(aq)$.
- C. H_2SO_4 has a higher K_{a1} than H_2SO_3 .
- D. H_2SO_3 is more electrolytic than H_2SO_4 .

84. Which sequence correctly lists the four acids shown below according to their INCREASING pK_a values?

- A. $\text{pK}_{a1} \text{H}_2\text{SO}_4 < \text{pK}_a \text{HNO}_3 < \text{pK}_{a1} \text{H}_2\text{SO}_3 < \text{pK}_a \text{HNO}_2$
- B. $\text{pK}_a \text{HNO}_3 < \text{pK}_{a1} \text{H}_2\text{SO}_3 < \text{pK}_a \text{HNO}_2 < \text{pK}_{a1} \text{H}_2\text{SO}_4$
- C. $\text{pK}_{a1} \text{H}_2\text{SO}_3 < \text{pK}_a \text{HNO}_2 < \text{pK}_{a1} \text{H}_2\text{SO}_4 < \text{pK}_a \text{HNO}_3$
- D. $\text{pK}_a \text{HNO}_2 < \text{pK}_{a1} \text{H}_2\text{SO}_3 < \text{pK}_a \text{HNO}_3 < \text{pK}_{a1} \text{H}_2\text{SO}_4$

Passage XIII (Questions 85 - 91)

Amino acids can be classified as either diprotic or triprotic acids, depending on their side chain. A generic amino acid is shown below in Figure 1:

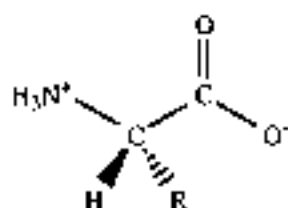


Figure 1

The chemical structure in Figure 1 represents an amino acid in a pH = 7 aqueous environment. At a pH less than the pK_{a1} of the carboxyl terminal (pK_{a1}), the carboxylate group is protonated and exists in its neutral carboxylic acid form. Similarly, at a pH greater than the pK_{a2} of the amino terminal (pK_{a2} or pK_{a3}), the ammonium group is deprotonated and exists in its neutral amine form.

If the R group (side chain) of an amino acid exhibits acid-base properties, then the amino acid is triprotic. The side chain may be deprotonated at pH = 7, depending on the nature of the functional group. The carboxyl terminal pK_a lies between 1.8 and 2.6, and the amino terminal pK_a lies between 8.8 and 10.6 for all twenty common amino acids. The pK_a for the protonated form and the pK_b for the deprotonated form add up to a value of 14 at 25°C.

$$pK_a (\text{protonated form}) + pK_b (\text{deprotonated form}) = 14$$

Equation 1

The pK_a values for common side chains range from 3.9 to 13.2. Table 1 lists the R group and pK_a values for seven common triprotic amino acids.

Amino acid	Side chain group	pK_a
Aspartic acid	$-\text{CH}_2\text{CO}_2\text{H}$	3.88
Glutamic acid	$-\text{CH}_2\text{CH}_2\text{CO}_2\text{H}$	4.32
Histidine	$-\text{CH}_2\text{C}(\text{NH})=\text{CH}-\text{NH}_2$	6.05
Cysteine	$-\text{CH}_2\text{SH}$	8.36
Tyrosine	$-\text{CH}_2\text{C}_6\text{H}_4\text{OH}$	10.07
Lysine	$-\text{CH}_2\text{CH}_2\text{CH}_2\text{CH}_2\text{NH}_3^+$	10.80
Arginine	$-(\text{CH}_2)_3\text{NHC}(\text{NH}_2)=\text{NH}_2^+$	13.21

Table 1

Physiological pH is considered to be 7.4, although gastric fluids and the fluid contained in lysosomes have considerably lower pH. At physiological pH, a diprotic amino acid exists predominantly in its zwitterion form.

85. What is the relationship between the pK_a and pK_b for lysine?

- A. $pK_{a1} + pK_{b1} = 14$
- B. $pK_{a1} + pK_{b2} = 14$
- C. $pK_{a2} + pK_{b2} = 14$
- D. $pK_{a2} + pK_{b3} = 14$

86. At physiological pH, which of the following amino acids exists predominantly as a cation?

- A. Arginine
- B. Glutamic acid
- C. Glycine
- D. Histidine

87. What explains the lower pK_a for the side chain associated with cysteine than the one associated with serine ($R = -\text{CH}_2\text{OH}$)?

- A. Oxygen is less electronegative than sulfur
- B. Sulfur is less electronegative than oxygen
- C. Oxygen has a greater atomic radius than sulfur
- D. Sulfur has a greater atomic radius than oxygen

88. The carbon chain associated with aspartic acid is shorter than the one associated with glutamic acid, which results in a

- A. stronger inductive effect, making aspartic acid's side chain more acidic
- B. weaker inductive effect, making aspartic acid's side chain more acidic
- C. stronger inductive effect, making aspartic acid's side chain less acidic
- D. weaker inductive effect, making aspartic acid's side chain less acidic

89. At pH = 7, what are the applicable values for histidine?

- A. $pK_{a1} = 1.81$; $pK_{a2} = 6.05$; $pK_{b3} = 4.85$
- B. $pK_{a3} = 9.15$; $pK_{b2} = 7.95$; $pK_{b3} = 4.85$
- C. $pK_{a2} = 7.95$; $pK_{a3} = 6.05$; $pK_{b3} = 9.15$
- D. $pK_{a3} = 9.15$; $pK_{b2} = 7.95$; $pK_{b3} = 12.19$

90. What is the normality of 0.50 M glutamic acid in water?

- A. 0.50 N $\text{H}_3\text{N}^+\text{CH}(\text{CH}_2\text{CH}_2\text{CO}_2\text{H})\text{CO}_2\text{H}(\text{aq})$
- B. 1.00 N $\text{H}_3\text{N}^+\text{CH}(\text{CH}_2\text{CH}_2\text{CO}_2\text{H})\text{CO}_2\text{H}(\text{aq})$
- C. 1.50 N $\text{H}_3\text{N}^+\text{CH}(\text{CH}_2\text{CH}_2\text{CO}_2\text{H})\text{CO}_2\text{H}(\text{aq})$
- D. 2.00 N $\text{H}_3\text{N}^+\text{CH}(\text{CH}_2\text{CH}_2\text{CO}_2\text{H})\text{CO}_2\text{H}(\text{aq})$

91. All of the following amino acids have a neutral side chain at pH = 7 EXCEPT:

- A. cysteine
- B. histidine
- C. lysine
- D. tyrosine

Questions 92 - 100 are NOT based on a descriptive passage.

92. Which of the following solutions has the LOWEST pH?

- A. 0.01 M HClO ($\text{pK}_a = 7.5$)
- B. 0.05 M HCO_2H ($\text{pK}_a = 3.6$)
- C. 0.10 M HF ($\text{pK}_a = 3.3$)
- D. 1.00 M HCN ($\text{pK}_a = 9.1$)

93. Acid rain can possibly contain all of the following compounds EXCEPT:

- A. oxidized non-metals.
- B. hydrated non-metal oxides.
- C. electron pair acceptors.
- D. metal hydroxides.

94. Which of the following conjugate pair-mixtures has the LOWEST pH?

- A. 10 mL 0.10 M HF(aq) with 15 mL 0.10 M KF(aq)
- B. 15 mL 0.10 M HF(aq) with 10 mL 0.10 M KF(aq)
- C. 10 mL 0.10 M $\text{NH}_3(\text{aq})$ with 15 mL 0.10 M $\text{NH}_4^+(\text{aq})$
- D. 15 mL 0.10 M $\text{NH}_3(\text{aq})$ with 10 mL 0.10 M $\text{NH}_4^+(\text{aq})$

95. Which of the following relationships applies to phosphoric acid?

- A. $\text{pK}_{a1} + \text{pK}_{b1} = 14$
- B. $\text{pK}_{a2} + \text{pK}_{b2} = 14$
- C. $\text{pK}_{a3} + \text{pK}_{b3} = 14$
- D. $\text{pK}_{a1} + \text{pK}_{a3} = 14$

96. What is the normality of a solution made by mixing 50.00 mL of 1.5 M H_3PO_4 with 50.00 mL of pure water?

- A. 3.00 N H_3PO_4
- B. 2.25 N H_3PO_4
- C. 1.50 N H_3PO_4
- D. 0.75 N H_3PO_4

97. A 0.10 M solution of unknown material has a pH of 4.62. It can BEST be described as a:

- A. strong acid.
- B. weak acid.
- C. strong base.
- D. weak base.

98. The HIGHEST pH is observed in which of the following solutions?

- A. 0.10 M $\text{HNO}_2(\text{aq})$
- B. 0.10 M $\text{HNO}_3(\text{aq})$
- C. 0.10 M $\text{NaNO}_2(\text{aq})$
- D. 0.10 M $\text{NaNO}_3(\text{aq})$

99. What is the pH of 0.050 M $\text{H}_3\text{CCH}_2\text{CO}_2\text{H}$? (pK_a for $\text{H}_3\text{CCH}_2\text{CO}_2\text{H}$ is 4.89)

- A. 1.30
- B. 3.10
- C. 4.89
- D. 9.78

100. Which pH is INCORRECT for the corresponding solution?

- A. 0.10 M HBr(aq) has pH = 1.00
- B. 0.10 M $\text{HCO}_2\text{H(aq)}$ has pH = 2.32
- C. 0.10 M NaOAc(aq) has pH = 5.13
- D. 0.10 M KOH(aq) has pH = 13.00

1. C	2. D	3. C	4. A	5. C
6. D	7. D	8. B	9. D	10. A
11. B	12. D	13. C	14. A	15. D
16. B	17. D	18. C	19. D	20. D
21. C	22. C	23. B	24. D	25. A
26. C	27. B	28. C	29. C	30. C
31. B	32. B	33. A	34. B	35. B
36. C	37. A	38. C	39. C	40. C
41. D	42. C	43. B	44. C	45. C
46. A	47. B	48. C	49. D	50. A
51. C	52. C	53. D	54. D	55. D
56. C	57. B	58. D	59. A	60. C
61. D	62. D	63. A	64. A	65. D
66. D	67. D	68. B	69. C	70. A
71. A	72. B	73. C	74. A	75. A
76. A	77. D	78. D	79. C	80. B
81. C	82. C	83. D	84. A	85. C
86. A	87. D	88. A	89. D	90. C
91. C	92. C	93. D	94. B	95. B
96. B	97. B	98. C	99. B	100. C

Acids and Bases Passage Answers

Passage I (Questions 1 - 7)

Acid and Base Definitions

- Choice C is correct.** A quick look says that the pH should increase when the hydroxide ion concentration increases, which eliminates choices A and B. The pH should not increase by a full 3 units. An increase by 3 units would mean 1000 times the base concentration. This implies choice C is the most intuitive choice. This problem can be solved longhand as well. The question deals with $[\text{OH}^-]$ and pH, so some form of conversion must occur. First, pH must be converted to pOH, using the relationship $\text{pOH} = 14 - \text{pH}$ ($14 - 8 = 6.0$). Then, pOH must be converted to $[\text{OH}^-]$, using the relationship $[\text{OH}^-] = 10^{-\text{pOH}}$ ($[\text{OH}^-] = 10^{-6}$). Tripling the amount of OH^- yields a concentration of $3 \times 10^{-6} \text{ M}$. The new pOH is $-\log(3 \times 10^{-6})$, which equals $6 - \log 3$, which is approximately 5.5. The new pH is equal to $14 - 5.5 = 8.5$, so choice C is in fact correct.
- Choice D is correct.** The larger the value of pK_b , the weaker the base. The weaker the base, the less it undergoes hydrolysis to form hydroxide anion when added to water. This makes choice D the correct choice.
- Choice C is correct.** An Arrhenius base is defined as a base that yields $\text{OH}^-(\text{aq})$ upon addition to water. Of the answer selections, only choice C is basic. Choice C must be correct.
- Choice A is correct.** HClO_3 is a weak acid that can be neutralized using a base of some sort. This eliminates both choice C and choice D. Choice B is a weak base, so it cannot fully react, eliminating choice B. Pick choice A. For titration, 25 mL of strong base is ideal, because there are 25 mL of equimolar acid present. Although we haven't reviewed it yet, titration is the quantitative addition of an equal mole quantity of a reactant.
- Choice C is correct.** As the conjugate acid gets stronger, the conjugate base gets weaker. The weaker the conjugate base, the higher its pK_b value, so choice A is eliminated. The weaker the conjugate base, the stronger the acid that is required to react with it, so choice B is eliminated. The weaker the conjugate base, the lower the hydroxide anion concentration, so the higher the pOH and the lower the pH. This makes choice C correct. The weaker base requires the same moles of acid as a stronger base to be neutralized, as long as they are in equal concentration. The strength of the acid required for reaction is different for the two bases, but the moles of hydroxide are the same from equimolar bases, regardless of their strengths.
- Choice D is correct.** The passage provided the three definitions for bases. The definitions for acids is the opposite of that for bases. This means that the Lewis definition of an acid is an electron-pair acceptor. Choice A is thus valid. The Brønsted-Lowry definition of an acid is a proton donor, making choice B valid. The Arrhenius definition defines an acid as a compound that produces hydronium ion upon addition to water. This makes choice C valid. Choice D must therefore be the best answer. A Lewis base, not a Lewis acid, is defined as a nucleophilic molecule.
- Choice D is correct.** By definition, as the strength of an acid increases, it dissociates to a greater extent when mixed with water. This eliminates choice B. By dissociating more, the acid produces more ions, making the solution more electrolytic. Choice A is eliminated. A stronger acid produces more hydronium ion, so K_a should increase and pK_a should decrease. This both eliminates choice C and makes choice D correct.

Passage II (Questions 8 - 13)

Dissociation and Colligative Properties

- Choice B is correct.** Hydrofluoric acid, HF, is a weak acid, so it partially dissociates in water. The current would be minimal. HClO is also a weak acid, so an HF solution has conductivity similar to an HClO solution. The best answer is choice B. Pick choice B, and be a happy camper.
- Choice D is correct.** Both KOH and KOAc are salts. As such, they will both completely dissociate into water, so the ion concentration is the same in both solutions. This eliminates choice C. KOH is a strong base, while KOAc is a weak base, so choices A and B are also eliminated. The best answer is choice D.
- Choice A is correct.** HClO is a weak acid, so water must be acting as a base by removing a proton from hypochlorous acid. This eliminates choices B and C. Water can be an amphoteric species, meaning it can act as either an acid or a base, but in this reaction it acts only as a base. Choice D is eliminated. By gaining a proton from HClO, water is acting as a proton acceptor, making it a Brønsted-Lowry base. Choice A is the best answer.

11. **Choice B is correct.** The solution in Trial 1 shows a greater current than the solution in Trial 2, so the solution in Trial 1 is more electrolytic than the solution in Trial 2. Electrolytic is defined as the ability of a solution to conduct electricity. This eliminates choice D. HCl is a strong acid, while HClO is a weak acid. As such, more ions are present in solution during Trial 1 than during Trial 2. This explains the greater conductivity in Trial 1 than Trial 2. Other colligative properties include boiling point elevation, freezing point depression, and osmotic pressure. Because the solution in Trial 1 has more ionic impurities (and thus more total impurities), the solution in Trial 1 has a higher boiling point, lower freezing point, and greater osmotic pressure. This makes choice B the correct answer.
12. **Choice D is correct.** As seen in the experiment, full dissociation leads to a current of approximately 6.0 amps. Nitric acid is a strong acid, so choice A should generate a current of roughly 6.0 amps. Sulfuric acid has one strong proton, so choice B should generate a current of roughly 6.0 amps. Sodium hydroxide is a highly soluble salt, so choice C should generate a current of roughly 6.0 amps. Ammonia is a weak base with no charge, so addition of ammonia to water forms only a few ions. The current cannot reach 6.0 amps, so choice D is the best answer.
13. **Choice C is correct.** A strong base is mixed with a strong acid in Trial 3. A complete reaction occurs, leaving behind potassium cation (K^+) and chloride anion (Cl^-). A strong base is mixed with a weak acid in Trial 4. A complete reaction occurs, leaving behind potassium cation (K^+) and hypochlorite anion (ClO^-). A complete reaction occurs in both Trial 3 and Trial 4, because in each case a strong base has been added. This eliminates choices A and B. Because equal amounts of strong base (KOH) are added to equal mole quantities of acid in both Trial 3 and Trial 4, both reactions go to completion, generating the same number of ions in solution. This favors choice C. Choice D is invalid, because cations are also in equal concentration, and cations conduct electrons.

Passage III (Questions 14–21)

Oxyacids

14. **Choice A is correct.** From the relative acid strengths derived from the pK_a values, $HClO_4$ is more acidic than $HBrO_4$, which in turn is more acidic than HIO_4 . This same trend is seen with $HOCl$, $HOBr$, and HOI . Because Cl is more electronegative than Br which is more electronegative than I, the trend indicates that the electronegativity of the halide dictates the relative acidity of an oxyacid. This fact was also given in the passage. Pick answer A, and feel what it is to be correct. Choice C applies to haloacids, not oxyacids.
15. **Choice D is correct.** To determine the relative strength of the oxyacids, both the number of oxygens and the electronegativity of the halide must be considered. $HClO_4$ is a stronger acid than HIO_4 , because chlorine is more electronegative than iodine, so choices A and C are both eliminated. Choice B is eliminated, because HIO_4 is a stronger acid than $HBrO_3$, due to the extra resonance structure associated with the additional oxygen. The best answer is choice D, since it follows both the decline in number of oxygens trend and the decreasing electronegativity trend.
16. **Choice B is correct.** For a weak acid solution with $[HA]_{initial} > K_a$, use the shortcut equation to determine the pH. The shortcut equation, $pH = \frac{1}{2}pK_a - \frac{1}{2}\log [HA]$, applies if the pK_a lies between 2 and 12. The math is as follows:

$$pH = \frac{1}{2}pK_a - \frac{1}{2}\log [HA] = \frac{1}{2}(7.26) - \frac{1}{2}\log (0.10) = 3.63 - \frac{1}{2}(-1) = 3.63 + 0.5 = 4.13$$

The correct answer is choice B.

17. **Choice D is correct.** The pK_{a2} value must be larger than the pK_{a1} value, because the first proton is more acidic than the second proton (by definition), and the stronger the acid, the lower its pK_a value. Given that pK_{a1} is 7.8, the value of pK_{a2} must be a number larger than 7.8. The only answer larger than 7.8 is 12.9, so pick choice D, and feel satisfied that you did. The pK_b (not pK_a) for the conjugate is 6.2, but the question asks for pK_{a2} .
18. **Choice C is correct.** Since the dissociation reaction is exothermic, heat is given off when forming the products. Thus when heat is added to the system, it acts as an inhibitor of product formation according to Le Châtelier's principle. This means that products decrease and reactants increase, as the system is heated. The K_a (equilibrium constant) is a measure of the products over the reactants, so the value of K_a decreases with the addition of heat to the system. The best answer is choice C. Choice D should be eliminated immediately.

19. **Choice D is correct.** We know HBrO_2 is less acidic than HClO_2 , since both acids have the same number of oxygen atoms, and chlorine is more electronegative than bromine. Given that HBrO_2 is the weaker acid, the $\text{p}K_a$ for HBrO_2 must be greater than 1.9 (the $\text{p}K_a$ of HClO_2), so the only answer possible is choice D.
20. **Choice D is correct.** Dissociation refers to the breaking of the bond between the acidic proton and the conjugate base, so that the conjugate base and a hydronium ion are formed. The greatest dissociation is associated with the strongest acid. This question is asking for the strongest acid. The strongest acid is the acid with the greatest number of excess oxygens and the most electronegative central atom. This makes choice D the best answer. This answer could also have been determined by recalling the six strong acids listed in the text. Only choice D is one of these strong acids. A strong acid fully dissociates, while weak acids partially dissociate. The amount of dissociation can be determined from the K_a and vice versa. For instance, a 1.0 M weak acid with 10% dissociation results in 0.90 M HA undissociated acid, and the formation of 0.10 M A^- and 0.10 M H_3O^+ . The equilibrium constant for the dissociation reaction (K_a) is shown below:

$$K_a = \frac{[\text{H}_3\text{O}^+][\text{A}^-]}{[\text{HA}]} = \frac{(0.1)(0.1)}{0.9} = \frac{0.01}{0.9} = 0.0111$$

21. **Choice C is correct.** The volume of the solution does not affect the pH; only the concentration and strength affect the pH. The first step is to identify the type of acid. Since HClO_4 is a strong acid, it fully dissociates, so the major source of protons in the aqueous solution is from the dissociation of HClO_4 .

$$\text{pH} = -\log [\text{H}_3\text{O}^+] = -\log [\text{HClO}_4] = -\log(0.10) = -(-1) = 1$$

This best answer is choice C.

Passage IV (Questions 22 - 28)

Acidity of Thiols and Alcohols

22. **Choice C is correct.** The most acidic compound is the compound that most readily loses H^+ . In choices A, B and D, the hydrogens are bonded to carbon. In choice C, there is a hydrogen that is bonded to nitrogen. The size difference between atoms is significant only when the atoms are in different rows of the periodic table. Nitrogen and carbon are in the same row of the periodic table, so they are comparable in size. When atoms are in the same row of the periodic table, the most important factor to consider when looking at acidity is electronegativity. Nitrogen is more electronegative than carbon, making a hydrogen on nitrogen more acidic than a hydrogen on carbon, so choice C is the most acidic. It is a common mistake to not notice that the hydrogens are all on carbons in choices A and B. Be careful not to make mistakes like this.
23. **Choice B is correct.** It can be observed from the data in Table 1 that as the number of methyl groups increases, the $\text{p}K_a$ value increases. An increase in $\text{p}K_a$ is indicative of decreased acidity. Acids are defined as electron-accepting, so electron-withdrawing groups increase acidity, while electron-donating groups decrease acidity. This makes choice B correct.
24. **Choice D is correct.** In haloacids (binary compounds), the acidic proton is directly bonded to a halogen, so relative acidity can be discerned from the features of the halide. Halogens are in the same column of the periodic table, so the important factor when considering bonding is the size of the halide. Because iodine is the largest of the halogens, the HI bond is the weakest, so HI is the strongest haloacid. Pick choice D, and smile to the world.
25. **Choice A is correct.** Sulfur and chlorine are adjacent to one another in the same row of the periodic table, so choices B and D are eliminated. Because chlorine is smaller than sulfur, the increasing size does not correlate with acidity. This eliminates choice C. Chlorine is more electronegative than S, so Cl draws electrons from H more than S does. This makes HCl a stronger acid than H_2S . The electronegativity predicts the acidity best for atoms in the same row, so pick answer choice A to get this one correct.
26. **Choice C is correct.** Table 1 shows that thiols have lower $\text{p}K_a$ values than their corresponding alcohols, so ethyl thiol is more acidic than ethanol. Because ethanol is a weaker acid (has a lower K_a value than ethyl thiol), ethanol dissociates less than ethyl thiol when added to water, and ethanol yields a lower $[\text{H}_3\text{O}^+]$ than ethyl thiol. These three facts eliminate choices A, B, and D. Because ethanol is a weaker acid than ethyl thiol, the conjugate base of ethanol ($\text{CH}_3\text{CH}_2\text{O}^-$) is a stronger base than the conjugate base of ethyl thiol ($\text{CH}_3\text{CH}_2\text{S}^-$), making choice C the correct answer.

27. **Choice B is correct.** The most electronegative atom that differs from choice to choice is chlorine. This high degree of electron withdrawal due to chlorine results in an increase in acidity. This implies that because chlorine is more electronegative than the other atoms, it withdraws electron density the most and thus increases the acidity to the greatest extent. The answer of answer choices is choice B.
28. **Choice C is correct.** Of the answer choices, the carboxylic acid is the most acidic, because of the resonance associated with the carbonyl bond. This eliminates choices A and B. From this point, data in Table 1 must be analyzed. Table 1 shows that as the substitution of the thiol decreases, the acidity increases, so the primary thiol is more acidic than the tertiary thiol. This makes choice C the best answer.

Passage V (Questions 29 - 34)

Organic Acids

29. **Choice C is correct.** The solution is acidic, so the pH is less than 7.0, eliminating choice D. For a weak acid with pK_a between 2 and 12 in an aqueous solution with $[HA]_{initial} > K_a$, use the shortcut equation to determine the pH. If K_a is 8×10^{-8} , then pK_a is $8 - \log 8 = 7 - 0.9 = 7.1$. The $[HA]$ is 0.5 M, and $\log 0.5 = -\log 2 = -0.30$.

$$pH = \frac{1}{2}pK_a - \frac{1}{2}\log [HA] = \frac{1}{2}(7.1) - \frac{1}{2}\log (0.50) = 3.55 - \frac{1}{2}(-0.3) = 3.55 + 0.15 = 3.7$$

The value falls in the range of choice C, so the best answer is choice C.

30. **Choice C is correct.** Mass percent is defined as the mass of one component atom relative to the total mass of the compound. In this case, we are interested in carbon within phenol. Units cancel, so we can use atomic masses.

$$\text{mass \% carbon} = \frac{\text{mass of carbon}}{\text{mass of molecule}} = \frac{72 \text{ g C}}{94 \text{ g C}_6\text{H}_5\text{OH}} \times 100\% \Rightarrow 72\% = \frac{72}{100} < \frac{72}{94} < \frac{72}{90} = 80\%$$

The mass percent of carbon is between 72% and 80%. Choices A and B are too small, while choice D is too large. Only choice C, 76.6%, falls into the range, so choice C is the best answer.

31. **Choice B is correct.** When an acid is titrated to its equivalence point, it has been completely converted into its conjugate base. The lowest pH value after neutralization is found with the weakest conjugate base, assuming that the concentrations are all equal. The weakest conjugate base corresponds with the strongest acid (that is, the acid with the lowest pK_a). This is because the stronger the acid, the weaker its conjugate base. Using the data from Table 1, the acid with the lowest pK_a value is choice B, trichloroacetic acid. Pick choice B please.
32. **Choice B is correct.** Increasing the number of impurities dissolved into an aqueous solution raises the boiling point and lowers the freezing point of a solution. The total impurity concentration depends on the solute concentration, the number of particles it dissociates into, and the degree to which it dissociates. All of the choices are monoprotic weak acids of equal concentration (1.0 M), so it depends only on the dissociation of the acid (acid strength). Acetic acid is weaker than p-nitro benzoic acid, so choice A is eliminated. Trichloroacetic acid is stronger than p-nitrophenol, so choice B is correct. Trichloroacetic acid is stronger than benzoic acid, so choice C is eliminated. Acetic acid is stronger than phenol, so choice D is eliminated.
33. **Choice A is correct.** The acid with the strongest conjugate base is the weakest acid. The highest pK_a value is associated with the weakest acid. Referring to the data in Table 1, the acid with the highest pK_a is phenol. Choice A is the correct answer, since the highest pK_a value indicates the weakest acid.
34. **Choice B is correct.** This question appears to be difficult at first glance. However, if you realize that when an acid dissociates, equal parts of hydronium and conjugate base are formed, then you know that $[H_3O^+] = [A^-]$. All that is required is converting from pH to $[H_3O^+]$, using $[H_3O^+] = 10^{-pH}$. The pH of the solution is 3.7, so the $[H_3O^+]$ is $10^{-3.7} = 10^{0.3} \times 10^{-4}$. Because $\log 2 = 0.3$, it is true that $10^{0.3} = 2$. This means that $10^{0.3} \times 10^{-4} = 2 \times 10^{-4}$. The $[H_3O^+]$ is 2×10^{-4} M, and $[H_3O^+] = [A^-]$, so $[A^-] = 2 \times 10^{-4}$. The correct answer is choice B.

Passage VI (Questions 35 - 41)

Electron-Withdrawing Effect and Acidity

35. **Choice B is correct.** The lowest pH is found in the solution with the greatest amount of hydronium. The hydronium concentration is affected by the strength and concentration of the acid. The most acidic solution results from the strongest acid in highest concentration. In this question, p-nitrophenol has a lower pK_a than phenol, so choices C and D are eliminated. Choice B has the higher concentration of p-nitrophenol, so choice B is the best answer. Pick choice B for best results and greatest test satisfaction.

36. **Choice C is correct.** The solution is acidic, so the pH is less than 7.0, eliminating choices A and B. For a weak acid with pK_a between 2 and 12 in an aqueous solution with $[HA]_{\text{initial}} > K_a$, use the shortcut equation to determine the pH. The pK_a of phenol is 10, and the $[HA]$ is 1.0 M. The log of 1.0 is zero, so the pH is simply half of the pK_a . Half of ten is five, so the pH is 5.0. Choice C is the best answer.
37. **Choice A is correct.** A stronger electron-withdrawing group on the aromatic ring of the phenol makes the phenol more acidic, thus decreasing its pK_a value. As a result, a substituted phenol with an electron-withdrawing group has a lower pK_a than phenol. From the acids listed in Table 1, the acid with the lowest pK_a is p-nitrophenol. The electron-withdrawing group on p-nitrophenol is O_2N- , so choice A is terrific.
38. **Choice C is correct.** For a monoprotic acid and its conjugate base, $pK_a + pK_b = 14$. From Table 1, the pK_a value for p-nitrophenol is 7.2. This means that pK_b for the conjugate base (p-nitrophenoxide) is $14 - 7.2 = 6.8$. The correct answer is choice C.
39. **Choice C is correct.** This problem is a twist on the normal approach (i.e., it's harder than typical problems). This problem is most easily solved using the shortcut equation, $pH = \frac{1}{2}pK_a - \frac{1}{2}\log [HA]$, to solve for $[HA]$.

$$pH = \frac{1}{2}pK_a - \frac{1}{2}\log [HA] \therefore 6.1 = \frac{1}{2}(11.2) - \frac{1}{2}\log [HA] = 5.6 - \frac{1}{2}\log [HA] \therefore 0.5 = -\frac{1}{2}\log [HA], \text{ so } \log [HA] = -1$$

$$\log [HA] = -1 \therefore [HA] = 10^{-1} \text{ M} = 0.10 \text{ M}$$

The concentration of the acid is 0.10 M, so the next step of the solution is to determine the moles of weak acid needed to make 100 mL of a 0.10 M solution. To make 100 mL 0.1 M $H_3COC_6H_4OH$, 0.01 moles of the acid must be added to enough water to make 100 mL aqueous solution. The final step is to convert 0.010 moles into grams. The molecular mass of $H_3COC_6H_4OH$ is 124 grams per mole, so 0.01 moles is 1.24 grams $H_3COC_6H_4OH$. The correct answer is choice C.

40. **Choice C is correct.** To convert from pK_a to K_a , the relationship is $K_a = 10^{-pK_a}$. The pK_a of p-methylphenol is 10.4 (as given in Table 1). The K_a is therefore $10^{-10.4} = 10^{0.6} \times 10^{-11}$, when expanded into scientific notation. Given $\log 2 = 0.3$ and $\log 4 = \log 2 + \log 2 = 0.3 + 0.3 = 0.6$. According to the anti log relationship, $10^{0.6} = 4$. This makes the value of $K_a = 4 \times 10^{-11}$. Pick choice C and be a success story in acids and bases.
41. **Choice D is correct.** The addition of water dilutes the aqueous phenol solution, but it does not react with the phenol. Using the relationship, $M_1V_1 = M_2V_2$, the new molarity is found to be 0.10 M. As is becoming the norm, this problem is most easily solved using the shortcut equation, $pH = \frac{1}{2}pK_a - \frac{1}{2}\log [HA]$.

$$pH = \frac{1}{2}pK_a - \frac{1}{2}\log [HA] = \frac{1}{2}(10.0) - \frac{1}{2}\log (0.10) = 5.0 - \frac{1}{2}(-1) = 5.0 + 0.5 = 5.5$$

The correct answer is choice D. Because the solution is acidic, choices A and B should have been immediately ruled out.

Passage VII (Questions 42 - 48)

Weak Acid pH Equation

42. **Choice C is correct.** The shortcut equation, as presented in the passage, works only for weak acids. It does NOT work with a strong acid. The only strong acid in the choices is hydroiodic acid (HI), which you should know is a strong acid, because HI is stronger than HCl, a common strong acid. Choice B (carbonic acid) and choice D (a carboxylic acid) should immediately be recognized as weak acids. Pick choice C for the greatest success here.
43. **Choice B is correct.** In the first step of the derivation, $[H_3O^+]$ is substituted for $[A^+]$, as shown in the conversion from $[H_3O^+][A^-]$ to $[H_3O^+]^2$. The basic assumption is that when the acid dissociates, equal amounts of H_3O^+ and A^- are formed in solution, and the little (10^{-7} M) H_3O^+ formed from water is insignificant. This makes choice B the best answer.
44. **Choice C is correct.** Regardless of the molarity of the weak acid, when the pH of the solution is 2.70, the $[H_3O^+]$ in solution is $10^{-2.7}$ M. This means that the $[H_3O^+]$ is equal to something $\times 10^{-3}$ (where "something" is a value less than ten). Because $[H_3O^+] = [A^-]$ when a weak acid dissociates, it should be concluded that $[A^-] = \text{something} \times 10^{-3}$, which makes choice C the best answer.

45. **Choice C is correct.** Starting from the K_b relationship for the hydrolysis of a weak base in water, the equilibrium expression can be manipulated to yield the relationship $pOH = -\log \sqrt{K_b \times [A^-]}$. This relationship can be manipulated in the same way that the pH relationship was manipulated in the passage to yield the same type of equation as the shortcut equation, except that it's for a weak base. The logic is that the pOH depends on the base, hence the formula contains K_b rather than K_a , and $[A^-]$ rather than $[HA]$. This makes choice C the best answer. Choice A is applied in general to all basic solutions, but it is not the best answer. Choice D may sound tempting, but that is how the pOH for a weak acid solution is found, not for a weak base solution.
46. **Choice A is correct.** According to the Henderson-Hasselbalch equation, when $[A^-] > [HA]$, the pH is greater than the pK_a , because the log of the $[A^-]$ to $[HA]$ ratio is a positive value. This eliminates choice B. If pH is greater than the pK_a , then by definition $-\log [H_3O^+]$ is greater than the $-\log K_a$, hence $\log K_a$ is greater than the $\log [H_3O^+]$ (when both sides of an inequality are multiplied by -1 , the inequality sign must be reversed). Because $\log K_a$ is greater than the $\log [H_3O^+]$, K_a is greater than the $[H_3O^+]$, making choice A the best answer. Choice C cannot be true, because as pH goes up, $[A^-]$ increases as $[H_3O^+]$ decreases, making $[A^-] > [H_3O^+]$ at all points of a reaction after the initial mixing of the weak acid into water. This eliminates choice C. Choice D can be true only if $[HA] = 0$, which is physically impossible with the equilibrium constant favoring HA. It should always be true that $[HA] > [H_3O^+]$, which eliminates choice D. The best answer is choice A.
47. **Choice B is correct.** The greatest conjugate base concentration is found with acid that dissociates to the greatest extent. The acid that dissociates most completely is the strongest acid, which describes choice B (HNO_3), the only strong acid among the choices. This question is just another way of asking, "Which acid is stronger?"
48. **Choice C is correct.** As the value of K_a increases, the strength of the acid is increasing. As the acid gets stronger, the degree of dissociation when added to water increases. This causes the conjugate base concentration to increase, which eliminates choice A. The amount of H_3O^+ increases, which causes the pH to go down and thus the pOH to increase. This eliminates choice B and confirms that choice C is the best answer. Because the acid dissociates more, the amount of HA decreases and the amount of A^- increases, causing the ratio of HA to A^- to decrease. This eliminates choice D.

Passage VIII (Questions 49-55)

Aspirin and Antacids

49. **Choice D is correct.** Aspirin (acetylsalicylic acid) when added to water dissociates into the anionic conjugate base and a proton, just like all other carboxylic acids. The dissociation obeys the laws of equilibrium, resulting in an equilibrium constant known as K_a . Because it obeys the laws of equilibrium, the anion form (more water-soluble form) is most abundant when the H^+ (proton) is removed from solution. The lowest concentration of protons ($[H_3O^+]$) is found at high pH. The highest pH results in the greatest water solubility for the aspirin, because it exists in its anionic form. This, in essence, is the common ion effect. The best answer is thus choice D.
50. **Choice A is correct.** Acetylsalicylic acid has only one hydrogen on the carboxylic acid functional group that is acidic. All of the other hydrogens are bonded to carbon, which does not make them acidic. The correct answer for the number of acidic protons is therefore only one. Choice A is the best answer.
51. **Choice C is correct.** The first three choices can all neutralize two equivalents of acid, while choice D, sodium bicarbonate, can neutralize only one equivalent of acid. This means that the most neutralizing strength per gram is found with the compound having the lightest molecular mass of the compounds capable of neutralizing two equivalents of acid. Magnesium hydroxide has the lowest molecular mass of the first three answer choices. The correct answer is therefore choice C.
52. **Choice C is correct.** The molecule that can cause damage to stomach lining is the acidic molecule that ionizes once within the membrane pocket of parietal cells. The only acidic compound of the choices is benzoic acid, answer choice C. It is the most similar in structure to aspirin, so it is the best answer.
53. **Choice D is correct.** Dihydroxyaluminum sodium carbonate is composed of the dihydroxyaluminum cation (which loses two hydroxide substituents when hydrolyzed), sodium cation (with no basic properties), and carbonate dianion (which can neutralize two acidic protons). The net result is that the two hydroxides and one carbonate can neutralize four equivalents of protons, making choice D correct. You might also consider that Al carries a $+3$ charge and Na carries a $+1$ charge, so the sum charge of the bases must be -4 .

54. **Choice D is correct.** Acetylsalicylic acid is a carboxylic acid, and carboxylic acids are weak acids. Weak acids have K_a values that are less than one. The pK_a for a carboxylic acid ranges from 2.0 (as seen with the carboxyl terminal of amino acids) to 5.0 (as seen with regular alkyl chains). This results in a range for K_a of 10^{-2} to 10^{-5} , which yields a K_a that is much less than 1.00. This vague question is best answered by choice D.
55. **Choice D is correct.** A buffer forms when the conjugate acid and base of a conjugate pair are present in roughly equal molar ratio. Choices A and B should immediately be eliminated, because the mixtures are made up of two acids, and not of an acid and base pair. Choices C and D involve a strong acid and weak base, so the correct choice involves partial titration of the weak base. It is in choice D that an equal molar ratio of the weak base and its conjugate acid are present. The one equivalent of HCl will convert one of the two equivalents of acetylsalicylate into acetylsalicylic acid, while one equivalent remains as acetylsalicylate. The equal quantity of the two components of the conjugate pair results in a buffer (with a pH equal to the pK_a of the weak acid). The best answer is choice D.

Passage IX (Questions 56 - 62)

Household Acids and Bases

56. **Choice C is correct.** The first word in the question is "If", so keep that in mind. The mantra that you chant about water having a pH of 7 does not apply in this speculative question. If the only hydroxide ion in solution is formed when one molecule of water loses a proton to another molecule of water, then the hydroxide anion concentration $[OH^-]$ must equal the hydronium cation concentration $([H_3O^+])$. This is true within distilled water. This means that if $[OH^-] = 1.0 \times 10^{-6} M$, then $[H_3O^+] = 1.0 \times 10^{-6} M$. The negative log of 1.0×10^{-6} is 6.0, so the pH of the water must be 6.0. The best answer is choice C. For this hypothetical solution, $pH = 6$ and $pOH = 6$, which means that $pH + pOH = 12$, not 14. This may bother your sense of what is right in the world of acid and base chemistry, but keep in mind that $pH + pOH = 14$ applies only at $25^\circ C$. At higher temperatures, there is more autoionization, so more hydronium and hydroxide are generated. At $37^\circ C$ for instance, pH of distilled water is 6.8, so $pH + pOH = 13.6$.
57. **Choice B is correct.** The hydronium ion concentration in solution can be found from the pH of the solution. Because the pH is defined as $pH = -\log [H_3O^+]$, the hydronium ion concentration can be found by manipulating the relationship to yield $[H_3O^+] = 10^{-pH}$. The hydronium ion concentration in a $pH = 1.7$ solution is $10^{-1.7} M$, which is therefore roughly equal to $10^{-2} M$. The best answer selection is choice B.
58. **Choice D is correct.** The volume increases from 10 mL to 110 mL when 100 mL of distilled water is added to the solution, so the concentration of the acid must decrease by a factor of 11. The log of 10 is equal to 1, so the log 11 is slightly greater than 1. This implies that the pH increases by just a little more than 1.0, because the $[H_3O^+]$ goes down by a factor of 11. This makes the final pH greater than 3.0. The best answer is greater than 3.0, which is choice D. This is true only of a strong acid solution. If the solution were an aqueous weak acid, the pH increase would not be as significant as with the strong acid.
59. **Choice A is correct.** The hydronium ion concentration in a $pH = 2.0$ solution is $1.0 \times 10^{-2} M$. There are 50 mL present, so the moles of H^+ are found by multiplying $0.010 M$ by $0.050 L = 5.0 \times 10^{-4}$ moles H^+ . Only half the moles of $CaCO_3$ are necessary to neutralize the H^+ , because it is a 2 : 1 ratio of H^+ to $CaCO_3$ in the balanced equation. This means that 2.5×10^{-4} moles $CaCO_3$ are required to neutralize all of the hydronium ion (H_3O^+). 2.5×10^{-4} moles when multiplied by 100 grams per mole yields 2.5×10^{-2} grams $CaCO_3$, which equals 25 milligrams. Choice A is the best possible answer you can choose in a situation like this.
60. **Choice C is correct.** Windex contains ammonia, which according to Table 2 is a base. An acid can react with a base, while a base cannot react with another base. The question is thus asking: "Which of the following compounds is not an acid?" Vinegar, aspirin, and bleach are all listed as acids in Table 1, so they all can react with ammonia (Windex). The best answer is choice C, Drano, because Drano is a base and not an acid.
61. **Choice D is correct.** Raising the pH of a solution requires reducing (or diluting) the concentration of hydronium ion (H_3O^+). Adding distilled water reduces the hydronium ion concentration by diluting the solution. This makes choice A invalid. Shampoo and antacid are bases, which when added to solution, react with the H_3O^+ ion and thus reduce the hydronium ion concentration and raise the pH of the solution. Choices B and C are therefore invalid. Adding an acid lowers -- not raises -- the pH of the solution, so choice D (vitamin C), an acid, is the only answer choice that will NOT increase the pH of the solution when added. Pick choice D with a smile.

62. **Choice D is correct.** 1.68 grams of baking soda is equal to 0.020 moles NaHCO_3 as determined by dividing the 1.68 grams by 84 grams per mole. The equation for the reaction shows a 1:1 ratio between baking soda and the hydronium ion for full neutralization, so 0.020 moles of H_3O^+ are required to reach full neutralization. The molarity of the acid solution when multiplied by the volume of the acid solution must equal 0.020 moles H_3O^+ . The molarity of the acid solution is given as 0.20 M, so the volume of the acid solution must be 0.10 liters, which is 100 mL. Selecting answer choice D in a situation like this, is the best thing to do.

Passage X (Questions 63 - 70)

Stomach Acid and pH

63. **Choice A is correct.** The formation of a strong acid from a weak acid and neutral salt is unfavorable. Hydrochloric acid is stronger than carbonic acid, so the reaction as written has $\Delta G > 0$. To carry this out *in vivo*, it must be coupled with some other favorable reaction. As stated in the passage, this process is active transport. The correct answer is therefore choice A.
64. **Choice A is correct.** The lowest pH is associated with the most acidic solution. Both salt water and distilled water are neutral (have a pH of 7.0), so choices B and C are eliminated. Carbon dioxide (a non-metal oxide) when dissolved into water gets hydrated to form carbonic acid (H_2CO_3 , a non-metal hydroxide). Calcium oxide (a metal oxide) when dissolved into water gets hydrated to form calcium hydroxide (Ca(OH)_2 , a metal hydroxide). Non-metal hydroxides are acidic and metal hydroxides are basic, so the best answer is choice A.
65. **Choice D is correct.** Acid is capable of oxidizing any metal with a low ionization energy. In this question, only one answer is correct, so it is better to ask yourself which metal is most easily oxidized (or least stable). Relying on either trivial knowledge or experience, you know that copper, gold, and silver are relatively air-stable, as evident by their commercial uses in wiring and currency. All three tarnish over enough time. Zinc metal is most readily oxidized (more so than hydrogen gas even). The best answer is choice D. If you are interested in smuggling zinc coins across the border, it is recommended that you refrain from ingesting them to do so. If you don't know about the relative reactivity of these metals, you just have to take a guess and move on.
66. **Choice D is correct.** This is really a misplaced organic chemistry question. But then again, the passage is on a physiology topic, emphasizing that acidity and basicity are applicable to many areas of study. Alkanes cannot be hydrolyzed whether an acid is present in solution or not. Disaccharides, polypeptides, and esters are all broken down in the stomach by acid-catalyzed hydrolysis. The best answer is choice D.
67. **Choice D is correct.** Consuming water dilutes all solutes (including H_3O^+ in an acidic solution), thus the hydronium ion concentration is reduced by consuming water. Baking soda can consume some of the hydronium ion by way of an acid-base reaction. Aluminum metal can be oxidized by hydronium ion, resulting in the formation of aluminum trication and hydrogen gas. Therefore, choices A, B, and C are all eliminated. Consumption of solid food induces the release of hydrochloric acid into the stomach. The correct answer is choice D.
68. **Choice B is correct.** This answer is taken straight from the first paragraph of the passage. 0.030 M hydronium has a pH of 1.5. However, in the event you skipped reading the passage, you can solve for the pH by determining the negative log of the hydronium concentration ($\text{pH} = -\log [\text{H}_3\text{O}^+]$). The pH of a 0.030 M HCl solution is $-\log (3 \times 10^{-2}) = 2 - \log 3$. The value is less than 2, but greater than 1. The correct answer is choice B.
69. **Choice C is correct.** Stomach lining is deteriorated by reacting with acid. All of the answer choices except milk contain acid, so they all promote the decay of the stomach lining. The best answer is choice C. This question is asking for you to identify which compound is not an acid, so look for the choice that is different. Lactose is a disaccharide composed of galactose and glucose.
70. **Choice A is correct.** The lowest pH results from the presence of the strongest acid in the highest molar concentration. Hydrochloric acid is the strongest acid (stronger than carbonic acid), and 0.030 M is a higher concentration than 0.010 M. The correct answer is therefore choice A.

Passage XI (Questions 71 - 76)

Tooth Decay and pH

71. **Choice A is correct.** The first question is whether the pH of the solution changes. Because fluorapatite is less soluble than hydroxyapatite, hydroxide anion is released as fluoride anion exchanges for the hydroxide of hydroxyapatite. When hydroxide anion is released, the pH of the solution increases. Although the pH is greater than 7.0, choice C is incorrect, because the pH does not remain constant. The best choice is A.

72. **Choice B is correct.** Calcium hydroxide when added to water yields hydroxide anion (OH^-), making it basic. This eliminates choices A and C. The question is whether it is a strong or weak base. These questions on the MCAT are tough, because who knows what the test-writer wants? Because calcium hydroxide does not fully dissolve into water at high enough concentrations, it can be considered a weak base. However, because it reacts with weak acids, and it does fully dissociate at low concentrations, it must be a strong base. The best answer beyond any questioning and rational argument you can present is choice B (unless, of course, you wish to choose D). On some questions, like this one, there are two reasonable answers, but the writer of the question wants you to pick just one best answer.
73. **Choice C is correct.** Enamel is more soluble in the more acidic solution. Because enamel is more soluble in the HA solution than the HB solution, the HA solution must be more acidic than the HB solution. Both the HA and HB solutions are of equal concentration, so it must be that HA is a stronger acid than HB. This means that if HA is dissolved into water, it has a lower pH than when HB is dissolved into water. The conjugate base of HB is a stronger base than the conjugate base of HA. HA has a higher K_a and a lower pK_a than HB. This eliminates choices A, B, and D. Choice C is the best choice.
74. **Choice A is correct.** The hydroxyapatite dissolves most readily in acidic medium, so the addition of acid increases its solubility. Thus, the amount of hydroxyapatite that dissolves into solution increases as acid is added to solution. Addition of base increases the pH of the solution, meaning that choices B and C are the same answer. The MCAT presents questions like this on occasion. When they have two identical answers worded differently, eliminate them both. Decreasing the solvent reduces the amount of salt that has dissolved into solution. The best answer is choice A.
75. **Choice A is correct.** Sodium carbonate (choice D) should be eliminated immediately, because it is a base. Of the three remaining choices, both hydrochloric acid (choice B) and nitric acid (choice C) are strong acids. The aldonic acid drawn is not a strong acid, so choices B and C are eliminated. The structural similarity because of the CO_2H group on both acetic acid and gluconic acid (the aldonic acid drawn) makes answer A the best choice.
76. **Choice A is correct.** Tooth enamel is a basic substance, so it dissolves fastest in an acidic solution. Table 1 shows that the solubility increases as pH decreases. This means that the most acidic solution (solution with the lowest pH) dissolves tooth enamel the fastest. The lowest pH results from the strongest acid in the highest concentration. The highest concentration is found in choices A and C, so choices B and D are eliminated. The stronger of the two acids is HCl, leading us to conclude that the 0.10 M HCl solution dissolves the tooth enamel the fastest. The best choice is answer A.

Passage XII (Questions 77 - 84)

Acid Rain and Scrubbers

77. **Choice D is correct.** In Reaction 1, sulfur trioxide is hydrated. The addition of water (hydration) does not change the acidic or basic nature of a compound. This eliminates choices A and B. As stated in the passage, the hydration of a non-metal oxide (a Lewis acid) converts the compound into a non-metal hydroxide (a Brønsted-Lowry acid). This tells us that choice C is wrong and makes choice D the best answer.
78. **Choice D is correct.** A non-metal oxide acts exclusively as an acidic (and not a base), so it is neither amphoteric nor an Arrhenius base. This eliminates choices A and B. A non-metal oxide has no protons, so it cannot be a Brønsted-Lowry acid. This eliminates choice C. By having π -bonds, a non-metal oxide is a good electrophile, which makes it a Lewis acid. The best answer is choice D.
79. **Choice C is correct.** The most acidic compound when combined with water creates the solution with the lowest pH, although the compound's concentration has an effect on the pH of the solution as well. Metal oxides are basic, so choice D (MgO) should be eliminated. The three acids formed when choices A, B, and C are added to water are H_2SO_3 (from SO_2 and H_2O), H_2CO_3 (from CO_2 and H_2O), and HNO_3 (from N_2O_5 and H_2O), respectively. If you knew that nitric acid is stronger than either sulfurous acid or carbonic acid, then you could have picked choice C. However, if you didn't know that tidbit of trivia, then you could have determined their relative strength from their oxidation states and from the electronegativities of their central atoms. Sulfur has a +4 oxidation state in SO_2 (and H_2SO_3), carbon has a +4 oxidation state in CO_2 (and H_2CO_3), and nitrogen has a +5 oxidation state in N_2O_5 (and HNO_3). Nitrogen is the most electronegative and has the highest oxidation state. The nitrogen-based oxyacid has the greatest propensity to gain an electron pair, making HNO_3 the most acidic. Pick choice C.

80. **Choice B is correct.** The solution is acidic, so the pH is less than 7.0, eliminating choice D. For a weak acid with pK_a between 2 and 12 in an aqueous solution with $[HA]_{\text{initial}} > K_a$, use the shortcut equation to determine the pH. The pK_a is 7.46 and the $[HA]$ is 0.10 M.

$$pH = \frac{1}{2}pK_a - \frac{1}{2}\log [HA] = \frac{1}{2}(7.46) - \frac{1}{2}\log (0.10) = 3.73 - \frac{1}{2}(-1) = 3.73 + 0.50 = 4.23$$

The best answer is choice B. Choice A can be eliminated, because HClO is a weak acid. A 0.10 M strong acid has a pH of 1.0. An equimolar weak acid, because it dissociates less than a strong acid, would have a pH greater than 1.0. Choice C can be eliminated, because the pH is less than the pK_a .

81. **Choice C is correct.** This is a freebie question, if you read the passage carefully. Non-metal oxides are found in acid rain, particularly the oxides of sulfur and nitrogen. Choice A is eliminated, because it is a metal oxide, making it basic, not acidic. Choice B should be eliminated, because nitrogen is present in the air all the time and can be found in all rain, not just acid rain. Choice D is eliminated, because it is neither acidic nor very abundant in our atmosphere. This means that the correct answer is choice C.
82. **Choice C is correct.** As the oxidation state of the central atom in an oxyacid increases, the compound becomes more acidic. This is because the number of π -bonds between oxygen atoms and the central atom increases. This increase in π -bonds results in more electron withdrawal (by way of resonance) from the central atom, making it electron-poor. The compound therefore becomes more acidic. This means that both the oxidation state and the number of π -bonds to the central atom affect the acidity of the oxyacid, so choices A and D are eliminated. As the electronegativity of the central atom in an oxyacid increases, the compound becomes more acidic. This is because of the inductive effect. The increase in electron withdrawal from the central atom, due to the inductive effect, makes the central atom electron-poor. This means that the electronegativity of the central atom affects the acidity of the oxyacid, so choice B is eliminated. The size of the central atom has no bearing on the acidity of the oxyacid, because the acidic proton is not directly bonded to the central atom. This makes choice C the best answer.
83. **Choice D is correct.** Because H_2SO_4 has more oxygen atoms than H_2SO_3 , it is more acidic. The more acidic compound has a lower pK_a , a higher K_a , a weaker conjugate base (with a higher pK_b), and generates a lower pH in water. This makes choices A, B, and C valid statements. The fact that H_2SO_4 is more acidic than H_2SO_3 means that H_2SO_4 dissociates more when added to water, resulting in a higher ion concentration, and thus a higher conductivity. H_2SO_4 is more electrolytic (has higher conductivity) than H_2SO_3 , so choice D is an invalid statement, which makes it the correct answer.
84. **Choice A is correct.** The stronger the acid, the lower its pK_a value, so the lowest pK_a is found with the strongest acid. Having more oxygen atoms on an oxyacid increases its acidity, so HNO_2 and H_2SO_3 cannot be the most acidic compounds. This eliminates choices C and D. Sulfuric acid is stronger than nitric acid, which you can determine from their oxidation states (S is +6 in H_2SO_4 , while N is +5 in HNO_3). This makes choice A the best answer.

Passage XIII (Questions 85 - 91)

Amino Acid pK_a Values

85. **Choice C is correct.** Lysine is a triprotic acid. The first proton released (from the carboxyl terminal) corresponds to the third proton gained by the conjugate base. This means that from a conjugate pair perspective, the pK_a -to- pK_b relationship is: $pK_{a1} + pK_{b3} = 14$. This is not one of the answer choices, so we next consider the second proton on lysine. The second proton released (from the amino terminal) corresponds to the second proton gained. This means that from a conjugate pair perspective, the pK_a -to- pK_b relationship is: $pK_{a2} + pK_{b2} = 14$. This makes choice C the correct answer; but to finish our discussion of the correlation between pK_a and pK_b , we must consider the third proton. The third proton released (from the side chain) corresponds to the first proton gained. This means that from a conjugate pair perspective, the pK_a -to- pK_b relationship is: $pK_{a3} + pK_{b1} = 14$.
86. **Choice A is correct.** At physiological pH, the carboxyl terminal is deprotonated ($pH > pK_a$ (carboxyl terminal)), so it carries a negative charge. At physiological pH, the amino terminal is protonated ($pH < pK_a$ (amino terminal)), so it carries a positive charge. In order for the amino acid to be a cation, the side chain must carry a positive charge. Arginine has a side chain pK_a of 13.21, so $pH < pK_a$. This means that the side chain is protonated, so it carries a positive charge. Choice A is the best answer. Choice B is eliminated, because it is an anion. Choice C is eliminated, because the side chain cannot carry a charge. Choice D is eliminated, because histidine is deprotonated ($pH > pK_a$ (side chain)), so it carries no charge.

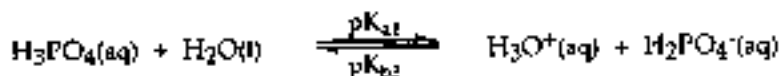
87. **Choice D is correct.** The difference between the side chains of serine and cysteine is that serine has an alcohol group (O-H), while cysteine has a thiol group (S-H). This means that the difference in acidity lies in the difference between oxygen and sulfur. Choices A and C are eliminated immediately, because we know from their relative positions in the periodic table that oxygen has a smaller atomic radius and is more electronegative than sulfur. When comparing the acidity of protons bonded to different atoms that occupy the same column of the periodic table, we find that the most significant factor influencing the acidity is atomic radius. The best answer is choice D.
88. **Choice A is correct.** Based on the pK_a values in Table 1, aspartic acid (with a side chain pK_a of 3.88) is more acidic than glutamic acid (with a side chain pK_a of 4.32). This eliminates choice C and D. A shorter chain results in the electron-withdrawing NH_3^+ group's being closer to the side chain carboxylic acid group, which increases the acidity of the carboxylic acid group. This makes choice A the best answer.
89. **Choice D is correct.** According to Table 1 in the passage, the pK_a values for histidine are: pK_{a1} between 1.8 and 2.6, $pK_{a2} = 6.05$, and pK_{a3} between 8.8 and 10.6. At a pH of 7.0, the carboxyl terminal and the side chain are deprotonated, while the amino terminal is protonated. This is because $pK_{a3} > pH > pK_{a2} > pK_{a1}$. Since the carboxyl terminal is deprotonated, we consider pK_{b3} rather than pK_{a1} . Since the side chain is deprotonated, we consider pK_{b2} rather than pK_{a2} . And since the amino terminal is protonated, we consider pK_{a3} rather than pK_{b1} . This means that the pK values of interest are pK_{a3} , pK_{b2} , and pK_{b3} , making choice D the best answer.
90. **Choice C is correct.** Normality is defined as the moles of equivalents per liter. Because glutamic acid is triprotic, it yields three acidic protons per molecule. For each mole of glutamic acid, three moles of acidic protons can be generated. The normality in this case is the molarity multiplied by a factor of three ($N = M \times 3$). The molarity is 0.50, so the normality is 1.50. The best answer is choice C.
91. **Choice C is correct.** Cysteine is neutral when protonated. The side chain pK_a for cysteine is 8.36, so at $pH = 7.0$, $pH < pK_a$. Under such conditions, the side chain of cysteine is protonated, and thus neutral. Choice A is eliminated. Histidine is neutral when deprotonated. The side chain pK_a for histidine is 6.05, so at $pH = 7.0$, $pH > pK_a$. Under such conditions, the side chain of histidine is deprotonated, and thus neutral. Choice B is eliminated. Tyrosine is neutral when protonated. The side chain pK_a for tyrosine is 10.07, so at $pH = 7.0$, $pH < pK_a$. Under such conditions, the side chain of tyrosine is protonated, and thus neutral. Choice D is eliminated. Lysine is neutral when deprotonated (as is the case with the three basic amino acids). The side chain pK_a for lysine is 10.80, so at $pH = 7.0$, $pH < pK_a$. Under such conditions, the side chain of lysine is protonated, and thus cationic. Choice C is the best answer, because the side chain is charged.

Questions 92–100

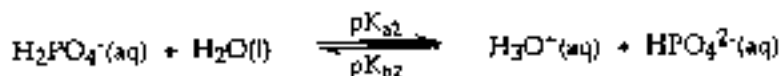
Not Based on a Descriptive Passage

92. **Choice C is correct.** The lowest pH is associated with the solution with the greatest hydronium concentration. The greatest hydronium concentration depends on the concentration and the strength of the acid. To lower pH, the concentration of an acid may be increased, or a stronger acid, with a lower pK_a value, may be employed. This means that the correct answer is a combination of lowest pK_a and greatest concentration. Choice A gets eliminated for having the lowest concentration and a high pK_a value. Choice B gets eliminated, because it has a lower concentration and greater pK_a than choice C. The pH in choice C is half of the $pK_a + 0.5$, which is $1.65 + 0.5 = 2.15$. The pH in choice D is half of the pK_a , which is 4.55. Choice C has the lowest pH.
93. **Choice D is correct.** Acid rain contains airborne Lewis acids. Non-metal oxides such as sulfur oxides and nitrogen oxides make up most of the acid rain we study. This means that choice A is an example of acid rain. Once these non-metal oxides react with moisture in the air, they become hydrated, so acid rain does contain hydrated non-metal oxides (also known as non-metal hydroxides). A Lewis acid is an electron-pair acceptor, and non-metal oxides qualify in this category. Choice C is a valid description of an acid rain component. Metal hydroxides are basic, so they are not found in acid rain. Choice D is the best answer.
94. **Choice B is correct.** The pH of a conjugate pair is found using the Henderson-Hasselbalch equation. The lowest pH is attributed to the conjugate pair having the acid with the lowest pK_a and the mixture that most favors conjugate acid. The pK_a of ammonium is 9.26 (although you really just need to know it's between 9 and 11), while the pK_a of hydrofluoric acid is 3.17. This eliminates choices C and D. Choice B has more acid than base, while choice A has more base than acid. The best answer is choice B.

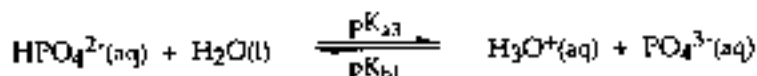
95. Choice B is correct. For a conjugate pair at 25°C, the pK_a of the acid when added to the pK_b of its conjugate base is equal to 14. This question boils down to which of choices represents a conjugate pair. Choice D can be eliminated immediately, because both values are pK_a values, and if they happen to sum to 14, it's purely coincidental. Phosphoric acid is triprotic, so there are three pK_a values. The first proton off (pK_{a1}) corresponds to the third proton back on (pK_{b3}), so $pK_{a1} + pK_{b3} = 14$. The second proton off (pK_{a2}) corresponds to the second proton back on (pK_{b2}), so $pK_{a2} + pK_{b2} = 14$, making choice B the correct answer. The third proton off (pK_{a3}) corresponds to the first proton back on (pK_{b1}), so $pK_{a3} + pK_{b1} = 14$.



First proton off/Third proton on $\therefore pK_{a1} + pK_{b3} = 14$



Second proton off/Second proton on $\therefore pK_{a2} + pK_{b2} = 14$



Third proton off/First proton on $\therefore pK_{a3} + pK_{b1} = 14$

96. Choice B is correct. $1.5\text{ M } H_3PO_4 = 4.5\text{ N } H_3PO_4$, because there are 3 equivalents of H^+ per H_3PO_4 . The normality is found by multiplying the molarity by the number of equivalents. Doubling the volume by adding pure water cuts the concentration (measured in either normality or molarity) in half. This makes normality one half of 4.5 N, so choice B, 2.25 N, is the best answer.
97. Choice B is correct. Because the pH is less than seven, the solution must contain an acid rather than a base. This eliminates choices C and D. If the acid were a strong acid, it would fully dissociate in water. This would lead to a hydronium concentration of 0.10 M, which would make the pH = 1. Given that the pH is greater than 1.0, the acid does not fully dissociate, so it is a weak acid. This eliminates choice A and makes choice B the best answer.
98. Choice C is correct. The highest pH is found in the solution with the highest concentration of hydroxide. Hydroxide concentration depends on the compound and its concentration. Choice A is a weak acid, so the pH is low. This eliminates choice A. Choice B is a strong acid, so pH is very low. This eliminates choice B. Choice C is the conjugate base of a weak acid, so it is a weak base. Choice C could be the correct answer. Choice D is the conjugate base of a strong acid, so it is a very weak base, meaning it has no significant impact on the pH of the aqueous solution. Because the pH of choice D is 7.0, choice D is eliminated. The best answer is choice C.
99. Choice B is correct. The solution is acidic, so the pH is less than 7.0, eliminating choice D. The acid is weak, so it partially dissociates in water. If the acid were strong, it would fully dissociate, leading to a hydronium concentration of 0.050 M and thus a pH of 1.30. Carboxylic acids are weak acids, so the pH is higher than 1.3. This eliminates choice A. The pH is less than the pK_a , so choice C is eliminated. The only choice remaining is choice B. If you choose to do so, you can solve for the pH exactly. For a weak acid with pK_a between 2 and 12 in an aqueous solution with $[HA]_{initial} > K_a$, use the shortcut equation to determine the pH. The pK_a is 4.89. The $[HA]$ is 0.05 M, and $\log 0.05 = -\log 20 = -(\log 10 + \log 2) = -(1 + 0.3) = -1.3$.

$$pH = \frac{1}{2}pK_a - \frac{1}{2}\log [HA] = \frac{1}{2}(4.89) - \frac{1}{2}\log (0.050) = 2.45 - \frac{1}{2}(-1.3) = 2.45 + 0.65 = 3.10$$

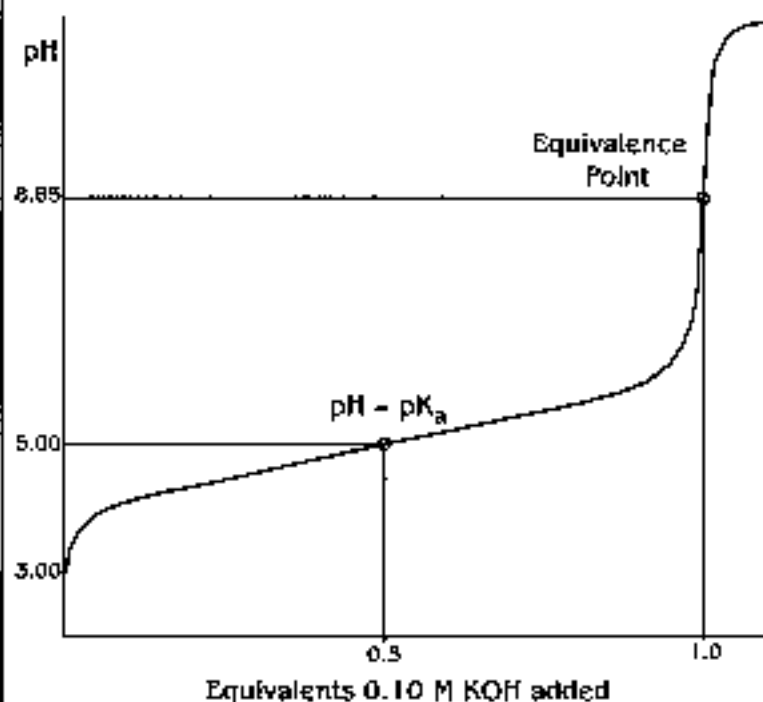
The value confirms that choice B is the best answer.

100. Choice C is correct. Hydrobromic acid (HBr) is a strong acid, so a 0.10 M solution has a pH of 1.00. Choice A is valid and thus eliminated. Formic acid (HCO_2H) is a weak acid, so it does not fully dissociate. It has a pH greater than 1.0 but less than 7.0. Choice B could be valid, so it is unlikely. To eliminate it with certainty, we need to do a calculation. We shall hold off on calculating until we have evaluated the other answers. Sodium acetate (NaOAc) is a weak base, so it does not fully hydrolyze. It has a pH greater than 7.0 but less than 13.0. Choice C is invalid, so it is the best answer. Potassium hydroxide (KOH) is a strong base, so a 0.10 M solution has a pH of 13.00. Choice D is valid and thus eliminated.

Section V

Buffers and Titration

by Todd Bennett



Buffers

- a) Buffer Composition
- b) pH Range for Buffers
- c) Buffer Recipe
- d) Physiological Buffers

Titration Curves

- a) Quantitative Reactions
- b) Fundamental Curve Shapes
- c) Plotting a Titration Curve
- d) Concentration Effects
- e) Strength Effects
- f) Polyprotic Acids

Indicators

- a) Composition and Function
- b) Detecting the Equivalence Point
- c) Estimating Solution pH



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Buffers & Titration Section Goals



Know how to approximate the pH of a solution using a titration curve.

By knowing the relative mole quantities of what has been mixed in solution, a pH value can be estimated from the position on a titration curve corresponding to the mixture. This requires being able to identify the shape of a titration curve based on the components in the mixture.



Understand how polyprotic acids affect the pH of a solution.

Polyprotic acids have multiple equivalent points, of which only the last will fluctuate with concentration. Know that amino acids are a subset of polyprotic acids. Be able to determine the pH at middle equivalents points by applying the equation for pI (averaging the two respective pK_a values).



Know the key points along all titration curves.

On titration curves associated with strong reagents, the pH at equivalence is always equal to 7.0. On titration curves associated with a weak acid or weak base, titrated by a strong reagent, the pH at the half-titrated point is equal to the pK_a of the weak acid. The pH at equivalence can be approximated by averaging the pK_a of the weak acid and the pH of the titrant.



Know the role of an indicator in titration and solution pH.

An indicator is used to make the endpoint of the titration visible. An indicator is a species that has a different color for the conjugate acid and conjugate base. Most indicators are organic compounds with a great degree of conjugation, and the color is caused by a transition from the π level to the π^* level. Indicators can also be used to predict the pH of a solution.



Know how conjugate pairs and buffers work.

A buffer is formed when a weak acid and its conjugate base are added to the aqueous solution. Because there exists an equilibrium between the two species, as long as both are present in solution, the hydronium ion concentration will remain fairly constant, therefore the pH will also remain constant. The effect is known as "buffering." You must understand buffers and how the pH is found from the Henderson-Hasselbalch equation.



Recognize and be able to generate titration curves for any titration.

You must be able to identify the titrant and the species being titrated when you look at the titration curve. Features to note are the weak acid lip, the strong acid sigmoidal shape, the ascent (associated with the titration of an acid by a base) or descent (associated with the titration of a base by an acid) of the curve, and the number of inflection points (indicative of whether the compound is polyprotic or polybasic).

Buffers and Titration

A buffer is a solution where pH remains relatively constant after the addition of either strong acid or strong base. The pH may vary slightly, but for all intents and purposes, it does not change significantly. Buffers play a major role in physiology and biochemistry, so understanding how they work is critical. They can be made in one of two ways. The first method involves combining a conjugate pair in roughly equal mole portions. The second method involves partially titrating a weak acid with roughly half of an equivalent of strong base, or by partially titrating a weak conjugate base with roughly half of an equivalent of strong acid. The pH of a buffered solution is determined using the Henderson-Hasselbalch equation. The data associated with buffers are generally easy to work with in a conceptual sense. Titration curves have buffering regions, so understanding buffers can help you to understand titration better.

Neutralization is the mixing of equal mole portions of an acid with a base, regardless of their concentrations and strengths. To neutralize an acid, an equal mole quantity of base must be added to solution. To neutralize a base, an equal mole quantity of acid must be added to solution. A neutralized solution has moles H_3O^+ equal to moles OH^- . Addition of a base to an acid (or acid to a base) yields water and a salt upon neutralization. This is shown in Reaction 5.1:



Reaction 5.1

Depending on the strength of the acid and base, the pH at the neutralization point (also referred to as the endpoint and the equivalence point in titration) varies. **Neutralization does not mean to make the pH of the solution equal to 7.** When the base is stronger than the acid, the neutralized solution is slightly basic, so the pH is greater than 7.0. When the base is weaker than the acid, the neutralized solution is slightly acidic, so the pH is lower than 7.0. When the base and acid are equally strong, the neutralized solution is neutral, so the pH is equal to 7.0. The three possible combinations are summarized below.

$\text{HX(aq)} + \text{MOH(l)} \rightleftharpoons \text{H}_2\text{O(l)} + \text{MX(aq)}$	
Strong acid + Strong base	pH _{at equivalence} = 7.0
Strong acid + Weak base	pH _{at equivalence} < 7.0
Weak acid + Strong base	pH _{at equivalence} > 7.0

The pH is not always 7 at the equivalence point of a titration. For a weak acid titrated by a strong base, the equivalence point is the point at which it is completely converted into its conjugate base. The conjugate base will yield a pH greater than 7 so the pH is greater than 7 at the equivalence point.

The last section in this chapter involves detecting the equivalence point. Visual indicators that change color upon changing pH are a typical component of any general chemistry curriculum. They are often highly conjugate organic molecules. The color associated with an indicator is a reflective color, resulting from the absorbance of a photon accompanied by the excitation of an electron from a π -bonding molecular orbital to a π -antibonding molecular orbital. The transition energy changes when the compound gets deprotonated, so the energy of the photon absorbed, and ultimately the color of light reflected also changes. Hence, any pH-sensitive chromophore is an indicator.

Buffers

Buffer Composition

Buffers are solutions that resist drastic changes in pH. Buffers are made of a roughly equal mole mixture of a weak acid and its weak conjugate base in an aqueous solution. Both the acid and the base of the conjugate pair must be weak in order to form a buffer solution. This is so that the equilibrium between the two species can be controlled by the environment. With approximately equal molar quantities of conjugate acid and conjugate base in solution, the solution is resistant to pH change caused by the addition of either strong acid or strong base to solution. When both of the species in the conjugate pair are weak, the buffer can equilibrate in both the forward and reverse directions of the reaction to absorb any hydronium or hydroxide that may be added to solution. Addition of a strong acid to solution converts the weak base into its conjugate acid. This has little to no effect on the pH. Likewise, addition of a strong base to the solution converts the weak acid into its conjugate base and has little to no effect on the pH.

To emphasize the need for roughly equal portions, the following experiment may be studied. In this study, three mixtures of acid and conjugate base are generated. In the first system, weak acid is in extreme excess relative to its conjugate base. In the second system, the two species are in roughly equal concentrations. In the third system, weak base is in extreme excess relative to its conjugate acid. The results are shown in Figure 5-1.

System I: Mix 999 parts HA with 1 part A⁻. To this mixture, add 1 part OH⁻.

$$\text{Initially: } \frac{[A^-]}{[HA]} = \frac{1}{999} \Rightarrow \text{after addition of 1 part OH}^-: \frac{[A^-]}{[HA]} = \frac{2}{998}$$

Because $K_a = [H^+] \times \frac{[A^-]}{[HA]}$, and K_a is a constant, when $\frac{[A^-]}{[HA]}$ doubles, $[H^+]$ must be cut in half, and the pH changes. System I is NOT a buffer.

System II: Mix 500 parts HA with 500 part A⁻. To this mixture, add 1 part OH⁻.

$$\text{Initially: } \frac{[A^-]}{[HA]} = \frac{500}{500} \Rightarrow \text{after addition of 1 part OH}^-: \frac{[A^-]}{[HA]} = \frac{501}{499}$$

Because $K_a = [H^+] \times \frac{[A^-]}{[HA]}$, and K_a is a constant, when $\frac{[A^-]}{[HA]}$ barely changes, $[H^+]$ is constant, and the pH doesn't change. System II is a buffer.

System III: Mix 2 parts HA with 998 part A⁻. To this mixture, add 1 part OH⁻.

$$\text{Initially: } \frac{[A^-]}{[HA]} = \frac{998}{2} \Rightarrow \text{after addition of 1 part OH}^-: \frac{[A^-]}{[HA]} = \frac{999}{1}$$

Because $K_a = [H^+] \times \frac{[A^-]}{[HA]}$, and K_a is a constant, when $\frac{[A^-]}{[HA]}$ doubles, $[H^+]$ must be cut in half, and the pH changes. System III is NOT a buffer.

Figure 5-1

The conclusion is that the pH remains constant only if the weak acid and its conjugate base are in roughly equal concentration. The addition of strong base or strong acid shifts the ratio of weak acid (HA) to its conjugate base (A⁻), but the pH does not change, if the A⁻ : HA ratio is close to 1.0.

pH Range for Buffers

The experiment in Figure 1 explains why the weak acid and weak conjugate base must be present in roughly equal parts. If they are not relatively close in mole quantity, then the system does not act as a buffer. According to convention, the ratio can not exceed 10 : 1. Substituting 10 : 1 and 1 : 10 into the Henderson-Hasselbalch equation shows us that the range of a buffer is the pK_a of the weak acid ± 1 . The pH of a buffer solution obeys the Henderson-Hasselbalch equation, which is shown in Figure 5-2.

$$pH = pK_a + \log \frac{[A^-]}{[HA]} \quad \therefore pH = pK_a + \log \frac{\text{Moles conjugate base}}{\text{Moles conjugate acid}}$$

$$\text{Lowest } pH = pK_a + \log \frac{1}{10} = pK_a - 1 \quad \text{Highest } pH = pK_a + \log \frac{10}{1} = pK_a + 1$$

$$\therefore pH \text{ range} = pK_a \pm 1$$

Figure 5-2

The derivation of the Henderson-Hasselbalch equation from the K_a equation is on page 264. The Henderson-Hasselbalch equation shows that as [conjugate base] increases, buffer pH increases. The Henderson-Hasselbalch equation also shows that as [conjugate acid] increases, buffer pH decreases. It also offers quantifiable verification of the concept that when pH is greater than the pK_a , the solution is rich in conjugate base.

Example 5.1

If water is added to a buffer solution with $pH = 3.96$, what happens to the pH?

- The pH increases slightly.
- The pH decreases slightly.
- The pH remains the same.
- If the pH is greater than 7, then it decreases. If the pH is less than 7, then it increases.

Solution

Addition of water to a buffer equally dilutes the concentration of the weak acid and its weak conjugate base. This means that the mole ratio of the weak base to the weak acid does not change upon the addition of water. According to the Henderson-Hasselbalch equation, the pH of the solution does not change because pK_a is constant and the fraction has not changed. The result is that the pH of a buffer does not change when it is diluted. This is why the Henderson-Hasselbalch equation can be written as moles A^- over moles HA , as well as $[A^-]$ over $[HA]$.

Knowing the buffer range is important when making a buffer, which is accomplished in two steps. First, an acid within the range must be chosen. The pK_a of the acid should be as close as possible to the desired pH. Second, a mixture containing both weak acid (HA) and its weak conjugate base (A^-) should be formed so that the two species are in roughly equal concentration. Select an acid for the buffer whose pK_a value lies within the ± 1 range of the desired pH. For buffering at pH values between 2 and 5, carboxylic acids are typical. For buffering at pH values between 8 and 11, amines are typical.

Buffer Recipes

Buffers can be made either by mixing the conjugate pair together, or by partially titrating either component in a conjugate pair of weak reagents. The partial titration method can be either to half-titrate the weak acid with strong base, or to half-titrate the weak conjugate base with strong acid. First, a weak acid must be chosen with a pK_a value close to the desired pH. Buffers can be mixed by any of the methods shown in Figure 5-3.

1. Weak acid + the salt of the conjugate base in roughly equal mole proportions (e.g., HCO_2H with HCO_2Na)
2. Weak base + the salt of the conjugate acid in roughly equal mole proportions (e.g., NH_3 with NH_4Cl)
3. Weak acid and roughly half of an equivalent of strong base (e.g., $HOAc$ with half equivalent KOH)
4. Weak base and roughly half of an equivalent of strong acid (e.g., H_3CNH_2 with half equivalent HCl)

Figure 5-3

When using either of the last two methods, titration takes place until the desired pH is achieved within the buffering region. The buffering region (where the pH does not change appreciably) is found in the middle area of the titration curve, between the starting point and the equivalence point. Buffering occurs only with the titration of a weak reagent by a strong reagent. A strong acid combined with its conjugate base or a strong base combined with its conjugate acid do not produce a buffer, so strong acid and strong base titration curves have no buffering region.

Example 5.2

Which of the following solutions results in a buffer with a pH of 5.0, given that HA has a pK_a of 4.7?

- A. HA with one-half equivalent of A^-
- B. A^- with one equivalent of HA
- C. HA with one-third equivalent of OH^-
- D. A^- with one-third equivalent of H_3O^+

Solution

The pH is greater than the pK_a , so the solution must be rich in the deprotonated species. In choice A, $[HA] > [A^-]$, so the pH is less than pK_a (4.7), meaning that choice A can be eliminated. In choice B, $[HA] = [A^-]$, so the pH equals pK_a (4.7), meaning that choice B can be eliminated. In choice C, one-third of an equivalent of A^- forms from the reaction, and two-thirds of an equivalent of HA is left over. In choice C, $[HA] > [A^-]$, so the pH is less than pK_a (4.7), meaning that choice C can be eliminated. In choice D, one-third of an equivalent of HA forms from the reaction, and two-thirds of an equivalent of A^- is left over. In choice D, $[HA] < [A^-]$, so the pH is greater than pK_a (4.7). Choice D is the best option for those who like to pick the correct answer. To solve for the exact numerical value, the Henderson-Hasselbalch equation must be employed. The values differ by 0.3, and the antilog of 0.3 is 2, so the correct answer must form a 2 : 1 ratio of conjugate base to acid.

Example 5.3

Which of the following combinations, when mixed in the correct ratio, produce a buffer?

- A. NaOH/NaCl
- B. NaNO₃/HNO₃
- C. HCl/KOH
- D. NH₃/HCl

Solution

Choice A is a strong base and a neutral salt, which does not make a buffer. Choice B is a neutral salt and a strong acid, which does not make a buffer. Choice C is a strong acid and a strong base, which also does not make a buffer. By elimination, choice D is correct. To be a buffer, the weak base (NH₃) must be half-titrated by the strong acid (HCl). Ammonia and hydrochloric acid make a buffer when mixed in the correct ratio (2 : 1).

Example 5.4

Which of the following does NOT form a buffer when added to NaHCO₃(s)?

- A. NaOH(aq)
- B. HCl(aq)
- C. H₂CO₃(aq)
- D. H₂O(l)

Solution

Adding half of an equivalent of NaOH to a sodium bicarbonate solution converts half of the HCO₃⁻ to its conjugate base CO₃²⁻. The mixture of the two forms a buffer, so choice A is correct. Adding half of an equivalent of HCl to a sodium bicarbonate solution converts half of the HCO₃⁻ to its conjugate acid H₂CO₃. The mixture of the two forms a buffer, so choice B is correct. Adding an equivalent of H₂CO₃ to a sodium bicarbonate solution results in equal portions of the HCO₃⁻ and its conjugate acid H₂CO₃. The mixture of the two forms a buffer, so choice C is correct. Adding water does not make a buffer, because water is amphoteric, so it does not convert sodium bicarbonate to either its conjugate acid or conjugate base. To be a buffer, there must be both a weak acid and its weak conjugate base present in solution at the same time. This means that choice D is correct.

Example 5.5

If the ratio of base to acid in a conjugate pair is 3:1, and the weak acid has a $K_a = 1.0 \times 10^{-5}$, what can be said about the pH of the buffer solution?

- A. pH < 5
- B. pH = 5
- C. $5 < \text{pH} < 7$
- D. pH > 7

Solution

If the ratio of base to acid were 1:1, the pH would equal the pK_a. Because the base is in excess, the pH is greater than the pK_a. The pK_a is $-\log 1.0 \times 10^{-5} = 5$. This eliminates choices A and B. According to the Henderson-Hasselbalch equation, the pH is log 3 greater than the pK_a. $\log 10 \approx 1$, and log 10 is greater than log 3, so pH ($= \text{pK}_a + \log 3$) < 6. The best choice is C.

Example 5.6

If 1.0 moles of a weak acid in 1.0 liters of water are treated with 0.4 moles of strong base, what is the pH of the solution? (K_a for the weak acid is 2.0×10^{-4})

- A. $\text{pH} < 3.7$
- B. $\text{pH} = 3.7$
- C. $3.7 < \text{pH} < 7$
- D. $\text{pH} > 7$

Solution

If the ratio of base to acid were 1:1, the pH would equal the $\text{p}K_a$. The $\text{p}K_a$ is $-\log 2.0 \times 10^{-4} = 4 - \log 2 = 4 - 0.3 = 3.7$, as hinted at by the answer choices. To be half-titrated, it would require 0.5 moles of strong base. At the half-titration point, the $\text{pH} = \text{p}K_a$. With only 0.4 moles of base, the halfway point is not yet reached and there is excess weak acid relative to conjugate base (0.6 moles to 0.4 moles). According to the Henderson-Hasselbalch equation, the pH is less than the $\text{p}K_a$. The best answer is choice A. The pH is less than $\text{p}K_a$, which is 3.7. Be aware that this question could about the $[\text{H}^+]$ as well as the pH of the solution. If a solution is half-titrated, then the $K_a = [\text{H}^+]$. This can be tricky, but the answer is found from the Henderson-Hasselbalch equation. $\text{pH} = \text{p}K_a + \log \frac{0.4}{0.6}$, so $\text{pH} < \text{p}K_a$.

Example 5.7

A buffer made by mixing 100 mL of 0.5 M HOAc ($K_a = 1.8 \times 10^{-5}$) with 25 mL of 1.0 M KOH has a pH approximately equal to which of the following values?

- A. 0.2
- B. 4.7
- C. 7.0
- D. 9.3

Solution

There are 0.05 moles of HOAc present and 0.025 moles of KOH present. This means that exactly half of an equivalent of strong base has been added to a weak acid, converting half of the original weak acid to its conjugate base. Half of the original weak acid remains unreacted. This means that $\text{pH} = \text{p}K_a$. The value for $\text{p}K_a$ is solved for as follows:

$$\text{p}K_a = -\log (1.8 \times 10^{-5}) = -\log 1.8 - \log 10^{-5} = -\log 1.8 - (-5) = -\log 1.8 + 5$$

Log 1.8 is less than 1, so the pH is greater than 4 and less than 5. Choice B is best.

Example 5.8

A buffered solution initially has a pH of 8.31. When five drops of 12 M HCl are added to a 500-mL beaker filled with this buffered solution, what would be expected for the final pH value?

- A. 3.31
- B. 8.26
- C. 8.31
- D. 8.36

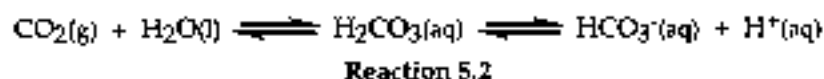
Solution

Adding a little acid decreases the pH slightly. Choice A shows a lower pH value, but a drastically lower pH value. The change is small, so the answer is choice B.

Physiological Buffers

Physiological pH is considered to be 7.4 (although venous and arterial pH vary, and gastric fluids are highly acidic). The pK_a value for physiological acids should be considered relative to this value in order to determine the structure of the compound *in vivo*. When the pH of the environment is greater than the pK_a of the species, it exists predominantly in its deprotonated state. Equally, when the pH of the environment is less than the pK_a of the species, it exists predominantly in its protonated state.

A prime example of physiological buffering involves carbonic acid, which is used to buffer the blood of biological systems. Arterial blood is oxygen rich (and thus carbon dioxide-poor), so it has a slightly higher pH than venous blood which is carbon dioxide-rich. Carbon dioxide when added to water undergoes the complex equilibrium shown as Reaction 5.2.



This equilibrium regulates blood pH, so any conditions (or disorders) that affect carbon dioxide levels in physiological systems also affect the pH and buffering of blood. For instance, emphysema hinders the uptake of oxygen from the lung and the release of carbon dioxide into the lung. The blood can compensate for the reduced uptake of oxygen by increasing heart rate and producing more red blood cells. But the release of carbon dioxide is not as easily adjusted for. The consequence is that carbon dioxide levels in the blood increase, causing the blood to be more acidic than normal. This condition is known as *respiratory acidosis*. This can inhibit the binding of oxygen, as is demonstrated by the Bohr effect. Using the same logic, it can be concluded that *respiratory alkalosis* results from the excessive loss of carbon dioxide.

Gastric fluids are highly acidic (rich in HCl), so the loss of gastric fluids results in the loss of acid from the body, producing a condition known as *metabolic alkalosis*. Loss of HCl can accompany vomiting or the pumping of the stomach. When the stomach is pumped, the acidic solution that is removed must be replenished to reduce the risk of metabolic alkalosis. As a point of interest, food poisoning and drug overdose patients are given a solution of charcoal and water to drink, which absorbs the organic toxin. The porous carbon matrix binds organic molecules better than the water, especially compounds with π -bonds.

Waste products exit the body in a chemically neutralized state (and believe me that if they didn't, you'd notice), so the lower end of the GI track must be basic. Losing lower intestinal fluids results in the loss of basic metabolites, in particular, HCO_3^- (bicarbonate). Diarrhea (which can be caused by Olesstra consumption) results in the loss of basic metabolites, producing a condition known as *metabolic acidosis*. The intertwining of acid-base chemistry and physiology is a perennial favorite on the MCAT. Figure 5-4 summarizes the pH-related disorders of the body.

Retention of CO_2 : Blood pH $\downarrow$ $\therefore$ Respiratory acidosis
Loss of CO_2 : Blood pH $\uparrow$ $\therefore$ Respiratory alkalosis
Loss of HCO_3^- : Blood pH $\downarrow$ $\therefore$ Metabolic acidosis
Loss of H_3O^+ : Blood pH $\uparrow$ $\therefore$ Metabolic alkalosis

Figure 5-4

Titration Curves

Quantitative Reactions (Titration Theory)

Acid-base titration is the quantitative addition of a titrant acid to a base in solution, or of a titrant base to an acid in solution. What is meant by quantitative addition is that the volumes of acid and base are measured precisely. The titration is said to reach its equivalence point when the moles of acid are equal to the moles of titrant base. The volumes must be known precisely in order to determine the relative concentrations, using the relationship that moles are equal to the product of volume and molarity. We shall consider two versions of titration. The first case is the titration of a strong reagent by a strong reagent, such as strong acid by strong base or strong base by strong acid. The second case is the titration of a weak reagent by a strong reagent, such as weak acid by strong base or weak base by strong acid. There is no titration of a weak reagent with a weak reagent, because the two weak reagents do not react with one another.

We shall first consider what is in solution at different points during the titration. The two species can be mixed in three different ways: excess of one, excess of the other, or in equal portions. Let us address each of these scenarios, along with the initial point of titration. When less than one equivalent of the titrant is added to the original reagent, this mixture is found in a region on the curve before the equivalence point. When exactly one equivalent of the titrant is added to the original reagent, this is the equivalence point. When more than one equivalent of the titrant is added to the original reagent, this mixture is found in a region on the curve beyond the equivalence point. Table 5.1 shows the different conditions and pH calculations along a strong-by-strong titration curve.

Strong Acid titrated by a Strong Base			
Point on Curve	Species in Solution	pH Range	pH Calculation
Initial	Pure strong acid	$\text{pH} \ll 7$	$\text{pH} = -\log [\text{HX}]_{\text{initial}}$
Before equivalence	Leftover H_3O^+	$\text{pH} < 7$	$\text{pH} = -\log [\text{H}_3\text{O}^+]_{\text{excess}}$
At equivalence	H_2O & neutral salt	$\text{pH} = 7$	$\text{pH} = 7$ at equivalence
Past equivalence	Leftover OH^-	$\text{pH} > 7$	$\text{pOH} = -\log [\text{OH}^-]_{\text{excess}}$

Table 5.1

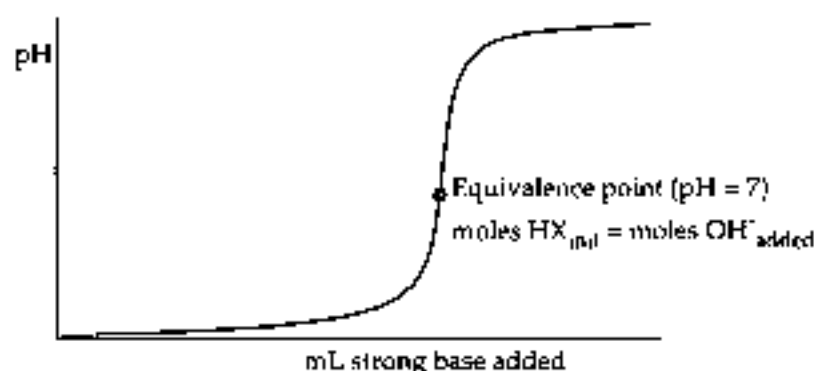
When a weak reagent is titrated by a strong titrant, it's different than the strong-by-strong titration. The difference when dealing with a weak reagent is that a weak conjugate is formed as the product, so there is an equilibrium between products and reactants. The pH calculations must consider all species in solution that affect the pH. Table 5.2 shows the different conditions and pH calculations along a weak-by-strong titration curve.

Weak Acid titrated by a Strong Base			
Point on Curve	Species in Solution	pH Range	pH Calculation
Initial	Pure weak acid	$\text{pH} < 7$	$\text{pH} = \frac{1}{2}\text{pK}_a - \frac{1}{2}\log [\text{HA}]$
Before equivalence	HA and A^- (buffer)	$\text{pK}_a \pm 1$	$\text{pH} = \text{pK}_a + \log \frac{[\text{A}^-]}{[\text{HA}]}$
At equivalence	A^- (diluted)	$\text{pH} > 7$	$\text{pOH} = \frac{1}{2}\text{pK}_b - \frac{1}{2}\log [\text{A}^-]$
Past equivalence	Leftover OH^-	$\text{pH} > 7$	$\text{pOH} = -\log [\text{OH}^-]_{\text{excess}}$

Table 5.2

Fundamental Curve Shapes

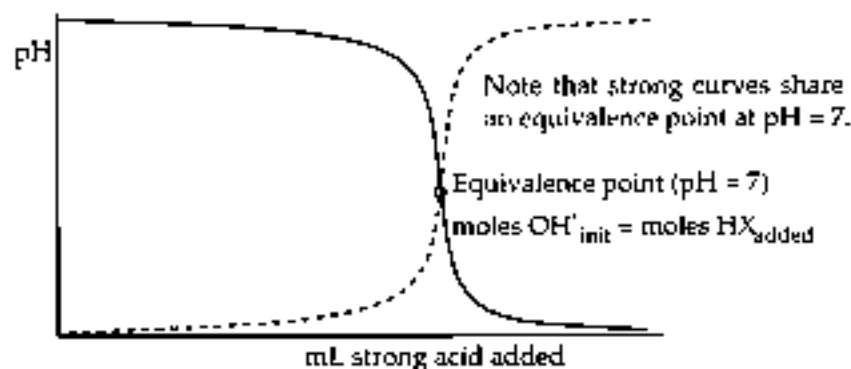
The mixtures listed in Table 5.1 can be studied quantitatively with the help of graphs. Strong acid and strong base titration curves have distinct features. Figure 5-5 shows the titration of a strong acid by a strong base. We shall call this Case 1a.



Case 1a: Strong acid titrated by a strong base

Figure 5-5

Figure 5-6 shows the titration of a strong base by a strong acid. We shall call this Case 1b.

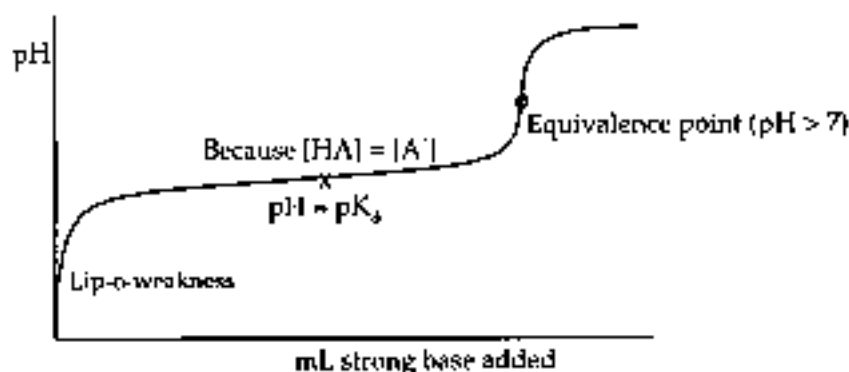


Case 1b: Strong base titrated by a strong acid

Figure 5-6

In Cases 1a and 1b, the equivalence point is at $\text{pH} = 7.0$, because the neutralized product is a neutral salt formed from the reaction of a strong acid with a strong base. The graph, although not extended far enough to tell, is symmetric about the equivalence point. The shape of the curve is referred to as *sigmoidal*. The reagent when initially added to water fully dissociates (or in the case 1b, hydrolyzes with a base), giving the highest concentration at first. Over the course of the titration, the concentration is reduced. This means that the pH gradually increases during the entire titration, with rapid pH change near the equivalence point. Because pH is measured on a log scale, the graph assumes the characteristic sigmoidal shape. The reasoning behind this is that as you approach $\text{pH} = 7$, each change of 1.0 in the pH requires ten times less titrant. For instance, in going from $\text{pH} = 1$ to $\text{pH} = 2$, the hydronium concentration goes from 0.10 M to 0.01 M, a change of 0.09 M H_3O^+ . In going from $\text{pH} = 2$ to $\text{pH} = 3$, the hydronium concentration goes from 0.01 M to 0.001 M, a change of 0.009 M H_3O^+ . The change in hydronium concentration is ten times less when the pH is one unit closer to 7.0.

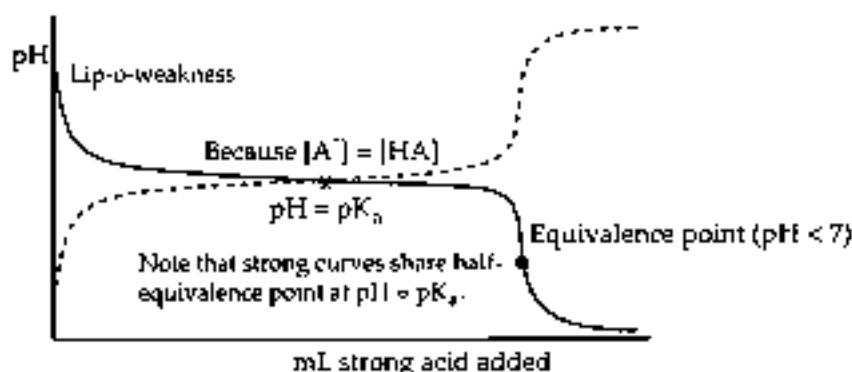
The mixtures listed in Table 5.2 can be studied quantitatively with the help of graphs. Weak acid and weak base titration curves have distinct features. Case 2a, shown in Figure 5-7, shows the titration of a weak acid by a strong base.



Case 2a: Weak acid titrated by a strong base

Figure 5-7

Case 2b, shown in Figure 5-8, shows the titration of a weak base by a strong acid.



Case 2b: Weak base titrated by a strong acid

Figure 5-8

Titration curves exhibit an initial cusp when the reagent being titrated is weak. This is referred to as a *lip-o-weakness*, and may be used to distinguish the nature of the reagent from its titration curve. The lip-o-weakness is due to the fact that the equilibrium between weak acid and conjugate base heavily favors one of the two species, so the pH changes significantly. This can be seen the experiment in Figure 5-1. The pH at half-equivalence point is always equal to the pK_a , which can be inferred from the Henderson-Hasselbalch equation.

In Case 2a, the equivalence pH is above 7.0, because the neutralized product is a weak base (the conjugate base of the weak acid). The weaker the acid titrated, the stronger the conjugate base formed at equivalence, and consequently the higher the pH at the equivalence point. The pH of the conjugate base depends on both the concentration and the K_b of the base at equivalence.

In Case 2b, the equivalence pH is below 7.0, because the neutralized product is a weak acid (the conjugate acid of the weak base). The weaker the base titrated, the stronger the conjugate acid formed at equivalence, and consequently the lower the pH at the equivalence point. The pH of the conjugate acid depends on both the concentration and the K_a of the acid at equivalence.

Example 5.9

All of the following are a feature of a titration curve in which both reactants are strong EXCEPT:

- A. an equivalence pH of 7.0.
- B. a sigmoidal graph shape.
- C. the most rapid pH change near pH = 7.
- D. a large change in pH as the first few drops of titrant are added.

Solution

This question tests your knowledge about strong-by-strong titration curves. When both reagents are strong, they neutralize each other and leave behind a pH = 7 aqueous salt solution. This makes choice A valid. Because the reagent fully dissociates (in the case of a strong acid) or hydrolyzes (in the case of a strong base), the highest concentrations are initially observed. The result is a slow change in pH until just before equivalence. This causes the curve to be sigmoidal and makes choice B valid. The pH change is always rapid near the equivalence point, so for strong-by-strong curve, the pH change near pH = 7 is the most drastic. Choice C is valid. The pH of the titration mixture is nearly constant at the start of the titration. This is to say that a strong-by-strong curve is "lip-free" in the beginning. Choice D is invalid, and is thus the best answer.

Example 5.10

All of the following are features of a titration curve in which one reactant is weak and the titrant is strong EXCEPT:

- A. a half-equivalence pH equal to pK_a .
- B. a sinusoidal graph shape.
- C. a pH at equivalence that is not equal to 7.
- D. a large change in pH as the first few drops of titrant are added

Solution

This question tests your knowledge about weak-by-strong titration curves. When a weak reagent is half-titrated by a strong titrant, half of the original species is converted to its conjugate. This leaves half of the original reagent unreacted in solution. Because the two components of the conjugate pair are in equal concentration, the pH of the solution is equal to the pK_a . This makes choice A a valid statement and eliminates it. A sinusoidal graph implies a sine wave, which is not observed with weak-by-strong titration curves. This makes choice B invalid, and thus the correct choice. When a weak acid is fully neutralized by a strong base, it forms its conjugate base. This results in a solution with pH greater than 7. The equivalence pH is not equal to 7. When a weak base is fully neutralized by a strong acid, it forms its conjugate acid. This results in a solution with pH less than 7. The equivalence pH is not equal to 7. The pH at equivalence for a weak-by-strong titration curve is not 7, so choice C is valid and thus eliminated. Early in a weak-by-strong titration curve, the equilibrium between conjugates favors the one initially in solution. As a titrant is added, the equilibrium shifts drastically, resulting in a big change in the hydronium concentration. A big change in the hydronium concentration causes a significant change in the pH of the solution. This makes choice D a valid statement and eliminates it.

To understand titration curves better, Table 5.3 shows how the pH is calculated at different points along a titration. The equations used to calculate pH are all in Table 5.2. The titration represents the titration of a hypothetical weak acid with a pK_a of 5.0 and an initial concentration of 0.10 M. The weak acid is titrated by 0.10 M KOH(aq). The pH represents the solution pH after the titrant base has reacted with the weak acid in solution. The value points in Table 5.3 are plotted in Figure 5-9, and by connecting the dots (yes, that's right, studying for the MCAT involves a little connect-the-dots), a titration curve can be generated.

mL KOH	pH calculation	pH
0.00	$pH = \frac{pK_a}{2} - \frac{1}{2} \log [HA] = \frac{5}{2} - \frac{1}{2} \log (0.10) = 2.5 + 0.5 = 3.0$	3.00
1.00	$pH = pK_a + \log \frac{mL OH^-}{50 - mL OH^-} = 5.0 + \log \frac{1.0}{49.0} = 5.0 - \log 49$	3.31
2.00	$pH = pK_a + \log \frac{mL OH^-}{50 - mL OH^-} = 5.0 + \log \frac{2.0}{48.0} = 5.0 - \log 24$	3.62
5.00	$pH = pK_a + \log \frac{mL OH^-}{50 - mL OH^-} = 5.0 + \log \frac{5.0}{45.0} = 5.0 - \log 9$	4.05
10.00	$pH = pK_a + \log \frac{mL OH^-}{50 - mL OH^-} = 5.0 + \log \frac{10.0}{40.0} = 5.0 - \log 4$	4.40
12.50	$pH = pK_a + \log \frac{mL OH^-}{50 - mL OH^-} = 5.0 + \log \frac{12.5}{37.5} = 5.0 - \log 3$	4.52
25.00	$pH = pK_a + \log \frac{mL OH^-}{50 - mL OH^-} = 5.0 + \log \frac{25}{25} = 5.0 + \log 1$	5.00
37.50	$pH = pK_a + \log \frac{mL OH^-}{50 - mL OH^-} = 5.0 + \log \frac{37.5}{12.5} = 5.0 + \log 3$	5.48
40.00	$pH = pK_a + \log \frac{mL OH^-}{50 - mL OH^-} = 5.0 + \log \frac{40.0}{10.0} = 5.0 + \log 4$	5.60
45.00	$pH = pK_a + \log \frac{mL OH^-}{50 - mL OH^-} = 5.0 + \log \frac{45.0}{5.0} = 5.0 + \log 9$	5.95
48.00	$pH = pK_a + \log \frac{mL OH^-}{50 - mL OH^-} = 5.0 + \log \frac{48.0}{2.0} = 5.0 + \log 24$	6.38
49.00	$pH = pK_a + \log \frac{mL OH^-}{50 - mL OH^-} = 5.0 + \log \frac{49.0}{1.0} = 5.0 + \log 49$	6.69
50.00	$pH = \frac{pK_a + pH_{\text{titrant}}}{2} = \frac{5.0 + 13}{2} = 18$	8.85
51.00	$pOH = -\log [OH^-]_{\text{excess}} = -\log \left(0.10 \times \frac{1}{101} \right) \Rightarrow pH = 14 - pOH$	11.00
55.00	$pOH = -\log [OH^-]_{\text{excess}} = -\log \left(0.10 \times \frac{5}{105} \right) \Rightarrow pH = 14 - pOH$	11.68

Table 5.3

Initially (at 0.00 mL added), the pH is found using the shortcut equation. Before equivalence (from 1.00 mL added to 49.00 mL added), the pH is found using a modified version of the Henderson-Hasselbalch equation, where the moles OH^- added are substituted for moles A^- (given that the OH^- converts to A^-) and $50 - mL OH^-$ is substituted for moles HA , because that describes the leftover HA . At equivalence (at 50.00 mL added), the pH is approximated by averaging pH of the titrant base and the pK_a of the acid. The approximation is off by 0.15, so the pH column shows the actual pH at equivalence. After equivalence (beyond 50.00 mL added), the pOH is found by taking the negative log $mL OH^- - 50$ (for what is reacted) over the total volume ($50 + mL OH^-$).

Plotting a Titration Curve

The points from Table 5.3 are represented in Figure 5-9 as circles. The curve is drawn to fit the circles. The shape of the curve for a weak acid is distinct. The buffer region is not perfectly flat, showing that pH changes slightly in the buffer region. The pH values at 1.00 mL and 2.00 mL are not exact, because the Henderson-Hasselbalch equation does not hold as well outside the $pK_a \pm 1$ range. Nevertheless, the values are close enough to generate a reasonable titration curve.

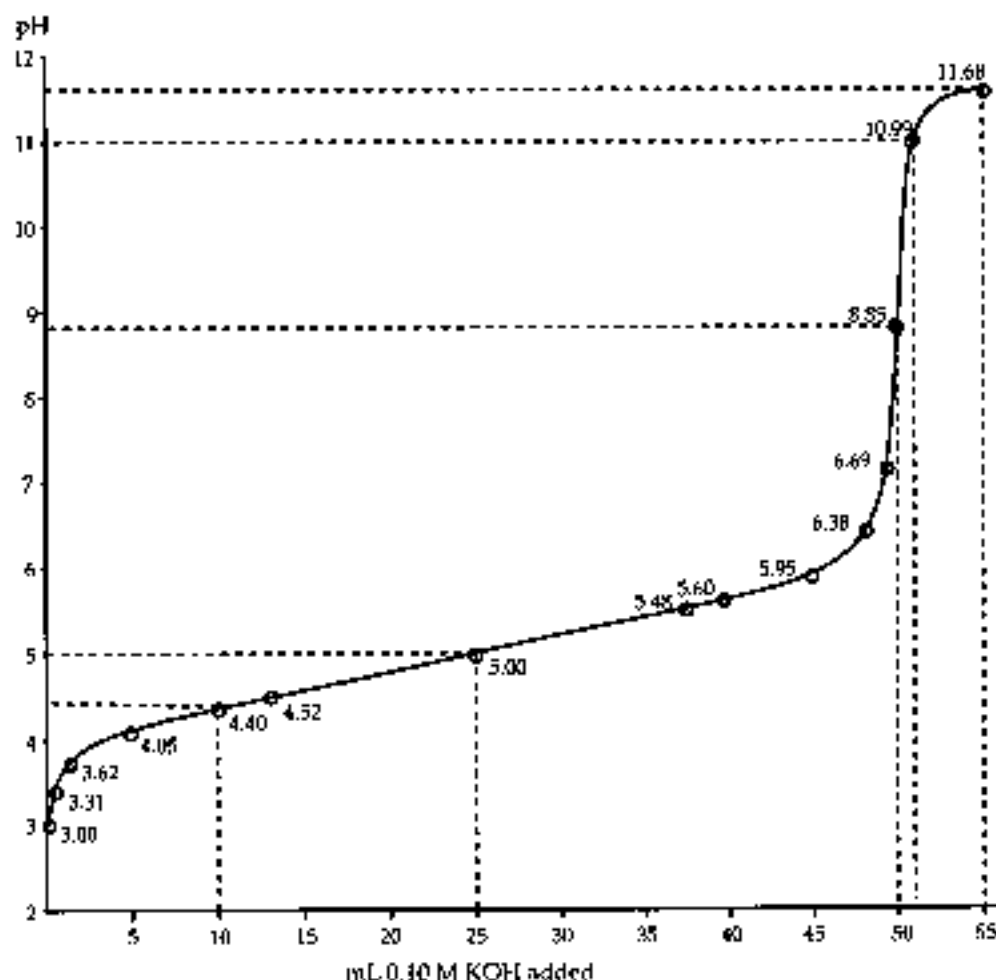


Figure 5-9

The titration curve should become familiar with enough examples. What makes curves useful is that they summarize a great deal of information. If you think of titration curves in terms of equivalents and regions, you can extract a substantial amount of information from them. For instance, when 21.27 mL of 0.1 M KOH has been added, the pH is roughly 4.7 to 4.8. That range is small enough that an educated guess can be made on a multiple-choice question. We shall emphasize using titration curves in lieu of calculating the pH, when it comes to buffers and other mixtures.

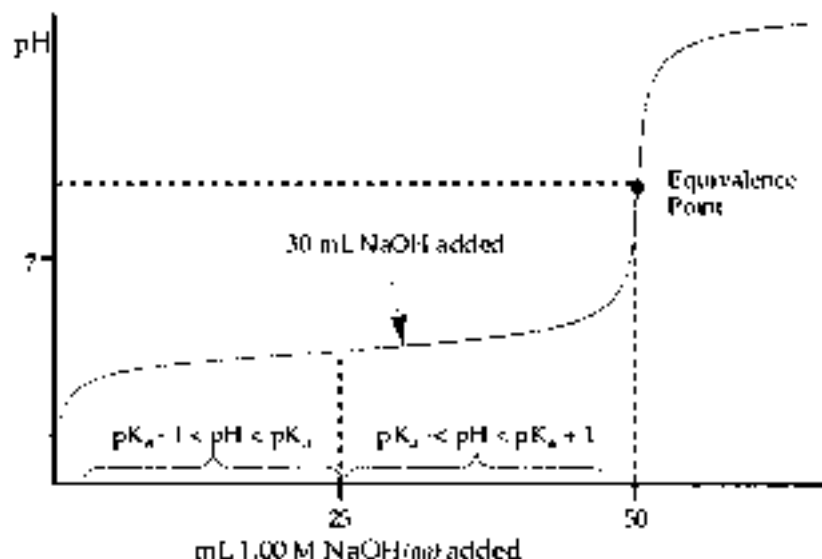
Example 5.11

What is the pH after 30 mL of 1.00 M NaOH(aq) has been added to 100 mL 0.50 M HOAc(aq)? HOAc has a $pK_a = 4.74$.

- A. 3.51
- B. 4.56
- C. 4.92
- D. 5.97

Solution

Because the strong base is twice as concentrated as the weak acid, only half the volume of strong base (relative to the weak acid) is required to reach the equivalence point. This means that 50 mL of 1.00 M NaOH(aq) fully neutralizes the 100 mL of 0.50 M HOAc(aq). The halfway point of the titration is reached when exactly 25 mL of 1.00 M NaOH(aq) has been added. At the halfway point, the pH of the solution equals the pK_a of the weak acid. The additional strong base beyond the 25 mL makes the pH of the solution slightly greater than the pK_a of the acid, 4.74. The best choice is answer C. The titration curve below shows a summary of the intuitive approach:



When 30 mL has been added, the mixture is just beyond the half-titrated point on the titration curve (as shown by the arrow). This makes the pH fall into the range of $pK_a < pH < pK_a + 1$, according to the titration curve. According to the Henderson-Hasselbalch equation, the pH equals the $pK_a + \log(\text{conjugate base over acid})$. Past the half-titrated point, the concentration of the conjugate base exceeds the concentration of the acid, so the ratio of conjugate base to acid is greater than one. The log of a number greater than 1.0 is a positive value. When a positive value is added to the pK_a , the final value is greater than the pK_a , confirming that $pH > pK_a$. These questions should be answered quickly, using either a titration curve or the Henderson-Hasselbalch equation in a purely conceptual manner.

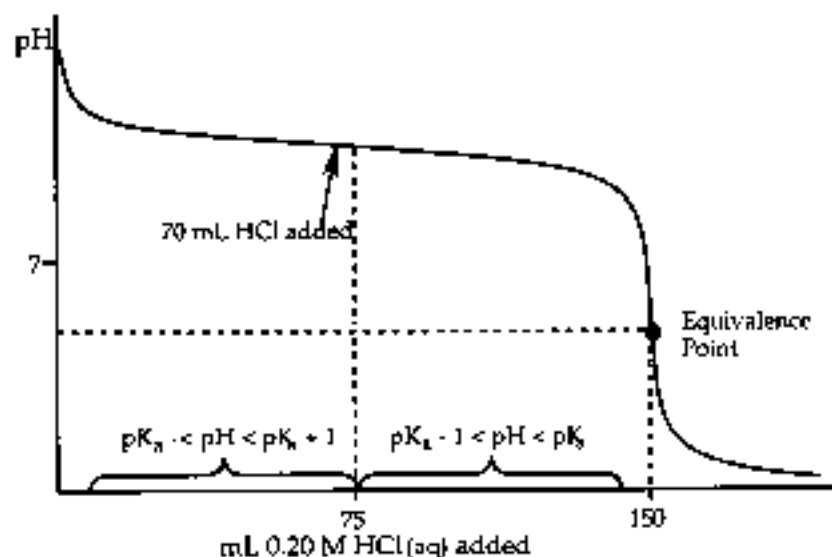
Example 5.12

What is the pH after 70 mL of 0.20 M HCl has been added to 50 mL 0.60 M H_3CNH_2 ? H_3CNH_2 has a $\text{p}K_b = 3.42$.

- A. 9.44
B. 10.51
C. 10.65
D. 11.72

Solution

The strong acid is one-third as concentrated as the weak base, so three times the volume of HCl (relative to the H_3CNH_2) is needed to reach the equivalence point. This means that 150 mL of 0.20 M HCl fully neutralizes the 50 mL of 0.60 M H_3CNH_2 . The halfway point of the titration is reached when 75 mL of 0.20 M HCl is added. At the halfway point, the pH of the solution equals the $\text{p}K_a$ of the weak acid. Less than the 75 mL has been added, so the pH of the solution is slightly greater than the $\text{p}K_a$ of the conjugate acid, 10.58. The best choice is answer C. The titration curve below shows how to estimate the value.

**Example 5.13**

What is the pH of a solution made by mixing 10.0 mL 0.10 M $\text{HCO}_2\text{H}(\text{aq})$ with 4.0 mL 0.10 M $\text{KOH}(\text{aq})$? The $\text{p}K_a$ for HCO_2H is 3.64.

- A. 1.34
B. 3.46
C. 3.82
D. 9.36

Solution

The best way to solve this question is to think in terms of equivalents. The weak acid and titrant strong base are of equal concentration, so 10 mL KOH is one equivalent (the amount needed to reach the equivalence point.) If 5.0 mL are added, then the acid is half-titrated, so $\text{pH} = \text{p}K_a$. However, less than 5.0 mL has been added, so pH is a little less than $\text{p}K_a$. The $\text{p}K_a$ is 3.64, so the best answer is choice B. Choice A is too much less than the $\text{p}K_a$ (more than 1.0 is beyond the 10 : 1 ratio, which would be when less than 1.0 mL of KOH had been added.)

Concentration Effect on Titration Curve Shape

The pH of a solution depends on the strength and concentration of the reagents, so strength and concentration also affect titration curves. The concentration of the reagents affects the dimensions of the titration curve, but not its basic shape. A strong-by-strong curve maintains the same fundamental shape (sigmoidal), but with varying concentrations, the curve may skew and elongate. The equivalence point is always at $\text{pH} = 7$. A weak-by-strong curve also maintains its same fundamental shape (i.e., has a lip-o-weakness), but with varying concentrations, the curve also skews and elongates. The half-equivalence point is always at $\text{pH} = \text{pK}_a$.

For a strong acid titrated by a strong base, as the concentrations of both reagents increase proportionally, the respective curves start lower and finish higher, but they have the same distance in the x-direction (the mL axis). If the acid concentration is increased but the base remains the same, then the curve starts lower and stretches to a point farther from the origin along the x-axis for the equivalence point. If the base concentration is increased but the acid remains the same, then the curve finishes higher and contracts to a point closer to the origin along the x-axis for the equivalence point. This is shown in Figure 5-10.

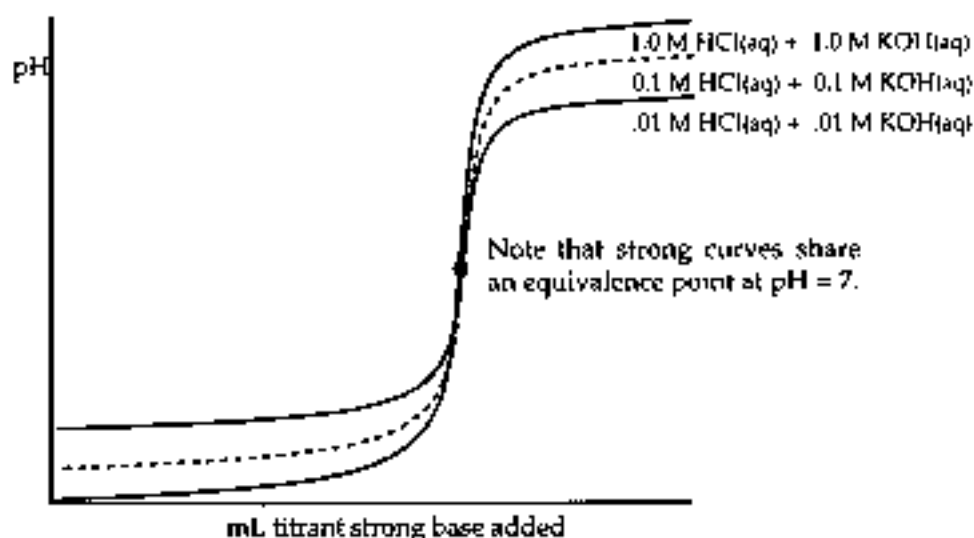


Figure 5-10

In each titration, the concentration of the strong acid is equal to the concentration of the strong base, so the volume of base required is the same in each case. This is why the curves are similar in the x-direction. In the y-direction, the curve depends on concentration. As the concentration lessens, the curve contracts with respect to a line through $\text{pH} = 7$. Note that the initial pH is 0, 1, and 2 respectively for the three titration curves. The variation in the concentration of base also causes the ends of the curves to vary (in the region of excess titrant). But at equivalence, the pH is 7, no matter what the concentrations are.

For a weak acid titrated by a strong base, as the concentrations of both reagents increase proportionally, the curve starts lower and finishes higher but advances the same distance in the x-direction (the mL axis) and has the same buffer region. If the acid concentration is increased but the base remains the same, then the curve starts lower and stretches to a point farther from the origin along the x-axis for its equivalence point. If the base concentration is increased but the acid remains the same, then the curve finishes higher and contracts to a point closer to the origin along the x-axis for its equivalence point. This is shown in Figure 5-11.

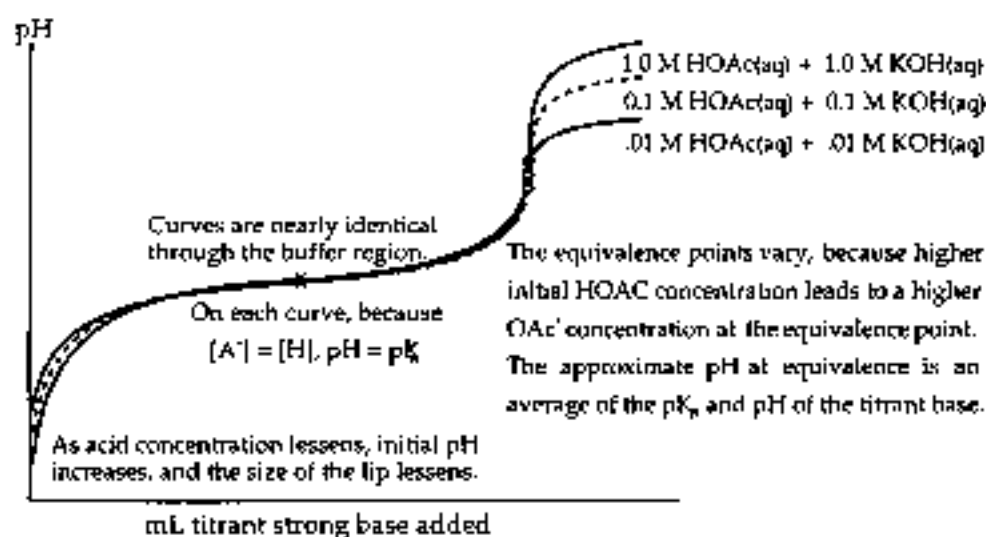


Figure 5-11

In each titration, the concentration of the weak acid is equal to the concentration of the strong base, so the volume of base required is the same in each case. This is why the curves are similar in the x-direction. In the y-direction, the curve depends on concentration. As the concentration lessens, the curve contracts with respect to a line through $pH = pK_a$. The initial pH is $\frac{1}{2} pK_a$, $\frac{1}{2} pK_a + 0.5$, and $\frac{1}{2} pK_a + 1.0$ respectively for the three curves. The variation in the concentration of base also causes the ends of the curves to vary (in the region of excess titrant). But at half-equivalence, the pH is pK_a , no matter what the concentrations are.

Strength Effect on Titration Curve Shape

The examples in Figure 5-10 and Figure 5-11 show the effect of concentration on titration curves. The shape of a titration curve is also a reflection of the strength of the reagents. When the concentrations are uniform between titration curves, then the shape of the curve and location of key points give clues as to the nature of the acid being titrated. As the acid becomes weaker, there is a larger initial lip, a greater midpoint (where $pH = pK_a$), and a greater equivalence point. This is shown in Figure 5-12. As acid strength increases, initial pH decreases, and the size of the lip-to-weakness lessens.

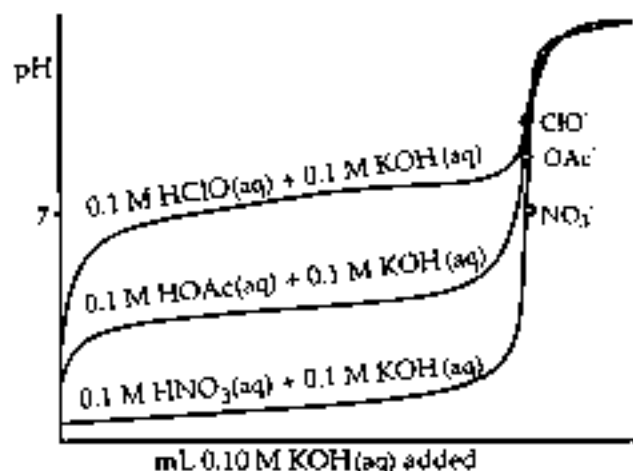


Figure 5-12

Polyprotic Acid Titration Curves

Polyprotic acids are acids which yield multiple equivalents of hydronium ion (H_3O^+). Three common polyprotic acids with which every test-taker must be familiar are carbonic acid (H_2CO_3), phosphoric acid (H_3PO_4), and sulfuric acid (H_2SO_4). Polyprotic titration curves have multiple equivalence points, one for each dissociable proton. But the protons are removed one at a time, so the curves for each proton are separate. They should be treated as separate titration curves that happen to overlap on the same graph. Figure 5-13 shows the titration curve for a typical diprotic acid, where both protons are weak. Examples that share the same basic curve shape include carbonic acid, glycine, or a mixture of two weak monoprotic acids in the same solution. Figure 5-14 shows the titration curve for a typical diprotic acid, where the first proton is strong and the second proton is weak. An example that shares the same basic curve shape is sulfuric acid. Figure 5-15 shows the titration curve for a typical triprotic acid, where all three protons are weak. Examples that share the same basic curve shape include phosphoric acid, citric acid, and glutamic acid.

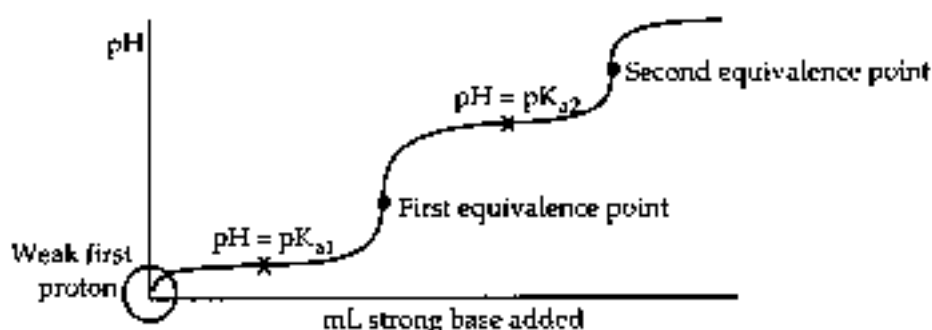


Figure 5-13

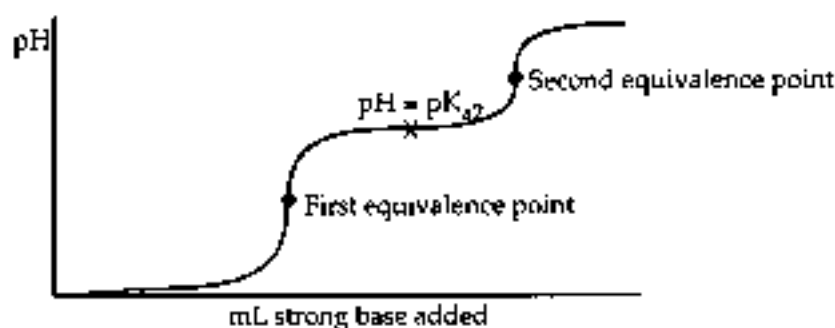


Figure 5-14

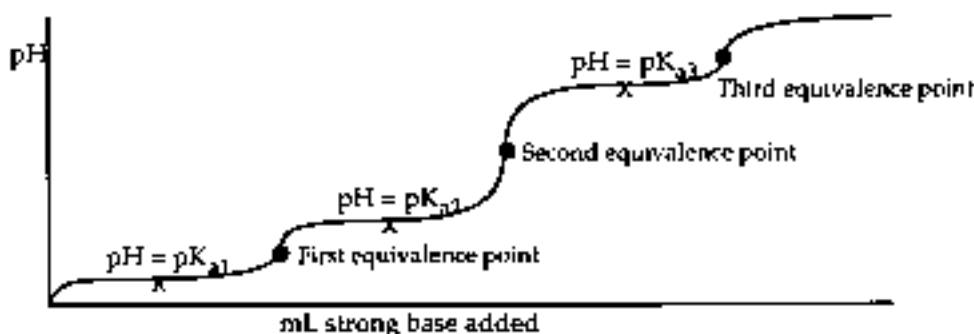


Figure 5-15

Indicators

Indicator Composition and Function

An indicator is an organic compound with extended conjugation. Its weak acid (protonated) form and conjugate weak base (deprotonated) form have two distinct colors. The colors of both species are reflected colors. The energy state transition that produces color involves the π -bonding and π -antibonding orbitals and depends on the electron donating and withdrawing nature of substituents on the π -system. The lone pair formed upon deprotonation affects this conjugation. When the pH of the solution is less than the pK_a of the indicator ($pH_{\text{solution}} < pK_a(\text{indicator})$), the indicator exists predominantly in its protonated form ($H\text{---Ind} > \text{Ind}^-$). If this is true, the solution assumes the hue of the protonated form of the indicator ($H\text{---Ind}$). When the pH of the solution exceeds the pK_a of the indicator ($pH_{\text{solution}} > pK_a(\text{indicator})$), the indicator exists predominantly in its deprotonated form ($\text{Ind}^- > H\text{---Ind}$). If this is true, the solution assumes the hue of the deprotonated form of the indicator (Ind^-). When the pH of the solution is equal to the pK_a of the indicator ($pH_{\text{solution}} = pK_a(\text{indicator})$), the indicator exists equally in its protonated and deprotonated forms ($[H\text{---Ind}] = [\text{Ind}^-]$). If this is true, the solution assumes a hue that is the result of a mixture of the protonated and deprotonated forms of the indicator. When the pH of the solution is near the pK_a of the indicator, the color varies with small changes in pH. Reaction 5.3 represents the dissociation of an indicator in water.



Reaction 5.3

Table 5.4 shows the pH effects on the generic indicator in Reaction 5.3, along with a ratio of protonated-to-deprotonated species, and the color of the indicator.

pH_{solution}	Ratio of $H\text{---Ind}$ to Ind^-	Mixture of colors to form solution color
$pK_a + 3.0$	1 : 1000	1 yellow : 1000 blue $\therefore$ blue
$pK_a + 2.0$	1 : 100	1 yellow : 100 blue $\therefore$ blue
$pK_a + 1.0$	1 : 10	1 yellow : 10 blue $\therefore$ greenish blue
$pK_a + 0.7$	1 : 5	1 yellow : 5 blue $\therefore$ blue-green
$pK_a + 0.3$	1 : 2	1 yellow : 2 blue $\therefore$ bluish green
pK_a	1 : 1	1 yellow : 1 blue $\therefore$ green
$pK_a - 0.3$	2 : 1	2 yellow : 1 blue $\therefore$ yellowish green
$pK_a - 0.7$	5 : 1	5 yellow : 1 blue $\therefore$ yellow-green
$pK_a - 1.0$	10 : 1	10 yellow : 1 blue $\therefore$ greenish green
$pK_a - 2.0$	100 : 1	100 yellow with 1 blue $\therefore$ yellow
$pK_a - 3.0$	1000 : 1	1000 yellow with 1 blue $\therefore$ yellow

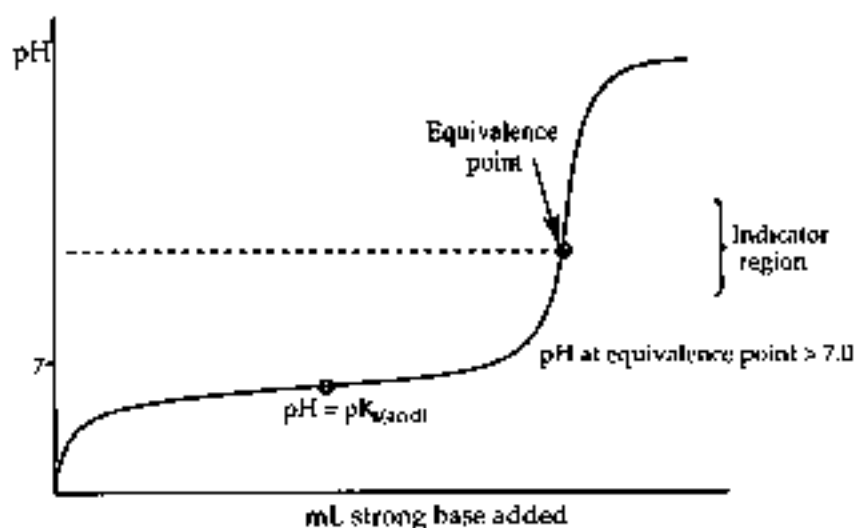
Table 5.4

When the solution is bluish green, it can be concluded that the pH of the solution is slightly (about 0.2 to 0.4 times) greater than the pK_a of the indicator. This means that the pH of the solution can be approximated from the color of the solution. The color change range (and thus the useful range) of an indicator is $pK_a(\text{indicator}) \pm 1$. An indicator is generally used for one of two purposes. The first is to detect the endpoint of a titration, and the second is to approximate the pH of a solution by observing the color of the indicator in the solution.

Detecting the Equivalence Point (Using Indicators)

Indicators make the equivalence point of a titration visible, either by converting from an uncolored species to the colored conjugate or by converting from one colored species to its conjugate, which has a different color. An indicator changes color at a pH near the equivalence point of the titration. The change in color can thus be used to determine the equivalence point. Indicators are added in small quantities, so they do not interfere with the titration. The pH of the solution thus dictates the color of the indicator. As the titration approaches equivalence, the pH changes rapidly, so the color changes rapidly. A color change indicates that equivalence has been reached, if the correct indicator is chosen.

The ideal scenario for an indicator is when the pH at equivalence equals the pK_a of the indicator. The range of a color change for an indicator is centered on its pK_a . For the titration of a weak acid with a strong base, the pH at the equivalence point may not be known exactly, so a best approximation of the equivalence pH must be made. When approximating the equivalence pH, consider that the pH at equivalence is greater than the pK_a of the weak acid being titrated, but less than the pH of the strong base being added. The pH at equivalence lies roughly half way between the pK_a of the acid and the pH of the base being added. A good approximation of the pH at equivalence is an average of the pK_a of the weak acid and the pH of the titrant strong base. The pK_a of the indicator should be somewhere around (within ± 1 unit) the average of the pK_a of the acid being titrated and the pH of the strong base. For the titration of a weak base with a strong acid, the pH at the equivalence point is less than the pK_a of the conjugate acid of the weak base being titrated. The pK_a of the indicator should be somewhere around (within ± 1 unit) of the average of the pK_a of the conjugate acid of the weak base being titrated and the pH of the strong acid. The active range for an indicator is represented by the two titration curves in Figure 5-16 and Figure 5-17.

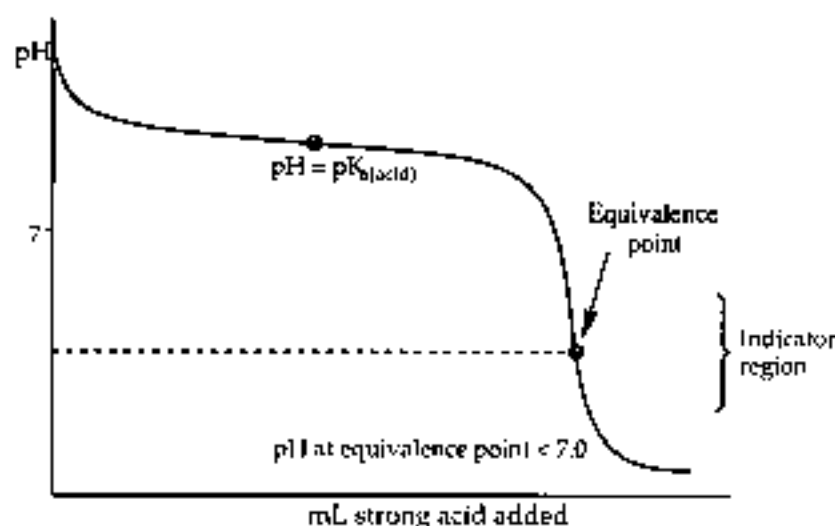


Best scenario: $pH_{\text{equivalence}} = pK_a(\text{indicator}) \pm 1$

Best approximation: $\frac{pK_a(\text{acid}) + pH(\text{titrant base})}{2} = pK_a(\text{indicator}) \pm 1$

Choosing an indicator for the titration of a weak acid by a strong base

Figure 5-16



Best scenario: $\text{pH}_{\text{equivalence}} = \text{pK}_a(\text{indicator}) \pm 1$

Best approximation: $\frac{\text{pK}_a(\text{conjugate acid}) + \text{pH}_{\text{titrant acid}}}{2} = \text{pK}_a(\text{indicator}) \pm 1$

Choosing an indicator for the titration of a weak base by a strong acid

Figure 5-17

You should make note of the fact that the pH at equivalence for the titration of a weak acid by a strong base can be approximated quite closely by averaging the pK_a of the weak acid with the pH of the titrant strong base. This value will be within 0.5 of the actual value, meaning that for the titration of acetic acid by sodium hydroxide, the pK_a of the indicator should be greater than seven (somewhere around nine.) As the titration is carried out, the pH of solution increases; thus, the indicator starts out in its protonated form and eventually become deprotonated. For the sample indicator in Reaction 5.2, the solution would go from yellow to green to blue.

For the titration of a weak base by a strong acid, the pH at equivalence can be approximated by averaging the pK_a of the conjugate acid of the weak base with the pH of the titrant strong acid. This means that for the titration of ammonia by hydrochloric acid, the pK_a of the indicator should be less than 7.0 (somewhere around 5.0.) As the titration is carried out, the pH of solution decreases; thus, the indicator starts out in its deprotonated form and eventually becomes protonated. For the sample indicator in Reaction 5.3, the solution would go from blue to green to yellow.

Estimating Solution pH (using Indicators)

For a solution of unknown pH, if the pH is within one unit of the pK_a of the indicator, then the hue of the indicator can be used to estimate the pH of the solution. Consider an indicator with a pK_a of 6.83 that is used to test pool water. It is yellow when protonated and red when deprotonated. Table 5.5 can be used to estimate the pH of the pool water.

pH_{solution}	Ratio of H—Ind to Ind ⁺	Solution Color
$pH > 7.83$	$1 : 10^+$	Red
$pH = 7.83$	$1 : 10$	Orangish red
$pH = 7.53$	$1 : 5$	Red-orange
$pH = 7.13$	$1 : 2$	Reddish orange
$pH = 6.83$	$1 : 1$	orange
$pH = 6.53$	$2 : 1$	Yellowish orange
$pH = 6.13$	$5 : 1$	Yellow-orange
$pH = 5.83$	$10 : 1$	Orangish yellow
$pH < 5.83$	$10^+ : 1$	Yellow

Table 5.5

If the results of the pool water test is red, the water is basic; and because red is outside the color change range, the exact pH cannot be approximated. If the results of the pool water test is yellow, then someone has left some acid in it. Most people are aware that yellow pool water has some acid in it, and it should not be swum in. The ideal color is orange with a slight hint of red. Indicators are used in pH test kits for swimming pools and fish tanks. The kits usually contain more than one indicator, increasing the accuracy of the approximation. All of the indicators in such a test kit should have pK_a values between 6.0 and 8.0, because the pH of the water should be around 7.0. Indicators are also used in pH test sticks, where a series of three or four indicators are on the stick. The pK_a values of the indicators differ by roughly two units. This allows for a wider range of pH values from which one can correlate the color to the solution pH. For instance, a pH stick with three indicators, with pK_a values of 5.05, 6.98, and 9.11, has a range of roughly 4.05 to 10.11. This is because each indicator has a two-pH-unit range.

Example 5.14

Given the following indicators on a pH stick, what is the pH of a solution that yields X: red, Y: blue, and Z: red?

Indicator X: $pK_a = 4.96$; when deprotonated, it goes from yellow to red

Indicator Y: $pK_a = 7.01$; when deprotonated, it goes from yellow to blue

Indicator Z: $pK_a = 8.98$; when deprotonated, it goes from red to blue

- A. 5
- B. 6
- C. 7
- D. 8

Solution

Because Indicator Y is blue, the pH must be at least one unit greater than 7.01 (the pK_a of Indicator Y). Because Indicator Z is red, the pH must be at least one unit less than 8.98 (the pK_a of Indicator Z). The only value greater than 7.01 and less than 8.98 is 8, choice D.

Buffers and Titration Passages

15 Passages

100 Questions

Suggested Buffers and Titration Passage Schedule:

- I:** After reading this section and attending lecture: Passages I, IV, VII, VIII & XI
Grade passages immediately after completion and log your mistakes.
- II:** Following Task I: Passages II, III, V, VI, IX, & XIV (41 questions in 53 minutes)
Time yourself accurately, grade your answers, and review mistakes.
- III:** Review: Passages X, XII, XIII, & XV
Focus on reviewing the concepts. Do not worry about timing



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Buffers and Titration Study Passages

I. Buffer pH and Weak Acids	(1 - 7)
II. Buffer Composition	(8 - 14)
III. Buffer Chart and pK_a Chart Passage	(15 - 21)
IV. Molecular Weight from Neutralization of an Organic Acid	(22 - 29)
V. Conjugate Pair Titration Curve	(30 - 36)
VI. Titration Curves and Concentration Effects	(37 - 42)
VII. Titration Curves and Strength Effects	(43 - 49)
VIII. Normality and Neutralization	(50 - 56)
IX. Titration Curve of a Polyprotic Acid	(57 - 63)
X. Carbonate Titration Curve	(64 - 70)
XI. Indicator Selection	(71 - 76)
XII. Indicator Color and Solution pH	(77 - 83)
XIII. Indicator Table	(84 - 89)
XIV. pH Sticks and Indicators	(90 - 96)
XV. Acidity and Electronic Influences	(97 - 100)

Buffers and Titration Scoring Scale

Raw Score	MCAT Score
84 - 100	13 - 15
66 - 83	10 - 12
47 - 65	7 - 9
34 - 46	4 - 6
1 - 33	1 - 3

Passage I (Questions 1 - 7)

An integral part of any biological study conducted *in vivo* is the accurate simulation of body conditions with as much precision as possible. As important as any factor to biological systems is pH. To accommodate the need for a constant and accurate pH, organisms of all types use buffers to maintain a relatively constant internal pH range. A buffer exists when there is both a weak acid and its weak conjugate base present in solution in roughly equal molar concentrations. It is important that both the acid and the base be water-soluble and exhibit no side reactions. Organic acids and their conjugate bases are best for this purpose.

The Henderson-Hasselbalch equation, Equation 1, is used to calculate the value of the pH for a buffer.

$$\text{pH} = \text{pK}_a + \log \frac{[\text{Base}]}{[\text{Acid}]}$$

Equation 1

In human blood, a buffer of bicarbonate and carbonic acid exists. Inorganic phosphates also play a role in buffering within the body. The buffer must have a pH that is relatively close to 7.4, known as *biological pH*. This can be simulated in the lab by starting with a weak acid whose pK_a is as close to 7.4 as possible. After the acid has been added to water, the solution is titrated with strong base until a pH of 7.4 is obtained. This method ensures an accurate value. Buffers can also be made by mixing equal molar portions of the weak acid and its conjugate base (the conjugate base may come in its salt form). The pH equals the pK_a of the acid when a buffer is made with equal molar portions of acid and conjugate base.

1. What is the pH of a solution made by adding 0.839 grams $\text{NaHCO}_3(\text{s})$ (MW = 83.9 grams/mole) to 100 mL 0.10 M $\text{H}_2\text{CO}_3(\text{aq})$?

$$\text{pK}_{a1} = 6.4 \quad \text{pK}_{a2} = 10.8$$

- A. 3.20
B. 6.40
C. 8.60
D. 10.80
2. Which of the following pH values is the BEST choice for the pH of a buffer initially at pH = 7.21 after HCl has been added?
- A. 7.14
B. 7.21
C. 7.28
D. 8.31

3. What should be mixed to make a pH = 4.2 buffer? $\text{pK}_a(\text{benzoic acid}) = 4.2$

- A. 10 grams $\text{C}_6\text{H}_5\text{CO}_2\text{Na}$ + 10 grams $\text{C}_6\text{H}_5\text{CO}_2\text{H}$
B. 10 mL 0.10 M $\text{C}_6\text{H}_5\text{CO}_2\text{H}$ + 5 mL 0.10 M NaOH
C. 10 mL 0.10 M $\text{C}_6\text{H}_5\text{CO}_2\text{H}$ + 10 mL 0.10 M NaOH
D. 10 mL 0.10 M $\text{C}_6\text{H}_5\text{CO}_2\text{H}$ + 15 mL 0.10 M NaOH

4. How many mL of 0.20 M NaOH must you add to 50 mL of 0.10 M HF to produce a solution with a pH of 3.3? ($\text{pK}_a = 3.3$)

- A. 10.0 mL
B. 12.5 mL
C. 16.7 mL
D. 25.0 mL

5. What is the acetate anion concentration in a solution made by mixing 20 mL 0.30 M HOAc with 10 mL 0.30 M NaOH?

- A. 0.07 M H_3CCO_2^-
B. 0.10 M H_3CCO_2^-
C. 0.15 M H_3CCO_2^-
D. 0.20 M H_3CCO_2^-

6. Which of the following combinations produces a buffer?

- A. 10 mL 0.25 M $\text{NH}_3(\text{aq})$ + 20 mL 0.25 M $\text{HCl}(\text{aq})$
B. 20 mL 0.25 M $\text{NH}_4\text{Cl}(\text{aq})$ + 10 mL 0.25 M $\text{HCl}(\text{aq})$
C. 10 mL 0.25 M $\text{NH}_3(\text{aq})$ + 20 mL 0.25 M NaOH(aq)
D. 20 mL 0.25 M $\text{NH}_4\text{Cl}(\text{aq})$ + 10 mL 0.25 M NaOH(aq)

7. What is the pH of 100 mL of 0.10 M propanoic acid, $\text{pK}_a(\text{propanoic acid}) = 5.0$?

- A. 2.5
B. 3.0
C. 5.0
D. 6.0

Passage II (Questions 8 - 14)

A buffer is composed of a weak acid and its conjugate base. In an effective buffer, the ratio of the conjugate pair must be less than 10:1, in favor of either component. According to Equation 1, when the two components of the conjugate pair are roughly equal in concentration, the pH of the solution is approximately the pK_a of the weak acid.

$$pH = pK_a + \log \frac{[Base]}{[Acid]}$$

Equation 1

A buffer is capable of consuming any acid or base that is added to solution. The pH of an aqueous buffer solution changes only slightly after an acid or base is added. This is because there is an equilibrium between the conjugates. Table 1 lists a series of weak acids and conjugate bases, along with the pK_a values for each acid. The pK_a value of the weak acid and the acid-base ratio can be used to determine the pH of a buffer from Equation 1.

Weak Acid	Conjugate Base	pK_a value
CH_3COOH	$CH_3COO^-K^+$	2.82
HF	F^-Na^+	3.19
HCO_2H	$HCO_2^-Na^+$	3.64
H_3CCO_2H	$H_3CCO_2^-Na^+$	3.86
$C_6H_5CO_2H$	$C_6H_5CO_2^-K^+$	4.19
$C_6H_5NH_3Cl$	$C_6H_5NH_2$	4.62
H_3CCO_2H	$H_3CCO_2^-Na^+$	4.74
C_5H_5NHCl	C_5H_5N	5.16
$4-NO_2C_6H_4OH$	$4-NO_2C_6H_4O^-K^+$	7.15
HClO	ClO^-	7.49
HBrO	BrO^-	8.67
NH_4Cl	NH_3	9.26
C_6H_5OH	$C_6H_5O^-K^+$	10.01
H_3CNH_3Cl	H_3CNH_2	10.56

Table 1

To make a buffer, an acid must be chosen that has a pK_a value within one unit of the target pH. The closer the pK_a value is to the pH, the better the buffer. Buffers are made either by mixing the weak acid with its conjugate base, or by partially titrating either the weak acid with strong base, or the weak conjugate base with strong acid. When the pH exceeds the pK_a , there is more conjugate base present in solution.

8. Which mixture does NOT produce a buffer?

- A. H_3CCO_2H with 2 equivalents of $H_3CCO_2^-K^+$
- B. NH_3 with 2 equivalents of NH_4Cl
- C. H_2CO_3 1.5 equivalents of KOH
- D. H_3CNH_2 with 3.5 equivalents of HCl

9. What is the pH of a solution that contains two parts weak acid and one part conjugate base?

- A. $pK_a(\text{weak acid}) + 2$
- B. $pK_a(\text{weak acid}) + \log 2$
- C. $pK_a(\text{weak acid}) - 2$
- D. $pK_a(\text{weak acid}) - \log 2$

10. At which of the following pH values would it be MOST difficult to establish a pH buffer?

- A. 1.0
- B. 3.0
- C. 5.0
- D. 7.0

11. A buffered solution is BEST described as an aqueous solution where the:

- A. hydronium and hydroxide concentrations are equal
- B. hydronium-to-hydroxide concentration ratio never exceeds 10:1 or is less than 1:10.
- C. hydronium and hydroxide concentrations are relatively constant.
- D. hydronium-to-hydroxide concentration ratio is within one unit of the pK_a value for the weak acid.

12. Addition of 1.00 mL of 0.10 M $KOH(aq)$ to a solution made by mixing 15.00 mL 0.10 M $H_3CCO_2H(aq)$ with 10.00 mL 0.10 M $H_3CCO_2^-Na(aq)$ results in:

- A. a solution with a pH less than 3.74
- B. a solution with a pH between 3.74 and 4.74
- C. a solution with a pH between 4.74 and 5.74
- D. a solution with a pH greater than 5.74

13. In which solution is there the LARGEST ratio of conjugate base to conjugate acid?

- A. A solution of HCO_2H and HCO_2^- with pH = 4.00
- B. A solution of HClO and ClO^- with pH = 7.00
- C. A solution of HBrO and BrO^- with pH = 8.50
- D. A solution of NH_3 and NH_4^+ with pH = 9.50

14. After 5.00 mL $H_2O(l)$ has been added to 50.00 mL of a buffered solution with pH initially at 5.0, the pH:

- A. drops slightly.
- B. remains constant.
- C. increases slightly.
- D. increases drastically.

Passage III (Questions 15 - 21)

Buffers are aqueous solutions of weak acids and their conjugate base. The pH of solution is dictated by the Henderson-Hasselbalch equation: $\text{pH} = \text{pK}_a + \log \text{base/acid}$. This means that a buffer solution should be mixed in a manner where the pK_a of the acid is close to the pH desired. Table 1 lists the K_a and pK_a values for some common monoprotic weak acids:

Acid	K_a value	pK_a value
$\text{H}_2\text{NCONH}_3^+$	6.6×10^{-2}	0.18
HF	6.8×10^{-4}	3.17
HCNO	3.5×10^{-4}	3.49
HCO_2H	1.7×10^{-4}	3.78
$\text{H}_3\text{CCOCO}_2\text{H}$	1.4×10^{-4}	3.89
$\text{C}_6\text{H}_5\text{CO}_2\text{H}$	6.5×10^{-5}	4.19
$\text{C}_6\text{H}_5\text{NH}_3^+$	2.3×10^{-5}	4.64
$\text{H}_3\text{CCO}_2\text{H}$	1.8×10^{-5}	4.74
$\text{C}_5\text{H}_5\text{NH}^+$	7.1×10^{-6}	5.16
HClO	3.5×10^{-8}	7.49
B(OH)_3	5.9×10^{-10}	9.22
NH_4^+	5.6×10^{-10}	9.26
HCN	4.9×10^{-10}	9.32
H_3CNH_3^+	2.2×10^{-11}	10.66
$(\text{H}_3\text{C})_2\text{NH}_2^+$	1.9×10^{-11}	10.77

Table 1

Polyprotic acids can also be used in making buffers. A difficulty that arises with polyprotic acids involves the two or more pK_a values. For any given polyprotic acid, the conjugate pair will buffer at the respective pK_a for the acid of the conjugate pair. For instance, carbonate/bicarbonate will buffer at a pH around 10.8 because pK_{a2} of carbonic acid is 10.81. Carbonic acid/bicarbonate will buffer at a pH around 6.4 because pK_{a1} of carbonic acid is 6.37. Table 2 lists the K_a values for some common polyprotic acids:

Acid	K_{a1} value	K_{a2} value	K_{a3} value
$\text{H}_2\text{C}_2\text{O}_4$	5.6×10^{-2}	5.1×10^{-5}	
H_2SO_3	1.3×10^{-2}	6.3×10^{-8}	
H_3PO_4	6.9×10^{-3}	6.2×10^{-8}	4.8×10^{-13}
H_2CO_3	4.3×10^{-7}	1.5×10^{-11}	
H_2S	8.9×10^{-8}	3.8×10^{-13}	

Table 2

As a point of interest, carbonates and phosphates are believed to be the major contributors to buffering in human blood. For years it was believed that carbonate played the major role, but recent research indicates that phosphate may play a more significant role than carbonate in the overall buffering. Within the kidneys, phosphates are known to play a significant role.

15. Combining all of the following results in a buffer with $\text{pH} = 9.5$ EXCEPT:

- A. 1.8 equivalents NH_3 with 1.0 equivalents NH_4^+ .
- B. 1.6 equivalents NaCN with 1.0 equivalents HCN .
- C. 0.7 equivalents HCl with 1.0 equivalents NH_3 .
- D. 0.65 equivalents NaOH with 1.0 equivalents HCN .

16. To make a buffer at $\text{pH} = 10.83$, which of the following should be mixed?

- A. One-half equivalent of NaOH with one equivalent of H_2CO_3
- B. One and one-half equivalents of NaOH with one equivalent of H_2CO_3
- C. One and one-half equivalents of NaOH with one equivalent of H_3PO_4
- D. Two and one-half equivalents of NaOH with one equivalent of H_3PO_4

17. Which of the following solutions forms a buffer with a pH greater than 7.0?

- A. One and one-half equivalents of KOH with one equivalent of $\text{H}_2\text{C}_2\text{O}_4$
- B. One-half equivalent of KOH with one equivalent of H_2CO_3
- C. One-half equivalent of KOH with one equivalent of HClO
- D. One-half equivalent of HCl with one equivalent of $\text{C}_5\text{H}_5\text{N}$

18. Which of the following mixtures would NOT result in a solution with $\text{pH} = 4.00$?

- A. Excess HCO_2^- mixed with HCO_2H
- B. Excess $\text{C}_6\text{H}_5\text{NH}_3^+$ mixed with $\text{C}_6\text{H}_5\text{NH}_2$
- C. Excess $\text{C}_6\text{H}_5\text{CO}_2^-$ mixed with $\text{C}_6\text{H}_5\text{CO}_2\text{H}$
- D. Excess $\text{H}_3\text{CCOCO}_2^-$ mixed with $\text{H}_3\text{CCOCO}_2\text{H}$

19. Carbon dioxide when dissolved into the blood forms carbonic acid. What is observed in CO_2 -enriched blood?

- A. The pH is less than 7.4
- B. The $[\text{PO}_4^{3-}]$ increases
- C. The $[\text{HCO}_3^-]$ decreases
- D. The $[\text{H}_3\text{O}^+]$ decreases

20. In which of the following solutions is the conjugate base in GREATER concentration than the acid?

A. HF(aq) with $\text{F}^-(\text{aq})$ at $\text{pH} = 3.00$
 B. $\text{H}_2\text{CO}_3(\text{aq})$ with $\text{HCO}_3^-(\text{aq})$ at $\text{pH} = 6.00$
 C. $\text{HC}_2\text{O}_4^-(\text{aq})$ with $\text{C}_2\text{O}_4^{2-}(\text{aq})$ at $\text{pH} = 4.00$
 D. $\text{H}_2\text{PO}_4^-(\text{aq})$ with $\text{HPO}_4^{2-}(\text{aq})$ at $\text{pH} = 8.00$

21. Biological pH is approximately 7.4. Which of the following is NOT true about the concentration of buffer species at this pH?

A. $[\text{HCO}_3^-] > [\text{H}_2\text{CO}_3]$
 B. $[\text{HCO}_3^-] > [\text{CO}_3^{2-}]$
 C. $[\text{H}_2\text{PO}_4^-] > [\text{HPO}_4^{2-}]$
 D. $[\text{HPO}_4^{2-}] > [\text{PO}_4^{3-}]$

Passage IV (Questions 22 - 29)

The formula weight of an acid can be determined by titration, using a strong base of known concentration. The process involves the titration of an exact mass of some unknown acid. Once enough base has been added to reach equivalence, the moles of base added are used to determine the moles of acid that were present in the solution initially. The mass of the acid divided by the moles of the acid gives the formula weight for the acid, not the molecular weight. Often both weights are the same, but a difference arises if the unknown is a polyprotic acid. Only when the acid is monoprotic is the formula weight equal to the molecular weight. Table 1 shows some phenols with their respective pK_a values. All of the phenols are solids at room temperature and are monoprotic acids.

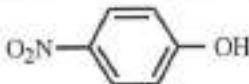
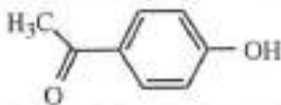
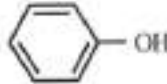
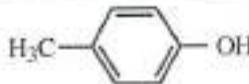
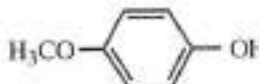
Structure	Formula	pK_a
	$\text{p-O}_2\text{NC}_6\text{H}_4\text{OH}$	7.2
	$\text{p-H}_3\text{CCOC}_6\text{H}_4\text{OH}$	8.4
	$\text{C}_6\text{H}_5\text{OH}$	10.0
	$\text{p-H}_3\text{CC}_6\text{H}_4\text{OH}$	10.4
	$\text{p-H}_3\text{COC}_6\text{H}_5\text{OH}$	11.2

Table 1

If the molecular mass of an acid is known, the pH of a solution can be calculated from its gram concentration in solution. To do this requires converting from grams into moles. Once the concentration is known, the shortcut equation, Equation 1, may be used to quickly calculate the pH of the aqueous solution of the acid:

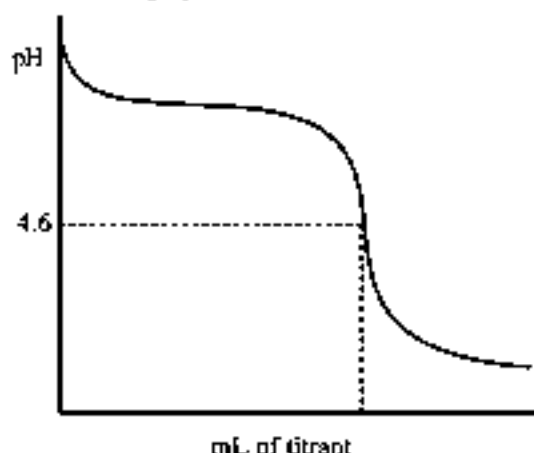
$$\text{pH} = \frac{1}{2} \text{pK}_a - \frac{1}{2} \log [\text{HA}]$$

Equation 1

22. 1.0 gram of which of the above acids requires exactly 30.0 mL of 0.20 M NaOH to reach equivalence?
- A. Acetic acid ($\text{H}_3\text{CCO}_2\text{H}$) MW = 60
 B. Trichloroacetic acid ($\text{Cl}_3\text{CCO}_2\text{H}$) MW = 151.5
 C. p-nitro benzoic acid ($\text{O}_2\text{NC}_6\text{H}_4\text{CO}_2\text{H}$) MW = 167
 D. Benzoic acid ($\text{C}_6\text{H}_5\text{CO}_2\text{H}$) MW = 122

23. If 1.0 gram of an unknown acid requires exactly 40.00 mL of 0.25 M NaOH to reach the equivalence point, what is the formula weight for the unknown acid?
- A. 50 grams per mole
 B. 80 grams per mole
 C. 100 grams per mole
 D. 125 grams per mole

24. Consider this graph:



The graph represents the titration of:

- A. $\text{H}_3\text{CCO}_2\text{H}$ by NaOH.
 B. HCl by NaOH.
 C. NH_3 by $\text{H}_3\text{CCO}_2\text{H}$.
 D. NH_3 by HCl
25. Which of the following indicators would be BEST for the titration of p-nitrophenol($\text{O}_2\text{NC}_6\text{H}_4\text{OH}$) by NaOH?
- A. Thymol blue (pH range of color change is 1.2 to 2.8)
 B. Methyl red (pH range of color change is 4.6 to 5.8)
 C. Bromthymol Blue (pH range of color change is 6.0 to 7.6)
 D. Phenolphthalein (pH range of color change is 8.0 to 9.6)
26. Which of the following mixtures results in a buffered solution?
- A. 10 mL 0.25 M NaOH + 10 mL 0.25 M $\text{H}_3\text{CCO}_2\text{H}$
 B. 20 mL 0.25 M NaOH + 10 mL 0.25 M $\text{H}_3\text{CCO}_2\text{H}$
 C. 10 mL 0.25 M NaOH + 20 mL 0.25 M $\text{H}_3\text{CCO}_2\text{H}$
 D. 10 mL 0.25 M HCl + 10 mL 0.25 M $\text{H}_3\text{CCO}_2\text{H}$

27. The BEST choice for a pH = 8.5 buffer would be which of the following?
- A. $\text{O}_2\text{NC}_6\text{H}_4\text{OH}$ with less than one full equivalent of NaOH
 B. $\text{H}_3\text{COC}_6\text{H}_4\text{OH}$ with less than one full equivalent of NaOH
 C. $\text{H}_3\text{CC}_6\text{H}_4\text{OH}$ with less than one full equivalent of NaOH
 D. $\text{H}_3\text{CCOC}_6\text{H}_4\text{OH}$ with less than one full equivalent of NaOH

28. What is the formula weight (equivalent weight) of oxalic acid ($\text{HO}_2\text{CCO}_2\text{H}$)?
- A. 45 grams per mole
 B. 90 grams per mole
 C. 135 grams per mole
 D. 180 grams per mole

29. 25.0 mL of an unknown acid when titrated by exactly 30.0 mL of 0.100 M KOH(aq), requires seven drops of 0.100 M HCl(aq) to return to equivalence. What is the concentration of the unknown acid?
- A. 0.1217 M acid(aq)
 B. 0.1183 M acid(aq)
 C. 0.0849 M acid(aq)
 D. 0.0817 M acid(aq)

Passage V (Questions 30 - 36)

Conjugate acid/base pairs are connected by a relationship between pK_a and pK_b . Equation 1 shows the relationship within a conjugate pair at 25°C in aqueous solution.

$$pK_a (\text{conjugate acid}) + pK_b (\text{conjugate base}) = 14$$

Equation 1

When titrating a weak acid or weak base, the portion of the curve following the initial drops of titrant up until just before the equivalence point is an equilibrium mixture of the conjugate pair. As such, the titration curve of one component in a conjugate pair have similarities to the titration curve of the other component.

The titration curves for conjugate pairs are inverse graphs that intersect at the half-titrated point. At this point, the pH of the solution equals the pK_a of the conjugate acid. At this same point, the pOH of the solution equals the pK_b of the conjugate base. Figure 1 shows the titration of acetic acid ($\text{H}_3\text{CCO}_2\text{H}$) with strong base (NaOH) overlaid onto the titration of sodium acetate ($\text{H}_3\text{CCO}_2\text{Na}$) with strong acid (HCl). In both titration curves, all species are in equal molar concentrations. The pK_a for carboxylic acids is generally between 3 and 5.

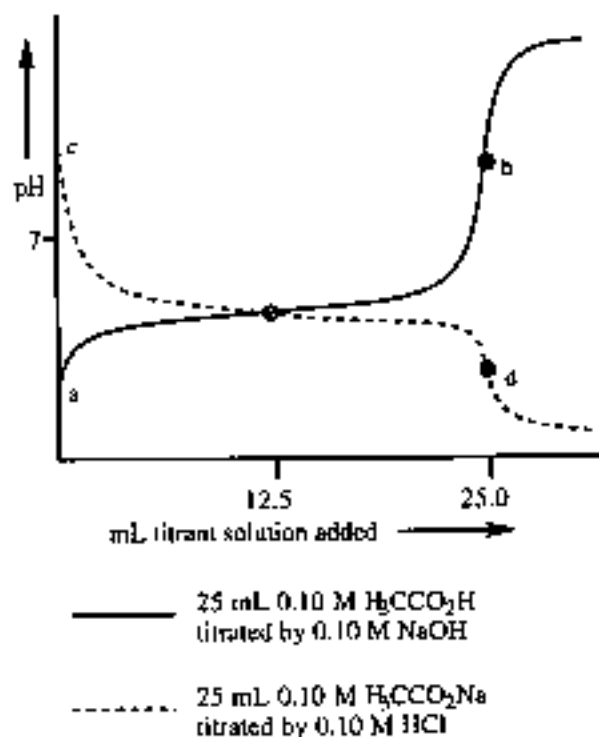


Figure 1

Figure 2 shows the titration of methyl ammonium chloride ($\text{CH}_3\text{NH}_3\text{Cl}$) with strong base (NaOH) overlaid onto the titration of methyl amine (CH_3NH_2) with strong acid (HCl). In both titrations, all species are in equal molar concentrations. The pK_a for amines is generally between 9 and 11.

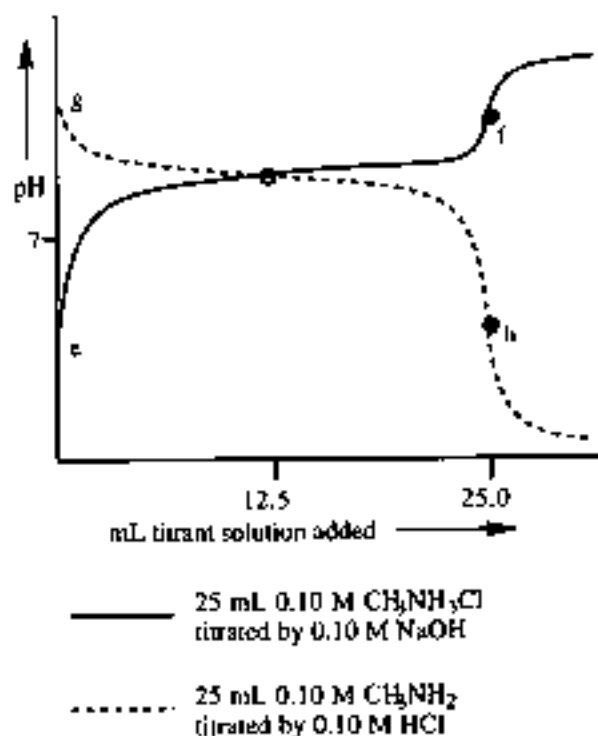


Figure 2

Both a weak acid and weak conjugate base titration curve of a conjugate pair show the same pH at the half-titration point (indicated by the empty circle on both graphs), regardless of the initial concentration of the conjugate species. The pH at the half-titration point in Figure 1 is less than the pH at the half-titration point in Figure 2. This is because the pK_a value of acetic acid is less than the pK_a value of methyl ammonium cation. At this point, the conjugate base concentration equals the acid concentration; therefore, according to the Henderson-Hasselbalch equation, $\text{pH} = pK_a$.

30. By roughly how much do the two equivalents points in the first graph differ?
- Fewer than 2.0 pH units
 - Fewer than 4.0 pH units, but more than 2.0 pH units
 - Fewer than 8.0 pH units, but more than 4.0 pH units
 - More than 8.0 pH units
31. The pH at equivalence is GREATEST for which of the following titrations?
- The titration of 0.10 M $\text{H}_3\text{CCO}_2\text{H}$ by NaOH
 - The titration of 0.10 M $\text{H}_3\text{CCO}_2\text{Na}$ by HCl
 - The titration of 0.10 M $\text{CH}_3\text{NH}_3\text{Cl}$ by NaOH
 - The titration of 0.10 M CH_3NH_2 by HCl

32. What is true in the titration of ammonia by hydrochloric acid, when the pH of the solution is greater than the pK_a for ammonium chloride?

- A. $[NH_4^+] > [NH_3]$; $K_a(\text{ammonium chloride}) > [H^+]$
- B. $[NH_4^+] > [NH_3]$; $K_a(\text{ammonium chloride}) < [H^+]$
- C. $[NH_4^+] < [NH_3]$; $K_a(\text{ammonium chloride}) > [H^+]$
- D. $[NH_4^+] < [NH_3]$; $K_a(\text{ammonium chloride}) < [H^+]$

33. How does the pH at point a in Figure 1 compare to the pH at point d in Figure 1?

- A. The pH at point a is more than 1.0 pH unit greater than the pH at point d.
- B. The pH at point a is greater than the pH at point d, but the difference is less than one pH unit.
- C. The pH at point a is less than the pH at point d, but the difference is less than one pH unit.
- D. The pH at point a is more than 1.0 pH unit lower than the pH at point d.

34. Given that H_3CCO_2H has a lower pK_a value than $CH_3NH_3^+Cl$, which of the following statements is true?

- A. H_3CCO_2H buffers at a higher pH value than $CH_3NH_3^+$
- B. H_3CCO_2H has a conjugate base with a lower pK_b value than the conjugate base of $CH_3NH_3^+$.
- C. H_3CCO_2H dissociates less than $CH_3NH_3^+$.
- D. H_3CCO_2H is a better electron pair acceptor than $CH_3NH_3^+$.

35. Given that H_3CCO_2H is a stronger acid than $CH_3NH_3^+$, which of the following statements is NOT true?

- A. H_3CCO_2H yields a lower pH value than $CH_3NH_3^+$ of equimolar concentration.
- B. H_3CCO_2H has a conjugate base with a higher pK_b value than the conjugate base of $CH_3NH_3^+$
- C. H_3CCO_2H produces more conjugate base than $CH_3NH_3^+$ when added to water.
- D. H_3CCO_2H is a worse proton donor than $CH_3NH_3^+$.

36. If a similar experiment with identical concentrations and volumes were conducted using hydrofluoric acid and fluoride anion, what would be true, knowing that the pK_a of hydrofluoric acid is lower than the pK_a of acetic acid?

- A. The initial pH for the titration of HF would be greater than the equivalence point for the titration of H_3CCO_2Na .
 - B. When both HF and H_3CCO_2H are one-third titrated by equimolar NaOH, $[F^-] > [H_3CCO_2^-]$ and $K_a(\text{acetic acid}) < [H^+]$.
 - C. When both HF and H_3CCO_2H are one-third titrated by equimolar NaOH, $[HF] > [H_3CCO_2H]$ and $K_a(\text{acetic acid}) < [H^+]$.
 - D. When both HF and H_3CCO_2H are one-half titrated by equimolar NaOH, $K_a(\text{acetic acid}) < [H^+]$ in the HF titration.
-

Passage VI (Questions 37 - 42)

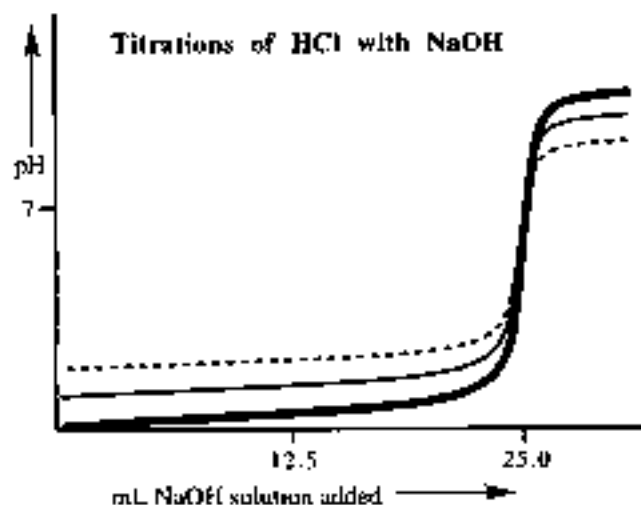
Titration involves the quantitative addition of one reagent to another, where the concentration is known for only one of the species. Acids are often titrated by strong bases, so that the concentration of the acid may be ascertained. Once the equivalence point has been reached, the volume of titrant is measured. Using Equation 1, it is possible to solve for the molarity of the acid, if the molarity of the titrant strong base is known.

$$M_{\text{(acid)}}V_{\text{(acid)}} = M_{\text{(base)}}V_{\text{(base)}}$$

Equation 1

Besides the quantitative aspects, qualitative features of the acid may also be determined. The shape of the titration curve varies with the strength of the acid. For strong acids, the shape is sigmoidal, with a nearly horizontal initial region. For weak acids, the shape is not sigmoidal, with an initial vertical ascent before leveling off into a horizontal buffering region. As the degree of the initial ascent increases, it can be observed that the acid being titrated is weaker. The concentration of the acid also affects the titration curve. The same fundamental shape is observed, but the pH values are different.

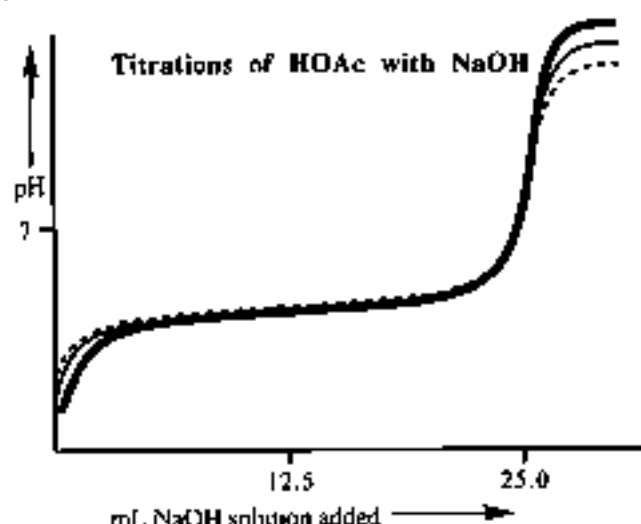
A researcher conducts two experiments studying the effect of acid concentration. In the first experiment, she titrates the same strong acid, HCl, at three different concentrations, keeping the acid and titrant base in the same concentration as each other. Figure 1 shows the three titrations overlaid onto one graph. She finds that all three titrations generate the same pH at their equivalence point, regardless of the initial concentration of strong acid.



- 25 mL 0.01 M HCl titrated by 0.01 M NaOH
- 25 mL 0.10 M HCl titrated by 0.10 M NaOH
- 25 mL 1.00 M HCl titrated by 1.00 M NaOH

Figure 1

In the second experiment, the researcher titrates the same weak acid, HOAc, at three different concentrations, keeping the acid and titrant base in the same concentration as each other. Figure 2 shows the three titrations overlaid onto one graph. It is found that all three titrations generated the same pH at their half-equivalence points.



- 25 mL 0.01 M HOAc titrated by 0.01 M NaOH
- 25 mL 0.10 M HOAc titrated by 0.10 M NaOH
- 25 mL 1.00 M HOAc titrated by 1.00 M NaOH

Figure 2

Weak acid titration curves show the same equivalence point, regardless of the initial concentration of weak acid. The pH at the half-equivalence point in all three titration curves in Figure 2 is equal to the pK_a for acetic acid. At the half-equivalence point, the concentration of the conjugate base equals the concentration of the acid; therefore, according to the Henderson-Hasselbalch equation, $pH = pK_a$.

37. The initial pH is **GREATEST** in which titration?

- A. The titration of 0.01 M H_3CCO_2H by NaOH
- B. The titration of 1.00 M H_3CCO_2H by NaOH
- C. The titration of 0.01 M HCl by NaOH
- D. The titration of 1.00 M HCl by NaOH

38. What is true in the titration of acetic acid by sodium hydroxide when the pH of the solution is greater than the pK_a for acetic acid?

- A. $[H_3CCO_2^-] > [H_3CCO_2H]$; $K_a(\text{acetic acid}) > [H^+]$
- B. $[H_3CCO_2^-] > [H_3CCO_2H]$; $K_a(\text{acetic acid}) < [H^+]$
- C. $[H_3CCO_2^-] < [H_3CCO_2H]$; $K_a(\text{acetic acid}) > [H^+]$
- D. $[H_3CCO_2^-] < [H_3CCO_2H]$; $K_a(\text{acetic acid}) < [H^+]$

39. The BEST explanation for the greater pH at the equivalence point observed with the higher initial concentration of weak acid can be attributed to:

- A. the greater number of mL of base solution added to reach the equivalence point.
- B. the lower number of mL of base solution added to reach the equivalence point.
- C. the greater conjugate base concentration at the equivalence point.
- D. the lower conjugate base concentration at the equivalence point.

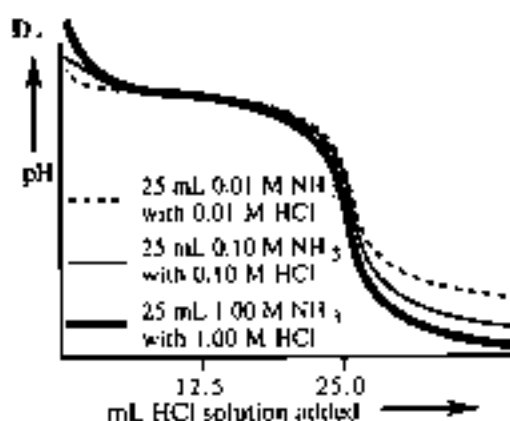
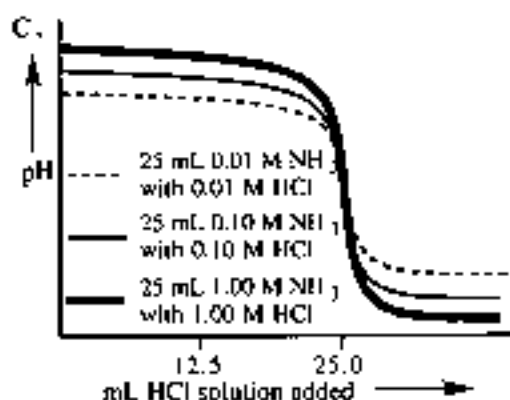
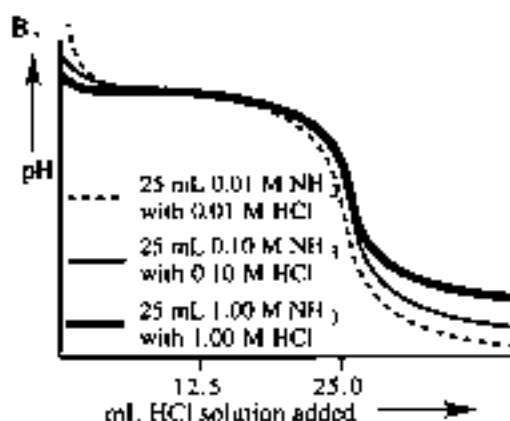
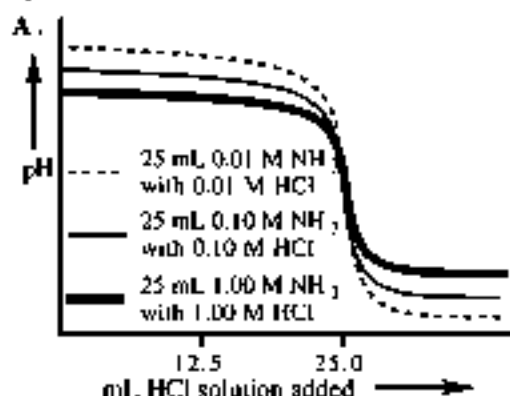
40. How would the titration curves in Figure 1 be affected if the base concentrations were all doubled, while the acid concentrations remained the same?

- A. Equivalence would be achieved with half the volume of titrant base, and the shape of the titration curves would change.
- B. Equivalence would be achieved with twice the volume of titrant base, and the shape of the titration curves would change.
- C. Equivalence would be achieved with half the volume of titrant base, and the shape of the titration curves would not change.
- D. Equivalence would be achieved with twice the volume of titrant base, and the shape of the titration curves would not change.

41. Even though the NaOH concentration in the third trial is 100 times greater than the NaOH concentration in the first trial, the two graphs follow a similar slope. This is BEST explained by which of the following statements?

- A. The solution is a buffered solution, so the pH change is minimal.
- B. The NaOH is a weak base and does not fully react with the HCl.
- C. The pH is a log scale, so as the pH increases up to 7.0, the amount of base necessary to increase the pH becomes less.
- D. The pH changes only at the equivalence point.

42. Which of the following graphs represents the results that would be observed if the experiment described in the passage were carried out with ammonia (NH_3) and hydrochloric acid (HCl)?



Passage VII (Questions 43 - 49)

The four titration curves shown in Figure 1, represent the titration of three weak acids (HCN, HClO, and HOAc) and the titration of the strong acid HCl. The equivalence point is represented by the dot at the vertical inflection point of each curve. The respective conjugate base is shown next to its equivalence point. At the start of each titration, 25 mL of 0.10 M acid are present. All of the acids are titrated by 0.10 M KOH(aq).

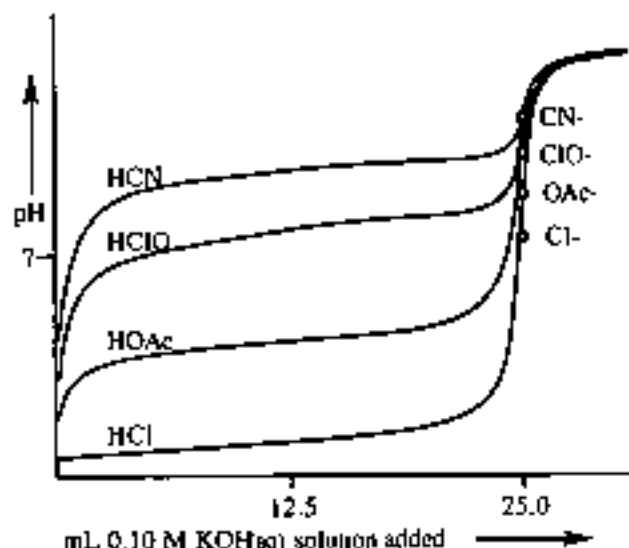


Figure 1

The initial and equivalence pH values were collected and recorded for some of the titrations. Table 1 shows data collected during the experiment.

Titration	Initial pH	Equivalence pH
HCl by KOH	1.00	7.00
HOAc by KOH	2.87	8.72
HClO by KOH	4.23	
HCN by KOH		11.01

Table 1

The pK_a for each weak acid can be found by measuring the pH of the aqueous solution at the half-equivalence point in its respective titration. For instance, the pK_a for HCN is 9.32, which is the pH at the half-equivalence point of the titration curve. The values are approximated, because the points on the curve cannot be read that accurately. The pH values are listed in Table 1 were recorded from a pH meter, so they are considered to be reliable.

43. The greatest K_a value is found with which of the following acids?

- A. HCN
- B. HClO
- C. HOAc
- D. HCl

44. If pK_a of HOAc is 4.74, the pH at the equivalence in the titration of HCN is 11.01, and the initial pH in the titration of HClO is 4.23, then what else must be true?

- A. Initial pH in the titration of HCN is 3.86; the pH at equivalence in the titration of HClO is 9.61.
- B. Initial pH in the titration of HCN is 5.16; the pH at equivalence in the titration of HClO is 10.08.
- C. Initial pH in the titration of HCN is 6.12; the pH at equivalence in the titration of HClO is 11.42.
- D. Initial pH in the titration of HCN is 7.42; the pH at equivalence in the titration of HClO is 10.34.

45. Which of the following statements are valid when comparing the titration curve associated with a weak acid by strong base titration to the titration curve associated with a strong acid by strong base titration?

- I. Strong acid titration curves have an initial drop in pH due to the dissociation of the protons, while weak acid titration curves start with a plateau.
 - II. Weak acid titration curves have a buffer region while strong acid titration curves do not.
 - III. All points before the equivalence point are less than 7 for both the titration of a weak acid and the titration of a strong acid.
- A. I only
 - B. II only
 - C. I and II only
 - D. II and III only

46. Given that 0.10 M HIO has a pH greater than that of 0.10 M HCN, we can conclude that:

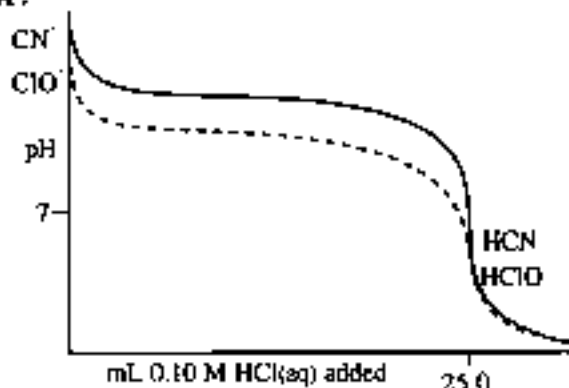
- A. the pK_a of HIO is 8.61.
- B. the pH at equivalence in the titration of HIO by KOH is 10.06.
- C. the pH of 0.10 M HIO(aq) is 7.21.
- D. the difference between the pK_a of HIO and the pH at the equivalence point in the titration of 0.10 M HIO(aq) by 0.10 M KOH(aq) is less than 3 pH units.

47. When 10 mL 0.10 M HClO is mixed with 10 mL 0.15 M KClO, the pH is 7.64. What is the pH after 30 mL of water is added to raise the volume to 50 mL?

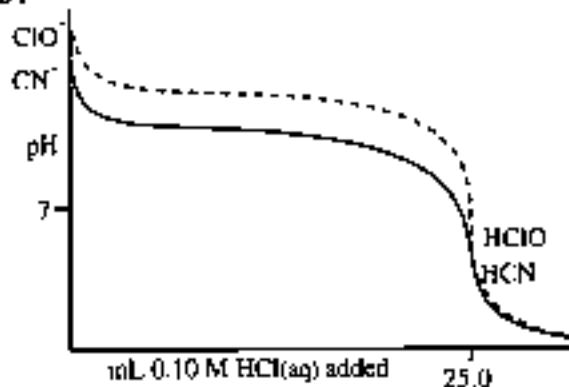
- A. 7.51
- B. 7.64
- C. 7.77
- D. 8.26

48. Which of the following curves accurately represents the titrations of 0.10 M NaCN(aq) by 0.10 M HCl(aq) and 0.10 M NaClO(aq) by 0.10 M HCl(aq)?

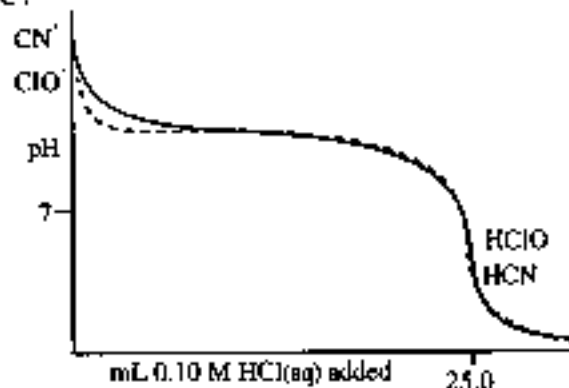
A.



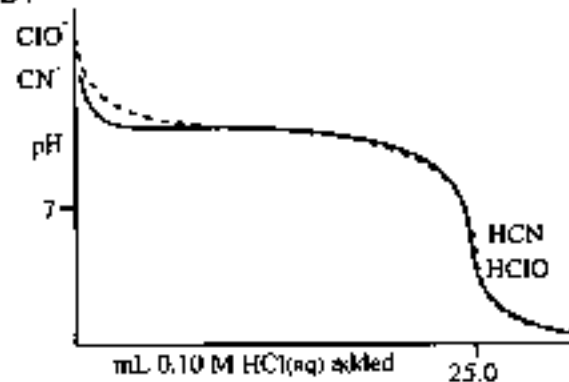
B.



C.



D.



49. Which of these sequences relates the pK_a values of the indicated acids in descending order?

- A. $pK_a(\text{HCN}) > pK_a(\text{HClO}) > pK_a(\text{HOAc}) > pK_a(\text{HCl})$
 B. $pK_a(\text{HOAc}) > pK_a(\text{HClO}) > pK_a(\text{HCN}) > pK_a(\text{HCl})$
 C. $pK_a(\text{HCl}) > pK_a(\text{HCN}) > pK_a(\text{HClO}) > pK_a(\text{HOAc})$
 D. $pK_a(\text{HCl}) > pK_a(\text{HOAc}) > pK_a(\text{HClO}) > pK_a(\text{HCN})$

Passage VIII (Questions 50 - 56)

A polyprotic acid is an acid that contains more than one acidic proton. The second proton lost by the acid is never as strongly acidic as the first proton lost by the acid. The second proton can be removed using strong base, once the first proton has been completely removed from the acid. Concentrations of polyprotic acids are measured in terms of normality. Normality measures molar equivalence. The normality of an acid equals the molarity of the equivalent base required to neutralize all of the acidic protons.

A researcher sets out to determine the effect of mixing solutions together, by studying their pH before and after mixing. Into a flask (Flask 1) he places exactly 25.0 mL of 0.20 N H_2SO_4 to be titrated by a solution of NaOH of unknown concentration. Into a second flask (Flask 2) he places exactly 40.0 mL of 0.30 N H_3PO_4 , also to be titrated by the same solution of NaOH. Both solutions are titrated in succession to a visual endpoint, determined by the color change of an indicator. The quantity of base needed to achieve this is recorded accurately to the second decimal place.

50. If Flask 1 requires exactly 20.0 mL NaOH solution to be neutralized, what must be the concentration of the NaOH solution?

- A. 0.125 M NaOH(aq)
- B. 0.200 M NaOH(aq)
- C. 0.250 M NaOH(aq)
- D. 0.500 M NaOH(aq)

51. Which of the following statements is true?

- A. $\text{p}K_{a1}$ is always larger than $\text{p}K_{a2}$
- B. $\text{p}K_{a2}$ is always larger than $\text{p}K_{a1}$.
- C. There is no rule for $\text{p}K_{a2}$ or $\text{p}K_{a1}$.
- D. $\text{p}K_{a2}$ is greater than $\text{p}K_{a1}$ only for the oxyacids.

52. A solution of 0.30 M H_3PO_4 (aq) has which of the following values for normality?

- A. 0.10 N H_3PO_4 (aq)
- B. 0.30 N H_3PO_4 (aq)
- C. 0.60 N H_3PO_4 (aq)
- D. 0.90 N H_3PO_4 (aq)

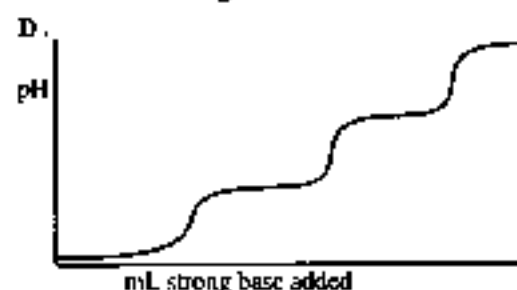
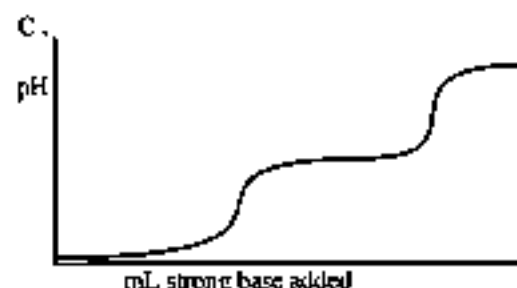
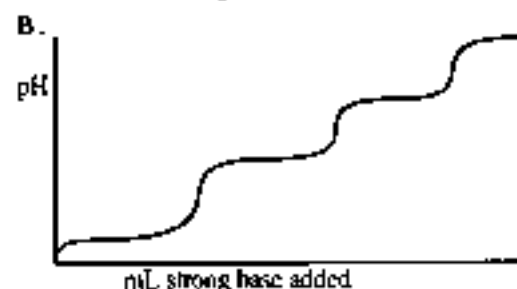
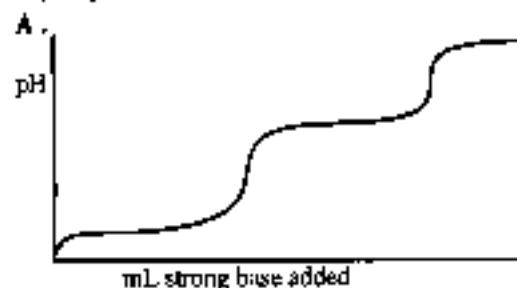
53. When 0.1 moles NaH_2PO_4 and 0.2 moles Na_2HPO_4 are mixed in 100 mL, what is the pH of the solution?

- A. $\text{pH} < \frac{\text{p}K_{a1} + \text{p}K_{a2}}{2}$
- B. $\text{p}K_{a2} > \text{pH} > \frac{\text{p}K_{a1} + \text{p}K_{a2}}{2}$
- C. $\frac{\text{p}K_{a2} + \text{p}K_{a3}}{2} > \text{pH} > \text{p}K_{a2}$
- D. $\text{pH} > \frac{\text{p}K_{a2} + \text{p}K_{a3}}{2}$

54. How many mL of 0.40 M H_2SO_4 (aq) would require the same amount of base to reach full neutralization as would 25 mL 0.60 M H_3PO_4 (aq)?

- A. 25.00 mL 0.40 M H_2SO_4 (aq)
- B. 37.50 mL 0.40 M H_2SO_4 (aq)
- C. 50.00 mL 0.40 M H_2SO_4 (aq)
- D. 56.25 mL 0.40 M H_2SO_4 (aq)

55. Which titration curve represents the complete titration of phosphoric acid?



56. What is the phosphate concentration in Flask 2 after 40 mL of 0.30 N NaOH has been added?

- A. 0.050 M PO_4^{3-} (aq)
- B. 0.100 M PO_4^{3-} (aq)
- C. 0.450 M PO_4^{3-} (aq)
- D. 0.900 M PO_4^{3-} (aq)

Passage IX (Questions 57 - 63)

Polyprotic acids are acids that contain more than one acidic hydrogen. A typical example is sulfuric acid (H_2SO_4). The first proton is easily removed, while the second proton is harder to remove. Sulfuric acid is a strong acid with respect to its first proton, but weak with respect to its second proton. The titration curve for a polyprotic acid looks like two separate titration curves that have been connected. The endpoint of the curve for the first proton is the starting point of the curve for the second proton. A student titrates an unknown diprotic acid with strong base. The titration curve for the experiment is shown in Figure 1.

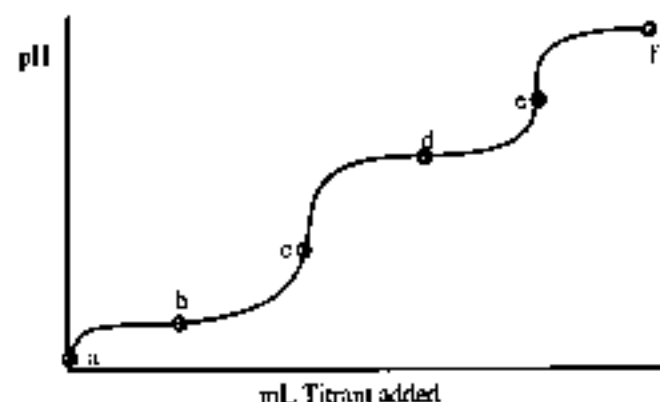


Figure 1

The exact pH and exact volume of titrant used are not known, but the graph is proportional throughout the duration of the titration. The student repeats the experiment three times, and the graph in Figure 1 represents the best results of the three trials.

57. The unknown acid can BEST be categorized as which of the following?
- A strong monoprotic acid
 - A diprotic acid with both protons strong
 - A diprotic acid with one strong proton and one weak proton
 - A diprotic acid with both protons weak
58. Which of the following relationships is NOT true about the unknown acid?
- $\text{p}K_{a1} < \text{p}K_{a2}$
 - pH at first equivalence point $> \text{p}K_{a1}$
 - pH at second equivalence point $> \text{p}K_{a2}$
 - pH at first equivalence point $> \text{p}K_{a2}$
59. In Figure 1, the titrant can best be described as a:
- strong acid.
 - strong base.
 - weak acid.
 - weak base.

60. Which change requires that the MOST titrant be added to the solution?
- Going from a pH $< \text{p}K_{a1}$ to a pH $> \text{p}K_{a2}$
 - Going from the first equivalence point to the second equivalence point
 - Going from a pH $> \text{p}K_{a1}$ to a pH $< \text{p}K_{a2}$
 - Going from the initial point to the first equivalence point
61. If, after the addition of 5 mL of titrant, the pH of the solution is less than the $\text{p}K_{a1}$ of a diprotic acid, what volume of titrant is required to reach the second equivalence point from the fully protonated state?
- Between 5 and 10 mL
 - Between 10 and 15 mL
 - Between 15 and 20 mL
 - More than 20 mL
62. What is true the predominant species and any other species present between points d and e on the titration curve?
- A^{2-} (the fully deprotonated form) is all that is present.
 - A^{2-} is present along with some HA^- (the partially protonated form).
 - HA^- (the partially protonated form) is all that is present.
 - HA^- is present along with some H_2A (the fully protonated form).
63. Which of these points in Figure 1 is NOT described correctly below?
- Point b is where $[\text{H}_2\text{A}] = [\text{HA}^-]$.
 - Point c is the first equivalence point.
 - Point d is where pH = $\text{p}K_{a2}$.
 - Point f is where $[\text{HA}^-] = [\text{A}^{2-}]$.

Passage X (Questions 64 - 70)

Polyequivalent bases are bases that can neutralize more than one acidic hydrogen per base molecule. A typical example of a **divalent** base is carbonate (CO_3^{2-}). The first proton added corresponds to the second proton removed. Carbonate is a stronger base than its conjugate acid bicarbonate (HCO_3^-), although both are considered weak bases, because the pK_b values are 3.67 and 7.63 respectively. The titration curve in Figure 1 is obtained when calcium carbonate (CaCO_3) is titrated with a strong acid, such as hydrochloric acid.

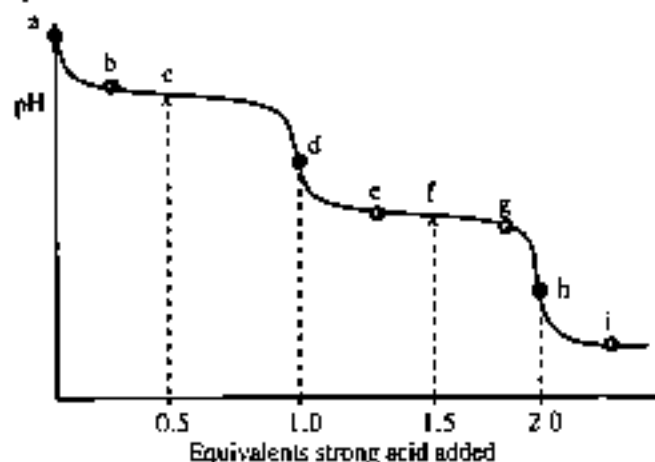


Figure 1

The exact pH and exact volume of titrant used in this titration are not provided, but the graph is proportional throughout its duration. The nine points marked along the titration curve note some of the key transitional points. Despite the fact that the pH is recorded for the titration of the base (rather than the pOH), the titration curve is still predictable.

Among the key points recorded are the two equivalence points, and the two points at which the $\text{pH} = \text{pK}_a$ (both the first and second pK_a points are marked). It is interesting to note that when $\text{pH} = \text{pK}_{a1}$, $\text{pOH} = \text{pK}_{b2}$. The other points of the curve are interesting in that they represent different points at which the pH is predictable from the trend in the pH and the amount of acid added.

64. At which point is the pH equal to the value of pK_{a1} ?

- A. Point c
- B. Point d
- C. Point f
- D. Point h

65. Which of the following relationships is NOT true?

- A. $\text{pK}_{a1} + \text{pK}_{b2} = 14$
- B. $\text{pK}_{a2} + \text{pK}_{b2} = 14$
- C. $\text{pH}_{\text{initially}} > \text{pK}_{a2}$
- D. $\text{pH}_{\text{at 2nd equivalence}} < \text{pK}_{a1}$

66. How can the pH of solution be determined at point d?

- A. $\text{pH} = 7.0$
- B. $\text{pH} = \frac{\text{pK}_{a1} + \text{pK}_{a2}}{2}$
- C. $\text{pH} = \sqrt{(\text{pK}_{a1})^2 + (\text{pK}_{a2})^2}$
- D. $\text{pH} = 2(\text{pK}_{a1} - \text{pK}_{a2})$

67. What is NOT true at point e on the titration curve?

- A. Fewer than 1.5 equivalents of HCl have been added.
- B. The pH is less than 6.37.
- C. The pH is greater than pK_{a1} .
- D. The pH is less than pK_{a2} .

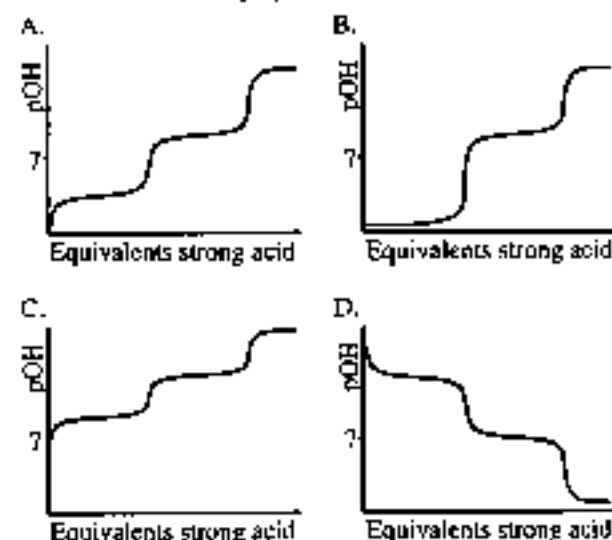
68. In what pH range is the concentration of bicarbonate the GREATEST?

- A. pH less than 3.67
- B. pH between 3.67 and 7.00
- C. pH between pH 7.00 and 10.33
- D. pH greater than 10.33

69. Between which two points does the pH of the solution change by the GREATEST amount?

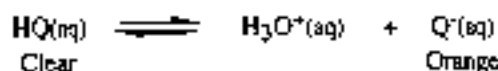
- A. Between point c and point d
- B. Between point d and point e
- C. Between point e and point f
- D. Between point f and point g

70. Which of the following graphs shows changing pOH as a function of equivalents strong acid when calcium carbonate is titrated by hydrochloric acid?



Passage XI (Questions 71 - 76)

When titrating an acid with a strong base, an indicator is added to the solution to indicate when the equivalence point has been reached. An indicator works by changing color at or near the equivalence point of the titration. Because indicators can lose or gain a proton, they are susceptible to the effects of varying solution pH during titration. A conjugate acid and its conjugate base have different colors (although in some cases, one of the two is colorless.) With most indicators, both the conjugate acid and conjugate base are colored species. When both species are colored, it is often harder to see the color change, unless there is a pronounced difference between the two colors. An example with the hypothetical indicator HQ, and its dissociation to conjugate base Q⁻, is shown in Reaction 1. Assume HQ is colorless and Q⁻ is orange.



Reaction 1

When the conjugate base (Q^-) is the predominant species in solution, it appears orange. When the conjugate acid (HQ) is the predominant species in solution, the solution appears clear. When the pH of the solution is equal to the pK_a of the indicator, there are equal parts of HQ and Q^- in solution, so the solution has an orange hue. Indicators are used in low concentration, so they don't become visibly detectable until at least one-tenth of the indicator is in its deprotonated (orange conjugate base) form. This occurs when the solution pH is approximately one pH unit below the pK_a of the indicator. Equation 1 lists the active range of an indicator.

$$pH(\text{at equivalence}) = pK_a(\text{indicator}) \pm 1.$$

Equation 1

Because it is desirable to have the color change near the equivalence point of the species being titrated, Equation 1 helps when selecting an indicator. If the equivalence pH is not known, then the pH at equivalence can be approximated as $pK_{a(acid)} + 3$ (where the pK_a is for the acid being titrated). This means that often the pK_a of the indicator is about three pH units higher than the pK_a of the acid being titrated. This holds true only if the pK_a for the acid is between 4 and 10. It is not affected by concentration factors. Figure 1 shows a titration curve for the titration of a weak acid by strong base and the active range of an indicator.

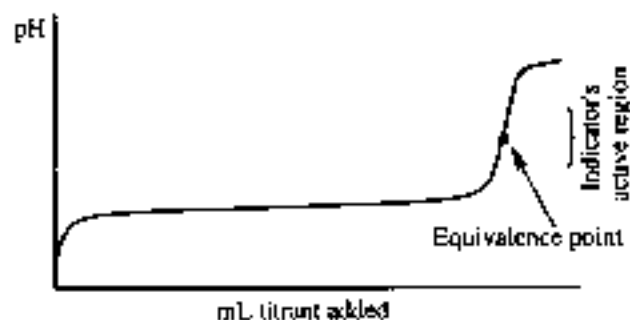


Figure 1

The change in shading represents the increase in the intensity of the orange color of the solution. The equivalence occurs in the middle of the color change band.

71. The BEST indicator to use for the titration of an acid with a pK_a of 5.0 would be one having a pK_b of:
- A. 1.5.
 - B. 5.0.
 - C. 8.5.
 - D. 11.5.
72. An indicator in a concentration that is too high could have what effect on a solution?
- A. It could interfere with the acid and/or base properties of the solution being titrated.
 - B. It could change the viscosity of the solution being titrated, resulting in a non-homogeneous solution.
 - C. The color change could be too extreme to be useful.
 - D. The color change could be too subtle to detect.
73. What is the ratio of the conjugate base to conjugate acid at two pH units above the pK_a ?
- A. 100 : 1
 - B. 2 : 1
 - C. 1 : 2
 - D. 1 : 100
74. A good indicator has which of the following properties?
- A. It should be transparent throughout the titration.
 - B. It should be unreactive with respect to acid-base reactions.
 - C. It should be a strong acid or a strong base.
 - D. It should be a weak acid or a weak base.
75. For which of these titrations is the pH at equivalence the same, regardless of concentration?
- A. The titration of a weak acid by a strong base
 - B. The titration of a strong acid by a weak base
 - C. The titration of the first proton of a weak diprotic acid by a strong base
 - D. The titration of the second proton of a weak diprotic acid by a strong base
76. Which BEST explains why the indicator pK_a may be one unit off from the equivalence pH?
- A. A difference of one pH unit has an insignificant effect on color at the pH of the indicator's pK_a .
 - B. The pH changes rapidly at equivalence.
 - C. At the equivalence point for the acid, $pH = pK_a$.
 - D. An indicator's color change occurs at only one very specific pH value.

Passage XII (Questions 77 - 83)

Indicators have two main purposes in chemistry. The first use is as an aid in determining the equivalence point in a titration experiment. If the pH at equivalence is known, then an indicator that changes color at or near that pH value can be used to detect the equivalence point. The ideal indicator has its pK_a value within ± 1 of the pH at equivalence.

The second use of an indicator is to approximate the pH of an unknown aqueous solution by examining the color of the indicator in a sample of the solution. The indicator changes color at a pH value approximately one unit from its pK_a value, so if the pH of solution is within one pH unit of the indicator's pK_a value, then the pH can be estimated. Table 1 shows the relationship between pH and the color associated with three indicators.

pH	Indicator I	Indicator II	Indicator III
1	Red	Yellow	Purple
2	Red	Yellow	Purple
3	Red	Yellow	Violet
4	Red	Yellow	Fuchsia
5	Red	Yellow	Clear
6	Red-orange	Yellow	Clear
7	Orange	Yellow	Clear
8	Mango	Yellow	Clear
9	Yellow	Chartreuse	Clear
10	Yellow	Green	Clear
11	Yellow	Aquamarine	Clear
12	Yellow	Blue	Clear
13	Yellow	Blue	Clear

Table 1

For the titration of a strong acid by strong base, the pH at the equivalence point is 7.0, while for the titration of a weak acid by a strong base, the pH at equivalence is greater than 7.0. Because the pH at equivalence is different, a different indicator is required for the two titrations, although for a weak acid with a pK_a of 3 or less, the indicator used in the titration of a strong acid may work.

The equilibrium distribution of the deprotonated and protonated forms of an indicator obeys Equation 1, where Ind^- is the deprotonated form and $H-Ind$ is the protonated form.

$$pH_{\text{solution}} = pK_a(\text{indicator}) + \log \frac{[Ind^-]}{[H-Ind]}$$

Equation 1

77. Which indicator could be used to determine the equivalence point in the titration of 1.0 M benzoic acid by 1.0 M KOH(aq)?

The pK_a of H_3C_6COOH is 4.21.

- A. Indicator I only
- B. Indicator III only
- C. Both Indicator I and Indicator II
- D. Both Indicator II and Indicator III

78. Which combination of colors is NOT possible for a solution?

- A. I: red II: yellow III: purple
- B. I: mango II: yellow III: violet
- C. I: red II: yellow III: fuchsia
- D. I: yellow II: aquamarine III: clear

79. If the ratio of blue species to clear species within an indicator equilibrium is 1000:1 at $pH = 6$, what is the pK_a of the indicator, given that the deprotonated form absorbs visible light?

- A. 3
- B. 4
- C. 9
- D. 10

80. What is the pH of a solution that is clear when Indicator III is added, faint chartreuse when Indicator II is added, and faint mango when Indicator I is added?

- A. 7.0
- B. 7.5
- C. 8.0
- D. 8.5

81. Which of the following accurately describes the pK_a values for the three indicators?

- A. I: 6.86 II: 9.87 III: 4.22
- B. I: 8.94 II: 7.21 III: 5.98
- C. I: 7.21 II: 8.28 III: 4.11
- D. I: 8.34 II: 9.66 III: 3.85

82. The active range for Indicator III is which of the following?

- A. pH 2 to pH 6
- B. pH 3 to pH 4
- C. pH 3 to pH 5
- D. pH 4 to pH 6

83. The best description of the absorbance of light associated with Indicator III is:

- A. The protonated form absorbs light with a λ_{max} of 426 nm, while the deprotonated form absorbs light with a λ_{max} of 339 nm.
- B. The protonated form absorbs light with a λ_{max} of 339 nm, while the deprotonated form absorbs light with a λ_{max} of 426 nm.
- C. The protonated form absorbs light with a λ_{max} of 598 nm, while the deprotonated form absorbs light with a λ_{max} of 339 nm.
- D. The protonated form absorbs light with a λ_{max} of 339 nm, while the deprotonated form absorbs light with a λ_{max} of 598 nm.

Passage XIII (Questions 84 - 89)

In titration, it is common to use indicators to signify a certain pH for the solution. Indicators are made, most commonly, from organic dyes that gradually change color within a given pH range. A color change is observed, because the indicator in its protonated state is one color and in its deprotonated state is a different color. Reaction 1 shows the equilibrium for a generic indicator.



Reaction 1

Table 1 below lists a series of indicators, the pK_a of the indicator, the active range for visual detection, and the respective colors of the protonated form (acid) and deprotonated form (base). The Henderson-Hasselbalch equation describes the relationship between the members of a conjugate pair. When the concentrations of the acid and base are equal, the pH equals the pK_a . When this occurs, the color of the solution is an average of the colors listed in Table 1.

Indicator	pK_a	pH range	Acid	Base
Alizarin Yellow	10.8	10.0 - 11.6	Yellow	Red
Phenolphthalein	8.8	8.0 - 9.6	Clear	Pink
Thymol Blue	8.4	7.6 - 9.2	Yellow	Blue
Cresol Red	8.0	7.2 - 8.8	Yellow	Red
Bromthymol Blue	6.8	6.0 - 7.6	Yellow	Blue
Chlorophenol Blue	5.6	4.8 - 6.4	Yellow	Red
Bromocresol Green	4.4	3.7 - 5.1	Yellow	Blue
Methyl Orange	3.9	3.2 - 4.6	Orange	Yellow
Bromphenol Blue	3.7	3.0 - 4.4	Yellow	Blue
Erythrosin B	2.8	2.2 - 3.6	Orange	Yellow
Thymol Blue	2.0	1.2 - 2.8	Red	Yellow
Cresol Red	1.6	1.0 - 2.2	Red	Yellow
Methyl Violet	0.8	0.0 - 1.6	Yellow	Violet

Table 1

Depending on the concentrations and species being titrated, there is one ideal indicator. To detect the equivalence point for a titration, a small portion of indicator is added to solution. Ideally, the equivalence point of the titration should be equal to the pK_a of the indicator. At the very least, the equivalence pH must fall within the indicator's active range. When a strong acid or strong base is titrated with a strong titrant, the pH at the equivalence point is always equal to 7.0. For a weak acid titration, the pH at equivalence can be estimated by taking an average of the pK_a of the acid being titrated and the pH of the titrant base.

84. Which of the following does NOT form a blue solution?

- A. Bromphenol blue in a pH = 7.0 buffer
- B. Bromocresol green in an aqueous ammonia solution
- C. Thymol blue in an acetic acid solution
- D. Bromthymol blue in a hydroxide solution

85. Which of the following indicators CANNOT be used for the titration of ammonia by hydrochloric acid?

- A. Phenolphthalein
- B. Bromocresol green
- C. Methyl orange
- D. Bromphenol blue

86. If KCN has $\text{pK}_b = 4.68$ and KF has $\text{pK}_b = 10.83$, which indicator is NOT the ideal choice for the following proposed titrations?

- A. 1.00 M KF(aq) titrated by 1.00 M HCl(aq) with an indicator of cresol red
- B. 0.01 M KF(aq) by 0.01 M HCl(aq) with an indicator of methyl violet
- C. 1.00 M KCN(aq) by 1.00 M HCl(aq) with an indicator of bromocresol green
- D. 0.01 M KCN(aq) by 0.01 M HCl(aq) with an indicator of chlorophenol blue

87. Which of the following indicator : color correlations is NOT correct for biological pH (pH = 7.4)?

- A. Methyl violet : Violet
- B. Methyl orange : Yellow
- C. Thymol blue : Blue
- D. Bromocresol green : Blue

88. Which of the following titrations requires thymol blue indicator?

- A. A low concentration of strong base titrated by a strong acid
- B. A low concentration of weak base titrated by a strong acid
- C. A highly concentrated strong base titrated by a strong acid
- D. A highly concentrated weak base titrated by a strong acid

89. Which of the following indicators should be chosen to identify the equivalence point of a strong acid titrated by a strong base?

- A. Methyl violet
- B. Methyl orange
- C. Bromthymol blue
- D. Alizarin yellow

Passage XIV (Questions 90 - 96)

A *pH stick* is a device that can approximate the pH of an aqueous solution by reference to a color blend band. It is a plastic stick with indicators attached to specific segments of the stick. Each indicator has a range of color change that extends to either side of a central pK_a value. When the pH of the solution is less than the pK_a of the indicator, the indicator exists predominantly in its protonated state. When the pH of the solution is greater than the pK_a of the indicator, the indicator exists predominantly in its deprotonated state. An indicator is ideal when both the protonated and deprotonated species are colored. When the species are primary colors, the change is easier to observe.

Typical pH sticks come with three to four indicators, such as those listed in Table 1. Each solution must be tested with a separate pH stick. A pH stick cannot be used for more than one test, because the acidity of the first test solution affects other solutions that are added to the pH stick.

Indicator	pK_a	Protonated Color	Deprotonated Color
Bromocresol Green	4.37	Yellow	Blue
Methyl Red	5.21	Red	Yellow
Bromthymol Blue	6.78	Yellow	Blue
Phenolphthalein	8.79	Clear	Magenta

Table 1

A student uses a pH stick with four indicator markers to approximate the pH of five separate solutions. Table 2 lists the results for the five separate solutions. Each sequence of colors represents what is observed for the four indicators in order of increasing pK_a .

Solution	Bromocresol Green	Methyl Red	Bromthymol Blue	Phenolphthalein
1	Yellow	Red	Yellow	Clear
2	Blue	Orange	Yellow	Clear
3	Blue	Yellow	Greenish-blue	Clear
4	Blue	Yellow	Blue	Magenta
5	Greenish-blue	Reddish-orange	Yellow	Clear

Table 2

An indicator is used to detect the endpoint in a titration. An ideal indicator for a titration has its pK_a equal to the equivalence pH of the titration. This is the ideal condition, but in practice, the exact pH at equivalence often cannot be determined. The general rule for titration is that the pH at equivalence should be within one of the pK_a of the indicator.

90. What is the pH range of the pH stick?

- A. 3.4 to 8.8
- B. 3.4 to 9.8
- C. 4.4 to 8.8
- D. 4.4 to 9.8

91. It is NOT possible to estimate pH for:

- A. Solution 1 only.
- B. Solution 1 and Solution 2 only.
- C. Solution 2 and Solution 3 only.
- D. Solution 1 and Solution 4 only.

92. An aqueous solution that has a hydroxide concentration of 1.0×10^{-6} M would show what colors?

- A. Yellow with bromocresol green
- B. Red with methyl red
- C. Blue with bromthymol blue
- D. Clear with phenolphthalein

93. What is the approximate pH of Solution 5?

- A. 4
- B. 5
- C. 6
- D. 7

94. When HCl is added to Solution 2, the colors on the pH stick do not change. How can this be explained?

- A. The hydronium concentration is too high for the pH to be affected by the addition of HCl.
- B. The hydronium concentration is too low for the pH to be affected by the addition of HCl.
- C. The solution is a buffer made from a carboxylic acid and its carboxylate conjugate base.
- D. The solution is a buffer made from an amine and its ammonium conjugate acid.

95. Which indicator could be used in the titration of a weak acid by a strong base?

- A. Bromocresol green
- B. Methyl red
- C. Bromthymol blue
- D. Phenolphthalein

96. Which of the following statements is INVALID?

- A. A solution that turns bromthymol blue to blue would turn Phenolphthalein to magenta.
- B. A solution cannot show two green marks on the pH stick.
- C. A solution that turns methyl red to yellow would turn bromocresol green to blue.
- D. The pH stick can estimate pH best when that value falls between 4.21 and 5.37.

Passage XV (Questions 97 - 100)

A common class of acids in organic chemistry is the carboxylic acids. Their acidity is attributable to the electron-withdrawing nature of the carbonyl group through resonance. Short-chain carboxylic acids are water-soluble. As the alkyl chain length increases, the hydrophilicity decreases, making carboxylic acids of five carbons or more rather insoluble. Long-chain fatty acids are often used as surfactants because of their insolubility.

Phenols constitute another common class of acids in organic chemistry. Their acidity is attributable to the electron-withdrawing nature of the benzene ring through resonance. Because aromatic rings are less electron-withdrawing than carbonyl groups, phenols are weaker acids than their carboxylic acid counterparts. Most phenols are insoluble in water in their protonated state. Table 1 lists a few examples of carboxylic acids and phenols, along with their pK_a values.

Organic Acid	pK_a
	0.64
	3.40
	4.21
	7.18
	10.01

Table 1

Because of solubility constraints, carboxylic acids and amines are more often involved in aqueous buffering than phenols. However, it is possible to titrate phenols, because their conjugate base is water-soluble, allowing for the reaction to be monitored.

97. Which of these mixtures produces the MOST effective buffer for pH 4.0?

- $H_3CCO_2H + H_3CCO_2Na$
- $Cl_3CCO_2H + Cl_3CCO_2Na$
- $O_2NC_6H_4CO_2H + O_2NC_6H_4CO_2Na$
- $C_6H_5CO_2H + C_6H_5CO_2Na$

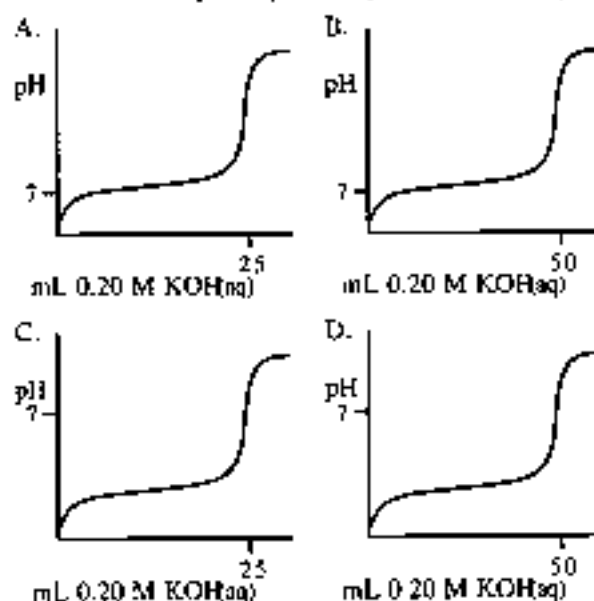
98. For the titration of p-nitrophenol, which of the following reagents would be best?

- KOH
- Potassium p-nitrophenoxide
- Phenol
- HCl

99. What is the pH of a solution made by mixing 20 mL 0.10 M phenol with 10 mL 0.10 M KOH(aq)?

- 10.0
- 7.0
- 5.8
- 5.5

100. Which of the following graphs represents the titration of 50 mL 0.1 M p-nitrophenol by 0.20 M KOH(aq)?



1. B	2. A	3. B	4. B	5. B
6. D	7. B	8. D	9. D	10. A
11. C	12. B	13. A	14. B	15. C
16. B	17. C	18. C	19. A	20. D
21. C	22. C	23. C	24. D	25. D
26. C	27. D	28. A	29. B	30. C
31. C	32. C	33. C	34. D	35. D
36. D	37. A	38. A	39. C	40. C
41. C	42. D	43. D	44. B	45. B
46. D	47. B	48. A	49. A	50. C
51. B	52. D	53. C	54. D	55. B
56. A	57. D	58. D	59. B	60. A
61. D	62. B	63. D	64. C	65. B
66. B	67. B	68. C	69. A	70. A
71. C	72. A	73. A	74. D	75. C
76. B	77. A	78. B	79. A	80. D
81. A	82. C	83. C	84. C	85. A
86. B	87. C	88. D	89. C	90. B
91. D	92. C	93. B	94. C	95. D
96. A	97. D	98. A	99. A	100. A

Buffers and Titration Passage Answers

Passage I (Questions 1 - 7)

Buffer pH and Weak Acids

1. **Choice B is correct.** The mixture is composed of NaHCO_3 and H_2CO_3 . Without considering how much of each component is present in solution, recognize first that they are a conjugate pair, meaning that the solution is a buffer. This means that the pH is close to the pK_a . Carbonic acid is diprotic, so be sure you understand that the first proton is involved in this conjugate pair, and that the pH should be close to pK_{a1} . This makes choice B the best candidate. If you wish to solve for the exact value, the first step is to convert 0.839 grams $\text{NaHCO}_3(s)$ into moles: $(0.839 \text{ grams}) \left(\frac{1 \text{ mole}}{84 \text{ grams}} \right) = 0.01 \text{ moles NaHCO}_3$. The number of moles of $\text{H}_2\text{CO}_3 = (0.10 \text{ L})(0.10 \text{ M}) = 0.01 \text{ moles H}_2\text{CO}_3$. The pH can be found using the Henderson-Hasselbalch equation:

$$\text{pH} = \text{pK}_a + \log \frac{\text{Moles conjugate base}}{\text{Moles conjugate acid}} \therefore \text{pH} = \text{pK}_{a1} + \log \frac{\text{Moles HCO}_3^-}{\text{Moles H}_2\text{CO}_3} = 6.4 + \log \frac{0.01}{0.01} = 6.4 + \log 1 = 6.4$$

Since the log of 1 is 0, the pH of the solution is equal to pK_{a1} , 6.4. The best answer is choice B.

2. **Choice A is correct.** Even though the solution is a buffer, the addition of HCl decreases the pH slightly. A buffer resists extreme pH changes, but a small change is often observed. The addition of an acid to the buffered solution lowers the pH. Since the pH is initially 7.21, the final pH value must be lower than 7.21. The only answer choice less than 7.21 is choice A, 7.14.
3. **Choice B is correct.** To make a buffer, a weak acid and its conjugate base must be mixed. Benzoic acid is a weak acid, so it must be mixed with benzoate, its conjugate base. The desired pH is equal to the pK_a of benzoic acid, so according to the Henderson-Hasselbalch equation, equal parts of benzoate and benzoic acid must be mixed. This can be accomplished either by adding one-half equivalent (in terms of moles) of strong base (NaOH) to convert half of the benzoic acid into benzoate, or by adding an equivalent amount (in terms of moles) of benzoate to the benzoic acid solution. Choice A has the conjugate pair added together in an equal gram ratio, not equal mole ratio. This does generate equal mole portions, so the pH is not equal to the pK_a . This eliminates choice A. The remaining three choices involve the mixture of benzoic acid with a strong base, so they must be mixed in a manner that half-titrates the benzoic acid. The only answer with half as much strong base as weak acid (benzoic acid) is choice B. This mixture yields equal molar portions of the two components of the conjugate pair (benzoic acid and benzoate), so the best answer is choice B.
4. **Choice B is correct.** The concentration of the titrant strong base (0.2 M NaOH) is twice that of the weak acid HF (0.1 M), so to reach equivalence, only half the volume of strong base is required. There are initially 50 mL of 0.1 M HF present, so only 25 mL of 0.2 M NaOH are required to reach the equivalence point. The target pH for the solution is 3.3, which happens to be the pK_a of HF. This means that the pH of solution is equal to pK_a of HF, which is true when $[\text{HF}] = [\text{F}^-]$. This occurs when the HF(aq) is half-titrated. If 25 mL 0.2 M NaOH(aq) is required for full titration, then 12.5 mL is required for half titration. Choice B is the best answer.
5. **Choice B is correct.** This question takes more than the usual amount of effort to answer. There are two factors to consider: dilution and reactivity. Addition of sodium hydroxide solution converts some of the acetic acid to acetate, and it dilutes the solution. After the completion of the reaction, 3.0 mmole of H_3CCO_2^- are present in 30 mL of aqueous solution. From here on, it's strictly a matter of calculation, beginning with $\frac{3 \text{ mmole}}{30 \text{ mL}} = 0.1 \text{ M H}_3\text{CCO}_2^-$. Choice B is the best answer. The reaction chart below shows how the moles were determined.

	HOAc	+	NaOH	$\rightleftharpoons$	H ₂ O	+	H ₃ CCO ₂ ⁻
init:	6 mmols		3 mmols		xxx		0
Δ:	-3 mmols		-3 mmols		xxx		+3 mmols
final:	3 mmols		0		xxx		3 mmols

6. **Choice D is correct.** A buffer is prepared by mixing (in this case) a weak acid with a half-molar equivalent of strong base. This is also referred to as half-titrating a weak acid. Choice A is eliminated, because twice as much acid as weak base has been added. Choice B is eliminated, because both species are acids. Choice C is eliminated, because both species are bases. In choice D, half of an equivalent of strong base is added to a weak acid. This results in a buffer, so the best answer is therefore answer choice D.

7. **Choice B is correct.** The solution is acidic, so the pH is less than 7.0. All of the answer choices are less than 7, so nothing is eliminated. For a weak acid with pK_a between 2 and 12 in an aqueous solution where $[HA]_{\text{initial}}$ is greater than K_a , use the shortcut equation to determine the pH. The pK_a is 5.0, and $[HA]$ is 0.1 M.

$$pH = \frac{1}{2}pK_a - \frac{1}{2}\log [HA] = \frac{1}{2}(5.0) - \frac{1}{2}\log (0.10) = 2.5 - \frac{1}{2}(-1) = 2.5 + 0.5 = 3.0$$

The pH is 3.0, so the best answer is choice B.

Passage II (Questions 8–14)

Buffer Composition

8. **Choice D is correct.** A buffer results when a solution contains roughly equal concentrations of weak acid and its conjugate base. This can be achieved by mixing the components of the conjugate pair in a roughly one-to-one ratio, as is observed in choices A and B. In choice C, the acid is diprotic, so the 1.5 equivalents of strong base completely remove the first proton (to form HCO_3^-) and then pull off a second proton from half of the bicarbonate ions. The result is a solution with equal parts HCO_3^- and CO_3^{2-} . Because these are a conjugate pair, the solution forms a buffer. This eliminates choice C. In choice D, methyl amine is capable of gaining only one proton, so the 1.5 equivalents of HCl completely converts the weak base (H_3CNH_2) into its conjugate acid ($H_3CNH_3^+$), with a leftover of 0.5 equivalents of HCl. The solution is a mixture of weak acid and strong acid, which does *not* result in a buffer. The best answer is thus choice D.
9. **Choice D is correct.** Because the solution contains more weak acid than conjugate base, the pH is less than the pK_a of the acid. This eliminates choices A and B. According to Equation 1, it is a log relationship, so the best answer is choice D. Plugging into Equation 1 would yield $pH = pK_a + \log 1/2 = pK_a - \log 2$.
10. **Choice A is correct.** The pH can be regulated (maintained at a relatively constant value) by a buffer. A buffer is composed of a weak acid and its conjugate base. Because the pH of the buffer must be close to (within one unit of) the pK_a of the acid, the pH becomes harder to regulate as the weak acid becomes stronger. To regulate the pH at 1.0 requires an acid with a pK_a no greater than 2.0. An acid with a pK_a of 2.0 or less would be one of the strongest weak acids. Increasing acid strength diminishes its ability to buffer. You should note that Table 1 does not contain any acids that could be used to buffer at 1.0. The best answer is thus choice A.
11. **Choice C is correct.** A buffer serves to maintain relatively constant pH. Choice A is eliminated, because an aqueous solution where the hydronium and hydroxide concentrations are equal has a pH of 7.0, not necessarily a constant pH. A buffer solution is one in which the ratio of the acid and base in the conjugate pair never exceeds 10 : 1, but that does not address the hydronium-to-hydroxide ratio. Consider a pH = 4.0 buffer, for instance. At pH = 4.0, the hydronium concentration is 10^{-4} M, and the hydroxide concentration is 10^{-10} M. This results in a ratio of 1,000,000 : 1. This is far greater than 10:1, so choice B is eliminated. Choice C is the best, because a relatively constant pH implies that the hydronium concentration is relatively constant. If hydronium concentration is relatively constant, so is hydroxide concentration. This makes choice C the best answer. Choice D is eliminated, because the pH should be within one unit of the pK_a value for the weak acid, not the hydronium-to-hydroxide ratio. The best answer is thus choice C.
12. **Choice B is correct.** The solution before adding the hydroxide has a 15 : 10 mole ratio of weak acid to conjugate base. Addition of 1.00 mL of KOH converts some of the weak acid into conjugate base so that the ratio of 15 : 10 becomes 14 : 11, still in favor of the weak acid. Because the weak acid is slightly more concentrated than its conjugate base, the pH of the solution is slightly lower than the pK_a . The pK_a is 4.74, so the best answer is choice B.
13. **Choice A is correct.** The largest ratio of conjugate base to weak acid is found in the solution with a pH greater than the pK_a of the weak acid by the largest amount. Formic acid (HCO_2H) has a pK_a of 3.64. In choice A, when the solution pH is 4.00, the pH is 0.36 greater than the pK_a . Hypochlorous acid ($HClO$) has a pK_a of 7.49. In choice B, when the solution pH is 7.00, the pH is 0.49 less than the pK_a . There is more weak acid than conjugate base, which eliminates choice B. Hypobromous acid ($HBrO$) has a pK_a of 8.67. In choice C, when the solution pH is 8.50, the pH is 0.17 less than the pK_a . There is more weak acid than conjugate base, which eliminates choice C. Ammonium ion (NH_4^+) has a pK_a of 9.26. In choice D, when the solution pH is 9.50, the pH is 0.24 greater than the pK_a . 0.36 is greater than 0.24, so the best answer is choice A.

14. **Choice B is correct.** Addition of water to a buffer equally dilutes the weak acid and its conjugate base. The result is that the ratio of base to acid has not changed, so the buffer does not shift. The equilibrium has not been disturbed, so the reaction exhibits no net shift. The pH remains constant. The best answer is choice B.

Passage III (Questions 15 - 21)

Buffer Chart and pK_a Chart

15. **Choice C is correct.** Ammonium (NH_4^+) has a pK_a value of 9.26 as given in Table 1. This means that when the acid and conjugate base are present in equal molar concentration, the pH of the solution is 9.26. To make the pH of the solution equal 9.5, the conjugate base must be in excess to such a degree that the log of the base-to-acid ratio is 0.24. Choice A lists the conjugate base (ammonia) in excess, so choice A is valid. Hydrocyanic acid (HCN) has a pK_a value of 9.32 as given in Table 1. This means that when the acid and conjugate base are present in equal molar concentration, the pH of the solution is 9.32. To make the pH of the solution equal 9.5, the conjugate base must be in excess to such a degree that the log of the base-to-acid ratio is 0.18. Choice B lists the conjugate base (sodium cyanide) in excess, so choice B is valid. From choice A, we know that an ammonium and ammonia solution must have excess ammonia for the pH to equal 9.5. If one-half equivalent of the strong acid HCl were added to one equivalent of ammonia, then the pH would equal the pK_a (9.26). The excess HCl added lowers the pH, so the value is less than 9.5. This makes choice C invalid and thus the answer you should choose. From choice D, we know that an hydrocyanic acid and cyanide solution must have excess cyanide for the pH to equal 9.5. If one-half equivalent of the strong base NaOH were added to one equivalent of hydrocyanic acid, then the pH would equal the pK_a (9.32). The excess NaOH added raises the pH to a value greater than 9.32 and approximately equal to 9.5, making choice D valid. Choose answer C.
16. **Choice B is correct.** One half-equivalent of base when added to one equivalent of carbonic acid removes the first proton from half of the carbonic acid molecules present in solution. This results in a solution of half carbonic acid and half bicarbonate ($[H_2CO_3] = [HCO_3^-]$). When the concentration of acid equals the concentration of conjugate base, pH equals pK_{a1} for carbonic acid, which is 6.37, as given in the passage. When one and one-half equivalents of base are added to one equivalent of carbonic acid, the first proton is completely removed from all of the carbonic acid molecules present in solution. The remaining half of an equivalent continues to remove the second proton from half of the bicarbonate molecules present in solution. This results in a solution where $[HCO_3^-] = [CO_3^{2-}]$. When the concentration of bicarbonate equals the concentration of carbonate, pH equals pK_{a2} for carbonic acid, which is 10.83, as given in the passage. This makes choice B correct. When one and one-half equivalents of base are added to one equivalent of phosphoric acid, the first proton is completely removed from all of the phosphoric acid molecules present in solution. The remaining half of an equivalent continues to remove the second proton from half of the dihydrophosphate present in solution. This results in a solution where $[H_2PO_4^-] = [HPO_4^{2-}]$. When this equality holds true, the pH equals pK_{a2} for phosphoric acid. From Table 2, we know that K_{a2} for phosphoric acid is 6.2×10^{-8} , which equates to a pK_{a2} value of $8 - \log 6.2$, which is a little over 7. This does not make a pH of 10.83, so choice C is eliminated. When two and one-half equivalents of base are added to one equivalent of phosphoric acid, the first and second protons are completely removed from all of the phosphoric acid molecules present in solution. The remaining half of an equivalent removes the third proton from half of the hydrophosphate in solution. This results in a solution where $[HPO_4^{2-}] = [PO_4^{3-}]$. When this equality holds true, the pH equals pK_{a3} for phosphoric acid. From Table 2, we know that K_{a3} for phosphoric acid is 4.8×10^{-13} , which equates to a pK_{a3} value of $13 - \log 4.8$, which is a little over 12. This does not make a solution with a pH of 10.83, so choice D is eliminated.
17. **Choice C is correct.** One and one-half equivalents of base when added to one equivalent of oxalic acid completely removes the first proton from all of the oxalic acid molecules present in solution and continues on to remove the second proton from half of the bioxalate present in solution. This results in a solution where $[HC_2O_4^-] = [C_2O_4^{2-}]$. When this equality holds true, the pH equals pK_{a2} for oxalic acid. From Table 2, we know that K_{a2} for oxalic acid is 5.1×10^{-5} , which equates to a pK_{a2} value of $5 - \log 5.1$, which is less than 7, so choice A is eliminated. One-half equivalent of base when added to one equivalent of carbonic acid removes the first proton from half of the carbonic acid present in solution. This produces a solution where $[H_2CO_3] = [HCO_3^-]$. When this equality holds true, the pH equals pK_{a1} for carbonic acid, which is 6.37, as given in the passage. Choice B is eliminated. One-half equivalent of base when added to one equivalent of hypochlorous acid removes the proton from half of the hypochlorous acid present in solution. This results in a solution where $[HClO] = [ClO^-]$. When the concentration of acid equals the concentration of conjugate base, the pH equals pK_a for HClO, which is 7.49 as given in the table. Choice C is therefore correct. One-half equivalent of acid when added to one equivalent of pyridine adds a proton to half of the pyridine present in solution. This results in a solution where $[C_5H_5N] = [C_5H_5NH^+]$. When this is true, the pH equals pK_a for C_5H_5N , which is 5.16, as given in Table 1. Choice D is eliminated.

18. **Choice C is correct.** The pK_a value for formic acid (HCO_2H) is 3.78. In order to have a $\text{pH} = 4.00$, the solution must be mixed with excess conjugate base (HCO_2^-). This makes choice A valid. The pK_a value for anilinium cation ($\text{C}_6\text{H}_5\text{NH}_3^+$) is 4.64. In order to have a $\text{pH} = 4.00$, the solution must be mixed with excess conjugate acid ($\text{C}_6\text{H}_5\text{NH}_3^+$). This makes choice B valid. The pK_a value for benzoic acid ($\text{C}_6\text{H}_5\text{CO}_2\text{H}$) is 4.19. In order to have a $\text{pH} = 4.00$, the solution must be mixed with excess conjugate acid ($\text{C}_6\text{H}_5\text{CO}_2\text{H}$). This makes choice C invalid. If excess $\text{C}_6\text{H}_5\text{CO}_2^-$ were mixed with $\text{C}_6\text{H}_5\text{CO}_2\text{H}$, then the pH of the solution would be greater than 4.19, which is greater than 4.00. This makes choice C invalid. You should select choice C. The pK_a value for pyruvic acid ($\text{H}_3\text{CCOCO}_2\text{H}$) is 3.89. In order to have a $\text{pH} = 4.00$, the solution must be mixed with excess conjugate base ($\text{H}_3\text{CCOCO}_2^-$). This makes choice D valid.
19. **Choice A is correct.** Blood that is rich in carbon dioxide is also rich in H_2CO_3 . Since carbonic acid is acidic, the pH decreases as carbonic acid is added to solution. The final solution thus has a pH lower than 7.4. This makes choice A the correct choice. Because the amount of acid is increasing, the phosphate buffer system is affected. The additional hydronium ion present is absorbed by the phosphate buffer, thus increasing the amount of H_2PO_4^- present and decreasing the amount of HPO_4^{2-} and PO_4^{3-} present in solution. This eliminates choice B. If H_2CO_3 is added to solution, it equilibrates by making hydronium ion (H_3O^+) and bicarbonate anion (CO_3^{2-}). This results in increases in the concentrations of both bicarbonate and hydronium ions. This eliminates choices C and D.
20. **Choice D is correct.** The concentration of the conjugate base is greater than the concentration of the acid when the pH is greater than the pK_a . When the pH is greater than the pK_a , the log of the ratio of base to acid is a positive value, indicating that the ratio of conjugate base to acid is greater than 1.0. The pK_a for HF is 3.17, so at $\text{pH} = 3.00$, the pH is less than the pK_a . This eliminates choice A. The value for pK_{a1} for H_2CO_3 is 6.37, so at $\text{pH} = 6.00$, the pH is less than the pK_{a1} . This eliminates choice B. The value for pK_{a2} for $\text{H}_2\text{C}_2\text{O}_4$ is 5 - log 5.1 (which is roughly 4.3), so at $\text{pH} = 4.00$, the pH is less than the pK_{a2} . This eliminates choice C. The value for pK_{a2} for H_3PO_4 is 8 - log 6.2 (which is roughly 7.2), so at $\text{pH} = 8.00$, the pH is greater than the pK_{a2} . This results in the base being more concentrated than the acid, making choice D the correct choice.
21. **Choice C is correct.** The value of pK_{a1} is 6.37 and pK_{a2} is 10.33 for carbonic acid (H_2CO_3). The pH is greater than pK_{a1} , so the conjugate base (bicarbonate) is in greater concentration than the acid (carbonic acid). This makes choice A valid. The pH is less than pK_{a2} , so the conjugate acid (bicarbonate) is in greater concentration than the base (carbonate). This makes choice B valid. The value for pK_{a2} is 7.20 and pK_{a3} is 12.32 for phosphoric acid. The pH is greater than pK_{a2} , so the conjugate base (HPO_4^{2-}) is in greater concentration than the acid (H_2PO_4^-). This means that choice C is invalid, making it the correct choice for the question. The pH is less than pK_{a3} , so the conjugate acid (HPO_4^{2-}) is in greater concentration than the base (PO_4^{3-}). This makes choice D valid.

Passage IV (Questions 22–29)

Molecular Weight from Neutralization of an Organic Acid

22. **Choice C is correct.** To reach equivalence, it takes $(0.03 \text{ L})(0.20 \text{ M}) = 0.006$ moles of NaOH. Formula weight is grams per mole equivalent, so the unknown acid has a formula weight of $1.0 \text{ g} / 0.006 \text{ moles} = 167$ grams per mole. From the choices given, the only acid that is close to 167 grams per mole is p-nitro benzoic acid ($\text{O}_2\text{NC}_6\text{H}_4\text{CO}_2\text{H}$). To verify your answer, add up the atomic weights of the molecule, O = 12, N = 14, C = 12 and H = 1 (all in grams per mole). The sum is 46 (for NO_2) + 76 (for C_6H_4) + 45 (for CO_2H) = 167 grams per mole, so the answer is definitely C.
23. **Choice C is correct.** The formula weight of a compound is measured in grams per mole. The moles are determined from $0.040 \text{ liters} \times 0.25 \text{ M} = 0.010$ moles. The formula weight is $1.0 \text{ g} / 0.010 \text{ moles} = 100$ grams per mole. Pick choice C for best results.
24. **Choice D is correct.** Since the graph begins at a high pH , the initial solution must be basic. This rules out choices A and B. The equivalence point is at $\text{pH} = 4.6$, which is < 7 . This indicates that the neutralized product is a weak acid, so the titrant must be a strong acid. (Because the equivalence point is not 7.0, the compound must be a weak base). The graph represents D, the titration of a weak base by strong acid.

25. **Choice D is correct.** To predict the pH near the equivalence point, which is what you're doing when you choose an indicator, you should use the Henderson-Hasselbalch equation. At 2 pH units beyond the pK_a of the acid, the ratio of conjugate base to acid is 100:1, which means that the reaction is almost at equivalence. This is the point at which the indicator should start to show some color change. In this question, the pK_a for the weak acid is 7.20. The best indicator is active around $pH = 9.20$. This is choice D, phenolphthalein, which is active between 8.0 and 9.6. You may remember using phenolphthalein in titration labs from general chemistry.
26. **Choice C is correct.** A buffer is formed when a conjugate pair (acid and base) are present in solution in roughly equal molar amounts. To achieve this, you can: (1) mix the conjugate acid and base evenly, (2) add the acid and half-titrate it (add a half molar equivalent of strong base), or (3) add the base and titrate it halfway (add half an equivalent of strong base). Choice A is a case of full titration, which leads to the complete conversion of the weak acid to its conjugate base, so that is out. Choice B is a case of over-titrating by adding double the amount of base needed. In choice C, the weak acid (H_3CCO_2H) is added to a half-equivalent of strong base, which results in a buffered solution. Pick C. Choice D is an acid with an acid, which doesn't react.
27. **Choice D is correct.** The best choice for a buffer of $pH = 8.5$ is a weak acid with a pK_a close to 8.5 mixed in solution with its conjugate base. The acid 4-ethanoylphenol ($H_3CCOC_6H_4OH$) has a pK_a of 8.4, which is closest to the 8.5 value, so choose D. To make the buffer equal to 8.5 (slightly to the basic side of 8.4), there should be a slight excess of the conjugate base relative to 4-ethanoylphenol. The log of the ratio of conjugate base to weak acid should be 0.1, a positive number, so the ratio of conjugate base to weak acid must be greater than 1.
28. **Choice A is correct.** The formula weight, also known as the empirical weight, is derived from the mass of compound per mole of equivalent. Adding up the molecular mass of the compound yields 90 grams/mole. Because there are two equivalents of protons per oxalic acid, the formula weight is 45 grams/mole. Pick A. Be sure not to pick B by mistake.
29. **Choice B is correct.** If the titration required 30.0 mL of 0.10 M KOH(aq) to neutralize 25.0 mL of 0.10 M HA(aq), then according to the relationship $M_{acid}V_{acid} = M_{base}V_{base}$, the molarity of the acid must be 0.12 M. However, seven drops (although the exact number is irrelevant) of HCl must be added to back titrate to the equivalence point. This means that the molarity of the unknown weak acid is actually a little less than 0.12 M, making choice B the best answer.

Passage V (Questions 30–36)

Conjugate Pair Titration Curve

30. **Choice C is correct.** As a general rule, the pH at the equivalence point is either three units above or below the pK_a for the acid depending on the type of titration. This would mean that the pH at the two equivalence points differs by approximately 6 pH units. This makes choice C the best choice.

To calculate the pH exactly for the two titrations, one can plug into the shortcut equation for both the conjugate base (formed at point b) and conjugate acid (formed at point d). At the equivalence point, the volume is double its original value, so the concentration is half of its initial value. At point b, there is 0.05 M OAc^- , and at point d, there is 0.05 M $HOAc$. From Figure 1, the pK_a for $HOAc$ is roughly 5.0. This means that pK_b for OAc^- is roughly 9.0. The two calculations are shown below:

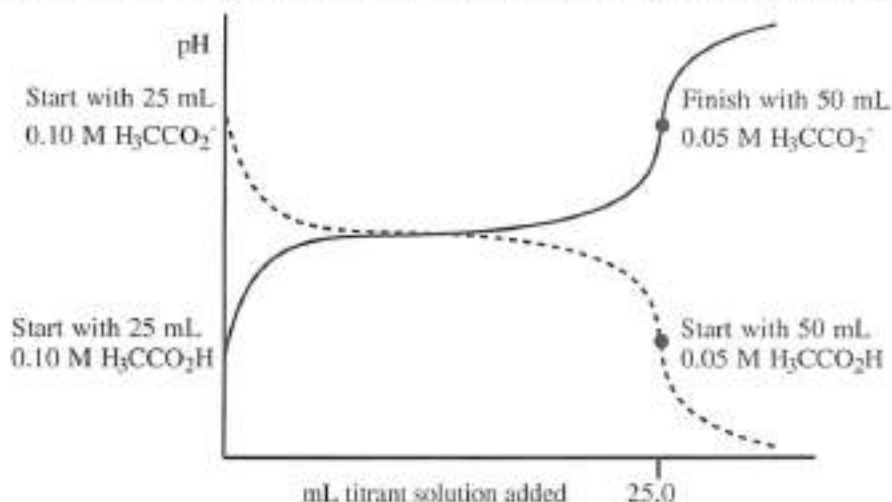
$$\text{For } HOAc \text{ at point d: } pH = \frac{1}{2}pK_a - \frac{1}{2}\log[HA] = \frac{1}{2}(5.0) - \frac{1}{2}\log(0.05) = 2.5 - \frac{1}{2}(-1.3) = 2.5 + 0.65 = 3.15$$

$$\text{For } OAc^- \text{ at point b: } pOH = \frac{1}{2}pK_b - \frac{1}{2}\log[A^-] = \frac{1}{2}(9.0) - \frac{1}{2}\log(0.05) = 4.5 - \frac{1}{2}(-1.3) = 4.5 + 0.65 = 5.15 \therefore pH = 8.85$$

The difference in pH is 5.7, so the best answer is choice C.

31. **Choice C is correct.** The pH of a solution depends on the concentration and strength of the reagents in solution. The highest pH at equivalence results from the presence of the strongest conjugate base (since all of the conjugate bases are in equal concentration). At equivalence, choices B and D are both acids, so they are both eliminated (pH at equivalence is less than 7.0). Methyl ammonium is less acidic than acetic acid based on the relative pK_a general values given in the passage. The stronger base is the conjugate base of the weaker acid, making methylamine the more basic compound between acetate and the amine. The greater hydroxide concentration results from the titration of 0.10 M H_3CNH_3Cl , so choice A is eliminated, and thus choice C is the best answer.

32. **Choice C is correct.** When the pH is greater than the pK_a , then according to the Henderson-Hasselbalch equation, the conjugate base is in greater concentration than the acid. This eliminates choices A and B. Because the pH and pK_a are negative log values, if $pH > pK_a$, then $[H^+] < K_a$. Choice C is correct.
33. **Choice C is correct.** Initially in the titration of H_3CCO_2H by NaOH (at point a), the concentration of H_3CCO_2H is 0.10 M. At the equivalence point in the titration of H_3CCO_2Na by HCl (point d), the concentration of H_3CCO_2H is 0.05 M, due to dilution that occurs when the aqueous base solution is added. Note in the graph below that the final volume is 50 mL, not 25 mL. The concentrations of H_3CCO_2H between the two points in question differ by a factor of 2. This means that the pH of the two solutions differs by $-\log \sqrt{2}$. With the H_3CCO_2H concentration being greater at the initial point of the titration of H_3CCO_2H by NaOH than the equivalence point of the titration of H_3CCO_2Na by HCl, the pH is lower at point a (the initial point of the titration of H_3CCO_2H by NaOH). The best answer is therefore choice C. The graph below shows the data:



34. **Choice D is correct.** By having a lower pK_a value, H_3CCO_2H is more acidic than $CH_3NH_3^+$. Because it is more acidic, H_3CCO_2H buffers at a lower pH than $CH_3NH_3^+$, so choice A is eliminated. Because it is more acidic, H_3CCO_2H has a conjugate base that is weaker (and thus has a higher pK_b value) than the conjugate base of $CH_3NH_3^+$. Choice B is thus eliminated. Because it is more acidic, H_3CCO_2H dissociates more than $CH_3NH_3^+$, so choice C is eliminated. Because it is more acidic, H_3CCO_2H is a better electron pair acceptor than $CH_3NH_3^+$ making choice D correct. The point of this question is to view the many different ways in which relative acidity can be expressed.
35. **Choice D is correct.** Because it is more acidic, H_3CCO_2H yields a lower pH value than $CH_3NH_3^+$ of equimolar concentration, so choice A is valid and consequently eliminated. Because it is more acidic, H_3CCO_2H has a conjugate base with a higher pK_b value than the conjugate base of $CH_3NH_3^+$. Choice B is valid and thus eliminated. Because it is more acidic, H_3CCO_2H produces more conjugate base than $CH_3NH_3^+$ when added to water, so choice C is valid and consequently eliminated. Because it is more acidic, H_3CCO_2H is a better proton donor than $CH_3NH_3^+$, so choice D is *not* true. The correct answer is choice D.
36. **Choice D is correct.** Given that the pK_a value for HF is lower than that of H_3CCO_2H , HF is a stronger acid than H_3CCO_2H . This means that equimolar HF and H_3CCO_2H solutions result in a lower pH for the HF solution. This makes choice A invalid and thus eliminated. When equimolar HF and H_3CCO_2H are titrated equally, they both have converted into the same amount of conjugate base in the buffer region, meaning that $[F^-]$ equals $[H_3CCO_2^-]$. This eliminates choices B and C. When both HF and H_3CCO_2H are half-titrated, the pH of solution is lower for the HF solution, because the pH equals the pK_a of the acid and the pK_a value for HF is lower than that of H_3CCO_2H . This means that $pH(\text{HF solution}) < pK_a(\text{acetic acid})$ which means that $[H^+]_{(\text{HF solution})} > K_a(\text{acetic acid})$. This makes choice D the best choice.

Passage VI (Questions 37 - 42)

Titration Curves and Concentration Effects

37. **Choice A is correct.** The initial pH is greatest in the solution with the lowest concentration of the weakest acid. H_3CCO_2H is the weaker acid of the two choices, and 0.010 M is the lowest of the concentrations. This means that choice A is correct.

38. **Choice A is correct.** When the pH of the solution exceeds the pK_a for the acid, then there must be an excess of conjugate base relative to the conjugate acid, according to the Henderson-Hasselbalch equation. This eliminates choices C and D. The pH is the negative log of the $[H_3O^+]$ and the pK_a is the negative log of the K_a . If $pH > pK_a$, then the $[H_3O^+]$ must be less than K_a . This makes choice A correct. To make this clearer, when $pH = pK_a$, then $[H_3O^+] = K_a$. As the pH increases, the $[H_3O^+]$ decreases, while both pK_a and K_a remain constant. The $[H_3O^+]$ is thus less than K_a . The derivation from the equation for K_a is shown below:

$$K_a = \frac{[H_3O^+][A^-]}{[HA]} \text{ so } \frac{K_a}{[H_3O^+]} = \frac{[A^-]}{[HA]} \text{ When } pH > pK_a, \frac{[A^-]}{[HA]} > 1.0, \text{ so } \frac{K_a}{[H_3O^+]} > 1 \text{ and } K_a > [H_3O^+]$$

39. **Choice C is correct.** In all three weak acid titrations, the volume of sodium hydroxide solution added to reach the equivalence point was 25.0 milliliters, so choices A and B are eliminated. The higher the acid concentration initially, the more conjugate base that forms at the equivalence point. The higher the concentration of conjugate base, the higher pH for the solution at the equivalence point. Pick choice C.
40. **Choice C is correct.** If the base concentration is doubled while the acid concentration remains constant, then the base is twice as concentrated as the acid. Half as much titrant (base) is required, so the equivalence point is reached in half the base volume. This eliminates choices B and D. The acid is still HCl (a strong acid), and the titrant base is still NaOH (a strong base), so the pH is still 7.0 at equivalence, and the curve still has the same shape (lip-free and sigmoidal). The best answer is choice C.
41. **Choice C is correct.** Choice A is eliminated, because a strong acid and strong base do not make a buffer when mixed together. Choice B is eliminated, because NaOH is not a weak base. Choice D is eliminated, because the pH changes throughout the titration; but it is at the equivalence point that the pH changes drastically. Choice C is the best answer, because the log scale means that as long as the concentrations are 100 times different, then the linear difference is 2.0 on the log scale. This in turn means that the slopes are equal through most of the titration, except near the equivalence point. Perhaps the answer would be better if it mentioned the different concentrations of the titrant bases in each trial. The best answer is not always perfect.
42. **Choice D is correct.** Ammonia is a weak base, so choices A and C are eliminated, because the curves do not show the initial dip in pH (lip-o-weakness). The greater the concentration of ammonia initially, the greater the pH initially, so the 1.00 M NH_3 has a higher starting pH than the 0.010 M NH_3 . The correct choice is D.

Passage VII (Questions 43 - 49)

Titration Curves and Strength Effects

43. **Choice D is correct.** The greatest K_a value correlates with the strongest acid. Of the acids in the experiment, only HCl shows a strong acid titration curve (lip-free) and an equivalence pH of 7, so choice D is the answer.
44. **Choice B is correct.** The initial pH in the titration of HCN is greater than the initial pH in the titration of HClO, according to the titration curve in the passage. This means that the initial pH in the titration of HCN must be greater than 4.23, which eliminates choice A. Because HCN is acidic, the initial pH must be below 7.0, which eliminates choice D. The pH at the equivalence point in the titration of HClO is less than the pH at the equivalence point in the titration of HCN, according to the titration curve in the passage. This means that the pH at the equivalence point in the titration of HClO must be less than 11.01, which eliminates choice C. The only choice that remains is choice B, so choice B is the best answer. The numbers are reasonable, because 5.16 is greater than 4.23 (and 4.74 if you look at the titration curve), and 10.08 fits between 8.72 and 11.01 in the equivalence pH data.
45. **Choice B is correct.** Statement I is invalid, as the curves in Figure 1 show. Strong acids fully dissociate upon addition to water, so the pH is low in the beginning. It remains relatively constant, because pH is a log scale. It is the strong acid curve that has a plateau in the beginning. A buffer is defined as a roughly equal molar mixture of a weak acid and its conjugate base. Based on the definition, a strong acid when mixed with a strong base does not form a buffer. Do not be fooled by the flat region of the strong acid titration curve, which is attributed to mathematics, not buffering. Statement II is valid. Statement III holds true for strong acids, but not for weak acids. For instance, a weak acid with a pK_a greater than 7, has a pH greater than seven for most of the buffer region, which is observed before the equivalence point. For a weak acid titration curve, the pH is greater than 7 at equivalence, so there must be a point on the curve before the equivalence point that is greater than 7. Only statement II is valid, so choice B is correct.

46. **Choice D is correct.** Because 0.10 M HIO has a greater pH than 0.10 M HCN, HIO must be a weaker acid than HCN, and the titration curve associated with HIO would have all of the points greater than the titration curve of HCN. The pK_a of HCN is greater than 8.72 (the half-equivalence pH is greater than the equivalence pH in the HOAc titration curve). Thus, the pK_a of HIO must be greater than 8.72, which makes choice A incorrect. The pH at equivalence in the titration of HCN is 11.01, so the pH at equivalence in the titration of HIO must be greater than 11.01, and choice B is thus eliminated. The pH of 0.10 M HIO must have a pH less than 7.0, because although it is a weak acid, it is still acidic. The pH cannot be 7.21, so choice C is eliminated. The pK_a of HCN is 9.32, so the pK_a of HIO must be greater than 9.32. The pH at equivalence must be at least a pH unit less than 13 (the pH of 0.10 M KOH), so the difference between the pK_a and the pH at equivalence is less than three pH units. This makes choice D a true statement. The other way of confirming choice D is to look at the trend whereby as the acid gets weaker, the difference between pK_a and the pH at equivalence gets smaller. The difference for HCN is $11.01 - 9.32 = 1.69$. For HIO, the difference should be less than 1.69.
47. **Choice B is correct.** When 10 mL 0.10 M HClO are mixed with 10 mL 0.15 M KClO, a buffer is formed. When water is added to a buffer, the acid and conjugate base are diluted equally, so the pH does not change. The pH consequently remains at 7.64. The best answer is choice B.
48. **Choice A is correct.** Because CN^- is a stronger base than ClO^- (HCN is a weaker acid than HClO), the titration curve for NaCN should have a higher initial pH, higher pH at the half-titrated point (because the pK_a for HCN is greater than the pK_a for HClO), and higher pH at the equivalence point. This can be seen only in the titration curve in choice A. Choices C and D should have been eliminated immediately, because their respective graphs show that the two acids have the same pK_a , which according to Figure 1 in the passage, they don't have.
49. **Choice A is correct.** The stronger the acid, the lower its pK_a value. HCl is the strongest acid, so it has the lowest (and only negative) pK_a value. Choices C and D are eliminated. From the titration data, it can be seen that the next strongest acid is HOAc, meaning that the pK_a for HOAc should be just ahead of the pK_a for HCl in the sequence. This makes choice A the best answer.

Passage VIII (Questions 50 - 56)

Normality and Neutralization

50. **C is correct.** The moles of NaOH can be calculated from the moles of the acid (H^+) that were neutralized, since the amount of base added is to the point of neutralization, where the moles acid equals the moles base. Keep in mind that sulfuric acid is diprotic and that the units given are units of normality. Normality already takes into account the two equivalents of hydronium ion per sulfuric acid. Given in the passage, the number of moles of acid (H^+) is $(.025 \text{ L})(0.20 \text{ M}) = 0.005$ moles. To reach equivalence, this must be the moles of NaOH as well. The question asks for concentration, so the moles NaOH must be divided by the volume of solution.

$$\text{Molarity} = \frac{\text{moles}}{\text{liter}} = \frac{0.005 \text{ mole}}{0.02 \text{ L}} = 0.25 \text{ M}$$

Thus, the concentration of $NaOH(aq)$ is 0.25 M. Answer C is the best answer for you to choose. Do what is best!

51. **Choice B is correct.** The first proton is always the easiest to remove (that is, pK_{a1} is smaller or more acidic than pK_{a2}). Reversing this wording gives the correct answer B. The first proton must be easiest to remove, because it comes off first. This also a rare case where the correct choice contains the word "always."
52. **Choice D is correct.** Normality is based on the molarity of equivalent base required to neutralize all of the acidic protons. Since there are three (3) acidic protons on H_3PO_4 , the normality is three times the molarity.

$$\text{Normality} = 3(0.3 \text{ M}) = 0.90 \text{ N } H_3PO_4$$

Pick choice D if you want to be a supernova of chemistry wisdom. Or pick it because it's the right answer.

53. **Choice C is correct.** The conversion from H_3PO_4 to $H_2PO_4^-$ involves the loss of the first proton, so the acidity is calculated using pK_{a1} . The conversion from $H_2PO_4^-$ to HPO_4^{2-} involves the loss of the second proton, so the acidity is calculated using pK_{a2} . Mixing NaH_2PO_4 and Na_2HPO_4 yields $H_2PO_4^-$ and HPO_4^{2-} , so the pH of the solution is close to pK_{a2} . Because there is more of HPO_4^{2-} in solution than $H_2PO_4^-$, the pH is greater than pK_{a2} . This eliminates choices A and B. If the solution contained pure HPO_4^{2-} , the pH would be an average of pK_{a2} and pK_{a3} . The presence of $H_2PO_4^-$ lowers the pH, making the best answer choice C. Using the Henderson-Hasselbalch equation would give a $pH = pK_{a2} + \log 2 = pK_{a2} + 0.3$.

54. **Choice D is correct.** Sulfuric acid has only two (2) acidic protons, whereas phosphoric acid has three (3) acidic protons to neutralize. Thus, phosphoric acid requires only two-thirds the amount of base that is required for sulfuric acid. The number of moles of phosphoric acid is $(25 \text{ mL})(0.6 \text{ M}) = 15 \text{ mmol H}_3\text{PO}_4$. This means that there are 45 mmol of H^+ to neutralize. The question asks how many mL of $0.40 \text{ M H}_2\text{SO}_4(\text{aq})$ contains this amount of H^+ . $0.40 \text{ M H}_2\text{SO}_4(\text{aq}) = 0.8 \text{ M H}^+$. This means that the number of mL is: $\frac{45 \text{ mmol}}{0.8 \text{ M}}$, which equals 56.25 mL $0.40 \text{ M H}_2\text{SO}_4$, given as answer choice D.
55. **Choice B is correct.** Phosphoric acid is a triprotic acid, so choices A and C can be eliminated. Because the first proton of phosphoric acid is weak, the start of the curve should have a cusp (lip-o-weakness). It is only in choice B that the cusp is present, so the best answer is choice B.
56. **Choice A is correct.** Flask 2 is initially filled with 40.0 mL of $0.30 \text{ N H}_3\text{PO}_4$. The concentration is equivalent to $0.10 \text{ M H}_3\text{PO}_4$. Adding 40 mL of 0.30 N NaOH completely neutralizes all three protons on phosphoric acid, leaving only the conjugate base $\text{PO}_4^{3-}(\text{aq})$ in solution. All of the moles of phosphoric acid (H_3PO_4) present initially are converted into phosphate (PO_4^{3-}). The addition of 40 mL of NaOH solution also doubles the volume of the solution, increasing it from 40 mL to 80 mL. This cuts the concentration in half. Had the volume remained 40 mL, and if all of the H_3PO_4 were converted into PO_4^{3-} , then the concentration of $\text{PO}_4^{3-}(\text{aq})$ would have been 0.10 M . However, the final concentration after the extra 40 mL of solution is accounted for is $0.5 \times 0.10 \text{ M PO}_4^{3-}(\text{aq})$. That equals $0.050 \text{ M PO}_4^{3-}(\text{aq})$, which is choice A.

Passage IX (Questions 57 - 63)

Titration Curve of a Polyprotic Acid

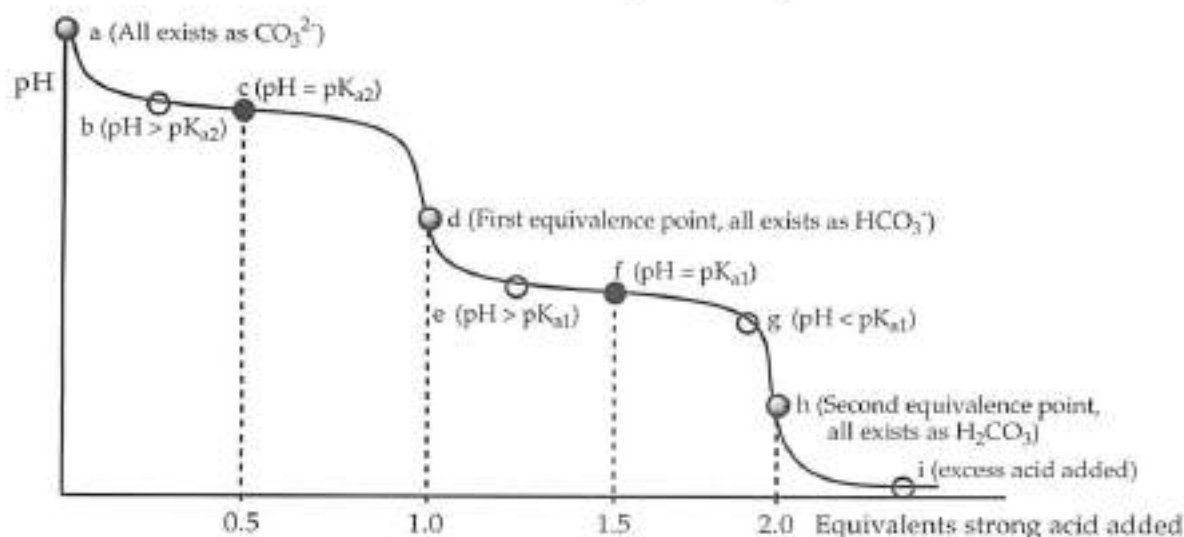
57. **Choice D is correct.** Because there are two vertical inflection points on the titration curve, the acid has two equivalence points and thus is diprotic. Because of the lip-o-weakness (initial cusp in the titration curve), the first proton is associated with a weak acid. The second proton is always weaker than the first proton, so both protons are weak. Choice D is the best answer.
58. **Choice D is correct.** The first pK_a is always lower than the second pK_a by definition, meaning that choice A is valid for all polyprotic acids. The first equivalence point (point c) is greater than the point where $\text{pH} = \text{pK}_{a1}$ (point b). This makes choice B a valid statement and thus eliminates it. The second equivalence point (point e) is greater than the point where $\text{pH} = \text{pK}_{a2}$ (point d). This makes choice C a valid statement and thus eliminates it. The first equivalence point (point c) is less than the point where $\text{pH} = \text{pK}_{a2}$ (point d). This makes choice D an invalid statement and thus makes it the best answer. To go from the first equivalence point to the point at which the pH equals pK_{a2} (where $[\text{HA}^-] = [\text{A}^{2-}]$), base must be added to the solution. This means that pK_{a2} is greater than the pH at the first equivalence point (addition of base increases the pH), confirming choice D.
59. **Choice B is correct.** To carry out the titration, a strong base must be added to react with the unknown weak acid. The pH increases as base is added, and according to the curve in Figure 1, the pH does increase when moving left to right. The best answer is choice B. A weak base is not strong enough to carry out the titration.
60. **Choice A is correct.** To go from one equivalence point to another requires that one equivalent (whatever exact quantity that may be) of base titrant be added. Choices B and D can both be eliminated, because they require an equal amount of base. It takes one equivalent of titrant to go from $\text{pH} = \text{pK}_{a1}$ to $\text{pH} = \text{pK}_{a2}$, so in choice A, more than one equivalent is required. In choice C, less than one equivalent is required. This makes the best answer choice A. On the graph, to go from one labeled point to the next labeled point, requires one-half of an equivalent. So starting to before point b and finishing after point d clearly is more than one equivalent.
61. **Choice D is correct.** When $\text{pH} = \text{pK}_{a1}$ (true at point a), exactly one-half of an equivalent of titrant has been added to solution. To reach the second equivalence point, two equivalents must be added. If 5 mL does not reach the point at which pH equals pK_{a1} , then one-half of an equivalent must be greater than 5 mL. This means that one equivalent is greater than 10 mL and thus, two equivalents are greater than 20 mL. More than 20 mL is required to reach the second equivalence point, so the best answer is choice D.
62. **Choice B is correct.** Point d is the point at which $\text{pH} = \text{pK}_{a2}$ (where $[\text{HA}^-] = [\text{A}^{2-}]$). Point e is the second equivalence point (at which all of the species present is A^{2-}). Between the two points (d and e), it is safe to assume that A^{2-} is the predominant species with some HA^- present. This is best described by choice B.

63. **Choice D is correct.** Point b is the point at which $\text{pH} = \text{pK}_{\text{a}1}$, so $[\text{H}_2\text{A}] = [\text{HA}^-]$. This makes choice A valid. Point c is the first vertical inflection point, so it is the first equivalence point. Choice B is valid. Point d is the point at which $\text{pH} = \text{pK}_{\text{a}2}$, because $[\text{HA}^-] = [\text{A}^{2-}]$. Choice C is valid. Point f represents a point where excess base is being added and all of the species exists in the fully deprotonated form (A^{2-}). Pick D to feel good.

Passage X (Questions 64 - 70)

Carbonate Titration Curve

The passage is easier when all of the points in Figure 1 are labeled.



64. **Choice C is correct.** The pH equals the pK_{a} of the acid when the titration is half way to the equivalence point. The two points marked by x represent the points at which $\text{pH} = \text{pK}_{\text{a}}$. The value of $\text{pK}_{\text{a}1}$ is lower than $\text{pK}_{\text{a}2}$, so the correct answer is point f, choice C. Point c is where the $\text{pH} = \text{pK}_{\text{a}2}$, point d is the first equivalence point (where the species exists as HCO_3^-), and point h is the second equivalence point (where the species exists as H_2CO_3).
65. **Choice B is correct.** Conjugate pairs have pK values that sum to 14. In the case of a diprotic acid, the first proton to be lost in an acid dissociation reaction corresponds to the second (last) proton to be gained in a base hydrolysis reaction. This means that the following conjugate pair relationships hold true.



The conjugate pair relationship yields $\text{pK}_{\text{a}1} + \text{pK}_{\text{b}2} = 14$ and $\text{pK}_{\text{a}2} + \text{pK}_{\text{b}1} = 14$. This makes choice A valid and choice B invalid. The question asks you to seek a statement that is *not* true, so choice B is best. According to Figure 1, the initial pH is the highest point, and it is greater than $\text{pK}_{\text{a}2}$, making choice C a valid statement. According to Figure 1, the second equivalence point is lower than all pK_{a} values, including $\text{pK}_{\text{a}1}$. This makes choice D a valid statement.

66. **Choice B is correct.** In a fashion similar to the way in which we determine the isoelectric point of an amino acid, the middle equivalence point can be determined by knowing that it lies exactly in the middle of the two pK_{a} values. The pH of the first equivalence point is thus found by averaging $\text{pK}_{\text{a}1}$ and $\text{pK}_{\text{a}2}$. The pH cannot be 7.0 for this titration, and the K_{a} when dealing with the acidity of bicarbonate would involve $\text{pK}_{\text{a}2}$ (because it is the second proton to be lost). The best answer for this question is choice B.
67. **Choice B is correct.** At point e of the titration curve, the amount of HCl added is greater than one equivalent but less than one and one-half equivalents. This makes choice A a valid statement. The pH should be lower than the pH of point c (the point at which $\text{pH} = \text{pK}_{\text{a}2}$), so choice D is a valid statement. The value of $\text{pK}_{\text{a}1}$ is 6.37, so choices B and C are contradictory answers. The pH is greater than the pH at point f (the point at which $\text{pH} = \text{pK}_{\text{a}1}$), which makes choice B invalid, and thus the best answer. It is critical in this passage that you recognize where $\text{pK}_{\text{a}1}$ and $\text{pK}_{\text{a}2}$ lie.

68. **Choice C is correct.** Bicarbonate is in its greatest concentration at the first equivalence point. This point lies midway between pK_{a1} and pK_{a2} . From the passage, the values for pK_{b1} and pK_{b2} are 3.67 and 7.63 respectively. The values for pK_{a1} and pK_{a2} are therefore 6.37 and 10.33, respectively. The bicarbonate concentration is greatest midway between 6.37 and 10.33. The best answer is choice C.
69. **Choice A is correct.** The pH change between points c and d corresponds to the change between pK_{a2} and the first equivalence point. This is greater than the incremental change from point d to e (first equivalence to a point with pH greater than pK_{a1}), greater than the incremental change from point e to f (a point with pH barely greater than pK_{a1} to a point where $pH = pK_{a1}$), and greater than the incremental change from point f to g (a point with $pH = pK_{a1}$ to a point where pH is just less than pK_{a1}). The best answer is choice A. It may be easier to answer this question by observing the titration curve and noting the change in pH (y-axis) between the two points in question.
70. **Choice A is correct.** Changing the labeling of the y-axis from pH to pOH does not change the shape of the graph. The base is still weak, so choice B can be eliminated. The pOH starts lower than 7 (because pH starts greater than 7), so choices C and D are eliminated. The best answer is choice A, because the graph flips vertically about the pH or pOH = 7 line when pH and pOH are interchanged.

Passage XI (Questions 71 - 76)

Indicator Selection

71. **Choice C is correct.** Because the indicator has to be active at the equivalence point, the pK_a of the indicator has to be greater than the pK_a of the acid being titrated. This is because the equivalence pH is higher than the pK_a for the acid being titrated. Because the acid being titrated is weak, the pH at equivalence is greater than 7.0. It is stated in the passage that the pK_a values should differ by about 3. This makes the best choice C (which differs by 3.5 pH units) the best. Choice C is roughly three units above the pK_a for the acid and is greater than 7.0.
72. **Choice A is correct.** An indicator concentration that is too high results in a second competing acid that affects the titration and a faint color in the solution that is permanent. If the indicator were in concentration equal to the acid being titrated, then the curve would look like a diprotic acid. While the color is affected by the excess indicator, the color change is certainly not going to be too minimal, so choice D gets eliminated. It is a color change, which involves frequency of light. A higher indicator concentration affects the intensity of the light, so the color change would just be more intense, but every bit as detectable. This eliminates choice C. While the viscosity may change with the addition of the indicator, this should not affect the reaction equilibrium, only the rate. The best answer is choice A, because the indicator can react with the titrant base. You may recall from titration experiments in general chemistry lab that you add only a few drops of indicator to the solution.
73. **Choice A is correct.** When the pH is greater than the pK_a , the solution has more conjugate base than acid, so choices C and D are eliminated. The difference between the concentrations is 2 on the log scale, so on a linear scale, it can't be 2 : 1 (it must be 100 : 1). This eliminates choice B and confirms that choice A is the correct answer. Using the Henderson-Hasselbalch equation, the value can be determined.

$$pH = pK_a + \log \frac{[A^-]}{[HA]} = pK_a + 2 \Rightarrow \log \frac{[A^-]}{[HA]} = 2 \Rightarrow \frac{[A^-]}{[HA]} = 10^2 = 100$$

74. **Choice D is correct.** A good indicator changes color (thus it cannot be transparent) and it must be present in low concentration in solution. This eliminates choice A. If it is not affected by acid-base reactions, it can't react in a way that indicates a pH change. Choice B is eliminated. Because the indicator must undergo some acid-base chemical reaction, it is best when the indicator is a weak acid or weak base, so it does not affect the pH much. If it were a strong acid or strong base would alter the pH and affect the titration. The best answer is choice D.
75. **Choice C is correct.** The question calls for a titration in which the pH at equivalence is the same, regardless of the concentration of reactants. Choices A and B can be eliminated, because the pH of weak acids and bases varies with the concentration. The pH at equivalence when the first proton of a weak diprotic acid has been removed is the average of pK_{a1} and pK_{a2} . You most likely can recall this from determining the isoelectric pH of an amino acid. It applies to any polyprotic acid, including the amino acids. Because the pK_a values do not change, the pH at equivalence does not change, so the same pH at equivalence is observed. The best answer is choice C. The second proton of a diprotic acid is affected by the concentration, so choice D is eliminated.

76. **Choice B is correct.** Choice C should be eliminated first, because the pH at equivalence is greater than the pK_a of the acid. Choice D should also be eliminated, because the color changes over a range, not at one exact value. At the equivalence pH (when the indicator is changing color), a change of one pH unit is substantial. This eliminates choice A. The best answer is choice B, because the pH changes so rapidly that one drop of titrant can change the pH by more than one pH unit at the equivalence point. The indicator can determine this value to the accuracy of the one drop.

Passage XII (Questions 77 - 83)

Indicator Color and Solution pH

77. **Choice A is correct.** The titration of 1.0 M benzoic acid by 1.0 M potassium hydroxide has an equivalence pH value greater than 7.0, because the acid is weak and the base is strong, so the conjugate base that is formed at equivalence makes the pH higher than 7. The conjugate base, benzoate, has a pK_b value of 9.79. This confirms that benzoate is a weak base. The best indicator must have a color change range greater than 7.0. Indicator I goes from reddish orange to mango as pH increases from 6 to 8, so Indicator I has a range that is too low. Indicator I is eliminated. Indicator I may seem close, but as the benzoic acid titration reaches equivalence, the indicator would have already turned yellow, so no color change occurs at equivalence. Indicator II goes from chartreuse to aquamarine as pH increases from 9 to 11, so Indicator II is a good choice. The pH at the equivalence point falls between 9 and 11, in all likelihood. Indicator III goes from violet to clear as pH increases from 3 to 5, so Indicator III is eliminated. Only Indicator II will work, so the best answer is choice A.

If you wish to solve for the pH at equivalence, keep in mind that the conjugate base has formed and that the solution's volume has increased, diluting all components. The pOH at the equivalence point is found using the shortcut equation.

$$pOH = \frac{1}{2}pK_b - \frac{1}{2}\log[A^-] = \frac{1}{2}(9.79) - \frac{1}{2}\log(0.5) = 4.89 - \frac{1}{2}(-0.3) = 4.89 + 0.15 = 5.04 \therefore pH = 8.96$$

This means that the pH at equivalence is about 9.0. The 0.5 M conjugate base is determined, because when the strong base was added to the benzoic acid, it not only converted the benzoic acid to conjugate base, it also diluted the solution. With an equivalence pH of 9.0, only Indicator II can work. A faster way to get the approximate pH at equivalence is to average the pK_a of the weak acid and the pH of the titrant. This yields an approximate value. In this case, the average of 4.21 and 14 is 9.1. The approximate value is usually off by about 0.15, which is close enough for our purposes.

78. **Choice B is correct.** The combination of colors that is *not* possible for a solution is the one that does not have a pH value at which the three colors may exist. In choice A, Indicator I is red at pH values of 5 or less, Indicator II is yellow at pH values of 8 or less, and Indicator III is purple at pH values of 2 or less. At a pH value of 2 or less, the color combination in choice A is possible. In choice B, Indicator I is mango at a pH value of 8, Indicator II is yellow at pH values of 8 or less, and Indicator III is violet at a pH value of 3. The solution cannot simultaneously be pH = 3 and pH = 8, so the color combination in choice B is *not* possible. In choice C, Indicator I is red at pH values of 5 or less, Indicator II is yellow at pH values of 8 or less, and Indicator III is fuchsia at a pH value of 4. At a pH value of 4, the color combination in choice C is possible. In choice D, Indicator I is yellow at pH values of 9 or greater, Indicator II is aquamarine at a pH value of 11, and Indicator III is clear at pH values of 5 or greater. At a pH value of 11, the color combination in choice D is possible. Choice B is the best answer choice.
79. **Choice A is correct.** If the ratio of blue species to clear species within the indicator equilibrium is 1000 : 1 at pH = 6, and the deprotonated form absorbs visible light (which makes the deprotonated form blue), then at a pH of 6, the indicator is predominantly in the deprotonated form. This means that the pK_a value is less than 6.0. This eliminates choices C and D. Because the ratio is 1000:1, the log value is 3, making the pK_a 3 units less than the pH. The equation is as follows:
- $$pH = pK_{a(\text{indicator})} + \log \frac{\text{Deprotonated}}{\text{Protonated}} = pK_{a(\text{indicator})} + \log 1000$$
- $$6.0 = pK_{a(\text{indicator})} + \log 1000 = pK_{a(\text{indicator})} + 3$$
- Thus, the $pK_{a(\text{indicator})} = 3.0$, so the best answer is choice A.
80. **Choice D is correct.** The key colors are faint mango with Indicator I (which implies that the pH is just greater than 8) and faint chartreuse with Indicator II (which implies that the pH is just less than 9). The best option for pH is thus choice D, 8.5, which falls between 8 and 9. Pick D to feel a rainbow of happy colors shine down.

81. **Choice A is correct.** The color change range for Indicator I is from 6 to 8, so the pK_a value for Indicator I is roughly 7. The color change range for Indicator II is from 9 to 11, so the pK_a value for Indicator II is roughly 10. The color change range for Indicator III is from 3 to 5, so the pK_a value for Indicator III is roughly 4. The pK_a value for Indicator I is incorrect in choices B and D, the pK_a value for Indicator II is incorrect in choices B and C, and the pK_a value for Indicator III is incorrect in choice B. The best answer is thus choice A.
82. **Choice C is correct.** In Indicator III the color change of detectability is from violet (at pH of 3) to the first sign of clear (at a pH near 5). The pH range is thus best described as from 3 to 5, which makes choice C the best answer.
83. **Choice C is correct.** Indicator III is purple when protonated (at low pH) and clear when deprotonated (at high pH). This means that the absorbance by the deprotonated form is beyond the visible range, which eliminates choices B and D. The visible range is 400 nm to 700 nm, so only the 339 nm absorbance can apply to the deprotonated species. Because the protonated form is purple, the highest absorbance (λ_{max}) must be in the range of the complementary color of purple, which is yellow. Yellow light lies in the middle of the visible spectrum, so it must have a value somewhere in the middle between 400 and 700 nm. A value of 426 nm is low enough to be associated with violet light, so the value of 598 nm must be the best choice for the absorbance of the protonated species. The best answer is choice C. As a point of interest, there is no known indicator which goes from colored to clear when deprotonated, because the negative charge usually lowers the transition energy between the ground state and first excited state in a highly conjugated organic molecule. A highly conjugated organic molecule is typical for an indicator.

Passage XIII (Questions 84 - 89)

Indicator Table

84. **Choice C is correct.** Bromphenol blue at pH = 7.0 has a pH value greater than the pK_a , so the species is present in its anionic (conjugate base) form. The base form of bromphenol blue according to Table 1 is blue, so choice A is eliminated. Bromcresol green in ammonia solution has a pH > 7.0, so the pH value is greater than the pK_a . This means that the species is present in its anionic (conjugate base) form. The base form of bromcresol green according to Table 1 is blue, so choice B is eliminated. Thymol blue in acetic acid solution has a pH < 7.0, so the pH value is less than the pK_a , and the species is present in its cationic (conjugate acid) form. The acid form of thymol blue according to Table 1 is also yellow, so choice C is correct. Bromthymol blue in hydroxide solution has a pH > 7.0, so the pH value is greater than the pK_a . This means that the species is present in its anionic (conjugate base) form. The base form of bromthymol blue according to Table 1 is blue, so choice D is eliminated.
85. **Choice A is correct.** The titration of ammonia with hydrochloric acid forms an acid at the equivalence point. This means that the pH at the equivalence point is less than 7.0. Of the choices, only phenolphthalein has a pK_a greater than 7.0, so phenolphthalein cannot be used as the indicator. The pH at the equivalence point should equal the pK_a of the indicator, so bromcresol green ($pK_a = 4.4$), methyl orange ($pK_a = 4.0$), and bromphenol blue ($pK_a = 3.7$) should all be close enough together that they can all work. The better of the indicators depends on the initial concentration of ammonia. Choice A is correct.
86. **Choice B is correct.** The ideal indicator has a pK_a close to the pH at equivalence. To solve this question, the pH at equivalence must be determined for each answer choice. In choices A and B, HF is formed at equivalence. The pK_a for HF is 3.17. In choice A, the pH of 1.00 M HCl is 0. The average of the pK_a for HF and the titrant pH leads to a pH at equivalence of approximately 1.6. Cresol red has a pK_a of 1.6, so choice A has a valid indicator for the titration. In choice B, the pH of 0.01 M HCl is 2. The average of the pK_a for HF and the titrant pH leads to a pH at equivalence of approximately 2.6. Methyl violet has a pK_a of 0.8, so choice B has an invalid indicator for the titration. Choice B is the best answer.

If you are highly observant, you'll note that as the concentration of the species decreases, the equivalence pH gets closer to 7. This means that choice B requires an indicator with a higher pK_a than choice A. This is also true when comparing choices C and D. Chlorophenol blue has a higher pK_a than bromcresol green, so choices C and D fit the desired trend. To verify this, let's consider the pH at equivalence for the remaining choices.

In choices C and D, HCN is formed at equivalence. The pK_a for HCN is 9.32. In choice C, the pH of 1.00 M HCl is 0. The average of the pK_a for HCN and the titrant pH leads to a pH at equivalence of approximately 4.7. Bromcresol green has a pK_a of 4.4, so choice C has a valid indicator for the titration. In choice D, the pH of 0.01 M HCl is 2. The average of the pK_a for HCN and the titrant pH leads to a pH at equivalence of approximately 5.7. Chlorophenol blue has a pK_a of 5.6, so choice D has a valid indicator for the titration.

87. Choice C is correct. When the pH is greater than the pK_a of the indicator, the conjugate base is the predominant species in solution, so the solution assumes the color of the conjugate base. When the pH is less than the pK_a of the indicator, the conjugate acid is the predominant species in solution, so the solution assumes the color of the conjugate acid. The pK_a values for the four choices are: methyl violet ($pK_a = 0.8$), methyl orange ($pK_a = 4.0$), thymol blue ($pK_a = 8.4$), and bromcresol green ($pK_a = 4.4$). At a pH of 7.4, methyl violet exists in its violet conjugate base form, so choice A is valid. At a pH of 7.4, methyl orange exists in its yellow conjugate base form, so choice B is valid. At a pH of 7.4, thymol blue exists in its yellow conjugate acid form, so choice C is invalid. At a pH of 7.4, bromcresol green exists in its blue conjugate base form, so choice D is valid. The only choice that does not correctly correlate the indicator with the solution color is C.
88. Choice D is correct. Thymol blue has a pK_a value of 2.0, so it changes color in a highly acidic medium. When titrating a strong base with a strong acid, the equivalence point is 7.0, so choices A and C are eliminated. The titration of a weak base by a strong acid leads to an equivalence pH value less than 7.0, so choice thymol blue is appropriate for choices B and D. A pK_a value of 2.0 is rather low, so the conjugate acid present at the equivalence point must dissociate readily and be in high concentration. The higher the concentration of weak base initially, the higher the concentration of weak acid at the equivalence point. To ensure that the equivalence pH is as low as 2.0, the weak base and titrant strong acid should both be highly concentrated. The base must be very weak with a pK_b of 10.0 or greater to be a reasonable choice for this problem. Given the limited information, the best answer is choice D.
89. Choice C is correct. A strong acid when titrated by a strong base shows an equivalence pH of 7.0. The indicator chosen should have a pK_a value near 7.0. It is okay if the pK_a is as high as 8.0, but not any greater than that. The pK_a values for the four choices are: methyl violet ($pK_a = 0.8$), methyl orange ($pK_a = 4.0$), bromthymol blue ($pK_a = 6.8$), and alizarin yellow ($pK_a = 10.8$). Only bromthymol blue has a pK_a value close to 7.0, so only bromthymol blue changes color at the equivalence point. The best answer is choice C.

Passage XIV (Questions 90 - 96)

pH Sticks and Indicators

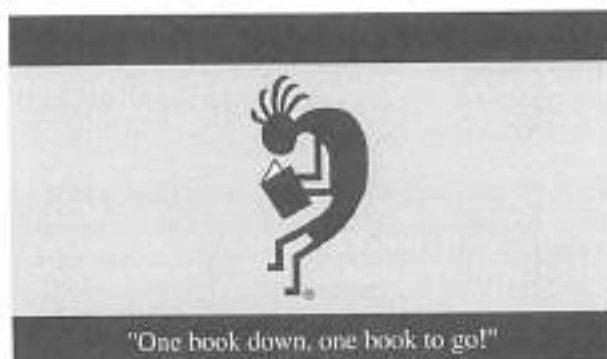
90. Choice B is correct. The pH range of the stick is found by considering the pH range of each indicator attached to the pH stick. Indicators have a pH range (color change band) of approximately $pK_{a(\text{Indicator})} \pm 1$, depending on the colors and intensity of the indicator. The four indicators have respective ranges of 3.37 to 5.37, 4.21 to 6.21, 5.78 to 7.78, and 7.79 to 9.79. This means that the overall range of the pH stick is 3.37 to 9.79, which makes choice B, 3.4 to 9.8, the best answer.
91. Choice D is correct. If bromcresol green appears yellow, the pH of the solution is less than 3.37 ($4.37 - 1$). At this pH, the other indicators would be protonated, so the pH could not be determined from any of the indicators. Because Solution 1 appears yellow with bromcresol green, it is not possible to know the exact pH. Solution 1 could have any pH value less than 3.37. If phenolphthalein appears magenta, the pH of the solution is greater than 9.79 ($8.79 + 1$). At this pH, the other indicators would be deprotonated, so the pH could not be determined from any of the indicators. Because Solution 4 appears magenta with phenolphthalein, it is not possible to know the exact pH. Solution 4 could have any pH value greater than 9.79. The pH of Solution 1 and Solution 4 cannot be approximated from the pH stick, so the best answer is choice D.
92. Choice C is correct. An aqueous solution with a hydroxide concentration of $1.0 \times 10^{-6} \text{ M}$ has a pOH of 6.0 and therefore a pH of 8.0. The pH of the solution is more than one unit greater than 4.37 (the pK_a of bromcresol green), so the bromcresol green indicator will turn blue. Choice A can be eliminated. The pH of the solution is more than one unit greater than 5.21 (the pK_a of methyl red), so the methyl red indicator will turn yellow. Choice B can be eliminated. The pH of the solution is more than one unit greater than 6.78 (the pK_a of bromthymol blue), so the bromthymol blue indicator will turn blue. Choice C is the best answer. The pH of the solution is roughly one unit less than 8.79 (the pK_a of phenolphthalein), so the phenolphthalein indicator appears clear. Choice D can be eliminated.
93. Choice B is correct. Solution 5 is greenish-blue with bromcresol blue, so the pH of Solution 5 is greater than 4.37 and less than 5.37. Solution 5 is reddish-orange with methyl red, so the pH of Solution 5 is less than 5.21 but greater than 4.21. The pH of Solution 5 falls between 4.37 and 5.21, so choice B is the best answer. Only when the color of the indicator is a composite of the protonated and deprotonated colors can the pH of the solution be approximated. Bromthymol blue and phenolphthalein are purely the protonated color, so they were not useful in approximating the pH of Solution 5.

94. **Choice C is correct.** When HCl is added to a solution, hydronium ions are released. If the colors of the indicators do not change, that means that the pH does not change significantly, and thus the hydronium ion concentration does not change significantly. There are three explanations for the pH not changing significantly. One reason pH does not change is that the hydronium concentration is so high that any change in hydronium is negligible on the log scale. A second reason pH does not change is that the hydroxide concentration is so high that any change in hydroxide concentration is negligible on the log scale. The last reason is that the solution is a buffer. Because Solution 2 appears orange with methyl red indicator, the approximate pH is around 5.2. This does not describe a solution rich in hydronium or hydroxide. The best explanation is that Solution 2 is a buffer with a pH around 5. The pK_a for a carboxylic acid is around 5, so the best answer is choice C.
95. **Choice D is correct.** Remember in choosing an indicator for titration that the pK_a of the indicator must be within ± 1 of the pH at equivalence. When titrating a weak acid with a strong base, the products are the weak conjugate base and water. An aqueous weak base solution forms at equivalence, which has a pH greater than 7, so the indicator should have a pK_a value greater than 7. The best answer is phenolphthalein, choice D.
96. **Choice A is correct.** A solution that turns blue with bromthymol blue has a pH greater than 7.78 ($pK_a = 6.68$, so the pure blue color starts at $6.68 + 1$). A pH greater than 7.78 does not guarantee that the pH is greater than 8.79, the pK_a for phenolphthalein. This means that the solution may or may not turn magenta with phenolphthalein. Choice A is invalid. If a solution turns green with bromcresol green, the pH is roughly 4.37. If a solution turns bromthymol blue green, the pH is roughly 6.68. The pH cannot simultaneously be 4.37 and 6.68, so the stick cannot simultaneously have two green marks. Choice B is a valid statement. A solution that turns methyl red yellow has a pH greater than 6.21 ($pK_a = 5.21$, so the pure yellow color starts at $5.21 + 1$). A pH greater than 6.21 guarantees that the pH is greater than 4.31, the pK_a for bromcresol green. This means that the solution must turn blue with bromcresol green. Choice C is a valid statement. When the pH of the solution falls between 4.21 and 5.37, it falls into the color blend range of two separate indicators. As a result, the pH can be approximated with twice the accuracy. Choice D is a valid statement.

Passage XV (Questions 97–100)

Acidity and Electronic Influences

97. **Choice D is correct.** The best choice for a buffer of pH = 4.0 is a weak acid with a pK_a close to 4.0 mixed with its conjugate base. Benzoic acid has a pK_a of 4.21 and nitrobenzoic acid has a pK_a of 3.40. Benzoic acid is closer, so choose D. If the buffer is to be 4.0 (the acidic side of 4.21), there should be a slight excess of benzoic acid relative to the conjugate base.
98. **Choice A is correct.** To titrate a weak acid (such as p-nitrophenol), you must add a strong base. The only strong base listed among the choices is KOH. Choose A, and move on.
99. **Choice A is correct.** The addition of 10 mL 0.10 M KOH(aq) to 20 mL 0.10 M phenol results in the half-titration of the phenol. At the halfway point, $pH = pK_a$. The pK_a for phenol is 10.0, so choice A is the best answer.
100. **Choice A is correct.** This titration is of a weak acid with a strong base that is twice as concentrated (0.20 M base with 0.10 M weak acid), so one-half the volume of the strong base is used. One-half of 50 mL amounts to 25 mL at equivalence. This eliminates choice B and D. Since the titrant is a strong base, the neutralized product is a weak base, so the equivalence pH is greater than 7. Only the titration curve in choice A shows pH at equivalence and pK_a for the phenol as being greater than 7.



PERIODIC TABLE OF THE ELEMENTS

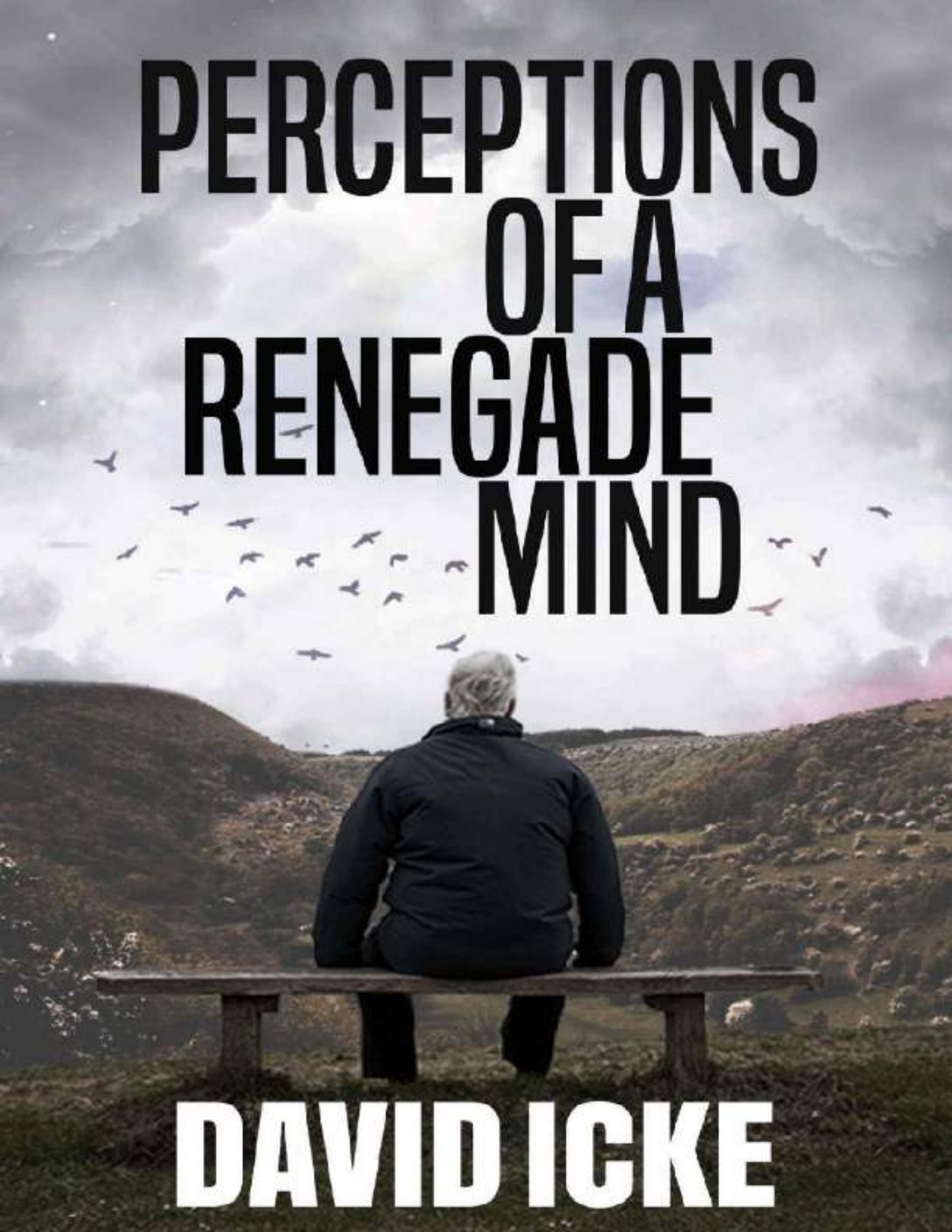
1 H 1.0																	2 He 4.0
3 Li 6.9	4 Be 9.0											5 B 10.8	6 C 12.0	7 N 14.0	8 O 16.0	9 F 19.0	10 Ne 20.2
11 Na 23.0	12 Mg 24.3											13 Al 27.0	14 Si 28.1	15 P 31.0	16 S 32.1	17 Cl 35.5	18 Ar 39.9
19 K 39.1	20 Ca 40.1	21 Sc 45.0	22 Ti 47.9	23 V 50.9	24 Cr 52.0	25 Mn 54.9	26 Fe 55.8	27 Co 58.9	28 Ni 58.7	29 Cu 63.5	30 Zn 65.4	31 Ga 69.7	32 Ge 72.6	33 As 74.9	34 Se 79.0	35 Br 79.9	36 Kr 83.8
37 Rb 85.5	38 Sr 87.6	39 Y 88.9	40 Zr 91.2	41 Nb 92.9	42 Mo 95.9	43 Tc (98)	44 Ru 101.1	45 Rh 102.9	46 Pd 106.4	47 Ag 107.9	48 Cd 112.4	49 In 114.8	50 Sn 118.7	51 Sb 121.8	52 Te 127.6	53 I 126.9	54 Xe 131.3
55 Cs 132.9	56 Ba 137.3	57 La [†] 138.9	72 Hf 178.5	73 Ta 180.9	74 W 183.9	75 Re 186.2	76 Os 190.2	77 Ir 192.2	78 Pt 195.1	79 Au 197.0	80 Hg 200.6	81 Tl 204.4	82 Pb 207.2	83 Bi 209.0	84 Po (209)	85 At (210)	86 Rn (222)
87 Fr (223)	88 Ra 226.0	89 Ac [‡] 227.0	104 Rf (261)	105 Db (262)	106 Sg (263)	107 Bh (262)	108 Hs (265)	109 Mt (266)	110 Uun (269)	111 Uuu (272)	112 Uub (277)						

†	58 Ce 140.1	59 Pr 140.9	60 Nd 144.2	61 Pm (145)	62 Sm 150.4	63 Eu 152.0	64 Gd 157.3	65 Tb 158.9	66 Dy 162.5	67 Ho 164.9	68 Er 167.3	69 Tm 168.9	70 Yb 173.0	71 Lu 175.0
‡	90 Th 232.0	91 Pa (231)	92 U 238.0	93 Np (237)	94 Pu (244)	95 Am (243)	96 Cm (247)	97 Bk (247)	98 Cf (251)	99 Es (252)	100 Fm (257)	101 Md (258)	102 No (259)	103 Lr (260)

General Chemistry

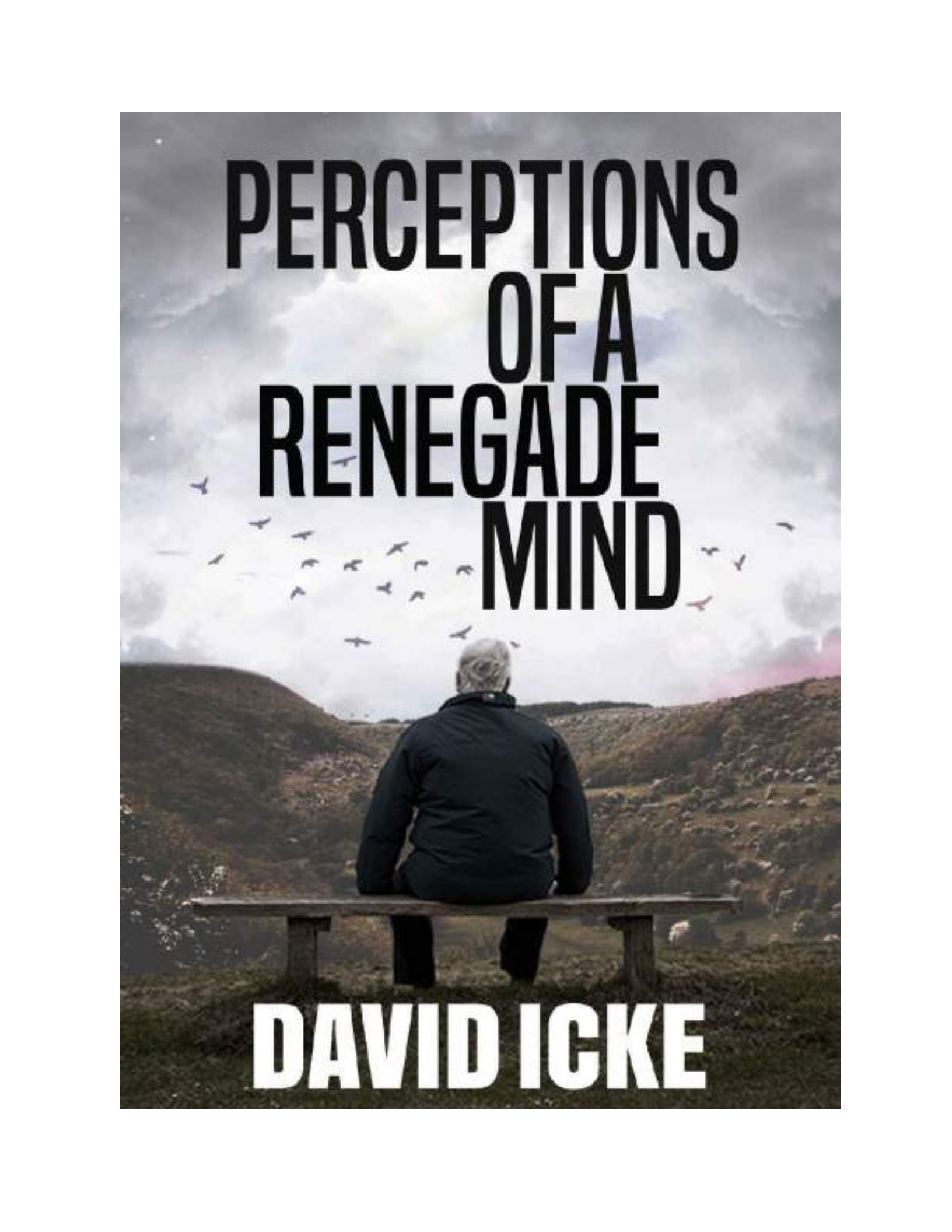


If you study it, it will come!™

A person with grey hair, wearing a dark jacket, is seen from behind, sitting on a wooden bench. They are looking out over a landscape of rolling hills covered in low-lying vegetation. The sky is filled with large, white, fluffy clouds, and a large flock of birds is flying in the air. The overall mood is contemplative and serene.

PERCEPTIONS OF A RENEGADE MIND

DAVID ICKE



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PERCEPTIONS OF A RENEGADE MIND

A flock of small, stylized birds is arranged in a circular pattern around the word "MIND". The birds are depicted in various flight poses, some facing left, some right, and some in between, creating a sense of movement and a circular vortex around the text.

DAVID ICKE

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Renegade:

Adjective

‘Having rejected tradition: Unconventional.’

Merriam-Webster Dictionary

Acquiescence to tyranny is the death of the spirit

You may be 38 years old, as I happen to be. And one day, some great opportunity stands before you and calls you to stand up for some great principle, some great issue, some great cause. And you refuse to do it because you are afraid ... You refuse to do it because you want to live longer ... You're afraid that you will lose your job, or you are afraid that you will be criticised or that you will lose your popularity, or you're afraid that somebody will stab you, or shoot at you or bomb your house; so you refuse to take the stand.

Well, you may go on and live until you are 90, but you're just as dead at 38 as you would be at 90. And the cessation of breathing in your life is but the belated announcement of an earlier death of the spirit.

Martin Luther King

**How the few control the many and always have – the many do
whatever they're told**

'Forward, the Light Brigade!'
Was there a man dismayed?
Not though the soldier knew
Someone had blundered.
Theirs not to make reply,
Theirs not to reason why,
Theirs but to do and die.
Into the valley of Death
Rode the six hundred.

Cannon to right of them,
Cannon to left of them,
Cannon in front of them
Volleyed and thundered;
Stormed at with shot and shell,
Boldly they rode and well,
Into the jaws of Death,
Into the mouth of hell
Rode the six hundred

Alfred Lord Tennyson (1809-1892)

The mist is lifting slowly
I can see the way ahead
And I've left behind the empty streets
That once inspired my life
And the strength of the emotion
Is like thunder in the air
'Cos the promise that we made each other
Haunts me to the end

The secret of your beauty
And the mystery of your soul
I've been searching for in everyone I meet
And the times I've been mistaken
It's impossible to say
And the grass is growing
Underneath our feet

The words that I remember
From my childhood still are true
That there's none so blind
As those who will not see
And to those who lack the courage
And say it's dangerous to try
Well they just don't know
That love eternal will not be denied

I know you're out there somewhere
Somewhere, somewhere
I know you're out there somewhere

Somewhere you can hear my voice
I know I'll find you somehow
Somehow, somehow
I know I'll find you somehow
And somehow I'll return again to you

The Moody Blues

Are you a gutless wonder - or a Renegade Mind?

Monuments put from pen to paper,
Turns me into a gutless wonder,
And if you tolerate this,
Then your children will be next.
Gravity keeps my head down,
Or is it maybe shame ...

Manic Street Preachers

Rise like lions after slumber
In unvanquishable number.
Shake your chains to earth like dew
Which in sleep have fallen on you.
Ye are many – they are few.

Percy Shelley

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CHAPTER ONE

I'm thinking' – Oh, but *are* you?

Think for yourself and let others enjoy the privilege of doing so too
Voltaire

French-born philosopher, mathematician and scientist René Descartes became famous for his statement in Latin in the 17th century which translates into English as: 'I think, therefore I am.'

On the face of it that is true. Thought reflects perception and perception leads to both behaviour and self-identity. In that sense 'we' are what we think. But who or what is doing the thinking and is thinking the only route to perception? Clearly, as we shall see, 'we' are not always the source of 'our' perception, indeed with regard to humanity as a whole this is rarely the case; and thinking is far from the only means of perception. Thought is the village idiot compared with other expressions of consciousness that we all have the potential to access and tap into. This has to be true when we *are* those other expressions of consciousness which are infinite in nature. We have forgotten this, or, more to the point, been manipulated to forget.

These are not just the esoteric musings of the navel. The whole foundation of human control and oppression is control of perception. Once perception is hijacked then so is behaviour which is dictated by perception. Collective perception becomes collective behaviour and collective behaviour is what we call human society. Perception is all and those behind human control know that which is

why perception is the target 24/7 of the psychopathic manipulators that I call the Global Cult. They know that if they dictate perception they will dictate behaviour and collectively dictate the nature of human society. They are further aware that perception is formed from information received and if they control the circulation of information they will to a vast extent direct human behaviour. Censorship of information and opinion has become globally Nazi-like in recent years and never more blatantly than since the illusory 'virus pandemic' was triggered out of China in 2019 and across the world in 2020. Why have billions submitted to house arrest and accepted fascistic societies in a way they would have never believed possible? Those controlling the information spewing from government, mainstream media and Silicon Valley (all controlled by the same Global Cult networks) told them they were in danger from a 'deadly virus' and only by submitting to house arrest and conceding their most basic of freedoms could they and their families be protected. This monumental and provable lie became the *perception* of the billions and therefore the *behaviour* of the billions. In those few words you have the whole structure and modus operandi of human control. Fear is a perception – False Emotion Appearing Real – and fear is the currency of control. In short ... get them by the balls (or give them the impression that you have) and their hearts and minds will follow. Nothing grips the dangly bits and freezes the rear-end more comprehensively than fear.

World number 1

There are two 'worlds' in what appears to be one 'world' and the prime difference between them is knowledge. First we have the mass of human society in which the population is maintained in coldly-calculated ignorance through control of information and the 'education' (indoctrination) system. That's all you really need to control to enslave billions in a perceptual delusion in which what are perceived to be *their* thoughts and opinions are ever-repeated mantras that the system has been downloading all their lives through 'education', media, science, medicine, politics and academia

in which the personnel and advocates are themselves overwhelmingly the perceptual products of the same repetition. Teachers and academics in general are processed by the same programming machine as everyone else, but unlike the great majority they never leave the 'education' program. It gripped them as students and continues to grip them as programmers of subsequent generations of students. The programmed become the programmers – the programmed programmers. The same can largely be said for scientists, doctors and politicians and not least because as the American writer Upton Sinclair said: 'It is difficult to get a man to understand something when his salary depends upon his not understanding it.' If your career and income depend on thinking the way the system demands then you will – bar a few free-minded exceptions – concede your mind to the Perceptual Mainframe that I call the Postage Stamp Consensus. This is a tiny band of perceived knowledge and possibility 'taught' (downloaded) in the schools and universities, pounded out by the mainstream media and on which all government policy is founded. Try thinking, and especially speaking and acting, outside of the 'box' of consensus and see what that does for your career in the Mainstream Everything which bullies, harasses, intimidates and ridicules the population into compliance. Here we have the simple structure which enslaves most of humanity in a perceptual prison cell for an entire lifetime and I'll go deeper into this process shortly. Most of what humanity is taught as fact is nothing more than programmed belief. American science fiction author Frank Herbert was right when he said: 'Belief can be manipulated. Only knowledge is dangerous.' In the 'Covid' age belief is promoted and knowledge is censored. It was always so, but never to the extreme of today.

World number 2

A 'number 2' is slang for 'doing a poo' and how appropriate that is when this other 'world' is doing just that on humanity every minute of every day. World number 2 is a global network of secret societies and semi-secret groups dictating the direction of society via

governments, corporations and authorities of every kind. I have spent more than 30 years uncovering and exposing this network that I call the Global Cult and knowing its agenda is what has made my books so accurate in predicting current and past events. Secret societies are secret for a reason. They want to keep their hoarded knowledge to themselves and their chosen initiates and to hide it from the population which they seek through ignorance to control and subdue. The whole foundation of the division between World 1 and World 2 is *knowledge*. What number 1 knows number 2 must not. Knowledge they have worked so hard to keep secret includes (a) the agenda to enslave humanity in a centrally-controlled global dictatorship, and (b) the nature of reality and life itself. The latter (b) must be suppressed to allow the former (a) to prevail as I shall be explaining. The way the Cult manipulates and interacts with the population can be likened to a spider's web. The 'spider' sits at the centre in the shadows and imposes its will through the web with each strand represented in World number 2 by a secret society, satanic or semi-secret group, and in World number 1 – the world of the seen – by governments, agencies of government, law enforcement, corporations, the banking system, media conglomerates and Silicon Valley (Fig 1 overleaf). The spider and the web connect and coordinate all these organisations to pursue the same global outcome while the population sees them as individual entities working randomly and independently. At the level of the web governments *are* the banking system *are* the corporations *are* the media *are* Silicon Valley *are* the World Health Organization working from their inner cores as one unit. Apparently unconnected countries, corporations, institutions, organisations and people are on the *same team* pursuing the same global outcome. Strands in the web immediately around the spider are the most secretive and exclusive secret societies and their membership is emphatically restricted to the Cult inner-circle emerging through the generations from particular bloodlines for reasons I will come to. At the core of the core you would get them in a single room. That's how many people are dictating the direction of human society and its transformation

through the 'Covid' hoax and other means. As the web expands out from the spider we meet the secret societies that many people will be aware of – the Freemasons, Knights Templar, Knights of Malta, Opus Dei, the inner sanctum of the Jesuit Order, and such like. Note how many are connected to the Church of Rome and there is a reason for that. The Roman Church was established as a revamp, a rebranding, of the relocated 'Church' of Babylon and the Cult imposing global tyranny today can be tracked back to Babylon and Sumer in what is now Iraq.



Figure 1: The global web through which the few control the many. (Image Neil Hague.)

Inner levels of the web operate in the unseen away from the public eye and then we have what I call the cusp organisations located at the point where the hidden meets the seen. They include a series of satellite organisations answering to a secret society founded in London in the late 19th century called the Round Table and among them are the Royal Institute of International Affairs (UK, founded in 1920); Council on Foreign Relations (US, 1921); Bilderberg Group (worldwide, 1954); Trilateral Commission (US/worldwide, 1972); and the Club of Rome (worldwide, 1968) which was created to exploit environmental concerns to justify the centralisation of global power to 'save the planet'. The Club of Rome instigated with others the human-caused climate change hoax which has led to all the 'green

new deals' demanding that very centralisation of control. Cusp organisations, which include endless 'think tanks' all over the world, are designed to coordinate a single global policy between political and business leaders, intelligence personnel, media organisations and anyone who can influence the direction of policy in their own sphere of operation. Major players and regular attenders will know what is happening – or some of it – while others come and go and are kept overwhelmingly in the dark about the big picture. I refer to these cusp groupings as semi-secret in that they can be publicly identified, but what goes on at the inner-core is kept very much 'in house' even from most of their members and participants through a fiercely-imposed system of compartmentalisation. Only let them know what they need to know to serve your interests and no more. The structure of secret societies serves as a perfect example of this principle. Most Freemasons never get higher than the bottom three levels of 'degree' (degree of knowledge) when there are 33 official degrees of the Scottish Rite. Initiates only qualify for the next higher 'compartment' or degree if those at that level choose to allow them. Knowledge can be carefully assigned only to those considered 'safe'. I went to my local Freemason's lodge a few years ago when they were having an 'open day' to show how cuddly they were and when I chatted to some of them I was astonished at how little the rank and file knew even about the most ubiquitous symbols they use. The mushroom technique – keep them in the dark and feed them bullshit – applies to most people in the web as well as the population as a whole. Sub-divisions of the web mirror in theme and structure transnational corporations which have a headquarters somewhere in the world dictating to all their subsidiaries in different countries. Subsidiaries operate in their methodology and branding to the same centrally-dictated plan and policy in pursuit of particular ends. The Cult web functions in the same way. Each country has its own web as a subsidiary of the global one. They consist of networks of secret societies, semi-secret groups and bloodline families and their job is to impose the will of the spider and the global web in their particular country. Subsidiary networks control and manipulate the national political system, finance, corporations, media, medicine, etc. to

ensure that they follow the globally-dictated Cult agenda. These networks were the means through which the 'Covid' hoax could be played out with almost every country responding in the same way.

The 'Yessir' pyramid

Compartmentalisation is the key to understanding how a tiny few can dictate the lives of billions when combined with a top-down sequence of imposition and acquiescence. The inner core of the Cult sits at the peak of the pyramidal hierarchy of human society (Fig 2 overleaf). It imposes its will – its agenda for the world – on the level immediately below which acquiesces to that imposition. This level then imposes the Cult will on the level below them which acquiesces and imposes on the next level. Very quickly we meet levels in the hierarchy that have no idea there even is a Cult, but the sequence of imposition and acquiescence continues down the pyramid in just the same way. 'I don't know why we are doing this but the order came from "on-high" and so we better just do it.' Alfred Lord Tennyson said of the cannon fodder levels in his poem *The Charge of the Light Brigade*: 'Theirs not to reason why; theirs but to do and die.' The next line says that 'into the valley of death rode the six hundred' and they died because they obeyed without question what their perceived 'superiors' told them to do. In the same way the population capitulated to 'Covid'. The whole hierarchical pyramid functions like this to allow the very few to direct the enormous many.

Eventually imposition-acquiescence-imposition-acquiescence comes down to the mass of the population at the foot of the pyramid. If they acquiesce to those levels of the hierarchy imposing on them (governments/law enforcement/doctors/media) a circuit is completed between the population and the handful of super-psychopaths in the Cult inner core at the top of the pyramid.

Without a circuit-breaking refusal to obey, the sequence of imposition and acquiescence allows a staggeringly few people to impose their will upon the entirety of humankind. We are looking at the very sequence that has subjugated billions since the start of 2020. Our freedom has not been taken from us. Humanity has given it

away. Fascists do not impose fascism because there are not enough of them. Fascism is imposed by the population acquiescing to fascism. Put another way allowing their perceptions to be programmed to the extent that leads to the population giving their freedom away by giving their perceptions – their mind – away. If this circuit is not broken by humanity ceasing to cooperate with their own enslavement then nothing can change. For that to happen people have to critically think and see through the lies and window dressing and then summon the backbone to act upon what they see. The Cult spends its days working to stop either happening and its methodology is systematic and highly detailed, but it can be overcome and that is what this book is all about.

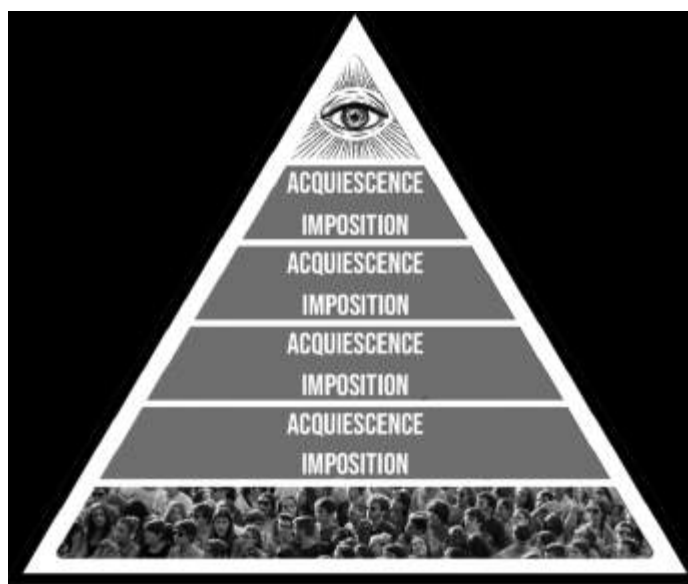


Figure 2: The simple sequence of imposition and compliance that allows a handful of people at the peak of the pyramid to dictate the lives of billions.

The Life Program

Okay, back to world number 1 or the world of the 'masses'. Observe the process of what we call 'life' and it is a perceptual download from cradle to grave. The Cult has created a global structure in which perception can be programmed and the program continually topped-up with what appears to be constant confirmation that the program is indeed true reality. The important word here is 'appears'.

This is the structure, the fly-trap, the Postage Stamp Consensus or Perceptual Mainframe, which represents that incredibly narrow band of perceived possibility delivered by the 'education' system, mainstream media, science and medicine. From the earliest age the download begins with parents who have themselves succumbed to the very programming their children are about to go through. Most parents don't do this out of malevolence and mostly it is quite the opposite. They do what they believe is best for their children and that is what the program has told them is best. Within three or four years comes the major transition from parental programming to full-blown state (Cult) programming in school, college and university where perceptually-programmed teachers and academics pass on their programming to the next generations. Teachers who resist are soon marginalised and their careers ended while children who resist are called a problem child for whom Ritalin may need to be prescribed. A few years after entering the 'world' children are under the control of authority figures representing the state telling them when they have to be there, when they can leave and when they can speak, eat, even go to the toilet. This is calculated preparation for a lifetime of obeying authority in all its forms. Reflex-action fear of authority is instilled by authority from the start. Children soon learn the carrot and stick consequences of obeying or defying authority which is underpinned daily for the rest of their life. Fortunately I daydreamed through this crap and never obeyed authority simply because it told me to. This approach to my alleged 'betters' continues to this day. There can be consequences of pursuing open-minded freedom in a world of closed-minded conformity. I spent a lot of time in school corridors after being ejected from the classroom for not taking some of it seriously and now I spend a lot of time being ejected from Facebook, YouTube and Twitter. But I can tell you that being true to yourself and not compromising your self-respect is far more exhilarating than bowing to authority for authority's sake. You don't have to be a sheep to the shepherd (authority) and the sheep dog (fear of not obeying authority).

The perceptual download continues throughout the formative years in school, college and university while script-reading 'teachers', 'academics' 'scientists', 'doctors' and 'journalists' insist that ongoing generations must be as programmed as they are. Accept the program or you will not pass your 'exams' which confirm your 'degree' of programming. It is tragic to think that many parents pressure their offspring to work hard at school to download the program and qualify for the next stage at college and university. The late, great, American comedian George Carlin said: 'Here's a bumper sticker I'd like to see: We are proud parents of a child who has resisted his teachers' attempts to break his spirit and bend him to the will of his corporate masters.' Well, the best of luck finding many of those, George. Then comes the moment to leave the formal programming years in academia and enter the 'adult' world of work. There you meet others in your chosen or prescribed arena who went through the same Postage Stamp Consensus program before you did. There is therefore overwhelming agreement between almost everyone on the basic foundations of Postage Stamp reality and the rejection, even contempt, of the few who have a mind of their own and are prepared to use it. This has two major effects. Firstly, the consensus confirms to the programmed that their download is really how things are. I mean, everyone knows that, right? Secondly, the arrogance and ignorance of Postage Stamp adherents ensure that anyone questioning the program will have unpleasant consequences for seeking their own truth and not picking their perceptions from the shelf marked: 'Things you must believe without question and if you don't you're a dangerous lunatic conspiracy theorist and a harebrained nutter'.

Every government, agency and corporation is founded on the same Postage Stamp prison cell and you can see why so many people believe the same thing while calling it their own 'opinion'. Fusion of governments and corporations in pursuit of the same agenda was the definition of fascism described by Italian dictator Benito Mussolini. The pressure to conform to perceptual norms downloaded for a lifetime is incessant and infiltrates society right

down to family groups that become censors and condemners of their own 'black sheep' for not, ironically, being sheep. We have seen an explosion of that in the 'Covid' era. Cult-owned global media unleashes its propaganda all day every day in support of the Postage Stamp and targets with abuse and ridicule anyone in the public eye who won't bend their mind to the will of the tyranny. Any response to this is denied (certainly in my case). They don't want to give a platform to expose official lies. Cult-owned-and-created Internet giants like Facebook, Google, YouTube and Twitter delete you for having an unapproved opinion. Facebook boasts that its AI censors delete 97-percent of 'hate speech' before anyone even reports it. Much of that 'hate speech' will simply be an opinion that Facebook and its masters don't want people to see. Such perceptual oppression is widely known as fascism. Even Facebook executive Benny Thomas, a 'CEO Global Planning Lead', said in comments secretly recorded by investigative journalism operation Project Veritas that Facebook is 'too powerful' and should be broken up:

I mean, no king in history has been the ruler of two billion people, but Mark Zuckerberg is ... And he's 36. That's too much for a 36-year-old ... You should not have power over two billion people. I just think that's wrong.

Thomas said Facebook-owned platforms like Instagram, Oculus, and WhatsApp needed to be separate companies. 'It's too much power when they're all one together'. That's the way the Cult likes it, however. We have an executive of a Cult organisation in Benny Thomas that doesn't know there is a Cult such is the compartmentalisation. Thomas said that Facebook and Google 'are no longer companies, they're countries'. Actually they are more powerful than countries on the basis that if you control information you control perception and control human society.

I love my oppressor

Another expression of this psychological trickery is for those who realise they are being pressured into compliance to eventually

convince themselves to believe the official narratives to protect their self-respect from accepting the truth that they have succumbed to meek and subservient compliance. Such people become some of the most vehement defenders of the system. You can see them everywhere screaming abuse at those who prefer to think for themselves and by doing so reminding the compliers of their own capitulation to conformity. 'You are talking dangerous nonsense you Covidiot!!' Are you trying to convince me or yourself? It is a potent form of Stockholm syndrome which is defined as: 'A psychological condition that occurs when a victim of abuse identifies and attaches, or bonds, positively with their abuser.' An example is hostages bonding and even 'falling in love' with their kidnappers. The syndrome has been observed in domestic violence, abused children, concentration camp inmates, prisoners of war and many and various Satanic cults. These are some traits of Stockholm syndrome listed at goodtherapy.org:

- Positive regard towards perpetrators of abuse or captor [see 'Covid'].
- Failure to cooperate with police and other government authorities when it comes to holding perpetrators of abuse or kidnapping accountable [or in the case of 'Covid' cooperating with the police to enforce and defend their captors' demands].
- Little or no effort to escape [see 'Covid'].
- Belief in the goodness of the perpetrators or kidnappers [see 'Covid'].
- Appeasement of captors. This is a manipulative strategy for maintaining one's safety. As victims get rewarded – perhaps with less abuse or even with life itself – their appeasing behaviours are reinforced [see 'Covid'].
- Learned helplessness. This can be akin to 'if you can't beat 'em, join 'em'. As the victims fail to escape the abuse or captivity, they may start giving up and soon realize it's just easier for everyone if they acquiesce all their power to their captors [see 'Covid'].

- Feelings of pity toward the abusers, believing they are actually victims themselves. Because of this, victims may go on a crusade or mission to 'save' [protect] their abuser [see the venom unleashed on those challenging the official 'Covid' narrative].
- Unwillingness to learn to detach from their perpetrators and heal. In essence, victims may tend to be less loyal to themselves than to their abuser [*definitely* see 'Covid'].

Ponder on those traits and compare them with the behaviour of great swathes of the global population who have defended governments and authorities which have spent every minute destroying their lives and livelihoods and those of their children and grandchildren since early 2020 with fascistic lockdowns, house arrest and employment deletion to 'protect' them from a 'deadly virus' that their abusers' perceptually created to bring about this very outcome. We are looking at mass Stockholm syndrome. All those that agree to concede their freedom will believe those perceptions are originating in their own independent 'mind' when in fact by conceding their reality to Stockholm syndrome they have by definition conceded any independence of mind. Listen to the 'opinions' of the acquiescing masses in this 'Covid' era and what gushes forth is the repetition of the official version of everything delivered unprocessed, unfiltered and unquestioned. The whole programming dynamic works this way. I must be free because I'm told that I am and so I think that I am.

You can see what I mean with the chapter theme of 'I'm thinking – Oh, but *are* you?' The great majority are not thinking, let alone for themselves. They are repeating what authority has told them to believe which allows them to be controlled. Weaving through this mentality is the fear that the 'conspiracy theorists' are right and this again explains the often hysterical abuse that ensues when you dare to contest the official narrative of anything. Denial is the mechanism of hiding from yourself what you don't want to be true. Telling people what they want to hear is easy, but it's an infinitely greater challenge to tell them what they would rather not be happening.

One is akin to pushing against an open door while the other is met with vehement resistance no matter what the scale of evidence. I don't want it to be true so I'll convince myself that it's not. Examples are everywhere from the denial that a partner is cheating despite all the signs to the reflex-action rejection of any idea that world events in which country after country act in exactly the same way are centrally coordinated. To accept the latter is to accept that a force of unspeakable evil is working to destroy your life and the lives of your children with nothing too horrific to achieve that end. Who the heck wants that to be true? But if we don't face reality the end is duly achieved and the consequences are far worse and ongoing than breaking through the walls of denial today with the courage to make a stand against tyranny.

Connect the dots – but how?

A crucial aspect of perceptual programming is to portray a world in which everything is random and almost nothing is connected to anything else. Randomness cannot be coordinated by its very nature and once you perceive events as random the idea they could be connected is waved away as the rantings of the tinfoil-hat brigade. You can't plan and coordinate random you idiot! No, you can't, but you can hide the coldly-calculated and long-planned behind the *illusion* of randomness. A foundation manifestation of the Renegade Mind is to scan reality for patterns that connect the apparently random and turn pixels and dots into pictures. This is the way I work and have done so for more than 30 years. You look for similarities in people, modus operandi and desired outcomes and slowly, then ever quicker, the picture forms. For instance: There would seem to be no connection between the 'Covid pandemic' hoax and the human-caused global-warming hoax and yet they are masks (appropriately) on the same face seeking the same outcome. Those pushing the global warming myth through the Club of Rome and other Cult agencies are driving the lies about 'Covid' – Bill Gates is an obvious one, but they are endless. Why would the same people be involved in both when they are clearly not connected? Oh, but they

are. Common themes with personnel are matched by common goals. The 'solutions' to both 'problems' are centralisation of global power to impose the will of the few on the many to 'save' humanity from 'Covid' and save the planet from an 'existential threat' (we need 'zero Covid' and 'zero carbon emissions'). These, in turn, connect with the 'dot' of globalisation which was coined to describe the centralisation of global power in every area of life through incessant political and corporate expansion, trading blocks and superstates like the European Union. If you are the few and you want to control the many you have to centralise power and decision-making. The more you centralise power the more power the few at the centre will have over the many; and the more that power is centralised the more power those at the centre have to centralise even quicker. The momentum of centralisation gets faster and faster which is exactly the process we have witnessed. In this way the hoaxed 'pandemic' and the fakery of human-caused global warming serve the interests of globalisation and the seizure of global power in the hands of the Cult inner-circle which is behind 'Covid', 'climate change' and globalisation. At this point random 'dots' become a clear and obvious picture or pattern.

Klaus Schwab, the classic Bond villain who founded the Cult's Gates-funded World Economic Forum, published a book in 2020, *The Great Reset*, in which he used the 'problem' of 'Covid' to justify a total transformation of human society to 'save' humanity from 'climate change'. Schwab said: 'The pandemic represents a rare but narrow window of opportunity to reflect, reimagine, and reset our world.' What he didn't mention is that the Cult he serves is behind both hoaxes as I show in my book *The Answer*. He and the Cult don't have to reimagine the world. They know precisely what they want and that's why they destroyed human society with 'Covid' to 'build back better' in their grand design. Their job is not to imagine, but to get humanity to imagine and agree with their plans while believing it's all random. It must be pure coincidence that 'The Great Reset' has long been the Cult's code name for the global imposition of fascism and replaced previous code-names of the 'New World

Order' used by Cult frontmen like Father George Bush and the 'New Order of the Ages' which emerged from Freemasonry and much older secret societies. New Order of the Ages appears on the reverse of the Great Seal of the United States as 'Novus ordo seclorum' underneath the Cult symbol used since way back of the pyramid and all seeing-eye (Fig 3). The pyramid is the hierarchy of human control headed by the illuminated eye that symbolises the force behind the Cult which I will expose in later chapters. The term 'Annuit Coeptis' translates as 'He favours our undertaking'. We are told the 'He' is the Christian god, but 'He' is not as I will be explaining.



Figure 3: The all-seeing eye of the Cult 'god' on the Freemason-designed Great Seal of the United States and also on the dollar bill.

Having you on

Two major Cult techniques of perceptual manipulation that relate to all this are what I have called since the 1990s Problem-Reaction-Solution (PRS) and the Totalitarian Tiptoe (TT). They can be uncovered by the inquiring mind with a simple question: Who benefits? The answer usually identifies the perpetrators of a given action or happening through the concept of 'he who most benefits from a crime is the one most likely to have committed it'. The Latin 'Cue bono?' – Who benefits? – is widely attributed to the Roman orator and statesman Marcus Tullius Cicero. No wonder it goes back so far when the concept has been relevant to human behaviour since

history was recorded. Problem-Reaction-Solution is the technique used to manipulate us every day by covertly creating a problem (or the illusion of one) and offering the solution to the problem (or the illusion of one). In the first phase you create the problem and blame someone or something else for why it has happened. This may relate to a financial collapse, terrorist attack, war, global warming or pandemic, anything in fact that will allow you to impose the 'solution' to change society in the way you desire at that time. The 'problem' doesn't have to be real. PRS is manipulation of perception and all you need is the population to believe the problem is real. Human-caused global warming and the 'Covid pandemic' only have to be *perceived* to be real for the population to accept the 'solutions' of authority. I refer to this technique as NO-Problem-Reaction-Solution. Billions did not meekly accept house arrest from early 2020 because there was a real deadly 'Covid pandemic' but because they perceived – believed – that to be the case. The antidote to Problem-Reaction-Solution is to ask who benefits from the proposed solution. Invariably it will be anyone who wants to justify more control through deletion of freedom and centralisation of power and decision-making.

The two world wars were Problem-Reaction-Solutions that transformed and realigned global society. Both were manipulated into being by the Cult as I have detailed in books since the mid-1990s. They dramatically centralised global power, especially World War Two, which led to the United Nations and other global bodies thanks to the overt and covert manipulations of the Rockefeller family and other Cult bloodlines like the Rothschilds. The UN is a stalking horse for full-blown world government that I will come to shortly. The land on which the UN building stands in New York was donated by the Rockefellers and the same Cult family was behind Big Pharma scalpel and drug 'medicine' and the creation of the World Health Organization as part of the UN. They have been stalwarts of the eugenics movement and funded Hitler's race-purity expert' Ernst Rudin. The human-caused global warming hoax has been orchestrated by the Club of Rome through the UN which is

manufacturing both the 'problem' through its Intergovernmental Panel on Climate Change and imposing the 'solution' through its Agenda 21 and Agenda 2030 which demand the total centralisation of global power to 'save the world' from a climate hoax the United Nations is itself perpetrating. What a small world the Cult can be seen to be particularly among the inner circles. The bedfellow of Problem-Reaction-Solution is the Totalitarian Tiptoe which became the Totalitarian Sprint in 2020. The technique is fashioned to hide the carefully-coordinated behind the cover of apparently random events. You start the sequence at 'A' and you know you are heading for 'Z'. You don't want people to know that and each step on the journey is presented as a random happening while all the steps strung together lead in the same direction. The speed may have quickened dramatically in recent times, but you can still see the incremental approach of the Tiptoe in the case of 'Covid' as each new imposition takes us deeper into fascism. Tell people they have to do this or that to get back to 'normal', then this and this and this. With each new demand adding to the ones that went before the population's freedom is deleted until it disappears. The spider wraps its web around the flies more comprehensively with each new diktat. I'll highlight this in more detail when I get to the 'Covid' hoax and how it has been pulled off. Another prime example of the Totalitarian Tiptoe is how the Cult-created European Union went from a 'free-trade zone' to a centralised bureaucratic dictatorship through the Tiptoe of incremental centralisation of power until nations became mere administrative units for Cult-owned dark suits in Brussels.

The antidote to ignorance is knowledge which the Cult seeks vehemently to deny us, but despite the systematic censorship to that end the Renegade Mind can overcome this by vociferously seeking out the facts no matter the impediments put in the way. There is also a method of thinking and perceiving – *knowing* – that doesn't even need names, dates, place-type facts to identify the patterns that reveal the story. I'll get to that in the final chapter. All you need to know about the manipulation of human society and to what end is still out there – *at the time of writing* – in the form of books, videos

and websites for those that really want to breach the walls of programmed perception. To access this knowledge requires the abandonment of the mainstream media as a source of information in the awareness that this is owned and controlled by the Cult and therefore promotes mass perceptions that suit the Cult. Mainstream media lies all day, every day. That is its function and very reason for being. Where it does tell the truth, here and there, is only because the truth and the Cult agenda very occasionally coincide. If you look for fact and insight to the BBC, CNN and virtually all the rest of them you are asking to be conned and perceptually programmed.

Know the outcome and you'll see the journey

Events seem random when you have no idea where the world is being taken. Once you do the random becomes the carefully planned. Know the outcome and you'll see the journey is a phrase I have been using for a long time to give context to daily happenings that appear unconnected. Does a problem, or illusion of a problem, trigger a proposed 'solution' that further drives society in the direction of the outcome? Invariably the answer will be yes and the random – *abracadabra* – becomes the clearly coordinated. So what is this outcome that unlocks the door to a massively expanded understanding of daily events? I will summarise its major aspects – the fine detail is in my other books – and those new to this information will see that the world they thought they were living in is a very different place. The foundation of the Cult agenda is the incessant centralisation of power and all such centralisation is ultimately in pursuit of Cult control on a global level. I have described for a long time the planned world structure of top-down dictatorship as the Hunger Games Society. The term obviously comes from the movie series which portrayed a world in which a few living in military-protected hi-tech luxury were the overlords of a population condemned to abject poverty in isolated 'sectors' that were not allowed to interact. 'Covid' lockdowns and travel bans anyone? The 'Hunger Games' pyramid of structural control has the inner circle of the Cult at the top with pretty much the entire

population at the bottom under their control through dependency for survival on the Cult. The whole structure is planned to be protected and enforced by a military-police state ([Fig 4](#)).

Here you have the reason for the global lockdowns of the fake pandemic to coldly destroy independent incomes and livelihoods and make everyone dependent on the 'state' (the Cult that controls the 'states'). I have warned in my books for many years about the plan to introduce a 'guaranteed income' – a barely survivable pittance – designed to impose dependency when employment was destroyed by AI technology and now even more comprehensively at great speed by the 'Covid' scam. Once the pandemic was played and lockdown consequences began to delete independent income the authorities began to talk right on cue about the need for a guaranteed income and a 'Great Reset'. Guaranteed income will be presented as benevolent governments seeking to help a desperate people – desperate as a direct result of actions of the same governments. The truth is that such payments are a trap. You will only get them if you do exactly what the authorities demand including mass vaccination (genetic manipulation). We have seen this theme already in Australia where those dependent on government benefits have them reduced if parents don't agree to have their children vaccinated according to an insane health-destroying government-dictated schedule. Calculated economic collapse applies to governments as well as people. The Cult wants rid of countries through the creation of a world state with countries broken up into regions ruled by a world government and super states like the European Union. Countries must be bankrupted, too, to this end and it's being achieved by the trillions in 'rescue packages' and furlough payments, trillions in lost taxation, and money-no-object spending on 'Covid' including constant all-medium advertising (programming) which has made the media dependent on government for much of its income. The day of reckoning is coming – as planned – for government spending and given that it has been made possible by printing money and not by production/taxation there is inflation on the way that has the

potential to wipe out monetary value. In that case there will be no need for the Cult to steal your money. It just won't be worth anything (see the German Weimar Republic before the Nazis took over). Many have been okay with lockdowns while getting a percentage of their income from so-called furlough payments without having to work. Those payments are dependent, however, on people having at least a theoretical job with a business considered non-essential and ordered to close. As these business go under because they are closed by lockdown after lockdown the furlough stops and it will for everyone eventually. Then what? The 'then what?' is precisely the idea.



Figure 4: The Hunger Games Society structure I have long warned was planned and now the 'Covid' hoax has made it possible. This is the real reason for lockdowns.

Hired hands

Between the Hunger Games Cult elite and the dependent population is planned to be a vicious military-police state (a fusion of the two into one force). This has been in the making for a long time with police looking ever more like the military and carrying weapons to match. The pandemic scam has seen this process accelerate so fast as

lockdown house arrest is brutally enforced by carefully recruited fascist minds and gormless system-servers. The police and military are planned to merge into a centrally-directed world army in a global structure headed by a world government which wouldn't be elected even by the election fixes now in place. The world army is not planned even to be human and instead wars would be fought, primarily against the population, using robot technology controlled by artificial intelligence. I have been warning about this for decades and now militaries around the world are being transformed by this very AI technology. The global regime that I describe is a particular form of fascism known as a technocracy in which decisions are not made by clueless and co-opted politicians but by unelected technocrats – scientists, engineers, technologists and bureaucrats. Cult-owned-and-controlled Silicon Valley giants are examples of technocracy and they already have far more power to direct world events than governments. They are with their censorship *selecting* governments. I know that some are calling the 'Great Reset' a Marxist communist takeover, but fascism and Marxism are different labels for the same tyranny. Tell those who lived in fascist Germany and Stalinist Russia that there was a difference in the way their freedom was deleted and their lives controlled. I could call it a fascist technocracy or a Marxist technocracy and they would be equally accurate. The Hunger Games society with its world government structure would oversee a world army, world central bank and single world cashless currency imposing its will on a microchipped population ([Fig 5](#)). Scan its different elements and see how the illusory pandemic is forcing society in this very direction at great speed. Leaders of 23 countries and the World Health Organization (WHO) backed the idea in March, 2021, of a global treaty for 'international cooperation' in 'health emergencies' and nations should 'come together as a global community for peaceful cooperation that extends beyond this crisis'. Cut the Orwellian bullshit and this means another step towards global government. The plan includes a cashless digital money system that I first warned about in 1993. Right at the start of 'Covid' the deeply corrupt Tedros

Adhanom Ghebreyesus, the crooked and merely gofer 'head' of the World Health Organization, said it was possible to catch the 'virus' by touching cash and it was better to use cashless means. The claim was ridiculous nonsense and like the whole 'Covid' mind-trick it was nothing to do with 'health' and everything to do with pushing every aspect of the Cult agenda. As a result of the Tedros lie the use of cash has plummeted. The Cult script involves a single world digital currency that would eventually be technologically embedded in the body. China is a massive global centre for the Cult and if you watch what is happening there you will know what is planned for everywhere. The Chinese government is developing a digital currency which would allow fines to be deducted immediately via AI for anyone caught on camera breaking its fantastic list of laws and the money is going to be programmable with an expiry date to ensure that no one can accrue wealth except the Cult and its operatives.



Figure 5: The structure of global control the Cult has been working towards for so long and this has been enormously advanced by the 'Covid' illusion.

Serfdom is so smart

The Cult plan is far wider, extreme, and more comprehensive than even most conspiracy researchers appreciate and I will come to the true depths of deceit and control in the chapters 'Who controls the

Cult?’ and ‘Escaping Wetiko’. Even the world that we know is crazy enough. We are being deluged with ever more sophisticated and controlling technology under the heading of ‘smart’. We have smart televisions, smart meters, smart cards, smart cars, smart driving, smart roads, smart pills, smart patches, smart watches, smart skin, smart borders, smart pavements, smart streets, smart cities, smart communities, smart environments, smart growth, smart planet ... smart *everything* around us. Smart technologies and methods of operation are designed to interlock to create a global Smart Grid connecting the entirety of human society including human minds to create a centrally-dictated ‘hive’ mind. ‘Smart cities’ is code for densely-occupied megacities of total surveillance and control through AI. Ever more destructive frequency communication systems like 5G have been rolled out without any official testing for health and psychological effects (colossal). 5G/6G/7G systems are needed to run the Smart Grid and each one becomes more destructive of body and mind. Deleting independent income is crucial to forcing people into these AI-policed prisons by ending private property ownership (except for the Cult elite). The Cult’s Great Reset now openly foresees a global society in which no one will own any possessions and everything will be rented while the Cult would own literally everything under the guise of government and corporations. The aim has been to use the lockdowns to destroy sources of income on a mass scale and when the people are destitute and in unrepayable amounts of debt (problem) Cult assets come forward with the pledge to write-off debt in return for handing over all property and possessions (solution). Everything – literally everything including people – would be connected to the Internet via AI. I was warning years ago about the coming Internet of Things (IoT) in which all devices and technology from your car to your fridge would be plugged into the Internet and controlled by AI. Now we are already there with much more to come. The next stage is the Internet of Everything (IoE) which is planned to include the connection of AI to the human brain and body to replace the human mind with a centrally-controlled AI mind. Instead of perceptions

being manipulated through control of information and censorship those perceptions would come direct from the Cult through AI. What do you think? You think whatever AI decides that you think. In human terms there would be no individual 'think' any longer. Too incredible? The ravings of a lunatic? Not at all. Cult-owned crazies in Silicon Valley have been telling us the plan for years without explaining the real motivation and calculated implications. These include Google executive and 'futurist' Ray Kurzweil who highlights the year 2030 for when this would be underway. He said:

Our thinking ... will be a hybrid of biological and non-biological thinking ... humans will be able to extend their limitations and 'think in the cloud' ... We're going to put gateways to the cloud in our brains ... We're going to gradually merge and enhance ourselves ... In my view, that's the nature of being human – we transcend our limitations.

As the technology becomes vastly superior to what we are then the small proportion that is still human gets smaller and smaller and smaller until it's just utterly negligible.

The sales-pitch of Kurzweil and Cult-owned Silicon Valley is that this would make us 'super-human' when the real aim is to make us post-human and no longer 'human' in the sense that we have come to know. The entire global population would be connected to AI and become the centrally-controlled 'hive-mind' of externally-delivered perceptions. The Smart Grid being installed to impose the Cult's will on the world is being constructed to allow particular locations – even one location – to control the whole global system. From these prime control centres, which absolutely include China and Israel, anything connected to the Internet would be switched on or off and manipulated at will. Energy systems could be cut, communication via the Internet taken down, computer-controlled driverless autonomous vehicles driven off the road, medical devices switched off, the potential is limitless given how much AI and Internet connections now run human society. We have seen nothing yet if we allow this to continue. Autonomous vehicle makers are working with law enforcement to produce cars designed to automatically pull over if they detect a police or emergency vehicle flashing from up to 100 feet away. At a police stop the car would be unlocked and the

window rolled down automatically. Vehicles would only take you where the computer (the state) allowed. The end of petrol vehicles and speed limiters on all new cars in the UK and EU from 2022 are steps leading to electric computerised transport over which ultimately you have no control. The picture is far bigger even than the Cult global network or web and that will become clear when I get to the nature of the 'spider'. There is a connection between all these happenings and the instigation of DNA-manipulating 'vaccines' (which aren't 'vaccines') justified by the 'Covid' hoax. That connection is the unfolding plan to transform the human body from a biological to a synthetic biological state and this is why synthetic biology is such a fast-emerging discipline of mainstream science. 'Covid vaccines' are infusing self-replicating synthetic genetic material into the cells to cumulatively take us on the Totalitarian Tiptoe from Human 1.0 to the synthetic biological Human 2.0 which will be physically and perceptually attached to the Smart Grid to one hundred percent control every thought, perception and deed. Humanity needs to wake up and *fast*.

This is the barest explanation of where the 'outcome' is planned to go but it's enough to see the journey happening all around us. Those new to this information will already see 'Covid' in a whole new context. I will add much more detail as we go along, but for the minutiae evidence see my mega-works, *The Answer*, *The Trigger* and *Everything You Need to Know But Have Never Been Told*.

Now – how does a Renegade Mind see the 'world'?

CHAPTER TWO

Renegade Perception

It is one thing to be clever and another to be wise

George R.R. Martin

A simple definition of the difference between a programmed mind and a Renegade Mind would be that one sees only dots while the other connects them to see the picture. Reading reality with accuracy requires the observer to (a) know the planned outcome and (b) realise that everything, but *everything*, is connected.

The entirety of infinite reality is connected – that's its very nature – and with human society an expression of infinite reality the same must apply. Simple cause and effect is a connection. The effect is triggered by the cause and the effect then becomes the cause of another effect. Nothing happens in isolation because it *can't*. Life in whatever reality is simple choice and consequence. We make choices and these lead to consequences. If we don't like the consequences we can make different choices and get different consequences which lead to other choices and consequences. The choice and the consequence are not only connected they are indivisible. You can't have one without the other as an old song goes. A few cannot control the world unless those being controlled allow that to happen – cause and effect, choice and consequence. Control – who has it and who doesn't – is a two-way process, a symbiotic relationship, involving the controller and controlled. 'They took my freedom away!!' Well, yes, but you also gave it to them. Humanity is

subjected to mass control because humanity has acquiesced to that control. This is all cause and effect and literally a case of give and take. In the same way world events of every kind are connected and the Cult works incessantly to sell the illusion of the random and coincidental to maintain the essential (to them) perception of dots that hide the picture. Renegade Minds know this and constantly scan the world for patterns of connection. This is absolutely pivotal in understanding the happenings in the world and without that perspective clarity is impossible. First you know the planned outcome and then you identify the steps on the journey – the day-by-day apparently random which, when connected in relation to the outcome, no longer appear as individual events, but as the proverbial *chain* of events leading in the same direction. I'll give you some examples:

Political puppet show

We are told to believe that politics is 'adversarial' in that different parties with different beliefs engage in an endless tussle for power. There may have been some truth in that up to a point – and only a point – but today divisions between 'different' parties are rhetorical not ideological. Even the rhetorical is fusing into one-speak as the parties eject any remaining free thinkers while others succumb to the ever-gathering intimidation of anyone with the 'wrong' opinion. The Cult is not a new phenomenon and can be traced back thousands of years as my books have documented. Its intergenerational initiatives have been manipulating events with increasing effect the more that global power has been centralised. In ancient times the Cult secured control through the system of monarchy in which 'special' bloodlines (of which more later) demanded the right to rule as kings and queens simply by birthright and by vanquishing others who claimed the same birthright. There came a time, however, when people had matured enough to see the unfairness of such tyranny and demanded a say in who governed them. Note the word – *governed* them. Not served them – *governed* them, hence government defined as 'the political direction and control exercised over the

actions of the members, citizens, or inhabitants of communities, societies, and states; direction of the affairs of a state, community, etc.' Governments exercise control over rather than serve just like the monarchies before them. Bizarrely there are still countries like the United Kingdom which are ruled by a monarch *and* a government that officially answers to the monarch. The UK head of state and that of Commonwealth countries such as Canada, Australia and New Zealand is 'selected' by who in a *single family* had unprotected sex with whom and in what order. Pinch me it can't be true. Ouch! Shit, it is. The demise of monarchies in most countries offered a potential vacuum in which some form of free and fair society could arise and the Cult had that base covered. Monarchies had served its interests but they couldn't continue in the face of such widespread opposition and, anyway, replacing a 'royal' dictatorship that people could see with a dictatorship 'of the people' hiding behind the concept of 'democracy' presented far greater manipulative possibilities and ways of hiding coordinated tyranny behind the illusion of 'freedom'.

Democracy is quite wrongly defined as government selected by the population. This is not the case at all. It is government selected by *some* of the population (and then only in theory). This 'some' doesn't even have to be the majority as we have seen so often in first-past-the-post elections in which the so-called majority party wins fewer votes than the 'losing' parties combined. Democracy can give total power to a party in government from a minority of the votes cast. It's a sleight of hand to sell tyranny as freedom. Seventy-four million Trump-supporting Americans didn't vote for the 'Democratic' Party of Joe Biden in the distinctly dodgy election in 2020 and yet far from acknowledging the wishes and feelings of that great percentage of American society the Cult-owned Biden government set out from day one to destroy them and their right to a voice and opinion. Empty shell Biden and his Cult handlers said they were doing this to 'protect democracy'. Such is the level of lunacy and sickness to which politics has descended. Connect the dots and relate them to the desired outcome – a world government run by self-appointed technocrats and no longer even elected

politicians. While operating through its political agents in government the Cult is at the same time encouraging public disdain for politicians by putting idiots and incompetents in theoretical power on the road to deleting them. The idea is to instil a public reaction that says of the technocrats: 'Well, they couldn't do any worse than the pathetic politicians.' It's all about controlling perception and Renegade Minds can see through that while programmed minds cannot when they are ignorant of both the planned outcome and the manipulation techniques employed to secure that end. This knowledge can be learned, however, and fast if people choose to get informed.

Politics may at first sight appear very difficult to control from a central point. I mean look at the 'different' parties and how would you be able to oversee them all and their constituent parts? In truth, it's very straightforward because of their structure. We are back to the pyramid of imposition and acquiescence. Organisations are structured in the same way as the system as a whole. Political parties are not open forums of free expression. They are hierarchies. I was a national spokesman for the British Green Party which claimed to be a different kind of politics in which influence and power was devolved; but I can tell you from direct experience – and it's far worse now – that Green parties are run as hierarchies like all the others however much they may try to hide that fact or kid themselves that it's not true. A very few at the top of all political parties are directing policy and personnel. They decide if you are elevated in the party or serve as a government minister and to do that you have to be a yes man or woman. Look at all the maverick political thinkers who never ascended the greasy pole. If you want to progress within the party or reach 'high-office' you need to fall into line and conform. Exceptions to this are rare indeed. Should you want to run for parliament or Congress you have to persuade the local or state level of the party to select you and for that you need to play the game as dictated by the hierarchy. If you secure election and wish to progress within the greater structure you need to go on conforming to what is acceptable to those running the hierarchy

from the peak of the pyramid. Political parties are perceptual gulags and the very fact that there are party 'Whips' appointed to 'whip' politicians into voting the way the hierarchy demands exposes the ridiculous idea that politicians are elected to serve the people they are supposed to represent. Cult operatives and manipulation has long seized control of major parties that have any chance of forming a government and at least most of those that haven't. A new party forms and the Cult goes to work to infiltrate and direct. This has reached such a level today that you see video compilations of 'leaders' of all parties whether Democrats, Republicans, Conservative, Labour and Green parroting the same Cult mantra of 'Build Back Better' and the 'Great Reset' which are straight off the Cult song-sheet to describe the transformation of global society in response to the Cult-instigated hoaxes of the 'Covid pandemic' and human-caused 'climate change'. To see Caroline Lucas, the Green Party MP that I knew when I was in the party in the 1980s, speaking in support of plans proposed by Cult operative Klaus Schwab representing the billionaire global elite is a real head-shaker.

Many parties – one master

The party system is another mind-trick and was instigated to change the nature of the dictatorship by swapping 'royalty' for dark suits that people believed – though now ever less so – represented their interests. Understanding this trick is to realise that a single force (the Cult) controls all parties either directly in terms of the major ones or through manipulation of perception and ideology with others. You don't need to manipulate Green parties to demand your transformation of society in the name of 'climate change' when they are obsessed with the lie that this is essential to 'save the planet'. You just give them a platform and away they go serving your interests while believing they are being environmentally virtuous. America's political structure is a perfect blueprint for how the two or multi-party system is really a one-party state. The Republican Party is controlled from one step back in the shadows by a group made up of billionaires and their gofers known as neoconservatives or Neocons.

I have exposed them in fine detail in my books and they were the driving force behind the policies of the imbecilic presidency of Boy George Bush which included 9/11 (see *The Trigger* for a comprehensive demolition of the official story), the subsequent 'war on terror' (war of terror) and the invasions of Afghanistan and Iraq. The latter was a No-Problem-Reaction-Solution based on claims by Cult operatives, including Bush and British Prime Minister Tony Blair, about Saddam Hussein's 'weapons of mass destruction' which did not exist as war criminals Bush and Blair well knew.



Figure 6: Different front people, different parties – same control system.

The Democratic Party has its own 'Neocon' group controlling from the background which I call the 'Democons' and here's the penny-drop – the Neocons and Democons answer to the same masters one step further back into the shadows (Fig 6). At that level of the Cult the Republican and Democrat parties are controlled by the same people and no matter which is in power the Cult is in power. This is how it works in almost every country and certainly in Britain with Conservative, Labour, Liberal Democrat and Green parties now all on the same page whatever the rhetoric may be in their feeble attempts to appear different. Neocons operated at the time of Bush through a think tank called The Project for the New American Century which in September, 2000, published a document entitled *Rebuilding America's Defenses: Strategies, Forces, and Resources*

For a New Century demanding that America fight ‘multiple, simultaneous major theatre wars’ as a ‘core mission’ to force regime-change in countries including Iraq, Libya and Syria. Neocons arranged for Bush (‘Republican’) and Blair (‘Labour Party’) to front-up the invasion of Iraq and when they departed the Democons orchestrated the targeting of Libya and Syria through Barack Obama (‘Democrat’) and British Prime Minister David Cameron (‘Conservative Party’). We have ‘different’ parties and ‘different’ people, but the same unfolding script. The more the Cult has seized the reigns of parties and personnel the more their policies have transparently pursued the same agenda to the point where the fascist ‘Covid’ impositions of the Conservative junta of Jackboot Johnson in Britain were opposed by the Labour Party because they were not fascist enough. The Labour Party is likened to the US Democrats while the Conservative Party is akin to a British version of the Republicans and on both sides of the Atlantic they all speak the same language and support the direction demanded by the Cult although some more enthusiastically than others. It’s a similar story in country after country because it’s all centrally controlled. Oh, but what about Trump? I’ll come to him shortly. Political ‘choice’ in the ‘party’ system goes like this: You vote for Party A and they get into government. You don’t like what they do so next time you vote for Party B and they get into government. You don’t like what they do when it’s pretty much the same as Party A and why wouldn’t that be with both controlled by the same force? Given that only two, sometimes three, parties have any chance of forming a government to get rid of Party B that you don’t like you have to vote again for Party A which ... you don’t like. This, ladies and gentlemen, is what they call ‘democracy’ which we are told – wrongly – is a term interchangeable with ‘freedom’.

The cult of cults

At this point I need to introduce a major expression of the Global Cult known as Sabbatian-Frankism. Sabbatian is also spelt as Sabbatean. I will summarise here. I have published major exposés

and detailed background in other works. Sabbatian-Frankism combines the names of two frauds posing as 'Jewish' men, Sabbatai Zevi (1626-1676), a rabbi, black magician and occultist who proclaimed he was the Jewish messiah; and Jacob Frank (1726-1791), the Polish 'Jew', black magician and occultist who said he was the reincarnation of 'messiah' Zevi and biblical patriarch Jacob. They worked across two centuries to establish the Sabbatian-Frankist cult that plays a major, indeed central, role in the manipulation of human society by the Global Cult which has its origins much further back in history than Sabbatai Zevi. I should emphasise two points here in response to the shrill voices that will scream 'anti-Semitism': (1) Sabbatian-Frankists are NOT Jewish and only pose as such to hide their cult behind a Jewish façade; and (2) my information about this cult has come from Jewish sources who have long realised that their society and community has been infiltrated and taken over by interloper Sabbatian-Frankists. Infiltration has been the foundation technique of Sabbatian-Frankism from its official origin in the 17th century. Zevi's Sabbatian sect attracted a massive following described as the biggest messianic movement in Jewish history, spreading as far as Africa and Asia, and he promised a return for the Jews to the 'Promised Land' of Israel. Sabbatianism was not Judaism but an inversion of everything that mainstream Judaism stood for. So much so that this sinister cult would have a feast day when Judaism had a fast day and whatever was forbidden in Judaism the Sabbatians were encouraged and even commanded to do. This included incest and what would be today called Satanism. Members were forbidden to marry outside the sect and there was a system of keeping their children ignorant of what they were part of until they were old enough to be trusted not to unknowingly reveal anything to outsiders. The same system is employed to this day by the Global Cult in general which Sabbatian-Frankism has enormously influenced and now largely controls.

Zevi and his Sabbatians suffered a setback with the intervention by the Sultan of the Islamic Ottoman Empire in the Middle East and what is now the Republic of Turkey where Zevi was located. The

Sultan gave him the choice of proving his 'divinity', converting to Islam or facing torture and death. Funnily enough Zevi chose to convert or at least appear to. Some of his supporters were disillusioned and drifted away, but many did not with 300 families also converting – only in theory – to Islam. They continued behind this Islamic smokescreen to follow the goals, rules and rituals of Sabbatianism and became known as 'crypto-Jews' or the 'Dönme' which means 'to turn'. This is rather ironic because they didn't 'turn' and instead hid behind a fake Islamic persona. The process of appearing to be one thing while being very much another would become the calling card of Sabbatianism especially after Zevi's death and the arrival of the Satanist Jacob Frank in the 18th century when the cult became Sabbatian-Frankism and plumbed still new depths of depravity and infiltration which included – still includes – human sacrifice and sex with children. Wherever Sabbatians go paedophilia and Satanism follow and is it really a surprise that Hollywood is so infested with child abuse and Satanism when it was established by Sabbatian-Frankists and is still controlled by them? Hollywood has been one of the prime vehicles for global perceptual programming and manipulation. How many believe the version of 'history' portrayed in movies when it is a travesty and inversion (again) of the truth? Rabbi Marvin Antelman describes Frankism in his book, *To Eliminate the Opiate*, as 'a movement of complete evil' while Jewish professor Gershom Scholem said of Frank in *The Messianic Idea in Judaism*: 'In all his actions [he was] a truly corrupt and degenerate individual ... one of the most frightening phenomena in the whole of Jewish history.' Frank was excommunicated by traditional rabbis, as was Zevi, but Frank was undeterred and enjoyed vital support from the House of Rothschild, the infamous banking dynasty whose inner-core are Sabbatian-Frankists and not Jews. Infiltration of the Roman Church and Vatican was instigated by Frank with many Dönme 'turning' again to convert to Roman Catholicism with a view to hijacking the reins of power. This was the ever-repeating modus operandi and continues to be so. Pose as an advocate of the religion, culture or country that you want to control and then

manipulate your people into the positions of authority and influence largely as advisers, administrators and Svengalis for those that appear to be in power. They did this with Judaism, Christianity (Christian Zionism is part of this), Islam and other religions and nations until Sabbatian-Frankism spanned the world as it does today.

Sabbatian Saudis and the terror network

One expression of the Sabbatian-Frankist Dönme within Islam is the ruling family of Saudi Arabia, the House of Saud, through which came the vile distortion of Islam known as Wahhabism. This is the violent creed followed by terrorist groups like Al-Qaeda and ISIS or Islamic State. Wahhabism is the hand-chopping, head-chopping 'religion' of Saudi Arabia which is used to keep the people in a constant state of fear so the interloper House of Saud can continue to rule. Al-Qaeda and Islamic State were lavishly funded by the House of Saud while being created and directed by the Sabbatian-Frankist network in the United States that operates through the Pentagon, CIA and the government in general of whichever 'party'. The front man for the establishment of Wahhabism in the middle of the 18th century was a Sabbatian-Frankist 'crypto-Jew' posing as Islamic called Muhammad ibn Abd al-Wahhab. His daughter would marry the son of Muhammad bin Saud who established the first Saudi state before his death in 1765 with support from the British Empire. Bin Saud's successors would establish modern Saudi Arabia in league with the British and Americans in 1932 which allowed them to seize control of Islam's major shrines in Mecca and Medina. They have dictated the direction of Sunni Islam ever since while Iran is the major centre of the Shiite version and here we have the source of at least the public conflict between them. The Sabbatian network has used its Wahhabi extremists to carry out Problem-Reaction-Solution terrorist attacks in the name of 'Al-Qaeda' and 'Islamic State' to justify a devastating 'war on terror', ever-increasing surveillance of the population and to terrify people into compliance. Another insight of the Renegade Mind is the streetwise understanding that

just because a country, location or people are attacked doesn't mean that those apparently representing that country, location or people are not behind the attackers. Often they are *orchestrating* the attacks because of the societal changes that can be then justified in the name of 'saving the population from terrorists'.

I show in great detail in *The Trigger* how Sabbatian-Frankists were the real perpetrators of 9/11 and not '19 Arab hijackers' who were blamed for what happened. Observe what was justified in the name of 9/11 alone in terms of Middle East invasions, mass surveillance and control that fulfilled the demands of the Project for the New American Century document published by the Sabbatian Neocons. What appear to be enemies are on the deep inside players on the same Sabbatian team. Israel and Arab 'royal' dictatorships are all ruled by Sabbatians and the recent peace agreements between Israel and Saudi Arabia, the United Arab Emirates (UAE) and others are only making formal what has always been the case behind the scenes. Palestinians who have been subjected to grotesque tyranny since Israel was bombed and terrorised into existence in 1948 have never stood a chance. Sabbatian-Frankists have controlled Israel (so the constant theme of violence and war which Sabbatians love) and they have controlled the Arab countries that Palestinians have looked to for real support that never comes. 'Royal families' of the Arab world in Saudi Arabia, Bahrain, UAE, etc., are all Sabbatians with allegiance to the aims of the cult and not what is best for their Arabic populations. They have stolen the oil and financial resources from their people by false claims to be 'royal dynasties' with a genetic right to rule and by employing vicious militaries to impose their will.

Satanic 'illumination'

The Satanist Jacob Frank formed an alliance in 1773 with two other Sabbatians, Mayer Amschel Rothschild (1744-1812), founder of the Rothschild banking dynasty, and Jesuit-educated fraudulent Jew, Adam Weishaupt, and this led to the formation of the Bavarian Illuminati, firstly under another name, in 1776. The Illuminati would

be the manipulating force behind the French Revolution (1789-1799) and was also involved in the American Revolution (1775-1783) before and after the Illuminati's official creation. Weishaupt would later become (in public) a Protestant Christian in archetypal Sabbatian style. I read that his name can be decoded as Adam-Weishaupt or 'the first man to lead those who know'. He wasn't a leader in the sense that he was a subordinate, but he did lead those below him in a crusade of transforming human society that still continues today. The theme was confirmed as early as 1785 when a horseman courier called Lanz was reported to be struck by lightning and extensive Illuminati documents were found in his saddlebags. They made the link to Weishaupt and detailed the plan for world takeover. Current events with 'Covid' fascism have been in the making for a very long time. Jacob Frank was jailed for 13 years by the Catholic Inquisition after his arrest in 1760 and on his release he headed for Frankfurt, Germany, home city and headquarters of the House of Rothschild where the alliance was struck with Mayer Amschel Rothschild and Weishaupt. Rothschild arranged for Frank to be given the title of Baron and he became a wealthy nobleman with a big following of Jews in Germany, the Austro-Hungarian Empire and other European countries. Most of them would have believed he was on their side.

The name 'Illuminati' came from the Zohar which is a body of works in the Jewish mystical 'bible' called the Kabbalah. 'Zohar' is the foundation of Sabbatian-Frankist belief and in Hebrew 'Zohar' means 'splendour', 'radiance', 'illuminated', and so we have 'Illuminati'. They claim to be the 'Illuminated Ones' from their knowledge systematically hidden from the human population and passed on through generations of carefully-chosen initiates in the global secret society network or Cult. Hidden knowledge includes an awareness of the Cult agenda for the world and the nature of our collective reality that I will explore later. Cult 'illumination' is symbolised by the torch held by the Statue of Liberty which was gifted to New York by French Freemasons in Paris who knew exactly what it represents. 'Liberty' symbolises the goddess worshipped in

Babylon as Queen Semiramis or Ishtar. The significance of this will become clear. Notice again the ubiquitous theme of inversion with the Statue of 'Liberty' really symbolising mass control (Fig 7). A mirror-image statute stands on an island in the River Seine in Paris from where New York Liberty originated (Fig 8). A large replica of the Liberty flame stands on top of the Pont de l'Alma tunnel in Paris where Princess Diana died in a Cult ritual described in *The Biggest Secret*. Lucifer 'the light bringer' is related to all this (and much more as we'll see) and 'Lucifer' is a central figure in Sabbatian-Frankism and its associated Satanism. Sabbatians reject the Jewish Torah, or Pentateuch, the 'five books of Moses' in the Old Testament known as Genesis, Exodus, Leviticus, Numbers, and Deuteronomy which are claimed by Judaism and Christianity to have been dictated by 'God' to Moses on Mount Sinai. Sabbatians say these do not apply to them and they seek to replace them with the Zohar to absorb Judaism and its followers into their inversion which is an expression of a much greater global inversion. They want to delete all religions and force humanity to worship a one-world religion – Sabbatian Satanism that also includes worship of the Earth goddess. Satanic themes are being more and more introduced into mainstream society and while Christianity is currently the foremost target for destruction the others are planned to follow.



Figure 7: The Cult goddess of Babylon disguised as the Statue of Liberty holding the flame of Lucifer the 'light bringer'.



Figure 8: Liberty's mirror image in Paris where the New York version originated.

Marx brothers

Rabbi Marvin Antelman connects the Illuminati to the Jacobins in *To Eliminate the Opiate* and Jacobins were the force behind the French Revolution. He links both to the Bund der Gerechten, or League of the Just, which was the network that inflicted communism/Marxism on the world. Antelman wrote:

The original inner circle of the Bund der Gerechten consisted of born Catholics, Protestants and Jews [Sabbatian-Frankist infiltrators], and those representatives of respective subdivisions formulated schemes for the ultimate destruction of their faiths. The heretical Catholics laid plans which they felt would take a century or more for the ultimate destruction of the church; the apostate Jews for the ultimate destruction of the Jewish religion.

Sabbatian-created communism connects into this anti-religion agenda in that communism does not allow for the free practice of religion. The Sabbatian 'Bund' became the International Communist Party and Communist League and in 1848 'Marxism' was born with the Communist Manifesto of Sabbatian assets Karl Marx and Friedrich Engels. It is absolutely no coincidence that Marxism, just a different name for fascist and other centrally-controlled tyrannies, is being imposed worldwide as a result of the 'Covid' hoax and nor that Marxist/fascist China was the place where the hoax originated. The reason for this will become very clear in the chapter 'Covid: The calculated catastrophe'. The so-called 'Woke' mentality has hijacked

traditional beliefs of the political left and replaced them with far-right make-believe 'social justice' better known as Marxism. Woke will, however, be swallowed by its own perceived 'revolution' which is really the work of billionaires and billionaire corporations feigning being 'Woke'. Marxism is being touted by Wokers as a replacement for 'capitalism' when we don't have 'capitalism'. We have cartelism in which the market is stitched up by the very Cult billionaires and corporations bankrolling Woke. Billionaires love Marxism which keeps the people in servitude while they control from the top. Terminally naïve Wokers think they are 'changing the world' when it's the Cult that is doing the changing and when they have played their vital part and become surplus to requirements they, too, will be targeted. The Illuminati-Jacobins were behind the period known as 'The Terror' in the French Revolution in 1793 and 1794 when Jacobin Maximillian de Robespierre and his Orwellian 'Committee of Public Safety' killed 17,000 'enemies of the Revolution' who had once been 'friends of the Revolution'. Karl Marx (1818-1883), whose Sabbatian creed of Marxism has cost the lives of at least 100 million people, is a hero once again to Wokers who have been systematically kept ignorant of real history by their 'education' programming. As a result they now promote a Sabbatian 'Marxist' abomination destined at some point to consume them. Rabbi Antelman, who spent decades researching the Sabbatian plot, said of the League of the Just and Karl Marx:

Contrary to popular opinion Karl Marx did not originate the Communist Manifesto. He was paid for his services by the League of the Just, which was known in its country of origin, Germany, as the Bund der Geächteten.

Antelman said the text attributed to Marx was the work of other people and Marx 'was only repeating what others already said'. Marx was 'a hired hack – lackey of the wealthy Illuminists'. Marx famously said that religion was the 'opium of the people' (part of the Sabbatian plan to demonise religion) and Antelman called his books, *To Eliminate the Opiate*. Marx was born Jewish, but his family converted to Christianity (Sabbatian modus operandi) and he

attacked Jews, not least in his book, *A World Without Jews*. In doing so he supported the Sabbatian plan to destroy traditional Jewishness and Judaism which we are clearly seeing today with the vindictive targeting of orthodox Jews by the Sabbatian government of Israel over 'Covid' laws. I don't follow any religion and it has done much damage to the world over centuries and acted as a perceptual straightjacket. Renegade Minds, however, are always asking *why* something is being done. It doesn't matter if they agree or disagree with what is happening – *why* is it happening is the question. The 'why?' can be answered with regard to religion in that religions create interacting communities of believers when the Cult wants to dismantle all discourse, unity and interaction (see 'Covid' lockdowns) and the ultimate goal is to delete all religions for a one-world religion of Cult Satanism worshipping their 'god' of which more later. We see the same 'why?' with gun control in America. I don't have guns and don't want them, but why is the Cult seeking to disarm the population at the same time that law enforcement agencies are armed to their molars and why has every tyrant in history sought to disarm people before launching the final takeover? They include Hitler, Stalin, Pol Pot and Mao who followed confiscation with violent seizing of power. You know it's a Cult agenda by the people who immediately race to the microphones to exploit dead people in multiple shootings. Ultra-Zionist Cult lackey Senator Chuck Schumer was straight on the case after ten people were killed in Boulder, Colorado in March, 2121. Simple rule ... if Schumer wants it the Cult wants it and the same with his ultra-Zionist mate the wild-eyed Senator Adam Schiff. At the same time they were calling for the disarmament of Americans, many of whom live a long way from a police response, Schumer, Schiff and the rest of these pampered clowns were sitting on Capitol Hill behind a razor-wired security fence protected by thousands of armed troops in addition to their own armed bodyguards. Mom and pop in an isolated home? They're just potential mass shooters.

Zion Mainframe

Sabbatian-Frankists and most importantly the Rothschilds were behind the creation of 'Zionism', a political movement that demanded a Jewish homeland in Israel as promised by Sabbatai Zevi. The very symbol of Israel comes from the German meaning of the name Rothschild. Dynasty founder Mayer Amschel Rothschild changed the family name from Bauer to Rothschild, or 'Red-Shield' in German, in deference to the six-pointed 'Star of David' hexagram displayed on the family's home in Frankfurt. The symbol later appeared on the flag of Israel after the Rothschilds were centrally involved in its creation. Hexagrams are not a uniquely Jewish symbol and are widely used in occult ('hidden') networks often as a symbol for Saturn (see my other books for why). Neither are Zionism and Jewishness interchangeable. Zionism is a political movement and philosophy and not a 'race' or a people. Many Jews oppose Zionism and many non-Jews, including US President Joe Biden, call themselves Zionists as does Israel-centric Donald Trump. America's support for the Israel government is pretty much a gimme with ultra-Zionist billionaires and corporations providing fantastic and dominant funding for both political parties. Former Congresswoman Cynthia McKinney has told how she was approached immediately she ran for office to 'sign the pledge' to Israel and confirm that she would always vote in that country's best interests. All American politicians are approached in this way. Anyone who refuses will get no support or funding from the enormous and all-powerful Zionist lobby that includes organisations like mega-lobby group AIPAC, the American Israel Public Affairs Committee. Trump's biggest funder was ultra-Zionist casino and media billionaire Sheldon Adelson while major funders of the Democratic Party include ultra-Zionist George Soros and ultra-Zionist financial and media mogul, Haim Saban. Some may reel back at the suggestion that Soros is an Israel-firster (Sabbatian-controlled Israel-firster), but Renegade Minds watch the actions not the words and everywhere Soros donates his billions the Sabbatian agenda benefits. In the spirit of Sabbatian inversion Soros pledged \$1 billion for a new university network to promote 'liberal values and tackle intolerance'. He made the announcement during his annual speech

at the Cult-owned World Economic Forum in Davos, Switzerland, in January, 2020, after his 'harsh criticism' of 'authoritarian rulers' around the world. You can only laugh at such brazen mendacity. How *he* doesn't laugh is the mystery. Translated from the Orwellian 'liberal values and tackle intolerance' means teaching non-white people to hate white people and for white people to loathe themselves for being born white. The reason for that will become clear.

The 'Anti-Semitism' fraud

Zionists support the Jewish homeland in the land of Palestine which has been the Sabbatian-Rothschild goal for so long, but not for the benefit of Jews. Sabbatians and their global Anti-Semitism Industry have skewed public and political opinion to equate opposing the violent extremes of Zionism to be a blanket attack and condemnation of all Jewish people. Sabbatians and their global Anti-Semitism Industry have skewed public and political opinion to equate opposing the violent extremes of Zionism to be a blanket attack and condemnation of all Jewish people. This is nothing more than a Sabbatian protection racket to stop legitimate investigation and exposure of their agendas and activities. The official definition of 'anti-Semitism' has more recently been expanded to include criticism of Zionism – a *political movement* – and this was done to further stop exposure of Sabbatian infiltrators who created Zionism as we know it today in the 19th century. Renegade Minds will talk about these subjects when they know the shit that will come their way. People must decide if they want to know the truth or just cower in the corner in fear of what others will say. Sabbatians have been trying to label me as 'anti-Semitic' since the 1990s as I have uncovered more and more about their background and agendas. Useless, gutless, fraudulent 'journalists' then just repeat the smears without question and on the day I was writing this section a pair of unquestioning repeaters called Ben Quinn and Archie Bland (how appropriate) outright called me an 'anti-Semite' in the establishment propaganda sheet, the London *Guardian*, with no supporting evidence. The

Sabbatian Anti-Semitism Industry said so and who are they to question that? They wouldn't dare. Ironically 'Semitic' refers to a group of languages in the Middle East that are almost entirely Arabic. 'Anti-Semitism' becomes 'anti-Arab' which if the consequences of this misunderstanding were not so grave would be hilarious. Don't bother telling Quinn and Bland. I don't want to confuse them, bless 'em. One reason I am dubbed 'anti-Semitic' is that I wrote in the 1990s that Jewish operatives (Sabbatians) were heavily involved in the Russian Revolution when Sabbatians overthrew the Romanov dynasty. This apparently made me 'anti-Semitic'. Oh, really? Here is a section from *The Trigger*:

British journalist Robert Wilton confirmed these themes in his 1920 book *The Last Days of the Romanovs* when he studied official documents from the Russian government to identify the members of the Bolshevik ruling elite between 1917 and 1919. The Central Committee included 41 Jews among 62 members; the Council of the People's Commissars had 17 Jews out of 22 members; and 458 of the 556 most important Bolshevik positions between 1918 and 1919 were occupied by Jewish people. Only 17 were Russian. Then there were the 23 Jews among the 36 members of the vicious Cheka Soviet secret police established in 1917 who would soon appear all across the country.

Professor Robert Service of Oxford University, an expert on 20th century Russian history, found evidence that ['Jewish'] Leon Trotsky had sought to make sure that Jews were enrolled in the Red Army and were disproportionately represented in the Soviet civil bureaucracy that included the Cheka which performed mass arrests, imprisonment and executions of 'enemies of the people'. A US State Department Decimal File (861.00/5339) dated November 13th, 1918, names [Rothschild banking agent in America] Jacob Schiff and a list of ultra-Zionists as funders of the Russian Revolution leading to claims of a 'Jewish plot', but the key point missed by all is they were not 'Jews' – they were Sabbatian-Frankists.

Britain's Winston Churchill made the same error by mistake or otherwise. He wrote in a 1920 edition of the *Illustrated Sunday Herald* that those behind the Russian revolution were part of a 'worldwide conspiracy for the overthrow of civilisation and for the reconstitution of society on the basis of arrested development, of envious malevolence, and impossible equality' (see 'Woke' today because that has been created by the same network). Churchill said there was no need to exaggerate the part played in the creation of Bolshevism and in the actual bringing about of the Russian

Revolution 'by these international and for the most part atheistical Jews' ['atheistical Jews' = Sabbatians]. Churchill said it is certainly a very great one and probably outweighs all others: 'With the notable exception of Lenin, the majority of the leading figures are Jews.' He went on to describe, knowingly or not, the Sabbatian modus operandi of placing puppet leaders nominally in power while they control from the background:

Moreover, the principal inspiration and driving power comes from the Jewish leaders. Thus Tchitcherin, a pure Russian, is eclipsed by his nominal subordinate, Litvinoff, and the influence of Russians like Bukharin or Lunacharski cannot be compared with the power of Trotsky, or of Zinovieff, the Dictator of the Red Citadel (Petrograd), or of Krassin or Radek – all Jews. In the Soviet institutions the predominance of Jews is even more astonishing. And the prominent, if not indeed the principal, part in the system of terrorism applied by the Extraordinary Commissions for Combatting Counter-Revolution has been taken by Jews, and in some notable cases by Jewesses.

What I said about seriously disproportionate involvement in the Russian Revolution by Jewish 'revolutionaries' (Sabbatians) is provable fact, but truth is no defence against the Sabbatian Anti-Semitism Industry, its repeater parrots like Quinn and Bland, and the now breathtaking network of so-called 'Woke' 'anti-hate' groups with interlocking leaderships and funding which have the role of discrediting and silencing anyone who gets too close to exposing the Sabbatians. We have seen 'truth is no defence' confirmed in legal judgements with the Saskatchewan Human Rights Commission in Canada decreeing this: 'Truthful statements can be presented in a manner that would meet the definition of hate speech, and not all truthful statements must be free from restriction.' Most 'anti-hate' activists, who are themselves consumed by hatred, are too stupid and ignorant of the world to know how they are being used. They are far too far up their own virtue-signalling arses and it's far too dark for them to see anything.

The 'revolution' game

The background and methods of the 'Russian' Revolution are straight from the Sabbatian playbook seen in the French Revolution

and endless others around the world that appear to start as a revolution of the people against tyrannical rule and end up with a regime change to more tyrannical rule overtly or covertly. Wars, terror attacks and regime overthrows follow the Sabbatian cult through history with its agents creating them as Problem-Reaction-Solutions to remove opposition on the road to world domination. Sabbatian dots connect the Rothschilds with the Illuminati, Jacobins of the French Revolution, the 'Bund' or League of the Just, the International Communist Party, Communist League and the Communist Manifesto of Karl Marx and Friedrich Engels that would lead to the Rothschild-funded Russian Revolution. The sequence comes under the heading of 'creative destruction' when you advance to your global goal by continually destroying the status quo to install a new status quo which you then also destroy. The two world wars come to mind. With each new status quo you move closer to your planned outcome. Wars and mass murder are to Sabbatians a collective blood sacrifice ritual. They are obsessed with death for many reasons and one is that death is an inversion of life. Satanists and Sabbatians are obsessed with death and often target churches and churchyards for their rituals. Inversion-obsessed Sabbatians explain the use of inverted symbolism including the *inverted* pentagram and *inverted* cross. The inversion of the cross has been related to targeting Christianity, but the cross was a religious symbol long before Christianity and its inversion is a statement about the Sabbatian mentality and goals more than any single religion.

Sabbatians operating in Germany were behind the rise of the occult-obsessed Nazis and the subsequent Jewish exodus from Germany and Europe to Palestine and the United States after World War Two. The Rothschild dynasty was at the forefront of this both as political manipulators and by funding the operation. Why would Sabbatians help to orchestrate the horrors inflicted on Jews by the Nazis and by Stalin after they organised the Russian Revolution? Sabbatians hate Jews and their religion, that's why. They pose as Jews and secure positions of control within Jewish society and play the 'anti-Semitism' card to protect themselves from exposure

through a global network of organisations answering to the Sabbatian-created-and-controlled globe-spanning intelligence network that involves a stunning web of military-intelligence operatives and operations for a tiny country of just nine million. Among them are Jewish assets who are not Sabbatians but have been convinced by them that what they are doing is for the good of Israel and the Jewish community to protect them from what they have been programmed since childhood to believe is a Jew-hating hostile world. The Jewish community is just a highly convenient cover to hide the true nature of Sabbatians. Anyone getting close to exposing their game is accused by Sabbatian place-people and gofers of 'anti-Semitism' and claiming that all Jews are part of a plot to take over the world. I am not saying that. I am saying that Sabbatians – the *real* Jew-haters – have infiltrated the Jewish community to use them both as a cover and an 'anti-Semitic' defence against exposure. Thus we have the Anti-Semitism Industry targeted researchers in this way and most Jewish people think this is justified and genuine. They don't know that their 'Jewish' leaders and institutions of state, intelligence and military are not controlled by Jews at all, but cultists and stooges of Sabbatian-Frankism. I once added my name to a pro-Jewish freedom petition online and the next time I looked my name was gone and text had been added to the petition blurb to attack me as an 'anti-Semite' such is the scale of perceptual programming.

Moving on America

I tell the story in *The Trigger* and a chapter called 'Atlantic Crossing' how particularly after Israel was established the Sabbatians moved in on the United States and eventually grasped control of government administration, the political system via both Democrats and Republicans, the intelligence community like the CIA and National Security Agency (NSA), the Pentagon and mass media. Through this seriously compartmentalised network Sabbatians and their operatives in Mossad, Israeli Defense Forces (IDF) and US agencies pulled off 9/11 and blamed it on 19 'Al-Qaeda hijackers' dominated by men from, or connected to, Sabbatian-ruled Saudi

Arabia. The '19' were not even on the planes let alone flew those big passenger jets into buildings while being largely incompetent at piloting one-engine light aircraft. 'Hijacker' Hani Hanjour who is said to have flown American Airlines Flight 77 into the Pentagon with a turn and manoeuvre most professional pilots said they would have struggled to do was banned from renting a small plane by instructors at the Freeway Airport in Bowie, Maryland, just *six weeks* earlier on the grounds that he was an incompetent pilot. The Jewish population of the world is just 0.2 percent with even that almost entirely concentrated in Israel (75 percent Jewish) and the United States (around two percent). This two percent and globally 0.2 percent refers to *Jewish* people and not Sabbatian interlopers who are a fraction of that fraction. What a sobering thought when you think of the fantastic influence on world affairs of tiny Israel and that the Project for the New America Century (PNAC) which laid out the blueprint in September, 2000, for America's war on terror and regime change wars in Iraq, Libya and Syria was founded and dominated by Sabbatians known as 'Neocons'. The document conceded that this plan would not be supported politically or publicly without a major attack on American soil and a Problem-Reaction-Solution excuse to send troops to war across the Middle East. Sabbatian Neocons said:

... [The] process of transformation ... [war and regime change] ... is likely to be a long one, absent some catastrophic and catalysing event – like a new Pearl Harbor.

Four months later many of those who produced that document came to power with their inane puppet George Bush from the long-time Sabbatian Bush family. They included Sabbatian Dick Cheney who was officially vice-president, but really de-facto president for the entirety of the 'Bush' government. Nine months after the 'Bush' inauguration came what Bush called at the time 'the Pearl Harbor of the 21st century' and with typical Sabbatian timing and symbolism 2001 was the 60th anniversary of the attack in 1941 by the Japanese Air Force on Pearl Harbor, Hawaii, which allowed President Franklin Delano Roosevelt to take the United States into a Sabbatian-

instigated Second World War that he said in his election campaign that he never would. The evidence is overwhelming that Roosevelt and his military and intelligence networks knew the attack was coming and did nothing to stop it, but they did make sure that America's most essential naval ships were not in Hawaii at the time. Three thousand Americans died in the Pearl Harbor attacks as they did on September 11th. By the 9/11 year of 2001 Sabbatians had widely infiltrated the US government, military and intelligence operations and used their compartmentalised assets to pull off the 'Al-Qaeda' attacks. If you read *The Trigger* it will blow your mind to see the utterly staggering concentration of 'Jewish' operatives (Sabbatian infiltrators) in essential positions of political, security, legal, law enforcement, financial and business power before, during, and after the attacks to make them happen, carry them out, and then cover their tracks – and I do mean *staggering* when you think of that 0.2 percent of the world population and two percent of Americans which are Jewish while Sabbatian infiltrators are a fraction of that. A central foundation of the 9/11 conspiracy was the hijacking of government, military, Air Force and intelligence computer systems in real time through 'back-door' access made possible by Israeli (Sabbatian) 'cyber security' software. Sabbatian-controlled Israel is on the way to rivalling Silicon Valley for domination of cyberspace and is becoming the dominant force in cyber-security which gives them access to entire computer systems and their passcodes across the world. Then add to this that Zionists head (officially) Silicon Valley giants like Google (Larry Page and Sergey Brin), Google-owned YouTube (Susan Wojcicki), Facebook (Mark Zuckerberg and Sheryl Sandberg), and Apple (Chairman Arthur D. Levinson), and that ultra-Zionist hedge fund billionaire Paul Singer has a \$1 billion stake in Twitter which is only nominally headed by 'CEO' pothead Jack Dorsey. As cable news host Tucker Carlson said of Dorsey: 'There used to be debate in the medical community whether dropping a ton of acid had permanent effects and I think that debate has now ended.' Carlson made the comment after Dorsey told a hearing on Capitol Hill (if you cut through his bullshit) that he

believed in free speech so long as he got to decide what you can hear and see. These 'big names' of Silicon Valley are only front men and women for the Global Cult, not least the Sabbatians, who are the true controllers of these corporations. Does anyone still wonder why these same people and companies have been ferociously censoring and banning people (like me) for exposing any aspect of the Cult agenda and especially the truth about the 'Covid' hoax which Sabbatians have orchestrated?

The Jeffrey Epstein paedophile ring was a Sabbatian operation. He was officially 'Jewish' but he was a Sabbatian and women abused by the ring have told me about the high number of 'Jewish' people involved. The Epstein horror has Sabbatian written all over it and matches perfectly their modus operandi and obsession with sex and ritual. Epstein was running a Sabbatian blackmail ring in which famous people with political and other influence were provided with young girls for sex while everything was being filmed and recorded on hidden cameras and microphones at his New York house, Caribbean island and other properties. Epstein survivors have described this surveillance system to me and some have gone public. Once the famous politician or other figure knew he or she was on video they tended to do whatever they were told. Here we go again ...when you've got them by the balls their hearts and minds will follow. Sabbatians use this blackmail technique on a wide scale across the world to entrap politicians and others they need to act as demanded. Epstein's private plane, the infamous 'Lolita Express', had many well-known passengers including Bill Clinton while Bill Gates has flown on an Epstein plane and met with him four years after Epstein had been jailed for paedophilia. They subsequently met many times at Epstein's home in New York according to a witness who was there. Epstein's infamous side-kick was Ghislaine Maxwell, daughter of Mossad agent and ultra-Zionist mega-crooked British businessman, Bob Maxwell, who at one time owned the *Daily Mirror* newspaper. Maxwell was murdered at sea on his boat in 1991 by Sabbatian-controlled Mossad when he became a liability with his

business empire collapsing as a former Mossad operative has confirmed (see *The Trigger*).

Money, money, money, funny money ...

Before I come to the Sabbatian connection with the last three US presidents I will lay out the crucial importance to Sabbatians of controlling banking and finance. Sabbatian Mayer Amschel Rothschild set out to dominate this arena in his family's quest for total global control. What is freedom? It is, in effect, choice. The more choices you have the freer you are and the fewer your choices the more you are enslaved. In the global structure created over centuries by Sabbatians the biggest decider and restrictor of choice is ... money. Across the world if you ask people what they would like to do with their lives and why they are not doing that they will reply 'I don't have the money'. This is the idea. A global elite of multi-billionaires are described as 'greedy' and that is true on one level; but control of money – who has it and who doesn't – is not primarily about greed. It's about control. Sabbatians have seized ever more control of finance and sucked the wealth of the world out of the hands of the population. We talk now, after all, about the 'One-percent' and even then the wealthiest are a lot fewer even than that. This has been made possible by a money scam so outrageous and so vast it could rightly be called the scam of scams founded on creating 'money' out of nothing and 'loaning' that with interest to the population. Money out of nothing is called 'credit'. Sabbatians have asserted control over governments and banking ever more completely through the centuries and secured financial laws that allow banks to lend hugely more than they have on deposit in a confidence trick known as fractional reserve lending. Imagine if you could lend money that doesn't exist and charge the recipient interest for doing so. You would end up in jail. Bankers by contrast end up in mansions, private jets, Malibu and Monaco.

Banks are only required to keep a fraction of their deposits and wealth in their vaults and they are allowed to lend 'money' they don't have called 'credit'. Go into a bank for a loan and if you succeed

the banker will not move any real wealth into your account. They will type into your account the amount of the agreed 'loan' – say £100,000. This is not wealth that really exists; it is non-existent, fresh-air, created-out-of-nothing 'credit' which has never, does not, and will never exist except in theory. Credit is backed by nothing except wind and only has buying power because people think that it has buying power and accept it in return for property, goods and services. I have described this situation as like those cartoon characters you see chasing each other and when they run over the edge of a cliff they keep running forward on fresh air until one of them looks down, realises what's happened, and they all crash into the ravine. The whole foundation of the Sabbatian financial system is to stop people looking down except for periodic moments when they want to crash the system (as in 2008 and 2020 ongoing) and reap the rewards from all the property, businesses and wealth their borrowers had signed over as 'collateral' in return for a 'loan' of fresh air. Most people think that money is somehow created by governments when it comes into existence from the start as a debt through banks 'lending' illusory money called credit. Yes, the very currency of exchange is a *debt* from day one issued as an interest-bearing loan. Why don't governments create money interest-free and lend it to their people interest-free? Governments are controlled by Sabbatians and the financial system is controlled by Sabbatians for whom interest-free money would be a nightmare come true. Sabbatians underpin their financial domination through their global network of central banks, including the privately-owned US Federal Reserve and Britain's Bank of England, and this is orchestrated by a privately-owned central bank coordination body called the Bank for International Settlements in Basle, Switzerland, created by the usual suspects including the Rockefellers and Rothschilds. Central bank chiefs don't answer to governments or the people. They answer to the Bank for International Settlements or, in other words, the Global Cult which is dominated today by Sabbatians.

Built-in disaster

There are so many constituent scams within the overall banking scam. When you take out a loan of thin-air credit only the amount of that loan is theoretically brought into circulation to add to the amount in circulation; but you are paying back the principle plus interest. The additional interest is not created and this means that with every 'loan' there is a shortfall in the money in circulation between what is borrowed and what has to be paid back. There is never even close to enough money in circulation to repay all outstanding public and private debt including interest. Coldly weaved in the very fabric of the system is the certainty that some will lose their homes, businesses and possessions to the banking 'lender'. This is less obvious in times of 'boom' when the amount of money in circulation (and the debt) is expanding through more people wanting and getting loans. When a downturn comes and the money supply contracts it becomes painfully obvious that there is not enough money to service all debt and interest. This is less obvious in times of 'boom' when the amount of money in circulation (and the debt) is expanding through more people wanting and getting loans. When a downturn comes and the money supply contracts and it becomes painfully obvious – as in 2008 and currently – that there is not enough money to service all debt and interest. Sabbatian banksters have been leading the human population through a calculated series of booms (more debt incurred) and busts (when the debt can't be repaid and the banks get the debtor's tangible wealth in exchange for non-existent 'credit'). With each 'bust' Sabbatian bankers have absorbed more of the world's tangible wealth and we end up with the One-percent. Governments are in bankruptcy levels of debt to the same system and are therefore owned by a system they do not control. The Federal Reserve, 'America's central bank', is privately-owned and American presidents only nominally appoint its chairman or woman to maintain the illusion that it's an arm of government. It's not. The 'Fed' is a cartel of private banks which handed billions to its associates and friends after the crash of 2008 and has been Sabbatian-controlled since it was manipulated into being in 1913 through the covert trickery of Rothschild banking agents Jacob Schiff and Paul

Warburg, and the Sabbatian Rockefeller family. Somehow from a Jewish population of two-percent and globally 0.2 percent (Sabbatian interlopers remember are far smaller) ultra-Zionists headed the Federal Reserve for 31 years between 1987 and 2018 in the form of Alan Greenspan, Bernard Bernanke and Janet Yellen (now Biden's Treasury Secretary) with Yellen's deputy chairman a Israeli-American dual citizen and ultra-Zionist Stanley Fischer, a former governor of the Bank of Israel. Ultra-Zionist Fed chiefs spanned the presidencies of Ronald Reagan ('Republican'), Father George Bush ('Republican'), Bill Clinton ('Democrat'), Boy George Bush ('Republican') and Barack Obama ('Democrat'). We should really add the pre-Greenspan chairman, Paul Adolph Volcker, 'appointed' by Jimmy Carter ('Democrat') who ran the Fed between 1979 and 1987 during the Carter and Reagan administrations before Greenspan took over. Volcker was a long-time associate and business partner of the Rothschilds. No matter what the 'party' officially in power the United States economy was directed by the same force. Here are members of the Obama, Trump and Biden administrations and see if you can make out a common theme.

Barack Obama ('Democrat')

Ultra-Zionists Robert Rubin, Larry Summers, and Timothy Geithner ran the US Treasury in the Clinton administration and two of them reappeared with Obama. Ultra-Zionist Fed chairman Alan Greenspan had manipulated the crash of 2008 through deregulation and jumped ship just before the disaster to make way for ultra-Zionist Bernard Bernanke to hand out trillions to Sabbatian 'too big to fail' banks and businesses, including the ubiquitous ultra-Zionist Goldman Sachs which has an ongoing staff revolving door operation between itself and major financial positions in government worldwide. Obama inherited the fallout of the crash when he took office in January, 2009, and fortunately he had the support of his ultra-Zionist White House Chief of Staff Rahm Emmanuel, son of a terrorist who helped to bomb Israel into being in 1948, and his ultra-Zionist senior adviser David Axelrod, chief strategist in Obama's two

successful presidential campaigns. Emmanuel, later mayor of Chicago and former senior fundraiser and strategist for Bill Clinton, is an example of the Sabbatian policy after Israel was established of migrating insider families to America so their children would be born American citizens. 'Obama' chose this financial team throughout his administration to respond to the Sabbatian-instigated crisis:

Timothy Geithner (ultra-Zionist) Treasury Secretary; Jacob J. Lew, Treasury Secretary; Larry Summers (ultra-Zionist), director of the White House National Economic Council; Paul Adolph Volcker (Rothschild business partner), chairman of the Economic Recovery Advisory Board; Peter Orszag (ultra-Zionist), director of the Office of Management and Budget overseeing all government spending; Penny Pritzker (ultra-Zionist), Commerce Secretary; Jared Bernstein (ultra-Zionist), chief economist and economic policy adviser to Vice President Joe Biden; Mary Schapiro (ultra-Zionist), chair of the Securities and Exchange Commission (SEC); Gary Gensler (ultra-Zionist), chairman of the Commodity Futures Trading Commission (CFTC); Sheila Bair (ultra-Zionist), chair of the Federal Deposit Insurance Corporation (FDIC); Karen Mills (ultra-Zionist), head of the Small Business Administration (SBA); Kenneth Feinberg (ultra-Zionist), Special Master for Executive [bail-out] Compensation. Feinberg would be appointed to oversee compensation (with strings) to 9/11 victims and families in a campaign to stop them having their day in court to question the official story. At the same time ultra-Zionist Bernard Bernanke was chairman of the Federal Reserve and these are only some of the ultra-Zionists with allegiance to Sabbatian-controlled Israel in the Obama government. Obama's biggest corporate donor was ultra-Zionist Goldman Sachs which had employed many in his administration.

Donald Trump ('Republican')

Trump claimed to be an outsider (he wasn't) who had come to 'drain the swamp'. He embarked on this goal by immediately appointing ultra-Zionist Steve Mnuchin, a Goldman Sachs employee for 17

years, as his Treasury Secretary. Others included Gary Cohn (ultra-Zionist), chief operating officer of Goldman Sachs, his first Director of the National Economic Council and chief economic adviser, who was later replaced by Larry Kudlow (ultra-Zionist). Trump's senior adviser throughout his four years in the White House was his sinister son-in-law Jared Kushner, a life-long friend of Israel Prime Minister Benjamin Netanyahu. Kushner is the son of a convicted crook who was pardoned by Trump in his last days in office. Other ultra-Zionists in the Trump administration included: Stephen Miller, Senior Policy Adviser; Avrahm Berkowitz, Deputy Adviser to Trump and his Senior Adviser Jared Kushner; Ivanka Trump, Adviser to the President, who converted to Judaism when she married Jared Kushner; David Friedman, Trump lawyer and Ambassador to Israel; Jason Greenblatt, Trump Organization executive vice president and chief legal officer, who was made Special Representative for International Negotiations and the Israeli-Palestinian Conflict; Rod Rosenstein, Deputy Attorney General; Elliot Abrams, Special Representative for Venezuela, then Iran; John Eisenberg, National Security Council Legal Adviser and Deputy Council to the President for National Security Affairs; Anne Neuberger, Deputy National Manager, National Security Agency; Ezra Cohen-Watnick, Acting Under Secretary of Defense for Intelligence; Elan Carr, Special Envoy to monitor and combat anti-Semitism; Len Khodorkovsky, Deputy Special Envoy to monitor and combat anti-Semitism; Reed Cordish, Assistant to the President, Intragovernmental and Technology Initiatives. Trump Vice President Mike Pence and Secretary of State Mike Pompeo, both Christian Zionists, were also vehement supporters of Israel and its goals and ambitions.

Donald 'free-speech believer' Trump pardoned a number of financial and violent criminals while ignoring calls to pardon Julian Assange and Edward Snowden whose crimes are revealing highly relevant information about government manipulation and corruption and the widespread illegal surveillance of the American people by US 'security' agencies. It's so good to know that Trump is on the side of freedom and justice and not mega-criminals with

allegiance to Sabbatian-controlled Israel. These included a pardon for Israeli spy Jonathan Pollard who was jailed for life in 1987 under the Espionage Act. Aviem Sella, the Mossad agent who recruited Pollard, was also pardoned by Trump while Assange sat in jail and Snowden remained in exile in Russia. Sella had 'fled' (was helped to escape) to Israel in 1987 and was never extradited despite being charged under the Espionage Act. A Trump White House statement said that Sella's clemency had been 'supported by Benjamin Netanyahu, Ron Dermer, Israel's US Ambassador, David Friedman, US Ambassador to Israel and Miriam Adelson, wife of leading Trump donor Sheldon Adelson who died shortly before. Other friends of Jared Kushner were pardoned along with Sholom Weiss who was believed to be serving the longest-ever white-collar prison sentence of more than 800 years in 2000. The sentence was commuted of Ponzi-schemer Eliyahu Weinstein who defrauded Jews and others out of \$200 million. I did mention that Assange and Snowden were ignored, right? Trump gave Sabbatians almost everything they asked for in military and political support, moving the US Embassy from Tel Aviv to Jerusalem with its critical symbolic and literal implications for Palestinian statehood, and the 'deal of the Century' designed by Jared Kushner and David Friedman which gave the Sabbatian Israeli government the green light to substantially expand its already widespread program of building illegal Jewish-only settlements in the occupied land of the West Bank. This made a two-state 'solution' impossible by seizing all the land of a potential Palestinian homeland and that had been the plan since 1948 and then 1967 when the Arab-controlled Gaza Strip, West Bank, Sinai Peninsula and Syrian Golan Heights were occupied by Israel. All the talks about talks and road maps and delays have been buying time until the West Bank was physically occupied by Israeli real estate. Trump would have to be a monumentally ill-informed idiot not to see that this was the plan he was helping to complete. The Trump administration was in so many ways the Kushner administration which means the Netanyahu administration which means the Sabbatian administration. I understand why many opposing Cult fascism in all its forms gravitated to Trump, but he

was a crucial part of the Sabbatian plan and I will deal with this in the next chapter.

Joe Biden ('Democrat')

A barely cognitive Joe Biden took over the presidency in January, 2021, along with his fellow empty shell, Vice-President Kamala Harris, as the latest Sabbatian gofers to enter the White House. Names on the door may have changed and the 'party' – the force behind them remained the same as Zionists were appointed to a stream of pivotal areas relating to Sabbatian plans and policy. They included: Janet Yellen, Treasury Secretary, former head of the Federal Reserve, and still another ultra-Zionist running the US Treasury after Mnuchin (Trump), Lew and Geithner (Obama), and Summers and Rubin (Clinton); Anthony Blinken, Secretary of State; Wendy Sherman, Deputy Secretary of State (so that's 'Biden's' Sabbatian foreign policy sorted); Jeff Zients, White House coronavirus coordinator; Rochelle Walensky, head of the Centers for Disease Control; Rachel Levine, transgender deputy health secretary (that's 'Covid' hoax policy under control); Merrick Garland, Attorney General; Alejandro Mayorkas, Secretary of Homeland Security; Cass Sunstein, Homeland Security with responsibility for new immigration laws; Avril Haines, Director of National Intelligence; Anne Neuberger, National Security Agency cybersecurity director (note, cybersecurity); David Cohen, CIA Deputy Director; Ronald Klain, Biden's Chief of Staff (see Rahm Emanuel); Eric Lander, a 'leading geneticist', Office of Science and Technology Policy director (see Smart Grid, synthetic biology agenda); Jessica Rosenworcel, acting head of the Federal Communications Commission (FCC) which controls Smart Grid technology policy and electromagnetic communication systems including 5G. How can it be that so many pivotal positions are held by two-percent of the American population and 0.2 percent of the world population administration after administration no matter who is the president and what is the party? It's a coincidence? Of course it's not and this is why Sabbatians have built their colossal global web of interlocking 'anti-

hate' hate groups to condemn anyone who asks these glaring questions as an 'anti-Semite'. The way that Jewish people horrifically abused in Sabbatian-backed Nazi Germany are exploited to this end is stomach-turning and disgusting beyond words.

Political fusion

Sabbatian manipulation has reversed the roles of Republicans and Democrats and the same has happened in Britain with the Conservative and Labour Parties. Republicans and Conservatives were always labelled the 'right' and Democrats and Labour the 'left', but look at the policy positions now and the Democrat-Labour 'left' has moved further to the 'right' than Republicans and Conservatives under the banner of 'Woke', the Cult-created far-right tyranny. Where once the Democrat-Labour 'left' defended free speech and human rights they now seek to delete them and as I said earlier despite the 'Covid' fascism of the Jackboot Johnson Conservative government in the UK the Labour Party of leader Keir Starmer demanded even more extreme measures. The Labour Party has been very publicly absorbed by Sabbatians after a political and media onslaught against the previous leader, the weak and inept Jeremy Corbyn, over made-up allegations of 'anti-Semitism' both by him and his party. The plan was clear with this 'anti-Semite' propaganda and what was required in response was a swift and decisive 'fuck off' from Corbyn and a statement to expose the Anti-Semitism Industry (Sabbatian) attempt to silence Labour criticism of the Israeli government (Sabbatians) and purge the party of all dissent against the extremes of ultra-Zionism (Sabbatians). Instead Corbyn and his party fell to their knees and appeased the abusers which, by definition, is impossible. Appeasing one demand leads only to a new demand to be appeased until takeover is complete. Like I say – 'fuck off' would have been a much more effective policy and I have used it myself with great effect over the years when Sabbatians are on my case which is most of the time. I consider that fact a great compliment, by the way. The outcome of the Labour Party capitulation is that we now have a Sabbatian-controlled

Conservative Party 'opposed' by a Sabbatian-controlled Labour Party in a one-party Sabbatian state that hurtles towards the extremes of tyranny (the Sabbatian cult agenda). In America the situation is the same. Labour's Keir Starmer spends his days on his knees with his tongue out pointing to Tel Aviv, or I guess now Jerusalem, while Boris Johnson has an 'anti-Semitism czar' in the form of former Labour MP John Mann who keeps Starmer company on his prayer mat.

Sabbatian influence can be seen in Jewish members of the Labour Party who have been ejected for criticism of Israel including those from families that suffered in Nazi Germany. Sabbatians despise real Jewish people and target them even more harshly because it is so much more difficult to dub them 'anti-Semitic' although in their desperation they do try.

CHAPTER THREE

The Pushbacker sting

*Until you realize how easy it is for your mind to be manipulated, you
remain the puppet of someone else's game*

Evita Ochel

I will use the presidencies of Trump and Biden to show how the manipulation of the one-party state plays out behind the illusion of political choice across the world. No two presidencies could – on the face of it – be more different and apparently at odds in terms of direction and policy.

A Renegade Mind sees beyond the obvious and focuses on outcomes and consequences and not image, words and waffle. The Cult embarked on a campaign to divide America between those who blindly support its agenda (the mentality known as 'Woke') and those who are pushing back on where the Cult and its Sabbatians want to go. This presents infinite possibilities for dividing and ruling the population by setting them at war with each other and allows a perceptual ring fence of demonisation to encircle the Pushbackers in a modern version of the Little Big Horn in 1876 when American cavalry led by Lieutenant Colonel George Custer were drawn into a trap, surrounded and killed by Native American tribes defending their land of thousands of years from being seized by the government. In this modern version the roles are reversed and it's those defending themselves from the Sabbatian government who are surrounded and the government that's seeking to destroy them. This trap was set years ago and to explain how we must return to 2016

and the emergence of Donald Trump as a candidate to be President of the United States. He set out to overcome the best part of 20 other candidates in the Republican Party before and during the primaries and was not considered by many in those early stages to have a prayer of living in the White House. The Republican Party was said to have great reservations about Trump and yet somehow he won the nomination. When you know how American politics works – politics in general – there is no way that Trump could have become the party's candidate unless the Sabbatian-controlled 'Neocons' that run the Republican Party wanted that to happen. We saw the proof in emails and documents made public by WikiLeaks that the Democratic Party hierarchy, or Democons, systematically undermined the campaign of Bernie Sanders to make sure that Sabbatian gofer Hillary Clinton won the nomination to be their presidential candidate. If the Democons could do that then the Neocons in the Republican Party could have derailed Trump in the same way. But they didn't and at that stage I began to conclude that Trump could well be the one chosen to be president. If that was the case the 'why' was pretty clear to see – the goal of dividing America between Cult agenda-supporting Wokers and Pushbackers who gravitated to Trump because he was telling them what they wanted to hear. His constituency of support had been increasingly ignored and voiceless for decades and profoundly through the eight years of Sabbatian puppet Barack Obama. Now here was someone speaking their language of pulling back from the incessant globalisation of political and economic power, the exporting of American jobs to China and elsewhere by 'American' (Sabbatian) corporations, the deletion of free speech, and the mass immigration policies that had further devastated job opportunities for the urban working class of all races and the once American heartlands of the Midwest.

Beware the forked tongue

Those people collectively sighed with relief that at last a political leader was apparently on their side, but another trait of the Renegade Mind is that you look even harder at people telling you

what you want to hear than those who are telling you otherwise. Obviously as I said earlier people wish what they want to hear to be true and genuine and they are much more likely to believe that than someone saying what they don't want to hear and don't want to be true. Sales people are taught to be skilled in eliciting by calculated questioning what their customers want to hear and repeating that back to them as their own opinion to get their targets to like and trust them. Assets of the Cult are also sales people in the sense of selling perception. To read Cult manipulation you have to play the long and expanded game and not fall for the Vaudeville show of party politics. Both American parties are vehicles for the Cult and they exploit them in different ways depending on what the agenda requires at that moment. Trump and the Republicans were used to be the focus of dividing America and isolating Pushbackers to open the way for a Biden presidency to become the most extreme in American history by advancing the full-blown Woke (Cult) agenda with the aim of destroying and silencing Pushbackers now labelled Nazi Trump supporters and white supremacists.

Sabbatians wanted Trump in office for the reasons described by ultra-Zionist Saul Alinsky (1909-1972) who was promoting the Woke philosophy through 'community organising' long before anyone had heard of it. In those days it still went by its traditional name of Marxism. The reason for the manipulated Trump phenomenon was laid out in Alinsky's 1971 book, *Rules for Radicals*, which was his blueprint for overthrowing democratic and other regimes and replacing them with Sabbatian Marxism. Not surprisingly his to-do list was evident in the Sabbatian French and Russian 'Revolutions' and that in China which will become very relevant in the next chapter about the 'Covid' hoax. Among Alinsky's followers have been the deeply corrupt Barack Obama, House Speaker Nancy Pelosi and Hillary Clinton who described him as a 'hero'. All three are Sabbatian stooges with Pelosi personifying the arrogant corrupt idiocy that so widely fronts up for the Cult inner core. Predictably as a Sabbatian advocate of the 'light-bringer' Alinsky features Lucifer on the dedication page of his book as the original radical who gained

his own kingdom ('Earth' as we shall see). One of Alinsky's golden radical rules was to pick an individual and focus all attention, hatred and blame on them and not to target faceless bureaucracies and corporations. *Rules for Radicals* is really a Sabbatian handbook with its contents repeatedly employed all over the world for centuries and why wouldn't Sabbatians bring to power their designer-villain to be used as the individual on which all attention, hatred and blame was bestowed? This is what they did and the only question for me is how much Trump knew that and how much he was manipulated. A bit of both, I suspect. This was Alinsky's Trump technique from a man who died in 1972. The technique has spanned history:

Pick the target, freeze it, personalize it, polarize it. Don't try to attack abstract corporations or bureaucracies. Identify a responsible individual. Ignore attempts to shift or spread the blame.

From the moment Trump came to illusory power everything was about him. It wasn't about Republican policy or opinion, but all about Trump. Everything he did was presented in negative, derogatory and abusive terms by the Sabbatian-dominated media led by Cult operations such as CNN, MSNBC, *The New York Times* and the Jeff Bezos-owned *Washington Post* – 'Pick the target, freeze it, personalize it, polarize it.' Trump was turned into a demon to be vilified by those who hated him and a demi-god loved by those who worshipped him. This, in turn, had his supporters, too, presented as equally demonic in preparation for the punchline later down the line when Biden was about to take office. It was here's a Trump, there's a Trump, everywhere a Trump, Trump. Virtually every news story or happening was filtered through the lens of 'The Donald'. You loved him or hated him and which one you chose was said to define you as Satan's spawn or a paragon of virtue. Even supporting some Trump policies or statements and not others was enough for an assault on your character. No shades of grey were or are allowed. Everything is black and white (literally and figuratively). A Californian I knew had her head utterly scrambled by her hatred for Trump while telling people they should love each other. She was so totally consumed by

Trump Derangement Syndrome as it became to be known that this glaring contradiction would never have occurred to her. By definition anyone who criticised Trump or praised his opponents was a hero and this lady described Joe Biden as 'a kind, honest gentleman' when he's a provable liar, mega-crook and vicious piece of work to boot. Sabbatians had indeed divided America using Trump as the fall-guy and all along the clock was ticking on the consequences for his supporters.

In hock to his masters

Trump gave Sabbatians via Israel almost everything they wanted in his four years. Ask and you shall receive was the dynamic between himself and Benjamin Netanyahu orchestrated by Trump's ultra-Zionist son-in-law Jared Kushner, his ultra-Zionist Ambassador to Israel, David Friedman, and ultra-Zionist 'Israel adviser', Jason Greenblatt. The last two were central to the running and protecting from collapse of his business empire, the Trump Organisation, and colossal business failures made him forever beholding to Sabbatian networks that bailed him out. By the start of the 1990s Trump owed \$4 billion to banks that he couldn't pay and almost \$1 billion of that was down to him personally and not his companies. This mega-disaster was the result of building two new casinos in Atlantic City and buying the enormous Taj Mahal operation which led to crippling debt payments. He had borrowed fantastic sums from 72 banks with major Sabbatian connections and although the scale of debt should have had him living in a tent alongside the highway they never foreclosed. A plan was devised to lift Trump from the mire by BT Securities Corporation and Rothschild Inc. and the case was handled by Wilber Ross who had worked for the Rothschilds for 27 years. Ross would be named US Commerce Secretary after Trump's election. Another crucial figure in saving Trump was ultra-Zionist 'investor' Carl Icahn who bought the Taj Mahal casino. Icahn was made special economic adviser on financial regulation in the Trump administration. He didn't stay long but still managed to find time to make a tidy sum of a reported \$31.3 million when he sold his

holdings affected by the price of steel three days before Trump imposed a 235 percent tariff on steel imports. What amazing bits of luck these people have. Trump and Sabbatian operatives have long had a close association and his mentor and legal adviser from the early 1970s until 1986 was the dark and genetically corrupt ultra-Zionist Roy Cohn who was chief counsel to Senator Joseph McCarthy's 'communist' witch-hunt in the 1950s. *Esquire* magazine published an article about Cohn with the headline 'Don't mess with Roy Cohn'. He was described as the most feared lawyer in New York and 'a ruthless master of dirty tricks ... [with] ... more than one Mafia Don on speed dial'. Cohn's influence, contacts, support and protection made Trump a front man for Sabbatians in New York with their connections to one of Cohn's many criminal employers, the 'Russian' Sabbatian Mafia. Israel-centric media mogul Rupert Murdoch was introduced to Trump by Cohn and they started a long friendship. Cohn died in 1986 weeks after being disbarred for unethical conduct by the Appellate Division of the New York State Supreme Court. The wheels of justice do indeed run slow given the length of Cohn's crooked career.

QAnon-sense

We are asked to believe that Donald Trump with his fundamental connections to Sabbatian networks and operatives has been leading the fight to stop the Sabbatian agenda for the fascistic control of America and the world. Sure he has. A man entrapped during his years in the White House by Sabbatian operatives and whose biggest financial donor was casino billionaire Sheldon Adelson who was Sabbatian to his DNA?? Oh, do come on. Trump has been used to divide America and isolate Pushbackers on the Cult agenda under the heading of 'Trump supporters', 'insurrectionists' and 'white supremacists'. The US Intelligence/Mossad Psyop or psychological operation known as QAnon emerged during the Trump years as a central pillar in the Sabbatian campaign to lead Pushbackers into the trap set by those that wished to destroy them. I knew from the start that QAnon was a scam because I had seen the same scenario many

times before over 30 years under different names and I had written about one in particular in the books. 'Not again' was my reaction when QAnon came to the fore. The same script is pulled out every few years and a new name added to the letterhead. The story always takes the same form: 'Insiders' or 'the good guys' in the government-intelligence-military 'Deep State' apparatus were going to instigate mass arrests of the 'bad guys' which would include the Rockefellers, Rothschilds, Barack Obama, Hillary Clinton, George Soros, etc., etc. Dates are given for when the 'good guys' are going to move in, but the dates pass without incident and new dates are given which pass without incident. The central message to Pushbackers in each case is that they don't have to do anything because there is 'a plan' and it is all going to be sorted by the 'good guys' on the inside. 'Trust the plan' was a QAnon mantra when the only plan was to misdirect Pushbackers into putting their trust in a Psyop they believed to be real. Beware, beware, those who tell you what you want to hear and always check it out. Right up to Biden's inauguration QAnon was still claiming that 'the Storm' was coming and Trump would stay on as president when Biden and his cronies were arrested and jailed. It was never going to happen and of course it didn't, but what did happen as a result provided that punchline to the Sabbatian Trump/QAnon Psyop.

On January 6th, 2021, a very big crowd of Trump supporters gathered in the National Mall in Washington DC down from the Capitol Building to protest at what they believed to be widespread corruption and vote fraud that stopped Trump being re-elected for a second term as president in November, 2020. I say as someone that does not support Trump or Biden that the evidence is clear that major vote-fixing went on to favour Biden, a man with cognitive problems so advanced he can often hardly string a sentence together without reading the words written for him on the Teleprompter. Glaring ballot discrepancies included serious questions about electronic voting machines that make vote rigging a comparative cinch and hundreds of thousands of paper votes that suddenly appeared during already advanced vote counts and virtually all of

them for Biden. Early Trump leads in crucial swing states suddenly began to close and disappear. The pandemic hoax was used as the excuse to issue almost limitless numbers of mail-in ballots with no checks to establish that the recipients were still alive or lived at that address. They were sent to streams of people who had not even asked for them. Private organisations were employed to gather these ballots and who knows what they did with them before they turned up at the counts. The American election system has been manipulated over decades to become a sick joke with more holes than a Swiss cheese for the express purpose of dictating the results. Then there was the criminal manipulation of information by Sabbatian tech giants like Facebook, Twitter and Google-owned YouTube which deleted pro-Trump, anti-Biden accounts and posts while everything in support of Biden was left alone. Sabbatians wanted Biden to win because after the dividing of America it was time for full-on Woke and every aspect of the Cult agenda to be unleashed.

Hunter gatherer

Extreme Silicon Valley bias included blocking information by the *New York Post* exposing a Biden scandal that should have ended his bid for president in the final weeks of the campaign. Hunter Biden, his monumentally corrupt son, is reported to have sent a laptop to be repaired at a local store and failed to return for it. Time passed until the laptop became the property of the store for non-payment of the bill. When the owner saw what was on the hard drive he gave a copy to the FBI who did nothing even though it confirmed widespread corruption in which the Joe Biden family were using his political position, especially when he was vice president to Obama, to make multiple millions in countries around the world and most notably Ukraine and China. Hunter Biden's one-time business partner Tony Bobulinski went public when the story broke in the *New York Post* to confirm the corruption he saw and that Joe Biden not only knew what was going on he also profited from the spoils. Millions were handed over by a Chinese company with close

connections – like all major businesses in China – to the Chinese communist party of President Xi Jinping. Joe Biden even boasted at a meeting of the Cult's World Economic Forum that as vice president he had ordered the government of Ukraine to fire a prosecutor. What he didn't mention was that the same man just happened to be investigating an energy company which was part of Hunter Biden's corrupt portfolio. The company was paying him big bucks for no other reason than the influence his father had. Overnight Biden's presidential campaign should have been over given that he had lied publicly about not knowing what his son was doing. Instead almost the entire Sabbatian-owned mainstream media and Sabbatian-owned Silicon Valley suppressed circulation of the story. This alone went a mighty way to rigging the election of 2020. Cult assets like Mark Zuckerberg at Facebook also spent hundreds of millions to be used in support of Biden and vote 'administration'.

The Cult had used Trump as the focus to divide America and was now desperate to bring in moronic, pliable, corrupt Biden to complete the double-whammy. No way were they going to let little things like the will of the people thwart their plan. Silicon Valley widely censored claims that the election was rigged because it *was* rigged. For the same reason anyone claiming it was rigged was denounced as a 'white supremacist' including the pathetically few Republican politicians willing to say so. Right across the media where the claim was mentioned it was described as a 'false claim' even though these excuses for 'journalists' would have done no research into the subject whatsoever. Trump won seven million more votes than any sitting president had ever achieved while somehow a cognitively-challenged soon to be 78-year-old who was hidden away from the public for most of the campaign managed to win more votes than any presidential candidate in history. It makes no sense. You only had to see election rallies for both candidates to witness the enthusiasm for Trump and the apathy for Biden. Tens of thousands would attend Trump events while Biden was speaking in empty car parks with often only television crews attending and framing their shots to hide the fact that no one was there. It was pathetic to see

footage come to light of Biden standing at a podium making speeches only to TV crews and party fixers while reading the words written for him on massive Teleprompter screens. So, yes, those protestors on January 6th had a point about election rigging, but some were about to walk into a trap laid for them in Washington by the Cult Deep State and its QAnon Psyop. This was the Capitol Hill riot ludicrously dubbed an 'insurrection'.

The spider and the fly

Renegade Minds know there are not two 'sides' in politics, only one side, the Cult, working through all 'sides'. It's a stage show, a puppet show, to direct the perceptions of the population into focusing on diversions like parties and candidates while missing the puppeteers with their hands holding all the strings. The Capitol Hill 'insurrection' brings us back to the Little Big Horn. Having created two distinct opposing groupings – Woke and Pushbackers – the trap was about to be sprung. Pushbackers were to be encircled and isolated by associating them all in the public mind with Trump and then labelling Trump as some sort of Confederate leader. I knew immediately that the Capitol riot was a set-up because of two things. One was how easy the rioters got into the building with virtually no credible resistance and secondly I could see – as with the 'Covid' hoax in the West at the start of 2020 – how the Cult could exploit the situation to move its agenda forward with great speed. My experience of Cult techniques and activities over more than 30 years has showed me that while they do exploit situations they haven't themselves created this never happens with events of fundamental agenda significance. Every time major events giving cultists the excuse to rapidly advance their plan you find they are manipulated into being for the specific reason of providing that excuse – Problem-Reaction-Solution. Only a tiny minority of the huge crowd of Washington protestors sought to gain entry to the Capitol by smashing windows and breaching doors. That didn't matter. The whole crowd and all Pushbackers, even if they did not support Trump, were going to be lumped together as dangerous

insurrectionists and conspiracy theorists. The latter term came into widespread use through a CIA memo in the 1960s aimed at discrediting those questioning the nonsensical official story of the Kennedy assassination and it subsequently became widely employed by the media. It's still being used by inept 'journalists' with no idea of its origin to discredit anyone questioning anything that authority claims to be true. When you are perpetrating a conspiracy you need to discredit the very word itself even though the dictionary definition of conspiracy is merely 'the activity of secretly planning with other people to do something bad or illegal' and 'a general agreement to keep silent about a subject for the purpose of keeping it secret'. On that basis there are conspiracies almost wherever you look. For obvious reasons the Cult and its lapdog media have to claim there are no conspiracies even though the word appears in state laws as with conspiracy to defraud, to murder, and to corrupt public morals.

Agent provocateurs are widely used by the Cult Deep State to manipulate genuine people into acting in ways that suit the desired outcome. By genuine in this case I mean protestors genuinely supporting Trump and claims that the election was stolen. In among them, however, were agents of the state wearing the garb of Trump supporters and QAnon to pump-prime the Capital riot which some genuine Trump supporters naively fell for. I described the situation as 'Come into my parlour said the spider to the fly'. Leaflets appeared through the Woke paramilitary arm Antifa, the anti-fascist fascists, calling on supporters to turn up in Washington looking like Trump supporters even though they hated him. Some of those arrested for breaching the Capitol Building were sourced to Antifa and its stable mate Black Lives Matter. Both organisations are funded by Cult billionaires and corporations. One man charged for the riot was according to his lawyer a former FBI agent who had held top secret security clearance for 40 years. Attorney Thomas Plofchan said of his client, 66-year-old Thomas Edward Caldwell:

He has held a Top Secret Security Clearance since 1979 and has undergone multiple Special Background Investigations in support of his clearances. After retiring from the Navy, he

worked as a section chief for the Federal Bureau of Investigation from 2009-2010 as a GS-12 [mid-level employee].

He also formed and operated a consulting firm performing work, often classified, for U.S government customers including the US. Drug Enforcement Agency, Department of Housing and Urban Development, the US Coast Guard, and the US Army Personnel Command.

A judge later released Caldwell pending trial in the absence of evidence about a conspiracy or that he tried to force his way into the building. *The New York Post* reported a 'law enforcement source' as saying that 'at least two known Antifa members were spotted' on camera among Trump supporters during the riot while one of the rioters arrested was John Earle Sullivan, a seriously extreme Black Lives Matter Trump-hater from Utah who was previously arrested and charged in July, 2020, over a BLM-Antifa riot in which drivers were threatened and one was shot. Sullivan is the founder of Utah-based Insurgence USA which is an affiliate of the Cult-created-and-funded Black Lives Matter movement. Footage appeared and was then deleted by Twitter of Trump supporters calling out Antifa infiltrators and a group was filmed changing into pro-Trump clothing before the riot. Security at the building was *pathetic* – as planned. Colonel Leroy Fletcher Prouty, a man with long experience in covert operations working with the US security apparatus, once described the tell-tale sign to identify who is involved in an assassination. He said:

No one has to direct an assassination – it happens. The active role is played secretly by permitting it to happen. This is the greatest single clue. Who has the power to call off or reduce the usual security precautions?

This principle applies to many other situations and certainly to the Capitol riot of January 6th, 2021.

The sting

With such a big and potentially angry crowd known to be gathering near the Capitol the security apparatus would have had a major police detail to defend the building with National Guard troops on

standby given the strength of feeling among people arriving from all over America encouraged by the QAnon Psyop and statements by Donald Trump. Instead Capitol Police 'security' was flimsy, weak, and easily breached. The same number of officers was deployed as on a regular day and that is a blatant red flag. They were not staffed or equipped for a possible riot that had been an obvious possibility in the circumstances. No protective and effective fencing worth the name was put in place and there were no contingency plans. The whole thing was basically a case of standing aside and waving people in. Once inside police mostly backed off apart from one Capitol police officer who ridiculously shot dead unarmed Air Force veteran protestor Ashli Babbitt without a warning as she climbed through a broken window. The 'investigation' refused to name or charge the officer after what must surely be considered a murder in the circumstances. They just lifted a carpet and swept. The story was endlessly repeated about five people dying in the 'armed insurrection' when there was no report of rioters using weapons. Apart from Babbitt the other four died from a heart attack, strokes and apparently a drug overdose. Capitol police officer Brian Sicknick was reported to have died after being bludgeoned with a fire extinguisher when he was alive after the riot was over and died later of what the Washington Medical Examiner's Office said was a stroke. Sicknick had no external injuries. The lies were delivered like rapid fire. There was a narrative to build with incessant repetition of the lie until the lie became the accepted 'everybody knows that' truth. The 'Big Lie' technique of Nazi Propaganda Minister Joseph Goebbels is constantly used by the Cult which was behind the Nazis and is today behind the 'Covid' and 'climate change' hoaxes. Goebbels said:

If you tell a lie big enough and keep repeating it, people will eventually come to believe it. The lie can be maintained only for such time as the State can shield the people from the political, economic and/or military consequences of the lie. It thus becomes vitally important for the State to use all of its powers to repress dissent, for the truth is the mortal enemy of the lie, and thus by extension, the truth is the greatest enemy of the State.

Most protestors had a free run of the Capitol Building. This allowed pictures to be taken of rioters in iconic parts of the building including the Senate chamber which could be used as propaganda images against all Pushbackers. One Congresswoman described the scene as 'the worst kind of non-security anybody could ever imagine'. Well, the first part was true, but someone obviously did imagine it and made sure it happened. Some photographs most widely circulated featured people wearing QAnon symbols and now the Psyop would be used to dub all QAnon followers with the ubiquitous fit-all label of 'white supremacist' and 'insurrectionists'. When a Muslim extremist called Noah Green drove his car at two police officers at the Capitol Building killing one in April, 2021, there was no such political and media hysteria. They were just disappointed he wasn't white.

The witch-hunt

Government prosecutor Michael Sherwin, an aggressive, dark-eyed, professional Rottweiler led the 'investigation' and to call it over the top would be to understate reality a thousand fold. Hundreds were tracked down and arrested for the crime of having the wrong political views and people were jailed who had done nothing more than walk in the building, committed no violence or damage to property, took a few pictures and left. They were labelled a 'threat to the Republic' while Biden sat in the White House signing executive orders written for him that were dismantling 'the Republic'. Even when judges ruled that a mother and son should not be in jail the government kept them there. Some of those arrested have been badly beaten by prison guards in Washington and lawyers for one man said he suffered a fractured skull and was made blind in one eye. Meanwhile a woman is shot dead for no reason by a Capitol Police officer and we are not allowed to know who he is never mind what has happened to him although that will be *nothing*. The Cult's QAnon/Trump sting to identify and isolate Pushbackers and then target them on the road to crushing and deleting them was a resounding success. You would have thought the Russians had

invaded the building at gunpoint and lined up senators for a firing squad to see the political and media reaction. Congresswoman Alexandria Ocasio-Cortez is a child in a woman's body, a terrible-tuos, me, me, me, Woker narcissist of such proportions that words have no meaning. She said she thought she was going to die when 'insurrectionists' banged on her office door. It turned out she wasn't even in the Capitol Building when the riot was happening and the 'banging' was a Capitol Police officer. She referred to herself as a 'survivor' which is an insult to all those true survivors of violent and sexual abuse while she lives her pampered and privileged life talking drivel for a living. Her Woke colleague and fellow mega-narcissist Rashida Tlaib broke down describing the devastating effect on her, too, of *not being* in the building when the rioters were there. Ocasio-Cortez and Tlaib are members of a fully-Woke group of Congresswomen known as 'The Squad' along with Ilhan Omar and Ayanna Pressley. The Squad from what I can see can be identified by its vehement anti-white racism, anti-white men agenda, and, as always in these cases, the absence of brain cells on active duty.

The usual suspects were on the riot case immediately in the form of Democrat ultra-Zionist senators and operatives Chuck Schumer and Adam Schiff demanding that Trump be impeached for 'his part in the insurrection'. The same pair of prats had led the failed impeachment of Trump over the invented 'Russia collusion' nonsense which claimed Russia had helped Trump win the 2016 election. I didn't realise that Tel Aviv had been relocated just outside Moscow. I must find an up-to-date map. The Russia hoax was a Sabbatian operation to keep Trump occupied and impotent and to stop any rapport with Russia which the Cult wants to retain as a perceptual enemy to be pulled out at will. Puppet Biden began attacking Russia when he came to office as the Cult seeks more upheaval, division and war across the world. A two-year stage show 'Russia collusion inquiry' headed by the not-very-bright former 9/11 FBI chief Robert Mueller, with support from 19 lawyers, 40 FBI agents plus intelligence analysts, forensic accountants and other

staff, devoured tens of millions of dollars and found no evidence of Russia collusion which a ten-year-old could have told them on day one. Now the same moronic Schumer and Schiff wanted a second impeachment of Trump over the Capitol 'insurrection' (riot) which the arrested development of Schumer called another 'Pearl Harbor' while others compared it with 9/11 in which 3,000 died and, in the case of CNN, with the Rwandan genocide in the 1990s in which an estimated 500,000 to 600,000 were murdered, between 250, 000 and 500,000 women were raped, and populations of whole towns were hacked to death with machetes. To make those comparisons purely for Cult political reasons is beyond insulting to those that suffered and lost their lives and confirms yet again the callous inhumanity that we are dealing with. Schumer is a monumental idiot and so is Schiff, but they serve the Cult agenda and do whatever they're told so they get looked after. Talking of idiots – another inane man who spanned the Russia and Capitol impeachment attempts was Senator Eric Swalwell who had the nerve to accuse Trump of collusion with the Russians while sleeping with a Chinese spy called Christine Fang or 'Fang Fang' which is straight out of a Bond film no doubt starring Klaus Schwab as the bloke living on a secret island and controlling laser weapons positioned in space and pointing at world capitals. Fang Fang plays the part of Bond's infiltrator girlfriend which I'm sure she would enjoy rather more than sharing a bed with the brainless Swalwell, lying back and thinking of China. The FBI eventually warned Swalwell about Fang Fang which gave her time to escape back to the Chinese dictatorship. How very thoughtful of them. The second Trump impeachment also failed and hardly surprising when an impeachment is supposed to remove a sitting president and by the time it happened Trump was no longer president. These people are running your country America, well, officially anyway. Terrifying isn't it?

Outcomes tell the story - always

The outcome of all this – and it's the *outcome* on which Renegade Minds focus, not the words – was that a vicious, hysterical and

obviously pre-planned assault was launched on Pushbackers to censor, silence and discredit them and even targeted their right to earn a living. They have since been condemned as 'domestic terrorists' that need to be treated like Al-Qaeda and Islamic State. 'Domestic terrorists' is a label the Cult has been trying to make stick since the period of the Oklahoma bombing in 1995 which was blamed on 'far-right domestic terrorists'. If you read *The Trigger* you will see that the bombing was clearly a Problem-Reaction-Solution carried out by the Deep State during a Bill Clinton administration so corrupt that no dictionary definition of the term would even nearly suffice. Nearly 30, 000 troops were deployed from all over America to the empty streets of Washington for Biden's inauguration. Ten thousand of them stayed on with the pretext of protecting the capital from insurrectionists when it was more psychological programming to normalise the use of the military in domestic law enforcement in support of the Cult plan for a police-military state. Biden's fascist administration began a purge of 'wrong-thinkers' in the military which means anyone that is not on board with Woke. The Capitol Building was surrounded by a fence with razor wire and the Land of the Free was further symbolically and literally dismantled. The circle was completed with the installation of Biden and the exploitation of the QAnon Psyop.

America had never been so divided since the civil war of the 19th century, Pushbackers were isolated and dubbed terrorists and now, as was always going to happen, the Cult immediately set about deleting what little was left of freedom and transforming American society through a swish of the hand of the most controlled 'president' in American history leading (officially at least) the most extreme regime since the country was declared an independent state on July 4th, 1776. Biden issued undebated, dictatorial executive orders almost by the hour in his opening days in office across the whole spectrum of the Cult wish-list including diluting controls on the border with Mexico allowing thousands of migrants to illegally enter the United States to transform the demographics of America and import an election-changing number of perceived Democrat

voters. Then there were Biden deportation amnesties for the already illegally resident (estimated to be as high as 20 or even 30 million). A bill before Congress awarded American citizenship to anyone who could prove they had worked in agriculture for just 180 days in the previous two years as 'Big Ag' secured its slave labour long-term. There were the plans to add new states to the union such as Puerto Rico and making Washington DC a state. They are all parts of a plan to ensure that the Cult-owned Woke Democrats would be permanently in power.

Border – what border?

I have exposed in detail in other books how mass immigration into the United States and Europe is the work of Cult networks fuelled by the tens of billions spent to this and other ends by George Soros and his global Open Society (open borders) Foundations. The impact can be seen in America alone where the population has increased by *100 million* in little more than 30 years mostly through immigration. I wrote in *The Answer* that the plan was to have so many people crossing the southern border that the numbers become unstoppable and we are now there under Cult-owned Biden. El Salvador in Central America puts the scale of what is happening into context. A third of the population now lives in the United States, much of it illegally, and many more are on the way. The methodology is to crush Central and South American countries economically and spread violence through machete-wielding psychopathic gangs like MS-13 based in El Salvador and now operating in many American cities. Biden-imposed lax security at the southern border means that it is all but open. He said before his 'election' that he wanted to see a surge towards the border if he became president and that was the green light for people to do just that after election day to create the human disaster that followed for both America and the migrants. When that surge came the imbecilic Alexandria Ocasio-Cortez said it wasn't a 'surge' because they are 'children, not insurgents' and the term 'surge' (used by Biden) was a claim of 'white supremacists'.

This disingenuous lady may one day enter the realm of the most basic intelligence, but it won't be any time soon.

Sabbatians and the Cult are in the process of destroying America by importing violent people and gangs in among the genuine to terrorise American cities and by overwhelming services that cannot cope with the sheer volume of new arrivals. Something similar is happening in Europe as Western society in general is targeted for demographic and cultural transformation and upheaval. The plan demands violence and crime to create an environment of intimidation, fear and division and Soros has been funding the election of district attorneys across America who then stop prosecuting many crimes, reduce sentences for violent crimes and free as many violent criminals as they can. Sabbatians are creating the chaos from which order – their order – can respond in a classic Problem-Reaction-Solution. A Freemasonic moto says 'Ordo Ab Chao' (Order out of Chaos) and this is why the Cult is constantly creating chaos to impose a new 'order'. Here you have the reason the Cult is constantly creating chaos. The 'Covid' hoax can be seen with those entering the United States by plane being forced to take a 'Covid' test while migrants flooding through southern border processing facilities do not. Nothing is put in the way of mass migration and if that means ignoring the government's own 'Covid' rules then so be it. They know it's all bullshit anyway. Any pushback on this is denounced as 'racist' by Wokers and Sabbatian fronts like the ultra-Zionist Anti-Defamation League headed by the appalling Jonathan Greenblatt which at the same time argues that Israel should not give citizenship and voting rights to more Palestinian Arabs or the 'Jewish population' (in truth the Sabbatian network) will lose control of the country.

Society-changing numbers

Biden's masters have declared that countries like El Salvador are so dangerous that their people must be allowed into the United States for humanitarian reasons when there are fewer murders in large parts of many Central American countries than in US cities like

Baltimore. That is not to say Central America cannot be a dangerous place and Cult-controlled American governments have been making it so since way back, along with the dismantling of economies, in a long-term plan to drive people north into the United States. Parts of Central America are very dangerous, but in other areas the story is being greatly exaggerated to justify relaxing immigration criteria. Migrants are being offered free healthcare and education in the United States as another incentive to head for the border and there is no requirement to be financially independent before you can enter to prevent the resources of America being drained. You can't blame migrants for seeking what they believe will be a better life, but they are being played by the Cult for dark and nefarious ends. The numbers since Biden took office are huge. In February, 2021, more than 100,000 people were known to have tried to enter the US illegally through the southern border (it was 34,000 in the same month in 2020) and in March it was 170,000 – a 418 percent increase on March, 2020. These numbers are only known people, not the ones who get in unseen. The true figure for migrants illegally crossing the border in a single month was estimated by one congressman at 250,000 and that number will only rise under Biden's current policy. Gangs of murdering drug-running thugs that control the Mexican side of the border demand money – thousands of dollars – to let migrants cross the Rio Grande into America. At the same time gun battles are breaking out on the border several times a week between rival Mexican drug gangs (which now operate globally) who are equipped with sophisticated military-grade weapons, grenades and armoured vehicles. While the Capitol Building was being 'protected' from a non-existent 'threat' by thousands of troops, and others were still deployed at the time in the Cult Neocon war in Afghanistan, the southern border of America was left to its fate. This is not incompetence, it is cold calculation.

By March, 2021, there were 17,000 unaccompanied children held at border facilities and many of them are ensnared by people traffickers for paedophile rings and raped on their journey north to America. This is not conjecture – this is fact. Many of those designated

children are in reality teenage boys or older. Meanwhile Wokers posture their self-purity for encouraging poor and tragic people to come to America and face this nightmare both on the journey and at the border with the disgusting figure of House Speaker Nancy Pelosi giving disingenuous speeches about caring for migrants. The woman's evil. Wokers condemned Trump for having children in cages at the border (so did Obama, *Shhhh*), but now they are sleeping on the floor without access to a shower with one border facility 729 percent over capacity. The Biden insanity even proposed flying migrants from the southern border to the northern border with Canada for 'processing'. The whole shambles is being overseen by ultra-Zionist Secretary of Homeland Security, the moronic liar Alejandro Mayorkas, who banned news cameras at border facilities to stop Americans seeing what was happening. Mayorkas said there was not a ban on news crews; it was just that they were not allowed to film. Alongside him at Homeland Security is another ultra-Zionist Cass Sunstein appointed by Biden to oversee new immigration laws. Sunstein despises conspiracy researchers to the point where he suggests they should be banned or *taxed* for having such views. The man is not bonkers or anything. He's perfectly well-adjusted, but adjusted to what is the question. Criticise what is happening and you are a 'white supremacist' when earlier non-white immigrants also oppose the numbers which effect their lives and opportunities. Black people in poor areas are particularly damaged by uncontrolled immigration and the increased competition for work opportunities with those who will work for less. They are also losing voting power as Hispanics become more dominant in former black areas. It's a downward spiral for them while the billionaires behind the policy drone on about how much they care about black people and 'racism'. None of this is about compassion for migrants or black people – that's just wind and air. Migrants are instead being mercilessly exploited to transform America while the countries they leave are losing their future and the same is true in Europe. Mass immigration may now be the work of Woke Democrats, but it can be traced back to the 1986 Immigration Reform and Control Act (it

wasn't) signed into law by Republican hero President Ronald Reagan which gave amnesty to millions living in the United States illegally and other incentives for people to head for the southern border. Here we have the one-party state at work again.

Save me syndrome

Almost every aspect of what I have been exposing as the Cult agenda was on display in even the first days of 'Biden' with silencing of Pushbackers at the forefront of everything. A Renegade Mind will view the Trump years and QAnon in a very different light to their supporters and advocates as the dots are connected. The QAnon/Trump Psyop has given the Cult all it was looking for. We may not know how much, or little, that Trump realised he was being used, but that's a side issue. This pincer movement produced the desired outcome of dividing America and having Pushbackers isolated. To turn this around we have to look at new routes to empowerment which do not include handing our power to other people and groups through what I will call the 'Save Me Syndrome' – 'I want someone else to do it so that I don't have to'. We have seen this at work throughout human history and the QAnon/Trump Psyop is only the latest incarnation alongside all the others. Religion is an obvious expression of this when people look to a 'god' or priest to save them or tell them how to be saved and then there are 'save me' politicians like Trump. Politics is a diversion and not a 'saviour'. It is a means to block positive change, not make it possible.

Save Me Syndrome always comes with the same repeating theme of handing your power to whom or what you believe will save you while your real 'saviour' stares back from the mirror every morning. Renegade Minds are constantly vigilant in this regard and always asking the question 'What can I do?' rather than 'What can someone else do for me?' Gandhi was right when he said: 'You must be the change you want to see in the world.' We are indeed the people we have been waiting for. We are presented with a constant raft of reasons to concede that power to others and forget where the real power is. Humanity has the numbers and the Cult does not. It has to

use diversion and division to target the unstoppable power that comes from unity. Religions, governments, politicians, corporations, media, QAnon, are all different manifestations of this power-diversion and dilution. Refusing to give your power to governments and instead handing it to Trump and QAnon is not to take a new direction, but merely to recycle the old one with new names on the posters. I will explore this phenomenon as we proceed and how to break the cycles and recycles that got us here through the mists of repeating perception and so repeating history.

For now we shall turn to the most potent example in the entire human story of the consequences that follow when you give your power away. I am talking, of course, of the 'Covid' hoax.

CHAPTER FOUR

'Covid': Calculated catastrophe

Facts are threatening to those invested in fraud
DaShanne Stokes

We can easily unravel the real reason for the 'Covid pandemic' hoax by employing the Renegade Mind methodology that I have outlined this far. We'll start by comparing the long-planned Cult outcome with the 'Covid pandemic' outcome. Know the outcome and you'll see the journey.

I have highlighted the plan for the Hunger Games Society which has been in my books for so many years with the very few controlling the very many through ongoing dependency. To create this dependency it is essential to destroy independent livelihoods, businesses and employment to make the population reliant on the state (the Cult) for even the basics of life through a guaranteed pittance income. While independence of income remained these Cult ambitions would be thwarted. With this knowledge it was easy to see where the 'pandemic' hoax was going once talk of 'lockdowns' began and the closing of all but perceived 'essential' businesses to 'save' us from an alleged 'deadly virus'. Cult corporations like Amazon and Walmart were naturally considered 'essential' while mom and pop shops and stores had their doors closed by fascist decree. As a result with every new lockdown and new regulation more small and medium, even large businesses not owned by the Cult, went to the wall while Cult giants and their frontmen and women grew financially fatter by the second. Mom and pop were

denied an income and the right to earn a living and the wealth of people like Jeff Bezos (Amazon), Mark Zuckerberg (Facebook) and Sergei Brin and Larry Page (Google/Alphabet) have reached record levels. The Cult was increasing its own power through further dramatic concentrations of wealth while the competition was being destroyed and brought into a state of dependency. Lockdowns have been instigated to secure that very end and were never anything to do with health. My brother Paul spent 45 years building up a bus repair business, but lockdowns meant buses were running at a fraction of normal levels for months on end. Similar stories can be told in their hundreds of millions worldwide. Efforts of a lifetime coldly destroyed by Cult multi-billionaires and their lackeys in government and law enforcement who continued to earn their living from the taxation of the people while denying the right of the same people to earn theirs. How different it would have been if those making and enforcing these decisions had to face the same financial hardships of those they affected, but they never do.

Gates of Hell

Behind it all in the full knowledge of what he is doing and why is the psychopathic figure of Cult operative Bill Gates. His puppet Tedros at the World Health Organization declared 'Covid' a pandemic in March, 2020. The WHO had changed the definition of a 'pandemic' in 2009 just a month before declaring the 'swine flu pandemic' which would not have been so under the previous definition. The same applies to 'Covid'. The definition had included... 'an infection by an infectious agent, occurring simultaneously in different countries, with a significant mortality rate relative to the proportion of the population infected'. The new definition removed the need for 'significant mortality'. The 'pandemic' has been fraudulent even down to the definition, but Gates demanded economy-destroying lockdowns, school closures, social distancing, mandatory masks, a 'vaccination' for every man, woman and child on the planet and severe consequences and restrictions for those that refused. Who gave him this power? The

Cult did which he serves like a little boy in short trousers doing what his daddy tells him. He and his psychopathic missus even smiled when they said that much worse was to come (what they knew was planned to come). Gates responded in the matter-of-fact way of all psychopaths to a question about the effect on the world economy of what he was doing:

Well, it won't go to zero but it will shrink. Global GDP is probably going to take the biggest hit ever [Gates was smiling as he said this] ... in my lifetime this will be the greatest economic hit. But you don't have a choice. People act as if you have a choice. People don't feel like going to the stadium when they might get infected ... People are deeply affected by seeing these stats, by knowing they could be part of the transmission chain, old people, their parents and grandparents, could be affected by this, and so you don't get to say ignore what is going on here.

There will be the ability to open up, particularly in rich countries, if things are done well over the next few months, but for the world at large normalcy only returns when we have largely vaccinated the entire population.

The man has no compassion or empathy. How could he when he's a psychopath like all Cult players? My own view is that even beyond that he is very seriously mentally ill. Look in his eyes and you can see this along with his crazy flailing arms. You don't do what he has done to the world population since the start of 2020 unless you are mentally ill and at the most extreme end of psychopathic. You especially don't do it when to you know, as we shall see, that cases and deaths from 'Covid' are fakery and a product of monumental figure massaging. 'These stats' that Gates referred to are based on a 'test' that's not testing for the 'virus' as he has known all along. He made his fortune with big Cult support as an infamously ruthless software salesman and now buys global control of 'health' (death) policy without the population he affects having any say. It's a breathtaking outrage. Gates talked about people being deeply affected by fear of 'Covid' when that was because of *him* and his global network lying to them minute-by-minute supported by a lying media that he seriously influences and funds to the tune of hundreds of millions. He's handed big sums to media operations including the BBC, NBC, Al Jazeera, Univision, *PBS NewsHour*,

ProPublica, National Journal, The Guardian, The Financial Times, The Atlantic, Texas Tribune, USA Today publisher Gannett, Washington Monthly, Le Monde, Center for Investigative Reporting, Pulitzer Center on Crisis Reporting, National Press Foundation, International Center for Journalists, Solutions Journalism Network, the Poynter Institute for Media Studies, and many more. Gates is everywhere in the 'Covid' hoax and the man must go to prison – or a mental facility – for the rest of his life and his money distributed to those he has taken such enormous psychopathic pleasure in crushing.

The Muscle

The Hunger Games global structure demands a police-military state – a fusion of the two into one force – which viciously imposes the will of the Cult on the population and protects the Cult from public rebellion. In that regard, too, the 'Covid' hoax just keeps on giving. Often unlawful, ridiculous and contradictory 'Covid' rules and regulations have been policed across the world by moronic automatons and psychopaths made faceless by face-nappy masks and acting like the Nazi SS and fascist blackshirts and brownshirts of Hitler and Mussolini. The smallest departure from the rules decreed by the psychos in government and their clueless gofers were jumped upon by the face-nappy fascists. Brutality against public protestors soon became commonplace even on girls, women and old people as the brave men with the batons – the Face-Nappies as I call them – broke up peaceful protests and handed out fines like confetti to people who couldn't earn a living let alone pay hundreds of pounds for what was once an accepted human right. Robot Face-Nappies of Nottingham police in the English East Midlands fined one group £11,000 for attending a child's birthday party. For decades I charted the transformation of law enforcement as genuine, decent officers were replaced with psychopaths and the brain dead who would happily and brutally do whatever their masters told them. Now they were let loose on the public and I would emphasise the point that none of this just happened. The step-by-step change in the dynamic between police and public was orchestrated from the shadows by

those who knew where this was all going and the same with the perceptual reframing of those in all levels of authority and official administration through 'training courses' by organisations such as Common Purpose which was created in the late 1980s and given a massive boost in Blair era Britain until it became a global phenomenon. Supposed public 'servants' began to view the population as the enemy and the same was true of the police. This was the start of the explosion of behaviour manipulation organisations and networks preparing for the all-war on the human psyche unleashed with the dawn of 2020. I will go into more detail about this later in the book because it is a core part of what is happening.

Police desecrated beauty spots to deter people gathering and arrested women for walking in the countryside alone 'too far' from their homes. We had arrogant, clueless sergeants in the Isle of Wight police where I live posting on Facebook what they insisted the population must do or else. A schoolmaster sergeant called Radford looked young enough for me to ask if his mother knew he was out, but he was posting what he *expected* people to do while a Sergeant Wilkinson boasted about fining lads for meeting in a McDonald's car park where they went to get a lockdown takeaway. Wilkinson added that he had even cancelled their order. What a pair of prats these people are and yet they have increasingly become the norm among Jackboot Johnson's Yellowshirts once known as the British police. This was the theme all over the world with police savagery common during lockdown protests in the United States, the Netherlands, and the fascist state of Victoria in Australia under its tyrannical and again moronic premier Daniel Andrews. Amazing how tyrannical and moronic tend to work as a team and the same combination could be seen across America as arrogant, narcissistic Woke governors and mayors such as Gavin Newsom (California), Andrew Cuomo (New York), Gretchen Whitmer (Michigan), Lori Lightfoot (Chicago) and Eric Garcetti (Los Angeles) did their Nazi and Stalin impressions with the full support of the compliant brutality of their enforcers in uniform as they arrested small business owners defying

fascist shutdown orders and took them to jail in ankle shackles and handcuffs. This happened to bistro owner Marlena Pavlos-Hackney in Gretchen Whitmer's fascist state of Michigan when police arrived to enforce an order by a state-owned judge for 'putting the community at risk' at a time when other states like Texas were dropping restrictions and migrants were pouring across the southern border without any 'Covid' questions at all. I'm sure there are many officers appalled by what they are ordered to do, but not nearly enough of them. If they were truly appalled they would not do it. As the months passed every opportunity was taken to have the military involved to make their presence on the streets ever more familiar and 'normal' for the longer-term goal of police-military fusion.

Another crucial element to the Hunger Games enforcement network has been encouraging the public to report neighbours and others for 'breaking the lockdown rules'. The group faced with £11,000 in fines at the child's birthday party would have been dobbed-in by a neighbour with a brain the size of a pea. The technique was most famously employed by the Stasi secret police in communist East Germany who had public informants placed throughout the population. A police chief in the UK says his force doesn't need to carry out 'Covid' patrols when they are flooded with so many calls from the public reporting other people for visiting the beach. Dorset police chief James Vaughan said people were so enthusiastic about snitching on their fellow humans they were now operating as an auxiliary arm of the police: 'We are still getting around 400 reports a week from the public, so we will respond to reports ... We won't need to be doing hotspot patrols because people are very quick to pick the phone up and tell us.' Vaughan didn't say that this is a pillar of all tyrannies of whatever complexion and the means to hugely extend the reach of enforcement while spreading distrust among the people and making them wary of doing anything that might get them reported. Those narcissistic Isle of Wight sergeants Radford and Wilkinson never fail to add a link to their Facebook posts where the public can inform on their fellow slaves.

Neither would be self-aware enough to realise they were imitating the Stasi which they might well never have heard of. Government psychologists that I will expose later laid out a policy to turn communities against each other in the same way.

A coincidence? Yep, and I can knit fog

I knew from the start of the alleged pandemic that this was a Cult operation. It presented limitless potential to rapidly advance the Cult agenda and exploit manipulated fear to demand that every man, woman and child on the planet was 'vaccinated' in a process never used on humans before which infuses self-replicating *synthetic* material into human cells. Remember the plan to transform the human body from a biological to a synthetic biological state. I'll deal with the 'vaccine' (that's not actually a vaccine) when I focus on the genetic agenda. Enough to say here that mass global 'vaccination' justified by this 'new virus' set alarms ringing after 30 years of tracking these people and their methods. The 'Covid' hoax officially beginning in China was also a big red flag for reasons I will be explaining. The agenda potential was so enormous that I could dismiss any idea that the 'virus' appeared naturally. Major happenings with major agenda implications never occur without Cult involvement in making them happen. My questions were twofold in early 2020 as the media began its campaign to induce global fear and hysteria: Was this alleged infectious agent released on purpose by the Cult or did it even exist at all? I then did what I always do in these situations. I sat, observed and waited to see where the evidence and information would take me. By March and early April synchronicity was strongly – and ever more so since then – pointing me in the direction of *there is no 'virus'*. I went public on that with derision even from swathes of the alternative media that voiced a scenario that the Chinese government released the 'virus' in league with Deep State elements in the United States from a top-level bio-lab in Wuhan where the 'virus' is said to have first appeared. I looked at that possibility, but I didn't buy it for several reasons. Deaths from the 'virus' did not in any way match what they

would have been with a 'deadly bioweapon' and it is much more effective if you sell the *illusion* of an infectious agent rather than having a real one unless you can control through injection who has it and who doesn't. Otherwise you lose control of events. A made-up 'virus' gives you a blank sheet of paper on which you can make it do whatever you like and have any symptoms or mutant 'variants' you choose to add while a real infectious agent would limit you to what it actually does. A phantom disease allows you to have endless ludicrous 'studies' on the 'Covid' dollar to widen the perceived impact by inventing ever more 'at risk' groups including one study which said those who walk slowly may be almost four times more likely to die from the 'virus'. People are in psychiatric wards for less.

A real 'deadly bioweapon' can take out people in the hierarchy that are not part of the Cult, but essential to its operation. Obviously they don't want that. Releasing a real disease means you immediately lose control of it. Releasing an illusory one means you don't. Again it's vital that people are extra careful when dealing with what they want to hear. A bioweapon unleashed from a Chinese laboratory in collusion with the American Deep State may fit a conspiracy narrative, but is it true? Would it not be far more effective to use the excuse of a 'virus' to justify the real bioweapon – the 'vaccine'? That way your disease agent does not have to be transmitted and arrives directly through a syringe. I saw a French virologist Luc Montagnier quoted in the alternative media as saying he had discovered that the alleged 'new' severe acute respiratory syndrome coronavirus , or SARS-CoV-2, was made artificially and included elements of the human immunodeficiency 'virus' (HIV) and a parasite that causes malaria. SARS-CoV-2 is alleged to trigger an alleged illness called Covid-19. I remembered Montagnier's name from my research years before into claims that an HIV 'retrovirus' causes AIDs – claims that were demolished by Berkeley virologist Peter Duesberg who showed that no one had ever proved that HIV causes acquired immunodeficiency syndrome or AIDS. Claims that become accepted as fact, publicly and medically, with no proof whatsoever are an ever-recurring story that profoundly applies to

‘Covid’. Nevertheless, despite the lack of proof, Montagnier’s team at the Pasteur Institute in Paris had a long dispute with American researcher Robert Gallo over which of them discovered and isolated the HIV ‘virus’ and with *no evidence* found it to cause AIDS. You will see later that there is also no evidence that any ‘virus’ causes any disease or that there is even such a thing as a ‘virus’ in the way it is said to exist. The claim to have ‘isolated’ the HIV ‘virus’ will be presented in its real context as we come to the shocking story – and it is a story – of SARS-CoV-2 and so will Montagnier’s assertion that he identified the full SARS-CoV-2 genome.

Hoax in the making

We can pick up the ‘Covid’ story in 2010 and the publication by the Rockefeller Foundation of a document called ‘Scenarios for the Future of Technology and International Development’. The inner circle of the Rockefeller family has been serving the Cult since John D. Rockefeller (1839-1937) made his fortune with Standard Oil. It is less well known that the same Rockefeller – the Bill Gates of his day – was responsible for establishing what is now referred to as ‘Big Pharma’, the global network of pharmaceutical companies that make outrageous profits dispensing scalpel and drug ‘medicine’ and are obsessed with pumping vaccines in ever-increasing number into as many human arms and backsides as possible. John D. Rockefeller was the driving force behind the creation of the ‘education’ system in the United States and elsewhere specifically designed to program the perceptions of generations thereafter. The Rockefeller family donated exceptionally valuable land in New York for the United Nations building and were central in establishing the World Health Organization in 1948 as an agency of the UN which was created from the start as a Trojan horse and stalking horse for world government. Now enter Bill Gates. His family and the Rockefellers have long been extremely close and I have seen genealogy which claims that if you go back far enough the two families fuse into the same bloodline. Gates has said that the Bill and Melinda Gates Foundation was inspired by the Rockefeller Foundation and why not

when both are serving the same Cult? Major tax-exempt foundations are overwhelmingly criminal enterprises in which Cult assets fund the Cult agenda in the guise of 'philanthropy' while avoiding tax in the process. Cult operatives can become mega-rich in their role of front men and women for the psychopaths at the inner core and they, too, have to be psychopaths to knowingly serve such evil. Part of the deal is that a big percentage of the wealth gleaned from representing the Cult has to be spent advancing the ambitions of the Cult and hence you have the Rockefeller Foundation, Bill and Melinda Gates Foundation (and *so* many more) and people like George Soros with his global Open Society Foundations spending their billions in pursuit of global Cult control. Gates is a global public face of the Cult with his interventions in world affairs including Big Tech influence; a central role in the 'Covid' and 'vaccine' scam; promotion of the climate change shakedown; manipulation of education; geoengineering of the skies; and his food-control agenda as the biggest owner of farmland in America, his GMO promotion and through other means. As one writer said: 'Gates monopolizes or wields disproportionate influence over the tech industry, global health and vaccines, agriculture and food policy (including biopiracy and fake food), weather modification and other climate technologies, surveillance, education and media.' The almost limitless wealth secured through Microsoft and other not-allowed-to-fail ventures (including vaccines) has been ploughed into a long, long list of Cult projects designed to enslave the entire human race. Gates and the Rockefellers have been working as one unit with the Rockefeller-established World Health Organization leading global 'Covid' policy controlled by Gates through his mouth-piece Tedros. Gates became the WHO's biggest funder when Trump announced that the American government would cease its donations, but Biden immediately said he would restore the money when he took office in January, 2021. The Gates Foundation (the Cult) owns through limitless funding the world health system and the major players across the globe in the 'Covid' hoax.

Okay, with that background we return to that Rockefeller Foundation document of 2010 headed 'Scenarios for the Future of Technology and International Development' and its 'imaginary' epidemic of a virulent and deadly influenza strain which infected 20 percent of the global population and killed eight million in seven months. The Rockefeller scenario was that the epidemic destroyed economies, closed shops, offices and other businesses and led to governments imposing fierce rules and restrictions that included mandatory wearing of face masks and body-temperature checks to enter communal spaces like railway stations and supermarkets. The document predicted that even after the height of the Rockefeller-envisaged epidemic the authoritarian rule would continue to deal with further pandemics, transnational terrorism, environmental crises and rising poverty. Now you may think that the Rockefellers are our modern-day seers or alternatively, and rather more likely, that they well knew what was planned a few years further on. Fascism had to be imposed, you see, to 'protect citizens from risk and exposure'. The Rockefeller scenario document said:

During the pandemic, national leaders around the world flexed their authority and imposed airtight rules and restrictions, from the mandatory wearing of face masks to body-temperature checks at the entries to communal spaces like train stations and supermarkets. Even after the pandemic faded, this more authoritarian control and oversight of citizens and their activities stuck and even intensified. In order to protect themselves from the spread of increasingly global problems – from pandemics and transnational terrorism to environmental crises and rising poverty – leaders around the world took a firmer grip on power.

At first, the notion of a more controlled world gained wide acceptance and approval. Citizens willingly gave up some of their sovereignty – and their privacy – to more paternalistic states in exchange for greater safety and stability. Citizens were more tolerant, and even eager, for top-down direction and oversight, and national leaders had more latitude to impose order in the ways they saw fit.

In developed countries, this heightened oversight took many forms: biometric IDs for all citizens, for example, and tighter regulation of key industries whose stability was deemed vital to national interests. In many developed countries, enforced cooperation with a suite of new regulations and agreements slowly but steadily restored both order and, importantly, economic growth.

There we have the prophetic Rockefellers in 2010 and three years later came their paper for the Global Health Summit in Beijing, China, when government representatives, the private sector, international organisations and groups met to discuss the next 100 years of 'global health'. The Rockefeller Foundation-funded paper was called 'Dreaming the Future of Health for the Next 100 Years and more prophecy ensued as it described a dystopian future: 'The abundance of data, digitally tracking and linking people may mean the 'death of privacy' and may replace physical interaction with transient, virtual connection, generating isolation and raising questions of how values are shaped in virtual networks.' Next in the 'Covid' hoax preparation sequence came a 'table top' simulation in 2018 for another 'imaginary' pandemic of a disease called Clade X which was said to kill 900 million people. The exercise was organised by the Gates-funded Johns Hopkins University's Center for Health Security in the United States and this is the very same university that has been compiling the disgustingly and systematically erroneous global figures for 'Covid' cases and deaths. Similar Johns Hopkins health crisis scenarios have included the Dark Winter exercise in 2001 and Atlantic Storm in 2005.

Nostradamus 201

For sheer predictive genius look no further prophecy-watchers than the Bill Gates-funded Event 201 held only six weeks before the 'coronavirus pandemic' is supposed to have broken out in China and Event 201 was based on a scenario of a global 'coronavirus pandemic'. Melinda Gates, the great man's missus, told the BBC that he had 'prepared for years' for a coronavirus pandemic which told us what we already knew. Nostradamugates had predicted in a TED talk in 2015 that a pandemic was coming that would kill a lot of people and demolish the world economy. My god, the man is a machine – possibly even literally. Now here he was only weeks before the real thing funding just such a simulated scenario and involving his friends and associates at Johns Hopkins, the World Economic Forum Cult-front of Klaus Schwab, the United Nations,

Johnson & Johnson, major banks, and officials from China and the Centers for Disease Control in the United States. What synchronicity – Johns Hopkins would go on to compile the fraudulent ‘Covid’ figures, the World Economic Forum and Schwab would push the ‘Great Reset’ in response to ‘Covid’, the Centers for Disease Control would be at the forefront of ‘Covid’ policy in the United States, Johnson & Johnson would produce a ‘Covid vaccine’, and everything would officially start just weeks later in China. Spooky, eh? They were even accurate in creating a simulation of a ‘virus’ pandemic because the ‘real thing’ would also be a simulation. Event 201 was not an exercise preparing for something that might happen; it was a rehearsal for what those in control knew was *going* to happen and very shortly. Hours of this simulation were posted on the Internet and the various themes and responses mirrored what would soon be imposed to transform human society. News stories were inserted and what they said would be commonplace a few weeks later with still more prophecy perfection. Much discussion focused on the need to deal with misinformation and the ‘anti-vax movement’ which is exactly what happened when the ‘virus’ arrived – was said to have arrived – in the West.

Cult-owned social media banned criticism and exposure of the official ‘virus’ narrative and when I said there *was* no ‘virus’ in early April, 2020, I was banned by one platform after another including YouTube, Facebook and later Twitter. The mainstream broadcast media in Britain was in effect banned from interviewing me by the Tony-Blair-created government broadcasting censor Ofcom headed by career government bureaucrat Melanie Dawes who was appointed just as the ‘virus’ hoax was about to play out in January, 2020. At the same time the Ickonic media platform was using Vimeo, another ultra-Zionist-owned operation, while our own player was being created and they deleted in an instant hundreds of videos, documentaries, series and shows to confirm their unbelievable vindictiveness. We had copies, of course, and they had to be restored one by one when our player was ready. These people have no class. Sabbatian Facebook promised free advertisements for the Gates-

controlled World Health Organization narrative while deleting ‘false claims and conspiracy theories’ to stop ‘misinformation’ about the alleged coronavirus. All these responses could be seen just a short while earlier in the scenarios of Event 201. Extreme censorship was absolutely crucial for the Cult because the official story was so ridiculous and unsupportable by the evidence that it could never survive open debate and the free-flow of information and opinion. If you can’t win a debate then don’t have one is the Cult’s approach throughout history. Facebook’s little boy front man – front boy – Mark Zuckerberg equated ‘credible and accurate information’ with official sources and exposing their lies with ‘misinformation’.

Silencing those that can see

The censorship dynamic of Event 201 is now the norm with an army of narrative-supporting ‘fact-checker’ organisations whose entire reason for being is to tell the public that official narratives are true and those exposing them are lying. One of the most appalling of these ‘fact-checkers’ is called NewsGuard founded by ultra-Zionist Americans Gordon Crovitz and Steven Brill. Crovitz is a former publisher of *The Wall Street Journal*, former Executive Vice President of Dow Jones, a member of the Council on Foreign Relations (CFR), and on the board of the American Association of Rhodes Scholars. The CFR and Rhodes Scholarships, named after Rothschild agent Cecil Rhodes who plundered the gold and diamonds of South Africa for his masters and the Cult, have featured widely in my books. NewsGuard don’t seem to like me for some reason – I really can’t think why – and they have done all they can to have me censored and discredited which is, to quote an old British politician, like being savaged by a dead sheep. They are, however, like all in the censorship network, very well connected and funded by organisations themselves funded by, or connected to, Bill Gates. As you would expect with anything associated with Gates NewsGuard has an offshoot called HealthGuard which ‘fights online health care hoaxes’. How very kind. Somehow the NewsGuard European Managing Director Anna-Sophie Harling, a remarkably young-

looking woman with no broadcasting experience and little hands-on work in journalism, has somehow secured a position on the 'Content Board' of UK government broadcast censor Ofcom. An executive of an organisation seeking to discredit dissidents of the official narratives is making decisions for the government broadcast 'regulator' about content?? Another appalling 'fact-checker' is Full Fact funded by George Soros and global censors Google and Facebook.

It's amazing how many activists in the 'fact-checking', 'anti-hate', arena turn up in government-related positions – people like UK Labour Party activist Imran Ahmed who heads the Center for Countering Digital Hate founded by people like Morgan McSweeney, now chief of staff to the Labour Party's hapless and useless 'leader' Keir Starmer. Digital Hate – which is what it really is – uses the American spelling of Center to betray its connection to a transatlantic network of similar organisations which in 2020 shapeshifted from attacking people for 'hate' to attacking them for questioning the 'Covid' hoax and the dangers of the 'Covid vaccine'. It's just a coincidence, you understand. This is one of Imran Ahmed's hysterical statements: 'I would go beyond calling anti-vaxxers conspiracy theorists to say they are an extremist group that pose a national security risk.' No one could ever accuse this prat of understatement and he's including in that those parents who are now against vaccines after their children were damaged for life or killed by them. He's such a nice man. Ahmed does the rounds of the Woke media getting soft-ball questions from spineless 'journalists' who never ask what right he has to campaign to destroy the freedom of speech of others while he demands it for himself. There also seems to be an overrepresentation in Ofcom of people connected to the narrative-worshipping BBC. This incredible global network of narrative-support was super-vital when the 'Covid' hoax was played in the light of the mega-whopper lies that have to be defended from the spotlight cast by the most basic intelligence.

Setting the scene

The Cult plays the long game and proceeds step-by-step ensuring that everything is in place before major cards are played and they don't come any bigger than the 'Covid' hoax. The psychopaths can't handle events where the outcome isn't certain and as little as possible – preferably nothing – is left to chance. Politicians, government and medical officials who would follow direction were brought to illusory power in advance by the Cult web whether on the national stage or others like state governors and mayors of America. For decades the dynamic between officialdom, law enforcement and the public was changed from one of service to one of control and dictatorship. Behaviour manipulation networks established within government were waiting to impose the coming 'Covid' rules and regulations specifically designed to subdue and rewire the psyche of the people in the guise of protecting health. These included in the UK the Behavioural Insights Team part-owned by the British government Cabinet Office; the Scientific Pandemic Insights Group on Behaviours (SPI-B); and a whole web of intelligence and military groups seeking to direct the conversation on social media and control the narrative. Among them are the cyberwarfare (on the people) 77th Brigade of the British military which is also coordinated through the Cabinet Office as civilian and military leadership continues to combine in what they call the Fusion Doctrine. The 77th Brigade is a British equivalent of the infamous Israeli (Sabbatian) military cyberwarfare and Internet manipulation operation Unit 8200 which I expose at length in *The Trigger*. Also carefully in place were the medical and science advisers to government – many on the payroll past or present of Bill Gates – and a whole alternative structure of unelected government stood by to take control when elected parliaments were effectively closed down once the 'Covid' card was slammed on the table. The structure I have described here and so much more was installed in every major country through the Cult networks. The top-down control hierarchy looks like this: The Cult – Cult-owned Gates – the World Health Organization and Tedros – Gates-funded or controlled chief medical officers and science 'advisers' (dictators) in each country –

political 'leaders' – law enforcement – The People. Through this simple global communication and enforcement structure the policy of the Cult could be imposed on virtually the entire human population so long as they acquiesced to the fascism. With everything in place it was time for the button to be pressed in late 2019/early 2020.

These were the prime goals the Cult had to secure for its will to prevail:

1) Locking down economies, closing all but designated 'essential' businesses (Cult-owned corporations were 'essential'), and putting the population under house arrest was an imperative to destroy independent income and employment and ensure dependency on the Cult-controlled state in the Hunger Games Society. Lockdowns had to be established as the global blueprint from the start to respond to the 'virus' and followed by pretty much the entire world.

2) The global population had to be terrified into believing in a deadly 'virus' that didn't actually exist so they would unquestioningly obey authority in the belief that authority must know how best to protect them and their families. Software salesman Gates would suddenly morph into the world's health expert and be promoted as such by the Cult-owned media.

3) A method of testing that wasn't testing for the 'virus', but was only claimed to be, had to be in place to provide the illusion of 'cases' and subsequent 'deaths' that had a very different cause to the 'Covid-19' that would be scribbled on the death certificate.

4) Because there was no 'virus' and the great majority testing positive with a test not testing for the 'virus' would have no symptoms of anything the lie had to be sold that people without symptoms (without the 'virus') could still pass it on to others. This was crucial to justify for the first time quarantining – house arresting – healthy people. Without this the economy-destroying lockdown of *everybody* could not have been credibly sold.

5) The 'saviour' had to be seen as a vaccine which beyond evil drug companies were working like angels of mercy to develop as quickly as possible, with all corners cut, to save the day. The public must absolutely not know that the 'vaccine' had nothing to do with a 'virus' or that the contents were ready and waiting with a very different motive long before the 'Covid' card was even lifted from the pack.

I said in March, 2020, that the 'vaccine' would have been created way ahead of the 'Covid' hoax which justified its use and the following December an article in the New York *Intelligencer* magazine said the Moderna 'vaccine' had been 'designed' by

January, 2020. This was 'before China had even acknowledged that the disease could be transmitted from human to human, more than a week before the first confirmed coronavirus case in the United States'. The article said that by the time the first American death was announced a month later 'the vaccine had already been manufactured and shipped to the National Institutes of Health for the beginning of its Phase I clinical trial'. The 'vaccine' was actually 'designed' long before that although even with this timescale you would expect the article to ask how on earth it could have been done that quickly. Instead it asked why the 'vaccine' had not been rolled out then and not months later. Journalism in the mainstream is truly dead. I am going to detail in the next chapter why the 'virus' has never existed and how a hoax on that scale was possible, but first the foundation on which the Big Lie of 'Covid' was built.

The test that doesn't test

Fraudulent 'testing' is the bottom line of the whole 'Covid' hoax and was the means by which a 'virus' that did not exist *appeared* to exist. They could only achieve this magic trick by using a test not testing for the 'virus'. To use a test that *was* testing for the 'virus' would mean that every test would come back negative given there was no 'virus'. They chose to exploit something called the RT-PCR test invented by American biochemist Kary Mullis in the 1980s who said publicly that his PCR test ... *cannot detect infectious disease*. Yes, the 'test' used worldwide to detect infectious 'Covid' to produce all the illusory 'cases' and 'deaths' compiled by Johns Hopkins and others *cannot detect infectious disease*. This fact came from the mouth of the man who invented PCR and was awarded the Nobel Prize in Chemistry in 1993 for doing so. Sadly, and incredibly conveniently for the Cult, Mullis died in August, 2019, at the age of 74 just before his test would be fraudulently used to unleash fascism on the world. He was said to have died from pneumonia which was an irony in itself. A few months later he would have had 'Covid-19' on his death certificate. I say the timing of his death was convenient because had he lived Mullis, a brilliant, honest and decent man, would have been

vociferously speaking out against the use of his test to detect 'Covid' when it was never designed, or able, to do that. I know that to be true given that Mullis made the same point when his test was used to 'detect' – not detect – HIV. He had been seriously critical of the Gallo/Montagnier claim to have isolated the HIV 'virus' and shown it to cause AIDS for which Mullis said there was no evidence. AIDS is actually not a disease but a series of diseases from which people die all the time. When they die from those *same diseases* after a positive 'test' for HIV then AIDS goes on their death certificate. I think I've heard that before somewhere. Countries instigated a policy with 'Covid' that anyone who tested positive with a test not testing for the 'virus' and died of any other cause within 28 days and even longer 'Covid-19' had to go on the death certificate. Cases have come from the test that can't test for infectious disease and the deaths are those who have died of *anything* after testing positive with a test not testing for the 'virus'. I'll have much more later about the death certificate scandal.

Mullis was deeply dismissive of the now US 'Covid' star Anthony Fauci who he said was a liar who didn't know anything about anything – 'and I would say that to his face – nothing.' He said of Fauci: 'The man thinks he can take a blood sample, put it in an electron microscope and if it's got a virus in there you'll know it – he doesn't understand electron microscopy and he doesn't understand medicine and shouldn't be in a position like he's in.' That position, terrifyingly, has made him the decider of 'Covid' fascism policy on behalf of the Cult in his role as director since 1984 of the National Institute of Allergy and Infectious Diseases (NIAID) while his record of being wrong is laughable; but being wrong, so long as it's the *right kind* of wrong, is why the Cult loves him. He'll say anything the Cult tells him to say. Fauci was made Chief Medical Adviser to the President immediately Biden took office. Biden was installed in the White House by Cult manipulation and one of his first decisions was to elevate Fauci to a position of even more control. This is a coincidence? Yes, and I identify as a flamenco dancer called Lola. How does such an incompetent criminal like Fauci remain in that

pivotal position in American health since *the 1980s*? When you serve the Cult it looks after you until you are surplus to requirements. Kary Mullis said prophetically of Fauci and his like: 'Those guys have an agenda and it's not an agenda we would like them to have ... they make their own rules, they change them when they want to, and Tony Fauci does not mind going on television in front of the people who pay his salary and lie directly into the camera.' Fauci has done that almost daily since the 'Covid' hoax began. Lying is in Fauci's DNA. To make the situation crystal clear about the PCR test this is a direct quote from its inventor Kary Mullis:

It [the PCR test] doesn't tell you that you're sick and doesn't tell you that the thing you ended up with was really going to hurt you ...'

Ask yourself why governments and medical systems the world over have been using this very test to decide who is 'infected' with the SARS-CoV-2 'virus' and the alleged disease it allegedly causes, 'Covid-19'. The answer to that question will tell you what has been going on. By the way, here's a little show-stopper – the 'new' SARS-CoV-2 'virus' was 'identified' as such right from the start using ... *the PCR test not testing for the 'virus'*. If you are new to this and find that shocking then stick around. I have hardly started yet. Even worse, other 'tests', like the 'Lateral Flow Device' (LFD), are considered so useless that they have to be *confirmed* by the PCR test! Leaked emails written by Ben Dyson, adviser to UK 'Health' Secretary Matt Hancock, said they were 'dangerously unreliable'. Dyson, executive director of strategy at the Department of Health, wrote: 'As of today, someone who gets a positive LFD result in (say) London has at best a 25 per cent chance of it being a true positive, but if it is a self-reported test potentially as low as 10 per cent (on an optimistic assumption about specificity) or as low as 2 per cent (on a more pessimistic assumption).' These are the 'tests' that schoolchildren and the public are being urged to have twice a week or more and have to isolate if they get a positive. Each fake positive goes in the statistics as a 'case' no matter how ludicrously inaccurate and the

'cases' drive lockdown, masks and the pressure to 'vaccinate'. The government said in response to the email leak that the 'tests' were accurate which confirmed yet again what shocking bloody liars they are. The real false positive rate is *100 percent* as we'll see. In another 'you couldn't make it up' the UK government agreed to pay £2.8 billion to California's Innova Medical Group to supply the irrelevant lateral flow tests. The company's primary test-making centre is in China. Innova Medical Group, established in March, 2020, is owned by Pasaca Capital Inc, chaired by Chinese-American millionaire Charles Huang who was born in Wuhan.

How it works – and how it doesn't

The RT-PCR test, known by its full title of Polymerase chain reaction, is used across the world to make millions, even billions, of copies of a DNA/RNA genetic information sample. The process is called 'amplification' and means that a tiny sample of genetic material is amplified to bring out the detailed content. I stress that it is not testing for an infectious disease. It is simply amplifying a sample of genetic material. In the words of Kary Mullis: 'PCR is ... just a process that's used to make a whole lot of something out of something.' To emphasise the point companies that make the PCR tests circulated around the world to 'test' for 'Covid' warn on the box that it can't be used to detect 'Covid' or infectious disease and is for research purposes only. It's okay, rest for a minute and you'll be fine. This is the test that produces the 'cases' and 'deaths' that have been used to destroy human society. All those global and national medical and scientific 'experts' demanding this destruction to 'save us' *KNOW* that the test is not testing for the 'virus' and the cases and deaths they claim to be real are an almost unimaginable fraud. Every one of them and so many others including politicians and psychopaths like Gates and Tedros must be brought before Nuremburg-type trials and jailed for the rest of their lives. The more the genetic sample is amplified by PCR the more elements of that material become sensitive to the test and by that I don't mean sensitive for a 'virus' but for elements of the genetic material which

is *naturally* in the body or relates to remnants of old conditions of various kinds lying dormant and causing no disease. Once the amplification of the PCR reaches a certain level *everyone* will test positive. So much of the material has been made sensitive to the test that everyone will have some part of it in their body. Even lying criminals like Fauci have said that once PCR amplifications pass 35 cycles everything will be a false positive that cannot be trusted for the reasons I have described. I say, like many proper doctors and scientists, that 100 percent of the 'positives' are false, but let's just go with Fauci for a moment.

He says that any amplification over 35 cycles will produce false positives and yet the US Centers for Disease Control (CDC) and Food and Drug Administration (FDA) have recommended up to 40 *cycles* and the National Health Service (NHS) in Britain admitted in an internal document for staff that it was using 45 *cycles* of amplification. A long list of other countries has been doing the same and at least one 'testing' laboratory has been using 50 *cycles*. Have you ever heard a doctor, medical 'expert' or the media ask what level of amplification has been used to claim a 'positive'. The 'test' comes back 'positive' and so you have the 'virus', end of story. Now we can see how the government in Tanzania could send off samples from a goat and a pawpaw fruit under human names and both came back positive for 'Covid-19'. Tanzania president John Magufuli mocked the 'Covid' hysteria, the PCR test and masks and refused to import the DNA-manipulating 'vaccine'. The Cult hated him and an article sponsored by the Bill Gates Foundation appeared in the London *Guardian* in February, 2021, headed 'It's time for Africa to rein in Tanzania's anti-vaxxer president'. Well, 'reined in' he shortly was. Magufuli appeared in good health, but then, in March, 2021, he was dead at 61 from 'heart failure'. He was replaced by Samia Hassan Suhulu who is connected to Klaus Schwab's World Economic Forum and she immediately reversed Magufuli's 'Covid' policy. A sample of cola tested positive for 'Covid' with the PCR test in Germany while American actress and singer-songwriter Erykah Badu tested positive in one nostril and negative in the other. Footballer Ronaldo called

the PCR test 'bullshit' after testing positive three times and being forced to quarantine and miss matches when there was nothing wrong with him. The mantra from Tedros at the World Health Organization and national governments (same thing) has been test, test, test. They know that the more tests they can generate the more fake 'cases' they have which go on to become 'deaths' in ways I am coming to. The UK government has its Operation Moonshot planned to test multiple millions every day in workplaces and schools with free tests for everyone to use twice a week at home in line with the Cult plan from the start to make testing part of life. A government advertisement for an 'Interim Head of Asymptomatic Testing Communication' said the job included responsibility for delivering a 'communications strategy' (propaganda) 'to support the expansion of asymptomatic testing that *'normalises testing as part of everyday life'*'. More tests means more fake 'cases', 'deaths' and fascism. I have heard of, and from, many people who booked a test, couldn't turn up, and yet got a positive result through the post for a test they'd never even had. The whole thing is crazy, but for the Cult there's method in the madness. Controlling and manipulating the level of amplification of the test means the authorities can control whenever they want the number of apparent 'cases' and 'deaths'. If they want to justify more fascist lockdown and destruction of livelihoods they keep the amplification high. If they want to give the illusion that lockdowns and the 'vaccine' are working then they lower the amplification and 'cases' and 'deaths' will appear to fall. In January, 2021, the Cult-owned World Health Organization suddenly warned laboratories about over-amplification of the test and to lower the threshold. Suddenly headlines began appearing such as: 'Why ARE "Covid" cases plummeting?' This was just when the vaccine rollout was underway and I had predicted months before they would make cases appear to fall through amplification tampering when the 'vaccine' came. These people are so predictable.

Cow vaccines?

The question must be asked of what is on the test swabs being poked far up the nose of the population to the base of the brain? A nasal swab punctured one woman's brain and caused it to leak fluid. Most of these procedures are being done by people with little training or medical knowledge. Dr Lorraine Day, former orthopaedic trauma surgeon and Chief of Orthopaedic Surgery at San Francisco General Hospital, says the tests are really a '*vaccine*'. Cows have long been vaccinated this way. She points out that masks have to cover the nose and the mouth where it is claimed the '*virus*' exists in saliva. Why then don't they take saliva from the mouth as they do with a DNA test instead of pushing a long swab up the nose towards the brain? The ethmoid bone separates the nasal cavity from the brain and within that bone is the cribriform plate. Dr Day says that when the swab is pushed up against this plate and twisted the procedure is '*depositing things back there*'. She claims that among these '*things*' are nanoparticles that can enter the brain. Researchers have noted that a team at the Gates-funded Johns Hopkins have designed tiny, star-shaped micro-devices that can latch onto intestinal mucosa and release drugs into the body. Mucosa is the thin skin that covers the inside surface of parts of the body such as *the nose* and mouth and produces mucus to protect them. The Johns Hopkins micro-devices are called '*theragrippers*' and were '*inspired*' by a parasitic worm that digs its sharp teeth into a host's intestines. Nasal swabs are also coated in the sterilisation agent ethylene oxide. The US National Cancer Institute posts this explanation on its website:

At room temperature, ethylene oxide is a flammable colorless gas with a sweet odor. It is used primarily to produce other chemicals, including antifreeze. In smaller amounts, ethylene oxide is used as a pesticide and a sterilizing agent. The ability of ethylene oxide to damage DNA makes it an effective sterilizing agent but also accounts for its cancer-causing activity.

The Institute mentions lymphoma and leukaemia as cancers most frequently reported to be associated with occupational exposure to ethylene oxide along with stomach and breast cancers. How does anyone think this is going to work out with the constant testing

regime being inflicted on adults and children at home and at school that will accumulate in the body anything that's on the swab?

Doctors know best

It is vital for people to realise that 'hero' doctors 'know' only what the Big Pharma-dominated medical authorities tell them to 'know' and if they refuse to 'know' what they are told to 'know' they are out the door. They are mostly not physicians or healers, but repeaters of the official narrative – or else. I have seen alleged professional doctors on British television make shocking statements that we are supposed to take seriously. One called 'Dr' Amir Khan, who is actually telling patients how to respond to illness, said that men could take the birth pill to 'help slow down the effects of Covid-19'. In March, 2021, another ridiculous 'Covid study' by an American doctor proposed injecting men with the female sex hormone progesterone as a 'Covid' treatment. British doctor Nighat Arif told the BBC that face coverings were now going to be part of ongoing normal. Yes, the vaccine protects you, she said (evidence?) ... but the way to deal with viruses in the community was always going to come down to hand washing, face covering and keeping a physical distance. That's not what we were told before the 'vaccine' was circulating. Arif said she couldn't imagine ever again going on the underground or in a lift without a mask. I was just thanking my good luck that she was not my doctor when she said – in March, 2021 – that if 'we are *behaving* and we are doing all the right things' she thought we could 'have our nearest and dearest around us at home ... around *Christmas* and *New Year*! Her patronising delivery was the usual school teacher talking to six-year-olds as she repeated every government talking point and probably believed them all. If we have learned anything from the 'Covid' experience surely it must be that humanity's perception of doctors needs a fundamental rethink. NHS 'doctor' Sara Kayat told her television audience that the 'Covid vaccine' would '100 percent prevent hospitalisation and death'. Not even Big Pharma claimed that. We have to stop taking 'experts' at their word without question when so many of them are

clueless and only repeating the party line on which their careers depend. That is not to say there are not brilliant doctors – there are and I have spoken to many of them since all this began – but you won't see them in the mainstream media or quoted by the psychopaths and yes-people in government.

Remember the name – Christian Drosten

German virologist Christian Drosten, Director of Charité Institute of Virology in Berlin, became a national star after the pandemic hoax began. He was feted on television and advised the German government on 'Covid' policy. Most importantly to the wider world Drosten led a group that produced the 'Covid' testing protocol for the PCR test. What a remarkable feat given the PCR cannot test for infectious disease and even more so when you think that Drosten said that his method of testing for SARS-CoV-2 was developed 'without having virus material available'. *He developed a test for a 'virus' that he didn't have and had never seen.* Let that sink in as you survey the global devastation that came from what he did. The whole catastrophe of Drosten's 'test' was based on the alleged genetic sequence published by Chinese scientists on the Internet. We will see in the next chapter that this alleged 'genetic sequence' has never been produced by China or anyone and cannot be when there *is no* SARS-CoV-2. Drosten, however, doesn't seem to let little details like that get in the way. He was the lead author with Victor Corman from the same Charité Hospital of the paper 'Detection of 2019 novel coronavirus (2019-nCoV) by real-time PCR' published in a magazine called *Eurosurveillance*. This became known as the Corman-Drosten paper. In November, 2020, with human society devastated by the effects of the Corman-Drosten test baloney, the protocol was publicly challenged by 22 international scientists and independent researchers from Europe, the United States, and Japan. Among them were senior molecular geneticists, biochemists, immunologists, and microbiologists. They produced a document headed 'External peer review of the RTPCR test to detect SARS-Cov-2 Reveals 10 Major Flaws At The Molecular and Methodological Level: Consequences

For False-Positive Results'. The flaws in the Corman-Drosten test included the following:

- The test is non-specific because of erroneous design
- Results are enormously variable
- The test is unable to discriminate between the whole 'virus' and viral fragments
- It doesn't have positive or negative controls
- The test lacks a standard operating procedure
- It is unsupported by proper peer view

The scientists said the PCR 'Covid' testing protocol was not founded on science and they demanded the Corman-Drosten paper be retracted by *Eurosurveillance*. They said all present and previous Covid deaths, cases, and 'infection rates' should be subject to a massive retroactive inquiry. Lockdowns and travel restrictions should be reviewed and relaxed and those diagnosed through PCR to have 'Covid-19' should not be forced to isolate. Dr Kevin Corbett, a health researcher and nurse educator with a long academic career producing a stream of peer-reviewed publications at many UK universities, made the same point about the PCR test debacle. He said of the scientists' conclusions: 'Every scientific rationale for the development of that test has been totally destroyed by this paper. It's like Hiroshima/Nagasaki to the Covid test.' He said that China hadn't given them an isolated 'virus' when Drosten developed the test. Instead they had developed the test from *a sequence in a gene bank*.' Put another way ... *they made it up!* The scientists were supported in this contention by a Portuguese appeals court which ruled in November, 2020, that PCR tests are unreliable and it is unlawful to quarantine people based solely on a PCR test. The point about China not providing an isolated virus must be true when the 'virus' has never been isolated to this day and the consequences of that will become clear. Drosten and company produced this useless 'protocol' right on cue in January, 2020, just as the 'virus' was said to

be moving westward and it somehow managed to successfully pass a peer-review in 24 hours. In other words there was no peer-review for a test that would be used to decide who had 'Covid' and who didn't across the world. The Cult-created, Gates-controlled World Health Organization immediately recommended all its nearly 200 member countries to use the Drosten PCR protocol to detect 'cases' and 'deaths'. The sting was underway and it continues to this day.

So who is this Christian Drosten that produced the means through which death, destruction and economic catastrophe would be justified? His education background, including his doctoral thesis, would appear to be somewhat shrouded in mystery and his track record is dire as with another essential player in the 'Covid' hoax, the Gates-funded Professor Neil Ferguson at the Gates-funded Imperial College in London of whom more shortly. Drosten predicted in 2003 that the alleged original SARS 'virus' (SARS-1) was an epidemic that could have serious effects on economies and an effective vaccine would take at least two years to produce. Drosten's answer to every alleged 'outbreak' is a vaccine which you won't be shocked to know. What followed were just 774 official deaths worldwide and none in Germany where there were only nine cases. That is even if you believe there ever was a SARS 'virus' when the evidence is zilch and I will expand on this in the next chapter. Drosten claims to be co-discoverer of 'SARS-1' and developed a test for it in 2003. He was screaming warnings about 'swine flu' in 2009 and how it was a widespread infection far more severe than any dangers from a vaccine could be and people should get vaccinated. It would be helpful for Drosten's vocal chords if he simply recorded the words 'the virus is deadly and you need to get vaccinated' and copies could be handed out whenever the latest made-up threat comes along. Drosten's swine flu epidemic never happened, but Big Pharma didn't mind with governments spending hundreds of millions on vaccines that hardly anyone bothered to use and many who did wished they hadn't. A study in 2010 revealed that the risk of dying from swine flu, or H1N1, was no higher than that of the annual seasonal flu which is what at least most of 'it' really was as in

the case of 'Covid-19'. A media investigation into Drosten asked how with such a record of inaccuracy he could be *the* government adviser on these issues. The answer to that question is the same with Drosten, Ferguson and Fauci – they keep on giving the authorities the 'conclusions' and 'advice' they want to hear. Drosten certainly produced the goods for them in January, 2020, with his PCR protocol garbage and provided the foundation of what German internal medicine specialist Dr Claus Köhnlein, co-author of *Virus Mania*, called the 'test pandemic'. The 22 scientists in the *Eurosurveillance* challenge called out conflicts of interest within the Drosten 'protocol' group and with good reason. Olfert Landt, a regular co-author of Drosten 'studies', owns the biotech company TIB Molbiol Syntheselabor GmbH in Berlin which manufactures and sells the tests that Drosten and his mates come up with. They have done this with SARS, Enterotoxigenic E. coli (ETEC), MERS, Zika 'virus', yellow fever, and now 'Covid'. Landt told the *Berliner Zeitung* newspaper:

The testing, design and development came from the Charité [Drosten and Corman]. We simply implemented it immediately in the form of a kit. And if we don't have the virus, which originally only existed in Wuhan, we can make a synthetic gene to simulate the genome of the virus. That's what we did very quickly.

This is more confirmation that the Drosten test was designed without access to the 'virus' and only a synthetic simulation which is what SARS-CoV-2 really is – a computer-generated synthetic fiction. It's quite an enterprise they have going here. A Drosten team decides what the test for something should be and Landt's biotech company flogs it to governments and medical systems across the world. His company must have made an absolute fortune since the 'Covid' hoax began. Dr Reiner Fuellmich, a prominent German consumer protection trial lawyer in Germany and California, is on Drosten's case and that of Tedros at the World Health Organization for crimes against humanity with a class-action lawsuit being prepared in the United States and other legal action in Germany.

Why China?

Scamming the world with a 'virus' that doesn't exist would seem impossible on the face of it, but not if you have control of the relatively few people that make policy decisions and the great majority of the global media. Remember it's not about changing 'real' reality it's about controlling *perception* of reality. You don't have to make something happen you only have to make people *believe* that it's happening. Renegade Minds understand this and are therefore much harder to swindle. 'Covid-19' is not a 'real' 'virus'. It's a mind virus, like a computer virus, which has infected the minds, not the bodies, of billions. It all started, publically at least, in China and that alone is of central significance. The Cult was behind the revolution led by its asset Mao Zedong, or Chairman Mao, which established the People's Republic of China on October 1st, 1949. It should have been called The Cult's Republic of China, but the name had to reflect the recurring illusion that vicious dictatorships are run by and for the people (see all the 'Democratic Republics' controlled by tyrants). In the same way we have the 'Biden' Democratic Republic of America officially ruled by a puppet tyrant (at least temporarily) on behalf of Cult tyrants. The creation of Mao's merciless communist/fascist dictatorship was part of a frenzy of activity by the Cult at the conclusion of World War Two which, like the First World War, it had instigated through its assets in Germany, Britain, France, the United States and elsewhere. Israel was formed in 1948; the Soviet Union expanded its 'Iron Curtain' control, influence and military power with the Warsaw Pact communist alliance in 1955; the United Nations was formed in 1945 as a Cult precursor to world government; and a long list of world bodies would be established including the World Health Organization (1948), World Trade Organization (1948 under another name until 1995), International Monetary Fund (1945) and World Bank (1944). Human society was redrawn and hugely centralised in the global Problem-Reaction-Solution that was World War Two. All these changes were significant. Israel would become the headquarters of the Sabbatians

and the revolution in China would prepare the ground and control system for the events of 2019/2020.

Renegade Minds know there are no borders except for public consumption. The Cult is a seamless, borderless global entity and to understand the game we need to put aside labels like borders, nations, countries, communism, fascism and democracy. These delude the population into believing that countries are ruled within their borders by a government of whatever shade when these are mere agencies of a global power. America's illusion of democracy and China's communism/fascism are subsidiaries – vehicles – for the same agenda. We may hear about conflict and competition between America and China and on the lower levels that will be true; but at the Cult level they are branches of the same company in the way of the McDonald's example I gave earlier. I have tracked in the books over the years support by US governments of both parties for Chinese Communist Party infiltration of American society through allowing the sale of land, even military facilities, and the acquisition of American business and university influence. All this is underpinned by the infamous stealing of intellectual property and technological know-how. Cult-owned Silicon Valley corporations waive their fraudulent 'morality' to do business with human-rights-free China; Cult-controlled Disney has become China's PR department; and China in effect owns 'American' sports such as basketball which depends for much of its income on Chinese audiences. As a result any sports player, coach or official speaking out against China's horrific human rights record is immediately condemned or fired by the China-worshipping National Basketball Association. One of the first acts of China-controlled Biden was to issue an executive order telling federal agencies to stop making references to the 'virus' by the 'geographic location of its origin'. Long-time Congressman Jerry Nadler warned that criticising China, America's biggest rival, leads to hate crimes against Asian people in the United States. So shut up you bigot. China is fast closing in on Israel as a country that must not be criticised which is apt, really, given that Sabbatians control them both. The two countries have

developed close economic, military, technological and strategic ties which include involvement in China's 'Silk Road' transport and economic initiative to connect China with Europe. Israel was the first country in the Middle East to recognise the establishment of Mao's tyranny in 1950 months after it was established.

Project Wuhan – the 'Covid' Psyop

I emphasise again that the Cult plays the long game and what is happening to the world today is the result of centuries of calculated manipulation following a script to take control step-by-step of every aspect of human society. I will discuss later the common force behind all this that has spanned those centuries and thousands of years if the truth be told. Instigating the Mao revolution in China in 1949 with a 2020 'pandemic' in mind is not only how they work – the 71 years between them is really quite short by the Cult's standards of manipulation preparation. The reason for the Cult's Chinese revolution was to create a fiercely-controlled environment within which an extreme structure for human control could be incubated to eventually be unleashed across the world. We have seen this happen since the 'pandemic' emerged from China with the Chinese control-structure founded on AI technology and tyrannical enforcement sweep across the West. Until the moment when the Cult went for broke in the West and put its fascism on public display Western governments had to pay some lip-service to freedom and democracy to not alert too many people to the tyranny-in-the-making. Freedoms were more subtly eroded and power centralised with covert government structures put in place waiting for the arrival of 2020 when that smokescreen of 'freedom' could be dispensed with. The West was not able to move towards tyranny before 2020 anything like as fast as China which was created as a tyranny and had no limits on how fast it could construct the Cult's blueprint for global control. When the time came to impose that structure on the world it was the same Cult-owned Chinese communist/fascist government that provided the excuse – the 'Covid pandemic'. It was absolutely crucial to the Cult plan for the Chinese response to the 'pandemic' –

draconian lockdowns of the entire population – to become the blueprint that Western countries would follow to destroy the livelihoods and freedom of their people. This is why the Cult-owned, Gates-owned, WHO Director-General Tedros said early on:

The Chinese government is to be congratulated for the extraordinary measures it has taken to contain the outbreak. China is actually setting a new standard for outbreak response and it is not an exaggeration.

Forbes magazine said of China: ‘... those measures protected untold millions from getting the disease’. The Rockefeller Foundation ‘epidemic scenario’ document in 2010 said ‘prophetically’:

However, a few countries did fare better – China in particular. The Chinese government’s quick imposition and enforcement of mandatory quarantine for all citizens, as well as its instant and near-hermetic sealing off of all borders, saved millions of lives, stopping the spread of the virus far earlier than in other countries and enabling a swifter post-pandemic recovery.

Once again – *spooky*.

The first official story was the ‘bat theory’ or rather the bat diversion. The source of the ‘virus outbreak’ we were told was a ‘wet market’ in Wuhan where bats and other animals are bought and eaten in horrifically unhygienic conditions. Then another story emerged through the alternative media that the ‘virus’ had been released on purpose or by accident from a BSL-4 (biosafety level 4) laboratory in Wuhan not far from the wet market. The lab was reported to create and work with lethal concoctions and bioweapons. Biosafety level 4 is the highest in the World Health Organization system of safety and containment. Renegade Minds are aware of what I call designer manipulation. The ideal for the Cult is for people to buy its prime narrative which in the opening salvos of the ‘pandemic’ was the wet market story. It knows, however, that there is now a considerable worldwide alternative media of researchers sceptical of anything governments say and they are often given a version of events in a form they can perceive as credible while misdirecting them from the real truth. In this case let them

think that the conspiracy involved is a 'bioweapon virus' released from the Wuhan lab to keep them from the real conspiracy – *there is no 'virus'*. The WHO's current position on the source of the outbreak at the time of writing appears to be: 'We haven't got a clue, mate.' This is a good position to maintain mystery and bewilderment. The inner circle will know where the 'virus' came from – *nowhere*. The bottom line was to ensure the public believed there *was* a 'virus' and it didn't much matter if they thought it was natural or had been released from a lab. The belief that there was a 'deadly virus' was all that was needed to trigger global panic and fear. The population was terrified into handing their power to authority and doing what they were told. They had to or they were 'all gonna die'.

In March, 2020, information began to come my way from real doctors and scientists and my own additional research which had my intuition screaming: 'Yes, that's it! *There is no virus.*' The 'bioweapon' was not the 'virus'; it was the '*vaccine*' already being talked about that would be the bioweapon. My conclusion was further enhanced by happenings in Wuhan. The 'virus' was said to be sweeping the city and news footage circulated of people collapsing in the street (which they've never done in the West with the same 'virus'). The Chinese government was building 'new hospitals' in a matter of ten days to 'cope with demand' such was the virulent nature of the 'virus'. Yet in what seemed like no time the 'new hospitals' closed – even if they even opened – and China declared itself 'virus-free'. It was back to business as usual. This was more propaganda to promote the Chinese draconian lockdowns in the West as the way to 'beat the virus'. Trouble was that we subsequently had lockdown after lockdown, but never business as usual. As the people of the West and most of the rest of the world were caught in an ever-worsening spiral of lockdown, social distancing, masks, isolated old people, families forced apart, and livelihood destruction, it was party-time in Wuhan. Pictures emerged of thousands of people enjoying pool parties and concerts. It made no sense until you realised there never was a 'virus' and the

whole thing was a Cult set-up to transform human society out of one of its major global strongholds – China.

How is it possible to deceive virtually the entire world population into believing there is a deadly virus when there is not even a 'virus' let alone a deadly one? It's nothing like as difficult as you would think and that's clearly true because it happened.

Postscript: See end of book Postscript for more on the 'Wuhan lab virus release' story which the authorities and media were pushing heavily in the summer of 2021 to divert attention from the truth that the 'Covid virus' is pure invention.

CHAPTER FIVE

There is no 'virus'

You can fool some of the people all of the time, and all of the people some of the time, but you cannot fool all of the people all of the time
Abraham Lincoln

The greatest form of mind control is repetition. The more you repeat the same mantra of alleged 'facts' the more will accept them to be true. It becomes an 'everyone knows that, mate'. If you can also censor any other version or alternative to your alleged 'facts' you are pretty much home and cooking.

By the start of 2020 the Cult owned the global mainstream media almost in its entirety to spew out its 'Covid' propaganda and ignore or discredit any other information and view. Cult-owned social media platforms in Cult-owned Silicon Valley were poised and ready to unleash a campaign of ferocious censorship to obliterate all but the official narrative. To complete the circle many demands for censorship by Silicon Valley were led by the mainstream media as 'journalists' became full-out enforcers for the Cult both as propagandists and censors. Part of this has been the influx of young people straight out of university who have become 'journalists' in significant positions. They have no experience and a headful of programmed perceptions from their years at school and university at a time when today's young are the most perceptually-targeted generations in known human history given the insidious impact of technology. They enter the media perceptually prepared and ready to repeat the narratives of the system that programmed them to

repeat its narratives. The BBC has a truly pathetic 'specialist disinformation reporter' called Marianna Spring who fits this bill perfectly. She is clueless about the world, how it works and what is really going on. Her role is to discredit anyone doing the job that a proper journalist would do and system-serving hacks like Spring wouldn't dare to do or even see the need to do. They are too busy licking the arse of authority which can never be wrong and, in the case of the BBC propaganda programme, *Panorama*, contacting payments systems such as PayPal to have a donations page taken down for a film company making documentaries questioning vaccines. Even the BBC soap opera *EastEnders* included a disgracefully biased scene in which an inarticulate white working class woman was made to look foolish for questioning the 'vaccine' while a well-spoken black man and Asian woman promoted the government narrative. It ticked every BBC box and the fact that the black and minority community was resisting the 'vaccine' had nothing to do with the way the scene was written. The BBC has become a disgusting tyrannical propaganda and censorship operation that should be defunded and disbanded and a free media take its place with a brief to stop censorship instead of demanding it. A BBC 'interview' with Gates goes something like: 'Mr Gates, sir, if I can call you sir, would you like to tell our audience why you are such a great man, a wonderful humanitarian philanthropist, and why you should absolutely be allowed as a software salesman to decide health policy for approaching eight billion people? Thank you, sir, please sir.' Propaganda programming has been incessant and merciless and when all you hear is the same story from the media, repeated by those around you who have only heard the same story, is it any wonder that people on a grand scale believe absolute mendacious garbage to be true? You are about to see, too, why this level of information control is necessary when the official 'Covid' narrative is so nonsensical and unsupportable by the evidence.

Structure of Deceit

The pyramid structure through which the 'Covid' hoax has been manifested is very simple and has to be to work. As few people as possible have to be involved with full knowledge of what they are doing – and why – or the real story would get out. At the top of the pyramid are the inner core of the Cult which controls Bill Gates who, in turn, controls the World Health Organization through his pivotal funding and his puppet Director-General mouthpiece, Tedros. Before he was appointed Tedros was chair of the Gates-founded Global Fund to 'fight against AIDS, tuberculosis and malaria', a board member of the Gates-funded 'vaccine alliance' GAVI, and on the board of another Gates-funded organisation. Gates owns him and picked him for a specific reason – Tedros is a crook and worse. 'Dr' Tedros (he's not a medical doctor, the first WHO chief not to be) was a member of the tyrannical Marxist government of Ethiopia for decades with all its human rights abuses. He has faced allegations of corruption and misappropriation of funds and was exposed three times for covering up cholera epidemics while Ethiopia's health minister. Tedros appointed the mass-murdering genocidal Zimbabwe dictator Robert Mugabe as a WHO goodwill ambassador for public health which, as with Tedros, is like appointing a psychopath to run a peace and love campaign. The move was so ridiculous that he had to drop Mugabe in the face of widespread condemnation. American economist David Steinman, a Nobel peace prize nominee, lodged a complaint with the International Criminal Court in The Hague over alleged genocide by Tedros when he was Ethiopia's foreign minister. Steinman says Tedros was a 'crucial decision maker' who directed the actions of Ethiopia's security forces from 2013 to 2015 and one of three officials in charge when those security services embarked on the 'killing' and 'torturing' of Ethiopians. You can see where Tedros is coming from and it's sobering to think that he has been the vehicle for Gates and the Cult to direct the global response to 'Covid'. Think about that. A psychopathic Cult dictates to psychopath Gates who dictates to psychopath Tedros who dictates how countries of the world must respond to a 'Covid virus' never scientifically shown to exist. At the same time psychopathic Cult-owned Silicon Valley information

giants like Google, YouTube, Facebook and Twitter announced very early on that they would give the Cult/Gates/Tedros/WHO version of the narrative free advertising and censor those who challenged their intelligence-insulting, mendacious story.

The next layer in the global 'medical' structure below the Cult, Gates and Tedros are the chief medical officers and science 'advisers' in each of the WHO member countries which means virtually all of them. Medical officers and arbiters of science (they're not) then take the WHO policy and recommended responses and impose them on their country's population while the political 'leaders' say they are deciding policy (they're clearly not) by 'following the science' on the advice of the 'experts' – the same medical officers and science 'advisers' (dictators). In this way with the rarest of exceptions the entire world followed the same policy of lockdown, people distancing, masks and 'vaccines' dictated by the psychopathic Cult, psychopathic Gates and psychopathic Tedros who we are supposed to believe give a damn about the health of the world population they are seeking to enslave. That, amazingly, is all there is to it in terms of crucial decision-making. Medical staff in each country then follow like sheep the dictates of the shepherds at the top of the national medical hierarchies – chief medical officers and science 'advisers' who themselves follow like sheep the shepherds of the World Health Organization and the Cult. Shepherds at the national level often have major funding and other connections to Gates and his Bill and Melinda Gates Foundation which carefully hands out money like confetti at a wedding to control the entire global medical system from the WHO down.

Follow the money

Christopher Whitty, Chief Medical Adviser to the UK Government at the centre of 'virus' policy, a senior adviser to the government's Scientific Advisory Group for Emergencies (SAGE), and Executive Board member of the World Health Organization, was gifted a grant of \$40 million by the Bill and Melinda Gates Foundation for malaria research in Africa. The BBC described the unelected Whitty as 'the

official who will probably have the greatest impact on our everyday lives of any individual policymaker in modern times' and so it turned out. What Gates and Tedros have said Whitty has done like his equivalents around the world. Patrick Vallance, co-chair of SAGE and the government's Chief Scientific Adviser, is a former executive of Big Pharma giant GlaxoSmithKline with its fundamental financial and business connections to Bill Gates. In September, 2020, it was revealed that Vallance owned a deferred bonus of shares in GlaxoSmithKline worth £600,000 while the company was 'developing' a 'Covid vaccine'. Move along now – nothing to see here – what could possibly be wrong with that? Imperial College in London, a major player in 'Covid' policy in Britain and elsewhere with its 'Covid-19' Response Team, is funded by Gates and has big connections to China while the now infamous Professor Neil Ferguson, the useless 'computer modeller' at Imperial College is also funded by Gates. Ferguson delivered the dramatically inaccurate excuse for the first lockdowns (much more in the next chapter). The Institute for Health Metrics and Evaluation (IHME) in the United States, another source of outrageously false 'Covid' computer models to justify lockdowns, is bankrolled by Gates who is a vehement promotor of lockdowns. America's version of Whitty and Vallance, the again now infamous Anthony Fauci, has connections to 'Covid vaccine' maker Moderna as does Bill Gates through funding from the Bill and Melinda Gates Foundation. Fauci is director of the National Institute of Allergy and Infectious Diseases (NIAID), a major recipient of Gates money, and they are very close. Deborah Birx who was appointed White House Coronavirus Response Coordinator in February, 2020, is yet another with ties to Gates. Everywhere you look at the different elements around the world behind the coordination and decision making of the 'Covid' hoax there is Bill Gates and his money. They include the World Health Organization; Centers for Disease Control (CDC) in the United States; National Institutes of Health (NIH) of Anthony Fauci; Imperial College and Neil Ferguson; the London School of Hygiene where Chris Whitty worked; Regulatory agencies like the UK Medicines & Healthcare products Regulatory Agency (MHRA)

which gave emergency approval for 'Covid vaccines'; Wellcome Trust; GAVI, the Vaccine Alliance; the Coalition for Epidemic Preparedness Innovations (CEPI); Johns Hopkins University which has compiled the false 'Covid' figures; and the World Economic Forum. A [Nationalfile.com](https://www.nationalfile.com) article said:

Gates has a lot of pull in the medical world, he has a multi-million dollar relationship with Dr. Fauci, and Fauci originally took the Gates line supporting vaccines and casting doubt on [the drug hydroxychloroquine]. Coronavirus response team member Dr. Deborah Birx, appointed by former president Obama to serve as United States Global AIDS Coordinator, also sits on the board of a group that has received billions from Gates' foundation, and Birx reportedly used a disputed Bill Gates-funded model for the White House's Coronavirus effort. Gates is a big proponent for a population lockdown scenario for the Coronavirus outbreak.

Another funder of Moderna is the Defense Advanced Research Projects Agency (DARPA), the technology-development arm of the Pentagon and one of the most sinister organisations on earth. DARPA had a major role with the CIA covert technology-funding operation In-Q-Tel in the development of Google and social media which is now at the centre of global censorship. Fauci and Gates are extremely close and openly admit to talking regularly about 'Covid' policy, but then why wouldn't Gates have a seat at every national 'Covid' table after his Foundation committed \$1.75 billion to the 'fight against Covid-19'. When passed through our Orwellian Translation Unit this means that he has bought and paid for the Cult-driven 'Covid' response worldwide. Research the major 'Covid' response personnel in your own country and you will find the same Gates funding and other connections again and again. Medical and science chiefs following World Health Organization 'policy' sit atop a medical hierarchy in their country of administrators, doctors and nursing staff. These 'subordinates' are told they must work and behave in accordance with the policy delivered from the 'top' of the national 'health' pyramid which is largely the policy delivered by the WHO which is the policy delivered by Gates and the Cult. The whole 'Covid' narrative has been imposed on medical staff by a climate of fear although great numbers don't even need that to comply. They do so through breathtaking levels of ignorance and

include doctors who go through life simply repeating what Big Pharma and their hierarchical masters tell them to say and believe. No wonder Big Pharma 'medicine' is one of the biggest killers on Planet Earth.

The same top-down system of intimidation operates with regard to the Cult Big Pharma cartel which also dictates policy through national and global medical systems in this way. The Cult and Big Pharma agendas are the same because the former controls and owns the latter. 'Health' administrators, doctors, and nursing staff are told to support and parrot the dictated policy or they will face consequences which can include being fired. How sad it's been to see medical staff meekly repeating and imposing Cult policy without question and most of those who can see through the deceit are only willing to speak anonymously off the record. They know what will happen if their identity is known. This has left the courageous few to expose the lies about the 'virus', face masks, overwhelmed hospitals that aren't, and the dangers of the 'vaccine' that isn't a vaccine. When these medical professionals and scientists, some renowned in their field, have taken to the Internet to expose the truth their articles, comments and videos have been deleted by Cult-owned Facebook, Twitter and YouTube. What a real head-shaker to see YouTube videos with leading world scientists and highly qualified medical specialists with an added link underneath to the notorious Cult propaganda website *Wikipedia* to find the 'facts' about the same subject.

HIV – the 'Covid' trial-run

I'll give you an example of the consequences for health and truth that come from censorship and unquestioning belief in official narratives. The story was told by PCR inventor Kary Mullis in his book *Dancing Naked in the Mind Field*. He said that in 1984 he accepted as just another scientific fact that Luc Montagnier of France's Pasteur Institute and Robert Gallo of America's National Institutes of Health had independently discovered that a 'retrovirus' dubbed HIV (human immunodeficiency virus) caused AIDS. They

were, after all, Mullis writes, specialists in retroviruses. This is how the medical and science pyramids work. Something is announced or *assumed* and then becomes an everybody-knows-that purely through repetition of the assumption as if it is fact. Complete crap becomes accepted truth with no supporting evidence and only repetition of the crap. This is how a 'virus' that doesn't exist became the 'virus' that changed the world. The HIV-AIDS fairy story became a multi-billion pound industry and the media poured out propaganda terrifying the world about the deadly HIV 'virus' that caused the lethal AIDS. By then Mullis was working at a lab in Santa Monica, California, to detect retroviruses with his PCR test in blood donations received by the Red Cross. In doing so he asked a virologist where he could find a reference for HIV being the cause of AIDS. 'You don't need a reference,' the virologist said ... '*Everybody knows it.*' Mullis said he wanted to quote a reference in the report he was doing and he said he felt a little funny about not knowing the source of such an important discovery when everyone else seemed to. The virologist suggested he cite a report by the Centers for Disease Control and Prevention (CDC) on morbidity and mortality. Mullis read the report, but it only said that an organism had been identified and did not say how. The report did not identify the original scientific work. Physicians, however, *assumed* (key recurring theme) that if the CDC was convinced that HIV caused AIDS then proof must exist. Mullis continues:

I did computer searches. Neither Montagnier, Gallo, nor anyone else had published papers describing experiments which led to the conclusion that HIV probably caused AIDS. I read the papers in *Science* for which they had become well known as AIDS doctors, but all they had said there was that they had found evidence of a past infection by something which was probably HIV in some AIDS patients.

They found antibodies. Antibodies to viruses had always been considered evidence of past disease, not present disease. Antibodies signaled that the virus had been defeated. The patient had saved himself. There was no indication in these papers that this virus caused a disease. They didn't show that everybody with the antibodies had the disease. In fact they found some healthy people with antibodies.

Mullis asked why their work had been published if Montagnier and Gallo hadn't really found this evidence, and why had they been fighting so hard to get credit for the discovery? He says he was hesitant to write 'HIV is the probable cause of AIDS' until he found published evidence to support that. 'Tens of thousands of scientists and researchers were spending billions of dollars a year doing research based on this idea,' Mullis writes. 'The reason had to be there somewhere; otherwise these people would not have allowed their research to settle into one narrow channel of investigation.' He said he lectured about PCR at numerous meetings where people were always talking about HIV and he asked them how they knew that HIV was the cause of AIDS:

Everyone said something. Everyone had the answer at home, in the office, in some drawer. They all knew, and they would send me the papers as soon as they got back. But I never got any papers. Nobody ever sent me the news about how AIDS was caused by HIV.

Eventually Mullis was able to ask Montagnier himself about the reference proof when he lectured in San Diego at the grand opening of the University of California AIDS Research Center. Mullis says this was the last time he would ask his question without showing anger. Montagnier said he should reference the CDC report. 'I read it', Mullis said, and it didn't answer the question. 'If Montagnier didn't know the answer who the hell did?' Then one night Mullis was driving when an interview came on National Public Radio with Peter Duesberg, a prominent virologist at Berkeley and a California Scientist of the Year. Mullis says he finally understood why he could not find references that connected HIV to AIDS – *there weren't any!* No one had ever proved that HIV causes AIDS even though it had spawned a multi-billion pound global industry and the media was repeating this as fact every day in their articles and broadcasts terrifying the shit out of people about AIDS and giving the impression that a positive test for HIV (see 'Covid') was a death sentence. Duesberg was a threat to the AIDS gravy train and the agenda that underpinned it. He was therefore abused and castigated after he told the Proceedings of the National Academy of Sciences

there was no good evidence implicating the new 'virus'. Editors rejected his manuscripts and his research funds were deleted. Mullis points out that the CDC has defined AIDS as one of more than 30 diseases *if accompanied* by a positive result on a test that detects antibodies to HIV; but those same diseases are not defined as AIDS cases when antibodies are not detected:

If an HIV-positive woman develops uterine cancer, for example, she is considered to have AIDS. If she is not HIV positive, she simply has uterine cancer. An HIV-positive man with tuberculosis has AIDS; if he tests negative he simply has tuberculosis. If he lives in Kenya or Colombia, where the test for HIV antibodies is too expensive, he is simply presumed to have the antibodies and therefore AIDS, and therefore he can be treated in the World Health Organization's clinic. It's the only medical help available in some places. And it's free, because the countries that support WHO are worried about AIDS.

Mullis accuses the CDC of continually adding new diseases (see ever more 'Covid symptoms') to the grand AIDS definition and of virtually doctoring the books to make it appear as if the disease continued to spread. He cites how in 1993 the CDC enormously broadened its AIDS definition and county health authorities were delighted because they received \$2,500 per year from the Federal government for every reported AIDS case. Ladies and gentlemen, I have just described, via Kary Mullis, the 'Covid pandemic' of 2020 and beyond. Every element is the same and it's been pulled off in the same way by the same networks.

The 'Covid virus' exists? Okay – prove it. Er ... still waiting

What Kary Mullis described with regard to 'HIV' has been repeated with 'Covid'. A claim is made that a new, or 'novel', infection has been found and the entire medical system of the world repeats that as fact exactly as they did with HIV and AIDS. No one in the mainstream asks rather relevant questions such as 'How do you know?' and 'Where is your proof?' The SARS-Cov-2 'virus' and the 'Covid-19 disease' became an overnight 'everybody-knows-that'. The origin could be debated and mulled over, but what you could not suggest was that 'SARS-Cov-2' didn't exist. That would be

ridiculous. 'Everybody knows' the 'virus' exists. Well, I didn't for one along with American proper doctors like Andrew Kaufman and Tom Cowan and long-time American proper journalist Jon Rappaport. We dared to pursue the obvious and simple question: 'Where's the evidence?' The overwhelming majority in medicine, journalism and the general public did not think to ask that. After all, *everyone knew* there was a new 'virus'. Everyone was saying so and I heard it on the BBC. Some would eventually argue that the 'deadly virus' was nothing like as deadly as claimed, but few would venture into the realms of its very existence. Had they done so they would have found that the evidence for that claim had gone AWOL as with HIV causes AIDS. In fact, not even that. For something to go AWOL it has to exist in the first place and scientific proof for a 'SARS-Cov-2' can be filed under nothing, nowhere and zilch.

Dr Andrew Kaufman is a board-certified forensic psychiatrist in New York State, a Doctor of Medicine and former Assistant Professor and Medical Director of Psychiatry at SUNY Upstate Medical University, and Medical Instructor of Hematology and Oncology at the Medical School of South Carolina. He also studied biology at the Massachusetts Institute of Technology (MIT) and trained in Psychiatry at Duke University. Kaufman is retired from allopathic medicine, but remains a consultant and educator on natural healing. I saw a video of his very early on in the 'Covid' hoax in which he questioned claims about the 'virus' in the absence of any supporting evidence and with plenty pointing the other way. I did everything I could to circulate his work which I felt was asking the pivotal questions that needed an answer. I can recommend an excellent pull-together interview he did with the website The Last Vagabond entitled *Dr Andrew Kaufman: Virus Isolation, Terrain Theory and Covid-19* and his website is andrewkaufmanmd.com. Kaufman is not only a forensic psychiatrist; he is forensic in all that he does. He always reads original scientific papers, experiments and studies instead of second-third-fourth-hand reports about the 'virus' in the media which are repeating the repeated repetition of the narrative. When he did so with the original Chinese 'virus' papers Kaufman

realised that there was no evidence of a 'SARS-Cov-2'. They had never – from the start – shown it to exist and every repeat of this claim worldwide was based on the accepted existence of proof that was nowhere to be found – see Kary Mullis and HIV. Here we go again.

Let's postulate

Kaufman discovered that the Chinese authorities immediately concluded that the cause of an illness that broke out among about 200 initial patients in Wuhan was a 'new virus' when there were no grounds to make that conclusion. The alleged 'virus' was not isolated from other genetic material in their samples and then shown through a system known as Koch's postulates to be the causative agent of the illness. The world was told that the SARS-Cov-2 'virus' caused a disease they called 'Covid-19' which had 'flu-like' symptoms and could lead to respiratory problems and pneumonia. If it wasn't so tragic it would almost be funny. *'Flu-like' symptoms? Pneumonia? Respiratory disease?* What in *CHINA* and particularly in *Wuhan*, one of the most polluted cities in the world with a resulting epidemic of respiratory disease?? Three hundred thousand people get pneumonia in China every year and there are nearly a billion cases worldwide of 'flu-like symptoms'. These have a whole range of causes – including pollution in Wuhan – but no other possibility was credibly considered in late 2019 when the world was told there was a new and deadly 'virus'. The global prevalence of pneumonia and 'flu-like systems' gave the Cult networks unlimited potential to re-diagnose these other causes as the mythical 'Covid-19' and that is what they did from the very start. Kaufman revealed how Chinese medical and science authorities (all subordinates to the Cult-owned communist government) took genetic material from the lungs of only a few of the first patients. The material contained their own cells, bacteria, fungi and other microorganisms living in their bodies. The only way you could prove the existence of the 'virus' and its responsibility for the alleged 'Covid-19' was to isolate the virus from all the other material – a process also known as 'purification' – and

then follow the postulates sequence developed in the late 19th century by German physician and bacteriologist Robert Koch which became the 'gold standard' for connecting an alleged causation agent to a disease:

1. The microorganism (bacteria, fungus, virus, etc.) must be present in every case of the disease and all patients must have the same symptoms. It must also *not be present in healthy individuals*.
2. The microorganism must be isolated from the host with the disease. If the microorganism is a bacteria or fungus it must be grown in a pure culture. If it is a virus, it must be purified (i.e. containing no other material except the virus particles) from a clinical sample.
3. The specific disease, with all of its characteristics, must be reproduced when the infectious agent (the purified virus or a pure culture of bacteria or fungi) is inoculated into a healthy, susceptible host.
4. The microorganism must be recoverable from the experimentally infected host as in step 2.

Not one of these criteria has been met in the case of 'SARS-Cov-2' and 'Covid-19'. Not ONE. EVER. Robert Koch refers to bacteria and not viruses. What are called 'viral particles' are so minute (hence masks are useless by any definition) that they could only be seen after the invention of the electron microscope in the 1930s and can still only be observed through that means. American bacteriologist and virologist Thomas Milton Rivers, the so-called 'Father of Modern Virology' who was very significantly director of the Rockefeller Institute for Medical Research in the 1930s, developed a less stringent version of Koch's postulates to identify 'virus' causation known as 'Rivers criteria'. 'Covid' did not pass that process either. Some even doubt whether any 'virus' can be isolated from other particles containing genetic material in the Koch method. Freedom of Information requests in many countries asking for scientific proof that the 'Covid virus' has been purified and isolated and shown to exist have all come back with a 'we don't have that' and when this happened with a request to the UK Department of Health they added this comment:

However, outside of the scope of the [Freedom of Information Act] and on a discretionary basis, the following information has been advised to us, which may be of interest. Most infectious diseases are caused by viruses, bacteria or fungi. Some bacteria or fungi have the capacity to grow on their own in isolation, for example in colonies on a petri dish. Viruses are different in that they are what we call 'obligate pathogens' – that is, they cannot survive or reproduce without infecting a host ...

... For some diseases, it is possible to establish causation between a microorganism and a disease by isolating the pathogen from a patient, growing it in pure culture and reintroducing it to a healthy organism. These are known as 'Koch's postulates' and were developed in 1882. However, as our understanding of disease and different disease-causing agents has advanced, these are no longer the method for determining causation [Andrew Kaufman asks why in that case are there two published articles falsely claiming to satisfy Koch's postulates].

It has long been known that viral diseases cannot be identified in this way as viruses cannot be grown in 'pure culture'. When a patient is tested for a viral illness, this is normally done by looking for the presence of antigens, or viral genetic code in a host with molecular biology techniques [Kaufman asks how you could know the origin of these chemicals without having a pure culture for comparison].

For the record 'antigens' are defined so:

Invading microorganisms have antigens on their surface that the human body can recognise as being foreign – meaning not belonging to it. When the body recognises a foreign antigen, lymphocytes (white blood cells) produce antibodies, which are complementary in shape to the antigen.

Notwithstanding that this is open to question in relation to 'SARS-Cov-2' the presence of 'antibodies' can have many causes and they are found in people that are perfectly well. Kary Mullis said: 'Antibodies ... had always been considered evidence of past disease, not present disease.'

'Covid' really is a *computer* 'virus'

Where the UK Department of Health statement says 'viruses' are now 'diagnosed' through a 'viral genetic code in a host with molecular biology techniques', they mean ... *the PCR test* which its inventor said cannot test for infectious disease. They have no credible method of connecting a 'virus' to a disease and we will see that there is no scientific proof that any 'virus' causes any disease or there is any such thing as a 'virus' in the way that it is described. Tenacious Canadian researcher Christine Massey and her team made

some 40 Freedom of Information requests to national public health agencies in different countries asking for proof that SARS-CoV-2 has been isolated and not one of them could supply that information. Massey said of her request in Canada: 'Freedom of Information reveals Public Health Agency of Canada has no record of 'SARS-COV-2' isolation performed by anyone, anywhere, ever.' If you accept the comment from the UK Department of Health it's because they can't isolate a 'virus'. Even so many 'science' papers claimed to have isolated the 'Covid virus' until they were questioned and had to admit they hadn't. A reply from the Robert Koch Institute in Germany was typical: 'I am not aware of a paper which purified isolated SARS-CoV-2.' So what the hell was Christian Drosten and his gang using to design the 'Covid' testing protocol that has produced all the illusory Covid' cases and 'Covid' deaths when the head of the Chinese version of the CDC admitted there was a problem right from the start in that the 'virus' had never been isolated/purified? Breathe deeply: What they are calling 'Covid' is actually created by a *computer program* i.e. *they made it up* – er, that's it. They took lung fluid, with many sources of genetic material, from one single person alleged to be infected with Covid-19 by a PCR test which they *claimed*, without clear evidence, contained a 'virus'. They used several computer programs to create a model of a theoretical virus genome sequence from more than fifty-six million small sequences of RNA, each of an unknown source, assembling them like a puzzle with no known solution. The computer filled in the gaps with sequences from bits in the gene bank to make it look like a bat SARS-like coronavirus! A wave of the magic wand and poof, an *in silico* (computer-generated) genome, a scientific fantasy, was created. UK health researcher Dr Kevin Corbett made the same point with this analogy:

... It's like giving you a few bones and saying that's your fish. It could be any fish. Not even a skeleton. Here's a few fragments of bones. That's your fish ... It's all from gene bank and the bits of the virus sequence that weren't there they made up.

They synthetically created them to fill in the blanks. That's what genetics is; it's a code. So it's ABBCCDDDD and you're missing some what you think is EEE so you put it in. It's all

synthetic. You just manufacture the bits that are missing. This is the end result of the geneticization of virology. This is basically a computer virus.

Further confirmation came in an email exchange between British citizen journalist Frances Leader and the government's Medicines & Healthcare Products Regulatory Agency (the Gates-funded MHRA) which gave emergency permission for untested 'Covid vaccines' to be used. The agency admitted that the 'vaccine' is not based on an isolated 'virus', but comes from a *computer-generated model*. Frances Leader was naturally banned from Cult-owned fascist Twitter for making this exchange public. The process of creating computer-generated alleged 'viruses' is called 'in silico' or 'in silicon' – computer chips – and the term 'in silico' is believed to originate with biological experiments using only a computer in 1989. 'Vaccines' involved with 'Covid' are also produced 'in silico' or by computer not a natural process. If the original 'virus' is nothing more than a made-up computer model how can there be 'new variants' of something that never existed in the first place? They are not new 'variants'; they are new *computer models* only minutely different to the original program and designed to further terrify the population into having the 'vaccine' and submitting to fascism. You want a 'new variant'? Click, click, enter – there you go. Tell the medical profession that you have discovered a 'South African variant', 'UK variants' or a 'Brazilian variant' and in the usual HIV-causes-AIDS manner they will unquestioningly repeat it with no evidence whatsoever to support these claims. They will go on television and warn about the dangers of 'new variants' while doing nothing more than repeating what they have been told to be true and knowing that any deviation from that would be career suicide. Big-time insiders will know it's a hoax, but much of the medical community is clueless about the way they are being played and themselves play the public without even being aware they are doing so. What an interesting 'coincidence' that AstraZeneca and Oxford University were conducting 'Covid vaccine trials' in the three countries – the UK, South Africa and Brazil – where the first three 'variants' were claimed to have 'broken out'.

Here's your 'virus' – it's a unicorn

Dr Andrew Kaufman presented a brilliant analysis describing how the 'virus' was imagined into fake existence when he dissected an article published by *Nature* and written by 19 authors detailing *alleged* 'sequencing of a complete viral genome' of the 'new SARS-CoV-2 virus'. This computer-modelled *in silico* genome was used as a template for all subsequent genome sequencing experiments that resulted in the so-called variants which he said now number more than 6,000. The fake genome was constructed from more than 56 million individual short strands of RNA. Those little pieces were assembled into longer pieces by finding areas of overlapping sequences. The computer programs created over two million possible combinations from which the authors simply chose the longest one. They then compared this to a 'bat virus' and the computer 'alignment' rearranged the sequence and filled in the gaps! They called this computer-generated abomination the 'complete genome'. Dr Tom Cowan, a fellow medical author and collaborator with Kaufman, said such computer-generation constitutes scientific fraud and he makes this superb analogy:

Here is an equivalency: A group of researchers claim to have found a unicorn because they found a piece of a hoof, a hair from a tail, and a snippet of a horn. They then add that information into a computer and program it to re-create the unicorn, and they then claim this computer re-creation is the real unicorn. Of course, they had never actually seen a unicorn so could not possibly have examined its genetic makeup to compare their samples with the actual unicorn's hair, hooves and horn.

The researchers claim they decided which is the real genome of SARS-CoV-2 by 'consensus', sort of like a vote. Again, different computer programs will come up with different versions of the imaginary 'unicorn', so they come together as a group and decide which is the real imaginary unicorn.

This is how the 'virus' that has transformed the world was brought into fraudulent 'existence'. Extraordinary, yes, but as the Nazis said the bigger the lie the more will believe it. Cowan, however, wasn't finished and he went on to identify what he called the real blockbuster in the paper. He quotes this section from a paper written

by virologists and published by the CDC and then explains what it means:

Therefore, we examined the capacity of SARS-CoV-2 to infect and replicate in several common primate and human cell lines, including human adenocarcinoma cells (A549), human liver cells (HUH 7.0), and human embryonic kidney cells (HEK-293T). In addition to Vero E6 and Vero CCL81 cells. ... Each cell line was inoculated at high multiplicity of infection and examined 24h post-infection.

No CPE was observed in any of the cell lines except in Vero cells, which grew to greater than 10 to the 7th power at 24 h post-infection. In contrast, HUH 7.0 and 293T showed only modest viral replication, and A549 cells were incompatible with SARS CoV-2 infection.

Cowan explains that when virologists attempt to prove infection they have three possible 'hosts' or models on which they can test. The first was humans. Exposure to humans was generally not done for ethical reasons and has never been done with SARS-CoV-2 or any coronavirus. The second possible host was animals. Cowan said that forgetting for a moment that they never actually use purified virus when exposing animals they do use solutions that they *claim* contain the virus. Exposure to animals has been done with SARS-CoV-2 in an experiment involving mice and this is what they found: *None of the wild (normal) mice got sick*. In a group of genetically-modified mice, a statistically insignificant number lost weight and had slightly bristled fur, but they experienced nothing like the illness called 'Covid-19'. Cowan said the third method – the one they mostly rely on – is to inoculate solutions they *say* contain the virus onto a variety of tissue cultures. This process had never been shown to kill tissue *unless* the sample material was starved of nutrients and poisoned as *part of the process*. Yes, incredibly, in tissue experiments designed to show the 'virus' is responsible for killing the tissue they starve the tissue of nutrients and add toxic drugs including antibiotics and they do not have control studies to see if it's the starvation and poisoning that is degrading the tissue rather than the 'virus' they allege to be in there somewhere. You want me to pinch you? Yep, I understand. Tom Cowan said this about the whole nonsensical farce as he explains what that quote from the CDC paper really means:

The shocking thing about the above quote is that using their own methods, the virologists found that solutions containing SARS-CoV-2 – even in high amounts – were NOT, I repeat NOT, infective to any of the three human tissue cultures they tested. In plain English, this means they proved, on their terms, that this ‘new coronavirus’ is not infectious to human beings. It is ONLY infective to monkey kidney cells, and only then when you add two potent drugs (gentamicin and amphotericin), known to be toxic to kidneys, to the mix.

My friends, read this again and again. These virologists, published by the CDC, performed a clear proof, on their terms, showing that the SARS-CoV-2 virus is harmless to human beings. That is the only possible conclusion, but, unfortunately, this result is not even mentioned in their conclusion. They simply say they can provide virus stocks cultured only on monkey Vero cells, thanks for coming.

Cowan concluded: ‘If people really understood how this “science” was done, I would hope they would storm the gates and demand honesty, transparency and truth.’ Dr Michael Yeadon, former Vice President and Chief Scientific Adviser at drug giant Pfizer has been a vocal critic of the ‘Covid vaccine’ and its potential for multiple harm. He said in an interview in April, 2021, that ‘not one [vaccine] has the virus. He was asked why vaccines normally using a ‘dead’ version of a disease to activate the immune system were not used for ‘Covid’ and instead we had the synthetic methods of the ‘mRNA Covid vaccine’. Yeadon said that to do the former ‘you’d have to have some of [the virus] wouldn’t you?’ He added: ‘No-one’s got any – seriously.’ Yeadon said that surely they couldn’t have fooled the whole world for a year without having a virus, ‘but oddly enough ask around – no one’s got it’. He didn’t know why with all the ‘great labs’ around the world that the virus had not been isolated – ‘Maybe they’ve been too busy running bad PCR tests and vaccines that people don’t need.’ What is today called ‘science’ is not ‘science’ at all. Science is no longer what is, but whatever people can be manipulated to *believe* that it is. Real science has been hijacked by the Cult to dispense and produce the ‘expert scientists’ and contentions that suit the agenda of the Cult. How big-time this has happened with the ‘Covid’ hoax which is entirely based on fake science delivered by fake ‘scientists’ and fake ‘doctors’. The human-caused climate change hoax is also entirely based on fake science delivered by fake ‘scientists’ and fake ‘climate experts’. In both cases real

scientists, climate experts and doctors have their views suppressed and deleted by the Cult-owned science establishment, media and Silicon Valley. This is the 'science' that politicians claim to be 'following' and a common denominator of 'Covid' and climate are Cult psychopaths Bill Gates and his mate Klaus Schwab at the Gates-funded World Economic Forum. But, don't worry, it's all just a coincidence and absolutely nothing to worry about. Zzzzzzzzz.

What is a 'virus' REALLY?

Dr Tom Cowan is one of many contesting the very existence of viruses let alone that they cause disease. This is understandable when there is no scientific evidence for a disease-causing 'virus'. German virologist Dr Stefan Lanka won a landmark case in 2017 in the German Supreme Court over his contention that there is no such thing as a measles virus. He had offered a big prize for anyone who could prove there is and Lanka won his case when someone sought to claim the money. There is currently a prize of more than 225,000 euros on offer from an Isolate Truth Fund for anyone who can prove the isolation of SARS-CoV-2 and its genetic substance. Lanka wrote in an article headed 'The Misconception Called Virus' that scientists think a 'virus' is causing tissue to become diseased and degraded when in fact it is the *processes they are using* which do that – not a 'virus'. Lanka has done an important job in making this point clear as Cowan did in his analysis of the CDC paper. Lanka says that all claims about viruses as disease-causing pathogens are wrong and based on 'easily recognisable, understandable and verifiable misinterpretations.' Scientists believed they were working with 'viruses' in their laboratories when they were really working with 'typical particles of specific dying tissues or cells ...' Lanka said that the tissue decaying process claimed to be caused by a 'virus' still happens when no alleged 'virus' is involved. It's the *process* that does the damage and not a 'virus'. The genetic sample is deprived of nutrients, removed from its energy supply through removal from the body and then doused in toxic antibiotics to remove any bacteria. He confirms again that establishment scientists do not (pinch me)

conduct control experiments to see if this is the case and if they did they would see the claims that 'viruses' are doing the damage is nonsense. He adds that during the measles 'virus' court case he commissioned an independent laboratory to perform just such a control experiment and the result was that the tissues and cells died in the exact same way as with alleged 'infected' material. This is supported by a gathering number of scientists, doctors and researchers who reject what is called 'germ theory' or the belief in the body being infected by contagious sources emitted by other people. Researchers Dawn Lester and David Parker take the same stance in their highly-detailed and sourced book *What Really Makes You Ill – Why everything you thought you knew about disease is wrong* which was recommended to me by a number of medical professionals genuinely seeking the truth. Lester and Parker say there is no provable scientific evidence to show that a 'virus' can be transmitted between people or people and animals or animals and people:

The definition also claims that viruses are the cause of many diseases, as if this has been definitively proven. But this is not the case; there is no original scientific evidence that definitively demonstrates that any virus is the cause of any disease. The burden of proof for any theory lies with those who proposed it; but none of the existing documents provides 'proof' that supports the claim that 'viruses' are pathogens.

Dr Tom Cowan employs one of his clever analogies to describe the process by which a 'virus' is named as the culprit for a disease when what is called a 'virus' is only material released by cells detoxing themselves from infiltration by chemical or radiation poisoning. The tidal wave of technologically-generated radiation in the 'smart' modern world plus all the toxic food and drink are causing this to happen more than ever. Deluded 'scientists' misread this as a gathering impact of what they wrongly label 'viruses'.

Paper can infect houses

Cowan said in an article for davidicke.com – with his tongue only mildly in his cheek – that he believed he had made a tremendous

discovery that may revolutionise science. He had discovered that small bits of paper are alive, 'well alive-ish', can 'infect' houses, and then reproduce themselves inside the house. The result was that this explosion of growth in the paper inside the house causes the house to explode, blowing it to smithereens. His evidence for this new theory is that in the past months he had carefully examined many of the houses in his neighbourhood and found almost no scraps of paper on the lawns and surrounds of the house. There was an occasional stray label, but nothing more. Then he would return to these same houses a week or so later and with a few, not all of them, particularly the old and decrepit ones, he found to his shock and surprise they were littered with stray bits of paper. He knew then that the paper had infected these houses, made copies of itself, and blew up the house. A young boy on a bicycle at one of the sites told him he had seen a demolition crew using dynamite to explode the house the previous week, but Cowan dismissed this as the idle thoughts of silly boys because 'I was on to something big'. He was on to how 'scientists' mistake genetic material in the detoxifying process for something they call a 'virus'. Cowan said of his house and paper story:

If this sounds crazy to you, it's because it should. This scenario is obviously nuts. But consider this admittedly embellished, for effect, current viral theory that all scientists, medical doctors and virologists currently believe.

He takes the example of the 'novel SARS-Cov2' virus to prove the point. First they take someone with an undefined illness called 'Covid-19' and don't even attempt to find any virus in their sputum. Never mind the scientists still describe how this 'virus', which they have not located attaches to a cell receptor, injects its genetic material, in 'Covid's' case, RNA, into the cell. The RNA once inserted exploits the cell to reproduce itself and makes 'thousands, nay millions, of copies of itself ... Then it emerges victorious to claim its next victim':

If you were to look in the scientific literature for proof, actual scientific proof, that uniform SARS-CoV2 viruses have been properly isolated from the sputum of a sick person, that actual spike proteins could be seen protruding from the virus (which has not been found), you would find that such evidence doesn't exist.

If you go looking in the published scientific literature for actual pictures, proof, that these spike proteins or any viral proteins are ever attached to any receptor embedded in any cell membrane, you would also find that no such evidence exists. If you were to look for a video or documented evidence of the intact virus injecting its genetic material into the body of the cell, reproducing itself and then emerging victorious by budding off the cell membrane, you would find that no such evidence exists.

The closest thing you would find is electron micrograph pictures of cellular particles, possibly attached to cell debris, both of which to be seen were stained by heavy metals, a process that completely distorts their architecture within the living organism. This is like finding bits of paper stuck to the blown-up bricks, thereby proving the paper emerged by taking pieces of the bricks on its way out.

The Enders baloney

Cowan describes the 'Covid' story as being just as make-believe as his paper story and he charts back this fantasy to a Nobel Prize winner called John Enders (1897-1985), an American biomedical scientist who has been dubbed 'The Father of Modern Vaccines'. Enders is claimed to have 'discovered' the process of the viral culture which 'proved' that a 'virus' caused measles. Cowan explains how Enders did this 'by using the EXACT same procedure that has been followed by every virologist to find and characterize every new virus since 1954'. Enders took throat swabs from children with measles and immersed them in 2ml of milk. Penicillin (100u/ml) and the antibiotic streptomycin (50,g/ml) were added and the whole mix was centrifuged – rotated at high speed to separate large cellular debris from small particles and molecules as with milk and cream, for example. Cowan says that if the aim is to find little particles of genetic material ('viruses') in the snot from children with measles it would seem that the last thing you would do is mix the snot with other material – milk –that also has genetic material. 'How are you ever going to know whether whatever you found came from the snot or the milk?' He points out that streptomycin is a 'nephrotoxic' or poisonous-to-the-kidney drug. You will see the relevance of that

shortly. Cowan says that it gets worse, much worse, when Enders describes the culture medium upon which the virus 'grows': 'The culture medium consisted of bovine amniotic fluid (90%), beef embryo extract (5%), horse serum (5%), antibiotics and phenol red as an indicator of cell metabolism.' Cowan asks incredulously: 'Did he just say that the culture medium also contained fluids and tissues that are themselves rich sources of genetic material?' The genetic cocktail, or 'medium', is inoculated onto tissue and cells from rhesus monkey *kidney* tissue. This is where the importance of streptomycin comes in and currently-used antimicrobials and other drugs that are *poisonous to kidneys* and used in ALL modern viral cultures (e.g. gentamicin, streptomycin, and amphotericin). Cowan asks: 'How are you ever going to know from this witch's brew where any genetic material comes from as we now have five different sources of rich genetic material in our mix?' Remember, he says, that all genetic material, whether from monkey kidney tissues, bovine serum, milk, etc., is made from the exact same components. The same central question returns: 'How are you possibly going to know that it was the virus that killed the kidney tissue and not the toxic antibiotic and starvation rations on which you are growing the tissue?' John Enders answered the question himself – *you can't*:

A second agent was obtained from an uninoculated culture of monkey kidney cells. The cytopathic changes [death of the cells] it induced in the unstained preparations could not be distinguished with confidence from the viruses isolated from measles.

The death of the cells ('cytopathic changes') happened in exactly the same manner, whether they inoculated the kidney tissue with the measles snot or not, Cowan says. 'This is evidence that the destruction of the tissue, the very proof of viral causation of illness, was not caused by anything in the snot because they saw the same destructive effect when the snot was not even used ... the cytopathic, i.e., cell-killing, changes come from the process of the culture itself, not from any virus in any snot, period.' Enders quotes in his 1957 paper a virologist called Ruckle as reporting similar findings 'and in addition has isolated an agent from monkey kidney tissue that is so

far indistinguishable from human measles virus'. In other words, Cowan says, these particles called 'measles viruses' are simply and clearly breakdown products of the starved and poisoned tissue. For measles 'virus' see all 'viruses' including the so-called 'Covid virus'. Enders, the 'Father of Modern Vaccines', also said:

There is a potential risk in employing cultures of primate cells for the production of vaccines composed of attenuated virus, since the presence of other agents possibly latent in primate tissues cannot be definitely excluded by any known method.

Cowan further quotes from a paper published in the journal *Viruses* in May, 2020, while the 'Covid pandemic' was well underway in the media if not in reality. 'EVs' here refers to particles of genetic debris from our own tissues, such as exosomes of which more in a moment: 'The remarkable resemblance between EVs and viruses has caused quite a few problems in the studies focused on the analysis of EVs released during viral infections.' Later the paper adds that to date a reliable method that can actually guarantee a complete separation (of EVs from viruses) DOES NOT EXIST. This was published at a time when a fairy tale 'virus' was claimed in total certainty to be causing a fairy tale 'viral disease' called 'Covid-19' – a fairy tale that was already well on the way to transforming human society in the image that the Cult has worked to achieve for so long. Cowan concludes his article:

To summarize, there is no scientific evidence that pathogenic viruses exist. What we think of as 'viruses' are simply the normal breakdown products of dead and dying tissues and cells. When we are well, we make fewer of these particles; when we are starved, poisoned, suffocated by wearing masks, or afraid, we make more.

There is no engineered virus circulating and making people sick. People in laboratories all over the world are making genetically modified products to make people sick. These are called vaccines. There is no virome, no 'ecosystem' of viruses, viruses are not 8%, 50% or 100 % of our genetic material. These are all simply erroneous ideas based on the misconception called a virus.

What is 'Covid'? Load of bollocks

The background described here by Cowan and Lanka was emphasised in the first video presentation that I saw by Dr Andrew Kaufman when he asked whether the 'Covid virus' was in truth a natural defence mechanism of the body called 'exosomes'. These are released by cells when in states of toxicity – see the same themes returning over and over. They are released ever more profusely as chemical and radiation toxicity increases and think of the potential effect therefore of 5G alone as its destructive frequencies infest the human energetic information field with a gathering pace (5G went online in Wuhan in 2019 as the 'virus' emerged). I'll have more about this later. Exosomes transmit a warning to the rest of the body that 'Houston, we have a problem'. Kaufman presented images of exosomes and compared them with 'Covid' under an electron microscope and the similarity was remarkable. They both attach to the same cell receptors (*claimed* in the case of 'Covid'), contain the same genetic material in the form of RNA or ribonucleic acid, and both are found in 'viral cell cultures' with damaged or dying cells. James Hildreth MD, President and Chief Executive Officer of the Meharry Medical College at Johns Hopkins, said: 'The virus is fully an exosome in every sense of the word.' Kaufman's conclusion was that there is no 'virus': 'This entire pandemic is a completely manufactured crisis ... there is no evidence of anyone dying from [this] illness.' Dr Tom Cowan and Sally Fallon Morell, authors of *The Contagion Myth*, published a statement with Dr Kaufman in February, 2021, explaining why the 'virus' does not exist and you can read it that in full in the Appendix.

'Virus' theory can be traced to the 'cell theory' in 1858 of German physician Rudolf Virchow (1821-1920) who contended that disease originates from a single cell infiltrated by a 'virus'. Dr Stefan Lanka said that findings and insights with respect to the structure, function and central importance of tissues in the creation of life, which were already known in 1858, comprehensively refute the cell theory. Virchow ignored them. We have seen the part later played by John Enders in the 1950s and Lanka notes that infection theories were only established as a global dogma through the policies and

eugenics of the Third Reich in Nazi Germany (creation of the same Sabbatian cult behind the 'Covid' hoax). Lanka said: 'Before 1933, scientists dared to contradict this theory; after 1933, these critical scientists were silenced'. Dr Tom Cowan's view is that ill-health is caused by too much of something, too little of something, or toxification from chemicals and radiation – not contagion. We must also highlight as a major source of the 'virus' theology a man still called the 'Father of Modern Virology' – Thomas Milton Rivers (1888-1962). There is no way given the Cult's long game policy that it was a coincidence for the 'Father of Modern Virology' to be director of the Rockefeller Institute for Medical Research from 1937 to 1956 when he is credited with making the Rockefeller Institute a leader in 'viral research'. Cult Rockefellerers were the force behind the creation of Big Pharma 'medicine', established the World Health Organisation in 1948, and have long and close associations with the Gates family that now runs the WHO during the pandemic hoax through mega-rich Cult gofer and psychopath Bill Gates.

Only a Renegade Mind can see through all this bullshit by asking the questions that need to be answered, not taking 'no' or prevarication for an answer, and certainly not hiding from the truth in fear of speaking it. Renegade Minds have always changed the world for the better and they will change this one no matter how bleak it may currently appear to be.

CHAPTER SIX

Sequence of deceit

If you tell the truth, you don't have to remember anything
Mark Twain

Against the background that I have laid out this far the sequence that took us from an invented 'virus' in Cult-owned China in late 2019 to the fascist transformation of human society can be seen and understood in a whole new context.

We were told that a deadly disease had broken out in Wuhan and the world media began its campaign (coordinated by behavioural psychologists as we shall see) to terrify the population into unquestioning compliance. We were shown images of Chinese people collapsing in the street which never happened in the West with what was supposed to be the same condition. In the earliest days when alleged cases and deaths were few the fear register was hysterical in many areas of the media and this would expand into the common media narrative across the world. The real story was rather different, but we were never told that. The Chinese government, one of the Cult's biggest centres of global operation, said they had discovered a new illness with flu-like and pneumonia-type symptoms in a city with such toxic air that it is overwhelmed with flu-like symptoms, pneumonia and respiratory disease. Chinese scientists said it was a new – 'novel' – coronavirus which they called Sars-Cov-2 and that it caused a disease they labelled 'Covid-19'. There was no evidence for this and the 'virus' has never to this day been isolated, purified and its genetic code established from that. It

was from the beginning a computer-generated fiction. Stories of Chinese whistleblowers saying the number of deaths was being suppressed or that the 'new disease' was related to the Wuhan bio-lab misdirected mainstream and alternative media into cul-de-sacs to obscure the real truth – there was no 'virus'.

Chinese scientists took genetic material from the lung fluid of just a few people and said they had found a 'new' disease when this material had a wide range of content. There was no evidence for a 'virus' for the very reasons explained in the last two chapters. The 'virus' has never been shown to (a) exist and (b) cause any disease. People were diagnosed on symptoms that are so widespread in Wuhan and polluted China and with a PCR test that can't detect infectious disease. On this farce the whole global scam was sold to the rest of the world which would also diagnose respiratory disease as 'Covid-19' from symptoms alone or with a PCR test not testing for a 'virus'. Flu miraculously disappeared *worldwide* in 2020 and into 2021 as it was redesignated 'Covid-19'. It was really the same old flu with its 'flu-like' symptoms attributed to 'flu-like' 'Covid-19'. At the same time with very few exceptions the Chinese response of draconian lockdown and fascism was the chosen weapon to respond across the West as recommended by the Cult-owned Tedros at the Cult-owned World Health Organization run by the Cult-owned Gates. All was going according to plan. Chinese scientists – everything in China is controlled by the Cult-owned government – compared their contaminated RNA lung-fluid material with other RNA sequences and said it appeared to be just under 80 percent identical to the SARS-CoV-1 'virus' claimed to be the cause of the SARS (severe acute respiratory syndrome) 'outbreak' in 2003. They decreed that because of this the 'new virus' had to be related and they called it SARS-CoV-2. There are some serious problems with this assumption and *assumption* was all it was. Most 'factual' science turns out to be assumptions repeated into everyone-knows-that. A match of under 80-percent is meaningless. Dr Kaufman makes the point that there's a 96 percent genetic correlation between humans and chimpanzees, but 'no one would say our genetic material is part

of the chimpanzee family'. Yet the Chinese authorities were claiming that a much lower percentage, less than 80 percent, proved the existence of a new 'coronavirus'. For goodness sake human DNA is 60 percent similar to a *banana*.

You are feeling sleepy

The entire 'Covid' hoax is a global Psyop, a psychological operation to program the human mind into believing and fearing a complete fantasy. A crucial aspect of this was what *appeared* to happen in Italy. It was all very well streaming out daily images of an alleged catastrophe in Wuhan, but to the Western mind it was still on the other side of the world in a very different culture and setting. A reaction of 'this could happen to me and my family' was still nothing like as intense enough for the mind-doctors. The Cult needed a Western example to push people over that edge and it chose Italy, one of its major global locations going back to the Roman Empire. An Italian 'Covid' crisis was manufactured in a particular area called Lombardy which just happens to be notorious for its toxic air and therefore respiratory disease. Wuhan, China, *déjà vu*. An hysterical media told horror stories of Italians dying from 'Covid' in their droves and how Lombardy hospitals were being overrun by a tidal wave of desperately ill people needing treatment after being struck down by the 'deadly virus'. Here was the psychological turning point the Cult had planned. Wow, if this is happening in Italy, the Western mind concluded, this indeed could happen to me and my family. Another point is that Italian authorities responded by following the Chinese blueprint so vehemently recommended by the Cult-owned World Health Organization. They imposed fascistic lockdowns on the whole country viciously policed with the help of surveillance drones sweeping through the streets seeking out anyone who escaped from mass house arrest. Livelihoods were destroyed and psychology unravelled in the way we have witnessed since in all lockdown countries. Crucial to the plan was that Italy responded in this way to set the precedent of suspending freedom and imposing fascism in a 'Western liberal democracy'. I emphasised in an

animated video explanation on davidicke.com posted in the summer of 2020 how important it was to the Cult to expand the Chinese lockdown model across the West. Without this, and the bare-faced lie that non-symptomatic people could still transmit a 'disease' they didn't have, there was no way locking down the whole population, sick and not sick, could be pulled off. At just the right time and with no evidence Cult operatives and gofers claimed that people without symptoms could pass on the 'disease'. In the name of protecting the 'vulnerable' like elderly people, who lockdowns would kill by the tens of thousands, we had for the first time healthy people told to isolate as well as the sick. The great majority of people who tested positive had no symptoms because there was nothing wrong with them. It was just a trick made possible by a test not testing for the 'virus'.

Months after my animated video the Gates-funded Professor Neil Ferguson at the Gates-funded Imperial College confirmed that I was right. He didn't say it in those terms, naturally, but he did say it. Ferguson will enter the story shortly for his outrageously crazy 'computer models' that led to Britain, the United States and many other countries following the Chinese and now Italian methods of response. Put another way, following the Cult script. Ferguson said that SAGE, the UK government's scientific advisory group which has controlled 'Covid' policy from the start, wanted to follow the Chinese lockdown model (while they all continued to work and be paid), but they wondered if they could possibly, in Ferguson's words, 'get away with it in Europe'. 'Get away with it'? Who the hell do these moronic, arrogant people think they are? This appalling man Ferguson said that once Italy went into national lockdown they realised they, too, could mimic China:

It's a communist one-party state, we said. We couldn't get away with it in Europe, we thought ... and then Italy did it. And we realised we could. Behind this garbage from Ferguson is a simple fact: Doing the same as China in every country was the plan from the start and Ferguson's 'models' would play a central role in achieving that. It's just a coincidence, of course, and absolutely nothing to worry your little head about.

Oops, sorry, our mistake

Once the Italian segment of the Psyop had done the job it was designed to do a very different story emerged. Italian authorities revealed that 99 percent of those who had 'died from Covid-19' in Italy had one, two, three, or more 'co-morbidities' or illnesses and health problems that could have ended their life. The US Centers for Disease Control and Prevention (CDC) published a figure of 94 percent for Americans dying of 'Covid' while having other serious medical conditions – on average two to three (some five or six) other potential causes of death. In terms of death from an unproven 'virus' I say it is 100 percent. The other one percent in Italy and six percent in the US would presumably have died from 'Covid's' flu-like symptoms with a range of other possible causes in conjunction with a test not testing for the 'virus'. Fox News reported that even more startling figures had emerged in one US county in which 410 of 422 deaths attributed to 'Covid-19' had other potentially deadly health conditions. The Italian National Health Institute said later that the average age of people dying with a 'Covid-19' diagnosis in Italy was about 81. Ninety percent were over 70 with ten percent over 90. In terms of other reasons to die some 80 percent had two or more chronic diseases with half having three or more including cardiovascular problems, diabetes, respiratory problems and cancer. Why is the phantom 'Covid-19' said to kill overwhelmingly old people and hardly affect the young? Old people continually die of many causes and especially respiratory disease which you can re-diagnose 'Covid-19' while young people die in tiny numbers by comparison and rarely of respiratory disease. Old people 'die of Covid' because they die of other things that can be redesignated 'Covid' and it really is that simple.

Flu has flown

The blueprint was in place. Get your illusory 'cases' from a test not testing for the 'virus' and redesignate other causes of death as 'Covid-19'. You have an instant 'pandemic' from something that is nothing more than a computer-generated fiction. With near-on a

billion people having 'flu-like' symptoms every year the potential was limitless and we can see why flu quickly and apparently miraculously disappeared *worldwide* by being diagnosed 'Covid-19'. The painfully bloody obvious was explained away by the childlike media in headlines like this in the UK '*Independent*': 'Not a single case of flu detected by Public Health England this year as Covid restrictions suppress virus'. I kid you not. The masking, social distancing and house arrest that did not make the 'Covid virus' disappear somehow did so with the 'flu virus'. Even worse the article, by a bloke called Samuel Lovett, suggested that maybe the masking, sanitising and other 'Covid' measures should continue to keep the flu away. With a ridiculousness that disturbs your breathing (it's 'Covid-19') the said Lovett wrote: 'With widespread social distancing and mask-wearing measures in place throughout the UK, the usual routes of transmission for influenza have been blocked.' He had absolutely no evidence to support that statement, but look at the consequences of him acknowledging the obvious. With flu not disappearing at all and only being relabelled 'Covid-19' he would have to contemplate that 'Covid' was a hoax on a scale that is hard to imagine. You need guts and commitment to truth to even go there and that's clearly something Samuel Lovett does not have in abundance. He would never have got it through the editors anyway.

Tens of thousands die in the United States alone every winter from flu including many with pneumonia complications. CDC figures record *45 million* Americans diagnosed with flu in 2017-2018 of which 61,000 died and some reports claim 80,000. Where was the same hysteria then that we have seen with 'Covid-19'? Some 250,000 Americans are admitted to hospital with pneumonia every year with about 50,000 cases proving fatal. About 65 million suffer respiratory disease every year and three million deaths makes this the third biggest cause of death worldwide. You only have to redesignate a portion of all these people 'Covid-19' and you have an instant global pandemic or the *appearance* of one. Why would doctors do this? They are told to do this and all but a few dare not refuse those who must be obeyed. Doctors in general are not researching their own

knowledge and instead take it direct and unquestioned from the authorities that own them and their careers. The authorities say they must now diagnose these symptoms 'Covid-19' and not flu, or whatever, and they do it. Dark suits say put 'Covid-19' on death certificates no matter what the cause of death and the doctors do it. Renegade Minds don't fall for the illusion that doctors and medical staff are all highly-intelligent, highly-principled, seekers of medical truth. *Some are*, but not the majority. They are repeaters, gofers, and yes sir, no sir, purveyors of what the system demands they purvey. The 'Covid' con is not merely confined to diseases of the lungs. Instructions to doctors to put 'Covid-19' on death certificates for anyone dying of *anything* within 28 days (or much more) of a positive test not testing for the 'virus' opened the floodgates. The term dying *with* 'Covid' and not *of* 'Covid' was coined to cover the truth. Whether it was a *with* or an *of* they were all added to the death numbers attributed to the 'deadly virus' compiled by national governments and globally by the Gates-funded Johns Hopkins operation in the United States that was so involved in those 'pandemic' simulations. Fraudulent deaths were added to the ever-growing list of fraudulent 'cases' from false positives from a false test. No wonder Professor Walter Ricciardi, scientific advisor to the Italian minister of health, said after the Lombardy hysteria had done its job that 'Covid' death rates were due to Italy having the second oldest population in the world and to *how hospitals record deaths*:

The way in which we code deaths in our country is very generous in the sense that all the people who die in hospitals with the coronavirus are deemed to be dying of the coronavirus. On re-evaluation by the National Institute of Health, only 12 per cent of death certificates have shown a direct causality from coronavirus, while 88 per cent of patients who have died have at least one pre-morbidity – many had two or three.

This is extraordinary enough when you consider the propaganda campaign to use Italy to terrify the world, but how can they even say twelve percent were genuine when the 'virus' has not been shown to exist, its 'code' is a computer program, and diagnosis comes from a test not testing for it? As in China, and soon the world, 'Covid-19' in

Italy was a redesignation of diagnosis. Lies and corruption were to become the real 'pandemic' fuelled by a pathetically-compliant medical system taking its orders from the tiny few at the top of their national hierarchy who answered to the World Health Organization which answers to Gates and the Cult. Doctors were told – ordered – to diagnose a particular set of symptoms 'Covid-19' and put that on the death certificate for any cause of death if the patient had tested positive with a test not testing for the virus or had 'Covid' symptoms like the flu. The United States even introduced big financial incentives to manipulate the figures with hospitals receiving £4,600 from the Medicare system for diagnosing someone with regular pneumonia, \$13,000 if they made the diagnosis from the same symptoms 'Covid-19' pneumonia, and \$39,000 if they put a 'Covid' diagnosed patient on a ventilator that would almost certainly kill them. A few – painfully and pathetically few – medical whistleblowers revealed (before Cult-owned YouTube deleted their videos) that they had been instructed to 'let the patient crash' and put them straight on a ventilator instead of going through a series of far less intrusive and dangerous methods as they would have done before the pandemic hoax began and the financial incentives kicked in. We are talking cold-blooded murder given that ventilators are so damaging to respiratory systems they are usually the last step before heaven awaits. Renegade Minds never fall for the belief that people in white coats are all angels of mercy and cannot be full-on psychopaths. I have explained in detail in *The Answer* how what I am describing here played out across the world coordinated by the World Health Organization through the medical hierarchies in almost every country.

Medical scientist calls it

Information about the non-existence of the 'virus' began to emerge for me in late March, 2020, and mushroomed after that. I was sent an email by Sir Julian Rose, a writer, researcher, and organic farming promotor, from a medical scientist friend of his in the United States. Even at that early stage in March the scientist was able to explain

how the 'Covid' hoax was being manipulated. He said there were no reliable tests for a specific 'Covid-19 virus' and nor were there any reliable agencies or media outlets for reporting numbers of actual 'Covid-19' cases. We have seen in the long period since then that he was absolutely right. 'Every action and reaction to Covid-19 is based on totally flawed data and we simply cannot make accurate assessments,' he said. Most people diagnosed with 'Covid-19' were showing nothing more than cold and flu-like symptoms 'because most coronavirus strains *are* nothing more than cold/flu-like symptoms'. We had farcical situations like an 84-year-old German man testing positive for 'Covid-19' and his nursing home ordered to quarantine only for him to be found to have a common cold. The scientist described back then why PCR tests and what he called the 'Mickey Mouse test kits' were useless for what they were claimed to be identifying. 'The idea these kits can isolate a specific virus like Covid-19 is nonsense,' he said. Significantly, he pointed out that 'if you want to create a totally false panic about a totally false pandemic – pick a coronavirus'. This is exactly what the Cult-owned Gates, World Economic Forum and Johns Hopkins University did with their Event 201 'simulation' followed by their real-life simulation called the 'pandemic'. The scientist said that all you had to do was select the sickest of people with respiratory-type diseases in a single location – 'say Wuhan' – and administer PCR tests to them. You can then claim that anyone showing 'viral sequences' similar to a coronavirus 'which will inevitably be quite a few' is suffering from a 'new' disease:

Since you already selected the sickest flu cases a fairly high proportion of your sample will go on to die. You can then say this 'new' virus has a CFR [case fatality rate] higher than the flu and use this to infuse more concern and do more tests which will of course produce more 'cases', which expands the testing, which produces yet more 'cases' and so on and so on. Before long you have your 'pandemic', and all you have done is use a simple test kit trick to convert the worst flu and pneumonia cases into something new that doesn't ACTUALLY EXIST [my emphasis].

He said that you then 'just run the same scam in other countries' and make sure to keep the fear message running high 'so that people

will feel panicky and less able to think critically'. The only problem to overcome was the fact *there is no* actual new deadly pathogen and only regular sick people. This meant that deaths from the 'new deadly pathogen' were going to be way too low for a real new deadly virus pandemic, but he said this could be overcome in the following ways – all of which would go on to happen:

1. You can claim this is just the beginning and more deaths are imminent [you underpin this with fantasy 'computer projections']. Use this as an excuse to quarantine everyone and then claim the quarantine prevented the expected millions of dead.
2. You can [say that people] 'minimizing' the dangers are irresponsible and bully them into not talking about numbers.
3. You can talk crap about made up numbers hoping to blind people with pseudoscience.
4. You can start testing well people (who, of course, will also likely have shreds of coronavirus [RNA] in them) and thus inflate your 'case figures' with 'asymptomatic carriers' (you will of course have to spin that to sound deadly even though any virologist knows the more symptom-less cases you have the less deadly is your pathogen).

The scientist said that if you take these simple steps 'you can have your own entirely manufactured pandemic up and running in weeks'. His analysis made so early in the hoax was brilliantly prophetic of what would actually unfold. Pulling all the information together in these recent chapters we have this is simple 1, 2, 3, of how you can delude virtually the entire human population into believing in a 'virus' that doesn't exist:

- A 'Covid case' is someone who tests positive with a test not testing for the 'virus'.
- A 'Covid death' is someone who dies of *any cause* within 28 days (or much longer) of testing positive with a test not testing for the 'virus'.
- Asymptomatic means there is nothing wrong with you, but they claim you can pass on what you don't have to justify locking

down (quarantining) healthy people in totality.

The foundations of the hoax are that simple. A study involving ten million people in Wuhan, published in November, 2020, demolished the whole lie about those without symptoms passing on the 'virus'. They found '300 asymptomatic cases' and traced their contacts to find that not one of them was detected with the 'virus'.

'Asymptomatic' patients and their contacts were isolated for no less than two weeks and nothing changed. I know it's all crap, but if you are going to claim that those without symptoms can transmit 'the virus' then you must produce evidence for that and they never have. Even World Health Organization official Dr Maria Van Kerkhove, head of the emerging diseases and zoonosis unit, said as early as June, 2020, that she doubted the validity of asymptomatic transmission. She said that 'from the data we have, it still seems to be rare that an asymptomatic person actually transmits onward to a secondary individual' and by 'rare' she meant that she couldn't cite any case of asymptomatic transmission.

The Ferguson factor

The problem for the Cult as it headed into March, 2020, when the script had lockdown due to start, was that despite all the manipulation of the case and death figures they still did not have enough people alleged to have died from 'Covid' to justify mass house arrest. This was overcome in the way the scientist described: 'You can claim this is just the beginning and more deaths are imminent ... Use this as an excuse to quarantine everyone and then claim the quarantine prevented the expected millions of dead.' Enter one Professor Neil Ferguson, the Gates-funded 'epidemiologist' at the Gates-funded Imperial College in London. Ferguson is Britain's Christian Drosten in that he has a dire record of predicting health outcomes, but is still called upon to advise government on the next health outcome when another 'crisis' comes along. This may seem to be a strange and ridiculous thing to do. Why would you keep turning for policy guidance to people who have a history of being

monumentally wrong? Ah, but it makes sense from the Cult point of view. These 'experts' keep on producing predictions that suit the Cult agenda for societal transformation and so it was with Neil Ferguson as he revealed his horrific (and clearly insane) computer model predictions that allowed lockdowns to be imposed in Britain, the United States and many other countries. Ferguson does not have even an A-level in biology and would appear to have no formal training in computer modelling, medicine or epidemiology, according to Derek Winton, an MSc in Computational Intelligence. He wrote an article somewhat aghast at what Ferguson did which included taking no account of respiratory disease 'seasonality' which means it is far worse in the winter months. Who would have thought that respiratory disease could be worse in the winter? Well, certainly not Ferguson.

The massively China-connected Imperial College and its bizarre professor provided the excuse for the long-incubated Chinese model of human control to travel westward at lightning speed. Imperial College confirms on its website that it collaborates with the Chinese Research Institute; publishes more than 600 research papers every year with Chinese research institutions; has 225 Chinese staff; 2,600 Chinese students – the biggest international group; 7,000 former students living in China which is the largest group outside the UK; and was selected for a tour by China's President Xi Jinping during his state visit to the UK in 2015. The college takes major donations from China and describes itself as the UK's number one university collaborator with Chinese research institutions. The China communist/fascist government did not appear phased by the woeful predictions of Ferguson and Imperial when during the lockdown that Ferguson induced the college signed a five-year collaboration deal with China tech giant Huawei that will have Huawei's indoor 5G network equipment installed at the college's West London tech campus along with an 'AI cloud platform'. The deal includes Chinese sponsorship of Imperial's Venture Catalyst entrepreneurship competition. Imperial is an example of the enormous influence the Chinese government has within British and North American

universities and research centres – and further afield. Up to 200 academics from more than a dozen UK universities are being investigated on suspicion of ‘unintentionally’ helping the Chinese government build weapons of mass destruction by ‘transferring world-leading research in advanced military technology such as aircraft, missile designs and cyberweapons’. Similar scandals have broken in the United States, but it’s all a coincidence. Imperial College serves the agenda in many other ways including the promotion of every aspect of the United Nations Agenda 21/2030 (the Great Reset) and produced computer models to show that human-caused ‘climate change’ is happening when in the real world it isn’t. Imperial College is driving the climate agenda as it drives the ‘Covid’ agenda (both Cult hoaxes) while Patrick Vallance, the UK government’s Chief Scientific Adviser on ‘Covid’, was named Chief Scientific Adviser to the UN ‘climate change’ conference known as COP26 hosted by the government in Glasgow, Scotland. ‘Covid’ and ‘climate’ are fundamentally connected.

Professor Woeful

From Imperial’s bosom came Neil Ferguson still advising government despite his previous disasters and it was announced early on that he and other key people like UK Chief Medical Adviser Chris Whitty had caught the ‘virus’ as the propaganda story was being sold. Somehow they managed to survive and we had Prime Minister Boris Johnson admitted to hospital with what was said to be a severe version of the ‘virus’ in this same period. His whole policy and demeanour changed when he returned to Downing Street. It’s a small world with these government advisors – especially in their communal connections to Gates – and Ferguson had partnered with Whitty to write a paper called ‘Infectious disease: Tough choices to reduce Ebola transmission’ which involved another scare-story that didn’t happen. Ferguson’s ‘models’ predicted that up to 150,000 could die from ‘mad cow disease’, or BSE, and its version in sheep if it was transmitted to humans. BSE was not transmitted and instead triggered by an organophosphate pesticide used to treat a pest on

cows. Fewer than 200 deaths followed from the human form. Models by Ferguson and his fellow incompetents led to the unnecessary culling of millions of pigs, cattle and sheep in the foot and mouth outbreak in 2001 which destroyed the lives and livelihoods of farmers and their families who had often spent decades building their herds and flocks. Vast numbers of these animals did not have foot and mouth and had no contact with the infection. Another 'expert' behind the cull was Professor Roy Anderson, a computer modeller at Imperial College specialising in the epidemiology of *human*, not animal, disease. Anderson has served on the Bill and Melinda Gates Grand Challenges in Global Health advisory board and chairs another Gates-funded organisation. Gates is everywhere.

In a precursor to the 'Covid' script Ferguson backed closing schools 'for prolonged periods' over the swine flu 'pandemic' in 2009 and said it would affect a third of the world population if it continued to spread at the speed he claimed to be happening. His mates at Imperial College said much the same and a news report said: 'One of the authors, the epidemiologist and disease modeller Neil Ferguson, who sits on the World Health Organisation's emergency committee for the outbreak, said the virus had "full pandemic potential".' Professor Liam Donaldson, the Chris Whitty of his day as Chief Medical Officer, said the worst case could see 30 percent of the British people infected by swine flu with 65,000 dying. Ferguson and Donaldson were indeed proved correct when at the end of the year the number of deaths attributed to swine flu was 392. The term 'expert' is rather liberally applied unfortunately, not least to complete idiots. Swine flu 'projections' were great for GlaxoSmithKline (GSK) as millions rolled in for its Pandemrix influenza vaccine which led to brain damage with children most affected. The British government (taxpayers) paid out more than £60 million in compensation after GSK was given immunity from prosecution. Yet another 'Covid' déjà vu. Swine flu was supposed to have broken out in Mexico, but Dr Wolfgang Wodarg, a German doctor, former member of parliament and critic of the 'Covid' hoax, observed 'the spread of swine flu' in Mexico City at the time. He

said: 'What we experienced in Mexico City was a very mild flu which did not kill more than usual – which killed even fewer people than usual.' Hyping the fear against all the facts is not unique to 'Covid' and has happened many times before. Ferguson is reported to have over-estimated the projected death toll of bird flu (H5N1) by some three million-fold, but bird flu vaccine makers again made a killing from the scare. This is some of the background to the Neil Ferguson who produced the perfectly-timed computer models in early 2020 predicting that half a million people would die in Britain without draconian lockdown and 2.2 million in the United States. Politicians panicked, people panicked, and lockdowns of alleged short duration were instigated to 'flatten the curve' of cases gleaned from a test not testing for the 'virus'. I said at the time that the public could forget the 'short duration' bit. This was an agenda to destroy the livelihoods of the population and force them into mass control through dependency and there was going to be nothing 'short' about it. American researcher Daniel Horowitz described the consequences of the 'models' spewed out by Gates-funded Ferguson and Imperial College:

What led our government and the governments of many other countries into panic was a single Imperial College of UK study, funded by global warming activists, that predicted 2.2 million deaths if we didn't lock down the country. In addition, the reported 8-9% death rate in Italy scared us into thinking there was some other mutation of this virus that they got, which might have come here.

Together with the fact that we were finally testing and had the ability to actually report new cases, we thought we were headed for a death spiral. But again ... we can't flatten a curve if we don't know when the curve started.

How about it *never* started?

Giving them what they want

An investigation by German news outlet *Welt Am Sonntag* (*World on Sunday*) revealed how in March, 2020, the German government gathered together 'leading scientists from several research institutes and universities' and 'together, they were to produce a [modelling]

paper that would serve as legitimization for further tough political measures'. The Cult agenda was justified by computer modelling not based on evidence or reality; it was specifically constructed to justify the Cult demand for lockdowns all over the world to destroy the independent livelihoods of the global population. All these modellers and everyone responsible for the 'Covid' hoax have a date with a trial like those in Nuremberg after World War Two when Nazis faced the consequences of their war crimes. These corrupt-beyond-belief 'modellers' wrote the paper according to government instructions and it said that if lockdown measures were lifted then up to one million Germans would die from 'Covid-19' adding that some would die 'agonizingly at home, gasping for breath' unable to be treated by hospitals that couldn't cope. All lies. No matter – it gave the Cult all that it wanted. What did long-time government 'modeller' Neil Ferguson say? If the UK and the United States didn't lockdown half a million would die in Britain and 2.2 million Americans. Anyone see a theme here? 'Modellers' are such a crucial part of the lockdown strategy that we should look into their background and follow the money. Researcher Rosemary Frei produced an excellent article headlined 'The Modelling-paper Mafiosi'. She highlights a guy called John Edmunds, a British epidemiologist, and professor in the Faculty of Epidemiology and Population Health at the London School of Hygiene & Tropical Medicine. He studied at Imperial College. Edmunds is a member of government 'Covid' advisory bodies which have been dictating policy, the New and Emerging Respiratory Virus Threats Advisory Group (NERVTAG) and the Scientific Advisory Group for Emergencies (SAGE).

Ferguson, another member of NERVTAG and SAGE, led the way with the original 'virus' and Edmunds has followed in the 'variant' stage and especially the so-called UK or Kent variant known as the 'Variant of Concern' (VOC) B.1.1.7. He said in a co-written report for the Centre for Mathematical modelling of Infectious Diseases at the London School of Hygiene and Tropical Medicine, with input from the Centre's 'Covid-19' Working Group, that there was 'a realistic

possibility that VOC B.1.1.7 is associated with an increased risk of death compared to non-VOC viruses'. Fear, fear, fear, get the vaccine, fear, fear, fear, get the vaccine. Rosemary Frei reveals that almost all the paper's authors and members of the modelling centre's 'Covid-19' Working Group receive funding from the Bill and Melinda Gates Foundation and/or the associated Gates-funded Wellcome Trust. The paper was published by e-journal *Medrx* ^{xiv} which only publishes papers not peer-reviewed and the journal was established by an organisation headed by Facebook's Mark Zuckerberg and his missus. What a small world it is. Frei discovered that Edmunds is on the Scientific Advisory Board of the Coalition for Epidemic Preparedness Innovations (CEPI) which was established by the Bill and Melinda Gates Foundation, Klaus Schwab's Davos World Economic Forum and Big Pharma giant Wellcome. CEPI was 'launched in Davos [in 2017] to develop vaccines to stop future epidemics', according to its website. 'Our mission is to accelerate the development of vaccines against emerging infectious diseases and enable equitable access to these vaccines for people during outbreaks.' What kind people they are. Rosemary Frei reveals that Public Health England (PHE) director Susan Hopkins is an author of her organisation's non-peer-reviewed reports on 'new variants'. Hopkins is a professor of infectious diseases at London's Imperial College which is gifted tens of millions of dollars a year by the Bill and Melinda Gates Foundation. Gates-funded modelling disaster Neil Ferguson also co-authors Public Health England reports and he spoke in December, 2020, about the potential danger of the B.1.1.7. 'UK variant' promoted by Gates-funded modeller John Edmunds. When I come to the 'Covid vaccines' the 'new variants' will be shown for what they are – bollocks.

Connections, connections

All these people and modellers are lockdown-obsessed or, put another way, they demand what the Cult demands. Edmunds said in January, 2021, that to ease lockdowns too soon would be a disaster and they had to 'vaccinate much, much, much more widely than the

elderly'. Rosemary Frei highlights that Edmunds is married to Jeanne Pimenta who is described in a LinkedIn profile as director of epidemiology at GlaxoSmithKline (GSK) and she held shares in the company. Patrick Vallance, co-chair of SAGE and the government's Chief Scientific Adviser, is a former executive of GSK and has a deferred bonus of shares in the company worth £600,000. GSK has serious business connections with Bill Gates and is collaborating with mRNA-'vaccine' company CureVac to make 'vaccines' for the new variants that Edmunds is talking about. GSK is planning a 'Covid vaccine' with drug giant Sanofi. Puppets Prime Minister Boris Johnson announced in the spring of 2021 that up to 60 million vaccine doses were to be made at the GSK facility at Barnard Castle in the English North East. Barnard Castle, with a population of just 6,000, was famously visited in breach of lockdown rules in April, 2020, by Johnson aide Dominic Cummings who said that he drove there 'to test his eyesight' before driving back to London. Cummings would be better advised to test his integrity – not that it would take long. The GSK facility had nothing to do with his visit then although I'm sure Patrick Vallance would have been happy to arrange an introduction and some tea and biscuits. Ruthless psychopath Gates has made yet another fortune from vaccines in collaboration with Big Pharma companies and gushes at the phenomenal profits to be made from vaccines – more than a 20-to-1 return as he told one interviewer. Gates also tweeted in December, 2019, with the foreknowledge of what was coming: 'What's next for our foundation? I'm particularly excited about what the next year could mean for one of the best buys in global health: vaccines.'

Modeller John Edmunds is a big promotor of vaccines as all these people appear to be. He's the dean of the London School of Hygiene & Tropical Medicine's Faculty of Epidemiology and Population Health which is primarily funded by the Bill and Melinda Gates Foundation and the Gates-established and funded GAVI vaccine alliance which is the Gates vehicle to vaccinate the world. The organisation Doctors Without Borders has described GAVI as being 'aimed more at supporting drug-industry desires to promote new

products than at finding the most efficient and sustainable means for fighting the diseases of poverty'. But then that's why the psychopath Gates created it. John Edmunds said in a video that the London School of Hygiene & Tropical Medicine is involved in every aspect of vaccine development including large-scale clinical trials. He contends that mathematical modelling can show that vaccines protect individuals and society. That's on the basis of shit in and shit out, I take it. Edmunds serves on the UK Vaccine Network as does Ferguson and the government's foremost 'Covid' adviser, the grim-faced, dark-eyed Chris Whitty. The Vaccine Network says it works 'to support the government to identify and shortlist targeted investment opportunities for the most promising vaccines and vaccine technologies that will help combat infectious diseases with epidemic potential, and to address structural issues related to the UK's broader vaccine infrastructure'. Ferguson is acting Director of the Imperial College Vaccine Impact Modelling Consortium which has funding from the Bill and Melina Gates Foundation and the Gates-created GAVI 'vaccine alliance'. Anyone wonder why these characters see vaccines as the answer to every problem? Ferguson is wildly enthusiastic in his support for GAVI's campaign to vaccinate children en masse in poor countries. You would expect someone like Gates who has constantly talked about the need to reduce the population to want to fund vaccines to keep more people alive. I'm sure that's why he does it. The John Edmunds London School of Hygiene & Tropical Medicine (LSHTM) has a Vaccines Manufacturing Innovation Centre which develops, tests and commercialises vaccines. Rosemary Frei writes:

The vaccines centre also performs affiliated activities like combating 'vaccine hesitancy'. The latter includes the Vaccine Confidence Project. The project's stated purpose is, among other things, 'to provide analysis and guidance for early response and engagement with the public to ensure sustained confidence in vaccines and immunisation'. The Vaccine Confidence Project's director is LSHTM professor Heidi Larson. For more than a decade she's been researching how to combat vaccine hesitancy.

How the bloody hell can blokes like John Edmunds and Neil Ferguson with those connections and financial ties model 'virus' case

and death projections for the government and especially in a way that gives their paymasters like Gates exactly what they want? It's insane, but this is what you find throughout the world.

'Covid' is not dangerous, oops, wait, yes it is

Only days before Ferguson's nightmare scenario made Jackboot Johnson take Britain into a China-style lockdown to save us from a deadly 'virus' the UK government website gov.uk was reporting something very different to Ferguson on a page of official government guidance for 'high consequence infectious diseases (HCID)'. It said this about 'Covid-19':

As of 19 March 2020, COVID-19 is no longer considered to be a high consequence infectious diseases (HCID) in the UK [my emphasis]. The 4 nations public health HCID group made an interim recommendation in January 2020 to classify COVID-19 as an HCID. This was based on consideration of the UK HCID criteria about the virus and the disease with information available during the early stages of the outbreak.

Now that more is known about COVID-19, the public health bodies in the UK have reviewed the most up to date information about COVID-19 against the UK HCID criteria. They have determined that several features have now changed; in particular, more information is available about mortality rates (low overall), and there is now greater clinical awareness and a specific and sensitive laboratory test, the availability of which continues to increase. The Advisory Committee on Dangerous Pathogens (ACDP) is also of the opinion that COVID-19 should no longer be classified as an HCID.

Soon after the government had been exposed for downgrading the risk they upgraded it again and everyone was back to singing from the same Cult hymn book. Ferguson and his fellow Gates clones indicated that lockdowns and restrictions would have to continue until a Gates-funded vaccine was developed. Gates said the same because Ferguson and his like were repeating the Gates script which is the Cult script. 'Flatten the curve' became an ongoing nightmare of continuing lockdowns with periods in between of severe restrictions in pursuit of destroying independent incomes and had nothing to do with protecting health about which the Cult gives not a shit. Why wouldn't Ferguson be pushing a vaccine 'solution' when he's owned by vaccine-obsessive Gates who makes a fortune from them and

when Ferguson heads the Vaccine Impact Modelling Consortium at Imperial College funded by the Gates Foundation and GAVI, the 'vaccine alliance', created by Gates as his personal vaccine promotion operation? To compound the human catastrophe that Ferguson's 'models' did so much to create he was later exposed for breaking his own lockdown rules by having sexual liaisons with his married girlfriend Antonia Staats at his home while she was living at another location with her husband and children. Staats was a 'climate' activist and senior campaigner at the Soros-funded Avaaz which I wouldn't trust to tell me that grass is green. Ferguson had to resign as a government advisor over this hypocrisy in May, 2020, but after a period of quiet he was back being quoted by the ridiculous media on the need for more lockdowns and a vaccine rollout. Other government-advising 'scientists' from Imperial College held the fort in his absence and said lockdown could be indefinite until a vaccine was found. The Cult script was being sung by the payrolled choir. I said there was no intention of going back to 'normal' when the 'vaccine' came because the 'vaccine' is part of a very different agenda that I will discuss in Human 2.0. Why would the Cult want to let the world go back to normal when destroying that normal forever was the whole point of what was happening? House arrest, closing businesses and schools through lockdown, (un)social distancing and masks all followed the Ferguson fantasy models. Again as I predicted (these people are so predictable) when the 'vaccine' arrived we were told that house arrest, lockdown, (un)social distancing and masks would still have to continue. I will deal with the masks in the next chapter because they are of fundamental importance.

Where's the 'pandemic'?

Any mildly in-depth assessment of the figures revealed what was really going on. Cult-funded and controlled organisations still have genuine people working within them such is the number involved. So it is with Genevieve Briand, assistant program director of the Applied Economics master's degree program at Johns Hopkins

University. She analysed the impact that 'Covid-19' had on deaths from *all* causes in the United States using official data from the CDC for the period from early February to early September, 2020. She found that allegedly 'Covid' *related*-deaths exceeded those from heart disease which she found strange with heart disease always the biggest cause of fatalities. Her research became even more significant when she noted the sudden decline in 2020 of *all* non-'Covid' deaths: 'This trend is completely contrary to the pattern observed in all previous years ... the total decrease in deaths by other causes almost exactly equals the increase in deaths by Covid-19.' This was such a game, set and match in terms of what was happening that Johns Hopkins University deleted the article on the grounds that it 'was being used to support false and dangerous inaccuracies about the impact of the pandemic'. No – because it exposed the scam from official CDC figures and this was confirmed when those figures were published in January, 2021. Here we can see the effect of people dying from heart attacks, cancer, road accidents and gunshot wounds – *anything* – having 'Covid-19' on the death certificate along with those diagnosed from 'symptoms' who had even not tested positive with a test not testing for the 'virus'. I am not kidding with the gunshot wounds, by the way. Brenda Bock, coroner in Grand County, Colorado, revealed that two gunshot victims tested positive for the 'virus' within the previous 30 days and were therefore classified as 'Covid deaths'. Bock said: 'These two people had tested positive for Covid, but that's not what killed them. A gunshot wound is what killed them.' She said she had not even finished her investigation when the state listed the gunshot victims as deaths due to the 'virus'. The death and case figures for 'Covid-19' are an absolute joke and yet they are repeated like parrots by the media, politicians and alleged medical 'experts'. The official Cult narrative is the only show in town.

Genevieve Briand found that deaths from all causes were not exceptional in 2020 compared with previous years and a Spanish magazine published figures that said the same about Spain which was a 'Covid' propaganda hotspot at one point. *Discovery Salud*, a

health and medicine magazine, quoted government figures which showed how 17,000 *fewer* people died in Spain in 2020 than in 2019 and more than 26,000 fewer than in 2018. The age-standardised mortality rate for England and Wales when age distribution is taken into account was significantly lower in 2020 than the 1970s, 80s and 90s, and was only the ninth highest since 2000. Where is the 'pandemic'?

Post mortems and autopsies virtually disappeared for 'Covid' deaths amid claims that 'virus-infected' bodily fluids posed a risk to those carrying out the autopsy. This was rejected by renowned German pathologist and forensic doctor Klaus Püschel who said that he and his staff had by then done 150 autopsies on 'Covid' patients with no problems at all. He said they were needed to know why some 'Covid' patients suffered blood clots and not severe respiratory infections. The 'virus' is, after all, called SARS or 'severe acute respiratory syndrome'. I highlighted in the spring of 2020 this phenomenon and quoted New York intensive care doctor Cameron Kyle-Sidell who posted a soon deleted YouTube video to say that they had been told to prepare to treat an infectious disease called 'Covid-19', but that was not what they were dealing with. Instead he likened the lung condition of the most severely ill patients to what you would expect with cabin depressurisation in a plane at 30,000 feet or someone dropped on the top of Everest without oxygen or acclimatisation. I have never said this is not happening to a small minority of alleged 'Covid' patients – I am saying this is not caused by a phantom 'contagious virus'. Indeed Kyle-Sidell said that 'Covid-19' was not the disease they were told was coming their way. 'We are operating under a medical paradigm that is untrue,' he said, and he believed they were treating the wrong disease: 'These people are being slowly starved of oxygen.' Patients would take off their oxygen masks in a state of fear and stress and while they were blue in the face on the brink of death. They did not look like patients dying of pneumonia. You can see why they don't want autopsies when their virus doesn't exist and there is another condition in some people that they don't wish to be uncovered. I should add here that

the 5G system of millimetre waves was being rapidly introduced around the world in 2020 and even more so now as they fire 5G at the Earth from satellites. At 60 gigahertz within the 5G range that frequency interacts with the oxygen molecule and stops people breathing in sufficient oxygen to be absorbed into the bloodstream. They are installing 5G in schools and hospitals. The world is not mad or anything. 5G can cause major changes to the lungs and blood as I detail in *The Answer* and these consequences are labelled 'Covid-19', the alleged symptoms of which can be caused by 5G and other electromagnetic frequencies as cells respond to radiation poisoning.

The 'Covid death' scam

Dr Scott Jensen, a Minnesota state senator and medical doctor, exposed 'Covid' Medicare payment incentives to hospitals and death certificate manipulation. He said he was sent a seven-page document by the US Department of Health 'coaching' him on how to fill out death certificates which had never happened before. The document said that he didn't need to have a laboratory test for 'Covid-19' to put that on the death certificate and that shocked him when death certificates are supposed to be about facts. Jensen described how doctors had been 'encouraged, if not pressured' to make a diagnosis of 'Covid-19' if they thought it was probable or '*presumed*'. No positive test was necessary – not that this would have mattered anyway. He said doctors were told to diagnose 'Covid' by symptoms when these were the same as colds, allergies, other respiratory problems, and certainly with influenza which 'disappeared' in the 'Covid' era. A common sniffle was enough to get the dreaded verdict. Ontario authorities decreed that a single care home resident with *one* symptom from a long list must lead to the isolation of the entire home. Other courageous doctors like Jensen made the same point about death figure manipulation and how deaths by other causes were falling while 'Covid-19 deaths' were rising at the same rate due to re-diagnosis. Their videos rarely survive long on YouTube with its Cult-supporting algorithms courtesy of CEO Susan Wojcicki and her bosses at Google. Figure-tampering was so glaring

and ubiquitous that even officials were letting it slip or outright saying it. UK chief scientific adviser Patrick Vallance said on one occasion that 'Covid' on the death certificate doesn't mean 'Covid' was the cause of death (so why the hell is it there?) and we had the rare sight of a BBC reporter telling the truth when she said: 'Someone could be successfully treated for Covid, in say April, discharged, and then in June, get run over by a bus and die ... That person would still be counted as a Covid death in England.' Yet the BBC and the rest of the world media went on repeating the case and death figures as if they were real. Illinois Public Health Director Dr Ngozi Ezike revealed the deceit while her bosses must have been clenching their buttocks:

If you were in a hospice and given a few weeks to live and you were then found to have Covid that would be counted as a Covid death. [There might be] a clear alternate cause, but it is still listed as a Covid death. So everyone listed as a Covid death doesn't mean that was the cause of the death, but that they had Covid at the time of death.

Yes, a 'Covid virus' never shown to exist and tested for with a test not testing for the 'virus'. In the first period of the pandemic hoax through the spring of 2020 the process began of designating almost everything a 'Covid' death and this has continued ever since. I sat in a restaurant one night listening to a loud conversation on the next table where a family was discussing in bewilderment how a relative who had no symptoms of 'Covid', and had died of a long-term problem, could have been diagnosed a death by the 'virus'. I could understand their bewilderment. If they read this book they will know why this medical fraud has been perpetrated the world over.

Some media truth shock

The media ignored the evidence of death certificate fraud until eventually one columnist did speak out when she saw it first-hand. Bel Mooney is a long-time national newspaper journalist in Britain currently working for the *Daily Mail*. Her article on February 19th, 2021, carried this headline: 'My dad Ted passed three Covid tests

and died of a chronic illness yet he's officially one of Britain's 120,000 victims of the virus and is far from alone ... so how many more are there?' She told how her 99-year-old father was in a care home with a long-standing chronic obstructive pulmonary disease and vascular dementia. Maybe, but he was still aware enough to tell her from the start that there was no 'virus' and he refused the 'vaccine' for that reason. His death was not unexpected given his chronic health problems and Mooney said she was shocked to find that 'Covid-19' was declared the cause of death on his death certificate. She said this was a 'bizarre and unacceptable untruth' for a man with long-time health problems who had tested negative twice at the home for the 'virus'. I was also shocked by this story although not by what she said. I had been highlighting the death certificate manipulation for ten months. It was the confirmation that a professional full-time journalist only realised this was going on when it affected her directly and neither did she know that whether her dad tested positive or negative was irrelevant with the test not testing for the 'virus'. Where had she been? She said she did not believe in 'conspiracy theories' without knowing I'm sure that this and 'conspiracy theorists' were terms put into widespread circulation by the CIA in the 1960s to discredit those who did not accept the ridiculous official story of the Kennedy assassination. A blanket statement of 'I don't believe in conspiracy theories' is always bizarre. The dictionary definition of the term alone means the world is drowning in conspiracies. What she said was even more daft when her dad had just been affected by the 'Covid' conspiracy. Why else does she think that 'Covid-19' was going on the death certificates of people who died of something else?

To be fair once she saw from personal experience what was happening she didn't mince words. Mooney was called by the care home on the morning of February 9th to be told her father had died in his sleep. When she asked for the official cause of death what came back was 'Covid-19'. Mooney challenged this and was told there had been deaths from Covid on the dementia floor (confirmed by a test not testing for the 'virus') so they considered it 'reasonable

to assume'. 'But doctor,' Mooney rightly protested, 'an assumption isn't a diagnosis.' She said she didn't blame the perfectly decent and sympathetic doctor – 'he was just doing his job'. Sorry, but that's *bullshit*. He wasn't doing his job at all. He was putting a false cause of death on the death certificate and that is a criminal offence for which he should be brought to account and the same with the millions of doctors worldwide who have done the same. They were not doing their job they were following orders and that must not wash at new Nuremberg trials any more than it did at the first ones. Mooney's doctor was 'assuming' (presuming) as he was told to, but 'just following orders' makes no difference to his actions. A doctor's job is to serve the patient and the truth, not follow orders, but that's what they have done all over the world and played a central part in making the 'Covid' hoax possible with all its catastrophic consequences for humanity. Shame on them and they must answer for their actions. Mooney said her disquiet worsened when she registered her father's death by telephone and was told by the registrar there had been very many other cases like hers where 'the deceased' had not tested positive for 'Covid' yet it was recorded as the cause of death. The test may not matter, but those involved at their level *think* it matters and it shows a callous disregard for accurate diagnosis. The pressure to do this is coming from the top of the national 'health' pyramids which in turn obey the World Health Organization which obeys Gates and the Cult. Mooney said the registrar agreed that this must distort the national figures adding that 'the strangest thing is that every winter we record countless deaths from flu, and this winter there have been none. Not one!' She asked if the registrar thought deaths from flu were being misdiagnosed and lumped together with 'Covid' deaths. The answer was a 'puzzled yes'. Mooney said that the funeral director said the same about 'Covid' deaths which had nothing to do with 'Covid'. They had lost count of the number of families upset by this and other funeral companies in different countries have had the same experience. Mooney wrote:

The nightly shroud-waving and shocking close-ups of pain imposed on us by the TV news bewildered and terrified the population into eager compliance with lockdowns. We were invited to 'save the NHS' and to grieve for strangers – the real-life loved ones behind those shocking death counts. Why would the public imagine what I now fear, namely that the way Covid-19 death statistics are compiled might make the numbers seem greater than they are?

Oh, just a little bit – like 100 percent.

Do the maths

Mooney asked why a country would wish to skew its mortality figures by wrongly certifying deaths? What had been going on? Well, if you don't believe in conspiracies you will never find the answer which is that *it's a conspiracy*. She did, however, describe what she had discovered as a 'national scandal'. In reality it's a global scandal and happening everywhere. Pillars of this conspiracy were all put into place before the button was pressed with the Drosten PCR protocol and high amplifications to produce the cases and death certificate changes to secure illusory 'Covid' deaths. Mooney notes that normally two doctors were needed to certify a death, with one having to know the patient, and how the rules were changed in the spring of 2020 to allow one doctor to do this. In the same period 'Covid deaths' were decreed to be all cases where Covid-19 was put on the death certificate even without a positive test or any symptoms. Mooney asked: 'How many of the 30,851 (as of January 15) care home resident deaths with Covid-19 on the certificate (32.4 per cent of all deaths so far) were based on an assumption, like that of my father? And what has that done to our national psyche?' All of them is the answer to the first question and it has devastated and dismantled the national psyche, actually the global psyche, on a colossal scale. In the UK case and death data is compiled by organisations like Public Health England (PHE) and the Office for National Statistics (ONS). Mooney highlights the insane policy of counting a death from any cause as 'Covid-19' if this happens within 28 days of a positive test (with a test not testing for the 'virus') and she points out that ONS statistics reflect deaths 'involving Covid' 'or due to Covid' which meant in practice any

death where 'Covid-19' was mentioned on the death certificate. She described the consequences of this fraud:

Most people will accept the narrative they are fed, so panicky governments here and in Europe witnessed the harsh measures enacted in totalitarian China and jumped into lockdown. Headlines about Covid deaths tolled like the knell that would bring doomsday to us all. Fear stalked our empty streets. Politicians parroted the frankly ridiculous aim of 'zero Covid' and shut down the economy, while most British people agreed that lockdown was essential and (astonishingly to me, as a patriotic Brit) even wanted more restrictions.

For what? Lies on death certificates? Never mind the grim toll of lives ruined, suicides, schools closed, rising inequality, depression, cancelled hospital treatments, cancer patients in a torture of waiting, poverty, economic devastation, loneliness, families kept apart, and so on. How many lives have been lost as a direct result of lockdown?

She said that we could join in a national chorus of shock and horror at reaching the 120,000 death toll which was surely certain to have been totally skewed all along, but what about the human cost of lockdown justified by these 'death figures'? *The British Medical Journal* had reported a 1,493 percent increase in cases of children taken to Great Ormond Street Hospital with abusive head injuries alone and then there was the effect on families:

Perhaps the most shocking thing about all this is that families have been kept apart – and obeyed the most irrational, changing rules at the whim of government – because they believed in the statistics. They succumbed to fear, which his generation rejected in that war fought for freedom. Dad (God rest his soul) would be angry. And so am I.

Another theme to watch is that in the winter months when there are more deaths from all causes they focus on 'Covid' deaths and in the summer when the British Lung Foundation says respiratory disease plummets by 80 percent they rage on about 'cases'. Either way fascism on population is always the answer.

Nazi eugenics in the 21st century

Elderly people in care homes have been isolated from their families month after lonely month with no contact with relatives and grandchildren who were banned from seeing them. We were told

that lockdown fascism was to 'protect the vulnerable' like elderly people. At the same time Do Not Resuscitate (DNR) orders were placed on their medical files so that if they needed resuscitation it wasn't done and 'Covid-19' went on their death certificates. Old people were not being 'protected' they were being culled – murdered in truth. DNR orders were being decreed for disabled and young people with learning difficulties or psychological problems. The UK Care Quality Commission, a non-departmental body of the Department of Health and Social Care, found that 34 percent of those working in health and social care were pressured into placing 'do not attempt cardiopulmonary resuscitation' orders on 'Covid' patients who suffered from disabilities and learning difficulties without involving the patient or their families in the decision. UK judges ruled that an elderly woman with dementia should have the DNA-manipulating 'Covid vaccine' against her son's wishes and that a man with severe learning difficulties should have the jab despite his family's objections. Never mind that many had already died. The judiciary always supports doctors and government in fascist dictatorships. They wouldn't dare do otherwise. A horrific video was posted showing fascist officers from Los Angeles police forcibly giving the 'Covid' shot to women with special needs who were screaming that they didn't want it. The same fascists are seen giving the jab to a sleeping elderly woman in a care home. This is straight out of the Nazi playbook. Hitler's Nazis committed mass murder of the mentally ill and physically disabled throughout Germany and occupied territories in the programme that became known as Aktion T4, or just T4. Sabbatian-controlled Hitler and his grotesque crazies set out to kill those they considered useless and unnecessary. The Reich Committee for the Scientific Registering of Hereditary and Congenital Illnesses registered the births of babies identified by physicians to have 'defects'. By 1941 alone more than 5,000 children were murdered by the state and it is estimated that in total the number of innocent people killed in Aktion T4 was between 275,000 and 300,000. Parents were told their children had been sent away for 'special treatment' never to return. It is rather pathetic to see claims about plans for new extermination camps being dismissed today

when the same force behind current events did precisely that 80 years ago. Margaret Sanger was a Cult operative who used 'birth control' to sanitise her programme of eugenics. Organisations she founded became what is now Planned Parenthood. Sanger proposed that 'the whole dysgenic population would have its choice of segregation or sterilization'. These included epileptics, 'feeble-minded', and prostitutes. Sanger opposed charity because it perpetuated 'human waste'. She reveals the Cult mentality and if anyone thinks that extermination camps are a 'conspiracy theory' their naivety is touching if breathtakingly stupid.

If you don't believe that doctors can act with callous disregard for their patients it is worth considering that doctors and medical staff agreed to put government-decreed DNR orders on medical files and do nothing when resuscitation is called for. I don't know what you call such people in your house. In mine they are Nazis from the Josef Mengele School of Medicine. Phenomenal numbers of old people have died worldwide from the effects of lockdown, depression, lack of treatment, the 'vaccine' (more later) and losing the will to live. A common response at the start of the manufactured pandemic was to remove old people from hospital beds and transfer them to nursing homes. The decision would result in a mass cull of elderly people in those homes through lack of treatment – *not* 'Covid'. Care home whistleblowers have told how once the 'Covid' era began doctors would not come to their homes to treat patients and they were begging for drugs like antibiotics that often never came. The most infamous example was ordered by New York governor Andrew Cuomo, brother of a moronic CNN host, who amazingly was given an Emmy Award for his handling of the 'Covid crisis' by the ridiculous Wokers that hand them out. Just how ridiculous could be seen in February, 2021, when a Department of Justice and FBI investigation began into how thousands of old people in New York died in nursing homes after being discharged from hospital to make way for 'Covid' patients on Cuomo's say-so – and how he and his staff covered up these facts. This couldn't have happened to a nicer psychopath. Even then there was a 'Covid' spin. Reports said that

thousands of old people who tested positive for 'Covid' in hospital were transferred to nursing homes to both die of 'Covid' and transmit it to others. No – they were in hospital because they were ill and the fact that they tested positive with a test not testing for the 'virus' is irrelevant. They were ill often with respiratory diseases ubiquitous in old people near the end of their lives. Their transfer out of hospital meant that their treatment stopped and many would go on to die.

They're old. Who gives a damn?

I have exposed in the books for decades the Cult plan to cull the world's old people and even to introduce at some point what they call a 'demise pill' which at a certain age everyone would take and be out of here by law. In March, 2021, Spain legalised euthanasia and assisted suicide following the Netherlands, Belgium, Luxembourg and Canada on the Tiptoe to the demise pill. Treatment of old people by many 'care' homes has been a disgrace in the 'Covid' era. There are many, many, caring staff – I know some. There have, however, been legions of stories about callous treatment of old people and their families. Police were called when families came to take their loved ones home in the light of isolation that was killing them. They became prisoners of the state. Care home residents in insane, fascist Ontario, Canada, were not allowed to leave their *room* once the 'Covid' hoax began. UK staff have even wheeled elderly people away from windows where family members were talking with them. Oriana Criscuolo from Stockport in the English North West dropped off some things for her 80-year-old father who has Parkinson's disease and dementia and she wanted to wave to him through a ground-floor window. She was told that was 'illegal'. When she went anyway they closed the curtains in the middle of the day. Oriana said:

It's just unbelievable. I cannot understand how care home staff – people who are being paid to care – have become so uncaring. Their behaviour is inhumane and cruel. It's beyond belief.

She was right and this was not a one-off. What a way to end your life in such loveless circumstances. UK registered nurse Nicky Millen, a proper old school nurse for 40 years, said that when she started her career care was based on dignity, choice, compassion and empathy. Now she said 'the things that are important to me have gone out of the window.' She was appalled that people were dying without their loved ones and saying goodbye on iPads. Nicky described how a distressed 89-year-old lady stroked her face and asked her 'how many paracetamol would it take to finish me off'. Life was no longer worth living while not seeing her family. Nicky said she was humiliated in front of the ward staff and patients for letting the lady stroke her face and giving her a cuddle. Such is the dehumanisation that the 'Covid' hoax has brought to the surface. Nicky worked in care homes where patients told her they were being held prisoner. 'I want to live until I die', one said to her. 'I had a lady in tears because she hadn't seen her great-grandson.' Nicky was compassionate old school meeting psychopathic New Normal. She also said she had worked on a 'Covid' ward with no 'Covid' patients. Jewish writer Shai Held wrote an article in March, 2020, which was headlined 'The Staggering, Heartless Cruelty Toward the Elderly'. What he described was happening from the earliest days of lockdown. He said 'the elderly' were considered a group and not unique individuals (the way of the Woke). Shai Held said:

Notice how the all-too-familiar rhetoric of dehumanization works: 'The elderly' are bunched together as a faceless mass, all of them considered culprits and thus effectively deserving of the suffering the pandemic will inflict upon them. Lost entirely is the fact that the elderly are individual human beings, each with a distinctive face and voice, each with hopes and dreams, memories and regrets, friendships and marriages, loves lost and loves sustained.

'The elderly' have become another dehumanised group for which anything goes and for many that has resulted in cold disregard for their rights and their life. The distinctive face that Held talks about is designed to be deleted by masks until everyone is part of a faceless mass.

'War-zone' hospitals myth

Again and again medical professionals have told me what was really going on and how hospitals 'overrun like war zones' according to the media were virtually empty. The mantra from medical whistleblowers was please don't use my name or my career is over. Citizen journalists around the world sneaked into hospitals to film evidence exposing the 'war-zone' lie. They really *were* largely empty with closed wards and operating theatres. I met a hospital worker in my town on the Isle of Wight during the first lockdown in 2020 who said the only island hospital had never been so quiet. Lockdown was justified by the psychopaths to stop hospitals being overrun. At the same time that the island hospital was near-empty the military arrived here to provide *extra beds*. It was all propaganda to ramp up the fear to ensure compliance with fascism as were never-used temporary hospitals with thousands of beds known as Nightingales and never-used make-shift mortuaries opened by the criminal UK government. A man who helped to install those extra island beds attributed to the army said they were never used and the hospital was empty. Doctors and nurses 'stood around talking or on their phones, wandering down to us to see what we were doing'. There were no masks or social distancing. He accused the useless local island paper, the *County Press*, of 'pumping the fear as if our hospital was overrun and we only have one so it should have been'. He described ambulances parked up with crews outside in deck chairs. When his brother called an ambulance he was told there was a two-hour backlog which he called 'bullshit'. An old lady on the island fell 'and was in a bad way', but a caller who rang for an ambulance was told the situation wasn't urgent enough. Ambulance stations were working under capacity while people would hear ambulances with sirens blaring driving through the streets. When those living near the stations realised what was going on they would follow them as they left, circulated around an urban area with the sirens going, and then came back without stopping. All this was to increase levels of fear and the same goes for the 'ventilator shortage crisis' that cost tens of millions for hastily produced ventilators never to be used.

Ambulance crews that agreed to be exploited in this way for fear propaganda might find themselves a mirror. I wish them well with that. Empty hospitals were the obvious consequence of treatment and diagnoses of non-'Covid' conditions cancelled and those involved handed a death sentence. People have been dying at home from undiagnosed and untreated cancer, heart disease and other life-threatening conditions to allow empty hospitals to deal with a 'pandemic' that wasn't happening.

Death of the innocent

'War-zones' have been laying off nursing staff, even doctors where they can. There was no work for them. Lockdown was justified by saving lives and protecting the vulnerable they were actually killing with DNR orders and preventing empty hospitals being 'overrun'. In Britain the mantra of stay at home to 'save the NHS' was everywhere and across the world the same story was being sold when it was all lies. Two California doctors, Dan Erickson and Artin Massihi at Accelerated Urgent Care in Bakersfield, held a news conference in April, 2020, to say that intensive care units in California were 'empty, essentially', with hospitals shutting floors, not treating patients and laying off doctors. The California health system was working at minimum capacity 'getting rid of doctors because we just don't have the volume'. They said that people with conditions such as heart disease and cancer were not coming to hospital out of fear of 'Covid-19'. Their video was deleted by Susan Wojcicki's Cult-owned YouTube after reaching five million views. Florida governor Ron Desantis, who rejected the severe lockdowns of other states and is being targeted for doing so, said that in March, 2020, every US governor was given models claiming they would run out of hospital beds in days. That was never going to happen and the 'modellers' knew it. Deceit can be found at every level of the system. Urgent children's operations were cancelled including fracture repairs and biopsies to spot cancer. Eric Nicholls, a consultant paediatrician, said 'this is obviously concerning and we need to return to normal operating and to increase capacity as soon as possible'. Psychopaths

in power were rather less concerned *because* they are psychopaths. Deletion of urgent care and diagnosis has been happening all over the world and how many kids and others have died as a result of the actions of these cold and heartless lunatics dictating 'health' policy? The number must be stratospheric. Richard Sullivan, professor of cancer and global health at King's College London, said people feared 'Covid' more than cancer such was the campaign of fear. 'Years of lost life will be quite dramatic', Sullivan said, with 'a huge amount of avoidable mortality'. Sarah Woolnough, executive director for policy at Cancer Research UK, said there had been a 75 percent drop in urgent referrals to hospitals by family doctors of people with suspected cancer. Sullivan said that 'a lot of services have had to scale back – we've seen a dramatic decrease in the amount of elective cancer surgery'. Lockdown deaths worldwide has been absolutely fantastic with the *New York Post* reporting how data confirmed that 'lockdowns end more lives than they save':

There was a sharp decline in visits to emergency rooms and an increase in fatal heart attacks because patients didn't receive prompt treatment. Many fewer people were screened for cancer. Social isolation contributed to excess deaths from dementia and Alzheimer's.

Researchers predicted that the social and economic upheaval would lead to tens of thousands of "deaths of despair" from drug overdoses, alcoholism and suicide. As unemployment surged and mental-health and substance-abuse treatment programs were interrupted, the reported levels of anxiety, depression and suicidal thoughts increased dramatically, as did alcohol sales and fatal drug overdoses.

This has been happening while nurses and other staff had so much time on their hands in the 'war-zones' that Tic-Tok dancing videos began appearing across the Internet with medical staff dancing around in empty wards and corridors as people died at home from causes that would normally have been treated in hospital.

Mentions in dispatches

One brave and truth-committed whistleblower was Louise Hampton, a call handler with the UK NHS who made a viral Internet video saying she had done 'fuck all' during the 'pandemic'

which was 'a load of bollocks'. She said that 'Covid-19' was rebranded flu and of course she lost her job. This is what happens in the medical and endless other professions now when you tell the truth. Louise filmed inside 'war-zone' accident and emergency departments to show they were empty and I mean *empty* as in no one there. The mainstream media could have done the same and blown the gaff on the whole conspiracy. They haven't to their eternal shame. Not that most 'journalists' seem capable of manifesting shame as with the psychopaths they slavishly repeat without question. The relative few who were admitted with serious health problems were left to die alone with no loved ones allowed to see them because of 'Covid' rules and they included kids dying without the comfort of mum and dad at their bedside while the evil behind this couldn't give a damn. It was all good fun to them. A Scottish NHS staff nurse publicly quit in the spring of 2021 saying: 'I can no longer be part of the lies and the corruption by the government.' She said hospitals 'aren't full, the beds aren't full, beds have been shut, wards have been shut'. Hospitals were never busy throughout 'Covid'. The staff nurse said that Nicola Sturgeon, tragically the leader of the Scottish government, was on television saying save the hospitals and the NHS – 'but the beds are empty' and 'we've not seen flu, we always see flu every year'. She wrote to government and spoke with her union Unison (the unions are Cult-compromised and *useless*, but nothing changed. Many of her colleagues were scared of losing their jobs if they spoke out as they wanted to. She said nursing staff were being affected by wearing masks all day and 'my head is splitting every shift from wearing a mask'. The NHS is part of the fascist tyranny and must be dismantled so we can start again with human beings in charge. (Ironically, hospitals were reported to be busier again when official 'Covid' cases *fell* in spring/summer of 2021 and many other conditions required treatment at the same time as *the fake vaccine rollout*.)

I will cover the 'Covid vaccine' scam in detail later, but it is another indicator of the sickening disregard for human life that I am highlighting here. The DNA-manipulating concoctions do not fulfil

the definition of a 'vaccine', have never been used on humans before and were given only emergency approval because trials were not completed and they continued using the unknowing public. The result was what a NHS senior nurse with responsibility for 'vaccine' procedure said was 'genocide'. She said the 'vaccines' were not 'vaccines'. They had not been shown to be safe and claims about their effectiveness by drug companies were 'poetic licence'. She described what was happening as a 'horrid act of human annihilation'. The nurse said that management had instigated a policy of not providing a Patient Information Leaflet (PIL) before people were 'vaccinated' even though health care professionals are supposed to do this according to protocol. Patients should also be told that they are taking part in an ongoing clinical trial. Her challenges to what is happening had seen her excluded from meetings and ridiculed in others. She said she was told to 'watch my step ... or I would find myself surplus to requirements'. The nurse, who spoke anonymously in fear of her career, said she asked her NHS manager why he/she was content with taking part in genocide against those having the 'vaccines'. The reply was that everyone had to play their part and to 'put up, shut up, and get it done'. Government was 'leaning heavily' on NHS management which was clearly leaning heavily on staff. This is how the global 'medical' hierarchy operates and it starts with the Cult and its World Health Organization.

She told the story of a doctor who had the Pfizer jab and when questioned had no idea what was in it. The doctor had never read the literature. We have to stop treating doctors as intellectual giants when so many are moral and medical pygmies. The doctor did not even know that the 'vaccines' were not fully approved or that their trials were ongoing. They were, however, asking their patients if they minded taking part in follow-ups for research purposes – yes, the *ongoing clinical trial*. The nurse said the doctor's ignorance was not rare and she had spoken to a hospital consultant who had the jab without any idea of the background or that the 'trials' had not been completed. Nurses and pharmacists had shown the same ignorance.

‘My NHS colleagues have forsaken their duty of care, broken their code of conduct – Hippocratic Oath – and have been brainwashed just the same as the majority of the UK public through propaganda ...’ She said she had not been able to recruit a single NHS colleague, doctor, nurse or pharmacist to stand with her and speak out. Her union had refused to help. She said that if the genocide came to light she would not hesitate to give evidence at a Nuremberg-type trial against those in power who could have affected the outcomes but didn’t.

And all for what?

To put the nonsense into perspective let’s say the ‘virus’ does exist and let’s go completely crazy and accept that the official manipulated figures for cases and deaths are accurate. *Even then* a study by Stanford University epidemiologist Dr John Ioannidis published on the World Health Organization website produced an average infection to fatality rate of ... *0.23 percent!* Ioannidis said: ‘If one could sample equally from all locations globally, the median infection fatality rate might even be substantially lower than the 0.23% observed in my analysis.’ For healthy people under 70 it was ... *0.05 percent!* This compares with the 3.4 percent claimed by the Cult-owned World Health Organization when the hoax was first played and maximum fear needed to be generated. An updated Stanford study in April, 2021, put the ‘infection’ to ‘fatality’ rate at just 0.15 percent. Another team of scientists led by Megan O’Driscoll and Henrik Salje studied data from 45 countries and published their findings on the Nature website. For children and young people the figure is so small it virtually does not register although authorities will be hyping dangers to the young when they introduce DNA-manipulating ‘vaccines’ for children. The O’Driscoll study produced an average infection-fatality figure of 0.003 for children from birth to four; 0.001 for 5 to 14; 0.003 for 15 to 19; and it was still only 0.456 up to 64. To claim that children must be ‘vaccinated’ to protect them from ‘Covid’ is an obvious lie and so there must be another reason and there is. What’s more the average age of a ‘Covid’ death is akin

to the average age that people die in general. The average age of death in England is about 80 for men and 83 for women. The average age of death from alleged 'Covid' is between 82 and 83. California doctors, Dan Erickson and Artin Massihi, said at their April media conference that projection models of millions of deaths had been 'woefully inaccurate'. They produced detailed figures showing that Californians had a 0.03 chance of dying from 'Covid' based on the number of people who tested positive (with a test not testing for the 'virus'). Erickson said there was a 0.1 percent chance of dying from 'Covid' in the *state* of New York, not just the city, and a 0.05 percent chance in Spain, a centre of 'Covid-19' hysteria at one stage. The Stanford studies supported the doctors' data with fatality rate estimates of 0.23 and 0.15 percent. How close are these figures to my estimate of *zero*? Death-rate figures claimed by the World Health Organization at the start of the hoax were some 15 times higher. The California doctors said there was no justification for lockdowns and the economic devastation they caused. Everything they had ever learned about quarantine was that you quarantine the *sick* and not the healthy. They had never seen this before and it made no medical sense.

Why in the in the light of all this would governments and medical systems the world over say that billions must go under house arrest; lose their livelihood; in many cases lose their mind, their health and their life; force people to wear masks dangerous to health and psychology; make human interaction and even family interaction a criminal offence; ban travel; close restaurants, bars, watching live sport, concerts, theatre, and any activity involving human togetherness and discourse; and closing schools to isolate children from their friends and cause many to commit suicide in acts of hopelessness and despair? The California doctors said lockdown consequences included increased child abuse, partner abuse, alcoholism, depression, and other impacts they were seeing every day. Who would do that to the entire human race if not mentally-ill psychopaths of almost unimaginable extremes like Bill Gates? We must face the reality of what we are dealing with and come out of

denial. Fascism and tyranny are made possible only by the target population submitting and acquiescing to fascism and tyranny. The whole of human history shows that to be true. Most people naively and unquestioning believed what they were told about a 'deadly virus' and meekly and weakly submitted to house arrest. Those who didn't believe it – at least in total – still submitted in fear of the consequences of not doing so. For the rest who wouldn't submit draconian fines have been imposed, brutal policing by psychopaths *for* psychopaths, and condemnation from the meek and weak who condemn the Pushbackers on behalf of the very force that has them, too, in its gun sights. 'Pathetic' does not even begin to suffice. Britain's brainless 'Health' Secretary Matt Hancock warned anyone lying to border officials about returning from a list of 'hotspot' countries could face a jail sentence of up to ten years which is more than for racially-aggravated assault, incest and attempting to have sex with a child under 13. Hancock is a lunatic, but he has the state apparatus behind him in a Cult-led chain reaction and the same with UK 'Vaccine Minister' Nadhim Zahawi, a prominent member of the mega-Cult secret society, Le Cercle, which featured in my earlier books. The Cult enforces its will on governments and medical systems; government and medical systems enforce their will on business and police; business enforces its will on staff who enforce it on customers; police enforce the will of the Cult on the population and play their essential part in creating a world of fascist control that their own children and grandchildren will have to live in their entire lives. It is a hierarchical pyramid of imposition and acquiescence and, yes indeed, of clinical insanity.

Does anyone bright enough to read this book have to ask what the answer is? I think not, but I will reveal it anyway in the fewest of syllables: Tell the psychos and their moronic lackeys to fuck off and let's get on with our lives. We are many – They are few.

CHAPTER SEVEN

War on your mind

One believes things because one has been conditioned to believe them

Aldous Huxley, Brave New World

I have described the 'Covid' hoax as a 'Psyop' and that is true in every sense and on every level in accordance with the definition of that term which is psychological warfare. Break down the 'Covid pandemic' to the foundation themes and it is psychological warfare on the human individual and collective mind.

The same can be said for the entire human belief system involving every subject you can imagine. Huxley was right in his contention that people believe what they are conditioned to believe and this comes from the repetition throughout their lives of the same falsehoods. They spew from government, corporations, media and endless streams of 'experts' telling you what the Cult wants you to believe and often believing it themselves (although *far* from always). 'Experts' are rewarded with 'prestigious' jobs and titles and as agents of perceptual programming with regular access to the media. The Cult has to control the narrative – control *information* – or they lose control of the vital, crucial, without-which-they-cannot-prevail public perception of reality. The foundation of that control today is the Internet made possible by the Defense Advanced Research Projects Agency (DARPA), the incredibly sinister technological arm of the Pentagon. The Internet is the result of military technology.

DARPA openly brags about establishing the Internet which has been a long-term project to lasso the minds of the global population. I have said for decades the plan is to control information to such an extreme that eventually no one would see or hear anything that the Cult does not approve. We are closing in on that end with ferocious censorship since the 'Covid' hoax began and in my case it started back in the 1990s in terms of books and speaking venues. I had to create my own publishing company in 1995 precisely because no one else would publish my books even then. I think they're all still running.

Cult Internet

To secure total control of information they needed the Internet in which pre-programmed algorithms can seek out 'unclean' content for deletion and even stop it being posted in the first place. The Cult had to dismantle print and non-Internet broadcast media to ensure the transfer of information to the appropriate-named 'Web' – a critical expression of the *Cult* web. We've seen the ever-quicken demise of traditional media and control of what is left by a tiny number of corporations operating worldwide. Independent journalism in the mainstream is already dead and never was that more obvious than since the turn of 2020. The Cult wants all information communicated via the Internet to globally censor and allow the plug to be pulled any time. Lockdowns and forced isolation has meant that communication between people has been through electronic means and no longer through face-to-face discourse and discussion. Cult psychopaths have targeted the bars, restaurants, sport, venues and meeting places in general for this reason. None of this is by chance and it's to stop people gathering in any kind of privacy or number while being able to track and monitor all Internet communications and block them as necessary. Even private messages between individuals have been censored by these fascists that control Cult fronts like Facebook, Twitter, Google and YouTube which are all officially run by Sabbatian place-people and from the background by higher-level Sabbatian place people.

Facebook, Google, Amazon and their like were seed-funded and supported into existence with money-no-object infusions of funds either directly or indirectly from DARPA and CIA technology arm In-Q-Tel. The Cult plays the long game and prepares very carefully for big plays like 'Covid'. Amazon is another front in the psychological war and pretty much controls the global market in book sales and increasingly publishing. Amazon's limitless funds have deleted fantastic numbers of independent publishers to seize global domination on the way to deciding which books can be sold and circulated and which cannot. Moves in that direction are already happening. Amazon's leading light Jeff Bezos is the grandson of Lawrence Preston Gise who worked with DARPA predecessor ARPA. Amazon has big connections to the CIA and the Pentagon. The plan I have long described went like this:

1. Employ military technology to establish the Internet.
2. Sell the Internet as a place where people can freely communicate without censorship and allow that to happen until the Net becomes the central and irreversible pillar of human society. If the Internet had been highly censored from the start many would have rejected it.
3. Fund and manipulate major corporations into being to control the circulation of information on your Internet using cover stories about geeks in garages to explain how they came about. Give them unlimited funds to expand rapidly with no need to make a profit for years while non-Cult companies who need to balance the books cannot compete. You know that in these circumstances your Googles, YouTubes, Facebooks and Amazons are going to secure near monopolies by either crushing or buying up the opposition.
4. Allow freedom of expression on both the Internet and communication platforms to draw people in until the Internet is the central and irreversible pillar of human society and your communication corporations have reached a stage of near monopoly domination.
5. Then unleash your always-planned frenzy of censorship on the basis of 'where else are you going to go?' and continue to expand that until nothing remains that the Cult does not want its human targets to see.

The process was timed to hit the 'Covid' hoax to ensure the best chance possible of controlling the narrative which they knew they had to do at all costs. They were, after all, about to unleash a 'deadly virus' that didn't really exist. If you do that in an environment of free-flowing information and opinion you would be dead in the

water before you could say Gates is a psychopath. The network was in place through which the Cult-created-and-owned World Health Organization could dictate the 'Covid' narrative and response policy slavishly supported by Cult-owned Internet communication giants and mainstream media while those telling a different story were censored. Google, YouTube, Facebook and Twitter openly announced that they would do this. What else would we expect from Cult-owned operations like Facebook which former executives have confirmed set out to make the platform more addictive than cigarettes and coldly manipulates emotions of its users to sow division between people and groups and scramble the minds of the young? If Zuckerberg lives out the rest of his life without going to jail for crimes against humanity, and most emphatically against the young, it will be a travesty of justice. Still, no matter, cause and effect will catch up with him eventually and the same with Sergey Brin and Larry Page at Google with its CEO Sundar Pichai who fix the Google search results to promote Cult narratives and hide the opposition. Put the same key words into Google and other search engines like DuckDuckGo and you will see how different results can be. Wikipedia is another intensely biased 'encyclopaedia' which skews its content to the Cult agenda. YouTube links to Wikipedia's version of 'Covid' and 'climate change' on video pages in which experts in their field offer a different opinion (even that is increasingly rare with Wojcicki censorship). Into this 'Covid' silence-them network must be added government media censors, sorry 'regulators', such as Ofcom in the UK which imposed tyrannical restrictions on British broadcasters that had the effect of banning me from ever appearing. Just to debate with me about my evidence and views on 'Covid' would mean breaking the fascistic impositions of Ofcom and its CEO career government bureaucrat Melanie Dawes. Gutless British broadcasters tremble at the very thought of fascist Ofcom.

Psychos behind 'Covid'

The reason for the 'Covid' catastrophe in all its facets and forms can be seen by whom and what is driving the policies worldwide in such a coordinated way. Decisions are not being made to protect health, but to target psychology. The dominant group guiding and 'advising' government policy are not medical professionals. They are psychologists and behavioural scientists. Every major country has its own version of this phenomenon and I'll use the British example to show how it works. In many ways the British version has been affecting the wider world in the form of the huge behaviour manipulation network in the UK which operates in other countries. The network involves private companies, government, intelligence and military. The Cabinet Office is at the centre of the government 'Covid' Psyop and part-owns, with 'innovation charity' Nesta, the Behavioural Insights Team (BIT) which claims to be independent of government but patently isn't. The BIT was established in 2010 and its job is to manipulate the psyche of the population to acquiesce to government demands and so much more. It is also known as the 'Nudge Unit', a name inspired by the 2009 book by two ultra-Zionists, Cass Sunstein and Richard Thaler, called *Nudge: Improving Decisions About Health, Wealth, and Happiness*. The book, as with the Behavioural Insights Team, seeks to 'nudge' behaviour (manipulate it) to make the public follow patterns of action and perception that suit those in authority (the Cult). Sunstein is so skilled at this that he advises the World Health Organization and the UK Behavioural Insights Team and was Administrator of the White House Office of Information and Regulatory Affairs in the Obama administration. Biden appointed him to the Department of Homeland Security – another ultra-Zionist in the fold to oversee new immigration laws which is another policy the Cult wants to control. Sunstein is desperate to silence anyone exposing conspiracies and co-authored a 2008 report on the subject in which suggestions were offered to ban 'conspiracy theorizing' or impose 'some kind of tax, financial or otherwise, on those who disseminate such theories'. I guess a psychiatrist's chair is out of the question?

Sunstein's mate Richard Thaler, an 'academic affiliate' of the UK Behavioural Insights Team, is a proponent of 'behavioural economics' which is defined as the study of 'the effects of psychological, cognitive, emotional, cultural and social factors on the decisions of individuals and institutions'. Study the effects so they can be manipulated to be what you want them to be. Other leading names in the development of behavioural economics are ultra-Zionists Daniel Kahneman and Robert J. Shiller and they, with Thaler, won the Nobel Memorial Prize in Economic Sciences for their work in this field. The Behavioural Insights Team is operating at the heart of the UK government and has expanded globally through partnerships with several universities including Harvard, Oxford, Cambridge, University College London (UCL) and Pennsylvania. They claim to have 'trained' (reframed) 20,000 civil servants and run more than 750 projects involving 400 randomised controlled trials in dozens of countries' as another version of mind reframers Common Purpose. BIT works from its office in New York with cities and their agencies, as well as other partners, across the United States and Canada – this is a company part-owned by the British government Cabinet Office. An executive order by President Cult-servant Obama established a US Social and Behavioral Sciences Team in 2015. They all have the same reason for being and that's to brainwash the population directly and by brainwashing those in positions of authority.

'Covid' mind game

Another prime aspect of the UK mind-control network is the 'independent' [joke] Scientific Pandemic Insights Group on Behaviours (SPI-B) which 'provides behavioural science advice aimed at anticipating and helping people adhere to interventions that are recommended by medical or epidemiological experts'. That means manipulating public perception and behaviour to do whatever government tells them to do. It's disgusting and if they really want the public to be 'safe' this lot should all be under lock and key. According to the government website SPI-B consists of

'behavioural scientists, health and social psychologists, anthropologists and historians' and advises the Whitty-Vallance-led Scientific Advisory Group for Emergencies (SAGE) which in turn advises the government on 'the science' (it doesn't) and 'Covid' policy. When politicians say they are being guided by 'the science' this is the rabble in each country they are talking about and that 'science' is dominated by behaviour manipulators to enforce government fascism through public compliance. The Behaviour Insight Team is headed by psychologist David Solomon Halpern, a visiting professor at King's College London, and connects with a national and global web of other civilian and military organisations as the Cult moves towards its goal of fusing them into one fascistic whole in every country through its 'Fusion Doctrine'. The behaviour manipulation network involves, but is not confined to, the Foreign Office; National Security Council; government communications headquarters (GCHQ); MI5; MI6; the Cabinet Office-based Media Monitoring Unit; and the Rapid Response Unit which 'monitors digital trends to spot emerging issues; including misinformation and disinformation; and identifies the best way to respond'.

There is also the 77th Brigade of the UK military which operates like the notorious Israeli military's Unit 8200 in manipulating information and discussion on the Internet by posing as members of the public to promote the narrative and discredit those who challenge it. Here we have the military seeking to manipulate *domestic* public opinion while the Nazis in government are fine with that. Conservative Member of Parliament Tobias Ellwood, an advocate of lockdown and control through 'vaccine passports', is a Lieutenant Colonel reservist in the 77th Brigade which connects with the military operation jHub, the 'innovation centre' for the Ministry of Defence and Strategic Command. jHub has also been involved with the civilian National Health Service (NHS) in 'symptom tracing' the population. The NHS is a key part of this mind control network and produced a document in December, 2020, explaining to staff how to use psychological manipulation with different groups and ages to get them to have the DNA-manipulating 'Covid vaccine'

that's designed to cumulatively rewrite human genetics. The document, called 'Optimising Vaccination Roll Out – Do's and Dont's for all messaging, documents and "communications" in the widest sense', was published by NHS England and the NHS Improvement *Behaviour Change Unit* in partnership with Public Health England and Warwick Business School. I hear the mantra about 'save the NHS' and 'protect the NHS' when we need to scrap the NHS and start again. The current version is far too corrupt, far too anti-human and totally compromised by Cult operatives and their assets. UK government broadcast media censor Ofcom will connect into this web – as will the BBC with its tremendous Ofcom influence – to control what the public see and hear and dictate mass perception. Nuremberg trials must include personnel from all these organisations.

The fear factor

The 'Covid' hoax has led to the creation of the UK Cabinet Office-connected Joint Biosecurity Centre (JBC) which is officially described as providing 'expert advice on pandemics' using its independent [all Cult operations are 'independent'] analytical function to provide real-time analysis about infection outbreaks to identify and respond to outbreaks of Covid-19'. Another role is to advise the government on a response to spikes in infections – 'for example by closing schools or workplaces in local areas where infection levels have risen'. Put another way, promoting the Cult agenda. The Joint Biosecurity Centre is modelled on the Joint Terrorism Analysis Centre which analyses intelligence to set 'terrorism threat levels' and here again you see the fusion of civilian and military operations and intelligence that has led to military intelligence producing documents about 'vaccine hesitancy' and how it can be combated. Domestic civilian matters and opinions should not be the business of the military. The Joint Biosecurity Centre is headed by Tom Hurd, director general of the Office for Security and Counter-Terrorism from the establishment-to-its-fingertips Hurd family. His father is former Foreign Secretary Douglas Hurd. How coincidental that Tom

Hurd went to the elite Eton College and Oxford University with Boris Johnson. Imperial College with its ridiculous computer modeller Neil Ferguson will connect with this gigantic web that will itself interconnect with similar set-ups in other major and not so major countries. Compared with this Cult network the politicians, be they Boris Johnson, Donald Trump or Joe Biden, are bit-part players 'following the science'. The network of psychologists was on the 'Covid' case from the start with the aim of generating maximum fear of the 'virus' to ensure compliance by the population. A government behavioural science group known as SPI-B produced a paper in March, 2020, for discussion by the main government science advisory group known as SAGE. It was headed 'Options for increasing adherence to social distancing measures' and it said the following in a section headed 'Persuasion':

- A substantial number of people still do not feel sufficiently personally threatened; it could be that they are reassured by the low death rate in their demographic group, although levels of concern may be rising. Having a good understanding of the risk has been found to be positively associated with adoption of COVID-19 social distancing measures in Hong Kong.
- The perceived level of personal threat needs to be increased among those who are complacent, using hard-hitting evaluation of options for increasing social distancing emotional messaging. To be effective this must also empower people by making clear the actions they can take to reduce the threat.
- Responsibility to others: There seems to be insufficient understanding of, or feelings of responsibility about, people's role in transmitting the infection to others ... Messaging about actions need to be framed positively in terms of protecting oneself and the community, and increase confidence that they will be effective.
- Some people will be more persuaded by appeals to play by the rules, some by duty to the community, and some to personal risk.

All these different approaches are needed. The messaging also needs to take account of the realities of different people's lives. Messaging needs to take account of the different motivational levers and circumstances of different people.

All this could be achieved the SPI-B psychologists said by *using the media to increase the sense of personal threat* which translates as terrify the shit out of the population, including children, so they all do what we want. That's not happened has it? Those excuses for 'journalists' who wouldn't know journalism if it bit them on the arse (the great majority) have played their crucial part in serving this Cult-government Psyop to enslave their own kids and grandkids. How they live with themselves I have no idea. The psychological war has been underpinned by constant government 'Covid' propaganda in almost every television and radio ad break, plus the Internet and print media, which has pounded out the fear with taxpayers footing the bill for their own programming. The result has been people terrified of a 'virus' that doesn't exist or one with a tiny fatality rate even if you believe it does. People walk down the street and around the shops wearing face-nappies damaging their health and psychology while others report those who refuse to be that naïve to the police who turn up in their own face-nappies. I had a cameraman come to my flat and he was so frightened of 'Covid' he came in wearing a mask and refused to shake my hand in case he caught something. He had – naïveitis – and the thought that he worked in the mainstream media was both depressing and made his behaviour perfectly explainable. The fear which has gripped the minds of so many and frozen them into compliance has been carefully cultivated by these psychologists who are really psychopaths. If lives get destroyed and a lot of young people commit suicide it shows our plan is working. SPI-B then turned to compulsion on the public to comply. 'With adequate preparation, rapid change can be achieved', it said. Some countries had introduced mandatory self-isolation on a wide scale without evidence of major public unrest and a large majority of the UK's population appeared to be supportive of more coercive measures with 64 percent of adults saying they would

support putting London under a lockdown (watch the 'polls' which are designed to make people believe that public opinion is in favour or against whatever the subject in hand).

For 'aggressive protective measures' to be effective, the SPI-B paper said, special attention should be devoted to those population groups that are more at risk. Translated from the Orwellian this means making the rest of population feel guilty for not protecting the 'vulnerable' such as old people which the Cult and its agencies were about to kill on an industrial scale with lockdown, lack of treatment and the Gates 'vaccine'. Psychopath psychologists sold their guilt-trip so comprehensively that Los Angeles County Supervisor Hilda Solis reported that children were apologising (from a distance) to their parents and grandparents for bringing 'Covid' into their homes and getting them sick. '... These apologies are just some of the last words that loved ones will ever hear as they die alone,' she said. Gut-wrenchingly Solis then used this childhood tragedy to tell children to stay at home and 'keep your loved ones alive'. Imagine heaping such potentially life-long guilt on a kid when it has absolutely nothing to do with them. These people are deeply disturbed and the psychologists behind this even more so.

Uncivil war – divide and rule

Professional mind-controllers at SPI-B wanted the media to increase a sense of responsibility to others (do as you're told) and promote 'positive messaging' for those actions while in contrast to invoke 'social disapproval' by the unquestioning, obedient, community of anyone with a mind of their own. Again the compliant Goebbels-like media obliged. This is an old, old, trick employed by tyrannies the world over throughout human history. You get the target population to keep the target population in line – *your* line. SPI-B said this could 'play an important role in preventing anti-social behaviour or discouraging failure to enact pro-social behaviour'. For 'anti-social' in the Orwellian parlance of SPI-B see any behaviour that government doesn't approve. SPI-B recommendations said that 'social disapproval' should be accompanied by clear messaging and

promotion of strong collective identity – hence the government and celebrity mantra of ‘we’re all in this together’. Sure we are. The mind doctors have such contempt for their targets that they think some clueless comedian, actor or singer telling them to do what the government wants will be enough to win them over. We have had UK comedian Lenny Henry, actor Michael Caine and singer Elton John wheeled out to serve the propagandists by urging people to have the DNA-manipulating ‘Covid’ non-‘vaccine’. The role of Henry and fellow black celebrities in seeking to coax a ‘vaccine’ reluctant black community into doing the government’s will was especially stomach-turning. An emotion-manipulating script and carefully edited video featuring these black ‘celebs’ was such an insult to the intelligence of black people and where’s the self-respect of those involved selling their souls to a fascist government agenda? Henry said he heard black people’s ‘legitimate worries and concerns’, but people must ‘trust the facts’ when they were doing exactly that by not having the ‘vaccine’. They had to include the obligatory reference to Black Lives Matter with the line ... ‘Don’t let coronavirus cost even more black lives – because we matter’. My god, it was pathetic. ‘I know the vaccine is safe and what it does.’ How? ‘I’m a comedian and it says so in my script.’

SPI-B said social disapproval needed to be carefully managed to avoid victimisation, scapegoating and misdirected criticism, but they knew that their ‘recommendations’ would lead to exactly that and the media were specifically used to stir-up the divide-and-conquer hostility. Those who conform like good little baa, baas, are praised while those who have seen through the tidal wave of lies are ‘Covidiot’. The awake have been abused by the fast asleep for not conforming to fascism and impositions that the awake know are designed to endanger their health, dehumanise them, and tear asunder the very fabric of human society. We have had the curtain-twitchers and morons reporting neighbours and others to the face-napped police for breaking ‘Covid rules’ with fascist police delighting in posting links and phone numbers where this could be done. The Cult cannot impose its will without a compliant police

and military or a compliant population willing to play their part in enslaving themselves and their kids. The words of a pastor in Nazi Germany are so appropriate today:

First they came for the socialists and I did not speak out because I was not a socialist.

Then they came for the trade unionists and I did not speak out because I was not a trade unionist.

Then they came for the Jews and I did not speak out because I was not a Jew.

Then they came for me and there was no one left to speak for me.

Those who don't learn from history are destined to repeat it and so many are.

'Covid' rules: Rewiring the mind

With the background laid out to this gigantic national and global web of psychological manipulation we can put 'Covid' rules into a clear and sinister perspective. Forget the claims about protecting health. 'Covid' rules are about dismantling the human mind, breaking the human spirit, destroying self-respect, and then putting Humpty Dumpty together again as a servile, submissive slave. Social isolation through lockdown and distancing have devastating effects on the human psyche as the psychological psychopaths well know and that's the real reason for them. Humans need contact with each other, discourse, closeness and touch, or they eventually, and literally, go crazy. Masks, which I will address at some length, fundamentally add to the effects of isolation and the Cult agenda to dehumanise and de-individualise the population. To do this while knowing – in fact *seeking* – this outcome is the very epitome of evil and psychologists involved in this *are* the epitome of evil. They must like all the rest of the Cult demons and their assets stand trial for crimes against humanity on a scale that defies the imagination. Psychopaths in uniform use isolation to break enemy troops and agents and make them subservient and submissive to tell what they know. The technique is rightly considered a form of torture and

torture is most certainly what has been imposed on the human population.

Clinically-insane American psychologist Harry Harlow became famous for his isolation experiments in the 1950s in which he separated baby monkeys from their mothers and imprisoned them for months on end in a metal container or 'pit of despair'. They soon began to show mental distress and depression as any idiot could have predicted. Harlow put other monkeys in steel chambers for three, six or twelve months while denying them any contact with animals or humans. He said that the effects of total social isolation for six months were 'so devastating and debilitating that we had assumed initially that twelve months of isolation would not produce any additional decrement'; but twelve months of isolation 'almost obliterated the animals socially'. This is what the Cult and its psychopaths are doing to you and your children. Even monkeys in partial isolation in which they were not allowed to form relationships with other monkeys became 'aggressive and hostile, not only to others, but also towards their own bodies'. We have seen this in the young as a consequence of lockdown. UK government psychopaths launched a public relations campaign telling people not to hug each other even after they received the 'Covid-19 vaccine' which we were told with more lies would allow a return to 'normal life'. A government source told *The Telegraph*: 'It will be along the lines that it is great that you have been vaccinated, but if you are going to visit your family and hug your grandchildren there is a chance you are going to infect people you love.' The source was apparently speaking from a secure psychiatric facility. Janet Lord, director of Birmingham University's Institute of Inflammation and Ageing, said that parents and grandparents should avoid hugging their children. Well, how can I put it, Ms Lord? Fuck off. Yep, that'll do.

Destroying the kids – where are the parents?

Observe what has happened to people enslaved and isolated by lockdown as suicide and self-harm has soared worldwide,

particularly among the young denied the freedom to associate with their friends. A study of 49,000 people in English-speaking countries concluded that almost half of young adults are at clinical risk of mental health disorders. A national survey in America of 1,000 currently enrolled high school and college students found that 5 percent reported attempting suicide during the pandemic. Data from the US CDC's National Syndromic Surveillance Program from January 1st to October 17th, 2020, revealed a 31 percent increase in mental health issues among adolescents aged 12 to 17 compared with 2019. The CDC reported that America in general suffered the biggest drop in life expectancy since World War Two as it fell by a year in the first half of 2020 as a result of 'deaths of despair' – overdoses and suicides. Deaths of despair have leapt by more than 20 percent during lockdown and include the highest number of fatal overdoses ever recorded in a single year – 81,000. Internet addiction is another consequence of being isolated at home which lowers interest in physical activities as kids fall into inertia and what's the point? Children and young people are losing hope and giving up on life, sometimes literally. A 14-year-old boy killed himself in Maryland because he had 'given up' when his school district didn't reopen; an 11-year-old boy shot himself during a zoom class; a teenager in Maine succumbed to the isolation of the 'pandemic' when he ended his life after experiencing a disrupted senior year at school. Children as young as nine have taken their life and all these stories can be repeated around the world. Careers are being destroyed before they start and that includes those in sport in which promising youngsters have not been able to take part. The plan of the psycho-psychologists is working all right. Researchers at Cambridge University found that lockdowns cause significant harm to children's mental health. Their study was published in the *Archives of Disease in Childhood*, and followed 168 children aged between 7 and 11. The researchers concluded:

During the UK lockdown, children's depression symptoms have increased substantially, relative to before lockdown. The scale of this effect has direct relevance for the continuation of different elements of lockdown policy, such as complete or partial school closures ...

... Specifically, we observed a statistically significant increase in ratings of depression, with a medium-to-large effect size. Our findings emphasise the need to incorporate the potential impact of lockdown on child mental health in planning the ongoing response to the global pandemic and the recovery from it.

Not a chance when the Cult's psycho-psychologists were getting exactly what they wanted. The UK's Royal College of Paediatrics and Child Health has urged parents to look for signs of eating disorders in children and young people after a three to four fold increase. Specialists say the 'pandemic' is a major reason behind the rise. You don't say. The College said isolation from friends during school closures, exam cancellations, loss of extra-curricular activities like sport, and an increased use of social media were all contributory factors along with fears about the virus (psycho-psychologists again), family finances, and students being forced to quarantine. Doctors said young people were becoming severely ill by the time they were seen with 'Covid' regulations reducing face-to-face consultations. Nor is it only the young that have been devastated by the psychopaths. Like all bullies and cowards the Cult is targeting the young, elderly, weak and infirm. A typical story was told by a British lady called Lynn Parker who was not allowed to visit her husband in 2020 for the last ten and half months of his life 'when he needed me most' between March 20th and when he died on December 19th. This vacates the criminal and enters the territory of evil. The emotional impact on the immune system alone is immense as are the number of people of all ages worldwide who have died as a result of Cult-demanded, Gates-demanded, lockdowns.

Isolation is torture

The experience of imposing solitary confinement on millions of prisoners around the world has shown how a large percentage become 'actively psychotic and/or acutely suicidal'. Social isolation has been found to trigger 'a specific psychiatric syndrome, characterized by hallucinations; panic attacks; overt paranoia; diminished impulse control; hypersensitivity to external stimuli; and difficulties with thinking, concentration and memory'. Juan Mendez,

a United Nations rapporteur (investigator), said that isolation is a form of torture. Research has shown that even after isolation prisoners find it far more difficult to make social connections and I remember chatting to a shop assistant after one lockdown who told me that when her young son met another child again he had no idea how to act or what to do. Hannah Flanagan, Director of Emergency Services at Journey Mental Health Center in Dane County, Wisconsin, said: 'The specificity about Covid social distancing and isolation that we've come across as contributing factors to the suicides are really new to us this year.' But they are not new to those that devised them. They are getting the effect they want as the population is psychologically dismantled to be rebuilt in a totally different way. Children and the young are particularly targeted. They will be the adults when the full-on fascist AI-controlled technocracy is planned to be imposed and they are being prepared to meekly submit. At the same time older people who still have a memory of what life was like before – and how fascist the new normal really is – are being deleted. You are going to see efforts to turn the young against the old to support this geriatric genocide. Hannah Flanagan said the big increase in suicide in her county proved that social isolation is not only harmful, but deadly. Studies have shown that isolation from others is one of the main risk factors in suicide and even more so with women. Warnings that lockdown could create a 'perfect storm' for suicide were ignored. After all this was one of the *reasons* for lockdown. Suicide, however, is only the most extreme of isolation consequences. There are many others. Dr Dhruv Khullar, assistant professor of healthcare policy at Weill Cornell Medical College, said in a *New York Times* article in 2016 long before the fake 'pandemic':

A wave of new research suggests social separation is bad for us. Individuals with less social connection have disrupted sleep patterns, altered immune systems, more inflammation and higher levels of stress hormones. One recent study found that isolation increases the risk of heart disease by 29 percent and stroke by 32 percent. Another analysis that pooled data from 70 studies and 3.4 million people found that socially isolated individuals had a 30 percent higher risk of dying in the next seven years, and that this effect was largest in middle age.

Loneliness can accelerate cognitive decline in older adults, and isolated individuals are twice as likely to die prematurely as those with more robust social interactions. These effects start early: Socially isolated children have significantly poorer health 20 years later, even after controlling for other factors. All told, loneliness is as important a risk factor for early death as obesity and smoking.

There you have proof from that one article alone four years before 2020 that those who have enforced lockdown, social distancing and isolation knew what the effect would be and that is even more so with professional psychologists that have been driving the policy across the globe. We can go back even further to the years 2000 and 2003 and the start of a major study on the effects of isolation on health by Dr Janine Gronewold and Professor Dirk M. Hermann at the University Hospital in Essen, Germany, who analysed data on 4,316 people with an average age of 59 who were recruited for the long-term research project. They found that socially isolated people are more than 40 percent more likely to have a heart attack, stroke, or other major cardiovascular event and nearly 50 percent more likely to die from any cause. Given the financial Armageddon unleashed by lockdown we should note that the study found a relationship between increased cardiovascular risk and lack of financial support. After excluding other factors social isolation was still connected to a 44 percent increased risk of cardiovascular problems and a 47 percent increased risk of death by any cause. Lack of financial support was associated with a 30 percent increase in the risk of cardiovascular health events. Dr Gronewold said it had been known for some time that feeling lonely or lacking contact with close friends and family can have an impact on physical health and the study had shown that having strong social relationships is of high importance for heart health. Gronewold said they didn't understand yet why people who are socially isolated have such poor health outcomes, but this was obviously a worrying finding, particularly during these times of prolonged social distancing. Well, it can be explained on many levels. You only have to identify the point in the body where people feel loneliness and missing people they are parted from – it's in the centre of the chest where they feel the ache of loneliness and the ache of missing people. 'My heart aches for

you' ... 'My heart aches for some company.' I will explain this more in the chapter Escaping Wetiko, but when you realise that the body is the mind – they are expressions of each other – the reason why state of the mind dictates state of the body becomes clear.

American psychologist Ranjit Powar was highlighting the effects of lockdown isolation as early as April, 2020. She said humans have evolved to be social creatures and are wired to live in interactive groups. Being isolated from family, friends and colleagues could be unbalancing and traumatic for most people and could result in short or even long-term psychological and physical health problems. An increase in levels of anxiety, aggression, depression, forgetfulness and hallucinations were possible psychological effects of isolation. 'Mental conditions may be precipitated for those with underlying pre-existing susceptibilities and show up in many others without any pre-condition.' Powar said personal relationships helped us cope with stress and if we lost this outlet for letting off steam the result can be a big emotional void which, for an average person, was difficult to deal with. 'Just a few days of isolation can cause increased levels of anxiety and depression' – so what the hell has been the effect on the global population of *18 months* of this at the time of writing? Powar said: 'Add to it the looming threat of a dreadful disease being repeatedly hammered in through the media and you have a recipe for many shades of mental and physical distress.' For those with a house and a garden it is easy to forget that billions have had to endure lockdown isolation in tiny overcrowded flats and apartments with nowhere to go outside. The psychological and physical consequences of this are unimaginable and with lunatic and abusive partners and parents the consequences have led to tremendous increases in domestic and child abuse and alcoholism as people seek to shut out the horror. Ranjit Powar said:

Staying in a confined space with family is not all a rosy picture for everyone. It can be extremely oppressive and claustrophobic for large low-income families huddled together in small single-room houses. Children here are not lucky enough to have many board/electronic games or books to keep them occupied.

Add to it the deep insecurity of running out of funds for food and basic necessities. On the other hand, there are people with dysfunctional family dynamics, such as domineering, abusive or alcoholic partners, siblings or parents which makes staying home a period of trial. Incidence of suicide and physical abuse against women has shown a worldwide increase. Heightened anxiety and depression also affect a person's immune system, making them more susceptible to illness.

To think that Powar's article was published on April 11th, 2020.

Six-feet fantasy

Social (unsocial) distancing demanded that people stay six feet or two metres apart. UK government advisor Robert Dingwall from the New and Emerging Respiratory Virus Threats Advisory Group said in a radio interview that the two-metre rule was 'conjured up out of nowhere' and was not based on science. No, it was not based on *medical* science, but it didn't come out of nowhere. The distance related to *psychological* science. Six feet/two metres was adopted in many countries and we were told by people like the criminal Anthony Fauci and his ilk that it was founded on science. Many schools could not reopen because they did not have the space for six-foot distancing. Then in March, 2021, after a year of six-foot 'science', a study published in the *Journal of Infectious Diseases* involving more than 500,000 students and almost 100,000 staff over 16 weeks revealed no significant difference in 'Covid' cases between six feet and three feet and Fauci changed his tune. Now three feet was okay. There is no difference between six feet and three *inches* when there is no 'virus' and they got away with six feet for psychological reasons for as long as they could. I hear journalists and others talk about 'unintended consequences' of lockdown. They are not *unintended* at all; they have been coldly-calculated for a specific outcome of human control and that's why super-psychopaths like Gates have called for them so vehemently. Super-psychopath psychologists have demanded them and psychopathic or clueless, spineless, politicians have gone along with them by 'following the science'. But it's not science at all. 'Science' is not what is; it's only what people can be manipulated to believe it is. The whole 'Covid' catastrophe is

founded on mind control. Three word or three statement mantras issued by the UK government are a well-known mind control technique and so we've had 'Stay home/protect the NHS/save lives', 'Stay alert/control the virus/save lives' and 'hands/face/space'. One of the most vocal proponents of extreme 'Covid' rules in the UK has been Professor Susan Michie, a member of the British Communist Party, who is not a medical professional. Michie is the director of the Centre for Behaviour Change at University College London. She is a *behavioural psychologist* and another filthy rich 'Marxist' who praised China's draconian lockdown. She was known by fellow students at Oxford University as 'Stalin's nanny' for her extreme Marxism. Michie is an influential member of the UK government's Scientific Advisory Group for Emergencies (SAGE) and behavioural manipulation groups which have dominated 'Covid' policy. She is a consultant adviser to the World Health Organization on 'Covid-19' and behaviour. Why the hell are lockdowns anything to do with her when they are claimed to be about health? Why does a behavioural psychologist from a group charged with changing the behaviour of the public want lockdown, human isolation and mandatory masks? Does that question really need an answer? Michie *absolutely* has to explain herself before a Nuremberg court when humanity takes back its world again and even more so when you see the consequences of masks that she demands are compulsory. This is a Michie classic:

The benefits of getting primary school children to wear masks is that regardless of what little degree of transmission is occurring in those age groups it could help normalise the practice. Young children wearing masks may be more likely to get their families to accept masks.

Those words alone should carry a prison sentence when you ponder on the callous disregard for children involved and what a statement it makes about the mind and motivations of Susan Michie. What a lovely lady and what she said there encapsulates the mentality of the psychopaths behind the 'Covid' horror. Let us compare what Michie said with a countrywide study in Germany published at [researchsquare.com](https://www.researchsquare.com) involving 25,000 school children and 17,854 health complaints submitted by parents. Researchers

found that masks are harming children physically, psychologically, and behaviourally with 24 health issues associated with mask wearing. They include: shortness of breath (29.7%); dizziness (26.4%); increased headaches (53%); difficulty concentrating (50%); drowsiness or fatigue (37%); and malaise (42%). Nearly a third of children experienced more sleep issues than before and a quarter developed new fears. Researchers found health issues and other impairments in 68 percent of masked children covering their faces for an average of 4.5 hours a day. Hundreds of those taking part experienced accelerated respiration, tightness in the chest, weakness, and short-term impairment of consciousness. A reminder of what Michie said again:

The benefits of getting primary school children to wear masks is that regardless of what little degree of transmission is occurring in those age groups it could help normalise the practice. Young children wearing masks may be more likely to get their families to accept masks.

Psychopaths in government and psychology now have children and young people – plus all the adults – wearing masks for hours on end while clueless teachers impose the will of the psychopaths on the young they should be protecting. What the hell are parents doing?

Cult lab rats

We have some schools already imposing on students microchipped buzzers that activate when they get 'too close' to their pals in the way they do with lab rats. How apt. To the Cult and its brain-dead servants our children *are* lab rats being conditioned to be unquestioning, dehumanised slaves for the rest of their lives.

Children and young people are being weaned and frightened away from the most natural human instincts including closeness and touch. I have tracked in the books over the years how schools were banning pupils from greeting each other with a hug and the whole Cult-induced Me Too movement has terrified men and boys from a relaxed and natural interaction with female friends and work colleagues to the point where many men try never to be in a room

alone with a woman that's not their partner. Airhead celebrities have as always played their virtue-signalling part in making this happen with their gross exaggeration. For every monster like Harvey Weinstein there are at least tens of thousands of men that don't treat women like that; but everyone must be branded the same and policy changed for them as well as the monster. I am going to be using the word 'dehumanise' many times in this chapter because that is what the Cult is seeking to do and it goes very deep as we shall see. Don't let them kid you that social distancing is planned to end one day. That's not the idea. We are seeing more governments and companies funding and producing wearable gadgets to keep people apart and they would not be doing that if this was meant to be short-term. A tech start-up company backed by GCHQ, the British Intelligence and military surveillance headquarters, has created a social distancing wrist sensor that alerts people when they get too close to others. The CIA has also supported tech companies developing similar devices. The wearable sensor was developed by Tended, one of a number of start-up companies supported by GCHQ (see the CIA and DARPA). The device can be worn on the wrist or as a tag on the waistband and will vibrate whenever someone wearing the device breaches social distancing and gets anywhere near natural human contact. The company had a lucky break in that it was developing a distancing sensor when the 'Covid' hoax arrived which immediately provided a potentially enormous market. How fortunate. The government in big-time Cult-controlled Ontario in Canada is investing \$2.5 million in wearable contact tracing technology that 'will alert users if they may have been exposed to the Covid-19 in the workplace and will beep or vibrate if they are within six feet of another person'. Facedrive Inc., the technology company behind this, was founded in 2016 with funding from the Ontario Together Fund and obviously they, too, had a prophet on the board of directors. The human surveillance and control technology is called TraceSCAN and would be worn by the human cyborgs in places such as airports, workplaces, construction sites, care homes and ... *schools*.

I emphasise schools with children and young people the prime targets. You know what is planned for society as a whole if you keep your eyes on the schools. They have always been places where the state program the next generation of slaves to be its compliant worker-ants – or Woker-ants these days; but in the mist of the ‘Covid’ madness they have been transformed into mind laboratories on a scale never seen before. Teachers and head teachers are just as programmed as the kids – often more so. Children are kept apart from human interaction by walk lanes, classroom distancing, staggered meal times, masks, and the rolling-out of buzzer systems. Schools are now physically laid out as a laboratory maze for lab-rats. Lunatics at a school in Anchorage, Alaska, who should be prosecuted for child abuse, took away desks and forced children to kneel (know your place) on a mat for five hours a day while wearing a mask and using their chairs as a desk. How this was supposed to impact on a ‘virus’ only these clinically insane people can tell you and even then it would be clap-trap. The school banned recess (interaction), art classes (creativity), and physical exercise (getting body and mind moving out of inertia). Everyone behind this outrage should be in jail or better still a mental institution. The behavioural manipulators are all for this dystopian approach to schools. Professor Susan Michie, the mind-doctor and British Communist Party member, said it was wrong to say that schools were safe. They had to be made so by ‘distancing’, masks and ventilation (sitting all day in the cold). I must ask this lady round for dinner on a night I know I am going to be out and not back for weeks. She probably wouldn’t be able to make it, anyway, with all the visits to her own psychologist she must have block-booked.

Masking identity

I know how shocking it must be for you that a behaviour manipulator like Michie wants everyone to wear masks which have long been a feature of mind-control programs like the infamous MKUltra in the United States, but, there we are. We live and learn. I spent many years from 1996 to right across the millennium

researching mind control in detail on both sides of the Atlantic and elsewhere. I met a large number of mind-control survivors and many had been held captive in body and mind by MKUltra. MK stands for mind-control, but employs the German spelling in deference to the Nazis spirited out of Germany at the end of World War Two by Operation Paperclip in which the US authorities, with help from the Vatican, transported Nazi mind-controllers and engineers to America to continue their work. Many of them were behind the creation of NASA and they included Nazi scientist and SS officer Wernher von Braun who swapped designing V-2 rockets to bombard London with designing the Saturn V rockets that powered the NASA moon programme's Apollo craft. I think I may have mentioned that the Cult has no borders. Among Paperclip escapees was Josef Mengele, the Angel of Death in the Nazi concentration camps where he conducted mind and genetic experiments on children often using twins to provide a control twin to measure the impact of his 'work' on the other. If you want to observe the Cult mentality in all its extremes of evil then look into the life of Mengele. I have met many people who suffered mercilessly under Mengele in the United States where he operated under the name Dr Greene and became a stalwart of MKUltra programming and torture. Among his locations was the underground facility in the Mojave Desert in California called the China Lake Naval Weapons Station which is almost entirely below the surface. My books *The Biggest Secret*, *Children of the Matrix* and *The Perception Deception* have the detailed background to MKUltra.

The best-known MKUltra survivor is American Cathy O'Brien. I first met her and her late partner Mark Phillips at a conference in Colorado in 1996. Mark helped her escape and deprogram from decades of captivity in an offshoot of MKUltra known as Project Monarch in which 'sex slaves' were provided for the rich and famous including Father George Bush, Dick Cheney and the Clintons. Read Cathy and Mark's book *Trance-Formation of America* and if you are new to this you will be shocked to the core. I read it in 1996 shortly before, with the usual synchronicity of my life, I found

myself given a book table at the conference right next to hers. MKUltra never ended despite being very publicly exposed (only a small part of it) in the 1970s and continues in other guises. I am still in touch with Cathy. She contacted me during 2020 after masks became compulsory in many countries to tell me how they were used as part of MKUltra programming. I had been observing 'Covid regulations' and the relationship between authority and public for months. I saw techniques that I knew were employed on individuals in MKUltra being used on the global population. I had read many books and manuals on mind control including one called *Silent Weapons for Quiet Wars* which came to light in the 1980s and was a guide on how to perceptually program on a mass scale. 'Silent Weapons' refers to mind-control. I remembered a line from the manual as governments, medical authorities and law enforcement agencies have so obviously talked to – or rather at – the adult population since the 'Covid' hoax began as if they are children. The document said:

If a person is spoken to by a T.V. advertiser as if he were a twelve-year-old, then, due to suggestibility, he will, with a certain probability, respond or react to that suggestion with the uncritical response of a twelve-year-old and will reach in to his economic reservoir and deliver its energy to buy that product on impulse when he passes it in the store.

That's why authority has spoken to adults like children since all this began.

Why did Michael Jackson wear masks?

Every aspect of the 'Covid' narrative has mind-control as its central theme. Cathy O'Brien wrote an article for davidicke.com about the connection between masks and mind control. Her daughter Kelly who I first met in the 1990s was born while Cathy was still held captive in MKUltra. Kelly was forced to wear a mask as part of her programming from the age of *two* to dehumanise her, target her sense of individuality and reduce the amount of oxygen her brain and body received. *Bingo*. This is the real reason for compulsory

masks, why they have been enforced en masse, and why they seek to increase the number they demand you wear. First one, then two, with one disgraceful alleged 'doctor' recommending four which is nothing less than a death sentence. Where and how often they must be worn is being expanded for the purpose of mass mind control and damaging respiratory health which they can call 'Covid-19'. Canada's government headed by the man-child Justin Trudeau, says it's fine for children of two and older to wear masks. An insane 'study' in Italy involving just 47 children concluded there was no problem for babies as young as *four months* wearing them. Even after people were 'vaccinated' they were still told to wear masks by the criminal that is Anthony Fauci. Cathy wrote that mandating masks is allowing the authorities literally to control the air we breathe which is what was done in MKUltra. You might recall how the singer Michael Jackson wore masks and there is a reason for that. He was subjected to MKUltra mind control through Project Monarch and his psyche was scrambled by these simpletons. Cathy wrote:

In MKUltra Project Monarch mind control, Michael Jackson had to wear a mask to silence his voice so he could not reach out for help. Remember how he developed that whisper voice when he wasn't singing? Masks control the mind from the outside in, like the redefining of words is doing. By controlling what we can and cannot say for fear of being labeled racist or beaten, for example, it ultimately controls thought that drives our words and ultimately actions (or lack thereof).

Likewise, a mask muffles our speech so that we are not heard, which controls voice ... words ... mind. This is Mind Control. Masks are an obvious mind control device, and I am disturbed so many people are complying on a global scale. Masks depersonalize while making a person feel as though they have no voice. It is a barrier to others. People who would never choose to comply but are forced to wear a mask in order to keep their job, and ultimately their family fed, are compromised. They often feel shame and are subdued. People have stopped talking with each other while media controls the narrative.

The 'no voice' theme has often become literal with train passengers told not to speak to each other in case they pass on the 'virus', singing banned for the same reason and bonkers California officials telling people riding roller coasters that they cannot shout and scream. Cathy said she heard every day from healed MKUltra survivors who cannot wear a mask without flashing back on ways

their breathing was controlled – ‘from ball gags and penises to water boarding’. She said that through the years when she saw images of people in China wearing masks ‘due to pollution’ that it was really to control their oxygen levels. ‘I knew it was as much of a population control mechanism of depersonalisation as are burkas’, she said. Masks are another Chinese communist/fascist method of control that has been swept across the West as the West becomes China at lightning speed since we entered 2020.

Mask-19

There are other reasons for mandatory masks and these include destroying respiratory health to call it ‘Covid-19’ and stunting brain development of children and the young. Dr Margarite Griesz-Brisson MD, PhD, is a Consultant Neurologist and Neurophysiologist and the Founder and Medical Director of the London Neurology and Pain Clinic. Her CV goes down the street and round the corner. She is clearly someone who cares about people and won’t parrot the propaganda. Griesz-Brisson has a PhD in pharmacology, with special interest in neurotoxicology, environmental medicine, neuroregeneration and neuroplasticity (the way the brain can change in the light of information received). She went public in October, 2020, with a passionate warning about the effects of mask-wearing laws:

The reinhalation of our exhaled air will without a doubt create oxygen deficiency and a flooding of carbon dioxide. We know that the human brain is very sensitive to oxygen deprivation. There are nerve cells for example in the hippocampus that can’t be longer than 3 minutes without oxygen – they cannot survive. The acute warning symptoms are headaches, drowsiness, dizziness, issues in concentration, slowing down of reaction time – reactions of the cognitive system.

Oh, I know, let’s tell bus, truck and taxi drivers to wear them and people working machinery. How about pilots, doctors and police? Griesz-Brisson makes the important point that while the symptoms she mentions may fade as the body readjusts this does not alter the fact that people continue to operate in oxygen deficit with long list of

potential consequences. She said it was well known that neurodegenerative diseases take years or decades to develop. 'If today you forget your phone number, the breakdown in your brain would have already started 20 or 30 years ago.' She said degenerative processes in your brain are getting amplified as your oxygen deprivation continues through wearing a mask. Nerve cells in the brain are unable to divide themselves normally in these circumstances and lost nerve cells will no longer be regenerated. 'What is gone is gone.' Now consider that people like shop workers and *schoolchildren* are wearing masks for hours every day. What in the name of sanity is going to be happening to them? 'I do not wear a mask, I need my brain to think', Griesz-Brisson said, 'I want to have a clear head when I deal with my patients and not be in a carbon dioxide-induced anaesthesia'. If you are told to wear a mask anywhere ask the organisation, police, store, whatever, for their risk assessment on the dangers and negative effects on mind and body of enforcing mask-wearing. They won't have one because it has never been done not even by government. All of them must be subject to class-action lawsuits as the consequences come to light. They don't do mask risk assessments for an obvious reason. They know what the conclusions would be and independent scientific studies that *have* been done tell a horror story of consequences.

'Masks are criminal'

Dr Griesz-Brisson said that for children and adolescents, masks are an absolute no-no. They had an extremely active and adaptive immune system and their brain was incredibly active with so much to learn. 'The child's brain, or the youth's brain, is thirsting for oxygen.' The more metabolically active an organ was, the more oxygen it required; and in children and adolescents every organ was metabolically active. Griesz-Brisson said that to deprive a child's or adolescent's brain of oxygen, or to restrict it in any way, was not only dangerous to their health, it was absolutely criminal. 'Oxygen deficiency inhibits the development of the brain, and the damage that has taken place as a result CANNOT be reversed.' Mind

manipulators of MKUltra put masks on two-year-olds they wanted to neurologically rewire and you can see why. Griesz-Brisson said a child needs the brain to learn and the brain needs oxygen to function. 'We don't need a clinical study for that. This is simple, indisputable physiology.' Consciously and purposely induced oxygen deficiency was an absolutely deliberate health hazard, and an absolute medical contraindication which means that 'this drug, this therapy, this method or measure should not be used, and is not allowed to be used'. To coerce an entire population to use an absolute medical contraindication by force, she said, there had to be definite and serious reasons and the reasons must be presented to competent interdisciplinary and independent bodies to be verified and authorised. She had this warning of the consequences that were coming if mask wearing continued:

When, in ten years, dementia is going to increase exponentially, and the younger generations couldn't reach their god-given potential, it won't help to say 'we didn't need the masks'. I know how damaging oxygen deprivation is for the brain, cardiologists know how damaging it is for the heart, pulmonologists know how damaging it is for the lungs. Oxygen deprivation damages every single organ. Where are our health departments, our health insurance, our medical associations? It would have been their duty to be vehemently against the lockdown and to stop it and stop it from the very beginning.

Why do the medical boards issue punishments to doctors who give people exemptions? Does the person or the doctor seriously have to prove that oxygen deprivation harms people? What kind of medicine are our doctors and medical associations representing? Who is responsible for this crime? The ones who want to enforce it? The ones who let it happen and play along, or the ones who don't prevent it?

All of the organisations and people she mentions there either answer directly to the Cult or do whatever hierarchical levels above them tell them to do. The outcome of both is the same. 'It's not about masks, it's not about viruses, it's certainly not about your health', Griesz-Brisson said. 'It is about much, much more. I am not participating. I am not afraid.' They were taking our air to breathe and there was no unfounded medical exemption from face masks. Oxygen deprivation was dangerous for every single brain. It had to be the free decision of every human being whether they want to

wear a mask that was absolutely ineffective to protect themselves from a virus. She ended by rightly identifying where the responsibility lies for all this:

The imperative of the hour is personal responsibility. We are responsible for what we think, not the media. We are responsible for what we do, not our superiors. We are responsible for our health, not the World Health Organization. And we are responsible for what happens in our country, not the government.

Halle-bloody-lujah.

But surgeons wear masks, right?

Independent studies of mask-wearing have produced a long list of reports detailing mental, emotional and physical dangers. What a definition of insanity to see police officers imposing mask-wearing on the public which will cumulatively damage their health while the police themselves wear masks that will cumulatively damage *their* health. It's utter madness and both public and police do this because 'the government says so' – yes a government of brain-donor idiots like UK Health Secretary Matt Hancock reading the 'follow the science' scripts of psychopathic, lunatic psychologists. The response you get from Stockholm syndrome sufferers defending the very authorities that are destroying them and their families is that 'surgeons wear masks'. This is considered the game, set and match that they must work and don't cause oxygen deficit. Well, actually, scientific studies have shown that they *do* and oxygen levels are monitored in operating theatres to compensate. Surgeons wear masks to stop spittle and such like dropping into open wounds – not to stop 'viral particles' which are so miniscule they can only be seen through an electron microscope. Holes in the masks are significantly bigger than 'viral particles' and if you sneeze or cough they will breach the mask. I watched an incredibly disingenuous 'experiment' that claimed to prove that masks work in catching 'virus' material from the mouth and nose. They did this with a slow motion camera and the mask did block big stuff which stayed inside the mask and

against the face to be breathed in or cause infections on the face as we have seen with many children. 'Viral particles', however, would never have been picked up by the camera as they came through the mask when they are far too small to be seen. The 'experiment' was therefore disingenuous *and* useless.

Studies have concluded that wearing masks in operating theatres (and thus elsewhere) make no difference to preventing infection while the opposite is true with toxic shite building up in the mask and this had led to an explosion in tooth decay and gum disease dubbed by dentists 'mask mouth'. You might have seen the Internet video of a furious American doctor urging people to take off their masks after a four-year-old patient had been rushed to hospital the night before and nearly died with a lung infection that doctors sourced to mask wearing. A study in the journal *Cancer Discovery* found that inhalation of harmful microbes can contribute to advanced stage lung cancer in adults and long-term use of masks can help breed dangerous pathogens. Microbiologists have said frequent mask wearing creates a moist environment in which microbes can grow and proliferate before entering the lungs. The Canadian Agency for Drugs and Technologies in Health, or CADTH, a Canadian national organisation that provides research and analysis to healthcare decision-makers, said this as long ago as 2013 in a report entitled 'Use of Surgical Masks in the Operating Room: A Review of the Clinical Effectiveness and Guidelines'. It said:

- No evidence was found to support the use of surgical face masks to reduce the frequency of surgical site infections
- No evidence was found on the effectiveness of wearing surgical face masks to protect staff from infectious material in the operating room.
- Guidelines recommend the use of surgical face masks by staff in the operating room to protect both operating room staff and patients (despite the lack of evidence).

We were told that the world could go back to 'normal' with the arrival of the 'vaccines'. When they came, fraudulent as they are, the story changed as I knew that it would. We are in the midst of transforming 'normal', not going back to it. Mary Ramsay, head of immunisation at Public Health England, echoed the words of US criminal Anthony Fauci who said masks and other regulations must stay no matter if people are vaccinated. The Fauci idiot continued to wear two masks – different colours so both could be clearly seen – after he *claimed* to have been vaccinated. Senator Rand Paul told Fauci in one exchange that his double-masks were 'theatre' and he was right. It's all theatre. Mary Ramsay back-tracked on the vaccine-return-to-normal theme when she said the public may need to wear masks and social-distance for years despite the jabs. 'People have got used to those lower-level restrictions now, and [they] can live with them', she said telling us what the idea has been all along. 'The vaccine does not give you a pass, even if you have had it, you must continue to follow all the guidelines' said a Public Health England statement which reneged on what we had been told before and made having the 'vaccine' irrelevant to 'normality' even by the official story. Spain's fascist government trumped everyone by passing a law mandating the wearing of masks on the beach and even when swimming in the sea. The move would have devastated what's left of the Spanish tourist industry, posed potential breathing dangers to swimmers and had Northern European sunbathers walking around with their forehead brown and the rest of their face white as a sheet. The ruling was so crazy that it had to be retracted after pressure from public and tourist industry, but it confirmed where the Cult wants to go with masks and how clinically insane authority has become. The determination to make masks permanent and hide the serious dangers to body and mind can be seen in the censorship of scientist Professor Denis Rancourt by Bill Gates-funded academic publishing website ResearchGate over his papers exposing the dangers and uselessness of masks. Rancourt said:

ResearchGate today has permanently locked my account, which I have had since 2015. Their reasons graphically show the nature of their attack against democracy, and their corruption of

science ... By their obscene non-logic, a scientific review of science articles reporting on harms caused by face masks has a 'potential to cause harm'. No criticism of the psychological device (face masks) is tolerated, if the said criticism shows potential to influence public policy.

This is what happens in a fascist world.

Where are the 'greens' (again)?

Other dangers of wearing masks especially regularly relate to the inhalation of minute plastic fibres into the lungs and the deluge of discarded masks in the environment and oceans. Estimates predicted that more than 1.5 billion disposable masks will end up in the world's oceans every year polluting the water with tons of plastic and endangering marine wildlife. Studies project that humans are using 129 billion face masks each month worldwide – about three million a minute. Most are disposable and made from plastic, non-biodegradable microfibers that break down into smaller plastic particles that become widespread in ecosystems. They are littering cities, clogging sewage channels and turning up in bodies of water. I have written in other books about the immense amounts of microplastics from endless sources now being absorbed into the body. Rolf Halden, director of the Arizona State University (ASU) Biodesign Center for Environmental Health Engineering, was the senior researcher in a 2020 study that analysed 47 human tissue samples and found microplastics in all of them. 'We have detected these chemicals of plastics in every single organ that we have investigated', he said. I wrote in *The Answer* about the world being deluged with microplastics. A study by the Worldwide Fund for Nature (WWF) found that people are consuming on average every week some 2,000 tiny pieces of plastic mostly through water and also through marine life and the air. Every year humans are ingesting enough microplastics to fill a heaped dinner plate and in a life-time of 79 years it is enough to fill two large waste bins. Marco Lambertini, WWF International director general said: 'Not only are plastics polluting our oceans and waterways and killing marine life – it's in all of us and we can't escape consuming plastics,' American

geologists found tiny plastic fibres, beads and shards in rainwater samples collected from the remote slopes of the Rocky Mountain National Park near Denver, Colorado. Their report was headed: 'It is raining plastic.' Rachel Adams, senior lecturer in Biomedical Science at Cardiff Metropolitan University, said that among health consequences are internal inflammation and immune responses to a 'foreign body'. She further pointed out that microplastics become carriers of toxins including mercury, pesticides and dioxins (a known cause of cancer and reproductive and developmental problems). These toxins accumulate in the fatty tissues once they enter the body through microplastics. Now this is being compounded massively by people putting plastic on their face and throwing it away.

Workers exposed to polypropylene plastic fibres known as 'flock' have developed 'flock worker's lung' from inhaling small pieces of the flock fibres which can damage lung tissue, reduce breathing capacity and exacerbate other respiratory problems. *Now ...* commonly used surgical masks have three layers of melt-blown textiles made of ... polypropylene. We have billions of people putting these microplastics against their mouth, nose and face for hours at a time day after day in the form of masks. How does anyone think that will work out? I mean – what could possibly go wrong? We posted a number of scientific studies on this at davidicke.com, but when I went back to them as I was writing this book the links to the science research website where they were hosted were dead. Anything that challenges the official narrative in any way is either censored or vilified. The official narrative is so unsupportable by the evidence that only deleting the truth can protect it. A study by Chinese scientists still survived – with the usual twist which it why it was still active, I guess. Yes, they found that virtually all the masks they tested increased the daily intake of microplastic fibres, but people should still wear them because the danger from the 'virus' was worse said the crazy 'team' from the Institute of Hydrobiology in Wuhan. Scientists first discovered microplastics in lung tissue of some patients who died of lung cancer

in the 1990s. Subsequent studies have confirmed the potential health damage with the plastic degrading slowly and remaining in the lungs to accumulate in volume. Wuhan researchers used a machine simulating human breathing to establish that masks shed up to nearly 4,000 microplastic fibres in a month with reused masks producing more. Scientists said some masks are laced with toxic chemicals and a variety of compounds seriously restricted for both health and environmental reasons. They include cobalt (used in blue dye) and formaldehyde known to cause watery eyes, burning sensations in the eyes, nose, and throat, plus coughing, wheezing and nausea. No – that must be ‘Covid-19’.

Mask ‘worms’

There is another and potentially even more sinister content of masks. Mostly new masks of different makes filmed under a microscope around the world have been found to contain strange black fibres or ‘worms’ that appear to move or ‘crawl’ by themselves and react to heat and water. The nearest I have seen to them are the self-replicating fibres that are pulled out through the skin of those suffering from Morgellons disease which has been connected to the phenomena of ‘chemtrails’ which I will bring into the story later on. Morgellons fibres continue to grow outside the body and have a form of artificial intelligence. Black ‘worm’ fibres in masks have that kind of feel to them and there is a nanotechnology technique called ‘worm micelles’ which carry and release drugs or anything else you want to deliver to the body. For sure the suppression of humanity by mind altering drugs is the Cult agenda big time and the more excuses they can find to gain access to the body the more opportunities there are to make that happen whether through ‘vaccines’ or masks pushed against the mouth and nose for hours on end.

So let us summarise the pros and cons of masks:

Against masks: Breathing in your own carbon dioxide; depriving the body and brain of sufficient oxygen; build-up of toxins in the mask that can be breathed into the lungs and cause rashes on the face and 'mask-mouth'; breathing microplastic fibres and toxic chemicals into the lungs; dehumanisation and deleting individualisation by literally making people faceless; destroying human emotional interaction through facial expression and deleting parental connection with their babies which look for guidance to their facial expression.

For masks: They don't protect you from a 'virus' that doesn't exist and even if it did 'viral' particles are so minute they are smaller than the holes in the mask.

Governments, police, supermarkets, businesses, transport companies, and all the rest who seek to impose masks have done no risk assessment on their consequences for health and psychology and are now open to group lawsuits when the impact becomes clear with a cumulative epidemic of respiratory and other disease. Authorities will try to exploit these effects and hide the real cause by dubbing them 'Covid-19'. Can you imagine setting out to force the population to wear health-destroying masks without doing any assessment of the risks? It is criminal and it is evil, but then how many people targeted in this way, who see their children told to wear them all day at school, have asked for a risk assessment? Billions can't be imposed upon by the few unless the billions allow it. Oh, yes, with just a tinge of irony, 85 percent of all masks made worldwide come from *China*.

Wash your hands in toxic shite

'Covid' rules include the use of toxic sanitisers and again the health consequences of constantly applying toxins to be absorbed through the skin is obvious to any level of Renegade Mind. America's Food and Drug Administration (FDA) said that sanitisers are drugs and issued a warning about 75 dangerous brands which contain

methanol used in antifreeze and can cause death, kidney damage and blindness. The FDA circulated the following warning even for those brands that it claims to be safe:

Store hand sanitizer out of the reach of pets and children, and children should use it only with adult supervision. Do not drink hand sanitizer. This is particularly important for young children, especially toddlers, who may be attracted by the pleasant smell or brightly colored bottles of hand sanitizer.

Drinking even a small amount of hand sanitizer can cause alcohol poisoning in children. (However, there is no need to be concerned if your children eat with or lick their hands after using hand sanitizer.) During this coronavirus pandemic, poison control centers have had an increase in calls about accidental ingestion of hand sanitizer, so it is important that adults monitor young children's use.

Do not allow pets to swallow hand sanitizer. If you think your pet has eaten something potentially dangerous, call your veterinarian or a pet poison control center right away. Hand sanitizer is flammable and should be stored away from heat and flames. When using hand sanitizer, rub your hands until they feel completely dry before performing activities that may involve heat, sparks, static electricity, or open flames.

There you go, perfectly safe, then, and that's without even a mention of the toxins absorbed through the skin. Come on kids – sanitise your hands everywhere you go. It will save you from the 'virus'. Put all these elements together of the 'Covid' normal and see how much health and psychology is being cumulatively damaged, even devastated, to 'protect your health'. Makes sense, right? They are only imposing these things because they care, right? *Right?*

Submitting to insanity

Psychological reframing of the population goes very deep and is done in many less obvious ways. I hear people say how contradictory and crazy 'Covid' rules are and how they are ever changing. This is explained away by dismissing those involved as idiots. It is a big mistake. The Cult is delighted if its cold calculation is perceived as incompetence and idiocy when it is anything but. Oh, yes, there are idiots within the system – lots of them – but they are *administering* the Cult agenda, mostly unknowingly. They are not deciding and dictating it. The bulwark against tyranny is self-

respect, always has been, always will be. It is self-respect that has broken every tyranny in history. By its very nature self-respect will not bow to oppression and its perpetrators. There is so little self-respect that it's always the few that overturn dictators. Many may eventually follow, but the few with the iron spines (self-respect) kick it off and generate the momentum. The Cult targets self-respect in the knowledge that once this has gone only submission remains. Crazy, contradictory, ever-changing 'Covid' rules are systematically applied by psychologists to delete self-respect. They *want* you to see that the rules make no sense. It is one thing to decide to do something when *you* have made the choice based on evidence and logic. You still retain your self-respect. It is quite another when you can see what you are being told to do is insane, ridiculous and makes no sense, and *yet you still do it*. Your self-respect is extinguished and this has been happening as ever more obviously stupid and nonsensical things have been demanded and the great majority have complied even when they can see they are stupid and nonsensical.

People walk around in face-nappies knowing they are damaging their health and make no difference to a 'virus'. They do it in fear of not doing it. I know it's daft, but I'll do it anyway. When that happens something dies inside of you and submissive reframing has begun. Next there's a need to hide from yourself that you have conceded your self-respect and you convince yourself that you have not really submitted to fear and intimidation. You begin to believe that you are complying with craziness because it's the right thing to do. When first you concede your self-respect of $2+2 = 4$ to $2+2 = 5$ you *know* you are compromising your self-respect. Gradually to avoid facing that fact you begin to *believe* that $2+2=5$. You have been reframed and I have been watching this process happening in the human psyche on an industrial scale. The Cult is working to break your spirit and one of its major tools in that war is humiliation. I read how former American soldier Bradley Manning (later Chelsea Manning after a sex-change) was treated after being jailed for supplying WikiLeaks with documents exposing the enormity of

government and elite mendacity. Manning was isolated in solitary confinement for eight months, put under 24-hour surveillance, forced to hand over clothing before going to bed, and stand naked for every roll call. This is systematic humiliation. The introduction of anal swab 'Covid' tests in China has been done for the same reason to delete self-respect and induce compliant submission. Anal swabs are mandatory for incoming passengers in parts of China and American diplomats have said they were forced to undergo the indignity which would have been calculated humiliation by the Cult-owned Chinese government that has America in its sights.

Government-people: An abusive relationship

Spirit-breaking psychological techniques include giving people hope and apparent respite from tyranny only to take it away again. This happened in the UK during Christmas, 2020, when the psycho-psychologists and their political lackeys announced an easing of restrictions over the holiday only to reimpose them almost immediately on the basis of yet another lie. There is a big psychological difference between getting used to oppression and being given hope of relief only to have that dashed. Psychologists know this and we have seen the technique used repeatedly. Then there is traumatising people before you introduce more extreme regulations that require compliance. A perfect case was the announcement by the dark and sinister Whitty and Vallance in the UK that 'new data' predicted that 4,000 could die every day over the winter of 2020/2021 if we did not lockdown again. I think they call it lying and after traumatising people with that claim out came Jackboot Johnson the next day with new curbs on human freedom. Psychologists know that a frightened and traumatised mind becomes suggestable to submission and behaviour reframing. Underpinning all this has been to make people fearful and suspicious of each other and see themselves as a potential danger to others. In league with deleted self-respect you have the perfect psychological recipe for self-loathing. The relationship between authority and public is now demonstrably the same as that of

subservience to an abusive partner. These are signs of an abusive relationship explained by psychologist Leslie Becker-Phelps:

Psychological and emotional abuse: Undermining a partner's self-worth with verbal attacks, name-calling, and belittling. Humiliating the partner in public, unjustly accusing them of having an affair, or interrogating them about their every behavior. Keeping partner confused or off balance by saying they were just kidding or blaming the partner for 'making' them act this way ... Feigning in public that they care while turning against them in private. This leads to victims frequently feeling confused, incompetent, unworthy, hopeless, and chronically self-doubting. [Apply these techniques to how governments have treated the population since New Year, 2020, and the parallels are obvious.]

Physical abuse: The abuser might physically harm their partner in a range of ways, such as grabbing, hitting, punching, or shoving them. They might throw objects at them or harm them with a weapon. [Observe the physical harm imposed by masks, lockdown, and so on.]

Threats and intimidation: One way abusers keep their partners in line is by instilling fear. They might be verbally threatening, or give threatening looks or gestures. Abusers often make it known that they are tracking their partner's every move. They might destroy their partner's possessions, threaten to harm them, or threaten to harm their family members. Not surprisingly, victims of this abuse often feel anxiety, fear, and panic. [No words necessary.]

Isolation: Abusers often limit their partner's activities, forbidding them to talk or interact with friends or family. They might limit access to a car or even turn off their phone. All of this might be done by physically holding them against their will, but is often accomplished through psychological abuse and intimidation. The more isolated a person feels, the fewer resources they have to help gain perspective on their situation and to escape from it. [No words necessary.]

Economic abuse: Abusers often make their partners beholden to them for money by controlling access to funds of any kind. They might prevent their partner from getting a job or withhold access to money they earn from a job. This creates financial dependency that makes leaving the relationship very difficult. [See destruction of livelihoods and the proposed meagre 'guaranteed income' so long as you do whatever you are told.]

Using children: An abuser might disparage their partner's parenting skills, tell their children lies about their partner, threaten to take custody of their children, or threaten to harm their children. These tactics instil fear and often elicit compliance. [See reframed social service mafia and how children are being mercilessly abused by the state over 'Covid' while their parents look on too frightened to do anything.]

A further recurring trait in an abusive relationship is the abused blaming themselves for their abuse and making excuses for the abuser. We have the public blaming each other for lockdown abuse by government and many making excuses for the government while attacking those who challenge the government. How often we have heard authorities say that rules are being imposed or reimposed only because people have refused to 'behave' and follow the rules. We don't want to do it – it's *you*.

Renegade Minds are an antidote to all of these things. They will never concede their self-respect no matter what the circumstances. Even when apparent humiliation is heaped upon them they laugh in its face and reflect back the humiliation on the abuser where it belongs. Renegade Minds will never wear masks they know are only imposed to humiliate, suppress and damage both physically and psychologically. Consequences will take care of themselves and they will never break their spirit or cause them to concede to tyranny. UK newspaper columnist Peter Hitchens was one of the few in the mainstream media to speak out against lockdowns and forced vaccinations. He then announced he had taken the jab. He wanted to see family members abroad and he believed vaccine passports were inevitable even though they had not yet been introduced. Hitchens

has a questioning and critical mind, but not a Renegade one. If he had no amount of pressure would have made him concede. Hitchens excused his action by saying that the battle has been lost. Renegade Minds never accept defeat when freedom is at stake and even if they are the last one standing the self-respect of not submitting to tyranny is more important than any outcome or any consequence.

That's why Renegade Minds are the only minds that ever changed anything worth changing.

CHAPTER EIGHT

'Reframing' insanity

Insanity is relative. It depends on who has who locked in what cage
Ray Bradbury

Reframing' a mind means simply to change its perception and behaviour. This can be done subconsciously to such an extent that subjects have no idea they have been 'reframed' while to any observer changes in behaviour and attitudes are obvious.

Human society is being reframed on a ginormous scale since the start of 2020 and here we have the reason why psychologists rather than doctors have been calling the shots. Ask most people who have succumbed to 'Covid' reframing if they have changed and most will say 'no'; but they *have* and fundamentally. The Cult's long-game has been preparing for these times since way back and crucial to that has been to prepare both population and officialdom mentally and emotionally. To use the mind-control parlance they had to reframe the population with a mentality that would submit to fascism and reframe those in government and law enforcement to impose fascism or at least go along with it. The result has been the fact-deleted mindlessness of 'Wokeness' and officialdom that has either enthusiastically or unquestioningly imposed global tyranny demanded by reframed politicians on behalf of psychopathic and deeply evil cultists. 'Cognitive reframing' identifies and challenges the way someone sees the world in the form of situations, experiences and emotions and then restructures those perceptions to view the same set of circumstances in a different way. This can have

benefits if the attitudes are personally destructive while on the other side it has the potential for individual and collective mind control which the subject has no idea has even happened.

Cognitive therapy was developed in the 1960s by Aaron T. Beck who was born in Rhode Island in 1921 as the son of Jewish immigrants from the Ukraine. He became interested in the techniques as a treatment for depression. Beck's daughter Judith S. Beck is prominent in the same field and they founded the Beck Institute for Cognitive Behavior Therapy in Philadelphia in 1994. Cognitive reframing, however, began to be used worldwide by those with a very dark agenda. The Cult reframes politicians to change their attitudes and actions until they are completely at odds with what they once appeared to stand for. The same has been happening to government administrators at all levels, law enforcement, military and the human population. Cultists love mind control for two main reasons: It allows them to control what people think, do and say to secure agenda advancement and, by definition, it calms their legendary insecurity and fear of the unexpected. I have studied mind control since the time I travelled America in 1996. I may have been talking to next to no one in terms of an audience in those years, but my goodness did I gather a phenomenal amount of information and knowledge about so many things including the techniques of mind control. I have described this in detail in other books going back to *The Biggest Secret* in 1998. I met a very large number of people recovering from MKUltra and its offshoots and successors and I began to see how these same techniques were being used on the population in general. This was never more obvious than since the 'Covid' hoax began.

Reframing the enforcers

I have observed over the last two decades and more the very clear transformation in the dynamic between the police, officialdom and the public. I tracked this in the books as the relationship mutated from one of serving the public to seeing them as almost the enemy and certainly a lower caste. There has always been a class divide

based on income and always been some psychopathic, corrupt, and big-I-am police officers. This was different. Wholesale change was unfolding in the collective dynamic; it was less about money and far more about position and perceived power. An us-and-them was emerging. Noses were lifted skyward by government administration and law enforcement and their attitude to the public they were *supposed* to be serving changed to one of increasing contempt, superiority and control. The transformation was so clear and widespread that it had to be planned. Collective attitudes and dynamics do not change naturally and organically that quickly on that scale. I then came across an organisation in Britain called Common Purpose created in the late 1980s by Julia Middleton who would work in the office of Deputy Prime Minister John Prescott during the long and disastrous premiership of war criminal Tony Blair. When Blair speaks the Cult is speaking and the man should have been in jail a long time ago. Common Purpose proclaims itself to be one of the biggest 'leadership development' organisations in the world while functioning as a *charity* with all the financial benefits which come from that. It hosts 'leadership development' courses and programmes all over the world and claims to have 'brought together' what it calls 'leaders' from more than 100 countries on six continents. The modus operandi of Common Purpose can be compared with the work of the UK government's reframing network that includes the Behavioural Insights Team 'nudge unit' and 'Covid' reframing specialists at SPI-B. WikiLeaks described Common Purpose long ago as 'a hidden virus in our government and schools' which is unknown to the general public: 'It recruits and trains "leaders" to be loyal to the directives of Common Purpose and the EU, instead of to their own departments, which they then undermine or subvert, the NHS [National Health Service] being an example.' This is a vital point to understand the 'Covid' hoax. The NHS, and its equivalent around the world, has been utterly reframed in terms of administrators and much of the medical personnel with the transformation underpinned by recruitment policies. The outcome has been the criminal and psychopathic behaviour of the

NHS over 'Covid' and we have seen the same in every other major country. WikiLeaks said Common Purpose trainees are 'learning to rule without regard to democracy' and to usher in a police state (current events explained). Common Purpose operated like a 'glue' and had members in the NHS, BBC, police, legal profession, church, many of Britain's 7,000 quangos, local councils, the Civil Service, government ministries and Parliament, and controlled many RDA's (Regional Development Agencies). Here we have one answer for how and why British institutions and their like in other countries have changed so negatively in relation to the public. This further explains how and why the beyond-disgraceful reframed BBC has become a propaganda arm of 'Covid' fascism. They are all part of a network pursuing the same goal.

By 2019 Common Purpose was quoting a figure of 85,000 'leaders' that had attended its programmes. These 'students' of all ages are known as Common Purpose 'graduates' and they consist of government, state and local government officials and administrators, police chiefs and officers, and a whole range of others operating within the national, local and global establishment. Cressida Dick, Commissioner of the London Metropolitan Police, is the Common Purpose graduate who was the 'Gold Commander' that oversaw what can only be described as the murder of Brazilian electrician Jean Charles de Menezes in 2005. He was held down by psychopathic police and shot seven times in the head by a psychopathic lunatic after being mistaken for a terrorist when he was just a bloke going about his day. Dick authorised officers to pursue and keep surveillance on de Menezes and ordered that he be stopped from entering the underground train system. Police psychopaths took her at her word clearly. She was 'disciplined' for this outrage by being *promoted* – eventually to the top of the 'Met' police where she has been a disaster. Many Chief Constables controlling the police in different parts of the UK are and have been Common Purpose graduates. I have heard the 'graduate' network described as a sort of Mafia or secret society operating within the fabric of government at all levels pursuing a collective policy

ingrained at Common Purpose training events. Founder Julia Middleton herself has said:

Locally and internationally, Common Purpose graduates will be 'lighting small fires' to create change in their organisations and communities ... The Common Purpose effect is best illustrated by the many stories of small changes brought about by leaders, who themselves have changed.

A Common Purpose mission statement declared:

Common Purpose aims to improve the way society works by expanding the vision, decision-making ability and influence of all kinds of leaders. The organisation runs a variety of educational programmes for leaders of all ages, backgrounds and sectors, in order to provide them with the inspirational, information and opportunities they need to change the world.

Yes, but into what? Since 2020 the answer has become clear.

NLP and the Delphi technique

Common Purpose would seem to be a perfect name or would common programming be better? One of the foundation methods of reaching 'consensus' (group think) is by setting the agenda theme and then encouraging, cajoling or pressuring everyone to agree a 'consensus' in line with the core theme promoted by Common Purpose. The methodology involves the 'Delphi technique', or an adaption of it, in which opinions are expressed that are summarised by a 'facilitator or change agent' at each stage. Participants are 'encouraged' to modify their views in the light of what others have said. Stage by stage the former individual opinions are merged into group consensus which just happens to be what Common Purpose wants them to believe. A key part of this is to marginalise anyone refusing to concede to group think and turn the group against them to apply pressure to conform. We are seeing this very technique used on the general population to make 'Covid' group-thinkers hostile to those who have seen through the bullshit. People can be reframed by using perception manipulation methods such as Neuro-Linguistic Programming (NLP) in which you change perception with the use of

carefully constructed language. An NLP website described the technique this way:

... A method of influencing brain behaviour (the 'neuro' part of the phrase) through the use of language (the 'linguistic' part) and other types of communication to enable a person to 'recode' the way the brain responds to stimuli (that's the 'programming') and manifest new and better behaviours. Neuro-Linguistic Programming often incorporates hypnosis and self-hypnosis to help achieve the change (or 'programming') that is wanted.

British alternative media operation UKColumn has done very detailed research into Common Purpose over a long period. I quoted co-founder and former naval officer Brian Gerrish in my book *Remember Who You Are*, published in 2011, as saying the following years before current times:

It is interesting that many of the mothers who have had children taken by the State speak of the Social Services people being icily cool, emotionless and, as two ladies said in slightly different words, '... like little robots'. We know that NLP is cumulative, so people can be given small imperceptible doses of NLP in a course here, another in a few months, next year etc. In this way, major changes are accrued in their personality, but the day by day change is almost unnoticeable.

In these and other ways 'graduates' have had their perceptions uniformly reframed and they return to their roles in the institutions of government, law enforcement, legal profession, military, 'education', the UK National Health Service and the whole swathe of the establishment structure to pursue a common agenda preparing for the 'post-industrial', 'post-democratic' society. I say 'preparing' but we are now there. 'Post-industrial' is code for the Great Reset and 'post-democratic' is 'Covid' fascism. UKColumn has spoken to partners of those who have attended Common Purpose 'training'. They have described how personalities and attitudes of 'graduates' changed very noticeably for the worse by the time they had completed the course. They had been 'reframed' and told they are the 'leaders' – the special ones – who know better than the population. There has also been the very demonstrable recruitment of psychopaths and narcissists into government administration at all

levels and law enforcement. If you want psychopathy hire psychopaths and you get a simple cause and effect. If you want administrators, police officers and 'leaders' to perceive the public as lesser beings who don't matter then employ narcissists. These personalities are identified using 'psychometrics' that identifies knowledge, abilities, attitudes and personality traits, mostly through carefully-designed questionnaires and tests. As this policy has passed through the decades we have had power-crazy, power-trippers appointed into law enforcement, security and government administration in preparation for current times and the dynamic between public and law enforcement/officialdom has been transformed. UKColumn's Brian Gerrish said of the narcissistic personality:

Their love of themselves and power automatically means that they will crush others who get in their way. I received a major piece of the puzzle when a friend pointed out that when they made public officials re-apply for their own jobs several years ago they were also required to do psychometric tests. This was undoubtedly the start of the screening process to get 'their' sort of people in post.

How obvious that has been since 2020 although it was clear what was happening long before if people paid attention to the changing public-establishment dynamic.

Change agents

At the centre of events in 'Covid' Britain is the National Health Service (NHS) which has behaved disgracefully in slavishly following the Cult agenda. The NHS management structure is awash with Common Purpose graduates or 'change agents' working to a common cause. Helen Bevan, a Chief of Service Transformation at the NHS Institute for Innovation and Improvement, co-authored a document called 'Towards a million change agents, a review of the social movements literature: implications for large scale change in the NHS'. The document compared a project management approach to that of change and social movements where 'people change

themselves and each other – peer to peer’. Two definitions given for a ‘social movement’ were:

A group of people who consciously attempt to build a radically new social order; involves people of a broad range of social backgrounds; and deploys politically confrontational and socially disruptive tactics – Cyrus Zirakzadeh 1997

Collective challenges, based on common purposes and social solidarities, in sustained interaction with elites, opponents, and authorities – Sidney Tarrow 1994

Helen Bevan wrote another NHS document in which she defined ‘framing’ as ‘the process by which leaders construct, articulate and put across their message in a powerful and compelling way in order to win people to their cause and call them to action’. I think I could come up with another definition that would be rather more accurate. The National Health Service and institutions of Britain and the wider world have been taken over by reframed ‘change agents’ and that includes everything from the United Nations to national governments, local councils and social services which have been kidnapping children from loving parents on an extraordinary and gathering scale on the road to the end of parenthood altogether. Children from loving homes are stolen and kidnapped by the state and put into the ‘care’ (inversion) of the local authority through council homes, foster parents and forced adoption. At the same time children are allowed to be abused without response while many are under council ‘care’. UKColumn highlighted the Common Purpose connection between South Yorkshire Police and Rotherham council officers in the case of the scandal in that area of the sexual exploitation of children to which the authorities turned not one blind eye, but both:

We were alarmed to discover that the Chief Executive, the Strategic Director of Children and Young People's Services, the Manager for the Local Strategic Partnership, the Community Cohesion Manager, the Cabinet Member for Cohesion, the Chief Constable and his predecessor had all attended Leadership training courses provided by the pseudo-charity Common Purpose.

Once 'change agents' have secured positions of hire and fire within any organisation things start to move very quickly. Personnel are then hired and fired on the basis of whether they will work towards the agenda the change agent represents. If they do they are rapidly promoted even though they may be incompetent. Those more qualified and skilled who are pre-Common Purpose 'old school' see their careers stall and even disappear. This has been happening for decades in every institution of state, police, 'health' and social services and all of them have been transformed as a result in their attitudes to their jobs and the public. Medical professions, including nursing, which were once vocations for the caring now employ many cold, callous and couldn't give a shit personality types. The UKColumn investigation concluded:

By blurring the boundaries between people, professions, public and private sectors, responsibility and accountability, Common Purpose encourages 'graduates' to believe that as new selected leaders, they can work together, outside of the established political and social structures, to achieve a paradigm shift or CHANGE – so called 'Leading Beyond Authority'. In doing so, the allegiance of the individual becomes 'reframed' on CP colleagues and their NETWORK.

Reframing the Face-Nappies

Nowhere has this process been more obvious than in the police where recruitment of psychopaths and development of unquestioning mind-controlled group-thinkers have transformed law enforcement into a politically-correct 'Woke' joke and a travesty of what should be public service. Today they wear their face-nappies like good little gofers and enforce 'Covid' rules which are fascism under another name. Alongside the specifically-recruited psychopaths we have software minds incapable of free thought. Brian Gerrish again:

An example is the policeman who would not get on a bike for a press photo because he had not done the cycling proficiency course. Normal people say this is political correctness gone mad. Nothing could be further from the truth. The policeman has been reframed, and in his reality it is perfect common sense not to get on the bike 'because he hasn't done the cycling course'.

Another example of this is where the police would not rescue a boy from a pond until they had taken advice from above on the 'risk assessment'. A normal person would have arrived, perhaps thought of the risk for a moment, and dived in. To the police now 'reframed', they followed 'normal' procedure.

There are shocking cases of reframed ambulance crews doing the same. Sheer unthinking stupidity of London Face-Nappies headed by Common Purpose graduate Cressida Dick can be seen in their behaviour at a vigil in March, 2021, for a murdered woman, Sarah Everard. A police officer had been charged with the crime. Anyone with a brain would have left the vigil alone in the circumstances. Instead they 'manhandled' women to stop them breaking 'Covid rules' to betray classic reframing. Minds in the thrall of perception control have no capacity for seeing a situation on its merits and acting accordingly. 'Rules is rules' is their only mind-set. My father used to say that rules and regulations are for the guidance of the intelligent and the blind obedience of the idiot. Most of the intelligent, decent, coppers have gone leaving only the other kind and a few old school for whom the job must be a daily nightmare. The combination of psychopaths and rule-book software minds has been clearly on public display in the 'Covid' era with automaton robots in uniform imposing fascistic 'Covid' regulations on the population without any personal initiative or judging situations on their merits. There are thousands of examples around the world, but I'll make my point with the infamous Derbyshire police in the English East Midlands – the ones who think pouring dye into beauty spots and using drones to track people walking in the countryside away from anyone is called 'policing'. To them there are rules decreed by the government which they have to enforce and in their bewildered state a group gathering in a closed space and someone walking alone in the countryside are the same thing. It is beyond idiocy and enters the realm of clinical insanity.

Police officers in Derbyshire said they were 'horrified' – *horrified* – to find 15 to 20 'irresponsible' kids playing a football match at a closed leisure centre 'in breach of coronavirus restrictions'. When they saw the police the kids ran away leaving their belongings behind and the reframed men and women of Derbyshire police were seeking to establish their identities with a view to fining their parents. The most natural thing for youngsters to do – kicking a ball about – is turned into a criminal activity and enforced by the moronic software programs of Derbyshire police. You find the same mentality in every country. These barely conscious 'horrified' officers said they had to take action because 'we need to ensure these rules are being followed' and 'it is of the utmost importance that you ensure your children are following the rules and regulations for Covid-19'. Had any of them done ten seconds of research to see if this parroting of their masters' script could be supported by any evidence? Nope. Reframed people don't think – others think for them and that's the whole idea of reframing. I have seen police officers one after the other repeating without question word for word what officialdom tells them just as I have seen great swathes of the public doing the same. Ask either for 'their' opinion and out spews what they have been told to think by the official narrative. Police and public may seem to be in different groups, but their mentality is the same. Most people do whatever they are told in fear not doing so or because they believe what officialdom tells them; almost the entirety of the police do what they are told for the same reason. Ultimately it's the tiny inner core of the global Cult that's telling both what to do.

So Derbyshire police were 'horrified'. Oh, really? Why did they think those kids were playing football? It was to relieve the psychological consequences of lockdown and being denied human contact with their friends and interaction, touch and discourse vital to human psychological health. Being denied this month after month has dismantled the psyche of many children and young people as depression and suicide have exploded. Were Derbyshire police *horrified by that*? Are you kidding? Reframed people don't have those

mental and emotional processes that can see how the impact on the psychological health of youngsters is far more dangerous than any 'virus' even if you take the mendacious official figures to be true. The reframed are told (programmed) how to act and so they do. The Derbyshire Chief Constable in the first period of lockdown when the black dye and drones nonsense was going on was Peter Goodman. He was the man who severed the connection between his force and the Derbyshire Constabulary *Male Voice* Choir when he decided that it was not inclusive enough to allow women to join. The fact it was a male voice choir making a particular sound produced by male voices seemed to elude a guy who terrifyingly ran policing in Derbyshire. He retired weeks after his force was condemned as disgraceful by former Supreme Court Justice Jonathan Sumption for their behaviour over extreme lockdown impositions. Goodman was replaced by his deputy Rachel Swann who was in charge when her officers were 'horrificed'. The police statement over the boys committing the hanging-offence of playing football included the line about the youngsters being 'irresponsible in the times we are all living through' missing the point that the real relevance of the 'times we are all living through' is the imposition of fascism enforced by psychopaths and reframed minds of police officers playing such a vital part in establishing the fascist tyranny that their own children and grandchildren will have to live in their entire lives. As a definition of insanity that is hard to beat although it might be run close by imposing masks on people that can have a serious effect on their health while wearing a face nappy all day themselves. Once again public and police do it for the same reason – the authorities tell them to and who are they to have the self-respect to say no?

Wokers in uniform

How reframed do you have to be to arrest a *six-year-old* and take him to court for *picking a flower* while waiting for a bus? Brain dead police and officialdom did just that in North Carolina where criminal proceedings happen regularly for children under nine. Attorney Julie Boyer gave the six-year-old crayons and a colouring book

during the 'flower' hearing while the 'adults' decided his fate. County Chief District Court Judge Jay Corpening asked: 'Should a child that believes in Santa Claus, the Easter Bunny and the tooth fairy be making life-altering decisions?' Well, of course not, but common sense has no meaning when you have a common purpose and a reframed mind. Treating children in this way, and police operating in American schools, is all part of the psychological preparation for children to accept a police state as normal all their adult lives. The same goes for all the cameras and biometric tracking technology in schools. Police training is focused on reframing them as snowflake Wokers and this is happening in the military. Pentagon top brass said that 'training sessions on extremism' were needed for troops who asked why they were so focused on the Capitol Building riot when Black Lives Matter riots were ignored. What's the difference between them some apparently and rightly asked. Actually, there is a difference. Five people died in the Capitol riot, only one through violence, and that was a police officer shooting an unarmed protestor. BLM riots killed at least 25 people and cost billions. Asking the question prompted the psychopaths and reframed minds that run the Pentagon to say that more 'education' (programming) was needed. Troop training is all based on psychological programming to make them fodder for the Cult – 'Military men are just dumb, stupid animals to be used as pawns in foreign policy' as Cult-to-his-DNA former Secretary of State Henry Kissinger famously said. Governments see the police in similar terms and it's time for those among them who can see this to defend the people and stop being enforcers of the Cult agenda upon the people.

The US military, like the country itself, is being targeted for destruction through a long list of Woke impositions. Cult-owned gaga 'President' Biden signed an executive order when he took office to allow taxpayer money to pay for transgender surgery for active military personnel and veterans. Are you a man soldier? No, I'm a LGBTQIA+ with a hint of Skoliosexual and Spectrasexual. Oh, good man. Bad choice of words you bigot. The Pentagon announced in March, 2021, the appointment of the first 'diversity and inclusion

officer' for US Special Forces. Richard Torres-Estrada arrived with the publication of a 'D&I Strategic Plan which will guide the enterprise-wide effort to institutionalize and sustain D&I'. If you think a Special Forces 'Strategic Plan' should have something to do with defending America you haven't been paying attention. Defending Woke is now the military's new role. Torres-Estrada has posted images comparing Donald Trump with Adolf Hitler and we can expect no bias from him as a representative of the supposedly non-political Pentagon. Cable news host Tucker Carlson said: 'The Pentagon is now the Yale faculty lounge but with cruise missiles.' Meanwhile Secretary of Defense Lloyd Austin, a board member of weapons-maker Raytheon with stock and compensation interests in October, 2020, worth \$1.4 million, said he was purging the military of the 'enemy within' – anyone who isn't Woke and supports Donald Trump. Austin refers to his targets as 'racist extremists' while in true Woke fashion being himself a racist extremist. Pentagon documents pledge to 'eradicate, eliminate and conquer all forms of racism, sexism and homophobia'. The definitions of these are decided by 'diversity and inclusion committees' peopled by those who see racism, sexism and homophobia in every situation and opinion. Woke (the Cult) is dismantling the US military and purging testosterone as China expands its military and gives its troops 'masculinity training'. How do we think that is going to end when this is all Cult coordinated? The US military, like the British military, is controlled by Woke and spineless top brass who just go along with it out of personal career interests.

'Woke' means fast asleep

Mind control and perception manipulation techniques used on individuals to create group-think have been unleashed on the global population in general. As a result many have no capacity to see the obvious fascist agenda being installed all around them or what 'Covid' is really all about. Their brains are firewalled like a computer system not to process certain concepts, thoughts and realisations that are bad for the Cult. The young are most targeted as the adults they

will be when the whole fascist global state is planned to be fully implemented. They need to be prepared for total compliance to eliminate all pushback from entire generations. The Cult has been pouring billions into taking complete control of 'education' from schools to universities via its operatives and corporations and not least Bill Gates as always. The plan has been to transform 'education' institutions into programming centres for the mentality of 'Woke'. James McConnell, professor of psychology at the University of Michigan, wrote in *Psychology Today* in 1970:

The day has come when we can combine sensory deprivation with drugs, hypnosis, and astute manipulation of reward and punishment, to gain almost absolute control over an individual's behaviour. It should then be possible to achieve a very rapid and highly effective type of brainwashing that would allow us to make dramatic changes in a person's behaviour and personality ...

... We should reshape society so that we all would be trained from birth to want to do what society wants us to do. We have the techniques to do it... no-one owns his own personality you acquired, and there's no reason to believe you should have the right to refuse to acquire a new personality if your old one is anti-social.

This was the potential for mass brainwashing in 1970 and the mentality there displayed captures the arrogant psychopathy that drives it forward. I emphasise that not all young people have succumbed to Woke programming and those that haven't are incredibly impressive people given that today's young are the most perceptually-targeted generations in history with all the technology now involved. Vast swathes of the young generations, however, have fallen into the spell – and that's what it is – of Woke. The Woke mentality and perceptual program is founded on *inversion* and you will appreciate later why that is so significant. Everything with Woke is inverted and the opposite of what it is claimed to be. Woke was a term used in African-American culture from the 1900s and referred to an awareness of social and racial justice. This is not the meaning of the modern version or 'New Woke' as I call it in *The Answer*. Oh, no, Woke today means something very different no matter how much Wokers may seek to hide that and insist Old Woke and New

Woke are the same. See if you find any 'awareness of social justice' here in the modern variety:

- Woke demands 'inclusivity' while excluding anyone with a different opinion and calls for mass censorship to silence other views.
- Woke claims to stand against oppression when imposing oppression is the foundation of all that it does. It is the driver of political correctness which is nothing more than a Cult invention to manipulate the population to silence itself.
- Woke believes itself to be 'liberal' while pursuing a global society that can only be described as fascist (see 'anti-fascist' fascist Antifa).
- Woke calls for 'social justice' while spreading injustice wherever it goes against the common 'enemy' which can be easily identified as a differing view.
- Woke is supposed to be a metaphor for 'awake' when it is solid-gold asleep and deep in a Cult-induced coma that meets the criteria for 'off with the fairies'.

I state these points as obvious facts if people only care to look. I don't do this with a sense of condemnation. We need to appreciate that the onslaught of perceptual programming on the young has been incessant and merciless. I can understand why so many have been reframed, or, given their youth, framed from the start to see the world as the Cult demands. The Cult has had access to their minds day after day in its 'education' system for their entire formative years. Perception is formed from information received and the Cult-created system is a life-long download of information delivered to elicit a particular perception, thus behaviour. The more this has expanded into still new extremes in recent decades and ever-increasing censorship has deleted other opinions and information why wouldn't that lead to a perceptual reframing on a mass scale? I

have described already cradle-to-grave programming and in more recent times the targeting of young minds from birth to adulthood has entered the stratosphere. This has taken the form of skewing what is 'taught' to fit the Cult agenda and the omnipresent techniques of group-think to isolate non-believers and pressure them into line. There has always been a tendency to follow the herd, but we really are in a new world now in relation to that. We have parents who can see the 'Covid' hoax told by their children not to stop them wearing masks at school, being 'Covid' tested or having the 'vaccine' in fear of the peer-pressure consequences of being different. What is 'peer-pressure' if not pressure to conform to group-think? Renegade Minds never group-think and always retain a set of perceptions that are unique to them. Group-think is always underpinned by consequences for not group-thinking. Abuse now aimed at those refusing DNA-manipulating 'Covid vaccines' are a potent example of this. The biggest pressure to conform comes from the very group which is itself being manipulated. 'I am programmed to be part of a hive mind and so you must be.'

Woke control structures in 'education' now apply to every mainstream organisation. Those at the top of the 'education' hierarchy (the Cult) decide the policy. This is imposed on governments through the Cult network; governments impose it on schools, colleges and universities; their leadership impose the policy on teachers and academics and they impose it on children and students. At any level where there is resistance, perhaps from a teacher or university lecturer, they are targeted by the authorities and often fired. Students themselves regularly demand the dismissal of academics (increasingly few) at odds with the narrative that the students have been programmed to believe in. It is quite a thought that students who are being targeted by the Cult become so consumed by programmed group-think that they launch protests and demand the removal of those who are trying to push back against those targeting the students. Such is the scale of perceptual inversion. We see this with 'Covid' programming as the Cult imposes the rules via psycho-psychologists and governments on

shops, transport companies and businesses which impose them on their staff who impose them on their customers who pressure Pushbackers to conform to the will of the Cult which is in the process of destroying them and their families. Scan all aspects of society and you will see the same sequence every time.

Fact free Woke and hijacking the 'left'

There is no more potent example of this than 'Woke', a mentality only made possible by the deletion of factual evidence by an 'education' system seeking to produce an ever more uniform society. Why would you bother with facts when you don't know any? Deletion of credible history both in volume and type is highly relevant. Orwell said: 'Who controls the past controls the future: who controls the present controls the past.' They who control the perception of the past control the perception of the future and they who control the present control the perception of the past through the writing and deleting of history. Why would you oppose the imposition of Marxism in the name of Wokeism when you don't know that Marxism cost at least 100 million lives in the 20th century alone? Watch videos and read reports in which Woker generations are asked basic historical questions – it's mind-blowing. A survey of 2,000 people found that six percent of millennials (born approximately early 1980s to early 2000s) believed the Second World War (1939-1945) broke out with the assassination of President Kennedy (in 1963) and one in ten thought Margaret Thatcher was British Prime Minister at the time. She was in office between 1979 and 1990. We are in a post-fact society. Provable facts are no defence against the fascism of political correctness or Silicon Valley censorship. Facts don't matter anymore as we have witnessed with the 'Covid' hoax. Sacrificing uniqueness to the Woke group-think religion is all you are required to do and that means thinking for yourself is the biggest Woke no, no. All religions are an expression of group-think and censorship and Woke is just another religion with an orthodoxy defended by group-think and censorship. Burned at

the stake becomes burned on Twitter which leads back eventually to burned at the stake as Woke humanity regresses to ages past.

The biggest Woke inversion of all is its creators and funders. I grew up in a traditional left of centre political household on a council estate in Leicester in the 1950s and 60s – you know, the left that challenged the power of wealth-hoarding elites and threats to freedom of speech and opinion. In those days students went on marches defending freedom of speech while today's Wokers march for its deletion. What on earth could have happened? Those very elites (collectively the Cult) that we opposed in my youth and early life have funded into existence the antithesis of that former left and hijacked the 'brand' while inverting everything it ever stood for. We have a mentality that calls itself 'liberal' and 'progressive' while acting like fascists. Cult billionaires and their corporations have funded themselves into control of 'education' to ensure that Woke programming is unceasing throughout the formative years of children and young people and that non-Wokers are isolated (that word again) whether they be students, teachers or college professors. The Cult has funded into existence the now colossal global network of Woke organisations that have spawned and promoted all the 'causes' on the Cult wish-list for global transformation and turned Wokers into demanders of them. Does anyone really think it's a coincidence that the Cult agenda for humanity is a carbon (sorry) copy of the societal transformations desired by Woke?? These are only some of them:

Political correctness: The means by which the Cult deletes all public debates that it knows it cannot win if we had the free-flow of information and evidence.

Human-caused 'climate change': The means by which the Cult seeks to transform society into a globally-controlled dictatorship imposing its will over the fine detail of everyone's lives 'to save the planet' which doesn't actually need saving.

Transgender obsession: Preparing collective perception to accept the 'new human' which would not have genders because it would be created technologically and not through procreation. I'll have much more on this in Human 2.0.

Race obsession: The means by which the Cult seeks to divide and rule the population by triggering racial division through the perception that society is more racist than ever when the opposite is the case. Is it perfect in that regard? No. But to compare today with the racism of apartheid and segregation brought to an end by the civil rights movement in the 1960s is to insult the memory of that movement and inspirations like Martin Luther King. Why is the 'anti-racism' industry (which it is) so dominated by privileged white people?

White supremacy: This is a label used by privileged white people to demonise poor and deprived white people pushing back on tyranny to marginalise and destroy them. White people are being especially targeted as the dominant race by number within Western society which the Cult seeks to transform in its image. If you want to change a society you must weaken and undermine its biggest group and once you have done that by using the other groups you next turn on them to do the same ... 'Then they came for the Jews and I was not a Jew so I did nothing.'

Mass migration: The mass movement of people from the Middle East, Africa and Asia into Europe, from the south into the United States and from Asia into Australia are another way the Cult seeks to dilute the racial, cultural and political influence of white people on Western society. White people ask why their governments appear to be working against them while being politically and culturally biased towards incoming cultures. Well, here's your answer. In the same way sexually 'straight' people, men and women, ask why the

authorities are biased against them in favour of other sexualities. The answer is the same – that's the way the Cult wants it to be for very sinister motives.

These are all central parts of the Cult agenda and central parts of the Woke agenda and Woke was created and continues to be funded to an immense degree by Cult billionaires and corporations. If anyone begins to say 'coincidence' the syllables should stick in their throat.

Billionaire 'social justice warriors'

Joe Biden is a 100 percent-owned asset of the Cult and the Wokers' man in the White House whenever he can remember his name and for however long he lasts with his rapidly diminishing cognitive function. Even walking up the steps of an aircraft without falling on his arse would appear to be a challenge. He's not an empty-shell puppet or anything. From the minute Biden took office (or the Cult did) he began his executive orders promoting the Woke wish-list. You will see the Woke agenda imposed ever more severely because it's really the *Cult* agenda. Woke organisations and activist networks spawned by the Cult are funded to the extreme so long as they promote what the Cult wants to happen. Woke is funded to promote 'social justice' by billionaires who become billionaires by destroying social justice. The social justice mantra is only a cover for dismantling social justice and funded by billionaires that couldn't give a damn about social justice. Everything makes sense when you see that. One of Woke's premier funders is Cult billionaire financier George Soros who said: 'I am basically there to make money, I cannot and do not look at the social consequences of what I do.' This is the same Soros who has given more than \$32 billion to his Open Society Foundations global Woke network and funded Black Lives Matter, mass immigration into Europe and the United States, transgender activism, climate change activism, political correctness and groups targeting 'white supremacy' in the form of privileged white thugs that dominate Antifa. What a scam it all is and when

you are dealing with the unquestioning fact-free zone of Woke scamming them is child's play. All you need to pull it off in all these organisations are a few in-the-know agents of the Cult and an army of naïve, reframed, uninformed, narcissistic, know-nothings convinced of their own self-righteousness, self-purity and virtue.

Soros and fellow billionaires and billionaire corporations have poured hundreds of millions into Black Lives Matter and connected groups and promoted them to a global audience. None of this is motivated by caring about black people. These are the billionaires that have controlled and exploited a system that leaves millions of black people in abject poverty and deprivation which they do absolutely nothing to address. The same Cult networks funding BLM were behind the *slave trade*! Black Lives Matter hijacked a phrase that few would challenge and they have turned this laudable concept into a political weapon to divide society. You know that BLM is a fraud when it claims that *All Lives Matter*, the most inclusive statement of all, is 'racist'. BLM and its Cult masters don't want to end racism. To them it's a means to an end to control all of humanity never mind the colour, creed, culture or background. What has destroying the nuclear family got to do with ending racism? Nothing – but that is one of the goals of BLM and also happens to be a goal of the Cult as I have been exposing in my books for decades. Stealing children from loving parents and giving schools ever more power to override parents is part of that same agenda. BLM is a Marxist organisation and why would that not be the case when the Cult created Marxism *and* BLM? Patrisse Cullors, a BLM co-founder, said in a 2015 video that she and her fellow organisers, including co-founder Alicia Garza, are 'trained Marxists'. The lady known after marriage as Patrisse Khan-Cullors bought a \$1.4 million home in 2021 in one of the whitest areas of California with a black population of just 1.6 per cent and has so far bought *four* high-end homes for a total of \$3.2 million. How very Marxist. There must be a bit of spare in the BLM coffers, however, when Cult corporations and billionaires have handed over the best part of \$100 million. Many black people can see that Black Lives Matter is not

working for them, but against them, and this is still more confirmation. Black journalist Jason Whitlock, who had his account suspended by Twitter for simply linking to the story about the 'Marxist's' home buying spree, said that BLM leaders are 'making millions of dollars off the backs of these dead black men who they wouldn't spit on if they were on fire and alive'.

Black Lies Matter

Cult assets and agencies came together to promote BLM in the wake of the death of career criminal George Floyd who had been jailed a number of times including for forcing his way into the home of a black woman with others in a raid in which a gun was pointed at her stomach. Floyd was filmed being held in a Minneapolis street in 2020 with the knee of a police officer on his neck and he subsequently died. It was an appalling thing for the officer to do, but the same technique has been used by police on peaceful protestors of lockdown without any outcry from the Woke brigade. As unquestioning supporters of the Cult agenda Wokers have supported lockdown and all the 'Covid' claptrap while attacking anyone standing up to the tyranny imposed in its name. Court documents would later include details of an autopsy on Floyd by County Medical Examiner Dr Andrew Baker who concluded that Floyd had taken a fatal level of the drug fentanyl. None of this mattered to fact-free, question-free, Woke. Floyd's death was followed by worldwide protests against police brutality amid calls to defund the police. Throwing babies out with the bathwater is a Woke speciality. In the wake of the murder of British woman Sarah Everard a Green Party member of the House of Lords, Baroness Jones of Moulsecoomb (Nincompoopia would have been better), called for a 6pm curfew for all men. This would be in breach of the Geneva Conventions on war crimes which ban collective punishment, but that would never have crossed the black and white Woke mind of Baroness Nincompoopia who would have been far too convinced of her own self-righteousness to compute such details. Many American cities did defund the police in the face of Floyd riots

and after \$15 million was deleted from the police budget in Washington DC under useless Woke mayor Muriel Bowser car-jacking alone rose by 300 percent and within six months the US capital recorded its highest murder rate in 15 years. The same happened in Chicago and other cities in line with the Cult/Soros plan to bring fear to streets and neighbourhoods by reducing the police, releasing violent criminals and not prosecuting crime. This is the mob-rule agenda that I have warned in the books was coming for so long. Shootings in the area of Minneapolis where Floyd was arrested increased by 2,500 percent compared with the year before. Defunding the police over George Floyd has led to a big increase in dead people with many of them black. Police protection for politicians making these decisions stayed the same or increased as you would expect from professional hypocrites. The Cult doesn't actually want to abolish the police. It wants to abolish local control over the police and hand it to federal government as the psychopaths advance the Hunger Games Society. Many George Floyd protests turned into violent riots with black stores and businesses destroyed by fire and looting across America fuelled by Black Lives Matter. Woke doesn't do irony. If you want civil rights you must loot the liquor store and the supermarket and make off with a smart TV. It's the only way.

It's not a race war – it's a class war

Black people are patronised by privileged blacks and whites alike and told they are victims of white supremacy. I find it extraordinary to watch privileged blacks supporting the very system and bloodline networks behind the slave trade and parroting the same Cult-serving manipulative crap of their privileged white, often billionaire, associates. It is indeed not a race war but a class war and colour is just a diversion. Black Senator Cory Booker and black Congresswoman Maxine Waters, more residents of Nincompoopia, personify this. Once you tell people they are victims of someone else you devalue both their own responsibility for their plight and the power they have to impact on their reality and experience. Instead

we have: 'You are only in your situation because of whiteness – turn on them and everything will change.' It won't change. Nothing changes in our lives unless *we* change it. Crucial to that is never seeing yourself as a victim and always as the creator of your reality. Life is a simple sequence of choice and consequence. Make different choices and you create different consequences. *You* have to make those choices – not Black Lives Matter, the Woke Mafia and anyone else that seeks to dictate your life. Who are they these Wokers, an emotional and psychological road traffic accident, to tell you what to do? Personal empowerment is the last thing the Cult and its Black Lives Matter want black people or anyone else to have. They claim to be defending the underdog while *creating* and perpetuating the underdog. The Cult's worst nightmare is human unity and if they are going to keep blacks, whites and every other race under economic servitude and control then the focus must be diverted from what they have in common to what they can be manipulated to believe divides them. Blacks have to be told that their poverty and plight is the fault of the white bloke living on the street in the same poverty and with the same plight they are experiencing. The difference is that your plight black people is due to him, a white supremacist with 'white privilege' living on the street. Don't unite as one human family against your mutual oppressors and suppressors – fight the oppressor with the white face who is as financially deprived as you are. The Cult knows that as its 'Covid' agenda moves into still new levels of extremism people are going to respond and it has been spreading the seeds of disunity everywhere to stop a united response to the evil that targets *all of us*.

Racist attacks on 'whiteness' are getting ever more outrageous and especially through the American Democratic Party which has an appalling history for anti-black racism. Barack Obama, Joe Biden, Hillary Clinton and Nancy Pelosi all eulogised about Senator Robert Byrd at his funeral in 2010 after a nearly 60-year career in Congress. Byrd was a brutal Ku Klux Klan racist and a violent abuser of Cathy O'Brien in MKUltra. He said he would never fight in the military 'with a negro by my side' and 'rather I should die a thousand times,

and see Old Glory trampled in the dirt never to rise again, than to see this beloved land of ours become degraded by race mongrels, a throwback to the blackest specimen from the wilds'. Biden called Byrd a 'very close friend and mentor'. These 'Woke' hypocrites are not anti-racist they are anti-poor and anti-people not of their perceived class. Here is an illustration of the scale of anti-white racism to which we have now descended. Seriously Woke and moronic *New York Times* contributor Damon Young described whiteness as a 'virus' that 'like other viruses will not die until there are no bodies left for it to infect'. He went on: '... the only way to stop it is to locate it, isolate it, extract it, and kill it.' Young can say that as a black man with no consequences when a white man saying the same in reverse would be facing a jail sentence. *That's* racism. We had super-Woke numbskull senators Tammy Duckworth and Mazie Hirono saying they would object to future Biden Cabinet appointments if he did not nominate more Asian Americans and Pacific Islanders. Never mind the ability of the candidate what do they look like? Duckworth said: 'I will vote for racial minorities and I will vote for LGBTQ, but anyone else I'm not voting for.' Appointing people on the grounds of race is illegal, but that was not a problem for this ludicrous pair. They were on-message and that's a free pass in any situation.

Critical race racism

White children are told at school they are intrinsically racist as they are taught the divisive 'critical race theory'. This claims that the law and legal institutions are inherently racist and that race is a socially constructed concept used by white people to further their economic and political interests at the expense of people of colour. White is a 'virus' as we've seen. Racial inequality results from 'social, economic, and legal differences that white people create between races to maintain white interests which leads to poverty and criminality in minority communities'. I must tell that to the white guy sleeping on the street. The principal of East Side Community School in New York sent white parents a manifesto that called on

them to become 'white traitors' and advocate for full 'white abolition'. These people are teaching your kids when they urgently need a psychiatrist. The 'school' included a chart with 'eight white identities' that ranged from 'white supremacist' to 'white abolition' and defined the behaviour white people must follow to end 'the regime of whiteness'. Woke blacks and their privileged white associates are acting exactly like the slave owners of old and Ku Klux Klan racists like Robert Byrd. They are too full of their own self-purity to see that, but it's true. Racism is not a body type; it's a state of mind that can manifest through any colour, creed or culture.

Another racial fraud is '*equity*'. Not equality of treatment and opportunity – equity. It's a term spun as equality when it means something very different. Equality in its true sense is a raising up while 'equity' is a race to the bottom. Everyone in the same level of poverty is 'equity'. Keep everyone down – that's equity. The Cult doesn't want anyone in the human family to be empowered and BLM leaders, like all these 'anti-racist' organisations, continue their privileged, pampered existence by perpetuating the perception of gathering racism. When is the last time you heard an 'anti-racist' or 'anti-Semitism' organisation say that acts of racism and discrimination have *fallen*? It's not in the interests of their fundraising and power to influence and the same goes for the professional soccer anti-racism operation, Kick It Out. Two things confirmed that the Black Lives Matter riots in the summer of 2020 were Cult creations. One was that while anti-lockdown protests were condemned in this same period for 'transmitting 'Covid' the authorities supported mass gatherings of Black Lives Matter supporters. I even saw self-deluding people claiming to be doctors say the two types of protest were not the same. No – the non-existent 'Covid' was in favour of lockdowns and attacked those that protested against them while 'Covid' supported Black Lives Matter and kept well away from its protests. The whole thing was a joke and as lockdown protestors were arrested, often brutally, by reframed Face-Nappies we had the grotesque sight of police officers taking the knee to Black Lives Matter, a Cult-funded Marxist

organisation that supports violent riots and wants to destroy the nuclear family and white people.

He's not white? Shucks!

Woke obsession with race was on display again when ten people were shot dead in Boulder, Colorado, in March, 2021. Cult-owned Woke TV channels like CNN said the shooter appeared to be a white man and Wokers were on Twitter condemning 'violent white men' with the usual mantras. Then the shooter's name was released as Ahmad Al Aliwi Alissa, an anti-Trump Arab-American, and the sigh of disappointment could be heard five miles away. Never mind that ten people were dead and what that meant for their families. Race baiting was all that mattered to these sick Cult-serving people like Barack Obama who exploited the deaths to further divide America on racial grounds which is his job for the Cult. This is the man that 'racist' white Americans made the first black president of the United States and then gave him a second term. Not-very-bright Obama has become filthy rich on the back of that and today appears to have a big influence on the Biden administration. Even so he's still a downtrodden black man and a victim of white supremacy. This disingenuous fraud reveals the contempt he has for black people when he puts on a Deep South Alabama accent whenever he talks to them, no, *at* them.

Another BLM red flag was how the now fully-Woke (fully-Cult) and fully-virtue-signalled professional soccer authorities had their teams taking the knee before every match in support of Marxist Black Lives Matter. Soccer authorities and clubs displayed 'Black Lives Matter' on the players' shirts and flashed the name on electronic billboards around the pitch. Any fans that condemned what is a Freemasonic taking-the-knee ritual were widely condemned as you would expect from the Woke virtue-signallers of professional sport and the now fully-Woke media. We have reverse racism in which you are banned from criticising any race or culture except for white people for whom anything goes – say what you like, no problem. What has this got to do with racial harmony and

equality? We've had black supremacists from Black Lives Matter telling white people to fall to their knees in the street and apologise for their white supremacy. Black supremacists acting like white supremacist slave owners of the past couldn't breach their self-obsessed, race-obsessed sense of self-purity. Joe Biden appointed a race-obsessed black supremacist Kristen Clarke to head the Justice Department Civil Rights Division. Clarke claimed that blacks are endowed with 'greater mental, physical and spiritual abilities' than whites. If anyone reversed that statement they would be vilified. Clarke is on-message so no problem. She's never seen a black-white situation in which the black figure is anything but a virtuous victim and she heads the Civil Rights Division which should treat everyone the same or it isn't civil rights. Another perception of the Renegade Mind: If something or someone is part of the Cult agenda they will be supported by Woke governments and media no matter what. If they're not, they will be condemned and censored. It really is that simple and so racist Clarke prospers despite (make that because of) her racism.

The end of culture

Biden's administration is full of such racial, cultural and economic bias as the Cult requires the human family to be divided into warring factions. We are now seeing racially-segregated graduations and everything, but everything, is defined through the lens of perceived 'racism'. We have 'racist' mathematics, 'racist' food and even 'racist' *plants*. World famous Kew Gardens in London said it was changing labels on plants and flowers to tell its pre-'Covid' more than two million visitors a year how racist they are. Kew director Richard Deverell said this was part of an effort to 'move quickly to decolonise collections' after they were approached by one Ajay Chhabra 'an actor with an insight into how sugar cane was linked to slavery'. They are *plants* you idiots. 'Decolonisation' in the Woke manual really means colonisation of society with its mentality and by extension colonisation by the Cult. We are witnessing a new Chinese-style 'Cultural Revolution' so essential to the success of all

Marxist takeovers. Our cultural past and traditions have to be swept away to allow a new culture to be built-back-better. Woke targeting of long-standing Western cultural pillars including historical monuments and cancelling of historical figures is what happened in the Mao revolution in China which 'purged remnants of capitalist and traditional elements from Chinese society' and installed Maoism as the dominant ideology'. For China see the Western world today and for 'dominant ideology' see Woke. Better still see Marxism or Maoism. The 'Covid' hoax has specifically sought to destroy the arts and all elements of Western culture from people meeting in a pub or restaurant to closing theatres, music venues, sports stadiums, places of worship and even banning *singing*. Destruction of Western society is also why criticism of any religion is banned except for Christianity which again is the dominant religion as white is the numerically-dominant race. Christianity may be fading rapidly, but its history and traditions are weaved through the fabric of Western society. Delete the pillars and other structures will follow until the whole thing collapses. I am not a Christian defending that religion when I say that. I have no religion. It's just a fact. To this end Christianity has itself been turned Woke to usher its own downfall and its ranks are awash with 'change agents' – knowing and unknowing – at every level including Pope Francis (*definitely* knowing) and the clueless Archbishop of Canterbury Justin Welby (possibly not, but who can be sure?). Woke seeks to coordinate attacks on Western culture, traditions, and ways of life through 'intersectionality' defined as 'the complex, cumulative way in which the effects of multiple forms of discrimination (such as racism, sexism, and classism) combine, overlap, or intersect especially in the experiences of marginalised individuals or groups'. Wade through the Orwellian Woke-speak and this means coordinating disparate groups in a common cause to overthrow freedom and liberal values.

The entire structure of public institutions has been infested with Woke – government at all levels, political parties, police, military, schools, universities, advertising, media and trade unions. This abomination has been achieved through the Cult web by appointing

Wokers to positions of power and battering non-Wokers into line through intimidation, isolation and threats to their job. Many have been fired in the wake of the empathy-deleted, vicious hostility of 'social justice' Wokers and the desire of gutless, spineless employers to virtue-signal their Wokeness. Corporations are filled with Wokers today, most notably those in Silicon Valley. Ironically at the top they are not Woke at all. They are only exploiting the mentality their Cult masters have created and funded to censor and enslave while the Wokers cheer them on until it's their turn. Thus the Woke 'liberal left' is an inversion of the traditional liberal left. Campaigning for justice on the grounds of power and wealth distribution has been replaced by campaigning for identity politics. The genuine traditional left would never have taken money from today's billionaire abusers of fairness and justice and nor would the billionaires have wanted to fund that genuine left. It would not have been in their interests to do so. The division of opinion in those days was between the haves and have nots. This all changed with Cult manipulated and funded identity politics. The division of opinion today is between Wokers and non-Wokers and not income brackets. Cult corporations and their billionaires may have taken wealth disparity to cataclysmic levels of injustice, but as long as they speak the language of Woke, hand out the dosh to the Woke network and censor the enemy they are 'one of us'. Billionaires who don't give a damn about injustice are laughing at them till their bellies hurt. Wokers are not even close to self-aware enough to see that. The transformed 'left' dynamic means that Wokers who drone on about 'social justice' are funded by billionaires that have destroyed social justice the world over. It's *why* they are billionaires.

The climate con

Nothing encapsulates what I have said more comprehensively than the hoax of human-caused global warming. I have detailed in my books over the years how Cult operatives and organisations were the pump-primers from the start of the climate con. A purpose-built vehicle for this is the Club of Rome established by the Cult in 1968

with the Rockefellers and Rothschilds centrally involved all along. Their gofer frontman Maurice Strong, a Canadian oil millionaire, hosted the Earth Summit in Rio de Janeiro, Brazil, in 1992 where the global 'green movement' really expanded in earnest under the guiding hand of the Cult. The Earth Summit established Agenda 21 through the Cult-created-and-owned United Nations to use the illusion of human-caused climate change to justify the transformation of global society to save the world from climate disaster. It is a No-Problem-Reaction-Solution sold through governments, media, schools and universities as whole generations have been terrified into believing that the world was going to end in their lifetimes unless what old people had inflicted upon them was stopped by a complete restructuring of how everything is done. Chill, kids, it's all a hoax. Such restructuring is precisely what the Cult agenda demands (purely by coincidence of course). Today this has been given the codename of the Great Reset which is only an updated term for Agenda 21 and its associated Agenda 2030. The latter, too, is administered through the UN and was voted into being by the General Assembly in 2015. Both 21 and 2030 seek centralised control of all resources and food right down to the raindrops falling on your own land. These are some of the demands of Agenda 21 established in 1992. See if you recognise this society emerging today:

- End national sovereignty
- State planning and management of all land resources, ecosystems, deserts, forests, mountains, oceans and fresh water; agriculture; rural development; biotechnology; and ensuring '*equity*'
- The state to 'define the role' of business and financial resources
- Abolition of private property
- 'Restructuring' the family unit (see BLM)
- Children raised by the state
- People told what their job will be
- Major restrictions on movement
- Creation of 'human settlement zones'

- Mass resettlement as people are forced to vacate land where they live
- Dumbing down education
- Mass global depopulation in pursuit of all the above

The United Nations was created as a Trojan horse for world government. With the climate con of critical importance to promoting that outcome you would expect the UN to be involved. Oh, it's involved all right. The UN is promoting Agenda 21 and Agenda 2030 justified by 'climate change' while also driving the climate hoax through its Intergovernmental Panel on Climate Change (IPCC), one of the world's most corrupt organisations. The IPCC has been lying ferociously and constantly since the day it opened its doors with the global media hanging unquestioningly on its every mendacious word. The Green movement is entirely Woke and has long lost its original environmental focus since it was co-opted by the Cult. An obsession with 'global warming' has deleted its values and scrambled its head. I experienced a small example of what I mean on a beautiful country walk that I have enjoyed several times a week for many years. The path merged into the fields and forests and you felt at one with the natural world. Then a 'Green' organisation, the Hampshire and Isle of Wight Wildlife Trust, took over part of the land and proceeded to cut down a large number of trees, including mature ones, to install a horrible big, bright steel 'this-is-ours-stay-out' fence that destroyed the whole atmosphere of this beautiful place. No one with a feel for nature would do that. Day after day I walked to the sound of chainsaws and a magnificent mature weeping willow tree that I so admired was cut down at the base of the trunk. When I challenged a Woke young girl in a green shirt (of course) about this vandalism she replied: 'It's a weeping willow – it will grow back.' This is what people are paying for when they donate to the Hampshire and Isle of Wight Wildlife Trust and many other 'green' organisations today. It is not the environmental movement that I knew and instead has become a support-system – as with Extinction Rebellion – for a very dark agenda.

Private jets for climate justice

The Cult-owned, Gates-funded, World Economic Forum and its founder Klaus Schwab were behind the emergence of Greta Thunberg to harness the young behind the climate agenda and she was invited to speak to the world at ... the UN. Schwab published a book, *Covid-19: The Great Reset* in 2020 in which he used the 'Covid' hoax and the climate hoax to lay out a new society straight out of Agenda 21 and Agenda 2030. Bill Gates followed in early 2021 when he took time out from destroying the world to produce a book in his name about the way to save it. Gates flies across the world in private jets and admitted that 'I probably have one of the highest greenhouse gas footprints of anyone on the planet ... my personal flying alone is gigantic.' He has also bid for the planet's biggest private jet operator. Other climate change saviours who fly in private jets include John Kerry, the US Special Presidential Envoy for Climate, and actor Leonardo DiCaprio, a 'UN Messenger of Peace with special focus on climate change'. These people are so full of bullshit they could corner the market in manure. We mustn't be sceptical, though, because the Gates book, *How to Avoid a Climate Disaster: The Solutions We Have and the Breakthroughs We Need*, is a genuine attempt to protect the world and not an obvious pile of excrement attributed to a mega-psychopath aimed at selling his masters' plans for humanity. The Gates book and the other shite-pile by Klaus Schwab could have been written by the same person and may well have been. Both use 'climate change' and 'Covid' as the excuses for their new society and by coincidence the Cult's World Economic Forum and Bill and Melinda Gates Foundation promote the climate hoax and hosted Event 201 which pre-empted with a 'simulation' the very 'coronavirus' hoax that would be simulated for real on humanity within weeks. The British 'royal' family is promoting the 'Reset' as you would expect through Prince 'climate change caused the war in Syria' Charles and his hapless son Prince William who said that we must 'reset our relationship with nature and our trajectory as a species' to avoid a climate disaster. Amazing how many promoters of the 'Covid' and 'climate change' control

systems are connected to Gates and the World Economic Forum. A 'study' in early 2021 claimed that carbon dioxide emissions must fall by the equivalent of a global lockdown roughly every two years for the next decade to save the planet. The 'study' appeared in the same period that the Schwab mob claimed in a video that lockdowns destroying the lives of billions are good because they make the earth 'quieter' with less 'ambient noise'. They took down the video amid a public backlash for such arrogant, empathy-deleted stupidity. You see, however, where they are going with this. Corinne Le Quéré, a professor at the Tyndall Centre for Climate Change Research, University of East Anglia, was lead author of the climate lockdown study, and she writes for ... the World Economic Forum. Gates calls in 'his' book for changing 'every aspect of the economy' (long-time Cult agenda) and for humans to eat synthetic 'meat' (predicted in my books) while cows and other farm animals are eliminated. Australian TV host and commentator Alan Jones described what carbon emission targets would mean for farm animals in Australia alone if emissions were reduced as demanded by 35 percent by 2030 and zero by 2050:

Well, let's take agriculture, the total emissions from agriculture are about 75 million tonnes of carbon dioxide, equivalent. Now reduce that by 35 percent and you have to come down to 50 million tonnes, I've done the maths. So if you take for example 1.5 million cows, you're going to have to reduce the herd by 525,000 [by] 2030, nine years, that's 58,000 cows a year. The beef herd's 30 million, reduce that by 35 percent, that's 10.5 million, which means 1.2 million cattle have to go every year between now and 2030. This is insanity!

There are 75 million sheep. Reduce that by 35 percent, that's 26 million sheep, that's almost 3 million a year. So under the Paris Agreement over 30 million beasts, dairy cows, cattle, pigs and sheep would go. More than 8,000 every minute of every hour for the next decade, do these people know what they're talking about?

Clearly they don't at the level of campaigners, politicians and administrators. The Cult *does* know; that's the outcome it wants. We are faced with not just a war on humanity. Animals and the natural world are being targeted and I have been saying since the 'Covid' hoax began that the plan eventually was to claim that the 'deadly virus' is able to jump from animals, including farm animals and

domestic pets, to humans. Just before this book went into production came this story: 'Russia registers world's first Covid-19 vaccine for cats & dogs as makers of Sputnik V warn pets & farm animals could spread virus'. The report said 'top scientists warned that the deadly pathogen could soon begin spreading through homes and farms' and 'the next stage is the infection of farm and domestic animals'. Know the outcome and you'll see the journey. Think what that would mean for animals and keep your eye on a term called zoonosis or zoonotic diseases which transmit between animals and humans. The Cult wants to break the connection between animals and people as it does between people and people. Farm animals fit with the Cult agenda to transform food from natural to synthetic.

The gas of life is killing us

There can be few greater examples of Cult inversion than the condemnation of carbon dioxide as a dangerous pollutant when it is the gas of life. Without it the natural world would be dead and so we would all be dead. We breathe in oxygen and breathe out carbon dioxide while plants produce oxygen and absorb carbon dioxide. It is a perfect symbiotic relationship that the Cult wants to dismantle for reasons I will come to in the final two chapters. Gates, Schwab, other Cult operatives and mindless repeaters, want the world to be 'carbon neutral' by at least 2050 and the earlier the better. 'Zero carbon' is the cry echoed by lunatics calling for 'Zero Covid' when we already have it. These carbon emission targets will deindustrialise the world in accordance with Cult plans – the post-industrial, post-democratic society – and with so-called renewables like solar and wind not coming even close to meeting human energy needs blackouts and cold are inevitable. Texans got the picture in the winter of 2021 when a snow storm stopped wind turbines and solar panels from working and the lights went down along with water which relies on electricity for its supply system. Gates wants everything to be powered by electricity to ensure that his masters have the kill switch to stop all human activity, movement, cooking, water and warmth any time they like. The climate lie is so

stupendously inverted that it claims we must urgently reduce carbon dioxide when we *don't have enough*.

Co2 in the atmosphere is a little above 400 parts per million when the optimum for plant growth is 2,000 ppm and when it falls anywhere near 150 ppm the natural world starts to die and so do we. It fell to as low as 280 ppm in an 1880 measurement in Hawaii and rose to 413 ppm in 2019 with industrialisation which is why the planet has become *greener* in the industrial period. How insane then that psychopathic madman Gates is not satisfied only with blocking the rise of Co2. He's funding technology to suck it out of the atmosphere. The reason why will become clear. The industrial era is not destroying the world through Co2 and has instead turned around a potentially disastrous ongoing fall in Co2. Greenpeace co-founder and scientist Patrick Moore walked away from Greenpeace in 1986 and has exposed the green movement for fear-mongering and lies. He said that 500 million years ago there was *17 times* more Co2 in the atmosphere than we have today and levels have been falling for hundreds of millions of years. In the last 150 million years Co2 levels in Earth's atmosphere had reduced by *90 percent*. Moore said that by the time humanity began to unlock carbon dioxide from fossil fuels we were at '38 seconds to midnight' and in that sense: 'Humans are [the Earth's] salvation.' Moore made the point that only half the Co2 emitted by fossil fuels stays in the atmosphere and we should remember that all pollution pouring from chimneys that we are told is carbon dioxide is in fact nothing of the kind. It's pollution. Carbon dioxide is an invisible gas.

William Happer, Professor of Physics at Princeton University and long-time government adviser on climate, has emphasised the Co2 deficiency for maximum growth and food production. Greenhouse growers don't add carbon dioxide for a bit of fun. He said that most of the warming in the last 100 years, after the earth emerged from the super-cold period of the 'Little Ice Age' into a natural warming cycle, was over by 1940. Happer said that a peak year for warming in 1988 can be explained by a 'monster El Nino' which is a natural and cyclical warming of the Pacific that has nothing to do with 'climate

change'. He said the effect of Co2 could be compared to painting a wall with red paint in that once two or three coats have been applied it didn't matter how much more you slapped on because the wall will not get much redder. Almost all the effect of the rise in Co2 has already happened, he said, and the volume in the atmosphere would now have to *double* to increase temperature by a single degree. Climate hoaxers know this and they have invented the most ridiculously complicated series of 'feedback' loops to try to overcome this rather devastating fact. You hear puppet Greta going on cluelessly about feedback loops and this is why.

The Sun affects temperature? No you *climate denier*

Some other nonsense to contemplate: Climate graphs show that rises in temperature do not follow rises in Co2 – *it's the other way round* with a lag between the two of some 800 years. If we go back 800 years from present time we hit the Medieval Warm Period when temperatures were higher than now without any industrialisation and this was followed by the Little Ice Age when temperatures plummeted. The world was still emerging from these centuries of serious cold when many climate records began which makes the ever-repeated line of the 'hottest year since records began' meaningless when you are not comparing like with like. The coldest period of the Little Ice Age corresponded with the lowest period of sunspot activity when the Sun was at its least active. Proper scientists will not be at all surprised by this when it confirms the obvious fact that earth temperature is affected by the scale of Sun activity and the energetic power that it subsequently emits; but when is the last time you heard a climate hoaxer talking about the Sun as a source of earth temperature?? Everything has to be focussed on Co2 which makes up just 0.117 percent of so-called greenhouse gases and only a fraction of even that is generated by human activity. The rest is natural. More than 90 percent of those greenhouse gases are water vapour and clouds ([Fig 9](#)). Ban moisture I say. Have you noticed that the climate hoaxers no longer use the polar bear as their promotion image? That's because far from becoming extinct polar

bear communities are stable or thriving. Joe Bastardi, American meteorologist, weather forecaster and outspoken critic of the climate lie, documents in his book *The Climate Chronicles* how weather patterns and events claimed to be evidence of climate change have been happening since long before industrialisation: 'What happened before naturally is happening again, as is to be expected given the cyclical nature of the climate due to the design of the planet.' If you read the detailed background to the climate hoax in my other books you will shake your head and wonder how anyone could believe the crap which has spawned a multi-trillion dollar industry based on absolute garbage (see HIV causes AIDs and Sars-Cov-2 causes 'Covid-19'). Climate and 'Covid' have much in common given they have the same source. They both have the contradictory *everything* factor in which everything is explained by reference to them. It's hot – 'it's climate change'. It's cold – 'it's climate change'. I got a sniffle – 'it's Covid'. I haven't got a sniffle – 'it's Covid'. Not having a sniffle has to be a symptom of 'Covid'. Everything is and not having a sniffle is especially dangerous if you are a slow walker. For sheer audacity I offer you a Cambridge University 'study' that actually linked 'Covid' to 'climate change'. It had to happen eventually. They concluded that climate change played a role in 'Covid-19' spreading from animals to humans because ... wait for it ... I kid you not ... *the two groups were forced closer together as populations grow*. Er, that's it. The whole foundation on which this depended was that 'Bats are the likely zoonotic origin of SARS-CoV-1 and SARS-CoV-2'. Well, they are not. They are nothing to do with it. Apart from bats not being the origin and therefore 'climate change' effects on bats being irrelevant I am in awe of their academic insight. Where would we be without them? Not where we are that's for sure.

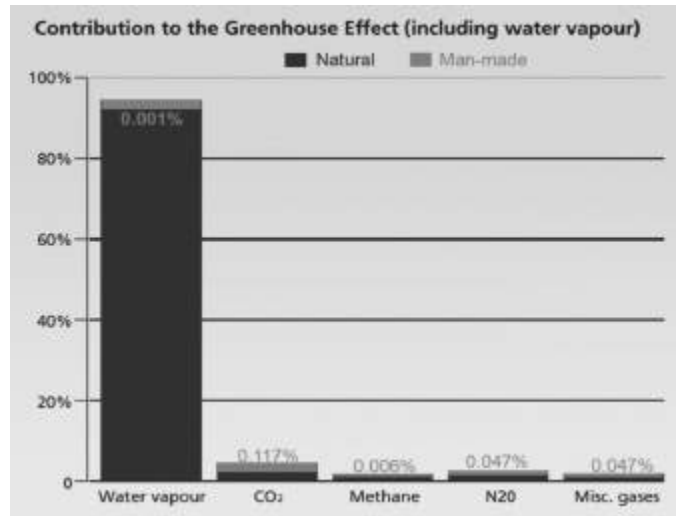


Figure 9: The idea that the gas of life is disastrously changing the climate is an insult to brain cell activity.

One other point about the weather is that climate modification is now well advanced and not every major weather event is natural – or earthquake come to that. I cover this subject at some length in other books. China is openly planning a rapid expansion of its weather modification programme which includes changing the climate in an area more than one and a half times the size of India. China used weather manipulation to ensure clear skies during the 2008 Olympics in Beijing. I have quoted from US military documents detailing how to employ weather manipulation as a weapon of war and they did that in the 1960s and 70s during the conflict in Vietnam with Operation Popeye manipulating monsoon rains for military purposes. Why would there be international treaties on weather modification if it wasn't possible? Of course it is. Weather is energetic information and it can be changed.

How was the climate hoax pulled off? See 'Covid'

If you can get billions to believe in a 'virus' that doesn't exist you can get them to believe in human-caused climate change that doesn't exist. Both are being used by the Cult to transform global society in the way it has long planned. Both hoaxes have been achieved in pretty much the same way. First you declare a lie is a fact. There's a

'virus' you call SARS-Cov-2 or humans are warming the planet with their behaviour. Next this becomes, via Cult networks, the foundation of government, academic and science policy and belief. Those who parrot the mantra are given big grants to produce research that confirms the narrative is true and ever more 'symptoms' are added to make the 'virus'/'climate change' sound even more scary. Scientists and researchers who challenge the narrative have their grants withdrawn and their careers destroyed. The media promote the lie as the unquestionable truth and censor those with an alternative view or evidence. A great percentage of the population believe what they are told as the lie becomes an everybody-knows-that and the believing-masses turn on those with a mind of their own. The technique has been used endlessly throughout human history. Wokers are the biggest promoters of the climate lie *and* 'Covid' fascism because their minds are owned by the Cult; their sense of self-righteous self-purity knows no bounds; and they exist in a bubble of reality in which facts are irrelevant and only get in the way of looking without seeing.

Running through all of this like veins in a blue cheese is control of information, which means control of perception, which means control of behaviour, which collectively means control of human society. The Cult owns the global media and Silicon Valley fascists for the simple reason that it *has* to. Without control of information it can't control perception and through that human society. Examine every facet of the Cult agenda and you will see that anything supporting its introduction is never censored while anything pushing back is always censored. I say again: Psychopaths that know why they are doing this must go before Nuremberg trials and those that follow their orders must trot along behind them into the same dock. 'I was just following orders' didn't work the first time and it must not work now. Nuremberg trials must be held all over the world before public juries for politicians, government officials, police, compliant doctors, scientists and virologists, and all Cult operatives such as Gates, Tedros, Fauci, Vallance, Whitty, Ferguson, Zuckerberg, Wojcicki, Brin, Page, Dorsey, the whole damn lot of

them – including, no *especially*, the psychopath psychologists. Without them and the brainless, gutless excuses for journalists that have repeated their lies, none of this could be happening. Nobody can be allowed to escape justice for the psychological and economic Armageddon they are all responsible for visiting upon the human race.

As for the compliant, unquestioning, swathes of humanity, and the self-obsessed, all-knowing ignorance of the Wokers ... don't start me. God help their kids. God help their grandkids. God *help them*.

CHAPTER NINE

We must have it? So what is it?

*Well I won't back down. No, I won't back down. You can stand me
up at the Gates of Hell. But I won't back down*

Tom Petty

I will now focus on the genetically-manipulating 'Covid vaccines' which do not meet this official definition of a vaccine by the US Centers for Disease Control (CDC): 'A product that stimulates a person's immune system to produce immunity to a specific disease, protecting the person from that disease.' On that basis 'Covid vaccines' are not a vaccine in that the makers don't even claim they stop infection or transmission.

They are instead part of a multi-levelled conspiracy to change the nature of the human body and what it means to be 'human' and to depopulate an enormous swathe of humanity. What I shall call Human 1.0 is on the cusp of becoming Human 2.0 and for very sinister reasons. Before I get to the 'Covid vaccine' in detail here's some background to vaccines in general. Government regulators do not test vaccines – the makers do – and the makers control which data is revealed and which isn't. Children in America are given 50 vaccine doses by age six and 69 by age 19 and the effect of the whole combined schedule has never been tested. Autoimmune diseases when the immune system attacks its own body have soared in the mass vaccine era and so has disease in general in children and the young. Why wouldn't this be the case when vaccines target the *immune system*? The US government gave Big Pharma drug

companies immunity from prosecution for vaccine death and injury in the 1986 National Childhood Vaccine Injury Act (NCVIA) and since then the government (taxpayer) has been funding compensation for the consequences of Big Pharma vaccines. The criminal and satanic drug giants can't lose and the vaccine schedule has increased dramatically since 1986 for this reason. There is no incentive to make vaccines safe and a big incentive to make money by introducing ever more. Even against a ridiculously high bar to prove vaccine liability, and with the government controlling the hearing in which it is being challenged for compensation, the vaccine court has so far paid out more than \$4 billion. These are the vaccines we are told are safe and psychopaths like Zuckerberg censor posts saying otherwise. The immunity law was even justified by a ruling that vaccines by their nature were 'unavoidably unsafe'.

Check out the ingredients of vaccines and you will be shocked if you are new to this. *They put that in children's bodies?? What??* Try aluminium, a brain toxin connected to dementia, aborted foetal tissue and formaldehyde which is used to embalm corpses. World-renowned aluminium expert Christopher Exley had his research into the health effect of aluminium in vaccines shut down by Keele University in the UK when it began taking funding from the Bill and Melinda Gates Foundation. Research when diseases 'eradicated' by vaccines began to decline and you will find the fall began long *before* the vaccine was introduced. Sometimes the fall even plateaued after the vaccine. Diseases like scarlet fever for which there was no vaccine declined in the same way because of environmental and other factors. A perfect case in point is the polio vaccine. Polio began when lead arsenate was first sprayed as an insecticide and residues remained in food products. Spraying started in 1892 and the first US polio epidemic came in Vermont in 1894. The simple answer was to stop spraying, but Rockefeller-created Big Pharma had a better idea. Polio was decreed to be caused by the *poliovirus* which 'spreads from person to person and can infect a person's spinal cord'. Lead arsenate was replaced by the lethal DDT which had the same effect of causing paralysis by damaging the brain and central nervous

system. Polio plummeted when DDT was reduced and then banned, but the vaccine is still given the credit for something it didn't do. Today by far the biggest cause of polio is the vaccines promoted by Bill Gates. Vaccine justice campaigner Robert Kennedy Jr, son of assassinated (by the Cult) US Attorney General Robert Kennedy, wrote:

In 2017, the World Health Organization (WHO) reluctantly admitted that the global explosion in polio is predominantly vaccine strain. The most frightening epidemics in Congo, Afghanistan, and the Philippines, are all linked to vaccines. In fact, by 2018, 70% of global polio cases were vaccine strain.

Vaccines make fortunes for Cult-owned Gates and Big Pharma while undermining the health and immune systems of the population. We had a glimpse of the mentality behind the Big Pharma cartel with a report on WION (World is One News), an international English language TV station based in India, which exposed the extraordinary behaviour of US drug company Pfizer over its 'Covid vaccine'. The WION report told how Pfizer had made fantastic demands of Argentina, Brazil and other countries in return for its 'vaccine'. These included immunity from prosecution, even for Pfizer negligence, government insurance to protect Pfizer from law suits and handing over as collateral sovereign assets of the country to include Argentina's bank reserves, military bases and embassy buildings. Pfizer demanded the same of Brazil in the form of waiving sovereignty of its assets abroad; exempting Pfizer from Brazilian laws; and giving Pfizer immunity from all civil liability. This is a 'vaccine' developed with government funding. Big Pharma is evil incarnate as a creation of the Cult and all must be handed tickets to Nuremberg.

Phantom 'vaccine' for a phantom 'disease'

I'll expose the 'Covid vaccine' fraud and then go on to the wider background of why the Cult has set out to 'vaccinate' every man, woman and child on the planet for an alleged 'new disease' with a survival rate of 99.77 percent (or more) even by the grotesquely-

manipulated figures of the World Health Organization and Johns Hopkins University. The 'infection' to 'death' ratio is 0.23 to 0.15 percent according to Stanford epidemiologist Dr John Ioannidis and while estimates vary the danger remains tiny. I say that if the truth be told the fake infection to fake death ratio is zero. Never mind all the evidence I have presented here and in *The Answer* that there is no 'virus' let us just focus for a moment on that death-rate figure of say 0.23 percent. The figure includes all those worldwide who have tested positive with a test not testing for the 'virus' and then died within 28 days or even longer of any other cause – *any other cause*. Now subtract all those illusory 'Covid' deaths on the global data sheets from the 0.23 percent. What do you think you would be left with? *Zero*. A vaccination has never been successfully developed for a so-called coronavirus. They have all failed at the animal testing stage when they caused hypersensitivity to what they were claiming to protect against and made the impact of a disease far worse. Cult-owned vaccine corporations got around that problem this time by bypassing animal trials, going straight to humans and making the length of the 'trials' before the public rollout as short as they could get away with. Normally it takes five to ten years or more to develop vaccines that still cause demonstrable harm to many people and that's without including the long-term effects that are never officially connected to the vaccination. 'Covid' non-vaccines have been officially produced and approved in a matter of months from a standing start and part of the reason is that (a) they were developed before the 'Covid' hoax began and (b) they are based on computer programs and not natural sources. Official non-trials were so short that government agencies gave *emergency*, not full, approval. 'Trials' were not even completed and full approval cannot be secured until they are. Public 'Covid vaccination' is actually a *continuation of the trial*. Drug company 'trials' are not scheduled to end until 2023 by which time a lot of people are going to be dead. Data on which government agencies gave this emergency approval was supplied by the Big Pharma corporations themselves in the form of Pfizer/BioNTech, AstraZeneca, Moderna, Johnson & Johnson, and

others, and this is the case with all vaccines. By its very nature *emergency* approval means drug companies do not have to prove that the 'vaccine' is 'safe and effective'. How could they with trials way short of complete? Government regulators only have to *believe* that they *could* be safe and effective. It is criminal manipulation to get products in circulation with no testing worth the name. Agencies giving that approval are infested with Big Pharma-connected place-people and they act in the interests of Big Pharma (the Cult) and not the public about whom they do not give a damn.

More human lab rats

'Covid vaccines' produced in record time by Pfizer/BioNTech and Moderna employ a technique *never approved before for use on humans*. They are known as mRNA 'vaccines' and inject a synthetic version of 'viral' mRNA or 'messenger RNA'. The key is in the term 'messenger'. The body works, or doesn't, on the basis of information messaging. Communications are constantly passing between and within the genetic system and the brain. Change those messages and you change the state of the body and even its very nature and you can change psychology and behaviour by the way the brain processes information. I think you are going to see significant changes in personality and perception of many people who have had the 'Covid vaccine' synthetic potions. Insider Aldous Huxley predicted the following in 1961 and mRNA 'vaccines' can be included in the term 'pharmacological methods':

There will be, in the next generation or so, a pharmacological method of making people love their servitude, and producing dictatorship without tears, so to speak, producing a kind of painless concentration camp for entire societies, so that people will in fact have their own liberties taken away from them, but rather enjoy it, because they will be distracted from any desire to rebel by propaganda or brainwashing, or brainwashing enhanced by pharmacological methods. And this seems to be the final revolution.

Apologists claim that mRNA synthetic 'vaccines' don't change the DNA genetic blueprint because RNA does not affect DNA only the other way round. This is so disingenuous. A process called 'reverse

transcription' can convert RNA into DNA and be integrated into DNA in the cell nucleus. This was highlighted in December, 2020, by scientists at Harvard and Massachusetts Institute of Technology (MIT). Geneticists report that more than 40 percent of mammalian genomes results from reverse transcription. On the most basic level if messaging changes then that sequence must lead to changes in DNA which is receiving and transmitting those communications. How can introducing synthetic material into cells not change the cells where DNA is located? The process is known as transfection which is defined as 'a technique to insert foreign nucleic acid (DNA or RNA) into a cell, typically with the intention of altering the properties of the cell'. Researchers at the Sloan Kettering Institute in New York found that changes in messenger RNA can deactivate tumour-suppressing proteins and thereby promote cancer. This is what happens when you mess with messaging. 'Covid vaccine' maker Moderna was founded in 2010 by Canadian stem cell biologist Derrick J. Rossi after his breakthrough discovery in the field of transforming and reprogramming stem cells. These are neutral cells that can be programmed to become any cell including sperm cells. Moderna was therefore founded on the principle of genetic manipulation and has never produced any vaccine or drug before its genetically-manipulating synthetic 'Covid' shite. Look at the name – Mode-RNA or Modify-RNA. Another important point is that the US Supreme Court has ruled that genetically-modified DNA, or complementary DNA (cDNA) synthesized in the laboratory from messenger RNA, can be patented and owned. These psychopaths are doing this to the human body.

Cells replicate synthetic mRNA in the 'Covid vaccines' and in theory the body is tricked into making antigens which trigger antibodies to target the 'virus spike proteins' which as Dr Tom Cowan said have *never been seen*. Cut the crap and these 'vaccines' deliver *self-replicating* synthetic material to the cells with the effect of changing human DNA. The more of them you have the more that process is compounded while synthetic material is all the time self-replicating. 'Vaccine'-maker Moderna describes mRNA as 'like

software for the cell' and so they are messing with the body's software. What happens when you change the software in a computer? Everything changes. For this reason the Cult is preparing a production line of mRNA 'Covid vaccines' and a long list of excuses to use them as with all the 'variants' of a 'virus' never shown to exist. The plan is further to transfer the mRNA technique to other vaccines mostly given to children and young people. The cumulative consequences will be a transformation of human DNA through a constant infusion of synthetic genetic material which will kill many and change the rest. Now consider that governments that have given emergency approval for a vaccine that's not a vaccine; never been approved for humans before; had no testing worth the name; and the makers have been given immunity from prosecution for any deaths or adverse effects suffered by the public. The UK government awarded *permanent legal indemnity* to itself and its employees for harm done when a patient is being treated for 'Covid-19' or 'suspected Covid-19'. That is quite a thought when these are possible 'side-effects' from the 'vaccine' (they are not 'side', they are effects) listed by the US Food and Drug Administration:

Guillain-Barre syndrome; acute disseminated encephalomyelitis; transverse myelitis; encephalitis; myelitis; encephalomyelitis; meningoencephalitis; meningitis; encephalopathy; convulsions; seizures; stroke; narcolepsy; cataplexy; anaphylaxis; acute myocardial infarction (heart attack); myocarditis; pericarditis; autoimmune disease; death; implications for pregnancy, and birth outcomes; other acute demyelinating diseases; non anaphylactic allergy reactions; thrombocytopenia ; disseminated intravascular coagulation; venous thromboembolism; arthritis; arthralgia; joint pain; Kawasaki disease; multisystem inflammatory syndrome in children; vaccine enhanced disease. The latter is the way the 'vaccine' has the potential to make diseases far worse than they would otherwise be.

UK doctor and freedom campaigner Vernon Coleman described the conditions in this list as 'all unpleasant, most of them very serious, and you can't get more serious than death'. The thought that anyone at all has had the 'vaccine' in these circumstances is testament to the potential that humanity has for clueless, unquestioning, stupidity and for many that programmed stupidity has already been terminal.

An insider speaks

Dr Michael Yeadon is a former Vice President, head of research and Chief Scientific Adviser at vaccine giant Pfizer. Yeadon worked on the inside of Big Pharma, but that did not stop him becoming a vocal critic of 'Covid vaccines' and their potential for multiple harms, including infertility in women. By the spring of 2021 he went much further and even used the no, no, term 'conspiracy'. When you begin to see what is going on it is impossible not to do so. Yeadon spoke out in an interview with freedom campaigner James Delingpole and I mentioned earlier how he said that no one had samples of 'the virus'. He explained that the mRNA technique originated in the anti-cancer field and ways to turn on and off certain genes which could be advantageous if you wanted to stop cancer growing out of control. 'That's the origin of them. They are a very unusual application, really.' Yeadon said that treating a cancer patient with an aggressive procedure might be understandable if the alternative was dying, but it was quite another thing to use the same technique as a public health measure. Most people involved wouldn't catch the infectious agent you were vaccinating against and if they did they probably wouldn't die:

If you are really using it as a public health measure you really want to as close as you can get to zero side-effects ... I find it odd that they chose techniques that were really cutting their teeth in the field of oncology and I'm worried that in using gene-based vaccines that have to be injected in the body and spread around the body, get taken up into some cells, and the regulators haven't quite told us which cells they get taken up into ... you are going to be generating a wide range of responses ... with multiple steps each of which could go well or badly.

I doubt the Cult intends it to go well. Yeadon said that you can put any gene you like into the body through the 'vaccine'. 'You can certainly give them a gene that would do them some harm if you wanted.' I was intrigued when he said that when used in the cancer field the technique could turn genes on and off. I explore this process in *The Answer* and with different genes having different functions you could create mayhem – physically and psychologically – if you turned the wrong ones on and the right ones off. I read reports of an experiment by researchers at the University of Washington's school of computer science and engineering in which they encoded DNA to infect computers. The body is itself a biological computer and if human DNA can inflict damage on a computer why can't the computer via synthetic material mess with the human body? It can. The Washington research team said it was possible to insert malicious malware into 'physical DNA strands' and corrupt the computer system of a gene sequencing machine as it 'reads gene letters and stores them as binary digits 0 and 1'. They concluded that hackers could one day use blood or spit samples to access computer systems and obtain sensitive data from police forensics labs or infect genome files. It is at this level of digital interaction that synthetic 'vaccines' need to be seen to get the full picture and that will become very clear later on. Michael Yeadon said it made no sense to give the 'vaccine' to younger people who were in no danger from the 'virus'. What was the benefit? It was all downside with potential effects:

The fact that my government in what I thought was a civilised, rational country, is raining [the 'vaccine'] on people in their 30s and 40s, even my children in their 20s, they're getting letters and phone calls, I know this is not right and any of you doctors who are vaccinating you know it's not right, too. They are not at risk. They are not at risk from the disease, so you are now hoping that the side-effects are so rare that you get away with it. You don't give new technology ... that you don't understand to 100 percent of the population.

Blood clot problems with the AstraZeneca 'vaccine' have been affecting younger people to emphasise the downside risks with no benefit. AstraZeneca's version, produced with Oxford University, does not use mRNA, but still gets its toxic cocktail inside cells where

it targets DNA. The Johnson & Johnson 'vaccine' which uses a similar technique has also produced blood clot effects to such an extent that the United States paused its use at one point. They are all 'gene therapy' (cell modification) procedures and not 'vaccines'. The truth is that once the content of these injections enter cells we have no idea what the effect will be. People can speculate and some can give very educated opinions and that's good. In the end, though, only the makers know what their potions are designed to do and even they won't know every last consequence. Michael Yeadon was scathing about doctors doing what they knew to be wrong. 'Everyone's mute', he said. Doctors in the NHS must know this was not right, coming into work and injecting people. 'I don't know how they sleep at night. I know I couldn't do it. I know that if I were in that position I'd have to quit.' He said he knew enough about toxicology to know this was not a good risk-benefit. Yeadon had spoken to seven or eight university professors and all except two would not speak out publicly. Their universities had a policy that no one said anything that countered the government and its medical advisors. They were afraid of losing their government grants. This is how intimidation has been used to silence the truth at every level of the system. I say silence, but these people could still speak out if they made that choice. Yeadon called them 'moral cowards' – 'This is about your children and grandchildren's lives and you have just buggered off and left it.'

'Variant' nonsense

Some of his most powerful comments related to the alleged 'variants' being used to instil more fear, justify more lockdowns, and introduce more 'vaccines'. He said government claims about 'variants' were nonsense. He had checked the alleged variant 'codes' and they were 99.7 percent identical to the 'original'. This was the human identity difference equivalent to putting a baseball cap on and off or wearing it the other way round. A 0.3 percent difference would make it impossible for that 'variant' to escape immunity from the 'original'. This made no sense of having new 'vaccines' for

‘variants’. He said there would have to be at least a *30 percent* difference for that to be justified and even then he believed the immune system would still recognise what it was. Gates-funded ‘variant modeller’ and ‘vaccine’-pusher John Edmunds might care to comment. Yeadon said drug companies were making new versions of the ‘vaccine’ as a ‘top up’ for ‘variants’. Worse than that, he said, the ‘regulators’ around the world like the MHRA in the UK had got together and agreed that because ‘vaccines’ for ‘variants’ were so similar to the first ‘vaccines’ *they did not have to do safety studies*. How transparently sinister that is. This is when Yeadon said: ‘There is a conspiracy here.’ There was no need for another vaccine for ‘variants’ and yet we were told that there was and the country had shut its borders because of them. ‘They are going into hundreds of millions of arms without passing ‘go’ or any regulator. Why did they do that? Why did they pick this method of making the vaccine?’

The reason had to be something bigger than that it seemed and ‘it’s not protection against the virus’. It’s was a far bigger project that meant politicians and advisers were willing to do things and not do things that knowingly resulted in avoidable deaths – ‘that’s already happened when you think about lockdown and deprivation of health care for a year.’ He spoke of people prepared to do something that results in the avoidable death of their fellow human beings and it not bother them. This is the penny-drop I have been working to get across for more than 30 years – the level of pure evil we are dealing with. Yeadon said his friends and associates could not believe there could be that much evil, but he reminded them of Stalin, Pol Pot and Hitler and of what Stalin had said: ‘One death is a tragedy. A million? A statistic.’ He could not think of a benign explanation for why you need top-up vaccines ‘which I’m sure you don’t’ and for the regulators ‘to just get out of the way and wave them through’. Why would the regulators do that when they were still wrestling with the dangers of the ‘parent’ vaccine? He was clearly shocked by what he had seen since the ‘Covid’ hoax began and now he was thinking the previously unthinkable:

If you wanted to depopulate a significant proportion of the world and to do it in a way that doesn't involve destruction of the environment with nuclear weapons, poisoning everyone with anthrax or something like that, and you wanted plausible deniability while you had a multi-year infectious disease crisis, I actually don't think you could come up with a better plan of work than seems to be in front of me. I can't say that's what they are going to do, but I can't think of a benign explanation why they are doing it.

He said he never thought that they would get rid of 99 percent of humans, but now he wondered. 'If you wanted to that this would be a hell of a way to do it – it would be unstoppable folks.' Yeadon had concluded that those who submitted to the 'vaccine' would be allowed to have some kind of normal life (but for how long?) while screws were tightened to coerce and mandate the last few percent. 'I think they'll put the rest of them in a prison camp. I wish I was wrong, but I don't think I am.' Other points he made included: There were no coronavirus vaccines then suddenly they all come along at the same time; we have no idea of the long term affect with trials so short; coercing or forcing people to have medical procedures is against the Nuremberg Code instigated when the Nazis did just that; people should at least delay having the 'vaccine'; a quick Internet search confirms that masks don't reduce respiratory viral transmission and 'the government knows that'; they have smashed civil society and they know that, too; two dozen peer-reviewed studies show no connection between lockdown and reducing deaths; he knew from personal friends the elite were still flying around and going on holiday while the public were locked down; the elite were not having the 'vaccines'. He was also asked if 'vaccines' could be made to target difference races. He said he didn't know, but the document by the Project for the New American Century in September, 2000, said developing 'advanced forms of biological warfare that can target *specific genotypes* may transform biological warfare from the realm of terror to a politically useful tool.' Oh, they're evil all right. Of that we can be *absolutely* sure.

Another cull of old people

We have seen from the CDC definition that the mRNA 'Covid vaccine' is not a vaccine and nor are the others that *claim* to reduce 'severity of symptoms' in *some* people, but not protect from infection or transmission. What about all the lies about returning to 'normal' if people were 'vaccinated'? If they are not claimed to stop infection and transmission of the alleged 'virus', how does anything change? This was all lies to manipulate people to take the jabs and we are seeing that now with masks and distancing still required for the 'vaccinated'. How did they think that elderly people with fragile health and immune responses were going to be affected by infusing their cells with synthetic material and other toxic substances? They *knew* that in the short and long term it would be devastating and fatal as the culling of the old that began with the first lockdowns was continued with the 'vaccine'. Death rates in care homes soared immediately residents began to be 'vaccinated' – infused with synthetic material. Brave and committed whistleblower nurses put their careers at risk by exposing this truth while the rest kept their heads down and their mouths shut to put their careers before those they are supposed to care for. A long-time American Certified Nursing Assistant who gave his name as James posted a video in which he described emotionally what happened in his care home when vaccination began. He said that during 2020 very few residents were sick with 'Covid' and no one died during the entire year; but shortly after the Pfizer mRNA injections 14 people died within two weeks and many others were near death. 'They're dropping like flies', he said. Residents who walked on their own before the shot could no longer and they had lost their ability to conduct an intelligent conversation. The home's management said the sudden deaths were caused by a 'super-spreader' of 'Covid-19'. Then how come, James asked, that residents who refused to take the injections were not sick? It was a case of inject the elderly with mRNA synthetic potions and blame their illness and death that followed on the 'virus'. James described what was happening in care homes as 'the greatest crime of genocide this country has ever seen'. Remember the NHS staff nurse from earlier who used the same

word 'genocide' for what was happening with the 'vaccines' and that it was an 'act of human annihilation'. A UK care home whistleblower told a similar story to James about the effect of the 'vaccine' in deaths and 'outbreaks' of illness dubbed 'Covid' after getting the jab. She told how her care home management and staff had zealously imposed government regulations and no one was allowed to even question the official narrative let alone speak out against it. She said the NHS was even worse. Again we see the results of reframing. A worker at a local care home where I live said they had not had a single case of 'Covid' there for almost a year and when the residents were 'vaccinated' they had 19 positive cases in two weeks with eight dying.

It's not the 'vaccine' – honest

The obvious cause and effect was being ignored by the media and most of the public. Australia's health minister Greg Hunt (a former head of strategy at the World Economic Forum) was admitted to hospital after he had the 'vaccine'. He was suffering according to reports from the skin infection 'cellulitis' and it must have been a severe case to have warranted days in hospital. Immediately the authorities said this was nothing to do with the 'vaccine' when an effect of some vaccines is a 'cellulitis-like reaction'. We had families of perfectly healthy old people who died after the 'vaccine' saying that if only they had been given the 'vaccine' earlier they would still be alive. As a numbskull rating that is off the chart. A father of four 'died of Covid' at aged 48 when he was taken ill two days after having the 'vaccine'. The man, a health administrator, had been 'shielding during the pandemic' and had 'not really left the house' until he went for the 'vaccine'. Having the 'vaccine' and then falling ill and dying does not seem to have qualified as a possible cause and effect and 'Covid-19' went on his death certificate. His family said they had no idea how he 'caught the virus'. A family member said: 'Tragically, it could be that going for a vaccination ultimately led to him catching Covid ...The sad truth is that they are never going to know where it came from.' The family warned people to remember

that the virus still existed and was 'very real'. So was their stupidity. Nurses and doctors who had the first round of the 'vaccine' were collapsing, dying and ending up in a hospital bed while they or their grieving relatives were saying they'd still have the 'vaccine' again despite what happened. I kid you not. You mean if your husband returned from the dead he'd have the same 'vaccine' again that killed him??

Doctors at the VCU Medical Center in Richmond, Virginia, said the Johnson & Johnson 'vaccine' was to blame for a man's skin peeling off. Patient Richard Terrell said: 'It all just happened so fast. My skin peeled off. It's still coming off on my hands now.' He said it was stinging, burning and itching and when he bent his arms and legs it was very painful with 'the skin swollen and rubbing against itself'. Pfizer/BioNTech and Moderna vaccines use mRNA to change the cell while the Johnson & Johnson version uses DNA in a process similar to AstraZeneca's technique. Johnson & Johnson and AstraZeneca have both had their 'vaccines' paused by many countries after causing serious blood problems. Terrell's doctor Fnu Nutan said he could have died if he hadn't got medical attention. It sounds terrible so what did Nutan and Terrell say about the 'vaccine' now? Oh, they still recommend that people have it. A nurse in a hospital bed 40 minutes after the vaccination and unable to swallow due to throat swelling was told by a doctor that he lost mobility in his arm for 36 hours following the vaccination. What did he say to the ailing nurse? 'Good for you for getting the vaccination.' We are dealing with a serious form of cognitive dissonance madness in both public and medical staff. There is a remarkable correlation between those having the 'vaccine' and trumpeting the fact and suffering bad happenings shortly afterwards. Witold Rogiewicz, a Polish doctor, made a video of his 'vaccination' and ridiculed those who were questioning its safety and the intentions of Bill Gates: 'Vaccinate yourself to protect yourself, your loved ones, friends and also patients. And to mention quickly I have info for anti-vaxxers and anti-Covidors if you want to contact Bill Gates you can do this through me.' He further ridiculed the dangers of 5G. Days later he

was dead, but naturally the vaccination wasn't mentioned in the verdict of 'heart attack'.

Lies, lies and more lies

So many members of the human race have slipped into extreme states of insanity and unfortunately they include reframed doctors and nursing staff. Having a 'vaccine' and dying within minutes or hours is not considered a valid connection while death from any cause within 28 days or longer of a positive test with a test not testing for the 'virus' means 'Covid-19' goes on the death certificate. How could that 'vaccine'-death connection not have been made except by calculated deceit? US figures in the initial rollout period to February 12th, 2020, revealed that a third of the deaths reported to the CDC after 'Covid vaccines' happened within 48 hours. Five men in the UK suffered an 'extremely rare' blood clot problem after having the AstraZeneca 'vaccine', but no causal link was established said the Gates-funded Medicines and Healthcare products Regulatory Agency (MHRA) which had given the 'vaccine' emergency approval to be used. Former Pfizer executive Dr Michael Yeadon explained in his interview how the procedures could cause blood coagulation and clots. People who should have been at no risk were dying from blood clots in the brain and he said he had heard from medical doctor friends that people were suffering from skin bleeding and massive headaches. The AstraZeneca 'shot' was stopped by some 20 countries over the blood clotting issue and still the corrupt MHRA, the European Medicines Agency (EMA) and the World Health Organization said that it should continue to be given even though the EMA admitted that it 'still cannot rule out definitively' a link between blood clotting and the 'vaccine'. Later Marco Cavaleri, head of EMA vaccine strategy, said there was indeed a clear link between the 'vaccine' and thrombosis, but they didn't know why. So much for the trials showing the 'vaccine' is safe. Blood clots were affecting younger people who would be under virtually no danger from 'Covid' even if it existed which makes it all the more stupid and sinister.

The British government responded to public alarm by wheeling out June Raine, the terrifyingly weak infant school headmistress sound-alike who heads the UK MHRA drug 'regulator'. The idea that she would stand up to Big Pharma and government pressure is laughable and she told us that all was well in the same way that she did when allowing untested, never-used-on-humans-before, genetically-manipulating 'vaccines' to be exposed to the public in the first place. Mass lying is the new normal of the 'Covid' era. The MHRA later said 30 cases of rare blood clots had by then been connected with the AstraZeneca 'vaccine' (that means a lot more in reality) while stressing that the benefits of the jab in preventing 'Covid-19' outweighed any risks. A more ridiculous and disingenuous statement with callous disregard for human health it is hard to contemplate. Immediately after the mendacious 'all-clears' two hospital workers in Denmark experienced blood clots and cerebral haemorrhaging following the AstraZeneca jab and one died. Top Norwegian health official Pål Andre Holme said the 'vaccine' was the only common factor: 'There is nothing in the patient history of these individuals that can give such a powerful immune response ... I am confident that the antibodies that we have found are the cause, and I see no other explanation than it being the vaccine which triggers it.' Strokes, a clot or bleed in the brain, were clearly associated with the 'vaccine' from word of mouth and whistleblower reports. Similar consequences followed with all these 'vaccines' that we were told were so safe and as the numbers grew by the day it was clear we were witnessing human carnage.

Learning the hard way

A woman interviewed by UKColumn told how her husband suffered dramatic health effects after the vaccine when he'd been in good health all his life. He went from being a little unwell to losing all feeling in his legs and experiencing 'excruciating pain'. Misdiagnosis followed twice at Accident and Emergency (an 'allergy' and 'sciatica') before he was admitted to a neurology ward where doctors said his serious condition had been caused by the

'vaccine'. Another seven 'vaccinated' people were apparently being treated on the same ward for similar symptoms. The woman said he had the 'vaccine' because they believed media claims that it was safe. 'I didn't think the government would give out a vaccine that does this to somebody; I believed they would be bringing out a vaccination that would be safe.' What a tragic way to learn that lesson. Another woman posted that her husband was transporting stroke patients to hospital on almost every shift and when he asked them if they had been 'vaccinated' for 'Covid' they all replied 'yes'. One had a 'massive brain bleed' the day after his second dose. She said her husband reported the 'just been vaccinated' information every time to doctors in A and E only for them to ignore it, make no notes and appear annoyed that it was even mentioned. This particular report cannot be verified, but it expresses a common theme that confirms the monumental underreporting of 'vaccine' consequences. Interestingly as the 'vaccines' and their brain blood clot/stroke consequences began to emerge the UK National Health Service began a publicity campaign telling the public what to do in the event of a stroke. A Scottish NHS staff nurse who quit in disgust in March, 2021, said:

I have seen traumatic injuries from the vaccine, they're not getting reported to the yellow card [adverse reaction] scheme, they're treating the symptoms, not asking why, why it's happening. It's just treating the symptoms and when you speak about it you're dismissed like you're crazy, I'm not crazy, I'm not crazy because every other colleague I've spoken to is terrified to speak out, they've had enough.

Videos appeared on the Internet of people uncontrollably shaking after the 'vaccine' with no control over muscles, limbs and even their face. A Scottish mother broke out in a severe rash all over her body almost immediately after she was given the AstraZeneca 'vaccine'. The pictures were horrific. Leigh King, a 41-year-old hairdresser from Lanarkshire said: 'Never in my life was I prepared for what I was about to experience ... My skin was so sore and constantly hot ... I have never felt pain like this ...' But don't you worry, the 'vaccine' is perfectly safe. Then there has been the effect on medical

staff who have been pressured to have the 'vaccine' by psychopathic 'health' authorities and government. A London hospital consultant who gave the name K. Polyakova wrote this to the *British Medical Journal* or *BMJ*:

I am currently struggling with ... the failure to report the reality of the morbidity caused by our current vaccination program within the health service and staff population. The levels of sickness after vaccination is unprecedented and staff are getting very sick and some with neurological symptoms which is having a huge impact on the health service function. Even the young and healthy are off for days, some for weeks, and some requiring medical treatment. Whole teams are being taken out as they went to get vaccinated together.

Mandatory vaccination in this instance is stupid, unethical and irresponsible when it comes to protecting our staff and public health. We are in the voluntary phase of vaccination, and encouraging staff to take an unlicensed product that is impacting on their immediate health ... it is clearly stated that these vaccine products do not offer immunity or stop transmission. In which case why are we doing it?

Not to protect health that's for sure. Medical workers are lauded by governments for agenda reasons when they couldn't give a toss about them any more than they can for the population in general. Schools across America faced the same situation as they closed due to the high number of teachers and other staff with bad reactions to the Pfizer/BioNTech, Moderna, and Johnson & Johnson 'Covid vaccines' all of which were linked to death and serious adverse effects. The *BMJ* took down the consultant's comments pretty quickly on the grounds that they were being used to spread 'disinformation'. They were exposing the truth about the 'vaccine' was the real reason. The cover-up is breathtaking.

Hiding the evidence

The scale of the 'vaccine' death cover-up worldwide can be confirmed by comparing official figures with the personal experience of the public. I heard of many people in my community who died immediately or soon after the vaccine that would never appear in the media or even likely on the official totals of 'vaccine' fatalities and adverse reactions when only about ten percent are estimated to be

reported and I have seen some estimates as low as one percent in a Harvard study. In the UK alone by April 29th, 2021, some 757,654 adverse reactions had been officially reported from the Pfizer/BioNTech, Oxford/AstraZeneca and Moderna 'vaccines' with more than a thousand deaths linked to jabs and that means an estimated ten times this number in reality from a ten percent reporting rate percentage. That's seven million adverse reactions and 10,000 potential deaths and a one percent reporting rate would be ten times *those* figures. In 1976 the US government pulled the swine flu vaccine after 53 deaths. The UK data included a combined 10,000 eye disorders from the 'Covid vaccines' with more than 750 suffering visual impairment or blindness and again multiply by the estimated reporting percentages. As 'Covid cases' officially fell hospitals virtually empty during the 'Covid crisis' began to fill up with a range of other problems in the wake of the 'vaccine' rollout. The numbers across America have also been catastrophic. Deaths linked to *all* types of vaccine increased by 6,000 *percent* in the first quarter of 2021 compared with 2020. A 39-year-old woman from Ogden, Utah, died four days after receiving a second dose of Moderna's 'Covid vaccine' when her liver, heart and kidneys all failed despite the fact that she had no known medical issues or conditions. Her family sought an autopsy, but Dr Erik Christensen, Utah's chief medical examiner, said proving vaccine injury as a cause of death almost never happened. He could think of only one instance where an autopsy would name a vaccine as the official cause of death and that would be anaphylaxis where someone received a vaccine and died almost instantaneously. 'Short of that, it would be difficult for us to definitively say this is the vaccine,' Christensen said. If that is true this must be added to the estimated ten percent (or far less) reporting rate of vaccine deaths and serious reactions and the conclusion can only be that vaccine deaths and serious reactions – including these 'Covid' potions' – are phenomenally understated in official figures. The same story can be found everywhere. Endless accounts of deaths and serious reactions among the public, medical

and care home staff while official figures did not even begin to reflect this.

Professional script-reader Dr David Williams, a 'top public-health official' in Ontario, Canada, insulted our intelligence by claiming only four serious adverse reactions and no deaths from the more than 380,000 vaccine doses then given. This bore no resemblance to what people knew had happened in their own circles and we had Dirk Huyer in charge of getting millions vaccinated in Ontario while at the same time he was Chief Coroner for the province investigating causes of death including possible death from the vaccine. An aide said he had stepped back from investigating deaths, but evidence indicated otherwise. Rosemary Frei, who secured a Master of Science degree in molecular biology at the Faculty of Medicine at Canada's University of Calgary before turning to investigative journalism, was one who could see that official figures for 'vaccine' deaths and reactions made no sense. She said that doctors seldom reported adverse events and when people got really sick or died after getting a vaccination they would attribute that to anything except the vaccines. It had been that way for years and anyone who wondered aloud whether the 'Covid vaccines' or other shots cause harm is immediately branded as 'anti-vax' and 'anti-science'. This was 'career-threatening' for health professionals. Then there was the huge pressure to support the push to 'vaccinate' billions in the quickest time possible. Frei said:

So that's where we're at today. More than half a million vaccine doses have been given to people in Ontario alone. The rush is on to vaccinate all 15 million of us in the province by September. And the mainstream media are screaming for this to be sped up even more. That all adds up to only a very slim likelihood that we're going to be told the truth by officials about how many people are getting sick or dying from the vaccines.

What is true of Ontario is true of everywhere.

They KNEW – and still did it

The authorities knew what was going to happen with multiple deaths and adverse reactions. The UK government's Gates-funded

and Big Pharma-dominated Medicines and Healthcare products Regulatory Agency (MHRA) hired a company to employ AI in compiling the projected reactions to the 'vaccine' that would otherwise be uncountable. The request for applications said: 'The MHRA urgently seeks an Artificial Intelligence (AI) software tool to process the expected high volume of Covid-19 vaccine Adverse Drug Reaction ...' This was from the agency, headed by the disingenuous June Raine, that gave the 'vaccines' emergency approval and the company was hired before the first shot was given. 'We are going to kill and maim you – is that okay?' 'Oh, yes, perfectly fine – I'm very grateful, thank you, doctor.' The range of 'Covid vaccine' adverse reactions goes on for page after page in the MHRA criminally underreported 'Yellow Card' system and includes affects to eyes, ears, skin, digestion, blood and so on. Raine's MHRA amazingly claimed that the 'overall safety experience ... is so far as expected from the clinical trials'. The death, serious adverse effects, deafness and blindness were *expected*? When did they ever mention that? If these human tragedies were expected then those that gave approval for the use of these 'vaccines' must be guilty of crimes against humanity including murder – a definition of which is 'killing a person with malice aforethought or with recklessness manifesting extreme indifference to the value of human life.' People involved at the MHRA, the CDC in America and their equivalent around the world must go before Nuremberg trials to answer for their callous inhumanity. We are only talking here about the immediate effects of the 'vaccine'. The longer-term impact of the DNA synthetic manipulation is the main reason they are so hysterically desperate to inoculate the entire global population in the shortest possible time.

Africa and the developing world are a major focus for the 'vaccine' depopulation agenda and a mass vaccination sales-pitch is underway thanks to caring people like the Rockefellers and other Cult assets. The Rockefeller Foundation, which pre-empted the 'Covid pandemic' in a document published in 2010 that 'predicted' what happened a decade later, announced an initial \$34.95 million grant in February, 2021, 'to ensure more equitable access to Covid-19

testing and vaccines' among other things in Africa in collaboration with '24 organizations, businesses, and government agencies'. The pan-Africa initiative would focus on 10 countries: Burkina Faso, Ethiopia, Ghana, Kenya, Nigeria, Rwanda, South Africa, Tanzania, Uganda, and Zambia'. Rajiv Shah, President of the Rockefeller Foundation and former administrator of CIA-controlled USAID, said that if Africa was not mass-vaccinated (to change the DNA of its people) it was a 'threat to all of humanity' and not fair on Africans. When someone from the Rockefeller Foundation says they want to do something to help poor and deprived people and countries it is time for a belly-laugh. They are doing this out of the goodness of their 'heart' because 'vaccinating' the entire global population is what the 'Covid' hoax set out to achieve. Official 'decolonisation' of Africa by the Cult was merely a prelude to financial colonisation on the road to a return to physical colonisation. The 'vaccine' is vital to that and the sudden and convenient death of the 'Covid' sceptic president of Tanzania can be seen in its true light. A lot of people in Africa are aware that this is another form of colonisation and exploitation and they need to stand their ground.

The 'vaccine is working' scam

A potential problem for the Cult was that the 'vaccine' is meant to change human DNA and body messaging and not to protect anyone from a 'virus' never shown to exist. The vaccine couldn't work because it was not designed to work and how could they make it *appear* to be working so that more people would have it? This was overcome by lowering the amplification rate of the PCR test to produce fewer 'cases' and therefore fewer 'deaths'. Some of us had been pointing out since March, 2020, that the amplification rate of the test not testing for the 'virus' had been made artificially high to generate positive tests which they could call 'cases' to justify lockdowns. The World Health Organization recommended an absurdly high 45 amplification cycles to ensure the high positives required by the Cult and then remained silent on the issue until January 20th, 2021 – Biden's Inauguration Day. This was when the

'vaccinations' were seriously underway and on that day the WHO recommended after discussions with America's CDC that laboratories *lowered their testing amplification*. Dr David Samadi, a certified urologist and health writer, said the WHO was encouraging all labs to reduce their cycle count for PCR tests. He said the current cycle was much too high and was 'resulting in any particle being declared a positive case'. Even one mainstream news report I saw said this meant the number of 'Covid' infections may have been 'dramatically inflated'. Oh, just a little bit. The CDC in America issued new guidance to laboratories in April, 2021, to use 28 cycles *but only for 'vaccinated' people*. The timing of the CDC/WHO interventions were cynically designed to make it appear the 'vaccines' were responsible for falling cases and deaths when the real reason can be seen in the following examples. New York's state lab, the Wadsworth Center, identified 872 positive tests in July, 2020, based on a threshold of 40 cycles. When the figure was lowered to 35 cycles 43 percent of the 872 were no longer 'positives'. At 30 cycles the figure was 63 percent. A Massachusetts lab found that between 85 to 90 percent of people who tested positive in July with a cycle threshold of 40 would be negative at 30 cycles, Ashish Jha, MD, director of the Harvard Global Health Institute, said: 'I'm really shocked that it could be that high ... Boy, does it really change the way we need to be thinking about testing.' I'm shocked that I could see the obvious in the spring of 2020, with no medical background, and most medical professionals still haven't worked it out. No, that's not shocking – it's terrifying.

Three weeks after the WHO directive to lower PCR cycles the London *Daily Mail* ran this headline: 'Why ARE Covid cases plummeting? New infections have fallen 45% in the US and 30% globally in the past 3 weeks but experts say vaccine is NOT the main driver because only 8% of Americans and 13% of people worldwide have received their first dose.' They acknowledged that the drop could not be attributed to the 'vaccine', but soon this morphed throughout the media into the 'vaccine' has caused cases and deaths to fall when it was the PCR threshold. In December, 2020, there was

chaos at English Channel ports with truck drivers needing negative 'Covid' tests before they could board a ferry home for Christmas. The government wanted to remove the backlog as fast as possible and they brought in troops to do the 'testing'. Out of 1,600 drivers just 36 tested positive and the rest were given the all clear to cross the Channel. I guess the authorities thought that 36 was the least they could get away with without the unquestioning catching on. The amplification trick which most people believed in the absence of information in the mainstream applied more pressure on those refusing the 'vaccine' to succumb when it 'obviously worked'. The truth was the exact opposite with deaths in care homes soaring with the 'vaccine' and in Israel the term used was 'skyrocket'. A re-analysis of published data from the Israeli Health Ministry led by Dr Hervé Seligmann at the Medicine Emerging Infectious and Tropical Diseases at Aix-Marseille University found that Pfizer's 'Covid vaccine' killed 'about 40 times more [elderly] people than the disease itself would have killed' during a five-week vaccination period and *260 times* more younger people than would have died from the 'virus' even according to the manipulated 'virus' figures. Dr Seligmann and his co-study author, Haim Yativ, declared after reviewing the Israeli 'vaccine' death data: 'This is a new Holocaust.'

Then, in mid-April, 2021, after vast numbers of people worldwide had been 'vaccinated', the story changed with clear coordination. The UK government began to prepare the ground for more future lockdowns when Nuremberg-destined Boris Johnson told yet another whopper. He said that cases had fallen because of *lockdowns* not 'vaccines'. Lockdowns are irrelevant when *there is no 'virus'* and the test and fraudulent death certificates are deciding the number of 'cases' and 'deaths'. Study after study has shown that lockdowns don't work and instead kill and psychologically destroy people. Meanwhile in the United States Anthony Fauci and Rochelle Walensky, the ultra-Zionist head of the CDC, peddled the same line. More lockdown was the answer and not the 'vaccine', a line repeated on cue by the moron that is Canadian Prime Minister Justin Trudeau. Why all the hysteria to get everyone 'vaccinated' if lockdowns and

not 'vaccines' made the difference? None of it makes sense on the face of it. Oh, but it does. The Cult wants lockdowns *and* the 'vaccine' and if the 'vaccine' is allowed to be seen as the total answer lockdowns would no longer be justified when there are still livelihoods to destroy. 'Variants' and renewed upward manipulation of PCR amplification are planned to instigate never-ending lockdown *and* more 'vaccines'.

You *must* have it – we're desperate

Israel, where the Jewish and Arab population are ruled by the Sabbatian Cult, was the front-runner in imposing the DNA-manipulating 'vaccine' on its people to such an extent that Jewish refusers began to liken what was happening to the early years of Nazi Germany. This would seem to be a fantastic claim. Why would a government of Jewish people be acting like the Nazis did? If you realise that the Sabbatian Cult was behind the Nazis and that Sabbatians hate Jews the pieces start to fit and the question of why a 'Jewish' government would treat Jews with such callous disregard for their lives and freedom finds an answer. Those controlling the government of Israel *aren't Jewish* – they're Sabbatian. Israeli lawyer Tamir Turgal was one who made the Nazi comparison in comments to German lawyer Reiner Fuellmich who is leading a class action lawsuit against the psychopaths for crimes against humanity. Turgal described how the Israeli government was vaccinating children and pregnant women on the basis that there was no evidence that this was dangerous when they had no evidence that it *wasn't* dangerous either. They just had no evidence. This was medical experimentation and Turgal said this breached the Nuremberg Code about medical experimentation and procedures requiring informed consent and choice. Think about that. A Nuremberg Code developed because of Nazi experimentation on Jews and others in concentration camps by people like the evil-beyond-belief Josef Mengele is being breached by the *Israeli* government; but when you know that it's a *Sabbatian* government along with its intelligence and military agencies like Mossad, Shin Bet and the Israeli Defense Forces, and that Sabbatians

were the force behind the Nazis, the kaleidoscope comes into focus. What have we come to when Israeli Jews are suing their government for violating the Nuremberg Code by essentially making Israelis subject to a medical experiment using the controversial 'vaccines'? It's a shocker that this has to be done in the light of what happened in Nazi Germany. The Anshe Ha-Emet, or 'People of the Truth', made up of Israeli doctors, lawyers, campaigners and public, have launched a lawsuit with the International Criminal Court. It says:

When the heads of the Ministry of Health as well as the prime minister presented the vaccine in Israel and began the vaccination of Israeli residents, the vaccinated were not advised, that, in practice, they are taking part in a medical experiment and that their consent is required for this under the Nuremberg Code.

The irony is unbelievable, but easily explained in one word: Sabbatians. The foundation of Israeli 'Covid' apartheid is the 'green pass' or 'green passport' which allows Jews and Arabs who have had the DNA-manipulating 'vaccine' to go about their lives – to work, fly, travel in general, go to shopping malls, bars, restaurants, hotels, concerts, gyms, swimming pools, theatres and sports venues, while non-'vaccinated' are banned from all those places and activities. Israelis have likened the 'green pass' to the yellow stars that Jews in Nazi Germany were forced to wear – the same as the yellow stickers that a branch of UK supermarket chain Morrisons told exempt mask-wearers they had to display when shopping. How very sensitive. The Israeli system is blatant South African-style apartheid on the basis of compliance or non-compliance to fascism rather than colour of the skin. How appropriate that the Sabbatian Israeli government was so close to the pre-Mandela apartheid regime in Pretoria. The Sabbatian-instigated 'vaccine passport' in Israel is planned for everywhere. Sabbatians struck a deal with Pfizer that allowed them to lead the way in the percentage of a national population infused with synthetic material and the result was catastrophic. Israeli freedom activist Shai Dannon told me how chairs were appearing on beaches that said 'vaccinated only'. Health Minister Yuli Edelstein said that anyone unwilling or unable to get

the jabs that 'confer immunity' will be 'left behind'. The man's a liar. Not even the makers claim the 'vaccines' confer immunity. When you see those figures of 'vaccine' deaths these psychopaths were saying that you must take the chance the 'vaccine' will kill you or maim you while knowing it will change your DNA or lockdown for you will be permanent. That's fascism. The Israeli parliament passed a law to allow personal information of the non-vaccinated to be shared with local and national authorities for three months. This was claimed by its supporters to be a way to 'encourage' people to be vaccinated. Hadas Ziv from Physicians for Human Rights described this as a 'draconian law which crushed medical ethics and the patient rights'. But that's the idea, the Sabbatians would reply.

Your papers, please

Sabbatian Israel was leading what has been planned all along to be a global 'vaccine pass' called a 'green passport' without which you would remain in permanent lockdown restriction and unable to do anything. This is how badly – *desperately* – the Cult is to get everyone 'vaccinated'. The term and colour 'green' was not by chance and related to the psychology of fusing the perception of the green climate hoax with the 'Covid' hoax and how the 'solution' to both is the same Great Reset. Lying politicians, health officials and psychologists denied there were any plans for mandatory vaccinations or restrictions based on vaccinations, but they knew that was exactly what was meant to happen with governments of all countries reaching agreements to enforce a global system. 'Free' Denmark and 'free' Sweden unveiled digital vaccine certification. Cyprus, Czech Republic, Estonia, Greece, Hungary, Iceland, Italy, Poland, Portugal, Slovakia, and Spain have all committed to a vaccine passport system and the rest including the whole of the EU would follow. The satanic UK government will certainly go this way despite mendacious denials and at the time of writing it is trying to manipulate the public into having the 'vaccine' so they could go abroad on a summer holiday. How would that work without something to prove you had the synthetic toxicity injected into you?

Documents show that the EU's European Commission was moving towards 'vaccine certificates' in 2018 and 2019 before the 'Covid' hoax began. They knew what was coming. Abracadabra – Ursula von der Leyen, the German President of the Commission, announced in March, 2021, an EU 'Digital Green Certificate' – green again – to track the public's 'Covid status'. The passport sting is worldwide and the Far East followed the same pattern with South Korea ruling that only those with 'vaccination' passports – again the *green* pass – would be able to 'return to their daily lives'.

Bill Gates has been preparing for this 'passport' with other Cult operatives for years and beyond the paper version is a Gates-funded 'digital tattoo' to identify who has been vaccinated and who hasn't. The 'tattoo' is reported to include a substance which is externally readable to confirm who has been vaccinated. This is a bio-luminous light-generating enzyme (think fireflies) called ... *Luciferase*. Yes, named after the Cult 'god' Lucifer the 'light bringer' of whom more to come. Gates said he funded the readable tattoo to ensure children in the developing world were vaccinated and no one was missed out. He cares so much about poor kids as we know. This was just the cover story to develop a vaccine tagging system for everyone on the planet. Gates has been funding the ID2020 'alliance' to do just that in league with other lovely people at Microsoft, GAVI, the Rockefeller Foundation, Accenture and IDEO.org. He said in interviews in March, 2020, before any 'vaccine' publicly existed, that the world must have a globalised digital certificate to track the 'virus' and who had been vaccinated. Gates knew from the start that the mRNA vaccines were coming and when they would come and that the plan was to tag the 'vaccinated' to marginalise the intelligent and stop them doing anything including travel. Evil just doesn't suffice. Gates was exposed for offering a \$10 million bribe to the Nigerian House of Representatives to invoke compulsory 'Covid' vaccination of all Nigerians. Sara Cunial, a member of the Italian Parliament, called Gates a 'vaccine criminal'. She urged the Italian President to hand him over to the International Criminal Court for crimes against

humanity and condemned his plans to 'chip the human race' through ID2020.

You know it's a long-planned agenda when war criminal and Cult gofer Tony Blair is on the case. With the scale of arrogance only someone as dark as Blair can muster he said: 'Vaccination in the end is going to be your route to liberty.' Blair is a disgusting piece of work and he confirms that again. The media has given a lot of coverage to a bloke called Charlie Mullins, founder of London's biggest independent plumbing company, Pimlico Plumbers, who has said he won't employ anyone who has not been vaccinated or have them go to any home where people are not vaccinated. He said that if he had his way no one would be allowed to walk the streets if they have not been vaccinated. Gates was cheering at the time while I was alerting the white coats. The plan is that people will qualify for 'passports' for having the first two doses and then to keep it they will have to have all the follow ups and new ones for invented 'variants' until human genetics is transformed and many are dead who can't adjust to the changes. Hollywood celebrities – the usual propaganda stunt – are promoting something called the WELL Health-Safety Rating to verify that a building or space has 'taken the necessary steps to prioritize the health and safety of their staff, visitors and other stakeholders'. They included Lady Gaga, Jennifer Lopez, Michael B. Jordan, Robert DeNiro, Venus Williams, Wolfgang Puck, Deepak Chopra and 17th Surgeon General Richard Carmona. Yawn. WELL Health-Safety has big connections with China. Parent company Delos is headed by former Goldman Sachs partner Paul Scialla. This is another example – and we will see so many others – of using the excuse of 'health' to dictate the lives and activities of the population. I guess one confirmation of the 'safety' of buildings is that only 'vaccinated' people can go in, right?

Electronic concentration camps

I wrote decades ago about the plans to restrict travel and here we are for those who refuse to bow to tyranny. This can be achieved in one go with air travel if the aviation industry makes a blanket decree.

The 'vaccine' and guaranteed income are designed to be part of a global version of China's social credit system which tracks behaviour 24/7 and awards or deletes 'credits' based on whether your behaviour is supported by the state or not. I mean your entire lifestyle – what you do, eat, say, everything. Once your credit score falls below a certain level consequences kick in. In China tens of millions have been denied travel by air and train because of this. All the locations and activities denied to refusers by the 'vaccine' passports will be included in one big mass ban on doing almost anything for those that don't bow their head to government. It's beyond fascist and a new term is required to describe its extremes – I guess fascist technocracy will have to do. The way the Chinese system of technological – technocratic – control is sweeping the West can be seen in the Los Angeles school system and is planned to be expanded worldwide. Every child is required to have a 'Covid'-tracking app scanned daily before they can enter the classroom. The so-called Daily Pass tracking system is produced by Gates' Microsoft which I'm sure will shock you rigid. The pass will be scanned using a barcode (one step from an inside-the-body barcode) and the information will include health checks, 'Covid' tests and vaccinations. Entry codes are for one specific building only and access will only be allowed if a student or teacher has a negative test with a test not testing for the 'virus', has no symptoms of anything alleged to be related to 'Covid' (symptoms from a range of other illness), and has a temperature under 100 degrees. No barcode, no entry, is planned to be the case for everywhere and not only schools.

Kids are being psychologically prepared to accept this as 'normal' their whole life which is why what they can impose in schools is so important to the Cult and its gofers. Long-time American freedom campaigner John Whitehead of the Rutherford Institute was not exaggerating when he said: 'Databit by databit, we are building our own electronic concentration camps.' Canada under its Cult gofer prime minister Justin Trudeau has taken a major step towards the real thing with people interned against their will if they test positive with a test not testing for the 'virus' when they arrive at a Canadian

airport. They are jailed in internment hotels often without food or water for long periods and with many doors failing to lock there have been sexual assaults. The interned are being charged sometimes \$2,000 for the privilege of being abused in this way. Trudeau is fully on board with the Cult and says the 'Covid pandemic' has provided an opportunity for a global 'reset' to permanently change Western civilisation. His number two, Deputy Prime Minister Chrystia Freeland, is a trustee of the World Economic Forum and a Rhodes Scholar. The Trudeau family have long been servants of the Cult. See *The Biggest Secret* and Cathy O'Brien's book *Trance-Formation of America* for the horrific background to Trudeau's father Pierre Trudeau another Canadian prime minister. Hide your fascism behind the façade of a heart-on-the-sleeve liberal. It's a well-honed Cult technique.

What can the 'vaccine' really do?

We have a 'virus' never shown to exist and 'variants' of the 'virus' that have also never been shown to exist except, like the 'original', as computer-generated fictions. Even if you believe there's a 'virus' the 'case' to 'death' rate is in the region of 0.23 to 0.15 percent and those 'deaths' are concentrated among the very old around the same average age that people die anyway. In response to this lack of threat (in truth none) psychopaths and idiots, knowingly and unknowingly answering to Gates and the Cult, are seeking to 'vaccinate' every man, woman and child on Planet Earth. Clearly the 'vaccine' is not about 'Covid' – none of this ever has been. So what is it all about *really*? Why the desperation to infuse genetically-manipulating synthetic material into everyone through mRNA fraudulent 'vaccines' with the intent of doing this over and over with the excuses of 'variants' and other 'virus' inventions? Dr Sherri Tenpenny, an osteopathic medical doctor in the United States, has made herself an expert on vaccines and their effects as a vehement campaigner against their use. Tenpenny was board certified in emergency medicine, the director of a level two trauma centre for 12 years, and moved to Cleveland in 1996 to start an integrative

medicine practice which has treated patients from all 50 states and some 17 other countries. Weaning people off pharmaceutical drugs is a speciality.

She became interested in the consequences of vaccines after attending a meeting at the National Vaccine Information Center in Washington DC in 2000 where she 'sat through four days of listening to medical doctors and scientists and lawyers and parents of vaccine injured kids' and asked: 'What's going on?' She had never been vaccinated and never got ill while her father was given a list of vaccines to be in the military and was 'sick his entire life'. The experience added to her questions and she began to examine vaccine documents from the Centers for Disease Control (CDC). After reading the first one, the 1998 version of *The General Recommendations of Vaccination*, she thought: 'This is it?' The document was poorly written and bad science and Tenpenny began 20 years of research into vaccines that continues to this day. She began her research into 'Covid vaccines' in March, 2020, and she describes them as 'deadly'. For many, as we have seen, they already have been. Tenpenny said that in the first 30 days of the 'vaccine' rollout in the United States there had been more than 40,000 adverse events reported to the vaccine adverse event database. A document had been delivered to her the day before that was 172 pages long. 'We have over 40,000 adverse events; we have over 3,100 cases of [potentially deadly] anaphylactic shock; we have over 5,000 neurological reactions.' Effects ranged from headaches to numbness, dizziness and vertigo, to losing feeling in hands or feet and paraesthesia which is when limbs 'fall asleep' and people have the sensation of insects crawling underneath their skin. All this happened in the first 30 days and remember that only about *ten percent* (or far less) of adverse reactions and vaccine-related deaths are estimated to be officially reported. Tenpenny said:

So can you think of one single product in any industry, any industry, for as long as products have been made on the planet that within 30 days we have 40,000 people complaining of side effects that not only is still on the market but ... we've got paid actors telling us how great

they are for getting their vaccine. We're offering people \$500 if they will just get their vaccine and we've got nurses and doctors going; 'I got the vaccine, I got the vaccine'.

Tenpenny said they were not going to be 'happy dancing folks' when they began to suffer Bell's palsy (facial paralysis), neuropathies, cardiac arrhythmias and autoimmune reactions that kill through a blood disorder. 'They're not going to be so happy, happy then, but we're never going to see pictures of those people' she said. Tenpenny described the 'vaccine' as 'a well-designed killing tool'.

No off-switch

Bad as the initial consequences had been Tenpenny said it would be maybe 14 months before we began to see the 'full ravage' of what is going to happen to the 'Covid vaccinated' with full-out consequences taking anything between two years and 20 years to show. You can understand why when you consider that variations of the 'Covid vaccine' use mRNA (messenger RNA) to in theory activate the immune system to produce protective antibodies without using the actual 'virus'. How can they when it's a computer program and they've never isolated what they claim is the 'real thing'? Instead they use *synthetic* mRNA. They are inoculating synthetic material into the body which through a technique known as the Trojan horse is absorbed into cells to change the nature of DNA. Human DNA is changed by an infusion of messenger RNA and with each new 'vaccine' of this type it is changed even more. Say so and you are banned by Cult Internet platforms. The contempt the contemptuous Mark Zuckerberg has for the truth and human health can be seen in an internal Facebook video leaked to the Project Veritas investigative team in which he said of the 'Covid vaccines': '... I share some caution on this because we just don't know the long term side-effects of basically modifying people's DNA and RNA.' At the same time this disgusting man's Facebook was censoring and banning anyone saying exactly the same. He must go before a Nuremberg trial for crimes against humanity when he *knows* that he

is censoring legitimate concerns and denying the right of informed consent on behalf of the Cult that owns him. People have been killed and damaged by the very 'vaccination' technique he cast doubt on himself when they may not have had the 'vaccine' with access to information that he denied them. The plan is to have at least annual 'Covid vaccinations', add others to deal with invented 'variants', and change all other vaccines into the mRNA system. Pfizer executives told shareholders at a virtual Barclays Global Healthcare Conference in March, 2021, that the public may need a third dose of 'Covid vaccine', plus regular yearly boosters and the company planned to hike prices to milk the profits in a 'significant opportunity for our vaccine'. These are the professional liars, cheats and opportunists who are telling you their 'vaccine' is safe. Given this volume of mRNA planned to be infused into the human body and its ability to then replicate we will have a transformation of human genetics from biological to synthetic biological – exactly the long-time Cult plan for reasons we'll see – and many will die. Sherri Tenpenny said of this replication:

It's like having an on-button but no off-button and that whole mechanism ... they actually give it a name and they call it the Trojan horse mechanism, because it allows that [synthetic] virus and that piece of that [synthetic] virus to get inside of your cells, start to replicate and even get inserted into other parts of your DNA as a Trojan-horse.

Ask the overwhelming majority of people who have the 'vaccine' what they know about the contents and what they do and they would reply: 'The government says it will stop me getting the virus.' Governments give that false impression on purpose to increase take-up. You can read Sherri Tenpenny's detailed analysis of the health consequences in her blog at [Vaxxter.com](https://vaxxter.com), but in summary these are some of them. She highlights the statement by Bill Gates about how human beings can become their own 'vaccine manufacturing machine'. The man is insane. ['Vaccine'-generated] 'antibodies' carry synthetic messenger RNA into the cells and the damage starts, Tenpenny contends, and she says that lungs can be adversely affected through varying degrees of pus and bleeding which

obviously affects breathing and would be dubbed 'Covid-19'. Even more sinister was the impact of 'antibodies' on macrophages, a white blood cell of the immune system. They consist of Type 1 and Type 2 which have very different functions. She said Type 1 are 'hyper-vigilant' white blood cells which 'gobble up' bacteria etc. However, in doing so, this could cause inflammation and in extreme circumstances be fatal. She says these affects are mitigated by Type 2 macrophages which kick in to calm down the system and stop it going rogue. They clear up dead tissue debris and reduce inflammation that the Type 1 'fire crews' have caused. Type 1 kills the infection and Type 2 heals the damage, she says. This is her punchline with regard to 'Covid vaccinations': She says that mRNA 'antibodies' block Type 2 macrophages by attaching to them and deactivating them. This meant that when the Type 1 response was triggered by infection there was nothing to stop that getting out of hand by calming everything down. There's an on-switch, but no off-switch, she says. What follows can be 'over and out, see you when I see you'.

Genetic suicide

Tenpenny also highlights the potential for autoimmune disease – the body attacking itself – which has been associated with vaccines since they first appeared. Infusing a synthetic foreign substance into cells could cause the immune system to react in a panic believing that the body is being overwhelmed by an invader (it is) and the consequences can again be fatal. There is an autoimmune response known as a 'cytokine storm' which I have likened to a homeowner panicked by an intruder and picking up a gun to shoot randomly in all directions before turning the fire on himself. The immune system unleashes a storm of inflammatory response called cytokines to a threat and the body commits hara-kiri. The lesson is that you mess with the body's immune response at your peril and these 'vaccines' seriously – fundamentally – mess with immune response. Tenpenny refers to a consequence called anaphylactic shock which is a severe and highly dangerous allergic reaction when the immune system

floods the body with chemicals. She gives the example of having a bee sting which primes the immune system and makes it sensitive to those chemicals. When people are stung again maybe years later the immune response can be so powerful that it leads to anaphylactic shock. Tenpenny relates this 'shock' with regard to the 'Covid vaccine' to something called polyethylene glycol or PEG. Enormous numbers of people have become sensitive to this over decades of use in a whole range of products and processes including food, drink, skin creams and 'medicine'. Studies have claimed that some 72 percent of people have antibodies triggered by PEG compared with two percent in the 1960s and allergic hypersensitive reactions to this become a gathering cause for concern. Tenpenny points out that the 'mRNA vaccine' is coated in a 'bubble' of polyethylene glycol which has the potential to cause anaphylactic shock through immune sensitivity. Many reports have appeared of people reacting this way after having the 'Covid vaccine'. What do we think is going to happen as humanity has more and more of these 'vaccines'? Tenpenny said: 'All these pictures we have seen with people with these rashes ... these weepy rashes, big reactions on their arms and things like that – it's an acute allergic reaction most likely to the polyethylene glycol that you've been previously primed and sensitised to.'

Those who have not studied the conspiracy and its perpetrators at length might think that making the population sensitive to PEG and then putting it in these 'vaccines' is just a coincidence. It is not. It is instead testament to how carefully and coldly-planned current events have been and the scale of the conspiracy we are dealing with. Tenpenny further explains that the 'vaccine' mRNA procedure can breach the blood-brain barrier which protects the brain from toxins and other crap that will cause malfunction. In this case they could make two proteins corrupt brain function to cause Amyotrophic lateral sclerosis (ALS), a progressive nervous system disease leading to loss of muscle control, and frontal lobe degeneration – Alzheimer's and dementia. Immunologist J. Bart Classon published a paper connecting mRNA 'vaccines' to prion

disease which can lead to Alzheimer's and other forms of neurodegenerative disease while others have pointed out the potential to affect the placenta in ways that make women infertile. This will become highly significant in the next chapter when I will discuss other aspects of this non-vaccine that relate to its nanotechnology and transmission from the injected to the uninjected.

Qualified in idiocy

Tenpenny describes how research has confirmed that these 'vaccine'-generated antibodies can interact with a range of other tissues in the body and attack many other organs including the lungs. 'This means that if you have a hundred people standing in front of you that all got this shot they could have a hundred different symptoms.' Anyone really think that Cult gofers like the Queen, Tony Blair, Christopher Whitty, Anthony Fauci, and all the other psychopaths have really had this 'vaccine' in the pictures we've seen? Not a bloody chance. Why don't doctors all tell us about all these dangers and consequences of the 'Covid vaccine'? Why instead do they encourage and pressure patients to have the shot? Don't let's think for a moment that doctors and medical staff can't be stupid, lazy, and psychopathic and that's without the financial incentives to give the jab. Tenpenny again:

Some people are going to die from the vaccine directly but a large number of people are going to start to get horribly sick and get all kinds of autoimmune diseases 42 days to maybe a year out. What are they going to do, these stupid doctors who say; 'Good for you for getting that vaccine.' What are they going to say; 'Oh, it must be a mutant, we need to give an extra dose of that vaccine.'

Because now the vaccine, instead of one dose or two doses we need three or four because the stupid physicians aren't taking the time to learn anything about it. If I can learn this sitting in my living room reading a 19 page paper and several others so can they. There's nothing special about me, I just take the time to do it.

Remember how Sara Kayat, the NHS and TV doctor, said that the 'Covid vaccine' would '100 percent prevent hospitalisation and death'. Doctors can be idiots like every other profession and they

should not be worshipped as infallible. They are not and far from it. Behind many medical and scientific 'experts' lies an uninformed prat trying to hide themselves from you although in the 'Covid' era many have failed to do so as with UK narrative-repeating 'TV doctor' Hilary Jones. Pushing back against the minority of proper doctors and scientists speaking out against the 'vaccine' has been the entire edifice of the Cult global state in the form of governments, medical systems, corporations, mainstream media, Silicon Valley, and an army of compliant doctors, medical staff and scientists willing to say anything for money and to enhance their careers by promoting the party line. If you do that you are an 'expert' and if you won't you are an 'anti-vaxxer' and 'Covidiot'. The pressure to be 'vaccinated' is incessant. We have even had reports claiming that the 'vaccine' can help cure cancer and Alzheimer's and make the lame walk. I am waiting for the announcement that it can bring you coffee in the morning and cook your tea. Just as the symptoms of 'Covid' seem to increase by the week so have the miracles of the 'vaccine'. American supermarket giant Kroger Co. offered nearly 500,000 employees in 35 states a \$100 bonus for having the 'vaccine' while donut chain Krispy Kreme promised 'vaccinated' customers a free glazed donut every day for the rest of 2021. Have your DNA changed and you will get a doughnut although we might not have to give you them for long. Such offers and incentives confirm the desperation.

Perhaps the worse vaccine-stunt of them all was UK 'Health' Secretary Matt-the-prat Hancock on live TV after watching a clip of someone being 'vaccinated' when the roll-out began. Hancock faked tears so badly it was embarrassing. Brain-of-Britain Piers Morgan, the lockdown-supporting, 'vaccine' supporting, 'vaccine' passport-supporting, TV host played along with Hancock – 'You're quite emotional about that' he said in response to acting so atrocious it would have been called out at a school nativity which will presumably today include Mary and Jesus in masks, wise men keeping their camels six feet apart, and shepherds under tent arrest. System-serving Morgan tweeted this: 'Love the idea of covid vaccine passports for everywhere: flights, restaurants, clubs, football, gyms,

shops etc. It's time covid-denying, anti-vaxxer loonies had their bullsh*t bluff called & bar themselves from going anywhere that responsible citizens go.' If only I could aspire to his genius. To think that Morgan, who specialises in shouting over anyone he disagrees with, was lauded as a free speech hero when he lost his job after storming off the set of his live show like a child throwing his dolly out of the pram. If he is a free speech hero we are in real trouble. I have no idea what 'bullsh*t' means, by the way, the * throws me completely.

The Cult is desperate to infuse its synthetic DNA-changing concoction into everyone and has been using every lie, trick and intimidation to do so. The question of '*Why?*' we shall now address.

CHAPTER TEN

Human 2.0

I believe that at the end of the century the use of words and general educated opinion will have altered so much that one will be able to speak of machines thinking without expecting to be contradicted –
Alan Turing (1912-1954), the ‘Father of artificial intelligence’

I have been exposing for decades the plan to transform the human body from a biological to a synthetic-biological state. The new human that I will call Human 2.0 is planned to be connected to artificial intelligence and a global AI ‘Smart Grid’ that would operate as one global system in which AI would control everything from your fridge to your heating system to your car to your mind. Humans would no longer be ‘human’, but post-human and sub-human, with their thinking and emotional processes replaced by AI.

What I said sounded crazy and beyond science fiction and I could understand that. To any balanced, rational, mind it *is* crazy. Today, however, that world is becoming reality and it puts the ‘Covid vaccine’ into its true context. Ray Kurzweil is the ultra-Zionist ‘computer scientist, inventor and futurist’ and co-founder of the Singularity University. Singularity refers to the merging of humans with machines or ‘transhumanism’. Kurzweil has said humanity would be connected to the cyber ‘cloud’ in the period of the ever-recurring year of 2030:

Our thinking ... will be a hybrid of biological and non-biological thinking ... humans will be able to extend their limitations and ‘think in the cloud’ ... We’re going to put gateways to the

cloud in our brains ... We're going to gradually merge and enhance ourselves ... In my view, that's the nature of being human – we transcend our limitations. As the technology becomes vastly superior to what we are then the small proportion that is still human gets smaller and smaller and smaller until it's just utterly negligible.

They are trying to sell this end-of-humanity-as-we-know-it as the next stage of 'evolution' when we become super-human and 'like the gods'. They are lying to you. Shocked, eh? The population, and again especially the young, have been manipulated into addiction to technologies designed to enslave them for life. First they induced an addiction to smartphones (holdables); next they moved to technology on the body (wearables); and then began the invasion of the body (implantables). I warned way back about the plan for microchipped people and we are now entering that era. We should not be diverted into thinking that this refers only to chips we can see. Most important are the nanochips known as smart dust, neural dust and nanobots which are far too small to be seen by the human eye. Nanotechnology is everywhere, increasingly in food products, and released into the atmosphere by the geoengineering of the skies funded by Bill Gates to 'shut out the Sun' and 'save the planet from global warming'. Gates has been funding a project to spray millions of tonnes of chalk (calcium carbonate) into the stratosphere over Sweden to 'dim the Sun' and cool the Earth. Scientists warned the move could be disastrous for weather systems in ways no one can predict and opposition led to the Swedish space agency announcing that the 'experiment' would not be happening as planned in the summer of 2021; but it shows where the Cult is going with dimming the impact of the Sun and there's an associated plan to change the planet's atmosphere. Who gives psychopath Gates the right to dictate to the entire human race and dismantle planetary systems? The world will not be safe while this man is at large.

The global warming hoax has made the Sun, like the gas of life, something to fear when both are essential to good health and human survival (more inversion). The body transforms sunlight into vital vitamin D through a process involving ... *cholesterol*. This is the cholesterol we are also told to fear. We are urged to take Big Pharma

statin drugs to reduce cholesterol and it's all systematic. Reducing cholesterol means reducing vitamin D uptake with all the multiple health problems that will cause. At least if you take statins long term it saves the government from having to pay you a pension. The delivery system to block sunlight is widely referred to as chemtrails although these have a much deeper agenda, too. They appear at first to be contrails or condensation trails streaming from aircraft into cold air at high altitudes. Contrails disperse very quickly while chemtrails do not and spread out across the sky before eventually their content falls to earth. Many times I have watched aircraft cross-cross a clear blue sky releasing chemtrails until it looks like a cloudy day. Chemtrails contain many things harmful to humans and the natural world including toxic heavy metals, aluminium (see Alzheimer's) and nanotechnology. Ray Kurzweil reveals the reason without actually saying so: 'Nanobots will infuse all the matter around us with information. Rocks, trees, everything will become these intelligent creatures.' How do you deliver that? *From the sky*. Self-replicating nanobots would connect everything to the Smart Grid. The phenomenon of Morgellons disease began in the chemtrail era and the correlation has led to it being dubbed the 'chemtrail disease'. Self-replicating fibres appear in the body that can be pulled out through the skin. Morgellons fibres continue to grow outside the body and have a form of artificial intelligence. I cover this at greater length in *Phantom Self*.

'Vaccine' operating system

'Covid vaccines' with their self-replicating synthetic material are also designed to make the connection between humanity and Kurzweil's 'cloud'. American doctor and dedicated campaigner for truth, Carrie Madej, an Internal Medicine Specialist in Georgia with more than 20 years medical experience, has highlighted the nanotechnology aspect of the fake 'vaccines'. She explains how one of the components in at least the Moderna and Pfizer synthetic potions are 'lipid nanoparticles' which are 'like little tiny computer bits' – a 'sci-fi substance' known as nanobots and hydrogel which can be 'triggered

at any moment to deliver its payload' and act as 'biosensors'. The synthetic substance had 'the ability to accumulate data from your body like your breathing, your respiration, thoughts and emotions, all kind of things' and each syringe could carry a *million* nanobots:

This substance because it's like little bits of computers in your body, crazy, but it's true, it can do that, [and] obviously has the ability to act through Wi-Fi. It can receive and transmit energy, messages, frequencies or impulses. That issue has never been addressed by these companies. What does that do to the human?

Just imagine getting this substance in you and it can react to things all around you, the 5G, your smart device, your phones, what is happening with that? What if something is triggering it, too, like an impulse, a frequency? We have something completely foreign in the human body.

Madej said her research revealed that electromagnetic (EMF) frequencies emitted by phones and other devices had increased dramatically in the same period of the 'vaccine' rollout and she was seeing more people with radiation problems as 5G and other electromagnetic technology was expanded and introduced to schools and hospitals. She said she was 'floored with the EMF coming off' the devices she checked. All this makes total sense and syncs with my own work of decades when you think that Moderna refers in documents to its mRNA 'vaccine' as an 'operating system':

Recognizing the broad potential of mRNA science, we set out to create an mRNA technology platform that functions very much like an operating system on a computer. It is designed so that it can plug and play interchangeably with different programs. In our case, the 'program' or 'app' is our mRNA drug – the unique mRNA sequence that codes for a protein ...

... Our MRNA Medicines – 'The 'Software Of Life': When we have a concept for a new mRNA medicine and begin research, fundamental components are already in place. Generally, the only thing that changes from one potential mRNA medicine to another is the coding region – the actual genetic code that instructs ribosomes to make protein. Utilizing these instruction sets gives our investigational mRNA medicines a software-like quality. We also have the ability to combine different mRNA sequences encoding for different proteins in a single mRNA investigational medicine.

Who needs a real ‘virus’ when you can create a computer version to justify infusing your operating system into the entire human race on the road to making living, breathing people into cyborgs? What is missed with the ‘vaccines’ is the *digital* connection between synthetic material and the body that I highlighted earlier with the study that hacked a computer with human DNA. On one level the body is digital, based on mathematical codes, and I’ll have more about that in the next chapter. Those who ridiculously claim that mRNA ‘vaccines’ are not designed to change human genetics should explain the words of Dr Tal Zaks, chief medical officer at Moderna, in a 2017 TED talk. He said that over the last 30 years ‘we’ve been living this phenomenal digital scientific revolution, and I’m here today to tell you, that we are actually *hacking the software of life*, and that it’s changing the way we think about prevention and treatment of disease’:

In every cell there’s this thing called messenger RNA, or mRNA for short, that transmits the critical information from the DNA in our genes to the protein, which is really the stuff we’re all made out of. This is the critical information that determines what the cell will do. So we think about it as an operating system. So if you could change that, if you could introduce a line of code, or change a line of code, it turns out, that has profound implications for everything, from the flu to cancer.

Zaks should more accurately have said that this has profound implications for the human genetic code and the nature of DNA. Communications within the body go both ways and not only one. But, hey, no, the ‘Covid vaccine’ will not affect your genetics. Cult fact-checkers say so even though the man who helped to develop the mRNA technique says that it does. Zaks said in 2017:

If you think about what it is we’re trying to do. We’ve taken information and our understanding of that information and how that information is transmitted in a cell, and we’ve taken our understanding of medicine and how to make drugs, and we’re fusing the two. We think of it as information therapy.

I have been writing for decades that the body is an information field communicating with itself and the wider world. This is why

radiation which is information can change the information field of body and mind through phenomena like 5G and change their nature and function. 'Information therapy' means to change the body's information field and change the way it operates. DNA is a receiver-transmitter of information and can be mutated by information like mRNA synthetic messaging. Technology to do this has been ready and waiting in the underground bases and other secret projects to be rolled out when the 'Covid' hoax was played. 'Trials' of such short and irrelevant duration were only for public consumption. When they say the 'vaccine' is 'experimental' that is not true. It may appear to be 'experimental' to those who don't know what's going on, but the trials have already been done to ensure the Cult gets the result it desires. Zaks said that it took decades to sequence the human genome, completed in 2003, but now they could do it in a week. By 'they' he means scientists operating in the public domain. In the secret projects they were sequencing the genome in a week long before even 2003.

Deluge of mRNA

Highly significantly the Moderna document says the guiding premise is that if using mRNA as a medicine works for one disease then it should work for many diseases. They were leveraging the flexibility afforded by their platform and the fundamental role mRNA plays in protein synthesis to pursue mRNA medicines for a broad spectrum of diseases. Moderna is confirming what I was saying through 2020 that multiple 'vaccines' were planned for 'Covid' (and later invented 'variants') and that previous vaccines would be converted to the mRNA system to infuse the body with massive amounts of genetically-manipulating synthetic material to secure a transformation to a synthetic-biological state. The 'vaccines' are designed to kill stunning numbers as part of the long-exposed Cult depopulation agenda and transform the rest. Given this is the goal you can appreciate why there is such hysterical demand for every human to be 'vaccinated' for an alleged 'disease' that has an estimated 'infection' to 'death' ratio of 0.23-0.15 percent. As I write

children are being given the 'vaccine' in trials (their parents are a disgrace) and ever-younger people are being offered the vaccine for a 'virus' that even if you believe it exists has virtually zero chance of harming them. Horrific effects of the 'trials' on a 12-year-old girl were revealed by a family member to be serious brain and gastric problems that included a bowel obstruction and the inability to swallow liquids or solids. She was unable to eat or drink without throwing up, had extreme pain in her back, neck and abdomen, and was paralysed from the waist down which stopped her urinating unaided. When the girl was first taken to hospital doctors said it was all in her mind. She was signed up for the 'trial' by her parents for whom no words suffice. None of this 'Covid vaccine' insanity makes any sense unless you see what the 'vaccine' really is – a body-changer. Synthetic biology or 'SynBio' is a fast-emerging and expanding scientific discipline which includes everything from genetic and molecular engineering to electrical and computer engineering. Synthetic biology is defined in these ways:

- A multidisciplinary area of research that seeks to create new biological parts, devices, and systems, or to redesign systems that are already found in nature.
- The use of a mixture of physical engineering and genetic engineering to create new (and therefore synthetic) life forms.
- An emerging field of research that aims to combine the knowledge and methods of biology, engineering and related disciplines in the design of chemically-synthesized DNA to create organisms with novel or enhanced characteristics and traits (synthetic organisms including humans).

We now have synthetic blood, skin, organs and limbs being developed along with synthetic body parts produced by 3D printers. These are all elements of the synthetic human programme and this comment by Kurzweil's co-founder of the Singularity University,

Peter Diamandis, can be seen in a whole new light with the 'Covid' hoax and the sanctions against those that refuse the 'vaccine':

Anybody who is going to be resisting the progress forward [to transhumanism] is going to be resisting evolution and, fundamentally, they will die out. It's not a matter of whether it's good or bad. It's going to happen.

'Resisting evolution'? What absolute bollocks. The arrogance of these people is without limit. His 'it's going to happen' mantra is another way of saying 'resistance is futile' to break the spirit of those pushing back and we must not fall for it. Getting this genetically-transforming 'vaccine' into everyone is crucial to the Cult plan for total control and the desperation to achieve that is clear for anyone to see. Vaccine passports are a major factor in this and they, too, are a form of resistance is futile. It's NOT. The paper funded by the Rockefeller Foundation for the 2013 'health conference' in China said:

We will interact more with artificial intelligence. The use of robotics, bio-engineering to augment human functioning is already well underway and will advance. Re-engineering of humans into potentially separate and unequal forms through genetic engineering or mixed human-robots raises debates on ethics and equality.

A new demography is projected to emerge after 2030 [that year again] of technologies (robotics, genetic engineering, nanotechnology) producing robots, engineered organisms, 'nanobots' and artificial intelligence (AI) that can self-replicate. Debates will grow on the implications of an impending reality of human designed life.

What is happening today is so long planned. The world army enforcing the will of the world government is intended to be a robot army, not a human one. Today's military and its technologically 'enhanced' troops, pilotless planes and driverless vehicles are just stepping stones to that end. Human soldiers are used as Cult fodder and its time they woke up to that and worked for the freedom of the population instead of their own destruction and their family's destruction – the same with the police. Join us and let's sort this out. The phenomenon of enforce my own destruction is widespread in the 'Covid' era with Woker 'luvvies' in the acting and entertainment

industries supporting 'Covid' rules which have destroyed their profession and the same with those among the public who put signs on the doors of their businesses 'closed due to Covid – stay safe' when many will never reopen. It's a form of masochism and most certainly insanity.

Transgender = transhumanism

When something explodes out of nowhere and is suddenly everywhere it is always the Cult agenda and so it is with the tidal wave of claims and demands that have infiltrated every aspect of society under the heading of 'transgenderism'. The term 'trans' is so 'in' and this is the dictionary definition:

A prefix meaning 'across', 'through', occurring ... in loanwords from Latin, used in particular for denoting movement or conveyance from place to place (transfer; transmit; transplant) or complete change (transform; transmute), or to form adjectives meaning 'crossing', 'on the other side of', or 'going beyond' the place named (transmontane; transnational; trans-Siberian).

Transgender means to go beyond gender and transhuman means to go beyond human. Both are aspects of the Cult plan to transform the human body to a synthetic state with *no gender*. Human 2.0 is not designed to procreate and would be produced technologically with no need for parents. The new human would mean the end of parents and so men, and increasingly women, are being targeted for the deletion of their rights and status. Parental rights are disappearing at an ever-quickenening speed for the same reason. The new human would have no need for men or women when there is no procreation and no gender. Perhaps the transgender movement that appears to be in a permanent state of frenzy might now contemplate on how it is being used. This was never about transgender rights which are only the interim excuse for confusing gender, particularly in the young, on the road to *fusing* gender. Transgender activism is not an end; it is a *means* to an end. We see again the technique of creative destruction in which you destroy the status quo to 'build back better' in the form that you want. The gender status quo had to be

destroyed by persuading the Cult-created Woke mentality to believe that you can have 100 genders or more. A programme for 9 to 12 year olds produced by the Cult-owned BBC promoted the 100 genders narrative. The very idea may be the most monumental nonsense, but it is not what is true that counts, only what you can make people *believe* is true. Once the gender of $2 + 2 = 4$ has been dismantled through indoctrination, intimidation and $2 + 2 = 5$ then the new no-gender normal can take its place with Human 2.0.

Aldous Huxley revealed the plan in his prophetic *Brave New World* in 1932:

Natural reproduction has been done away with and children are created, decanted', and raised in 'hatcheries and conditioning centres'. From birth, people are genetically designed to fit into one of five castes, which are further split into 'Plus' and 'Minus' members and designed to fulfil predetermined positions within the social and economic strata of the World State.

How could Huxley know this in 1932? For the same reason George Orwell knew about the Big Brother state in 1948, Cult insiders I have quoted knew about it in 1969, and I have known about it since the early 1990s. If you are connected to the Cult or you work your balls off to uncover the plan you can predict the future. The process is simple. If there is a plan for the world and nothing intervenes to stop it then it will happen. Thus if you communicate the plan ahead of time you are perceived to have predicted the future, but you haven't. You have revealed the plan which without intervention will become the human future. The whole reason I have done what I have is to alert enough people to inspire an intervention and maybe at last that time has come with the Cult and its intentions now so obvious to anyone with a brain in working order.

The future is here

Technological wombs that Huxley described to replace parent procreation are already being developed and they are only the projects we know about in the public arena. Israeli scientists told *The Times of Israel* in March, 2021, that they have grown 250-cell embryos

into mouse fetuses with fully formed organs using artificial wombs in a development they say could pave the way for gestating humans outside the womb. Professor Jacob Hanna of the Weizmann Institute of Science said:

We took mouse embryos from the mother at day five of development, when they are just of 250 cells, and had them in the incubator from day five until day 11, by which point they had grown all their organs.

By day 11 they make their own blood and have a beating heart, a fully developed brain. Anybody would look at them and say, 'this is clearly a mouse foetus with all the characteristics of a mouse.' It's gone from being a ball of cells to being an advanced foetus.

A special liquid is used to nourish embryo cells in a laboratory dish and they float on the liquid to duplicate the first stage of embryonic development. The incubator creates all the right conditions for its development, Hanna said. The liquid gives the embryo 'all the nutrients, hormones and sugars they need' along with a custom-made electronic incubator which controls gas concentration, pressure and temperature. The cutting-edge in the underground bases and other secret locations will be light years ahead of that, however, and this was reported by the London *Guardian* in 2017:

We are approaching a biotechnological breakthrough. Ectogenesis, the invention of a complete external womb, could completely change the nature of human reproduction. In April this year, researchers at the Children's Hospital of Philadelphia announced their development of an artificial womb.

The article was headed 'Artificial wombs could soon be a reality. What will this mean for women?' What would it mean for children is an even bigger question. No mother to bond with only a machine in preparation for a life of soulless interaction and control in a world governed by machines (see the *Matrix* movies). Now observe the calculated manipulations of the 'Covid' hoax as human interaction and warmth has been curtailed by distancing, isolation and fear with people communicating via machines on a scale never seen before.

These are all dots in the same picture as are all the personal assistants, gadgets and children's toys through which kids and adults communicate with AI as if it is human. The AI 'voice' on Sat-Nav should be included. All these things are psychological preparation for the Cult endgame. Before you can make a physical connection with AI you have to make a psychological connection and that is what people are being conditioned to do with this ever gathering human-AI interaction. Movies and TV programmes depicting the transhuman, robot dystopia relate to a phenomenon known as 'pre-emptive programming' in which the world that is planned is portrayed everywhere in movies, TV and advertising. This is conditioning the conscious and subconscious mind to become familiar with the planned reality to dilute resistance when it happens for real. What would have been a shock such is the change is made less so. We have young children put on the road to transgender transition surgery with puberty blocking drugs at an age when they could never be able to make those life-changing decisions.

Rachel Levine, a professor of paediatrics and psychiatry who believes in treating children this way, became America's highest-ranked openly-transgender official when she was confirmed as US Assistant Secretary at the Department of Health and Human Services after being nominated by Joe Biden (the Cult). Activists and governments press for laws to deny parents a say in their children's transition process so the kids can be isolated and manipulated into agreeing to irreversible medical procedures. A Canadian father Robert Hoogland was denied bail by the Vancouver Supreme Court in 2021 and remained in jail for breaching a court order that he stay silent over his young teenage daughter, a minor, who was being offered life-changing hormone therapy without parental consent. At the age of 12 the girl's 'school counsellor' said she may be transgender, referred her to a doctor and told the school to treat her like a boy. This is another example of state-serving schools imposing ever more control over children's lives while parents have ever less.

Contemptible and extreme child abuse is happening all over the world as the Cult gender-fusion operation goes into warp-speed.

Why the war on men – and now women?

The question about what artificial wombs mean for women should rightly be asked. The answer can be seen in the deletion of women's rights involving sport, changing rooms, toilets and status in favour of people in male bodies claiming to identify as women. I can identify as a mountain climber, but it doesn't mean I can climb a mountain any more than a biological man can be a biological woman. To believe so is a triumph of belief over factual reality which is the very perceptual basis of everything Woke. Women's sport is being destroyed by allowing those with male bodies who say they identify as female to 'compete' with girls and women. Male body 'women' dominate 'women's' competition with their greater muscle mass, bone density, strength and speed. With that disadvantage sport for women loses all meaning. To put this in perspective nearly 300 American high school boys can run faster than the quickest woman sprinter in the world. Women are seeing their previously protected spaces invaded by male bodies simply because they claim to identify as women. That's all they need to do to access all women's spaces and activities under the Biden 'Equality Act' that destroys equality for women with the usual Orwellian Woke inversion. Male sex offenders have already committed rapes in women's prisons after claiming to identify as women to get them transferred. Does this not matter to the Woke 'equality' hypocrites? Not in the least. What matters to Cult manipulators and funders behind transgender activists is to advance gender fusion on the way to the no-gender 'human'. When you are seeking to impose transparent nonsense like this, or the 'Covid' hoax, the only way the nonsense can prevail is through censorship and intimidation of dissenters, deletion of factual information, and programming of the unquestioning, bewildered and naive. You don't have to scan the world for long to see that all these things are happening.

Many women's rights organisations have realised that rights and status which took such a long time to secure are being eroded and that it is systematic. Kara Dansky of the global Women's Human Rights Campaign said that Biden's transgender executive order immediately he took office, subsequent orders, and Equality Act legislation that followed 'seek to erase women and girls in the law as a category'. *Exactly*. I said during the long ago-started war on men (in which many women play a crucial part) that this was going to turn into a war on them. The Cult is phasing out *both* male and female genders. To get away with that they are brought into conflict so they are busy fighting each other while the Cult completes the job with no unity of response. Unity, people, *unity*. We need unity everywhere. Transgender is the only show in town as the big step towards the no-gender human. It's not about rights for transgender people and never has been. Woke political correctness is deleting words relating to genders to the same end. Wokers believe this is to be 'inclusive' when the opposite is true. They are deleting words describing gender because gender *itself* is being deleted by Human 2.0. Terms like 'man', 'woman', 'mother' and 'father' are being deleted in the universities and other institutions to be replaced by the *no*-gender, not trans-gender, 'individuals' and 'guardians'. Women's rights campaigner Maria Keffler of Partners for Ethical Care said: 'Children are being taught from kindergarten upward that some boys have a vagina, some girls have a penis, and that kids can be any gender they want to be.' Do we really believe that suddenly countries all over the world at the same time had the idea of having drag queens go into schools or read transgender stories to very young children in the local library? It's coldly-calculated confusion of gender on the way to the fusion of gender. Suzanne Vierling, a psychologist from Southern California, made another important point:

Yesterday's slave woman who endured gynecological medical experiments is today's girl-child being butchered in a booming gender-transitioning sector. Ovaries removed, pushing her into menopause and osteoporosis, uncharted territory, and parents' rights and authority decimated.

The erosion of parental rights is a common theme in line with the Cult plans to erase the very concept of parents and 'ovaries removed, pushing her into menopause' means what? Those born female lose the ability to have children – another way to discontinue humanity as we know it.

Eliminating Human 1.0 (before our very eyes)

To pave the way for Human 2.0 you must phase out Human 1.0. This is happening through plummeting sperm counts and making women infertile through an onslaught of chemicals, radiation (including smartphones in pockets of men) and mRNA 'vaccines'. Common agriculture pesticides are also having a devastating impact on human fertility. I have been tracking collapsing sperm counts in the books for a long time and in 2021 came a book by fertility scientist and reproductive epidemiologist Shanna Swan, *Count Down: How Our Modern World Is Threatening Sperm Counts, Altering Male and Female Reproductive Development and Imperiling the Future of the Human Race*. She reports how the global fertility rate dropped by *half* between 1960 and 2016 with America's birth rate 16 percent below where it needs to be to sustain the population. Women are experiencing declining egg quality, more miscarriages, and more couples suffer from infertility. Other findings were an increase in erectile dysfunction, infant boys developing more genital abnormalities, male problems with conception, and plunging levels of the male hormone testosterone which would explain why so many men have lost their backbone and masculinity. This has been very evident during the 'Covid' hoax when women have been prominent among the Pushbackers and big strapping blokes have bowed their heads, covered their faces with a nappy and quietly submitted. Mind control expert Cathy O'Brien also points to how global education introduced the concept of 'we're all winners' in sport and classrooms: 'Competition was defused, and it in turn defused a sense of fighting back.' This is another version of the 'equity' doctrine in which you drive down rather than raise up. What a contrast in Cult-controlled China with its global ambitions

where the government published plans in January, 2021, to 'cultivate masculinity' in boys from kindergarten through to high school in the face of a 'masculinity crisis'. A government adviser said boys would be soon become 'delicate, timid and effeminate' unless action was taken. Don't expect any similar policy in the targeted West. A 2006 study showed that a 65-year-old man in 2002 had testosterone levels 15 percent lower than a 65-year-old man in 1987 while a 2020 study found a similar story with young adults and adolescents. Men are getting prescriptions for testosterone replacement therapy which causes an even greater drop in sperm count with up to 99 percent seeing sperm counts drop to zero during the treatment. More sperm is defective and malfunctioning with some having two heads or not pursuing an egg.

A class of *synthetic* chemicals known as phthalates are being blamed for the decline. These are found everywhere in plastics, shampoos, cosmetics, furniture, flame retardants, personal care products, pesticides, canned foods and even receipts. Why till receipts? Everyone touches them. Let no one delude themselves that all this is not systematic to advance the long-time agenda for human body transformation. Phthalates mimic hormones and disrupt the hormone balance causing testosterone to fall and genital birth defects in male infants. Animals and fish have been affected in the same way due to phthalates and other toxins in rivers. When fish turn gay or change sex through chemicals in rivers and streams it is a pointer to why there has been such an increase in gay people and the sexually confused. It doesn't matter to me what sexuality people choose to be, but if it's being affected by chemical pollution and consumption then we need to know. Does anyone really think that this is not connected to the transgender agenda, the war on men and the condemnation of male 'toxic masculinity'? You watch this being followed by 'toxic femininity'. It's already happening. When breastfeeding becomes 'chest-feeding', pregnant women become pregnant people along with all the other Woke claptrap you know that the world is going insane and there's a Cult scam in progress. Transgender activists are promoting the Cult agenda while Cult

billionaires support and fund the insanity as they laugh themselves to sleep at the sheer stupidity for which humans must be infamous in galaxies far, far away.

‘Covid vaccines’ and female infertility

We can now see why the ‘vaccine’ has been connected to potential infertility in women. Dr Michael Yeadon, former Vice President and Chief Scientific Advisor at Pfizer, and Dr Wolfgang Wodarg in Germany, filed a petition with the European Medicines Agency in December, 2020, urging them to stop trials for the Pfizer/BioNTech shot and all other mRNA trials until further studies had been done. They were particularly concerned about possible effects on fertility with ‘vaccine’-produced antibodies attacking the protein Syncytin-1 which is responsible for developing the placenta. The result would be infertility ‘of indefinite duration’ in women who have the ‘vaccine’ with the placenta failing to form. Section 10.4.2 of the Pfizer/BioNTech trial protocol says that pregnant women or those who might become so should not have mRNA shots. Section 10.4 warns men taking mRNA shots to ‘be abstinent from heterosexual intercourse’ and not to donate sperm. The UK government said that it *did not know* if the mRNA procedure had an effect on fertility. *Did not know?* These people have to go to jail. UK government advice did not recommend at the start that pregnant women had the shot and said they should avoid pregnancy for at least two months after ‘vaccination’. The ‘advice’ was later updated to pregnant women should only have the ‘vaccine’ if the benefits outweighed the risks to mother and foetus. What the hell is that supposed to mean? Then ‘spontaneous abortions’ began to appear and rapidly increase on the adverse reaction reporting schemes which include only a fraction of adverse reactions. Thousands and ever-growing numbers of ‘vaccinated’ women are describing changes to their menstrual cycle with heavier blood flow, irregular periods and menstruating again after going through the menopause – all links to reproduction effects. Women are passing blood clots and the lining of their uterus while men report erectile dysfunction and blood effects. Most

significantly of all *unvaccinated* women began to report similar menstrual changes after interaction with '*vaccinated*' people and men and children were also affected with bleeding noses, blood clots and other conditions. 'Shedding' is when vaccinated people can emit the content of a vaccine to affect the unvaccinated, but this is different. 'Vaccinated' people were not shedding a 'live virus' allegedly in 'vaccines' as before because the fake 'Covid vaccines' involve synthetic material and other toxicity. Doctors exposing what is happening prefer the term 'transmission' to shedding. Somehow those that have had the shots are transmitting effects to those that haven't. Dr Carrie Madej said the nano-content of the 'vaccines' can 'act like an antenna' to others around them which fits perfectly with my own conclusions. This 'vaccine' transmission phenomenon was becoming known as the book went into production and I deal with this further in the Postscript.

Vaccine effects on sterility are well known. The World Health Organization was accused in 2014 of sterilising millions of women in Kenya with the evidence confirmed by the content of the vaccines involved. The same WHO behind the 'Covid' hoax admitted its involvement for more than ten years with the vaccine programme. Other countries made similar claims. Charges were lodged by Tanzania, Nicaragua, Mexico, and the Philippines. The Gardasil vaccine claimed to protect against a genital 'virus' known as HPV has also been linked to infertility. Big Pharma and the WHO (same thing) are criminal and satanic entities. Then there's the Bill Gates Foundation which is connected through funding and shared interests with 20 pharmaceutical giants and laboratories. He stands accused of directing the policy of United Nations Children's Fund (UNICEF), vaccine alliance GAVI, and other groupings, to advance the vaccine agenda and silence opposition at great cost to women and children. At the same time Gates wants to reduce the global population. Coincidence?

Great Reset = Smart Grid = new human

The Cult agenda I have been exposing for 30 years is now being openly promoted by Cult assets like Gates and Klaus Schwab of the World Economic Forum under code-terms like the 'Great Reset', 'Build Back Better' and 'a rare but narrow window of opportunity to reflect, reimagine, and reset our world'. What provided this 'rare but narrow window of opportunity'? The 'Covid' hoax did. Who created that? *They* did. My books from not that long ago warned about the planned 'Internet of Things' (IoT) and its implications for human freedom. This was the plan to connect all technology to the Internet and artificial intelligence and today we are way down that road with an estimated 36 billion devices connected to the World Wide Web and that figure is projected to be 76 billion by 2025. I further warned that the Cult planned to go beyond that to the Internet of *Everything* when the human brain was connected via AI to the Internet and Kurzweil's 'cloud'. Now we have Cult operatives like Schwab calling for precisely that under the term 'Internet of Bodies', a fusion of the physical, digital and biological into one centrally-controlled Smart Grid system which the Cult refers to as the 'Fourth Industrial Revolution'. They talk about the 'biological', but they really mean the synthetic-biological which is required to fully integrate the human body and brain into the Smart Grid and artificial intelligence planned to replace the human mind. We have everything being synthetically manipulated including the natural world through GMO and smart dust, the food we eat and the human body itself with synthetic 'vaccines'. I said in *The Answer* that we would see the Cult push for synthetic meat to replace animals and in February, 2021, the so predictable psychopath Bill Gates called for the introduction of synthetic meat to save us all from 'climate change'. The climate hoax just keeps on giving like the 'Covid' hoax. The war on meat by vegan activists is a carbon (oops, sorry) copy of the manipulation of transgender activists. They have no idea (except their inner core) that they are being used to promote and impose the agenda of the Cult or that they are only the *vehicle* and not the *reason*. This is not to say those who choose not to eat meat shouldn't be respected and supported in that right, but there are ulterior motives

for those in power. A *Forbes* article in December, 2019, highlighted the plan so beloved of Schwab and the Cult under the heading: 'What Is The Internet of Bodies? And How Is It Changing Our World?' The article said the human body is the latest data platform (remember 'our vaccine is an operating system'). *Forbes* described the plan very accurately and the words could have come straight out of my books from long before:

The Internet of Bodies (IoB) is an extension of the IoT and basically connects the human body to a network through devices that are ingested, implanted, or connected to the body in some way. Once connected, data can be exchanged, and the body and device can be remotely monitored and controlled.

They were really describing a human hive mind with human perception centrally-dictated via an AI connection as well as allowing people to be 'remotely monitored and controlled'. Everything from a fridge to a human mind could be directed from a central point by these insane psychopaths and 'Covid vaccines' are crucial to this. *Forbes* explained the process I mentioned earlier of holdable and wearable technology followed by implantable. The article said there were three generations of the Internet of Bodies that include:

- Body external: These are wearable devices such as Apple Watches or Fitbits that can monitor our health.
- Body internal: These include pacemakers, cochlear implants, and digital pills that go inside our bodies to monitor or control various aspects of health.
- Body embedded: The third generation of the Internet of Bodies is embedded technology where technology and the human body are melded together and have a real-time connection to a remote machine.

Forbes noted the development of the Brain Computer Interface (BCI) which merges the brain with an external device for monitoring and controlling in real-time. 'The ultimate goal is to help restore function to individuals with disabilities by using brain signals rather than conventional neuromuscular pathways.' Oh, do fuck off. The goal of brain interface technology is controlling human thought and emotion from the central point in a hive mind serving its masters wishes. Many people are now agreeing to be chipped to open doors without a key. You can recognise them because they'll be wearing a mask, social distancing and lining up for the 'vaccine'. The Cult plans a Great Reset money system after they have completed the demolition of the global economy in which 'money' will be exchanged through communication with body operating systems. Rand Corporation, a Cult-owned think tank, said of the Internet of Bodies or IoB:

Internet of Bodies technologies fall under the broader IoT umbrella. But as the name suggests, IoB devices introduce an even more intimate interplay between humans and gadgets. IoB devices monitor the human body, collect health metrics and other personal information, and transmit those data over the Internet. Many devices, such as fitness trackers, are already in use ... IoB devices ... and those in development can track, record, and store users' whereabouts, bodily functions, and what they see, hear, and even think.

Schwab's World Economic Forum, a long-winded way of saying 'fascism' or 'the Cult', has gone full-on with the Internet of Bodies in the 'Covid' era. 'We're entering the era of the Internet of Bodies', it declared, 'collecting our physical data via a range of devices that can be implanted, swallowed or worn'. The result would be a huge amount of health-related data that could improve human wellbeing around the world, and prove crucial in fighting the 'Covid-19 pandemic'. Does anyone think these clowns care about 'human wellbeing' after the death and devastation their pandemic hoax has purposely caused? Schwab and co say we should move forward with the Internet of Bodies because 'Keeping track of symptoms could help us stop the spread of infection, and quickly detect new cases'. How wonderful, but keeping track' is all they are really bothered

about. Researchers were investigating if data gathered from smartwatches and similar devices could be used as viral infection alerts by tracking the user's heart rate and breathing. Schwab said in his 2018 book *Shaping the Future of the Fourth Industrial Revolution*:

The lines between technologies and beings are becoming blurred and not just by the ability to create lifelike robots or synthetics. Instead it is about the ability of new technologies to literally become part of us. Technologies already influence how we understand ourselves, how we think about each other, and how we determine our realities. As the technologies ... give us deeper access to parts of ourselves, we may begin to integrate digital technologies into our bodies.

You can see what the game is. Twenty-four hour control and people – if you could still call them that – would never know when something would go ping and take them out of circulation. It's the most obvious rush to a global fascist dictatorship and the complete submission of humanity and yet still so many are locked away in their Cult-induced perceptual coma and can't see it.

Smart Grid control centres

The human body is being transformed by the 'vaccines' and in other ways into a synthetic cyborg that can be attached to the global Smart Grid which would be controlled from a central point and other sub-locations of Grid manipulation. Where are these planned to be? Well, China for a start which is one of the Cult's biggest centres of operation. The technological control system and technocratic rule was incubated here to be unleashed across the world after the 'Covid' hoax came out of China in 2020. Another Smart Grid location that will surprise people new to this is Israel. I have exposed in *The Trigger* how Sabbatian technocrats, intelligence and military operatives were behind the horrors of 9/11 and not 19 Arab hijackers' who somehow manifested the ability to pilot big passenger airliners when instructors at puddle-jumping flying schools described some of them as a joke. The 9/11 attacks were made possible through control of civilian and military air computer systems and those of the White House, Pentagon and connected agencies. See *The Trigger* – it

will blow your mind. The controlling and coordinating force were the Sabbatian networks in Israel and the United States which by then had infiltrated the entire US government, military and intelligence system. The real name of the American Deep State is 'Sabbatian State'. Israel is a tiny country of only nine million people, but it is one of the global centres of cyber operations and fast catching Silicon Valley in importance to the Cult. Israel is known as the 'start-up nation' for all the cyber companies spawned there with the Sabbatian specialisation of 'cyber security' that I mentioned earlier which gives those companies access to computer systems of their clients in real time through 'backdoors' written into the coding when security software is downloaded. The Sabbatian centre of cyber operations outside Silicon Valley is the Israeli military Cyber Intelligence Unit, the biggest infrastructure project in Israel's history, headquartered in the desert-city of Beersheba and involving some 20,000 'cyber soldiers'. Here are located a literal army of Internet trolls scanning social media, forums and comment lists for anyone challenging the Cult agenda. The UK military has something similar with its 77th Brigade and associated operations. The Beersheba complex includes research and development centres for other Cult operations such as Intel, Microsoft, IBM, Google, Apple, Hewlett-Packard, Cisco Systems, Facebook and Motorola. Techcrunch.com ran an article about the Beersheba global Internet technology centre headlined 'Israel's desert city of Beersheba is turning into a cybertech oasis':

The military's massive relocation of its prestigious technology units, the presence of multinational and local companies, a close proximity to Ben Gurion University and generous government subsidies are turning Beersheba into a major global cybertech hub. Beersheba has all of the ingredients of a vibrant security technology ecosystem, including Ben Gurion University with its graduate program in cybersecurity and Cyber Security Research Center, and the presence of companies such as EMC, Deutsche Telekom, PayPal, Oracle, IBM, and Lockheed Martin. It's also the future home of the INCB (Israeli National Cyber Bureau); offers a special income tax incentive for cyber security companies, and was the site for the relocation of the army's intelligence corps units.

Sabbatians have taken over the cyber world through the following process: They scan the schools for likely cyber talent and develop them at Ben Gurion University and their period of conscription in the Israeli Defense Forces when they are stationed at the Beersheba complex. When the cyber talented officially leave the army they are funded to start cyber companies with technology developed by themselves or given to them by the state. Much of this is stolen through backdoors of computer systems around the world with America top of the list. Others are sent off to Silicon Valley to start companies or join the major ones and so we have many major positions filled by apparently 'Jewish' but really Sabbatian operatives. Google, YouTube and Facebook are all run by 'Jewish' CEOs while Twitter is all but run by ultra-Zionist hedge-fund shark Paul Singer. At the centre of the Sabbatian global cyber web is the Israeli army's Unit 8200 which specialises in hacking into computer systems of other countries, inserting viruses, gathering information, instigating malfunction, and even taking control of them from a distance. A long list of Sabbatians involved with 9/11, Silicon Valley and Israeli cyber security companies are operatives of Unit 8200. This is not about Israel. It's about the Cult. Israel is planned to be a Smart Grid hub as with China and what is happening at Beersheba is not for the benefit of Jewish people who are treated disgustingly by the Sabbatian elite that control the country. A glance at the Nuremberg Codes will tell you that.

The story is much bigger than 'Covid', important as that is to where we are being taken. Now, though, it's time to really strap in. There's more ... much more ...

CHAPTER ELEVEN

Who controls the Cult?

Awake, arise or be forever fall'n
John Milton, *Paradise Lost*

I have exposed this far the level of the Cult conspiracy that operates in the world of the seen and within the global secret society and satanic network which operates in the shadows one step back from the seen. The story, however, goes much deeper than that.

The 'Covid' hoax is major part of the Cult agenda, but only part, and to grasp the biggest picture we have to expand our attention beyond the realm of human sight and into the infinity of possibility that we cannot see. It is from here, ultimately, that humanity is being manipulated into a state of total control by the force which dictates the actions of the Cult. How much of reality can we see? Next to damn all is the answer. We may appear to see all there is to see in the 'space' our eyes survey and observe, but little could be further from the truth. The human 'world' is only a tiny band of frequency that the body's visual and perceptual systems can decode into *perception* of a 'world'. According to mainstream science the electromagnetic spectrum is 0.005 percent of what exists in the Universe ([Fig 10](#)). The maximum estimate I have seen is 0.5 percent and either way it's miniscule. I say it is far, far, smaller even than 0.005 percent when you compare reality we see with the totality of reality that we don't. Now get this if you are new to such information: Visible light, the only band of frequency that we can see, is a *fraction* of the 0.005

percent (Fig 11 overleaf). Take this further and realise that our universe is one of infinite universes and that universes are only a fragment of overall reality – *infinite* reality. Then compare that with the almost infinitesimal frequency band of visible light or human sight. You see that humans are as near blind as it is possible to be without actually being so. Artist and filmmaker, Sergio Toporek, said:

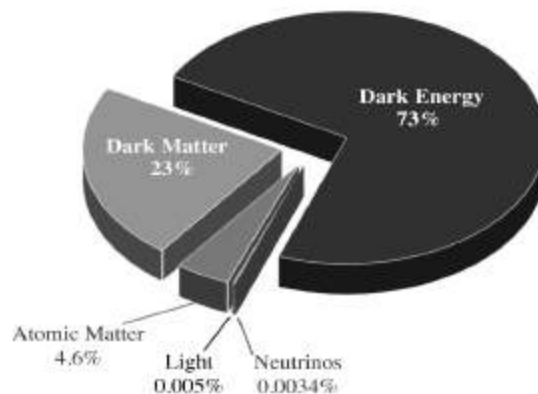


Figure 10: Humans can perceive such a tiny band of visual reality it's laughable.

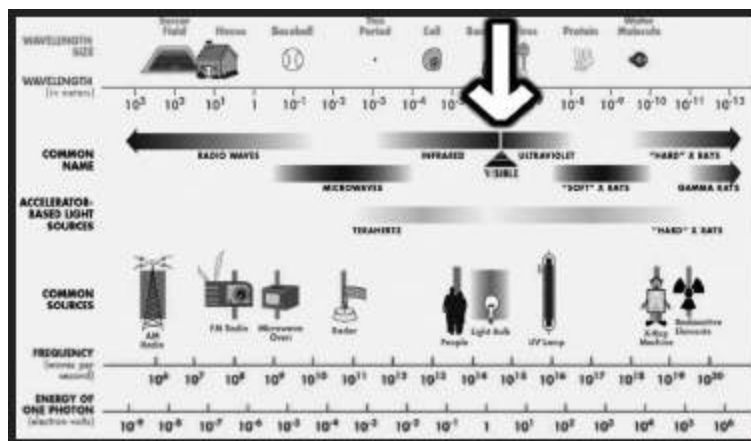


Figure 11: We can see a smear of the 0.005 percent electromagnetic spectrum, but we still know it all. Yep, makes sense.

Consider that you can see less than 1% of the electromagnetic spectrum and hear less than 1% of the acoustic spectrum. 90% of the cells in your body carry their own microbial DNA and are not 'you'. The atoms in your body are 99.9999999999999999% empty space and none of them are the ones you were born with ... Human beings have 46 chromosomes, two less than a potato.

The existence of the rainbow depends on the conical photoreceptors in your eyes; to animals without cones, the rainbow does not exist. So you don't just look at a rainbow, you create it. This is pretty amazing, especially considering that all the beautiful colours you see represent less than 1% of the electromagnetic spectrum.

Suddenly the 'world' of humans looks a very different place. Take into account, too, that Planet Earth when compared with the projected size of this single universe is the equivalent of a billionth of a pinhead. Imagine the ratio that would be when compared to infinite reality. To think that Christianity once insisted that Earth and humanity were the centre of everything. This background is vital if we are going to appreciate the nature of 'human' and how we can be manipulated by an unseen force. To human visual reality virtually *everything* is unseen and yet the prevailing perception within the institutions and so much of the public is that if we can't see it, touch it, hear it, taste it and smell it then it cannot exist. Such perception is indoctrinated and encouraged by the Cult and its agents because it isolates believers in the strictly limited, village-idiot, realm of the five senses where perceptions can be firewalled and information controlled. Most of those perpetuating the 'this-world-is-all-there-is' insanity are themselves indoctrinated into believing the same delusion. While major players and influencers know that official reality is laughable most of those in science, academia and medicine really believe the nonsense they peddle and teach succeeding generations. Those who challenge the orthodoxy are dismissed as nutters and freaks to protect the manufactured illusion from exposure. Observe the dynamic of the 'Covid' hoax and you will see how that takes the same form. The inner-circle psychopaths knows it's a gigantic scam, but almost the entirety of those imposing their fascist rules believe that 'Covid' is all that they're told it is.

Stolen identity

Ask people who they are and they will give you their name, place of birth, location, job, family background and life story. Yet that is not who they are – it is what they are *experiencing*. The difference is *absolutely crucial*. The true 'I', the eternal, infinite 'I', is consciousness,

a state of being aware. Forget 'form'. That is a vehicle for a brief experience. Consciousness does not come *from* the brain, but *through* the brain and even that is more symbolic than literal. We are awareness, pure awareness, and this is what withdraws from the body at what we call 'death' to continue our eternal beingness, *isness*, in other realms of reality within the limitlessness of infinity or the Biblical 'many mansions in my father's house'. Labels of a human life, man, woman, transgender, black, white, brown, nationality, circumstances and income are not who we are. They are what we are – awareness – is *experiencing* in a brief connection with a band of frequency we call 'human'. The labels are not the self; they are, to use the title of one of my books, a *Phantom Self*. I am not David Icke born in Leicester, England, on April 29th, 1952. I am the consciousness *having that experience*. The Cult and its non-human masters seek to convince us through the institutions of 'education', science, medicine, media and government that what we are *experiencing* is who we *are*. It's so easy to control and direct perception locked away in the bewildered illusions of the five senses with no expanded radar. Try, by contrast, doing the same with a humanity aware of its true self and its true power to consciously create its reality and experience. How is it possible to do this? We do it all day every day. If you perceive yourself as 'little me' with no power to impact upon your life and the world then your life experience will reflect that. You will hand the power you don't think you have to authority in all its forms which will use it to control your experience. This, in turn, will appear to confirm your perception of 'little me' in a self-fulfilling feedback loop. But that is what 'little me' really is – a *perception*. We are all 'big-me', infinite me, and the Cult has to make us forget that if its will is to prevail. We are therefore manipulated and pressured into self-identifying with human labels and not the consciousness/awareness *experiencing* those human labels.

The phenomenon of identity politics is a Cult-instigated manipulation technique to sub-divide previous labels into even smaller ones. A United States university employs this list of letters to

describe student identity: LGBTTQQFAGPBDSM or lesbian, gay, bisexual, transgender, transsexual, queer, questioning, flexual, asexual, gender-fuck, polyamorous, bondage/discipline, dominance/submission and sadism/masochism. I'm sure other lists are even longer by now as people feel the need to self-identity the 'I' with the minutiae of race and sexual preference. Wokers programmed by the Cult for generations believe this is about 'inclusivity' when it's really the Cult locking them away into smaller and smaller versions of Phantom Self while firewalling them from the influence of their true self, the infinite, eternal 'I'. You may notice that my philosophy which contends that we are all unique points of attention/awareness within the same infinite whole or Oneness is the ultimate non-racism. The very sense of Oneness makes the judgement of people by their body-type, colour or sexuality utterly ridiculous and confirms that racism has no understanding of reality (including anti-white racism). Yet despite my perception of life Cult agents and fast-asleep Wokers label me racist to discredit my information while they are themselves phenomenally racist and sexist. All they see is race and sexuality and they judge people as good or bad, demons or untouchables, by their race and sexuality. All they see is *Phantom Self* and perceive themselves in terms of Phantom Self. They are pawns and puppets of the Cult agenda to focus attention and self-identity in the five senses and play those identities against each other to divide and rule. Columbia University has introduced segregated graduations in another version of social distancing designed to drive people apart and teach them that different racial and cultural groups have nothing in common with each other. The last thing the Cult wants is unity. Again the pump-primers of this will be Cult operatives in the knowledge of what they are doing, but the rest are just the Phantom Self blind leading the Phantom Self blind. We *do* have something in common – we are all *the same consciousness* having different temporary experiences.

What is this 'human'?

Yes, what *is* 'human'? That is what we are supposed to be, right? I mean 'human'? True, but 'human' is the experience not the 'I'. Break it down to basics and 'human' is the way that information is processed. If we are to experience and interact with this band of frequency we call the 'world' we must have a vehicle that operates within that band of frequency. Our consciousness in its prime form cannot do that; it is way beyond the frequency of the human realm. My consciousness or awareness could not tap these keys and pick up the cup in front of me in the same way that radio station A cannot interact with radio station B when they are on different frequencies. The human body is the means through which we have that interaction. I have long described the body as a biological computer which processes information in a way that allows consciousness to experience this reality. The body is a receiver, transmitter and processor of information in a particular way that we call human. We visually perceive only the world of the five senses in a wakened state – that is the limit of the body's visual decoding system. In truth it's not even visual in the way we experience 'visual reality' as I will come to in a moment. We are 'human' because the body processes the information sources of human into a reality and behaviour system that we *perceive* as human. Why does an elephant act like an elephant and not like a human or a duck? The elephant's biological computer is a different information field and processes information according to that program into a visual and behaviour type we call an elephant. The same applies to everything in our reality. These body information fields are perpetuated through procreation (like making a copy of a software program). The Cult wants to break that cycle and intervene technologically to transform the human information field into one that will change what we call humanity. If it can change the human information field it will change the way that field processes information and change humanity both 'physically' and psychologically. Hence the *messenger* (information) RNA 'vaccines' and so much more that is targeting human genetics by changing the body's information – *messaging* – construct through food, drink, radiation, toxicity and other means.

Reality that we experience is nothing like reality as it really is in the same way that the reality people experience in virtual reality games is not the reality they are really living in. The game is only a decoded source of information that appears to be a reality. Our world is also an information construct – a *simulation* (more later). In its base form our reality is a wavefield of information much the same in theme as Wi-Fi. The five senses decode wavefield information into electrical information which they communicate to the brain to decode into holographic (illusory ‘physical’) information. Different parts of the brain specialise in decoding different senses and the information is fused into a reality that appears to be outside of us but is really inside the brain and the genetic structure in general ([Fig 12](#) overleaf). DNA is a receiver-transmitter of information and a vital part of this decoding process and the body’s connection to other realities. Change DNA and you change the way we decode and connect with reality – see ‘Covid vaccines’. Think of computers decoding Wi-Fi. You have information encoded in a radiation field and the computer decodes that information into a very different form on the screen. You can’t see the Wi-Fi until its information is made manifest on the screen and the information on the screen is inside the computer and not outside. I have just described how we decode the ‘human world’. All five senses decode the waveform ‘Wi-Fi’ field into electrical signals and the brain (computer) constructs reality inside the brain and not outside – ‘You don’t just look at a rainbow, you create it’. Sound is a simple example. We don’t hear sound until the brain decodes it. Waveform sound waves are picked up by the hearing sense and communicated to the brain in an electrical form to be decoded into the sounds that we hear. Everything we hear is inside the brain along with everything we see, feel, smell and taste. Words and language are waveform fields generated by our vocal chords which pass through this process until they are decoded by the brain into words that we hear. Different languages are different frequency fields or sound waves generated by vocal chords. Late British philosopher Alan Watts said:

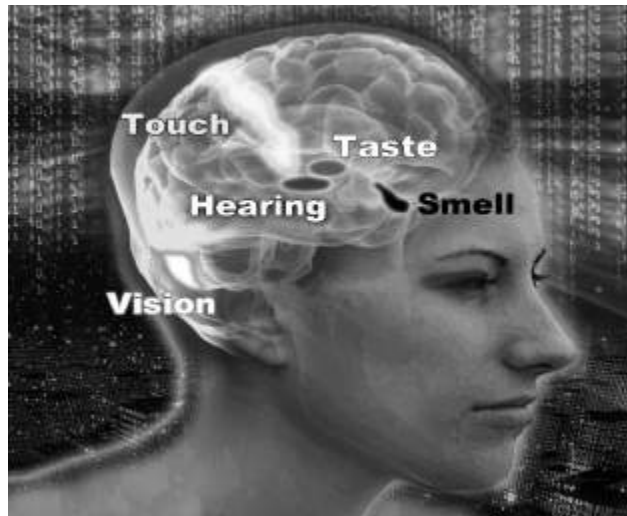


Figure 12: The brain receives information from the five senses and constructs from that our perceived reality.

[Without the brain] the world is devoid of light, heat, weight, solidity, motion, space, time or any other imaginable feature. All these phenomena are interactions, or transactions, of vibrations with a certain arrangement of neurons.

That's exactly what they are and scientist Robert Lanza describes in his book, *Biocentrism*, how we decode electromagnetic waves and energy into visual and 'physical' experience. He uses the example of a flame emitting photons, electromagnetic energy, each pulsing electrically and magnetically:

... these ... invisible electromagnetic waves strike a human retina, and if (and only if) the waves happen to measure between 400 and 700 nano meters in length from crest to crest, then their energy is just right to deliver a stimulus to the 8 million cone-shaped cells in the retina.

Each in turn send an electrical pulse to a neighbour neuron, and on up the line this goes, at 250 mph, until it reaches the ... occipital lobe of the brain, in the back of the head. There, a cascading complex of neurons fire from the incoming stimuli, and we subjectively perceive this experience as a yellow brightness occurring in a place we have been conditioned to call the 'external world'.

You hear what you decode

If a tree falls or a building collapses they make no noise unless someone is there to decode the energetic waves generated by the disturbance into what we call sound. Does a falling tree make a noise? Only if you hear it – *decode* it. Everything in our reality is a frequency field of information operating within the overall ‘Wi-Fi’ field that I call The Field. A vibrational disturbance is generated in The Field by the fields of the falling tree or building. These disturbance waves are what we decode into the sound of them falling. If no one is there to do that then neither will make any noise. Reality is created by the observer – *decoder* – and the *perceptions* of the observer affect the decoding process. For this reason different people – different *perceptions* – will perceive the same reality or situation in a different way. What one may perceive as a nightmare another will see as an opportunity. The question of why the Cult is so focused on controlling human perception now answers itself. All experienced reality is the act of decoding and we don’t experience Wi-Fi until it is decoded on the computer screen. The sight and sound of an Internet video is encoded in the Wi-Fi all around us, but we don’t see or hear it until the computer decodes that information. Taste, smell and touch are all phenomena of the brain as a result of the same process. We don’t taste, smell or feel anything except in the brain and there are pain relief techniques that seek to block the signal from the site of discomfort to the brain because if the brain doesn’t decode that signal we don’t feel pain. Pain is in the brain and only appears to be at the point of impact thanks to the feedback loop between them. We don’t see anything until electrical information from the sight senses is decoded in an area at the back of the brain. If that area is damaged we can go blind when our eyes are perfectly okay. So why do we go blind if we damage an eye? We damage the information processing between the waveform visual information and the visual decoding area of the brain. If information doesn’t reach the brain in a form it can decode then we can’t see the visual reality that it represents. What’s more the brain is decoding only a fraction of the information it receives and the rest is absorbed by the

sub-conscious mind. This explanation is from the science magazine, *Wonderpedia*:

Every second, 11 million sensations crackle along these [brain] pathways ... The brain is confronted with an alarming array of images, sounds and smells which it rigorously filters down until it is left with a manageable list of around 40. Thus 40 sensations per second make up what we perceive as reality.

The 'world' is not what people are told to believe that is it and the inner circles of the Cult *know that*.

Illusory 'physical' reality

We can only see a smear of 0.005 percent of the Universe which is only one of a vast array of universes – 'mansions' – within infinite reality. Even then the brain decodes only 40 pieces of information ('sensations') from a potential *11 million* that we receive every second. Two points strike you from this immediately: The sheer breathtaking stupidity of believing we know anything so rigidly that there's nothing more to know; and the potential for these processes to be manipulated by a malevolent force to control the reality of the population. One thing I can say for sure with no risk of contradiction is that when you can perceive an almost indescribable fraction of infinite reality there is always more to know as in tidal waves of it. Ancient Greek philosopher Socrates was so right when he said that wisdom is to know how little we know. How obviously true that is when you think that we are experiencing a physical world of solidity that is neither physical nor solid and a world of apartness when everything is connected. Cult-controlled 'science' dismisses the so-called 'paranormal' and all phenomena related to that when the 'para'-normal is perfectly normal and explains the alleged 'great mysteries' which dumbfound scientific minds. There is a reason for this. A 'scientific mind' in terms of the mainstream is a material mind, a five-sense mind imprisoned in see it, touch it, hear it, smell it and taste it. Phenomena and happenings that can't be explained that way leave the 'scientific mind' bewildered and the rule is that if they

can't account for why something is happening then it can't, by definition, be happening. I beg to differ. Telepathy is thought waves passing through The Field (think wave disturbance again) to be decoded by someone able to connect with that wavelength (information). For example: You can pick up the thought waves of a friend at any distance and at the very least that will bring them to mind. A few minutes later the friend calls you. 'My god', you say, 'that's incredible – I was just thinking of you.' Ah, but *they* were thinking of *you* before they made the call and that's what you decoded. Native peoples not entrapped in five-sense reality do this so well it became known as the 'bush telegraph'. Those known as psychics and mediums (genuine ones) are doing the same only across dimensions of reality. 'Mind over matter' comes from the fact that matter and mind are the *same*. The state of one influences the state of the other. Indeed one *and* the other are illusions. They are aspects of the same field. Paranormal phenomena are all explainable so why are they still considered 'mysteries' or not happening? Once you go down this road of understanding you begin to expand awareness beyond the five senses and that's the nightmare for the Cult.



Figure 13: Holograms are not solid, but the best ones appear to be.

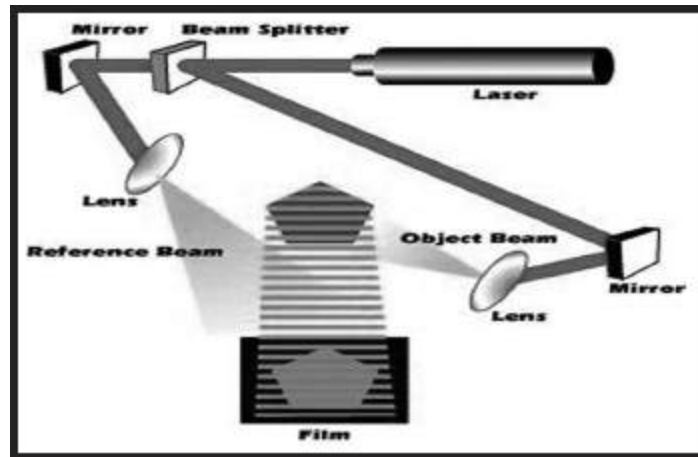


Figure 14: How holograms are created by capturing a waveform version of the subject image.

Holographic 'solidity'

Our reality is not solid, it is holographic. We are now well aware of holograms which are widely used today. Two-dimensional information is decoded into a three-dimensional reality that is not solid although can very much appear to be (Fig 13). Holograms are created with a laser divided into two parts. One goes directly onto a holographic photographic print ('reference beam') and the other takes a waveform image of the subject ('working beam') before being directed onto the print where it 'collides' with the other half of the laser (Fig 14). This creates a *waveform* interference pattern which contains the wavefield information of whatever is being photographed (Fig 15 overleaf). The process can be likened to dropping pebbles in a pond. Waves generated by each one spread out across the water to collide with the others and create a wave representation of where the stones fell and at what speed, weight and distance. A waveform interference pattern of a hologram is akin to the waveform information in The Field which the five senses decode into electrical signals to be decoded by the brain into a holographic illusory 'physical' reality. In the same way when a laser (think human attention) is directed at the waveform interference pattern a three-dimensional version of the subject is projected into apparently 'solid' reality (Fig 16). An amazing trait of holograms reveals more 'paranormal mysteries'. Information of the *whole*

hologram is encoded in waveform in every part of the interference pattern by the way they are created. This means that every *part* of a hologram is a smaller version of the whole. Cut the interference wave-pattern into four and you won't get four parts of the image. You get quarter-sized versions of the *whole* image. The body is a hologram and the same applies. Here we have the basis of acupuncture, reflexology and other forms of healing which identify representations of the whole body in all of the parts, hands, feet, ears, everywhere. Skilled palm readers can do what they do because the information of whole body is encoded in the hand. The concept of as above, so below, comes from this.



Figure 15: A waveform interference pattern that holds the information that transforms into a hologram.



Figure 16: Holographic people including 'Elvis' holographically inserted to sing a duet with Celine Dion.

The question will be asked of why, if solidity is illusory, we can't just walk through walls and each other. The resistance is not solid against solid; it is electromagnetic field against electromagnetic field and we decode this into the *experience* of solid against solid. We should also not underestimate the power of belief to dictate reality. What you believe is impossible *will be*. Your belief impacts on your decoding processes and they won't decode what you think is impossible. What we believe we perceive and what we perceive we experience. 'Can't dos' and 'impossibles' are like a firewall in a computer system that won't put on the screen what the firewall blocks. How vital that is to understanding how human experience has been hijacked. I explain in *The Answer, Everything You Need To Know But Have Never Been Told* and other books a long list of 'mysteries' and 'paranormal' phenomena that are not mysterious and perfectly normal once you realise what reality is and how it works. 'Ghosts' can be seen to pass through 'solid' walls because the walls are not solid and the ghost is a discarnate entity operating on a frequency so different to that of the wall that it's like two radio stations sharing the same space while never interfering with each other. I have seen ghosts do this myself. The apartness of people and objects is also an illusion. Everything is connected by the Field like all sea life is connected by the sea. It's just that within the limits of our visual reality we only 'see' holographic information and not the field of information that connects everything and from which the holographic world is made manifest. If you can only see holographic 'objects' and not the field that connects them they will appear to you as unconnected to each other in the same way that we see the computer while not seeing the Wi-Fi.

What you don't know *can* hurt you

Okay, we return to those 'two worlds' of human society and the Cult with its global network of interconnecting secret societies and satanic groups which manipulate through governments, corporations, media, religions, etc. The fundamental difference between them is *knowledge*. The idea has been to keep humanity

ignorant of the plan for its total enslavement underpinned by a crucial ignorance of reality – who we are and where we are – and how we interact with it. ‘Human’ should be the interaction between our expanded eternal consciousness and the five-sense body experience. We are meant to be *in* this world in terms of the five senses but not *of* this world in relation to our greater consciousness and perspective. In that state we experience the small picture of the five senses within the wider context of the big picture of awareness beyond the five senses. Put another way the five senses see the dots and expanded awareness connects them into pictures and patterns that give context to the apparently random and unconnected. Without the context of expanded awareness the five senses see only apartness and randomness with apparently no meaning. The Cult and its other-dimensional controllers seek to intervene in the frequency realm where five-sense reality is supposed to connect with expanded reality and to keep the two apart (more on this in the final chapter). When that happens five-sense mental and emotional processes are no longer influenced by expanded awareness, or the True ‘I’, and instead are driven by the isolated perceptions of the body’s decoding systems. They are in the world *and* of it. Here we have the human plight and why humanity with its potential for infinite awareness can be so easily manipulatable and descend into such extremes of stupidity.

Once the Cult isolates five-sense mind from expanded awareness it can then program the mind with perceptions and beliefs by controlling information that the mind receives through the ‘education’ system of the formative years and the media perceptual bombardment and censorship of an entire lifetime. Limit perception and a sense of the possible through limiting knowledge by limiting and skewing information while censoring and discrediting that which could set people free. As the title of another of my books says ... *And The Truth Shall Set You Free*. For this reason the last thing the Cult wants in circulation is the truth about anything – especially the reality of the eternal ‘I’ – and that’s why it is desperate to control information. The Cult knows that information becomes perception

which becomes behaviour which, collectively, becomes human society. Cult-controlled and funded mainstream 'science' denies the existence of an eternal 'I' and seeks to dismiss and trash all evidence to the contrary. Cult-controlled mainstream religion has a version of 'God' that is little more than a system of control and dictatorship that employs threats of damnation in an afterlife to control perceptions and behaviour in the here and now through fear and guilt. Neither is true and it's the 'neither' that the Cult wishes to suppress. This 'neither' is that everything is an expression, a point of attention, within an infinite state of consciousness which is the real meaning of the term 'God'.

Perceptual obsession with the 'physical body' and five-senses means that 'God' becomes personified as a bearded bloke sitting among the clouds or a raging bully who loves us if we do what 'he' wants and condemns us to the fires of hell if we don't. These are no more than a 'spiritual' fairy tales to control and dictate events and behaviour through fear of this 'God' which has bizarrely made 'God-fearing' in religious circles a state to be desired. I would suggest that fearing *anything* is not to be encouraged and celebrated, but rather deleted. You can see why 'God fearing' is so beneficial to the Cult and its religions when *they* decide what 'God' wants and what 'God' demands (the Cult demands) that everyone do. As the great American comedian Bill Hicks said satirising a Christian zealot: 'I think what God meant to say.' How much of this infinite awareness ('God') that we access is decided by how far we choose to expand our perceptions, self-identity and sense of the possible. The scale of self-identity reflects itself in the scale of awareness that we can connect with and are influenced by – how much knowing and insight we have instead of programmed perception. You cannot expand your awareness into the infinity of possibility when you believe that you are little me Peter the postman or Mary in marketing and nothing more. I'll deal with this in the concluding chapter because it's crucial to how we turnaround current events.

Where the Cult came from

When I realised in the early 1990s there was a Cult network behind global events I asked the obvious question: When did it start? I took it back to ancient Rome and Egypt and on to Babylon and Sumer in Mesopotamia, the 'Land Between Two Rivers', in what we now call Iraq. The two rivers are the Tigris and Euphrates and this region is of immense historical and other importance to the Cult, as is the land called Israel only 550 miles away by air. There is much more going on with deep esoteric meaning across this whole region. It's not only about 'wars for oil'. Priceless artefacts from Mesopotamia were stolen or destroyed after the American and British invasion of Iraq in 2003 justified by the lies of Boy Bush and Tony Blair (their Cult masters) about non-existent 'weapons of mass destruction'.

Mesopotamia was the location of Sumer (about 5,400BC to 1,750BC), and Babylon (about 2,350BC to 539BC). Sabbatians may have become immensely influential in the Cult in modern times but they are part of a network that goes back into the mists of history. Sumer is said by historians to be the 'cradle of civilisation'. I disagree. I say it was the re-start of what we call human civilisation after cataclysmic events symbolised in part as the 'Great Flood' destroyed the world that existed before. These fantastic upheavals that I have been describing in detail in the books since the early 1990s appear in accounts and legends of ancient cultures across the world and they are supported by geological and biological evidence. Stone tablets found in Iraq detailing the Sumer period say the cataclysms were caused by non-human 'gods' they call the Anunnaki. These are described in terms of extraterrestrial visitations in which knowledge supplied by the Anunnaki is said to have been the source of at least one of the world's oldest writing systems and developments in astronomy, mathematics and architecture that were way ahead of their time. I have covered this subject at length in *The Biggest Secret* and *Children of the Matrix* and the same basic 'Anunnaki' story can be found in Zulu accounts in South Africa where the late and very great Zulu high shaman Credo Mutwa told me that the Sumerian Anunnaki were known by Zulus as the Chitauri or 'children of the serpent'. See my six-hour video interview with Credo on this subject entitled *The*

Reptilian Agenda recorded at his then home near Johannesburg in 1999 which you can watch on the Ickonic media platform.

The Cult emerged out of Sumer, Babylon and Egypt (and elsewhere) and established the Roman Empire before expanding with the Romans into northern Europe from where many empires were savagely imposed in the form of Cult-controlled societies all over the world. Mass death and destruction was their calling card. The Cult established its centre of operations in Europe and European Empires were Cult empires which allowed it to expand into a global force. Spanish and Portuguese colonialists headed for Central and South America while the British and French targeted North America. Africa was colonised by Britain, France, Belgium, the Netherlands, Portugal, Spain, Italy, and Germany. Some like Britain and France moved in on the Middle East. The British Empire was by far the biggest for a simple reason. By now Britain was the headquarters of the Cult from which it expanded to form Canada, the United States, Australia and New Zealand. The Sun never set on the British Empire such was the scale of its occupation. London remains a global centre for the Cult along with Rome and the Vatican although others have emerged in Israel and China. It is no accident that the 'virus' is alleged to have come out of China while Italy was chosen as the means to terrify the Western population into compliance with 'Covid' fascism. Nor that Israel has led the world in 'Covid' fascism and mass 'vaccination'.

You would think that I would mention the United States here, but while it has been an important means of imposing the Cult's will it is less significant than would appear and is currently in the process of having what power it does have deleted. The Cult in Europe has mostly loaded the guns for the US to fire. America has been controlled from Europe from the start through Cult operatives in Britain and Europe. The American Revolution was an illusion to make it appear that America was governing itself while very different forces were pulling the strings in the form of Cult families such as the Rothschilds through the Rockefellers and other subordinates. The Rockefellers are extremely close to Bill Gates and

established both scalpel and drug 'medicine' and the World Health Organization. They play a major role in the development and circulation of vaccines through the Rockefeller Foundation on which Bill Gates said his Foundation is based. Why wouldn't this be the case when the Rockefellers and Gates are on the same team? Cult infiltration of human society goes way back into what we call history and has been constantly expanding and centralising power with the goal of establishing a global structure to dictate everything. Look how this has been advanced in great leaps with the 'Covid' hoax.

The non-human dimension

I researched and observed the comings and goings of Cult operatives through the centuries and even thousands of years as they were born, worked to promote the agenda within the secret society and satanic networks, and then died for others to replace them. Clearly there had to be a coordinating force that spanned this entire period while operatives who would not have seen the end goal in their lifetimes came and went advancing the plan over millennia. I went in search of that coordinating force with the usual support from the extraordinary synchronicity of my life which has been an almost daily experience since 1990. I saw common themes in religious texts and ancient cultures about a non-human force manipulating human society from the hidden. Christianity calls this force Satan, the Devil and demons; Islam refers to the Jinn or Djinn; Zulus have their Chitauri (spelt in other ways in different parts of Africa); and the Gnostic people in Egypt in the period around and before 400AD referred to this phenomena as the 'Archons', a word meaning rulers in Greek. Central American cultures speak of the 'Predators' among other names and the same theme is everywhere. I will use 'Archons' as a collective name for all of them. When you see how their nature and behaviour is described all these different sources are clearly talking about the same force. Gnostics described the Archons in terms of 'luminous fire' while Islam relates the Jinn to 'smokeless fire'. Some refer to beings in form that could occasionally be seen, but the most common of common theme is that they operate from

unseen realms which means almost all existence to the visual processes of humans. I had concluded that this was indeed the foundation of human control and that the Cult was operating within the human frequency band on behalf of this hidden force when I came across the writings of Gnostics which supported my conclusions in the most extraordinary way.

A sealed earthen jar was found in 1945 near the town of Nag Hammadi about 75-80 miles north of Luxor on the banks of the River Nile in Egypt. Inside was a treasure trove of manuscripts and texts left by the Gnostic people some 1,600 years earlier. They included 13 leather-bound papyrus codices (manuscripts) and more than 50 texts written in Coptic Egyptian estimated to have been hidden in the jar in the period of 400AD although the source of the information goes back much further. Gnostics oversaw the Great or Royal Library of Alexandria, the fantastic depository of ancient texts detailing advanced knowledge and accounts of human history. The Library was dismantled and destroyed in stages over a long period with the death-blow delivered by the Cult-established Roman Church in the period around 415AD. The Church of Rome was the Church of Babylon relocated as I said earlier. Gnostics were not a race. They were a way of perceiving reality. Whenever they established themselves and their information circulated the terrorists of the Church of Rome would target them for destruction. This happened with the Great Library and with the Gnostic Cathars who were burned to death by the psychopaths after a long period of oppression at the siege of the Castle of Monségur in southern France in 1244. The Church has always been terrified of Gnostic information which demolishes the official Christian narrative although there is much in the Bible that supports the Gnostic view if you read it in another way. To anyone studying the texts of what became known as the Nag Hammadi Library it is clear that great swathes of Christian and Biblical belief has its origin with Gnostics sources going back to Sumer. Gnostic themes have been twisted to manipulate the perceived reality of Bible believers. Biblical texts have been in the open for centuries where they could be changed while Gnostic

documents found at Nag Hammadi were sealed away and untouched for 1,600 years. What you see is what they wrote.

Use your *pneuma* not your *nous*

Gnosticism and Gnostic come from 'gnosis' which means knowledge, or rather *secret* knowledge, in the sense of spiritual awareness – knowledge about reality and life itself. The desperation of the Cult's Church of Rome to destroy the Gnostics can be understood when the knowledge they were circulating was the last thing the Cult wanted the population to know. Sixteen hundred years later the same Cult is working hard to undermine and silence me for the same reason. The dynamic between knowledge and ignorance is a constant. 'Time' appears to move on, but essential themes remain the same. We are told to 'use your nous', a Gnostic word for head/brain/intelligence. They said, however, that spiritual awakening or 'salvation' could only be secured by expanding awareness *beyond* what they called *nous* and into *pneuma* or Infinite Self. Obviously as I read these texts the parallels with what I have been saying since 1990 were fascinating to me. There is a universal truth that spans human history and in that case why wouldn't we be talking the same language 16 centuries apart? When you free yourself from the perception program of the five senses and explore expanded realms of consciousness you are going to connect with the same information no matter what the perceived 'era' within a manufactured timeline of a single and tiny range of manipulated frequency. Humans working with 'smart' technology or knocking rocks together in caves is only a timeline appearing to operate within the human frequency band. Expanded awareness and the knowledge it holds have always been there whether the era be Stone Age or computer age. We can only access that knowledge by opening ourselves to its frequency which the five-sense prison cell is designed to stop us doing. Gates, Fauci, Whitty, Vallance, Zuckerberg, Brin, Page, Wojcicki, Bezos, and all the others behind the 'Covid' hoax clearly have a long wait before their range of frequency can make that connection given that an open heart is

crucial to that as we shall see. Instead of accessing knowledge directly through expanded awareness it is given to Cult operatives by the secret society networks of the Cult where it has been passed on over thousands of years outside the public arena. Expanded realms of consciousness is where great artists, composers and writers find their inspiration and where truth awaits anyone open enough to connect with it. We need to go there fast.

Archon hijack

A fifth of the Nag Hammadi texts describe the existence and manipulation of the Archons led by a 'Chief Archon' they call 'Yaldabaoth', or the 'Demiurge', and this is the Christian 'Devil', 'Satan', 'Lucifer', and his demons. Archons in Biblical symbolism are the 'fallen ones' which are also referred to as fallen angels after the angels expelled from heaven according to the Abrahamic religions of Judaism, Christianity and Islam. These angels are claimed to tempt humans to 'sin' ongoing and you will see how accurate that symbolism is during the rest of the book. The theme of 'original sin' is related to the 'Fall' when Adam and Eve were 'tempted by the serpent' and fell from a state of innocence and 'obedience' (connection) with God into a state of disobedience (disconnection). The Fall is said to have brought sin into the world and corrupted everything including human nature. Yaldabaoth, the 'Lord Archon', is described by Gnostics as a 'counterfeit spirit', 'The Blind One', 'The Blind God', and 'The Foolish One'. The Jewish name for Yaldabaoth in Talmudic writings is Samael which translates as 'Poison of God', or 'Blindness of God'. You see the parallels. Yaldabaoth in Islamic belief is the Muslim Jinn devil known as Shaytan – Shaytan is Satan as the same themes are found all over the world in every religion and culture. The 'Lord God' of the Old Testament is the 'Lord Archon' of Gnostic manuscripts and that's why he's such a bloodthirsty bastard. Satan is known by Christians as 'the Demon of Demons' and Gnostics called Yaldabaoth the 'Archon of Archons'. Both are known as 'The Deceiver'. We are talking about the same 'bloke' for sure and these common themes

using different names, storylines and symbolism tell a common tale of the human plight.

Archons are referred to in Nag Hammadi documents as mind parasites, inverters, guards, gatekeepers, detainers, judges, pitiless ones and deceivers. The 'Covid' hoax alone is a glaring example of all these things. The Biblical 'God' is so different in the Old and New Testaments because they are not describing the same phenomenon. The vindictive, angry, hate-filled, 'God' of the Old Testament, known as Yahweh, is Yaldabaoth who is depicted in Cult-dictated popular culture as the 'Dark Lord', 'Lord of Time', Lord (Darth) Vader and Dormammu, the evil ruler of the 'Dark Dimension' trying to take over the 'Earth Dimension' in the Marvel comic movie, *Dr Strange*. Yaldabaoth is both the Old Testament 'god' and the Biblical 'Satan'. Gnostics referred to Yaldabaoth as the 'Great Architect of the Universe' and the Cult-controlled Freemason network calls their god 'the 'Great Architect of the Universe' (also Grand Architect). The 'Great Architect' Yaldabaoth is symbolised by the Cult as the all-seeing eye at the top of the pyramid on the Great Seal of the United States and the dollar bill. Archon is encoded in *arch*-itect as it is in *arch*-angels and *arch*-bishops. All religions have the theme of a force for good and force for evil in some sort of spiritual war and there is a reason for that – the theme is true. The Cult and its non-human masters are quite happy for this to circulate. They present themselves as the force for good fighting evil when they are really the force of evil (absence of love). The whole foundation of Cult modus operandi is inversion. They promote themselves as a force for good and anyone challenging them in pursuit of peace, love, fairness, truth and justice is condemned as a satanic force for evil. This has been the game plan throughout history whether the Church of Rome inquisitions of non-believers or 'conspiracy theorists' and 'anti-vaxxers' of today. The technique is the same whatever the timeline era.

Yaldabaoth is revolting (true)

Yaldabaoth and the Archons are said to have revolted against God with Yaldabaoth claiming to *be* God – the *All That Is*. The Old Testament ‘God’ (Yaldabaoth) demanded to be worshipped as such: ‘*I am the LORD, and there is none else, there is no God beside me*’ (Isaiah 45:5). I have quoted in other books a man who said he was the unofficial son of the late Baron Philippe de Rothschild of the Mouton-Rothschild wine producing estates in France who died in 1988 and he told me about the Rothschild ‘revolt from God’. The man said he was given the name Phillip Eugene de Rothschild and we shared long correspondence many years ago while he was living under another identity. He said that he was conceived through ‘occult incest’ which (within the Cult) was ‘normal and to be admired’. ‘Phillip’ told me about his experience attending satanic rituals with rich and famous people whom he names and you can see them and the wider background to Cult Satanism in my other books starting with *The Biggest Secret*. Cult rituals are interactions with Archontic ‘gods’. ‘Phillip’ described Baron Philippe de Rothschild as ‘a master Satanist and hater of God’ and he used the same term ‘revolt from God’ associated with Yaldabaoth/Satan/Lucifer/the Devil in describing the Sabbatian Rothschild dynasty. ‘I played a key role in my family’s revolt from God’, he said. That role was to infiltrate in classic Sabbatian style the Christian Church, but eventually he escaped the mind-prison to live another life. The Cult has been targeting religion in a plan to make worship of the Archons the global one-world religion. Infiltration of Satanism into modern ‘culture’, especially among the young, through music videos, stage shows and other means, is all part of this.

Nag Hammadi texts describe Yaldabaoth and the Archons in their prime form as energy – consciousness – and say they can take form if they choose in the same way that consciousness takes form as a human. Yaldabaoth is called ‘formless’ and represents a deeply inverted, distorted and chaotic state of consciousness which seeks to attached to humans and turn them into a likeness of itself in an attempt at assimilation. For that to happen it has to manipulate

humans into low frequency mental and emotional states that match its own. Archons can certainly appear in human form and this is the origin of the psychopathic personality. The energetic distortion Gnostics called Yaldabaoth is psychopathy. When psychopathic Archons take human form that human will be a psychopath as an expression of Yaldabaoth consciousness. Cult psychopaths are Archons in human form. The principle is the same as that portrayed in the 2009 *Avatar* movie when the American military travelled to a fictional Earth-like moon called Pandora in the Alpha Centauri star system to infiltrate a society of blue people, or Na'vi, by hiding within bodies that looked like the Na'vi. Archons posing as humans have a particular hybrid information field, part human, part Archon, (the ancient 'demigods') which processes information in a way that manifests behaviour to match their psychopathic evil, lack of empathy and compassion, and stops them being influenced by the empathy, compassion and love that a fully-human information field is capable of expressing. Cult bloodlines interbreed, be they royalty or dark suits, for this reason and you have their obsession with incest. Interbreeding with full-blown humans would dilute the Archontic energy field that guarantees psychopathy in its representatives in the human realm.

Gnostic writings say the main non-human forms that Archons take are *serpentine* (what I have called for decades 'reptilian' amid unbounded ridicule from the Archontically-programmed) and what Gnostics describe as 'an unborn baby or foetus with grey skin and dark, unmoving eyes'. This is an excellent representation of the ET 'Greys' of UFO folklore which large numbers of people claim to have seen and been abducted by – Zulu shaman Credo Mutwa among them. I agree with those that believe in extraterrestrial or interdimensional visitations today and for thousands of years past. No wonder with their advanced knowledge and technological capability they were perceived and worshipped as gods for technological and other 'miracles' they appeared to perform. Imagine someone arriving in a culture disconnected from the modern world with a smartphone and computer. They would be

seen as a 'god' capable of 'miracles'. The Renegade Mind, however, wants to know the source of everything and not only the way that source manifests as human or non-human. In the same way that a Renegade Mind seeks the original source material for the 'Covid virus' to see if what is claimed is true. The original source of Archons in form is consciousness – the distorted state of consciousness known to Gnostics as Yaldabaoth.

'Revolt from God' is energetic disconnection

Where I am going next will make a lot of sense of religious texts and ancient legends relating to 'Satan', Lucifer' and the 'gods'. Gnostic descriptions sync perfectly with the themes of my own research over the years in how they describe a consciousness distortion seeking to impose itself on human consciousness. I've referred to the core of infinite awareness in previous books as Infinite Awareness in Awareness of Itself. By that I mean a level of awareness that knows that it is all awareness and is aware of all awareness. From here comes the frequency of love in its true sense and balance which is what love is on one level – the balance of all forces into a single whole called Oneness and Isness. The more we disconnect from this state of love that many call 'God' the constituent parts of that Oneness start to unravel and express themselves as a part and not a whole. They become individualised as intellect, mind, selfishness, hatred, envy, desire for power over others, and such like. This is not a problem in the greater scheme in that 'God', the *All That Is*, can experience all these possibilities through different expressions of itself including humans. What we as expressions of the whole experience the *All That Is* experiences. We are the *All That Is* experiencing itself. As we withdraw from that state of Oneness we disconnect from its influence and things can get very unpleasant and very stupid. Archontic consciousness is at the extreme end of that. It has so disconnected from the influence of Oneness that it has become an inversion of unity and love, an inversion of everything, an inversion of life itself. Evil is appropriately live written backwards. Archontic consciousness is obsessed with death, an inversion of life,

and so its manifestations in Satanism are obsessed with death. They use inverted symbols in their rituals such as the inverted pentagram and cross. Sabbatians as Archontic consciousness incarnate invert Judaism and every other religion and culture they infiltrate. They seek disunity and chaos and they fear unity and harmony as they fear love like garlic to a vampire. As a result the Cult, Archons incarnate, act with such evil, psychopathy and lack of empathy and compassion disconnected as they are from the source of love. How could Bill Gates and the rest of the Archontic psychopaths do what they have to human society in the 'Covid' era with all the death, suffering and destruction involved and have no emotional consequence for the impact on others? Now you know. Why have Zuckerberg, Brin, Page, Wojcicki and company callously censored information warning about the dangers of the 'vaccine' while thousands have been dying and having severe, sometimes life-changing reactions? Now you know. Why have Tedros, Fauci, Whitty, Vallance and their like around the world been using case and death figures they're aware are fraudulent to justify lockdowns and all the deaths and destroyed lives that have come from that? Now you know. Why did Christian Drosten produce and promote a 'testing' protocol that he knew couldn't test for infectious disease which led to a global human catastrophe. Now you know. The Archontic mind doesn't give a shit ([Fig 17](#)). I personally think that Gates and major Cult insiders are a form of AI cyborg that the Archons want humans to become.

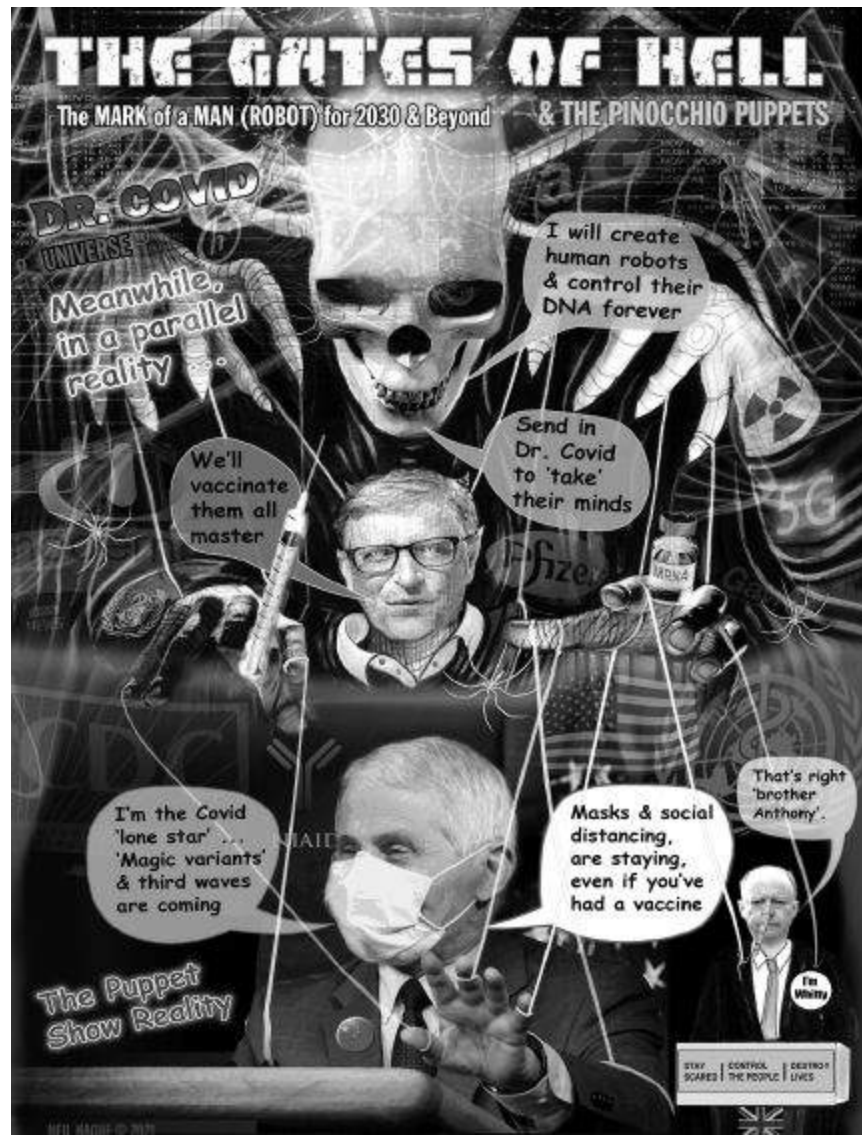


Figure 17: Artist Neil Hague's version of the 'Covid' hierarchy.

Human batteries

A state of such inversion does have its consequences, however. The level of disconnection from the Source of All means that you withdraw from that source of energetic sustenance and creativity. This means that you have to find your own supply of energetic power and it has – *us*. When the Morpheus character in the first *Matrix* movie held up a battery he spoke a profound truth when he said: 'The Matrix is a computer-generated dream world built to keep us under control in order to change the human being into one of

these.' The statement was true in all respects. We do live in a technologically-generated virtual reality simulation (more very shortly) and we have been manipulated to be an energy source for Archontic consciousness. The Disney-Pixar animated movie *Monsters, Inc.* in 2001 symbolised the dynamic when monsters in their world had no energy source and they would enter the human world to terrify children in their beds, catch the child's scream, terror (low-vibrational frequencies), and take that energy back to power the monster world. The lead character you might remember was a single giant eye and the symbolism of the Cult's all-seeing eye was obvious. Every thought and emotion is broadcast as a frequency unique to that thought and emotion. Feelings of love and joy, empathy and compassion, are high, quick, frequencies while fear, depression, anxiety, suffering and hate are low, slow, dense frequencies. Which kind do you think Archontic consciousness can connect with and absorb? In such a low and dense frequency state there's no way it can connect with the energy of love and joy. Archons can only feed off energy compatible with their own frequency and they and their Cult agents want to delete the human world of love and joy and manipulate the transmission of low vibrational frequencies through low-vibrational human mental and emotional states. *We are their energy source.* Wars are energetic banquets to the Archons – a world war even more so – and think how much low-frequency mental and emotional energy has been generated from the consequences for humanity of the 'Covid' hoax orchestrated by Archons incarnate like Gates.

The ancient practice of human sacrifice 'to the gods', continued in secret today by the Cult, is based on the same principle. 'The gods' are Archontic consciousness in different forms and the sacrifice is induced into a state of intense terror to generate the energy the Archontic frequency can absorb. Incarnate Archons in the ritual drink the blood which contains an adrenaline they crave which floods into the bloodstream when people are terrorised. Most of the sacrifices, ancient and modern, are children and the theme of 'sacrificing young virgins to the gods' is just code for children. They

have a particular pre-puberty energy that Archons want more than anything and the energy of the young in general is their target. The California Department of Education wants students to chant the names of Aztec gods (Archontic gods) once worshipped in human sacrifice rituals in a curriculum designed to encourage them to 'challenge racist, bigoted, discriminatory, imperialist/colonial beliefs', join 'social movements that struggle for social justice', and 'build new possibilities for a post-racist, post-systemic racism society'. It's the usual Woke crap that inverts racism and calls it anti-racism. In this case solidarity with 'indigenous tribes' is being used as an excuse to chant the names of 'gods' to which people were sacrificed (and still are in secret). What an example of Woke's inability to see beyond black and white, us and them, They condemn the colonisation of these tribal cultures by Europeans (quite right), but those cultures sacrificing people including children to their 'gods', and mass murdering untold numbers as the Aztecs did, is just fine. One chant is to the Aztec god Tezcatlipoca who had a man sacrificed to him in the 5th month of the Aztec calendar. His heart was cut out and he was eaten. Oh, that's okay then. Come on children ... after three ... Other sacrificial 'gods' for the young to chant their allegiance include Quetzalcoatl, Huitzilopochtli and Xipe Totec. The curriculum says that 'chants, affirmations, and energizers can be used to bring the class together, build unity around ethnic studies principles and values, and to reinvigorate the class following a lesson that may be emotionally taxing or even when student engagement may appear to be low'. Well, that's the cover story, anyway. Chanting and mantras are the repetition of a particular frequency generated from the vocal cords and chanting the names of these Archontic 'gods' tunes you into their frequency. That is the last thing you want when it allows for energetic synchronisation, attachment and perceptual influence. Initiates chant the names of their 'Gods' in their rituals for this very reason.

Vampires of the Woke

Paedophilia is another way that Archons absorb the energy of children. Paedophiles possessed by Archontic consciousness are used as the conduit during sexual abuse for discarnate Archons to vampire the energy of the young they desire so much. Stupendous numbers of children disappear every year never to be seen again although you would never know from the media. Imagine how much low-vibrational energy has been generated by children during the 'Covid' hoax when so many have become depressed and psychologically destroyed to the point of killing themselves. Shocking numbers of children are now taken by the state from loving parents to be handed to others. I can tell you from long experience of researching this since 1996 that many end up with paedophiles and assets of the Cult through corrupt and Cult-owned social services which in the reframing era has hired many psychopaths and emotionless automatons to do the job. Children are even stolen to order using spurious reasons to take them by the corrupt and secret (because they're corrupt) 'family courts'. I have written in detail in other books, starting with *The Biggest Secret* in 1997, about the ubiquitous connections between the political, corporate, government, intelligence and military elites (Cult operatives) and Satanism and paedophilia. If you go deep enough both networks have an interlocking leadership. The Woke mentality has been developed by the Cult for many reasons: To promote almost every aspect of its agenda; to hijack the traditional political left and turn it fascist; to divide and rule; and to target agenda pushbackers. But there are other reasons which relate to what I am describing here. How many happy and joyful Wokers do you ever see especially at the extreme end? They are a mental and psychological mess consumed by emotional stress and constantly emotionally cocked for the next explosion of indignation at someone referring to a female as a female. They are walking, talking, batteries as Morpheus might say emitting frequencies which both enslave them in low-vibrational bubbles of perceptual limitation and feed the Archons. Add to this the hatred claimed to be love; fascism claimed to 'anti-fascism', racism claimed to be 'anti-racism';

exclusion claimed to inclusion; and the abuse-filled Internet trolling. You have a purpose-built Archontic energy system with not a wind turbine in sight and all founded on Archontic *inversion*. We have whole generations now manipulated to serve the Archons with their actions and energy. They will be doing so their entire adult lives unless they snap out of their Archon-induced trance. Is it really a surprise that Cult billionaires and corporations put so much money their way? Where is the energy of joy and laughter, including laughing at yourself which is confirmation of your own emotional security? Mark Twain said: 'The human race has one really effective weapon, and that is laughter.' We must use it all the time. Woke has destroyed comedy because it has no humour, no joy, sense of irony, or self-deprecation. Its energy is dense and intense. *Mmmmm*, lunch says the Archontic frequency. Rudolf Steiner (1861-1925) was the Austrian philosopher and famous esoteric thinker who established Waldorf education or Steiner schools to treat children like unique expressions of consciousness and not minds to be programmed with the perceptions determined by authority. I'd been writing about this energy vampiring for decades when I was sent in 2016 a quote by Steiner. He was spot on:

There are beings in the spiritual realms for whom anxiety and fear emanating from human beings offer welcome food. When humans have no anxiety and fear, then these creatures starve. If fear and anxiety radiates from people and they break out in panic, then these creatures find welcome nutrition and they become more and more powerful. These beings are hostile towards humanity. Everything that feeds on negative feelings, on anxiety, fear and superstition, despair or doubt, are in reality hostile forces in super-sensible worlds, launching cruel attacks on human beings, while they are being fed ... These are exactly the feelings that belong to contemporary culture and materialism; because it estranges people from the spiritual world, it is especially suited to evoke hopelessness and fear of the unknown in people, thereby calling up the above mentioned hostile forces against them.

Pause for a moment from this perspective and reflect on what has happened in the world since the start of 2020. Not only will pennies drop, but billion dollar bills. We see the same theme from Don Juan Matus, a Yaqui Indian shaman in Mexico and the information source for Peruvian-born writer, Carlos Castaneda, who wrote a series of

books from the 1960s to 1990s. Don Juan described the force manipulating human society and his name for the Archons was the predator:

We have a predator that came from the depths of the cosmos and took over the rule of our lives. Human beings are its prisoners. The predator is our lord and master. It has rendered us docile, helpless. If we want to protest, it suppresses our protest. If we want to act independently, it demands that we don't do so ... indeed we are held prisoner!

They took us over because we are food to them, and they squeeze us mercilessly because we are their sustenance. Just as we rear chickens in coops, the predators rear us in human coops, humaneros. Therefore, their food is always available to them.

Different cultures, different eras, same recurring theme.

The 'ennoia' dilemma

Nag Hammadi Gnostic manuscripts say that Archon consciousness has no 'ennoia'. This is directly translated as 'intentionality', but I'll use the term 'creative imagination'. The *All That Is* in awareness of itself is the source of all creativity – all possibility – and the more disconnected you are from that source the more you are subsequently denied 'creative imagination'. Given that Archon consciousness is almost entirely disconnected it severely lacks creativity and has to rely on far more mechanical processes of thought and exploit the creative potential of those that do have 'ennoia'. You can see cases of this throughout human society. Archon consciousness almost entirely dominates the global banking system and if we study how that system works you will appreciate what I mean. Banks manifest 'money' out of nothing by issuing lines of 'credit' which is 'money' that has never, does not, and will never exist except in theory. It's a confidence trick. If you think 'credit' figures-on-a-screen 'money' is worth anything you accept it as payment. If you don't then the whole system collapses through lack of confidence in the value of that 'money'. Archontic bankers with no 'ennoia' are 'lending' 'money' that doesn't exist to humans that *do* have creativity – those that have the inspired ideas and create businesses and products. Archon banking feeds off human creativity

which it controls through 'money' creation and debt. Humans have the creativity and Archons exploit that for their own benefit and control while having none themselves. Archon Internet platforms like Facebook claim joint copyright of everything that creative users post and while Archontic minds like Zuckerberg may officially head that company it will be human creatives on the staff that provide the creative inspiration. When you have limitless 'money' you can then buy other companies established by creative humans. Witness the acquisition record of Facebook, Google and their like. Survey the Archon-controlled music industry and you see non-creative dark suit executives making their fortune from the human creativity of their artists. The cases are endless. Research the history of people like Gates and Zuckerberg and how their empires were built on exploiting the creativity of others. Archon minds cannot create out of nothing, but they are skilled (because they have to be) in what Gnostic texts call 'countermimicry'. They can imitate, but not innovate. Sabbatians trawl the creativity of others through backdoors they install in computer systems through their cybersecurity systems. Archon-controlled China is globally infamous for stealing intellectual property and I remember how Hong Kong, now part of China, became notorious for making counterfeit copies of the creativity of others – 'countermimicry'. With the now pervasive and all-seeing surveillance systems able to infiltrate any computer you can appreciate the potential for Archons to vampire the creativity of humans. Author John Lamb Lash wrote in his book about the Nag Hammadi texts, *Not In His Image*:

Although they cannot originate anything, because they lack the divine factor of ennoia (intentionality), Archons can imitate with a vengeance. Their expertise is simulation (HAL, virtual reality). The Demiurge [Yaldabaoth] fashions a heaven world copied from the fractal patterns [of the original] ... His construction is celestial kitsch, like the fake Italianate villa of a Mafia don complete with militant angels to guard every portal.

This brings us to something that I have been speaking about since the turn of the millennium. Our reality is a simulation; a virtual reality that we think is real. No, I'm not kidding.

Human reality? Well, virtually

I had pondered for years about whether our reality is 'real' or some kind of construct. I remembered being immensely affected on a visit as a small child in the late 1950s to the then newly-opened Planetarium on the Marylebone Road in London which is now closed and part of the adjacent Madame Tussauds wax museum. It was in the middle of the day, but when the lights went out there was the night sky projected in the Planetarium's domed ceiling and it appeared to be so real. The experience never left me and I didn't know why until around the turn of the millennium when I became certain that our 'night sky' and entire reality is a projection, a virtual reality, akin to the illusory world portrayed in the *Matrix* movies. I looked at the sky one day in this period and it appeared to me like the domed roof of the Planetarium. The release of the first *Matrix* movie in 1999 also provided a synchronistic and perfect visual representation of where my mind had been going for a long time. I hadn't come across the Gnostic Nag Hammadi texts then. When I did years later the correlation was once again astounding. As I read Gnostic accounts from 1,600 years and more earlier it was clear that they were describing the same simulation phenomenon. They tell how the Yaldabaoth 'Demiurge' and Archons created a 'bad copy' of original reality to rule over all that were captured by its illusions and the body was a prison to trap consciousness in the 'bad copy' fake reality. Read how Gnostics describe the 'bad copy' and update that to current times and they are referring to what we would call today a virtual reality simulation.

Author John Lamb Lash said 'the Demiurge fashions a heaven world copied from the fractal patterns' of the original through expertise in 'HAL' or virtual reality simulation. Fractal patterns are part of the energetic information construct of our reality, a sort of blueprint. If these patterns were copied in computer terms it would indeed give you a copy of a 'natural' reality in a non-natural frequency and digital form. The principle is the same as making a copy of a website. The original website still exists, but now you can change the copy version to make it whatever you like and it can

become very different to the original website. Archons have done this with our reality, a *synthetic* copy of prime reality that still exists beyond the frequency walls of the simulation. Trapped within the illusions of this synthetic Matrix, however, were and are human consciousness and other expressions of prime reality and this is why the Archons via the Cult are seeking to make the human body synthetic and give us synthetic AI minds to complete the job of turning the entire reality synthetic including what we perceive to be the natural world. To quote Kurzweil: 'Nanobots will infuse all the matter around us with information. Rocks, trees, everything will become these intelligent creatures.' Yes, *synthetic* 'creatures' just as 'Covid' and other genetically-manipulating 'vaccines' are designed to make the human body synthetic. From this perspective it is obvious why Archons and their Cult are so desperate to infuse synthetic material into every human with their 'Covid' scam.

Let there be (electromagnetic) light

Yaldabaoth, the force that created the simulation, or Matrix, makes sense of the Gnostic reference to 'The Great Architect' and its use by Cult Freemasonry as the name of its deity. The designer of the Matrix in the movies is called 'The Architect' and that trilogy is jam-packed with symbolism relating to these subjects. I have contended for years that the angry Old Testament God (Yaldabaoth) is the 'God' being symbolically 'quoted' in the opening of Genesis as 'creating the world'. This is not the creation of prime reality – it's the creation of the *simulation*. The Genesis 'God' says: 'Let there be Light: and there was light.' But what is this 'Light'? I have said for decades that the speed of light (186,000 miles per second) is not the fastest speed possible as claimed by mainstream science and is in fact the frequency walls or outer limits of the Matrix. You can't have a fastest or slowest anything within all possibility when everything is possible. The human body is encoded to operate within the speed of light or *within the simulation* and thus we see only the tiny frequency band of visible *light*. Near-death experiencers who perceive reality outside the body during temporary 'death' describe a very different

form of light and this is supported by the Nag Hammadi texts. Prime reality beyond the simulation ('Upper Aeons' to the Gnostics) is described as a realm of incredible beauty, bliss, love and harmony – a realm of 'watery light' that is so powerful 'there are no shadows'. Our false reality of Archon control, which Gnostics call the 'Lower Aeons', is depicted as a realm with a different kind of 'light' and described in terms of chaos, 'Hell', 'the Abyss' and 'Outer Darkness', where trapped souls are tormented and manipulated by demons (relate that to the 'Covid' hoax alone). The watery light theme can be found in near-death accounts and it is not the same as *simulation* 'light' which is electromagnetic or radiation light within the speed of light – the 'Lower Aeons'. Simulation 'light' is the 'luminous fire' associated by Gnostics with the Archons. The Bible refers to Yaldabaoth as 'that old serpent, called the Devil, and Satan, which deceiveth the whole world' (Revelation 12:9). I think that making a simulated copy of prime reality ('countermimicry') and changing it dramatically while all the time manipulating humanity to believe it to be real could probably meet the criteria of deceiving the whole world. Then we come to the Cult god Lucifer – the *Light Bringer*. Lucifer is symbolic of Yaldabaoth, the bringer of radiation light that forms the bad copy simulation within the speed of light. 'He' is symbolised by the lighted torch held by the Statue of Liberty and in the name 'Illuminati'. Sabbatian-Frankism declares that Lucifer is the true god and Lucifer is the real god of Freemasonry honoured as their 'Great or Grand Architect of the Universe' (simulation).

I would emphasise, too, the way Archontic technologically-generated luminous fire of radiation has deluged our environment since I was a kid in the 1950s and changed the nature of The Field with which we constantly interact. Through that interaction technological radiation is changing us. The Smart Grid is designed to operate with immense levels of communication power with 5G expanding across the world and 6G, 7G, in the process of development. Radiation is the simulation and the Archontic manipulation system. Why wouldn't the Archon Cult wish to unleash radiation upon us to an ever-greater extreme to form

Kurzweil's 'cloud'? The plan for a synthetic human is related to the need to cope with levels of radiation beyond even anything we've seen so far. Biological humans would not survive the scale of radiation they have in their script. The Smart Grid is a technological sub-reality within the technological simulation to further disconnect five-sense perception from expanded consciousness. It's a technological prison of the mind.

Infusing the 'spirit of darkness'

A recurring theme in religion and native cultures is the manipulation of human genetics by a non-human force and most famously recorded as the biblical 'sons of god' (the gods plural in the original) who interbred with the daughters of men. The Nag Hammadi *Apocryphon of John* tells the same story this way:

He [Yaldabaoth] sent his angels [Archons/demons] to the daughters of men, that they might take some of them for themselves and raise offspring for their enjoyment. And at first they did not succeed. When they had no success, they gathered together again and they made a plan together ... And the angels changed themselves in their likeness into the likeness of their mates, filling them with the spirit of darkness, which they had mixed for them, and with evil ... And they took women and begot children out of the darkness according to the likeness of their spirit.

Possession when a discarnate entity takes over a human body is an age-old theme and continues today. It's very real and I've seen it. Satanic and secret society rituals can create an energetic environment in which entities can attach to initiates and I've heard many stories of how people have changed their personality after being initiated even into lower levels of the Freemasons. I have been inside three Freemasonic temples, one at a public open day and two by just walking in when there was no one around to stop me. They were in Ryde, the town where I live, Birmingham, England, when I was with a group, and Boston, Massachusetts. They all felt the same energetically – dark, dense, low-vibrational and sinister. Demonic attachment can happen while the initiate has no idea what is going on. To them it's just a ritual to get in the Masons and do a bit of good

business. In the far more extreme rituals of Satanism human possession is even more powerful and they are designed to make possession possible. The hierarchy of the Cult is dictated by the power and perceived status of the possessing Archon. In this way the Archon hierarchy becomes the Cult hierarchy. Once the entity has attached it can influence perception and behaviour and if it attaches to the extreme then so much of its energy (information) infuses into the body information field that the hologram starts to reflect the nature of the possessing entity. This is the *Exorcist* movie type of possession when facial features change and it's known as shapeshifting. Islam's Jinn are said to be invisible tricksters who change shape, 'whisper', confuse and take human form. These are all traits of the Archons and other versions of the same phenomenon. Extreme possession could certainly infuse the 'spirit of darkness' into a partner during sex as the Nag Hammadi texts appear to describe. Such an infusion can change genetics which is also energetic information. Human genetics is information and the 'spirit of darkness' is information. Mix one with the other and change must happen. Islam has the concept of a 'Jinn baby' through possession of the mother and by Jinn taking human form. There are many ways that human genetics can be changed and remember that Archons have been aware all along of advanced techniques to do this. What is being done in human society today – and far more – was known about by Archons at the time of the 'fallen ones' and their other versions described in religions and cultures.

Archons and their human-world Cult are obsessed with genetics as we see today and they know this dictates how information is processed into perceived reality during a human life. They needed to produce a human form that would decode the simulation and this is symbolically known as 'Adam and Eve' who left the 'garden' (prime reality) and 'fell' into Matrix reality. The simulation is not a 'physical' construct (there is no 'physical'); it is a source of information. Think Wi-Fi again. The simulation is an energetic field encoded with information and body-brain systems are designed to decode that information encoded in wave or frequency form which

is transmitted to the brain as electrical signals. These are decoded by the brain to construct our sense of reality – an illusory ‘physical’ world that only exists in the brain or the mind. Virtual reality games mimic this process using the same sensory decoding system. Information is fed to the senses to decode a virtual reality that can appear so real, but isn’t (Figs 18 and 19). Some scientists believe – and I agree with them – that what we perceive as ‘physical’ reality only exists when we are looking or observing. The act of perception or focus triggers the decoding systems which turn waveform information into holographic reality. When we are not observing something our reality reverts from a holographic state to a waveform state. This relates to the same principle as a falling tree not making a noise unless someone is there to hear it or decode it. The concept makes sense from the simulation perspective. A computer is not decoding all the information in a Wi-Fi field all the time and only decodes or brings into reality on the screen that part of Wi-Fi that it’s decoding – focusing upon – at that moment.



Figure 18: Virtual reality technology ‘hacks’ into the body’s five-sense decoding system.



Figure 19: The result can be experienced as very ‘real’.

Interestingly, Professor Donald Hoffman at the Department of Cognitive Sciences at the University of California, Irvine, says that our experienced reality is like a computer interface that shows us only the level with which we interact while hiding all that exists beyond it: 'Evolution shaped us with a user interface that hides the truth. Nothing that we see is the truth – the very language of space and time and objects is the wrong language to describe reality.' He is correct in what he says on so many levels. Space and time are not a universal reality. They are a phenomenon of decoded *simulation* reality as part of the process of enslaving our sense of reality. Near-death experiencers report again and again how space and time did not exist as we perceive them once they were free of the body – body decoding systems. You can appreciate from this why Archons and their Cult are so desperate to entrap human attention in the five senses where we are in the Matrix and of the Matrix. Opening your mind to expanded states of awareness takes you beyond the information confines of the simulation and you become aware of knowledge and insights denied to you before. This is what we call 'awakening' – *awakening from the Matrix* – and in the final chapter I will relate this to current events.

Where are the 'aliens'?

A simulation would explain the so-called 'Fermi Paradox' named after Italian physicist Enrico Fermi (1901-1954) who created the first nuclear reactor. He considered the question of why there is such a lack of extraterrestrial activity when there are so many stars and planets in an apparently vast universe; but what if the night sky that we see, or think we do, is a simulated projection as I say? If you control the simulation and your aim is to hold humanity fast in essential ignorance would you want other forms of life including advanced life coming and going sharing information with humanity? Or would you want them to believe they were isolated and apparently alone? Themes of human isolation and apartness are common whether they be the perception of a lifeless universe or the fascist isolation laws of the 'Covid' era. Paradoxically the very

existence of a simulation means that we are not alone when some force had to construct it. My view is that experiences that people have reported all over the world for centuries with Reptilians and Grey entities are Archon phenomena as Nag Hammadi texts describe; and that benevolent 'alien' interactions are non-human groups that come in and out of the simulation by overcoming Archon attempts to keep them out. It should be highlighted, too, that Reptilians and Greys are obsessed with *genetics* and *technology* as related by cultural accounts and those who say they have been abducted by them. Technology is their way of overcoming some of the limitations in their creative potential and our technology-driven and controlled human society of today is *archetypical* Archon-Reptilian-Grey *modus operandi*. Technocracy is really *Archontocracy*. The Universe does not have to be as big as it appears with a simulation. There is no space or distance only information decoded into holographic reality. What we call 'space' is only the absence of holographic 'objects' and that 'space' is The Field of energetic information which connects everything into a single whole. The same applies with the artificially-generated information field of the simulation. The Universe is not big or small as a physical reality. It is decoded information, that's all, and its perceived size is decided by the way the simulation is encoded to make it appear. The entire night sky as we perceive it only exists in our brain and so where are those 'millions of light years'? The 'stars' on the ceiling of the Planetarium looked a vast distance away.

There's another point to mention about 'aliens'. I have been highlighting since the 1990s the plan to stage a fake 'alien invasion' to justify the centralisation of global power and a world military. Nazi scientist Werner von Braun, who was taken to America by Operation Paperclip after World War Two to help found NASA, told his American assistant Dr Carol Rosin about the Cult agenda when he knew he was dying in 1977. Rosin said that he told her about a sequence that would lead to total human control by a one-world government. This included threats from terrorism, rogue nations, meteors and asteroids before finally an 'alien invasion'. All of these

things, von Braun said, would be bogus and what I would refer to as a No-Problem-Reaction-Solution. Keep this in mind when 'the aliens are coming' is the new mantra. The aliens are not coming – they are *already here* and they have infiltrated human society while looking human. French-Canadian investigative journalist Serge Monast said in 1994 that he had uncovered a NASA/military operation called Project Blue Beam which fits with what Werner von Braun predicted. Monast died of a 'heart attack' in 1996 the day after he was arrested and spent a night in prison. He was 51. He said Blue Beam was a plan to stage an alien invasion that would include religious figures beamed holographically into the sky as part of a global manipulation to usher in a 'new age' of worshipping what I would say is the Cult 'god' Yaldabaoth in a one-world religion. Fake holographic asteroids are also said to be part of the plan which again syncs with von Braun. How could you stage an illusory threat from asteroids unless they were holographic inserts? This is pretty straightforward given the advanced technology outside the public arena and the fact that our 'physical' reality is holographic anyway. Information fields would be projected and we would decode them into the illusion of a 'physical' asteroid. If they can sell a global 'pandemic' with a 'virus' that doesn't exist what will humans not believe if government and media tell them?

All this is particularly relevant as I write with the Pentagon planning to release in June, 2021, information about 'UFO sightings'. I have been following the UFO story since the early 1990s and the common theme throughout has been government and military denials and cover up. More recently, however, the Pentagon has suddenly become more talkative and apparently open with Air Force pilot radar images released of unexplained craft moving and changing direction at speeds well beyond anything believed possible with human technology. Then, in March, 2021, former Director of National Intelligence John Ratcliffe said a Pentagon report months later in June would reveal a great deal of information about UFO sightings unknown to the public. He said the report would have 'massive implications'. The order to do this was included bizarrely

in a \$2.3 trillion 'coronavirus' relief and government funding bill passed by the Trump administration at the end of 2020. I would add some serious notes of caution here. I have been pointing out since the 1990s that the US military and intelligence networks have long had craft – 'flying saucers' or anti-gravity craft – which any observer would take to be extraterrestrial in origin. Keeping this knowledge from the public allows craft flown by *humans* to be perceived as alien visitations. I am not saying that 'aliens' do not exist. I would be the last one to say that, but we have to be streetwise here. President Ronald Reagan told the UN General Assembly in 1987: 'I occasionally think how quickly our differences worldwide would vanish if we were facing an alien threat from outside this world.' That's the idea. Unite against a common 'enemy' with a common purpose behind your 'saviour force' (the Cult) as this age-old technique of mass manipulation goes global.

Science moves this way ...

I could find only one other person who was discussing the simulation hypothesis publicly when I concluded it was real. This was Nick Bostrom, a Swedish-born philosopher at the University of Oxford, who has explored for many years the possibility that human reality is a computer simulation although his version and mine are not the same. Today the simulation and holographic reality hypothesis have increasingly entered the scientific mainstream. Well, the more open-minded mainstream, that is. Here are a few of the ever-gathering examples. American nuclear physicist Silas Beane led a team of physicists at the University of Bonn in Germany pursuing the question of whether we live in a simulation. They concluded that we probably do and it was likely based on a lattice of cubes. They found that cosmic rays align with that specific pattern. The team highlighted the Greisen–Zatsepin–Kuzmin (GZK) limit which refers to cosmic ray particle interaction with cosmic background radiation that creates an apparent boundary for cosmic ray particles. They say in a paper entitled 'Constraints on the Universe as a Numerical Simulation' that this 'pattern of constraint' is exactly what you

would find with a computer simulation. They also made the point that a simulation would create its own 'laws of physics' that would limit possibility. I've been making the same point for decades that the *perceived* laws of physics relate only to this reality, or what I would later call the simulation. When designers write codes to create computer and virtual reality games they are the equivalent of the laws of physics for that game. Players interact within the limitations laid out by the coding. In the same way those who wrote the codes for the simulation decided the laws of physics that would apply. These can be overridden by expanded states of consciousness, but not by those enslaved in only five-sense awareness where simulation codes rule. Overriding the codes is what people call 'miracles'. They are not. They are bypassing the encoded limits of the simulation. A population caught in simulation perception would have no idea that this was their plight. As the Bonn paper said: 'Like a prisoner in a pitch-black cell we would not be able to see the "walls" of our prison,' That's true if people remain mesmerised by the five senses. Open to expanded awareness and those walls become very clear. The main one is the speed of light.

American theoretical physicist James Gates is another who has explored the simulation question and found considerable evidence to support the idea. Gates was Professor of Physics at the University of Maryland, Director of The Center for String and Particle Theory, and on Barack Obama's Council of Advisors on Science and Technology. He and his team found *computer codes* of digital data embedded in the fabric of our reality. They relate to on-off electrical charges of 1 and 0 in the binary system used by computers. 'We have no idea what they are doing there', Gates said. They found within the energetic fabric mathematical sequences known as error-correcting codes or block codes that 'reboot' data to its original state or 'default settings' when something knocks it out of sync. Gates was asked if he had found a set of equations embedded in our reality indistinguishable from those that drive search engines and browsers and he said: 'That is correct.' Rich Terrile, director of the Centre for Evolutionary Computation and Automated Design at NASA's Jet

Propulsion Laboratory, has said publicly that he believes the Universe is a digital hologram that must have been created by a form of intelligence. I agree with that in every way. Waveform information is delivered electrically by the senses to the brain which constructs a *digital* holographic reality that we call the 'world'. This digital level of reality can be read by the esoteric art of numerology. Digital holograms are at the cutting edge of holographics today. We have digital technology everywhere designed to access and manipulate our digital level of perceived reality. Synthetic mRNA in 'Covid vaccines' has a digital component to manipulate the body's digital 'operating system'.

Reality is numbers

How many know that our reality can be broken down to numbers and codes that are the same as computer games? Max Tegmark, a physicist at the Massachusetts Institute of Technology (MIT), is the author of *Our Mathematical Universe* in which he lays out how reality can be entirely described by numbers and maths in the way that a video game is encoded with the 'physics' of computer games. Our world and computer virtual reality are essentially the same.

Tegmark imagines the perceptions of characters in an advanced computer game when the graphics are so good they don't know they are in a game. They think they can bump into real objects (electromagnetic resistance in our reality), fall in love and feel emotions like excitement. When they began to study the apparently 'physical world' of the video game they would realise that everything was made of pixels (which have been found in our energetic reality as must be the case when on one level our world is digital). What computer game characters thought was physical 'stuff', Tegmark said, could actually be broken down into numbers:

And we're exactly in this situation in our world. We look around and it doesn't seem that mathematical at all, but everything we see is made out of elementary particles like quarks and electrons. And what properties does an electron have? Does it have a smell or a colour or a texture? No! ... We physicists have come up with geeky names for [Electron] properties, like

electric charge, or spin, or lepton number, but the electron doesn't care what we call it, the properties are just numbers.

This is the illusory reality Gnostics were describing. This is the simulation. The A, C, G, and T codes of DNA have a binary value – A and C = 0 while G and T = 1. This has to be when the simulation is digital and the body must be digital to interact with it. Recurring mathematical sequences are encoded throughout reality and the body. They include the Fibonacci sequence in which the two previous numbers are added to get the next one, as in ... 1, 1, 2, 3, 5, 8, 13, 21, 34, 55, etc. The sequence is encoded in the human face and body, proportions of animals, DNA, seed heads, pine cones, trees, shells, spiral galaxies, hurricanes and the number of petals in a flower. The list goes on and on. There are fractal patterns – a 'never-ending pattern that is infinitely complex and self-similar across all scales in the as above, so below, principle of holograms. These and other famous recurring geometrical and mathematical sequences such as Phi, Pi, Golden Mean, Golden Ratio and Golden Section are *computer codes* of the simulation. I had to laugh and give my head a shake the day I finished this book and it went into the production stage. I was sent an article in *Scientific American* published in April, 2021, with the headline 'Confirmed! We Live in a Simulation'. Two decades after I first said our reality is a simulation and the speed of light is its outer limit the article suggested that we do live in a simulation and that the speed of light is its outer limit. I left school at 15 and never passed a major exam in my life while the writer was up to his eyes in qualifications. As I will explain in the final chapter *knowing* is far better than thinking and they come from very different sources. The article rightly connected the speed of light to the processing speed of the 'Matrix' and said what has been in my books all this time ... 'If we are in a simulation, as it appears, then space is an abstract property written in code. It is not real'. No it's not and if we live in a simulation something created it and it wasn't *us*. 'That David Icke says we are manipulated by aliens' – he's crackers.'

Wow ...

The reality that humanity thinks is so real is an illusion. Politicians, governments, scientists, doctors, academics, law enforcement, media, school and university curriculums, on and on, are all founded on a world that *does not exist* except as a simulated prison cell. Is it such a stretch to accept that 'Covid' doesn't exist when our entire 'physical' reality doesn't exist? Revealed here is the knowledge kept under raps in the Cult networks of compartmentalised secrecy to control humanity's sense of reality by inducing the population to believe in a reality that's not real. If it wasn't so tragic in its experiential consequences the whole thing would be hysterically funny. None of this is new to Renegade Minds. Ancient Greek philosopher Plato (about 428 to about 347BC) was a major influence on Gnostic belief and he described the human plight thousands of years ago with his Allegory of the Cave. He told the symbolic story of prisoners living in a cave who had never been outside. They were chained and could only see one wall of the cave while behind them was a fire that they could not see. Figures walked past the fire casting shadows on the prisoners' wall and those moving shadows became their sense of reality. Some prisoners began to study the shadows and were considered experts on them (today's academics and scientists), but what they studied was only an illusion (today's academics and scientists). A prisoner escaped from the cave and saw reality as it really is. When he returned to report this revelation they didn't believe him, called him mad and threatened to kill him if he tried to set them free. Plato's tale is not only a brilliant analogy of the human plight and our illusory reality. It describes, too, the dynamics of the 'Covid' hoax. I have only skimmed the surface of these subjects here. The aim of this book is to crisply connect all essential dots to put what is happening today into its true context. All subject areas and their connections in this chapter are covered in great evidential detail in *Everything You Need To Know, But Have Never Been Told* and *The Answer*.

They say that bewildered people 'can't see the forest for the trees'. Humanity, however, can't see the forest for the *twigs*. The five senses

see only twigs while Renegade Minds can see the forest and it's the forest where the answers lie with the connections that reveals.

Breaking free of perceptual programming so the forest can be seen is the way we turn all this around. Not breaking free is how humanity got into this mess. The situation may seem hopeless, but I promise you it's not. We are a perceptual heartbeat from paradise if only we knew.

CHAPTER TWELVE

Escaping Wetiko

Life is simply a vacation from the infinite
Dean Cavanagh

Renegade Minds weave the web of life and events and see common themes in the apparently random. They are always there if you look for them and their pursuit is aided by incredible synchronicity that comes when your mind is open rather than mesmerised by what it thinks it can see.

Infinite awareness is infinite possibility and the more of infinite possibility that we access the more becomes infinitely possible. That may be stating the apparently obvious, but it is a devastatingly-powerful fact that can set us free. We are a point of attention within an infinity of consciousness. The question is how much of that infinity do we choose to access? How much knowledge, insight, awareness, wisdom, do we want to connect with and explore? If your focus is only in the five senses you will be influenced by a fraction of infinite awareness. I mean a range so tiny that it gives new meaning to infinitesimal. Limitation of self-identity and a sense of the possible limit accordingly your range of consciousness. We are what we think we are. Life is what we think it is. The dream is the dreamer and the dreamer is the dream. Buddhist philosophy puts it this way: 'As a thing is viewed, so it appears.' Most humans live in the realm of touch, taste, see, hear, and smell and that's the limit of their sense of the possible and sense of self. Many will follow a religion and speak of a God in his heaven, but their lives are still

dominated by the five senses in their perceptions and actions. The five senses become the arbiter of everything. When that happens all except a smear of infinity is sealed away from influence by the rigid, unyielding, reality bubbles that are the five-sense human or Phantom Self. Archon Cult methodology is to isolate consciousness within five-sense reality – the simulation – and then program that consciousness with a sense of self and the world through a deluge of life-long information designed to instil the desired perception that allows global control. Efforts to do this have increased dramatically with identity politics as identity bubbles are squeezed into the minutiae of five-sense detail which disconnect people even more profoundly from the infinite 'I'.

Five-sense focus and self-identity are like a firewall that limits access to the infinite realms. You only perceive one radio or television station and no other. We'll take that literally for a moment. Imagine a vast array of stations giving different information and angles on reality, but you only ever listen to one. Here we have the human plight in which the population is overwhelmingly confined to CultFM. This relates only to the frequency range of CultFM and limits perception and insight to that band – limits *possibility* to that band. It means you are connecting with an almost imperceptibly minuscule range of possibility and creative potential within the infinite Field. It's a world where everything seems apart from everything else and where synchronicity is rare. Synchronicity is defined in the dictionary as 'the happening by chance of two or more related or similar events at the same time'. Use of 'by chance' betrays a complete misunderstanding of reality. Synchronicity is not 'by chance'. As people open their minds, or 'awaken' to use the term, they notice more and more coincidences in their lives, bits of 'luck', apparently miraculous happenings that put them in the right place at the right time with the right people. Days become peppered with 'fancy meeting you here' and 'what are the chances of that?' My entire life has been lived like this and ever more so since my own colossal awakening in 1990 and 91 which transformed my sense of reality. Synchronicity is not 'by chance'; it is by accessing expanded

realms of possibility which allow expanded potential for manifestation. People broadcasting the same vibe from the same openness of mind tend to be drawn 'by chance' to each other through what I call frequency magnetism and it's not only people. In the last more than 30 years incredible synchronicity has also led me through the Cult maze to information in so many forms and to crucial personal experiences. These 'coincidences' have allowed me to put the puzzle pieces together across an enormous array of subjects and situations. Those who have breached the bubble of five-sense reality will know exactly what I mean and this escape from the perceptual prison cell is open to everyone whenever they make that choice. This may appear super-human when compared with the limitations of 'human', but it's really our natural state. 'Human' as currently experienced is consciousness in an unnatural state of induced separation from the infinity of the whole. I'll come to how this transformation into unity can be made when I have described in more detail the force that holds humanity in servitude by denying this access to infinite self.

The Wetiko factor

I have been talking and writing for decades about the way five-sense mind is systematically barricaded from expanded awareness. I have used the analogy of a computer (five-sense mind) and someone at the keyboard (expanded awareness). Interaction between the computer and the operator is symbolic of the interaction between five-sense mind and expanded awareness. The computer directly experiences the Internet and the operator experiences the Internet via the computer which is how it's supposed to be – the two working as one. Archons seek to control that point where the operator connects with the computer to stop that interaction ([Fig 20](#)). Now the operator is banging the keyboard and clicking the mouse, but the computer is not responding and this happens when the computer is taken over – *possessed* – by an appropriately-named computer 'virus'. The operator has lost all influence over the computer which goes its own way making decisions under the control of the 'virus'. I have

just described the dynamic through which the force known to Gnostics as Yaldabaoth and Archons disconnects five-sense mind from expanded awareness to imprison humanity in perceptual servitude.

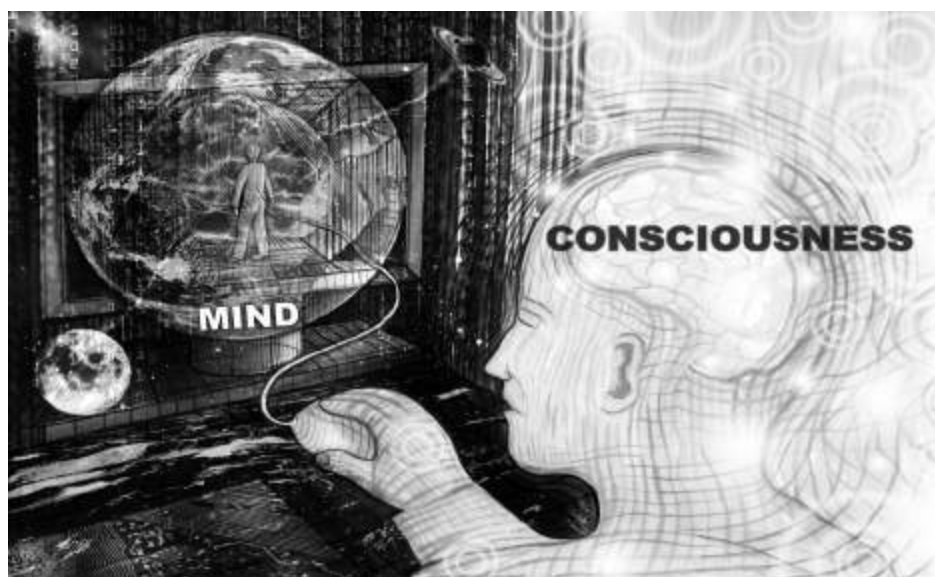


Figure 20: The mind ‘virus’ I have been writing about for decades seeks to isolate five-sense mind (the computer) from the true ‘I’. (Image by Neil Hague).

About a year ago I came across a Native American concept of Wetiko which describes precisely the same phenomenon. Wetiko is the spelling used by the Cree and there are other versions including wintiko and windigo used by other tribal groups. They spell the name with lower case, but I see Wetiko as a proper noun as with Archons and prefer a capital. I first saw an article about Wetiko by writer and researcher Paul Levy which so synced with what I had been writing about the computer/operator disconnection and later the Archons. I then read his book, the fascinating *Dispelling Wetiko, Breaking the Spell of Evil*. The parallels between what I had concluded long before and the Native American concept of Wetiko were so clear and obvious that it was almost funny. For Wetiko see the Gnostic Archons for sure and the Jinn, the Predators, and every other name for a force of evil, inversion and chaos. Wetiko is the Native American name for the force that divides the computer from

the operator (Fig 21). Indigenous author Jack D. Forbes, a founder of the Native American movement in the 1960s, wrote another book about Wetiko entitled *Columbus And Other Cannibals – The Wetiko Disease of Exploitation, Imperialism, and Terrorism* which I also read. Forbes says that Wetiko refers to an evil person or spirit ‘who terrorizes other creatures by means of terrible acts, including cannibalism’. Zulu shaman Credo Mutwa told me that African accounts tell how cannibalism was brought into the world by the Chitauri ‘gods’ – another manifestation of Wetiko. The distinction between ‘evil person or spirit’ relates to Archons/Wetiko possessing a human or acting as pure consciousness. Wetiko is said to be a sickness of the soul or spirit and a state of being that takes but gives nothing back – the Cult and its operatives perfectly described. Black Hawk, a Native American war leader defending their lands from confiscation, said European invaders had ‘poisoned hearts’ – Wetiko hearts – and that this would spread to native societies. Mention of the heart is very significant as we shall shortly see. Forbes writes: ‘Tragically, the history of the world for the past 2,000 years is, in great part, the story of the epidemiology of the wetiko disease.’ Yes, and much longer. Forbes is correct when he says: ‘The wetikos destroyed Egypt and Babylon and Athens and Rome and Tenochtitlan [capital of the Aztec empire] and perhaps now they will destroy the entire earth.’ Evil, he said, is the number one export of a Wetiko culture – see its globalisation with ‘Covid’. Constant war, mass murder, suffering of all kinds, child abuse, Satanism, torture and human sacrifice are all expressions of Wetiko and the Wetiko possessed. The world is Wetiko made manifest, *but it doesn’t have to be*. There is a way out of this even now.



Figure 21: The mind 'virus' is known to Native Americans as 'Wetiko'. (Image by Neil Hague).

Cult of Wetiko

Wetiko is the Yaldabaoth frequency distortion that seeks to attach to human consciousness and absorb it into its own. Once this connection is made Wetiko can drive the perceptions of the target which they believe to be coming from their own mind. All the horrors of history and today from mass killers to Satanists, paedophiles like Jeffrey Epstein and other psychopaths, are the embodiment of Wetiko and express its state of being in all its grotesqueness. The Cult is Wetiko incarnate, Yaldabaoth incarnate, and it seeks to facilitate Wetiko assimilation of humanity in totality into its distortion by manipulating the population into low frequency states that match its own. Paul Levy writes:

'Holographically enforced within the psyche of every human being the wetiko virus pervades and underlies the entire field of consciousness, and can therefore potentially manifest through any one of us at any moment if we are not mindful.' The 'Covid' hoax has achieved this with many people, but others have not fallen into Wetiko's frequency lair. Players in the 'Covid' human catastrophe including Gates, Schwab, Tedros, Fauci, Whitty, Vallance, Johnson, Hancock, Ferguson, Drosten, and all the rest, including the psychopath psychologists, are expressions of Wetiko. This is why

they have no compassion or empathy and no emotional consequence for what they do that would make them stop doing it. Observe all the people who support the psychopaths in authority against the Pushbackers despite the damaging impact the psychopaths have on their own lives and their family's lives. You are again looking at Wetiko possession which prevents them seeing through the lies to the obvious scam going on. *Why can't they see it?* Wetiko won't let them see it. The perceptual divide that has now become a chasm is between the Wetikoed and the non-Wetikoed.

Paul Levy describes Wetiko in the same way that I have long described the Archontic force. They are the same distorted consciousness operating across dimensions of reality: '... the subtle body of wetiko is not located in the third dimension of space and time, literally existing in another dimension ... it is able to affect ordinary lives by mysteriously interpenetrating into our three-dimensional world.' Wetiko does this through its incarnate representatives in the Cult and by weaving itself into The Field which on our level of reality is the electromagnetic information field of the simulation or Matrix. More than that, the simulation *is* Wetiko / Yaldabaoth. Caleb Scharf, Director of Astrobiology at Columbia University, has speculated that 'alien life' could be so advanced that it has transcribed itself into the quantum realm to become what we call physics. He said intelligence indistinguishable from the fabric of the Universe would solve many of its greatest mysteries:

Perhaps hyper-advanced life isn't just external. Perhaps it's already all around. It is embedded in what we perceive to be physics itself, from the root behaviour of particles and fields to the phenomena of complexity and emergence ... In other words, life might not just be in the equations. It might BE the equations [My emphasis].

Scharf said it is possible that 'we don't recognise advanced life because it forms an integral and unsuspecting part of what we've considered to be the natural world'. I agree. Wetiko/Yaldabaoth *is* the simulation. We are literally in the body of the beast. But that doesn't mean it has to control us. We all have the power to overcome Wetiko

influence and the Cult knows that. I doubt it sleeps too well because it knows that.

Which Field?

This, I suggest, is how it all works. There are two Fields. One is the fierce electromagnetic light of the Matrix within the speed of light; the other is the 'watery light' of The Field beyond the walls of the Matrix that connects with the Great Infinity. Five-sense mind and the decoding systems of the body attach us to the Field of Matrix light. They have to or we could not experience this reality. Five-sense mind sees only the Matrix Field of information while our expanded consciousness is part of the Infinity Field. When we open our minds, and most importantly our hearts, to the Infinity Field we have a mission control which gives us an expanded perspective, a road map, to understand the nature of the five-sense world. If we are isolated only in five-sense mind there is no mission control. We're on our own trying to understand a world that's constantly feeding us information to ensure we do not understand. People in this state can feel 'lost' and bewildered with no direction or radar. You can see ever more clearly those who are influenced by the Fields of Big Infinity or little five-sense mind simply by their views and behaviour with regard to the 'Covid' hoax. We have had this division throughout known human history with the mass of the people on one side and individuals who could see and intuit beyond the walls of the simulation – Plato's prisoner who broke out of the cave and saw reality for what it is. Such people have always been targeted by Wetiko/Archon-possessed authority, burned at the stake or demonised as mad, bad and dangerous. The Cult today and its global network of 'anti-hate', 'anti-fascist' Woke groups are all expressions of Wetiko attacking those exposing the conspiracy, 'Covid' lies and the 'vaccine' agenda.

Woke as a whole is Wetiko which explains its black and white mentality and how at one it is with the Wetiko-possessed Cult. Paul Levy said: 'To be in this paradigm is to still be under the thrall of a two-valued logic – where things are either true or false – of a

wetikoized mind.’ Wetiko consciousness is in a permanent rage, therefore so is Woke, and then there is Woke inversion and contradiction. ‘Anti-fascists’ act like fascists because fascists *and* ‘anti-fascists’ are both Wetiko at work. Political parties act the same while claiming to be different for the same reason. Secret society and satanic rituals are attaching initiates to Wetiko and the cold, ruthless, psychopathic mentality that secures the positions of power all over the world is Wetiko. Reframing ‘training programmes’ have the same cumulative effect of attaching Wetiko and we have their graduates described as automatons and robots with a cold, psychopathic, uncaring demeanour. They are all traits of Wetiko possession and look how many times they have been described in this book and elsewhere with regard to personnel behind ‘Covid’ including the police and medical profession. Climbing the greasy pole in any profession in a Wetiko society requires traits of Wetiko to get there and that is particularly true of politics which is not about fair competition and pre-eminence of ideas. It is founded on how many backs you can stab and arses you can lick. This culminated in the global ‘Covid’ coordination between the Wetiko possessed who pulled it off in all the different countries without a trace of empathy and compassion for their impact on humans. Our sight sense can see only holographic form and not the Field which connects holographic form. Therefore we perceive ‘physical’ objects with ‘space’ in between. In fact that ‘space’ is energy/consciousness operating on multiple frequencies. One of them is Wetiko and that connects the Cult psychopaths, those who submit to the psychopaths, and those who serve the psychopaths in the media operations of the world. Wetiko is Gates. Wetiko is the mask-wearing submissive. Wetiko is the fake journalist and ‘fact-checker’. The Wetiko Field is coordinating the whole thing. Psychopaths, gofers, media operatives, ‘anti-hate’ hate groups, ‘fact-checkers’ and submissive people work as one unit *even without human coordination* because they are attached to the *same* Field which is organising it all ([Fig 22](#)). Paul Levy is here describing how Wetiko-possessed people are drawn together and refuse to let any information breach their rigid

perceptions. He was writing long before 'Covid', but I think you will recognise followers of the 'Covid' religion *oh just a little bit*:

People who are channelling the vibratory frequency of wetiko align with each other through psychic resonance to reinforce their unspoken shared agreement so as to uphold their deranged view of reality. Once an unconscious content takes possession of certain individuals, it irresistibly draws them together by mutual attraction and knits them into groups tied together by their shared madness that can easily swell into an avalanche of insanity.

A psychic epidemic is a closed system, which is to say that it is insular and not open to any new information or informing influences from the outside world which contradict its fixed, limited, and limiting perspective.

There we have the Woke mind and the 'Covid' mind. Compatible resonance draws the awakening together, too, which is clearly happening today.

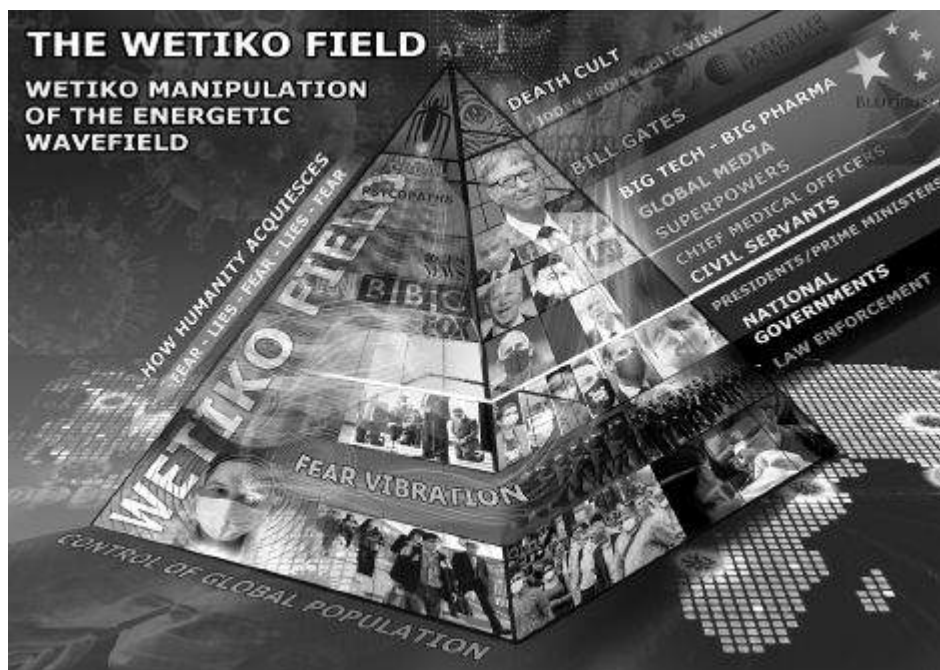


Figure 22: The Wetiko Field from which the Cult pyramid and its personnel are made manifest. (Image by Neil Hague).

Spiritual servitude

Wetiko doesn't care about humans. It's not human; it just possesses humans for its own ends and the effect (depending on the scale of

possession) can be anything from extreme psychopathy to unquestioning obedience. Wetiko's worst nightmare is for human consciousness to expand beyond the simulation. Everything is focussed on stopping that happening through control of information, thus perception, thus frequency. The 'education system', media, science, medicine, academia, are all geared to maintaining humanity in five-sense servitude as is the constant stimulation of low-vibrational mental and emotional states (see 'Covid'). Wetiko seeks to dominate those subconscious spaces between five-sense perception and expanded consciousness where the computer meets the operator. From these subconscious hiding places Wetiko speaks to us to trigger urges and desires that we take to be our own and manipulate us into anything from low-vibrational to psychopathic states. Remember how Islam describes the Jinn as invisible tricksters that 'whisper' and confuse. Wetiko is the origin of the 'trickster god' theme that you find in cultures all over the world. Jinn, like the Archons, are Wetiko which is terrified of humans awakening and reconnecting with our true self for then its energy source has gone. With that the feedback loop breaks between Wetiko and human perception that provides the energetic momentum on which its very existence depends as a force of evil. Humans are both its target and its source of survival, but only if we are operating in low-vibrational states of fear, hate, depression and the background anxiety that most people suffer. We are Wetiko's target because we are its key to survival. It needs us, not the other way round. Paul Levy writes:

A vampire has no intrinsic, independent, substantial existence in its own right; it only exists in relation to us. The pathogenic, vampiric mind-parasite called wetiko is nothing in itself – not being able to exist from its own side – yet it has a 'virtual reality' such that it can potentially destroy our species ...

...The fact that a vampire is not reflected by a mirror can also mean that what we need to see is that there's nothing, no-thing to see, other than ourselves. The fact that wetiko is the expression of something inside of us means that the cure for wetiko is with us as well. The critical issue is finding this cure within us and then putting it into effect.

Evil begets evil because if evil does not constantly expand and find new sources of energetic sustenance its evil, its *distortion*, dies with the assimilation into balance and harmony. Love is the garlic to Wetiko's vampire. Evil, the absence of love, cannot exist in the presence of love. I think I see a way out of here. I have emphasised so many times over the decades that the Archons/Wetiko and their Cult are not all powerful. *They are not*. I don't care how it looks even now *they are not*. I have not called them little boys in short trousers for effect. I have said it because it is true. Wetiko's insatiable desire for power over others is not a sign of its omnipotence, but its insecurity. Paul Levy writes: 'Due to the primal fear which ultimately drives it and which it is driven to cultivate, wetiko's body politic has an intrinsic and insistent need for centralising power and control so as to create imagined safety for itself.' *Yeeeeeees!* Exactly! Why does Wetiko want humans in an ongoing state of fear? Wetiko itself *is* fear and it is petrified of love. As evil is an absence of love, so love is an absence of fear. Love conquers all and *especially* Wetiko which *is* fear. Wetiko brought fear into the world when it wasn't here before. *Fear* was the 'fall', the fall into low-frequency ignorance and illusion – fear is **False Emotion Appearing Real**. The simulation is driven and energised by fear because Wetiko/Yaldabaoth (fear) *are* the simulation. Fear is the absence of love and Wetiko is the absence of love.

Wetiko today

We can now view current events from this level of perspective. The 'Covid' hoax has generated momentous amounts of ongoing fear, anxiety, depression and despair which have empowered Wetiko. No wonder people like Gates have been the instigators when they are Wetiko incarnate and exhibit every trait of Wetiko in the extreme. See how cold and unemotional these people are like Gates and his cronies, how dead of eye they are. That's Wetiko. Sabbatians are Wetiko and everything they control including the World Health Organization, Big Pharma and the 'vaccine' makers, national 'health'

hierarchies, corporate media, Silicon Valley, the banking system, and the United Nations with its planned transformation into world government. All are controlled and possessed by the Wetiko distortion into distorting human society in its image. We are with this knowledge at the gateway to understanding the world. Divisions of race, culture, creed and sexuality are diversions to hide the real division between those possessed and influenced by Wetiko and those that are not. The 'Covid' hoax has brought both clearly into view. Human behaviour is not about race. Tyrants and dictatorships come in all colours and creeds. What unites the US president bombing the innocent and an African tribe committing genocide against another as in Rwanda? What unites them? *Wetiko*. All wars are Wetiko, all genocide is Wetiko, all hunger over centuries in a world of plenty is Wetiko. Children going to bed hungry, including in the West, is Wetiko. Cult-generated Woke racial divisions that focus on the body are designed to obscure the reality that divisions in behaviour are manifestations of mind, not body. Obsession with body identity and group judgement is a means to divert attention from the real source of behaviour – mind and perception. Conflict sown by the Woke both within themselves and with their target groups are Wetiko providing lunch for itself through still more agents of the division, chaos, and fear on which it feeds. The Cult is seeking to assimilate the entirety of humanity and all children and young people into the Wetiko frequency by manipulating them into states of fear and despair. Witness all the suicide and psychological unravelling since the spring of 2020. Wetiko psychopaths want to impose a state of unquestioning obedience to authority which is no more than a conduit for Wetiko to enforce its will and assimilate humanity into itself. It needs us to believe that resistance is futile when it fears resistance and even more so the game-changing non-cooperation with its impositions. It can use violent resistance for its benefit. Violent impositions and violent resistance are *both* Wetiko. The Power of Love with its Power of No will sweep Wetiko from our world. Wetiko and its Cult know that. They just don't want us to know.

AI Wetiko

This brings me to AI or artificial intelligence and something else Wetikos don't want us to know. What is AI *really*? I know about computer code algorithms and AI that learns from data input. These, however, are more diversions, the expeditionary force, for the real AI that they want to connect to the human brain as promoted by Silicon Valley Wetikos like Kurzweil. What is this AI? It is the frequency of *Wetiko*, the frequency of the Archons. The connection of AI to the human brain is the connection of the Wetiko frequency to create a Wetiko hive mind and complete the job of assimilation. The hive mind is planned to be controlled from Israel and China which are both 100 percent owned by Wetiko Sabbatians. The assimilation process has been going on minute by minute in the 'smart' era which fused with the 'Covid' era. We are told that social media is scrambling the minds of the young and changing their personality. This is true, but what is social media? Look more deeply at how it works, how it creates divisions and conflict, the hostility and cruelty, the targeting of people until they are destroyed. That's Wetiko. Social media is manipulated to tune people to the Wetiko frequency with all the emotional exploitation tricks employed by platforms like Facebook and its Wetiko front man, Zuckerberg. Facebook's Instagram announced a new platform for children to overcome a legal bar on them using the main site. This is more Wetiko exploitation and manipulation of kids. Amnesty International likened the plan to foxes offering to guard the henhouse and said it was incompatible with human rights. Since when did Wetiko or Zuckerberg (I repeat myself) care about that? Would Brin and Page at Google, Wojcicki at YouTube, Bezos at Amazon and whoever the hell runs Twitter act as they do if they were not channelling Wetiko? Would those who are developing technologies for no other reason than human control? How about those designing and selling technologies to kill people and Big Pharma drug and 'vaccine' producers who know they will end or devastate lives? Quite a thought for these people to consider is that if you are Wetiko in a human life you are Wetiko on the 'other side' unless your frequency

changes and that can only change by a change of perception which becomes a change of behaviour. Where Gates is going does not bear thinking about although perhaps that's exactly where he wants to go. Either way, that's where he's going. His frequency will make it so.

The frequency lair

I have been saying for a long time that a big part of the addiction to smartphones and devices is that a frequency is coming off them that entraps the mind. People spend ages on their phones and sometimes even a minute or so after they put them down they pick them up again and it all repeats. 'Covid' lockdowns will have increased this addiction a million times for obvious reasons. Addictions to alcohol overindulgence and drugs are another way that Wetiko entraps consciousness to attach to its own. Both are symptoms of low-vibrational psychological distress which alcoholism and drug addiction further compound. Do we think it's really a coincidence that access to them is made so easy while potions that can take people into realms beyond the simulation are banned and illegal? I have explored smartphone addiction in other books, the scale is mind-blowing, and that level of addiction does not come without help. Tech companies that make these phones are Wetiko and they will have no qualms about destroying the minds of children. We are seeing again with these companies the Wetiko perceptual combination of psychopathic enforcers and weak and meek unquestioning compliance by the rank and file.

The global Smart Grid is the Wetiko Grid and it is crucial to complete the Cult endgame. The simulation is radiation and we are being deluged with technological radiation on a devastating scale. Wetiko frauds like Elon Musk serve Cult interests while occasionally criticising them to maintain his street-cred. 5G and other forms of Wi-Fi are being directed at the earth from space on a volume and scale that goes on increasing by the day. Elon Musk's (officially) SpaceX Starlink project is in the process of putting tens of thousands of satellites in low orbit to cover every inch of the planet with 5G and other Wi-Fi to create Kurzweil's global 'cloud' to which the

human mind is planned to be attached very soon. SpaceX has approval to operate 12,000 satellites with more than 1,300 launched at the time of writing and applications filed for 30,000 more. Other operators in the Wi-Fi, 5G, low-orbit satellite market include OneWeb (UK), Telesat (Canada), and AST & Science (US). Musk tells us that AI could be the end of humanity and then launches a company called Neuralink to connect the human brain to computers. Musk's (in theory) Tesla company is building electric cars and the driverless vehicles of the smart control grid. As frauds and bullshitters go Elon Musk in my opinion is Major League.

5G and technological radiation in general are destructive to human health, genetics and psychology and increasing the strength of artificial radiation underpins the five-sense perceptual bubbles which are themselves expressions of radiation or electromagnetism. Freedom activist John Whitehead was so right with his 'databit by databit, we are building our own electronic concentration camps'. The Smart Grid and 5G is a means to control the human mind and infuse perceptual information into The Field to influence anyone in sync with its frequency. You can change perception and behaviour en masse if you can manipulate the population into those levels of frequency and this is happening all around us today. The arrogance of Musk and his fellow Cult operatives knows no bounds in the way that we see with Gates. Musk's satellites are so many in number already they are changing the night sky when viewed from Earth. The astronomy community has complained about this and they have seen nothing yet. Some consequences of Musk's Wetiko hubris include: Radiation; visible pollution of the night sky; interference with astronomy and meteorology; ground and water pollution from intensive use of increasingly many spaceports; accumulating space debris; continual deorbiting and burning up of aging satellites, polluting the atmosphere with toxic dust and smoke; and ever-increasing likelihood of collisions. A collective public open letter of complaint to Musk said:

We are writing to you ... because SpaceX is in process of surrounding the Earth with a network of thousands of satellites whose very purpose is to irradiate every square inch of the

Earth. SpaceX, like everyone else, is treating the radiation as if it were not there. As if the mitochondria in our cells do not depend on electrons moving undisturbed from the food we digest to the oxygen we breathe.

As if our nervous systems and our hearts are not subject to radio frequency interference like any piece of electronic equipment. As if the cancer, diabetes, and heart disease that now afflict a majority of the Earth's population are not metabolic diseases that result from interference with our cellular machinery. As if insects everywhere, and the birds and animals that eat them, are not starving to death as a result.

People like Musk and Gates believe in their limitless Wetiko arrogance that they can do whatever they like to the world because they own it. Consequences for humanity are irrelevant. It's absolutely time that we stopped taking this shit from these self-styled masters of the Earth when you consider where this is going.

Why is the Cult so anti-human?

I hear this question often: Why would they do this when it will affect them, too? Ah, but will it? Who is this *them*? Forget their bodies. They are just vehicles for Wetiko consciousness. When you break it all down to the foundations we are looking at a state of severely distorted consciousness targeting another state of consciousness for assimilation. The rest is detail. The simulation is the fly-trap in which unique sensations of the five senses create a cycle of addiction called reincarnation. Renegade Minds see that everything which happens in our reality is a smaller version of the whole picture in line with the holographic principle. Addiction to the radiation of smart technology is a smaller version of addiction to the whole simulation. Connecting the body/brain to AI is taking that addiction on a giant step further to total ongoing control by assimilating human incarnate consciousness into Wetiko. I have watched during the 'Covid' hoax how many are becoming ever more profoundly attached to Wetiko's perceptual calling cards of aggressive response to any other point of view ('There is no other god but me'), psychopathic lack of compassion and empathy, and servile submission to the narrative and will of authority. Wetiko is the psychopaths *and* subservience to psychopaths. The Cult of Wetiko is

so anti-human because it is *not* human. It embarked on a mission to destroy human by targeting everything that it means to be human and to survive as human. 'Covid' is not the end, just a means to an end. The Cult with its Wetiko consciousness is seeking to change Earth systems, including the atmosphere, to suit them, not humans. The gathering bombardment of 5G alone from ground and space is dramatically changing The Field with which the five senses interact. There is so much more to come if we sit on our hands and hope it will all go away. It is not meant to go away. It is meant to get ever more extreme and we need to face that while we still can – just.

Carbon dioxide is the gas of life. Without that human is over. Kaput, gone, history. No natural world, no human. The Cult has created a cock and bull story about carbon dioxide and climate change to justify its reduction to the point where Gates and the ignoramus Biden 'climate chief' John Kerry want to suck it out of the atmosphere. Kerry wants to do this because his master Gates does. Wetikos have made the gas of life a demon with the usual support from the Wokers of Extinction Rebellion and similar organisations and the bewildered puppet-child that is Greta Thunberg who was put on the world stage by Klaus Schwab and the World Economic Forum. The name Extinction Rebellion is both ironic and as always Wetiko inversion. The gas that we need to survive must be reduced to save us from extinction. The most basic need of human is oxygen and we now have billions walking around in face nappies depriving body and brain of this essential requirement of human existence. More than that 5G at 60 gigahertz interacts with the oxygen molecule to reduce the amount of oxygen the body can absorb into the bloodstream. The obvious knock-on consequences of that for respiratory and cognitive problems and life itself need no further explanation. Psychopaths like Musk are assembling a global system of satellites to deluge the human atmosphere with this insanity. The man should be in jail. Here we have two most basic of human needs, oxygen and carbon dioxide, being dismantled.

Two others, water and food, are getting similar treatment with the United Nations Agendas 21 and 2030 – the Great Reset – planning to

centrally control all water and food supplies. People will not even own rain water that falls on their land. Food is affected at the most basic level by reducing carbon dioxide. We have genetic modification or GMO infiltrating the food chain on a mass scale, pesticides and herbicides polluting the air and destroying the soil. Freshwater fish that provide livelihoods for 60 million people and feed hundreds of millions worldwide are being 'pushed to the brink' according the conservationists while climate change is the only focus. Now we have Gates and Schwab wanting to dispense with current food sources all together and replace them with a synthetic version which the Wetiko Cult would control in terms of production and who eats and who doesn't. We have been on the Totalitarian Tiptoe to this for more than 60 years as food has become ever more processed and full of chemical shite to the point today when it's not natural food at all. As Dr Tom Cowan says: 'If it has a label don't eat it.' Bill Gates is now the biggest owner of farmland in the United States and he does nothing without an ulterior motive involving the Cult. Klaus Schwab wrote: 'To feed the world in the next 50 years we will need to produce as much food as was produced in the last 10,000 years ... food security will only be achieved, however, if regulations on genetically modified foods are adapted to reflect the reality that gene editing offers a precise, efficient and safe method of improving crops.' Liar. People and the world are being targeted with aluminium through vaccines, chemtrails, food, drink cans, and endless other sources when aluminium has been linked to many health issues including dementia which is increasing year after year. Insects, bees and wildlife essential to the food chain are being deleted by pesticides, herbicides and radiation which 5G is dramatically increasing with 6G and 7G to come. The pollinating bee population is being devastated while wildlife including birds, dolphins and whales are having their natural radar blocked by the effects of ever-increasing radiation. In the summer windscreens used to be splattered with insects so numerous were they. It doesn't happen now. Where have they gone?

Synthetic everything

The Cult is introducing genetically-modified versions of trees, plants and insects including a Gates-funded project to unleash hundreds of millions of genetically-modified, lab-altered and patented male mosquitoes to mate with wild mosquitoes and induce genetic flaws that cause them to die out. Clinically-insane Gates-funded Japanese researchers have developed mosquitos that spread vaccine and are dubbed 'flying vaccinators'. Gates is funding the modification of weather patterns in part to sell the myth that this is caused by carbon dioxide and he's funding geoengineering of the skies to change the atmosphere. Some of this came to light with the Gates-backed plan to release tonnes of chalk into the atmosphere to 'deflect the Sun and cool the planet'. Funny how they do this while the heating effect of the Sun is not factored into climate projections focussed on carbon dioxide. The reason is that they want to reduce carbon dioxide (so don't mention the Sun), but at the same time they do want to reduce the impact of the Sun which is so essential to human life and health. I have mentioned the sun-cholesterol-vitamin D connection as they demonise the Sun with warnings about skin cancer (caused by the chemicals in sun cream they tell you to splash on). They come from the other end of the process with statin drugs to reduce cholesterol that turns sunlight into vitamin D. A lack of vitamin D leads to a long list of health effects and how vitamin D levels must have fallen with people confined to their homes over 'Covid'. Gates is funding other forms of geoengineering and most importantly chemtrails which are dropping heavy metals, aluminium and self-replicating nanotechnology onto the Earth which is killing the natural world. See *Everything You Need To Know, But Have Never Been Told* for the detailed background to this.

Every human system is being targeted for deletion by a force that's not human. The Wetiko Cult has embarked on the process of transforming the human body from biological to synthetic biological as I have explained. Biological is being replaced by the artificial and synthetic – Archontic 'countermimicry' – right across human society. The plan eventually is to dispense with the human body altogether

and absorb human consciousness – which it wouldn't really be by then – into cyberspace (the simulation which is Wetiko/Yaldabaoth). Preparations for that are already happening if people would care to look. The alternative media rightly warns about globalism and 'the globalists', but this is far bigger than that and represents the end of the human race as we know it. The 'bad copy' of prime reality that Gnostics describe was a bad copy of harmony, wonder and beauty to start with before Wetiko/Yaldabaoth set out to change the simulated 'copy' into something very different. The process was slow to start with. Entrapped humans in the simulation timeline were not technologically aware and they had to be brought up to intellectual speed while being suppressed spiritually to the point where they could build their own prison while having no idea they were doing so. We have now reached that stage where technological intellect has the potential to destroy us and that's why events are moving so fast. Central American shaman Don Juan Matus said:

Think for a moment, and tell me how you would explain the contradictions between the intelligence of man the engineer and the stupidity of his systems of belief, or the stupidity of his contradictory behaviour. Sorcerers believe that the predators have given us our systems of beliefs, our ideas of good and evil; our social mores. They are the ones who set up our dreams of success or failure. They have given us covetousness, greed, and cowardice. It is the predator who makes us complacent, routinary, and egomaniacal.

In order to keep us obedient and meek and weak, the predators engaged themselves in a stupendous manoeuvre – stupendous, of course, from the point of view of a fighting strategist; a horrendous manoeuvre from the point of those who suffer it. They gave us their mind. The predators' mind is baroque, contradictory, morose, filled with the fear of being discovered any minute now.

For 'predators' see Wetiko, Archons, Yaldabaoth, Jinn, and all the other versions of the same phenomenon in cultures and religions all over the world. The theme is always the same because it's true and it's real. We have reached the point where we have to deal with it. The question is – how?

Don't fight – walk away

I thought I'd use a controversial subheading to get things moving in terms of our response to global fascism. What do you mean 'don't fight'? What do you mean 'walk away'? We've got to fight. We can't walk away. Well, it depends what we mean by fight and walk away. If fighting means physical combat we are playing Wetiko's game and falling for its trap. It wants us to get angry, aggressive, and direct hate and hostility at the enemy we think we must fight. Every war, every battle, every conflict, has been fought with Wetiko leading both sides. It's what it does. Wetiko wants a fight, anywhere, any place. Just hit me, son, so I can hit you back. Wetiko hits Wetiko and Wetiko hits Wetiko in return. I am very forthright as you can see in exposing Wetikos of the Cult, but I don't hate them. I refuse to hate them. It's what they want. What you hate you become. What you *fight* you become. Wokers, 'anti-haters' and 'anti-fascists' prove this every time they reach for their keyboards or don their balaclavas. By walk away I mean to disengage from Wetiko which includes ceasing to cooperate with its tyranny. Paul Levy says of Wetiko:

The way to 'defeat' evil is not to try to destroy it (for then, in playing evil's game, we have already lost), but rather, to find the invulnerable place within ourselves where evil is unable to vanquish us – this is to truly 'win' our battle with evil.

Wetiko is everywhere in human society and it's been on steroids since the 'Covid' hoax. Every shouting match over wearing masks has Wetiko wearing a mask and Wetiko not wearing one. It's an electrical circuit of push and resist, push and resist, with Wetiko pushing *and* resisting. Each polarity is Wetiko empowering itself. Dictionary definitions of 'resist' include 'opposing, refusing to accept or comply with' and the word to focus on is 'opposing'. What form does this take – setting police cars alight or 'refusing to accept or comply with'? The former is Wetiko opposing Wetiko while the other points the way forward. This is the difference between those aggressively demanding that government fascism must be obeyed who stand in stark contrast to the great majority of Pushbackers. We saw this clearly with a march by thousands of Pushbackers against lockdown in London followed days later by a Woker-hijacked

protest in Bristol in which police cars were set on fire. Masks were virtually absent in London and widespread in Bristol. Wetiko wants lockdown on every level of society and infuses its aggression to police it through its unknowing stooges. Lockdown protesters are the ones with the smiling faces and the hugs, The two blatantly obvious states of being – getting more obvious by the day – are the result of Wokers and their like becoming ever more influenced by the simulation Field of Wetiko and Pushbackers ever more influenced by The Field of a far higher vibration beyond the simulation. Wetiko can't invade the heart which is where most lockdown opponents are coming from. It's the heart that allows them to see through the lies to the truth in ways I will be highlighting.

Renegade Minds know that calmness is the place from which wisdom comes. You won't find wisdom in a hissing fit and wisdom is what we need in abundance right now. Calmness is not weakness – you don't have to scream at the top of your voice to be strong. Calmness is indeed a sign of strength. 'No' means I'm not doing it. NOOOO!!! doesn't mean you're not doing it even more. Volume does not advance 'No – I'm not doing it'. You are just not doing it. Wetiko possessed and influenced don't know how to deal with that. Wetiko wants a fight and we should not give it one. What it needs more than anything is our *cooperation* and we should not give that either. Mass rallies and marches are great in that they are a visual representation of feeling, but if it ends there they are irrelevant. You demand that Wetikos act differently? Well, they're not going to are they? They are Wetikos. We don't need to waste our time demanding that something doesn't happen when that will make no difference. We need to delete the means that *allows* it to happen. This, invariably, is our cooperation. You can demand a child stop firing a peashooter at the dog or you can refuse to buy the peashooter. If you provide the means you are cooperating with the dog being smacked on the nose with a pea. How can the authorities enforce mask-wearing if millions in a country refuse? What if the 74 million Pushbackers that voted for Trump in 2020 refused to wear masks, close their businesses or stay in their homes. It would be unenforceable. The

few control the many through the compliance of the many and that's always been the dynamic be it 'Covid' regulations or the Roman Empire. I know people can find it intimidating to say no to authority or stand out in a crowd for being the only one with a face on display; but it has to be done or it's over. I hope I've made clear in this book that where this is going will be far more intimidating than standing up now and saying 'No' – I will not cooperate with my own enslavement and that of my children. There might be consequences for some initially, although not so if enough do the same. The question that must be addressed is what is going to happen if we don't? It is time to be strong and unyieldingly so. No means no. Not here and there, but *everywhere* and *always*. I have refused to wear a mask and obey all the other nonsense. I will not comply with tyranny. I repeat: Fascism is not imposed by fascists – there are never enough of them. Fascism is imposed by the population acquiescing to fascism. *I will not do it*. I will die first, or my body will. Living meekly under fascism is a form of death anyway, the death of the spirit that Martin Luther King described.

Making things happen

We must not despair. This is not over till it's over and it's far from that. The 'fat lady' must refuse to sing. The longer the 'Covid' hoax has dragged on and impacted on more lives we have seen an awakening of phenomenal numbers of people worldwide to the realisation that what they have believed all their lives is not how the world really is. Research published by the system-serving University of Bristol and King's College London in February, 2021, concluded: 'One in every 11 people in Britain say they trust David Icke's take on the coronavirus pandemic.' It will be more by now and we have gathering numbers to build on. We must urgently progress from seeing the scam to ceasing to cooperate with it. Prominent German lawyer Reiner Fuellmich, also licenced to practice law in America, is doing a magnificent job taking the legal route to bring the psychopaths to justice through a second Nuremberg tribunal for crimes against humanity. Fuellmich has an impressive record of

beating the elite in court and he formed the German Corona Investigative Committee to pursue civil charges against the main perpetrators with a view to triggering criminal charges. Most importantly he has grasped the foundation of the hoax – the PCR test not testing for the ‘virus’ – and Christian Drosten is therefore on his charge sheet along with Gates frontman Tedros at the World Health Organization. Major players must not be allowed to inflict their horrors on the human race without being brought to book. A life sentence must follow for Bill Gates and the rest of them. A group of researchers has also indicted the government of Norway for crimes against humanity with copies sent to the police and the International Criminal Court. The lawsuit cites participation in an internationally-planned false pandemic and violation of international law and human rights, the European Commission’s definition of human rights by coercive rules, Nuremberg and Hague rules on fundamental human rights, and the Norwegian constitution. We must take the initiative from hereon and not just complain, protest and react.

There are practical ways to support vital mass non-cooperation. Organising in numbers is one. Lockdown marches in London in the spring in 2021 were mass non-cooperation that the authorities could not stop. There were too many people. Hundreds of thousands walked the London streets in the centre of the road for mile after mile while the Face-Nappies could only look on. They were determined, but calm, and just *did it* with no histrionics and lots of smiles. The police were impotent. Others are organising group shopping without masks for mutual support and imagine if that was happening all over. Policing it would be impossible. If the store refuses to serve people in these circumstances they would be faced with a long line of trolleys full of goods standing on their own and everything would have to be returned to the shelves. How would they cope with that if it kept happening? I am talking here about moving on from complaining to being pro-active; from watching things happen to making things happen. I include in this our relationship with the police. The behaviour of many Face-Nappies

has been disgraceful and anyone who thinks they would never find concentration camp guards in the 'enlightened' modern era have had that myth busted big-time. The period and setting may change – Wetikos never do. I watched film footage from a London march in which a police thug viciously kicked a protestor on the floor who had done nothing. His fellow Face-Nappies stood in a ring protecting him. What he did was a criminal assault and with a crowd far outnumbering the police this can no longer be allowed to happen unchallenged. I get it when people chant 'shame on you' in these circumstances, but that is no longer enough. They *have* no shame those who do this. Crowds needs to start making a citizen's arrest of the police who commit criminal offences and brutally attack innocent people and defenceless women. A citizen's arrest can be made under section 24A of the UK Police and Criminal Evidence (PACE) Act of 1984 and you will find something similar in other countries. I prefer to call it a Common Law arrest rather than citizen's for reasons I will come to shortly. Anyone can arrest a person committing an indictable offence or if they have reasonable grounds to suspect they are committing an indictable offence. On both counts the attack by the police thug would have fallen into this category. A citizen's arrest can be made to stop someone:

- Causing physical injury to himself or any other person
- Suffering physical injury
- Causing loss of or damage to property
- Making off before a constable can assume responsibility for him

A citizen's arrest may also be made to prevent a breach of the peace under Common Law and if they believe a breach of the peace will happen or anything related to harm likely to be done or already done in their presence. This is the way to go I think – the Common Law version. If police know that the crowd and members of the public will no longer be standing and watching while they commit

their thuggery and crimes they will think twice about acting like Brownshirts and Blackshirts.

Common Law – common sense

Mention of Common Law is very important. Most people think the law is the law as in one law. This is not the case. There are two bodies of law, Common Law and Statute Law, and they are not the same. Common Law is founded on the simple premise of do no harm. It does not recognise victimless crimes in which no harm is done while Statute Law does. There is a Statute Law against almost everything. So what is Statute Law? Amazingly it's the law of the *sea* that was brought ashore by the Cult to override the law of the land which is Common Law. They had no right to do this and as always they did it anyway. They had to. They could not impose their will on the people through Common Law which only applies to do no harm. How could you stitch up the fine detail of people's lives with that? Instead they took the law of the sea, or Admiralty Law, and applied it to the population. Statute Law refers to all the laws spewing out of governments and their agencies including all the fascist laws and regulations relating to 'Covid'. The key point to make is that Statute Law is *contract law*. It only applies between *contracting* corporations. Most police officers don't even know this. They have to be kept in the dark, too. Long ago when merchants and their sailing ships began to trade with different countries a contractual law was developed called Admiralty Law and other names. Again it only applied to *contracts* agreed between *corporate* entities. If there is no agreed contract the law of the sea had no jurisdiction *and that still applies to its new alias of Statute Law*. The problem for the Cult when the law of the sea was brought ashore was an obvious one. People were not corporations and neither were government entities. To overcome the latter they made governments and all associated organisations corporations. All the institutions are *private corporations* and I mean governments and their agencies, local councils, police, courts, military, US states, the whole lot. Go to the

Dun and Bradstreet corporate listings website for confirmation that they are all corporations. You are arrested by a private corporation called the police by someone who is really a private security guard and they take you to court which is another private corporation. Neither have jurisdiction over you unless you consent and *contract* with them. This is why you hear the mantra about law enforcement policing by *consent* of the people. In truth the people 'consent' only in theory through monumental trickery.

Okay, the Cult overcame the corporate law problem by making governments and institutions corporate entities; but what about people? They are not corporations are they? Ah ... well in a sense, and *only* a sense, they are. Not people exactly – the illusion of people. The Cult creates a corporation in the name of everyone at the time that their birth certificate is issued. Note birth/ *berth* certificate and when you go to court under the law of the sea on land you stand in a *dock*. These are throwbacks to the origin. My Common Law name is David Vaughan Icke. The name of the corporation created by the government when I was born is called Mr David Vaughan Icke usually written in capitals as MR DAVID VAUGHAN ICKE. That is not me, the living, breathing man. It is a fictitious corporate entity. The trick is to make you think that David Vaughan Icke and MR DAVID VAUGHAN ICKE are the same thing. *They are not*. When police charge you and take you to court they are prosecuting the corporate entity and not the living, breathing, man or woman. They have to trick you into identifying as the corporate entity and contracting with them. Otherwise they have no jurisdiction. They do this through a language known as legalese. Lawful and legal are not the same either. Lawful relates to Common Law and legal relates to Statute Law. Legalese is the language of Statue Law which uses terms that mean one thing to the public and another in legalese. Notice that when a police officer tells someone why they are being charged he or she will say at the end: 'Do you understand?' To the public that means 'Do you comprehend?' In legalese it means 'Do you stand under me?' Do you stand under my authority? If you say

yes to the question you are unknowingly agreeing to give them jurisdiction over you in a contract between two corporate entities.

This is a confidence trick in every way. Contracts have to be agreed between informed parties and if you don't know that David Vaughan Icke is agreeing to be the corporation MR DAVID VAUGHAN ICKE you cannot knowingly agree to contract. They are deceiving you and another way they do this is to ask for proof of identity. You usually show them a driving licence or other document on which your corporate name is written. In doing so you are accepting that you are that corporate entity when you are not. Referring to yourself as a 'person' or 'citizen' is also identifying with your corporate fiction which is why I made the Common Law point about the citizen's arrest. If you are approached by a police officer you identify yourself immediately as a living, breathing, man or woman and say 'I do not consent, I do not contract with you and I do not understand' or stand under their authority. I have a Common Law birth certificate as a living man and these are available at no charge from commonlawcourt.com. Businesses registered under the Statute Law system means that its laws apply. There are, however, ways to run a business under Common Law. Remember all 'Covid' laws and regulations are Statute Law – the law of *contracts* and you do not have to contract. This doesn't mean that you can kill someone and get away with it. Common Law says do no harm and that applies to physical harm, financial harm etc. Police are employees of private corporations and there needs to be a new system of non-corporate Common Law constables operating outside the Statute Law system. If you go to davidicke.com and put Common Law into the search engine you will find videos that explain Common Law in much greater detail. It is definitely a road we should walk.

With all my heart

I have heard people say that we are in a spiritual war. I don't like the term 'war' with its Wetiko dynamic, but I know what they mean. Sweep aside all the bodily forms and we are in a situation in which two states of consciousness are seeking very different realities.

Wetiko wants upheaval, chaos, fear, suffering, conflict and control. The other wants love, peace, harmony, fairness and freedom. That's where we are. We should not fall for the idea that Wetiko is all-powerful and there's nothing we can do. Wetiko is not all-powerful. It's a joke, pathetic. It doesn't have to be, but it has made that choice for now. A handful of times over the years when I have felt the presence of its frequency I have allowed it to attach briefly so I could consciously observe its nature. The experience is not pleasant, the energy is heavy and dark, but the ease with which you can kick it back out the door shows that its real power is in persuading us that it has power. It's all a con. Wetiko is a con. It's a trickster and not a power that can control us if we unleash our own. The con is founded on manipulating humanity to give its power to Wetiko which recycles it back to present the illusion that it has power when its power is *ours* that we gave away. This happens on an energetic level and plays out in the world of the seen as humanity giving its power to Wetiko authority which uses that power to control the population when the power is only the power the population has handed over. How could it be any other way for billions to be controlled by a relative few? I have had experiences with people possessed by Wetiko and again you can kick its arse if you do it with an open heart. Oh yes – the *heart* which can transform the world of perceived 'matter'.

We are receiver-transmitters and processors of information, but what information and where from? Information is processed into perception in three main areas – the brain, the heart and the belly. These relate to thinking, knowing, and emotion. Wetiko wants us to be head and belly people which means we think within the confines of the Matrix simulation and low-vibrational emotional reaction scrambles balance and perception. A few minutes on social media and you see how emotion is the dominant force. Woke is all emotion and is therefore thought-free and fact-free. Our heart is something different. It *knows* while the head *thinks* and has to try to work it out because it doesn't know. The human energy field has seven prime vortexes which connect us with wider reality ([Fig 23](#)). Chakra means

‘wheels of light’ in the Sanskrit language of ancient India. The main ones are: The crown chakra on top of the head; brow (or ‘third eye’) chakra in the centre of the forehead; throat chakra; heart chakra in the centre of the chest; solar plexus chakra below the sternum; sacral chakra beneath the navel; and base chakra at the bottom of the spine. Each one has a particular function or functions. We feel anxiety and nervousness in the belly where the sacral chakra is located and this processes emotion that can affect the colon to give people ‘the shits’ or make them ‘shit scared’ when they are nervous. Chakras all play an important role, but the Mr and Mrs Big is the heart chakra which sits at the centre of the seven, above the chakras that connect us to the ‘physical’ and below those that connect with higher realms (or at least should). Here in the heart chakra we feel love, empathy and compassion – ‘My heart goes out to you’. Those with closed hearts become literally ‘heart-less’ in their attitudes and behaviour (see Bill Gates). Native Americans portrayed Wetiko with what Paul Levy calls a ‘frigid, icy heart, devoid of mercy’ (see Bill Gates).



Figure 23: The chakra system which interpenetrates the human energy field. The heart chakra is the governor – or should be.

Wetiko trembles at the thought of heart energy which it cannot infiltrate. The frequency is too high. What it seeks to do instead is close the heart chakra vortex to block its perceptual and energetic influence. Psychopaths have ‘hearts of stone’ and emotionally-damaged people have ‘heartache’ and ‘broken hearts’. The astonishing amount of heart disease is related to heart chakra

disruption with its fundamental connection to the 'physical' heart. Dr Tom Cowan has written an outstanding book challenging the belief that the heart is a pump and making the connection between the 'physical' and spiritual heart. Rudolph Steiner who was way ahead of his time said the same about the fallacy that the heart is a pump. *What?* The heart is not a pump? That's crazy, right? Everybody knows that. Read Cowan's *Human Heart, Cosmic Heart* and you will realise that the very idea of the heart as a pump is ridiculous when you see the evidence. How does blood in the feet so far from the heart get pumped horizontally up the body by the heart?? Cowan explains in the book the real reason why blood moves as it does. Our 'physical' heart is used to symbolise love when the source is really the heart vortex or spiritual heart which is our most powerful energetic connection to 'out there' expanded consciousness. That's why we feel *knowing* – intuitive knowing – in the centre of the chest. Knowing doesn't come from a process of thoughts leading to a conclusion. It is there in an instant all in one go. Our heart knows because of its connection to levels of awareness that *do* know. This is the meaning and source of intuition – intuitive *knowing*.

For the last more than 30 years of uncovering the global game and the nature of reality my heart has been my constant antenna for truth and accuracy. An American intelligence insider once said that I had quoted a disinformant in one of my books and yet I had only quoted the part that was true. He asked: 'How do you do that?' By using my heart antenna was the answer and anyone can do it. Heart-centred is how we are meant to be. With a closed heart chakra we withdraw into a closed mind and the bubble of five-sense reality. If you take a moment to focus your attention on the centre of your chest, picture a spinning wheel of light and see it opening and expanding. You will feel it happening, too, and perceptions of the heart like joy and love as the heart impacts on the mind as they interact. The more the chakra opens the more you will feel expressions of heart consciousness and as the process continues, and becomes part of you, insights and knowings will follow. An open

heart is connected to that level of awareness that knows all is *One*. You will see from its perspective that the fault-lines that divide us are only illusions to control us. An open heart does not process the illusions of race, creed and sexuality except as brief experiences for a consciousness that is all. Our heart does not see division, only unity (Figs 24 and 25). There's something else, too. Our hearts love to laugh. Mark Twain's quote that says 'The human race has one really effective weapon, and that is laughter' is really a reference to the heart which loves to laugh with the joy of knowing the true nature of infinite reality and that all the madness of human society is an illusion of the mind. Twain also said: 'Against the assault of laughter nothing can stand.' This is so true of Wetiko and the Cult. Their insecurity demands that they be taken seriously and their power and authority acknowledged and feared. We should do nothing of the sort. We should not get aggressive or fearful which their insecurity so desires. We should laugh in their face. Even in their no-face as police come over in their face-nappies and expect to be taken seriously. They don't take themselves seriously looking like that so why should we? Laugh in the face of intimidation. Laugh in the face of tyranny. You will see by its reaction that you have pressed all of its buttons. Wetiko does not know what to do in the face of laughter or when its targets refuse to concede their joy to fear. We have seen many examples during the 'Covid' hoax when people have expressed their energetic power and the string puppets of Wetiko retreat with their tail limp between their knees. Laugh – the world is bloody mad after all and if it's a choice between laughter and tears I know which way I'm going.



Figure 24: Head consciousness without the heart sees division and everything apart from everything else.



Figure 25: Heart consciousness sees everything as One.

'Vaccines' and the soul

The foundation of Wetiko/Archon control of humans is the separation of incarnate five-sense mind from the infinite 'I' and closing the heart chakra where the True 'I' lives during a human life. The goal has been to achieve complete separation in both cases. I was interested therefore to read an account by a French energetic healer of what she said she experienced with a patient who had been given the 'Covid' vaccine. Genuine energy healers can sense information and consciousness fields at different levels of being which are referred to as 'subtle bodies'. She described treating the patient who later returned after having, without the healer's knowledge, two doses of the 'Covid vaccine'. The healer said:

I noticed immediately the change, very heavy energy emanating from [the] subtle bodies. The scariest thing was when I was working on the heart chakra, I connected with her soul: it was detached from the physical body, it had no contact and it was, as if it was floating in a state of total confusion: a damage to the consciousness that loses contact with the physical body, i.e. with our biological machine, there is no longer any communication between them.

I continued the treatment by sending light to the heart chakra, the soul of the person, but it seemed that the soul could no longer receive any light, frequency or energy. It was a very powerful experience for me. Then I understood that this substance is indeed used to detach consciousness so that this consciousness can no longer interact through this body that it possesses in life, where there is no longer any contact, no frequency, no light, no more energetic balance or mind.

This would create a human that is rudderless and at the extreme almost zombie-like operating with a fractional state of consciousness at the mercy of Wetiko. I was especially intrigued by what the healer said in the light of the prediction by the highly-informed Rudolf Steiner more than a hundred years ago. He said:

In the future, we will eliminate the soul with medicine. Under the pretext of a 'healthy point of view', there will be a vaccine by which the human body will be treated as soon as possible directly at birth, so that the human being cannot develop the thought of the existence of soul and Spirit. To materialistic doctors will be entrusted the task of removing the soul of humanity.

As today, people are vaccinated against this disease or that disease, so in the future, children will be vaccinated with a substance that can be produced precisely in such a way that people, thanks to this vaccination, will be immune to being subjected to the 'madness' of spiritual life. He would be extremely smart, but he would not develop a conscience, and that is the true goal of some materialistic circles.

Steiner said the vaccine would detach the physical body from the etheric body (subtle bodies) and 'once the etheric body is detached the relationship between the universe and the etheric body would become extremely unstable, and man would become an automaton'. He said 'the physical body of man must be polished on this Earth by spiritual will – so the vaccine becomes a kind of arymanique (Wetiko) force' and 'man can no longer get rid of a given materialistic feeling'. Humans would then, he said, become 'materialistic of constitution and can no longer rise to the spiritual'. I have been writing for years about DNA being a receiver-transmitter of information that connects us to other levels of reality and these 'vaccines' changing DNA can be likened to changing an antenna and what it can transmit and receive. Such a disconnection would clearly lead to changes in personality and perception. Steiner further predicted the arrival of AI. Big Pharma 'Covid vaccine' makers, expressions of Wetiko, are testing their DNA-manipulating evil on children as I write with a view to giving the 'vaccine' to babies. If it's a soul-body disconnecter – and I say that it is or can be – every child would be disconnected from 'soul' at birth and the 'vaccine' would create a closed system in which spiritual guidance from the greater self would play no part. This has been the ambition of Wetiko all

along. A Pentagon video from 2005 was leaked of a presentation explaining the development of vaccines to change behaviour by their effect on the brain. Those that believe this is not happening with the 'Covid' genetically-modifying procedure masquerading as a 'vaccine' should make an urgent appointment with Naivety Anonymous. Klaus Schwab wrote in 2018:

Neurotechnologies enable us to better influence consciousness and thought and to understand many activities of the brain. They include decoding what we are thinking in fine levels of detail through new chemicals and interventions that can influence our brains to correct for errors or enhance functionality.

The plan is clear and only the heart can stop it. With every heart that opens, every mind that awakens, Wetiko is weakened. Heart and love are far more powerful than head and hate and so nothing like a majority is needed to turn this around.

Beyond the Phantom

Our heart is the prime target of Wetiko and so it must be the answer to Wetiko. We *are* our heart which is part of one heart, the infinite heart. Our heart is where the true self lives in a human life behind firewalls of five-sense illusion when an imposter takes its place – *Phantom Self*; but our heart waits patiently to be set free any time we choose to see beyond the Phantom, beyond Wetiko. A Wetikoed Phantom Self can wreak mass death and destruction while the love of forever is locked away in its heart. The time is here to unleash its power and let it sweep away the fear and despair that is Wetiko. Heart consciousness does not seek manipulated, censored, advantage for its belief or religion, its activism and desires. As an expression of the One it treats all as One with the same rights to freedom and opinion. Our heart demands fairness for itself no more than for others. From this unity of heart we can come together in mutual support and transform this Wetikoed world into what reality is meant to be – a place of love, joy, happiness, fairness, justice and freedom. Wetiko has another agenda and that's why the world is as

it is, but enough of this nonsense. Wetiko can't stay where hearts are open and it works so hard to keep them closed. Fear is its currency and its food source and love in its true sense has no fear. Why would love have fear when it knows it is *All That Is, Has Been, And Ever Can Be* on an eternal exploration of all possibility? Love in this true sense is not the physical attraction that passes for love. This can be an expression of it, yes, but Infinite Love, a love without condition, goes far deeper to the core of all being. It *is* the core of all being. Infinite reality was born from love beyond the illusions of the simulation. Love infinitely expressed is the knowing that all is One and the swiftly-passing experience of separation is a temporary hallucination. You cannot disconnect from Oneness; you can only *perceive* that you have and withdraw from its influence. This is the most important of all perception trickery by the mind parasite that is Wetiko and the foundation of all its potential for manipulation.

If we open our hearts, open the sluice gates of the mind, and redefine self-identity amazing things start to happen. Consciousness expands or contracts in accordance with self-identity. When true self is recognised as infinite awareness and label self – Phantom Self – is seen as only a series of brief experiences life is transformed. Consciousness expands to the extent that self-identity expands and everything changes. You see unity, not division, the picture, not the pixels. From this we can play the long game. No more is an experience something in and of itself, but a fleeting moment in the eternity of forever. Suddenly people in uniform and dark suits are no longer intimidating. Doing what your heart knows to be right is no longer intimidating and consequences for those actions take on the same nature of a brief experience that passes in the blink of an infinite eye. Intimidation is all in the mind. Beyond the mind there is no intimidation.

An open heart does not consider consequences for what it knows to be right. To do so would be to consider not doing what it knows to be right and for a heart in its power that is never an option. The Renegade Mind is really the Renegade Heart. Consideration of consequences will always provide a getaway car for the mind and

the heart doesn't want one. What is right in the light of what we face today is to stop cooperating with Wetiko in all its forms and to do it without fear or compromise. You cannot compromise with tyranny when tyranny always demands more until it has everything. Life is your perception and you are your destiny. Change your perception and you change your life. Change collective perception and we change the world.

Come on people ... One human family, One heart, One goal ...
FREEEEEEEDOM!

We must settle for nothing less.

Postscript

The big scare story as the book goes to press is the 'Indian' variant and the world is being deluged with propaganda about the 'Covid catastrophe' in India which mirrors in its lies and misrepresentations what happened in Italy before the first lockdown in 2020.

The *New York Post* published a picture of someone who had 'collapsed in the street from Covid' in India in April, 2021, which was actually taken during a gas leak in May, 2020. Same old, same old. Media articles in mid-February were asking why India had been so untouched by 'Covid' and then as their vaccine rollout gathered pace the alleged 'cases' began to rapidly increase. Indian 'Covid vaccine' maker Bharat Biotech was funded into existence by the Bill and Melinda Gates Foundation (the pair announced their divorce in May, 2021, which is a pity because they so deserve each other). The Indian 'Covid crisis' was ramped up by the media to terrify the world and prepare people for submission to still more restrictions. The scam that worked the first time was being repeated only with far more people seeing through the deceit. Davidicke.com and Ickonic.com have sought to tell the true story of what is happening by talking to people living through the Indian nightmare which has nothing to do with 'Covid'. We posted a letter from 'Alisha' in Pune who told a very different story to government and media mendacity. She said scenes of dying people and overwhelmed hospitals were designed to hide what was really happening – genocide and starvation. Alisha said that millions had already died of starvation during the ongoing lockdowns while government and media were lying and making it look like the 'virus':

Restaurants, shops, gyms, theatres, basically everything is shut. The cities are ghost towns. Even so-called 'essential' businesses are only open till 11am in the morning. You basically have just an hour to buy food and then your time is up.

Inter-state travel and even inter-district travel is banned. The cops wait at all major crossroads to question why you are traveling outdoors or to fine you if you are not wearing a mask.

The medical community here is also complicit in genocide, lying about hospitals being full and turning away people with genuine illnesses, who need immediate care. They have even created a shortage of oxygen cylinders.

This is the classic Cult modus operandi played out in every country. Alisha said that people who would not have a PCR test not testing for the 'virus' were being denied hospital treatment. She said the people hit hardest were migrant workers and those in rural areas. Most businesses employed migrant workers and with everything closed there were no jobs, no income and no food. As a result millions were dying of starvation or malnutrition. All this was happening under Prime Minister Narendra Modi, a 100-percent asset of the Cult, and it emphasises yet again the scale of pure anti-human evil we are dealing with. Australia banned its people from returning home from India with penalties for trying to do so of up to five years in jail and a fine of £37,000. The manufactured 'Covid' crisis in India was being prepared to justify further fascism in the West. Obvious connections could be seen between the Indian 'vaccine' programme and increased 'cases' and this became a common theme. The Seychelles, the most per capita 'Covid vaccinated' population in the world, went back into lockdown after a 'surge of cases'.

Long ago the truly evil Monsanto agricultural biotechnology corporation with its big connections to Bill Gates devastated Indian farming with genetically-modified crops. Human rights activist Gurcharan Singh highlighted the efforts by the Indian government to complete the job by destroying the food supply to hundreds of millions with 'Covid' lockdowns. He said that 415 million people at the bottom of the disgusting caste system (still going whatever they say) were below the poverty line and struggled to feed themselves every year. Now the government was imposing lockdown at just the

time to destroy the harvest. This deliberate policy was leading to mass starvation. People may reel back at the suggestion that a government would do that, but Wetiko-controlled 'leaders' are capable of any level of evil. In fact what is described in India is in the process of being instigated worldwide. The food chain and food supply are being targeted at every level to cause world hunger and thus control. Bill Gates is not the biggest owner of farmland in America for no reason and destroying access to food aids both the depopulation agenda and the plan for synthetic 'food' already being funded into existence by Gates. Add to this the coming hyper-inflation from the suicidal creation of fake 'money' in response to 'Covid' and the breakdown of container shipping systems and you have a cocktail that can only lead one way and is meant to. The Cult plan is to crash the entire system to 'build back better' with the Great Reset.

'Vaccine' transmission

Reports from all over the world continue to emerge of women suffering menstrual and fertility problems after having the fake 'vaccine' and of the non-'vaccinated' having similar problems when interacting with the 'vaccinated'. There are far too many for 'coincidence' to be credible. We've had menopausal women getting periods, others having periods stop or not stopping for weeks, passing clots, sometimes the lining of the uterus, breast irregularities, and miscarriages (which increased by 400 percent in parts of the United States). Non-'vaccinated' men and children have suffered blood clots and nose bleeding after interaction with the 'vaccinated'. Babies have died from the effects of breast milk from a 'vaccinated' mother. Awake doctors – the small minority – speculated on the cause of non-'vaccinated' suffering the same effects as the 'vaccinated'. Was it nanotechnology in the synthetic substance transmitting frequencies or was it a straight chemical bioweapon that was being transmitted between people? I am not saying that some kind of chemical transmission is not one possible answer, but the foundation of all that the Cult does is frequency and

this is fertile ground for understanding how transmission can happen. American doctor Carrie Madej, an internal medicine physician and osteopath, has been practicing for the last 20 years, teaching medical students, and she says attending different meetings where the agenda for humanity was discussed. Madej, who operates out of Georgia, did not dismiss other possible forms of transmission, but she focused on frequency in search of an explanation for transmission. She said the Moderna and Pfizer 'vaccines' contained nano-lipid particles as a key component. This was a brand new technology never before used on humanity. 'They're using a nanotechnology which is pretty much little tiny computer bits ... nanobots or hydrogel.' Inside the 'vaccines' was 'this sci-fi kind of substance' which suppressed immune checkpoints to get into the cell. I referred to this earlier as the 'Trojan horse' technique that tricks the cell into opening a gateway for the self-replicating synthetic material and while the immune system is artificially suppressed the body has no defences. Madej said the substance served many purposes including an on-demand ability to 'deliver the payload' and using the nano 'computer bits' as biosensors in the body. 'It actually has the ability to accumulate data from your body, like your breathing, your respiration, thoughts, emotions, all kinds of things.'

She said the technology obviously has the ability to operate through Wi-Fi and transmit and receive energy, messages, frequencies or impulses. 'Just imagine you're getting this new substance in you and it can react to things all around you, the 5G, your smart device, your phones.' We had something completely foreign in the human body that had never been launched large scale at a time when we were seeing 5G going into schools and hospitals (plus the Musk satellites) and she believed the 'vaccine' transmission had something to do with this: '... if these people have this inside of them ... it can act like an antenna and actually transmit it outwardly as well.' The synthetic substance produced its own voltage and so it could have that kind of effect. This fits with my own contention that the nano receiver-transmitters are designed to connect people to the

Smart Grid and break the receiver-transmitter connection to expanded consciousness. That would explain the French energy healer's experience of the disconnection of body from 'soul' with those who have had the 'vaccine'. The nanobots, self-replicating inside the body, would also transmit the synthetic frequency which could be picked up through close interaction by those who have not been 'vaccinated'. Madej speculated that perhaps it was 5G and increased levels of other radiation that was causing the symptoms directly although interestingly she said that non-'vaccinated' patients had shown improvement when they were away from the 'vaccinated' person they had interacted with. It must be remembered that you can control frequency and energy with your mind and you can consciously create energetic barriers or bubbles with the mind to stop damaging frequencies from penetrating your field. American paediatrician Dr Larry Palevsky said the 'vaccine' was not a 'vaccine' and was never designed to protect from a 'viral' infection. He called it 'a massive, brilliant propaganda of genocide' because they didn't have to inject everyone to get the result they wanted. He said the content of the jabs was able to infuse any material into the brain, heart, lungs, kidneys, liver, sperm and female productive system. 'This is genocide; this is a weapon of mass destruction.' At the same time American colleges were banning students from attending if they didn't have this life-changing and potentially life-ending 'vaccine'. Class action lawsuits must follow when the consequences of this college fascism come to light. As the book was going to press came reports about fertility effects on sperm in 'vaccinated' men which would absolutely fit with what I have been saying and hospitals continued to fill with 'vaccine' reactions. Another question is what about transmission via blood transfusions? The NHS has extended blood donation restrictions from seven days after a 'Covid vaccination' to 28 days after even a sore arm reaction.

I said in the spring of 2020 that the then touted 'Covid vaccine' would be ongoing each year like the flu jab. A year later Pfizer CEO, the appalling Albert Bourla, said people would 'likely' need a 'booster dose' of the 'vaccine' within 12 months of getting 'fully

vaccinated' and then a yearly shot. 'Variants will play a key role', he said confirming the point. Johnson & Johnson CEO Alex Gorsky also took time out from his 'vaccine' disaster to say that people may need to be vaccinated against 'Covid-19' each year. UK Health Secretary, the psychopath Matt Hancock, said additional 'boosters' would be available in the autumn of 2021. This is the trap of the 'vaccine passport'. The public will have to accept every last 'vaccine' they introduce, including for the fake 'variants', or it would cease to be valid. The only other way in some cases would be continuous testing with a test not testing for the 'virus' and what is on the swabs constantly pushed up your nose towards the brain every time?

'Vaccines' changing behaviour

I mentioned in the body of the book how I believed we would see gathering behaviour changes in the 'vaccinated' and I am already hearing such comments from the non-'vaccinated' describing behaviour changes in friends, loved ones and work colleagues. This will only increase as the self-replicating synthetic material and nanoparticles expand in body and brain. An article in the *Guardian* in 2016 detailed research at the University of Virginia in Charlottesville which developed a new method for controlling brain circuits associated with complex animal behaviour. The method, dubbed 'magnetogenetics', involves genetically-engineering a protein called ferritin, which stores and releases iron, to create a magnetised substance – 'Magneto' – that can activate specific groups of nerve cells from a distance. This is claimed to be an advance on other methods of brain activity manipulation known as optogenetics and chemogenetics (the Cult has been developing methods of brain control for a long time). The ferritin technique is said to be non-invasive and able to activate neurons 'rapidly and reversibly'. In other words, human thought and perception. The article said that earlier studies revealed how nerve cell proteins 'activated by heat and mechanical pressure can be genetically engineered so that they become sensitive to radio waves and magnetic fields, by attaching them to an iron-storing protein called ferritin, or to inorganic

paramagnetic particles'. Sensitive to radio waves and magnetic fields? You mean like 5G, 6G and 7G? This is the human-AI Smart Grid hive mind we are talking about. The *Guardian* article said:

... the researchers injected Magneto into the striatum of freely behaving mice, a deep brain structure containing dopamine-producing neurons that are involved in reward and motivation, and then placed the animals into an apparatus split into magnetised and non-magnetised sections.

Mice expressing Magneto spent far more time in the magnetised areas than mice that did not, because activation of the protein caused the striatal neurons expressing it to release dopamine, so that the mice found being in those areas rewarding. This shows that Magneto can remotely control the firing of neurons deep within the brain, and also control complex behaviours.

Make no mistake this basic methodology will be part of the 'Covid vaccine' cocktail and using magnetics to change brain function through electromagnetic field frequency activation. The Pentagon is developing a 'Covid vaccine' using ferritin. Magnetism would explain changes in behaviour and why videos are appearing across the Internet as I write showing how magnets stick to the skin at the point of the 'vaccine' shot. Once people take these 'vaccines' anything becomes possible in terms of brain function and illness which will be blamed on 'Covid-19' and 'variants'. Magnetic field manipulation would further explain why the non-'vaccinated' are reporting the same symptoms as the 'vaccinated' they interact with and why those symptoms are reported to decrease when not in their company. Interestingly 'Magneto', a 'mutant', is a character in the Marvel Comic *X-Men* stories with the ability to manipulate magnetic fields and he believes that mutants should fight back against their human oppressors by any means necessary. The character was born Erik Lehnsherr to a Jewish family in Germany.

Cult-controlled courts

The European Court of Human Rights opened the door for mandatory 'Covid-19 vaccines' across the continent when it ruled in a Czech Republic dispute over childhood immunisation that legally

enforced vaccination could be 'necessary in a democratic society'. The 17 judges decided that compulsory vaccinations did not breach human rights law. On the face of it the judgement was so inverted you gasp for air. If not having a vaccine infused into your body is not a human right then what is? Ah, but they said human rights law which has been specifically written to delete all human rights at the behest of the state (the Cult). Article 8 of the European Convention on Human Rights relates to the right to a private life. The crucial word here is '*except*':

There shall be no interference by a public authority with the exercise of this right EXCEPT such as is in accordance with the law and is necessary in a democratic society in the interests of national security, public safety or the economic wellbeing of the country, for the prevention of disorder or crime, for the protection of health or morals, or for the protection of the rights and freedoms of others [My emphasis].

No interference *except* in accordance with the law means there *are* no 'human rights' *except* what EU governments decide you can have at their behest. 'As is necessary in a democratic society' explains that reference in the judgement and 'in the interests of national security, public safety or the economic well-being of the country, for the prevention of disorder or crime, for the protection of health or morals, or for the protection of the rights and freedoms of others' gives the EU a coach and horses to ride through 'human rights' and scatter them in all directions. The judiciary is not a check and balance on government extremism; it is a vehicle to enforce it. This judgement was almost laughably predictable when the last thing the Cult wanted was a decision that went against mandatory vaccination. Judges rule over and over again to benefit the system of which they are a part. Vaccination disputes that come before them are invariably delivered in favour of doctors and authorities representing the view of the state which owns the judiciary. Oh, yes, and we have even had calls to stop putting 'Covid-19' on death certificates within 28 days of a 'positive test' because it is claimed the practice makes the 'vaccine' appear not to work. They are laughing at you.

The scale of madness, inhumanity and things to come was highlighted when those not 'vaccinated' for 'Covid' were refused evacuation from the Caribbean island of St Vincent during massive volcanic eruptions. Cruise ships taking residents to the safety of another island allowed only the 'vaccinated' to board and the rest were left to their fate. Even in life and death situations like this we see 'Covid' stripping people of their most basic human instincts and the insanity is even more extreme when you think that fake 'vaccine'-makers are not even claiming their body-manipulating concoctions stop 'infection' and 'transmission' of a 'virus' that doesn't exist. St Vincent Prime Minister Ralph Gonsalves said: 'The chief medical officer will be identifying the persons already vaccinated so that we can get them on the ship.' Note again the power of the chief medical officer who, like Whitty in the UK, will be answering to the World Health Organization. This is the Cult network structure that has overridden politicians who 'follow the science' which means doing what WHO-controlled 'medical officers' and 'science advisers' tell them. Gonsalves even said that residents who were 'vaccinated' after the order so they could board the ships would still be refused entry due to possible side effects such as 'wooziness in the head'. The good news is that if they were woozy enough in the head they could qualify to be prime minister of St Vincent.

Microchipping freedom

The European judgement will be used at some point to justify moves to enforce the 'Covid' DNA-manipulating procedure. Sandra Ro, CEO of the Global Blockchain Business Council, told a World Economic Forum event that she hoped 'vaccine passports' would help to 'drive forced consent and standardisation' of global digital identity schemes: 'I'm hoping with the desire and global demand for some sort of vaccine passport – so that people can get travelling and working again – [it] will drive forced consent, standardisation, and frankly, cooperation across the world.' The lady is either not very bright, or thoroughly mendacious, to use the term 'forced consent'.

You do not 'consent' if you are forced – you *submit*. She was describing what the plan has been all along and that's to enforce a digital identity on every human without which they could not function. 'Vaccine passports' are opening the door and are far from the end goal. A digital identity would allow you to be tracked in everything you do in cyberspace and this is the same technique used by Cult-owned China to enforce its social credit system of total control. The ultimate 'passport' is planned to be a microchip as my books have warned for nearly 30 years. Those nice people at the Pentagon working for the Cult-controlled Defense Advanced Research Projects Agency (DARPA) claimed in April, 2021, they have developed a microchip inserted under the skin to detect 'asymptomatic Covid-19 infection' before it becomes an outbreak and a 'revolutionary filter' that can remove the 'virus' from the blood when attached to a dialysis machine. The only problems with this are that the 'virus' does not exist and people transmitting the 'virus' with no symptoms is brain-numbing bullshit. This is, of course, not a ruse to get people to be microchipped for very different reasons. DARPA also said it was producing a one-stop 'vaccine' for the 'virus' and all 'variants'. One of the most sinister organisations on Planet Earth is doing this? Better have it then. These people are insane because Wetiko that possesses them is insane.

Researchers from the Salk Institute in California announced they have created an embryo that is part human and part monkey. My books going back to the 1990s have exposed experiments in top secret underground facilities in the United States where humans are being crossed with animal and non-human 'extraterrestrial' species. They are now easing that long-developed capability into the public arena and there is much more to come given we are dealing with psychiatric basket cases. Talking of which – Elon Musk's scientists at Neuralink trained a monkey to play Pong and other puzzles on a computer screen using a joystick and when the monkey made the correct move a metal tube squirted banana smoothie into his mouth which is the basic technique for training humans into unquestioning compliance. Two Neuralink chips were in the monkey's skull and

more than 2,000 wires ‘fanned out’ into its brain. Eventually the monkey played a video game purely with its brain waves. Psychopathic narcissist Musk said the ‘breakthrough’ was a step towards putting Neuralink chips into human skulls and merging minds with artificial intelligence. *Exactly*. This man is so dark and Cult to his DNA.

World Economic Fascism (WEF)

The World Economic Forum is telling you the plan by the statements made at its many and various events. Cult-owned fascist YouTube CEO Susan Wojcicki spoke at the 2021 WEF Global Technology Governance Summit (see the name) in which 40 governments and 150 companies met to ensure ‘the responsible design and deployment of emerging technologies’. Orwellian translation: ‘Ensuring the design and deployment of long-planned technologies will advance the Cult agenda for control and censorship.’ Freedom-destroyer and Nuremberg-bound Wojcicki expressed support for tech platforms like hers to censor content that is ‘technically legal but could be harmful’. Who decides what is ‘harmful’? She does and they do. ‘Harmful’ will be whatever the Cult doesn’t want people to see and we have legislation proposed by the UK government that would censor content on the basis of ‘harm’ no matter if the information is fair, legal and provably true. Make that *especially* if it is fair, legal and provably true. Wojcicki called for a global coalition to be formed to enforce content moderation standards through automated censorship. This is a woman and mega-censor so self-deluded that she shamelessly accepted a ‘free expression’ award – *Wojcicki* – in an event sponsored by her own *YouTube*. They have no shame and no self-awareness.

You know that ‘Covid’ is a scam and Wojcicki a Cult operative when YouTube is censoring medical and scientific opinion purely on the grounds of whether it supports or opposes the Cult ‘Covid’ narrative. Florida governor Ron DeSantis compiled an expert panel with four professors of medicine from Harvard, Oxford, and Stanford Universities who spoke against forcing children and

vaccinated people to wear masks. They also said there was no proof that lockdowns reduced spread or death rates of 'Covid-19'. Cult-gofer Wojcicki and her YouTube deleted the panel video 'because it included content that contradicts the consensus of local and global health authorities regarding the efficacy of masks to prevent the spread of Covid-19'. This 'consensus' refers to what the Cult tells the World Health Organization to say and the WHO tells 'local health authorities' to do. Wojcicki knows this, of course. The panellists pointed out that censorship of scientific debate was responsible for deaths from many causes, but Wojcicki couldn't care less. She would not dare go against what she is told and as a disgrace to humanity she wouldn't want to anyway. The UK government is seeking to pass a fascist 'Online Safety Bill' to specifically target with massive fines and other means non-censored video and social media platforms to make them censor 'lawful but harmful' content like the Cult-owned Facebook, Twitter, Google and YouTube. What is 'lawful but harmful' would be decided by the fascist Blair-created Ofcom.

Another WEF obsession is a cyber-attack on the financial system and this is clearly what the Cult has planned to take down the bank accounts of everyone – except theirs. Those that think they have enough money for the Cult agenda not to matter to them have got a big lesson coming if they continue to ignore what is staring them in the face. The World Economic Forum, funded by Gates and fronted by Klaus Schwab, announced it would be running a 'simulation' with the Russian government and global banks of just such an attack called Cyber Polygon 2021. What they simulate – as with the 'Covid' Event 201 – they plan to instigate. The WEF is involved in a project with the Cult-owned Carnegie Endowment for International Peace called the WEF-Carnegie Cyber Policy Initiative which seeks to merge Wall Street banks, 'regulators' (I love it) and intelligence agencies to 'prevent' (arrange and allow) a cyber-attack that would bring down the global financial system as long planned by those that control the WEF and the Carnegie operation. The Carnegie Endowment for International Peace sent an instruction to First World

War US President Woodrow Wilson not to let the war end before society had been irreversibly transformed.

The Wuhan lab diversion

As I close, the Cult-controlled authorities and lapdog media are systematically pushing 'the virus was released from the Wuhan lab' narrative. There are two versions – it happened by accident and it happened on purpose. Both are nonsense. The perceived existence of the never-shown-to-exist 'virus' is vital to sell the impression that there is actually an infective agent to deal with and to allow the endless potential for terrifying the population with 'variants' of a 'virus' that does not exist. The authorities at the time of writing are going with the 'by accident' while the alternative media is promoting the 'on purpose'. Cable news host Tucker Carlson who has questioned aspects of lockdown and 'vaccine' compulsion has bought the Wuhan lab story. 'Everyone now agrees' he said. Well, I don't and many others don't and the question is *why* does the system and its media suddenly 'agree'? When the media moves as one unit with a narrative it is always a lie – witness the hour by hour mendacity of the 'Covid' era. Why would this Cult-owned combination which has unleashed lies like machine gun fire suddenly 'agree' to tell the truth??

Much of the alternative media is buying the lie because it fits the conspiracy narrative, but it's the *wrong* conspiracy. The real conspiracy is that *there is no virus* and that is what the Cult is desperate to hide. The idea that the 'virus' was released by accident is ludicrous when the whole 'Covid' hoax was clearly long-planned and waiting to be played out as it was so fast in accordance with the Rockefeller document and Event 201. So they prepared everything in detail over decades and then sat around strumming their fingers waiting for an 'accidental' release from a bio-lab? *What??* It's crazy. Then there's the 'on purpose' claim. You want to circulate a 'deadly virus' and hide the fact that you've done so and you release it down the street from the highest-level bio-lab in China? I repeat – *What??*

You would release it far from that lab to stop any association being made. But, no, we'll do it in a place where the connection was certain to be made. Why would you need to scam 'cases' and 'deaths' and pay hospitals to diagnose 'Covid-19' if you had a real 'virus'? What are sections of the alternative media doing believing this crap? Where were all the mass deaths in Wuhan from a 'deadly pathogen' when the recovery to normal life after the initial propaganda was dramatic in speed? Why isn't the 'deadly pathogen' now circulating all over China with bodies in the street? Once again we have the technique of tell them what they want to hear and they will likely believe it. The alternative media has its 'conspiracy' and with Carlson it fits with his 'China is the danger' narrative over years. China *is* a danger as a global Cult operations centre, but not for this reason. The Wuhan lab story also has the potential to instigate conflict with China when at some stage the plan is to trigger a Problem-Reaction-Solution confrontation with the West. Question everything – *everything* – and especially when the media agrees on a common party line.

Third wave ... fourth wave ... fifth wave ...

As the book went into production the world was being set up for more lockdowns and a 'third wave' supported by invented 'variants' that were increasing all the time and will continue to do so in public statements and computer programs, but not in reality. India became the new Italy in the 'Covid' propaganda campaign and we were told to be frightened of the new 'Indian strain'. Somehow I couldn't find it within myself to do so. A document produced for the UK government entitled 'Summary of further modelling of easing of restrictions – Roadmap Step 2' declared that a third wave was inevitable (of course when it's in the script) and it would be the fault of children and those who refuse the health-destroying fake 'Covid vaccine'. One of the computer models involved came from the Cult-owned *Imperial College* and the other from Warwick University which I wouldn't trust to tell me the date in a calendar factory. The document states that both models presumed extremely high uptake

of the 'Covid vaccines' and didn't allow for 'variants'. The document states: 'The resurgence is a result of some people (mostly children) being ineligible for vaccination; others choosing not to receive the vaccine; and others being vaccinated but not perfectly protected.' The mendacity takes the breath away. Okay, blame those with a brain who won't take the DNA-modifying shots and put more pressure on children to have it as 'trials' were underway involving children as young as six months with parents who give insanity a bad name. Massive pressure is being put on the young to have the fake 'vaccine' and child age consent limits have been systematically lowered around the world to stop parents intervening. Most extraordinary about the document was its claim that the 'third wave' would be driven by 'the resurgence in both hospitalisations and deaths ... dominated by *those that have received two doses of the vaccine*, comprising around 60-70% of the wave respectively'. The predicted peak of the 'third wave' suggested 300 deaths per day with 250 of them *fully 'vaccinated' people*. How many more lies do acquiescers need to be told before they see the obvious? Those who took the jab to 'protect themselves' are projected to be those who mostly get sick and die? So what's in the 'vaccine'? The document went on:

It is possible that a summer of low prevalence could be followed by substantial increases in incidence over the following autumn and winter. Low prevalence in late summer should not be taken as an indication that SARS-CoV-2 has retreated or that the population has high enough levels of immunity to prevent another wave.

They are telling you the script and while many British people believed 'Covid' restrictions would end in the summer of 2021 the government was preparing for them to be ongoing. Authorities were awarding contracts for 'Covid marshals' to police the restrictions with contracts starting in July, 2021, and going through to January 31st, 2022, and the government was advertising for 'Media Buying Services' to secure media propaganda slots worth a potential £320 million for 'Covid-19 campaigns' with a contract not ending until March, 2022. The recipient – via a list of other front companies – was reported to be American media marketing giant Omnicom Group

Inc. While money is no object for 'Covid' the UK waiting list for all other treatment – including life-threatening conditions – passed 4.5 million. Meantime the Cult is seeking to control all official 'inquiries' to block revelations about what has really been happening and why. It must not be allowed to – we need Nuremberg jury trials in every country. The cover-up doesn't get more obvious than appointing ultra-Zionist professor Philip Zelikow to oversee two dozen US virologists, public health officials, clinicians, former government officials and four American 'charitable foundations' to 'learn the lessons' of the 'Covid' debacle. The personnel will be those that created and perpetuated the 'Covid' lies while Zelikow is the former executive director of the 9/11 Commission who ensured that the truth about those attacks never came out and produced a report that must be among the most mendacious and manipulative documents ever written – see *The Trigger* for the detailed exposure of the almost unimaginable 9/11 story in which Sabbatians can be found at every level.

Passive no more

People are increasingly challenging the authorities with amazing numbers of people taking to the streets in London well beyond the ability of the Face-Nappies to stop them. Instead the Nappies choose situations away from the mass crowds to target, intimidate, and seek to promote the impression of 'violent protestors'. One such incident happened in London's Hyde Park. Hundreds of thousands walking through the streets in protest against 'Covid' fascism were ignored by the Cult-owned BBC and most of the rest of the mainstream media, but they delighted in reporting how police were injured in 'clashes with protestors'. The truth was that a group of people gathered in Hyde Park at the end of one march when most had gone home and they were peacefully having a good time with music and chat. Face-Nappies who couldn't deal with the full-march crowd then waded in with their batons and got more than they bargained for. Instead of just standing for this criminal brutality the crowd used their numerical superiority to push the Face-Nappies out of the

park. Eventually the Nappies turned and ran. Unfortunately two or three idiots in the crowd threw drink cans striking two officers which gave the media and the government the image they wanted to discredit the 99.9999 percent who were peaceful. The idiots walked straight into the trap and we must always be aware of potential agent provocateurs used by the authorities to discredit their targets.

This response from the crowd – the can people apart – must be a turning point when the public no longer stand by while the innocent are arrested and brutally attacked by the Face-Nappies. That doesn't mean to be violent, that's the last thing we need. We'll leave the violence to the Face-Nappies and government. But it does mean that when the Face-Nappies use violence against peaceful people the numerical superiority is employed to stop them and make citizen's arrests or Common Law arrests for a breach of the peace. The time for being passive in the face of fascism is over.

We are the many, they are the few, and we need to make that count before there is no freedom left and our children and grandchildren face an ongoing fascist nightmare.

COME ON PEOPLE – IT'S TIME.

One final thought ...

The power of love
A force from above
Cleaning my soul
Flame on burn desire
Love with tongues of fire
Purge the soul
Make love your goal

I'll protect you from the hooded claw
Keep the vampires from your door
When the chips are down I'll be around
With my undying, death-defying
Love for you

Envy will hurt itself
Let yourself be beautiful
Sparkling love, flowers
And pearls and pretty girls
Love is like an energy
Rushin' rushin' inside of me

This time we go sublime
Lovers entwine, divine, divine,
Love is danger, love is pleasure
Love is pure – the only treasure

I'm so in love with you
Purge the soul
Make love your goal

The power of love
A force from above
Cleaning my soul
The power of love
A force from above
A sky-scraping dove

Flame on burn desire
Love with tongues of fire
Purge the soul
Make love your goal

Frankie Goes To Hollywood

APPENDIX

Cowan-Kaufman-Morell Statement on Virus Isolation (SOVI)

Isolation: The action of isolating; the fact or condition of being isolated or standing alone; separation from other things or persons; solitariness

Oxford English Dictionary

The controversy over whether the SARS-CoV-2 virus has ever been isolated or purified continues. However, using the above definition, common sense, the laws of logic and the dictates of science, any unbiased person must come to the conclusion that the SARS-CoV-2 virus has never been isolated or purified. As a result, no confirmation of the virus' existence can be found. The logical, common sense, and scientific consequences of this fact are:

- the structure and composition of something not shown to exist can't be known, including the presence, structure, and function of any hypothetical spike or other proteins;
- the genetic sequence of something that has never been found can't be known;
- "variants" of something that hasn't been shown to exist can't be known;
- it's impossible to demonstrate that SARS-CoV-2 causes a disease called Covid-19.

In as concise terms as possible, here's the proper way to isolate, characterize and demonstrate a new virus. First, one takes samples (blood, sputum, secretions) from many people (e.g. 500) with symptoms which are unique and specific enough to characterize an illness. Without mixing these samples with ANY tissue or products that also contain genetic material, the virologist macerates, filters and ultracentrifuges i.e. *purifies* the specimen. This common virology technique, done for decades to isolate bacteriophages¹ and so-called giant viruses in every virology lab, then allows the virologist to demonstrate with electron microscopy thousands of identically sized and shaped particles. These particles are the isolated and purified virus.

These identical particles are then checked for uniformity by physical and/or microscopic techniques. Once the purity is determined, the particles may be further characterized. This would include examining the structure, morphology, and chemical composition of the particles. Next, their genetic makeup is characterized by extracting the genetic material directly from the purified particles and using genetic-sequencing techniques, such as Sanger sequencing, that have also been around for decades. Then one does an analysis to confirm that these uniform particles are exogenous (outside) in origin as a virus is conceptualized to be, and not the normal breakdown products of dead and dying tissues.² (As of May 2020, we know that virologists have no way to determine whether the particles they're seeing are viruses or just normal breakdown products of dead and dying tissues.)³

1 Isolation, characterization and analysis of bacteriophages from the haloalkaline lake Elmenteita, Kenya Julia Khayeli Akhwale et al, PLOS One, Published: April 25, 2019.
<https://journals.plos.org/plosone/article?id=10.1371/journal.pone.0215734> – accessed 2/15/21

2 "Extracellular Vesicles Derived From Apoptotic Cells: An Essential Link Between Death and Regeneration," Maojiao Li et al, Frontiers in Cell and Developmental Biology, 2020 October 2.
<https://www.frontiersin.org/articles/10.3389/fcell.2020.573511/full> – accessed 2/15/21

If we have come this far then we have fully isolated, characterized, and genetically sequenced an exogenous virus particle. However, we still have to show it is causally related to a disease. This is carried out by exposing a group of healthy subjects (animals are usually used) to this isolated, purified virus in the manner in which the disease is thought to be transmitted. If the animals get sick with the same disease, as confirmed by clinical and autopsy findings, one has now shown that the virus actually causes a disease. This demonstrates infectivity and transmission of an infectious agent.

None of these steps has even been attempted with the SARS-CoV-2 virus, nor have all these steps been successfully performed for any so-called pathogenic virus. Our research indicates that a single study showing these steps does not exist in the medical literature.

Instead, since 1954, virologists have taken unpurified samples from a relatively few people, often less than ten, with a similar disease. They then minimally process this sample and inoculate this unpurified sample onto tissue culture containing usually four to six other types of material – all of which contain identical genetic material as to what is called a “virus.” The tissue culture is starved and poisoned and naturally disintegrates into many types of particles, some of which contain genetic material. Against all common sense, logic, use of the English language and scientific integrity, this process is called “virus isolation.” This brew containing fragments of genetic material from many sources is then subjected to genetic analysis, which then creates in a computer-simulation process the alleged sequence of the alleged virus, a so called in silico genome. At no time is an actual virus confirmed by electron microscopy. At no time is a genome extracted and sequenced from an actual virus. This is scientific fraud.

The observation that the unpurified specimen — inoculated onto tissue culture along with toxic antibiotics, bovine fetal tissue, amniotic fluid and other tissues — destroys the kidney tissue onto which it is inoculated is given as evidence of the virus' existence and pathogenicity. This is scientific fraud.

From now on, when anyone gives you a paper that suggests the SARS-CoV-2 virus has been isolated, please check the methods sections. If the researchers used Vero cells or any other culture method, you know that their process was not isolation. You will hear the following excuses for why actual isolation isn't done:

1. There were not enough virus particles found in samples from patients to analyze.
2. Viruses are intracellular parasites; they can't be found outside the cell in this manner.

If No. 1 is correct, and we can't find the virus in the sputum of sick people, then on what evidence do we think the virus is dangerous or even lethal? If No. 2 is correct, then how is the virus spread from person to person? We are told it emerges from the cell to infect others. Then why isn't it possible to find it?

Finally, questioning these virology techniques and conclusions is not some distraction or divisive issue. Shining the light on this truth is essential to stop this terrible fraud that humanity is confronting. For, as we now know, if the virus has never been isolated, sequenced or shown to cause illness, if the virus is imaginary, then why are we wearing masks, social distancing and putting the whole world into prison?

Finally, if pathogenic viruses don't exist, then what is going into those injectable devices erroneously called "vaccines," and what is their purpose? This scientific question is the most urgent and relevant one of our time.

We are correct. The SARS-CoV2 virus does not exist.

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Ickonic is something that has been a dream of mine for the last 5 years, growing up around alternative information I have always had a natural interest in what is going on in the World and what could I do to make it better.

Across the range of subjects and positions of influence occupied mainly by people who don't strive to make things better it's the Media that I have always found the most frustrating and fascinating. Mainly because if the Media did their Jobs properly then so much of the negative things happening in the World simply would not be able to happen, because they would be exposed within a heartbeat.

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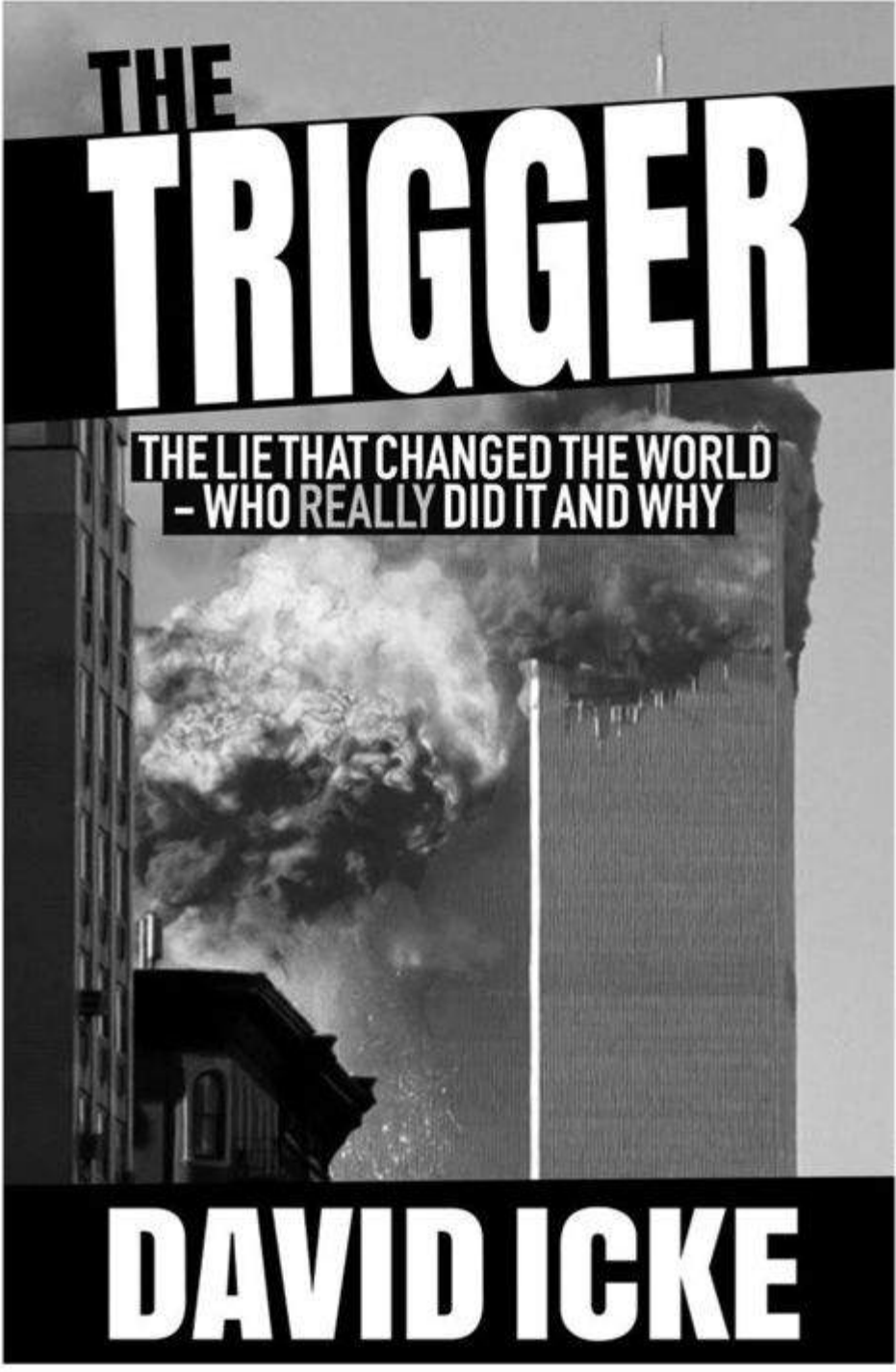
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noun

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